Layered Textbook

Mathematics · Computation · Machine Learning · Security

Why Machines Can Be Wrong?

From Linear Algebra to Machine-Learning Security

First Edition

Six parts, thirty-two chapters, four reading tracks

for readers aged 10, students, lecturers, and researchers

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A textbook about how we know that an answer is true — and how machines fail to know. Full sources are in the References.

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Preface

The first edition of this book grew out of one simple problem: the collection of textbooks I had gathered ran to more than ten thousand pages, but those pages were never read in sequence. Opening Rudin at chapter three without ever touching the definition of a supremum produces nothing but frustration. What was missing was not material, but order.

Before this manuscript there was a thirteen-chapter version. Read again, it showed two weaknesses. First, thirteen chapters went too fast, so the hardest parts went by in a glance. Second, that book could be read by only one kind of reader: a student already used to formal notation. A curious middle-school student had no door in, and a lecturer found no teaching notes inside it.

This version repairs both. Its scope has been widened to six parts and thirty-two chapters, with material the thirteen-chapter version did not have: sets and cardinality, infinite series, systems of linear equations and determinants, multiple integrals together with the vector field theorems, differential equations and dynamical systems, regression, and model generalization and regularization. Every chapter is now built in layers.

The first layer is the *For younger readers* box. It holds analogies to everyday objects, without a single mathematical symbol, and closes with one question that makes the reader curious. The second layer is the body of the chapter itself: definitions, theorems, proofs, worked examples, and exercises, as in an ordinary textbook. The third layer is the *Note for instructors* box: the teaching order I recommend, the misconceptions that almost always surface, and one classroom activity. The fourth layer is the *For researchers* box: what is really proved, what is only popularized, the primary sources, and the questions that are still open.

The four layers do not replace one another: a younger reader can read a chapter from the orange box and the summary alone, an undergraduate reads everything except the researcher box, an instructor uses the green box as a plan for the meeting, and a researcher starts from the gray box. One book, four paths.

Those layers set the *depth*. What sets the *way in* is the second set of four boxes: *Read it aloud* for readers who understand by hearing, *Do it by hand* for those who understand by moving and touching, *Step-by-step recipe* for those who understand by memorizing a procedure first, and *Challenge* for those who understand precisely when the problem is not yet clear. The two sets live together: one and the same chapter, with six doors.

Three principles of writing have not changed. *No claim without a testing tool*: every theorem comes with a proof, or with an honest admission that the proof is skipped together with where it is skipped; every number comes from a real execution that the reader can repeat. *Notation is introduced before it is used. Honesty in evaluation is part of mathematics*: the last chapters do not stop at how to build a model, but at how to know whether a claim about that model is proved.

One thing is deliberately absent from this book: complete answer keys. Giving a final number without a trace of the check would violate the first principle. In its place, Appendix C gives grading criteria and a verification tool for every kind of exercise, so that readers can check their own work in the same way the author checked this manuscript.

This book is not a substitute for the source texts it points to, but an ordered way in: if one chapter makes a reader open Rudin, Axler, Boyd, or a paper by Athalye and then read the section it names, this book has succeeded. Whatever errors remain belong to the author, not to those texts.

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How to Read This Book: Four Paths

Every chapter has the same shape, so the reader knows what to expect and may skip ahead without losing the way.

Path	What is read	What you get
Younger reader (10-14)	the *For younger readers* box, the Summary, one worked example	the core idea without notation; you know what is being computed
Undergraduate (S1)	the whole body of the chapter, the Computational lab, the Exercises; skip the researcher box	definitions, proofs, the ability to compute and to check
Instructor	the *Note for instructors* box, the source map, the Exercises	a plan for the meeting, misconceptions to anticipate, blackboard material
Researcher	the *For researchers* box, the source map, Appendices A and B	the limits of the claims, primary sources, open questions, consistent notation

The LEVEL mark at the start of every section shows how hard the reading is: *basic* can be read without any earlier chapter other than the ones named; *intermediate* demands the algebra and the notation of this book; *advanced* may be skipped on a first reading without breaking the flow of the chapter.

Six ways of learning, six doors in. People do not learn in the same way. So every chapter carries more than one route to the same idea, and no route is compulsory: pick the one that makes you understand fastest, then go back to the body of the chapter to see why that way works.

Type	**Look for this in every chapter**	**How to use it**
Visual	2-3 numbered figures (in the List of Figures), tables, and the box colors	read the figure before its text; the figure carries the same numbers as the computation in the text
Auditory	the *Read it aloud* box (indigo)	read the text aloud or record your voice; repeat the passages that have a rhythm
Kinesthetic	the *Do it by hand* box (green)	do it with real objects: paper, cards, the floor, money, or your own body
Concept-first	definitions, theorems, proofs, the *For researchers* box	do not use a formula before you know where it comes from; the proof is where that origin is written
Procedural memorizer	the *Step-by-step recipe* box (brown) and the Exercises	memorize the order of the recipe cards, then apply it to different numbers until it is automatic
Problem solver	the *Challenge* box (purple) and the starred exercise items	take the challenge first; the hints are tiered, do not open them all at once

The four paths below set *how much* is read; the six types above set *which door* you come in through. The two may be combined: a visual reader who pursues the researcher path still reads every proof, only starting from the figure.

Three learning paths that have already been tried, from the lightest:

1. **Curiosity path (8-12 weeks).** Read the orange box, the Summary, and one worked example from every chapter, in order from Chapter 1 to Chapter 32. No exercises. The aim is one thing: to know the map of the terrain.
2. **Student path (2 semesters).** First semester: Chapters 1-10 and Chapter 17. Second semester: Chapters 11-16, then 18-32. That is twenty-one chapters, more than an ordinary semester; the way to trim it without breaking the prerequisites is to postpone Chapters 31-32 (cryptography and privacy) to an elective course, so that nineteen chapters remain. Do the Computational lab of every chapter; at least four exercise items per chapter. This order keeps the promise about prerequisites: Chapter 18 demands Chapters 13 and 17, so the three must not be far apart; Chapter 21 demands Chapters 10 and 20; Chapter 23 demands Chapter 12.
3. **Machine-learning security researcher path (9 weeks).** Nineteen chapters, not seven: Chapters 3, 4, 9, 10, 12, 13, 18, 19, 20, 22, 23, 25, then 26-30, closing with Chapters 31 and 32. Chapters 13, 18, 22, and 23 are included because each is a prerequisite of Chapter 20 (hypothesis testing), Chapter 29 (adversarial), and Chapter 25 (neural networks); Chapters 31 and 32 complete the cryptography and privacy side that the title of the book demands. Reading this path in six weeks means skipping the prerequisites and losing the reason why the claims hold.

Symbols are used consistently throughout the book: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ for the number sets; $\mathbf{x}$ for a vector and x_i for its components; $\|\mathbf{x}\|$ for the Euclidean norm; ∇f for the gradient; ε and δ always small positive numbers; $\mathbb{E}[X]$ for the expected value; $\mathcal{L}(V, W)$ for the space of linear transformations; $\top$ for the transpose. Appendix A contains the complete list.

Chapter Dependency Map

This book may be read out of order, but not every chapter stands on its own. The table below names the minimum prerequisites of each chapter; every other prerequisite may be ignored on a first reading.

Chapter	Title	Prerequisites
1	The Language of Mathematics and the Art of Proof	—
2	Sets, Relations, Functions, and Cardinality	1
3	Axioms of the Real Numbers and Supremum	1, 2
4	Sequences and Limits	3
5	Infinite Series	4
6	Vector Spaces and Subspaces	2
7	Basis, Dimension, and Linear Transformations	6
8	Linear Systems, Determinants, and Inverses	6
9	Eigenvalues and Diagonalization	7, 8
10	Orthogonality, SVD, and Least Squares	7, 9
11	Continuity and Uniform Continuity	3, 4
12	Derivatives: One and Several Variables	11, 7
13	Riemann Integrals and the Fundamental Theorem of Calculus	11
14	Series of Functions and the Exchange of Limits	5, 11, 13
15	Multiple Integrals, Change of Variables, and Vector Field Theorems	10, 12, 13
16	Differential Equations and Dynamical Systems	12, 14, 15
17	Probability and Conditional Probability	2, 5
18	Random Variables and Distributions	17, 13
19	The Law of Large Numbers and the Central Limit Theorem	18, 5
20	Estimation, Confidence Intervals, and Hypothesis Testing	19
21	Regression and Applied Statistics	10, 20
22	Floating-Point Arithmetic and Error Analysis	3, 5
23	Convex Optimization and Duality	6, 12
24	Linear and Integer Programming	8, 23
25	Neural Networks and Backpropagation	10, 12, 23
26	Generalization, Regularization, and Training	18, 25
27	Honest Model Evaluation	20, 26
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Part I

Foundations: Language, Sets, and Numbers

This part answers a question that is usually skipped: what the words “holds for all” mean, and how a sentence can be proved rather than merely believed. The only prerequisite is school algebra. After five chapters you can write direct proofs, proofs by contradiction, and induction, and you know when an argument is still incomplete.

Chapter 1

The Language of Mathematics and the Art of Proof

In a classroom, a student writes on the board: "$n^2 + n + 41$ is prime for every integer $n \geq 0$". He then checks them one at a time. For $n = 0$ he obtains 41; for $n = 1$ he obtains 43; for $n = 2$ he obtains 47. He continues up to $n = 39$, obtains 1601, and 1601 is indeed prime. Forty values, forty failures that never came. At $n = 40$ the claim breaks: $40^2 + 40 + 41 = 1681 = 41^2$. One extra line cancels the whole preceding page.

That event is not an arithmetic accident but a grammatical mistake. "For every $n \geq 0$" speaks about infinitely many numbers, whereas a list touches only finitely many; no rule of logic moves truth from the finite to the infinite. This chapter therefore offers a way of speaking such that checking is not needed.

The need is not a luxury. In Chapter 29 we will conclude that a classifier can be encircled by shifting one input by ε, and that conclusion holds for all inputs. In Chapter 27 a good accuracy report on thousands of test data proves nothing about tomorrow's data. This chapter prepares the apparatus for both.

For younger readers (age 10 and up)

On the shore of a lake lived a flock of geese. A child came every morning for almost a year and counted the geese he saw: white, all of them, for two hundred and seventy mornings. So he wrote in his notebook: all geese on this lake are white.

His grandfather asked how many geese the child had never seen. The child admitted he did not know, and his grandfather said that the notebook told about the geese already seen, not about all the geese on the lake.

This story is not about geese but about the difference between "I have checked everything" and "it holds for everything". The first is only as strong as our memory. The second does not depend on who does the checking. One black goose suffices to bring down a notebook of any thickness, whereas a thousand white geese are never enough to set it up.

The black goose of this chapter is called forty. At that number the formula that looks tidy suddenly yields a number that can be split. In this chapter we will learn why one example brings down a claim, why many examples do not establish it, and how the gap is closed without inspecting the last goose.

1.1 Statements, connectives, and truth tables

Level: basic

A sentence can be judged only after we know what it promises. "Seven is a prime number"

promises something that can be checked, and that promise is true. "Close the window" promises nothing. "$x + 1 = 3$" cannot yet be judged, because its value depends on x. The three look the same to spelling, but only the first counts as mathematical material.

Definition 1.1 (Statement)**.** A *statement* is a sentence that is either true or false, but not both at once.

A single statement becomes useful once it is combined. Each connective has one meaning that is stipulated, not inferred from habits of speech.

logical connective (*penghubung logika*) — like a button on a calculator: two lamps in, one lamp out, and the relation between the lamps is fixed at the factory, not guessed by its user.

With the picture of that button in mind, we stipulate the meaning of the five connectives officially.

Definition 1.2 (Logical connectives)**.** Let P and Q be statements.

We use the following five connectives.

- $\neg P$ (*negation*, *not*): true exactly when P is false.
- $P \wedge Q$ (*conjunction*, *and*): true exactly when both are true.
- $P \vee Q$ (*disjunction*, *or*): true exactly when at least one is true; the word "or" here is inclusive.
- $P \Rightarrow Q$ (*implication*, *if–then*): false exactly when P is true and Q is false; true in the other three cases.
- $P \Leftrightarrow Q$ (*biconditional*, *iff*): true exactly when P and Q have the same value.

All the content of that definition can be compressed into a single table: the rows are all combinations of values, the columns are the value of each connective.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

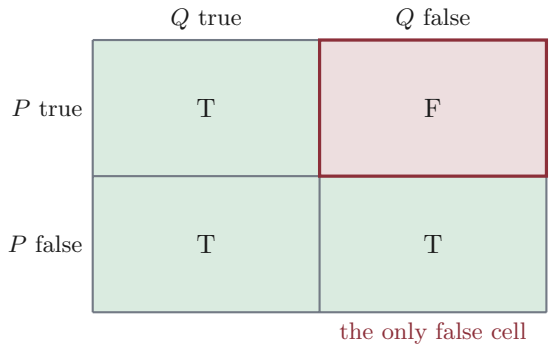


Figure 1.1: The truth grid of $P \Rightarrow Q$: rows hold the value of P, columns hold the value of Q. Three cells hold T and one cell holds F, namely when P is true while Q is false. Anyone who wants to bring the implication down need only point at the cell framed in thick lines.

The third and fourth rows are the ones that most often surprise. In the fourth row, P is false and Q is false, and the implication is judged true. In the third row, P is false and Q is true, and the implication is true as well. So the only way to bring down $P \Rightarrow Q$ is to exhibit P true while Q is false; the grid in Figure 1.1 contains that cell as the only false cell.

That stipulation is not a matter of style. Consider "if n is divisible by 4, then n is even", which must hold for every integer n, including $n = 3$. There the antecedent is false; if the

implication were judged false, the statement that is plainly true would collapse because of one irrelevant number. An implication records a relation of values, not cause and effect.

implication (*implikasi*) — like the promise "if it rains, the road is wet": the promise is broken only when rain falls and the road stays dry; when there is no rain, the one who made the promise has done nothing wrong.

Two compound statements can have the same value in every case; such an equality has a name of its own.

Definition 1.3 (Tautology and equivalence)**.** A compound statement is called a *tautology* if it is true under every assignment of values to its constituent statements. Two compound statements A and B are called *logically equivalent*, written $A \equiv B$, if $A \Leftrightarrow B$ is a tautology.

For two constituent statements there are $2^2 = 4$ assignments of values; for three constituents there are $2^3 = 8$. An equivalence that involves only two or three statements can therefore be proved by checking every row, and that check is complete. This is the only place in this chapter where exhaustive checking is accepted as proof, because the set of cases is finite and has been counted.

Proposition 1.1. *For every statement P and Q we have $P \Rightarrow Q \equiv \neg P \vee Q$.*

Proof. We compare the column $P \Rightarrow Q$ with the column $\neg P \vee Q$ in the four rows of the truth table. In the first row both are true. In the second row $\neg P$ is false and Q is false, so both are false. In the third and fourth rows $\neg P$ is true, so both are true. The four rows give the same value, hence the two statements are equivalent. ■

Two derived equivalences are used throughout this book. The first lets us reverse the direction of the talk without changing the value.

Corollary 1.2 (Contraposition)**.** *For every statement P and Q we have $P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$.*

Proof. By the preceding proposition, both sides equal $\neg P \vee Q$ and $\neg(\neg Q) \vee \neg P$. Since $\neg(\neg Q) \equiv Q$ and a disjunction pays no attention to order, the two forms are identical. ■

What is *not* equivalent is the converse, namely $Q \Rightarrow P$. In the third row of the table, $Q \Rightarrow P$ is true while $P \Rightarrow Q$ is false. Swapping antecedent for consequent is the most frequent mistake, and it is often invisible because the sentence still reads smoothly.

Proposition 1.3. *For every statement P and Q we have $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$.*

Proof. By the definition of negation and the first proposition, $\neg(P \Rightarrow Q) \equiv \neg(\neg P \vee Q)$. De Morgan's law, proved in Chapter 2, gives $\neg(\neg P \vee Q) \equiv \neg(\neg P) \wedge \neg Q$, and $\neg(\neg P) \equiv P$. Hence $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$. A reader who has not yet accepted De Morgan's law can check the four rows of the table directly. ■

Do it by hand (kinesthetic learner)

Prepare eight small cards: four bearing T, four bearing F. Lay out four rows, each row with one card for P and one for Q: T-T, T-F, F-T, F-F. To the right of each row place a third card that you write yourself: T if you judge "if P then Q" to be true, F if false. Work without looking at the table, then compare. Only one row gets the F card, namely the row whose P is T and whose Q is F. Slide that card up and down; it remains the only one. Now swap the P and Q cards in the T-F row so that it reads F-T, then evaluate again: the card changes from F to T. That one-card move is the converse.

1.2 Quantifiers and how to negate them

Level: basic

Most mathematical sentences speak about many objects, and that multiplicity is marked by quantifiers. There are two quantifiers, and both read order as part of the meaning.

quantifier (*kuantor*) — like two kinds of announcement at school: "all pupils must bring a book" and "some pupil brings a book"; the two sound similar but demand very different proofs.

We write down their meaning officially.

Definition 1.4 (Universal and existential quantifiers)**.** Let $P(x)$ be a statement that depends on x inside the set A.

- $(\forall x \in A)\, P(x)$ means $P(x)$ is true for every $x \in A$.
- $(\exists x \in A)\, P(x)$ means there is at least one $x \in A$ with $P(x)$ true.

The first statement is false if there is one $x \in A$ with $P(x)$ false; one such object is called a *counterexample*.

The existential quantifier demands at least one, not exactly one; adding a demand that is not written in the definition is the most frequent source of error.

The order of two quantifiers of different kinds may not be swapped. Inside $\mathbb{R}$, the statement

$$(\forall x)(\exists y)\ y > x$$

is true, whereas

$$(\exists y)(\forall x)\ y > x$$

is false. The proofs of both are short.

Proposition 1.4. *Inside $\mathbb{R}$ we have $(\forall x)(\exists y)\, y > x$, but we do not have $(\exists y)(\forall x)\, y > x$.*

Proof. For the first part, take any $x \in \mathbb{R}$ and choose $y = x + 1$. Then $y > x$, so y is a witness for that x. Since x was arbitrary, every x has a witness; the witness may differ for different x.

For the second part, suppose there is $y_0 \in \mathbb{R}$ with $y_0 > x$ for every $x \in \mathbb{R}$. Take $x = y_0 + 1$, which is also a real number. Then $y_0 > y_0 + 1$, that is $0 > 1$, which is impossible. So no such y_0 exists. ∎

Negation works from the outside in: every quantifier changes kind, and the negation travels inward until it sits on the innermost statement. For compound statements, De Morgan's laws move the negation past conjunctions and disjunctions.

Statement	Its negation
$(\forall x \in A)\, P(x)$	$(\exists x \in A)\, \neg P(x)$
$(\exists x \in A)\, P(x)$	$(\forall x \in A)\, \neg P(x)$
$P \wedge Q$	$\neg P \vee \neg Q$
$P \vee Q$	$\neg P \wedge \neg Q$
$P \Rightarrow Q$	$P \wedge \neg Q$

The last row is the one most often violated. Denying "if it rains then the road is wet" is not "if it rains then the road is dry", but "it rains and the road is not wet". The same mistake appears when people deny a definition: the verb inside it is negated, the quantifier is left intact.

Proposition 1.5. *For every statement $P(x)$ and $Q(x)$ we have*

$$\neg(\forall x)\big(P(x) \Rightarrow Q(x)\big) \ \equiv\ (\exists x)\big(P(x) \wedge \neg Q(x)\big).$$

Proof. A universal quantifier is negated into an existential quantifier over the negation of the body:

$$\neg(\forall x)R(x) \;\equiv\; (\exists x)\neg R(x).$$

The body $R(x)$ here is $P(x) \Rightarrow Q(x)$, whose negation is $P(x) \wedge \neg Q(x)$ by the proposition of the previous part. So the negation has the form $(\exists x)(P(x) \wedge \neg Q(x))$. ■

That form sets a clear task when one wants to reject a quantified statement: exhibit one object that satisfies the antecedent and fails the consequent.

The next two negations are used repeatedly; the first is the material of Chapter 4.

Example 1.1 (Negation of the definition of a limit). The definition of a limit reads: the sequence (a_n) tends to L if

$$(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)\ |a_n - L| < \varepsilon.$$

Negating from the outside in gives

$$(\exists \varepsilon > 0)(\forall N \in \mathbb{N})(\exists n \geq N)\ |a_n - L| \geq \varepsilon.$$

The reading of the second sentence: there is one tolerance ε that no N can repair, because there is always a term after N that deviates by at least ε. To prove that (a_n) does not tend to L, we must find that ε, then for every N find $n \geq N$ that violates it; n may depend on N.

The second negation concerns the existence of primes. Euclid proved that for every natural number n there is a prime larger than n:

$$(\forall n \in \mathbb{N})(\exists p)\big(p \text{ prime} \wedge p > n\big).$$

Its negation is

$$(\exists n \in \mathbb{N})(\forall p)\big(p \text{ prime} \Rightarrow p \leq n\big),$$

that is: there is one bound n such that all primes do not exceed n. Denying Euclid's claim demands such a bound, not merely failing to find a large prime.

For younger readers (age 10 and up)

Picture a class with thirty pupils. The teacher writes on the board: "all pupils bring a pencil." To prove that sentence true, the teacher must check thirty desks; to prove it false, one pupil without a pencil is enough.

Now the teacher writes another sentence: "some pupil brings a pencil." This time the situation is reversed. One pupil with a pencil is already enough to prove that sentence. But to prove it false, the teacher must check thirty desks and make sure there is not a single pencil.

The two sentences only swap a word at the front, but the work of checking them swaps entirely. Mathematicians are careful with "all" and "some" because both count how many things must be checked.

There is one more trap. If some pupil brings two pencils, is "some pupil brings a pencil" still true? Of course it is true: that sentence demands at least one, not exactly one. In this chapter we will learn why a demand that is not written may not be added on one's own.

1.3 Reading a definition literally

Level: intermediate

A mathematical definition is not a short explanation of a word; it is a contract that holds in both directions. If a number is called odd when it has the form $2k + 1$ for some integer k, we are bound to two things: every number of the form $2k + 1$ must be called odd, and every number called odd must have the form $2k + 1$.

To read a definition means to read a single sentence of the form "x is called B if and only if $C(x)$", even though the words "only if" are rarely written; the phrase "is called" already contains both directions. The first direction grants permission to use the name B, the second grants permission to use the condition $C(x)$, and reversing either one wrecks the whole argument.

necessary condition (*syarat perlu*) — like the condition for entering a building: without a membership card we may not enter, but having a membership card does not yet mean we are already seated inside.

Compare two statements about an integer n: "if 6 divides n then 3 divides n" is true, whereas "if 3 divides n then 6 divides n" is false, with counterexample $n = 3$. So "3 divides n" is a necessary condition for "6 divides n", but not a sufficient one. A good definition names both, since only in that way can the name and the condition be exchanged.

Read it aloud (auditory learner)

Read it aloud as a dialogue for two voices, alternating, with one breath at every full stop. First voice: "If it rains, the road is wet." Second voice: "The road is wet. So it rains." First voice: "Not necessarily. The road is wet when it is hosed down, too." Second voice: "Still wet. Still raining." First voice: "What I promised was one direction only. Rain first, then wet. The reverse direction was never promised."
Repeat that dialogue once more, faster, until the ear catches the difference. The first sentence puts the rain in front; the second moves the rain to the back. The sound turns around, the promise does not.
The line to repeat three times out loud: "if it rains then it is wet, not if it is wet then it rains."

Now consider the definition used most often in this book.

Definition 1.5 (Prime number)**.** An integer p is called *prime* if $p \geq 2$ and the only positive divisors of p are 1 and p itself.

In everyday words: a prime is larger than one and cannot be split into a product of two smaller numbers; the two clauses are tested separately.

Proposition 1.6. *The number* 1 *is not prime.*

Proof. The definition demands $p \geq 2$. For $p = 1$ that condition fails, so 1 does not satisfy the definition, whatever we can say about its divisors. Note that 1 in fact passes the second clause: the only positive divisors of 1 are indeed 1 and 1 itself. So what sets it aside is the first clause, not the second. ■

This kind of event explains why a definition must be read to the last clause. A reader who skips the first clause will conclude that 1 is prime, and the uniqueness of the factorization $12 = 2^2 \cdot 3$ in Chapter 28 will be lost, because 12 may also be written $1 \cdot 2^2 \cdot 3$.

A definition also determines how to deny membership. To show that a number is not prime, we must find a positive divisor that is neither 1 nor the number itself. Here is one example that often escapes a quick check.

Proposition 1.7. *The number* 91 *is not prime. More generally, the statement "every odd number that is not divisible by* 3 *and not divisible by* 5 *is prime" is false.*

Proof. In fact $91 = 7 \cdot 13$, and both factors lie between 1 and 91, so 91 has a positive divisor other than 1 and itself. The number 91 is odd, is not divisible by 3 because $9 + 1 = 10$ is not a multiple of 3, and is not divisible by 5 because its last digit is neither 0 nor 5. So 91 is a counterexample to that general statement. For comparison, 97 is indeed prime; among the integers 2 and 99 there are exactly 25 primes, a count produced by the program in the lab section. ∎

Definition 1.6 (Perfect square)**.** An integer n is called a *perfect square* if there is an integer k with $n = k^2$.

By this definition, 0 is a perfect square, because $0 = 0^2$, and so is 1, because $1 = 1^2$. Meanwhile 2 is not a perfect square, and that statement needs a proof, not an admission that we failed to find k.

Proposition 1.8. *The number* 2 *is not a perfect square.*

Proof. Suppose $2 = k^2$ for some $k \in \mathbb{Z}$. Since $k^2 = 2 > 0$, we have $k \neq 0$. If $|k| = 1$ then $k^2 = 1 \neq 2$. If $|k| \geq 2$ then $k^2 \geq 4 > 2$. Since $|k|$ is a positive integer, only those two possibilities remain, and both fail. Hence there is no such k. ∎

A habit worth keeping: every time you meet a definition, ask three questions. What are its clauses? Which objects satisfy all of them? Which objects are often thought to satisfy them although they fail one clause? The third question produces counterexamples, and a counterexample can be neither argued with nor debated.

1.4 Four patterns of proof

Level: intermediate

A proof is a transport of truth: from statements we have already accepted to the statement we want to accept, through steps that can each be checked. The four patterns below cover most needs, and Figure 1.2 maps the first three routes. The fifth common pattern, proof by cases, is merely the first pattern applied to cases that exclude one another.

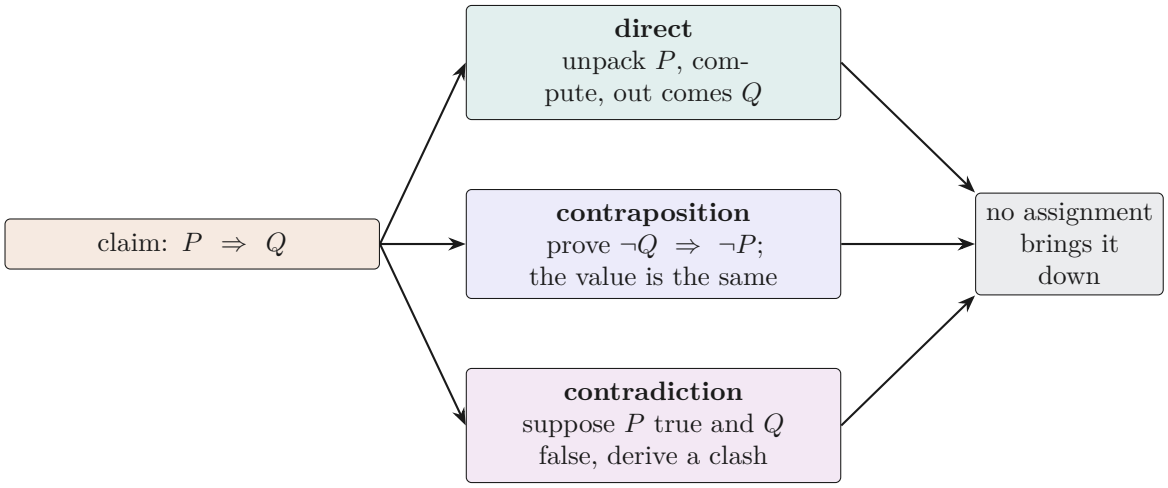


Figure 1.2: Road map of three forms of proof for an implication. All routes start from the same claim and end at the fact that no assignment of values makes P true while Q is false. A proof by contraposition is valid because $P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$. The fourth form, a constructive proof with a witness, does not enter this map because it proves a statement of the form $\exists$, not an implication.

Pattern one: direct proof. We start from the hypothesis, unpack it by definition, then compute until the conclusion appears. The next example is chosen because its conclusion is stronger than the one people usually prove.

Theorem 1.9. *For every odd integer n, the number $n^2 - 1$ is divisible by 8.*

Proof. Since n is odd, $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$n^2 - 1 = (2k + 1)^2 - 1 = 4k^2 + 4k = 4k(k + 1).$$

Among two consecutive integers k and $k + 1$, exactly one is even, so $k(k + 1) = 2m$ for some $m \in \mathbb{Z}$. Substitution gives $n^2 - 1 = 4 \cdot 2m = 8m$, which means $n^2 - 1$ is divisible by 8. ∎

An immediate consequence: the square of an odd number always leaves remainder 1 when divided by 8. For $n = 1, 3, 5, 7, 9, 11$ we obtain in turn $n^2 - 1 = 0, 8, 24, 48, 80, 120$, and all six are divisible by 8. That check is not a proof, but it makes sure that the proof above contains no typo.

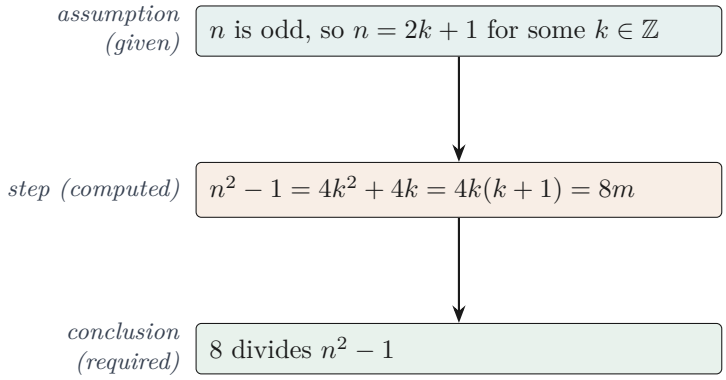


Figure 1.3: Anatomy of one direct proof of the statement "if n is odd then 8 divides $n^2 - 1$". The top box is the only one that may be used without reason; the middle box holds its reason, namely the definition of an odd number and algebra; the bottom box is what is required. The arrows may only run downward.

Figure 1.3 names the three parts of that proof: assumption, step, conclusion.

Pattern two: proof by contraposition. We prove $\neg Q \Rightarrow \neg P$ instead of $P \Rightarrow Q$. By the corollary in the first section the two are equivalent, so this is valid. The pattern is chosen when the negation of the conclusion is easier to hold than its hypothesis.

Proposition 1.10. *For every integer n, if $5n + 3$ is even then n is odd.*

Proof. The contrapositive reads: if n is even then $5n + 3$ is odd. Let $n = 2k$. Then $5n + 3 = 10k + 3 = 2(5k + 1) + 1$, which has the form two times an integer plus one, hence is odd. Since the contrapositive is proved, the original implication is proved. ∎

A direct proof would demand dividing $5n + 3 = 2m$ by 5; not impossible, but longer. The pattern of contraposition is chosen because it saves work.

Before the third pattern, we need one lemma about parity. This lemma will be used twice: once here, once in the proof that the square root of two is irrational.

Lemma 1.11. *For every integer n, if n^2 is even then n is even.*

Proof. We prove the contrapositive: if n is odd then n^2 is odd. By the first theorem of this section, for odd n we have $n^2 = 8m + 1$ for some m, so n^2 leaves remainder 1 when divided by 2, that is, is odd. ∎

Pattern three: proof by contradiction. We suppose the conclusion false, then derive two statements that cannot be true together. This pattern is used when the object being denied has no convenient form to compute with, for instance when what is denied is the existence of something.

Theorem 1.12. *There is no rational number r with $r^2 = 2$.*

Proof. Suppose there is $r \in \mathbb{Q}$ with $r^2 = 2$. Every rational number can be written as p/q with p, q integers, $q \neq 0$, and p, q having no common factor other than 1; we remove that common factor by dividing both, so a form without a common factor can always be reached.

From $p^2/q^2 = 2$ we obtain $p^2 = 2q^2$, so p^2 is even. By the lemma above, p is even; write $p = 2m$ with $m \in \mathbb{Z}$. Substitution gives $4m^2 = 2q^2$, that is $q^2 = 2m^2$, so q^2 is even and, once again by the lemma, q is even. Now p and q are both even, so both are divisible by 2, contradicting the choice of p/q without a common factor. The supposition is false, hence there is no rational r with $r^2 = 2$. ∎

The contradiction is not between $r^2 = 2$ and ordinary arithmetic, but between "p, q both even" and the opening promise "p, q without a common factor". That is why the step of removing common factors is not decoration; it is part of the hypothesis that is later violated.

Pattern four: constructive proof with a witness. To prove a statement of the form ∃, we point at one object and check that the object satisfies the condition. The object pointed at is called a *witness*.

Theorem 1.13. *For every natural number n there are n consecutive integers that are all not prime.*

Proof. Take $N = (n + 1)!$ and consider the n numbers

$$N + 2,\ N + 3,\ \ldots,\ N + (n + 1).$$

For every k with $2 \leq k \leq n + 1$, the number k divides $(n + 1)!$ because k is one of its factors, so k also divides $N + k$. Moreover $N + k > k > 1$, so $N + k$ has a positive divisor k that is neither 1 nor itself. By the definition, $N + k$ is not prime. We have pointed at the n numbers required, and there is no need to look for the smallest ones. ∎

For $n = 5$ that proof points at $6! = 720$, then $722, 723, 724, 725, 726$. All five indeed split: $722 = 2 \cdot 19^2$, $723 = 3 \cdot 241$, $724 = 2^2 \cdot 181$, $725 = 5^2 \cdot 29$, and $726 = 2 \cdot 3 \cdot 11^2$. The number $721 = 7 \cdot 103$ splits as well, but it is not on our list.

A constructive proof does not promise the smallest or the fastest witness; what is judged is only whether the object pointed at satisfies the condition. Denying a statement of the form ∃ demands showing that no object satisfies it, and that is where a contradiction is needed.

Step-by-step recipe (procedural learner)

Four standard steps to prove "if P then Q".

1. Write P and Q by their definitions, one line each.
2. Open with "suppose P is true". Only P may be supposed; Q may not yet be used.
3. Derive Q by substitution and algebra, one step per line, and name the definitions used.
4. Close with a sentence that states Q exactly as it appears in step 1.

A warning about order: step 2 may not be swapped with step 3. Scratch work from the conclusion backward is allowed, but the proof submitted must run from P to Q; otherwise what is proved is the converse.

1.5 Induction and strong induction

Level: intermediate

Induction is a proof pattern for statements that come in a row, one for each natural number. We set the first card upright and make sure that every card that falls knocks down the next card; after both, no card may remain standing.

induction (*induksi*) — like a row of dominoes: it is enough to make sure the first card falls and that every card knocks down the card in front of it, and then the whole row falls without needing to be pushed one by one.

We take the principle of induction as a basic property of $\mathbb{N} = \{1, 2, 3, \dots\}$, equivalent to the statement that every non-empty subset of $\mathbb{N}$ has a least member; nothing is proved circularly, we merely choose a starting point as in Chapter 3.

Note 1.1 (Principle of induction). Let $P(n)$ be a statement about natural numbers. If $P(1)$ is true and, for every $k \in \mathbb{N}$, the truth of $P(k)$ forces the truth of $P(k+1)$, then $P(n)$ is true for every $n \in \mathbb{N}$.

With that principle, we prove the formula that was already guessed in the opening of this section.

Theorem 1.14. *For every natural number n we have*

$$1 + 3 + 5 + \cdots + (2n-1) = n^2.$$

Proof. Base: for $n = 1$ the left side equals 1 and the right side equals $1^2 = 1$. Step: suppose the formula is true for $n = k$, that is $1 + 3 + \cdots + (2k-1) = k^2$. The next term is $2(k+1) - 1 = 2k + 1$, so

$$1 + 3 + \cdots + (2k-1) + (2k+1) = k^2 + 2k + 1 = (k+1)^2.$$

Hence the formula is true for $k + 1$. By the principle of induction, the formula is true for every $n \in \mathbb{N}$. ∎

The second example has the form of an inequality. From $2^k \geq k + 1$ and the condition $k \geq 0$ we obtain $2^{k+1} = 2 \cdot 2^k \geq 2(k+1) \geq k + 2$, so with base $n = 0$ we have $2^n \geq n + 1$ for every $n \geq 0$; the hypothesis and the side condition are used at once.

Ordinary induction demands one step back. Sometimes the conclusion for n follows only from many earlier values at once, and there ordinary induction is not enough.

Note 1.2 (Principle of strong induction). If for every $n \in \mathbb{N}$ the truth of $P(1), P(2), \dots, P(n-1)$ forces the truth of $P(n)$, then $P(n)$ is true for every $n \in \mathbb{N}$.

strong induction (*induksi kuat*) — like building a tower of blocks: to place the tenth block we may use every block beneath it, not only the ninth block.

With that permission we can prove something ordinary induction cannot reach.

Theorem 1.15. *Every integer $n \geq 2$ can be written as a product of prime numbers.*

Proof. We use strong induction on n. For $n = 2$: the number 2 is itself prime, so it is a product with one factor. Now take $n \geq 3$ and suppose the statement true for all integers m with $2 \leq m < n$. If n is prime, we are done. If n is not prime, by the definition there is a positive divisor a with $a \neq 1$ and $a \neq n$; write $n = ab$ with $b = n/a$. Since $a \neq 1$ and $a \neq n$, we have $2 \leq a \leq n - 1$, and likewise $2 \leq b \leq n - 1$. The induction hypothesis applies to a and to b, so both are products of primes, and the product of both is n. ∎

The strong induction hypothesis is used twice, on a and on b. That is where ordinary induction is awkward: both may be much smaller than $n - 1$, so the statement for $n - 1$ alone does not cover them.

That theorem closes the promise of the previous section: a list of primes is never complete.

Corollary 1.16 (Infinitely many primes)**.** *For every finite list of primes $p_1, p_2, \ldots, p_k$ there is a prime that is not on that list.*

Proof. Let $N = p_1 p_2 \cdots p_k + 1$. Since $N \geq 2$, the previous theorem gives that N is a product of primes; take one prime factor q of N. Suppose q is on the list, say $q = p_i$. Then q divides the product $p_1 \cdots p_k$, and since q also divides N, the number q divides the difference $N - p_1 \cdots p_k = 1$. No prime divides 1, so the supposition is false. Hence q is not on the list. ■

That proof does not prove that N is prime, only that N has a prime factor outside the list: for $2, 3, 5, 7$ we obtain $N = 211$, which happens to be prime, for $2, 3, 5, 7, 11, 13$ we obtain $N = 30031 = 59 \cdot 509$. The proof is constructive; "suppose there were only finitely many primes" is merely its rhetorical wrapping.

For younger readers (age 10 and up)

Picture a row of a thousand boxes, and we want to prove that every box contains a marble. We do not need to open a thousand boxes. It is enough to open the first box, then prove one thing: if a box contains a marble, the box beside it contains a marble too. If both things are true, a thousand boxes are checked without being opened one by one. This is induction, and it feels like magic. The key lies in the second promise: if the relation between boxes leaks in even one place, the whole row collapses.

Now picture the first box turning out empty. The relation between boxes remains true, but it does not help, because there is nothing first to fall. This is the mistake that happens most often.

In this chapter we will learn why the two steps may not be swapped, and why checking many boxes is no substitute for either of them.

1.6 When a proof fails

Level: advanced

Back to the blackboard from the opening of the chapter. The first forty values are prime, and at $n = 40$ the pattern breaks:

$$40^2 + 40 + 41 = 1600 + 40 + 41 = 1681 = 41^2.$$

That failure is no coincidence, and its shape can be explained without a computer. For every integer $p \geq 2$, the substitution $n = p - 1$ gives

$$(p-1)^2 + (p-1) + p = p^2 - 2p + 1 + p - 1 + p = p^2,$$

whereas the substitution $n = p$ gives $p^2 + p + p = p(p+2)$. So every number $p \geq 2$ produces two points of failure, namely $n = p - 1$ and $n = p$. With $p = 41$ both produce $1681 = 41^2$ and $1763 = 41 \cdot 43$.

The same pattern appears in smaller polynomials. For $n^2 + n + 11$ the first point of failure is $n = 10$, which gives $121 = 11^2$; for $n^2 + n + 17$ the first point of failure is $n = 16$, which gives $289 = 17^2$; for $n^2 + n + 29$ the failure comes at $n = 2$, because $35 = 5 \cdot 7$ is not prime. The larger the number in the constant position, the longer the clean initial stretch; that is the whole secret of the reputation of 41.

That failure yields three lessons that are often confused.

The first lesson is asymmetric. One counterexample brings down a statement of the form "for all", whereas forty successful examples prove nothing about the forty-first value. A computer check is therefore a tool for finding conjectures, not a tool for proving.

The second lesson concerns the location of the defect. The first defect that most often escapes notice appears in an induction proof. Consider the following false claim: for every $n \geq 1$ we have $1 + 3 + \cdots + (2n - 1) = n^2 + 1$. Its induction step, when inspected, turns out to be valid:

$$(n^2 + 1) + (2n + 1) = n^2 + 2n + 2 = (n + 1)^2 + 1.$$

So from the truth of the formula at n the truth at $n + 1$ does indeed flow. What fails is the base: for $n = 1$ the left side equals 1 while the right side equals 2. The induction step moves truth, it does not create it.

The third lesson concerns the limits of checking. The usual divisibility test: $m > 1$ is prime if it is not divisible by any prime smaller than $\sqrt{m}$. For $m = 1681$ we get $\sqrt{1681} = 41$, and the primes smaller than 41 are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37$; not one of them divides 1681. By that rule 1681 passes as prime. The error lies in the word "smaller": the correct rule uses "smaller than or equal to", because $1681 = 41 \cdot 41$ and the divisor is exactly the size of the root. One loose inequality sign produces one false prime, precisely at $n = 40$.

Three other defects recur in longer writing. The defect of circular reasoning: "since $n^2 + n + 41$ is prime for all n, it is also prime for $n = 40$" uses what must be proved as its reason. The defect of dividing by zero: from $a = b$ people conclude $a + b = b$ after dividing by $a - b$, although $a - b = 0$. The defect of swapping direction: from $P \Rightarrow Q$ and Q people conclude P; in the truth table, the third row is where that error hides.

All four defects share one root: a step is used outside its condition. A useful habit is not to memorize patterns but to check the condition of every step: in a contradiction the supposition must be the negation of the conclusion, in induction there is a base and a step, in a division the divisor is not zero.

Challenge (problem solver)

Claim: for every integer $k \geq 0$, the number $2^{2^k} + 1$ is prime. For $k = 0, 1, 2, 3, 4$ the values are 3, 5, 17, 257, 65537, and all five are indeed prime. Find the smallest k that brings the claim down, then exhibit a divisor; the claim is false, and the error can be shown without a computer.
Hint 1: five values are not a proof. Check $k = 5$; the number is 4294967297, and its smallest divisor is far smaller than its square root.
Hint 2: that divisor has the form $5 \cdot 2^7 + 1 = 641$, and $4294967297 = 641 \cdot 6700417$.
If you get stuck, go back to the section "When a proof fails".

1.7 Worked examples

Level: intermediate

The three computations below support this chapter and can be repeated on paper. The first is the sum of the first odd numbers, term by term from $n = 1$.

n	terms	sum	n^2
1	1	1	1
2	$1 + 3$	4	4
3	$1 + 3 + 5$	9	9
4	$1 + 3 + 5 + 7$	16	16
5	$1 + 3 + 5 + 7 + 9$	25	25
6	$1 + 3 + 5 + 7 + 9 + 11$	36	36

The continuation: $36 + 13 = 49 = 7^2$, then $49 + 15 = 64$, $64 + 17 = 81$, $81 + 19 = 100$. The difference of two consecutive squares is indeed $(n + 1)^2 - n^2 = 2n + 1$, and that is where induction takes its stand.

The second is a primality test on $n^2 + n + 41$, only at the interesting values.

n	$n^2 + n + 41$	remark
0	41	prime
1	43	prime
39	1601	prime
40	$1681 = 41^2$	not prime
41	$1763 = 41 \cdot 43$	not prime

An exhaustive check for $n = 0$ up to $n = 40$ gives 40 prime values out of 41; the only one that fails is $n = 40$, and $\sqrt{1681} = 41$ exactly, so the decisive divisor sits right at the boundary of the check.

The third is the contraposition check on the truth table, written out in full.

P	Q	$P \Rightarrow Q$	$\neg Q \Rightarrow \neg P$	equal
T	T	T	T	yes
T	F	F	F	yes
F	T	T	T	yes
F	F	T	T	yes

The four rows give equality: the contrapositive pattern is not a rhetorical trick but a replacement of form that does not change the value. The column $Q \Rightarrow P$ would differ in the third row, and that difference is what makes the converse unusable.

Compute $1 + 3 + \cdots + (2n - 1)$ for $n = 11$ and $n = 12$ by starting from $100 = 10^2$, that is by adding the next two odd numbers one at a time. Then compute $n^2 + n + 17$ for $n = 16$ and check by division whether 289 equals 17^2. Finally, find the first point of failure of $n^2 + n + 11$ by testing $n = 0, 1, 2, \ldots$ and explain why that point can be predicted from the shape of its polynomial.

1.8 Computational lab

Level: advanced

The following script uses only the Python standard library and can be rerun without any installation; every output line is genuine output.

Computational lab

```
from itertools import product

def prime(n):
    if n < 2:
        return False
    d = 2
    while d * d <= n:
        if n % d == 0:
            return False
        d += 1
    return True

pairs = product([False, True], repeat=2)
print("contraposition:", all((not P or Q) == (Q or not P)
                             for P, Q in pairs))

D = range(1, 11)
print("y>x:", sum(y > x for x in D for y in D), "of 100")
print("x without a witness:", [x for x in D if all(y <= x for y in D)])

values = [(n, n*n + n + 41) for n in range(41)]
print("prime:", sum(prime(v) for _, v in values), "of 41")
print("failed:", [(n, v, int(v**0.5)) for n, v in values if not prime(v)])
```

```
print("41*41, 41*43:", 41*41, 41*43)

residues = {(n*n - 1) % 8 for n in range(1, 40, 2)}
print("residues mod 8:", residues)

step = all(n*n + 1 + 2*n + 1 == (n+1)**2 + 1 for n in range(1, 500))
print("false step:", step)
```

The output obtained:

```
contraposition: True
y>x: 45 of 100
x without a witness: [10]
prime: 40 of 41
failed: [(40, 1681, 41)]
41*41, 41*43: 1681 1763
residues mod 8: {0}
false step: True
```

The first line checks all four rows of the table at once. The second and third lines compute what is awkward to do on paper: out of a hundred pairs (x, y) in $\{1, \ldots, 10\}$, 45 satisfy $y > x$, and the only x without a witness is 10; that is the failure of $(\forall x)(\exists y)$ on a finite set.

The fourth and fifth lines give 40 primes out of 41 values, failing only at $(40, 1681, 41)$; that last 41 is $\sqrt{1681}$ truncated. The sixth line confirms $1681 = 41^2$ and $1763 = 41 \cdot 43$. The seventh states that every odd n below 40 gives residue 0 modulo 8, in agreement with the theorem of the first pattern. The last line is a warning: the induction step of that false claim is valid for the first five hundred values, so a numerical check does not find its defect.

Note for instructors

A teaching order that has worked: start from $n = 40$ and 1681, not from the definitions. After the students are annoyed that forty checks were in vain, then enter the truth table, with emphasis on the third and fourth rows. Continue to the negation of quantifiers through the definition of the limit of a sequence, then to literal definitions with the example $91 = 7 \cdot 13$; close with induction, a failing base, and the lab.

Two misconceptions are most frequent. First, an implication is read as cause and effect, so a row with a false antecedent is judged false and a vacuous case proof is rejected. Second, induction is thought finished once the step is proved, and strong induction is thought to be ordinary induction repeated; as a result the base is not checked and the hypothesis is used without naming its range.

Blackboard activity: split the class into two groups. The first group writes a quantified claim, the second group writes its negation, then each names one object that refutes the other side's claim. The game stops when a negation can no longer be refuted.

Students are ready to go on if they can negate the definition of a limit without opening their notes and can explain why the base in induction may not be skipped.

1.9 Summary

LEVEL: BASIC

Summary.

- A statement is true or false; $P \Rightarrow Q$ is false only when P is true and Q is false.
- $P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$, but the converse is not equivalent; $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$.

- Negation works from the outside in: every quantifier changes kind, so denying “for all” means pointing at one counterexample.
- A definition holds in both directions: the name may be used only if every clause is satisfied, and every clause is used to check it.
- Four proof patterns: direct, contraposition, contradiction, constructive. Choose the shortest, not the most magnificent.
- Induction demands a base and a step; strong induction demands all earlier values. A step without a base proves nothing.

1.10 Exercises

Level: intermediate

Exercise 1.1.

1. Write the negation of the following statement, then turn the result into a plain sentence that states what must be shown in order to refute it: “for every $\varepsilon > 0$ there is a natural number N such that for every $n \geq N$ we have $|a_n - L| < \varepsilon$”.
2. Write the negation of “every prime larger than 2 is odd”, then name one object that refutes the original statement. Do the same for “there is a natural number n such that $n^2 + n + 41$ is not prime”.
3. Prove that for every integer n, the number $n^2 + n$ is always even. Do it in two ways: by cases on parity, and by contraposition.
4. Prove by induction that for every natural number n we have

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

   Check the formula for $n = 1, 2, 3$ before writing the proof.
5. Prove by strong induction that every natural number n can be written in the form $n = 2^k m$ with $k \geq 0$ an integer and m an odd number. Show where the strong induction hypothesis is used.
6. Judge the following claims: true, false, or not enough evidence.
   (a) If $P(1)$ is true and $P(k) \Rightarrow P(k+1)$ for every $k \in \mathbb{N}$, then $P(n)$ is true for every $n \in \mathbb{N}$.
   (b) If $P(n)$ is true for $n = 1, 2, \ldots, 10^6$, then $P(n)$ is true for every $n \in \mathbb{N}$.
   (c) If $n^2 + n + 41$ is prime for $n = 0, 1, \ldots, 39$, then the claim is true for every $n \geq 0$.
   (d) If 3 divides n, then 6 divides n.

   For every item that is false, give one concrete counterexample.
7. Using Python, count how many primes occur among n^2+n+41 for $n = 0$ up to $n = 100$, then determine all n in that range whose value is not prime. Check whether each point of failure has the form $n = p - 1$ or $n = p$ for some prime p.
8. For $n = 1$ up to $n = 12$, compute 2^n and n^2, then determine the smallest natural number n_0 such that $2^n > n^2$ holds for every $n \geq n_0$. Prove that claim by induction, with base n_0.
9. Prove that the statement “there is an integer $n > 1$ with $n^2 - 1$ prime” is false. First state the factorization of $n^2 - 1$ for $n > 1$, then explain why the factors cannot equal 1.
10. Show that the statement “if n^2 is divisible by 4 then n is divisible by 4” is false, then repair the conclusion into a true claim and prove it.

Hints for the three hardest items. Item 5: separate the case of odd n, which is settled at once with $k = 0$ and $m = n$, from the case of even n, which uses the induction hypothesis on

$n/2$. Item 7: also compute $\sqrt{n^2+n+41}$ for every point of failure, and compare it with the smallest divisor. Item 9: $n^2-1=(n-1)(n+1)$; for $n>1$ both factors lie between 1 and n^2-1.

For researchers: limits and next steps

What is genuinely proved in this chapter is three classical facts that survive without a computer: the irrationality of $\sqrt{2}$ by a parity proof, the infinitude of primes through Euclid's construction, and the uniqueness of the form of every natural number as a product of primes. What is merely popularized is the story that 41 is "almost" a prime formula; what happens is only a long initial stretch, and its length can be predicted from n^2+n+p.

Primary sources: Abbott, *Understanding Analysis*, Chapter 1 Section 1.1 (irrationality of $\sqrt{2}$) and Section 1.2 (introduction, including induction); Avigad and Massot, *Mathematics in Lean*, Chapter 3 (logic) and Chapter 5 Sections 5.1 and 5.2 (irrational roots, induction and recursion); Euclid, *Elements*, Book IX Proposition 20 for infinitely many primes.

Open questions: there is no polynomial of positive degree with integer coefficients that is prime for all $n \geq 0$ (a one-line proof: $f(0)$ divides $f(t\,f(0))$), but how long a prime stretch a polynomial of high degree can reach is still being studied, and the distribution of primes is tied to the Riemann hypothesis, which remains unproved.

Limits of applicability: all results in this chapter are elementary and do not depend on the choice of set axioms. Formal verification in Lean (Chapter 28) checks the validity of every deductive step, but it does not check whether the formal statement proved is the one its author intended.

Map to source texts: Chapter 1 of Understanding Analysis *(Abbott), in particular Section 1.1 on the irrationality of* $\sqrt{2}$ *and Section 1.2 on the introduction; Chapter 3 (logic), Chapter 5 Sections 5.1 and 5.2 (irrational roots as well as induction and recursion) of* Mathematics in Lean *(Avigad and Massot); and Book IX Proposition 20 of* Elements *(Euclid) for infinitely many primes.*

Chapter 2

Sets, Relations, Functions, and Cardinality

A security tester receives a model specification: a classifier with thirty binary features, each taking only the value zero or one. To test exhaustively, it must try every possible input. Each feature doubles the number of possibilities, so the answer is $2^{30} = 1{,}073{,}741{,}824$. At one million inputs per second, that test takes 1073.74 seconds, about 17.9 minutes. Add ten more features: the number becomes $2^{40} = 1{,}099{,}511{,}627{,}776$, and a machine a thousand times faster needs 1099.51 seconds, about 18.3 minutes. We multiply the problem by a thousand and speed the machine up a thousandfold, and we are back at the starting point.

The same difficulty appears more sharply once features may take an arbitrary real value. The number of inputs stops being "many" and starts being "infinitely many", and the question changes: can two infinite sets still be compared in size? Instinct says there are more rational numbers than natural numbers, since between any two natural numbers there always lie infinitely many rationals. That instinct is wrong, and this chapter shows why.

The previous chapter gave us a language: quantifiers, implication, proof patterns. What was missing is the object being talked about. This chapter installs its basic vocabulary, namely sets and their operations, relations, functions, and one way of comparing the size of sets that does not depend on the ability to count. That vocabulary is used throughout the book, from subspaces in Chapter 6 to formal proofs in Chapter 28.

For younger readers (age 10 and up)

Picture two baskets of marbles, one red and one blue. We want to know which basket holds more, but we are not allowed to count. Take one red marble, pair it with one blue marble, set both aside, and repeat until one basket is empty: if both empty at the same moment they hold the same number, if blue is left over then blue holds more. This method never mentions a number; it only pairs.

Now suppose the baskets never end, so that we can never finish moving marbles. Can pairing still be used? It can, and that is the surprise of this chapter: with the right pairing rule, all the counting numbers can be paired off with all the even numbers, with nothing left over on either side, and pairs of numbers can be paired with a single number. We will understand why pairing is stronger than counting, and why some endless sets still cannot be paired with the counting numbers, however hard we try.

2.1 Operations on sets and De Morgan's laws

Level: basic

A single set is of no use until we can build new sets from it. Four ways of building are defined by the same pattern: state a membership condition, and the new set is the collection of all objects satisfying it. Let X be a universal set and $A, B \subseteq X$; from Chapter 1 we recall that $A \subseteq B$ means every member of A is also a member of B, and that equality of sets is proved by two inclusions.

Definition 2.1 (Union, intersection, difference, complement)**.** The union $A \cup B$ is the set of all $x \in X$ that are members of A or members of B. The intersection $A \cap B$ is the set of all $x \in X$ that are members of A and at the same time members of B. The difference $A \setminus B$ is the set of all $x \in X$ that are members of A but not members of B. The complement $A^c = X \setminus A$ is the set of all $x \in X$ that are not members of A.

The word "or" in a union is inclusive: x may lie in both sets, and that is the condition under which $A \cap B \subseteq A \cup B$ is always true, as Hoffman and Kunze note in the Appendix. The empty set $\varnothing$ is needed so that the intersection of two disjoint sets is still a set.

Proposition 2.1 (De Morgan's laws)**.** *For any $A, B \subseteq X$,*

$$X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B), \qquad X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B).$$

Proof. Take any $x \in X$. By the definition of complement and union,

$$x \in X \setminus (A \cup B) \iff x \in X \text{ and not } (x \in A \text{ or } x \in B).$$

By De Morgan's law for statements from Chapter 1, the negation of "at least one" is "both fail", so

$$x \in X \setminus (A \cup B) \iff x \in X \text{ and } x \notin A \text{ and } x \notin B \iff x \in (X \setminus A) \cap (X \setminus B).$$

Since x was arbitrary, the two sets are equal. The second equality is obtained by replacing every "and" by "or" in the same chain. ∎

The law holds for two sets; what we need later is the version for any number of sets.

Definition 2.2 (Indexed family)**.** Let I be an index set and let $A_i \subseteq X$ be given for each $i \in I$.

$$\bigcup_{i \in I} A_i = \{\, x \in X : x \in A_i \text{ for some } i \in I \,\},$$
$$\bigcap_{i \in I} A_i = \{\, x \in X : x \in A_i \text{ for every } i \in I \,\}.$$

The general version reads $X \setminus \bigcup_i A_i = \bigcap_i (X \setminus A_i)$ and $X \setminus \bigcap_i A_i = \bigcup_i (X \setminus A_i)$. Its proof adds no new idea: only "for some" turns into "for every" when the negation is pulled inside, exactly as in Chapter 1.

Numbers make the definition checkable. For $X = \{1, \dots, 10\}$, $A = \{1, \dots, 5\}$, and $B = \{4, \dots, 8\}$, Figure 2.1 shows the three operations on the same sets: the union has $3 + 2 + 3 = 8$ members, the intersection has 2, and the complement of B has $3 + 2 = 5$.

Do it by hand (kinesthetic learner)

Materials: twenty used name cards, two pieces of string two meters long, an open floor. Form two string loops overlapping by about a quarter of their width. Write the names of five friends who like coffee and five friends who like tea. Put the coffee cards in the first loop, the tea cards in the second, then move the card of a friend who likes both into

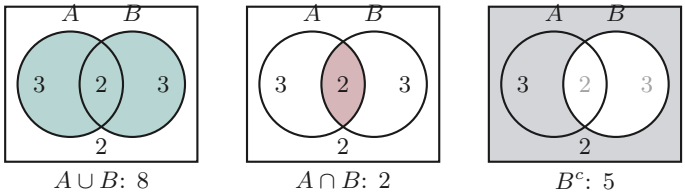


Figure 2.1: Three Venn diagrams over the same sets, with the number of members in each region. Left: $A \cup B$ has $3 + 2 + 3 = 8$ members; middle: $A \cap B$ has 2 members; right: B^c has $3 + 2 = 5$ members, namely the shaded region, while the members inside B are printed faded.

the overlap: that is the intersection. Pick one name who likes neither and put that card outside both loops.
Count three numbers: cards inside both loops, cards in the overlap, and cards outside the second loop. What must become visible: the number of cards in the union is not the number of coffee cards plus the number of tea cards, because the cards in the overlap are counted twice, and the excess is exactly the intersection.

The set we most often build from A is the set of all its subsets, and its size grows far faster than A itself.

Definition 2.3 (Power set)**.** The power set $\mathcal{P}(A)$ is the set of all subsets of A, that is, $\mathcal{P}(A) = \{\, B : B \subseteq A \,\}$.

power set (*himpunan kuasa*) — A list of every way of choosing part of the members of a bag, including choosing nothing and choosing everything.

For example, $\mathcal{P}(\varnothing) = \{\varnothing\}$ has one member, and $\mathcal{P}(\{1,2\}) = \{\varnothing, \{1\}, \{2\}, \{1,2\}\}$ has four.

Proposition 2.2. *If A is finite with $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.*

Proof. For $B \subseteq A$ define the characteristic function $\chi_B : A \to \{0,1\}$ by $\chi_B(a) = 1$ if $a \in B$ and $\chi_B(a) = 0$ if $a \notin B$. The map $B \mapsto \chi_B$ is a bijection between $\mathcal{P}(A)$ and the set of all functions $A \to \{0,1\}$: from χ we recover $B = \{a : \chi(a) = 1\}$, and two different subsets give two different functions. Hence $|\mathcal{P}(A)|$ equals the number of functions $A \to \{0,1\}$.

We count those functions by induction on n. For $n = 0$ there is only the empty function, and $2^0 = 1$. Suppose $|A| = n+1$, choose $a_0 \in A$, and write $A' = A \backslash \{a_0\}$. Every $f : A \to \{0,1\}$ is determined uniquely by the restriction $f|_{A'}$ and the value $f(a_0)$; the induction hypothesis gives 2^n choices of restriction and 2 choices of value, 2^{n+1} in all. ∎

That proof never mentions combinations: the bijection moves the count, and induction splits it into two free choices.

Read it aloud (auditory learner)

Read in pairs, one voice for member and one voice for subset. Start slowly, then speed up.
Member: One orange goes into the basket. I am inside. Subset: One basket goes into the store. I am inside. Member: I am cargo. Subset: I am container. Member: I am pointed at one by one. Subset: I am proved all at once. Member: When I go in, I am a single mark. Subset: When I go in, I am a single proof. Member: I am small and many. Subset: I am large and sometimes only one.
Repeat until the last line comes out without thinking. The line repeated three times

together: cargo goes into bag, bag goes into bag, and the two use different marks.

2.2 Cartesian products and relations

Level: basic

Order matters. The set $\{1, 2\}$ equals $\{2, 1\}$, but the coordinates of a point in the plane may not be swapped without moving the point. We therefore need an object that remembers order, and its definition must be built from sets alone.

Definition 2.4 (Ordered pair and Cartesian product)**.** The ordered pair (a, b) is defined as $\{\{a\}, \{a, b\}\}$. The Cartesian product $A \times B$ is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$.

The definition looks strange, but only one property is needed, namely $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$. Checking that property requires four cases, and we leave it as Exercise 4.

The size of a product is easy to guess, and its proof uses a cut along the first coordinate.

Proposition 2.3. *If A and B are finite, then $|A \times B| = |A| \cdot |B|$.*

Proof. For each $a \in A$ the set $\{a\} \times B$ has $|B|$ members, since $b \mapsto (a, b)$ is a bijection between B and $\{a\} \times B$. The family $\{\{a\} \times B\}_{a \in A}$ consists of mutually disjoint sets, because $(a, b) = (a', b')$ forces $a = a'$. Their union is $A \times B$, so $|A \times B|$ is the sum of $|A|$ terms each equal to $|B|$. ∎

Now the object that gives this chapter its title. A relation is merely a statement that may be true or false for each pair of objects, and so it can be stored as the set of pairs for which the statement is true.

Definition 2.5 (Relation)**.** A relation from A to B is any $R \subseteq A \times B$; when $A = B$ we call it a relation on A. We write $a \, R \, b$ for $(a, b) \in R$. The domain of R is $\{\, a \in A : (a, b) \in R \text{ for some } b \in B \,\}$ and its range is $\{\, b \in B : (a, b) \in R \text{ for some } a \in A \,\}$.

Since a relation on A is a subset of $A \times A$, the number of relations on a set with n members is 2^{n^2}. For $n = 3$ the number is $2^9 = 512$ and for $n = 4$ it becomes $2^{16} = 65{,}536$; both are checked by the script in the lab section.

Four properties filter them into useful objects. A relation R on A is called reflexive if $a \, R \, a$ for every $a \in A$; symmetric if $a \, R \, b$ implies $b \, R \, a$; transitive if $a \, R \, b$ and $b \, R \, c$ imply $a \, R \, c$; and antisymmetric if $a \, R \, b$ together with $b \, R \, a$ is possible only when $a = b$.

Definition 2.6 (Partial order and equivalence relation)**.** A relation on A that is reflexive, antisymmetric, and transitive is called a partial order. A relation on A that is reflexive, symmetric, and transitive is called an equivalence relation.

The relation $\leq$ on $\mathbb{R}$ is a partial order, as is the relation "divides" on the positive natural numbers, and $\subseteq$ on $\mathcal{P}(A)$ gives an example that will later become the framework of subspaces in Chapter 6. The word "partial" is used because two members need not be comparable: on $\mathcal{P}(\{1, 2, 3\})$ the sets $\{1\}$ and $\{2\}$ are not contained in one another. An order in which every two members are comparable is called a total order.

2.3 Equivalence relations and partitions

Level: intermediate

An equivalence relation cuts a set into groups that do not overlap. Before reaching the theorem, we need a name for the groups.

Definition 2.7 (Equivalence class)**.** Let $\sim$ be an equivalence relation on A and $x \in A$. The equivalence class of x is $[x] = \{\, y \in A : y \sim x \,\}$.

Since $x \sim x$, no class is empty. The more interesting property: two classes cannot overlap even slightly.

Lemma 2.4. *Let $\sim$ be an equivalence relation on A and $x, y \in A$. Then (i) $x \in [x]$; (ii) $[x] = [y]$ if and only if $x \sim y$; and (iii) if $[x] \cap [y] \neq \varnothing$, then $[x] = [y]$.*

Proof. Part (i) is the statement $x \sim x$, true because the relation is reflexive. For (ii), suppose $[x] = [y]$. By (i), $y \in [y] = [x]$, so $y \sim x$ and by symmetry $x \sim y$. Conversely, suppose $x \sim y$ and take $z \in [x]$, meaning $z \sim x$. From $z \sim x$ and $x \sim y$ we get $z \sim y$, so $z \in [y]$; this gives $[x] \subseteq [y]$, and the argument with the roles exchanged gives $[y] \subseteq [x]$. For (iii), suppose $z \in [x] \cap [y]$. Then $z \sim x$ and $z \sim y$; from $z \sim x$ and symmetry we get $x \sim z$, and together with $z \sim y$ and transitivity we get $x \sim y$, so $[x] = [y]$ by (ii). ■

Consequently the set of all classes divides A into pieces that are mutually disjoint and cover all of A.

Definition 2.8 (Partition)**.** A partition of a set A is a family $\mathcal{B}$ of nonempty subsets of A such that every $x \in A$ lies in exactly one member of $\mathcal{B}$.

Theorem 2.5. *Every equivalence relation on A determines a partition of A, every partition of A determines an equivalence relation on A, and the two processes invert each other.*

Proof. Let $\sim$ be an equivalence relation and $\mathcal{B} = \{\, [x] : x \in A \,\}$. By the Lemma, each $[x]$ is nonempty and two of its members are either disjoint or identical; every x lies in $[x]$. Hence $\mathcal{B}$ is a partition.

Conversely, let $\mathcal{B}$ be a partition of A and set $x \sim y$ when x and y lie in the same member of $\mathcal{B}$. Reflexivity and symmetry are clear. For transitivity, suppose $x \sim y$ and $y \sim z$, and let B be the member of $\mathcal{B}$ containing y; then $x \in B$ and $z \in B$, so $x \sim z$.

The two processes invert each other: from $\sim$ we obtain the partition into classes, and that partition returns the original $\sim$ because $x \sim y$ exactly when $x \in [y]$; from a partition we obtain $\sim$, and the class $[x]$ is the member of $\mathcal{B}$ containing x. ■

The number of classes is called the index. The set of all classes is written $A/\!\sim$ and called the quotient set, while $q : A \to A/\!\sim$ with $q(x) = [x]$ is called the quotient map. Hoffman and Kunze note that every equivalence relation can be viewed as coming from a function: two members are equivalent exactly when that function gives them the same value.

Example 2.1. On the integers, set $x \equiv y \pmod{n}$ when n divides $x - y$. This relation is reflexive because $n \mid 0$, symmetric because $n \mid (x - y)$ implies $n \mid (y - x)$, and transitive because $n \mid (x - y)$ together with $n \mid (y - z)$ implies $n \mid (x - z)$. Its classes are the sets of numbers with the same remainder, and there are exactly n classes, namely $[0], \dots, [n-1]$, since every integer can be written $x = qn + r$ with $0 \le r \le n - 1$ uniquely.

An example closer to our own work: for $f : A \to B$, set $x \sim y$ when $f(x) = f(y)$. The class $[x]$ is the set of inputs with the same output; in machine learning that is "one prediction file", and the number of classes equals the number of distinct labels.

For younger readers (age 10 and up)

The wardrobe at home is usually not sorted by any official rule. White shirts pile up on one shelf, printed shirts on another, sports shirts in a third basket. Nobody writes the rule down, yet every shirt sits in exactly one place.

That is what an equivalence relation does: it decides when two items count as roommates, then groups all the items. The result is a tidy division, every item lands in exactly one group, and different groups share no members.

Now the question that makes one curious. What if one shirt belongs to the white group under one rule, but to the sports group under another rule? Is the grouping still tidy? The tidiness does not depend on the rule chosen, but on three properties: every item is a roommate of itself, roommatehood works both ways, and roommatehood can be passed along through a third item.

2.4 Functions: injective, surjective, bijective

Level: intermediate

We now fix the meaning of the word "function", because this chapter depends on the fine distinction between codomain and range.

Definition 2.9 (Function)**.** A function from A to B is a rule f that pairs every $x \in A$ with exactly one $f(x) \in B$. The set A is called the domain, B is called the codomain, and $f(A) = \{\, f(x) : x \in A \,\}$ is called the range.

Codomain and range are different, and the difference matters. The function $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$ has codomain $\mathbb{R}$ but range $[0, \infty)$. Hoffman and Kunze use three names interchangeably for the same object: function, mapping, transformation.

For $S \subseteq B$ we define the preimage $f^{-1}(S) = \{\, x \in A : f(x) \in S \,\}$. The notation f^{-1} here does not presuppose that f has an inverse: a preimage is always defined, an inverse need not exist.

Definition 2.10 (Injective, surjective, bijective)**.** A function $f : A \to B$ is called injective if $f(x) = f(y)$ is possible only when $x = y$; surjective if $f(A) = B$; and bijective if both.

bijection (*bijeksi*) — A perfect matching: every item in the first box has exactly one partner in the second box, with nothing left over on either side.

Figure 2.2 compares the three on two columns of nodes: the left panel leaves no node over on either side, the middle panel leaves d over on the right, and the right panel sends two left nodes to one right node.

The word "bijective" is only a short name for the combination of the two previous properties. Its power appears in the following proposition: a bijection is the exact condition for a function to be invertible.

Proposition 2.6. *$f : A \to B$ is bijective if and only if there is $g : B \to A$ with $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$. If g exists, it is unique.*

Proof. Suppose g satisfies both equalities. If $f(x) = f(y)$ then $x = g(f(x)) = g(f(y)) = y$, so f is injective. For any $b \in B$ we have $b = f(g(b))$, so f is surjective.

Conversely, suppose f is bijective. For each $b \in B$ there is exactly one $x \in A$ with $f(x) = b$: existence from surjectivity, uniqueness from injectivity. Define $g(b) = x$ for that x, then $g(f(x)) = x$ and $f(g(b)) = b$.

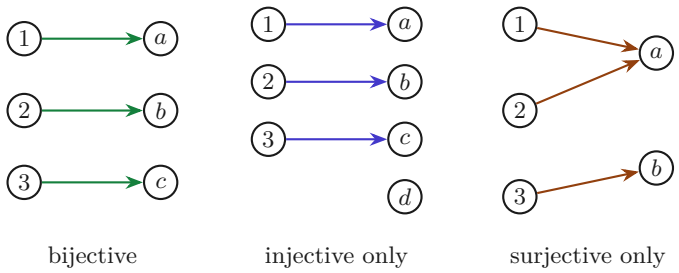


Figure 2.2: Three mappings between two columns of nodes. Left: every node is used exactly once, hence bijective. Middle: the right column has four members and the node d receives no incoming arrow, hence injective but not surjective. Right: two left nodes go to a, hence surjective but not injective.

For uniqueness, suppose g and h satisfy both equalities. Then

$$g = g \circ \mathrm{id}_B = g \circ (f \circ h) = (g \circ f) \circ h = \mathrm{id}_A \circ h = h,$$

which uses only the associativity of composition and the two equalities. ■

Composition behaves well with respect to all three properties: if f and g are injective then $g \circ f$ is injective, and similarly for surjective and bijective. The converse fails: with $f : \{1\} \to \{1, 2\}$, $f(1) = 1$, and $g : \{1, 2\} \to \{1\}$, $g(1) = g(2) = 1$, the composition $g \circ f$ is bijective, even though g is not injective and f is not surjective.

On finite sets the three properties lock together, and that is why "one-to-one" and "onto" can be distinguished only on infinite sets.

Lemma 2.7. *Let A and B be finite, $f : A \to B$, and $C \subseteq B$. Then $|f(A)| \leq |A|$ with equality if and only if f is injective, and $|C| \leq |B|$ with equality if and only if $C = B$.*

Proof. We prove the second part by induction on $n = |B|$. For $n = 0$ we have $B = \varnothing$, hence $C = B$. Assume the claim for sets with n members and suppose $|B| = n + 1$; choose $b \in B$ and write $B' = B \setminus \{b\}$.

If $b \notin C$, then $C \subseteq B'$ and the induction hypothesis gives $|C| \leq |B'| = n < n + 1$, with equality impossible. If $b \in C$, write $C' = C \setminus \{b\} \subseteq B'$. Then $|C| = |C'| + 1 \leq |B'| + 1 = |B|$; equality holds exactly when $|C'| = |B'|$, which by the induction hypothesis means $C' = B'$, that is $C = B$.

The first part is obtained by the same induction on $|A|$: write $A = A' \cup \{a\}$ with $a \notin A'$ and observe that $f(A) = f(A') \cup \{f(a)\}$, so $|f(A)| \leq |A'| + 1 = |A|$. Equality demands $f(a) \notin f(A')$ and $|f(A')| = |A'|$; the latter means $f|_{A'}$ is injective by the induction hypothesis, and the former means f does not repeat a value at a. ■

Proposition 2.8. *Let A and B be finite with $|A| = |B|$, and let $f : A \to B$. Then f injective, f surjective, and f bijective are three equivalent statements.*

Proof. If f is injective, the Lemma gives $|f(A)| = |A| = |B|$; since $f(A) \subseteq B$ and both have the same size, the Lemma gives $f(A) = B$, so f is surjective. If f is surjective, then $f(A) = B$ so that $|f(A)| = |B| = |A|$, and the Lemma forces f to be injective. Since bijective means both properties at once, the equivalence of the three is complete. ■

A warning to carry into the next chapter: that proposition uses the word "finite" essentially. On $\mathbb{N}$ the function $f(x) = 2x$ is injective but not surjective, and the function mapping 1 to 1 and $n \geq 2$ to $n - 1$ is surjective but not injective.

2.5 Finite, countable, and uncountable sets

Level: advanced

Now we return to the opening question. To compare two sets without counting we pair their members, and all we need to require is that the pairing uses an invertible function.

Definition 2.11 (Equinumerous)**.** The sets A and B are equinumerous, written $A \sim B$, if there is a bijection $A \to B$.

The relation $\sim$ is reflexive (the identity is a bijection), symmetric (the inverse of a bijection is a bijection), and transitive (the composition of two bijections is a bijection). So it behaves as an equivalence relation, but not on "the set of all sets", because that collection is not a set; this is discussed in the box for researchers.

Definition 2.12 (Finite, countable, uncountable)**.** A set A is finite if $A = \varnothing$ or there is $n \in \mathbb{N}$ and a bijection $A \to \{1, 2, \dots, n\}$; in the second case we write $|A| = n$. A set A is countable if $A \sim \mathbb{N}$. An infinite set that is not countable is called uncountable.

countable (*terhitung*) — All its members can be given serial numbers with none skipped, as when every passenger receives a distinct ticket number.

The name "countable" is apt: if $A \sim \mathbb{N}$, the bijection is a list. We can name the first member, the second, and so on, and every member appears exactly once. What is surprising is how many sets can be listed that way. Pairs of numbers, for instance.

Proposition 2.9 (Cantor pairing)**.** $\mathbb{N} \times \mathbb{N}$ *is countable.*

Proof. Define $\pi(m, n) = \frac{(m+n)(m+n+1)}{2} + n$ for $m, n \in \mathbb{N} = \{0, 1, 2, \dots\}$. Write $T_s = \frac{s(s+1)}{2}$, the number of pairs with $m + n < s$. For a pair with $m + n = s$ we have $0 \le n \le s$, so $\pi(m, n) = T_s + n$ takes the values $T_s, \dots, T_s + s$ exactly once each. Since $T_s + s = T_{s+1} - 1$, the block of values for diagonal s is $\{T_s, \dots, T_{s+1} - 1\}$. The blocks join without gaps and without overlap starting from $T_0 = 0$. Hence every counting number lies in exactly one block, and within its block comes from exactly one pair; therefore π is bijective. ∎

The inverse is explicit: $s = \lfloor (\sqrt{8q + 1} - 1)/2 \rfloor$, $n = q - T_s$, and $m = s - n$. That square root is the one place where machine arithmetic can deceive us, and the example is in the lab section.

Figure 2.3 draws that grid: the cell in row m and column n is labelled $\pi(m, n)$, and the color marks the diagonal $m + n = s$, so that reading by color gives 0 through 14 without gaps and without repetition. The right panel applies the same reading order to a list of four real numbers, with row d flipping the diagonal digits so that d is not in the list.

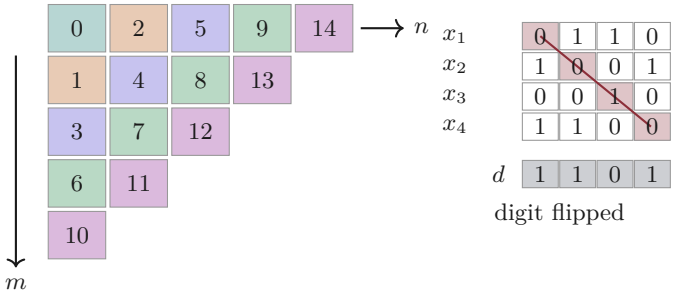


Figure 2.3: Left: the grid of pairs (m, n) with $m + n \le 4$, each cell labelled $\pi(m, n)$, with color marking the diagonal $m + n = s$; the first five diagonals carry the values 0 through 14. Right: the digit table of four real numbers, with the diagonal cells colored and row d flipping the diagonal digits $0, 0, 1, 0$ into $1, 1, 0, 1$.

Since every positive rational is a pair of natural numbers already in lowest terms, the set of rationals is countable too. All we must guard against is listing a number twice.

Proposition 2.10. *The set $\mathbb{Q}_{>0}$ of positive rational numbers is countable.*

Proof. Every $q \in \mathbb{Q}_{>0}$ can be written uniquely as $q = p/r$ with p, r natural numbers and $\gcd(p, r) = 1$. Define the height of q as $p + r$. For each $k \geq 2$ there are only finitely many rationals of height k, since p and r lie between 1 and $k - 1$. The following list therefore enumerates $\mathbb{Q}_{>0}$ without repetition: for $k = 2, 3, 4, \ldots$ write down all p/r of height k with $\gcd(p, r) = 1$, increasing in p. Every positive rational appears exactly once, namely at level $k = p + r$ in its unique reduced form, so that list is a bijection $\mathbb{N} \to \mathbb{Q}_{>0}$. ■

Corollary 2.11. *$\mathbb{Q}$ and $\mathbb{Z}$ are countable.*

Proof. The list $0, 1, -1, 2, -2, \ldots$ is a bijection $\mathbb{N} \to \mathbb{Z}$. For $\mathbb{Q}$, enumerate each nonzero rational as $\pm p/r$ in reduced form by $(p + r)$ with alternating signs, then place 0 at the front; one extra member does not destroy countability. ■

So $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ are equinumerous, even though $\mathbb{Q}$ feels far more crowded. Density is not a measure of size; the measure is whether or not a list exists.

Challenge (problem solver)

A hotel on the seafront has rooms numbered $1, 2, 3, \ldots$ without end, and that night every room holds exactly one guest. A traveler arrives without a reservation. The receptionist says the hotel is full; the manager says there is still a room for him without sending a single guest home.
Show a rule for moving guests that keeps the manager's promise, then explain why that rule is a bijection between the old guests plus one new guest and all the rooms. Why can the same hotel not accommodate as many new guests as there are real numbers?
Hint 1: do not look for an empty room, because there is none. Create an empty room by moving its occupant.
Hint 2: shift the occupant of room n to room $n + 1$ for every n, then put the traveler in room 1; check that every room holds exactly one person and that no guest is lost.
If you get stuck, return to the section Finite, countable, and uncountable sets, in particular the definition of equinumerous.

Cantor showed that $\mathcal{P}(A)$ is always larger than A, by a trick that does not depend on the size of A.

Theorem 2.12 (Cantor)**.** *For every set A there is no surjective function $A \to \mathcal{P}(A)$.*

Proof. Let $f : A \to \mathcal{P}(A)$ be arbitrary and define $D = \{\, a \in A : a \notin f(a) \,\}$. Then $D \subseteq A$, so $D \in \mathcal{P}(A)$. Suppose $D = f(a_0)$ for some $a_0 \in A$. If $a_0 \in D$, then by the definition of D we have $a_0 \notin f(a_0) = D$, a contradiction. If $a_0 \notin D$, then $a_0 \notin f(a_0)$, which is precisely the membership condition for D, so $a_0 \in D$, a contradiction again. Hence D is not in the range of f. ■

The set D is formed by asking, for each a, whether a is a member of the set paired with it, then flipping the answer; that flip is what makes D differ from every value of f. For $A = \mathbb{N}$ the trick appears as reading the diagonal of a table with every entry changed, which is why it is called the diagonal argument.

Corollary 2.13. *$\mathcal{P}(\mathbb{N})$ is uncountable, and $\mathbb{R}$ is uncountable.*

Proof. By Cantor's Theorem with $A = \mathbb{N}$, there is no surjection $\mathbb{N} \to \mathcal{P}(\mathbb{N})$, so $\mathcal{P}(\mathbb{N})$ is uncountable.

For $\mathbb{R}$ it suffices to show that $[0,1]$ is uncountable. Every $x \in [0,1]$ has a binary expansion $x = \sum_{k \geq 1} b_k 2^{-k}$ with $b_k \in \{0,1\}$; its existence is proved in Chapter 3 and Chapter 5. The map from the set of binary sequences to $[0,1]$ is therefore surjective, and the set of binary sequences is equinumerous with $\mathcal{P}(\mathbb{N})$ because a binary sequence is a function $\mathbb{N} \to \{0,1\}$. If $[0,1]$ were countable, its preimage in the set of binary sequences would be countable as well, contradicting the previous paragraph. ■

One note of honesty: binary expansions are not unique, since $0.0111\cdots = 0.1000\ldots$ in base two. The argument above is undisturbed because it uses only the surjectivity, but $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$ demands the following tool.

Proposition 2.14 (Cantor–Bernstein–Schroeder)**.** *If there is an injective function $A \to B$ and an injective function $B \to A$, then $A \sim B$.*

Proof sketch. Let $f : A \to B$ and $g : B \to A$ be injective. Let A_0 be the set of points of A that have no predecessor in B, then $A_{k+1} = g(f(A_k))$, and write A^+ for the union of all A_k with k even. Define $h(x) = f(x)$ if $x \in A^+$ and $h(x) = g^{-1}(x)$ if $x \notin A^+$. On the first part h is injective because f is injective, on the second because g^{-1} is injective, and the images of the two parts are separated by the way A^+ is built, so h is injective. For surjectivity of the image, take $y \in B$. If $g(y) \in A^+$, then $g(y)$ lies in some A_k with k even so that $g(y) = g(f(x))$ for $x \in A_{k-1}$ and $h(x) = y$; if $g(y) \notin A^+$, then $h(g(y)) = y$. Hence h is surjective.

The bookkeeping of chain depth is the delicate part, and its details are skipped here; the full formal version is in *Mathematics in Lean*, section 4.3. ■

With that tool we may conclude $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$: an injection from $\mathcal{P}(\mathbb{N})$ to $\mathbb{R}$ is given by binary expansion, and the injection the other way by the tangent function. The size of $\mathbb{R}$ is usually written $2^{\aleph_0}$, with $\aleph_0$ the size of $\mathbb{N}$.

For younger readers (age 10 and up)

Picture the ticket machine at the entrance to an exhibition. Every visitor receives one numbered ticket: 1, 2, 3, and so on, no number is used twice, and every possible number will eventually be given out. A collection of things that can be ticketed like that is called countable.

It looks as though every endless collection can be ticketed. But not so. Some collections have so many members that however many tickets we print, there is always a member left without one. The ticket machine is not broken; that collection really is bigger than the supply of numbers.

The question that makes one curious: how can we know that a collection cannot be ticketed, when not all methods have been tried? The answer is Cantor's flipping trick. Each time someone claims to have ticketed all the members, we build one new member deliberately different from every ticketed member; that member is certainly left out.

2.6 Inequalities, absolute value, and upper bounds

Level: intermediate

The last section installs two tools for Chapter 3. The first measures the distance between two numbers regardless of direction, namely the absolute value.

Definition 2.13 (Absolute value)**.** For $x \in \mathbb{R}$, $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$.

A route through a third point is never shorter than the direct route.

Proposition 2.15 (Triangle inequality)**.** *For every* $x, y \in \mathbb{R}$, $|x + y| \leq |x| + |y|$ *and* $||x| - |y|| \leq |x - y|$.

Proof. Note that $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$. Adding the two gives $-(|x| + |y|) \leq x + y \leq |x| + |y|$, which by the definition of absolute value means $|x + y| \leq |x| + |y|$. For the second, write $x = (x - y) + y$ and apply the first result: $|x| \leq |x - y| + |y|$, so $|x| - |y| \leq |x - y|$. Exchanging the roles of x and y gives $|y| - |x| \leq |x - y|$, hence $||x| - |y|| \leq |x - y|$. ■

Now bounds. A set of numbers can be bounded above without having a largest member, and precisely that case is the most important one.

Definition 2.14 (Upper bound and supremum)**.** Let $S \subseteq \mathbb{R}$ be nonempty. A number u is called an upper bound of S if $s \leq u$ for every $s \in S$; if one exists, S is called bounded above. A number u is called the supremum of S, written $u = \sup S$, if u is an upper bound of S and $u \leq v$ for every upper bound v of S.

supremum (*batas atas terkecil*) — The tightest fence: the highest boundary that cannot be lowered any further without leaving a member above it.

The supremum, when it exists, is unique, since two least upper bounds dominate each other. What the definition does not guarantee is existence: $\{x \in \mathbb{Q} : x^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no supremum there.

Note 2.1 (Completeness axiom). Every nonempty set $S \subseteq \mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. This axiom is not proved here; its construction together with its consequences is the content of Chapter 3, including the epsilon characterization that turns a statement about a least bound into a statement about ε, and the example of a set that has a supremum without a maximum.

On a finite set the supremum and the maximum coincide, and swapping the two often slips through on small examples.

2.7 Worked examples

Level: intermediate

The following four computations are done by hand down to the number, then checked again with a script. All of them are small, so arithmetic errors show up immediately.

Step-by-step recipe (procedural learner)

For two finite sets A and B:

1. Compute $|A|$, the number of members of A.
2. Compute $|B|$ the same way.
3. Compute $|A \cap B|$, the members appearing in both lists.
4. Add and then subtract: $|A \cup B| = |A| + |B| - |A \cap B|$.

A warning about order: step 3 may not be postponed until after step 4. If the addition is done first, the members of the intersection are counted twice and the result overshoots by exactly $|A \cap B|$. For $A = \{1, \ldots, 5\}$ and $B = \{4, \ldots, 8\}$: $5 + 5 - 2 = 8$, the same number as in the Venn diagram above.

First, the power set. For $A = \{1, 2, 3\}$ the subsets have sizes $0, 1, 2, 3$ with counts $1, 3, 3, 1$, in total $1 + 3 + 3 + 1 = 8 = 2^3$. The row $n = 4$ gives $1 + 4 + 6 + 4 + 1 = 16$, $n = 2$ gives $1 + 2 + 1 = 4$, and $n = 1$ gives $1 + 1 = 2$: the same Pascal triangle as in the Proposition about 2^n.

Second, the Cantor pairing. With $\pi(m,n) = T_{m+n} + n$ and $T_0 = 0$, $T_1 = 1$, $T_2 = 3$, $T_3 = 6$: diagonal $s = 0$ contains $(0,0) \mapsto 0$; $s = 1$ contains $(1,0) \mapsto 1$ and $(0,1) \mapsto 2$; $s = 2$ contains $(2,0) \mapsto 3$, $(1,1) \mapsto 4$, $(0,2) \mapsto 5$; $s = 3$ contains $(3,0) \mapsto 6$, $(2,1) \mapsto 7$, $(1,2) \mapsto 8$, $(0,3) \mapsto 9$. Those ten values are exactly $0, \ldots, 9$ without repetition and without gaps.

The opposite direction is easier to get wrong. Take $q = 1000$. We look for s with $T_s \le 1000 < T_{s+1}$. Since $T_{44} = 44 \cdot 45/2 = 990$ and $T_{45} = 1035$, we get $s = 44$, then $n = 1000 - 990 = 10$ and $m = 44 - 10 = 34$. Check: $\pi(34,10) = 44 \cdot 45/2 + 10 = 990 + 10 = 1000$, correct.

Third, the rational numbers. By the height rule, level $k = 2$ contains $1/1$; $k = 3$ contains $1/2, 2/1$; $k = 5$ contains $1/4, 2/3, 3/2, 4/1$. The number of terms at each level is the value of Euler's totient function, $1, 2, 2, 4, 2, 6$ for $k = 2, \ldots, 7$. The cumulative count up to $k = 7$ is $1 + 2 + 2 + 4 + 2 + 6 = 17$, so $6/1$ occupies position 17. Since the cumulative count up to $k = 9$ is 27 and level $k = 10$ contains $1/9, 3/7, 7/3, 9/1$, the number $7/3$ occupies position 30.

Fourth, the number of relations. On a set with 3 members there are $2^9 = 512$ relations, and only 5 are equivalence relations: 1 with one class, 3 with two classes, 1 with three classes. On a set with 4 members there are $2^{16} = 65{,}536$ relations and 15 that are equivalence relations. Those four numbers appear again as the last four lines of the script output; the sequence $1, 2, 5, 15, 52$ is known as the Bell numbers.

Four numbers already checked again. (i) $|\mathcal{P}(A)| = 2^n$ for $n = 0, 1, 2, 3, 4$ gives $1, 2, 4, 8, 16$. (ii) $\pi(34,10) = 990 + 10 = 1000$, and π on diagonals $s = 0, 1, 2, 3$ gives $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$. (iii) The number of positive rationals of height k is the totient value at k; the cumulative count up to $k = 9$ is 27, so $7/3$ sits at position 30. (iv) Relations on a set with 3 members: 512 in all, 5 equivalence relations.

2.8 Computational lab

Level: advanced

The following script runs the four computations and adds one check that cannot be done by hand: it compares the inverse of the Cantor pairing computed with machine floating point against the one computed with integer square roots. Both should give the same answer; at one point the answers differ.

Computational lab

```
from itertools import combinations
from math import gcd, isqrt, sqrt, floor

# (1) number of subsets for small n
for n in range(5):
    A = range(n)
    subs = [S for k in range(n + 1) for S in combinations(A, k)]
    print("n =", n, "|P(A)| =", len(subs), "2^n =", 2 ** n)

# (2) Cantor pairing pi(m, n) = (m+n)(m+n+1)/2 + n
def pi(m, n):
    s = m + n
    return s * (s + 1) // 2 + n

pasangan = [(0, 0), (1, 0), (0, 1), (2, 0), (1, 1),
            (0, 2), (3, 0), (2, 1), (1, 2), (0, 3)]
print("pi:", [pi(m, n) for m, n in pasangan])
keys = {pi(m, n) for m in range(61) for n in range(61)}
print("unique keys over 61x61 pairs:", len(keys) == 61 * 61)

# (3) inverse: float vs isqrt
```

```
s, n = 2 * 10**8, 2 * 10**8 - 1
q = s * (s + 1) // 2 + n
print("q =", q, "s correct =", s)
print("s from float:", int(floor((sqrt(8 * q + 1) - 1) / 2)))
print("s from isqrt:", (isqrt(8 * q + 1) - 1) // 2)

# (4) enumerate positive Q by height k = p + q
seq = []
for k in range(2, 10):
    row = [(p, k - p) for p in range(1, k) if gcd(p, k - p) == 1]
    print("k =", k, "count =", len(row), row)
    seq += row
print("terms 1..8:", seq[:8])
print("total up to k = 9:", len(seq))

# (5) relations on an n-element set: 2^(n*n) in all, Bell numbers equivalence
def ekuivalen(R, n):
    refl = all((a, a) in R for a in range(n))
    sim = all((b, a) in R for a, b in R)
    trans = all((a, c) in R for a, b in R for c in range(n)
                if (b, c) in R)
    return refl and sim and trans

for n in range(1, 5):
    semua = 2 ** (n * n)
    hitung = 0
    for bit in range(semua):
        R = {(a, b) for a in range(n) for b in range(n)
             if bit >> (a * n + b) & 1}
        if ekuivalen(R, n):
            hitung += 1
    print("n =", n, "relations =", semua, "equivalences =", hitung)
```

The output as printed, verbatim:

```
n = 0 |P(A)| = 1 2^n = 1
n = 1 |P(A)| = 2 2^n = 2
n = 2 |P(A)| = 4 2^n = 4
n = 3 |P(A)| = 8 2^n = 8
n = 4 |P(A)| = 16 2^n = 16
pi: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
unique keys over 61x61 pairs: True
q = 20000000299999999 s correct = 200000000
s from float: 200000001
s from isqrt: 200000000
k = 2 count = 1 [(1, 1)]
k = 3 count = 2 [(1, 2), (2, 1)]
k = 4 count = 2 [(1, 3), (3, 1)]
k = 5 count = 4 [(1, 4), (2, 3), (3, 2), (4, 1)]
k = 6 count = 2 [(1, 5), (5, 1)]
k = 7 count = 6 [(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)]
k = 8 count = 4 [(1, 7), (3, 5), (5, 3), (7, 1)]
k = 9 count = 6 [(1, 8), (2, 7), (4, 5), (5, 4), (7, 2), (8, 1)]
terms 1..8: [(1, 1), (1, 2), (2, 1), (1, 3), (3, 1), (1, 4), (2, 3), (3, 2)]
total up to k = 9: 27
n = 1 relations = 2 equivalences = 1
n = 2 relations = 16 equivalences = 2
n = 3 relations = 512 equivalences = 5
n = 4 relations = 65536 equivalences = 15
```

What must be read from that output: the first five lines confirm 2^n for $n = 0$ through 4; the ten values of π are consecutive without gaps, and $61 \times 61 = 3721$ pairs give 3721 unique

keys, which is numerical evidence, not a mathematical proof, because the mathematical proof is the T_s block argument. The last four lines sweep all 2^{n^2} relations for $n = 1$ through 4 and count those that are reflexive, symmetric, and transitive: the result is $1, 2, 5, 15$. There is no line for $n = 5$ because sweeping 2^{25} relations takes too long; its value is 52.

Third, the inverse with faulty floating point. At $q = 20{,}000{,}000{,}299{,}999{,}999$ the formula $\lfloor(\sqrt{8q+1}-1)/2\rfloor$ with 64-bit machine floating point gives $s = 200{,}000{,}001$, whereas the true value is 200,000,000. The error is one, and that is enough to return a completely different pair: with $s = 200{,}000{,}001$ the value $n = q - T_s$ becomes negative. The number $8q + 1$ is of order 10^{17}, whereas 64-bit floating point stores only about 16 decimal digits exactly; `isqrt` works on integers and gives 200,000,000. The lesson is general: inverting an exact numbering formula demands integer arithmetic, as discussed in Chapter 22.

Note for instructors

Four meetings. First: set operations and De Morgan, with each proof written as a chain of equivalences, not as two separate inclusions. Second: Cartesian products, relations, then the filtering into partial orders and equivalence relations. Third: classes, partitions, and functions, with the Lemma on classes proved at the board. Fourth: cardinality up to Cantor's Theorem, plus a sketch of Cantor–Bernstein if time allows.

Two misconceptions appear most often. First, students think a symmetric and transitive relation is automatically reflexive; the counterexample $R = \{(1,1),(2,2),(1,2),(2,1)\}$ on $\{1,2,3\}$ is usually enough, and Exercise 2 asks for a verdict on that claim. Second, students think $\mathbb{Q}$ is larger than $\mathbb{N}$ because of its density; ask for the first ten terms of the height enumeration at the board, then ask for the position of $7/3$.

An activity that works: divide the board into "natural numbers" and "positive rationals", then have two volunteers pair members for two minutes. The naive pairing will fail, and then together build the pairing by height. Students are ready to move on when they can explain why that list misses no rational, and why a similar list fails for $[0,1]$.

2.9 Summary

Level: basic

Summary.

- Union, intersection, difference, and complement are defined through membership conditions; De Morgan's laws are laws of logic transferred to sets, and their version holds for any indexed family.
- $|\mathcal{P}(A)| = 2^{|A|}$ for finite A, via a bijection to the characteristic functions $A \to \{0,1\}$.
- A relation is a subset of $A \times B$; on a set with n members there are 2^{n^2} relations, and four properties filter them into equivalence relations and partial orders.
- Equivalence relations and partitions are two faces of the same object; the classes are always nonempty, mutually disjoint, and cover the whole set.
- A bijective function is exactly one that has a unique two-sided inverse, and on finite sets of equal size the three properties lock together.
- $\mathbb{N} \times \mathbb{N}$ and $\mathbb{Q}$ are countable through explicit numberings, whereas $\mathcal{P}(\mathbb{N})$ and $\mathbb{R}$ are uncountable through Cantor's diagonal argument.
- The supremum is the least upper bound; its existence is guaranteed by the completeness axiom, while the epsilon characterization and its construction continue in Chapter 3.

2.10 Exercises

Level: intermediate

Exercise 2.1.

1. Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C)$ with chains of membership equivalences.
2. Assess this claim: true, false, or not enough evidence. "Every relation on a nonempty set that is symmetric and transitive is necessarily reflexive." If it is false, give the smallest possible counterexample; if there is not enough evidence, state what further evidence would be needed.
3. Prove that if $f : A \to B$ is injective, then $S \mapsto f(S)$ is an injective function from $\mathcal{P}(A)$ to $\mathcal{P}(B)$. Explain where injectivity is used and why the conclusion fails without it.
4. Prove Kuratowski's ordered pair lemma: $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$ if and only if $a = c$ and $b = d$. Handle the case $a = b$ separately.
5. Complete the first part of the counting Lemma: for $f : A \to B$ with A finite, $|f(A)| \leq |A|$ and equality holds only if f is injective.
6. Count the equivalence relations on a set with 5 members that have exactly two classes, then check the answer with Python.
7. Compute $\pi(7, 5)$ with the Cantor pairing formula, then find (m, n) with $\pi(m, n) = 1000$. Check both answers with Python.
8. Determine the position of $7/3$ in the enumeration of positive rationals by height, then determine the number at positions 30 and 41. Check with Python.
9. Prove that if $A_1, A_2, A_3, \ldots$ are each countable and the numbering of every A_n is given explicitly, then $\bigcup_{n \geq 1} A_n$ is countable.
10. Show that $(0, 1)$ is equinumerous with $\mathbb{R}$ through one explicit bijection, and explain why the function we choose must have its domain restricted.

Hints for the three hardest items. Exercise 5: in the induction step write $A = A' \cup \{a\}$ with $a \notin A'$, then compare $f(A)$ with $f(A') \cup \{f(a)\}$; equality of sizes demands $f(a) \notin f(A')$ and $|f(A')| = |A'|$ at the same time. Exercise 6: the number of ways to split n members into two nonempty parts is $(2^n - 2)/2$, since choosing the first part or the second gives the same split. Exercise 9: arrange the numbering of all the A_n as a two-column table, read it diagonal by diagonal as in the proof that $\mathbb{N} \times \mathbb{N}$ is countable, then discard repeated terms.

For researchers: limits and next steps

Four things are genuinely proved here. Cantor's Theorem uses no axiom of choice and is constructive: from any f we build D that lies outside its range. The uncountability of $\mathbb{R}$ depends on the binary expansion of every point of $[0, 1]$; that fact requires completeness and here is only referred to Chapter 3 and Chapter 5.

The Cantor–Bernstein–Schroeder theorem in this chapter is only a sketch. A full proof requires careful bookkeeping of chain depth; the complete formal version is in Avigad–Massot, *Mathematics in Lean*, section 4.3, while the classical treatment of sets, functions, and equivalence relations is in Hoffman and Kunze, *Linear Algebra*, Appendix A.1–A.3.

The primary references for the diagonal and for uncountability are Cantor, "Ueber eine Eigenschaft des Inbegriffes aller reellen algebraischen Zahlen" (1874) and "Ueber eine elementare Frage der Mannigfaltigkeitslehre" (1891).

The most famous open question: is there a cardinal between $|\mathbb{N}|$ and $|\mathbb{R}|$? The continuum hypothesis states that there is none. Godel (1940) showed that the hypothesis is consistent with ZFC, Cohen (1963) showed that its negation is consistent too. So the question cannot be answered inside ZFC.

Two technical limits. First, $\sim$ may not be called an equivalence relation on "the set of all sets", because that collection is not a set; the correct treatment uses classes as metatheoretic objects. Second, the corollary "a countable union of countable sets is countable" requires countable choice when the numberings are not given, and Exercise 9 deliberately requires an explicit numbering.

Map to source texts: Chapter 2 maps to Abbott, Understanding Analysis*, sections "Some Preliminaries" (sets, functions, cardinality) and "Cantor's Theorem"; Hoffman and Kunze,* Linear Algebra*, Appendix A.1 (sets), A.2 (functions and inverses), A.3 (equivalence relations, classes, quotients); Avigad–Massot,* Mathematics in Lean*, Chapter 4 ("Sets and Functions") for extensionality, preimages, and Cantor–Bernstein–Schroeder, and Chapter 6.1 for finite sets.*

Chapter 3

Axioms of the Real Numbers and Supremum

A program searches for the length of the diagonal of a square with side one. It keeps two numbers, $a = 1$ and $b = 2$, with $a^2 - 2 < 0 < b^2 - 2$. At every step it computes the midpoint $m = (a + b)/2$, evaluates $m^2 - 2$, and then replaces a or b by m so that the signs at the two ends stay different. After 40 steps the width of the interval is below 10^{-12}, after 100 steps below 10^{-30}. But the program never produces an answer: no step yields $x^2 - 2 = 0$ exactly, and the two endpoints keep squeezing something that is not a rational number.

That failure is not a mistake in the algorithm. It shows that the rational numbers have holes, and that the set $\{q \in \mathbb{Q} : q^2 < 2\}$ has no least upper bound inside $\mathbb{Q}$.

Chapter 1 taught us to write proofs. Chapter 2 gave us the language of sets and functions, and used $|x|$ and sup as computing tools without asking where they come from. What is still missing is the object that makes those tools legitimate. This chapter builds it, not out of decimals, but out of three groups of statements accepted without proof: the field laws, the order laws, and completeness. Everything else, including the existence of $\sqrt{2}$, is derived from there.

For younger readers (age 10 and up)

Imagine a ruler that has only fraction marks: one half, one quarter, one eighth, and ever closer together. There are no other numbers on that ruler. Now measure the diagonal of a square whose side is one hand-span. The end of the diagonal falls between two marks, and it always does, no matter how close the marks are. We can get as near as we like, but never exactly.

The question is: does that length really exist? The ruler clearly has no answer. Yet the diagonal, if you carry it around or lay it on the floor, has a definite length. That length does not disappear merely because our measuring tool does not contain it.

The same story happens with numbers. If we have only fractional numbers, there is a point on the line that has no name. Mathematicians close that hole with a single promise: every collection of numbers that is bounded above has a lowest roof. In this chapter we will learn why that promise is needed, what its consequences are, and how it gives a name to the diagonal length above, namely root two.

3.1 Field axioms

Level: basic

We do not define the real numbers as endless decimals: that approach makes addition

complicated, and makes the definition of a least upper bound circular, since "the supremum of all decimal cuts" already uses the numbers being built. It is cleaner to state the properties first and then derive the rest.

The first layer is algebraic. We assume two operations that obey the rules we have used since school; the following definition only gives those rules a name.

Definition 3.1 (Field)**.** A set F is called a *field* if addition and multiplication are defined on its members so that: the result of both operations always lies in F; both operations are commutative and associative; there is an element 0 with $0 + x = x$ and an element $1 \neq 0$ with $1 \cdot x = x$; every x has a negative $-x$ with $x + (-x) = 0$; every $x \neq 0$ has a reciprocal $1/x$ with $x \cdot (1/x) = 1$; and $x(y + z) = xy + xz$ holds.

Those axioms mention neither order, nor distance, nor completeness. Other sets therefore satisfy them too. The rational numbers with ordinary addition and multiplication form a field. A reasonable question: what is already contained inside those rules?

Three algebraic statements we usually memorize turn out to be consequences, not axioms; the one most often used is multiplication by zero, which destroys everything.

Theorem 3.1. *In an arbitrary field, for all x, y: (a) $0 \cdot x = 0$; (b) $(-x)y = -(xy)$; (c) $(-x)(-y) = xy$; (d) if $xy = 0$ then $x = 0$ or $y = 0$.*

Proof. (a) From $0 + 0 = 0$ and the distributive law, $0 \cdot x = (0 + 0)x = 0 \cdot x + 0 \cdot x$. Add $-(0 \cdot x)$ to both sides. The left side becomes 0, the right side becomes $0 \cdot x$. Hence $0 = 0 \cdot x$.

(b) Compute $(-x)y + xy = (-x + x)y = 0 \cdot y = 0$, using (a). Hence $(-x)y$ is the negative of xy, and that negative is unique.

(c) Apply (b) twice: $(-x)(-y) = -(x(-y)) = -(-(xy)) = xy$. The last step uses $--a = a$, which is also a consequence of the uniqueness of the negative.

(d) Suppose $x \neq 0$ and $xy = 0$. Multiply by $1/x$: $(1/x)(xy) = 0$, so that $y = 0$. Hence one of the factors is zero. □

Not one of those four statements is assumed; all of them descend from the distributive law and the existence of negatives. This pattern is used throughout the book: state as few rules as possible, then harvest the consequences.

field (*lapangan*) — A set with two arithmetic operations whose rules stay fixed, like a toolbox whose lid always shuts tight: the result of a computation never leaves the box.

Chapter 2 used $\mathbb{Q}$ as a countable set. Here it appears in a different role: as an ordered field. Those two properties are not yet enough, and that is the content of the next section.

3.2 Order axioms

Level: basic

A field alone cannot compare two numbers. We need a relation "less than" that respects addition and multiplication, because without it inequalities cannot be manipulated. The order axioms say three things: any two numbers can be compared, the comparison is transitive, and adding or multiplying by a positive number does not break it.

Definition 3.2 (Ordered field)**.** A field F is called *ordered* if there is a relation $<$ on F such that for all a, b, c: exactly one of $a < b$, $a = b$, $b < a$ holds; if $a < b$ and $b < c$ then $a < c$; if $a < b$ then $a + c < b + c$; and if $a < b$ and $c > 0$ then $ac < bc$.

Almost every sign rule we memorize is a consequence of that definition, not an addition to it. The following proof derives five of them at once; the pattern is the same: add something to both sides, or multiply by a number whose sign is known.

Proposition 3.2. *In every ordered field: (a) if $x > 0$ then $-x < 0$, and conversely; (b) if $x > 0$ and $y < z$ then $xy < xz$; (c) if $x < 0$ and $y < z$ then $xy > xz$; (d) if $x \neq 0$ then $x^2 > 0$, in particular $1 > 0$; (e) if $0 < x < y$ then $0 < 1/y < 1/x$.*

Proof. (a) If $x > 0$, add $-x$ to both sides: $0 > -x$. Conversely, if $x < 0$ then $0 - x > 0$, that is, $-x > 0$.

(b) Since $z > y$, we have $z - y > 0$. Multiply by $x > 0$: $x(z - y) > 0$. Hence $xz = x(z - y) + xy > xy$.

(c) From (a), $-x > 0$. Apply (b) to $-x$ and $y < z$: $(-x)y < (-x)z$. By the previous Theorem, the left side is $-(xy)$ and the right side is $-(xz)$. Hence $-(xy) < -(xz)$, and by (a) again $xy > xz$.

(d) If $x > 0$, the order axiom applied to $y = x > 0$ gives $x^2 > 0$. If $x < 0$ then $-x > 0$, so that $(-x)^2 > 0$; yet $(-x)^2 = x^2$. For $x = 0$ clearly $x^2 = 0$. Since $1 = 1^2$, we obtain $1 > 0$.

(e) Since $y > 0$ and $1/y \neq 0$, from (d) $1/y > 0$; the same holds for $1/x$. Multiply $x < y$ by the positive number $(1/x)(1/y)$: the result is $1/y < 1/x$. □

Two consequences that are used often: $a < b$ is equivalent to $b - a > 0$, and $a < b$ is equivalent to $-b < -a$; the second explains why multiplying by a negative number reverses the direction, because multiplying by -1 reflects the number line about zero. Another consequence: if $0 < a < b$ then $a^2 < b^2$, simply by multiplying $a < b$ by $a > 0$ and then by $b > 0$ and adding the two. Hence comparing squares is equivalent to comparing the original positive numbers; every time we write $q^2 < 2$ for a rational q, we are comparing q with something that need not be rational.

ordered field (*lapangan terurut*) — A field whose members can be arranged like the steps of a staircase, and whose arithmetic rules do not disturb that arrangement.

Note what we do not have yet. We do not know whether there is a positive number smaller than every $1/n$, and we do not know whether $\{q \in \mathbb{Q} : q^2 < 2\}$ has a roof. That is not a question of order but a question of completeness.

3.3 Absolute value and the triangle inequality

Level: intermediate

In Chapter 2 we wrote $|x - y|$ as a distance, and used it to compare errors, without proving that the symbol behaves like a distance. Now its properties are derived from the order. Absolute value discards the sign and leaves the distance to the point zero; the following definition writes two branches, and it is the second branch that is usually used.

Definition 3.3 (Absolute value)**.** For $x \in \mathbb{R}$, $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$. Equivalently, $|x| = \max\{x, -x\}$, and for real numbers $|x| = \sqrt{x^2}$ with the root discussed in the next section.

From that definition it follows directly: $|x| \geq 0$, $|-x| = |x|$, $|x|^2 = x^2$, and $|xy| = |x|\,|y|$. The last is proved by checking the four sign combinations, or by noting that both sides are nonnegative and their squares agree.

Inequalities containing an absolute value are easiest to handle after being converted into two ordinary inequalities. The following lemma is the reason that conversion is legitimate.

Lemma 3.3. *For $c \geq 0$, $|x| \leq c$ if and only if $-c \leq x \leq c$.*

Proof. If $|x| \leq c$: for $x \geq 0$ we have $x = |x| \leq c$ and $-c \leq 0 \leq x$; for $x < 0$ we have $-x = |x| \leq c$, so that $x \geq -c$, and $x < 0 \leq c$. Hence $-c \leq x \leq c$.

Conversely, if $-c \leq x \leq c$: for $x \geq 0$, $|x| = x \leq c$; for $x < 0$, $|x| = -x \leq c$ because $x \geq -c$. In both cases $|x| \leq c$. □

The triangle inequality is the only tool we need to control the difference of two numbers. Its name comes from a picture: the length of one side of a triangle does not exceed the sum of the lengths of the other two sides.

Theorem 3.4 (Triangle inequality)**.** *For all $x, y \in \mathbb{R}$, $|x + y| \le |x| + |y|$. Moreover $||x| - |y|| \le |x - y|$.*

Proof. From the definition, $-|x| \le x \le |x|$ and $-|y| \le y \le |y|$. Add the two pairs: $-(|x| + |y|) \le x + y \le |x| + |y|$. By the previous lemma with $c = |x| + |y|$, we obtain $|x + y| \le |x| + |y|$.

For the second part, write $x = (x - y) + y$ and apply the first part: $|x| \le |x - y| + |y|$, so that $|x| - |y| \le |x - y|$. Swapping the roles of x and y gives $|y| - |x| \le |x - y|$. Since $c \le d$ and $-c \le d$ is equivalent to $|c| \le d$, we have $||x| - |y|| \le |x - y|$. □

As a check, take $x = 3$ and $y = -5$: $|x + y| = 2$ while $|x| + |y| = 8$, so the inequality is slack. For $x = 3$ and $y = 5$ both sides equal 8 and the inequality becomes an equation; that happens exactly when the signs agree or one of them is zero.

Its practical use: bounds on distance become bounds on functions. If $|x - \sqrt{2}| < \delta$, then

$$|x^2 - 2| = |x - \sqrt{2}|\,|x + \sqrt{2}| < \delta\,(2\sqrt{2} + \delta),$$

because $|x + \sqrt{2}| \le |x - \sqrt{2}| + 2\sqrt{2}$. For $\delta = 10^{-3}$ that bound equals 0.002829427, and the true value at $x = \sqrt{2} + 10^{-3}$ is also 0.002829427 since $x^2 - 2 = 2\sqrt{2}\,\delta + \delta^2$ exactly: the bound is tight. For $\delta = 10^{-6}$ the same bound shrinks to 2.828428×10^{-6}.

absolute value (*nilai mutlak*) — The distance of a point from zero on the number line, regardless of which side of zero the point lies on.

The triangle inequality uses only the order axioms, not completeness, so it also holds in $\mathbb{Q}$ and cannot distinguish the two. The distinguishing tool appears in the next section.

3.4 Completeness: the least upper bound

Level: intermediate

Order gives comparison, but it does not give a roof. The set $A = \{1 - 1/n : n \in \mathbb{N}\}$ contains numbers that come ever closer to one without ever equaling one: one is not a member, yet it is the tightest bound. The set $B = \{q \in \mathbb{Q} : q^2 < 2\}$ is bounded, yet it has no tight bound in $\mathbb{Q}$. Both show that a slack bound and a tightest bound need different names.

For younger readers (age 10 and up)

Imagine we must put up a tent roof for a group of children. A roof that is too low: some head sticks out. A roof that is too high: the tent wastes material and wobbles. What we are looking for is the lowest roof that still lets all the children fit. Call it the right roof. Now something strange happens. In that group there is no tallest child; there is always someone a little taller. Yet the right roof still exists, because their heights are bounded: nobody is as tall as a tree. The right roof is determined by the whole group, not by a single child.

In this chapter the right roof has a name: the least upper bound. The property that guarantees a right roof always exists is called completeness. The fractional numbers are incomplete: there are groups of bounded height whose right roof is not among the fractional numbers. A question you can take home: if there is no tallest child, how do you prove that the right roof really is the lowest one?

An upper bound is a ceiling that contains every member; the least upper bound is the lowest ceiling among them. The following definition names both, together with their mirror image below.

Definition 3.4 (Upper bound and supremum)**.** Let $A \subseteq \mathbb{R}$ and $A \neq \varnothing$. A number u is called an *upper bound* of A if $a \leq u$ for every $a \in A$. If A has at least one upper bound, A is called bounded above. A number s is called the *supremum* of A, written $s = \sup A$, if s is an upper bound of A and $s \leq u$ for every upper bound u. Dually, t is called the *infimum* of A, written $t = \inf A$, if t is a lower bound and $t \geq v$ for every lower bound v.

The two conditions in the definition of supremum are often confused: the first says that s is still an upper bound, the second that no upper bound is smaller. For $A = \{1 - 1/n\}$, the number 1 satisfies both while 2 satisfies only the first; that set also has no largest member, so "maximum" and "supremum" are not synonyms.

Read it aloud (auditory learner)

Two voices. The first voice asks, the second answers. Read them aloud alternately with a friend.
"What is that fence?" "The fence is an upper bound: every member of the collection lies below it." "How many fences are there?" "Many. Every higher fence is also a fence." "Then which one do we use?" "The lowest one. Its name is the least upper bound." "What if that lowest fence happens to be occupied by a member of the collection?" "Then it is also a maximum. If it is not occupied, it is still the least upper bound. Two names, two jobs."
Repeat three times, faster each time: a tall fence is not the answer, the lowest fence is the answer.
The line the reader must repeat, once, in a clear voice: an upper bound is a fence; the least upper bound is the lowest fence that still holds all the members in.

Figure 3.1 places both ideas on a single line. The members of $A = \{1 - 1/n\}$ crowd toward one direction without ever reaching the end; every number to the right of one is an upper bound, and the leftmost among them is the supremum.

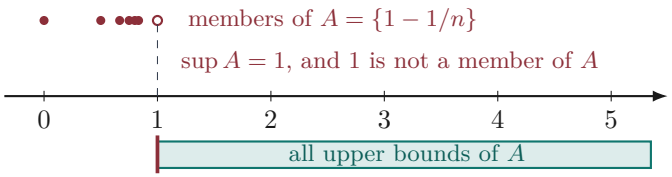


Figure 3.1: The set $A = \{1 - 1/n : n \in \mathbb{N}\}$ on the number line. Every $u \geq 1$ is an upper bound; the smallest, $u = 1$, is $\sup A$. The number 1 is not a member of A, so A has no maximum.

supremum (*supremum (batas atas terkecil)*) — The lowest roof that still covers every member of the set; that roof need not belong to the set.

A supremum, if it exists, is unique; the second condition can be written with epsilon, and that form is the one most often used in proofs.

Lemma 3.5. *(i) If A has a supremum, then its supremum is unique. (ii) A number s is the supremum of A if and only if s is an upper bound of A and for every $\varepsilon > 0$ there exists $a \in A$ with $a > s - \varepsilon$.*

Proof. (i) Suppose s and s' are both suprema. Since s is an upper bound and s' is a supremum, $s' \leq s$. Since s' is an upper bound and s is a supremum, $s \leq s'$. Hence $s = s'$.

(ii) Suppose s is an upper bound and the epsilon condition holds, but there is an upper bound $u < s$. Take $\varepsilon = s - u > 0$. By the epsilon condition there is $a \in A$ with $a > s - \varepsilon = u$, contradicting that u is an upper bound. Conversely, if s is a supremum and there is $\varepsilon > 0$ such that $a \leq s - \varepsilon$ for all $a \in A$, then $s - \varepsilon$ is an upper bound smaller than s, contradicting that s is the least upper bound. □

Step-by-step recipe (procedural learner)

To prove that a number s is the supremum of a set A:

1. Show $A \neq \varnothing$ by naming one of its members.
2. Prove that s is an upper bound: take arbitrary $a \in A$, show $a \leq s$.
3. Prove that no upper bound is smaller: take $\varepsilon > 0$, produce $a \in A$ with $a > s - \varepsilon$, for instance $a = 1 - 1/n$ with $n > 1/\varepsilon$.
4. Conclude: s is an upper bound and every number below s is not an upper bound, so $s = \sup A$.

Order warning: step 2 must be finished before step 3. Choosing $a > s - \varepsilon$ first proves nothing, since s is not yet known to be an upper bound. For $A = \{1 - 1/n\}$, the number $1/2$ passes step 3 but fails step 2.

The following axiom distinguishes $\mathbb{R}$ from $\mathbb{Q}$. It cannot be proved from the field and order laws, because it is a statement about the completeness of the line.

Theorem 3.6 (Least-upper-bound property)**.** *Every nonempty set $A \subseteq \mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.*

Proof. This statement is accepted as the completeness axiom. Other equivalent formulations: nested interval intersections, and convergence of Cauchy sequences. We choose the supremum because it is the most directly usable; the construction of $\mathbb{R}$ from $\mathbb{Q}$ that proves it exists is in the Appendix to Chapter 1 of Rudin. □

The infimum need not be axiomatized: it is the supremum seen in a mirror.

Proposition 3.7. *If A is nonempty and bounded below, then $\inf A = -\sup(-A)$, with $-A = \{-a : a \in A\}$.*

Proof. The set $-A$ is nonempty and bounded above, so $s = \sup(-A)$ exists. For every $a \in A$ we have $-a \leq s$, hence $a \geq -s$: so $-s$ is a lower bound of A. Now take $t > -s$. Then $-t < s$, so $-t$ is not an upper bound of $-A$; there is $a \in A$ with $-a > -t$, that is, $a < t$. Hence t is not a lower bound of A. Therefore $-s$ is the greatest lower bound. □

First consequence of completeness: the natural numbers have no ceiling. It sounds trivial, but without completeness it cannot be proved.

Theorem 3.8 (Archimedean property)**.** *For every $x > 0$ and every $y \in \mathbb{R}$ there exists $n \in \mathbb{N}$ with $nx > y$.*

Proof. Suppose the contrary: there are $x > 0$ and y such that $nx \leq y$ for all n. Then the set $S = \{nx : n \in \mathbb{N}\}$ is nonempty and bounded above by y, so $\alpha = \sup S$ exists. Since $x > 0$, the number $\alpha - x$ is smaller than α and therefore not an upper bound of S. Hence there is m with $mx > \alpha - x$, which gives $(m+1)x > \alpha$. Yet $(m+1)x \in S$, and α is an upper bound of S. Contradiction. □

The reciprocal version is the one most often used: for every $\varepsilon > 0$ there is n with $1/n < \varepsilon$, simply by applying the theorem to $x = \varepsilon$ and $y = 1$. The phrase "for n sufficiently large" is shorthand for that consequence, and now we know it is legitimate.

That property also destroys very small positive numbers: there is no $\delta > 0$ that is smaller than $1/n$ for all n, because $1/\delta$ would become an upper bound of $\mathbb{N}$. Real analysis therefore needs no "infinitesimals".

Second consequence: every open interval, however small, contains a rational number.

Theorem 3.9 (Density of $\mathbb{Q}$)**.** *If $x < y$ are real numbers, then there exists $p \in \mathbb{Q}$ with $x < p < y$.*

Proof. Since $y - x > 0$, the Archimedean property gives $n \in \mathbb{N}$ with $n(y - x) > 1$. Apply Archimedes again to obtain integers $m_1 > nx$ and $m_2 > -nx$; so $-m_2 < nx < m_1$. Since nx is trapped between two integers, there is an integer m with $m - 1 \le nx < m$. Then $nx < m$, and from $m - 1 \le nx$ we get $m \le nx + 1 < ny$. Since $n > 0$, division gives $x < m/n < y$. □

That proof is constructive: it gives a recipe for finding p, namely choosing n large enough and then rounding nx upward. It does not give a small denominator, so efficient approximation still demands other ideas.

3.5 Why the rational numbers are not enough

Level: advanced

The program in the opening fails because the set it squeezes has no roof in $\mathbb{Q}$. Let us prove that, not merely display the numbers. Take $A = \{q \in \mathbb{Q} : q^2 < 2\}$. This set is nonempty, since $1 \in A$, and it is bounded above inside $\mathbb{Q}$: if $q \ge 3/2$ then $q^2 \ge 9/4 > 2$, so $3/2$ is an upper bound.

The intervals that program keeps are drawn as a stack of ever shorter bars (Figure 3.2). Each bar still contains root two and its length halves, while all its endpoints stay rational; the bars narrow onto a single point without ever touching it.

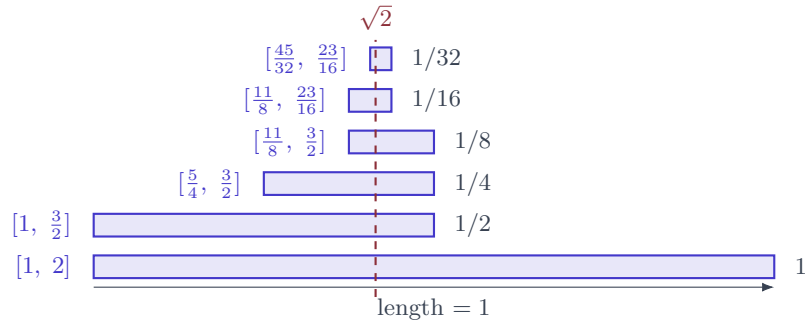


Figure 3.2: The intervals kept by the opening program: $[1, 2]$, $[1, \frac{3}{2}]$, $[\frac{5}{4}, \frac{3}{2}]$, $[\frac{11}{8}, \frac{3}{2}]$, $[\frac{11}{8}, \frac{23}{16}]$, $[\frac{45}{32}, \frac{23}{16}]$. Their lengths fall from 1 to $1/32$; continued without end, the intersection shrinks to a single point, namely $\sqrt{2}$, which is not a rational number.

First step: there is no largest member. Every rational with $q^2 < 2$ can still be nudged a little upward without passing two, and the size of the nudge has a formula.

Lemma 3.10. *If $q \in \mathbb{Q}$, $q > 0$, and $q^2 < 2$, then there exists $h \in \mathbb{Q}$, $h > 0$, with $(q + h)^2 < 2$. In particular $q + h \in A$ and $q + h > q$.*

Proof. Take $h = \dfrac{2 - q^2}{2q + 2}$. Since $q^2 < 2$ and $2q + 2 > 0$, we have $h > 0$ and h rational. Also $h < 1$, because $2 - q^2 < 2q + 2$ is equivalent to $q^2 + 2q > 0$, which is true for $q > 0$. Then $h^2 \le h$, so

$$(q + h)^2 = q^2 + 2qh + h^2 \le q^2 + h(2q + 1) = q^2 + (2 - q^2)\frac{2q + 1}{2q + 2} < q^2 + (2 - q^2) = 2,$$

because $\frac{2q+1}{2q+2} < 1$. □

That formula can be checked numerically. For $q = 1393/985$ we have $2 - q^2 = 1/970225$, so $h = 1/4684660$ and

$$q + h = \frac{6625109}{4684660} = 1.414213411432206\ldots,$$

whose square is $1.999999573074719\ldots < 2$. So $1393/985$ is not the largest member, even though its squared error is already about one part in a million. This procedure can be repeated without end: there is no last step.

Do it by hand (kinesthetic learner)

Materials: 1.5 m of rope, six pieces of tape, a tape measure.
Stick the tape on the floor perpendicular to the wall, at distances $60/n$ cm for $n = 1, \ldots, 6$: 60, 30, 20, 15, 12, and 10 cm. You have only stuck six, but the rule does not stop. Stretch the rope across as a fence, then slide it toward the wall. Each time you stop, check one thing: are all the tape marks on the same side of the rope as the wall? As long as they are, the rope is still a fence.
The rope can be brought arbitrarily close to the wall and still be a fence: 5 cm, 1 cm, 1 mm. What stops it is not the tape but the wall. The collection of tape marks has no smallest member, and not one of them touches the wall. The lowest fence still exists — the wall — and it is not a member of the collection. That is the supremum.

It is not only the maximum that is missing. The least upper bound is missing from the rational numbers as well, and the proof requires one fact about root two.

Proposition 3.11. *The set $A = \{q \in \mathbb{Q} : q^2 < 2\}$ has no supremum in $\mathbb{Q}$.*

Proof. Suppose $p \in \mathbb{Q}$ is the supremum of A. Since $1 \in A$, we have $p \ge 1 > 0$. We show that $p^2 = 2$, which is impossible because no rational number has square two.

If $p^2 < 2$, the previous lemma gives $q = p + h \in A$ with $q > p$, contradicting that p is an upper bound. If $p^2 > 2$, take $k = (p^2 - 2)/(2p) > 0$, which is rational and smaller than p since $p^2 - 2 < 2p^2$. For every $q \in A$,

$$(p - k)^2 = p^2 - 2pk + k^2 > p^2 - 2pk = 2 > q^2,$$

so $q < p - k$ for $q > 0$, and for $q \le 0$ this is clear because $p - k > 0$: the number $p - k$ is still an upper bound of A, but smaller than p. That contradicts p being the least upper bound. Hence $p^2 = 2$.

Now $p = m/n$ with m, n integers having no common factor. From $m^2 = 2n^2$, m^2 is even, so m is even; write $m = 2r$. Then $4r^2 = 2n^2$, that is, $n^2 = 2r^2$, so n is even as well. Both are even, contradicting the choice of m, n. □

Inside $\mathbb{R}$ that does not happen. The same set, viewed this time as a set of real numbers, is bounded above by 2, so by the completeness axiom it has a supremum. That supremum is the number we name $\sqrt{2}$. The following proof closes the two remaining possibilities, and its shape imitates the earlier lemma with q replaced by a real number.

Theorem 3.12 (Existence of root two)**.** *There is exactly one $s > 0$ with $s^2 = 2$.*

Proof. Let $E = \{t \in \mathbb{R} : t > 0,\ t^2 < 2\}$. This set is nonempty because $1 \in E$, and it is bounded above by 2: if $t > 2$ then $t^2 > 4 > 2$, so $t \notin E$. By the least-upper-bound property, $s = \sup E$ exists, and $s \geq 1 > 0$ because $1 \in E$.

Suppose $s^2 < 2$. Take $h = \frac{1}{2}\min\{1, (2 - s^2)/(2s + 1)\}$. Then $0 < h < 1$ and $h(2s + 1) < 2 - s^2$: if the first value is the minimum, $h = \frac{1}{2}$ and $\frac{1}{2}(2s + 1) \leq \frac{1}{2}(2 - s^2) < 2 - s^2$; if the second is, $h(2s + 1) = \frac{1}{2}(2 - s^2)$. Since $h < 1$, we have $h^2 < h$, so

$$(s + h)^2 = s^2 + 2sh + h^2 < s^2 + h(2s + 1) < s^2 + (2 - s^2) = 2.$$

Hence $s + h \in E$ although $s + h > s$, contradicting that s is an upper bound of E. Therefore $s^2 \geq 2$.

Suppose $s^2 > 2$. Take $k = (s^2 - 2)/(2s) > 0$; then $k < s$ because $s^2 - 2 < 2s^2$. For every $t \in E$ with $t \geq s - k$ we have

$$t^2 \geq (s - k)^2 = s^2 - 2sk + k^2 > s^2 - 2sk = 2,$$

so $t \notin E$. Hence $s - k$ is an upper bound of E smaller than s, contradicting that s is the least upper bound. Since $s^2 < 2$ and $s^2 > 2$ are both impossible, $s^2 = 2$.

Uniqueness: if $s, t > 0$ and $s^2 = t^2 = 2$, then $(s - t)(s + t) = 0$. Since $s + t > 0$, $s = t$. □

The number s is irrational, by the parity argument above: the hole in $\mathbb{Q}$ is real, and $\mathbb{R}$ closes it with exactly one number. What closes it is not order and not the field laws, but completeness. The opening program fails not from lack of precision, but because its number system does not contain the object it is looking for.

One thing must be stated honestly: we have not built $\mathbb{R}$ from $\mathbb{Q}$, we have only assumed that $\mathbb{R}$ satisfies three groups of axioms. The construction is known. Dedekind cuts $\mathbb{Q}$ in two: a *cut* is a subset $\alpha \subset \mathbb{Q}$ that is nonempty, not all of $\mathbb{Q}$, closed downward, and without a largest member. That last property is exactly our lemma above. Cantor uses equivalence classes of rational Cauchy sequences; both appeared in 1872. The details are in the Appendix to Chapter 1 of Rudin.

That construction explains the standing of decimals. For $x > 0$, take n_0 the largest integer with $n_0 \leq x$, then n_1 the largest digit with $n_0 + n_1/10 \leq x$, then n_2 with $n_0 + n_1/10 + n_2/100 \leq x$, and so on. That cut is bounded above by x, so its supremum exists and equals x. A decimal is not the raw material of the real numbers but a way of naming a supremum; completeness guarantees that the name exists, not that the name can be written with fractions.

completeness axiom (*aksioma kelengkapan*) — The promise that the number line has no holes: every collection that is bounded above always has a lowest roof.

The existence of $\sqrt{2}$ makes the computational problem sensible: s can be computed without knowing its digits in advance. Newton's iteration, $x_{k+1} = \frac{1}{2}(x_k + 2/x_k)$, stays rational when x_k is rational, so the whole process runs inside $\mathbb{Q}$. From $s^2 = 2$ we get

$$x_{k+1} - s = \frac{(x_k - s)^2}{2x_k}, \qquad e_{k+1} = e_k^2 \cdot \frac{s}{2x_k},$$

with $e_k = |x_k - s|/s$. The right side of the first form is nonnegative, so every step stays above s; there the factor $\frac{s}{2x_k}$ does not exceed $1/2$, so $e_{k+1} \leq e_k^2/2$ and the number of correct digits roughly doubles at each step.

One warning: we have not proved that $x_k \to s$. That needs the tools of Chapter 4. What can be said now: since $e_1 < 0.061$, from $e_{k+1} \leq e_k^2/2$ one can derive $e_k \leq 2^{-(2^k - 1)}$ for $k \geq 1$, and that is enough to design an honest numerical test.

3.6 Intervals, accumulation points, and open sets

Level: advanced

With the supremum in hand, we can name the pieces of the number line, like the finish rules of a race: the line is the same, only whether the line itself counts differs. Those names are used again in Chapter 11 and Chapter 13, so we take only what we need without building a topology in full.

An interval is a piece of the line with no holes. The following definition writes the nine possible forms; all are determined by two endpoints and by the decision whether each endpoint counts.

Definition 3.5 (Interval)**.** Let $a < b$ be real numbers. The set $(a, b) = \{x : a < x < b\}$ is called an open interval, $[a, b] = \{x : a \le x \le b\}$ a closed interval, and $[a, b)$, $(a, b]$ are mixed forms. The sets $(a, \infty) = \{x : x > a\}$, $[a, \infty)$, $(-\infty, b)$, $(-\infty, b]$, and $\mathbb{R}$ itself are called unbounded intervals. All nine forms are called intervals.

The feature that marks them all: if two points lie in the interval, the whole segment between them lies there too, so intervals are called hole-free. The list above is not proved complete; the classification only inspects upper bound, lower bound, and whether that bound is included. The notion of a point being "near" is expressed by a neighborhood, that is, a small open interval around one point.

Definition 3.6 (Neighborhood, interior point, open, closed)**.** For $\varepsilon > 0$, the neighborhood of a point x is $V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon)$. A point x is called an interior point of $S \subseteq \mathbb{R}$ if $V_\varepsilon(x) \subseteq S$ for some $\varepsilon > 0$; S is called open if every member is an interior point. A point x is called an accumulation point of S if every $V_\varepsilon(x)$ contains a member of S different from x. A set S is called closed if its complement is open.

A few examples organize the intuition. The interval $(0, 1)$ is open and not closed, because every neighborhood of 0 meets it. The interval $[0, 1]$ is closed and not open, while $[0, 1)$ is neither. The set $\mathbb{Q}$ is also neither, because every neighborhood contains rationals and irrationals. The set $\{1/n : n \in \mathbb{N}\}$ is not closed: 0 lies in its complement, but every neighborhood of 0 contains some $1/n$. The set $\mathbb{Z}$ is closed, because its complement is the union of the open intervals $(k, k + 1)$.

So "closed" is not the negation of "open": $\mathbb{R}$ and $\varnothing$ are both open and closed at once, and the complement of an open set is always closed. The density of $\mathbb{Q}$ makes these examples sharper. Every neighborhood contains rationals and irrationals, so no point is an interior point of $\mathbb{Q}$ or of its complement: that set is dense and hole-ridden everywhere at once. Completeness therefore cannot be replaced by density.

A neighborhood is itself open, and that is what makes the definition of an interior point reusable.

Proposition 3.13. *(i) Every neighborhood $V_\varepsilon(x)$ is open. (ii) The union of an arbitrary family of open sets is open. (iii) The intersection of finitely many open sets is open.*

Proof. (i) Take $y \in V_\varepsilon(x)$, so $|y - x| < \varepsilon$. Set $\delta = \varepsilon - |y - x| > 0$. For every $z \in V_\delta(y)$, $|z - x| \le |z - y| + |y - x| < \delta + |y - x| = \varepsilon$ by the triangle inequality. Hence $V_\delta(y) \subseteq V_\varepsilon(x)$.

(ii) If x lies in the union, it lies in one of its members, and since that member is open there is a neighborhood of x contained in it, hence also contained in the union.

(iii) If x lies in the finite intersection $S_1 \cap \cdots \cap S_n$, then for each i there is $\varepsilon_i > 0$ with $V_{\varepsilon_i}(x) \subseteq S_i$. Take $\varepsilon = \min\{\varepsilon_1, \ldots, \varepsilon_n\} > 0$. Then $V_\varepsilon(x)$ is contained in every S_i, hence contained in the intersection. □

Item (iii) holds only for finite intersections: $\bigcap_n (-1/n, 1/n) = \{0\}$ is not open, because for $x \neq 0$ the Archimedean property gives n with $1/n < |x|$.

3.7 Worked examples

Level: intermediate

Three computations promised earlier are carried out here: Newton's iteration, rational approximation to error below 10^{-6}, and a check that $\{q : q^2 < 2\}$ has no maximum. All the numbers are computed with exact fractions.

Starting from $x_0 = 1$, the iteration $x_{k+1} = (x_k + 2/x_k)/2$ gives the following fractions; the denominators grow, and that is the price paid for accuracy.

k	x_k	decimal	relative error
0	1	1	2.92893×10^{-1}
1	3/2	1.5	6.06602×10^{-2}
2	17/12	1.41666666667	1.73461×10^{-3}
3	577/408	1.41421568627	1.50183×10^{-6}
4	665857/470832	1.41421356237	1.12774×10^{-12}
5	886731088897/627013566048	1.41421356237	6.35896×10^{-25}

The relative error is measured against $\sqrt{2} = 1.414213562373095\ldots$. The reduction factor climbs by itself: from step 0 to 1 about 4.8, then 35, 1155, and 1.3×10^6 — that is the signature of squaring. The ratio e_{k+1}/e_k^2 takes the values 0.7071, 0.4714, 0.4991, 0.49999925, and 0.5, in agreement with the formula $e_{k+1} = e_k^2 \cdot \frac{s}{2x_k}$ because x_k approaches s from above. At steps 4 and 5 the decimals agree to eleven digits; the next digits do not appear because we print twelve significant digits, not because the error stops falling. The fraction at step 5 has twelve digits with error 6.36×10^{-25}, and one additional digit demands roughly one more step.

The second method uses rational approximations that spend denominators more economically, namely the continued fraction of $\sqrt{2}$:

$$1, \quad \frac{3}{2}, \quad \frac{7}{5}, \quad \frac{17}{12}, \quad \frac{41}{29}, \quad \ldots$$

with $p_{k+1} = p_k + 2q_k$ and $q_{k+1} = p_k + q_k$ starting from $(p_0, q_0) = (1, 1)$. The following table lists their absolute errors.

p/q	decimal	absolute error	$p^2 - 2q^2$
1/1	1	4.14214×10^{-1}	-1
3/2	1.5	8.57864×10^{-2}	1
7/5	1.4	1.42136×10^{-2}	-1
17/12	1.41666666667	2.45310×10^{-3}	1
41/29	1.41379310345	4.20459×10^{-4}	-1
99/70	1.41428571429	7.21519×10^{-5}	1
239/169	1.41420118343	1.23789×10^{-5}	-1
577/408	1.41421568627	2.12390×10^{-6}	1
1393/985	1.41421319797	3.64404×10^{-7}	-1
3363/2378	1.41421362489	6.25218×10^{-8}	1

The first approximation whose error is below 10^{-6} is 1393/985 with error 3.64×10^{-7}; the one before it, 577/408, is still 2.12×10^{-6}. The last column explains why: since $p^2 - 2q^2 = \pm 1$,

$$\left| \frac{p}{q} - \sqrt{2} \right| = \frac{|p^2 - 2q^2|}{q^2 \left(\frac{p}{q} + \sqrt{2} \right)} < \frac{1}{2\sqrt{2}\, q^2},$$

so the error falls like the inverse square of the denominator. For $q = 985$ that bound is 3.644×10^{-7} and the true error is 3.64404×10^{-7}: almost exactly, because $\frac{p}{q}$ is already

very close to $\sqrt{2}$. The denominator 985 is far smaller than 627013566048 at Newton step 5, although its error is only about a million times larger — that is the difference between a method chasing quadratic error and one chasing the smallest rational error.

Figure 3.3 displays that error column as a descending staircase. The sign of $p^2 - 2q^2$ alternates, so the fractions jump from below to above root two, and the ratio of two successive errors approaches $5.8284 = (1+\sqrt{2})^2$.

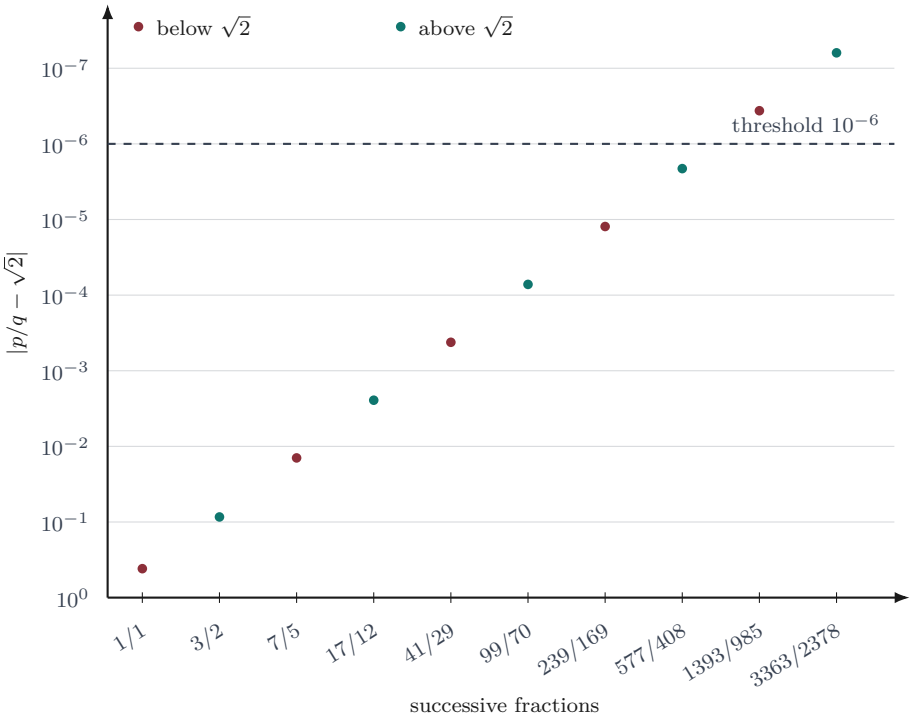


Figure 3.3: A staircase of rational approximations: the error $|p/q - \sqrt{2}|$ for ten successive fractions, logarithmic scale. The 10^{-6} threshold is first crossed by 1393/985, with error 3.644×10^{-7}; after that each step falls by a factor of 5.8284.

The third check concerns the absence of a maximum. One more step from the previous q gives

$$q' = \frac{149856914148961}{105964844887080} = 1.41421349985133266\ldots,$$

and its square is still below two. Both numbers are rational, both lie in A, and each is larger than the one before: there is no reason for this process to stop.

Challenge (problem solver)

Consider $S = \{q \in \mathbb{Q} : q \geq 0,\ q^2 \leq 2\}$. At first glance this set has a largest member: the number whose square is exactly two, since the sign $\leq$ seems to include that bound in the set. Prove that S has no maximum, then determine $\sup S$ in $\mathbb{R}$ and explain why the answer is not a member of S.

Hint 1: members of S must be rational. First check whether your candidate for largest is even allowed to be a member of the set.

Hint 2: for every $q \in S$ with $q > 0$ we have $q^2 < 2$ strictly, because $q^2 = 2$ is impossible for a rational. So the shift lemma $h = (2 - q^2)/(2q + 2)$ applies and produces a larger member of S; the unattainable bound is $\sqrt{2}$.

If you get stuck, go back to the section “Why the rational numbers are not enough” and to the lemma on shifting.

Finally, the Archimedean property in numbers. For $\varepsilon = 10^{-6}$ we need $n > 10^6$. At $n = 10^6$ the reciprocal is exactly 10^{-6}, which is not less than ε; at $n = 10^6 + 1$ the reciprocal is 9.99999×10^{-7}, which does qualify. One unit in n decides whether the choice is legitimate, and that is why strict inequalities must be respected.

For $x = 1393/985$, compute $h = (2 - x^2)/(2x + 2)$ and check by hand that $(x + h)^2 < 2$. Then compute the relative error $|x - \sqrt{2}|/\sqrt{2}$ and compare it with $1/(2\sqrt{2}\,x^2)$. Explain why the two are almost equal, and why that fact does not contradict the absence of a maximum in A.

3.8 Computational lab

Level: advanced

The following code runs with Python 3 and `mpmath` for high-precision arithmetic; Newton's iteration uses `Fraction`, so there is no rounding before printing.

Computational lab

```
from fractions import Fraction as F
import mpmath as mp
mp.mp.dps = 40
root2 = mp.sqrt(2)

print("Newton for root two, x0 = 1")
x = F(1)
for k in range(6):
    v = mp.mpf(x.numerator) / x.denominator
    print(k, f"{x.numerator}/{x.denominator}", mp.nstr(v, 12),
          mp.nstr(abs(v - root2) / root2, 6))
    x = (x + F(2) / x) / 2

print()
print("Successive rational approximations, error < 1e-6 ?")
p, q = 1, 1
for k in range(10):
    v = mp.mpf(p) / q
    print(f"{p}/{q}", mp.nstr(v, 12), mp.nstr(abs(v - root2), 6),
          abs(v - root2) < mp.mpf(10) ** -6)
    p, q = p + 2 * q, p + q

print()
print("Test: {rational q : q^2 < 2} has no maximum")
q = F(1393, 985)
for k in range(2):
    h = (2 - q * q) / (2 * q + 2)
    q = q + h
    print(q, mp.nstr(mp.mpf(q.numerator) / q.denominator, 18), q * q < 2)
```

The output obtained, verbatim:

```
Newton for root two, x0 = 1
0 1/1 1.0 0.292893
1 3/2 1.5 0.0606602
2 17/12 1.41666666667 0.00173461
3 577/408 1.41421568627 1.50183e-6
4 665857/470832 1.41421356237 1.12774e-12
5 886731088897/627013566048 1.41421356237 6.35896e-25

Successive rational approximations, error < 1e-6 ?
1/1 1.0 0.414214 False
```

```
3/2 1.5 0.0857864 False
7/5 1.4 0.0142136 False
17/12 1.41666666667 0.0024531 False
41/29 1.41379310345 0.000420459 False
99/70 1.41428571429 7.21519e-5 False
239/169 1.41420118343 1.23789e-5 False
577/408 1.41421568627 2.1239e-6 False
1393/985 1.41421319797 3.64404e-7 True
3363/2378 1.41421362489 6.25218e-8 True

Test: {rational q : q^2 < 2} has no maximum
6625109/4684660 1.41421341143220639 True
149856914148961/105964844887080 1.41421349985133266 True
```

Three things are readable from that output. The Newton error stops being printed at twelve digits at step 4, even though the true value still falls to 6.4×10^{-25}: what limits it is the print width, not the mathematics. The 10^{-6} threshold is crossed exactly at 1393/985, where the column of `True` begins. Both numbers in the maximum test are also `True`: both are rational and both have squares below two.

That code proves nothing. It checks that the formulas in this chapter give the numbers stated and that the direction of the inequalities is right; the proofs remain in the lemmas and theorems. Program output is data, not argument.

Note for instructors

The teaching order I suggest: the root-finding program in the opening, then the field axioms with $0 \cdot x = 0$ proved on the board, then order, absolute value, and completeness. Do not postpone completeness; the entire content of the chapter depends on it.
Two misconceptions appear often. First, students think a supremum must be a member of the set, so they reject $\sup\{1 - 1/n\} = 1$. Second, students think completeness can be proved from the field and order laws, whereas $\mathbb{Q}$ satisfies both and fails completeness. Both disappear once students see those two examples side by side.
A blackboard activity that works: each student takes $q = 1.4$, computes $h = (2-q^2)/(2q+2)$ and checks $(q+h)^2 < 2$; the differing results are precisely what shows the formula works for every q. A readiness signal: students can explain why A has no maximum without computing.

3.9 Summary

Level: basic

Summary.

- The real numbers rest on three groups of axioms: field, order, completeness. The first gives algebra, the second comparison, the third closes the holes.
- From the field laws we derive $0 \cdot x = 0$, $(-x)y = -(xy)$, and the absence of zero divisors.
- From the order laws we derive the sign rules, including nonnegative squares and $1 > 0$.
- Absolute value turns distance into inequality: $|x| \le c$ is equivalent to $-c \le x \le c$, and $|x + y| \le |x| + |y|$.
- Completeness guarantees a supremum for every nonempty set that is bounded above, and then yields the Archimedean property and the density of $\mathbb{Q}$.
- $\mathbb{Q}$ satisfies the field and order laws but is not complete: $\{q : q^2 < 2\}$ has no maximum; in $\mathbb{R}$ its supremum is $\sqrt{2}$.

3.10 Exercises

Level: intermediate

Exercise 3.1.

1. Prove that in an ordered field, $a < b$ is equivalent to $-b < -a$, and conclude that multiplying an inequality by -1 reverses its direction.
2. Prove $a < \frac{a+b}{2} < b$ for $a < b$, then explain why that result does not require completeness whereas $\sup\{q : q^2 < 2\}$ does.
3. Compute sup and inf of the set $S = \{2 - 1/n : n \in \mathbb{N}\}$, then check both conditions of the definition. Also determine whether S has a maximum.
4. Assess this claim: "Since every positive number is greater than $1/n$ for some n, there is no smallest positive real number." True, false, or not yet proved? Name the part that the proof supports.
5. With $h = (2 - q^2)/(2q + 2)$, compute $q + h$ for $q = 1$ and $q = 7/5$, then check that both still lie in $A = \{q \in \mathbb{Q} : q^2 < 2\}$. Compare the sizes of the increases.
6. Show that if $|x| \le 2$ and $|y| \le 2$ then $|x^3 - y^3| \le 12\,|x - y|$, and say where the triangle inequality is used.
7. Prove that if S is open then its complement is closed, and conversely, using the definition of a neighborhood.
8. Compute five steps of Newton's iteration for $\sqrt{3}$ starting from $x_0 = 2$, and report the relative error at each step. Check whether the ratio e_{k+1}/e_k^2 approaches $1/2$.
9. Show $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$ and explain why that does not contradict the closedness of finite intersections.
10. Prove that between any two real numbers there is always an irrational number.

Hints. Item 4: the claim contains two different statements, one about every ε and one about the existence of a minimum. Item 8: use $e_{k+1} = e_k^2 \cdot \frac{\sqrt{3}}{2x_k}$ and $x_k \ge \sqrt{3}$. Item 10: apply the density of $\mathbb{Q}$ to $x/\sqrt{2}$ and $y/\sqrt{2}$, then multiply back.

For researchers: limits and next steps

What is genuinely proved: the completeness axiom does not follow from the field and order laws, because $\mathbb{Q}$ satisfies both and fails the third. The existence of a complete ordered field is proved by Rudin via Dedekind cuts, and there it is also stated without proof that any two complete ordered fields are isomorphic; that claim is not used here. Primary sources: Rudin, *Principles of Mathematical Analysis*, Chapter 1 (Definition 1.12, Proposition 1.18, Theorems 1.19–1.21, the Appendix on cuts); Abbott, *Understanding Analysis*, Chapter 1 (The Axiom of Completeness and its consequences).
What remains open: the effective content of Newton's iteration. Without an explicit initial bound, $e_{k+1} \le e_k^2/2$ does not yet give a step count; that requires Chapter 4 and, for general functions, Kantorovich theory. The chapter's scope: topology only as far as intervals and neighborhoods, and $\mathbb{R}$ assumed rather than built.

Map to source texts: Chapter 3 in Rudin, Principles of Mathematical Analysis*: Chapter 1 (Ordered Sets, Fields, The Real Field, The Least-Upper-Bound Property, the Archimedean property, density of $\mathbb{Q}$, existence of n-th roots, Appendix on Dedekind cuts). In Abbott,* Understanding Analysis*: Chapter 1 (The Axiom of Completeness and Consequences of Completeness); open sets and accumulation points from Chapter 3, not developed here.*

Chapter 4

Sequences and Limits

A small program prints one number per line. The rule is only one: the next term is the square root of two plus the previous term, starting from one. What appears on the screen: 1.000000, then 1.732051, 1.931852, 1.982890, 1.995718, 1.998929, and so on. On the twelfth line the screen shows 1.999999738545. That number appears to stop moving. But appearing and being true are two different things: the program does not know whether the thousandth line will still shift, and it also does not know which number this run is heading toward.

The second example is more mischievous. The sequence $(1 + 1/n)^n$ for $n = 10$ has value 2.5937424601, for $n = 100$ has value 2.7048138294, for $n = 1000$ has value 2.7169239322, and for $n = 10^7$ has value 2.7182816941. The first six digits appear settled at 2.71828. But the true value at $n = 10^7$ is 2.7182816925, so the last seven digits in that machine output are rounding error, not a sign of closeness. A table can deceive in two directions at once: it makes a divergent sequence look calm, and it makes a convergent sequence look off course. Chapter 22 will explain why machine arithmetic behaves that way; this chapter prepares the tools that let us not depend on tables.

Chapter 3 already supplied the raw material: the field axioms, the order axioms, and the supremum property. But the supremum only works once the set of terms has been collected; it does not tell us whether the sequence stops moving. What we need is a definition that can be checked on paper, and then theorems that turn that check into estimated work: the ε-N definition, the algebraic properties of limits, the squeeze theorem, monotone sequences, subsequences, the Cauchy criterion, and the limit superior.

For younger readers (age 10 and up)

Have you ever walked toward a door under a strange rule: each step covers only half of the distance that remains? The first step takes you halfway. The second cuts half of what is left, so you are already three quarters of the way. The third adds another eighth. Every step moves you forward, yet there is always a remainder you have not reached. If that rule is kept up, you never actually touch the door.

Even so, there are questions that can be answered with certainty. How many steps are needed so that the distance to the door is less than a hand's width? The answer can be computed before you set out, and it does not change when the room becomes twice as long: you only need a few extra steps. So there are two different things here. "Never arrives" and "as close as one likes can be approached" are not two contradictory statements. In fact both are true at once.

Now the question that makes one curious. What if the rule is changed slightly, say each step covers a third of the remaining distance, or half plus a little? Can you still guess where that journey ends, and can its speed still be computed? In this chapter we will

know why such questions have definite answers, and why we never obtain the answers merely by watching numbers on a screen.

4.1 The definition of a limit and how to use it

LEVEL: BASIC

sequence (*barisan*) — a run of numbers given consecutive numbers, like the house numbers along one side of a street that never ends

A limit answers one question: how close the later terms gather, and from when on. In everyday language, a sequence tends to L when, after a few initial terms are discarded, all the remaining terms are as close as one likes to L.

Definition 4.1 (Convergence)**.** A sequence (x_n) of real numbers *converges* to $L \in \mathbb{R}$ if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that for every $n > N$ we have $|x_n - L| < \varepsilon$. We write $x_n \to L$, and L is called the *limit* of the sequence. A sequence that does not converge to any real number is called *divergent*.

The order of the quantifiers in that definition decides everything. The value ε is chosen first, then N follows and may depend on ε. Reversing that order produces a far stronger and usually false statement: there are sequences that satisfy "for every n there is N" vacuously, because $N = n$ can always be used. A healthy way to read it is as a game between two parties. The challenger names an ε, however small. The responder must answer with an N that works for all later terms. The sequence converges when the responder always has an answer, for every challenge.

Figure 4.1 captures that game for $x_n = 1 + 1/n$. Two bands centered at $L = 1$ are drawn with different widths, and each has its own threshold: $N = 2$ for $\varepsilon = 0.5$, $N = 10$ for $\varepsilon = 0.1$. Nothing about the sequence changes; only the challenger's demand changes.

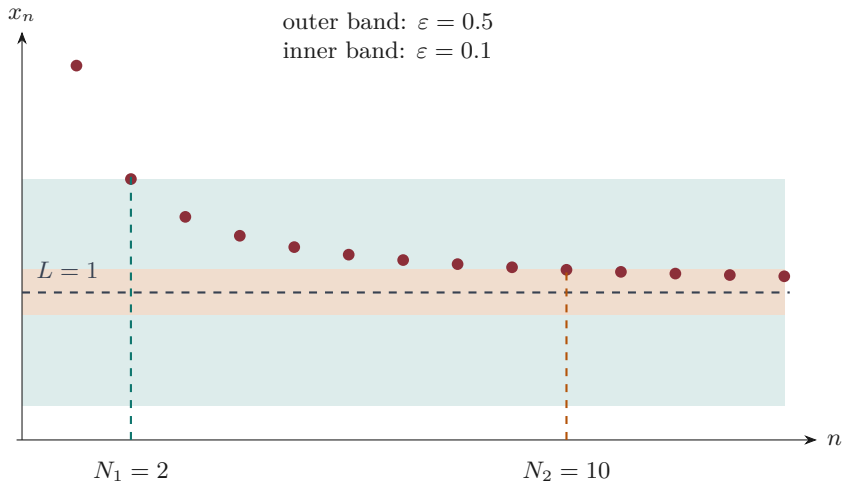


Figure 4.1: The sequence $x_n = 1 + 1/n$ entering the ε band around $L = 1$. With $\varepsilon = 0.5$ every term after $n = 2$ is already inside the band; with $\varepsilon = 0.1$ the threshold shifts to $n = 10$. A narrower band demands a more distant threshold.

Read it aloud (auditory learner)

Two voices take turns. The first asks, the second answers. Read it aloud slowly, pause at every period, and do not add words in between.

"Longer and longer?" "Closer and closer." "How close?" "As close as requested." "What if one thousandth is requested?" "Wait a thousand steps, then measure." "What if one hundredth of a millionth?" "Wait one hundred million steps. Patience costs nothing." "So it will arrive?" "No. It never arrives." "Then what is the use of waiting?" "Because the distance shrinks, and the one who asks is free to choose how small." "And if the requests stop?" "Then there is nothing to prove."
The line repeated three times, more slowly each time: close is not arrived; close is a distance we are allowed to set ourselves.

For $x_n = 1/n$ the answer is immediate: take N a natural number with $N > 1/\varepsilon$, which exists by the Archimedean property in Chapter 3. For $n > N$ we have $|1/n - 0| = 1/n < 1/N < \varepsilon$. If the challenger asks for $\varepsilon = 10^{-3}$, the answer is $N = 1000$; at $n = 1000$ the error is exactly 10^{-3} and does not yet meet the strict condition, whereas at $n = 1001$ the error is 0.000999. This one-term difference is not pedantry: the definition uses $n > N$, not $n \geq N$, so the N that is chosen must genuinely pass the threshold.

Do it by hand (kinesthetic learner)

Materials: duct tape, a tape measure, chalk, and one sheet of notes. Mark a number line on the floor two meters long: zero at the left end, two at the right end, and the number one exactly in the middle. Stick down two horizontal strips as wide as a foot, centered at the number one: a wide band from one half to one and a half meters, a narrow band from nine tenths to one point one meters.
Now walk to the position one plus one over n, for n equal to one, two, three, up to fourteen. Each time, note whether your heel lands in the wide band, in the narrow band, or outside both, then walk back so the marks are not stepped on.
What should be visible: the first two steps are still outside the narrow band, the tenth step stops exactly at its edge, and after that no heel comes out again. A narrower band demands more steps, while the number one is never touched by a heel.

The use of a definition is felt most strongly when we want to prove that something does *not* converge. The negation of the definition reads: there is $\varepsilon > 0$ such that for every N there is $n > N$ with $|x_n - L| \geq \varepsilon$. Note that L is still a free name there; to refute convergence we must do it for every L.

Example 4.1. The sequence $x_n = (-1)^n$ is divergent.

Proof. Suppose $x_n \to L$ for some L. Take $\varepsilon = 1$; there is N such that $|x_n - L| < 1$ for all $n > N$. Choose an even $n > N$; then $x_n = 1$ and $|1 - L| < 1$. The next term $n + 1$ also exceeds N and is odd, so $x_{n+1} = -1$ and $|-1 - L| < 1$. The triangle inequality gives

$$2 = |1 - (-1)| \leq |1 - L| + |L - (-1)| < 1 + 1 = 2,$$

an impossibility. So no L satisfies the definition. □

The proof just given shows a pattern that will appear often: to refute, we choose an ε smaller than the distance between two groups of terms. The even and odd groups are a fixed distance 2 apart, so no single N can separate them when $\varepsilon < 1$.

The first property worth recording is uniqueness. In everyday words: a sequence cannot stop in two different places at once.

Theorem 4.1 (Uniqueness of the limit)**.** *If $x_n \to L$ and $x_n \to M$, then $L = M$.*

Proof. Suppose $L \neq M$ and let $\varepsilon = |L-M|/2 > 0$. There is N_1 with $|x_n - L| < \varepsilon$ for $n > N_1$, and N_2 with $|x_n - M| < \varepsilon$ for $n > N_2$. For $n > \max(N_1, N_2)$ both inequalities hold at once, so

$$|L - M| \leq |L - x_n| + |x_n - M| < 2\varepsilon = |L - M|.$$

The inequality $|L - M| < |L - M|$ is impossible, so the supposition is false. □

The phrase "for n sufficiently large" means: there is N such that the stated property holds for all $n > N$. With that vocabulary, the definition of a limit in brief: x_n is as close as one likes to L when n is sufficiently large.

The second property: a convergent sequence cannot run off to infinity.

Theorem 4.2. *Every convergent sequence is bounded.*

Proof. Suppose $x_n \to L$. Take $\varepsilon = 1$; there is N with $|x_n - L| < 1$ for $n > N$, so $|x_n| \leq |x_n - L| + |L| < |L| + 1$. Only the first term up to term N is still uncontrolled, and there are finitely many of those. Hence

$$M = \max\{|x_1|, |x_2|, \ldots, |x_N|, |L| + 1\}$$

is a bound that works for every n. □

The converse does not hold: $(-1)^n$ is bounded but divergent. Boundedness is a necessary condition, not a sufficient one; from a table that looks calm we may only conclude boundedness.

4.2 Algebraic properties of limits

Level: intermediate

In everyday language: a limit is polite toward addition, multiplication, and division, as long as the divisor does not tend to zero.

Before proving those properties we need one small control: a sequence tending to a nonzero number eventually stays away from zero.

Lemma 4.3. *If $y_n \to y$ with $y \neq 0$, then there is N such that $|y_n| > |y|/2$ for all $n > N$.*

Proof. Take $\varepsilon = |y|/2 > 0$. There is N with $|y_n - y| < |y|/2$ for $n > N$. Then $|y_n| \geq |y| - |y_n - y| > |y| - |y|/2 = |y|/2$. □

The consequence for computation: limits can be computed term by term, and the only exception is division by zero.

Theorem 4.4 (Algebraic properties of limits)**.** *If $x_n \to x$ and $y_n \to y$, then $x_n + y_n \to x + y$, $x_n y_n \to xy$, and $cx_n \to cx$ for every $c \in \mathbb{R}$. If $y \neq 0$ and $y_n \neq 0$ for all n, then $x_n / y_n \to x/y$.*

Proof. For the sum, take $\varepsilon > 0$. Choose N_1 with $|x_n - x| < \varepsilon/2$ for $n > N_1$ and N_2 with $|y_n - y| < \varepsilon/2$ for $n > N_2$. For $n > \max(N_1, N_2)$, $|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \varepsilon$. The scalar multiple case is similar and shorter: for $c = 0$ the sequence is constant; for $c \neq 0$ use $\varepsilon/|c|$.

For the product, write the difference as two parts,

$$x_n y_n - xy = x_n(y_n - y) + (x_n - x)y.$$

Because (x_n) converges it is bounded: there is $M > 0$ with $|x_n| \leq M$ for all n. Choose N so that for $n > N$ we have $|y_n - y| < \varepsilon/(2M)$ and $|x_n - x| < \varepsilon/(2(|y| + 1))$. Then

$$|x_n y_n - xy| \leq M|y_n - y| + |x_n - x|\,|y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The denominator $|y|+1$ is used so that it is never zero when $y=0$; it stays a positive constant, so it does not change the conclusion.

For the quotient it suffices to show $1/y_n \to 1/y$, then use the product result. By the lemma, for $n > N_1$ we have $|y_n| > |y|/2$, so

$$\left|\frac{1}{y_n}-\frac{1}{y}\right| = \frac{|y-y_n|}{|y_n y|} \le \frac{2}{|y|^2}\,|y_n-y|.$$

The right side is smaller than ε as long as $|y_n-y| < \varepsilon|y|^2/2$, and that can be guaranteed by choosing N_2. $\square$

Note the technique that has just appeared: adding and subtracting the same term to split a difference into two small parts. Almost every convergence proof involving a product uses it, and the threshold taken is always the largest one.

The next property concerns order, not operations. In everyday language: a limit cannot jump over a fence that all terms obey.

Theorem 4.5 (Order property)**.** *If $x_n \to x$ and $x_n \ge 0$ for all n, then $x \ge 0$. More generally, if $x_n \le y_n$ for all sufficiently large n, $x_n \to x$, and $y_n \to y$, then $x \le y$.*

Proof. Suppose $x < 0$ and take $\varepsilon = -x > 0$. There is N with $|x_n - x| < \varepsilon$ for $n > N$, so in particular $x_n < x + \varepsilon = 0$. That contradicts $x_n \ge 0$. For the second statement, apply the first part to $y_n - x_n \ge 0$, whose limit is $y - x$ by the previous theorem. $\square$

Note that the legitimate conclusion is $x \ge 0$, not $x > 0$: the sequence $1/n$ is entirely positive but its limit is zero. The order property also must not be used backwards. From $x_n < y_n$ we only obtain $x \le y$; a strict gap can vanish in the limit.

With the algebraic properties, limits of rational sequences can be computed as usual. For

$$x_n = \frac{3n^2+2n}{n^2-5},$$

divide numerator and denominator by n^2: $x_n = \dfrac{3+2/n}{1-5/n^2}$. Because $1/n \to 0$ and $1/n^2 \to 0$, the numerator tends to 3 and the denominator to $1 \neq 0$, so $x_n \to 3$. The numerical values do creep there: $x_{10} = 3.368421$, $x_{100} = 3.021511$. Another example,

$$y_n = \frac{2n^2-n}{3n^2+7} \longrightarrow \frac{2}{3},$$

with $y_{10} = 0.618893$ and $y_{100} = 0.663179$; both are still far from $2/3$ at $n = 100$ because the error is about $1/n$.

The algebraic properties speak only about limits that already exist; applying them to a sequence not yet known to converge is an expensive mistake.

4.3 The squeeze theorem and comparison sequences

Level: intermediate

Sometimes the limit of a sequence cannot be computed directly, but it can be trapped from above and from below by two sequences with the same limit. In everyday language: when two guards walk close together with a child between them, the child must stop in the same place as they do.

Theorem 4.6 (Squeeze theorem)**.** *Suppose $a_n \le x_n \le b_n$ for all sufficiently large n. If $a_n \to L$ and $b_n \to L$, then $x_n \to L$.*

Proof. Take $\varepsilon > 0$. Choose N_1 so that $|a_n - L| < \varepsilon$ for $n > N_1$, that is $a_n > L - \varepsilon$, and N_2 so that $b_n < L + \varepsilon$ for $n > N_2$. Also take N_3 so that the squeezing holds for $n > N_3$. For $n > \max(N_1, N_2, N_3)$,

$$L - \varepsilon < a_n \le x_n \le b_n < L + \varepsilon,$$

so $|x_n - L| < \varepsilon$. □

Before using it we need one comparison sequence that will keep appearing: powers that shrink to zero.

Lemma 4.7. *If $0 < q < 1$, then $q^n \to 0$.*

Proof. Write $1/q = 1 + h$ with $h > 0$. Bernoulli's inequality, proved by induction in Chapter 1, gives $(1 + h)^n \ge 1 + nh$. Since both sides are positive,

$$0 < q^n = \frac{1}{(1+h)^n} \le \frac{1}{1+nh}.$$

Given $\varepsilon > 0$, the Archimedean property gives n with $1 + nh > 1/\varepsilon$, that is $1/(1 + nh) < \varepsilon$. By the squeeze theorem, $q^n \to 0$. □

Now an example that cannot be settled by dividing by the highest power. For $x_n = (2^n + 3^n)^{1/n}$, the largest term inside the brackets is 3^n; factor that power out:

$$x_n = 3\left(1 + \left(\tfrac{2}{3}\right)^n\right)^{1/n}.$$

Because $(2/3)^n \to 0$ and $1 \le 1 + (2/3)^n \le 2$, we obtain $1 \le x_n/3 \le 2^{1/n}$. It remains to show $2^{1/n} \to 1$: write $2^{1/n} = 1 + a_n$ with $a_n \ge 0$, then $2 = (1 + a_n)^n \ge 1 + na_n$, so $0 \le a_n \le 1/n$. The squeeze theorem closes both ends, and $x_n \to 3$. Python numbers confirm the direction: $x_1 = 5$, $x_2 = 3.605551$, $x_5 = 3.075152$, $x_{10} = 3.005162$, and $x_{100} = 3$.

For younger readers (age 10 and up)

Two school guards walk side by side along a corridor, and a pupil walks between them. The guard on the left steps toward the exit, the guard on the right does too, and the two get closer until their shoulders almost touch. The pupil squeezed in the middle does not need to know the way; he only needs to stay between the two guards. Wherever those guards stop, there the pupil stops as well.

Now the question that makes one curious. What if the left guard stops earlier than the right guard: can we still guess where the pupil stops? And what if their distance is never zero, only smaller from day to day? Does the pupil still have a definite place to stop, or does he keep walking with no end?

It turns out the answer does not depend on the faces of the guards, only on two things: where the left guard is heading, and where the right guard is heading. If both head for the same place, the pupil in the middle has no other choice. That way of thinking, "squeezed between two parties", is what mathematicians use for sequences whose formula is too complicated to compute directly.

4.4 Monotone sequences and the monotone convergence theorem

Level: intermediate

In everyday language: a sequence that moves in only one direction, and never crosses a certain fence, must stop somewhere. This is where completeness works most cheaply, because the limit need not be guessed in advance.

monotone sequence (*barisan monoton*) — a staircase whose steps only go up, never down

The everyday language is simple: one direction only, never turning back.

Definition 4.2 (Monotone sequence)**.** A sequence (x_n) is called *increasing* if $x_n \le x_{n+1}$ for all n, and *decreasing* if $x_n \ge x_{n+1}$ for all n. A sequence is called monotone if it is increasing or decreasing.

A rough picture: what moves in one direction without crossing a fence must stop, and it stops exactly at the nearest fence it never crosses.

Theorem 4.8 (Monotone convergence)**.** *An increasing sequence bounded above converges, and its limit is* $\sup\{x_n : n \in \mathbb{N}\}$.

Proof. Suppose (x_n) is increasing and bounded above. Its set of terms is nonempty and bounded above, so by the supremum property in Chapter 3 it has a supremum; call it $s = \sup\{x_n\}$. Take $\varepsilon > 0$. Because $s - \varepsilon$ is not an upper bound, there is N with $x_N > s - \varepsilon$. For $n > N$, monotonicity gives $x_n \ge x_N > s - \varepsilon$, while $x_n \le s$ by the definition of supremum. So $|x_n - s| < \varepsilon$. □

The immediate consequence: for such a sequence the limit can be read off from its set of terms, without computing term by term.

Corollary 4.9. *A decreasing sequence bounded below converges, and its limit is the infimum of its set of terms.*

Proof. Apply the theorem to $(-x_n)$, which is increasing and bounded above, then use the scalar multiple property of limits and the identity $\sup\{-x_n\} = -\inf\{x_n\}$. □

Note that this theorem does not say the limit equals the bound we use. The bound 3 in the following example is only a proof device; the limit is in fact e.

The first application is the recursive sequence that opens this chapter. Suppose $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$.

First, all terms lie between 1 and 2. For $n = 1$ this is clear. If $1 \le x_n \le 2$, then $3 \le 2 + x_n \le 4$, so $1 \le \sqrt{3} \le x_{n+1} \le 2$. By induction, $1 \le x_n \le 2$ for all n.

Second, the sequence is increasing. Because $x_n \le 2$,

$$x_{n+1} \ge x_n \iff 2 + x_n \ge x_n^2 \iff (2 - x_n)(1 + x_n) \ge 0,$$

and the last inequality is true since $x_n \le 2$ and $x_n \ge 1$.

By the monotone convergence theorem, (x_n) converges; call its limit L. Because $x_n \le 2$ for all n, the order property gives $L \le 2$. Now square the relation: $x_{n+1}^2 = 2 + x_n$. The left side tends to L^2 by the product property, the right side to $2 + L$, so $L^2 = 2 + L$, or $(L - 2)(L + 1) = 0$. Since $L \ge 1$, the only possibility is $L = 2$.

Note that we never use continuity of the square root; the algebraic properties of limits and the order property are enough, so the machinery of Chapter 11 is not needed here.

The second application is the sequence $y_n = (1 + 1/n)^n$. The binomial expansion gives

$$y_n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^{n} \frac{1}{k!} \prod_{j=1}^{k-1} \left(1 - \frac{j}{n}\right),$$

with the empty product equal to 1 for $k = 0$ and $k = 1$. Every factor $1 - j/n$ increases as n grows, so every term with index k increases; the sum also gains one new positive term. Hence (y_n) is increasing. Every factor is at most 1, so

$$y_n \le \sum_{k=0}^{n} \frac{1}{k!} \le 1 + 1 + \frac{1}{2} + \frac{1}{4} + \cdots = 3,$$

because $1/k! \le 1/2^{k-1}$ for $k \ge 1$, which is checked by induction. The sequence is increasing and bounded above by 3, so it converges. We give its limit the name e.

Figure 4.2 displays that sequence for $n = 1$ up to $n = 30$. The points rise without ever turning back, the dashed line at 3 is the fence used in the proof, and the dotted line at 2.71828 is where it actually stops. The gap between fence and destination stays there throughout the figure; the fence works because it is never crossed, not because it is reached.

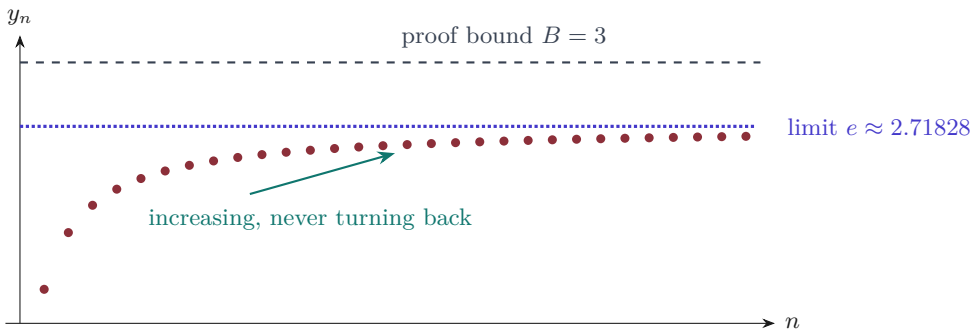


Figure 4.2: The sequence $y_n = (1 + 1/n)^n$ for $n = 1$ up to $n = 30$. The points rise and never turn back; the fence $B = 3$ is never touched; the value approached is $e \approx 2.71828$, not the fence.

4.5 Subsequences, Bolzano–Weierstrass, and the Cauchy criterion

Level: advanced

In everyday language: from a bounded sequence one can always pick a run of children that stops somewhere, even when the parent run is chaotic.

subsequence (*subbarisan*) — sorting out some letters from a pile without changing their order

A subsequence is formed by choosing only some terms: the indices used grow larger, but any number of terms may be skipped.

Definition 4.3 (Subsequence)**.** Let $n_1 < n_2 < n_3 < \cdots$ be increasing natural numbers. The sequence $(x_{n_k})_{k \in \mathbb{N}}$ is called a *subsequence* of (x_n).

Because $n_k \ge k$ for every k, which is checked by induction, a subsequence cannot be "slower" than its parent sequence in terms of index.

In everyday language: choosing some terms does not change the stopping place, it only slows down or speeds up its arrival.

Proposition 4.10. *If $x_n \to L$, then every subsequence (x_{n_k}) also converges to L.*

Proof. Take $\varepsilon > 0$ and N with $|x_n - L| < \varepsilon$ for $n > N$. For $k > N$ we have $n_k \ge k > N$, so $|x_{n_k} - L| < \varepsilon$. □

The consequence is simple but most often used to refute: two different stopping places are already enough.

Corollary 4.11. *If (x_n) has two subsequences with different limits, then (x_n) is divergent.*

Proof. If (x_n) converged, all its subsequences would converge to the same limit, and uniqueness of the limit forbids two different values. □

This corollary explains the example $(-1)^n$ without an argument by impossibility: the even subsequence tends to 1, the odd subsequence to -1. The way it works also explains why divergence is more often provable than convergence.

To prove Bolzano–Weierstrass we need the following lemma. Its content: every sequence, however chaotic, contains a monotone subsequence.

Lemma 4.12. *Every real sequence has a monotone subsequence.*

Proof. Call an index n a *peak* if $x_n \geq x_m$ for all $m > n$, that is, if no later term is larger. There are two possibilities.

If there are infinitely many peaks, list them as $n_1 < n_2 < \cdots$. Because n_1 is a peak, $x_{n_1} \geq x_{n_2}$; because n_2 is a peak, $x_{n_2} \geq x_{n_3}$; and so on. The result is a decreasing subsequence.

If there are only finitely many peaks, take n_1 beyond the last peak. Because n_1 is not a peak, there is $n_2 > n_1$ with $x_{n_2} > x_{n_1}$. Because n_2 also lies beyond the last peak, it is not a peak either, so there is $n_3 > n_2$ with $x_{n_3} > x_{n_2}$. The process continues without end and produces an increasing subsequence. □

The content of this theorem: from a trapped sequence one can always pick some terms that genuinely tend to a single point.

Theorem 4.13 (Bolzano–Weierstrass)**.** *Every bounded real sequence has a convergent subsequence.*

Proof. By the lemma, (x_n) has a monotone subsequence. A subsequence of a bounded sequence is also bounded. A bounded monotone sequence converges by the monotone convergence theorem. Hence that subsequence converges. □

An alternative proof uses repeated bisection of intervals and the nested interval property; both proofs lean on completeness, which we borrowed above through the supremum.

Cauchy sequence (*barisan Cauchy*) — passengers crowding ever tighter in a train corridor, without knowing where the exit door is

Now consider a different problem. The definition of a limit requires us to name L first, whereas in many computations it is L that we are seeking. All we have is a list of numbers. We therefore need a measure of closeness between the terms themselves, without naming a destination.

Definition 4.4 (Cauchy sequence)**.** A sequence (x_n) is called *Cauchy* if for every $\varepsilon > 0$ there is N such that for all $m, n > N$ we have $|x_n - x_m| < \varepsilon$.

Note the difference in the order of ideas. In the definition of a limit all terms are compared with one fixed number L; in the Cauchy definition terms are compared with other terms, with no outside number. That is why the Cauchy definition makes sense in any space that has a distance, whereas the limit definition demands a candidate destination.

Theorem 4.14 (Cauchy criterion)**.** *A real sequence converges if and only if it is Cauchy.*

Proof. Suppose $x_n \to L$ and take $\varepsilon > 0$. There is N with $|x_n - L| < \varepsilon/2$ for $n > N$. For $m, n > N$,

$$|x_n - x_m| \leq |x_n - L| + |L - x_m| < \varepsilon.$$

So convergence gives Cauchy.

Conversely, suppose (x_n) is Cauchy. Take $\varepsilon = 1$; there is N_0 such that $|x_n - x_m| < 1$ for $m, n > N_0$. Choosing $m = N_0 + 1$ gives $|x_n| < |x_{N_0+1}| + 1$ for $n > N_0$, so (x_n) is bounded. By Bolzano–Weierstrass there is a subsequence $x_{n_k} \to L$. Take $\varepsilon > 0$; choose N_1 with $|x_n - x_m| < \varepsilon/2$ for $m, n > N_1$, and K with $|x_{n_k} - L| < \varepsilon/2$ for $k > K$. Choose $k > K$ that also satisfies $n_k > N_1$; then for $n > N_1$,

$$|x_n - L| \le |x_n - x_{n_k}| + |x_{n_k} - L| < \varepsilon.$$

So $x_n \to L$. □

The second direction uses Bolzano–Weierstrass, which uses the monotone convergence theorem, which uses the supremum. That chain cannot be cut: inside $\mathbb{Q}$ this theorem fails. The sequence of decimal truncations 1; 1.4; 1.41; 1.414; 1.4142; ... tending to the square root of two is Cauchy, since the difference of two terms is at most 10^{-k} when both pass term k, yet no rational number is its limit. If that sequence converged to L, then $a_k^2 \to L^2$; on the other hand $a_k^2 \to 2$, so $L^2 = 2$. Chapter 3 already showed that no rational has square 2. So "Cauchy" and "convergent" are two different properties; what makes them agree is completeness, and that is the content of the supremum axiom.

Challenge (problem solver)

The sequence $h_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ grows very slowly: $h_{10} = 2.928968$, $h_{100} = 5.187378$, $h_{1000} = 7.485471$, and $h_{10^6} = 14.392727$. The difference of two consecutive terms, namely $1/(n+1)$, tends to zero, so this sequence passes the shallowest test, the one that most often deceives: its terms get closer together. The question is whether (h_n) converges. Prove your answer, and explain why $h_{10^6} < 15$ is not a proof that the sequence is bounded.

Hint 1: do not inspect terms one by one. Sum in blocks whose lengths double, that is one term, two terms, four terms, eight terms, and so on.

Hint 2: the k-th block contains 2^{k-1} terms, and the smallest term in it is at least $1/2^k$; so every block contributes at least $1/2$. Compare the result with h_{2^m} and see what happens as m grows.

If you are stuck, return to the Monotone Convergence Theorem: there an increasing sequence is proved convergent precisely because a fence exists, and here it is precisely the fence that is missing.

4.6 Limit superior and limit inferior

Level: advanced

In everyday language: for a sequence that oscillates, we can still name the highest and the lowest place it oscillates to, even though the sequence itself has no limit.

limit superior (*limit superior*) — the highest peak still visited by the sequence as often as one likes after passing any point

Throughout this section (x_n) is bounded. For every n form the tail set $\{x_k : k \ge n\}$ and write

$$s_n = \sup\{x_k : k \ge n\}, \qquad t_n = \inf\{x_k : k \ge n\}.$$

Because the tails shrink, (s_n) is decreasing and (t_n) is increasing; both are bounded, so both converge by the corollary to the monotone convergence theorem.

In everyday language: what we track is not where the sequence is, but how high and how low it still visits after any point.

Definition 4.5 (Limit superior and inferior)**.** For a bounded sequence (x_n),

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} s_n \qquad \text{and} \qquad \liminf_{n\to\infty} x_n = \lim_{n\to\infty} t_n.$$

The first number is called the *limit superior*, the second the *limit inferior*.

Because $s_n \geq t_n$ for every n and limits preserve order, we always have $\liminf x_n \leq \limsup x_n$. The two can differ greatly, and precisely that difference is the measure of the chaos of the sequence.

Example 4.2. For $x_n = (-1)^n + 1/n$, the tail contains the even terms $1 + 1/k$ and the odd terms $-1 + 1/k$. The supremum of the n-th tail comes from the smallest even term still available, so $s_1 = 1.5$, $s_2 = 1.5$, $s_3 = 1.25$, $s_4 = 1.25$, $s_5 = 1.166667$, and so on decreasing toward 1. The infimum of the tail behaves differently: the odd terms $-1 + 1/k$ are always present in every tail and decrease toward -1 without ever reaching it, so $t_n = -1$ for all n, even though not a single term equals -1. Hence $\limsup x_n = 1$ and $\liminf x_n = -1$, while the sequence is divergent.

Figure 4.3 plots the n-th term against n for the same sequence. The run of even points clings near 1, the run of odd points clings near -1, and the distance between the two groups is fixed at 2. No single horizontal line can be approached by all the points, however far we walk.

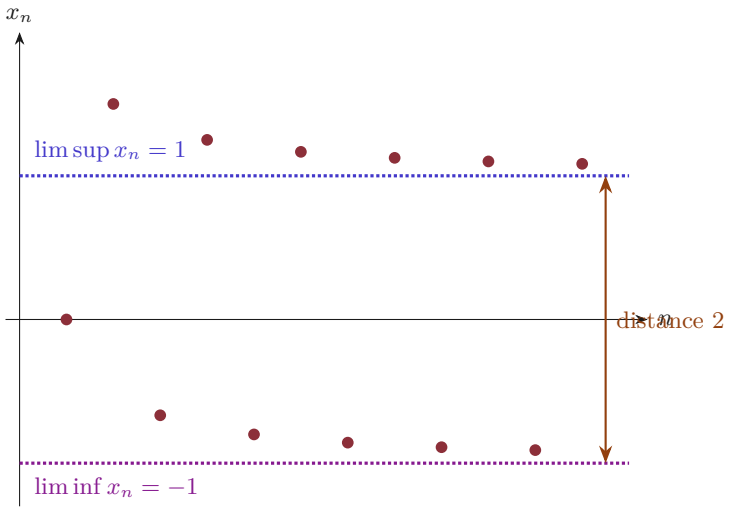


Figure 4.3: Graph of the n-th term against n for $x_n = (-1)^n + 1/n$. The even points cling near 1, the odd points near -1, and the distance between the two groups is fixed at 2. No horizontal band contains all the terms, so this sequence is divergent.

In everyday language: the highest value still visited by the sequence as often as one likes really is visited by some of its terms, not merely approached from afar.

Theorem 4.15. *For a bounded sequence (x_n), its limit superior is the largest subsequential limit:* $\limsup x_n$ *equals the supremum of the set of all limits of convergent subsequences, and that value itself is attained by some subsequence.*

Proof. Write $S = \limsup x_n$. First, every subsequential limit is at most S. If $x_{n_k} \to L$, then for every k we have $x_{n_k} \leq s_{n_k}$, and because $n_k \geq k$ and (s_n) is decreasing, $s_{n_k} \leq s_k$. So $x_{n_k} \leq s_k$, and taking limits gives $L \leq S$.

Second, we build a subsequence tending to S, starting from $n_0 = 0$. For every k, because $s_n \to S$ and (s_n) is decreasing, there is $M \geq k$ with $s_M < S + 1/k$. The tail $\{x_n : n \geq M\}$

has supremum s_M and contains indices as large as we like, so there is $n_k > \max(k, n_{k-1})$ with $n_k \geq M$ and $x_{n_k} > s_M - 1/k$. For that choice

$$S - \frac{1}{k} \leq s_M - \frac{1}{k} < x_{n_k} \leq s_M < S + \frac{1}{k},$$

so $|x_{n_k} - S| < 1/k$. Hence $x_{n_k} \to S$. □

The consequence: equality of limit inferior and limit superior becomes the cheapest way to check convergence without guessing the limit in advance.

Corollary 4.16. *A bounded sequence (x_n) converges to L if and only if* $\liminf x_n = \limsup x_n = L$.

Proof. If $x_n \to L$, every subsequence also tends to L, so the set of subsequential limits contains only L and its supremum is L; in the same way its infimum is L. Conversely, if both values equal L, then $t_n \leq x_n \leq s_n$ and both fences tend to L, so the squeeze theorem gives $x_n \to L$. □

For unbounded sequences the convention $\limsup x_n = +\infty$ is used when (x_n) is not bounded above; the arithmetic laws for ∞ do not hold automatically, and we do not use them here.

4.7 Worked examples

Level: intermediate

We return to the two opening sequences, this time with numbers that can be rechecked and with the intermediate steps written down.

Step-by-step recipe (procedural learner)

1. Write the difference. Form $|x_n - L|$, then simplify until a single fraction in n is left; for the example below, $|x_n - 3| = 5/(n+2)$.
2. Solve for n. Solve $5/(n+2) < \varepsilon$ for $n > 5/\varepsilon - 2$, and name that threshold $g(\varepsilon)$.
3. Choose N. Take $N = \lfloor g(\varepsilon) \rfloor$, the smallest admissible integer, then check the error at $n = N$ and $n = N + 1$ before writing anything.
4. Close with the sentence of the definition: for every $\varepsilon > 0$ that N is chosen for, every $n > N$ then satisfies $|x_n - 3| < \varepsilon$.

Order warning: ε is named first, N answers. Setting up N and then looking for a matching ε reverses the game; the result is usually an N that depends on n, not on ε, and at that point the proof collapses.

For $x_n = (3n+1)/(n+2)$,

$$|x_n - 3| = \left| \frac{3n + 1 - 3n - 6}{n+2} \right| = \frac{5}{n+2}.$$

The inequality $5/(n+2) < \varepsilon$ is equivalent to $n > 5/\varepsilon - 2$. Because the condition is strict, the largest integer still below that threshold is already admissible: $N = \lfloor 5/\varepsilon - 2 \rfloor$ is the smallest choice that satisfies the definition. For $\varepsilon = 10^{-2}$ the threshold is 498 and $N = 498$. Test the boundary: at $n = 498$ the error is $5/500 = 0.01$, which is *not* smaller than ε; at $n = 499$ the error is $5/501 = 0.00998004$, which just meets it. For $\varepsilon = 10^{-4}$ we obtain $N = 49998$, with error 0.0001 at $n = 49998$ and 9.9998×10^{-5} at $n = 49999$.

The terms move very slowly: $n = 10$ gives 2.583333, $n = 1000$ gives 2.995010, and an error of 10^{-4} is only reached at term 49999. Four digits before the decimal point are already correct long before the error is small.

For $x_n = (3n + 1)/(n + 2)$, compute a valid N for $\varepsilon = 10^{-3}$ with the formula $N = \lfloor 5/\varepsilon - 2 \rfloor$, then check the error at $n = N$ and $n = N + 1$. Explain why the error at $n = N$ may be exactly equal to ε while the definition is still satisfied.

For $y_n = (1 + 1/n)^n$, 40-digit arithmetic gives error 0.124539 at $n = 10$, 0.00135790 at $n = 10^3$, and 1.35914×10^{-7} at $n = 10^7$. The product of n with its error is nearly constant: 1.245394, 1.357896, then creeping to $1.3591409 = e/2$. So the error falls like $1/n$, not $1/n^2$; one extra digit after the decimal point demands an n ten times larger. This is where the machine table at the opening of the chapter breaks: at $n = 10^7$ float64 reports 1.34327×10^{-7}, because a small relative error in $1 + 1/n$ is amplified by the power n.

For the recursion $x_{n+1} = \sqrt{2 + x_n}$ with $x_1 = 1$, the error relative to 2 is 2.679×10^{-1} at $n = 2$, 1.711×10^{-2} at $n = 4$, and 2.615×10^{-7} at $n = 12$. The ratio of two consecutive errors settles at 0.250000.

The number 1/4 is no coincidence. From the relation,

$$x_{n+1} - 2 = \frac{(2 + x_n) - 4}{\sqrt{2 + x_n} + 2} = \frac{x_n - 2}{\sqrt{2 + x_n} + 2},$$

so

$$|x_{n+1} - 2| = \frac{|x_n - 2|}{\sqrt{2 + x_n} + 2} \leq \frac{|x_n - 2|}{2 + \sqrt{2}} = 0.292893\,|x_n - 2|,$$

because $x_n \geq 1$ gives $\sqrt{2 + x_n} \geq \sqrt{3}$. The bound 0.292893 holds for all n, and the actual ratio tends to 1/4 because the denominator tends to 4. The practical consequence: error 10^{-6} is reached at $n = 12$, 10^{-9} at $n = 17$, and machine precision around $n = 27$; far cheaper than bisecting an interval again and again.

4.8 Computational lab: four checks

Level: advanced

truncation error (*galat pemotongan*) — the difference between a portrait and the actual face

Four claims in this chapter can be checked with one short script: the explicit threshold N, the error rate of $(1 + 1/n)^n$, machine precision, and the error ratio of the recursion.

Computational lab

```
import math, mpmath as mp

def N_for(eps):
    N = int(5/eps - 2)
    while 5/(N+2) >= eps: N += 1
    while N > 0 and 5/(N+2) < eps: N -= 1
    return N

print("== 1. explicit N for (3n+1)/(n+2) -> 3 ==")
for eps in (1e-2, 1e-4):
    N = N_for(eps)
    print("eps=%.0e  N=%d  err(N)=%.6g  err(N+1)=%.6g"
          % (eps, N, abs((3*N+1)/(N+2)-3), abs((3*N+4)/(N+3)-3)))

print("== 2. (1+1/n)^n in float64 ==")
for k in range(1, 8):
    n = 10**k
```

```
    print("n=1e%d  %.16f  e-y=%.3e" % (k, (1+1/n)**n, math.e-(1+1/n)**n))

print("== 3. true error (mpmath, 40 digits) ==")
mp.mp.dps = 40
for k in (5, 6, 7):
    n = 10**k
    y = (1+mp.mpf(1)/n)**n
    print("n=1e%d  e-y=%s  (float64: %.3e)"
          % (k, mp.nstr(mp.e-y, 6), math.e-(1+1/n)**n))

print("== 4. recursion x_{n+1}=sqrt(2+x_n), x_1=1 ==")
x, xs = 1.0, []
for _ in range(30):
    xs.append(x); x = math.sqrt(2+x)
for n in (1, 2, 3, 4, 5, 6, 11, 12, 13, 14):
    e = abs(xs[n-1]-2)
    r = "" if n == 1 else "  ratio=%.6f" % (e/abs(xs[n-2]-2))
    print("n=%2d  x=%.12f  error=%.3e%s" % (n, xs[n-1], e, r))
e6 = next(n for n in range(1, 31) if abs(xs[n-1]-2) < 1e-6)
e9 = next(n for n in range(1, 31) if abs(xs[n-1]-2) < 1e-9)
print("error < 1e-6 first at n=%d ; error < 1e-9 at n=%d" % (e6, e9))
```

The output, as it stands:

```
== 1. explicit N for (3n+1)/(n+2) -> 3 ==
eps=1e-02  N=498  err(N)=0.01  err(N+1)=0.00998004
eps=1e-04  N=49998  err(N)=0.0001  err(N+1)=9.9998e-05
== 2. (1+1/n)^n in float64 ==
n=1e1  2.5937424601000023  e-y=1.245e-01
n=1e2  2.7048138294215285  e-y=1.347e-02
n=1e3  2.7169239322355936  e-y=1.358e-03
n=1e4  2.7181459268249255  e-y=1.359e-04
n=1e5  2.7182682371922975  e-y=1.359e-05
n=1e6  2.7182804690957534  e-y=1.359e-06
n=1e7  2.7182816941320818  e-y=1.343e-07
== 3. true error (mpmath, 40 digits) ==
n=1e5  e-y=1.35913e-5  (float64: 1.359e-05)
n=1e6  e-y=1.35914e-6  (float64: 1.359e-06)
n=1e7  e-y=1.35914e-7  (float64: 1.343e-07)
== 4. recursion x_{n+1}=sqrt(2+x_n), x_1=1 ==
n= 1  x=1.000000000000  error=1.000e+00
n= 2  x=1.732050807569  error=2.679e-01  ratio=0.267949
n= 3  x=1.931851652578  error=6.815e-02  ratio=0.254333
n= 4  x=1.982889722748  error=1.711e-02  ratio=0.251074
n= 5  x=1.995717846477  error=4.282e-03  ratio=0.250268
n= 6  x=1.998929174953  error=1.071e-03  ratio=0.250067
n=11  x=1.999998954179  error=1.046e-06  ratio=0.250000
n=12  x=1.999999738545  error=2.615e-07  ratio=0.250000
n=13  x=1.999999934636  error=6.536e-08  ratio=0.250000
n=14  x=1.999999983659  error=1.634e-08  ratio=0.250000
error < 1e-6 first at n=12 ; error < 1e-9 at n=17
```

Three things need to be read from that output. First, the error at $n = N$ is exactly equal to ε in both lines; that is why the definition uses the strict condition $n > N$. Second, the $e - y$ column shrinks by a factor of ten each time n grows tenfold up to $n = 10^6$, then stops following the pattern; part 3 shows that what stops is float64 precision, not the convergence of the sequence. The truncation error keeps shrinking like $e/(2n)$, but the rounding error takes over as soon as the two meet. Third, the error ratio in part 4 falls to 0.250000 and stays there, exactly as in the hand computation before. The program proves nothing: it only checks claims we have already proved.

Note for instructors

Teaching order: start with the ε-N game in class, not with the theorems. Point to two rows of students; one names ε, one answers N for $x_n = (3n + 1)/(n + 2)$, and the instructor writes it down until the formula appears by itself. After that come uniqueness, the algebraic properties, the squeeze theorem, monotone sequences, then subsequences and Cauchy. The limit superior is placed at the end, as "a limit for sequences that have no limit".

Two typical misconceptions. First, students think N may depend on n, then write "for every n there is $N = n$" and believe the definition is met; the symptom shows when they cannot produce a single N for the whole tail. Second, they identify "terms getting closer together" with "convergent" and use the Cauchy criterion without checking the space; the decimal truncations of the square root of two inside $\mathbb{Q}$ are an effective cure.

Board activity: ask students to write the negation of the definition of a limit, then prove the divergence of $(-1)^n$ with that negation, not by contradiction. The class is ready to move on when they can produce an explicit N without being given a formula and name one bounded sequence that is divergent.

4.9 Summary

Level: basic

Summary.

- $x_n \to L$ means that for every $\varepsilon > 0$ there is N such that $|x_n - L| < \varepsilon$ for all $n > N$. The ε is chosen first; N answers, and may depend on ε alone.
- The limit is unique, and a convergent sequence is bounded. The converse fails: boundedness is only a necessary condition.
- Limits respect $+$, $\times$, and division by a nonzero denominator; order is never jumped over, only weakly preserved.
- The squeeze theorem turns trapping into a limit, and the lemma $q^n \to 0$ supplies a comparison used repeatedly. A bounded monotone sequence always converges, and its limit is the supremum or the infimum.
- Every bounded sequence has a convergent subsequence; a real sequence converges if and only if it is Cauchy, and that equivalence is completeness.
- $\liminf \le \limsup$, and the two are equal to L exactly when the sequence converges to L.

4.10 Exercises

Level: intermediate

Exercise 4.1.

1. Prove from the definition that $x_n = (2n-1)/(3n+4)$ converges to $2/3$, and give an explicit N as a function of ε. Compare it with the smallest n that satisfies the condition.
2. Prove: if $x_n \to L$ then $|x_n| \to |L|$. Show by example that the converse is false.
3. Compute $\limsup$ and $\liminf$ of $x_n = 1 + (-1)^n + 1/n$ and of $y_n = (-1)^n(1 + 1/n)$. Check your answers with Python for n up to 1000.
4. Evaluate this claim: "Because $(1 + 1/n)^n$ is increasing and bounded above by 3, its limit is 3." True, false, or not yet enough evidence? Say which part is right and which part is wrong.
5. Prove that $(2^n + 3^n)^{1/n} \to 3$ by writing out both fences completely, including a proof that $2^{1/n} \to 1$.
6. For the recursion $x_{n+1} = \sqrt{2 + x_n}$ with $x_1 = 1$, use Python to find the smallest term with error below 10^{-6} and below 10^{-12}. Compare with the guess from the ratio $1/4$.
7. Prove that a Cauchy sequence is bounded, without using Bolzano–Weierstrass. Then prove that a bounded monotone sequence is Cauchy.
8. For $x_n = (3n + 1)/(n + 2)$ and $\varepsilon = 10^{-3}$, determine the smallest admissible N, then compute the error at $n = N$, $N + 1$, and $N + 2$ with Python. Explain why $N - 1$ is not admissible.
9. Prove: if every subsequence of (x_n) has a subsequence converging to L, then $x_n \to L$.

Hints for the three hardest items. Item 5: use $1 \le (1 + (2/3)^n)^{1/n} \le 2^{1/n}$ and then the lemma $2^{1/n} \to 1$ via Bernoulli's inequality. Item 7: for the second part, compare $|x_n - x_m|$ with $x_m - x_n$ when $m > n$ and use monotone convergence. Item 9: suppose x_n does not tend to L, build a subsequence staying a fixed distance from L, then use Bolzano–Weierstrass.

For researchers: limits and next steps

What is really proved here is only the following equivalence, and all of it rests on one axiom: the supremum property. In an ordered field, the following statements are equivalent: every nonempty set bounded above has a supremum; every bounded increasing sequence

converges; every Cauchy sequence converges; every bounded sequence has a convergent subsequence. The chain of proofs is in Rudin, *Principles of Mathematical Analysis*, Chapter 3 (the sections Convergent Sequences, Subsequences, Cauchy Sequences) and Abbott, *Understanding Analysis*, Chapter 2 (sections 2.2 to 2.6). What is popular but not proved: "a Cauchy sequence always converges". That is false without completeness, and the decimal truncations of the square root of two inside $\mathbb{Q}$ are the counterexample. An open question: how fast a sequence converges, and whether that rate can always be computed. The answer is no. A classical result of Specker (1949) gives a computable, increasing, bounded sequence of rationals whose limit is not computable; so "convergent" does not automatically give an error bound that can be programmed.
Limits of validity: this whole chapter works in $\mathbb{R}$ with the usual distance. In a general metric space, Bolzano–Weierstrass becomes a statement about compactness, and the Cauchy criterion remains a necessary condition but no longer sufficient. The squeeze theorem is also silent when the two fences do not tend to the same value; there we need a finer comparison, and that is the material of Chapter 5.

Map to source texts: Chapter 4 in Rudin, Principles of Mathematical Analysis*: Chapter 3 (Numerical Sequences and Series, with the sections Convergent Sequences, Subsequences, Cauchy Sequences, Upper and Lower Limits, and Some Special Sequences). In Abbott,* Understanding Analysis*: Chapter 2 (The Limit of a Sequence, The Algebraic and Order Limit Theorems, The Monotone Convergence Theorem and a First Look at Infinite Series, Subsequences and the Bolzano–Weierstrass Theorem, The Cauchy Criterion). The axioms and completeness consequences used here are built in Chapter 3 of this book, not repeated.*

Chapter 5

Infinite Series

A team trains a classifier on one million examples. The objective function is the sum of the per-example losses, and training stops automatically when the increase in the total falls below 10^{-7} per epoch. At the fortieth epoch the process stops by itself. The model remains bad: each new example contributes only about 10^{-6} to the total, so the total hardly moves, while that total itself is not necessarily heading toward a finite value.

The suspicion can be tested with the simplest series. The sum $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ at $n = 10^6$ equals 14.3927267229. Adding another million terms raises it only to 15.0858736534; the addition is 0.6931469306, almost exactly $\ln 2 = 0.6931471806$. Each time the number of terms doubles, the harmonic series rises by about $\ln 2$, without bound: the estimate $\ln n + \gamma$ gives $H_{10^9} \approx 21.3004815$. A stopping rule based on "the total has stopped moving" therefore cannot tell convergence from divergence; what tells them apart is the rate at which the terms shrink.

Chapter 4 already supplied limits, the Cauchy criterion, monotone convergence, and the limit superior. Series add no new idea, because a partial sum is merely a sequence. But partial sums rarely have a closed formula, and computing s_{10^6} does not tell us whether s_{10^7} is still moving. This chapter turns that question into one that can be answered by inspecting a_n alone.

For younger readers (age 10 and up)

Imagine two jars of coins. In the first jar you save under a strange rule: one coin on the first day, half a coin on the second, a quarter on the third, and so on, always half of the day before. You save every day, never taking a day off, for as long as you like. How full is that jar after a hundred years? The answer is surprising: never more than two coins. Every addition does shrink, and the sum stops at a number you can name exactly.

In the second jar the rule is different. The coin on the first day is large, then smaller, and smaller still. After a month the jar holds only a small heap. After a year it is a little thicker. You might conclude that this jar too will stop at some number, because the additions grow thinner. It does not. If you save long enough, the contents will pass any number you name, even though climbing by a single unit takes years. Two jars, two rules that look alike, two opposite fates.

The question: without waiting for years, how do we know which jar stops and which does not? Is there a sign in the coins themselves, rather than in the final result? This chapter answers it, and the answer is used every day by people who write programs.

5.1 Series, partial sums, and series we can compute directly

Level: basic

A series is a sum that never finishes. Its definition therefore does not use the word "sum" directly, but rather the sequence of running totals. We read the result after some number of terms, then ask where that reading is heading.

Definition 5.1. Given a sequence (a_n) in $\mathbb{R}$. The n-th partial sum is

$$s_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k.$$

The symbol $\sum_{n=1}^{\infty} a_n$ is called a series, and that series *converges* to s when the sequence (s_n) converges to s; in that case we write $\sum_{n=1}^{\infty} a_n = s$ and call s the sum of the series. If (s_n) diverges, the series is called *divergent*.

series (*deret*) — a sum that never finishes, whose value is read from the running total after some number of terms.

Note that $s_n - s_{n-1} = a_n$. So a series and a sequence are the same object; only the question changes: for a sequence the formula for s_n is known, for a series we hold only a_n.

The Cauchy criterion from Chapter 4 can be translated directly, and its translation is the most basic tool in this chapter.

Theorem 5.1 (Cauchy criterion for series)**.** *$\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is an integer N such that*

$$m \geq n \geq N \quad \Longrightarrow \quad \left| \sum_{k=n}^{m} a_k \right| < \varepsilon.$$

Proof. Since $\sum_{k=n}^{m} a_k = s_m - s_{n-1}$, the condition above is nothing but the Cauchy condition for the sequence (s_n). By Chapter 4, a real sequence converges if and only if it is a Cauchy sequence.

This criterion speaks about a cut-off tail of the series, not about a single term. Taking $m = n$ gives a necessary condition that is very cheap to check.

Corollary 5.2 (The n-th term test)**.** *If $\sum a_n$ converges, then $a_n \to 0$.*

Proof. Take $m = n$ in the Cauchy criterion: for every $\varepsilon > 0$ there is N such that $n \geq N$ implies $|a_n| < \varepsilon$. That is exactly the definition of $a_n \to 0$.

This test only filters. It says that a term that does not tend to zero certainly wrecks convergence, and it says nothing about terms that do tend to zero. Two quick examples for the easy side: $\sum \frac{n}{n+1}$ diverges because $\frac{n}{n+1} \to 1$, and $\sum (-1)^n$ diverges because its terms do not tend to zero. The hard side is the harmonic series: $\frac{1}{n} \to 0$ but $\sum \frac{1}{n}$ diverges. We prove that in the second section.

Read it aloud (auditory learner)

Two voices. Read them in turn, then swap roles.
Voice A: I shrink. Every day I am smaller than yesterday.
Voice B: Good. But shrinking does not yet mean stopping.
A: I shrink by half. Half of half. Half again.
B: Then you have a fence. Your sum will stop.
A: I shrink, but only a little. One third. One quarter. One fifth.

B: Careful. Thin does not mean used up.
A: However small my additions, I keep adding.
B: And however long it takes, your sum keeps rising.
Repeat three times, faster each time: shrinking is the business of terms, stopping is the business of sums. Memorize: what decides is not the term that shrinks, but the remainder that never runs out.

The first series that can be computed directly is the geometric series. Each of its terms is obtained by multiplying the previous term by a fixed number.

Theorem 5.3 (Geometric series)**.** *For $|r| < 1$,*

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r},$$

and for $|r| \geq 1$ the series diverges.

Proof. For $r \neq 1$ the identity

$$(1-r)\,(1 + r + \cdots + r^n) = 1 - r^{n+1},$$

holds, so $s_n = \frac{1-r^{n+1}}{1-r}$. If $|r| < 1$ then $r^{n+1} \to 0$ (Chapter 4, since $|r|^n \to 0$ for $0 \leq |r| < 1$), hence $s_n \to \frac{1}{1-r}$. If $|r| \geq 1$ then $|r|^k \geq 1$ for all k, so s_n cannot converge: for $r = 1$ clearly $s_n = n + 1$, and for $|r| \geq 1$ with $r \neq 1$ the term r^k does not tend to zero.

With $r = \frac{1}{2}$ we get $1 + \frac{1}{2} + \frac{1}{4} + \cdots = 2$, and with $r = -\frac{1}{2}$ we get $1 - \frac{1}{2} + \frac{1}{4} - \cdots = \frac{2}{3}$. The second is interesting: its terms alternate in sign and the sum is smaller than twice the first term, because opposite signs wipe each other out.

The case $r = \frac{1}{2}$ has a picture that explains itself: cut the unit square in half, cut the remainder again, and keep going. Figure 5.1 shows the pieces $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$ and the remainder $2^{-5} = 0.03125$ after five cuts, with used area $1 - 2^{-5} = 0.96875$. That remainder shrinks without vanishing; that is the geometric face of the sentence “the sum stops”.

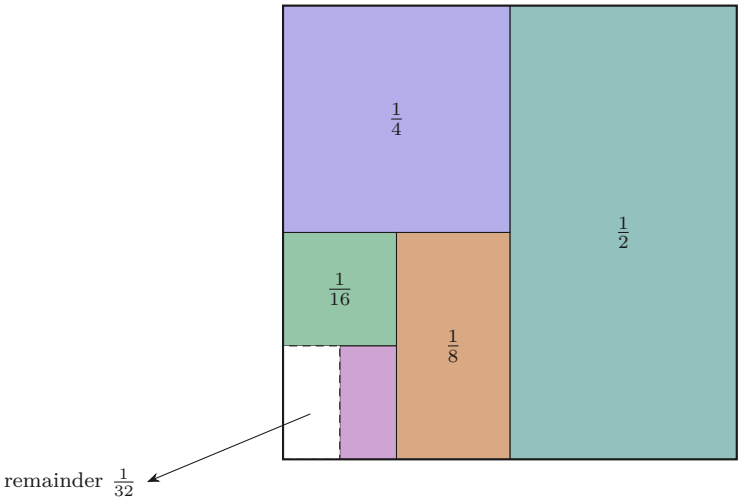


Figure 5.1: The geometric series of one half as pieces of the unit square. Pieces of size $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$ are laid down in order, and the remainder still uncovered after four pieces is $\frac{1}{16}$; after five pieces the remainder is $2^{-5} = 0.03125$ and the used area $1 - 2^{-5} = 0.96875$. The remainder always shrinks, it never vanishes.

Do it by hand (kinesthetic learner)

Materials: one sheet of square paper, a pencil, a ruler.

1. Fold the paper in half, open it, shade one half. That is the first piece.
2. Fold the unshaded part in half again, open it, shade half of it.
3. Repeat three more times, always only on the part not yet shaded.
4. Check the white part that remains: after five folds only one thirty-second of the original paper is left.

What you should see: every new shading is always half of the previous one, and the white part never vanishes, it only shrinks toward one corner. If the paper is thin enough to fold ten times, the white part is only $\frac{1}{1024} = 0.0009765625$; by the computation it tends to zero, so the shading finally covers the whole sheet. Now change the rule: shade $\frac{1}{2}$ of the paper, then $\frac{1}{3}$, then $\frac{1}{4}$. The folds can no longer follow, and the paper runs out before the series stops.

The second family that can be computed directly is the telescoping series, that is, a series whose terms can be written as the difference of two consecutive terms of another sequence, so that almost all subtractions cancel each other.

Example 5.1. For $n \geq 1$,

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Hence

$$s_N = \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{N+1},$$

because all the intermediate terms cancel and only the first and last terms remain. So $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$. At $N = 100$ that formula gives $s_{100} = \frac{100}{101} = 0.9900990099$, and the remaining tail is $\frac{1}{101} = 0.0099009901$; the two add up to exactly 1.

Notice that in the telescoping example we need no test at all. A closed formula for s_N gives the limit directly. This is what rarely happens. For almost every series, including the harmonic series and $\sum 1/n^2$, no such formula exists, and we must decide convergence without ever knowing the sum exactly.

Before turning to the tests, we record that series behave linearly, just as limits of sequences do.

Proposition 5.4. *If $\sum a_n = A$ and $\sum b_n = B$ converge, then for every $c, d \in \mathbb{R}$ the series $\sum (ca_n + db_n)$ converges with sum $cA + dB$. Adding or discarding finitely many terms does not change convergence, it only changes the sum.*

Proof. The partial sums of the combined series are $cs_n^{(a)} + ds_n^{(b)}$, and by the algebraic properties of limits in Chapter 4 that sequence converges to $cA + dB$. For the second statement, suppose terms are discarded starting at index M. The partial sums of the new series are $s_N - s_{M-1}$ for some N, that is, the difference of a convergent sequence and a constant, so the convergence is the same.

As an application, $\sum_{n=1}^{\infty} \left(\frac{1}{2^n} + \frac{1}{n(n+1)} \right) = 1 + 1 = 2$, because the first part is a geometric series with $r = \frac{1}{2}$ starting at $n = 1$.

5.2 Series of nonnegative terms: comparison and the p-series

LEVEL: INTERMEDIATE

When all terms are nonnegative, the partial sums can only rise or stay put. The question of convergence then becomes a far simpler one: do the partial sums stop below some fence?

Theorem 5.5. *If $a_n \geq 0$ for all n, then $\sum a_n$ converges if and only if its sequence of partial sums is bounded.*

Proof. With $a_n \geq 0$ we have $s_{n+1} = s_n + a_{n+1} \geq s_n$, so (s_n) is nondecreasing. By the monotone convergence theorem in Chapter 4, a monotone sequence bounded above converges, and a monotone sequence not bounded above diverges to $+\infty$.

This theorem looks simple, but this whole section is only a variation on it. All we need is a way to compare an unknown series with a known one, and the stock of known series is still thin: the geometric series.

Theorem 5.6 (Comparison test)**.** *Suppose $0 \leq a_n \leq c_n$ for $n \geq N_0$.*

1. *If $\sum c_n$ converges, then $\sum a_n$ converges.*
2. *If $a_n \geq d_n \geq 0$ for $n \geq N_0$ and $\sum d_n$ diverges, then $\sum a_n$ diverges.*

Proof. For (1), take $\varepsilon > 0$. By the Cauchy criterion there is $N \geq N_0$ such that $m \geq n \geq N$ implies $\sum_{k=n}^{m} c_k < \varepsilon$. Since every term is nonnegative and $a_k \leq c_k$,

$$\left| \sum_{k=n}^{m} a_k \right| = \sum_{k=n}^{m} a_k \leq \sum_{k=n}^{m} c_k < \varepsilon,$$

so $\sum a_n$ satisfies the Cauchy criterion. For (2), if $\sum a_n$ converged then by (1) and $d_n \leq a_n$ the series $\sum d_n$ would converge as well, contradicting the hypothesis.

The comparison test demands that we pick the right comparison. For terms in the form of rational fractions, the natural comparison is the largest term in the numerator divided by the largest term in the denominator. For instance $\frac{1}{n^2+3n} \leq \frac{1}{n^2}$, so $\sum \frac{1}{n^2+3n}$ converges provided $\sum \frac{1}{n^2}$ converges. Hence the whole question moves to a single family of series.

That family is the p-series, and the tool for handling it is grouping the terms into blocks whose size is a power of two. This grouping idea is used repeatedly, including in the Cauchy condensation test below.

Lemma 5.7 (Cauchy condensation)**.** *Suppose $a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.*

Proof. Since the terms are nonnegative, both series converge exactly when their partial sums are bounded (the first theorem of this section). We compare blocks. For $k \geq 0$, the block $a_{2^k+1} + \cdots + a_{2^{k+1}}$ contains 2^k terms, each no larger than a_{2^k} and no smaller than $a_{2^{k+1}}$. Thus

$$\sum_{n=1}^{2^m} a_n \;\leq\; a_1 + \sum_{k=0}^{m-1} 2^k a_{2^k}, \qquad \sum_{n=1}^{2^m} a_n \;\geq\; \frac{1}{2} \sum_{k=1}^{m} 2^k a_{2^k}.$$

The left inequality shows that the partial sums of $\sum a_n$ are bounded when $\sum 2^k a_{2^k}$ is bounded. The right inequality shows the converse, because the partial sums of $\sum 2^k a_{2^k}$ are bounded when $\sum a_n$ is bounded. Both directions are done.

Theorem 5.8 (The p-series)**.** $\sum_{n=1}^{\infty} \frac{1}{n^p}$ *converges if $p > 1$ and diverges if $p \leq 1$.*

Proof. For $p \leq 0$ the term $\frac{1}{n^p} = n^{-p}$ does not tend to zero, so the series diverges by the n-th term test. For $p > 0$ the sequence $a_n = n^{-p}$ decreases to zero, so the condensation lemma applies and

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = \sum_{k=0}^{\infty} 2^k \cdot 2^{-kp} = \sum_{k=0}^{\infty} \left(2^{1-p}\right)^k.$$

That is a geometric series with ratio $r = 2^{1-p}$. Since $p > 1$ exactly when $1 - p < 0$, that is exactly when $r < 1$, the geometric series theorem gives: convergence for $p > 1$, divergence for $p \le 1$.

The case $p = 1$ is the harmonic series, and its proof is now a single line: the geometric ratio is $2^{1-1} = 1$, so the condensed series $\sum 1$ diverges. We can read its fragility straight off the blocks: every block of size 2^k contributes at least $2^k \cdot \frac{1}{2^{k+1}} = \frac{1}{2}$. Hence

$$H_{2^m} \;\ge\; 1 + \frac{m}{2},$$

so $H_n \to \infty$, but raising H by one requires doubling n many times over: the growth rate is logarithmic. This is why the number 14.39 at the opening of this chapter looks like a bound although it is not.

Figure 5.2 puts the two kinds of terms side by side. The harmonic bars stay plump: the tenth term is still 0.1, whereas the tenth geometric term is $2^{-10} = 0.0009765625$. Sum of the first ten terms: $H_{10} = 2.928968254$ against $1 - 2^{-10} = 0.9990234375$. The harmonic term is larger at every index, so the size of the term misleads; what decides is whether the shrinking leaves a finite tail.

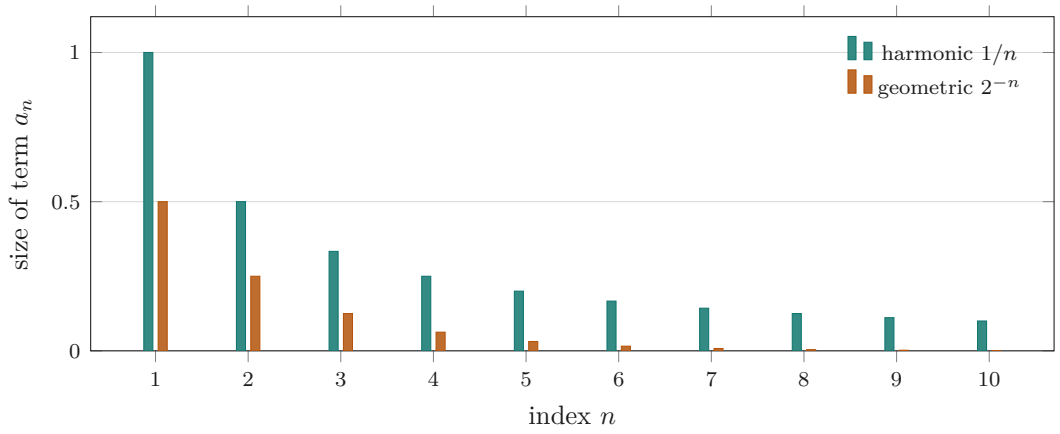


Figure 5.2: Bars of the harmonic terms $1/n$ and the geometric terms 2^{-n} for $n = 1, \dots, 10$. The harmonic bars are taller at every index and the tenth term is still 0.1, whereas the tenth geometric term is $2^{-10} = 0.0009765625$. Sum of the first ten terms: $H_{10} = 2.928968254$ against $1 - 2^{-10} = 0.9990234375$.

harmonic series (*deret harmonik*) — a pile that keeps growing although each addition is thinner than the last, so the total height climbs very slowly without bound.

The p-series is not the final boundary. With the same condensation, Rudin shows that $\sum \frac{1}{n(\log n)^p}$ converges for $p > 1$ and diverges for $p < 1$, and that adding one more logarithm shifts the boundary. Those series differ very little yet lie on opposite sides: there is no tidy boundary line, only tests that hold on a particular class.

Challenge (problem solver)

Take $k(n)$ the integer with $2^{k-1} \le n < 2^k$, then set $a_n = \frac{1}{k(n)}$. So $a_1 = 1$, $a_2 = a_3 = \frac{1}{2}$, $a_4 = \cdots = a_7 = \frac{1}{3}$, and so on. Show that $a_n \to 0$ but $\sum a_n$ diverges, then explain why the ratio test gives limit superior 1 here and is therefore useless.

Hint 1. Do not add the terms one by one. Add one block, and first count how many terms that block contains.

Hint 2. The k-th block contains 2^{k-1} terms, all of them equal to $\frac{1}{k}$, so its contribution is $\frac{2^{k-1}}{k}$. The first ten blocks already add up to 118.653968.

If you are stuck, go back to the section on series of nonnegative terms, where blocks of power-of-two size are used in the Cauchy condensation lemma.

For younger readers (age 10 and up)

Try to imagine stacking coins on a table. The first coin is thick. The second is half as thick. The third a third, the fourth a quarter, and so on: the millionth coin is very thin, as thin as one millionth of the first. If you stack a million coins, the height of the stack is about fourteen times the first coin, not a million times. It looks as though the stack will stop growing, because the next coin adds almost nothing.

Now the question that trips up many people: how many coins does it take for the stack to reach twenty times the first coin? The answer is about two hundred and seventy-two million coins. For twenty-one times, one needs about nine billion coins. Each level demands many times more coins, but no level is impossible. The pile never stops, it only slows down.

This is why computer programs are easily fooled. If you stop stacking after a million coins, the stack looks as though it no longer moves, when in fact it is not finished. In this chapter we will know why slow does not mean stopped, and how to tell the difference before spending the time to stack.

5.3 The ratio test and the root test

Level: intermediate

The comparison test demands that we already have a comparison series. For terms containing factorials or powers of n, such a comparison is hard to guess, but the comparison between consecutive terms is easy to compute. The idea: if every term is at most β times the previous one with $\beta < 1$, the series shrinks at least as fast as a geometric series, so it converges.

Theorem 5.9 (Ratio test)**.** *Suppose $a_n \neq 0$ for all sufficiently large n.*

1. *If $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$, then $\sum a_n$ converges, and in fact converges absolutely.*
2. *If there is n_0 such that $\left|\frac{a_{n+1}}{a_n}\right| \geq 1$ for all $n \geq n_0$, then $\sum a_n$ diverges.*

Proof. For (1), write $L = \limsup |a_{n+1}/a_n|$ and choose β with $L < \beta < 1$. By the properties of the limit superior in Chapter 4, only finitely many n satisfy $|a_{n+1}/a_n| > \beta$; so there is N such that $n \geq N$ implies $|a_{n+1}| \leq \beta |a_n|$. By induction,

$$|a_{N+k}| \leq \beta^k |a_N| \qquad (k \geq 0).$$

The series $\sum \beta^k |a_N|$ converges because $0 < \beta < 1$, so by the comparison test $\sum |a_n|$ converges, and by the absolute convergence theorem in the fifth section of this chapter, absolute convergence implies convergence.

For (2), the condition $|a_{n+1}| \geq |a_n|$ for $n \geq n_0$ makes $(|a_n|)$ nondecreasing and nonnegative, so it cannot tend to zero. Hence a_n does not tend to zero, and the n-th term test rules out convergence.

ratio test (*uji rasio*) — measures whether each term shrinks by at least a fixed fraction of the previous one, so that the series shrinks like a geometric series.

The bound 1 in the ratio test is a blind spot. For $1/n$ the ratio is $\frac{n}{n+1} \to 1$ and the series diverges; for $1/n^2$ the ratio is $\left(\frac{n}{n+1}\right)^2 \to 1$ and the series converges. So a ratio tending to 1 gives no information; shrinking rates that differ greatly can produce the same ratio.

The next two series show the ratio test working in opposite directions. For $a_n = \frac{n!}{10^n}$ we have $\frac{a_{n+1}}{a_n} = \frac{n+1}{10}$, which tends to $+\infty$; the series diverges. For $a_n = \frac{10^n}{n!}$ we have $\frac{a_{n+1}}{a_n} = \frac{10}{n+1}$, which tends to 0; the series converges. The sum of the second series is exactly $e^{10} = 22026.4657948$, whereas the sum of its first 20 terms is only 21950.3788494.

The root test uses a different quantity: the n-th root of the n-th term. Intuitively it measures the "average geometric ratio" of all the terms, not the ratio between two consecutive terms.

Theorem 5.10 (Root test)**.** *1. If* $\limsup_{n\to\infty} \sqrt[n]{|a_n|} < 1$*, then* $\sum a_n$ *converges, and in fact converges absolutely.*

2. If $\limsup_{n\to\infty} \sqrt[n]{|a_n|} > 1$*, then* $\sum a_n$ *diverges.*

Proof. For (1), choose β with $\limsup \sqrt[n]{|a_n|} < \beta < 1$. Only finitely many n satisfy $\sqrt[n]{|a_n|} > \beta$, so there is N such that $|a_n| \le \beta^n$ for $n \ge N$. The comparison test with the geometric series $\sum \beta^n$ closes this case.

For (2), if the limit superior is larger than 1, then $\sqrt[n]{|a_n|} > 1$, that is $|a_n| > 1$, for infinitely many n. Those terms do not tend to zero, so the series diverges by the n-th term test.

The root test is stronger than the ratio test. Let $a_n = 2^{-n}$ for even n and $a_n = 3^{-n}$ for odd n. Then $\sqrt[n]{a_n}$ equals $\frac{1}{2}$ or $\frac{1}{3}$, so its limit superior is $\frac{1}{2}$ and the root test declares the series convergent. The ratio swings: at $n = 100$, $\frac{a_{101}}{a_{100}}$ equals about 3.05×10^{17}, with limit superior $+\infty$ and limit inferior 0. The ratio test concludes nothing exactly when the terms swing on different scales.

The ratio test stays popular because factorials and powers of n cancel in the ratio, whereas the n-th root of a factorial demands Stirling's approximation. Order of use: the n-th term test, comparison with a p-series or a geometric series, then the ratio test; the root test is used when the ratio swings or gives 1.

Now the tests pile up, and the order of checking becomes a skill of its own.

Step-by-step recipe (procedural learner)

1. Check $a_n \to 0$. If not, the series diverges; done.
2. Recognize the shape: factorials call for the ratio test, powers of n call for the root test, rational fractions call for comparison with a p-series.
3. If the test you chose gives the bound 1, conclude nothing. Move to comparison or condensation.
4. If the signs alternate and $|a_n|$ decreases to zero, use Leibniz and report the error bound b_{N+1}.

Warning on order: do not swap step 1 with step 2. The ratio test on a series whose terms do not tend to zero can give the bound 1 and mislead, whereas the n-th term test settles it with no computation.

5.4 Alternating series and the Leibniz test

Level: intermediate

Up to here all series have nonnegative terms, and there convergence means the partial sums stop rising. Once the signs alternate, the partial sums can rise and fall, and cancellation between signs can produce convergence that the size of the terms does not guarantee.

Theorem 5.11 (Leibniz test)**.** *Suppose* (b_n) *is a nonnegative nonincreasing sequence, that is* $b_1 \ge b_2 \ge \cdots \ge 0$*, and* $b_n \to 0$*. Then* $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ *converges.*

Proof. Write s_N for the N-th partial sum. Consider the partial sums with even index and with odd index separately. Since

$$s_{2m+2} - s_{2m} = b_{2m+1} - b_{2m+2} \geq 0,$$

the sequence (s_{2m}) is nondecreasing. Since

$$s_{2m+1} - s_{2m-1} = -b_{2m} + b_{2m+1} \leq 0,$$

the sequence (s_{2m-1}) is nonincreasing. Besides, $s_{2m} \leq s_{2m-1}$ and $s_{2m-1} - s_{2m} = b_{2m} \to 0$. So (s_{2m}) is nondecreasing and bounded above by $s_1 = b_1$, hence converges to some s; (s_{2m-1}) is nonincreasing and bounded below by s_2, hence converges to some t; and $s - t = \lim(s_{2m} - s_{2m-1}) = 0$. Both subsequences tend to the same value, and since the even and odd subsequences together cover all indices, the whole sequence (s_N) converges to that s.

The Leibniz test gives more than convergence: an error bound that can be used without knowing the true sum, and that is what makes alternating series useful for computing constants.

Theorem 5.12 (Leibniz error bound)**.** *Under the hypotheses of the Leibniz test and with* $s = \sum(-1)^{n+1}b_n$*,*

$$|s - s_N| \leq b_{N+1}.$$

Sharper: s lies between s_N and s_{N+1}, so the sign of the error $s - s_N$ is the sign of the first discarded term, namely $(-1)^{N+2}$.

Proof. From the earlier proof, (s_{2m}) rises toward s and (s_{2m-1}) falls toward s. So for every N, the number s lies between s_N and s_{N+1}. Since $|s_{N+1} - s_N| = b_{N+1}$, the distance from s to s_N does not exceed b_{N+1}, and the direction of the shift is set by the sign of $s_{N+1} - s_N = (-1)^{N+2}b_{N+1}$.

The classic example is the alternating harmonic series,

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2.$$

The equality with $\ln 2$ is proved via the Taylor series for $\ln(1 + x)$ in Chapter 14; here we only use its convergence, which the Leibniz test guarantees because $b_n = \frac{1}{n}$ decreases to zero. The number $\ln 2$ appears again, now as the sum of a convergent series; the difference $H_{2n} - H_n \to \ln 2$ is another version of the same computation.

Figure 5.3 draws the partial sums of that series as two staircases: the odd ones descend from $s_1 = 1$, the even ones ascend from $s_2 = 0.5$, and both bracket $\ln 2 = 0.6931471806$. At $n = 10$ and $n = 11$ the bracket is $0.6456349206 < \ln 2 < 0.7365440115$, of width $b_{11} = \frac{1}{11} = 0.0909090909$, exactly as the error bound theorem promises.

The Leibniz bound is useful and is often estimated too optimistically. With 1000 terms, the partial sum is 0.7851481635 whereas $\pi/4 = 0.7853981634$; the actual error is 2.5000×10^{-4}, the promised bound $b_{1001} = \frac{1}{2001} = 4.9975 \times 10^{-4}$. The bound is correct but twice the real error, because the actual error is about $\frac{1}{4N}$ whereas the bound is $\frac{1}{2N}$. For an error below 10^{-6} we need $N \geq 500000$, and at that N the error is 5.0000×10^{-7}.

5.5 Absolute convergence, grouping, and rearrangement

Level: advanced

Series that converge because their signs cancel are far more fragile than series that converge because their terms shrink fast enough. That fragility has a name, and this chapter owes one explanation of it.

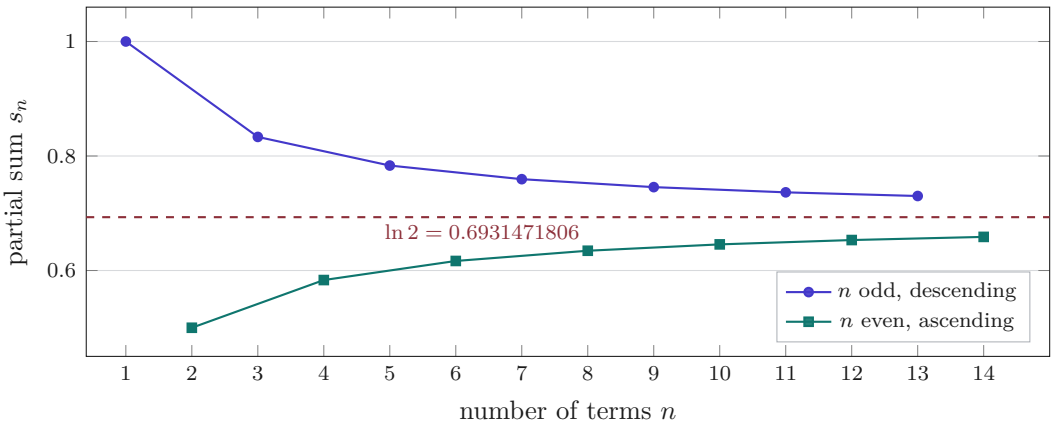


Figure 5.3: Partial sums of the alternating harmonic series. The odd subsequence descends, the even subsequence ascends, and both bracket the limit line $\ln 2 = 0.6931471806$. At $n = 10$ and $n = 11$: $s_{10} = 0.6456349206$ and $s_{11} = 0.7365440115$, of width $b_{11} = \frac{1}{11} = 0.0909090909$.

Definition 5.2. The series $\sum a_n$ *converges absolutely* if $\sum |a_n|$ converges. If $\sum a_n$ converges but $\sum |a_n|$ diverges, the series is called *conditionally convergent*.

absolute convergence (*konvergensi mutlak*) — a series that still converges after all its signs are forced positive, so it does not depend on cancellation.

Theorem 5.13. *If $\sum a_n$ converges absolutely, then $\sum a_n$ converges and $|\sum a_n| \le \sum |a_n|$.*

Proof. For every $n \le m$ we have $|\sum_{k=n}^{m} a_k| \le \sum_{k=n}^{m} |a_k|$. Since $\sum |a_n|$ converges, the rightmost expression can be made smaller than ε by the Cauchy criterion, so $\sum a_n$ satisfies the Cauchy criterion as well. The absolute value inequality for the sum is obtained by taking limits in $|s_n| \le \sum_{k=1}^{n} |a_k|$.

The converse does not hold: the alternating harmonic series converges conditionally, because $\sum \frac{1}{n}$ diverges. For such series, the order of summation begins to matter. Before going there, we first look at an operation that is safe, namely grouping.

Proposition 5.14 (Grouping)**.** *Suppose $\sum a_n$ converges to s, and $n_1 < n_2 < n_3 < \cdots$ is a sequence of indices. Then the series whose terms are*

$$A_k = a_{n_{k-1}+1} + a_{n_{k-1}+2} + \cdots + a_{n_k}$$

(with $n_0 = 0$) also converges to s. The converse fails: a convergent grouping does not guarantee that the original series converges.

Proof. The partial sums of the grouped series are $S_K = A_1 + \cdots + A_K = s_{n_K}$, that is, a subsequence of (s_n). A subsequence of a convergent sequence tends to the same limit (Chapter 4), so $S_K \to s$.

For the converse, take $1 - 1 + 1 - 1 + \cdots$. This series diverges because its terms do not tend to zero, but grouping in pairs gives $(1 - 1) + (1 - 1) + \cdots = 0 + 0 + \cdots$, which converges to 0.

That counterexample matters in practice: parentheses may be inserted into a series already known to converge, but they may not be used to prove it.

Now we change the order, not merely the grouping: a rearrangement renumbers the same terms so that each term appears exactly once, but at a different position.

Definition 5.3. Let σ be a bijection from the set of positive integers to itself. The series $\sum a_{\sigma(n)}$ is called a *rearrangement* of $\sum a_n$.

rearrangement (*susunan ulang*) — swapping the order in which goods arrive without changing the goods, so the running total can differ even though the warehouse contents are the same.

For an absolutely convergent series, rearrangement is harmless.

Theorem 5.15. *If $\sum a_n$ converges absolutely, then every rearrangement of it converges, and all of them tend to the same sum.*

Proof. Write $s = \sum a_n$ and let $\sum a_{\sigma(n)}$ be a rearrangement with partial sums s'_N. Take $\varepsilon > 0$. By the Cauchy criterion there is N_0 such that $m \geq n \geq N_0$ implies $\sum_{k=n}^{m} |a_k| < \varepsilon$; in particular $\sum_{k=N_0}^{\infty} |a_k| \leq \varepsilon$.

Choose P so that the numbers $1, 2, \ldots, N_0 - 1$ all appear among $\sigma(1), \ldots, \sigma(P)$. For $N > P$, the terms $a_1, \ldots, a_{N_0-1}$ cancel each other in the difference $s_N - s'_N$ (both contain those terms), so all that remains are terms with index $k \geq N_0$. Hence

$$|s_N - s'_N| \;\leq\; \sum_{k \geq N_0} |a_k| \;\leq\; \varepsilon \qquad (N > P).$$

Since $s_N \to s$ and $|s'_N - s| \leq |s'_N - s_N| + |s_N - s|$, we get $s'_N \to s$.

For series that converge only conditionally, that conclusion collapses completely. The next theorem is stated without proof; the proof is long and technical, and can be read in Rudin, in the chapter on sequences and series of numbers, the section “Rearrangements”.

Theorem 5.16 (Riemann rearrangement theorem, statement)**.** *Suppose $\sum a_n$ converges but does not converge absolutely, and suppose $\alpha \leq \beta$ are two real numbers (possibly $\pm\infty$). Then there is a rearrangement $\sum a_{\sigma(n)}$ whose partial sums have limit inferior α and limit superior β. In particular, for every $c \in \mathbb{R}$ there is a rearrangement that converges to c, and there is a rearrangement that diverges.*

The idea of the proof is this. If $\sum a_n$ converges conditionally, the series of its positive terms and the series of absolute values of its negative terms both diverge. We rearrange by rule: add positive terms until we pass β, then negative terms until we drop below α, then repeat. Because both series diverge, that step can always be carried out, and because $a_n \to 0$, the overshoot past the boundary gets smaller. The result oscillates with limit inferior α and limit superior β. The complete construction is in Rudin, Theorem 3.54.

The best-known case is the alternating harmonic series rearranged in the pattern two positive terms then one negative term,

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \cdots$$

This series converges, but not to $\ln 2$: its sum is $\frac{3}{2}\ln 2 = 1.0397207708$, and the computation is in the next section. Changing the order changes the answer even though the terms are exactly the same. Addition of real numbers is not commutative without conditions, and that is a concrete warning for anyone who sums fractions on a computer.

This also explains why the tests in the second and third sections require nonnegative terms. When the signs alternate, shrinking is not enough: the shrinking must be fast enough to beat the cancellation.

For younger readers (age 10 and up)

Imagine two piles of cards: green marked “in”, red marked “out”, each with a number. You move the cards one at a time, and the balance on the screen changes: green adds, red subtracts. If you change the order of the moves, does the final balance change?
If the greens and the reds are both large so that they cancel each other, the order matters.

Moving the greens first makes the balance spike before it falls; moving the reds first makes the balance go negative for a while. The final balance can differ too.
There is one safe situation: if the total of all the numbers, ignoring color, stays bounded, the order has no effect. That is absolute convergence. In this chapter we will know why, and why a computer can give different answers for sums that look the same.

5.6 Worked examples

Level: intermediate

This section works through three computations that form the backbone of the chapter. All the numbers are checked with Python in the lab section; here the intermediate steps are written so that they can be repeated with paper and pencil.

1. The harmonic series at one million terms

We want H_{10^6} without adding a million terms. The tool is the asymptotic form

$$H_n = \ln n + \gamma + \frac{1}{2n} - \frac{1}{12n^2} + O\left(\frac{1}{n^4}\right),$$

where $\gamma = 0.5772156649$ is the Euler–Mascheroni constant. For $n = 10^6$:

$$\ln 10^6 = 13.8155105580, \qquad \ln 10^6 + \gamma = 14.3927262229.$$

The first correction term $\frac{1}{2n} = 5 \times 10^{-7}$ gives 14.3927267229, and the second correction $-\frac{1}{12n^2} = -8.33 \times 10^{-14}$ is far below the precision we need. So

$$H_{10^6} = 14.3927267228657\ldots$$

Direct summation with `math.fsum` gives 14.392726722865724, in agreement to 10^{-15}.

More important is how many terms are needed to reach a given bound. From $H_n \approx \ln n + \gamma$ we get $n \approx e^{H-\gamma}$. For $H = 20$:

$$n \approx e^{20 - 0.5772156649} = e^{19.4227843351} = 2.724006 \times 10^8.$$

A direct search gives $n = 272400600$ as the first index with $H_n > 20$, and H at that point is 20.0000000016. For $H = 21$ one needs about 7.4×10^8 terms; the estimate at $n = 10^9$ gives 21.3004815. Each added unit demands multiplying n by $e \approx 2.718$.

2. The sum $\sum 1/n^2$ and $\pi^2/6$

With $N = 10^6$ terms, direct summation gives

$$s_{10^6} = 1.6449330668487263, \qquad \frac{\pi^2}{6} = 1.6449340668482264.$$

The difference is 9.9999950×10^{-7}, very close to $\frac{1}{N} = 10^{-6}$; that makes sense because the tail $\sum_{n>N} n^{-2}$ is about $\frac{1}{N}$. The next order correction is $-\frac{1}{2N^2} = -5 \times 10^{-13}$, so

$$s_{10^6} + \frac{1}{N} - \frac{1}{2N^2} = 1.6449340668482262,$$

which matches $\pi^2/6$ to 2×10^{-16}, the precision limit of 64-bit fractions. One million terms plus one correction already gives machine accuracy.

3. Rearranging the alternating harmonic series

Take the pattern two positive terms and one negative: in the k-th block we add $\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k}$. After K blocks, all the positive terms with denominator $1, 3, \ldots, 4K-1$ have been used, and the negative terms with denominator $2, 4, \ldots, 2K$ have been used as well. So

$$S_K = \sum_{j=1}^{2K} \frac{1}{2j-1} - \sum_{j=1}^{K} \frac{1}{2j} = \left(H_{4K} - \frac{1}{2}H_{2K}\right) - \frac{1}{2}H_K.$$

With $H_m \approx \ln m + \gamma$,

$$S_K \approx \left(\ln 4K - \tfrac{1}{2}\ln 2K\right) - \tfrac{1}{2}\ln K = \ln \frac{4K}{\sqrt{2K}\sqrt{K}} = \ln\left(2\sqrt{2}\right) = \tfrac{3}{2}\ln 2.$$

Every term γ cancels, and the result does not contain K. So $S_K \to 1.0397207708$, not $\ln 2 = 0.6931471806$. At $K = 10^6$ blocks the numerical value is 1.0397205208, an error of 2.5×10^{-7} against $\frac{3}{2}\ln 2$, consistent with order $\frac{1}{4K}$ as for the Leibniz series.

Numerical summary. $H_{10^6} = 14.3927267229$ and $\ln 10^6 + \gamma = 14.3927262229$, the difference is $\frac{1}{2n} = 5 \times 10^{-7}$. $H_n > 20$ is first reached at $n = 272400600$, close to $e^{20-\gamma} = 2.724006 \times 10^8$. The Leibniz series with 1000 terms gives an error of 2.5000×10^{-4} against $\pi/4$, with bound $\frac{1}{2001} = 4.9975 \times 10^{-4}$. The sum $\sum 1/n^2$ up to 10^6 terms falls short of $\pi^2/6$ by 9.9999950×10^{-7}; one Euler–Maclaurin correction closes it to machine precision. The rearrangement two positive one negative tends to $\frac{3}{2}\ln 2 = 1.0397207708$, not $\ln 2$.

5.7 Computational lab

Level: advanced

The following code runs as is on Python 3.14.4 with NumPy 2.5.2. The first part tests the accuracy of the harmonic sum and of its asymptotic estimate.

Computational lab

```
import math
import numpy as np

n = 10**6
a = 1.0 / np.arange(1, n + 1)
vals = a.tolist()
naive = 0.0
for x in vals:
    naive += x

print("np.sum   ", repr(float(np.sum(a))))
print("for-loop ", repr(naive))
print("fsum     ", repr(math.fsum(vals)))
small = a.astype(np.float32)
print("float32  ", repr(float(np.sum(small, dtype=np.float32))))
g = 0.5772156649015328606
print("ln n + g ", repr(math.log(n) + g))
print("1/(2n)   ", 1 / (2 * n))
```

The output obtained:

```
np.sum    14.392726722865726
for-loop  14.392726722864989
fsum      14.392726722865724
float32   14.392724990844727
ln n + g  14.392726222865807
1/(2n)    5e-07
```

The first four results should be the same, and three of them differ in the last digit: rounding error. Adding a million numbers of size 10^{-6} into an accumulator of size 14 throws away low bits. `float32` loses 1.7×10^{-6}, larger than the correction $\frac{1}{12n^2}$ we want to measure; when the error we are after is smaller than the error of the summation itself, the result means nothing.

The second part checks three numbers from the previous section.

Computational lab

```
import math
import numpy as np

pi4 = math.pi / 4
s = 0.0
for k in range(1000):
    s += (-1)**k / (2*k + 1)
print("Leibniz 1000 terms", repr(s))
print("error", repr(pi4 - s))
print("bound 1/2001", repr(1/2001))

N = 10**6
m = np.arange(1, N + 1, dtype=np.float64)
S = float(np.sum(1.0 / m**2))
print("sum 1/n^2", repr(S))
print("pi^2/6", repr(math.pi**2/6))
print("corrected", repr(S + 1/N - 1/(2*N**2)))

S = 0.0
for k in range(1, 10**6 + 1):
    S += 1.0/(4*k-3) + 1.0/(4*k-1) - 1.0/(2*k)
print("rearrangement", repr(S))
print("1.5 ln 2", repr(1.5*math.log(2)))
```

Its output:

```
Leibniz 1000 terms 0.7851481634599485
error 0.00024999993749974525
bound 1/2001 0.0004997501249375312
sum 1/n^2 1.6449330668487263
pi^2/6 1.6449340668482264
corrected 1.6449340668482262
rearrangement 1.0397205208399414
1.5 ln 2 1.0397207708399179
```

Three things need to be read from that output. First, the Leibniz error 2.5000×10^{-4} is exactly half the bound 4.9975×10^{-4}; that is no accident, because the remainder of a monotone alternating series approaches half of the first discarded term. Second, the Euler–Maclaurin correction closes the gap to 2×10^{-16}, one rounding unit at magnitude 1.64. Third, the rearrangement gives 1.0397205, not 0.6931472, and its error against $\frac{3}{2}\ln 2$ is only 2.5×10^{-7}.

Note for instructors

The teaching order I recommend: start from the epoch problem in the opening, then the definition of partial sums and the two series that can be computed (geometric, telescoping). The tests come in as a sequence of checks, not a list to memorize: the n-th term, comparison, ratio, root. The p-series follows after condensation, which explains why the harmonic series diverges and why it does so slowly.

Two misconceptions appear most often. First, students conclude convergence from $a_n \to 0$; the harmonic series is the cure, and it is best shown with a table $H_{10} = 2.9290$,

$H_{100} = 5.1874$, and $H_{1000} = 7.4855$ so that the slowness is visible. Second, students take "convergent series" to mean "partial sums stop changing", when the change per term can be far below machine precision while the series diverges. This is the right place to mention rounding error.
A class activity that works: split the class into groups, give each group four unnamed series, and the only task is to decide convergence or divergence together with the test, in ten minutes. The chosen series must include one trap with ratio = 1. Discuss groups that use different tests and are both right. Students are ready to move on when they can explain why $\sum 1/n^2$ converges while $\sum 1/n$ does not, using only block grouping.

5.8 Summary

Level: basic

Summary.

- A series is a sequence of partial sums; converging means that sequence converges.
- For nonnegative terms, convergence is equivalent to bounded partial sums. From this come the comparison test, condensation, and the classification of p-series: convergent exactly when $p > 1$.
- The ratio test and the root test measure the speed of shrinking relative to a geometric series. Both are blind at the bound 1, and the root test is stronger than the ratio test.
- The Leibniz test gives convergence for alternating series that decrease to zero, along with an error bound b_{N+1} whose sign is known.
- Absolute convergence is immune to grouping and rearrangement. Conditional convergence is not: a Riemann rearrangement can turn the sum into any number, or make it diverge.
- The harmonic series diverges logarithmically: $H_{10^6} \approx 14.39$ and it takes 272 million terms to reach 20. A stopping rule based on "the total no longer changes" cannot be trusted.

5.9 Exercises

Level: intermediate

Exercise 5.1.

1. Compute s_{20} for the geometric series $\sum_{k=0}^{\infty} \left(\frac{3}{5}\right)^k$, then compare it with the true sum. What is the error, and what error bound can you promise from the formula $\frac{r^{21}}{1-r}$?
2. Prove that $\sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{3}{4}$ by splitting the term into partial fractions and showing that the series telescopes.
3. Prove: if $a_n \geq 0$ and $\sum a_n$ converges, then $\sum a_n^2$ converges. Show by example that the converse is false.
4. Evaluate this claim: true, false, or not yet enough evidence. "Because $\frac{1}{n} \to 0$ and every term is smaller than the terms of the geometric series with ratio $\frac{1}{2}$ for n sufficiently large, the harmonic series converges." State exactly where the reasoning breaks.
5. Decide convergence or divergence, naming the test used: (a) $\sum \frac{n^2}{2^n}$, (b) $\sum \frac{2^n}{n^2}$, (c) $\sum \frac{n!}{n^n}$, (d) $\sum \frac{1}{n^{1/3}}$.
6. Compute the sum $\sum_{n=1}^{10^6} \frac{1}{n^2}$ with NumPy, then estimate the tail with $\frac{1}{N} - \frac{1}{2N^2}$ and compare the result with $\pi^2/6$. How many digits after the decimal point are correct before the correction, and after it?

7. The series $\sum \frac{(-1)^{n+1}}{\sqrt{n}}$ converges. Determine the smallest N such that the Leibniz bound guarantees an error below 10^{-3}, then compute the actual error at that N with Python. Is the bound loose or sharp?
8. Prove that if $\sum a_n$ converges absolutely and (b_n) is bounded, then $\sum a_n b_n$ converges absolutely.
9. Rearrange the alternating harmonic series in the pattern three positive terms and one negative. Guess the sum with the method of the worked examples section, then check it numerically. Is the result still a multiple of $\ln 2$?
10. Prove that if $\sum a_n$ converges conditionally, then the series of its positive terms diverges and the series of absolute values of its negative terms diverges as well. Explain why this fact is the fuel of the Riemann rearrangement theorem.

Hints for the three hardest items. Item 7: the Leibniz bound uses the first discarded term, so solve $\frac{1}{\sqrt{N+1}} < 10^{-3}$ and check whether that inequality is sharp enough. Item 9: after K blocks, the positive terms are used up to denominator $6K-1$, the negative ones up to $\frac{1}{2K}$; write the partial sums in terms of H_{6K}, H_{3K}, and H_K, then take the limit. Item 10: write $p_n = \max(a_n, 0)$ and $q_n = \max(-a_n, 0)$; then $a_n = p_n - q_n$ and $|a_n| = p_n + q_n$, so convergence of one of those series says something about $\sum |a_n|$.

For researchers: limits and next steps

What is really proved in this chapter are the statements with complete proofs: the Cauchy criterion for series, the comparison test, Cauchy condensation, the classification of p-series, the ratio test and the root test in limit superior form, the Leibniz test together with its error bound, and the closure of absolutely convergent series under rearrangement. I quote the Riemann rearrangement statement without proof. The primary source is Rudin, *Principles of Mathematical Analysis*, chapter "Numerical Sequences and Series": Theorems 3.22–3.23, 3.25, 3.27–3.28, 3.33–3.34, 3.42–3.43, 3.45–3.46, and 3.54–3.55. Abbott, *Understanding Analysis*, chapter "Sequences and Series", discusses the same material with emphasis on rearrangement in the opening section of that chapter.

What is often popularized beyond what is proved is the idea of a threshold between convergent and divergent series. Rudin shows that no such threshold exists: two series that differ very little can lie on opposite sides, and the procedure can be continued without end.

An open question: how reliable is a convergence decision when a_n is available only as a noisy estimate, such as the sum of losses in machine learning? For such series there is not yet a tidy theory.

Limits of validity: the whole chapter treats series of real numbers. In other normed spaces the definition is the same but the tests differ; series of functions require uniform convergence (Chapter 14). All numbers come from NumPy 2.5.2, 64-bit arithmetic.

Map to source texts: Chapter 5 in Rudin, Principles of Mathematical Analysis*, chapter "Numerical Sequences and Series": series and partial sums (3.21–3.24), series of nonnegative terms and comparison (3.25–3.29), the root and ratio tests (3.33–3.37), absolute convergence and rearrangement (3.45–3.55). In Abbott,* Understanding Analysis*, chapter "Sequences and Series": rearrangement in the opening section of that chapter. This chapter does not reuse the prose of Chapter 4; from there it refers only to the definition of a limit, the Cauchy criterion, and monotone convergence.*

Part II

Linear Algebra

This part installs the framework used by almost every later chapter: vector spaces, bases, determinants, eigenvalues, and the singular value decomposition. The prerequisite is Chapters 1–2. After five chapters you can turn a problem into a matrix, solve it, and explain why its answer is unique or not.

Chapter 6

Vector Spaces and Subspaces

The following table holds three records of network traffic; each row is one record, each column one feature (incoming packets, outgoing packets, duration, failed connections). Read from top to bottom, the columns are

$$\mathbf{c}_1 = (1, 2, 3), \quad \mathbf{c}_2 = (2, 4, 6), \quad \mathbf{c}_3 = (0, 1, 2), \quad \mathbf{c}_4 = (1, 3, 5),$$

while the rows are $\mathbf{r}_1 = (1, 2, 0, 1)$, $\mathbf{r}_2 = (2, 4, 1, 3)$, and $\mathbf{r}_3 = (3, 6, 2, 5)$.

How many columns actually carry new information? Counting the nonzero columns gives four, and that count is wrong: the second column is twice the first, the fourth is the sum of the first and the third. The table holds only two independent directions of data. The sentence "can be recomputed" acquires meaning only once we can measure how few vectors build it up.

Chapters 1 through 5 prepared the language of sets, the axioms of the real numbers, and then limits of sequences and series; all of them work with closeness. The problem above contains no closeness at all: its entries are exact, and the question is purely algebraic. Which sets are closed under both operations, and how few vectors build them. One habit from Chapter 5 stays in use: here every combination involves finitely many terms.

There is a second question that demands the same tool. A linear scorer assigns the score $s(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x}$ with $\mathbf{w} = (1, 1, 1)$. A direction $\boldsymbol{\delta}$ that does not change the score satisfies $\mathbf{w} \cdot \boldsymbol{\delta} = 0$, which is a plane; every perturbation splits into a part that changes the score and a part that does not. Is that split unique? Yes, and the property is called a direct sum.

For younger readers (age 10 and up)

Picture a box of poster paint holding four cans: red, yellow, blue, and orange. We may mix the contents of the cans in any proportion, and the mixture may itself be used to mix again. Orange can be made from red and yellow. So the orange can adds not a single capability that the box did not already have; it only takes up space.

The nagging question: how few cans must remain so that every color we can make stays a color we can make? The answer is not the same as the number of cans left after we throw one away, and the order of removal must not change that number. That is no accident, and proving it is not as simple as it looks.

There is a stranger question. Suppose two people mix according to different recipes, and the results are exactly the same. Are the recipes the same? Sometimes yes, sometimes no. In this chapter we will learn why that number has a name of its own, why it does not depend on the order of removal, and why two different mixtures sometimes yield the same color.

6.1 The vector space axioms and their examples

Level: basic

A vector space is a set whose members may be added and may be enlarged or shrunk by a number, with the result always staying inside that set, and with the same rules of computation as for ordinary pairs of numbers.

Definition 6.1 (Vector space)**.** A nonempty set V together with an addition $V \times V \to V$ and a scalar multiplication $\mathbb{R} \times V \to V$ is called a *vector space over* $\mathbb{R}$ if for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $a, b \in \mathbb{R}$ the following hold:

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;
3. there is $\mathbf{0} \in V$ with $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for every $\mathbf{v}$;
4. there is $-\mathbf{v} \in V$ with $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$;
5. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$;
6. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$;
7. $a(b\mathbf{v}) = (ab)\mathbf{v}$;
8. $1\mathbf{v} = \mathbf{v}$.

Both operations carry the closure requirement inside their type: the result always lies in V, so this definition contains no separate closure axiom. The eight axioms are not independent of one another either, and we do not chase the shortest list; what we need are their small consequences, all derived without looking at the shape of the members of V.

Lemma 6.1. *In any vector space, the zero vector is unique, the additive inverse of every vector is unique,* $0\mathbf{v} = \mathbf{0}$*,* $(-1)\mathbf{v} = -\mathbf{v}$*, and* $c\mathbf{v} = \mathbf{0}$ *implies* $c = 0$ *or* $\mathbf{v} = \mathbf{0}$*.*

Proof. For uniqueness of zero, suppose $\mathbf{0}$ and $\mathbf{0}'$ are both neutral. Then $\mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}'$. For uniqueness of inverses, suppose $\mathbf{v} + \mathbf{a} = \mathbf{0}$ and $\mathbf{v} + \mathbf{b} = \mathbf{0}$. Then

$$\mathbf{a} = \mathbf{a} + \mathbf{0} = \mathbf{a} + (\mathbf{v} + \mathbf{b}) = (\mathbf{a} + \mathbf{v}) + \mathbf{b} = \mathbf{0} + \mathbf{b} = \mathbf{b}.$$

Next $0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$. Adding $-(0\mathbf{v})$ to both sides gives $\mathbf{0} = 0\mathbf{v}$. From there $\mathbf{v} + (-1)\mathbf{v} = 1\mathbf{v} + (-1)\mathbf{v} = (1 - 1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$, so $(-1)\mathbf{v}$ is an inverse of $\mathbf{v}$, and since inverses are unique, $(-1)\mathbf{v} = -\mathbf{v}$. Finally, if $c\mathbf{v} = \mathbf{0}$ and $c \neq 0$, then

$$\mathbf{v} = 1\mathbf{v} = (c^{-1}c)\mathbf{v} = c^{-1}(c\mathbf{v}) = c^{-1}\mathbf{0} = \mathbf{0}.$$

□

The following four examples suffice for this entire book.

(i) Coordinate space. $\mathbb{R}^n$ with addition and scalar multiplication coordinate by coordinate. Here $\mathbf{0} = (0, \dots, 0)$, and the eight axioms reduce to the field properties of the real numbers already discussed in Chapter 3.

(ii) Matrix space. $M_{2\times 3}(\mathbb{R})$, the set of 2×3 matrices with real entries, with operations entry by entry and the zero matrix as $\mathbf{0}$. Nothing changes if 2×3 is replaced by $m \times n$.

(iii) Polynomial space. P_3 is the set of polynomials with real coefficients of degree at most 3, together with the zero polynomial, with operations term by term. A trap worth noting: polynomials of degree *exactly* 3 do not form a vector space, since x^3 and $-x^3 + x$ both have degree three but their sum x has degree one. The words "at most" are not decoration.

(iv) Function space. The set of all functions $\mathbb{R} \to \mathbb{R}$, written $\mathbb{R}^{\mathbb{R}}$, with $(f + g)(x) = f(x) + g(x)$ and $(af)(x) = af(x)$. The zero function is $\mathbf{0}$, and the vector space axioms are inherited from $\mathbb{R}$ at each point x.

Example (iv) yields a subtler example. The set $C[0,1]$ of continuous functions on $[0,1]$ is a vector space, because sums and scalar multiples of continuous functions remain continuous. So is the set of convergent real sequences, using the limit-of-sums property from Chapter 4. Both are parts of a larger space, and parts of that kind we now define.

Back to $\mathbb{R}^2$, where both operations are visible directly. Figure 6.1 uses $\mathbf{u} = (2,1)$ and $\mathbf{v} = (1,2)$: the sum $\mathbf{u}+\mathbf{v} = (3,3)$ is reached along two orders, and $2\mathbf{u}$ as well as $-\mathbf{u}$ stay collinear with $\mathbf{u}$.

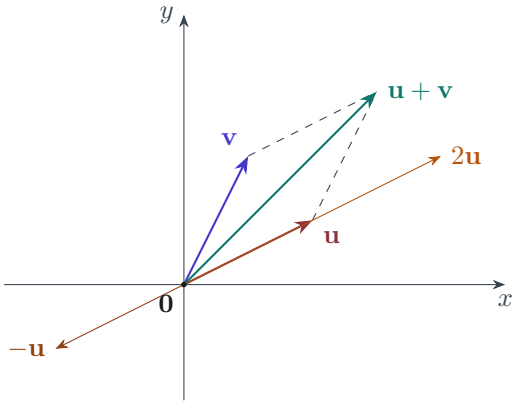


Figure 6.1: The two operations in $\mathbb{R}^2$. The thick arrows show $\mathbf{u} = (2,1)$ and $\mathbf{v} = (1,2)$; the dashed lines complete the parallelogram, so that $\mathbf{u}+\mathbf{v} = (3,3)$ is reached along two identical orders. The thin arrows show $2\mathbf{u} = (4,2)$ and $-\mathbf{u} = (-2,-1)$, both staying on one line with $\mathbf{u}$.

vector space (*ruang vektor*) — a mixing box: its contents may be added and scaled, and the result never leaves the box.

Read it aloud (auditory learner)

Eight axioms, read as a checklist. Two voices, alternating; the second voice always answers in three words.
First voice: Swap the places. Second voice: The result is the same. First voice: Move the brackets. Second voice: The result is the same. First voice: There is a zero vector. Second voice: It adds nothing. First voice: There is an opposite vector. Second voice: Back to zero. First voice: A scalar spreads over two terms. Second voice: Left and right agree. First voice: Two scalars add up. Second voice: One vector carries it. First voice: Two scalars multiply. Second voice: One multiplication only. First voice: The scalar one. Second voice: Silent, unchanged.
The last two lines sound almost alike although they differ: one adds proportions, the other multiplies proportions. That is where the most frequent arithmetic error is born. Say it once more, slowly, in a single breath: swap the places, move the brackets, there is a zero, there is an opposite, it spreads, it adds up, it multiplies, it is silent.

6.2 Subspaces and the subspace test

LEVEL: BASIC

A subspace is a part of a vector space that still behaves as a vector space even when we use only the members of that part.

Definition 6.2 (Subspace)**.** A set $W \subseteq V$ is called a *subspace* of V if W is a vector space with the same operations as on V.

That definition demands eight axioms, and checking eight things every time is too expensive. Six of them are free: commutativity, associativity, distributivity hold in W because they hold in V. Three things remain, and they are the content of the next theorem.

If W is nonempty and closed under both operations, then W is automatically a subspace.

Theorem 6.2 (Subspace test)**.** *Let V be a vector space and $W \subseteq V$ with $W \neq \varnothing$. Then W is a subspace of V if and only if W is closed under addition and scalar multiplication.*

Proof. If W is a subspace, the result of both operations on members of W must be a member of W, so both closures hold. Conversely, suppose W is nonempty and closed under both operations. Take $\mathbf{w} \in W$. By scalar closure, $\mathbf{0} = 0\mathbf{w} \in W$, so W contains the neutral element. Also $-\mathbf{w} = (-1)\mathbf{w} \in W$, so every member of W has its inverse inside W. The remaining axioms hold because they hold in V and the operations on W are merely the restrictions of the operations on V. □

In practice we check three things: $\mathbf{0} \in W$ and the two closures. All three are necessary; the next three examples show that failure of one condition cannot be covered up by another.

Step-by-step recipe (procedural learner)

1. Write the membership condition of W as an equation, not as a picture or a name.
2. Substitute $\mathbf{v} = \mathbf{0}$ into that condition. If it fails, stop: not a subspace.
3. Take arbitrary $\mathbf{u}, \mathbf{v} \in W$, check $\mathbf{u}+\mathbf{v}$ using the same condition, not using numerical examples.
4. Take arbitrary $c \in \mathbb{R}$ and $\mathbf{u} \in W$, check $c\mathbf{u}$; include $c < 0$ and $0 < c < 1$.

A warning about order: do not swap step 2 with steps 3 and 4. A set that does not contain $\mathbf{0}$ need not be tested for closure, but stopping after step 2 is also wrong: containing $\mathbf{0}$ is not enough, as the following examples show.

A line not through the origin. $W = \{(x, y) \in \mathbb{R}^2 : x + y = 1\}$ does not contain $\mathbf{0}$, so it is not a subspace; it is also closed under neither addition nor scalar multiplication, since $(1, 0) + (0, 1)$ and $2(1, 0)$ both leave W.

The union of two axes. $W = \{(x, 0)\} \cup \{(0, y)\}$ contains $\mathbf{0}$ and is closed under scalar multiplication, but $(1, 0) + (0, 1) = (1, 1)$ leaves W.

The first quadrant. $W = \{(x, y) : x \geq 0,\ y \geq 0\}$ is closed under addition, but $(-1)(1, 1) = (-1, -1)$ leaves W.

All three examples fail a different condition; Figure 6.2 shows the simplest pair, a line through the origin and a line that misses it.

Conversely, the next two classes of examples always pass.

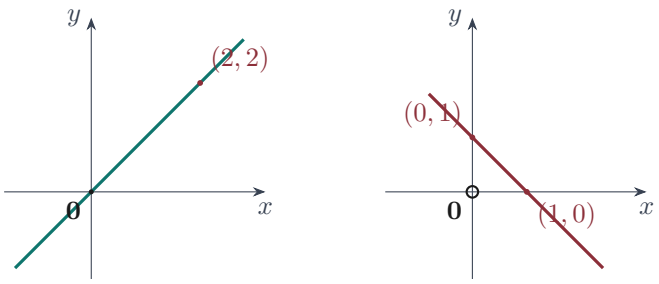
Null space. For an $m \times n$ matrix A, the set $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$ is a subspace, since $A\mathbf{0} = \mathbf{0}$, $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0}$, and $A(c\mathbf{x}) = cA\mathbf{x} = \mathbf{0}$. That set is called the null space or kernel of A, written $\ker A$, and is discussed further in Chapter 7.

Polynomials vanishing at one point. The set $\{p \in P_3 : p(1) = 0\}$ is a subspace of P_3, because evaluation at the point 1 is linear: $(p + q)(1) = p(1) + q(1)$ and $(cp)(1) = c\,p(1)$.

Both examples preserve the solution set of a homogeneous equation. That is no accident.

Challenge (problem solver)

The first-quadrant example fails because of one negative scalar. Find a set $W \subseteq \mathbb{R}^2$ that contains $\mathbf{0}$, is closed under addition, is not closed under scalar multiplication, and is neither the union of two axes, nor a quadrant, nor a set with integer coordinates.
Hint 1: do not regulate the direction of the arrows; regulate which numbers may appear



a line through the origin *a line without the origin*

Figure 6.2: Two sets in $\mathbb{R}^2$. Left: $\{(x, y) : y = x\}$ contains $\mathbf{0}$ and $(2, 2)$, so it is closed under both operations. Right: $\{(x, y) : x + y = 1\}$ contains $(1, 0)$ and $(0, 1)$, but the origin is marked with an empty circle because it is not included; a single missing point is enough to disqualify a subspace.

in the sum of the two coordinates.
Hint 2: require $x + y$ to be an integer. Then $(1, 0) + (0, 1) = (1, 1)$ stays inside, whereas $\frac{1}{2}(1, 0)$ leaves.
If you get stuck, go back to the section "Subspaces and the subspace test" and compare with the line example $x + y = 1$; the only difference is the right-hand side of the equation.

subspace (*subruang*) — a corridor inside a building: we may walk back and forth in that corridor, but we never leave it.

For younger readers (age 10 and up)

Picture two straight corridors in a building that meet at a single point, like the letter X. Each corridor, if we walk inside it, is still a corridor: we can go forward, backward, and stop anywhere without leaving it. The same holds if we walk twice as far along the same corridor; we are still in that corridor.
Now consider the union of the two corridors. Take one step in the first corridor, then one step in the second. The place where we stop lies in neither corridor. So the union of two corridors is not a corridor, even though each corridor is a corridor. This is the thing most often forgotten about the word "closed": what is tested is not the members one by one, but the result of mixing them.
The nagging question: if three corridors meet at one point, is their union a corridor? Sometimes yes, when all three are the same corridor. In this chapter we will learn why "closed under mixing" is far stronger than "contains the meeting point", and why a single example of a step that leaves is enough to disqualify everything.

6.3 Linear combinations and span

Level: intermediate

A linear combination is the result of mixing several vectors in certain proportions. The span, in turn, is the set of all those mixtures, for every conceivable proportion.

Definition 6.3 (Linear combination and span)**.** A vector $\mathbf{v}$ is a *linear combination* of $\mathbf{v}_1, \dots, \mathbf{v}_k$ if there are scalars $c_1, \dots, c_k$ with $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$. The *span* of a set $S \subseteq V$, written $\operatorname{span}(S)$, is the set of all finite linear combinations of members of S. For $S = \varnothing$ we stipulate $\operatorname{span}(S) = \{\mathbf{0}\}$.

A span is always a subspace, and it is the smallest subspace containing S. The first statement says the span deserves to be called a space; the second, that no other space is more economical.

Lemma 6.3. *For every $S \subseteq V$, $\operatorname{span}(S)$ is a subspace of V, and every subspace containing S also contains $\operatorname{span}(S)$.*

Proof. For $S = \varnothing$ the statement is true because $\{\mathbf{0}\}$ is a subspace. For S nonempty, take $\mathbf{u} = \sum a_i \mathbf{s}_i$ and $\mathbf{v} = \sum b_j \mathbf{t}_j$, two finite linear combinations of members of S. Then $\mathbf{u} + \mathbf{v} = \sum a_i \mathbf{s}_i + \sum b_j \mathbf{t}_j$ and $c\mathbf{u} = \sum (ca_i)\mathbf{s}_i$ are also finite linear combinations of members of S, so the span is closed under both operations; it also contains $\mathbf{0} = 0\mathbf{s}$ for any $\mathbf{s} \in S$. If W is a subspace and $S \subseteq W$, then every $\mathbf{s} \in S$ lies in W, and closure of W forces every finite linear combination of members of S to lie in W as well. $\square$

Therefore $\operatorname{span}(S)$ is the intersection of all subspaces containing S, and adding a vector already inside the span changes nothing: $\operatorname{span}(\operatorname{span}(S)) = \operatorname{span}(S)$.

If $\mathbf{b}$ lies in the span of the columns of A, then the system $A\mathbf{x} = \mathbf{b}$ has a solution.

Proposition 6.4. *Let A be an $m \times n$ matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ and let $\mathbf{b} \in \mathbb{R}^m$. The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b} \in \operatorname{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$.*

Proof. Writing $A\mathbf{x} = \mathbf{b}$ coordinate by coordinate is the same as writing $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$. So a solution $\mathbf{x}$ exists exactly when $\mathbf{b}$ is a linear combination of the columns of A with coefficients $x_1, \dots, x_n$. $\square$

The set $\operatorname{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ is called the *column space* of A, and it is a subspace of $\mathbb{R}^m$ by the lemma above. For the opening table, that column space is built by four columns, and two of them are redundant.

As a small example, in $\mathbb{R}^3$ consider $\mathbf{s}_1 = (1, 0, 1)$ and $\mathbf{s}_2 = (0, 1, 1)$. Every linear combination has the form $(a, b, a + b)$, so the span is the plane $z = x + y$. The vector $(1, 2, 4)$ lies outside that plane because $4 \neq 1 + 2$, whereas $(1, 2, 3)$ lies inside it with $a = 1$, $b = 2$.

Do it by hand (kinesthetic learner)

Materials: two pencils of equal length and one flat table. Mark a point on the table with a fingertip; call it the origin.

Place the first pencil with its tail at the origin. Place the second pencil with its tail at the tip of the first, at a right angle to it. Now slide the second pencil without rotating it until its tail is back at the origin. Its tip moves to another place, and that place is the sum of the two pencils: two orders of addition stop at the same point.

Now the scalars. Reverse one pencil, and its direction becomes negative. Join two identical pencils from the origin, and the length doubles. What the reader must see: all those tips stay on one straight line, however far the pencils are joined. One pencil covers only one line on the table; two nonparallel pencils cover the entire table surface. That is the difference between a line and a plane, and no third pencil adds a new surface.

6.4 Linear independence and dependence

Level: intermediate

A set of vectors is called linearly independent if not one of its members can be mixed from the other members. It is linearly dependent if some member is redundant.

Definition 6.4 (Linear independence and dependence)**.** A set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subseteq V$ is *linearly independent* if the only choice of scalars satisfying $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$ is $c_1 = \cdots = c_k = 0$. If some other choice exists, the set is *linearly dependent*.

Two small conventions: the empty set is linearly independent because there are no coefficients to check, and $\{\mathbf{0}\}$ is linearly dependent because $1 \cdot \mathbf{0} = \mathbf{0}$ is a nontrivial relation.

In a linearly dependent set there is always one vector that lies in the span of the preceding vectors.

Lemma 6.5 (Dependence)**.** *Let $\mathbf{v}_1 \neq \mathbf{0}$ and $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$. The set S is linearly dependent if and only if there is an index $j \geq 2$ with $\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$.*

Proof. If $\mathbf{v}_j = \sum_{i<j} c_i\mathbf{v}_i$, then $\sum_{i<j} c_i\mathbf{v}_i - \mathbf{v}_j = \mathbf{0}$ is a nontrivial relation, because the coefficient of $\mathbf{v}_j$ is -1; so S is dependent. Conversely, if S is dependent then there is a nontrivial relation $\sum_{i=1}^{k} c_i\mathbf{v}_i = \mathbf{0}$. Choose the largest j with $c_j \neq 0$. Since $\mathbf{v}_1 \neq \mathbf{0}$, the index j is not 1, so $j \geq 2$ and

$$\mathbf{v}_j = -\frac{1}{c_j} \sum_{i<j} c_i\mathbf{v}_i \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}).$$

□

Three consequences we use constantly: a set containing $\mathbf{0}$ is always dependent; two vectors are dependent exactly when one is a multiple of the other; and every subset of a linearly independent set is also linearly independent.

As an example, consider $\mathbf{v}_1 = (1, 2, -1)$, $\mathbf{v}_2 = (0, 1, 3)$, and $\mathbf{v}_3 = (2, 5, 1)$ in $\mathbb{R}^3$. Since $2\mathbf{v}_1 + \mathbf{v}_2 = (2, 5, 1) = \mathbf{v}_3$, the relation $2\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$ is nontrivial and the three are dependent. Not one of those vectors is zero, and no two are multiples; the dependence appears only at three vectors.

Figure 6.3 places the two side by side: the span of two vectors as a plane, and three vectors one of which already lies in the span of the other two.

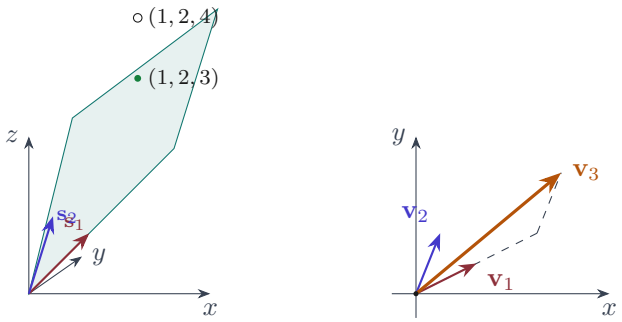


Figure 6.3: Left: $\operatorname{span}\{(1, 0, 1), (0, 1, 1)\}$ is the plane $z = x + y$; the point $(1, 2, 3)$ lies inside it, the point $(1, 2, 4)$ outside it. Right: a two-dimensional sketch of $\mathbf{v}_1 = (1, 2, -1)$ and $\mathbf{v}_2 = (0, 1, 3)$ with $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2 = (2, 5, 1)$; the dashed lines show $2\mathbf{v}_1$ plus $\mathbf{v}_2$ landing exactly at the tip of $\mathbf{v}_3$, so that the three are dependent.

In $\mathbb{R}^2$ any three vectors are always dependent, without computing anything.

Proposition 6.6. *Every set of three vectors in $\mathbb{R}^2$ is linearly dependent.*

Proof. Let $S = \{\mathbf{u}, \mathbf{v}, \mathbf{w}\} \subseteq \mathbb{R}^2$. If two of them are multiples, S is dependent. If not, $\mathbf{u}$ and $\mathbf{v}$ are linearly independent. Take $\mathbf{w} = (w_1, w_2)$ and look for a, b with $a\mathbf{u} + b\mathbf{v} = \mathbf{w}$. Writing

out the components gives two linear equations in a and b, and since $\mathbf{u}$, $\mathbf{v}$ are not multiples, that system has a unique solution. Hence $\mathbf{w} \in \operatorname{span}(\mathbf{u}, \mathbf{v})$. $\square$

In $\mathbb{R}^3$ a similar statement holds for four vectors, but the method above would become a long case check; we need a uniform argument.

To test independence computationally, arrange the vectors as the columns of a matrix and then solve $A\mathbf{c} = \mathbf{0}$; the set is linearly independent exactly when that system has only the trivial solution. The determinant offers a shortcut in $\mathbb{R}^3$, but that tool is only built in Chapter 8.

6.5 Basis and dimension

Level: advanced

A basis is a list of vectors that is enough to build the whole space and contains not one redundant vector. The words “enough” and “no redundancy” are exactly the two conditions we already have.

Definition 6.5 (Basis and dimension)**.** A set $B \subseteq V$ is called a *basis* of V if B is linearly independent and $\operatorname{span}(B) = V$. If V has a finite basis, V is called *finite-dimensional*; if not, V is infinite-dimensional.

The definition of dimension cannot yet be written, because it depends on a fact not yet proved: two bases of the same space always have the same size.

First step: a representation in a basis is always unique.

Lemma 6.7 (Uniqueness of coordinates)**.** *If $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis of V, then every $\mathbf{v} \in V$ can be written uniquely as $\mathbf{v} = a_1\mathbf{b}_1 + \cdots + a_n\mathbf{b}_n$.*

Proof. Existence of the representation comes from $\operatorname{span}(B) = V$. Suppose there are two representations with coefficients a_i and a_i'. Subtracting them gives $\sum_i (a_i - a_i')\mathbf{b}_i = \mathbf{0}$. Because B is linearly independent, all those difference coefficients are zero, so $a_i = a_i'$ for every i. $\square$

The numbers $a_1, \dots, a_n$ are called the coordinates of $\mathbf{v}$ with respect to B; change of basis is the subject of Chapter 7, and here we only need uniqueness.

Second step: one vector can be exchanged for another without destroying the spanning property. Replacing one vector in a spanning set by a combination that contains it does not reduce what can be built.

Lemma 6.8 (Exchange)**.** *Let $\mathbf{u} = \sum_{\mathbf{s} \in S} a_{\mathbf{s}}\mathbf{s}$ be a finite sum with $a_{\mathbf{s}_0} \neq 0$ for some $\mathbf{s}_0 \in S$. If $S \cup \{\mathbf{u}\}$ spans V, then $(S \setminus \{\mathbf{s}_0\}) \cup \{\mathbf{u}\}$ also spans V.*

Proof. From that relation, $\mathbf{s}_0 = a_{\mathbf{s}_0}^{-1}\big(\mathbf{u} - \sum_{\mathbf{s} \neq \mathbf{s}_0} a_{\mathbf{s}}\mathbf{s}\big)$, so $\mathbf{s}_0$ is a linear combination of members of the smaller set $T = (S \setminus \{\mathbf{s}_0\}) \cup \{\mathbf{u}\}$. Every member of $S \cup \{\mathbf{u}\}$ therefore lies in $\operatorname{span}(T)$, so $\operatorname{span}(S \cup \{\mathbf{u}\}) \subseteq \operatorname{span}(T)$. The reverse inclusion is clear because $T \subseteq S \cup \{\mathbf{u}\}$. Hence both spans equal V. $\square$

The third step is the heart of this chapter.

If a space can be built by n vectors, then it is impossible to find $n+1$ linearly independent vectors inside it.

Theorem 6.9 (Exchange)**.** *If V is spanned by n vectors and $\mathbf{w}_1, \dots, \mathbf{w}_m$ are linearly independent in V, then $m \leq n$.*

Proof. Let $S_0 = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ span V. For $j = 1$: $\mathbf{w}_1 \neq \mathbf{0}$ is a linear combination of members of S_0 with at least one nonzero coefficient, so the exchange lemma gives $T_1 \subseteq S_0$ of size $n-1$ with $\{\mathbf{w}_1\} \cup T_1$ spanning.

Suppose $\{\mathbf{w}_1, \dots, \mathbf{w}_j\} \cup T_j$ spans with $|T_j| = n-j$. The vector $\mathbf{w}_{j+1}$ is a linear combination of that set. If all coefficients on T_j are zero, then $\mathbf{w}_{j+1}$ becomes a linear combination of $\mathbf{w}_1, \dots, \mathbf{w}_j$, which is impossible because that set is linearly independent. So some member of T_j has a nonzero coefficient, and the exchange lemma gives T_{j+1} of size $n-j-1$ that still spans together with $\mathbf{w}_1, \dots, \mathbf{w}_{j+1}$.

After m steps we obtain a spanning set of size $n-m$; the size of a set cannot be negative, so $m \leq n$. □

Now the size of a space can be written: the number of vectors in its most economical list, and that number does not depend on the list chosen.

Two bases of a finite-dimensional space always have the same number of elements.

Corollary 6.10. *If V is finite-dimensional with a basis B_1 of size n and a basis B_2 of size m, then $n = m$.*

Proof. B_1 spans and B_2 is linearly independent, so the exchange theorem gives $m \leq n$. Conversely B_2 spans and B_1 is linearly independent, hence $n \leq m$. □

The number of vectors in that most economical list is important enough to deserve a name of its own.

Definition 6.6 (Dimension)**.** For finite-dimensional V, $\dim V$ is the number of members of any basis of V. The null space $\{\mathbf{0}\}$ is stipulated to have dimension 0.

The size of a subspace never exceeds the size of its space; a list whose length matches the dimension is immediately a basis; and a linearly independent list can always be completed to full length.

Corollary 6.11. *Let $\dim V = n$. (a) Every linearly independent set of n vectors is a basis. (b) Every spanning set of n vectors is a basis. (c) Every linearly independent set can be extended to a basis. (d) If U is a subspace of V, then $\dim U \leq \dim V$, and equality holds only when $U = V$.*

Proof. For (a), a linearly independent set of size n that does not span can be enlarged by one vector outside its span, and the result is still linearly independent; so there are $n+1$ linearly independent vectors, and the exchange theorem forbids that. For (b), a spanning set of size n that is dependent can lose one vector through the dependence lemma and still span with $n-1$ vectors; again this contradicts exchange. For (c), add a vector outside the span as long as one exists; the process stops after at most n steps, and when it stops the span is already all of V. For (d), a basis of U is a linearly independent set in V, so its size is at most n; if its size is n, part (a) makes it a basis of V, hence $U = V$. □

Thus $\dim \mathbb{R}^n = n$, $\dim P_n = n+1$ with basis $1, x, \dots, x^n$, $\dim M_{m\times n}(\mathbb{R}) = mn$ with basis the matrices having exactly one entry equal to 1, and $\dim\{\mathbf{0}\} = 0$. For function spaces such as $\mathbb{R}^{\mathbb{R}}$ there is no finite basis; the proof requires the axiom of choice and lies outside the scope of this book.

basis (*basis*) — the most economical list of paint cans: enough to make every color, and not one can in it can be thrown away.

dimension (*dimensi*) — the number of cans in that most economical list; the number belongs to the space, not to the list we happen to choose.

6.6 Direct sums and complementary subspaces

LEVEL: ADVANCED

The sum of two subspaces is the set of all results of adding one member of each subspace.

Definition 6.7 (Sum of subspaces)**.** For subspaces $U, W \subseteq V$, the *sum* $U + W$ is $\{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U,\ \mathbf{w} \in W\}$.

It differs from the union, which is almost never a subspace, as the two-axes example shows.

The sum is itself a subspace, and no smaller subspace contains both.

Proposition 6.12. *For subspaces $U, W \subseteq V$, the set $U + W$ is a subspace of V, and it is contained in every subspace that contains $U \cup W$.*

Proof. The set $U + W$ contains $\mathbf{0} = \mathbf{0} + \mathbf{0}$. If $\mathbf{u}_1 + \mathbf{w}_1$ and $\mathbf{u}_2 + \mathbf{w}_2$ are two of its members, their sum $(\mathbf{u}_1 + \mathbf{u}_2) + (\mathbf{w}_1 + \mathbf{w}_2)$ has its first term in U and its second in W, and $c(\mathbf{u}_1 + \mathbf{w}_1) = c\mathbf{u}_1 + c\mathbf{w}_1$ is of the same form; so $U + W$ is a subspace. If Z is a subspace containing U and W, then Z contains every $\mathbf{u} + \mathbf{w}$ because it is closed under addition, so $U + W \subseteq Z$. □

The size of a sum is not the sum of the sizes, because the part that overlaps is counted twice.

$\dim(U + W)$ equals $\dim U$ plus $\dim W$ minus the dimension of the intersection.

Theorem 6.13 (Dimension of a sum)**.** *If U and W are subspaces of a finite-dimensional space V, then*

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof. The intersection $U \cap W$ is a subspace, so it is finite-dimensional. Take a basis $\{\mathbf{z}_1, \ldots, \mathbf{z}_m\}$ of $U \cap W$. Since it is linearly independent in U, that set is extended to a basis of U by adding $\mathbf{u}_1, \ldots, \mathbf{u}_j$, so that $\dim U = m + j$; in the same way, adding $\mathbf{w}_1, \ldots, \mathbf{w}_k$ yields a basis of W, so that $\dim W = m + k$.

Claim: $\{\mathbf{z}_1, \ldots, \mathbf{z}_m, \mathbf{u}_1, \ldots, \mathbf{u}_j, \mathbf{w}_1, \ldots, \mathbf{w}_k\}$ is a basis of $U + W$. The set contains a basis of U and a basis of W, so its span contains $U + W$; the reverse inclusion is clear. For independence, suppose

$$\sum_i a_i \mathbf{z}_i + \sum_r b_r \mathbf{u}_r + \sum_s c_s \mathbf{w}_s = \mathbf{0}.$$

Then $\sum_s c_s \mathbf{w}_s = -\sum_i a_i \mathbf{z}_i - \sum_r b_r \mathbf{u}_r$ lies in U and at the same time in W; so it is a member of $U \cap W$ and can be written $\sum_i d_i \mathbf{z}_i$. Then $\sum_s c_s \mathbf{w}_s - \sum_i d_i \mathbf{z}_i = \mathbf{0}$ is a linear relation in the basis of W, so all c_s and d_i are zero; the original equation shrinks to a linear relation in the basis of U, so that a_i and b_r are also zero.

Hence $\dim(U + W) = m + j + k = (m + j) + (m + k) - m = \dim U + \dim W - \dim(U \cap W)$. □

Now the second question from the opening of the chapter: when is the decomposition into two terms unique?

The sum $U + W$ is called a *direct sum*, written $U \oplus W$, if every member of $U + W$ can be written as $\mathbf{u} + \mathbf{w}$ in exactly one way.

Definition 6.8 (Direct sum)**.** If every $\mathbf{x} \in U + W$ has exactly one representation $\mathbf{x} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in W$, then $U + W$ is called a direct sum and is written $U \oplus W$.

Uniqueness of all representations can be checked by looking only at the representation of zero.

$U + W$ is a direct sum exactly when $U \cap W = \{\mathbf{0}\}$.

Theorem 6.14 (Criterion for a direct sum)**.** *Let U and W be subspaces of V. Then $U + W = U \oplus W$ if and only if $U \cap W = \{\mathbf{0}\}$.*

Proof. Suppose $U + W$ is a direct sum and take $\mathbf{x} \in U \cap W$. Then $\mathbf{0} = \mathbf{x} + (-\mathbf{x})$ with $\mathbf{x} \in U$ and $-\mathbf{x} \in W$, and also $\mathbf{0} = \mathbf{0} + \mathbf{0}$. Uniqueness gives $\mathbf{x} = \mathbf{0}$.

Conversely, suppose $U \cap W = \{\mathbf{0}\}$. Take $\mathbf{x} \in U + W$ with two representations $\mathbf{x} = \mathbf{u}_1 + \mathbf{w}_1 = \mathbf{u}_2 + \mathbf{w}_2$. Then $\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{w}_2 - \mathbf{w}_1$ lies in U and in W, so it equals $\mathbf{0}$. Consequently $\mathbf{u}_1 = \mathbf{u}_2$ and $\mathbf{w}_1 = \mathbf{w}_2$. $\square$

For two subspaces that criterion is enough. For three subspaces or more, checking pairwise intersections is not enough, and the next example shows why.

In $\mathbb{R}^3$ take $V_1 = \operatorname{span}\{(1,0,0)\}$, $V_2 = \operatorname{span}\{(0,1,0)\}$, and $V_3 = \operatorname{span}\{(1,1,0)\}$. Every pairwise intersection contains only $\mathbf{0}$, but $V_1 + V_2 + V_3$ is not a direct sum: $(1,1,0)$ can be written $(1,0,0) + (0,1,0) + \mathbf{0}$ and also $\mathbf{0} + \mathbf{0} + (1,1,0)$. Even zero has two representations, namely $\mathbf{0} + \mathbf{0} + \mathbf{0}$ and $(1,0,0) + (0,1,0) + (-1,-1,0)$.

For many terms, the criterion returns to the definition: $\mathbf{0} = \mathbf{v}_1 + \cdots + \mathbf{v}_k$ with $\mathbf{v}_i \in V_i$ is possible only when all the $\mathbf{v}_i$ are zero.

Finally, every subspace can be paired with another subspace so that the two decompose the whole space.

Theorem 6.15 (Complement)**.** *If V is finite-dimensional and U is a subspace of V, then there is a subspace $W \subseteq V$ with $V = U \oplus W$. In that case* $\dim V = \dim U + \dim W$.

Proof. If $U = \{\mathbf{0}\}$, take $W = V$. If not, take a basis $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ of U and extend it to a basis $\{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_k\}$ of V; this extension is possible by consequence (c) in the previous section. Set $W = \operatorname{span}\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$.

Every $\mathbf{x} \in V$ is a linear combination of the entire basis, and separating the $\mathbf{u}$ terms from the $\mathbf{w}$ terms exhibits it as a member of $U + W$. For the intersection, take $\mathbf{x} \in U \cap W$. Then $\mathbf{x} = \sum a_i \mathbf{u}_i = \sum b_s \mathbf{w}_s$, so that $\sum a_i \mathbf{u}_i - \sum b_s \mathbf{w}_s = \mathbf{0}$. This is a linear relation in the basis of V, so all its coefficients are zero, in particular $\mathbf{x} = \mathbf{0}$. By the criterion above, $V = U \oplus W$, and from the dimension-of-a-sum theorem, $\dim V = \dim U + \dim W$.

The complement is not unique. In $\mathbb{R}^2$ with U the x-axis, every line through the origin other than the x-axis is a complement of U; what is unique is not the choice but the dimension. In Chapter 10, orthogonal complements make that decomposition the shortest one as well.

Now the opening question can be answered. For $\mathbf{w} = (1,1,1)$, let $U = \{\boldsymbol{\delta} : \mathbf{w} \cdot \boldsymbol{\delta} = 0\}$ and $W = \operatorname{span}\{\mathbf{w}\}$. Their intersection is only $\{\mathbf{0}\}$, since $\mathbf{w} \cdot (t\mathbf{w}) = 3t$ is zero only when $t = 0$; with $\dim U = 2$ and $\dim W = 1$ we obtain $U \oplus W = \mathbb{R}^3$. Every perturbation splits uniquely into a part that does not change the score, namely a member of U, and a part that changes it, a member of W.

direct sum (*jumlah langsung*) — two corridors that meet only at the origin, so that every place in the building is reachable by one route that cannot be disputed.

6.7 Worked examples

Level: intermediate

The following four computations use the entire content of the chapter; all of them are rechecked with Python in the lab section.

(a) Rank and a basis of the column space

Back to the opening table. Row elimination: $R_2 \leftarrow R_2 - 2R_1$ turns the second row into $(0,0,1,1)$, and $R_3 \leftarrow R_3 - 3R_1$ turns the third row into $(0,0,2,2)$. Then $R_3 \leftarrow R_3 - 2R_2$ eliminates the third row, so that what remains is $\mathbf{r}_1 = (1,2,0,1)$ and $(0,0,1,1)$.

That form is already reduced echelon, with pivot columns 1 and 3. Hence $\operatorname{rank} A = 2$ and its column space has dimension 2 with basis $\{\mathbf{c}_1, \mathbf{c}_3\} = \{(1,2,3),(0,1,2)\}$.

A direct check against the discarded columns:

$$\mathbf{c}_2 = (2,4,6) = 2\mathbf{c}_1, \qquad \mathbf{c}_4 = (1,3,5) = \mathbf{c}_1 + \mathbf{c}_3.$$

Both are exact, with no rounding. This is the answer "two" to the opening question: the pivot columns give a linearly independent set that spans the column space, and a column without a pivot is a combination of the pivot columns to its left.

The null space of A is also readable from the reduced form. From $x_1 + 2x_2 + x_4 = 0$ and $x_3 + x_4 = 0$ we obtain $x_1 = -2x_2 - x_4$ and $x_3 = -x_4$, so that

$$\mathbf{x} = x_2\,(-2,1,0,0) + x_4\,(-1,0,-1,1),$$

written as row vectors for brevity. Hence $\dim\ker A = 2$, and $\operatorname{rank} A + \dim\ker A = 2 + 2 = 4$ equals the number of columns; the general statement is rank-nullity, proved in Chapter 7.

(b) Three dependent vectors

Consider $\mathbf{v}_1 = (1,2,-1)$, $\mathbf{v}_2 = (0,1,3)$, $\mathbf{v}_3 = (2,5,1)$ as the columns of V. Cofactor expansion along the first row:

$$\det V = 1(1\cdot 1 - 3\cdot 5) + 0 + 2\big(2\cdot 3 - 1\cdot(-1)\big) = (1-15) + 2(6+1) = 0.$$

A zero determinant means the columns are dependent; the relation is guessed and then checked: $2\mathbf{v}_1 + \mathbf{v}_2 = (2,5,1) = \mathbf{v}_3$, so $2\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$ and $\operatorname{rank} V = 2$.

(c) Dimension of a polynomial space

$\dim P_3 = 4$, because $\{1, x, x^2, x^3\}$ is a basis. The set $B = \{1,\ 1+x,\ 1+x+x^2,\ 1+x+x^2+x^3\}$ is also a basis: its coordinate matrix with respect to the standard basis is upper triangular with diagonal 1, so B is linearly independent, and since its size is $4 = \dim P_3$, consequence (a) makes it a basis.

The coordinates of $p = 2 - x + 3x^3$ with respect to B are computed from the highest degree downward: $a_4 = 3$, then $a_3 + a_4 = 0$ gives $a_3 = -3$, $a_2 + a_3 + a_4 = -1$ gives $a_2 = -1$, and the constant term gives $a_1 = 3$. Hence $p = 3 - (1+x) - 3(1+x+x^2) + 3(1+x+x^2+x^3)$, which upon expansion gives $2 - x + 3x^3$.

(d) Decomposition as a direct sum

Take $U = \{(x,y,z) : x + y + z = 0\}$ and $W = \operatorname{span}\{(1,1,1)\}$ in $\mathbb{R}^3$. A basis of U is $\{(1,-1,0),(1,0,-1)\}$, so $\dim U = 2$; with $\dim W = 1$ and $U \cap W = \{\mathbf{0}\}$ we obtain $\mathbb{R}^3 = U \oplus W$.

Decompose $\mathbf{v} = (3,0,1)$ as $\mathbf{u} + t(1,1,1)$. The condition that the sum of the coordinates of $\mathbf{u}$ be zero gives $3 + 0 + 1 - 3t = 0$, so $t = 4/3$. Then

$$\mathbf{v} = \underbrace{\left(\tfrac{5}{3}, -\tfrac{4}{3}, -\tfrac{1}{3}\right)}_{\in U} + \underbrace{\left(\tfrac{4}{3}, \tfrac{4}{3}, \tfrac{4}{3}\right)}_{\in W},$$

and $\frac{5}{3} - \frac{4}{3} - \frac{1}{3} = 0$ confirms that the first term really lies in U. By the criterion theorem, there is no other decomposition.

Redo (a) without looking: carry out the elimination on A, mark the pivot columns, then write $\mathbf{c}_2$ and $\mathbf{c}_4$ as combinations of $\mathbf{c}_1$ and $\mathbf{c}_3$. Then compute $\dim \ker A$ from the reduced form and check that its sum with the rank equals the number of columns. Finally, for the matrix V in (b), replace entry $(3,1)$ by 0 and decide whether the columns are still dependent.

6.8 Computational lab

Level: advanced

The following code runs as is with numpy. The function `rref` is written by hand so that the pivot columns are visible; its last line cleans up leftover rounding and negative zeros.

Computational lab

```
import numpy as np

np.set_printoptions(precision=3, suppress=True)

def rref(M):
    M = M.astype(float).copy()
    r = 0
    for c in range(M.shape[1]):
        i = np.argmax(np.abs(M[r:, c])) + r
        if abs(M[i, c]) < 1e-12:
            continue
        M[[r, i]] = M[[i, r]]
        M[r] = M[r] / M[r, c]
        for j in range(M.shape[0]):
            if j != r:
                M[j] = M[j] - M[j, c] * M[r]
        r = r + 1
        if r == M.shape[0]:
            break
    M[np.abs(M) < 1e-12] = 0.0
    return M

A = np.array([[1, 2, 0, 1],
              [2, 4, 1, 3],
              [3, 6, 2, 5]], float)
print("RREF(A):"); print(rref(A))
print("rank(A) =", np.linalg.matrix_rank(A))
print("k2-2k1 =", A[:, 1] - 2 * A[:, 0])
print("k4-k1-k3 =", A[:, 3] - A[:, 0] - A[:, 2])

V = np.array([[1, 0, 2],
              [2, 1, 5],
              [-1, 3, 1]], float)
print("det(V) =", np.linalg.det(V))
print("2v1+v2-v3 =", 2 * V[:, 0] + V[:, 1] - V[:, 2])

x = np.array([0., 1, 2, 3])
M = np.vander(x, 4, increasing=True)
print("det(M) =", np.linalg.det(M))
print("rank(M) =", np.linalg.matrix_rank(M))

B = np.array([[1, 1, 1, 1],
              [0, 1, 1, 1],
              [0, 0, 1, 1],
```

```
              [0, 0, 0, 1]], float)
p = np.array([2, -1, 0, 3], float)
k = np.linalg.solve(B, p)
print("koordinat p =", k)
print("sisa =", B @ k - p)
```

The printed output:

```
RREF(A):
[[1. 2. 0. 1.]
 [0. 0. 1. 1.]
 [0. 0. 0. 0.]]
rank(A) = 2
k2-2k1 = [0. 0. 0.]
k4-k1-k3 = [0. 0. 0.]
det(V) = 0.0
2v1+v2-v3 = [0. 0. 0.]
det(M) = 12.0
rank(M) = 4
koordinat p = [ 3. -1. -3.  3.]
sisa = [0. 0. 0. 0.]
```

Four things are readable from that output. The pivot columns 1 and 3 as well as $\operatorname{rank} A = 2$ agree with the hand computation. The differences $\mathbf{c}_2 - 2\mathbf{c}_1$ and $\mathbf{c}_4 - \mathbf{c}_1 - \mathbf{c}_3$ are exactly zero; on real data those differences are small but nonzero, and that is where the limits of applying rank numerically lie. $\det V = 0$ confirms the dependence in (b), and the Vandermonde matrix of the points $0, 1, 2, 3$ has determinant 12 and rank 4. The coordinates of p are printed as $(3, -1, -3, 3)$, matching the hand computation.

Note for instructors

Start from the opening table, not from the axioms: ask the students to guess how many columns are independent, then show that the guess "four" is wrong. The axioms come in after that, as an answer to a need. The subspace test and span in the second meeting, linear independence and basis in the third, direct sums in the fourth.
Two typical misconceptions. First, testing a subspace only by "does it contain zero" and then stopping; the first quadrant contains zero and is closed under addition, but fails on negative multiples. Second, taking basis to mean standard basis, so that $(5, 7)$ and $(4, 3)$ are judged not to be a basis of $\mathbb{R}^2$; stress that being a basis is a property of the set.
Class activity: each group writes three vectors in $\mathbb{R}^3$ and decides independence or dependence by elimination; another group changes one entry and asks for a new decision. Students are ready to move on when they can explain why three vectors in $\mathbb{R}^2$ are always dependent.

6.9 Summary

LEVEL: BASIC

Summary.

- Vector space: a set with addition and scalar multiplication satisfying the eight axioms; its members may be vectors, matrices, polynomials, or functions.
- A subspace is checked with just three things: it contains $\mathbf{0}$ and is closed under both operations.
- $\operatorname{span}(S)$ is the smallest subspace containing S; $\mathbf{b}$ lies in the column space of A exactly when $A\mathbf{x} = \mathbf{b}$ is consistent.

- Linear dependence: one member lies in the span of the preceding members; three vectors in $\mathbb{R}^2$ are always dependent.
- The exchange theorem traps $n+1$ linearly independent vectors in a space spanned by n vectors, so that all bases have the same size and $\dim V$ is defined.
- $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$; $U+W$ is direct exactly when $U \cap W = \{\mathbf{0}\}$; every finite-dimensional subspace has a complement.

6.10 Exercises

Level: basic

Exercise 6.1.

1. Decide whether $W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0 \text{ and } x + y = 0\}$ is a subspace of $\mathbb{R}^3$. If so, find a basis and its dimension.
2. Prove that the set of all 2×2 matrices whose main diagonal entries are zero is a subspace of $M_{2\times 2}(\mathbb{R})$, then find a basis and its dimension.
3. Show that $S = \{1+x,\ 1+x^2,\ 2+x+x^2\}$ is linearly dependent in P_2. Find $\dim \operatorname{span}(S)$, then extend a largest linearly independent subset of S to a basis of P_2.
4. Prove: if $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent in V, then $\{\mathbf{v}_1 + \mathbf{v}_2,\ \mathbf{v}_2 + \mathbf{v}_3,\ \mathbf{v}_3 + \mathbf{v}_1\}$ is also linearly independent.
5. Compute $\dim(U \cap W)$ and $\dim(U + W)$ for the two subspaces
$$U = \operatorname{span}\{(1,0,0), (0,1,0)\},$$
$$W = \operatorname{span}\{(1,1,0), (0,0,1)\}$$
in $\mathbb{R}^3$. Check the dimension-of-a-sum formula, then compute them with numpy as well.
6. For the 3×4 matrix with rows $(1,2,3,4)$, $(2,4,6,9)$, and $(0,1,1,1)$, find the rank, one basis of the column space, and one basis of the null space. Check with the function `matrix_rank` from numpy and `null_space` from scipy.
7. Assess this claim: *if U_1, U_2, U_3 are subspaces of V and every pairwise intersection equals $\{\mathbf{0}\}$, then $U_1 + U_2 + U_3 = U_1 \oplus U_2 \oplus U_3$.* State true, false, or not enough evidence, and include a reason or a counterexample.
8. Prove that if $\dim V = n$ and U is a subspace with $\dim U = n - 1$, then there is a vector $\mathbf{v} \in V$ such that $V = U \oplus \operatorname{span}\{\mathbf{v}\}$.
9. Find the dimension of the space of all polynomials of degree at most 4 that vanish at $x = 0$ and $x = 1$ simultaneously, then give one basis of it. Check your answer by computing the rank of the 2×5 evaluation matrix at those two points.
10. Given $B = \{1,\ 1+x,\ 1+x+x^2,\ 1+x+x^2+x^3\}$ a basis of P_3. Find the coordinates of $q = 4 + 2x - 5x^2 + x^3$ with respect to B, then check by solving the linear system in numpy.

Hints for the three hardest items. Item 4: suppose $\sum c_i(\dots) = \mathbf{0}$, collect the coefficient of each $\mathbf{v}_i$, then use the independence of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. Item 8: take a basis of U, extend it by one vector, then use consequence (d). Item 9: a polynomial of degree at most 4 that vanishes at 0 and 1 is divisible by $x(x-1)$; count the free parameters.

For researchers: limits and next steps

Three things are genuinely proved in this chapter, and all three hold over any field: the uniqueness of the size of a basis through the exchange theorem (Steinitz's result), the formula $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$, and the existence of a complement for every finite-dimensional subspace. The popular claim that "rank equals the number of informative features" is meaningful only if "informative" means linearly independent,

and on real data that condition is never satisfied exactly.
Primary sources: Anton, *Elementary Linear Algebra* 11th edition, Chapter 4 sections 4.1–4.5; Axler, *Linear Algebra Done Right* 4th edition, sections 1B, 1C, 2A–2C; Hoffman and Kunze, *Linear Algebra* 2nd edition, Chapter 2.
Open question: how far does $\dim(U \cap W)$ shift as a result of perturbing the entries? Limits of validity: all results assume the vector space axioms; for infinite-dimensional spaces, the existence of a basis requires Zorn's lemma. A rank computed by a computer depends on a threshold.

Map to source texts: This chapter follows Anton, Elementary Linear Algebra *11th edition, Chapter 4 (General Vector Spaces) section 4.1 (Real Vector Spaces) for the axioms and examples, 4.2 (Subspaces) for the subspace test and span, 4.3 (Linear Independence) for independence and the dependence lemma, and 4.5 (Dimension) for the exchange theorem and its consequences; other editions number these differently. A more abstract counterpart is in Axler,* Linear Algebra Done Right *4th edition, Chapter 1 sections 1B and 1C (including sums and direct sums of subspaces) and Chapter 2 sections 2A, 2B, and 2C, where the exchange theorem appears as a comparison of the length of a linearly independent list with that of a spanning list. The treatment of coordinates, linear transformations, and rank-nullity is not in this chapter but in Chapter 7.*

Chapter 7

Basis, Dimension, and Linear Transformations

An anomaly detector does not read raw network traffic. It summarizes it first. Four counters, say failed connections, outbound packets, session length, and the number of unique ports, are compressed into three features by the linear map $F\colon \mathbb{R}^4 \to \mathbb{R}^3$. The decision threshold only looks at $F(\mathbf{x})$.

The decisive question: is there $\boldsymbol{\delta} \neq \mathbf{0}$ with $F(\mathbf{x}+\boldsymbol{\delta}) = F(\mathbf{x})$? Because F is linear, that condition is equivalent to $F(\boldsymbol{\delta}) = \mathbf{0}$. The answer is a subspace of dimension $4 - \operatorname{rank} F$; if F has rank three, there is a whole line of blind directions, and an attacker can shift all four counters along that line without shifting a single one of the features.

To answer it we need three things Chapter 6 has not yet given: the list of coordinates with respect to a basis, the rule for moving that list when the basis changes, and the translation of a linear map into a matrix so that the kernel can be computed by elimination. Chapter 6 already established that a basis exists and that its size is unique; its axioms are not repeated here.

For younger readers (age 10 and up)

Imagine a bookcase with many shelves. Every book has a fixed place. If we rearrange the bookcase, the books do not change, but each book's address does. To say where a book is, we must name the arrangement currently in use. A friend who uses another arrangement will name a different address for the same book, and the two of you are both right.

There is also a special shelf behind the bookcase. A book moved there cannot be seen from the front. As long as books only move within that shelf, the list of books in front does not change at all.

Have you ever moved something to a place nobody inspects? We are going to find out why the list of addresses can change while the contents of the bookcase stay the same, why that hidden shelf always has a definite size, and why some changes do not shift the summary at all.

7.1 Coordinates and transition matrices

Level: intermediate

Vectors in $\mathbb{R}^3$ are already lists of three numbers; vectors in P_2 or $M_{2\times 2}(\mathbb{R})$ are not. So that both can be computed in the same way, we need a translation that turns every vector into a list of numbers. That translation depends on the basis, and that is where the difficulty

of this section begins.

Uniqueness we already have: the uniqueness lemma for coordinates in Chapter 6 states that the expression of a vector in a basis is always unique. Therefore the following definition is legitimate without choosing among several expressions.

coordinate vector (*koordinat*) — the address of a book in the bookcase: its contents stay fixed, its address depends on how the shelves are arranged

Definition 7.1 (Coordinate vector)**.** Let $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ be an ordered basis of the vector space V over $\mathbb{R}$. For every $\mathbf{v} \in V$ there are unique scalars $a_1, \dots, a_n$ with $\mathbf{v} = a_1\mathbf{b}_1 + \cdots + a_n\mathbf{b}_n$. The column

$$[\mathbf{v}]_B = (a_1, \dots, a_n)^\top$$

is called the *coordinate vector* of $\mathbf{v}$ with respect to B.

The word "ordered" is not decoration. If two basis vectors are exchanged, two coordinate entries are exchanged with them while the vector stays put. A basis set alone does not determine the coordinate list; its order does as well.

One warning: $[\mathbf{v}]_B$ is not $\mathbf{v}$. Those numbers live in $\mathbb{R}^n$, whereas $\mathbf{v}$ lives in V. For $M_{2\times 2}(\mathbb{R})$ the difference is palpable: the vector holds four numbers, and its coordinates are four numbers too, but arranged as a column that depends on the basis.

The first important property of this translation: it respects both operations of the vector space.

Proposition 7.1. *For every basis B of a space V of dimension n, the map $\varphi_B \colon V \to \mathbb{R}^n$ with $\varphi_B(\mathbf{v}) = [\mathbf{v}]_B$ is linear, injective, and surjective.*

Proof. Take $\mathbf{v} = \sum_i a_i\mathbf{b}_i$ and $\mathbf{w} = \sum_i c_i\mathbf{b}_i$. Then $\mathbf{v} + \mathbf{w} = \sum_i (a_i + c_i)\mathbf{b}_i$. Uniqueness of coordinates forces $[\mathbf{v} + \mathbf{w}]_B = (a_i + c_i) = [\mathbf{v}]_B + [\mathbf{w}]_B$. In the same way, $[c\mathbf{v}]_B = c[\mathbf{v}]_B$.

For injectivity, suppose $[\mathbf{v}]_B = \mathbf{0}$. Then all its coefficients are zero, so $\mathbf{v} = \sum_i 0 \cdot \mathbf{b}_i = \mathbf{0}$. For surjectivity, take any $(a_1, \dots, a_n) \in \mathbb{R}^n$ and form $\mathbf{v} = \sum_i a_i\mathbf{b}_i$; its coordinates are exactly that list. □

The map φ_B turns a question about V into a question about columns in $\mathbb{R}^n$, then turns it back; that transfer depends on B. Coordinates are written as columns so that matrices act from the left: coordinates in, multiplied by the matrix, coordinates out. With row coordinates, every formula would have to be transposed.

Read it aloud (auditory learner)

Two voices stand in front of the class.
"How long is this table?" "Three." "Three what?" "Three rulers. Or three spans. Or three paces. Choose the measuring instrument first."
It is the same with a vector space. A vector has no address until we choose a basis. The basis is the ruler: it supplies a scale, a direction, and an order. Change the ruler and you change the numbers. The object being measured does not change with them.
One ruler, one number. Two rulers, two numbers. Without a ruler, no numbers at all.
Be careful with a bad ruler: two of its scale marks are almost parallel. Numbers still come out, but the numbers wobble, and two people measuring patiently will obtain different numbers.
Repeat together: a basis is not the object, a basis is the instrument. What changes is the address, what stays is the content.

Now two bases are used on the same space. Every vector has two coordinate lists, and we want a formula that turns the first list into the second. The matrix that performs that task is built from the coordinates of the old basis vectors in the new basis.

Definition 7.2 (Transition matrix)**.** Let $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ and $C = (\mathbf{c}_1, \dots, \mathbf{c}_n)$ be two bases of V. The *transition matrix* from B to C is the $n \times n$ matrix

$$P_{C \leftarrow B} = \Big([\mathbf{b}_1]_C \;\Big|\; \cdots \;\Big|\; [\mathbf{b}_n]_C \Big),$$

that is, the matrix whose j-th column is $[\mathbf{b}_j]_C$.

What still needs proof: multiplying the old coordinate list by this matrix produces the new coordinate list.

Theorem 7.2. *For every $\mathbf{v} \in V$ we have $[\mathbf{v}]_C = P_{C \leftarrow B}[\mathbf{v}]_B$.*

Proof. Write $\mathbf{v} = \sum_j a_j \mathbf{b}_j$, so that $[\mathbf{v}]_B = (a_1, \dots, a_n)^{\top}$. Linearity of φ_C gives $[\mathbf{v}]_C = \sum_j a_j [\mathbf{b}_j]_C$. That right-hand side is exactly the matrix product $P_{C \leftarrow B}$ with the column (a_j), by the definition of matrix-vector multiplication. □

A corollary that makes the matrix usable in both directions.

Corollary 7.3. *The transition matrix is always invertible, and $P_{B \leftarrow C} = P_{C \leftarrow B}^{-1}$. Moreover $P_{C \leftarrow B} = P_{C \leftarrow D}\, P_{D \leftarrow B}$ for three bases B, C, D of V.*

Proof. The previous theorem gives $P_{B \leftarrow C} P_{C \leftarrow B} [\mathbf{v}]_B = [\mathbf{v}]_B$ for all $\mathbf{v}$. Because φ_B is surjective, that product equals the identity matrix. The same argument applies in the opposite order, so the two matrices are mutually inverse. For the chain rule, apply the theorem twice: $[\mathbf{v}]_C = P_{C \leftarrow D}[\mathbf{v}]_D = P_{C \leftarrow D} P_{D \leftarrow B} [\mathbf{v}]_B$, then compare with $[\mathbf{v}]_C = P_{C \leftarrow B}[\mathbf{v}]_B$ at $\mathbf{v} = \mathbf{b}_j$. □

Note that $P_{C \leftarrow B}$ and $P_{B \leftarrow C}$ are not transposes of each other; they are mutual inverses, and that is a different thing. The direction of the subscript is read from right to left, like the composition of maps: take the j-th basis vector of B, then ask for its address according to C. When B is the standard basis, the columns of $P_{B \leftarrow C}$ are the vectors of C themselves.

Take $B = (\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ and $C = (\mathbf{b}'_1, \mathbf{b}'_2, \mathbf{b}'_3)$ with

$$\mathbf{b}'_1 = (1, 1, 0), \qquad \mathbf{b}'_2 = (1, 0, 1), \qquad \mathbf{b}'_3 = (0, 1, 1).$$

Because B is standard, the matrix $P_{B \leftarrow C}$ contains those three vectors as columns:

$$P_{B \leftarrow C} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \det P_{B \leftarrow C} = -2 \neq 0.$$

Its inverse is

$$P_{C \leftarrow B} = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix}.$$

For $\mathbf{v} = (3, 8, 11)$ we obtain

$$[\mathbf{v}]_C = P_{C \leftarrow B}[\mathbf{v}]_B = \frac{1}{2} \begin{pmatrix} 3 + 8 - 11 \\ 3 - 8 + 11 \\ -3 + 8 + 11 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 8 \end{pmatrix}.$$

The check is immediate: $3\mathbf{b}'_2 + 8\mathbf{b}'_3 = (3, 8, 11)$. Conversely $\mathbf{w} = (1, 0, 0)$ has coordinates $(1/2,\ 1/2,\ -1/2)$; a fractional entry is natural, because coordinates are nothing but an address.

7.2 Linear transformations, kernel, and range

Level: basic

Chapter 6 discussed spaces and subspaces, but not maps between them. Yet the main work of linear algebra is moving vectors: rotating, projecting, differentiating, summarizing.

A rule that respects addition and scalar multiplication is called a linear transformation: adding first and then mapping gives the same result as mapping first and then adding.

Definition 7.3 (Linear transformation)**.** A map $T\colon V \to W$ between two vector spaces over $\mathbb{R}$ is called a *linear transformation* if $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{v}) = cT(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$ and $c \in \mathbb{R}$.

The two conditions are equivalent to a single condition, $T(c\mathbf{u}+\mathbf{v}) = cT(\mathbf{u}) + T(\mathbf{v})$. The first consequence, $T(\mathbf{0}) = \mathbf{0}$, is used to reject linearity quickly: the map $T(\mathbf{x}) = \mathbf{x} + \mathbf{a}$ with $\mathbf{a} \neq \mathbf{0}$ fails at once.

Conversely $T(\mathbf{0}) = \mathbf{0}$ is not enough. The map $T(x, y) = (x^2, y)$ satisfies it, but $T(2,0) = (4,0)$ whereas $2T(1,0) = (2,0)$. Squaring breaks linearity.

A 90° rotation, $R(x, y) = (-y, x)$, is linear: $R(3,1) = (-1,3)$. The derivative $D\colon P_3 \to P_2$, $D(p) = p'$, is linear because $(p+q)' = p' + q'$. The map $T(x, y) = (x + y,\ 2x + 2y)$ is linear too, but all of $\mathbb{R}^2$ falls onto a single line; it is linear and simultaneously destroys information.

Take $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $T(\mathbf{x}) = A\mathbf{x}$. The vector $(1,0)$ goes to $(2,1)$ and $(0,1)$ to $(1,2)$, so the unit square becomes a parallelogram with vertices $(0,0)$, $(2,1)$, $(3,3)$, $(1,2)$ of area 3, since $|2 \cdot 2 - 1 \cdot 1| = 3$. Figure 7.1 shows the grid before and after the map: lines stay lines, the grid stays a grid, but distances and areas are no longer the same.

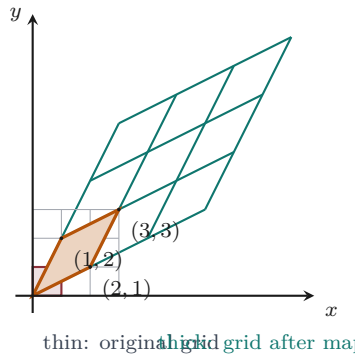


Figure 7.1: The unit grid and its image under $T(\mathbf{x}) = A\mathbf{x}$ with $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. One cell becomes a parallelogram with vertices $(0,0)$, $(2,1)$, $(3,3)$, and $(1,2)$, so its area grows from 1 to 3. The old grid is drawn thin, the new grid thick; both remain grids.

Do it by hand (kinesthetic learner)

Take a sheet of graph paper, a pencil, and a ruler. Draw a horizontal axis and a vertical axis, then mark their intersection as the center. Draw a small house: a square five cells wide, a triangular roof, a door one cell wide.

Now translate. Move every point of the house three cells to the right and two cells up. Do not guess; count. A point that was one cell right and one cell up is now four cells right and three cells up. Redraw the house. Its shape and size are the same, only its place has moved.

Now rotate. Turn the paper a quarter turn clockwise, then redraw the same house. The

length of each side stays, the direction of each side changes.
What should be visible: under translation, all directions are preserved but the center moves with the figure; under rotation, the center stays and all directions change. Because a translation moves the center, a translation is not a linear transformation. A rotation is.

The good news: to know a linear map, it is enough to know its values on the basis vectors.

Proposition 7.4. *Let $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ be a basis of V and let $\mathbf{w}_1, \dots, \mathbf{w}_n \in W$ be arbitrary. There is exactly one linear map $T\colon V \to W$ with $T(\mathbf{b}_j) = \mathbf{w}_j$ for every j.*

Proof. For existence, define $T(\sum_j a_j \mathbf{b}_j) = \sum_j a_j \mathbf{w}_j$. This definition does not depend on the expression, because coordinates with respect to a basis are unique. For linearity, use the linearity of coordinates from the proposition in the previous section: $T(\mathbf{u} + \mathbf{v}) = \sum_j (a_j + c_j)\mathbf{w}_j = T(\mathbf{u}) + T(\mathbf{v})$, and similarly for scalar multiplication.

For uniqueness, suppose S is linear and $S(\mathbf{b}_j) = \mathbf{w}_j$. For $\mathbf{v} = \sum_j a_j \mathbf{b}_j$ we have $S(\mathbf{v}) = \sum_j a_j S(\mathbf{b}_j) = \sum_j a_j \mathbf{w}_j = T(\mathbf{v})$. □

The practical corollary: n vectors in W determine one linear map from an n-dimensional V. When $W = \mathbb{R}^m$, that data is nothing other than an $m \times n$ matrix.

Every linear map carries two sets worth inspecting: the vectors mapped to zero, and the values that are actually attained.

kernel or null space (*inti*) — the list of changes that do not shift the summary at all

Definition 7.4 (Kernel, range, rank, and nullity)**.** For a linear $T\colon V \to W$, the *kernel* is $\ker T = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}$ and the *range* is $\operatorname{im} T = \{T(\mathbf{v}) : \mathbf{v} \in V\}$. When V is finite-dimensional, $\operatorname{nullity}(T) = \dim \ker T$ and $\operatorname{rank}(T) = \dim \operatorname{im} T$.

Both are subspaces, not merely sets.

Lemma 7.5. *$\ker T$ is a subspace of V and $\operatorname{im} T$ is a subspace of W.*

Proof. Both contain $\mathbf{0}$ because $T(\mathbf{0}) = \mathbf{0}$. If $\mathbf{u}, \mathbf{v} \in \ker T$ and $c \in \mathbb{R}$, then $T(\mathbf{u} + \mathbf{v}) = \mathbf{0}$ and $T(c\mathbf{u}) = \mathbf{0}$, so $\ker T$ is closed. If $\mathbf{u} = T(\mathbf{x})$ and $\mathbf{v} = T(\mathbf{y})$, then $\mathbf{u} + \mathbf{v} = T(\mathbf{x} + \mathbf{y})$ and $c\mathbf{u} = T(c\mathbf{x})$ belong to $\operatorname{im} T$. □

The kernel measures the failure of injectivity, and that is all that needs checking.

Proposition 7.6. *T is injective if and only if $\ker T = \{\mathbf{0}\}$.*

Proof. If T is injective, only $\mathbf{0}$ maps to $\mathbf{0}$. Conversely, suppose $\ker T = \{\mathbf{0}\}$ and $T(\mathbf{u}) = T(\mathbf{v})$. Then $T(\mathbf{u} - \mathbf{v}) = \mathbf{0}$, so $\mathbf{u} - \mathbf{v} \in \ker T$, hence $\mathbf{u} = \mathbf{v}$. □

Rank counts the directions that survive, nullity counts the directions that vanish. Together they divide the dimension of the source space.

rank (*rank*) — the number of independent directions that still survive after the map

A small example shows both fates. Take $B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. Then $B\mathbf{e}_1 = (1, 2)$ and $B\mathbf{e}_2 = (2, 4)$, two collinear vectors, so $\operatorname{rank}(B) = 1$. The direction $(1, 2)$ survives and stretches: $B(1, 2) = (5, 10)$. The direction $(2, -1)$ vanishes: $B(2, -1) = (0, 0)$. Figure 7.2 draws both, and rank-nullity closes the count: $2 = 1 + 1$.

Now the central theorem of this chapter. Its content: no dimension is lost along the way.

Theorem 7.7 (Rank-nullity)**.** *If V is finite-dimensional and $T\colon V \to W$ is linear, then*

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

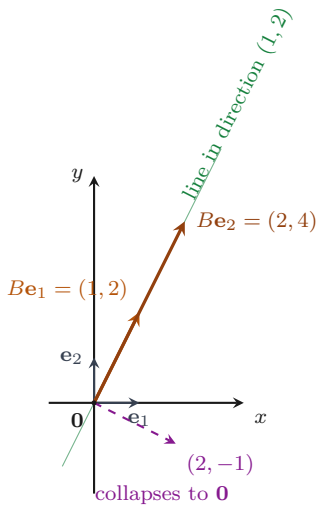


Figure 7.2: Two directions under $B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. The direction $(1, 2)$ survives and stretches into $(5, 10)$; the direction $(2, -1)$ collapses to $\mathbf{0}$. The images of the two basis vectors, $B\mathbf{e}_1 = (1, 2)$ and $B\mathbf{e}_2 = (2, 4)$, lie on a single line, so $\operatorname{rank}(B) = 1$ and $\operatorname{nullity}(B) = 1$.

Proof. Let $\dim V = n$ and $\dim \ker T = k$. Take a basis $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ of $\ker T$. By the basis extension corollary in Chapter 6, that basis can be completed to a basis of V by adding $n - k$ vectors $\mathbf{w}_1, \dots, \mathbf{w}_{n-k}$.

First, the set $\{T(\mathbf{w}_1), \dots, T(\mathbf{w}_{n-k})\}$ spans $\operatorname{im} T$. For an arbitrary $T(\mathbf{v})$, write $\mathbf{v} = \sum_i \alpha_i \mathbf{u}_i + \sum_j \beta_j \mathbf{w}_j$. The terms $\mathbf{u}_i$ vanish because $T(\mathbf{u}_i) = \mathbf{0}$, so $T(\mathbf{v}) = \sum_j \beta_j T(\mathbf{w}_j)$.

Second, that set is linearly independent. Suppose $\sum_j \beta_j T(\mathbf{w}_j) = \mathbf{0}$. Then $T\big(\sum_j \beta_j \mathbf{w}_j\big) = \mathbf{0}$, so $\sum_j \beta_j \mathbf{w}_j \in \ker T$, and there are scalars α_i with $\sum_j \beta_j \mathbf{w}_j = \sum_i \alpha_i \mathbf{u}_i$. Move all terms to one side:

$$\sum_j \beta_j \mathbf{w}_j - \sum_i \alpha_i \mathbf{u}_i = \mathbf{0}.$$

That relation lies inside a basis of V, so all its coefficients are zero; in particular every $\beta_j = 0$. Hence $\{T(\mathbf{w}_j)\}$ is a basis of $\operatorname{im} T$, and $\operatorname{rank}(T) = n - k$. □

Figure 7.3 arranges that theorem as two bars. For a space of dimension 4 the total length is always 4: the identity gives $4 + 0$, the projection that discards one coordinate gives $3 + 1$, and the feature map G later gives $2 + 2$. Only the position of the divider shifts.

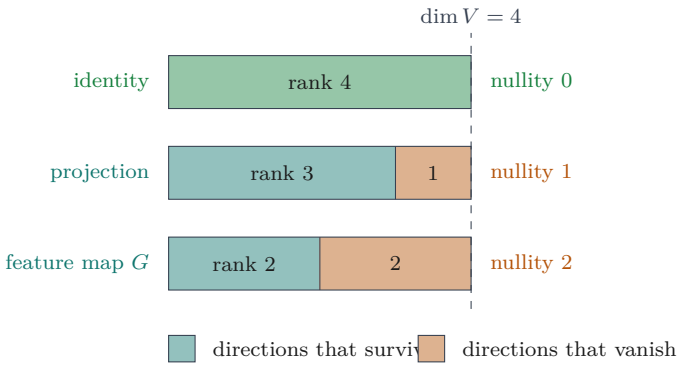


Figure 7.3: Rank-nullity as two bars whose sum is fixed. For $\dim V = 4$: the identity map gives $4 + 0$, the projection that discards one coordinate gives $3 + 1$, and the feature map G gives $2 + 2$. Only the position of the divider shifts; the total length stays 4.

Two corollaries that are used most often.

Corollary 7.8. *Let* $\dim V = \dim W$ *be finite and let* $T\colon V \to W$ *be linear. Then* T *is injective if and only if* T *is surjective. Moreover* $\operatorname{rank}(T) \le \min(\dim V, \dim W)$.

Proof. By rank-nullity, $\dim \ker T = n - \operatorname{rank} T$ with $n = \dim V$. Injectivity holds exactly when $\dim \ker T = 0$, that is $\operatorname{rank} T = n = \dim W$, which means $\operatorname{im} T = W$ because a subspace of full dimension fills the whole space. The reverse implication is obtained by reversing the order of the argument. The upper bound on the rank is clear from $\operatorname{im} T \subseteq W$ and from rank-nullity. □

As a numerical illustration, take $T\colon \mathbb{R}^3 \to \mathbb{R}^2$ with $T(x, y, z) = (x - y,\ y - z)$. Because $T(t, t, t) = (0, 0)$, its kernel is the line $\{(t, t, t)\}$ and its nullity is 1. The images $T(1, 0, 0) = (1, 0)$ and $T(0, 0, 1) = (0, -1)$ span $\mathbb{R}^2$, so the rank is 2; rank-nullity: $3 = 1 + 2$.

For machine learning, that corollary is harsh. A feature map $\mathbb{R}^n \to \mathbb{R}^m$ with $n > m$ cannot possibly be injective, and its kernel has dimension at least $n - m$. Every dimensional compression stores blind directions; the question is not whether such a direction exists, but whether an attacker can find it.

The standard procedure for a basis of the kernel: bring the matrix to reduced row echelon form, mark the columns without a pivot as free variables, then in turn give each free variable the value one and the other free variables zero. The pivot variables are read off from the rows. The result is one vector per column without a pivot, linearly independent because each has a one in a position no other one has.

For the range, the procedure is different and is often confused with the kernel procedure above. The following four steps are worth memorizing.

Step-by-step recipe (procedural learner)

1. Assemble the matrix A, then reduce it to reduced row echelon form by elimination.
2. Mark the columns that contain a pivot. A column without a pivot is not a candidate basis member.
3. Go back to the original matrix A, not to its echelon form. Take the original columns whose positions coincide with the pivot columns.
4. Check: the number of selected vectors equals the number of nonzero rows, and those vectors are linearly independent.

A warning about order: step three must not take columns from the echelon form. Row operations change the columns, whereas the column space does not change. Taking columns from the echelon matrix gives a wrong answer even when the count is right.

The kernel also answers a more general question: once a value has been attained, all its preimages are neatly arranged. Here $T^{-1}(\mathbf{w})$ denotes the set of preimages, not the inverse map.

Proposition 7.9. *Let* $T\colon V \to W$ *be linear and* $T(\mathbf{v}_0) = \mathbf{w}$. *Then* $T^{-1}(\mathbf{w}) = \{\mathbf{v}_0 + \mathbf{k} : \mathbf{k} \in \ker T\}$.

Proof. If $\mathbf{k} \in \ker T$, then $T(\mathbf{v}_0 + \mathbf{k}) = \mathbf{w} + \mathbf{0} = \mathbf{w}$, so every $\mathbf{v}_0 + \mathbf{k}$ belongs to the preimage of $\mathbf{w}$. Conversely, if $T(\mathbf{v}) = \mathbf{w}$, then $T(\mathbf{v} - \mathbf{v}_0) = \mathbf{0}$, so $\mathbf{v} - \mathbf{v}_0 \in \ker T$ and $\mathbf{v} = \mathbf{v}_0 + (\mathbf{v} - \mathbf{v}_0)$. □

The set of inputs that produce the same features is a shifted copy of the kernel: when the kernel has dimension one, the family is a line, namely the blind direction of the chapter opening; when it has dimension k, an attacker has k directions.

7.3 Representation matrices and composition

Level: intermediate

Now the two earlier tools are combined: coordinates turn a vector into a column, and a linear map is determined by its values on a basis. The combination produces a matrix, where all the computation takes place.

The representation matrix stores the image of each basis vector as one column, written in the coordinates of the target basis.

Definition 7.5 (Representation matrix)**.** Let $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ be a basis of V, $C = (\mathbf{c}_1, \dots, \mathbf{c}_m)$ a basis of W, and $T\colon V \to W$ linear. The *representation matrix* of T with respect to B and C is the $m \times n$ matrix

$$[T]_{C \leftarrow B} = \Big([T(\mathbf{b}_1)]_C \;\Big|\; \cdots \;\Big|\; [T(\mathbf{b}_n)]_C \Big).$$

Those columns are obtained by solving systems, not by guessing: for each j, solve $\sum_i x_i \mathbf{c}_i = T(\mathbf{b}_j)$, then write the solution as column j. When C is the standard basis, that solving need not be carried out, because coordinates with respect to the standard basis are the vector itself.

The following claim explains why the matrix is useful: it works in the expected way.

Theorem 7.10. *For every $\mathbf{v} \in V$ we have $[T(\mathbf{v})]_C = [T]_{C \leftarrow B}\, [\mathbf{v}]_B$.*

Proof. Write $\mathbf{v} = \sum_j a_j \mathbf{b}_j$, so $[\mathbf{v}]_B = (a_j)$. Linearity of T gives $T(\mathbf{v}) = \sum_j a_j T(\mathbf{b}_j)$. Taking coordinates is also linear, so $[T(\mathbf{v})]_C = \sum_j a_j [T(\mathbf{b}_j)]_C$, which is exactly the matrix product with the column (a_j). □

Two immediate corollaries. A linear map and its matrix determine each other, because the values on a basis already suffice. And the rank does not change under that translation.

Proposition 7.11. $\operatorname{rank}(T)$ *equals the rank of the matrix $[T]_{C \leftarrow B}$, and* $\ker T$ *corresponds to the null space of that matrix. The rank of a matrix here means the dimension of its column space, as already used in Chapter 6.*

Proof. Column j of the matrix is the coordinate vector of $T(\mathbf{b}_j)$, and $\{T(\mathbf{b}_j)\}$ spans $\operatorname{im} T$. Taking coordinates preserves linear combinations, so the dimension of the range equals the dimension of the column space. The previous theorem shows that $T(\mathbf{v}) = \mathbf{0}$ exactly when $[T(\mathbf{v})]_C = [T]_{C \leftarrow B}[\mathbf{v}]_B = \mathbf{0}$, that is, when $[\mathbf{v}]_B$ lies in the null space of that matrix. □

Composition of maps corresponds to matrix multiplication. That is no coincidence; it is the reason matrix multiplication is defined the way it is.

Theorem 7.12. *Let $T\colon V \to W$ and $S\colon W \to U$ be linear, with bases B, C, E on the three spaces. Then $[S \circ T]_{E \leftarrow B} = [S]_{E \leftarrow C}\, [T]_{C \leftarrow B}$.*

Proof. For every $\mathbf{v} \in V$,

$$[(S \circ T)(\mathbf{v})]_E = [S]_{E \leftarrow C} [T(\mathbf{v})]_C = [S]_{E \leftarrow C} [T]_{C \leftarrow B} [\mathbf{v}]_B.$$

So the matrix $[S]_{E \leftarrow C}[T]_{C \leftarrow B}$ satisfies the definition of the representation matrix for $S \circ T$. The representation matrix is unique: if two matrices give the same result on all $\mathbf{v}$, their results agree in particular at $\mathbf{v} = \mathbf{b}_j$, so their columns agree. □

Composition also provides a test for invertibility through matrices.

Proposition 7.13. *Let B and C be bases of a space V of dimension n, and let $T\colon V \to V$ be linear. Then T is invertible if and only if $[T]_{C\leftarrow B}$ is invertible, and in that case $[T^{-1}]_{B\leftarrow C} = ([T]_{C\leftarrow B})^{-1}$.*

Proof. If T is invertible, then $T^{-1}\circ T = I$ and $T\circ T^{-1} = I$. The composition theorem gives $[T^{-1}]_{B\leftarrow C}[T]_{C\leftarrow B} = I$ and $[T]_{C\leftarrow B}[T^{-1}]_{B\leftarrow C} = I$, so the two matrices are mutually inverse.

Conversely, suppose $A = [T]_{C\leftarrow B}$ is invertible. If $T(\mathbf{v}) = \mathbf{0}$, then $A[\mathbf{v}]_B = \mathbf{0}$, so $[\mathbf{v}]_B = \mathbf{0}$ and $\mathbf{v} = \mathbf{0}$. Hence T is injective. Because the source and target spaces both have dimension n, the rank-nullity corollary makes T surjective, so T is invertible. The equation for the matrix of T^{-1} then follows from the first part. □

With that, a question about the inverse of an operator becomes a question about the inverse of a matrix; the complete answer is in Chapter 8.

The special case used most often is an operator on a single space, $T\colon V \to V$, with two different bases. Here the two matrices are linked by the transition matrix.

Theorem 7.14 (Change of basis)**.** *Let $T\colon V \to V$ be linear, B and B' two bases of V, and $P = P_{B\leftarrow B'}$. Then*

$$[T]_{B'} = P^{-1}[T]_B P.$$

Proof. Write $T = I\circ T\circ I$ with I the identity map. By the composition theorem,

$$[T]_{B'\leftarrow B'} = [I]_{B'\leftarrow B}\,[T]_{B\leftarrow B}\,[I]_{B\leftarrow B'}.$$

The identity matrix with respect to two bases is the transition matrix, so $[I]_{B'\leftarrow B} = P^{-1}$ and $[I]_{B\leftarrow B'} = P$. □

Two square matrices linked in that way describe the same operator from different viewpoints, so they are given a name of their own.

Definition 7.6 (Similar matrices)**.** Matrices A and B of size $n\times n$ are called *similar* if there is an invertible matrix P with $B = P^{-1}AP$.

Similarity is therefore another name for the same map seen from different bases.

Proposition 7.15. *Two $n\times n$ matrices are similar if and only if they are the representation matrices of one and the same linear operator with respect to two bases.*

Proof. If $B = P^{-1}AP$, take the standard basis B_0 and the basis B' whose columns are the columns of P. Then $P = P_{B_0\leftarrow B'}$, and the change of basis theorem gives $B = [T]_{B'}$ for the operator $T(\mathbf{x}) = A\mathbf{x}$. Conversely, the same theorem gives that relation for every pair of bases. □

Rank is among the quantities that do not change: the rank of $[T]_{B'}$ equals the rank of T, and likewise the rank of $[T]_B$. Determinant and trace are also invariant, but both are only defined in Chapter 8 and discussed in Chapter 9.

A small example for the derivative operator: $D\colon P_2 \to P_2$, $D(p) = p'$, with basis $B = (1, x, x^2)$. Because $D(1) = 0$, $D(x) = 1$, and $D(x^2) = 2x$,

$$[D]_B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

For $p = x^2 + 3x$ the coordinates are $(0, 3, 1)^{\top}$, and the matrix gives $(3,\ 2,\ 0)^{\top}$, namely $3 + 2x$, exactly the derivative of p.

7.4 Isomorphisms, dimensional invariance, and spaces of linear maps

Level: advanced

Two different spaces can behave the same way. P_2 and $\mathbb{R}^3$ contain objects of different kinds, but both have three independent directions and addition rules of the same form.

A correspondence that preserves the whole structure is called an isomorphism: bijective, linear, and therefore preserving bases, dimension, and kernels.

isomorphism (*isomorfisme*) — two floor plans of a house whose drawings differ but whose number of rooms and connections between rooms are the same

Definition 7.7 (Isomorphism)**.** A linear map $T\colon V \to W$ is called an *isomorphism* if T is bijective. Two spaces are called *isomorphic* if there is an isomorphism between them, written $V \cong W$.

One lemma before the main theorem: an isomorphism carries bases to bases.

Lemma 7.16. *If $T\colon V \to W$ is an isomorphism and B is a basis of V, then $T(B)$ is a basis of W.*

Proof. Let $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$. For linear independence, suppose $\sum_j c_j T(\mathbf{b}_j) = \mathbf{0}$. Because T is linear, the vector $\sum_j c_j \mathbf{b}_j$ is mapped to $\mathbf{0}$, so that vector lies in the kernel. The kernel is only $\{\mathbf{0}\}$ because T is injective, so all $c_j = 0$. For spanning, take $\mathbf{w} \in W$. There is $\mathbf{v} \in V$ with $T(\mathbf{v}) = \mathbf{w}$ because T is surjective, and $\mathbf{v} = \sum_j a_j \mathbf{b}_j$ because B spans. Then $\mathbf{w} = \sum_j a_j T(\mathbf{b}_j)$. □

With that lemma, equality of dimension turns out to be the only obstacle.

Theorem 7.17. *Two finite-dimensional spaces over $\mathbb{R}$ are isomorphic if and only if their dimensions are equal.*

Proof. If $V \cong W$ via T, the previous lemma turns a basis of V into a basis of W with the same number of members, so $\dim V = \dim W$.

Conversely, suppose $\dim V = \dim W = n$. Choose a basis B of V and C of W. The coordinate maps $\varphi_B\colon V \to \mathbb{R}^n$ and $\varphi_C\colon W \to \mathbb{R}^n$ are bijective and linear. Then $\varphi_C^{-1} \circ \varphi_B$ is a bijective linear map from V to W, because the composition of bijective linear maps is also bijective and linear. □

That conclusion is felt at once in the spaces we know.

Corollary 7.18. *Every space of dimension n is isomorphic to $\mathbb{R}^n$. In particular $P_{n-1} \cong \mathbb{R}^n$ and $M_{2\times 2}(\mathbb{R}) \cong \mathbb{R}^4$.*

An isomorphism preserves everything defined from the operations: linear independence, span, kernel, range, rank, and nullity. Two isomorphic spaces cannot be distinguished by the tools of this chapter; dimension becomes the only number that needs computing in order to recognize a finite-dimensional space.

There is a trap in interpretation. The isomorphism above depends on the choice of basis; there is no canonical isomorphism between V and $\mathbb{R}^n$. Hence "V equals $\mathbb{R}^n$" is false, while "V is isomorphic to $\mathbb{R}^n$" is true. Polynomials of degree less than three and triples of real numbers are objects of different kinds, but the two are isomorphic because every polynomial is determined by its three coefficients.

Now the collection of all linear maps between two spaces; that set is itself a vector space.

Theorem 7.19. *The set $\mathcal{L}(V,W)$ of all linear maps $V \to W$, with the operations $(S+T)(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v})$ and $(cT)(\mathbf{v}) = cT(\mathbf{v})$, is a vector space. If $\dim V = n$ and $\dim W = m$, then $\dim \mathcal{L}(V,W) = mn$. The mapping $T \mapsto [T]_{C \leftarrow B}$ is an isomorphism $\mathcal{L}(V,W) \to M_{m\times n}(\mathbb{R})$.*

Proof. $\mathcal{L}(V,W)$ sits inside the space of all functions $V \to W$, which is a vector space by the example in Chapter 6. The zero map is linear. The sum of two linear maps is linear, and a scalar multiple of a linear map is also linear; so $\mathcal{L}(V,W)$ is a subspace, and therefore a vector space.

For the isomorphism claim, let $\Phi(T) = [T]_{C \leftarrow B}$. Linearity of Φ: column j of $\Phi(S + cT)$ is $[S(\mathbf{b}_j) + cT(\mathbf{b}_j)]_C$, which equals $[S(\mathbf{b}_j)]_C + c[T(\mathbf{b}_j)]_C$ because coordinates are linear. Hence $\Phi(S + cT) = \Phi(S) + c\Phi(T)$.

Injectivity: if $\Phi(T)$ is the zero matrix, then all $[T(\mathbf{b}_j)]_C = \mathbf{0}$, so $T(\mathbf{b}_j) = \mathbf{0}$ for all j, and $T = \mathbf{0}$ because a linear map is determined by its values on a basis.

Surjectivity: given a matrix $A = (a_{ij})$ of size $m \times n$. Define $T(\mathbf{b}_j) = \sum_i a_{ij}\mathbf{c}_i$ and extend linearly; by the proposition on values on a basis, that extension exists and is unique. Column j of $\Phi(T)$ is $(a_{ij})_i$, so $\Phi(T) = A$.

Because Φ is an isomorphism and $\dim M_{m\times n}(\mathbb{R}) = mn$, we get $\dim \mathcal{L}(V,W) = mn$. $\square$

That count gives an immediate corollary.

Corollary 7.20. $\dim \mathcal{L}(V,V) = n^2$ *if* $\dim V = n$. *For example,* $\dim \mathcal{L}(P_2, P_2) = 9$ *and* $\dim \mathcal{L}(P_2, \mathbb{R}^2) = 6$.

That last result explains why a matrix is more than a computational device. The space of operators and the space of matrices have the same dimension and are linked by an isomorphism that respects composition, so every statement about operators has a counterpart about matrices. What is not inherited is simplicity of form: the same operator can be complicated in one basis and simple in another; the search for the best basis is the program of Chapter 9.

Challenge (problem solver)

Find one linear map $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ that maps the unit square in the xy plane to another unit square of area still 1, yet $\operatorname{rank}(T) = 2$. Explain why an area preserved in one plane does not force full rank.

Hint 1: examine one plane, not the whole space. After that, ask where the third axis goes.

Hint 2: place a 90° rotation on the first two coordinates and send the third coordinate to zero, so that $T(\mathbf{e}_3) = \mathbf{0}$. Compute the rank from the images of $\mathbf{e}_1$ and $\mathbf{e}_2$.

If you get stuck, return to the section Linear transformations, kernel, and range, especially the definition of rank and the rank-nullity corollary.

For younger readers (age 10 and up)

Two classes at school can differ in size, yet have the same number of chairs. If the number is the same, every pupil in the first class can be paired with one chair in the second, with no empty chair and no pupil getting two chairs. As far as counting chairs goes, the two classes are equivalent.

What is not equivalent is the pupils' names and faces. The chair pairing does not say who sits where; it only pairs. Different people can make different pairings, and all of them are legitimate.

In this chapter we will learn when two spaces that look different may be regarded as the same for computational purposes, and when that assumption is misleading. The

equivalence always depends on our choice, not on something already present in the spaces. If two classes have the same number of chairs, may the headmaster simply swap the class names just like that? The answer is no, and the reason is in this chapter.

7.5 A worked example

LEVEL: INTERMEDIATE

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 7 & 4 \end{pmatrix}, \qquad \text{third row} = \text{first row} + \text{second row}.$$

We determine its rank, its nullity, and both bases, then examine the same operator in another basis.

Row elimination:

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 7 & 4 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then $R_1 \leftarrow R_1 - 2R_2$ gives the reduced echelon form

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

There are two nonzero rows, so $\operatorname{rank} A = 2$ and $\det A = 0$; the pivot columns are the first and the second.

Kernel: from $x_1 - x_3 = 0$ and $x_2 + x_3 = 0$ we obtain $x_1 = x_3$ and $x_2 = -x_3$. Hence

$$\ker A = \operatorname{span}\{(1, -1, 1)^{\top}\}, \qquad \operatorname{nullity}(A) = 1.$$

Rank-nullity is satisfied: $3 = 2 + 1$.

Range: the columns of A are $\mathbf{c}_1 = (1, 2, 3)^{\top}$, $\mathbf{c}_2 = (2, 5, 7)^{\top}$, and $\mathbf{c}_3 = (1, 3, 4)^{\top}$. The first and the second pivot columns give the basis $\{\mathbf{c}_1, \mathbf{c}_2\}$. The third column is not independent: $\mathbf{c}_2 - \mathbf{c}_1 = \mathbf{c}_3$, and that relation is exactly the kernel vector $(1, -1, 1)$: $\mathbf{c}_1 - \mathbf{c}_2 + \mathbf{c}_3 = \mathbf{0}$.

Now consider $T(\mathbf{x}) = A\mathbf{x}$ and the basis $B' = (\mathbf{b}'_1, \mathbf{b}'_2, \mathbf{b}'_3)$ with $\mathbf{b}'_1 = (1, 1, 0)$, $\mathbf{b}'_2 = (1, 0, 1)$, $\mathbf{b}'_3 = (0, 1, 1)$. Its transition matrix from the standard basis B is

$$P = P_{B \leftarrow B'} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix}.$$

The images of the new basis vectors are computed first:

$$A\mathbf{b}'_1 = \begin{pmatrix} 3 \\ 7 \\ 10 \end{pmatrix}, \quad A\mathbf{b}'_2 = \begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix}, \quad A\mathbf{b}'_3 = \begin{pmatrix} 3 \\ 8 \\ 11 \end{pmatrix}.$$

Hence

$$AP = \begin{pmatrix} 3 & 2 & 3 \\ 7 & 5 & 8 \\ 10 & 7 & 11 \end{pmatrix}, \qquad [T]_{B'} = P^{-1}AP = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 2 & 3 \\ 7 & 5 & 8 \end{pmatrix}.$$

Check one entry: first row, first column $\frac{1}{2}(3 + 7 - 10) = 0$; second row, third column $\frac{1}{2}(3 - 8 + 11) = 3$. The rest follows the same pattern.

A first row that is entirely zero is no accident: all three images lie inside $\operatorname{span}(\mathbf{b}'_2, \mathbf{b}'_3)$, for instance $A\mathbf{b}'_3 = (3, 8, 11) = 3\mathbf{b}'_2 + 8\mathbf{b}'_3$. Hence the first coordinate of every image is zero, and the range of T is contained in the coordinate plane of the first coordinate.

The pattern is general. A zero row signals a range contained in a coordinate subspace of lower dimension; a zero column signals a basis vector that goes into the kernel. Both symptoms are visible only in the chosen basis, whereas rank and nullity do not change.

A complete check:

$$\operatorname{rank}[T]_{B'} = 2, \qquad \operatorname{trace}[T]_{B'} = 0 + 2 + 8 = 10 = \operatorname{trace} A, \qquad \det[T]_{B'} = 0 = \det A.$$

The equality of traces is not an arithmetic accident. The j-th diagonal entry is the coefficient of $\mathbf{b}'_j$ in the image $T(\mathbf{b}'_j)$, that is, how much T preserves that direction itself. The sum does not depend on the directions chosen, only on the operator, so the trace serves as a cheap fingerprint: different traces mean the matrices are not similar, equal traces guarantee nothing.

For $\mathbf{v} = (3, 8, 11)$ we have $[\mathbf{v}]_{B'} = (0, 3, 8)^\top$ and

$$[T]_{B'}[\mathbf{v}]_{B'} = \begin{pmatrix} 0 \\ 30 \\ 79 \end{pmatrix},$$

whereas $T(\mathbf{v}) = A\mathbf{v} = (30, 79, 109)^\top$ and its coordinates are $(0, 30, 79)^\top$. The two agree.

The feature map $G\colon \mathbb{R}^4 \to \mathbb{R}^3$ is given by the matrix

$$G = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 3 & 1 & 2 \end{pmatrix},$$

whose third row is the sum of the first two rows. Elimination gives the reduced echelon form

$$\begin{pmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

so $\operatorname{rank} G = 2$ and $\operatorname{nullity}(G) = 4 - 2 = 2$. From $x_1 = 2x_2 + x_4$ and $x_3 = -x_2 - x_4$:

$$\mathbf{k}_1 = (-2, 1, -1, 0)^\top, \qquad \mathbf{k}_2 = (-1, 0, -1, 1)^\top.$$

Both satisfy $G\mathbf{k}_i = \mathbf{0}$ and are linearly independent, so the changes that the three features do not see form a two-dimensional plane, not a single line: an attacker can shift the input in any direction of that plane without moving the score.

> Repeat the computation with the basis $B'' = (\mathbf{b}'_2, \mathbf{b}'_3, \mathbf{b}'_1)$, that is, with B' permuted. Predict how the zero row and the other entries move, then compute $[T]_{B''}$ via $Q^{-1}[T]_{B'}Q$ with Q the corresponding permutation matrix. Check that rank and trace do not change, and explain why $\det A = 0$ can be concluded directly from "the third row is the sum of the first two rows".

All the numbers above were checked with exact arithmetic (SymPy); the numpy and scipy checks are in the next section, including one number that is not exactly zero because of machine rounding.

7.6 Computational lab

Level: advanced

Computational lab

```
import numpy as np
from scipy.linalg import null_space

A = np.array([[1., 2., 1.],
              [2., 5., 3.],
              [3., 7., 4.]])
print("rank A  :", np.linalg.matrix_rank(A))
print("det A   :", np.linalg.det(A))
K = null_space(A)
print("dim ker :", K.shape[1])
print("basis ker:\n", np.round(K, 4))
print("A @ K   :\n", np.round(A @ K, 12))

P = np.array([[1., 1., 0.],
              [1., 0., 1.],
              [0., 1., 1.]])
B = np.linalg.inv(P) @ A @ P
print("[T]_B'  :\n", np.round(B, 10))
print("difference :", np.linalg.norm(B - np.linalg.inv(P) @ A @ P))
print("trace   :", round(np.trace(B), 10), round(np.trace(A), 10))
print("rank B  :", np.linalg.matrix_rank(B))
print("cond P  :", np.linalg.cond(P))

v = np.array([3., 8., 11.])
print("[v]_B'  :", np.round(np.linalg.solve(P, v), 4))
print("[Av]_B' :", np.round(np.linalg.solve(P, A @ v), 4))
print("B[v]_B' :", np.round(B @ np.linalg.solve(P, v), 4))

x0 = np.array([1., 0., 0.])
d = 0.5 * K[:, 0]
print("A@d     :", np.round(A @ d, 12))
print("shift   :", np.linalg.norm(A @ (x0 + d) - A @ x0))
```

Actual output:

```
rank A  : 2
det A   : 6.66133814775094e-16
dim ker : 1
basis ker:
 [[ 0.5774]
 [-0.5774]
 [ 0.5774]]
A @ K   :
 [[-0.]
 [ 0.]
 [ 0.]]
[T]_B'  :
 [[0. 0. 0.]
 [3. 2. 3.]
 [7. 5. 8.]]
difference : 0.0
trace   : 10.0 10.0
rank B  : 2
cond P  : 2.0000000000000004
[v]_B'  : [0. 3. 8.]
[Av]_B' : [ 0. 30. 79.]
```

```
B[v]_B' : [ 0. 30. 79.]
A@d     : [-0.  0.  0.]
shift   : 6.280369834735101e-16
```

Four things are worth reading from that output.

First, $\det A$ is not zero but 6.66×10^{-16}. This matrix is singular exactly, but binary fraction arithmetic knows no exact zero. `matrix_rank` compares the largest and smallest singular values, not a simple threshold, so the rank is still reported as 2; Chapter 22 discusses why such a threshold is necessary.

The choice of singular values rather than the determinant: the determinant measures volume, and a volume can be very small without the rank changing. What determines the rank is the ratio of the smallest singular value to the largest.

Second, `null_space` returns a unit vector with entries 0.5774, -0.5774, and 0.5774, that is $(1, -1, 1)^\top$ divided by $\sqrt{3}$, and $1/\sqrt{3} = 0.57735$ agrees. The direction is the same as in the hand computation; its length is normalized because scipy chooses an orthonormal basis. A kernel basis is not unique, its dimension is.

Third, the $[T]_{B'}$ reported by numpy matches the hand computation exactly, and the traces of both matrices agree, 10. The difference of $P^{-1}AP$ computed twice is 0.0 because P contains small integers. The condition number of P is 2.0000000000000004, quite small; a bad basis inflates the error, and that is discussed in Chapter 10.

Fourth, the kernel direction is genuinely blind to T. Pushing $\mathbf{x}_0 = (1, 0, 0)$ by half the kernel vector changes $\mathbf{x}_0$ visibly, yet $A\mathbf{x}$ shifts only 6.28×10^{-16}, that is, zero at machine precision.

A final warning concerns the choice of basis. A nearly dependent basis makes coordinates explode, and the reported numbers lose their meaning.

Computational lab

```
Pb = np.array([[1., 1., 0.],
               [1., 1.001, 0.],
               [0., 0., 1.]])
print("cond Pb:", np.linalg.cond(Pb))
print("coord   :", np.linalg.inv(Pb) @ v)
print("error   :", np.linalg.norm(np.linalg.inv(Pb) @ (Pb @ v) - v))
```

Actual output:

```
cond Pb: 4002.00075012484
coord   : [-4997.  5000.    11.]
error   : 4.823324033076557e-13
```

The first two basis vectors differ only in the third digit, and the condition number of P_b reaches 4002. The coordinates $(3, 8, 11)$ explode into $(-4997, 5000, 11)$. The reconstruction error is still 4.8×10^{-13}, but the relative error in the coordinates grows by a factor of the condition number; with 16 digits of precision, about 12 remain. In a worse basis that remainder is used up and the reported rank cannot be trusted.

Note for instructors

A teaching order that works in practice: coordinates first, linear maps second. Students who know coordinates before the abstract definition are readier to accept the representation matrix, because the matrix appears as a list of coordinates, not as a new object. After rank-nullity, move to change of basis.

Two typical misconceptions. First, equating a vector with its coordinate list, then getting confused when two bases give two lists for the same vector. Second, the direction of the

transition matrix backwards. A quick check: the columns must contain the addresses of the basis vectors, and the subscript is read right to left.
Blackboard activity: give one operator on $\mathbb{R}^2$ and two bases. Ask half the class to compute $[T]_{B'}$ via the column definition, half via $P^{-1}AP$, then compare on the board; the noise that arises is usually exactly about the direction of P.
A sign of readiness for the next step: students can explain why a zero row appears in the new matrix, without recomputing. If they connect it with a range contained in a coordinate subspace, they are ready for Chapter 9.

7.7 Summary

Level: basic

Summary.

- The coordinates $[\mathbf{v}]_B$ are the unique list of coefficients with respect to an ordered basis; the map $\mathbf{v} \mapsto [\mathbf{v}]_B$ is linear and bijective, but depends on the basis.
- The transition matrix contains the coordinates of the old basis in the new basis: $[\mathbf{v}]_C = P_{C\leftarrow B}[\mathbf{v}]_B$, with $P_{B\leftarrow C} = P_{C\leftarrow B}^{-1}$.
- A linear map is determined by its values on a basis; T is injective exactly when $\ker T = \{\mathbf{0}\}$.
- Rank-nullity: $\dim V = \dim \ker T + \dim \operatorname{im} T$; in equal dimensions, injective is equivalent to surjective.
- The representation matrix satisfies $[T(\mathbf{v})]_C = [T]_{C\leftarrow B}[\mathbf{v}]_B$, composition becomes matrix multiplication, and $[T]_{B'} = P^{-1}[T]_B P$.
- $\dim \mathcal{L}(V, W) = mn$ if $\dim V = n$, $\dim W = m$; two finite-dimensional spaces are isomorphic exactly when their dimensions are equal.

7.8 Exercises

Level: intermediate

Exercise 7.1.

1. With $B' = (\mathbf{b}'_1, \mathbf{b}'_2, \mathbf{b}'_3)$ as in the worked example, determine $[\mathbf{w}]_{B'}$ for $\mathbf{w} = (2, -1, 3)$, then check the result by multiplying back.
2. Prove that every isomorphism $V \to \mathbb{R}^n$ has the form φ_B for some basis B, and explain why an isomorphism to $\mathbb{R}^n$ is not unique. (Hint: take the preimages of the standard basis, then show that this set is a basis.)
3. For $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$, determine the rank, the nullity, a basis of the kernel, and a basis of the range. Explain the connection with the derivative operator on P_2.
4. Prove $\operatorname{rank}(S \circ T) \leq \min(\operatorname{rank} S, \operatorname{rank} T)$ for linear maps that are defined. Use $\operatorname{im}(S \circ T) \subseteq \operatorname{im} S$ and compare the kernels. (Hint: for the second bound, compare $\ker T$ with $\ker(S \circ T)$ via rank-nullity.)
5. Assess this claim: *two square matrices of the same size that have the same rank are necessarily similar.* Decide whether it is true, false, or not supported by enough evidence, with reasons and a counterexample if needed.
6. Let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$ with $T(x, y, z) = (x + 2y,\ y + z)$. Compute $[T]_{C\leftarrow B}$ for $B = (\,(1,0,0), (1,1,0), (1,1,1)\,)$ and $C = (\,(1,1), (1,0)\,)$, then check the formula at $\mathbf{v} = (1, 2, 3)$. (Hint: $[T]_{C\leftarrow B} = C^{-1}AB$; compute AB first, then solve $C\mathbf{x} = \mathbf{y}$ for each of its columns.)

7. Prove: if U is a subspace of a finite-dimensional V and $\dim U = \dim V$, then $U = V$.
8. Prove that if A and B are similar then $\text{nullity}(A) = \text{nullity}(B)$. Explain why that statement is equivalent to "the kernel does not depend on the basis".
9. The feature map $F\colon \mathbb{R}^4 \to \mathbb{R}^3$ is given by the matrix

$$F = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 3 \\ 1 & 1 & 3 & 4 \end{pmatrix}.$$

Determine the rank and nullity of F, give one nonzero kernel vector, and explain why that vector is a direction of change that does not shift a single feature.

For researchers: limits and next steps

What is proved in this chapter are statements about finite dimension over the field $\mathbb{R}$: rank-nullity, uniqueness of the representation matrix, the equivalence of isomorphism with equality of dimension, and $\dim \mathcal{L}(V, W) = mn$. All of them are theorems, not conventions. The claim "a matrix is its transformation" is merely popular; without a choice of basis that claim is meaningless, and the zero-row example above shows that the form of a matrix depends on the basis.

Primary sources: Axler, *Linear Algebra Done Right* 4th edition, sections 3A to 3D, contain spaces of linear maps, kernel and range, matrices, isomorphisms, and change of basis; rank-nullity there is called the fundamental theorem for linear maps. Hoffman and Kunze, *Linear Algebra* 2nd edition, Chapter 3, is denser but equivalent.

An honest open question. The classification of operators up to similarity is not given here; Chapter 9 stops at diagonalization. Numerically, the rank decision depends on a threshold and condition numbers; this chapter uses library output without discussing its stability, a debt paid in Chapter 22. Limit of validity: in infinite-dimensional spaces the whole argument collapses, rank-nullity fails, and isomorphism is no longer equivalent to equality of dimension.

Map to source texts: This chapter follows Anton, Elementary Linear Algebra *11th edition, sections 4.4, 4.6, 4.8, and 8.1 to 8.5, with the abstract counterpart in Axler,* Linear Algebra Done Right *4th edition, Chapter 3, and Hoffman and Kunze,* Linear Algebra *2nd edition, Chapter 3.*

Chapter 8

Linear Systems, Determinants, and Inverses

A threat scorer adds three features with unknown weights, $s = w_1 f_1 + w_2 f_2 + w_3 f_3$. Calibration on three labeled examples gives three linear equations at once:

$$2w_1 + w_2 - w_3 = 8, \qquad -3w_1 - w_2 + 2w_3 = -11, \qquad -2w_1 + w_2 + 2w_3 = -3.$$

The answer is $w = (2, 3, -1)$, and nothing about it is surprising.

Then the coefficient matrix is replaced by the result of another measurement,

$$S = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9.001 \end{pmatrix}.$$

The second row minus the first gives $(3, 3, 3)$, while the third row minus the second gives $(3, 3, 3.001)$: the three rows are nearly dependent. The weights $w = (1, 1, 1)$ produce the right-hand side (6, 15, 24.001), so the system is consistent and the answer is exactly $w = (1, 1, 1)$. Raise the third measurement by 0.001, and the answer moves to $(2, -1, 2)$. A relative change in the data of 3.46×10^{-5} produces a relative change in the answer of 1.41.

There are four questions here, not one: does the system have an answer, how many answers does it have, which number decides that without solving anything, and how far does the answer shift when the data shift. Chapter 7 already supplied rank and nullity for the first. Three are still missing: a procedure that is written down and can be checked by someone else, one number that decides invertibility and at the same time yields a formula for the answer, and a measure of sensitivity; those three are the content of this chapter.

The determinant is usually introduced as the cofactor formula, and then its properties are proved one by one. We reverse that: definition first, through three properties, uniqueness proved, the cofactor formula appearing as a consequence. The reason is not style: the cofactor formula hides the fact that the determinant is the only normalized alternating function, and that fact is what makes the proofs in this chapter short.

For younger readers (age 10 and up)

Three friends weigh three boxes on three different scales. Each friend writes down three numbers, and from those numbers we want to know the weight of each box. As long as the three scales measure different things, the answer is easy: every new number supplies one new piece of information.

Here is the trouble. The second scale turns out to nearly copy the first one. The numbers are not exactly the same, only slightly different, perhaps because of dust on the pan.

The third scale also nearly copies the other two. Now the three pieces of information have nearly the same content. We feel we hold three pieces of information, when only one or two of them are really new.

This is where the strangeness begins. If one number changes by a thousandth, the box weights we compute can change by almost half. Not because the arithmetic is wrong, but because the information really is nearly identical. There is one number that can tell us in advance whether this will happen, before we compute anything at all.

Have you ever asked the same question of three people, received three answers with similar content, and still felt unsure? In this chapter we will learn why similar answers actually weaken the information, and how to recognize that from the numbers.

8.1 Gaussian elimination and three fates of a system

Level: basic

A linear system is written as a single unit, not as a list of separate equations. The way to do it is to attach the right-hand side column to the coefficient matrix. The solution is then carried out on that arrangement, and every step replaces one system by another system with the same solution set.

elimination (*eliminasi*) — Replacing a question by a simpler question without changing the answer, like simplifying a fraction without changing its value.

Definition 8.1 (Augmented matrix and elementary row operations)**.** For the system $A\mathbf{x} = \mathbf{b}$ with A of size $m \times n$, the *augmented matrix* is the matrix $(A \mid \mathbf{b})$ of size $m \times (n+1)$. The three *elementary row operations* are: interchanging two rows; multiplying one row by a nonzero scalar; adding a multiple of one row to another row. Two matrices are called *row equivalent* if one can be reached from the other by those operations.

Those operations are chosen because each of them can be undone. Interchanging two rows undoes itself, multiplying by $c \neq 0$ is undone by multiplying by $1/c$, and adding c times row i to row j is undone by subtracting c times row i. No information is therefore lost.

Theorem 8.1. *If two linear systems have row equivalent augmented matrices, the two systems have the same solution set.*

Proof. It is enough to check one operation, because repeated operations follow. Let S be the original system and S' the system after one operation. Every equation of S' is a linear combination of the equations of S, so every answer of S also answers S'. Conversely, because the operation can be undone by an operation of the same kind, every equation of S is a linear combination of the equations of S', so every answer of S' also answers S. The two solution sets therefore contain each other. □

The form we aim at is the staircase form. Every nonzero row starts further to the right than the nonzero row above it, and all entries below that first position are zero. The column where the first number of a nonzero row sits is called the pivot column, and that number itself is called a pivot.

Definition 8.2 (Row echelon form)**.** A matrix is in *row echelon form* if all zero rows lie at the bottom, and the first nonzero entry of every nonzero row is located further right than the first nonzero entry of the row above it. The form is called *reduced* if every pivot equals 1 and is the only nonzero entry in its column.

The reduced form is unique. This fact is used every time we speak of “the rank computed by elimination”, and its proof is not as short as it looks.

Theorem 8.2 (Uniqueness of the reduced echelon form)**.** *Every matrix is row equivalent to exactly one matrix in reduced row echelon form.*

Proof. The proof is skipped here. The statement is proved in full in Hoffman and Kunze, *Linear Algebra* 2nd edition, section 1.4, by showing that the solution space of $A\mathbf{x} = \mathbf{0}$ determines the position of the pivots uniquely. What we use in this chapter is only the consequence: the number of pivots and their positions do not depend on the order of operations chosen. □

With the reduced form, every question about the number of answers can be settled by counting pivots. These are the three fates of a system.

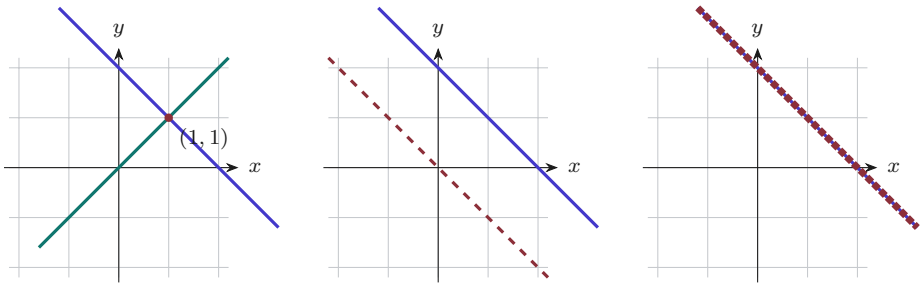
Theorem 8.3 (Three cases)**.** *Suppose* $(A \mid \mathbf{b})$ *is row equivalent to the reduced echelon form* $(R \mid \mathbf{c})$*, with* A *of size* $m \times n$*.*

1. *The system is* inconsistent *exactly when there is a pivot in column* $(n+1)$*.*
2. *If it is consistent, the system has a* unique answer *exactly when every column of* A *contains a pivot.*
3. *If it is consistent, the system has* many answers *exactly when some column of* A *has no pivot, and the number of free parameters equals the number of columns without a pivot.*

Proof. A pivot in the last column means there is a row of the form $0 =$ nonzero constant, which cannot possibly be satisfied. If there is no such pivot, every nonzero row gives one equation expressing a pivot variable in terms of the other variables, so the system is consistent.

If every column contains a pivot, the number of pivots equals n, so there is no free variable and every pivot variable is uniquely determined. If some column has no pivot, the variable in that column is free: it may take any value, and every choice determines the pivot variables uniquely. The answer is therefore infinite in number, and the number of free parameters is exactly the number of columns without a pivot. □

Figure 8.1 places those three fates in a single plane. The line $x + y = 2$ does not change; only the second line changes. Left panel: $x - y = 0$ cuts it at $(1, 1)$, two pivots, a unique answer. Middle panel: $x + y = 0$ is parallel, and the last column of the augmented matrix contains a pivot. Right panel: $2x + 2y = 4$ is the same line written with different numbers, one pivot and one free parameter.



(a) intersecting (b) parallel (c) coincident

Figure 8.1: Three fates of a 2×2 system with the first line $x + y = 2$ held fixed. Left: the second line $x - y = 0$ cuts it at $(1, 1)$, two pivots, a unique answer. Middle: the second line $x + y = 0$ is parallel, the right-hand side column contains a pivot, there is no answer. Right: the second line $2x + 2y = 4$ coincides with the first line, one pivot, infinitely many answers.

Read it aloud (auditory learner)

Two voices. The first voice names the situation, the second voice names the consequence. Read them alternately, then swap roles.
One system. Two lines. Three fates.
Intersecting: one point, one answer.
Parallel: they never meet, there is no answer.
Coincident: one line disguised as two, infinitely many answers.
Who decides, before we compute anything? The pivots decide.
Two pivots, two columns: a unique answer.
A pivot in the last column: some row reads zero equals nonzero.
A column without a pivot: one free parameter, and the answers line up in one direction.
Repeat the last line together: count the pivots first, then talk about answers.

The matrix from Chapter 7 displays two fates at once,

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 7 & 4 \end{pmatrix}, \qquad \text{third row} = \text{first row} + \text{second row}.$$

For $\mathbf{b} = (4, 9, 13)$ elimination gives

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 5 & 3 & 9 \\ 3 & 7 & 4 & 13 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

There are two pivots and three columns, so there is one free variable. The answer is $(2, 1, 0) + t\,(1, -1, 1)$ for $t \in \mathbb{R}$, and the check is immediate: $A(2, 1, 0)^{\top} = (4, 9, 13)^{\top}$ while $A(1, -1, 1)^{\top} = \mathbf{0}$, so the direction of that shift really is the blind direction of the matrix from Chapter 7.

For $\mathbf{b} = (4, 9, 14)$ the last row becomes $0 = 1$, and the system is inconsistent. The coefficient matrix is exactly the same; one number on the right-hand side moves the fate of the system from "many answers" to "no answer". The question of consistency therefore cannot be answered from A alone.

The case that appears most often in practice is $m = n$. There the three fates rest on a single number, and that number is what we look for in the next section.

Corollary 8.4. *Suppose A is of size $n \times n$. The system $A\mathbf{x} = \mathbf{b}$ has a unique answer for every $\mathbf{b} \in \mathbb{R}^n$ exactly when* $\operatorname{rank} A = n$.

Proof. If $\operatorname{rank} A = n$, the reduced echelon form of A is I_n, so elimination on $(A \mid \mathbf{b})$ always ends with a pivot in every column of A and no pivot in the last column; the previous theorem gives a unique answer. Conversely, if $\operatorname{rank} A < n$, some column has no pivot, and for a consistent $\mathbf{b}$ the answer is not unique. Since the rank is less than n, the column space of A does not fill $\mathbb{R}^n$, so some $\mathbf{b}$ lies outside it; for that $\mathbf{b}$ the system is inconsistent. □

8.2 The determinant as an alternating function

Level: intermediate

We want one number that answers the question of the preceding corollary without solving any system. That number must be zero exactly when the columns of A are not linearly independent, must change sign when two columns are interchanged, and must equal one for the identity matrix. Those three conditions turn out to determine the number uniquely.

determinant (*determinan*) — The volume factor: how many times space is stretched or flattened by the matrix.

Definition 8.3 (Determinant)**.** A function D from $n \times n$ matrices to $\mathbb{R}$ is called a *determinant* if

1. *multilinear in each row*: for scalars α, β and rows $\mathbf{u}, \mathbf{v}$, $D(\dots, \alpha\mathbf{u} + \beta\mathbf{v}, \dots) = \alpha D(\dots, \mathbf{u}, \dots) + \beta D(\dots, \mathbf{v}, \dots)$;
2. *alternating*: $D(\dots, \mathbf{u}, \dots, \mathbf{u}, \dots) = 0$ if two rows are equal;
3. *normalized*: $D(I_n) = 1$.

The value $D(A)$ is written $\det A$.

Multilinearity means the determinant is linear in one row while the other rows are held fixed. The alternating property is the strongest condition, and its consequences are long.

Lemma 8.5. *Interchanging two rows changes the sign of the determinant. If one row is a scalar multiple of another row, or if there is a zero row, the determinant is zero.*

Proof. Suppose two adjacent rows both contain $\mathbf{u} + \mathbf{v}$. Since the two rows are equal, the value is zero. Multilinearity splits that zero into four terms,

$$0 = D(\dots, \mathbf{u}, \mathbf{u}, \dots) + D(\dots, \mathbf{u}, \mathbf{v}, \dots) + D(\dots, \mathbf{v}, \mathbf{u}, \dots) + D(\dots, \mathbf{v}, \mathbf{v}, \dots),$$

and the first and fourth terms vanish. Interchanging two adjacent rows therefore reverses the sign:

$$D(\dots, \mathbf{v}, \mathbf{u}, \dots) = -D(\dots, \mathbf{u}, \mathbf{v}, \dots).$$

An interchange of rows that are not adjacent is achieved by several adjacent interchanges, and every step reverses the sign.

For a multiple, $D(\dots, c\mathbf{u}, \dots, \mathbf{u}, \dots) = c\, D(\dots, \mathbf{u}, \dots, \mathbf{u}, \dots) = 0$ by multilinearity and the alternating property. A zero row is the case $c = 0$ of the same form. □

Now uniqueness. The formula that appears in the middle of the proof is the Leibniz formula, and that formula will later yield the cofactor expansion as a consequence.

Theorem 8.6 (Uniqueness and the Leibniz formula)**.** *There is only one function satisfying the three conditions of the definition, and its value is*

$$\det A = \sum_{\sigma} \operatorname{sgn}(\sigma)\, a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n},$$

where the sum is taken over all permutations σ of $\{1, \dots, n\}$ and $\operatorname{sgn}(\sigma) = (-1)^k$ if σ is reached by k interchanges.

Proof. Write row i as $\mathbf{a}_i = \sum_j a_{ij}\mathbf{e}_j$. Multilinearity allows every row to be pulled out into a sum,

$$D(A) = \sum_{j_1, \dots, j_n} a_{1j_1} a_{2j_2} \cdots a_{nj_n}\, D(\mathbf{e}_{j_1}, \dots, \mathbf{e}_{j_n}).$$

Terms with $j_p = j_q$ for $p \neq q$ vanish because D is alternating, so only $(j_1, \dots, j_n)$ that form a permutation σ survive. By the previous lemma, $D(\mathbf{e}_{\sigma(1)}, \dots, \mathbf{e}_{\sigma(n)}) = \operatorname{sgn}(\sigma) D(I_n) = \operatorname{sgn}(\sigma)$.

For uniqueness, suppose D satisfies the three conditions. The same computation gives $D(A) = \left(\sum_\sigma \operatorname{sgn}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n}\right) D(I_n) = \det A$. Every function satisfying the three conditions therefore equals that formula. □

The Leibniz formula is correct but impractical: it sums $n!$ terms, and for $n = 12$ there are 479 001 600 of them. What people use is another form of the same formula.

Corollary 8.7 (Cofactor expansion)**.** *For every fixed row i,*

$$\det A = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij},$$

where M_{ij} is the determinant of the $(n-1) \times (n-1)$ submatrix obtained by deleting row i and column j. The number $C_{ij} = (-1)^{i+j} M_{ij}$ is called the cofactor *of the entry a_{ij}. The same formula holds if the expansion is carried out along a column.*

Proof. In the Leibniz formula, group the terms by the value of j at position row i. Every term containing a_{ij} has the form a_{ij} times a sign times the product of $n-1$ other entries, one from each row other than i and one from each column other than j. That product is exactly the Leibniz term for the minor M_{ij}, and its sign is $(-1)^{i+j}$ because moving row i to the last position requires $n-i$ interchanges and moving column j to the last position requires $n-j$ interchanges, so the sign is $(-1)^{2n-i-j} = (-1)^{i+j}$. Summing over j gives the formula as written.

For expansion along a column, we use $\det A = \det A^{\top}$. That equality itself follows from the Leibniz formula: every term for $A^{\top}$ has the form $a_{\sigma(1)1} \cdots a_{\sigma(n)n}$ with the index order reversed, and the map $\sigma \mapsto \sigma^{-1}$ preserves the sign because the number of interchanges is the same. □

Example 8.1. For $A = \begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix}$ expansion along the first row gives the cofactors

$$C_{11} = \begin{vmatrix} -1 & 2 \\ 1 & 2 \end{vmatrix} = -4, \qquad C_{12} = -\begin{vmatrix} -3 & 2 \\ -2 & 2 \end{vmatrix} = 2, \qquad C_{13} = \begin{vmatrix} -3 & -1 \\ -2 & 1 \end{vmatrix} = -5,$$

so that $\det A = 2(-4) + 1(2) + (-1)(-5) = -8 + 2 + 5 = -1$. The alternating sign is not decoration: without the sign, the result would be $-8 + 2 - 5 = -11$, and every conclusion afterwards would be wrong.

The determinant is useful for product matrices because it is multiplicative.

Theorem 8.8. *For $n \times n$ matrices,* $\det(AB) = \det A \det B$.

Proof. Fix A and define $f(B) = \det(AB)$. Row i of AB is a linear combination of the rows of B with coefficients from row i of A, so f is multilinear in the rows of B. If two rows of B are equal, two rows of AB are also equal, so f is alternating. By uniqueness, every multilinear alternating function f satisfies $f(B) = \det(B)\, f(I_n)$. Here $f(I_n) = \det A$, so $f(B) = \det B \det A$. □

Corollary 8.9. *A is invertible if and only if* $\det A \neq 0$. *If it is invertible,* $\det(A^{-1}) = 1/\det A$. *Moreover* $\det(cA) = c^n \det A$, *and the determinant of a triangular matrix is the product of its diagonal entries.*

Proof. If A is invertible, $\det A \det(A^{-1}) = \det I_n = 1$, so $\det A \neq 0$ and its inverse is $1/\det A$. If A is not invertible, its columns are linearly dependent by Chapter 7, so one column is a linear combination of the others. Multilinearity splits $\det A$ into a sum of determinants with two equal columns, so all of them are zero. The formula $\det(cA) = c^n \det A$ is obtained by pulling a factor c out of every row. For a triangular matrix, cofactor expansion along a row that borders only one diagonal entry, repeated n times, leaves the product of the diagonal. □

That corollary makes elimination and the determinant meet. Elimination brings A to an upper triangular matrix U by interchanges and additions of row multiples. Additions do not change the determinant; interchanges reverse its sign. Hence $\det A = (-1)^s \det U$ with s the number of interchanges, and $\det U$ is the product of the pivots: the determinant comes free after elimination, and that is how libraries compute it.

The other side of the determinant, just as important, is area. The columns of $B = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ are two arrows, $(3, 1)$ and $(1, 2)$. The unit square whose sides are carried by those two arrows turns into a parallelogram, and its area is five, exactly $\det B = 3 \cdot 2 - 1 \cdot 1 = 5$. Figure 8.2 shows that mapping. If the two columns are collinear, the parallelogram flattens into a single segment, its area is zero, and the determinant is zero as well.

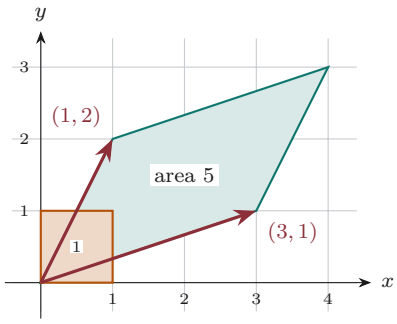


Figure 8.2: The determinant as area. The unit square (orange, area 1) is mapped by $B = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ to the parallelogram (teal) spanned by $(3, 1)$ and $(1, 2)$. Its area is 5, the same as $\det B = 3 \cdot 2 - 1 \cdot 1 = 5$: the determinant multiplies area by five, and its sign records whether the direction of rotation is reversed.

Do it by hand (kinesthetic learner)

Materials: graph paper, scissors, ruler, glue.

1. Draw a rectangle three squares by one square. Its area is three squares.
2. Cut out the rectangle. From the left end, cut off a right triangle with base two squares and height one square, then glue that piece onto the right end.
3. The new shape is a slanted parallelogram. Its area is still three squares, because no paper was thrown away and none was added.
4. Repeat with a slant of one square, then two, then three. Each time compute base times height, not the length of the slanted side.

What the reader should see: the shape changes, the area does not. A slant only shifts, it does not stretch, so the determinant does not grow.

For younger readers (age 10 and up)

Take a rectangular sheet of paper, then slant it by pushing its top edge sideways without shortening or lengthening anything. Its height stays the same, its width stays the same, its area stays the same, even though the shape is no longer a rectangle. Only the direction of rotation changes: trace the bottom edge and then the right edge starting from the lower left corner, and the slanted sheet still turns the same way; swap the two edges, and the direction reverses like a left shoe on a right foot.
In this chapter we will learn why a single number can report both area and orientation,

and why that number becomes zero exactly when the sheet is folded into a line. In your opinion, if the sheet is folded in half, is there still a way to compute its area with the same number?

8.3 Cramer's rule and the adjugate

Level: intermediate

The determinant answers whether a unique answer exists. The next question: can that answer be written as a formula, rather than as the result of steps? For small systems it can, and the formula is tidy.

adjugate (*adjoin*) — The cofactor map arranged as a transpose, used to write the inverse as a single formula.

Definition 8.4 (Adjugate)**.** For A of size $n \times n$ with cofactors C_{ij}, the *cofactor matrix* is the matrix with entry C_{ij} at position (i, j), and the *adjugate* $\operatorname{adj} A$ is its transpose, that is, the matrix with entry C_{ji} at position (i, j).

Note the reversal of the indices: the cofactor C_{ij} comes from deleting row i and column j, but is placed at position (j, i). This is the most frequent source of error when computing an adjugate by hand.

Theorem 8.10. *For every $n \times n$ matrix, $A \operatorname{adj} A = (\det A) I_n = \operatorname{adj} A \, A$.*

Proof. Entry (i, j) of $A \operatorname{adj} A$ is $\sum_k a_{ik} C_{jk}$. If $i = j$, that sum is exactly the cofactor expansion along row i, namely $\det A$. If $i \neq j$, that sum is the expansion along row j of the matrix obtained from A by replacing row j with row i. That matrix has two equal rows, so its determinant is zero. Hence $A \operatorname{adj} A = (\det A) I_n$. The second equality is obtained by the same argument on columns, or from $\operatorname{adj}(A^{\top}) = (\operatorname{adj} A)^{\top}$. □

Corollary 8.11. *If $\det A \neq 0$, then $A^{-1} = \dfrac{1}{\det A} \operatorname{adj} A$.*

Proof. From the previous theorem, $A(\operatorname{adj} A / \det A) = I_n$ and $(\operatorname{adj} A / \det A) A = I_n$. The inverse of a matrix is unique, so that matrix is A^{-1}. □

That formula holds for all n, but the price is high: $\operatorname{adj} A$ contains n^2 cofactors, and every cofactor is an $(n-1) \times (n-1)$ determinant.

With that inverse formula, the answer of a system can be written as a ratio of two determinants.

Theorem 8.12 (Cramer's rule)**.** *Suppose A is of size $n \times n$ with $\det A \neq 0$, and $\mathbf{b} \in \mathbb{R}^n$. The unique answer of $A\mathbf{x} = \mathbf{b}$ is*

$$x_j = \frac{\det A_j}{\det A}, \qquad j = 1, \ldots, n,$$

where A_j is the matrix obtained from A by replacing column j with $\mathbf{b}$.

Proof. Since $\det A \neq 0$, the previous corollary gives $\mathbf{x} = A^{-1}\mathbf{b} = (\operatorname{adj} A)\,\mathbf{b} / \det A$. Entry j of that product is $\sum_i b_i C_{ji} / \det A$. The sum $\sum_i b_i C_{ji}$ is the cofactor expansion along column j of the matrix whose column j has been replaced by $\mathbf{b}$, namely A_j. Hence $x_j = \det A_j / \det A$. □

Example 8.2. Back to the opening system, $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} = (8, -11, -3)^{\top}$. We already computed $\det A = -1$, and

$$\det A_1 = \begin{vmatrix} 8 & 1 & -1 \\ -11 & -1 & 2 \\ -3 & 1 & 2 \end{vmatrix} = -2, \qquad \det A_2 = -3, \qquad \det A_3 = 1.$$

So $x_1 = (-2)/(-1) = 2$, $x_2 = (-3)/(-1) = 3$, and $x_3 = 1/(-1) = -1$. Check: $A(2, 3, -1)^{\top} = (4 + 3 + 1,\ -6 - 3 - 2,\ -4 + 3 - 2)^{\top} = (8, -11, -3)^{\top}$.

The adjugate of that matrix, with the cofactors of its first row already computed above, is

$$\operatorname{adj} A = \begin{pmatrix} -4 & -3 & 1 \\ 2 & 2 & -1 \\ -5 & -4 & 1 \end{pmatrix}, \qquad A^{-1} = \frac{1}{-1} \operatorname{adj} A = \begin{pmatrix} 4 & 3 & -1 \\ -2 & -2 & 1 \\ 5 & 4 & -1 \end{pmatrix}.$$

The first row of A^{-1} times $\mathbf{b}$ gives $4(8) + 3(-11) - 1(-3) = 32 - 33 + 3 = 2$, matching x_1.

Cramer's rule is beautiful and rarely used for computing. Its value lies elsewhere: the formula shows that the answer depends smoothly on $\mathbf{b}$ and on the entries of A, and that dependence is what we measure through cost and sensitivity.

8.4 Elementary matrices, inverses, and equivalent conditions

Level: advanced

Row elimination can be seen as matrix multiplication. That view turns the question "is A invertible" into a question about a list of factors, and that list of factors yields an algorithm for the inverse.

elementary matrix (*matriks elementer*) — The identity matrix with one of its steps already carried out, so that it records a single row operation.

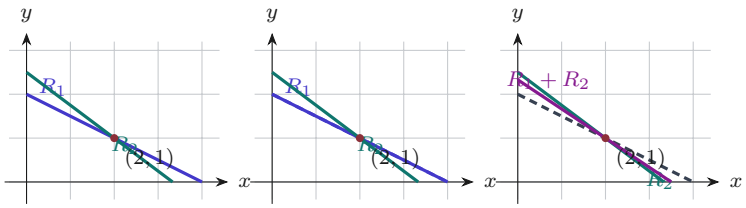
Definition 8.5 (Elementary matrix)**.** An *elementary matrix* is a matrix obtained from I_n by one elementary row operation. There are three kinds, matching the three operations: interchanging two rows, multiplying one row by a nonzero scalar, and adding a multiple of one row to another row.

Lemma 8.13. *If E is an $m \times m$ elementary matrix and A an $m \times n$ matrix, then EA is the result of performing the same row operation on A. Every elementary matrix is invertible, and its inverse is an elementary matrix of the same kind.*

Proof. Row i of EA is a linear combination of the rows of A with coefficients from row i of E. For E of the third kind, that row is $\mathbf{a}_i + c\mathbf{a}_j$, exactly as in the row operation; for the other two kinds the check is equally direct. The inverse is obtained by undoing the operation: interchanging rows undoes itself, and $\operatorname{diag}(1, \dots, c, \dots, 1)^{-1} = \operatorname{diag}(1, \dots, 1/c, \dots, 1)$. □

Figure 8.3 runs the three operations on the system $x + 2y = 4$, $3x + 4y = 10$, which intersect at $(2, 1)$. Interchanging rows only interchanges names; multiplying the first row by 2 writes the same line with numbers twice as large; adding the second row to the first gives $4x + 6y = 14$, which still passes through $(2, 1)$. The intersection point does not move, but the determinant of the coefficient matrix changes sign, doubles, then stays fixed. The solution set is preserved; the number that measures area is not.

Theorem 8.14. *An $n \times n$ matrix A is invertible if and only if A is a product of elementary matrices.*



(i) interchange rows (ii) multiply a row (iii) add a multiple

Figure 8.3: The three elementary row operations on the system $x + 2y = 4$, $3x + 4y = 10$. Interchanging rows only swaps the names R_1 and R_2; multiplying R_1 by 2 writes the same line; adding R_2 to R_1 rotates the first line into $4x + 6y = 14$. In all three panels, both lines pass through the same intersection point, $(2, 1)$.

Proof. A product of invertible matrices is also invertible, so one direction is clear.

For the other direction, suppose A is invertible and bring A to reduced echelon form by row operations, which is the same as multiplying from the left by elementary matrices $E_1, \dots, E_k$. By the theorem of three cases, A invertible means every column of A contains a pivot, so the reduced form of A is I_n, since A is square with n pivots. Hence $E_k \cdots E_1 A = I_n$, and therefore $A = E_1^{-1} \cdots E_k^{-1}$, a product of elementary matrices. □

That proof is also an algorithm: performing the same operations on I_n gives A^{-1}, so $(A \mid I_n) \to (I_n \mid A^{-1})$ produces the inverse without a single determinant, far cheaper than the adjugate.

We collect what has been proved into one list. This is the list people use every day, and its content is another face of a single statement.

Theorem 8.15 (Equivalent conditions)**.** *For an $n \times n$ matrix A, the following statements are equivalent.*

1. *A is invertible.*
2. *$A\mathbf{x} = \mathbf{0}$ has only the trivial answer $\mathbf{x} = \mathbf{0}$.*
3. *The reduced echelon form of A is I_n.*
4. *A is a product of elementary matrices.*
5. *$A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^n$.*
6. *$A\mathbf{x} = \mathbf{b}$ has exactly one answer for every $\mathbf{b} \in \mathbb{R}^n$.*
7. *$\det A \neq 0$.*

Proof. We chain the implications (1) ⇒ (2) ⇒ (3) ⇒ (4) ⇒ (1), then (1) ⇒ (5) ⇒ (6) ⇒ (1) and (1) ⇔ (7).

(1) ⇒ (2): multiply $A\mathbf{x} = \mathbf{0}$ from the left by A^{-1}. (2) ⇒ (3): the kernel of A is only $\{\mathbf{0}\}$, which means nullity zero, so by rank-nullity in Chapter 7, $\operatorname{rank} A = n$; hence there are n pivots and the reduced form is I_n. (3) ⇒ (4): the elimination steps give $E_k \cdots E_1 A = I_n$. (4) ⇒ (1): a product of invertible matrices is also invertible. (1) ⇒ (5): take $\mathbf{x} = A^{-1}\mathbf{b}$. (5) ⇒ (6): suppose some $\mathbf{b}$ has two different answers $\mathbf{x} \neq \mathbf{y}$. Then $\mathbf{x} - \mathbf{y} \neq \mathbf{0}$ and $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$, so the columns of A are linearly dependent and $\operatorname{rank} A < n$. The column space of A therefore has dimension less than n, so some $\mathbf{b}'$ is out of reach, contradicting (5). (6) ⇒ (1): if A is not invertible, its columns are dependent, so some $\mathbf{x} \neq \mathbf{0}$ satisfies $A\mathbf{x} = \mathbf{0}$, and the answer for $\mathbf{b} = \mathbf{0}$ is not unique. Finally (1) ⇔ (7) is the corollary of the previous section. □

That list explains why four questions that look different, about solutions, echelon form, factorization, and the determinant, are in fact one question; only the measuring tool changes. But the list is silent about cost and sensitivity, and those two things decide whether the answer obtained can be trusted.

Challenge (problem solver)

The following system looks ordinary: three equations for three numbers, no conspicuous duplicate row, so the answer seems to be unique.

$$x + y + z = 3, \qquad x + 2y + 3z = 6, \qquad 2x + 3y + 4z = 9.$$

Find all its answers. How many answers are there, and why is counting equations misleading here?
Hint 1: add the first two equations, then compare the result with the third equation.
Hint 2: the third equation adds no information at all, so only two pieces of information are independent. Once z is chosen, y and then x are determined; write the answer as one family with one parameter.
If you get stuck, go back to the section on three fates and count the pivots.

Cost and sensitivity

Elimination on an $n \times n$ matrix uses about $n^3/3$ multiplications: for $n = 1000$ about 3.33×10^8, which is still fast. Cramer's rule with cofactors uses $n \cdot n!$ multiplications, about 5.75×10^9 for $n = 12$ and above 10^{19} for $n = 20$; that is why it is used only for small n or for theoretical purposes.

The subtler matter is sensitivity. Suppose A is invertible and $\mathbf{b} \neq \mathbf{0}$, and suppose $\mathbf{b}$ changes to $\mathbf{b} + \delta\mathbf{b}$ so that the answer changes to $\mathbf{x} + \delta\mathbf{x}$.

Theorem 8.16. *With the notation above, for any subordinate matrix norm,*

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \kappa(A)\,\frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}, \qquad \kappa(A) = \|A\|\,\|A^{-1}\|.$$

Proof. From $A\,\delta\mathbf{x} = \delta\mathbf{b}$ we get $\delta\mathbf{x} = A^{-1}\delta\mathbf{b}$, so $\|\delta\mathbf{x}\| \leq \|A^{-1}\|\|\delta\mathbf{b}\|$. From $\mathbf{b} = A\mathbf{x}$ we get $\|\mathbf{b}\| \leq \|A\|\|\mathbf{x}\|$, that is, $1/\|\mathbf{x}\| \leq \|A\|/\|\mathbf{b}\|$. Multiplying the two inequalities gives the required result. □

The number $\kappa(A)$ is called the *condition number*. It does not depend on scale: multiplying A by c does not change it, because $\|cA\|\|(cA)^{-1}\| = |c|\|A\| \cdot |c|^{-1}\|A^{-1}\|$. The determinant has no such property, and this is why we do not use the determinant as a measure of sensitivity. A small determinant only means a small volume; it can be small because the whole matrix was scaled down, while the sensitivity does not change.

For the opening system, $\kappa(S) \approx 1.01 \times 10^5$. With machine precision around 10^{-16}, the digits that can be trusted in the answer are about 10^{-11} relative: enough for this example, exhausted for a worse matrix. Chapter 22 discusses thresholds and how to recognize a matrix that can no longer be trusted.

For younger readers (age 10 and up)

Imagine a long plank used as a base for measuring the height of a hill. The plank is almost horizontal, only slightly tilted. If the lower end is lifted by a finger, the other end rises far more, because the plank rests on a very small slope. A finger in one place becomes a hand-span in another.
Now imagine the plank is made very steep. Lift its lower end by a finger, and its upper end rises only a tiny bit. The same work produces two very different effects, only because the base is different.
Our computations work like that plank. There are arrangements of numbers in which

a small change in the input produces a small change in the answer; there are also arrangements in which a change the size of a speck of dust produces a completely different answer. What decides is not how large the numbers are, but the shape of the arrangement.
In this chapter we will learn why one number can predict that before we compute, and why that number does not change when all the numbers are enlarged or reduced by the same ratio.
If you weigh on a scale whose base is tilted, is the result wrong, or merely untrustworthy?

8.5 A worked example

Level: intermediate

We work one system in two ways, compute one 4×4 determinant, then measure the sensitivity of two matrices.

Step-by-step recipe (procedural learner)

1. Write the augmented matrix $(A \mid \mathbf{b})$: coefficient columns first, the right-hand side column last.
2. Work on the leftmost column. Choose one nonzero entry as the pivot; if the top entry is zero, interchange rows first.
3. Zero out all entries below the pivot: for each row j below it, $R_j \leftarrow R_j - \frac{a_{ji}}{a_{ii}} R_i$.
4. Shift one column to the right and repeat; stop when there are no nonzero rows left, then read the fate of the system from the pivot positions.

Warning about the order: the pivots are zeroed from top to bottom, column by column, before back substitution begins. Reversing that order destroys the pivots already obtained and forces the work to be repeated.

The system is $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 8 \\ -11 \\ -3 \end{pmatrix}.$$

First way, elimination, with the steps written in full:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 + \frac{3}{2} R_1} \left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 2 & 1 & 5 \end{array}\right),$$

then $R_3 \leftarrow R_3 - 4R_2$ gives

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 0 & -1 & 1 \end{array}\right).$$

Back substitution: from the third row $-x_3 = 1$, so $x_3 = -1$. The second row $\frac{1}{2}x_2 + \frac{1}{2}(-1) = 1$ gives $x_2 = 3$. The first row $2x_1 + 3 + 1 = 8$ gives $x_1 = 2$. The result is $\mathbf{x} = (2, 3, -1)$. The pivots that appear are 2, $\frac{1}{2}$, and -1, so the product of the pivots is $\det A = 2 \cdot \frac{1}{2} \cdot (-1) = -1$, matching the cofactor expansion above. Here there was no row interchange, so the sign did not change.

The second way, Cramer's rule, we already ran on the same system in the previous section. The three replacement determinants give $\det A_1 = -2$, $\det A_2 = -3$, and $\det A_3 = 1$, so $\mathbf{x} = (-2, -3, 1)/(-1) = (2, 3, -1)$. Two roads, one answer; a check like this is cheap and should always be carried out.

Now the 4×4 determinant. Take

$$M = \begin{pmatrix} 2 & 0 & 1 & 3 \\ 1 & 4 & 0 & 2 \\ 3 & 1 & 2 & 1 \\ 0 & 2 & 1 & 5 \end{pmatrix}.$$

Expansion along the first row uses four minors, and one of them we compute in full:

$$M_{11} = \begin{vmatrix} 4 & 0 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 5 \end{vmatrix} = 4(10 - 1) - 0(5 - 2) + 2(1 - 4) = 36 - 6 = 30.$$

In a similar way $M_{12} = 15$, $M_{13} = -45$, and $M_{14} = -15$. Hence

$$\det M = 2(30) - 0(15) + 1(-45) - 3(-15) = 60 - 45 + 45 = 60.$$

The number of terms summed is $4! = 24$, and for 5×5 it is already 120: the growth is factorial. This is why libraries do not use this method.

Now sensitivity. The 4×4 Hilbert matrix is defined by $h_{ij} = 1/(i + j - 1)$. Its determinant is exactly $1/6\,048\,000 \approx 1.65 \times 10^{-7}$, very small, so one easily concludes that this matrix is dangerous. Its condition number is $15\,513.74$. The opening matrix had $\det S = -0.003$ and $\kappa(S) \approx 1.01 \times 10^{5}$, so its determinant is eighteen thousand times larger, while its sensitivity is about six and a half times worse. The determinant is not a measure of sensitivity; the condition number is.

The jump test for S uses $\mathbf{x} = (1, 1, 1)$, so that $\mathbf{b} = (6,\ 15,\ 24.001)$. Changing the third component to 24.002 changes the answer to $(2, -1, 2)$. The relative change in the data is 3.46×10^{-5}, the relative change in the answer is $\sqrt{2} \approx 1.414$, and the ratio is $40\,916$: below $\kappa(S)$ as the theorem promises, but only by a factor of two and a half.

Which is more sensitive, M or the 4×4 Hilbert matrix? Predict by comparing their determinants, then compute the condition number of both with `numpy.linalg.cond`. Compare also with $10^{-3} I_4$, whose determinant is far smaller, then explain why the rule "small determinant means sensitive" fails. Finally, multiply M by 10^{-4}: its condition number does not change, while its determinant shrinks by a factor of 10^{-16}.

8.6 Computational lab

Level: advanced

Computational lab

```
import numpy as np, math

def det_kofaktor(A):
    n = A.shape[0]
    if n == 1:
        return A[0, 0]
    return sum((-1) ** j * A[0, j] *
               det_kofaktor(np.delete(np.delete(A, 0, 0), j, 1))
               for j in range(n))
```

```
A = np.array([[2., 1., -1.], [-3., -1., 2.], [-2., 1., 2.]])
b = np.array([8., -11., -3.])
x = np.linalg.solve(A, b)
print("det A    :", np.linalg.det(A))
print("det cof  :", det_kofaktor(A))
print("rank A   :", np.linalg.matrix_rank(A))
print("x        :", x)
print("residual :", np.linalg.norm(A @ x - b))
for j in range(3):
    Aj = A.copy(); Aj[:, j] = b
    print("det A%d   : %.1f  -> x%d = %.6f" % (j + 1, np.linalg.det(Aj),
          j + 1, np.linalg.det(Aj) / np.linalg.det(A)))

M = np.array([[2., 0., 1., 3.], [1., 4., 0., 2.],
              [3., 1., 2., 1.], [0., 2., 1., 5.]])
print("det M    :", det_kofaktor(M), np.linalg.det(M))

H = np.array([[1. / (i + j + 1) for j in range(4)] for i in range(4)])
print("det H4   :", np.linalg.det(H))
print("cond H4  : %.2f" % np.linalg.cond(H))

S = np.array([[1., 2., 3.], [4., 5., 6.], [7., 8., 9.001]])
bs = S @ np.array([1., 1., 1.])
print("det S    :", np.linalg.det(S), " cond S: %.1f" % np.linalg.cond(S))
print("x        :", np.linalg.solve(S, bs))
bs2 = bs.copy(); bs2[2] += 0.001
print("x'       :", np.linalg.solve(S, bs2))
print("cost     :", math.factorial(12) * 12, round(1000 ** 3 / 3))
```

Actual output:

```
det A    : -1.0
det cof  : -1.0
rank A   : 3
x        : [ 2.  3. -1.]
residual : 1.9860273225978185e-15
det A1   : -2.0  -> x1 = 2.000000
det A2   : -3.0  -> x2 = 3.000000
det A3   : 1.0  -> x3 = -1.000000
det M    : 60.0 60.000000000000036
det H4   : 1.6534391534390412e-07
cond H4  : 15513.74
det S    : -0.0030000000000036655  cond S: 101072.3
x        : [1. 1. 1.]
x'       : [ 2. -1.  2.]
cost     : 5748019200 333333333
```

Four things must be read from that output.

First, $x = (2, 3, -1)$ appears with residual 1.99×10^{-15}, which is zero to machine precision. The residual is computed on purpose: a library answer is never exact, and the residual is the cheapest way to check whether that answer satisfies its system. For a bad matrix, a small residual does not guarantee a small error; that is the topic of Chapter 22.

Second, the determinant from the cofactor expansion and the one from the library are both -1, and for M both are 60. The library does not use cofactors; it uses a factorization and multiplies the pivots. The difference of 3.6×10^{-14} on $\det M$ is rounding, not a difference of definition.

Third, $\det H_4$ is reported as 1.65×10^{-7} while its exact value is $1/6\,048\,000$, so the machine number is correct to seven digits. For higher orders, $1/\det$ passes 10^{12} quickly, and that is where the determinant stops being useful as a marker.

Fourth, the jump of the answer for S shows up intact: from $(1, 1, 1)$ to $(2, -1, 2)$ only because one number changed by a thousandth. The number reported for $\det S$ is not exactly -0.003 either; the difference 3.7×10^{-15} appears because 9.001 is not represented exactly in binary fractions.

Finally, cost: 5.748×10^9 multiplications for Cramer's rule at $n = 12$ against 3.33×10^8 for elimination at $n = 1000$. A far larger matrix is handled with far less work; that is the difference between a cube and a factorial.

Note for instructors

A teaching order that has worked: elimination first, with the three cases shown on one fixed coefficient matrix and three different right-hand sides. Students who see consistency change because of one number in $\mathbf{b}$ usually accept at once that the question about A and the question about $\mathbf{b}$ must be separated. Only after that does the determinant enter, as the answer to the question about A alone.

Two typical misconceptions. First, "a small determinant means the system is unstable". The matrix $10^{-6}I$ has determinant 10^{-18} and condition number one; conversely H_4 has determinant 1.65×10^{-7} and condition 1.5×10^4. This mistake appears every year, and the way to dismantle it is to ask students to multiply one matrix by a scalar and then compare the two numbers. Second, "Cramer's rule is the way to solve a system". Theoretically true; in cost, false. Give one 12×12 system and ask them to estimate the cost of both ways before computing.

A useful class activity: hand out three systems with the same coefficient matrix and right-hand sides $(4, 9, 13)$, $(4, 9, 14)$, and $(0, 0, 0)$. Ask each group to determine the fate of its system from the echelon form alone, then to conclude what can and cannot be concluded from A by itself.

A sign of readiness for advanced work: students can list the seven equivalent conditions without memorizing the order, and can explain why the condition about the determinant belongs in the same list as the conditions about pivots.

8.7 Summary

Level: basic

Summary.

- Elementary row operations do not change the solution set, because every operation can be undone. Because the reduced form is unique, the pivot positions determine three fates: inconsistent if a pivot appears in the last column, unique if every coefficient column has a pivot, many if some column has no pivot.
- The determinant is the only multilinear, alternating, normalized function. The Leibniz formula and the cofactor expansion are consequences of it, not its definition.
- $\det(AB) = \det A \det B$, so A is invertible exactly when $\det A \neq 0$; and $\det A$ equals the product of the pivots after elimination, times the sign of the interchanges.
- The adjugate gives $A \operatorname{adj} A = (\det A)I$ and $A^{-1} = \operatorname{adj} A / \det A$; Cramer's rule writes $x_j = \det A_j / \det A$.
- Seven conditions for invertibility are equivalent: trivial kernel, reduced form I_n, product of elementary matrices, consistent and unique for all $\mathbf{b}$, and $\det A \neq 0$.
- The cost of elimination is proportional to $n^3/3$, the cost of cofactors to $n \cdot n!$. Sensitivity is measured by $\kappa(A) = \|A\|\|A^{-1}\|$, not by $\det A$.

8.8 Exercises

Level: intermediate

Exercise 8.1.

1. Solve by elimination, then check with Cramer's rule:
$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 9 \\ 8 \\ 26 \end{pmatrix}.$$
Write down the pivots that appear, record how many times rows were interchanged, then conclude $\det A$ from the product of the pivots.
2. Compute $\det M$ for $M = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}$ by cofactor expansion along one row, then compute it with `numpy.linalg.det`. The two results are not exactly equal: the first is exact, the second 5.3×10^{-15}. Explain why. Afterwards find a nonzero vector $\mathbf{v}$ with $M\mathbf{v} = \mathbf{0}$ and conclude $\det M = 0$ without any expansion.
3. Prove that if two columns of A are equal, then $\det A = 0$, without using $\det A = \det A^{\top}$. (Hint: use the Leibniz formula, group the terms by pairs of permutations that swap on those two columns, and show that their signs are opposite.)
4. Prove that $\det(\operatorname{adj} A) = (\det A)^{n-1}$ for $n \geq 2$. (Hint: take determinants in $A \operatorname{adj} A = (\det A) I$ and use $\det(cI) = c^n$.)
5. Evaluate this claim: *if $\det A$ is very small, then the answer of $A\mathbf{x} = \mathbf{b}$ is very sensitive to changes in $\mathbf{b}$*. Decide true, false, or not yet enough evidence, and include a counterexample or an additional reason that makes it true.
6. For $A = \begin{pmatrix} 1 & 1 \\ 1 & 1+\varepsilon \end{pmatrix}$ with $\varepsilon > 0$, compute $\kappa(A)$ in the 1-norm as a function of ε. Determine the value of ε that makes $\kappa(A) > 10^6$, then check it with `numpy.linalg.cond`.
7. Prove: if A is invertible and B is obtained from A by adding a multiple of one row to another row, then $\det B = \det A$ and B^{-1} can be expressed through A^{-1} and one elementary matrix.
8. Suppose $D = \operatorname{diag}(d_1, \ldots, d_n)$ and A is of size $n \times n$. Show that $\det(A + AD) = \det(A) \prod_{i=1}^{n} (1 + d_i)$, then determine the condition on $d_1, \ldots, d_n$ for $A + AD$ to be invertible. Check your answer with a 3×3 matrix in Python.
9. Choose a 3×3 matrix with small integer entries and $\det A = 1$, compute A^{-1} through the adjugate and through `numpy.linalg.inv`, then compare the two. Repeat for a matrix with $\det A = 7$ and explain why the entries of A^{-1} become fractions.

Hints for the three hardest items. Item 4: take determinants of both sides, remember $\det(cI) = c^n$, and note that $\det A$ may be zero, so the conclusion about $\operatorname{adj} A$ must be read as a matrix, not as an inverse. Item 6: in the 1-norm, $\|A\|_1$ is the largest column sum, and for this matrix the second column is the largest; compute $\|A^{-1}\|_1$ from the 2×2 formula and note that the answer is proportional to $1/\varepsilon$. Item 8: write $A + AD$ as $A(I + D)$ before pulling out the determinant, and remember that $I + D$ is diagonal, so its determinant is the product of its diagonal entries.

For researchers: limits and next steps

What is really proved in this chapter, for the field $\mathbb{R}$ and finite dimension: the uniqueness of the determinant as a normalized alternating n-linear function together with the permutation formula, in Hoffman and Kunze, *Linear Algebra* 2nd edition, sections 5.2

and 5.3, which hold more generally over a commutative ring with identity (section 5.1); multiplicativity and the equivalence of invertibility with $\det \neq 0$, in Axler, *Linear Algebra Done Right* 4th edition, section 9C, statements 9.49 and 9.50, with a basis-free determinant of an operator; and the list of seven equivalent conditions, in Anton 11th edition, Theorem 2.3.8.

What is popularized: "the determinant is volume". What is proved is that $|\det A|$ equals the volume of the parallelepiped spanned by its columns; without the absolute value, the claim is false for a negative determinant. Also wrong numerically: Cramer's rule as a way to solve systems above $n = 11$, where the cost is factorial.

An open question. The sensitivity bound in the theorem above is normwise and worst-case, often too pessimistic compared with a componentwise analysis. More fundamentally: the maximum growth factor of Gaussian elimination with partial pivoting is not known for general n. The proved bound is 2^{n-1}, while numerical experiments give far less, and that gap is open.

Limits of validity. The argument collapses when the field is not a field, because modules do not always have a basis and elimination cannot always divide. For very large n, the determinant overflows or underflows long before the rank is lost, so libraries report the logarithm of the determinant. The sensitivity statements also depend on the norm, and no choice is canonical.

Map to source texts: This chapter follows Anton, Elementary Linear Algebra *11th edition, sections 1.1 through 1.6 (linear systems, Gauss-Jordan elimination, inverses, elementary matrices) and 2.1 through 2.3 (cofactors, the determinant through row reduction, Cramer's rule, including Theorem 2.3.8 on equivalent conditions). The axiomatic treatment follows Hoffman and Kunze,* Linear Algebra *2nd edition, sections 5.2 and 5.3, with section 1.4 for the uniqueness of the reduced echelon form. The basis-free view is in Axler,* Linear Algebra Done Right *4th edition, section 9C. For elimination as factorization and the view $A = CR$, see Strang,* Introduction to Linear Algebra *6th edition, Chapter 2 and the note "Elimination and Factorization". The numerical side, namely condition numbers, rounding error, and cost, is treated in Chapter 22 of this book.*

Chapter 9

Eigenvalues and Diagonalization

Every minute, a network monitor splits all incoming requests into three buckets: host A, host B, and host C. From old logs it builds a transition rule. Of the requests heading to A, 80 percent stay at A, 10 percent move to B, and 10 percent move to C. From B, 30 percent move to A, 60 percent stay, and 10 percent move to C. From C, 20 percent move to A, 30 percent to B, and 50 percent stay. If the share of requests in the three buckets at minute k is written as the column $\mathbf{x}_k$, that rule takes the form $\mathbf{x}_{k+1} = M\mathbf{x}_k$ with

$$M = \begin{pmatrix} 0.8 & 0.3 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0.1 & 0.1 & 0.5 \end{pmatrix}.$$

Starting from $\mathbf{x}_0 = (1, 0, 0)$, that is, all requests initially go to A, the monitor computes ten steps and obtains $\mathbf{x}_{10} = (0.5672,\ 0.2661,\ 0.1666)$. Thirty steps later the numbers are $(0.5667,\ 0.2667,\ 0.1667)$. The three entries still sum to one, and on the second reading the numbers stop moving at four decimal places.

The question worth asking is not what the result is after forty minutes; repeated multiplication has already answered that for one starting point. The more useful question is: is there a vector that M only enlarges without changing its direction? Here that direction is $(17/30,\ 4/15,\ 1/6) = (0.5667,\ 0.2667,\ 0.1667)$, exactly the numbers that appear after forty steps, and the multiplier is 1.

Chapter 8 supplied the determinant: $\det M = 1/5$, so M is invertible and every system $M\mathbf{y} = \mathbf{b}$ has a unique solution. That answers whether the process can be reversed, not which way it is heading, and not how fast. The number that governs the rate here is $1/2$: the largest eigenvalue after 1. Since $(1/2)^{40} \approx 9.1 \times 10^{-13}$, the initial error stays behind the first six digits, and that is why the second reading looks still.

This chapter prepares the tools for reading those two numbers straight from the matrix: how to compute them, the condition under which the preserved directions are numerous enough to form a basis, what changes when the basis is changed, and then three uses: matrix powers, linear differential systems, and Markov chains.

For younger readers (age 10 and up)

Take a sheet of rubber marked with a square grid, then pull it from two sides. The squares turn into diamonds, and almost every line becomes slanted. But one line stays straight and stays parallel to itself: the line along the pull. Its length changes, its direction does not. Sometimes there is also another line that does not slant but reverses its direction, like an arrow turned half a circle.

If we are told only how many times each of those special lines stretches, we already know the whole pull, without having to draw the grid again. That is a very economical method:

three or four numbers replace thousands of points on the sheet.
The trouble is that not every pull is that tidy. Try twisting the sheet slightly while pulling. Now no line is really straight any longer, and the economical method fails completely. Some pulls can be described entirely by a few stretch numbers, others cannot.
The question: from the picture of the pull alone, can we guess whether there are special lines that do not slant, and how many? In this chapter we will learn why some pulls have such lines and some do not, why some pulls can be replaced entirely by just a few numbers, and why a computing engine sometimes guesses wrong on pulls that are almost a twist.

9.1 Directions that do not rotate

Level: basic

A square matrix acts on vectors: it rotates, stretches, or flattens. Almost every vector comes out with a direction different from the one it went in with. A small minority do not. A vector that comes out in the same direction it went in, only with a different length, marks a direction that the matrix preserves. It is that kind of direction that we name first, before any computing tool exists.

eigenvalue (*nilai eigen*) — The multiplier along a direction that is not rotated: the direction stays the same, its length is multiplied by that number.

Definition 9.1 (Eigenvalues and eigenvectors)**.** Let A be an $n \times n$ matrix with real entries. A scalar λ is called an *eigenvalue* of A if there is a nonzero vector $\mathbf{v} \in \mathbb{R}^n$ with $A\mathbf{v} = \lambda\mathbf{v}$. The vector $\mathbf{v}$ is called an *eigenvector* for λ.

The condition $\mathbf{v} \neq \mathbf{0}$ is not decoration. Without it, $\mathbf{0}$ satisfies $A\mathbf{0} = \lambda\mathbf{0}$ for every λ, so the term loses its meaning. What we are looking for is precisely a direction, and the zero point has no direction.

The equation $A\mathbf{v} = \lambda\mathbf{v}$ can be rewritten as $(A - \lambda I)\mathbf{v} = \mathbf{0}$. So λ is an eigenvalue exactly when the matrix $A - \lambda I$ has a nontrivial kernel, that is, exactly when $A - \lambda I$ is singular. A question about eigenvalues can therefore always be translated into a question about rank deficiency, and Chapter 8 already supplied the tool for that.

Before computing, one important warning. Not every real matrix has real eigenvalues. The rotation matrix

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

sends $(1, 0)$ to $(0, 1)$. Every real vector rotates ninety degrees, so no line is preserved; $R^2 = -I$ turns half a circle after two steps. The absence of real eigenvalues here is not a failure of computation but a geometric statement.

To find eigenvalues without guessing, we use the determinant: the eigenvalues are the numbers that make the matrix lose its inverse once that number is subtracted from the diagonal.

characteristic polynomial (*polinomial karakteristik*) — The list of numbers that make the matrix singular, written as a polynomial in one variable.

Definition 9.2 (Characteristic polynomial)**.** For an $n \times n$ matrix, the *characteristic polynomial* is defined as $p_A(\lambda) = \det(\lambda I - A)$.

We use $\lambda I - A$, not $A - \lambda I$. The two differ by a factor $(-1)^n$, so their roots are the same, but the first choice makes the polynomial monic, and that makes the other coefficients easier to read, as the next corollary shows.

Theorem 9.1. *A scalar λ is an eigenvalue of A if and only if $p_A(\lambda) = 0$.*

Proof. Suppose $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $(A - \lambda I)\mathbf{v} = \mathbf{0}$, so $A - \lambda I$ has a nontrivial kernel and is therefore singular. By Chapter 8, singularity is equivalent to a zero determinant, so $\det(A - \lambda I) = 0$, which differs by a factor $(-1)^n$ from $\det(\lambda I - A) = 0$.

Conversely, if $\det(\lambda I - A) = 0$ then $\lambda I - A$ is singular. Its kernel is nontrivial, so there is $\mathbf{v} \neq \mathbf{0}$ with $(\lambda I - A)\mathbf{v} = \mathbf{0}$, that is, $A\mathbf{v} = \lambda\mathbf{v}$. □

The polynomial p_A has degree n and is monic, so it has at most n distinct roots. Over $\mathbb{C}$ it always factors into n linear factors, so every complex square matrix has exactly n eigenvalues when counted with multiplicity. Over $\mathbb{R}$ that can fail: $p_R(\lambda) = \lambda^2 + 1$ has no real root, but has a pair of complex roots $\pm i$.

The top two coefficients of the characteristic polynomial carry two quantities we already know.

Corollary 9.2. *With eigenvalues counted together with their multiplicities,* $\operatorname{tr}(A) = \sum_i \lambda_i$ *and* $\det(A) = \prod_i \lambda_i$.

Proof. The coefficient of λ^{n-1} in $\det(\lambda I - A)$ is $-\operatorname{tr}(A)$. That term is produced only by the identity permutation in the Leibniz formula, because every other permutation loses at least two diagonal entries. The constant term is $p_A(0) = \det(-A) = (-1)^n \det(A)$. Comparing those two coefficients with the form $\prod_i(\lambda - \lambda_i) = \lambda^n - (\sum_i \lambda_i)\lambda^{n-1} + \cdots + (-1)^n \prod_i \lambda_i$ gives both identities. □

That corollary gives a cheap check: after computing eigenvalues with great effort, their sum and their product must match the trace and the determinant. If they do not, there is an arithmetic mistake.

Read it aloud (auditory learner)

Two voices, taking turns.
First voice: what does a matrix do?
Second voice: it rotates, stretches, flattens.
First voice: to whom?
Second voice: to almost every vector.
First voice: and then some do not rotate?
Second voice: some do not. One direction. Sometimes two. Never more than the number of columns.
First voice: does the length change?
Second voice: it changes. The direction stays.
First voice: by how much?
Second voice: that is the number. That number is the eigenvalue.
First voice: and the direction?
Second voice: that is the eigenvector. One direction, one number.
First voice: what if the direction flips?
Second voice: the number is negative. The line stays the same, it only faces backwards.
First voice: what if the number is zero?
Second voice: that direction disappears, and the matrix loses its inverse.
Together: a direction that does not change stores a number that changes its length.
Repeat once in a normal voice: a direction that does not change stores a number that changes its length.

Step-by-step recipe (procedural learner)

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ whose entries are already known:

1. Subtract λ from both diagonal entries, leaving $A - \lambda I$.
2. Compute $\det(A - \lambda I)$, or use $\lambda^2 - (a+d)\lambda + (ad - bc)$ directly.
3. Solve that quadratic equation; check that the sum of the roots is $a + d$ and the product is $ad - bc$.
4. For each root λ, solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ and take one nonzero solution.

Steps 3 and 4 must not be swapped: an eigenvector only has meaning after λ is known, and the column order of P must follow the diagonal order of D.

Now one example by hand. Take

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}, \qquad p_A(\lambda) = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2).$$

The trace is $7 = 5 + 2$ and the determinant is $10 = 5 \cdot 2$, which matches. For $\lambda = 5$ we solve $(A - 5I)\mathbf{v} = \mathbf{0}$, that is $-v_1 + 2v_2 = 0$ and $v_1 - 2v_2 = 0$; both rows say the same thing, so $\mathbf{v} = (2, 1)$. For $\lambda = 2$ we obtain $2v_1 + 2v_2 = 0$, so $\mathbf{v} = (1, -1)$. Direct check:

$$A\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} = 5\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Both of those checks can be seen in the plane. Figure 9.1 draws both eigen directions together with their images, and one vector that is not an eigenvector as a comparison.

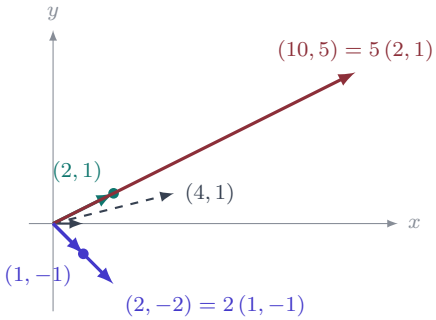


Figure 9.1: Eigen directions of the matrix $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$. The vector $(2, 1)$ comes out as $(10, 5)$, that is, 5 times as long without rotating; the vector $(1, -1)$ comes out as $(2, -2)$, that is, 2 times as long. The vector $(1, 0)$ comes out as $(4, 1)$ and rotates, so it is not an eigenvector.

An eigenvector is determined only up to a multiplier: $(4, 2)$ is also an eigenvector for 5. What is unique is the direction, not the length. We will use that freedom later to tidy up the matrix.

For a 2×2 matrix the roots can be written down directly. Since $p_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A)$, the quadratic formula gives

$$\lambda_{1,2} = \frac{\operatorname{tr}(A) \pm \sqrt{\operatorname{tr}(A)^2 - 4\det(A)}}{2},$$

so the sign of $\Delta = \operatorname{tr}(A)^2 - 4\det(A)$ determines the whole spectrum. For A above, $\Delta = 49 - 40 = 9$, and the roots are 5 and 2. If $\Delta = 0$ the two roots coincide, and if $\Delta < 0$ there is no real root: the rotation matrix above gives $\Delta = 0 - 4 = -4$.

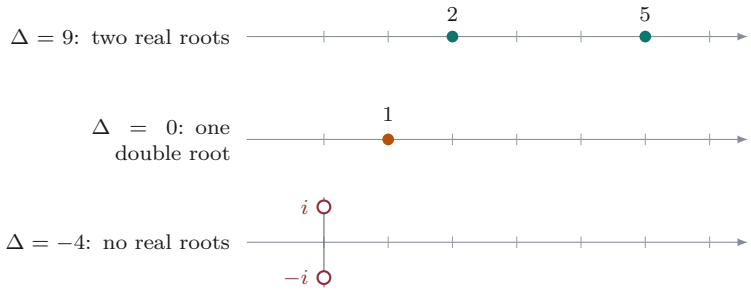


Figure 9.2: The spectrum of a 2×2 matrix on the number line, separated by the sign of $\Delta = \operatorname{tr}(A)^2 - 4\det(A)$. Top row: A with $\Delta = 49 - 40 = 9$ gives the roots 2 and 5. Middle row: a shear matrix with $\Delta = 0$ gives the root 1 counted twice. Bottom row: the rotation matrix with $\Delta = 0 - 4 = -4$ does not meet the real line, and both of its roots, $\pm i$, lie off that line.

Figure 9.2 arranges those three possibilities on a single number line: two points, one point counted twice, or no point at all. The last is not a shortcoming of the picture but a statement that the real line contains no preserved direction.

Do it by hand (kinesthetic learner)

Materials: one rubber band, two pencils, a sheet of paper, a ruler.
Place the two pencils parallel, then stretch the rubber band around both so that a rectangle forms. On the paper, draw three arrows from the lower left corner of that rectangle, each at an angle of 30, 45, and 60 degrees to the bottom side. Label all three. Slide the right pencil twice as far, so that the rubber stretches. Compare each arrow with its original length. The arrow parallel to the pull side keeps its direction; the other two slant, and the one that slants most is the one farthest from the pull side.
Now rotate the whole frame ninety degrees and repeat. The surviving direction rotates along with the rubber, because what is special is the direction relative to the rubber, not relative to the table.
What should be visible: one direction survives, two directions rotate, and the stretch factor along the surviving direction is the multiplier we call the eigenvalue.

9.2 Eigen spaces and two multiplicities

Level: intermediate

One eigenvalue usually comes with more than one eigenvector. The set of all eigenvectors for one eigenvalue, plus the zero vector, is not a random collection: it is a subspace. Its dimension will later decide whether the matrix can be simplified.

Definition 9.3 (Eigen space)**.** For an eigenvalue λ of A, the *eigen space* is $E_\lambda = \operatorname{null}(A - \lambda I) = \{\mathbf{v} : A\mathbf{v} = \lambda\mathbf{v}\}$.

eigen space (*ruang eigen*) — All directions that the matrix does not rotate, gathered into a single subspace.

An eigen space is always a subspace because it is the kernel of a matrix (Chapter 7). Its dimension is at least one exactly when λ really is an eigenvalue.

Eigen spaces for different eigenvalues intersect only at the zero point.

Proposition 9.3. *If $\lambda \neq \mu$, then $E_\lambda \cap E_\mu = \{\mathbf{0}\}$.*

Proof. Take $\mathbf{v}$ in that intersection. Then $A\mathbf{v} = \lambda\mathbf{v}$ and $A\mathbf{v} = \mu\mathbf{v}$, so $(\lambda - \mu)\mathbf{v} = \mathbf{0}$. Since $\lambda - \mu \neq 0$, we must have $\mathbf{v} = \mathbf{0}$. □

That property extends to any finite number of eigenvectors, by induction. The result is the lemma that forms the backbone of this chapter.

Lemma 9.4. *Eigenvectors corresponding to distinct eigenvalues are linearly independent of one another.*

Proof. We prove this by induction on the number of vectors. For one vector the statement is true because an eigenvector is nonzero. Suppose the statement holds for $m - 1$ vectors, and let

$$c_1\mathbf{v}_1 + \cdots + c_m\mathbf{v}_m = \mathbf{0}$$

with $\lambda_1, \ldots, \lambda_m$ distinct. Multiply that equation by A, and use $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$:

$$c_1\lambda_1\mathbf{v}_1 + \cdots + c_m\lambda_m\mathbf{v}_m = \mathbf{0}.$$

Subtract λ_m times the first equation from the second:

$$c_1(\lambda_1 - \lambda_m)\mathbf{v}_1 + \cdots + c_{m-1}(\lambda_{m-1} - \lambda_m)\mathbf{v}_{m-1} = \mathbf{0}.$$

By the induction hypothesis, $\mathbf{v}_1, \ldots, \mathbf{v}_{m-1}$ are independent, so $c_i(\lambda_i - \lambda_m) = 0$ for $i < m$. Since $\lambda_i \neq \lambda_m$, all $c_i = 0$ for $i < m$. The original equation is left with $c_m\mathbf{v}_m = \mathbf{0}$, so $c_m = 0$. □

That lemma gives a tidy upper bound: an $n \times n$ matrix cannot have more than n independent eigenvectors. When is that bound attained?

To answer, we need two ways of counting how often one eigenvalue occurs: from the polynomial and from the dimension of the eigen space. The two often differ, and it is the difference that decides.

multiplicity (*multiplisitas*) — How many times an eigenvalue occurs, counted in two ways that are not always equal.

Definition 9.4 (Algebraic and geometric multiplicity)**.** Let λ be an eigenvalue of A. The *algebraic multiplicity* of λ is the power of the factor $(\lambda' - \lambda)$ in $p_A(\lambda')$. The *geometric multiplicity* of λ is $\dim E_\lambda = \text{nullity}(A - \lambda I)$.

Geometric multiplicity is easy to compute: subtract λ from the diagonal, then compute the nullity of the resulting matrix. Algebraic multiplicity demands factoring a polynomial. The relation between the two always runs in one direction.

Lemma 9.5. *For every eigenvalue λ we have $1 \leq \dim E_\lambda \leq$ the algebraic multiplicity of λ.*

Proof. The left inequality is true because λ is an eigenvalue, so E_λ contains a nonzero vector.

For the right inequality, write $k = \dim E_\lambda$ and choose a basis $\mathbf{u}_1, \ldots, \mathbf{u}_k$ of E_λ. Extend it to a basis $\mathbf{u}_1, \ldots, \mathbf{u}_n$ of $\mathbb{R}^n$, which is possible by Chapter 7. In that basis, column j of the matrix A for $j \leq k$ is $A\mathbf{u}_j = \lambda\mathbf{u}_j$, that is, λ in the diagonal position and zero everywhere else. So the matrix A in that basis has the form

$$\begin{pmatrix} \lambda I_k & C \\ 0 & D \end{pmatrix}$$

with C of size $k \times (n-k)$ and D of size $(n-k) \times (n-k)$. The determinant of $\lambda' I - A$ in that basis, either through cofactor expansion along the first k columns or through the determinant formula for a block triangular matrix, equals $(\lambda' - \lambda)^k \det(\lambda' I - D)$. So $(\lambda' - \lambda)^k$ divides p_A, hence the algebraic multiplicity of λ is at least k. □

Now look at both possibilities in a single matrix. Take

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \qquad p_A(\lambda) = (\lambda - 1)^2(\lambda - 4).$$

The eigenvalues are 1 and 4, with algebraic multiplicities 2 and 1. The trace is $6 = 4 + 1 + 1$ and the determinant is $4 = 4 \cdot 1 \cdot 1$, which matches. For $\lambda = 4$, the equation $(A - 4I)\mathbf{v} = \mathbf{0}$ gives $v_1 = v_2 = v_3$, so E_4 is spanned by $(1, 1, 1)$. For $\lambda = 1$, the matrix $A - I$ is the matrix all of whose entries are 1, so its kernel is the set of vectors whose entries sum to zero. That set is two-dimensional, with basis $(1, -1, 0)$ and $(1, 0, -1)$. So the geometric multiplicity of 1 is 2, equal to its algebraic multiplicity.

A repeated eigenvalue is not automatically a problem; what becomes a problem is an eigen space that is too small, and that happens with the following matrix.

9.3 When a matrix can be diagonalized

Level: intermediate

If we may choose the basis, we want the matrix as simple as possible: diagonal, so that every basis vector is merely scaled and not mixed with the others. Is there always such a basis?

Definition 9.5 (Diagonalizable)**.** An $n \times n$ matrix is called *diagonalizable* if there is an invertible matrix P and a diagonal matrix D with $P^{-1}AP = D$, or equivalently $A = PDP^{-1}$.

diagonalization (*diagonalisasi*) — Changing the measuring instrument until the whole work of the matrix is a few multiplications of numbers.

The columns of P will contain eigenvectors, and D contains the eigenvalues on its diagonal. A question about diagonalization is therefore really a question about having enough eigenvectors.

Theorem 9.6. *An $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors.*

Proof. Suppose $P^{-1}AP = D$ with $D = \operatorname{diag}(d_1, \ldots, d_n)$. Multiply from the left by P: $AP = PD$. Column j of both sides gives $A\mathbf{p}_j = d_j\mathbf{p}_j$, with $\mathbf{p}_j$ the j-th column of P. Since P is invertible, each of its columns is nonzero, so $\mathbf{p}_j$ is an eigenvector. The columns of P are also linearly independent because P is invertible. So A has n independent eigenvectors.

Conversely, suppose $\mathbf{p}_1, \ldots, \mathbf{p}_n$ are independent eigenvectors with $A\mathbf{p}_j = \lambda_j\mathbf{p}_j$. Assemble P from those vectors as columns and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then $AP = PD$, and since P is invertible we obtain $P^{-1}AP = D$. □

The corollary most often used in hand computation: enough distinct eigenvalues is already sufficient.

Corollary 9.7. *If an $n \times n$ matrix has n distinct eigenvalues, it is diagonalizable.*

Proof. Take one eigenvector for each eigenvalue. By the earlier lemma, n eigenvectors with n distinct eigenvalues are independent of one another. The theorem above then gives a diagonalization. □

The converse does not hold: the 3×3 example above had only two distinct eigenvalues, yet is still diagonalizable because the eigen space for $\lambda = 1$ is two-dimensional. The exact condition is not the number of eigenvalues but the total dimension of the eigen spaces.

Theorem 9.8 (Multiplicity condition)**.** *An $n \times n$ matrix is diagonalizable if and only if for each of its eigenvalues the algebraic multiplicity equals the geometric multiplicity. That condition is equivalent to $\sum_\lambda \dim E_\lambda = n$.*

Proof. First, suppose the total dimension of all eigen spaces equals n. For each λ choose a basis of E_λ, then combine all those bases into a list $\mathbf{v}_1, \ldots, \mathbf{v}_n$. That list is linearly independent: if some linear combination equals zero, group the terms according to their eigenvalues to get $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_k \in E_{\lambda_k}$. By the lemma on distinct eigenvectors, every $\mathbf{u}_k = \mathbf{0}$; since each $\mathbf{u}_k$ is a combination of the basis of E_{λ_k}, all its coefficients are zero. That list also contains n vectors in a space of dimension n, so it is a basis of eigenvectors, and the previous theorem closes the argument.

Second, suppose A is diagonalizable, so there is a basis of eigenvectors $\mathbf{p}_1, \ldots, \mathbf{p}_n$. In that basis the matrix A is diagonal, so $p_A(\lambda') = \prod_j (\lambda' - \lambda_j)$ and the algebraic multiplicity of each eigenvalue equals the number of j with $\lambda_j = \lambda$. That number of j does not exceed $\dim E_\lambda$, because those vectors lie inside E_λ and are independent. So the algebraic multiplicity does not exceed the geometric one. The earlier lemma gives the opposite direction, so the two are equal. Finally, $\sum_\lambda \dim E_\lambda = n$ because the basis of eigenvectors fills the whole space. □

Now an example that fails. Take

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad p_S(\lambda) = (\lambda - 1)^2.$$

The only eigenvalue is 1, with algebraic multiplicity 2. But $S - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has rank one, so $\dim E_1 = 2 - 1 = 1$. Its geometric multiplicity is one short, and by the theorem above S is not diagonalizable.

Geometrically S is a shear: it preserves the direction $(1, 0)$ and slides $(0, 1)$ to $(1, 1)$. No basis makes this map a mere scaling, because a scaling never slides something that was previously parallel. That failure is a property of the map, not a choice of basis.

For younger readers (age 10 and up)

Have you ever tried to tidy something that turned out not to be tidyable? Take a stack of cards, then slide the top part slightly sideways. The picture on each card stays the same, but every point shifts by its distance from the base. There is no way to rotate that stack so that the shift becomes a mere lengthening or shortening of something. A shift remains a shift.

There are other objects that are more accommodating. A rubber sheet pulled straight can be described only by how many times each direction stretches. If we know that number for a few important directions, we know the whole pull, including for points we never measured.

The question: how do we tell the two apart from the picture alone, without trying every possibility? In this chapter we will learn why a shear always stores one direction that cannot be tidied up, why some pulls have enough special directions, and why that difference decides whether the tenth power from a computing engine can still be trusted.

A shear is easy to recognize because of the triangle. The same failure appears in matrices that do not look triangular. Take

$$B = \begin{pmatrix} -1 & 4 & 0 \\ -1 & 3 & 0 \\ 1 & -2 & 2 \end{pmatrix}, \qquad p_B(\lambda) = (\lambda - 1)^2(\lambda - 2).$$

For $\lambda = 1$, the matrix $B - I$ has rank two, so $\dim E_1 = 3 - 2 = 1$, while its algebraic multiplicity is 2. The only eigenvector for 1 is $(2, 1, 0)$. For $\lambda = 2$ its eigen space is spanned by $(0, 0, 1)$ and is one-dimensional, matching its algebraic multiplicity. The total dimension of all eigen spaces is only $2 < 3$, so B is not diagonalizable. A matrix of this kind is called *defective*: it is short of eigenvectors because its eigen spaces are narrow.

defective (*defective*) — A matrix whose eigenvalues repeat but whose eigen spaces are too narrow to form a basis.

One note before we go on. Whether a matrix is diagonalizable depends on the field of scalars: R is not diagonalizable over $\mathbb{R}$, but over $\mathbb{C}$ it has two distinct eigenvalues, $\pm i$, so it is diagonalizable. A statement without a field qualifier is therefore incomplete. For the rest of this chapter the field is $\mathbb{R}$ unless stated otherwise.

9.4 Similar matrices and invariants

Level: advanced

Diagonalization is a change of basis that produces the simplest possible form. But not every matrix can reach that form, and the next question is natural: what stays the same across all matrices connected by a change of basis?

Definition 9.6 (Similar matrices)**.** Two $n \times n$ matrices A and B are called *similar* if there is an invertible matrix P with $B = P^{-1}AP$.

similar (*serupa*) — Two reports about the same map from two different measuring spaces.

Similarity is an equivalence relation: every matrix is similar to itself, the relation is symmetric because $(P^{-1})^{-1}BP^{-1} = A$, and transitive because $(PQ)^{-1}A(PQ) = Q^{-1}(P^{-1}AP)Q$. The classes divide up neatly, and diagonalization means asking whether a class contains a diagonal matrix.

Proposition 9.9. *If A and B are similar, then $p_A = p_B$. Consequently A and B have the same eigenvalues together with their algebraic multiplicities, and also the same trace, determinant, rank, and nullity.*

Proof. Suppose $B = P^{-1}AP$. Then

$$\lambda I - B = \lambda I - P^{-1}AP = P^{-1}(\lambda I - A)P,$$

so $\det(\lambda I - B) = \det(P^{-1})\det(\lambda I - A)\det(P) = \det(\lambda I - A)$ because $\det(P^{-1})\det(P) = 1$. So the characteristic polynomials are equal, and all data derived from their coefficients are equal as well. Equality of rank and nullity follows from $B - \lambda I = P^{-1}(A - \lambda I)P$: multiplication by an invertible matrix does not change the rank, and it is that rank which determines the nullity through rank-nullity. □

The converse does not hold, and the counterexample is simple: I_2 and the shear S have the same characteristic polynomial, $(\lambda - 1)^2$, but are not similar. If $P^{-1}I_2P = S$, the left side equals I_2 for every P. So the characteristic polynomial is not a complete invariant; it only filters.

What changes and what does not can be separated neatly. The eigenvalues do not change. The eigenvectors do change, and they change in a way that is easy to remember.

Proposition 9.10. *If $B = P^{-1}AP$ and $A\mathbf{v} = \lambda\mathbf{v}$, then $B(P^{-1}\mathbf{v}) = \lambda(P^{-1}\mathbf{v})$.*

Proof. Compute $B(P^{-1}\mathbf{v}) = P^{-1}APP^{-1}\mathbf{v} = P^{-1}A\mathbf{v} = P^{-1}(\lambda\mathbf{v}) = \lambda(P^{-1}\mathbf{v})$. □

So eigenvectors move along with their coordinates, while the multipliers stay put. That makes sense: a preserved direction is a property of the map, while its list of coordinates depends on the basis. That is why eigenvalues are used as the fingerprint of a matrix, and eigenvectors are not.

On the other hand, the freedom to choose P explains why two diagonalizations can look different although both are correct: the columns of P may be multiplied by a nonzero scalar or permuted as long as D is permuted accordingly. What must be the same is the set of eigenvalues together with their multiplicities.

9.5 A worked example

Level: intermediate

Every number in this section is computed by hand first, then checked with Python in the lab section. We start from the 3×3 matrix that already appeared in the discussion of eigen spaces,

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Its characteristic polynomial is computed from $\lambda I - A$, whose diagonal entries are all $\lambda - 2$ and whose other entries are all -1. Writing $t = \lambda - 2$ and expanding along the first row,

$$p_A(\lambda) = t\,(t^2 - 1) - (-1)\,(-t - 1) + (-1)\,(1 + t) = t^3 - 3t - 2.$$

That form factors because $t = -1$ is a root: $t^3 - 3t - 2 = (t + 1)^2(t - 2)$. Back to λ,

$$p_A(\lambda) = (\lambda - 1)^2(\lambda - 4) = \lambda^3 - 6\lambda^2 + 9\lambda - 4.$$

The trace is 6 and the determinant is 4, matching $4 + 1 + 1$ and $4 \cdot 1 \cdot 1$.

Eigenvectors for $\lambda = 4$: the matrix $A - 4I$ has diagonal entries -2 and other entries 1. Its rows say $-2v_1 + v_2 + v_3 = 0$ and so on; all three equations hold exactly when $v_1 = v_2 = v_3$. So E_4 is spanned by $(1, 1, 1)$.

Eigenvectors for $\lambda = 1$: the matrix $A - I$ contains only 1s, so its kernel is the set of vectors with $v_1 + v_2 + v_3 = 0$. A basis is two vectors, for instance $(1, -1, 0)$ and $(1, 0, -1)$. Hence

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}, \qquad P^{-1} = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}, \qquad D = \operatorname{diag}(4, 1, 1).$$

The determinant of P is 3, and that is the denominator in P^{-1}. We check one entry by hand: the entry in the first row and first column of PDP^{-1} is $\frac{1}{3}(4 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 1) = 2$, matching the entry of A. A full check with SymPy gives $PDP^{-1} - A = 0$ exactly.

Now the tenth power. Multiplying nine times gives no insight at all; diagonalization provides it. Since $A = I + J$ with J the matrix all of whose entries are 1, and $J^2 = 3J$, every power of J returns as a multiple of J:

$$J^2 = 3J, \quad J^3 = 9J, \quad \ldots, \quad J^k = 3^{k-1}J.$$

Then

$$A^k = (I + J)^k = I + \Big(\sum_{j=1}^{k} \binom{k}{j} 3^{j-1}\Big) J = I + \frac{4^k - 1}{3}\, J,$$

with the last step using the geometric series formula $\sum_{j=1}^{k}\binom{k}{j}3^{j-1} = \big((1+3)^k-1\big)/3$. For $k=10$ we have $4^{10}=1048576$ and $(1048576-1)/3 = 349525$, so that

$$A^{10} = I + 349525\,J = \begin{pmatrix} 349526 & 349525 & 349525 \\ 349525 & 349526 & 349525 \\ 349525 & 349525 & 349526 \end{pmatrix}.$$

Every entry can be checked: $1+349525$ on the diagonal, 349525 off it. Direct multiplication with Python gives the same matrix, with largest difference 1.05×10^{-9} because of floating-point arithmetic.

The form of that formula explains what is not visible in the final result. The term 4^k dominates the direction $(1,1,1)$; the constant term 1 survives in the direction whose entries sum to zero. The eigenvalues 4 and 1 are two growth rates, and a matrix power is merely a mixture of the two. If the matrix is not diagonalizable, that mixture changes shape: a term $k\lambda^k$ appears or, for $\lambda=1$, a pure factor k.

The defective example above shows that. For the B that failed to be diagonalizable,

$$B^k = \begin{pmatrix} 1-2k & 4k & 0 \\ -k & 1+2k & 0 \\ 2^k-1 & -2(2^k-1) & 2^k \end{pmatrix},$$

which can be proved by induction and holds for $k=1$. At $k=10$ that formula gives

$$B^{10} = \begin{pmatrix} -19 & 40 & 0 \\ -10 & 21 & 0 \\ 1023 & -2046 & 1024 \end{pmatrix},$$

the same as repeated multiplication. The top-left entry, $-19 = 1-2\cdot10$, grows linearly, not exponentially: the eigenvalue 1 controls that direction, and the shortage of eigenvectors adds one factor of k.

Now the first application: a process that settles. For the Markov chain matrix M from the opening,

$$p_M(\lambda) = \lambda^3 - \tfrac{19}{10}\lambda^2 + \tfrac{11}{10}\lambda - \tfrac{1}{5} = (\lambda-1)(\lambda-\tfrac12)(\lambda-\tfrac25).$$

The trace is $19/10$ and the determinant is $1/5$, matching $1+\frac12+\frac25$ and $1\cdot\frac12\cdot\frac25$. The eigenvector for $\lambda=1$ is found from $M\mathbf{v}=\mathbf{v}$, that is $(M-I)\mathbf{v}=\mathbf{0}$, together with the condition that the entries sum to one. The solution gives $\boldsymbol{\pi} = (17/30,\ 4/15,\ 1/6)$. One coordinate checked by hand:

$$\tfrac{8}{10}\cdot\tfrac{17}{30} + \tfrac{3}{10}\cdot\tfrac{8}{30} + \tfrac{2}{10}\cdot\tfrac{5}{30} = \tfrac{13.6+2.4+1.0}{30} = \tfrac{17}{30}.$$

So $\boldsymbol{\pi}$ really is a fixed point.

Since M has three distinct eigenvalues, it is diagonalizable. Every starting point therefore decomposes as

$$\mathbf{x}_k = \boldsymbol{\pi} + c_2\big(\tfrac12\big)^k\mathbf{v}_2 + c_3\big(\tfrac25\big)^k\mathbf{v}_3,$$

with c_2, c_3 depending on $\mathbf{x}_0$. The second and third terms shrink, and the rate is governed by $1/2$. For $\mathbf{x}_0=(1,0,0)$ we obtain $c_2\mathbf{v}_2$ of order 10^{-1}, so the error after ten steps is of order $10^{-1}\cdot(1/2)^{10}\approx10^{-4}$, and after forty steps of order $(1/2)^{40}\approx9.1\times10^{-13}$. The computed result agrees: $\mathbf{x}_{10}$ differs from $\boldsymbol{\pi}$ in the fourth digit, while $\mathbf{x}_{40}$ already agrees to six digits.

Second application: differential systems. If $\mathbf{x}(t)$ satisfies $\mathbf{x}' = A\mathbf{x}$ with A above, the solution is $\mathbf{x}(t) = e^{At}\mathbf{x}(0)$. Since $A = I + J$ and $J^2 = 3J$,

$$e^{At} = e^t e^{Jt} = e^t\Big(I + \frac{e^{3t}-1}{3}J\Big) = e^tI + \frac{e^{4t}-e^t}{3}J.$$

At $t = 0$ that formula gives I. For $t = 0.5$ its off-diagonal entries are $(e^2 - e^{0.5})/3 = 1.9134$, and Python gives the same number to four digits. The component along $(1, 1, 1)$ grows like e^{4t}; the component in the plane where the entries sum to zero grows like e^t, so the solution quickly becomes parallel to $(1, 1, 1)$. A positive real eigenvalue means growth; a purely imaginary eigenvalue such as $\pm i$ means oscillation without growth.

Third application: power iteration. Multiplying repeatedly and then normalizing the result, $\mathbf{x}_{k+1} = A\mathbf{x}_k / \|A\mathbf{x}_k\|$, makes the direction turn toward the dominant eigenvector, because λ_1^k defeats λ_2^k when $|\lambda_1| > |\lambda_2|$. For A above starting from $(1, 0)$, the angle runs $0°$, $14.04°$, $21.25°$, $24.39°$, $25.69°$, $26.21°$, and heads toward $\arctan(1/2) = 26.57°$; the Rayleigh quotient $\mathbf{x}^\top A\mathbf{x}/\mathbf{x}^\top\mathbf{x}$ runs 4.00, 4.65, 4.88, 4.96, 4.98, 4.99, heading toward 5. Figure 9.3 shows that turning.

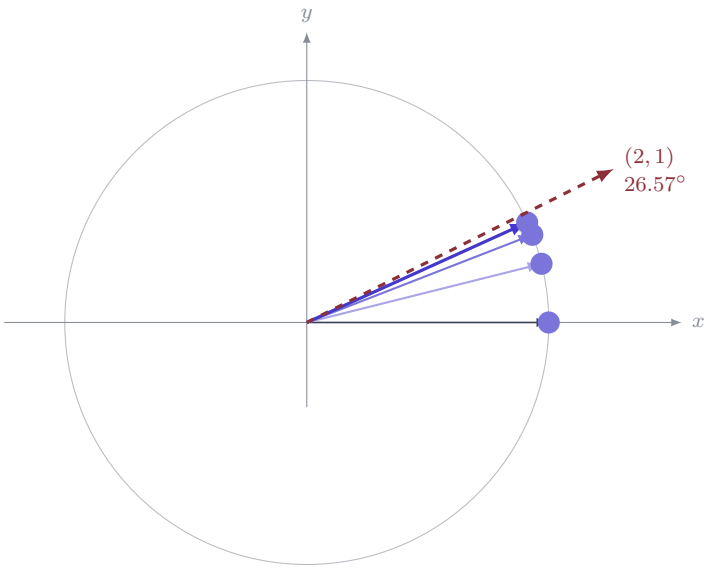


Figure 9.3: Power iteration on $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$ starting from $(1, 0)$. The solid arrows are the direction after $k = 0, 1, 2, 3$ steps, with angles $0°$, $14.04°$, $21.25°$, and $24.39°$. The dashed arrow is the dominant eigen direction $(2, 1)$ at $26.57°$, which is approached but never passed; the last two arrows coincide because the angles are getting tighter, and that is precisely the point of the picture.

For J the matrix all of whose entries are 1, take $B = J - I$. Determine the eigenvalues of B first, then predict B^{10} using the pattern $I + \frac{\mu^{10} - (-1)^{10}}{3} J$ with μ the nonzero eigenvalue of B. Check that prediction with Python, then explain why B is diagonalizable although it has a repeated eigenvalue. As an additional exercise, compute e^{Bt} and determine the limit $t \to \infty$ of $e^{-2t} e^{Bt}$.

9.6 Computational lab

LEVEL: ADVANCED

Computational lab

```
import numpy as np
from scipy.linalg import expm

np.set_printoptions(precision=4, suppress=True)

A = np.array([[2., 1., 1.],
              [1., 2., 1.],
```

```
             [1., 1., 2.]])
lam, V = np.linalg.eig(A)
print("eigenvalues A :", np.round(np.sort(lam.real), 10))
for i in range(3):
    sisa = A @ V[:, i] - lam[i] * V[:, i]
    print("  residual Av-lv :", round(float(np.linalg.norm(sisa)), 16))
print("A^10 via diagonalization :")
print(np.round((V @ np.diag(lam ** 10) @ np.linalg.inv(V)).real).astype(int))
print("A^10 via ordinary power :")
print(np.linalg.matrix_power(A, 10).astype(int))
print("max difference :", np.abs((V @ np.diag(lam ** 10)
                                  @ np.linalg.inv(V)).real
                                 - np.linalg.matrix_power(A, 10)).max())
print("cond V       :", round(float(np.linalg.cond(V)), 4))

A2 = np.array([[4., 2.],
               [1., 3.]])
x = np.array([1., 0.])
print("power iteration on A2:")
for k in range(6):
    sudut = np.degrees(np.arctan2(x[1], x[0]))
    rq = float(x @ A2 @ x / (x @ x))
    print("  k=%d angle=%7.2f rayleigh=%.4f" % (k, sudut, rq))
    x = A2 @ x / np.linalg.norm(A2 @ x)
print("limit angle  :", round(float(np.degrees(np.arctan2(1., 2.))), 2))

B = np.array([[-1., 4., 0.],
              [-1., 3., 0.],
              [1., -2., 2.]])
print("eigen B      :", np.round(np.sort(np.linalg.eigvals(B).real), 10))
print("rank(B - I)  :", np.linalg.matrix_rank(B - np.eye(3)))
print("B^10         :")
print(np.linalg.matrix_power(B, 10).astype(int))

M = np.array([[0.8, 0.3, 0.2],
              [0.1, 0.6, 0.3],
              [0.1, 0.1, 0.5]])
print("column sums  :", M.sum(axis=0))
print("eigen M      :", np.round(np.sort(np.linalg.eigvals(M).real), 10))
x = np.array([1., 0., 0.])
for _ in range(10):
    x = M @ x
print("x after 10   :", np.round(x, 6))
for _ in range(30):
    x = M @ x
print("x after 40   :", np.round(x, 6))

C = np.array([[1., 1000.], [0., 2.]])
Cp = C.copy()
Cp[1, 0] = 1e-6
print("eigen C      :", np.round(np.sort(np.linalg.eigvals(C).real), 6))
print("eigen C+1e-6 :", np.round(np.sort(np.linalg.eigvals(Cp).real), 6))
print("cond V(C)    :", round(float(np.linalg.cond(np.linalg.eig(C)[1])), 1))
print("expm(0.5A)   :", round(float(expm(0.5 * A)[0, 1]), 4))
print("formula      :", round(float((np.exp(2) - np.exp(0.5)) / 3), 4))
```

Actual output:

```
eigenvalues A : [1. 1. 4.]
  residual Av-lv : 8e-16
  residual Av-lv : 1.1e-15
  residual Av-lv : 3e-16
A^10 via diagonalization :
```

```
[[349526 349525 349525]
 [349525 349526 349525]
 [349525 349525 349526]]
A^10 via ordinary power :
[[349526 349525 349525]
 [349525 349526 349525]
 [349525 349525 349526]]
max difference : 1.0477378964424133e-09
cond V       : 1.5227
power iteration on A2:
  k=0 angle=   0.00 rayleigh=4.0000
  k=1 angle=  14.04 rayleigh=4.6471
  k=2 angle=  21.25 rayleigh=4.8820
  k=3 angle=  24.39 rayleigh=4.9578
  k=4 angle=  25.69 rayleigh=4.9840
  k=5 angle=  26.21 rayleigh=4.9938
limit angle  : 26.57
eigen B      : [1. 1. 2.]
rank(B - I)  : 2
B^10         :
[[  -19    40     0]
 [  -10    21     0]
 [ 1023 -2046  1024]]
column sums  : [1. 1. 1.]
eigen M      : [0.4 0.5 1. ]
x after 10   : [0.5672 0.2661 0.1666]
x after 40   : [0.5667 0.2667 0.1667]
eigen C      : [1. 2.]
eigen C+1e-6 : [0.999 2.001]
cond V(C)    : 2000.0
expm(0.5A)   : 1.9134
formula      : 1.9134
```

The first line of that output lists the eigenvalues in increasing order, $1, 1, 4$. The next three lines are the residuals $A\mathbf{v} - \lambda\mathbf{v}$ for the three eigenvectors that the library returned. A residual of order 10^{-16} means that the equation is satisfied to the limit of machine precision; no residual is exactly zero because eigenvectors are normalized and computed iteratively. The library eigenvectors are also not $(1, 1, 1)$ and $(1, -1, 0)$ but normalized multiples: the direction is the same, the length differs, and diagonalization cares only about direction.

The two A^{10} blocks give identical integer entries, with largest difference 1.05×10^{-9}. That difference is not an algebraic error: entries of size 3.5×10^5 are computed through V, V^{-1}, and powers of real numbers, so rounding errors of order 10^{-16} grow along with them.

The value `cond V` of 1.5227 is close to one, and that is good news: A is symmetric, so its eigenvectors can be chosen mutually perpendicular and P is nearly orthogonal. Diagonalization for such matrices is far more stable.

The power iteration block repeats the computation in Figure 9.3: the angle rises from 0 to 26.21 degrees, staying below the limit 26.57, while the Rayleigh quotient rises from 4.0000 to 4.9938 toward 5.

The B block shows the other side: its eigenvalues are $1, 1, 2$ and `rank(B - I)` equals 2, so the nullity is 1. The geometric multiplicity for $\lambda = 1$ is only one, while the algebraic one is two; the matrix is defective. The B^{10} block shows the consequence: the entries -19 and 1024 sit side by side, the first growing linearly and the second exponentially.

The M block verifies that every column sums to one, so that $\mathbf{x}_k$ stays stochastic. Its eigenvalues are 0.4, 0.5, and 1; after forty steps the numbers stop moving at six digits, matching the rate $(1/2)^k$ derived by hand.

The last block gives one warning: the matrix C has eigenvalues 1 and 2, and if one zero entry below the diagonal is replaced by 10^{-6}, its eigenvalues shift to 0.999 and 2.001; a shift of

10^{-3}, a thousand times larger than the perturbation. That number can be computed exactly: the roots of $\lambda^2 - 3\lambda + 1.999$ are $1.5 \mp \sqrt{2510}/100$, that is 0.999000998 and 2.000999002. The shift is proportional to $10^{-6} \cdot 1000/(2-1)$: the perturbation times the large entry divided by the gap between the two eigenvalues. The condition number 2000 captures the same amplification. Eigenvalues are still continuous in the entries, but they need not be Lipschitz continuous with constant one.

Challenge (problem solver)

Find a 2×2 matrix whose two eigenvalues are real and distinct, but whose two eigenvectors are not perpendicular. Show that your choice is correct by computing the dot product, then explain why a symmetric matrix cannot be used (the answer waits in Chapter 10).
Hint 1: two vectors are perpendicular exactly when their dot product is zero. Start from a simple upper triangular matrix, then compute both eigenvectors without rounding.
Hint 2: the matrix $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ is enough. The eigenvector for 3 is $(1, 1)$ and the one for 2 is $(1, 0)$; their dot product is 1, not zero. Compare with the symmetric matrix A in the lab, which gives `cond V` ≈ 1.5.
If you get stuck, go back to the section Similar matrices and invariants, then to the paragraph on the condition number in Computational lab.

Note for instructors

A teaching order that has worked: eigenvalues first, as preserved directions, then the characteristic polynomial as a computational tool. If the polynomial comes first, students see roots of a polynomial and forget that what is being sought is a direction. After eigen spaces, introduce geometric multiplicity through nullity, then the condition for diagonalization. The shear example should come before the condition theorem, so that the theorem answers a question that is already felt.
Two typical misconceptions. First, students assume that a repeated eigenvalue always means not diagonalizable. The matrix $(2, 1, 1; 1, 2, 1; 1, 1, 2)$ in the example section refutes that with geometric multiplicity two. Second, they write eigenvectors as rows of P, then obtain a $P^{-1}AP$ that is not diagonal and blame the formula. The rule to repeat: eigenvectors are columns, and the column order of P must be the same as the diagonal order of D.
Class activity: split the class into two groups for the matrix $(2, 1, 1; 1, 2, 1; 1, 1, 2)$. The first group computes A^{10} through diagonalization, the second through repeated powers or fast repeated multiplication. Compare the results on the board, then ask the class to explain why the two roads must meet. The noise usually arises over the denominator 3 in P^{-1}.
Sign of readiness for what follows: students can say, without computing, why the shear matrix is not diagonalizable, and can guess the form of S^k for a 2×2 shear. If they mention the appearance of a factor k, they are ready for Chapter 10.

9.7 Summary

LEVEL: BASIC

Summary.

- λ is an eigenvalue of A if there is $\mathbf{v} \neq \mathbf{0}$ with $A\mathbf{v} = \lambda\mathbf{v}$. Eigenvalues are the roots of $p_A(\lambda) = \det(\lambda I - A)$, and $E_\lambda = \operatorname{null}(A - \lambda I)$ is the corresponding subspace.
- The degree of p_A is n, so there are at most n distinct eigenvalues. Over $\mathbb{C}$ there are

always n when counted with multiplicity; over $\mathbb{R}$ there need not be, as with the rotation matrix.

- $\operatorname{tr}(A) = \sum_i \lambda_i$ and $\det(A) = \prod_i \lambda_i$, both counted with multiplicity. This is a cheap check for every hand computation.
- For every eigenvalue, $1 \le \dim E_\lambda \le$ the algebraic multiplicity. Eigenvectors with distinct eigenvalues are always linearly independent.
- A is diagonalizable exactly when there are n independent eigenvectors, and exactly when the algebraic multiplicity equals the geometric one for every eigenvalue. Otherwise A is called defective, and A^k in general contains a factor k, not only λ^k.
- Similar matrices have the same characteristic polynomial, eigenvalues, trace, and determinant; the converse does not hold. Eigenvectors change along with the basis.

9.8 Exercises

Level: intermediate

Exercise 9.1. 1. Determine the eigenvalues and eigenvectors of $A_1 = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$, then assemble P and D and check $P^{-1}A_1P = D$ by multiplying back.

2. Compute G^{10} for $G = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$ through diagonalization, then check the result with the NumPy function `matrix_power`. Write down the general form of G^k as well.
3. Prove that if λ is an eigenvalue of A then λ^2 is an eigenvalue of A^2, and in general λ^k is an eigenvalue of A^k for every integer $k \ge 1$. Explain why the converse fails: give an A with $A^2 = -I$, then compare the eigenvalues of A^2 with the real eigenvalues of A. (Hint: use the rotation matrix from the start of the chapter; it has no real eigenvalue.)
4. Prove that A is invertible if and only if 0 is not an eigenvalue of A, and that in that case the eigenvalues of A^{-1} are $1/\lambda$ for every eigenvalue λ of A. (Hint: start from $A\mathbf{v} = \lambda\mathbf{v}$, multiply on the left by A^{-1}, and make sure that $\lambda \neq 0$ so that the division is legitimate.)
5. Assess this claim: *two square matrices of the same size that have the same characteristic polynomial are necessarily similar.* Decide whether it is true, false, or not supported by enough evidence, and include a reason or a counterexample.
6. Prove that A and $A^\top$ have the same characteristic polynomial, using $\det(\lambda I - A) = \det((\lambda I - A)^\top)$ and the properties of the determinant from Chapter 8. Explain why this means that the transpose does not change the eigenvalues but usually changes the eigenvectors.
7. The matrix $K = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ has eigenvalues $2, 2, 1$. Determine the rank of $K - 2I$, then decide whether K is diagonalizable. Justify your answer with the multiplicity condition.
8. A two-state Markov chain has matrix $T = \begin{pmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{pmatrix}$. Determine the eigenvalues of T, its fixed point, and the number of steps needed for the initial error to shrink below 10^{-6}. Check your answer with Python. (Hint: once the eigenvalues are known, the error shrinks like $|\lambda_2|^k$; solve $|\lambda_2|^k < 10^{-6}$ with logarithms.)
9. Prove: if A is diagonalizable with $A = PDP^{-1}$, then $\operatorname{tr}(A^k) = \sum_i \lambda_i^k$ for every $k \ge 1$, using $\operatorname{tr}(XY) = \operatorname{tr}(YX)$. For the defective B above, compute $\operatorname{tr}(B^{10})$ from the printed

matrix, then compare it with $1^{10} + 1^{10} + 2^{10}$. Explain why that equality still holds although B is defective, and at which point the proof above used diagonalization. (Hint: the trace does not change under a change of basis; use the form of B^k already derived.)

For researchers: limits and next steps

What is really proved in this chapter are algebraic statements over any field and in finite dimension: the equivalence of eigenvalues with roots of the characteristic polynomial, the independence of eigenvectors for distinct eigenvalues, the inequality $1 \leq \dim E_\lambda \leq$ the algebraic multiplicity, and the equivalence of diagonalization with the two conditions used here. The standard references are Axler, *Linear Algebra Done Right* 4th edition, sections 5A and 5D (5.55, the four equivalent conditions, and 5.54); a matrix version is in Anton, *Elementary Linear Algebra* 11th edition, sections 5.1 and 5.2.

Two things are more often popularized than proved. First, the claim that eigenvalues "measure importance", the basis of the PageRank explanation: the principal eigenvector of a stochastic matrix gives steady weights only if the chain is irreducible and aperiodic, and that is the content of the Perron-Frobenius theorem, which is not proved here; a full proof is in Strang, *Introduction to Linear Algebra* 6th edition, Appendix 8. Second, the claim that the characteristic polynomial is a practical way to compute eigenvalues: for $n \geq 5$ there is no general root formula, and libraries use the QR algorithm on matrices, not polynomial factoring. This chapter also stops at the diagonalizable case; Jordan form and the minimal polynomial are not discussed, and are in Hoffman and Kunze, *Linear Algebra* 2nd edition, sections 6.3 and 7.3.

Open questions: how far eigenvalues shift when the entries are perturbed, whose quantitative answer requires the Bauer-Fike theorem or pseudospectra; and whether defectiveness itself is numerically stable. Limits of validity: finite dimension, real or complex field, with no claim about the stability of algorithms.

Map to source texts: This chapter follows Anton, Elementary Linear Algebra *11th edition, Chapter 5: section 5.1 (Eigenvalues and Eigenvectors), 5.2 (Diagonalization), 5.3 (Complex Vector Spaces), 5.4 (Application: Differential Equations), and 5.5 (Application: Dynamical Systems and Markov Chains). A more abstract counterpart is in Axler,* Linear Algebra Done Right *4th edition, Chapter 5: 5A (eigenvalues and eigen spaces), 5B (existence of eigenvalues over complex vector spaces), and 5D (diagonalizable operators, including 5.55). For Markov chains and the Perron-Frobenius theorem, see Strang,* Introduction to Linear Algebra *6th edition, sections 6.1, 6.2, 6.5, and Appendix 8. Canonical forms for defective matrices are in Hoffman and Kunze,* Linear Algebra *2nd edition, Chapter 6 (Elementary Canonical Forms), section 6.3 for the minimal polynomial, and Chapter 7 (The Rational and Jordan Forms), section 7.3 for the Jordan form.*

Chapter 10

Orthogonality, SVD, and Least Squares

Five readings from one temperature probe: 100, 101, 102, 103, and 104 degrees. The recorded readings are 302.0, 305.1, 307.9, 311.2, and 313.8 in that order. The calibration model is a single straight line $y = a + bT$, so there are five equations for two unknowns:

$$a + 100b = 302.0, \qquad a + 101b = 305.1, \qquad \ldots, \qquad a + 104b = 313.8.$$

No pair (a, b) satisfies all five; the five points are never collinear, and the residual of each equation has a different sign. There is still one pair that makes the most sense. In what sense is that pair best, and how far does it move if one reading changes by a tenth?

The same question arrives from the opposite side. A detector computes a score $s = \mathbf{w}^{\top}\mathbf{x}$ from three counters, and an attacker wants to shift s by exactly Δ with as small a change in $\mathbf{x}$ as possible, because a large change is easy to see. There are three unknowns and one condition, so the solution set is too large rather than empty: among all the changes that give the same shift, which one is smallest?

Those two problems are one problem. In the first case the set of candidate answers is empty and we look for the nearest one; in the second it is too large and we look for the shortest. Both are settled by orthogonal projection. Chapter 9 supplied eigenvalues, but only for square matrices with enough eigenvectors. The calibration matrix above is 5×2, and the detector matrix is 2×3. Forcing both to become square through $\mathbf{A}^{\top}\mathbf{A}$ does work, but that road squares the sensitivity to error. The sixth section shows the price in numbers.

For younger readers (age 10 and up)

Have you ever stood under a lamp and seen your shadow on the floor? That shadow is not you: it is shorter, flattened, and it cannot show your face. But it is the best the floor can do: of all the points on the floor, your shadow is the one closest to the top of your head. If the lamp is moved, the shadow moves with it, even though you have not moved. Now picture two rulers that differ by one hundredth of a centimeter. If you measure a table with both and then subtract the results, the leftover number comes almost entirely from reading mistakes. A small difference drowns easily.

In this chapter we will learn why that "closest" shadow can be computed cleanly, why among the many roads to the same destination there is one that is shortest, and why two nearly twin measuring rules make our answer wobble.

10.1 Inner product, norm, and the Cauchy-Schwarz inequality

Level: basic

Length and angle are two words we have used since school without ever defining them. For vectors in $\mathbb{R}^n$ both come from a single operation that eats two vectors and returns one number; that operation is what gives a vector space a size, because the axioms of a vector space in Chapter 6 say nothing about distance.

The rule: multiply component by component, then add. The result is zero exactly when the two vectors are at right angles, so this operation is used in place of angle even in a space of a hundred dimensions.

inner product (*hasilkali dalam*) — Two lists of numbers folded into a single number that measures how much they point the same way.

Definition 10.1 (Inner product and norm)**.** The *inner product* on a real vector space V is a function $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{R}$ that satisfies three conditions:

1. symmetric, $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$;
2. linear in the first argument, $\langle c\mathbf{u} + \mathbf{w}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle$;
3. positive definite, $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ with equality only when $\mathbf{v} = \mathbf{0}$.

On $\mathbb{R}^n$ the standard inner product is $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_i u_i v_i = \mathbf{u}^\top \mathbf{v}$. The *norm* it induces is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

The third condition is what makes $\|\mathbf{v}\| = 0$ only for the zero vector, so the norm deserves the name length. The norm also satisfies $\|c\mathbf{v}\| = |c|\,\|\mathbf{v}\|$ and $\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$; the second is not automatic, but follows from the next inequality.

In everyday language: the inner product of two vectors is never larger than the product of their lengths. Two vectors can be very long, but if their directions are at right angles the product stays small.

Theorem 10.1 (Cauchy-Schwarz)**.** *For all $\mathbf{u}, \mathbf{v} \in V$, $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\,\|\mathbf{v}\|$, with equality exactly when one vector is a scalar multiple of the other.*

Proof. If $\mathbf{v} = \mathbf{0}$, both sides are zero and there is nothing to prove. Suppose $\mathbf{v} \neq \mathbf{0}$ and $t \in \mathbb{R}$ is arbitrary. Positive definiteness gives

$$0 \leq \|\mathbf{u} - t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2.$$

The right-hand side is a quadratic polynomial in t that is never negative. The coefficient of t^2 is positive, so its discriminant cannot be positive:

$$4\langle \mathbf{u}, \mathbf{v} \rangle^2 - 4\|\mathbf{u}\|^2 \|\mathbf{v}\|^2 \leq 0.$$

After dividing by four and taking square roots, the required inequality follows. Equality holds exactly when that polynomial has a double root, that is, when $\mathbf{u} - t\mathbf{v} = \mathbf{0}$ for some t. □

That proof used no coordinates, so the inequality holds in every inner-product space, including function spaces.

Corollary 10.2 (Triangle inequality)**.** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Proof. Expand $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$. Cauchy-Schwarz bounds the middle term by $2\|\mathbf{u}\|\|\mathbf{v}\|$, so

$$\|\mathbf{u} + \mathbf{v}\|^2 \leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 = (\|\mathbf{u}\| + \|\mathbf{v}\|)^2.$$

Both sides are nonnegative, so the square root may be taken. □

Now we have length. Angle follows by definition: two vectors are perpendicular when their inner product is zero. That makes sense because Cauchy-Schwarz guarantees that $\langle \mathbf{u}, \mathbf{v} \rangle / (\|\mathbf{u}\| \|\mathbf{v}\|)$ lies between -1 and 1, so it may be read as a cosine.

orthogonal (*ortogonal*) — Two directions that contribute nothing to each other; the shadow of one direction on the other has length zero.

Definition 10.2 (Orthogonal and orthonormal)**.** $\mathbf{u}$ and $\mathbf{v}$ are *orthogonal* when $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, written $\mathbf{u} \perp \mathbf{v}$. A set of vectors is called *orthogonal* when every two distinct members are orthogonal, and *orthonormal* when all its members also have norm one.

One small consequence follows directly from expanding the norm: if $\mathbf{u} \perp \mathbf{v}$ then $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$. The Pythagorean theorem is a special case of this definition, not the other way round.

10.2 Orthogonal sets, projection, and Gram-Schmidt

Level: intermediate

For an arbitrary basis we must solve a linear system; for an orthonormal basis we need not, because each coefficient is read off with one inner product. One fact makes that legitimate: an orthogonal set with no zero vector is always linearly independent.

Lemma 10.3. *If $\mathbf{v}_1, \dots, \mathbf{v}_k$ are orthogonal and none of them is the zero vector, then that set is linearly independent.*

Proof. Suppose $\sum_i c_i \mathbf{v}_i = \mathbf{0}$. Take the inner product with $\mathbf{v}_j$. All terms with $i \neq j$ vanish because of orthogonality, so only $c_j \langle \mathbf{v}_j, \mathbf{v}_j \rangle = 0$ is left. Since $\mathbf{v}_j \neq \mathbf{0}$, its norm is nonzero, hence $c_j = 0$. This holds for every j. □

Now the statement about coefficients, in the form most often used.

Proposition 10.4. *Suppose $(\mathbf{q}_1, \dots, \mathbf{q}_n)$ is an orthonormal basis of V. Then for every $\mathbf{v} \in V$,*

$$\mathbf{v} = \sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{q}_i \rangle \, \mathbf{q}_i, \qquad \|\mathbf{v}\|^2 = \sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{q}_i \rangle^2.$$

Proof. Write $\mathbf{v} = \sum_i a_i \mathbf{q}_i$; those coefficients exist and are unique because this is a basis. The inner product with $\mathbf{q}_j$ gives $\langle \mathbf{v}, \mathbf{q}_j \rangle = \sum_i a_i \langle \mathbf{q}_i, \mathbf{q}_j \rangle = a_j$, since $\langle \mathbf{q}_i, \mathbf{q}_j \rangle$ equals one when $i = j$ and zero otherwise. For the norm formula, use orthogonality on the expansion $\|\mathbf{v}\|^2 = \langle \sum_i a_i \mathbf{q}_i, \sum_j a_j \mathbf{q}_j \rangle$; the cross terms all vanish. □

The second equality is called Parseval's identity: in an orthonormal basis the total length is the square root of the sum of squared coordinates; in a skew basis that property fails.

The next question comes from the opening problem: $\mathbf{b}$ lies outside the subspace W, and we want the point of W closest to it. That point is called the projection, and the way to find it is to add up the shadow of $\mathbf{b}$ on each orthonormal direction of W.

projection (*proyeksi*) — The shadow of a stick on the floor: the point on the floor that is closest to the tip of the stick.

Definition 10.3 (Orthogonal projection)**.** Suppose $W \subseteq \mathbb{R}^m$ is a subspace with orthonormal basis $(\mathbf{q}_1, \dots, \mathbf{q}_k)$. The *orthogonal projection* of $\mathbf{b}$ onto W is

$$\operatorname{proj}_W \mathbf{b} = \sum_{i=1}^{k} \langle \mathbf{b}, \mathbf{q}_i \rangle \mathbf{q}_i, \qquad \mathbf{r} = \mathbf{b} - \operatorname{proj}_W \mathbf{b}.$$

The vector $\mathbf{r}$ is called the *residual*.

That residual is orthogonal to all of W, not merely to the basis vectors: because $\langle \mathbf{r}, \mathbf{q}_j \rangle = \langle \mathbf{b}, \mathbf{q}_j \rangle - \langle \mathbf{b}, \mathbf{q}_j \rangle = 0$ and every vector of W is a combination of the $\mathbf{q}_j$.

What makes the projection useful: it is not just one point in W, it is the closest point. That is a claim about every other point in W, so it needs a proof.

Read it aloud (auditory learner)

Read aloud, two voices taking turns.
Voice one: There is a point hanging above the floor, and the floor is asked to produce its stand-in.
Voice two: The floor cannot rise. It can only drop a shadow, and that shadow is the floor point closest to the point above.
Voice one: Then where does the part that never reaches the floor go?
Voice two: That part stands upright. It has no floor to stand on, so no path on the floor can shorten it.
Repeat this line until it sits on the tongue: *what falls to the floor is what is closest, what stands upright is what cannot be brought back down.*

Theorem 10.5 (Best approximation)**.** *For every* $\mathbf{w} \in W$, $\|\mathbf{b} - \operatorname{proj}_W \mathbf{b}\| \le \|\mathbf{b} - \mathbf{w}\|$, *with equality only when* $\mathbf{w} = \operatorname{proj}_W \mathbf{b}$.

Proof. Write $\mathbf{p} = \operatorname{proj}_W \mathbf{b}$. For arbitrary $\mathbf{w} \in W$,

$$\mathbf{b} - \mathbf{w} = (\mathbf{b} - \mathbf{p}) + (\mathbf{p} - \mathbf{w}).$$

The two terms are orthogonal: the first is orthogonal to all of W, and $\mathbf{p} - \mathbf{w} \in W$. Therefore the squared norm splits:

$$\|\mathbf{b} - \mathbf{w}\|^2 = \|\mathbf{b} - \mathbf{p}\|^2 + \|\mathbf{p} - \mathbf{w}\|^2.$$

The second term is nonnegative, so the right-hand side is smallest when $\mathbf{w} = \mathbf{p}$. □

The geometric reading is in Figure 10.1.

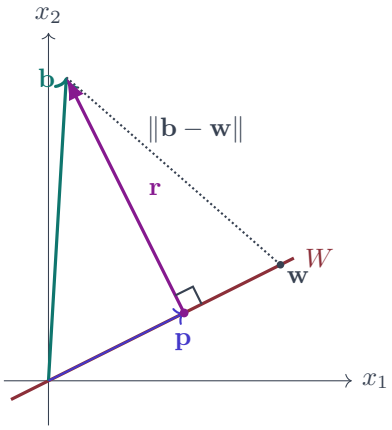


Figure 10.1: Orthogonal projection of $\mathbf{b} = (0.2; 3.4)$ onto the line W spanned by $\mathbf{q} = (2, 1)/\sqrt{5}$. Its perpendicular foot is $\mathbf{p} = (1.52; 0.76)$, marked as a right angle with W, and the residual $\mathbf{r} = \mathbf{b} - \mathbf{p}$ is perpendicular to all of W. The point $\mathbf{w} = (2.6; 1.3)$, which also lies in W, gives distance $\|\mathbf{b} - \mathbf{w}\| \approx 3.19$, longer than $\|\mathbf{b} - \mathbf{p}\| \approx 2.95$.

Do it by hand (kinesthetic learner)

Materials: one pencil, one desk lamp, one sheet of paper, and a ruler.
Place the paper on the table, then move the lamp to the far end of the room until its light is almost parallel. A lamp that is too close lengthens the shadow, and what is measured is then no longer an orthogonal projection.
Hold the pencil upright with its tip touching the paper, and mark that point. Tilt the pencil to an angle of sixty degrees with the paper, mark the end of the shadow, then repeat at forty-five and thirty degrees. Measure the three shadows with the ruler.
What to look for: the shadow is longest when the pencil almost lies on the paper, and shortest when the pencil is upright. The length of the shadow does not care how high the tip of the pencil is above the paper; it records only the horizontal part. That upright part is the residual the floor cannot reach.

If the columns of $\mathbf{Q}$ are $\mathbf{q}_1, \dots, \mathbf{q}_k$, the projection formula becomes a single matrix product: $\mathbf{p} = \mathbf{Q}\mathbf{Q}^{\top}\mathbf{b}$. The matrix $\mathbf{P} = \mathbf{Q}\mathbf{Q}^{\top}$ satisfies $\mathbf{P}^2 = \mathbf{P}$ and $\mathbf{P}^{\top} = \mathbf{P}$: projecting twice is the same as once. Its trace is k, the dimension of W; for the example in the seventh section the trace is 2.

Now we know how to project, but we do not yet have a way to obtain an orthonormal basis. The Gram-Schmidt algorithm builds one vector at a time: normalize the first vector, then strip from each following vector the part parallel to those already present. Each step uses only the projection we have just proved.

Theorem 10.6 (Gram-Schmidt and the QR factorization)**.** *Suppose $\mathbf{a}_1, \dots, \mathbf{a}_n$ are linearly independent in $\mathbb{R}^m$. Define in order*

$$\mathbf{v}_1 = \mathbf{a}_1, \qquad \mathbf{v}_j = \mathbf{a}_j - \sum_{i<j} \langle \mathbf{a}_j, \mathbf{q}_i \rangle \mathbf{q}_i, \qquad \mathbf{q}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}$$

for $j = 2, \dots, n$. Then $\mathbf{q}_1, \dots, \mathbf{q}_n$ are orthonormal and $\operatorname{span}(\mathbf{q}_1, \dots, \mathbf{q}_j) = \operatorname{span}(\mathbf{a}_1, \dots, \mathbf{a}_j)$ for every j. Consequently $\mathbf{A} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q}$ having orthonormal columns and $\mathbf{R}$ upper triangular with positive diagonal.

Proof. We prove this by induction on j. For $j = 1$, $\mathbf{q}_1$ is $\mathbf{a}_1$ normalized, so the claim holds. Assume the claim holds up to $j-1$. The vector $\mathbf{v}_j$ is $\mathbf{a}_j$ minus its combination with $\mathbf{q}_1, \dots, \mathbf{q}_{j-1}$, so $\mathbf{v}_j$ still lies in $\operatorname{span}(\mathbf{a}_1, \dots, \mathbf{a}_j)$. For every $i < j$,

$$\langle \mathbf{v}_j, \mathbf{q}_i \rangle = \langle \mathbf{a}_j, \mathbf{q}_i \rangle - \langle \mathbf{a}_j, \mathbf{q}_i \rangle = 0,$$

because $\mathbf{q}_i$ is orthonormal. So $\mathbf{v}_j$ is orthogonal to all the previous vectors. That vector is not zero: if it were, $\mathbf{a}_j$ would become a combination of $\mathbf{a}_1, \dots, \mathbf{a}_{j-1}$, contradicting linear independence. Therefore $\mathbf{q}_j$ is defined and the set stays orthonormal as it grows.

For the factorization, write each $\mathbf{a}_j$ as $\|\mathbf{v}_j\|\mathbf{q}_j + \sum_{i<j} \langle \mathbf{a}_j, \mathbf{q}_i \rangle \mathbf{q}_i$. Those coefficients fill column j of $\mathbf{R}$; the entries above the diagonal are $\langle \mathbf{a}_j, \mathbf{q}_i \rangle$ and the diagonal entries are $\|\mathbf{v}_j\| > 0$. $\square$

One warning: classical Gram-Schmidt computes each coefficient against the original vector $\mathbf{a}_j$, not against the vector already reduced. If the columns of $\mathbf{A}$ are nearly dependent, the result stops being orthonormal; subtracting from $\mathbf{v}_j$ first repairs it at almost no cost. The eighth section shows the numbers: on a matrix with $\kappa \approx 6.8 \times 10^6$, the classical version gives $\|\mathbf{Q}^{\top}\mathbf{Q} - \mathbf{I}\| = 2.05 \times 10^{-2}$, the modified version 3.40×10^{-10}, Householder 1.19×10^{-15}; the first number means two columns that should be at right angles differ by an angle of about one degree.

10.3 Least squares and the normal equations

Level: intermediate

Back to the five probe readings. The system $\mathbf{A}\mathbf{x} = \mathbf{b}$ has no solution, but we can demand that the residual $\mathbf{b} - \mathbf{A}\mathbf{x}$ be as small as possible: demanding a zero residual amounts to demanding a solution, while demanding minimum length always has an answer. The measure used is the sum of squared differences, not the sum of absolute values, because the sum of squares is born from the inner product, which puts the whole projection machinery at our disposal; the sum of absolute values does not, and its answer may fail to be unique.

least squares (*kuadrat terkecil*) — When no line passes through all the points, choose the line whose sum of squared errors is smallest.

Definition 10.4 (Least-squares solution)**.** For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ is called a *least-squares solution* if $\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\| \le \|\mathbf{A}\mathbf{x} - \mathbf{b}\|$ for every $\mathbf{x} \in \mathbb{R}^n$.

That definition does not mention derivatives. What is used is geometry: $\mathbf{A}\mathbf{x}$ always lies in $\operatorname{col}(\mathbf{A})$, so the smallest residual is the point of $\operatorname{col}(\mathbf{A})$ closest to $\mathbf{b}$; that is exactly the projection problem from before.

Theorem 10.7 (Normal equations)**.** *The vector $\hat{\mathbf{x}}$ minimizes $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|$ if and only if*

$$\mathbf{A}^{\top}\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^{\top}\mathbf{b}.$$

Proof. Take arbitrary $\mathbf{x}$ and arbitrary $\mathbf{h} \in \mathbb{R}^n$, then write $\mathbf{x} + \mathbf{h}$ as a neighbor of $\mathbf{x}$. Then

$$\|\mathbf{A}(\mathbf{x}+\mathbf{h}) - \mathbf{b}\|^2 = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + 2\mathbf{h}^{\top}\mathbf{A}^{\top}(\mathbf{A}\mathbf{x} - \mathbf{b}) + \|\mathbf{A}\mathbf{h}\|^2.$$

First, suppose $\mathbf{A}^{\top}(\mathbf{A}\mathbf{x} - \mathbf{b}) = \mathbf{0}$. The middle term vanishes for every $\mathbf{h}$, and the last term is nonnegative, so $\|\mathbf{A}(\mathbf{x}+\mathbf{h}) - \mathbf{b}\|^2 \ge \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$. Hence $\mathbf{x}$ minimizes.

Second, suppose $\mathbf{A}^{\top}(\mathbf{A}\mathbf{x} - \mathbf{b}) = \mathbf{z} \neq \mathbf{0}$. Choose $\mathbf{h} = -\epsilon\mathbf{z}$ with small $\epsilon > 0$. The middle term becomes $-2\epsilon\|\mathbf{z}\|^2$, while the last term grows by $\epsilon^2\|\mathbf{A}\mathbf{z}\|^2$. For ϵ small enough, the negative part beats the positive part, so $\mathbf{x}$ is not a minimizer. Hence the condition is necessary. □

The system $\mathbf{A}^{\top}\mathbf{A}\mathbf{x} = \mathbf{A}^{\top}\mathbf{b}$ is called the *normal equations*. The condition $\mathbf{A}^{\top}(\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) = \mathbf{0}$ says that the residual $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$ is orthogonal to every column of $\mathbf{A}$, hence to all of $\operatorname{col}(\mathbf{A})$. In other words $\mathbf{A}\hat{\mathbf{x}}$ is the projection of $\mathbf{b}$ onto the column space.

From here one practical question follows: when is the answer unique? The matrix $\mathbf{A}^{\top}\mathbf{A}$ is always square and symmetric, but not necessarily invertible.

Corollary 10.8. *If the columns of $\mathbf{A}$ are linearly independent, then $\mathbf{A}^{\top}\mathbf{A}$ is invertible and the least-squares solution is unique, namely $\hat{\mathbf{x}} = (\mathbf{A}^{\top}\mathbf{A})^{-1}\mathbf{A}^{\top}\mathbf{b}$.*

Proof. Suppose $\mathbf{A}^{\top}\mathbf{A}\mathbf{z} = \mathbf{0}$. Multiply from the left by $\mathbf{z}^{\top}$, which gives $\|\mathbf{A}\mathbf{z}\|^2 = \mathbf{z}^{\top}\mathbf{A}^{\top}\mathbf{A}\mathbf{z} = 0$, hence $\mathbf{A}\mathbf{z} = \mathbf{0}$. Linear independence of the columns forces $\mathbf{z} = \mathbf{0}$. So the kernel of $\mathbf{A}^{\top}\mathbf{A}$ contains only the zero vector, and a square matrix with trivial kernel is invertible. Uniqueness of $\hat{\mathbf{x}}$ then follows from the normal equations theorem. □

Step-by-step recipe (procedural learner)

1. Build $\mathbf{A}$ and $\mathbf{b}$: one row of $\mathbf{A}$ per equation, with the columns being the functions used.
2. Form $\mathbf{A}^{\top}\mathbf{A}$, which is $n \times n$, and $\mathbf{A}^{\top}\mathbf{b}$, which is $n \times 1$.
3. Solve $\mathbf{A}^{\top}\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^{\top}\mathbf{b}$; check $\det(\mathbf{A}^{\top}\mathbf{A}) \neq 0$ first, since zero means the answer is

not unique.

4. Compute $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$, check $\mathbf{A}^\top \mathbf{r} = \mathbf{0}$, then report $\|\mathbf{r}\|$.

Warning about order: do not invert anything before the third step. $\mathbf{A}$ is not square, so $\mathbf{A}^{-1}$ does not exist, and turning $\mathbf{A}^\top \mathbf{b}$ into $\mathbf{b}^\top \mathbf{A}$ gives a $1 \times n$ row vector, not a system of equations.

Conversely, if the columns of $\mathbf{A}$ are dependent, $\mathbf{A}^\top \mathbf{A}$ is singular and there are infinitely many solutions; the fifth section chooses one of them. The normal equations turn an $m \times n$ problem into an $n \times n$ one, a large saving for $m \gg n$, and the price shows up in the sixth section.

10.4 Singular value decomposition

Level: advanced

Chapter 9 turned square matrices into diagonal form by choosing one basis. The condition is heavy: the matrix must have enough eigenvectors, and for a real matrix its eigenvalues must be real. Our calibration matrix is 5×2, so it is not even square, and there is no such thing as an eigenvalue for it.

The way out: allow two different bases, one in the domain and one in the codomain, both orthonormal. With that relaxation every matrix is diagonal; that is the singular value decomposition. The numbers on the diagonal are the lengths of the axes of the ellipse produced from the unit ball, always real and nonnegative even when the matrix is not symmetric.

singular value (*nilai singular*) — How far a direction is stretched by the matrix; zero means that direction is flattened until it disappears.

Definition 10.5 (Singular values)**.** Suppose $\mathbf{A} \in \mathbb{R}^{m \times n}$. The *singular values* of $\mathbf{A}$ are the square roots of the eigenvalues of $\mathbf{A}^\top \mathbf{A}$, sorted in decreasing order: $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$ with $\sigma_i = \sqrt{\lambda_i(\mathbf{A}^\top \mathbf{A})}$.

That definition is legitimate if $\mathbf{A}^\top \mathbf{A}$ has n real eigenvalues with orthonormal eigenvectors; for a symmetric matrix that is true, and we state it as a lemma, with one step honestly labelled as not done.

Lemma 10.9 (Spectral theorem for symmetric matrices)**.** *If $\mathbf{S} \in \mathbb{R}^{n \times n}$ is symmetric, then there is an orthonormal basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$ of $\mathbb{R}^n$ and real numbers $\lambda_1, \ldots, \lambda_n$ with $\mathbf{S}\mathbf{v}_i = \lambda_i \mathbf{v}_i$.*

Sketch. The function $\mathbf{x} \mapsto \mathbf{x}^\top \mathbf{S} \mathbf{x}$ is continuous on the closed and bounded unit sphere, so it attains a maximum at some point $\mathbf{v}_1$ of norm one. By the Lagrange multiplier method, that point satisfies $\mathbf{S}\mathbf{v}_1 = \lambda_1 \mathbf{v}_1$; the value λ_1 is that maximum. Next, $\mathbf{v}_1^\perp$ is also invariant under $\mathbf{S}$, because $\langle \mathbf{S}\mathbf{w}, \mathbf{v}_1 \rangle = \langle \mathbf{w}, \mathbf{S}\mathbf{v}_1 \rangle = \lambda_1 \langle \mathbf{w}, \mathbf{v}_1 \rangle = 0$ for $\mathbf{w} \perp \mathbf{v}_1$. The restriction of $\mathbf{S}$ to $\mathbf{v}_1^\perp$ is symmetric and of dimension $n - 1$, so induction supplies the rest of the orthonormal basis.

The step not carried out here is the proof that a continuous function on a closed and bounded set attains a maximum. That is the extreme value theorem, proved in Chapter 11, and a full proof of this lemma is in Axler, *Linear Algebra Done Right* 4th edition, section 7B, or in Hoffman and Kunze's chapter on symmetric operators. □

With that lemma, $\mathbf{A}^\top \mathbf{A}$ is symmetric and $\mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2 \geq 0$, so its eigenvalues are nonnegative and their roots are real. Now the main theorem.

Theorem 10.10 (Singular value decomposition)**.** *Every* $\mathbf{A} \in \mathbb{R}^{m\times n}$ *of rank* r *can be written as*

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top,$$

with $\mathbf{U} \in \mathbb{R}^{m\times m}$ *and* $\mathbf{V} \in \mathbb{R}^{n\times n}$ *matrices with orthonormal columns, and* $\mathbf{\Sigma} \in \mathbb{R}^{m\times n}$ *diagonal with entries* $\sigma_1 \geq \cdots \geq \sigma_r > 0$ *and the rest zero.*

Proof. Because $\mathbf{A}^\top\mathbf{A}$ is symmetric, the spectral lemma gives an orthonormal basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$ with $\mathbf{A}^\top\mathbf{A}\mathbf{v}_i = \sigma_i^2\mathbf{v}_i$ and $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. The nonzero eigenvalues number r; call their indices $1, \ldots, r$.

For $i \leq r$ define $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i/\sigma_i$, which has norm one because $\|\mathbf{A}\mathbf{v}_i\|^2 = \mathbf{v}_i^\top\mathbf{A}^\top\mathbf{A}\mathbf{v}_i = \sigma_i^2$. For $i, j \leq r$,

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \frac{1}{\sigma_i\sigma_j}\mathbf{v}_i^\top\mathbf{A}^\top\mathbf{A}\mathbf{v}_j = \frac{\sigma_j^2}{\sigma_i\sigma_j}\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij},$$

so $\mathbf{u}_1, \ldots, \mathbf{u}_r$ are orthonormal. For $i > r$ we have $\|\mathbf{A}\mathbf{v}_i\|^2 = \sigma_i^2 = 0$, hence $\mathbf{A}\mathbf{v}_i = \mathbf{0}$.

Those vectors do not necessarily fill $\mathbb{R}^m$; complete them to an orthonormal basis with Gram-Schmidt on any basis of their orthogonal complement. Define $\mathbf{B} = \sum_{i\leq r} \sigma_i\mathbf{u}_i\mathbf{v}_i^\top$. For $j \leq r$ we have $\mathbf{B}\mathbf{v}_j = \sigma_j\mathbf{u}_j = \mathbf{A}\mathbf{v}_j$, and for $j > r$ both are zero. The two matrices agree on a basis, so $\mathbf{B} = \mathbf{A}$. □

The geometric reading: the unit ball is mapped by $\mathbf{A}$ to an ellipsoid. The columns $\mathbf{v}_i$ point along the axes of the ball, $\mathbf{u}_i$ along the axes of the ellipsoid, and σ_i is the length of that axis: $\mathbf{V}^\top$ rotates, $\mathbf{\Sigma}$ stretches, $\mathbf{U}$ rotates again.

Figure 10.2 carries out that reading on $\mathbf{A} = \begin{pmatrix} 0 & 3 \\ 4 & 1 \end{pmatrix}$: the axis $\mathbf{v}_1$, tilted 26.57°, ends up as the axis $\mathbf{u}_1$ of the ellipse, tilted 71.57°.



unit circle

its image

Figure 10.2: The action of $\mathbf{A} = \begin{pmatrix} 0 & 3 \\ 4 & 1 \end{pmatrix}$ on the unit circle. Left: two orthonormal axes of the circle, $\mathbf{v}_1 = (2,1)/\sqrt{5}$ and $\mathbf{v}_2 = (-1,2)/\sqrt{5}$. Right: the image is an ellipse with semi-axes $\sigma_1 = 3\sqrt{2} \approx 4.243$ (numpy prints 4.242641) along $\mathbf{u}_1 = (1,3)/\sqrt{10}$, and $\sigma_2 = 2\sqrt{2} \approx 2.828$ (numpy 2.828427) along $\mathbf{u}_2 = (3,-1)/\sqrt{10}$.

The sum form $\mathbf{A} = \sum_i \sigma_i\mathbf{u}_i\mathbf{v}_i^\top$ is the most honest way to read it: the matrix is broken into r layers of rank one, ordered from the largest contribution. Truncating after k terms gives the best rank-k approximation in the Frobenius norm; that is the Eckart-Young theorem, whose proof is not carried out here and can be read in Golub and Van Loan, section 2.4.

Figure 10.3 displays the singular value portrait of the same matrix: truncating after the first term kills σ_2, and the remaining error is exactly $\sigma_2 = 2.828427$.

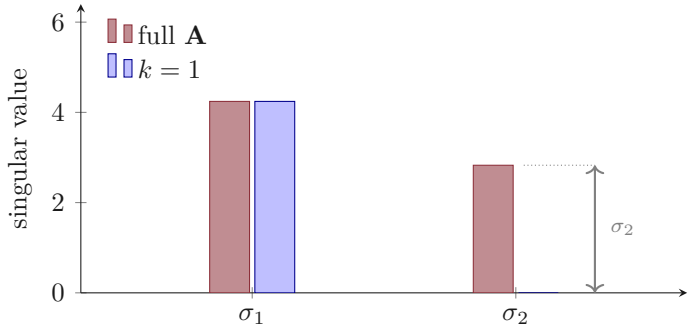


Figure 10.3: Singular value portrait of $\mathbf{A} = \begin{pmatrix} 0 & 3 \\ 4 & 1 \end{pmatrix}$. Red bars: full $\mathbf{A}$, $\sigma_1 = 4.243$ and $\sigma_2 = 2.828$. Blue bars: the rank-one approximation $\sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$, which leaves zero in the second slot. The difference of their tips, namely $\sigma_2 = 2.828$, is the Frobenius error of that approximation.

Several consequences follow directly, and all of them are used in the next section.

Corollary 10.11. *For $\mathbf{A} \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$:*

1. $\operatorname{rank} \mathbf{A} = r$*, the number of nonzero singular values;*
2. $\|\mathbf{A}\mathbf{x}\| \leq \sigma_1 \|\mathbf{x}\|$ *for every* $\mathbf{x}$*, and that bound is attained;*
3. $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$ *and* $\|\mathbf{A}\|_2 = \sigma_1$*;*
4. *if* $\mathbf{A}$ *is square and invertible,* $|\det \mathbf{A}| = \prod_i \sigma_i$ *and* $\kappa(\mathbf{A}) = \sigma_1 / \sigma_n$*.*

Proof. First item: $\mathbf{A}\mathbf{x} = \sum_i \sigma_i \mathbf{u}_i (\mathbf{v}_i^\top \mathbf{x})$, so the range of $\mathbf{A}$ is spanned by $\mathbf{u}_1, \ldots, \mathbf{u}_r$, which are linearly independent. Second item: $\|\mathbf{A}\mathbf{x}\|^2 = \sum_i \sigma_i^2 (\mathbf{v}_i^\top \mathbf{x})^2 \leq \sigma_1^2 \|\mathbf{x}\|^2$, with equality at $\mathbf{x} = \mathbf{v}_1$. Third item: the cross terms vanish, so the squared Frobenius norm is the sum of σ_i^2. Fourth item: $|\det \mathbf{A}| = |\det \mathbf{U} \det \mathbf{\Sigma} \det \mathbf{V}^\top| = \prod_i \sigma_i$, since the determinant of an orthogonal matrix is ± 1. $\square$

The number κ in the fourth item is the *condition number*, the ratio of the largest singular value to the smallest; for a matrix of rank less than n the ratio is infinite. A large κ means one direction is almost flat, so a small error in the data becomes a large error in the answer; the sixth section uses it to compare two ways of solving least squares.

One warning: singular values are not eigenvalues. The matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has eigenvalues 0 and 0, while its singular values are 1 and 0. The two agree only when the matrix is symmetric and positive semidefinite.

10.5 Pseudoinverse and minimum-norm solutions

Level: advanced

Now we return to the attacker's question. The score is one number, the counters are three, and the set of changes that give the same shift contains infinitely many answers. All those answers are correct; what separates them is the size of the change, and the attacker wants the smallest.

For younger readers (age 10 and up)

Imagine you have to go to the shop at the end of the street. There are many roads: through the market, across the field, or around by the bridge. All of them reach the same shop. If all you count is "arrive or not", every road is equally good. But if you count footsteps, only one road is shortest, and that road is usually the straightest.
Remember the two rulers that differ by one hundredth of a centimeter? A difference that small drowns easily in reading mistakes. That is why two roads that are the same mathematically can give very different answers when the numbers are computed by a machine.
In this chapter we will learn why among many roads there is one that is shortest, why that shortest road is always perpendicular to every other road that does not change the destination, and why a machine sometimes chooses a road that is not the most direct.

The tool that answers this is the generalized inverse. $\mathbf{A}$ is not necessarily square, so it does not necessarily have an inverse. The singular value decomposition gives a way to invert "as much as can be inverted": invert every σ_i that is nonzero, and let a direction that is already lost stay lost.

pseudoinverse (*pseudoinvers*) — An inverse that still works when some directions are already flat: what can be inverted is inverted, what is lost is left lost.

Definition 10.6 (Pseudoinverse)**.** Suppose $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ has rank r. The *pseudoinverse* $\mathbf{A}^+ \in \mathbb{R}^{n\times m}$ is defined as

$$\mathbf{A}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top, \qquad \mathbf{\Sigma}^+ = \operatorname{diag}(1/\sigma_1, \dots, 1/\sigma_r, 0, \dots, 0),$$

that is, the matrix $\mathbf{\Sigma}$ transposed and then with every nonzero entry inverted.

$\mathbf{A}^+$ always exists, even for the zero matrix. If $\mathbf{A}$ is square and invertible, all $\sigma_i > 0$ and $\mathbf{A}^+ = \mathbf{A}^{-1}$; this definition extends the inverse, it does not replace it.

The vector $\mathbf{A}^+\mathbf{b}$ answers two demands at once: the residual is smallest, and among those with the smallest residual, its length is shortest.

Theorem 10.12. *For every $\mathbf{b} \in \mathbb{R}^m$, the vector $\mathbf{x}^+ = \mathbf{A}^+\mathbf{b}$ satisfies $\|\mathbf{A}\mathbf{x}^+ - \mathbf{b}\| \le \|\mathbf{A}\mathbf{x} - \mathbf{b}\|$ for all $\mathbf{x}$, and among all vectors satisfying that inequality, $\mathbf{x}^+$ has the smallest norm.*

Proof. Write $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and let $\mathbf{y} = \mathbf{V}^\top\mathbf{x}$. Because $\mathbf{V}$ is square with orthonormal columns, $\|\mathbf{x}\| = \|\mathbf{y}\|$ and the map $\mathbf{x} \mapsto \mathbf{y}$ is bijective. Expand $\mathbf{b}$ along the columns of $\mathbf{U}$: $\mathbf{b} = \sum_{i=1}^{r} b_i\mathbf{u}_i + \mathbf{b}_\perp$ with $b_i = \langle \mathbf{u}_i, \mathbf{b}\rangle$ and $\mathbf{b}_\perp$ orthogonal to all $\mathbf{u}_i$. Then

$$\mathbf{A}\mathbf{x} - \mathbf{b} = \sum_{i=1}^{r}(\sigma_i y_i - b_i)\mathbf{u}_i + \sum_{i=r+1}^{m} 0 \cdot \mathbf{u}_i - \mathbf{b}_\perp,$$

and the three groups of terms are mutually orthogonal. Therefore

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^{r}(\sigma_i y_i - b_i)^2 + \|\mathbf{b}_\perp\|^2.$$

The last term does not depend on $\mathbf{x}$, so the minimum is attained with $y_i = b_i/\sigma_i$ for $i \le r$. The coordinates $y_{r+1}, \dots, y_n$ appear in no formula: whatever their values, the squared residual is the same. But $\|\mathbf{y}\|^2 = \sum_{i\le r}(b_i/\sigma_i)^2 + \sum_{i>r} y_i^2$, so the minimum norm is attained when $y_i = 0$ for $i > r$. The resulting vector is $\mathbf{x}^+ = \sum_{i\le r}(b_i/\sigma_i)\mathbf{v}_i = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top\mathbf{b}$. □

That proof contains the answer to the attacker's question. On an underdetermined system, $n > r$, and the coordinates $y_{r+1}, \dots, y_n$ do not affect the residual at all; those coordinates lie along $\mathbf{v}_{r+1}, \dots, \mathbf{v}_n$, exactly the vectors that span the kernel of $\mathbf{A}$. So the set of all solutions is $\mathbf{x}^+ + \ker \mathbf{A}$, an affine space of dimension $n - r$, and the pseudoinverse picks the point with zero kernel component: the only one orthogonal to the kernel, that is, in the row space of $\mathbf{A}$. Every other solution is $\mathbf{x}^+$ plus a nonzero vector in the kernel, and that addition increases the length by Pythagoras: $\|\mathbf{x}^+ + \mathbf{k}\|^2 = \|\mathbf{x}^+\|^2 + \|\mathbf{k}\|^2$.

For the two extreme cases there are closed formulas.

Corollary 10.13. *If* $\mathbf{A}$ *has full column rank,* $\mathbf{A}^+ = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$. *If* $\mathbf{A}$ *has full row rank,* $\mathbf{A}^+ = \mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^{-1}$.

Proof. For the first case we repeat the argument with the SVD. Take $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ with $\mathbf{V}$ square and orthonormal; then

$$(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top = \mathbf{V}\mathbf{\Sigma}^{-2}\mathbf{V}^\top \mathbf{V}\mathbf{\Sigma}\mathbf{U}^\top = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^\top = \mathbf{A}^+.$$

The second case is obtained by applying the first to $\mathbf{A}^\top$ and then transposing: $(\mathbf{A}^\top)^+ = (\mathbf{A}\mathbf{A}^\top)^{-1}\mathbf{A}$, and its transpose is $\mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}$. Since $\mathbf{A}^+ = (\mathbf{A}^\top)^{+\top}$ by definition, the result is correct. □

That second consequence is what answers the attacker's problem. The 2×3 detector matrix has full row rank, so the minimum change is $\mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}\boldsymbol{\delta}$, and that formula needs only the inverse of a 2×2 matrix. Note that $\mathbf{A}\mathbf{A}^\top$ is invertible precisely because its rows are linearly independent, not its columns.

Challenge (problem solver)

Take

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathbf{b} = (1, 1, 3).$$

The system $\mathbf{A}\mathbf{x} = \mathbf{b}$ has no solution: the first two rows force $\mathbf{x} = (1, 1)$, while the third row demands $x_1 + x_2 = 3$.

There is still one least-squares solution, because the columns are linearly independent. Determine $\hat{\mathbf{x}}$, then find all values of c such that for $\mathbf{b} = (1, 1, c)$ the length of the least-squares residual is exactly 1. There are two answers, and both are symmetric about $c = 2$.

Hint 1: the residual is always parallel to one fixed direction; find that direction first.

Hint 2: the distance from $\mathbf{b}$ to the column plane is $|c - 2|/\sqrt{3}$, so only one equation remains to be solved.

If you get stuck, go back to the section Least squares and the normal equations.

10.6 Why SVD is safer than the normal equations

Level: advanced

The normal equations and the pseudoinverse give the same answer mathematically. Numerically the two differ widely, and the reason is one line: the normal equations work on $\mathbf{A}^\top\mathbf{A}$, whose singular values are σ_i^2, not σ_i. Since the condition number is a ratio of singular values, it is squared as well.

Corollary 10.14. *If* $\mathbf{A}$ *has full column rank,* $\kappa(\mathbf{A}^\top\mathbf{A}) = \kappa(\mathbf{A})^2$.

Proof. From $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ we get $\mathbf{A}^\top\mathbf{A} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top$, with $\mathbf{V}$ orthogonal. So the singular values of $\mathbf{A}^\top\mathbf{A}$ are σ_i^2, and the ratio of largest to smallest becomes $\sigma_1^2/\sigma_n^2 = (\sigma_1/\sigma_n)^2$. $\square$

It means this. In binary fractions with about sixteen digits, the relative error of the answer is roughly κ times the relative error of the data: at $\kappa = 10^7$ nine digits remain, at $\kappa^2 = 10^{14}$ two remain, and at $\kappa = 10^8$ every digit is gone.

The numbers in the following table were computed with numpy and scipy, against a 50-digit reference from mpmath. The first row is the probe calibration from the chapter opening; the second row is a degree-nine polynomial fit on twelve points, a problem that looks harmless.

case	$\kappa(\mathbf{A})$	relative error normal equations	relative error SVD
calibration 5×2	7.36×10^{3}	9.10×10^{-11}	4.72×10^{-13}
polynomial 12×10	6.84×10^{6}	2.87×10^{-4}	4.90×10^{-12}

Both rows follow the same pattern. In the first row, $\kappa = 7.36 \times 10^3$ and $\kappa^2 = 5.42 \times 10^7$; the normal equations differ from the reference by 9.10×10^{-11}, the SVD by 4.72×10^{-13}, a ratio of 193. In the second row, $\kappa(\mathbf{A}) = 6.84 \times 10^6$ and $\kappa(\mathbf{A}^\top\mathbf{A}) = 4.68 \times 10^{13}$; the normal equations differ by 2.87×10^{-4}, the SVD by 4.90×10^{-12}, a ratio of 5.9×10^7. That difference grows together with κ, far faster for the normal equations; both are still below the bound $\kappa^2\epsilon$ because that bound is a worst case, not a prediction. What matters is the order, not the absolute numbers.

The squared residuals of the two roads are nearly the same, 4.47019×10^{-6} versus 4.47018×10^{-6}, even though the coefficients differ widely: a small residual does not prove the coefficients are right.

The honest conclusion: SVD does not remove the sensitivity, it only removes one source of magnification, namely the squaring, so the error grows like κ and not like κ^2. When κ approaches 10^{16}, no double precision saves you; what remains is regularization, that is, deliberately discarding directions with small σ_i, equivalent to approximating $\mathbf{A}$ by a matrix of lower rank. Chapter 26 discusses how to choose the threshold.

10.7 Worked examples

Level: intermediate

The four computations below use simple fractions, so they can be checked again by hand without a machine.

First, projection onto a plane. Take $\mathbf{a}_1 = (1,1,0)$, $\mathbf{a}_2 = (1,0,1)$, and $\mathbf{b} = (3,1,3)$ in $\mathbb{R}^3$. The subspace $W = \operatorname{span}(\mathbf{a}_1, \mathbf{a}_2)$ has dimension two. Gram-Schmidt gives

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1,1,0), \qquad \mathbf{u}_2 = \mathbf{a}_2 - \frac{1}{\sqrt{2}}\mathbf{q}_1 = \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right), \qquad \mathbf{q}_2 = \frac{1}{\sqrt{6}}(1,-1,2),$$

because $\|\mathbf{u}_2\| = \sqrt{3/2} = \sqrt{6}/2$. The projection coefficients are $\langle \mathbf{b}, \mathbf{q}_1 \rangle = 4/\sqrt{2} = 2\sqrt{2}$ and $\langle \mathbf{b}, \mathbf{q}_2 \rangle = 8/\sqrt{6}$. So

$$\mathbf{p} = 2\sqrt{2}\,\mathbf{q}_1 + \frac{8}{\sqrt{6}}\,\mathbf{q}_2 = (2,2,0) + \frac{4}{3}(1,-1,2) = \left(\tfrac{10}{3}, \tfrac{2}{3}, \tfrac{8}{3}\right).$$

The residual is $\mathbf{r} = \mathbf{b} - \mathbf{p} = (-\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3})$, with $\|\mathbf{r}\| = 1/\sqrt{3} = 0.577350$. Check: $\mathbf{a}_1^\top\mathbf{r} = 0$ and $\mathbf{a}_2^\top\mathbf{r} = 0$, so the residual is indeed perpendicular to both spanning vectors. The distance from $\mathbf{b}$ to the plane is 0.577; no other point in W is closer.

Second, least squares 5×2. For the chapter's opening data, $\mathbf{A}$ has columns $(1,1,1,1,1)^{\top}$ and $(100,101,102,103,104)^{\top}$, so

$$\mathbf{A}^{\top}\mathbf{A} = \begin{pmatrix} 5 & 510 \\ 510 & 52030 \end{pmatrix}, \qquad \mathbf{A}^{\top}\mathbf{y} = \begin{pmatrix} 1540 \\ 157109.7 \end{pmatrix}, \qquad \det(\mathbf{A}^{\top}\mathbf{A}) = 5 \cdot 52030 - 510^2 = 50.$$

Cramer's rule gives

$$a = \frac{1540 \cdot 52030 - 510 \cdot 157109.7}{50} = \frac{253}{50} = 5.06,$$
$$b = \frac{5 \cdot 157109.7 - 510 \cdot 1540}{50} = \frac{148.5}{50} = 2.97.$$

The predicted readings become 302.06, 305.03, 308.00, 310.97, and 313.94, whereas the actual data are 302.0, 305.1, 307.9, 311.2, 313.8. The residual is

$$\mathbf{r} = (-0.06,\ 0.07,\ -0.10,\ 0.23,\ -0.14), \qquad \|\mathbf{r}\|^2 = 0.0910, \qquad \|\mathbf{r}\| = 0.301662.$$

Two checks must pass if the computation is right: $\sum_i r_i = 0$ and $\sum_i t_i r_i = 0$. The first because $\mathbf{r}$ is perpendicular to the column of ones, the second because it is perpendicular to the column t. Both are indeed zero here.

Third, SVD of a 3×2 *matrix.* Take $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{pmatrix}$. Then

$$\mathbf{A}^{\top}\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \qquad \lambda^2 - 6\lambda + 8 = 0 \implies \lambda = 4,\ 2,$$

so $\sigma_1 = 2$ and $\sigma_2 = \sqrt{2} = 1.41421$, with $\kappa(\mathbf{A}) = \sqrt{2}$. Its right singular vectors are $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)$ and $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$; the left singular vectors are computed from $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i/\sigma_i$, namely $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1,0)$ and $\mathbf{u}_2 = (0,0,1)$. Check one entry: $(2\,\mathbf{u}_1\mathbf{v}_1^{\top} + \sqrt{2}\,\mathbf{u}_2\mathbf{v}_2^{\top})_{32} = -\sqrt{2}/\sqrt{2} = -1$, which matches $\mathbf{A}$.

Fourth, a rank-one matrix. Take $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$, whose second column is twice the first column. Then $\mathbf{B}^{\top}\mathbf{B} = 14\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ with eigenvalues 70 and 0, so $\sigma_1 = \sqrt{70} = 8.36660$ and $\sigma_2 = 0$. Its rank is one, and κ is infinite. For $\mathbf{b} = (1,0,0)$, the projection onto the column space is $\frac{1}{14}(1,2,3)$, so $\mathbf{B}\mathbf{x}$ must equal that vector. Since $\mathbf{B}(s,2s) = 5s(1,2,3)$, we need $5s = 1/14$, that is, $s = 1/70$. The minimum-norm solution is

$$\mathbf{x}^{+} = \left(\tfrac{1}{70}, \tfrac{2}{70}\right) = (0.0142857,\ 0.0285714), \qquad \|\mathbf{x}^{+}\| = \frac{\sqrt{5}}{70} = 0.031944.$$

The least-squares residual:

$$\mathbf{r} = \mathbf{b} - \mathbf{B}\mathbf{x}^{+} = \left(\tfrac{13}{14}, -\tfrac{1}{7}, -\tfrac{3}{14}\right), \qquad \|\mathbf{r}\| = \sqrt{13/14} = 0.963624.$$

That residual cannot be zero: $\mathbf{b}$ lies outside the column space. Other solutions still give the same residual, for instance $\mathbf{x}^{+} + (-\frac{2}{7}, \frac{1}{7})$, but their norm is larger.

Repeat the second computation with the last reading 314.3 and not 313.8. Predict first whether a goes up or down, then compute. Note that the point with the largest t shifts the line more than a point in the middle; explain this through the columns of $\mathbf{A}$ and the formula $(\mathbf{A}^{\top}\mathbf{A})^{-1}\mathbf{A}^{\top}$. Compute the new $\|\mathbf{r}\|$ as well and compare it with 0.301662.

10.8 Computational lab

Level: advanced

The two blocks below were run with numpy 2.5.2, scipy 1.18.0, Python 3.14.4; the output is copied as it stands.

Computational lab

```
import numpy as np
from scipy.linalg import lstsq, qr

np.set_printoptions(precision=6, suppress=True, linewidth=72)

# 1. projection onto a plane
a1 = np.array([1., 1., 0.])
a2 = np.array([1., 0., 1.])
b = np.array([3., 1., 3.])
Q, _ = qr(np.column_stack([a1, a2]), mode="economic")
P = Q @ Q.T
print("Q =\n", Q)
print("P b =", P @ b, " residual =", b - P @ b)
nr = np.linalg.norm(b - P @ b)
print("residual norm =", nr, " 1/sqrt3 =", 1/np.sqrt(3))
print("P^2 = P:", np.allclose(P @ P, P), " P^T = P:", np.allclose(P.T, P))

# 2. least squares 5x2
t = np.array([100., 101., 102., 103., 104.])
A = np.column_stack([np.ones(5), t])
y = np.array([302.0, 305.1, 307.9, 311.2, 313.8])
print("\nA^T A =\n", A.T @ A, " det =", np.linalg.det(A.T @ A))
print("A^T y =", A.T @ y)
xn = np.linalg.solve(A.T @ A, A.T @ y)
xs = lstsq(A, y)[0]
print("normal equations:", xn)
print("lstsq           :", xs)
print("solution gap    :", np.linalg.norm(xn - xs))
cA = np.linalg.cond(A)
print("cond A =", cA, " cond A^T A =", np.linalg.cond(A.T @ A))
r = y - A @ xn
print("residual =", r, " norm =", np.linalg.norm(r))
print("A^T r =", A.T @ r)

# 3. SVD of a 3x2 matrix
S = np.array([[1., 1.], [1., 1.], [1., -1.]])
U, s, Vt = np.linalg.svd(S, full_matrices=False)
print("\nsingular:", s)
print("rank", np.linalg.matrix_rank(S), " cond", np.linalg.cond(S))
print("U =\n", U, "\nV^T =\n", Vt)
print("reconstruction error:", np.linalg.norm(U @ np.diag(s) @ Vt - S))
```

Actual output:

```
Q =
 [[-0.707107  0.408248]
 [-0.707107 -0.408248]
 [-0.        0.816497]]
P b = [3.333333 0.666667 2.666667]  residual = [-0.333333  0.333333  0.333333]
residual norm = 0.5773502691896251  1/sqrt3 = 0.5773502691896258
P^2 = P: True  P^T = P: True

A^T A =
 [[   5.   510.]
```

```
 [  510. 52030.]]  det = 49.999999999983494
A^T y = [  1540.  157109.7]
normal equations: [5.06 2.97]
lstsq           : [5.06 2.97]
solution gap    : 5.315035090945416e-10
cond A = 7358.860135917719  cond A^T A = 54152822.48728771
residual = [-0.06  0.07 -0.1   0.23 -0.14]  norm = 0.30166206257998296
A^T r = [0. 0.]

singular: [2.       1.414214]
rank 2  cond 1.414213562373095
U =
 [[-0.707107 -0.      ]
 [-0.707107 -0.      ]
 [-0.        1.      ]]
V^T =
 [[-0.707107 -0.707107]
 [ 0.707107 -0.707107]]
reconstruction error: 6.568167990716596e-16
```

Three things need reading. First, $\mathbf{Q}$ from scipy differs in sign from the hand result: its columns are $-(1,1,0)/\sqrt{2}$ and $-(1,-1,2)/\sqrt{6}$. The sign of the columns in QR is not unique, and $\mathbf{QQ}^\top$ does not change because of it; the entries of $\mathbf{P}$ stay $(10/3, 2/3, 8/3)$, with residual $(-0.333333, 0.333333, 0.333333)$.

Second, the value of $\det(\mathbf{A}^\top\mathbf{A})$ is reported as 49.999999999983494 and not 50; the difference 1.7×10^{-11} is purely rounding. What matters is $\kappa(\mathbf{A}^\top\mathbf{A}) = 5.415 \times 10^7$, the square of $\kappa(\mathbf{A}) = 7.359 \times 10^3$.

Third, $\mathbf{A}^\top\mathbf{r}$ is reported as $(0, 0)$ to six digits and the difference between the two solutions is 5.32×10^{-10}: they only differ in the ninth digit, and that is no guarantee.

Computational lab

```
import numpy as np
import mpmath as mp
from scipy.linalg import lstsq

mp.mp.dps = 50
np.set_printoptions(precision=4, suppress=True, linewidth=72)

# 4. rank one: nonzero residual, minimum-norm solution
B = np.array([[1., 2.], [2., 4.], [3., 6.]])
b2 = np.array([1., 0., 0.])
sB = np.linalg.svd(B, compute_uv=False)
print("singular B:", sB, " rank", np.linalg.matrix_rank(B))
xb = np.linalg.pinv(B) @ b2
print("x+ =", xb, " B x+ =", B @ xb)
nr = np.linalg.norm(B @ xb - b2)
print("residual norm =", nr, " sqrt(13/14) =", np.sqrt(13/14))

# 5. underdetermined system: minimum norm
C = np.array([[1., 1., 1.], [0., 1., 1.]])
b3 = np.array([3., 1.])
xp = np.linalg.pinv(C) @ b3
print("\nx+ =", xp, " C x+ =", C @ xp, " norm =", np.linalg.norm(xp))
print("norm (2,1,0) =", np.linalg.norm([2., 1., 0.]))

# 6. least squares: normal equations vs lstsq vs 50-digit reference
x = np.linspace(0, 1, 12)
V = np.vander(x, 10, increasing=True)
b = V @ (0.1 * np.arange(1, 11)) + 0.001 * np.sin(np.arange(12))
```

```
Vm, bm = mp.matrix(V.tolist()), mp.matrix(b.tolist())
ref = np.array([float(v) for v in mp.lu_solve(Vm.T * Vm, Vm.T * bm)])
xne = np.linalg.solve(V.T @ V, V.T @ b)
xls = lstsq(V, b)[0]
rel = lambda u: np.linalg.norm(u - ref) / np.linalg.norm(ref)
print("\ncond V     :", np.linalg.cond(V))
print("cond V^T V :", np.linalg.cond(V.T @ V))
print("relative error normal equations:", rel(xne))
print("relative error lstsq           :", rel(xls))
print("residual normal equations:", np.linalg.norm(V @ xne - b))
print("residual lstsq           :", np.linalg.norm(V @ xls - b))

# 7. classical vs modified Gram-Schmidt vs Householder
def cgs(M):
    m, n = M.shape
    Q = np.zeros((m, n)); R = np.zeros((n, n))
    for j in range(n):
        v = M[:, j].astype(float).copy()
        for i in range(j):
            R[i, j] = Q[:, i] @ M[:, j]
            v -= R[i, j] * Q[:, i]
        R[j, j] = np.linalg.norm(v)
        Q[:, j] = v / R[j, j]
    return Q

def mgs(M):
    m, n = M.shape
    Q = np.zeros((m, n)); R = np.zeros((n, n)); W = M.astype(float).copy()
    for j in range(n):
        for i in range(j):
            R[i, j] = Q[:, i] @ W[:, j]
            W[:, j] -= R[i, j] * Q[:, i]
        R[j, j] = np.linalg.norm(W[:, j])
        Q[:, j] = W[:, j] / R[j, j]
    return Q

ort = lambda Q: np.linalg.norm(Q.T @ Q - np.eye(Q.shape[1]))
Qh, _ = np.linalg.qr(V)
print("\n||Q^T Q - I||  classical  :", ort(cgs(V)))
print("||Q^T Q - I||  modified   :", ort(mgs(V)))
print("||Q^T Q - I||  Householder:", ort(Qh))
```

Actual output:

```
singular B: [8.3666 0.    ]  rank 1
x+ = [0.0143 0.0286]  B x+ = [0.0714 0.1429 0.2143]
residual norm = 0.9636241116594315  sqrt(13/14) = 0.9636241116594315

x+ = [2.  0.5 0.5]  C x+ = [3. 1.]  norm = 2.121320343559642
norm (2,1,0) = 2.23606797749979

cond V     : 6841129.486331141
cond V^T V : 46788641042301.25
relative error normal equations: 0.0002869502788738492
relative error lstsq           : 4.899523677448492e-12
residual normal equations: 4.4701872339775605e-06
residual lstsq           : 4.47017846708843e-06

|||Q^T Q - I||  classical  : 0.020506481578936916
|||Q^T Q - I||  modified   : 3.397900311236548e-10
|||Q^T Q - I||  Householder: 1.1933579412809552e-15
```

First reading: the least-squares residual of the rank-one matrix is exactly $\sqrt{13/14}$; numpy

gives the same number to sixteen digits, so the fractional formula in the seventh section is not an approximation.

Second reading: $\mathbf{Cx} = (3, 1)$ gives $\mathbf{x}^+ = (2,\ 0.5,\ 0.5)$ with norm 2.12132, whereas $(2, 1, 0)$ gives 2.23607; the difference of squared norms is 0.5, exactly $2t^2$ at $t = 0.5$ for $\mathbf{k} = (0, 1, -1)$. The pseudoinverse chooses $t = 0$.

Third and fourth readings: on the 12×10 Vandermonde, the normal equations differ by 2.87×10^{-4} from the 50-digit reference and `lstsq` by 4.90×10^{-12}, even though the residuals are practically the same (4.47019×10^{-6} versus 4.47018×10^{-6}). The coefficients are wrong, the residual is right. For Gram-Schmidt, classical gives 2.05×10^{-2}, modified 3.40×10^{-10}, Householder 1.19×10^{-15}.

Note for instructors

The order that has worked: projection first, the definition of the inner product afterwards. Students who see the shadow first accept $\mathbf{QQ}^\top$ as the answer to the question "which point is closest", not as a magic formula; after that Gram-Schmidt becomes a need, not a memorized fact.

Two typical misconceptions. First, $\mathbf{A}^\top\mathbf{A}$ is assumed always invertible because "squaring a matrix never gives zero"; use the corollary in the third section, then ask for the kernel of $\mathbf{A}^\top\mathbf{A}$ for a rank-one 2×2 matrix. Second, the SVD is assumed to give eigenvalues; the matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is enough to break that.

Class activity: give five clearly non-collinear points, ask half the class to project $\mathbf{b}$ by hand and half to use `lstsq`; compare the residual vectors, not the coefficients.

A sign of readiness to go further: a student can explain why adding one equation does not always improve the answer, and name one situation in which the pseudoinverse is better than the inverse.

10.9 Summary

Summary.

- The inner product supplies length and angle; Cauchy-Schwarz supports the triangle inequality. In an orthonormal basis $\mathbf{v} = \sum_i \langle \mathbf{v}, \mathbf{q}_i \rangle \mathbf{q}_i$ and $\|\mathbf{v}\|^2 = \sum_i \langle \mathbf{v}, \mathbf{q}_i \rangle^2$.
- The projection $\mathbf{p} = \mathbf{QQ}^\top\mathbf{b}$ is the closest point in the subspace, and the residual is perpendicular to the whole subspace. Gram-Schmidt builds an orthonormal basis and gives $\mathbf{A} = \mathbf{QR}$.
- Least squares satisfies $\mathbf{A}^\top\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^\top\mathbf{b}$; if the columns of $\mathbf{A}$ are linearly independent, the answer is unique.
- Every matrix has $\mathbf{A} = \mathbf{U\Sigma V}^\top$: the rank is the number of $\sigma_i > 0$, $\|\mathbf{A}\|_2 = \sigma_1$, and $\kappa(\mathbf{A}) = \sigma_1/\sigma_n$.
- The pseudoinverse $\mathbf{A}^+ = \mathbf{V\Sigma}^+\mathbf{U}^\top$ gives the smallest residual and at the same time the smallest norm, and on an underdetermined system it chooses the answer perpendicular to the kernel.
- The normal equations work at κ^2, the SVD at κ; that is why a badly conditioned problem loses digits twice as fast through $\mathbf{A}^\top\mathbf{A}$.

10.10 Exercises

Exercise 10.1.

1. Project $\mathbf{b} = (1, 2, 2)$ onto $W = \text{span}\{(1, 0, 0), (0, 1, 1)\}$ by hand, check with `numpy.linalg.qr`, then compute the distance from $\mathbf{b}$ to W.

2. Prove $\|\mathbf{b}\|^2 = \|\operatorname{proj}_W \mathbf{b}\|^2 + \|\mathbf{r}\|^2$ for arbitrary W and $\mathbf{b}$, then explain why that equality fails when $\mathbf{p}$ is not the orthogonal projection. (Hint: take a skew basis in $\mathbb{R}^2$ for the second part.)
3. If the columns of $\mathbf{A}$ are orthonormal, prove $\mathbf{A}^\top\mathbf{A} = \mathbf{I}$, $\hat{\mathbf{x}} = \mathbf{A}^\top\mathbf{b}$, and $\kappa(\mathbf{A}) = 1$.
4. Evaluate this claim: *if $\mathbf{A}^\top\mathbf{A}$ is invertible then the rows of $\mathbf{A}$ are linearly independent.* Decide whether it is true, false, or not enough evidence to decide, and include a counterexample if needed.
5. Compute the SVD of $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ by hand: singular values, $\mathbf{U}$, $\mathbf{V}$, rank. Determine $\mathbf{A}^+\mathbf{b}$ for $\mathbf{b} = (1, 3)$, compute the residual, check with `numpy.linalg.pinv`.
6. Prove that the least-squares residual is always perpendicular to $\operatorname{col}(\mathbf{A})$ and conversely, then state where linear independence of the columns is used. (Hint: use the splitting of the norm as in the proof of the normal equations; the "necessary" argument chooses the direction $\mathbf{h}$.)
7. Run Gram-Schmidt on $\{(1,1,1), (1,1,0), (1,0,0)\}$ by hand, compare with `numpy.linalg.qr`, compute $\|\mathbf{Q}^\top\mathbf{Q} - \mathbf{I}\|$ for both, and explain why their signs may differ.
8. Suppose $m < n$ and the rows of $\mathbf{A}$ are linearly independent. Prove that $\mathbf{A}\mathbf{A}^\top$ is invertible, and that
$$\mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}\mathbf{b}$$
is the solution perpendicular to the kernel of $\mathbf{A}$. (Hint: imitate the proof of the corollary for the full-column case with $\mathbf{A}^\top$ as the matrix.)
9. With the calibration data in the seventh section, change the last reading to 314.3. Recompute a and b through Cramer's rule, compare with `scipy.linalg.lstsq`, and explain why a point at the end shifts the line more than a point in the middle.
10. For $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$ and $\mathbf{b} = (1, 0, 0)$, write all least-squares solutions in the form $\mathbf{x}^+ + t\mathbf{k}$, determine $\mathbf{k}$, then show that $\|\mathbf{x}^+ + t\mathbf{k}\|^2$ is minimal at $t = 0$.

For researchers: limits and next steps

What is proved in this chapter: Cauchy-Schwarz and the triangle inequality for an arbitrary inner-product space; uniqueness of coefficients in an orthonormal basis; the best-approximation property of the projection; the correctness of Gram-Schmidt; the equivalence of least squares with the normal equations; the existence of the SVD; and the pseudoinverse theorem. What is borrowed is only the spectral lemma, whose proof uses the extreme value theorem on the unit sphere (Chapter 11); here it is only sketched. Two claims are often popularized. First, the claim that "the SVD is always stable". What is true: the SVD algorithm is backward stable, meaning the computed answer is the exact answer for a matrix that differs slightly from $\mathbf{A}$; the distance to the answer for $\mathbf{A}$ itself can still grow by a factor of κ. Second, the claim that truncating small singular values "removes noise". Truncation removes a direction, and whether that direction is noise or signal cannot be decided from $\mathbf{A}$ alone.

Primary sources: Axler 4th edition, sections 6A through 6C and 7E through 7F; Anton 11th edition, sections 6.1 through 6.4 and 9.4 and 9.5; for the numerical side, Trefethen and Bau, *Numerical Linear Algebra*, and Golub and Van Loan, *Matrix Computations* 4th edition, sections 2.4 and 5.3.

Open questions. The threshold at which a singular value counts as zero has no parameter-

free answer; the choice depends on the scale of the data and on the error that can be accepted (Chapter 22). For large matrices, randomized algorithms only guarantee anything probabilistically. The limit of this chapter: real matrices of finite dimension; for complex matrices the transpose is replaced by the conjugate transpose, and for infinite-dimensional spaces the SVD is replaced by the decomposition of compact operators, whose singular values can accumulate at zero.

Map to source texts: This chapter follows Anton 11th edition, sections 6.1 (Inner Products), 6.2 (Angle and Orthogonality), 6.3 (Gram-Schmidt; QR-Decomposition), 6.4 (Best Approximation; Least Squares), 9.4 (Singular Value Decomposition), and 9.5 (Data Compression Using SVD). The abstract counterpart is Axler 4th edition, Chapter 6 sections 6A through 6C and Chapter 7 sections 7E through 7F, with the spectral theorem in 7B. Strang 6th edition gives the geometric road, but the copy in the source directory contains only the cover and the preface. The numerical side: Trefethen and Bau; Golub and Van Loan 2.4, 5.3.

Part III

Analysis and Calculus

This part turns the words “close”, “smooth”, and “shrinking” into statements that can be checked with numbers. The prerequisites are Chapters 3–5 and 7. After six chapters you can prove continuity, differentiate and integrate, exchange a limit with an integral, and solve first-order differential equations and two-dimensional systems.

Chapter 11

Continuity and Uniform Continuity

A program samples $f(x) = 1/x$ on a grid of points spaced 10^{-4} inside $(0, 1]$. Near $x = 10^{-4}$ two neighboring points, namely 10^{-4} and 2×10^{-4}, give outputs 10000 and 5000; the difference is 5000. Near $x = 1$ two neighboring points give 1 and 1.0001; the difference is about 10^{-4}. The same function, the same grid spacing, two differences seven powers apart.

The question that arises is not "is $1/x$ continuous", since the answer is yes at every point of $(0, 1]$. The question is a different one: how much grid spacing must be used so that *every* output on that interval stays inside a single tolerance? For tolerance 10^{-2}, the point $x = 1$ demands only a spacing below 9.901×10^{-3}, whereas the point $x = 10^{-4}$ demands a spacing below 1.000×10^{-10}. The ratio of the two demands is 9.9×10^{7}. No single spacing serves both, and that is not a defect of the program.

Chapter 4 supplied the limit of a sequence, Chapter 3 supplied the supremum, and neither is enough: the limit of a function compares f with one fixed number, so it cannot compare two outputs from two nearby inputs. What is needed is one threshold for the whole set, together with a way to prove that the threshold exists. This chapter builds that tool, from the ε-δ definition to compact sets and the class of Lipschitz functions, then closes with a function that is continuous at one point only.

For younger readers (age 10 and up)

Imagine a fire escape whose steps grow taller toward the top. At the bottom, one stride the width of a foot raises us a little. At the top, the same stride raises us a hand's width, then a forearm, then almost the height of a body. Our older sibling stands beside us and asks one thing: how wide is a safe stride for the whole staircase? The honest answer for a staircase like that does not exist. The width that is safe below is too wide above, and the width that is safe above makes us crawl below.

Now the question that makes one curious. Is every staircase like that? There are staircases whose steps are constant, and staircases whose steps change but never pass a certain bound. For the second kind, one step width works from bottom to top. The difference is not in the height of the staircase, but in how steeply its slope changes.

The same thing happens with the graph of a function. There are graphs that can be controlled by one distance rule over their whole domain, and graphs that demand a different rule at every point. In this chapter we will know why that difference can be proved with certainty, and why a domain that is closed and bounded always belongs to the first kind.

11.1 Limits of functions and the delta threshold

Level: basic

The word "approaches" cannot be used as a definition, because everyone may mean something different by it. What can be used is a wager: however much precision is demanded, the output will fall inside that precision as long as the input is close enough. In everyday language: we install a narrow window around L, and demand that f enter that window for all x close enough to c.

functional limit (*limit fungsi*) — a promise that the output can be forced into a narrow window as long as the input is close enough to the door

Before the definition, one technical term is needed. The point c is called a *limit point* of a set A if every open interval around c contains a point of A other than c. Without that condition, the definition of limit becomes vacuous at an isolated point.

Definition 11.1 (Limit of a function)**.** Let $A \subseteq \mathbb{R}$, c a limit point of A, and $f\colon A \to \mathbb{R}$. We write $\lim_{x\to c} f(x) = L$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every $x \in A$ with $0 < |x - c| < \delta$ we have $|f(x) - L| < \varepsilon$.

Two things are easily missed. First, the condition $0 < |x - c|$ means the value of f at c is not counted; the function may be undefined at c and the limit still exists. Second, the order of the quantifiers: δ may depend on ε, but not on x. In the section on uniform continuity, it is precisely that freedom which is at issue.

This definition is a checking tool, not a recipe for finding δ; δ is found by working backward from $|f(x) - L|$. For

$$f(x) = \frac{x^2 - 1}{x - 1} \quad (x \neq 1), \qquad L = 2,$$

algebra gives $f(x) = x + 1$, hence $|f(x) - 2| = |x - 1|$. So $\delta = \varepsilon$ works. For $\varepsilon = 10^{-2}$ take $\delta = 10^{-2}$: at $x = 1 + 10^{-2}$ the difference is exactly 10^{-2}, and that does *not* meet the strict condition; at $x = 1 + 5 \times 10^{-3}$ the difference is 5×10^{-3}. Therefore $\delta = \varepsilon/2$ is the safe choice, and $\delta = \varepsilon$ is the cheapest choice whose bound is never attained from inside.

One thing that may reasonably be demanded of a limit is its uniqueness: a function cannot stop at two different places at once.

Theorem 11.1 (Uniqueness of the limit)**.** *If* $\lim_{x\to c} f(x) = L_1$ *and* $\lim_{x\to c} f(x) = L_2$*, then* $L_1 = L_2$.

Proof. Suppose $L_1 \neq L_2$ and set $\varepsilon = |L_1 - L_2|/2 > 0$. There is $\delta_1 > 0$ such that $0 < |x - c| < \delta_1$ gives $|f(x) - L_1| < \varepsilon$, and there is $\delta_2 > 0$ such that $0 < |x - c| < \delta_2$ gives $|f(x) - L_2| < \varepsilon$. Take x with $0 < |x - c| < \min(\delta_1, \delta_2)$; such a point exists because c is a limit point. Then

$$|L_1 - L_2| \leq |L_1 - f(x)| + |f(x) - L_2| < 2\varepsilon = |L_1 - L_2|,$$

an impossibility. Hence $L_1 = L_2$. □

We also use one bridge that is not proved here: the limit of a function exists and equals L exactly when every sequence $x_n \to c$ with $x_n \neq c$ satisfies $f(x_n) \to L$. Its name is the sequence criterion; its proof is in Abbott, *Understanding Analysis*, Chapter 4, section Functional Limits, and in Rudin, *Principles of Mathematical Analysis*, Chapter 4, section Continuous Functions. That bridge is what allows the sequences of Chapter 4 to be used for dismantling limits that do not exist. For $g(x) = x/|x|$, the sequence $1/n$ gives $g \to 1$ while $-1/n$ gives $g \to -1$; two different limits, so $\lim_{x\to 0} g(x)$ does not exist. That pattern is used again for uniform continuity, with two sequences whose distance tends to zero.

11.2 Continuity and its algebraic properties

Level: intermediate

A limit compares f with a number L that stands on its own. Continuity asks for something stricter: that number must be the value of the function at its point. In everyday language: the graph does not jump at that point, and its value there really is the value approached from the left and from the right.

continuous (*kontinu*) — a graph that can be drawn without lifting the pen at that point

Definition 11.2 (Continuity at a single point)**.** Let $f\colon A \to \mathbb{R}$ and $c \in A$. The function f is *continuous at* c if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every $x \in A$ with $|x - c| < \delta$ we have $|f(x) - f(c)| < \varepsilon$. The function is called continuous on A if it is continuous at every point of A.

Note that $x = c$ is now allowed, and what is demanded is not merely that the limit exists, but that the limit equals the value of the function. Continuity is therefore local: it speaks only about the behavior of f around one point.

For $f(x) = x^2$ at $c = 2$ and $\varepsilon > 0$ we have $|x^2 - 4| = |x - 2|\,|x + 2|$, and if $|x - 2| < 1$ then $|x + 2| < 5$. So $\delta = \min(1, \varepsilon/5)$ works, whereas the largest δ that is still valid is the positive root of $\delta(4 + \delta) = \varepsilon$, namely $2.49984376953 \times 10^{-4}$ for $\varepsilon = 10^{-3}$. The gap between the two numbers is the price of safety paid for not wanting to compute a root. Figure 11.1 shows the two thresholds for $\varepsilon = 1$.

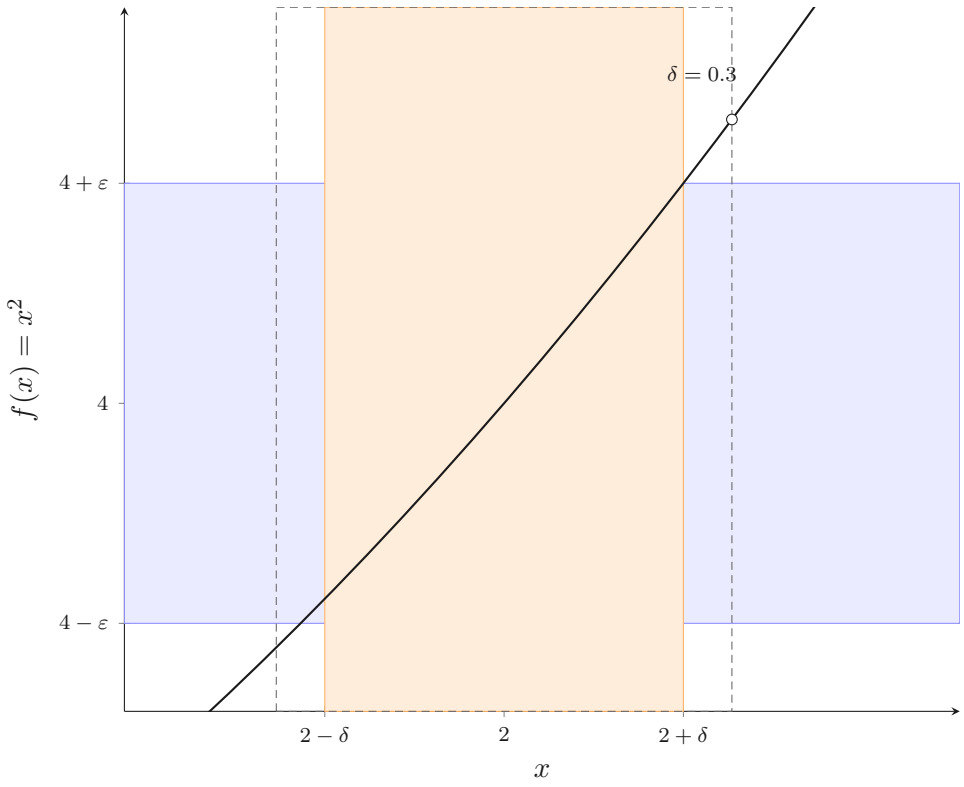


Figure 11.1: The ε band and the δ band for $f(x) = x^2$ at $a = 2$ with $\varepsilon = 1$. The δ bound uses the formula $\delta = \sqrt{4 + \varepsilon} - 2$ in this section, so $\delta = \sqrt{5} - 2 \approx 0.236$; the dashed band uses $\delta = 0.3$, and at $x = 2.3$ the value $f(2.3) = 5.29$ has already left the band.

Proposition 11.2. *Let c be a limit point of A. The function f is continuous at c if and only if $\lim_{x \to c} f(x) = f(c)$.*

Proof. If f is continuous at c and $\varepsilon > 0$, take δ from the definition of continuity. For $0 < |x - c| < \delta$ we have $|f(x) - f(c)| < \varepsilon$, and that is exactly the definition of limit with $L = f(c)$. Conversely, if the limit is $f(c)$, take δ from the definition of limit; the condition $0 < |x - c|$ can be dropped because for $x = c$ the inequality becomes $0 < \varepsilon$, which is true. □

Continuity is closed under the operations usually used. In everyday language: adding, multiplying, and dividing functions that do not jump produces functions that also do not jump, as long as the divisor is not zero.

Proposition 11.3 (Algebraic properties and composition)**.** *If f and g are continuous at c, then $f + g$, fg, and cf are continuous at c; if $g(c) \neq 0$, then f/g is also continuous at c. If f is continuous at c and g is continuous at $f(c)$, then $g \circ f$ is continuous at c.*

Proof. The algebraic part follows the same limit rules as for sequences, with $x_n \to c$ replaced by $x \to c$; since c is a limit point, the sequence criterion carries every statement about a limit of a function over to a statement about a limit of a sequence, and the algebraic rules for sequences were proved in Chapter 4. For composition, take $\varepsilon > 0$. Since g is continuous at $f(c)$, there is $\eta > 0$ such that $|y - f(c)| < \eta$ implies $|g(y) - g(f(c))| < \varepsilon$. Since f is continuous at c, for that η there is $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \eta$. That chain of implications gives $|g(f(x)) - g(f(c))| < \varepsilon$ whenever $|x - c| < \delta$. □

The pattern of the composition proof is worth remembering: two ε's are used in succession, and η plays the role of ε for the inner function. As a consequence, polynomials and rational functions are continuous on their whole domain. What is not guaranteed is a function defined by cases, and from there counterexamples appear at the end of the chapter.

The crudest form of failure is a jump. The function in Figure 11.2 equals x to the left of 1, equals 2 at the point 1, and equals $x + 1$ to its right. At $x = 1$ its value is 2, whereas the values from the left approach 1. That difference of 1 cannot be closed by any δ, however small, because for every δ there is still a point on the left with $|f(x) - f(1)|$ close to 1.

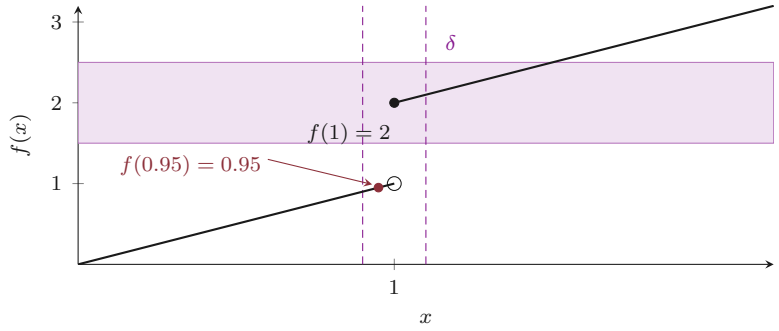


Figure 11.2: A jump at $x = 1$: $f(x) = x$ for $x < 1$, $f(1) = 2$, $f(x) = x + 1$ for $x > 1$. The band $\varepsilon = \frac{1}{2}$ around $f(1) = 2$ is the horizontal strip between 1.5 and 2.5, while the two dashed lines mark $|x - 1| < 0.1$. The point $x = 0.95$ is already inside that δ band, but its value 0.95 is still 0.55 below the lower edge of the ε band. The empty circle at $(1, 1)$ is the value approached from the left, not the value of the function. Shrinking δ does not help: a point on the left closer to 1 gives a value closer to 1, not to 2.

11.3 The intermediate value theorem

Level: intermediate

In everyday language: to move from the lower room to the upper room, a climber has to pass through every height in between. The graph of a continuous function cannot jump over a value.

Theorem 11.4 (Intermediate value)**.** *Let f be continuous on $[a,b]$ and $f(a) < 0 < f(b)$. Then there is $c \in (a,b)$ with $f(c) = 0$. More generally, for every v between $f(a)$ and $f(b)$ there is $c \in [a,b]$ with $f(c) = v$.*

Proof. It is enough to prove the case $f(a) < 0 < f(b)$, because the general case is obtained by applying that theorem to $f - v$. Form $S = \{x \in [a,b] : f(x) < 0\}$. That set is not empty because $a \in S$ and it is bounded above by b, so by the supremum property of Chapter 3, $c = \sup S$ exists and $a \le c \le b$.

First, $f(c) \le 0$. Since c is a supremum, there is a sequence $x_n \in S$ with $x_n \to c$, and $f(x_n) < 0$ for every n. Continuity gives $f(x_n) \to f(c)$, and a limit preserves weak inequalities, so $f(c) \le 0$.

Second, $f(c) \ge 0$. Suppose $f(c) < 0$. Since $f(b) > 0$, the point c is not b, so $c < b$. Continuity at c gives $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < -f(c)$, in particular $f(x) < 0$. For $\delta < 2(b - c)$ the point $c + \delta/2$ lies in $[a,b]$ and satisfies $f(c + \delta/2) < 0$, so it is a member of S; yet $c + \delta/2 > c = \sup S$. That impossibility gives $f(c) \ge 0$.

The two inequalities together give $f(c) = 0$, and since $f(a) < 0 < f(b)$, that point lies in (a,b). □

Note that the proof gives no way of finding c. It only shows that the set of negative points has a roof, and that the roof must be a root. The gap between "exists" and "can be computed" is what makes the intermediate value theorem used as a guarantee of existence, not as an algorithm.

For younger readers (age 10 and up)

A climber starts in a valley and stops at a summit. They may rest, may walk backward for a while, may stand still for a long time. What they do not do is jump. Therefore, if there is a river in the valley and no river at the summit, there must be one point on that path where they are for the first time exactly at the height of the river.

What makes one curious: how many such points are there? There may be one, there may be three, there may be infinitely many, and the climber's story does not say which. It only guarantees at least one. Mathematicians use that guarantee for something else: to prove that an equation certainly has an answer, even though the answer cannot always be written with a simple formula.

The next question is harder. What if the path is not connected, for example if there is a chasm in the middle? The guarantee vanishes at once, and that is no prank: there are functions whose value jumps from negative to positive without ever touching zero. In this chapter we will know why the property "does not jump" is already enough, and why that property has to be stated first before the theorem is used.

Two immediate consequences are worth recording. First, the image of an interval under a continuous function is an interval: $f([a,b])$ contains every value between $f(a)$ and $f(b)$. Second, every continuous function $f\colon [0,1] \to [0,1]$ has a fixed point. Consider $h(x) = f(x) - x$; at $x = 0$ we have $h(0) = f(0) \ge 0$ and at $x = 1$ we have $h(1) = f(1) - 1 \le 0$. If either value is zero, that point is a fixed point. If not, the intermediate value theorem gives c with $h(c) = 0$, that is $f(c) = c$.

The numbers in a concrete case can be checked. The equation $x^3 + x - 1 = 0$ has a root in $(0,1)$ because $f(0) = -1 < 0 < 1 = f(1)$. After twenty-one bisections, the root is trapped between 0.682327747 and 0.682328224, with interval width $2^{-21} = 4.768 \times 10^{-7}$; the true value is 0.6823278038280193. Bisection is not a proof, but the numbers can be compared with the theoretical bound.

11.4 The extreme value theorem

Level: intermediate

In everyday language: on a road that is closed and unbroken, there is certainly a highest point and a lowest point. Before proving it, we need one control: such a function cannot run off to infinity.

Lemma 11.5. *If f is continuous on $[a, b]$, then f is bounded on $[a, b]$.*

Proof. Suppose f is unbounded. Then for every n there is $x_n \in [a, b]$ with $|f(x_n)| > n$. The sequence (x_n) is bounded, so by the Bolzano–Weierstrass theorem of Chapter 4 there is a subsequence $x_{n_k} \to x$. Since $a \le x_{n_k} \le b$ for all k, a limit preserves inequalities and $x \in [a, b]$. Continuity at x gives $f(x_{n_k}) \to f(x)$, so that sequence is bounded. Yet $|f(x_{n_k})| > n_k \to \infty$. That impossibility closes the supposition. □

Now the main statement.

Theorem 11.6 (Extreme value)**.** *If f is continuous on $[a, b]$, then there are $c, d \in [a, b]$ such that $f(c) = \sup f([a, b])$ and $f(d) = \inf f([a, b])$. In other words, f attains its maximum and its minimum.*

Proof. By the lemma, the set $f([a, b])$ is bounded and not empty, so $M = \sup f([a, b])$ exists and is finite. For every n there is $x_n \in [a, b]$ with $f(x_n) > M - 1/n$; such a sequence exists because M is the least upper bound. By Bolzano–Weierstrass there is a subsequence $x_{n_k} \to c$, and since $a \le x_{n_k} \le b$ we get $c \in [a, b]$. Continuity at c gives $f(x_{n_k}) \to f(c)$, while the trapping above gives $f(x_{n_k}) \to M$. Uniqueness of the limit gives $f(c) = M$. For the minimum, apply that result to $-f$. □

Both proofs use the same bridge: a bounded sequence inside a closed interval has a subsequence that tends to a point inside that interval. Closedness is what ensures the limit does not escape, and that is what is lost on an open interval.

The counterexample is simple. The function $f(x) = x$ is continuous on $(0, 1)$ but attains neither a maximum nor a minimum, because 0 and 1 are not members of its domain. The function $g(x) = 1/x$ is continuous on $(0, 1]$ but unbounded there: $g(1/n) = n$. So closedness and boundedness of the domain are not decoration; they are the entire reason the theorem works.

As a consequence, for f continuous on $[a, b]$ with $m = \inf f$ and $M = \sup f$, we have $f([a, b]) = [m, M]$. That equality combines two theorems: the first says m and M are attained, the second says all values in between are used.

11.5 Uniform continuity

Level: advanced

In the definition of continuity, δ may depend on ε *and* on the point c. For $f(x) = 1/x$ on $(0, 1]$, every point has its own δ, and that δ shrinks without bound as c approaches zero. Locally everything is fine, but there is never one δ for the whole interval, and that is what the grid in the chapter opening needs.

uniform continuity (*kekontinuan seragam*) — one step width that works from the foot of the staircase to its top, not a different width on every step

Definition 11.3 (Uniform continuity)**.** A function $f\colon A \to \mathbb{R}$ is called *uniformly continuous* on A if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in A$ with $|x - y| < \delta$ we have $|f(x) - f(y)| < \varepsilon$.

Compare the two orders of quantifiers side by side:

$$\text{continuous: } \forall c \in A\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A, \qquad \text{uniform: } \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A.$$

In the uniform version, δ appears before x and y are introduced, so it may not exploit knowledge of any point. That is the whole content of the difference, and because of it uniformity is always relative to the set A: the same function can be uniform on one set and not on another.

Read it aloud (auditory learner)

Read aloud, two voices taking turns.
Voice one: Close at one point. However narrow the window installed, as long as the input is close enough, the output is inside the window.
Voice two: Close at all points. It is not a question of which door. One window width is installed first, then demanded to hold in the whole corridor, from the first door to the last.
Voice one: If the corridor narrows at the end, that width has to be cut again and again.
Voice two: And the cutting must not stop at one number. Stopping means one width for the whole corridor; shrinking forever means a promise in installments.
Repeat this line until it feels right on the tongue: *one point close enough, all points one width.*

The statement that uniformity is stronger than continuity is proved directly from its definition: uniformity gives one δ for all pairs of points, and continuity only asks for a δ for pairs one of whose members is a fixed point.

Proposition 11.7. *If f is uniformly continuous on A, then f is continuous at every point of A.*

Proof. Take $c \in A$ and $\varepsilon > 0$. Since f is uniformly continuous, there is $\delta > 0$ such that $|x-y| < \delta$ implies $|f(x)-f(y)| < \varepsilon$ for all $x, y \in A$. Choosing $y = c$, the condition $|x-c| < \delta$ gives $|f(x) - f(c)| < \varepsilon$. That is exactly the definition of continuity at c. □

The second way, used repeatedly in the rest of this chapter, works by contraposition: if f fails to be continuous at some point, two sequences whose distance tends to zero can be built so that their outputs stay apart, and that refutes uniformity. The proof for $1/x$ below uses that pattern in full.

Now an example showing that uniformity depends on the domain, not on the formula of the function.

Proposition 11.8. *$f(x) = x^2$ is uniformly continuous on $[0, 1]$, with $\delta = \varepsilon/2$.*

Proof. For $x, y \in [0, 1]$ we have $|x + y| \le 2$, hence

$$|x^2 - y^2| = |x - y|\,|x + y| \le 2|x - y|.$$

If $|x - y| < \varepsilon/2$, then $|x^2 - y^2| < \varepsilon$. □

That uniformity does not depend on the two endpoints: on $(0, 1)$ the same proof still works, because the bound $|x + y| \le 2$ uses no endpoint at all. The number $\varepsilon/2$ is safe but not the best. For $|x - y| = \delta$ the largest difference occurs at the end of the interval and equals $\delta(2 - \delta)$, so δ is valid provided $\delta \le 1 - \sqrt{1 - \varepsilon}$. For $\varepsilon = 10^{-2}$ that bound is 5.012563×10^{-3}, and with $\delta = \varepsilon/2$ the largest difference is 9.975×10^{-3}.

Proposition 11.9. *$f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.*

Proof. Take $\varepsilon = 1$. Suppose there is $\delta > 0$ that works. Choose a natural number $n > 1/\delta$ and set $x_n = n$, $y_n = n + 1/n$. Then $|x_n - y_n| = 1/n < \delta$, but

$$|x_n^2 - y_n^2| = \left| n^2 - \left(n + \tfrac{1}{n}\right)^2 \right| = 2 + \frac{1}{n^2} > 2 > 1 = \varepsilon.$$

So no δ works for $\varepsilon = 1$. □

The number 2 in that computation is the point: the difference of values does not shrink, it approaches 2, even though the difference of inputs tends to zero. What fails is the rate of change of x^2, which grows without bound, not its continuity. With $\delta = 10^{-4}$ and the pair $x = 10^4$, $y = 10^4 + 5 \times 10^{-5}$, the difference of outputs is 1.0000000149; with $\delta = 10^{-2}$ and $x = 100$, the difference is 1.000025. That number barely moves when δ shrinks ten thousand times, and it is precisely that immobility which proves the failure of uniformity.

An equally important example is $1/x$, because here pointwise continuity never fails.

Proposition 11.10. *$f(x) = 1/x$ is not uniformly continuous on $(0, 1]$, but is uniformly continuous on $[1, \infty)$ with $\delta = \varepsilon$.*

Proof of the first part, with two sequences. Take $x_n = 1/n$ and $y_n = 1/(n + 1)$. Then

$$|x_n - y_n| = \frac{1}{n(n+1)} \longrightarrow 0, \qquad \text{but} \qquad \left| \frac{1}{x_n} - \frac{1}{y_n} \right| = |n - (n+1)| = 1$$

for every n. If f were uniformly continuous, take δ for $\varepsilon = 1$; for large n we have $|x_n - y_n| < \delta$, so the difference of outputs would have to be less than 1. That contradicts the result above. □

Proof of the second part. For $x, y \geq 1$ we have $xy \geq 1$, hence

$$\left| \frac{1}{x} - \frac{1}{y} \right| = \frac{|x - y|}{xy} \leq |x - y|.$$

So $\delta = \varepsilon$ works, and the uniformity comes from one fact only: the product xy is never smaller than 1. □

The pattern is clear. On $[1, \infty)$ the denominator xy is kept away from zero by the lower bound of the domain; on $(0, 1]$ that denominator can be made as small as we like, and that is where uniformity is lost. Uniformity is a property of the pair of function and set, not of the function alone: $1/x$ appears in both propositions with two different answers.

The largest threshold $\delta(c)$ at each point shows the difference as a picture. For $c \in (0, 1]$ and $\varepsilon > 0$, the condition $1/(c - \delta) - 1/c \leq \varepsilon$ is equivalent to $\delta \leq \varepsilon c^2/(1 + \varepsilon c)$, so the largest threshold at c is $\delta(c) = \varepsilon c^2/(1 + \varepsilon c)$, and Figure 11.3 plots it together with one fixed value $\delta = \varepsilon/2$ that works for x^2 on $[0, 1]$. The first curve falls along a straight line with slope two on the log-log scale, from 9.901×10^{-3} at $c = 1$ to 1.000×10^{-10} at $c = 10^{-4}$; the second line is horizontal. Uniformity on $[0, 1]$ therefore reads as a horizontal line, and its failure on $(0, 1]$ as a curve that never touches the floor.

Do it by hand (kinesthetic learner)

Materials: a corridor two meters long, chalk or paper tape, and a ruler.
Mark the floor at the point 0, then walk with a stride that is continually halved: half a meter, a quarter of a meter, an eighth of a meter, and so on. Mark every footfall. After six or seven steps, stop and look at the marks from the end of the corridor.
What needs to be seen: the marks cluster near the end, and the remaining distance

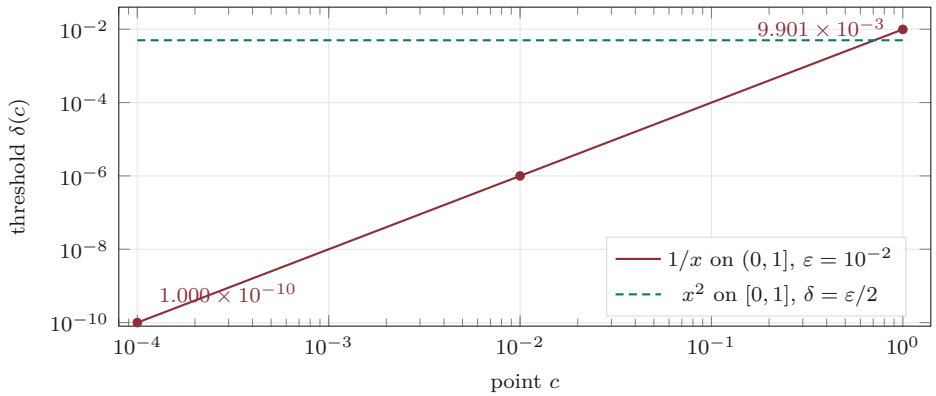


Figure 11.3: The largest threshold δ at each point c for $1/x$ on $(0,1]$ with $\varepsilon = 10^{-2}$, namely $\delta(c) = \varepsilon c^2/(1+\varepsilon c)$, on a log-log scale. The value falls from 9.901×10^{-3} at $c = 1$ to 9.999×10^{-7} at $c = 10^{-2}$ and 1.000×10^{-10} at $c = 10^{-4}$, so the graph is a straight line with slope two. The dashed line $\delta = 5 \times 10^{-3}$ is the safe choice $\varepsilon/2$ from the proof for x^2 on $[0,1]$: one and the same number works on that whole interval.

shrinks without ever becoming zero. That rule never finishes the journey, although every step is valid according to its own rule.
Repeat with another rule: a fixed stride of half a meter all the way to the end. The journey is finished in four steps, and one rule holds from the first mark to the last. That is the difference between a rule that shrinks forever and one rule for the whole corridor.

11.6 Compactness, Lipschitz, and the Heine–Cantor theorem

Level: advanced

In everyday language: on a domain that is closed and does not stretch to infinity, ordinary continuity is already enough to give one threshold for all points. To state it, we need one term about the domain.

compact (*kompak*) — a box that is closed tight and does not stick out without end

Definition 11.4 (Compact)**.** A set $K \subseteq \mathbb{R}$ is called *compact* if every family of open sets whose union contains K has a finite subfamily whose union also contains K.

That definition is not practical to use directly. What saves us is its equivalence with two properties that are easy to check: for sets in $\mathbb{R}$, compact is equivalent to closed and bounded. That equivalence is known as the Heine–Borel theorem; we skip its proof, and it is in Rudin, *Principles of Mathematical Analysis*, Chapter 2, section Compact Sets.

Theorem 11.11 (Heine–Cantor)**.** *If f is continuous on a compact set K, then f is uniformly continuous on K.*

Proof. Suppose f is not uniformly continuous. Then there is $\varepsilon > 0$ such that for every n there are $x_n, y_n \in K$ with $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \geq \varepsilon$. Since K is bounded, (x_n) is bounded, and by Bolzano–Weierstrass there is a subsequence $x_{n_k} \to x$. Since K is closed, $x \in K$. From $|x_{n_k} - y_{n_k}| < 1/n_k$ we get $y_{n_k} \to x$ as well, and $y_{n_k} \in K$. Continuity of f at x gives $f(x_{n_k}) \to f(x)$ and $f(y_{n_k}) \to f(x)$, so $|f(x_{n_k}) - f(y_{n_k})| \to 0$. That contradicts $|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon$. So the supposition is false. □

That proof uses the two properties of compactness separately: boundedness to trigger Bolzano–Weierstrass, and closedness to ensure the limit point is still inside K. Both are

needed. On $(0,1)$, the function $1/x$ is continuous but not uniform, and its domain is not closed. On $\mathbb{R}$, the function x^2 is continuous but not uniform, and its domain is not bounded.

The following class of functions gives a shortcut that is often faster than checking the definition.

Definition 11.5 (Lipschitz)**.** A function $f\colon A \to \mathbb{R}$ is called *Lipschitz* if there is a constant $L \geq 0$ such that $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in A$.

Proposition 11.12. *If f is Lipschitz on A, then f is uniformly continuous on A.*

Proof. Take $\varepsilon > 0$. If $L = 0$, the function is constant and any choice of δ works. If $L > 0$, take $\delta = \varepsilon/L$. For $|x - y| < \delta$ we have $|f(x) - f(y)| \leq L|x - y| < \varepsilon$. □

The converse does not hold, and the example shows that uniformity is weaker than Lipschitz. The function $f(x) = \sqrt{x}$ is continuous on $[0,1]$, hence uniformly continuous by the Heine–Cantor theorem. But

$$\frac{|\sqrt{x} - \sqrt{0}|}{|x - 0|} = \frac{1}{\sqrt{x}} \longrightarrow \infty \quad (x \to 0^+),$$

so no L satisfies the Lipschitz definition. Thus Lipschitz is a sufficient condition, not a necessary one. Conversely x^2 on $[0,1]$ is Lipschitz with $L = 2$ and $1/x$ on $[1,\infty)$ with $L = 1$, exactly the numbers in the preceding proofs.

Challenge (problem solver)

The two examples of failure of uniformity in this chapter, x^2 on $\mathbb{R}$ and $1/x$ on $(0,1]$, are both unbounded on their domains. Find a function that is *bounded* on $(0,1)$, continuous at every point, but not uniformly continuous there, then prove it.
Hint 1: failure does not require values that grow; values that oscillate are enough, with an oscillation that gets denser as x approaches the end of the interval.
Hint 2: try $f(x) = \sin(1/x)$ with $x_n = 1/(2\pi n + \pi/2)$ and $y_n = 1/(2\pi n - \pi/2)$. The function values are 1 and -1, while $|x_n - y_n| = \pi/((2\pi n)^2 - (\pi/2)^2)$ tends to zero.
If you are stuck, go back to the section Uniform continuity, the proof for $1/x$ with two sequences.

11.7 A function that is continuous at one point only

Level: advanced

Continuity is not built into all functions; it can fail on a very large set of points. The cheapest example is built from the density of the rationals in Chapter 2.

Example 11.1. Define $f\colon \mathbb{R} \to \mathbb{R}$ by $f(x) = x$ if x is rational and $f(x) = 0$ if x is irrational. Then f is continuous at 0 and not continuous at any other point.

Proof. At $x = 0$ we have $|f(x) - f(0)| = |f(x)| \leq |x|$, because $f(x)$ always equals either x or 0. So $\delta = \varepsilon$ works and f is continuous at 0.

Now take $c \neq 0$. Since $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are both dense, there is a rational sequence $r_n \to c$ and an irrational sequence $s_n \to c$. Then $f(r_n) = r_n \to c$ while $f(s_n) = 0 \to 0$. Two sequence limits differ, so by the sequence criterion $\lim_{x \to c} f(x)$ does not exist, and f is not continuous at c. □

A close relative of that example is the indicator of the rationals, which equals 1 at rationals and 0 at irrationals; that function is continuous at no point at all. The next example is more interesting, because its set of continuity points is in fact large; it is known as a modification of the Dirichlet function.

Example 11.2 (Thomae's function, a Dirichlet modification). Define $t\colon [0,1] \to \mathbb{R}$ by $t(x) = 0$ if x is irrational, and $t(x) = 1/q$ if $x = p/q$ in lowest terms with $q \geq 1$. Then t is continuous at every irrational point and not continuous at every rational point.

Proof. Take a irrational and $\varepsilon > 0$. Choose N with $1/N < \varepsilon$. The rational numbers in $[0,1]$ with denominator $q \leq N$ are finite in number, so among them there is one closest to a; call its distance $\eta > 0$. Since a is irrational, that distance is not zero. For $|x - a| < \eta$ there are two possibilities. If x is irrational, then $|t(x) - t(a)| = 0 < \varepsilon$. If $x = p/q$ with $q > N$, then $|t(x) - t(a)| = 1/q < 1/N < \varepsilon$. So t is continuous at a.

Now take $a = p/q$ rational. An irrational sequence $s_n \to a$ gives $t(s_n) = 0 \to 0$, while $t(a) = 1/q > 0$. The limit along that sequence is not equal to the value of the function, so t is not continuous at a. □

Both examples show that the set of discontinuity points can be the whole line, the set of rationals, or a single point. The full classification, which states that this set is always a countable union of closed sets, is in Rudin, Chapter 4, section Discontinuities. Continuity that fails at many points is not an oddity but a structured symptom.

11.8 A worked example: a delta that depends on the point

Level: intermediate

We return to $f(x) = x^2$ at $x_0 = 2$ with $\varepsilon = 10^{-3}$, this time down to the last digit.

Step-by-step recipe (procedural learner)

Three steps for writing an ε-δ proof of $|f(x) - f(c)| < \varepsilon$:

1. Work backward. Write $|f(x) - f(c)|$ as $|x - c|$ times a remainder, then find an upper bound for that remainder on the domain in use.
2. Bound $|x - c| < 1$ first (or $< r$) so that the remainder is bounded. Call its upper bound M, then set $\delta = \min(1, \varepsilon/M)$.
3. Write the proof forward. Take $\varepsilon > 0$, choose that δ, suppose $|x - c| < \delta$, then derive $|f(x) - f(c)| < \varepsilon$ line by line.

Steps 1 and 3 must not be swapped. The sentences of the proof are read from ε to δ, but δ is found in the opposite direction; writing directly from ε usually produces a δ that still contains x, and a choice like that is not a valid δ.

The condition that must be satisfied is

$$|x^2 - 4| = |x - 2|\,|x + 2| < \varepsilon$$

for all x with $|x - 2| < \delta$. If $\delta < 1$, then $|x + 2| < 4 + \delta$, so it is enough to demand $\delta(4 + \delta) = \varepsilon$. That quadratic equation has the positive root

$$\delta = \sqrt{4 + \varepsilon} - 2.$$

For $\varepsilon = 10^{-3}$ that formula gives $\delta = 2.49984376953 \times 10^{-4}$. A check at three points:

$$\begin{aligned} |x - 2| &= 2.497343925761 \times 10^{-4} \quad \Rightarrow \quad |x^2 - 4| = 9.989999375719 \times 10^{-4}, \\ |x - 2| &= 2.499843769530 \times 10^{-4} \quad \Rightarrow \quad |x^2 - 4| = 1.000000000000 \times 10^{-3}, \\ |x - 2| &= 2.502343613300 \times 10^{-4} \quad \Rightarrow \quad |x^2 - 4| = 1.001000062555 \times 10^{-3}. \end{aligned}$$

The second row shows that the largest δ produces a difference of exactly ε; the third row, that going slightly past it already violates the condition. The popular choice $\delta = \varepsilon/4 = 2.5 \times 10^{-4}$ turns out to *fail*: the difference is 1.0000625×10^{-3}, larger than ε by 6.25×10^{-8}. A failure that small is easily missed when checked with three-digit truncation.

Now notice how δ depends on the point. For $\varepsilon = 10^{-2}$ and $f(x) = x^2$, the positive root of $\delta(2x_0 + \delta) = \varepsilon$ is $\delta = \sqrt{x_0^2 + \varepsilon} - x_0$.

x_0	largest δ	δ/x_0	$\varepsilon/(2x_0)$
1	4.987562×10^{-3}	4.987562×10^{-3}	5.000000×10^{-3}
2	2.498439×10^{-3}	1.249220×10^{-3}	2.500000×10^{-3}
10	4.999875×10^{-4}	4.999875×10^{-5}	5.000000×10^{-4}
100	4.999999×10^{-5}	4.999999×10^{-7}	5.000000×10^{-5}
1000	5.000000×10^{-6}	5.000000×10^{-9}	5.000000×10^{-6}
10^4	5.000002×10^{-7}	5.000002×10^{-11}	5.000000×10^{-7}

The third column shrinks as fast as $1/x_0^2$, and for large x_0 the value of δ is practically the same as $\varepsilon/(2x_0)$. So there is no positive lower bound for δ as x_0 grows: the rate of change of x^2 at x_0 is $2x_0$, and that rate is what sets the threshold, not rounding.

The failure of uniformity of x^2 on $\mathbb{R}$ was proved in the section Uniform continuity and is repeated by machine in the Computational lab, part 3, so the numbers are not repeated here. What has not yet appeared as a picture is the dependence on the domain, and that is shown by $1/x$. On $[1, \infty)$ with $\delta = \varepsilon$ the largest difference occurs at $x = 1$ and equals $\varepsilon/(1 + \varepsilon)$, namely 9.900990×10^{-3} for $\varepsilon = 10^{-2}$. On $(0, 1]$ the pair $x = 1/n$, $y = 1/(n + 1)$ gives an output difference of exactly 1 while the input distance shrinks:

n	$\lvert x - y \rvert$	$\lvert 1/x - 1/y \rvert$
10	9.091×10^{-3}	1
100	9.901×10^{-5}	1
1000	9.990×10^{-7}	1
10^4	9.999×10^{-9}	1

For $f(x) = x^2$ and $\varepsilon = 10^{-2}$, compute the largest δ at $x_0 = 1000$ from the formula $\sqrt{x_0^2 + \varepsilon} - x_0$, then compare with $\varepsilon/(2x_0)$. Check that the difference at $x = x_0 + \delta$ is exactly ε, and explain why that formula cannot be used to prove uniformity on $\mathbb{R}$.

11.9 Computational lab: measuring uniformity

Level: advanced

The eight checks below turn the previous sections into numbers that can be rechecked: explicit δ and its bound, its dependence on the point, x^2 and $1/x$ on two domains, the largest threshold at each point, and bisection for the intermediate value theorem.

Computational lab

```
import math

# 1. optimal delta for f(x)=x^2 at x0=2, eps=1e-3
def delta_sq(x0, eps):
    return math.sqrt(x0*x0 + eps) - x0

eps = 1e-3
d = delta_sq(2.0, eps)
print("== 1. explicit delta, x0=2, eps=1e-3 ==")
print("optimal delta = %.12f" % d)
```

```
for t in (d*0.999, d, d*1.001):
    print("  x-2=%.12e  |x^2-4|=%.12e" % (t, abs((2+t)**2-4)))
print("  delta=eps/4 -> |x^2-4|=%.12e" % abs((2+eps/4)**2-4))

# 2. delta depending on the point
print("== 2. delta(x0) for eps=1e-2 ==")
print("      x0       delta     delta/x0")
for x0 in (1.0, 2.0, 10.0, 100.0, 1000.0, 1e4):
    d = delta_sq(x0, 1e-2)
    print("%8.0f  %.6e  %.6e" % (x0, d, d/x0))

# 3. failure of uniformity of x^2 on R
print("== 3. x^2 on R, eps=1 ==")
print("   delta   x=1/delta   |x^2-(x+d/2)^2|")
for delta in (1e-1, 1e-2, 1e-3, 1e-4):
    x = 1.0/delta
    print("%8.0e %9.0f   %.10f" % (delta, x, abs(x*x-(x+delta/2)**2)))

# 4. x^2 on [0,1]
print("== 4. x^2 on [0,1], |x-y|=delta ==")
print("   delta    2d-d^2       grid")
for delta in (1e-1, 1e-2, 1e-3):
    n = int(round(1/delta))
    g = max(abs((i*delta/n)**2-(i*delta/n+delta)**2)
            for i in range(0, int(n*(1-delta)/delta)+1))
    print("%8.0e  %.10e %.10e" % (delta, 2*delta-delta*delta, g))

# 5. 1/x on [1,inf) and on (0,1]
print("== 5. 1/x on [1,inf), delta=eps ==")
for eps in (1e-1, 1e-2, 1e-3):
    s = max(abs(1/x-1/(x+eps)) for x in [1+i*eps/500 for i in range(501)])
    print("eps=%.0e  sup=%.10e  eps/(1+eps)=%.10e" % (eps, s, eps/(1+eps)))
print("== 6. 1/x on (0,1]: x=1/n, y=1/(n+1) ==")
print("      n      |x-y|      |1/x-1/y|")
for n in (10, 100, 1000, 10000):
    print("%7d   %.3e       %.1f" % (n, 1/n-1/(n+1), 1.0))

# 7. largest valid delta at c, f=1/x, eps=1e-2
print("== 7. delta(c) for 1/x, eps=1e-2 ==")
print("       c      delta(c)    delta/c")
for c in (1.0, 1e-1, 1e-2, 1e-3, 1e-4):
    dd = 1e-2*c*c/(1+1e-2*c)
    print("%8.0e  %.4e  %.4e" % (c, dd, dd/c))

# 8. numerical IVT for x^3+x-1=0
print("== 8. root of x^3+x-1=0 by bisection ==")
lo, hi = 0.0, 1.0
for k in range(1, 22):
    mid = (lo+hi)/2
    if mid**3+mid-1 < 0:
        lo = mid
    else:
        hi = mid
    if k in (1, 5, 10, 15, 20, 21):
        print("k=%2d  [%.12f, %.12f]  width=%.3e"
              % (k, lo, hi, hi-lo))
print("f(lo)=%.3e  f(hi)=%.3e" % (lo**3+lo-1, hi**3+hi-1))
```

The output, as it stands:

```
== 1. explicit delta, x0=2, eps=1e-3 ==
optimal delta = 0.000249984377
  x-2=2.497343925761e-04  |x^2-4|=9.989999375719e-04
  x-2=2.499843769530e-04  |x^2-4|=1.000000000000e-03
  x-2=2.502343613300e-04  |x^2-4|=1.001000062555e-03
  delta=eps/4 -> |x^2-4|=1.000062499999e-03
== 2. delta(x0) for eps=1e-2 ==
      x0       delta     delta/x0
       1  4.987562e-03  4.987562e-03
       2  2.498439e-03  1.249220e-03
      10  4.999875e-04  4.999875e-05
     100  4.999999e-05  4.999999e-07
    1000  5.000000e-06  5.000000e-09
   10000  5.000002e-07  5.000002e-11
== 3. x^2 on R, eps=1 ==
   delta   x=1/delta   |x^2-(x+d/2)^2|
   1e-01         10   1.0025000000
   1e-02        100   1.0000250000
   1e-03       1000   1.0000002499
   1e-04      10000   1.0000000149
== 4. x^2 on [0,1], |x-y|=delta ==
   delta    2d-d^2       grid
   1e-01  1.9000000000e-01 1.9000000000e-01
   1e-02  1.9900000000e-02 1.9900000000e-02
   1e-03  1.9990000000e-03 1.9990000000e-03
== 5. 1/x on [1,inf), delta=eps ==
eps=1e-01  sup=9.0909090909e-02  eps/(1+eps)=9.0909090909e-02
eps=1e-02  sup=9.9009900990e-03  eps/(1+eps)=9.9009900990e-03
eps=1e-03  sup=9.9900099900e-04  eps/(1+eps)=9.9900099900e-04
== 6. 1/x on (0,1]: x=1/n, y=1/(n+1) ==
      n      |x-y|      |1/x-1/y|
     10   9.091e-03       1.0
    100   9.901e-05       1.0
   1000   9.990e-07       1.0
  10000   9.999e-09       1.0
== 7. delta(c) for 1/x, eps=1e-2 ==
       c      delta(c)    delta/c
   1e+00  9.9010e-03  9.9010e-03
   1e-01  9.9900e-05  9.9900e-04
   1e-02  9.9990e-07  9.9990e-05
   1e-03  9.9999e-09  9.9999e-06
   1e-04  1.0000e-10  1.0000e-06
== 8. root of x^3+x-1=0 by bisection ==
k= 1  [0.500000000000, 1.000000000000]  width=5.000e-01
k= 5  [0.656250000000, 0.687500000000]  width=3.125e-02
k=10  [0.681640625000, 0.682617187500]  width=9.766e-04
k=15  [0.682312011719, 0.682342529297]  width=3.052e-05
k=20  [0.682327270508, 0.682328224182]  width=9.537e-07
k=21  [0.682327747345, 0.682328224182]  width=4.768e-07
f(lo)=-1.354e-07  f(hi)=1.007e-06
```

There are four things to read from that output, and all of them were predicted before the program ran. Part 1 confirms that the optimal δ produces a difference of exactly 10^{-3}, whereas $\delta = \varepsilon/4$ produces 1.0000625×10^{-3}: that popular choice does fail, not merely look untidy. Part 2 confirms that the column δ/x_0 falls as fast as $1/x_0^2$ up to $x_0 = 1000$, then stops following the pattern at $x_0 = 10^4$. At the last point the true value is $4.99999999987 \times 10^{-7}$, while float64 prints 5.000002×10^{-7}; that is cancellation error in $\sqrt{x_0^2 + \varepsilon} - x_0$, not a property of the function, and that is why such a formula is not used at large scale.

Parts 3 through 7 show two faces of the same function. On $\mathbb{R}$ the difference for x^2 stays above 1 for every δ tried; on $[0,1]$ the largest difference is $2\delta - \delta^2$, and the grid prints the same number to ten digits because the largest difference occurs at $x = 1 - \delta$, a point that is in the grid list. On $[1,\infty)$ the difference for $1/x$ is exactly $\varepsilon/(1+\varepsilon)$, whereas on $(0,1]$ it stays 1.0 while the input distance shrinks to 9.999×10^{-9}. Part 8 closes with interval width 2^{-21} after twenty-one bisections, and the root of $x^3 + x - 1$ is trapped inside it.

The program does not prove a single one of those things. It checks claims that have already been proved, and in two places it shows the limits of itself: the failure of $\delta = \varepsilon/4$ only becomes visible at the eighth digit, and cancellation error only becomes visible at $x_0 = 10^4$.

Note for instructors

The teaching order I suggest: start from the grid in the chapter opening, not from the definition. Write 10000 and 5000 on the board, ask what grid spacing is needed at that point, then compare with the answer at $x = 1$; from that gap the ε-δ definition appears as the answer to a question that has already been felt, not as a sentence to memorize. After that continuity, the algebraic properties, the intermediate value theorem, the extreme value theorem, then uniform continuity and Heine–Cantor.

Two misconceptions appear most often. First, students believe δ may depend on x inside the definition of continuity, then "prove" uniformity by choosing a different δ at every point; the symptom shows when they cannot name one δ for two distant points. Second, they think the intermediate value theorem gives a way of computing a root, whereas it only guarantees existence.

A blackboard activity that works: ask two students to play challenger and responder for $f(x) = x^2$ at $x_0 = 2$ and at $x_0 = 1000$ with the same ε, then have the class guess at which point the responder runs out of numbers. The sign that the class is ready to move on: they can name one explicit δ for $1/x$ on $[1,\infty)$ and explain why the same δ fails on $(0,1]$.

11.10 Summary

Level: basic

Summary.

- $\lim_{x \to c} f(x) = L$ means that for every $\varepsilon > 0$ there is $\delta > 0$ such that $0 < |x - c| < \delta$ gives $|f(x) - L| < \varepsilon$. The value at c itself is not counted, and the limit is unique.
- f is continuous at c if the limit exists and equals $f(c)$. Continuity is local, closed under algebraic operations, and preserved under composition.
- On a closed bounded interval, a continuous function is bounded, attains its maximum and minimum, and takes all values in between. All three can fail on an open interval.
- f is uniformly continuous if one δ works for all pairs of points. Uniformity is stronger than continuity, and uniformity is always relative to its domain.
- x^2 is uniform on $[0,1]$ but not on $\mathbb{R}$; $1/x$ is uniform on $[1,\infty)$ but not on $(0,1]$. A function continuous on a compact set is always uniform, and a Lipschitz function is always uniform.
- Continuity can fail on a large set of points: the rational indicator fails everywhere, x at rationals and 0 at irrationals fails at every point except zero.

11.11 Exercises

Level: intermediate

Exercise 11.1.

1. Prove from the definition that $\lim_{x\to 3}(2x-1)=5$, and give the smallest explicit δ as a function of ε. Explain why $\delta=\varepsilon/2$ is the largest valid choice, whereas $\delta=\varepsilon/3$ is valid but smaller than necessary.
2. Prove that $f(x)=x^2$ is continuous at every point $c\in\mathbb{R}$ with $\delta=\min(1,\varepsilon/(2|c|+1))$, then check that formula at $c=2$ and $c=100$ for $\varepsilon=10^{-2}$.
3. Assess this claim: "Because $1/x$ is continuous at every point of $(0,1]$, that function is uniformly continuous on $(0,1]$." True, false, or not enough evidence? Name the part that is correct and the part that is wrong.
4. Prove that if f and g are uniformly continuous on A, then $f+g$ is uniformly continuous on A. Show by example that fg need not be uniformly continuous on an unbounded A.
5. Prove that $f(x)=\sqrt{x}$ is uniformly continuous on $[0,\infty)$ by using $|\sqrt{x}-\sqrt{y}|\le\sqrt{|x-y|}$. Explain why the proof does not require the domain to be closed.
6. Compute the largest δ for $f(x)=x^3$ at $x_0=1$ with $\varepsilon=10^{-2}$ by solving $(1+\delta)^3-1=\varepsilon$, then compare with $\delta=\varepsilon/3$. Check both values in Python.
7. For $f(x)=x^2$ on $[0,1]$ and $\varepsilon=10^{-2}$, compute the largest difference for $\delta=\varepsilon/2$ and for $\delta=1-\sqrt{1-\varepsilon}$. Show that the first choice is safe but not optimal, and that the second choice produces a difference of exactly ε.
8. Prove that if f is continuous on $[a,b]$ and $f(x)\neq 0$ for all $x\in[a,b]$, then there is $m>0$ with $|f(x)|\ge m$ for all x. Use the extreme value theorem, and explain why the conclusion fails for $f(x)=x$ on $(0,1)$.
9. Define $f(x)=x$ if x is rational and $f(x)=-x$ if x is irrational. Prove that f is continuous only at 0, then show that $|f|$ is continuous at every point. Explain why the second result does not contradict the first.

Hints for the three hardest items. Item 4: for the second part, take $f(x)=g(x)=x$ on $\mathbb{R}$ and use the pair $x_n=n$, $y_n=n+1/n$. Item 5: square $|\sqrt{x}-\sqrt{y}|\le\sqrt{|x-y|}$ to see where it comes from. Item 9: $|f|$ is none other than $x\mapsto|x|$, which is continuous everywhere; for $c\neq 0$ use rational and irrational sequences tending to c.

For researchers: limits and next steps

What is proved in this chapter: uniformity is stronger than continuity; continuity on a compact set gives uniformity; continuity on a closed bounded interval gives boundedness, extreme values, and the intermediate value property. What is popular but not proved is the impression that the intermediate value theorem gives an algorithm: it only gives existence, and the rate 2^{-n} comes from the procedure, not from the theorem.

Primary sources: Rudin, *Principles of Mathematical Analysis*, Chapter 4, section Continuity and Compactness, which proves Heine–Cantor in the framework of general metric spaces; and Abbott, *Understanding Analysis*, Chapter 4, sections Continuous Functions on Compact Sets and Sets of Discontinuity. The proof of Heine–Borel is deliberately skipped here and is in Rudin, Chapter 2, section Compact Sets.

Limit of validity: the whole chapter works in $\mathbb{R}$ with the usual distance. In a metric space, the definition of uniformity stays the same word for word, but the Heine–Cantor theorem demands compactness, not merely closedness and boundedness; in an infinite-dimensional space those two properties separate. In a complete metric space, uniformity on a dense set is enough to extend a function uniquely, and that is the door into the extension theorems not discussed here.

A question still open for the reader: how fast the modulus of continuity $\omega(\delta)=\sup_{|x-y|<\delta}|f(x)-f(y)|$ may fall. For a function continuous on a closed interval, $\omega(\delta)\to 0$ is guaranteed, but the rate of its decrease is not: there are continuous functions whose modulus falls more slowly than every power δ^{α}. The Lipschitz guarantee that is often

assumed on adversarial attack bounds in Chapter 29 is therefore sufficient, not necessary.

Map to source texts: Chapter 11 in Rudin, Principles of Mathematical Analysis*: Chapter 4 (Continuous Functions, Continuity and Compactness, Continuity and Connectedness, Discontinuities), with the Heine–Borel theorem taken from Chapter 2, section Compact Sets. In Abbott,* Understanding Analysis*: Chapter 4 (Discussion: Examples of Dirichlet and Thomae, Functional Limits, Combinations of Continuous Functions, Continuous Functions on Compact Sets, The Intermediate Value Theorem, Sets of Discontinuity). The ε-δ definition and the sequence criterion are built from the sequences in Chapter 4 of this book, whereas the supremum used to prove the intermediate value theorem comes from Chapter 3. The derivative in Chapter 12 continues this chapter by using the difference quotient as a limit, and Chapter 27 uses uniformity as one of the guarantees that is often assumed without being checked.*

Chapter 12

Derivatives: One and Several Variables

A position logger on a bicycle writes one point every second. To obtain the velocity, we divide the difference in distance by the difference in time. Two readings one second apart and four meters apart give four meters per second, even though that is only an average velocity. If the separation is one hundredth of a second, the difference in distance is a few centimeters while the instrument noise is several meters; the quotient jumps to hundreds of meters per second, even with a negative sign. The same division, two different answers.

The derivative resolves that tension by moving the question to the limit, where noise does not enter the count. But the limit cannot be measured: what is measurable is only the quotient at a finite separation, and it carries two errors at once, truncation because the separation is not zero and rounding because the machine stores only so many digits. The derivative is the exact mathematical object; the quotient is its approximation, with an error that can be computed and controlled.

The second difficulty appears when there is more than one input. A credit scoring model receives twelve inputs, and shifting one input gives one number for the rate of change. Shifting two inputs at once is not the same as adding those two numbers, unless the function behaves well; later we shall see an example that fails, and that function is still continuous. The correct answer is not a list of twelve numbers, but a single linear map on the shift vector: that is the true derivative, and its matrix is the Jacobian. The continuity of Chapter 11 does not say how fast something changes.

For younger readers (age 10 and up)

Have you ever stood on a curved roof? If you step forward, you climb. If you step sideways, you can stay at the same height. And yet you are standing at the same point, on the same roof. So the question “how steep is this place” is not complete before we name the direction of the step.

There is another problem. If you look at only two points on the roof, and then divide the difference in height by the difference in distance, you get an average slope, not the slope at a single point. To get the slope exactly under your feet, the two points must be brought closer and closer. The closer they are the more exact it is, but if they are too close your measurement numbers fall apart because the measuring instrument is not perfect.

In this chapter we shall learn why the slope at a single point needs a direction, why the steepest direction always exists, and why a computing tool whose step is too tight is precisely wrong.

12.1 The derivative as a limit and its two meanings

LEVEL: BASIC

The instantaneous rate of change is the number approached by the quotient of two differences as the difference in the input shrinks. The following definition merely writes that sentence down carefully, because the quotient itself is undefined when its denominator is zero.

derivative (*turunan*) — The slope of the line that hugs the graph most tightly at a single point, which is also the instantaneous rate of change at that point.

Definition 12.1 (The derivative at a single point)**.** Suppose f is defined on an open interval containing c. The derivative of f at c is

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h},$$

provided that limit exists and is finite. In this case f is called *differentiable* at c. Other notation used in this book: $Df(c)$ and $\dfrac{df}{dx}(c)$.

That quotient is defined for every $h \neq 0$, so there is no division by zero; what may fail is the limit. The failure is real: for $f(x) = |x|$ at $c = 0$ the quotient is 1 for $h > 0$ and -1 for $h < 0$, so the two-sided limit does not exist.

Physically, the derivative is the instantaneous rate of change: $s'(t)$ is velocity when $s(t)$ is position, $Q'(t)$ is current when $Q(t)$ is the charge that flows, and $C'(q)$ is marginal cost when $C(q)$ is the cost of producing q units.

Geometrically, the quotient is the slope of the secant from $(c, f(c))$ to $(c+h, f(c+h))$. As $h \to 0$, that secant rotates about the first point toward a single position; the line in the limiting position is called the tangent line, and its slope is $f'(c)$. The equation of the tangent line at c is

$$y = f(c) + f'(c)(x - c).$$

The tangent line is not "the line that touches the graph exactly once", for with $f(x) = x^3$ at $c = 0$ the tangent line $y = 0$ crosses the graph at the origin. What is correct: its difference from the graph shrinks faster than the distance does.

That rotation can be followed numerically. For $f(x) = x^2$ at $c = 2$, the secant with $h = 1$ has slope 5, with $h = 0.5$ the slope is 4.5, and the tangent line has slope 4. Figure 12.1 draws all three: from 5 to 4.5, settling at 4.

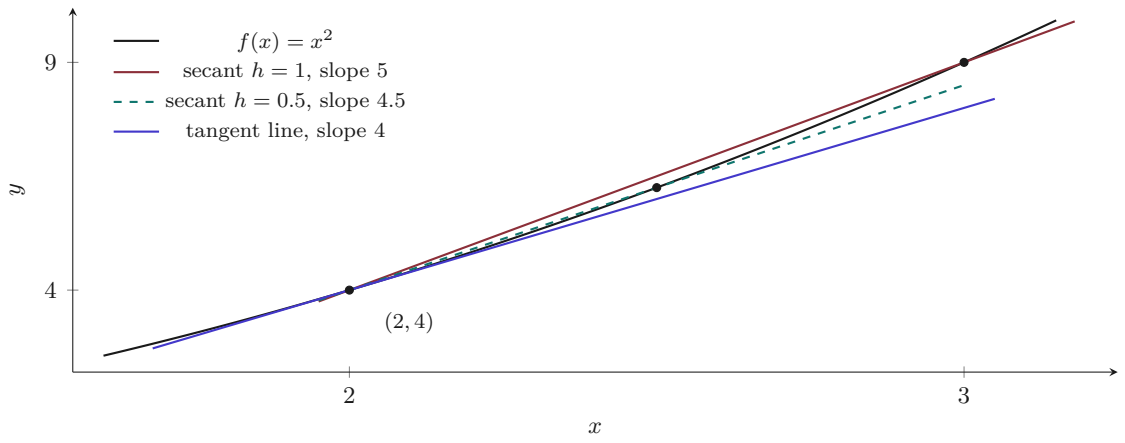


Figure 12.1: Secants rotating into the tangent line of $f(x) = x^2$ at $(2, 4)$. Their slopes drop from 5 (for $h = 1$) to 4.5 (for $h = 0.5$) and settle at 4 as $h \to 0$.

Read it aloud (auditory learner)

Two voices take turns; the answer gets shorter every time.
— How fast is it moving? — From which point to which point?
— From the first second to the second second. — Three meters.
— From the first second to second one and a half. — Two and a half meters.
— From the first second to second one and one tenth. — Two and one fifth.
— From the first second to second one and one hundredth. — Almost two.
— From the first second to the first second. — I cannot. The denominator is zero.
Stop there. What changes is not the answer, but the distance between the two points being compared. The distance shrinks, the secant rotates, the number settles: a value that is never reached yet always approached, that is the derivative.
Repeat together: the distance shrinks, the line rotates, the number settles.

Let us compute one derivative from the definition. For $f(x) = x^2$ at $c = 2$,

$$\frac{f(2+h) - f(2)}{h} = \frac{(4 + 4h + h^2) - 4}{h} = \frac{h(4+h)}{h} = 4 + h.$$

Since $h \neq 0$, that division is legitimate, and the limit is 4. For $h = 0.01$ the quotient is 4.01; the difference 0.01 is not an arithmetic mistake but the order-h term that disappears only in the limit. For $f(x) = 1/x$ at $c = 2$,

$$\frac{1/(2+h) - 1/2}{h} = \frac{2 - (2+h)}{2h(2+h)} = \frac{-1}{2(2+h)} \longrightarrow -\frac{1}{4}.$$

In both computations the algebra removes the factor h from the denominator, so that the form $0/0$ becomes a function continuous at $h = 0$.

That quotient form is important, but it does not survive when the variables become many. The form that survives is the linear approximation.

Definition 12.2 (Derivative as the coefficient of a linear approximation)**.** A function f is differentiable at c with derivative L if

$$f(c+h) = f(c) + Lh + r(h), \qquad \text{with} \qquad \lim_{h \to 0} \frac{r(h)}{h} = 0.$$

The remainder $r(h)$ is smaller than h; that is the content of the definition. For one variable this form feels excessive, but only this form survives up to the Jacobian section.

Proposition 12.1. *The two definitions of the derivative above are equivalent, and the number L in the second definition is unique.*

Proof. If the limit of the quotient equals L, set $r(h) = f(c+h) - f(c) - Lh$. Then $r(h)/h$ is exactly the quotient minus L, hence tends to 0. Conversely, if r with that property exists, the quotient equals $L + r(h)/h$, which tends to L.

For uniqueness, suppose L_1 and L_2 both satisfy it. Then $(L_1 - L_2)h = r_2(h) - r_1(h)$, and after dividing by h we obtain $L_1 - L_2 = r_2(h)/h - r_1(h)/h$. The right-hand side tends to 0, whereas the left-hand side does not depend on h. Hence $L_1 = L_2$. □

The first property to hold on to: differentiability is stronger than continuity.

Proposition 12.2. *If f is differentiable at c, then f is continuous at c.*

Proof. For $h \neq 0$ we have

$$f(c+h) - f(c) = h \cdot \frac{f(c+h) - f(c)}{h}.$$

The second factor tends to $f'(c)$ and the first factor tends to 0, so the product tends to 0. In other words $\lim_{h \to 0} f(c+h) = f(c)$. □

The converse does not hold. The function $|x|$ is continuous at 0 but not differentiable there, because the two one-sided quotients have opposite signs. The function $x \sin(1/x)$, equal to 0 at $x = 0$, is also continuous but not differentiable there: its quotient is $\sin(1/x)$, which oscillates without approaching a single value.

Challenge (problem solver)

Suppose $f(0) = 0$ and, for $x \neq 0$,

$$f(x) = |x|\bigl(2 + \sin(1/x)\bigr).$$

Show that f is continuous at 0 but has no derivative there. The two properties do not contradict each other: continuity demands that the values of the function approach $f(0)$, the derivative demands that the difference quotient approach a single number.
Hint 1: for continuity, one inequality that traps $|f(x)|$ between 0 and something tending to 0 is enough.
Hint 2: for the second, compute $f(h)/h$ separately for $h > 0$ and $h < 0$. For $h > 0$ the quotient is $2 + \sin(1/h)$, which oscillates between 1 and 3; for $h < 0$ it oscillates between -3 and -1. The two sets of values do not overlap.
If you get stuck, go back to the definition of the derivative at the beginning of this section and compute the quotient for $h > 0$ and then $h < 0$ for the function $|x|$.

The following computation rules free us from the definition every time.

Theorem 12.3 (Basic rules)**.** *Suppose f and g are differentiable at c. Then*

1. $(f+g)'(c) = f'(c) + g'(c)$ *and* $(af)'(c) = af'(c)$ *for* $a \in \mathbb{R}$;
2. $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$;
3. $\left(\dfrac{f}{g}\right)'(c) = \dfrac{g(c)f'(c) - f(c)g'(c)}{g(c)^2}$ *if* $g(c) \neq 0$;
4. *if n is an integer, the derivative of x^n at $x \neq 0$ is nx^{n-1}.*

Proof. Part (1) is immediate from the limit of a sum. For (2), write

$$f(x)g(x) - f(c)g(c) = f(x)\bigl[g(x) - g(c)\bigr] + g(c)\bigl[f(x) - f(c)\bigr].$$

After dividing by $x - c$ and letting $x \to c$, the factor $f(x)$ tends to $f(c)$ because f is continuous at c (the previous proposition), and both quotients tend to their derivatives.

For (3), it suffices to show that $1/g$ is differentiable with derivative $-g'(c)/g(c)^2$, then apply (2) to $f \cdot (1/g)$. The quotient for $1/g$ is

$$\frac{1/g(x) - 1/g(c)}{x - c} = -\frac{1}{g(x)g(c)} \cdot \frac{g(x) - g(c)}{x - c},$$

which tends to $-g'(c)/g(c)^2$ because g is continuous and $g(c) \neq 0$. Part (4) follows from (2) by induction on $n \geq 0$, then from (3) for $n < 0$. □

12.2 The chain rule

Level: intermediate

If one quantity depends on a second quantity, and the second quantity depends on time, then the rate of change of the first quantity is the product of the two rates. That sentence sounds like everyday arithmetic, and that is indeed the content of the chain rule.

Figure 12.2 uses two gears: the inner wheel turns three times per turn of the middle wheel, and the middle wheel twice per turn of the outer wheel. One turn of the outer wheel turns the inner wheel $3 \times 2 = 6$ times; adding the two ratios gives 5, a number that appears in no turn at all.

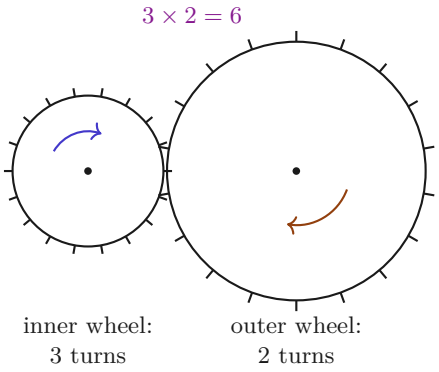


Figure 12.2: The chain rule as two gears. The inner wheel turns 3 times for one turn of the outer wheel, and the outer wheel 2 times for one turn of its output; the total ratio is $3 \times 2 = 6$, not $3 + 2 = 5$.

chain rule (*aturan rantai*) — The rate at the end of the chain equals the product of the rates at every link.

Theorem 12.4 (Chain rule)**.** *Suppose g is differentiable at c and f is differentiable at $g(c)$. Then $f \circ g$ is differentiable at c and*

$$(f \circ g)'(c) = f'(g(c))\, g'(c).$$

Proof. Write $y = g(c)$. Since f is differentiable at y, by the linear approximation definition there is a function v with $v(s) \to 0$ as $s \to y$ and

$$f(s) - f(y) = (s - y)\,[f'(y) + v(s)].$$

In the same way, since g is differentiable at c, there is u with $u(t) \to 0$ as $t \to c$ and

$$g(t) - g(c) = (t - c)\,[g'(c) + u(t)].$$

Substitute $s = g(t)$ in the first equation, then use the second:

$$f(g(t)) - f(g(c)) = (t - c)\,[g'(c) + u(t)]\,[f'(y) + v(g(t))].$$

For $t \neq c$, divide by $t - c$. Since g is continuous at c, $g(t) \to y$, so that $v(g(t)) \to 0$, and the quotient tends to $g'(c) f'(y)$. □

The naive proof that divides by $g(t) - g(c)$ fails when $g(t) = g(c)$ for t near c but $t \neq c$; a locally constant function, for instance, makes that denominator zero at every point. The buffer v above avoids that division altogether.

Short example: for $h(x) = (x^2 + 1)^5$, take $g(x) = x^2 + 1$ and $f(u) = u^5$. Then $h'(x) = 5(x^2 + 1)^4 \cdot 2x$ and $h'(1) = 5 \cdot 16 \cdot 2 = 160$; direct expansion is possible, but unpleasant.

Step-by-step recipe (procedural learner)

1. **Separate the two layers.** Point out which is inside and which is outside. For $k(x) = \sin(x^3 + 2x)$: the inner layer is $g(x) = x^3 + 2x$, the outer layer $f(u) = \sin u$.
2. **Differentiate the outer layer, leave its content intact.** $f'(u) = \cos u$ becomes $\cos(x^3 + 2x)$; do not simplify the content yet.
3. **Multiply by the derivative of the inner layer.** $g'(x) = 3x^2 + 2$, so that $k'(x) = (3x^2 + 2)\cos(x^3 + 2x)$.
4. **Test one point.** At $x = 0$ the formula gives 2, and the central difference with $h = 10^{-3}$ gives 1.9999997.

Warning about order: the outer layer is evaluated at $g(x)$, not at x. Swapping the place of evaluation gives $\cos(x)\,(3x^2 + 2)$, whose value at $x = 1$ is 2.7015, whereas the true value is -4.9499.

Rate example: the area of a circle $A = \pi r^2$ with $\dot r = 0.5$ cm per second. At $r = 10$ cm,

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi \cdot 10 \cdot 0.5 = 10\pi \approx 31.4159 \text{ cm}^2 \text{ per second.}$$

The number 10π is exact; at $r = 20$ cm the same rate gives 20π: the answer depends on r.

12.3 The mean value theorem and its consequences

Level: intermediate

Between any two points on a smooth graph there is always a point whose slope equals the slope of the secant joining those two points. That statement looks like geometric fantasy, yet it is the bridge between the derivative and the function.

The key to its proof is the special case in which the two ends have equal value.

Theorem 12.5 (Rolle)**.** *If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there is $c \in (a, b)$ with $f'(c) = 0$.*

Proof. If f is constant, every $c \in (a, b)$ works. If not, f attains a maximum or a minimum that differs from the value at the ends; since f is continuous on a closed bounded interval, that value is attained at some point, and that point cannot be an endpoint, so $c \in (a, b)$ (extreme value theorem, Chapter 11). At an interior extreme point, the quotient $(f(c+h) - f(c))/h$ is ≤ 0 for small $h > 0$ and ≥ 0 for small $h < 0$. The two inequalities together force its two-sided limit, namely $f'(c)$, to be 0. □

Theorem 12.6 (Mean value)**.** *If f is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Set $s = (f(b) - f(a))/(b - a)$ and $g(x) = f(x) - f(a) - s(x - a)$. Then g is continuous on $[a, b]$, differentiable on (a, b), and $g(a) = g(b) = 0$. By Rolle's theorem there is $c \in (a, b)$ with $g'(c) = 0$, that is, $f'(c) = s$. □

Numerical example: $f(x) = x^3$ on $[0, 3]$ has secant slope $27/3 = 9$, and $3c^2 = 9$ gives $c = \sqrt{3} \approx 1.7321$; there is no guarantee that c sits in the middle.

Do it by hand (kinesthetic learner)

Prepare a thin 30 cm board, two books, a ruler, and a pencil. Prop up the left end of the board to a height of 4 cm, then press the middle slightly so that it curves upward. Hold the ruler upright at 5 cm and at 25 cm from the raised end; the difference in height divided by 20 cm gives one slope number. Repeat with 5 cm and 10 cm: the number changes, because what is being measured is the average slope over different intervals. Now lay the ruler flat at the 15 cm point and rotate it until only one point touches. Lift the other end slowly: the ruler rotates about that point, exactly like the secant in Figure 12.1. The slope of the ruler that touches a single point is the slope at that point, and that is where the mean value theorem promises that such a point exists.

Corollary 12.7 (Monotonicity test)**.** *Suppose f is continuous on $[a,b]$ and differentiable on (a,b). If $f'(x) \geq 0$ for all $x \in (a,b)$, then f is nondecreasing on $[a,b]$. If $f'(x) > 0$ for all x, then f is strictly increasing. If $f'(x) = 0$ for all x, then f is constant.*

Proof. Take $a \leq x < y \leq b$. The mean value theorem on $[x,y]$ gives $c \in (x,y)$ with $f(y) - f(x) = f'(c)(y-x)$. Since $y - x > 0$, the sign of the difference in values follows the sign of $f'(c)$. If f' is zero everywhere, the difference is zero for every pair, so f is constant. □

The direction of the logic deserves attention: this corollary turns information about the derivative into information about the function. The converse does not hold: a strictly increasing function may have zero derivative at some points, such as x^3 at 0. Nor does the conclusion run backward from a single point: $f'(c) = 0$ does not mean that c is an extreme point.

Corollary 12.8 (Lipschitz continuity)**.** *If f is differentiable on an interval I and $|f'(x)| \leq M$ for all $x \in I$, then $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in I$.*

Proof. For $x \neq y$, the mean value theorem on the interval between them gives c with $f(x) - f(y) = f'(c)(x - y)$. Take the absolute value of both sides and use $|f'(c)| \leq M$. □

Therefore a function with bounded derivative is automatically uniformly continuous (take $\delta = \varepsilon / M$), but the converse does not hold: $\sqrt{x}$ is uniformly continuous on $[0,1]$ with an unbounded derivative near zero.

12.4 Taylor polynomials and their remainder

Level: advanced

The tangent line is a polynomial of degree one that clings to the graph at a single point. If we want it to cling longer, we add terms of degree two, three, and so on. The question is not whether the additions improve the approximation, but how large the remainder is.

Definition 12.3 (Taylor polynomial)**.** Suppose f has derivatives up to order n at a. The Taylor polynomial of order n of f at a is

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k.$$

The requirement: P_n and f agree in value and in their first n derivatives at a; since $n+1$ coefficients determine one polynomial of degree n, P_n is unique.

Theorem 12.9 (Taylor with Lagrange remainder)**.** *Suppose f has derivatives up to order $n-1$ that are continuous on $[a,b]$ and a derivative of order n on (a,b). For every $\beta \in (a,b]$ there is $\xi \in (a,\beta)$ with*

$$f(\beta) = P_{n-1}(\beta) + \frac{f^{(n)}(\xi)}{n!}(\beta - a)^n.$$

Proof. Define M by $f(\beta) = P_{n-1}(\beta) + M(\beta - a)^n$, then set

$$g(t) = f(t) - P_{n-1}(t) - M(t-a)^n, \qquad a \le t \le b.$$

P_{n-1} is built from the derivatives of f at a, so $g(a) = \cdots = g^{(n-1)}(a) = 0$, and the choice of M gives $g(\beta) = 0$. Rolle gives $x_1 \in (a,\beta)$ with $g'(x_1) = 0$; since $g'(a) = 0$, Rolle applied to g' on $[a, x_1]$ gives x_2 with $g''(x_2) = 0$. Repeat through n steps: there is $x_n \in (a,\beta)$ with $g^{(n)}(x_n) = 0$. The derivative of order n of P_{n-1} is zero and that of $(t-a)^n$ is $n!$, so $0 = f^{(n)}(x_n) - n!M$, that is, $n!M = f^{(n)}(\xi)$ with $\xi = x_n$. □

For $n = 1$ this theorem is exactly the mean value theorem; what is new is the ability to compute errors of higher order.

Example 12.1. Approximate e by $P_3(1)$ from $f(x) = e^x$ at $a = 0$, then compare with the remainder bound.

Here $f^{(k)}(0) = 1$ for every k, so that

$$P_3(1) = 1 + 1 + \frac{1}{2} + \frac{1}{6} = \frac{8}{3} = 2.6666\ldots$$

The true value is $e = 2.71828\ldots$, so the error is 0.0516152. The Lagrange remainder gives $|e - P_3(1)| = e^{\xi}/24$ for some $\xi \in (0,1)$, and since $e^{\xi} \le e < 3$, the bound is $3/24 = 0.125$. That bound is correct, but loose by a factor of about two and a half. Using order five, $P_5(1) = 2.71666\ldots$ with error 0.0016152, whereas the bound is $3/720 = 0.0041667$.

For younger readers (age 10 and up)

Imagine that you are guessing your friend's route from a few clues only. First clue: where he is now. Your guess is a single point, and that point does not move. Second clue: which way he is stepping. Your guess is a straight line, already closer, but a real route usually curves. Third clue: how fast the direction changes. Your guess is a piece of a parabola, which clings longer.

The more clues, the longer your guess clings, but never exactly. There is always a remainder. That remainder can be bounded without knowing the true route, only from how large the change is at the next level. In this chapter we shall learn why a straight line is the guess of order one, why a parabola is of order two, and why a remainder bound is more useful than an answer that claims to be exact.

12.5 Partial derivatives, the gradient, and directional derivatives

Level: intermediate

When there is more than one input, "rate of change" is not necessarily a single number. It depends on the direction in which we step. The simplest way to obtain a single number is to freeze every coordinate except one, and then use the definition we already have.

Definition 12.4 (Partial derivative)**.** Suppose $f\colon U \to \mathbb{R}$ is defined on an open set $U \subseteq \mathbb{R}^n$ and $a \in U$. The partial derivative of f with respect to x_j at a is

$$\frac{\partial f}{\partial x_j}(a) = \lim_{h \to 0} \frac{f(a + h\,e_j) - f(a)}{h},$$

with e_j the unit vector in the direction of x_j, provided the limit exists.

Geometrically, the partial derivative with respect to x is the slope of the slice of the graph by the plane y constant, hence an ordinary derivative of a function of one variable.

gradient (*gradien*) — An arrow pointing in the steepest direction on a hill, with length equal to the steepness in that direction.

Definition 12.5 (Gradient)**.** If all partial derivatives of f at a exist, the vector

$$\nabla f(a) = \left(\frac{\partial f}{\partial x_1}(a), \dots, \frac{\partial f}{\partial x_n}(a) \right)$$

is called the gradient of f at a.

For $f(x, y) = x^2 y^3$ we have $\partial f/\partial x = 2xy^3$ and $\partial f/\partial y = 3x^2 y^2$, so that $\nabla f(2, -1) = (-4, 12)$.

Definition 12.6 (Directional derivative)**.** For a unit vector $u \in \mathbb{R}^n$, the directional derivative of f at a along u is

$$D_u f(a) = \lim_{t \to 0} \frac{f(a + tu) - f(a)}{t},$$

provided the limit exists.

The partial derivatives are the special case $u = e_j$. For other directions $D_u f$ cannot be predicted from the list of partials, except in one well-behaved situation.

Proposition 12.10. *If f is differentiable at a (in the sense of the linear approximation of the next section), then for every unit vector u we have $D_u f(a) = \nabla f(a) \cdot u$.*

Proof. With $v = tu$ in the linear approximation definition, $f(a+tu) = f(a)+t\,\nabla f(a)\cdot u+r(tu)$, so that

$$\frac{f(a + tu) - f(a)}{t} = \nabla f(a) \cdot u + \frac{r(tu)}{t}.$$

Since $\|r(tu)\|/\|tu\| \to 0$ and $\|tu\| = |t|$, the last term tends to 0. □

Proposition 12.11 (Direction of steepest ascent)**.** *If $\nabla f(a) \neq 0$, then among all unit vectors u the value $D_u f(a)$ is largest at $u = \nabla f(a)/\|\nabla f(a)\|$, and the largest value is $\|\nabla f(a)\|$.*

Proof. By the Cauchy–Schwarz inequality (Chapter 10),

$$|D_u f(a)| = |\nabla f(a) \cdot u| \le \|\nabla f(a)\|\,\|u\| = \|\nabla f(a)\|.$$

Equality holds exactly when u is parallel to $\nabla f(a)$. Since u is a unit vector, only two signs are possible along that direction, and the positive sign gives the largest value. □

For $f(x, y) = x^2 y^3$ at $(2, -1)$, the length of the gradient is $\sqrt{160} \approx 12.6491$, so the direction of steepest ascent is $(-4, 12)/\sqrt{160} \approx (-0.3162,\ 0.9487)$ with rate 12.6491 per unit of distance. The x direction gives -4, the y direction gives 12; both are smaller.

If a curve $t \mapsto \gamma(t)$ stays on the level set $f = \text{constant}$, then $f(\gamma(t))$ is constant and its derivative is zero; the chain rule gives $\nabla f(\gamma(t)) \cdot \gamma'(t) = 0$. Hence the gradient is perpendicular

to the level curve, and the direction of steepest ascent is perpendicular to the contour lines on a map.

For example $f(x, y) = x^2 + 2y^2$ at $(2, -1)$, a point with $f = 6$, so it sits on the level ellipse $x^2 + 2y^2 = 6$. Its gradient is $(2x, 4y) = (4, -4)$, of length $4\sqrt{2} \approx 5.6569$, whereas the tangent direction of the ellipse there is $(1, 1)$ and $(4, -4) \cdot (1, 1) = 0$. Figure 12.3 draws all three: the level ellipse, the gradient arrow, and the tangent line perpendicular to it. The gradient arrow does not run along the contour; it pierces the contour in the direction in which the value of f increases.

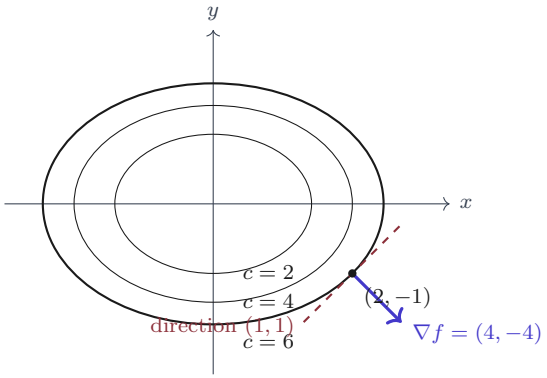


Figure 12.3: Level curves $x^2 + 2y^2 = c$ for $c = 2, 4, 6$ and the gradient at $(2, -1)$. The gradient arrow $(4, -4)$ is perpendicular to the tangent direction $(1, 1)$ of the level curve $c = 6$, because $(4, -4) \cdot (1, 1) = 0$.

12.6 Differentiability, the Jacobian, and the multivariable chain rule

Level: advanced

The true derivative is not a list of partials, but a single linear map that imitates the function near a point; the list of partials can mislead.

Definition 12.7 (Differentiability)**.** Suppose $U \subseteq \mathbb{R}^n$ is open and $F\colon U \to \mathbb{R}^m$. F is differentiable at $a \in U$ if there is a linear map $L\colon \mathbb{R}^n \to \mathbb{R}^m$ with

$$F(a + v) = F(a) + Lv + r(v), \qquad \lim_{v \to 0} \frac{\|r(v)\|}{\|v\|} = 0.$$

The map L is called the derivative of F at a and is written $F'(a)$.

Proposition 12.12. *The map L in that definition is unique if it exists.*

Proof. If L_1 and L_2 both work, then $(L_1 - L_2)v = r_2(v) - r_1(v)$. Take $v = tw$ with w fixed and $t > 0$, divide by t, then let $t \to 0$: this gives $(L_1 - L_2)w = 0$. Since this holds for every w, the two maps are equal. □

Theorem 12.13. *If F is differentiable at a, then all its partial derivatives at a exist, and the matrix of L with respect to the standard bases is*

$$J_F(a) = \begin{pmatrix} \dfrac{\partial F_1}{\partial x_1}(a) & \cdots & \dfrac{\partial F_1}{\partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \dfrac{\partial F_m}{\partial x_1}(a) & \cdots & \dfrac{\partial F_m}{\partial x_n}(a) \end{pmatrix}.$$

Proof. Take $v = te_j$ with $t \neq 0$. The definition gives $F(a + te_j) - F(a) = tLe_j + r(te_j)$, and after dividing by t and letting $t \to 0$, the j-th column of L appears as the limit of a quotient, namely the vector of partial derivatives with respect to x_j. □

Jacobian matrix (*Jacobian*) — A table of all the slopes of a function of several variables: each row is one output, each column is one input.

That $m \times n$ matrix is called the Jacobian matrix of F at a. For $m = 1$ it becomes a single row, namely the transpose of the gradient. Example: $F(x, y) = (x^2 y,\ x + y^2)$ at $(1, 2)$ gives

$$J_F(1,2) = \begin{pmatrix} 2xy & x^2 \\ 1 & 2y \end{pmatrix}_{(1,2)} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix},$$

so that $F(1.01,\ 1.98) \approx (2, 5) + J_F(1, 2)(0.01, -0.02) = (2.02,\ 4.93)$, whereas the exact value is $(2.019798,\ 4.9304)$; the approximation is off by 2×10^{-4} in the first component.

Existence of the partials does not guarantee differentiability; the next two examples show two levels of failure.

Example 12.2 (Partials exist, function discontinuous). Let $f(x, y) = \dfrac{xy}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$. At the origin $f(h, 0) = 0$, so both partial derivatives vanish there. But along the line $y = kx$ the value of the function is $k/(1 + k^2)$: for $k = 1$ the value is $1/2$, for $k = 0$ the value is 0. The limit does not exist, and f is not continuous at the origin.

Example 12.3 (Continuous, partials exist, not differentiable). Let $g(x, y) = \dfrac{xy}{\sqrt{x^2 + y^2}}$ for $(x, y) \neq (0, 0)$ and $g(0, 0) = 0$. Since $2|xy| \leq x^2 + y^2$, we have $|g(x, y)| \leq \tfrac{1}{2}\sqrt{x^2 + y^2}$, so g is continuous at the origin, and both of its partials vanish there. If g were differentiable, its linear map would have to be zero, so the remainder $r(v) = g(v)$ would have to satisfy $\|r(v)\|/\|v\| \to 0$. Take $v = t(1, 1)/\sqrt{2}$ with $t > 0$: then $g(v) = (t^2/2)/t = t/2$, so $g(v)/\|v\| = 1/2$ for every t. That ratio tends to $1/2$, not to 0; hence g is not differentiable at the origin.

The consequence matters: in the second example $\nabla g(0, 0) = 0$ yet the diagonal directional derivative does not exist, because its quotient is $\pm 1/2$ depending on the sign of t. Thus $D_u f = \nabla f \cdot u$ requires differentiability.

A condition that is easy to check and sufficient for differentiability is continuity of the partials.

Theorem 12.14. *If the partial derivatives of F exist and are continuous near a, then F is differentiable at a.*

Proof. It suffices to prove this for $m = 1$ and $n = 2$; the other cases follow component by component with increasing indices. Write $a = (p, q)$ and $v = (h, k)$. Then

$$f(p + h, q + k) - f(p, q) = \big[f(p + h, q + k) - f(p, q + k)\big] + \big[f(p, q + k) - f(p, q)\big].$$

By the mean value theorem in one variable, the first term equals $h\,\partial_x f(\xi, q + k)$ for some ξ between p and $p + h$, and the second term equals $k\,\partial_y f(p, \eta)$ for some η between q and $q + k$. Therefore

$$f(a + v) - f(a) - \nabla f(a) \cdot v = h\big[\partial_x f(\xi, q + k) - \partial_x f(a)\big] + k\big[\partial_y f(p, \eta) - \partial_y f(a)\big].$$

Divide by $\|v\| = \sqrt{h^2 + k^2}$: since $|h|/\|v\| \leq 1$ and $|k|/\|v\| \leq 1$, and both differences in brackets tend to 0 (this is where continuity of the partials is used), the right-hand side divided by $\|v\|$ tends to 0. □

Now composition.

Theorem 12.15 (Multivariable chain rule)**.** *Suppose $H\colon \mathbb{R}^n \to \mathbb{R}^k$ is differentiable at a and $G\colon \mathbb{R}^k \to \mathbb{R}^m$ is differentiable at $H(a)$. Then $G \circ H$ is differentiable at a and*

$$(G \circ H)'(a) = G'(H(a))\, H'(a),$$

that is, the composition of two linear maps; in matrices, $J_{G\circ H}(a) = J_G(H(a))\, J_H(a)$.

Proof. Write $A = H'(a)$, $B = G'(y_0)$ with $y_0 = H(a)$, and

$$H(a+v) = H(a) + Av + r(v), \qquad G(y_0 + w) = G(y_0) + Bw + q(w),$$

with $\|r(v)\| \le \varepsilon(v)\|v\|$ and $\|q(w)\| \le \eta(w)\|w\|$, where $\varepsilon(v) \to 0$ and $\eta(w) \to 0$. For $w = H(a+v) - H(a) = Av + r(v)$ we have $\|w\| \le (\|A\| + \varepsilon(v))\|v\|$, so $w \to 0$ as $v \to 0$. Then

$$G(H(a+v)) - G(H(a)) - BAv = B\big(H(a+v) - H(a) - Av\big) + q(w) = B\,r(v) + q(w).$$

Divide by $\|v\|$:

$$\frac{\|B\,r(v) + q(w)\|}{\|v\|} \le \|B\|\,\varepsilon(v) + (\|A\| + \varepsilon(v))\eta(w),$$

and the right-hand side tends to 0. Hence BA satisfies the definition of the derivative of $G \circ H$ at a. □

The matrix form explains why the order of multiplication may not be reversed: J_H is $k \times n$ and J_G is $m \times k$, so only one order can be multiplied. For scalar functions the result is $\frac{d}{dt} f(\gamma(t)) = \nabla f(\gamma(t)) \cdot \gamma'(t)$.

Finally, second derivatives of a function of several variables are stored in a single matrix.

Hessian matrix (*Hessian*) — A table of curvature: whether the surface bends upward, downward, or in both directions at once.

Definition 12.8 (Hessian matrix)**.** If all second partial derivatives of $f\colon U \to \mathbb{R}$ exist at a, the matrix

$$H_f(a) = \left(\frac{\partial^2 f}{\partial x_i\, \partial x_j}(a) \right)_{i,j=1}^{n}$$

is called the Hessian matrix of f at a.

For a function whose second partials are continuous, $H_f(a)$ is symmetric (Clairaut's theorem, proved in Stewart in the section on partial derivatives). Without continuity, that conclusion can fail.

The Hessian is the derivative of the gradient. If the second partials are continuous near a critical point a ($\nabla f(a) = 0$), then

$$f(a+v) = f(a) + \tfrac{1}{2}\, v^{\top} H_f(a)\, v + o(\|v\|^2),$$

so that the sign of the quadratic form $v \mapsto v^{\top} H v$ determines the type of the critical point.

Proposition 12.16 (Second derivative test)**.** *Suppose f has continuous second partial derivatives near a critical point a ($\nabla f(a) = 0$).*

1. *If $H_f(a)$ is positive definite, a is a strict local minimum.*
2. *If $H_f(a)$ is negative definite, a is a strict local maximum.*
3. *If $H_f(a)$ has eigenvalues of different signs, a is a saddle point.*
4. *If $H_f(a)$ is semidefinite but singular, this test gives no conclusion.*

The first three parts follow from the quadratic expansion above: along a direction with $v^{\top} H v > 0$ the function increases, along a direction with $v^{\top} H v < 0$ it decreases. The complete proof is in Stewart in the section on maximum and minimum values. The fourth part is genuinely empty: $x^4 + y^4$ and $x^4 - y^4$ both have $H = 0$ at the origin, but the first is a minimum and the second a saddle. For two variables, $AC - B^2 > 0$ with $A > 0$ means positive definite, $AC - B^2 < 0$ means indefinite.

12.7 Worked examples

Level: intermediate

The following three computations use the same numbers as the code in the lab section.

First, finite differences for $f(x) = \sin x$ at $x_0 = 1$; the exact value is $\cos 1$, whose digits appear in the code output in the lab section. The two formulas:

$$\text{forward: } \frac{f(x_0+h)-f(x_0)}{h}, \qquad \text{central: } \frac{f(x_0+h)-f(x_0-h)}{2h}.$$

h	forward difference	relative error	central difference	relative error
10^{-1}	0.497363752535	7.947×10^{-2}	0.539402252170	1.666×10^{-3}
10^{-2}	0.536085981012	7.804×10^{-3}	0.540293300875	1.667×10^{-5}
10^{-4}	0.540260231419	7.787×10^{-5}	0.540302304968	1.667×10^{-9}
10^{-6}	0.540301885121	7.787×10^{-7}	0.540302305896	5.130×10^{-11}
10^{-8}	0.540302302898	5.497×10^{-9}	0.540302308449	4.777×10^{-9}
10^{-10}	0.540302247387	1.082×10^{-7}	0.540302247387	1.082×10^{-7}

The pattern of the errors is no accident. For the forward difference, the Taylor expansion gives

$$\frac{f(x_0+h)-f(x_0)}{h} = f'(x_0) + \frac{h}{2}f''(x_0) + \mathcal{O}(h^2),$$

so its relative error is of order h and drops by a factor of ten each time h is divided by ten: 7.947×10^{-2}, 7.804×10^{-3}, 7.787×10^{-5}. The numbers can be matched: $\frac{h}{2}f''(1) = -\frac{1}{2} \cdot 10^{-2} \sin 1 = -4.207 \times 10^{-3}$, and $4.207 \times 10^{-3}/0.5403 = 7.787 \times 10^{-3}$. The central difference gives

$$\frac{f(x_0+h)-f(x_0-h)}{2h} = f'(x_0) + \frac{h^2}{6}f'''(x_0) + \mathcal{O}(h^4),$$

so its error is of order h^2 and drops by a factor of a hundred: 1.666×10^{-3}, 1.667×10^{-5}, 1.667×10^{-9}. Here $\frac{10^{-4}}{6}\cos 1 = 9.005 \times 10^{-6}$, and $9.005 \times 10^{-6}/0.5403$ again gives 1.667×10^{-5}.

At $h = 10^{-8}$ and 10^{-10} the central difference deteriorates again: the truncation error is already smaller than the rounding error, which grows like ε/h, with $\varepsilon \approx 2.2 \times 10^{-16}$. The best entry in the table, $h = 10^{-6}$ with error 5×10^{-11}, has an order consistent with $\varepsilon^{1/3} \approx 6 \times 10^{-6}$.

Second, the gradient of $f(x, y) = x^2 y^3$ at $(2, -1)$ with analytic value $(-4, 12)$. The central difference gives an error of 3.333×10^{-5} in the y component for $h = 10^{-2}$, dropping to 3.333×10^{-9} for $h = 10^{-4}$: order h^2. The x component is exact to machine error, because f is quadratic in x.

Third, the chain rule on a composition of different dimensions. Let

$$H(x, y) = (x^2 + y,\ xy,\ y^3), \qquad G(u, v, w) = (uv + w,\ u^2 - vw),$$

and consider $F = G \circ H$ at $a = (1, 2)$. Here $H(a) = (3, 2, 8)$, and

$$J_H(1,2) = \begin{pmatrix} 2 & 1 \\ 2 & 1 \\ 0 & 12 \end{pmatrix}_{3\times 2}, \qquad J_G(3,2,8) = \begin{pmatrix} 2 & 3 & 1 \\ 6 & -8 & -2 \end{pmatrix}_{2\times 3}.$$

Their product is 2×2:

$$J_G(3,2,8)\, J_H(1,2) = \begin{pmatrix} 2 & 3 & 1 \\ 6 & -8 & -2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 2 & 1 \\ 0 & 12 \end{pmatrix} = \begin{pmatrix} 10 & 17 \\ -4 & -26 \end{pmatrix}.$$

For the composition above, compute J_F at $(2,1)$ in two ways: the product $J_G(H(2,1))J_H(2,1)$ and the direct derivative of the explicit form. Then check the direction $v=(1,3)$ by comparing $J_F(2,1)v$ against $\big(F(a+sv)-F(a)\big)/s$ for $s=10^{-2}$ and $s=10^{-4}$. Why does the error drop only by a factor of 10, not 100?

12.8 Computational lab

Level: advanced

The following code is run as is with numpy 2.5.2 on Python 3: the first part repeats the three computations above, then one experiment that cannot be done by hand, finite differences under measurement noise.

Computational lab

```
import numpy as np

# 1. finite differences for f(x) = sin x at x0 = 1
x0 = 1.0
exact = float(np.cos(x0))
print("cos(1) =", exact)
for h in [1e-1, 1e-2, 1e-4, 1e-6, 1e-8, 1e-10]:
    fwd = (np.sin(x0 + h) - np.sin(x0)) / h
    cen = (np.sin(x0 + h) - np.sin(x0 - h)) / (2*h)
    em = abs(fwd - exact) / abs(exact)
    ep = abs(cen - exact) / abs(exact)
    print(f"{h:8.0e} fwd={fwd:.12f} {em:.3e} "
          f"cen={cen:.12f} {ep:.3e}")
```

Actual output:

```
cos(1) = 0.5403023058681398
   1e-01 fwd=0.497363752535 7.947e-02 cen=0.539402252170 1.666e-03
   1e-02 fwd=0.536085981012 7.804e-03 cen=0.540293300875 1.667e-05
   1e-04 fwd=0.540260231419 7.787e-05 cen=0.540302304968 1.667e-09
   1e-06 fwd=0.540301885121 7.787e-07 cen=0.540302305896 5.130e-11
   1e-08 fwd=0.540302302898 5.497e-09 cen=0.540302308449 4.777e-09
   1e-10 fwd=0.540302247387 1.082e-07 cen=0.540302247387 1.082e-07
```

The last line deserves a second reading: at $h=10^{-10}$ both formulas give identical digits, because they are both computing the same rounding error, not the derivative.

Computational lab

```
def f(p):
    return p[0]**2 * p[1]**3

def grad_fd(fun, a, h):
    g = np.zeros_like(a)
    for j in range(a.size):
        e = np.zeros_like(a)
        e[j] = h
        g[j] = (fun(a + e) - fun(a - e)) / (2*h)
    return g

a = np.array([2.0, -1.0])
g_an = np.array([2*a[0]*a[1]**3, 3*a[0]**2*a[1]**2])
print("gradient =", g_an)
for h in [1e-2, 1e-4, 1e-6]:
    g = grad_fd(f, a, h)
    print(f"{h:8.0e} {g[0]:.12f} {g[1]:.12f} "
          f"{abs(g[0]+4)/4:.3e} {abs(g[1]-12)/12:.3e}")
```

Actual output:

```
gradient = [-4. 12.]
   1e-02 -4.000000000000 12.000400000000 1.577e-14 3.333e-05
   1e-04 -4.000000000004 12.000000039998 1.000e-12 3.333e-09
   1e-06 -4.000000000115 11.999999999679 2.876e-11 2.676e-11
```

Computational lab

```
def hh(p):
    x, y = p
    return np.array([x*x + y, x*y, y**3])

def gg(q):
    u, v, w = q
    return np.array([u*v + w, u*u - v*w])

def F(p):
    return gg(hh(p))

def jac(fun, p, step):
    n = p.size
    J = np.zeros((fun(p).size, n))
    for j in range(n):
        e = np.zeros(n)
        e[j] = step
        J[:, j] = (fun(p + e) - fun(p - e)) / (2*step)
    return J

a = np.array([1.0, 2.0])
q = hh(a)
Jh = np.array([[2*a[0], 1.0], [a[1], a[0]], [0.0, 3*a[1]**2]])
Jg = np.array([[q[1], q[0], 1.0], [2*q[0], -q[2], -q[1]]])
J_chain = Jg @ Jh
J_num = jac(F, a, 1e-6)
print("h(1,2) =", q)
print("Jg @ Jh =\n", J_chain)
print("max difference =", f"{np.max(np.abs(J_num - J_chain)):.3e}")

v = np.array([0.7, -1.3])
d_chain = J_chain @ v
d_num = (F(a + 1e-6*v) - F(a - 1e-6*v)) / 2e-6
print("J v =", d_chain)
print("finite difference =", d_num)
print("relative error =",
      f"{np.max(np.abs(d_num - d_chain)/np.abs(d_chain)):.3e}")
```

Actual output:

```
h(1,2) = [3. 2. 8.]
Jg @ Jh =
 [[ 10.  17.]
 [ -4. -26.]]
max difference = 1.858e-09
J v = [-15.1  31. ]
finite difference = [-15.1  31. ]
relative error = 9.860e-11
```

The difference 1.858×10^{-9} between the chain-rule Jacobian and the finite-difference Jacobian is the truncation error of the central difference, not an algebraic mistake; the directional check gives 9.86×10^{-11}, because both use the same point.

Computational lab

```
rng = np.random.default_rng(7)
sigma = 1e-3
for h in [1e-1, 1e-2, 1e-3, 1e-4, 1e-5]:
    err = []
    for _ in range(2000):
        s1 = np.sin(x0 + h) + sigma*rng.standard_normal()
        s0 = np.sin(x0) + sigma*rng.standard_normal()
        err.append(abs((s1 - s0)/h - exact))
    print(f"{h:8.0e} mean error = {np.mean(err):.3e}")
```

Actual output:

```
1e-01 mean error = 4.203e-02
1e-02 mean error = 1.112e-01
1e-03 mean error = 1.160e+00
1e-04 mean error = 1.117e+01
1e-05 mean error = 1.088e+02
```

The last experiment is the heart of this chapter in numbers. With noise $\sigma = 10^{-3}$, the quotient carries two errors: a truncation bias of $\approx 0.42h$ and noise that amplifies like $\sqrt{2}\sigma/h$. At $h = 10^{-1}$ the bias dominates (0.042), at $h = 10^{-4}$ the noise dominates (11.2), and the two balance when $0.42h = \sqrt{2} \cdot 10^{-3}/h$, that is $h \approx 0.058$: with an instrument that bad, the best step is about six hundredths, not one hundred-millionth.

That is the honest answer to the opening question. The derivative is a limit, but its best approximation is not the closest possible limit. The best is a compromise between two errors that move in opposite directions, and that compromise can be computed before a single number is measured.

Note for instructors

The order that works: the quotient definition, then the linear approximation form, then the computation rules. That form feels excessive for one variable, but only it survives up to the Jacobian; if it is introduced later, students memorize the chain rule as matrix multiplication without knowing why the order is what it is.

Two typical misconceptions. First, "the partial derivatives exist means the function is smooth"; the example $xy/\sqrt{x^2+y^2}$ on the board breaks it, provided the facts are computed together: partials zero, continuous, yet the diagonal direction gives $1/2$. Second, the direction of multiplication in the chain rule: students write $J_H J_G$ because the order in which the functions are written is the reverse of the order in which they are applied; the way to check is the size of the matrices.

Blackboard activity: ask the class to compute both partials at the origin for the two counterexamples, then to decide whether that function has a tangent plane. Sign of readiness to go further: students explain why $D_u f = \nabla f \cdot u$ fails in the second example, and use the word "remainder", not "formula".

12.9 Summary

LEVEL: BASIC

Summary.

- The derivative $f'(c)$ is the limit of a difference quotient, and also the unique coefficient of the linear approximation $f(c+h) = f(c) + Lh + r(h)$ with $r(h)/h \to 0$. Differentiability guarantees continuity, not the other way around.
- The chain rule $(f \circ g)' = f'(g(c))g'(c)$ is proved through the linear approximation, and the proof survives into many dimensions: $J_{G\circ H}(a) = J_G(H(a))J_H(a)$.

- The mean value theorem links the sign of the derivative to the behavior of the function: $f' \geq 0$ means nondecreasing, $|f'| \leq M$ means Lipschitz; it fails for vector functions.
- Taylor with Lagrange remainder uses an unknown ξ, so only an upper bound for $|f^{(n)}|$ is used.
- For several variables, the derivative is a linear map with a Jacobian matrix. Partials may exist without continuity, and may exist together with continuity without differentiability.
- The gradient points in the direction of steepest ascent with rate $\|\nabla f\|$, perpendicular to the contours; a definite Hessian determines the type of a critical point.
- Finite differences are of order h (forward) or h^2 (central), and the rounding error grows like ε/h. The smallest step is not the best step.

12.10 Exercises

Level: intermediate

Exercise 12.1.

1. Compute $f'(2)$ from the definition for $f(x) = x^3$ and $f(x) = \sqrt{x}$, until the factor h disappears from the denominator.
2. Prove that if $f'(c) > 0$, there is $\delta > 0$ such that $f(c+h) > f(c)$ for every $h \in (0, \delta)$, and explain why this does not mean that f is increasing on some interval.
3. Prove the quotient rule by differentiating $1/g$ first, then using the product rule. State at which step the continuity of g is used.
4. Assess this claim: *if f has all its partial derivatives at a point, then f is continuous at that point.* State whether it is true, false, or not yet supported by proof, and include a counterexample if needed.
5. For $f(x, y) = x^3 y - x y^3$, compute the gradient at $(2, 1)$, determine the direction of steepest ascent, and compute its maximum rate. Check with a central difference $h = 10^{-4}$ and compare the relative error with the order h^2.
6. Prove that if $f' = 0$ everywhere on an interval, then f is constant. Explain why that proof fails on a disconnected set, and give an example.
7. Suppose $H(x, y) = (e^x y,\ x + y^2)$ and $G(u, v) = (u^2 v,\ u - v)$. Compute $J_{G \circ H}$ at $(0, 1)$ in two ways: the matrix product and the direct derivative. (Hint: $H(0, 1) = (1, 1)$.)
8. Find all critical points of $f(x, y) = x^3 + y^3 - 3xy$, then classify them with the Hessian; explain why the sign of the determinant is not enough when the determinant is zero.
9. Estimate $\sqrt{1.04}$ with the Taylor polynomial of order two at $a = 1$, then bound its error with the Lagrange remainder. Compare that bound with the actual error: is it tight? (Hint: $f''(x) = -\frac{1}{4} x^{-3/2}$ and f''' is positive on that interval.)

For researchers: limits and next steps

What is proved in this chapter are the classical statements for one and several variables: the linear approximation and the uniqueness of its map, the chain rule, the mean value theorem and its consequences, Taylor with Lagrange remainder, the equivalence of differentiability with the Jacobian, and the sufficient condition of continuous partials. All of them are theorems, not conventions, and they hold in finite-dimensional spaces over $\mathbb{R}$. What is often popularized is the claim "the gradient is the derivative". Without differentiability that claim is false, and $xy/\sqrt{x^2 + y^2}$ is a counterexample that stays continuous. The claim "the Hessian determines the type of a critical point" holds only when the Hessian is nonsingular; the singular case requires higher derivatives or direct analysis.
Primary sources. Rudin, *Principles of Mathematical Analysis*, the chapters *Differentiation* and *Functions of Several Variables*, contains all the statements above. Spivak,

Calculus on Manifolds, the chapter *Differentiation*, takes the same route and is shorter. Stewart, *Calculus: Early Transcendentals* ninth edition, sections 14.3 to 14.7, supplies the computational counterparts, including Clairaut's theorem.
Limits of validity: all arguments assume that the domain is open and that the derivatives used exist. For functions that are differentiable only almost everywhere, the monotonicity and Taylor conclusions collapse, and that is territory of measure theory which this book does not cover. For vector functions, the mean value theorem does not hold; what replaces it is the mean value inequality, which requires an operator norm. The best order of a finite difference depends on the scale of the function, the noise, and the arithmetic; this chapter has only shown the order empirically.

Map to source texts: This chapter follows Rudin, Principles of Mathematical Analysis*, the chapters* Differentiation *and* Functions of Several Variables*; its computational version is Stewart,* Calculus: Early Transcendentals *ninth edition, sections 14.3 to 14.7; its abstract counterpart is Spivak,* Calculus on Manifolds*, the chapter* Differentiation*. Continuity and the extreme value theorem are used from Chapter 11, the Cauchy–Schwarz inequality from Chapter 10, and the Jacobian norm returns in Chapter 23 and Chapter 28.*

Chapter 13

Riemann Integrals and the Fundamental Theorem of Calculus

A program computes the area under $y = x^2$ on $[0, 1]$ with 1000 slices of equal width. If the height of each rectangle is taken at the left endpoint, the result is 0.3328335; if at the right endpoint, 0.3338335. The difference is 0.001, and there is no reason to prefer either one. The correct value is $1/3$, which does not appear in either computation. Increasing the slices to a million does not help: both answers still bracket the value, they never catch it.

What is needed is not a finer partition, but a way to read two approximations at once. If the difference can be made smaller than any positive number, then there is one number trapped between them, and that is the integral. This entire chapter rests on that simple idea.

Chapter 12 taught us to take change apart into rates: from position to velocity, from a function to its derivative. What is still missing is the way home. We know that the velocity at second t is $v(t)$, but we do not yet know how much distance was covered in one minute. The intuition is clear: distance is the sum of velocity times small slices of time. What is not clear is how to add up infinitely many such terms correctly. This chapter answers that, and the answer is surprising: the way home is the antiderivative.

For younger readers (age 10 and up)

Imagine we want to know the area of a lake whose shape is irregular. There is no formula for such a shape. All we have is transparent graph paper. We lay that paper over the drawing of the lake, then count how many squares lie entirely inside the water. The result is too small. We count again how many squares are touched by water, even slightly. The result is too large. The true area lies between those two numbers.

Now cut the squares into ten times smaller ones. The lower number rises, the upper number falls, and the two grow closer. If the squares are very small, the two numbers are almost equal, and we can state the area of the lake to any accuracy we like.

There is one condition that is easy to overlook. If the lake has very thin coral islands scattered everywhere, the squares that touch them can keep the upper estimate far from the lower estimate. Mathematicians call that condition integrability. In this chapter we shall learn why two bracketing estimates can be trusted, and when they fail.

13.1 Partitions, Upper Sums, and Lower Sums

Level: basic

We want to give meaning to the area under a graph without assuming that the area already has a meaning: enclose the region with rectangles from inside and from outside, then

shrink the difference.

Definition 13.1 (Partitions and upper-lower sums)**.** Let $[a, b]$ be a closed interval. A *partition* $\mathcal{P}$ of $[a, b]$ is a finite set of points

$$a = x_0 < x_1 < \cdots < x_n = b.$$

The width of the i-th slice is $\Delta x_i = x_i - x_{i-1}$, and the *mesh* of that partition is the widest slice width, $\|\mathcal{P}\| = \max_i \Delta x_i$.

Let f be bounded on $[a, b]$. On the i-th slice write

$$M_i = \sup\{f(x) : x_{i-1} \le x \le x_i\}, \qquad m_i = \inf\{f(x) : x_{i-1} \le x \le x_i\}.$$

The *upper sum* and the *lower sum* of the partition $\mathcal{P}$ are

$$U(\mathcal{P}, f) = \sum_{i=1}^{n} M_i \, \Delta x_i, \qquad L(\mathcal{P}, f) = \sum_{i=1}^{n} m_i \, \Delta x_i. \tag{13.1}$$

We use supremum and infimum, not the largest and smallest values, because f need not attain its extremes on a slice: the function equal to x for $x < 1$ and to 0 at $x = 1$ is bounded on $[0, 1]$, but its largest value on the last slice does not exist, whereas its supremum does, namely 1.

The upper sum is always at least equal to the lower sum, because $M_i \ge m_i$ for every i. For $f(x) = x^2$ on $[0, 1]$ with four slices of equal width, the lower sum is

$$L = \frac{1}{4}\Big(0 + \frac{1}{16} + \frac{4}{16} + \frac{9}{16}\Big) = \frac{7}{32},$$

and the upper sum is $\frac{15}{32}$. Both bracket the value sought, with a difference of $\frac{1}{4}$ that is still large. If f takes negative values, these sums compute signed area, and the minus sign is part of the definition, not an error.

Figure 13.1 shows two values of n at once, with the right endpoint as height. At $n = 4$ the staircase is coarse and $U_4 = 0.46875$; at $n = 16$ the serration is fine and $U_{16} = 0.3652343750$. The distance from the staircase to the line $1/3$ shrinks from 0.1354 to 0.0319.

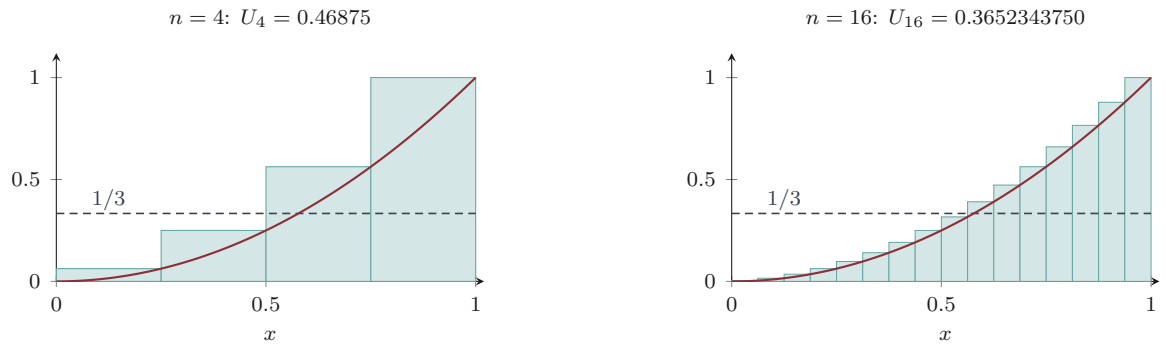


Figure 13.1: Rectangular approximation of $f(x) = x^2$ on $[0, 1]$ with right-endpoint heights. At $n = 4$ the upper sum is $U_4 = 0.46875$, at $n = 16$ the upper sum is $U_{16} = 0.3652343750$, and both are still above the true value $1/3 = 0.333\ldots$ (dashed line). The difference $U_n - L_n$ is exactly $1/n$, so going from $n = 4$ to $n = 16$ cuts the distance from the staircase to that line down to a quarter of it.

There are two facts that will be used again and again. The first: refining a partition does not make the estimate worse.

Lemma 13.1 (Refinement)**.** *If $\mathcal{P}^*$ is a refinement of $\mathcal{P}$, meaning that $\mathcal{P} \subseteq \mathcal{P}^*$, then*

$$L(\mathcal{P}, f) \le L(\mathcal{P}^*, f) \le U(\mathcal{P}^*, f) \le U(\mathcal{P}, f).$$

Proof. It suffices to prove this for a refinement that adds one point c inside the slice $[x_{i-1}, x_i]$, since a general refinement is obtained by repeating that step. Let m' and m'' be the infima of f on $[x_{i-1}, c]$ and on $[c, x_i]$. Both intervals are contained in $[x_{i-1}, x_i]$, so that $m' \geq m_i$ and $m'' \geq m_i$. Consequently

$$m'(c - x_{i-1}) + m''(x_i - c) \;\geq\; m_i(c - x_{i-1}) + m_i(x_i - c) = m_i\,\Delta x_i,$$

so the lower sum does not decrease. The same argument with suprema gives $M' \leq M_i$ and $M'' \leq M_i$, so that the upper sum does not increase. □

The second fact is stronger: two different partitions can still be compared, as long as what is compared is a lower sum with an upper sum.

Lemma 13.2. *For any two partitions $\mathcal{P}_1$ and $\mathcal{P}_2$ of $[a, b]$ we have*

$$L(\mathcal{P}_1, f) \leq U(\mathcal{P}_2, f).$$

Proof. Take $\mathcal{P}$ the common refinement, that is, the union of the points of the two partitions. By the previous lemma, $L(\mathcal{P}_1, f) \leq L(\mathcal{P}, f)$ and $U(\mathcal{P}, f) \leq U(\mathcal{P}_2, f)$. Since $L(\mathcal{P}, f) \leq U(\mathcal{P}, f)$ always holds, that chain of inequalities gives the result asked for. □

That lemma ensures that every lower sum is a lower bound for every upper sum. Hence the set of all lower sums is bounded above, and the set of all upper sums is bounded below; the following two numbers always exist.

Definition 13.2 (Riemann integrable)**.** For f bounded on $[a, b]$, the *lower integral* and the *upper integral* are defined as

$$\underline{I}(f) = \sup_{\mathcal{P}} L(\mathcal{P}, f), \qquad \overline{I}(f) = \inf_{\mathcal{P}} U(\mathcal{P}, f),$$

with the supremum and infimum taken over all partitions of $[a, b]$. Since $\underline{I}(f) \leq \overline{I}(f)$, the function f is called *Riemann integrable* on $[a, b]$ when those two numbers are equal. Their common value is written $\int_a^b f$ or $\int_a^b f(x)\,dx$.

Notice what is absent from that definition: there is no sample point and no limit over n. The measure of success of an approximation is the difference between its two brackets, not its closeness to a guess.

partition (*partisi*) — A way of cutting an interval into several slices, like dividing a bar of chocolate with a knife, not measuring exactly how long each slice is.

upper sum (*jumlah atas*) — An estimate that is deliberately inflated, like weighing goods together with their box.

Read it aloud (auditory learner)

Two voices. Call the first *Cutter*, the second *Adder*.
Cutter: I have a strip of ribbon. I cut it into four. Adder: I add up those four pieces. The result is too small. Cutter: I cut again. So, eight. Adder: The result goes up a little. Still too small. Cutter: Sixteen. Adder: Up again. Closer, not there yet. Cutter: A hundred. Adder: Very close. Cutter: A thousand. Adder: Closer still, still not there. Cutter: When do we get there? Adder: Never, if you only add slices. Stop chasing the number, start chasing the bracket. Cutter: What bracket? Adder: Thin slices give a lower bound. Fat slices give an upper bound. If those two bounds meet at one point, that point is the answer. Cutter: So my slices are never enough? Adder: Never. Slices

only bring the two bounds close. What answers is not the last slice, but the meeting of the two bounds.
Now repeat one line: cut thin, bracket tight, bounds meet.

13.2 The Integrability Criterion

Level: intermediate

The definition just given speaks of suprema and infima over all partitions, and there are infinitely many partitions. The following criterion replaces them with a single inequality that can be checked on one partition alone.

Theorem 13.3 (Integrability criterion)**.** *A bounded function f is Riemann integrable on $[a,b]$ if and only if for every $\varepsilon > 0$ there is a partition $\mathcal{P}$ with*

$$U(\mathcal{P}, f) - L(\mathcal{P}, f) < \varepsilon.$$

Proof. For every partition, $L(\mathcal{P},f) \le \underline{I}(f) \le \overline{I}(f) \le U(\mathcal{P},f)$; the first inequality comes from the definition of supremum, the last from the definition of infimum.

($\Rightarrow$) Suppose f is integrable with value I, and take $\varepsilon > 0$. By the definition of supremum, there is $\mathcal{P}_1$ with $L(\mathcal{P}_1,f) > I - \varepsilon/2$. By the definition of infimum, there is $\mathcal{P}_2$ with $U(\mathcal{P}_2,f) < I + \varepsilon/2$. Take $\mathcal{P}$ the common refinement of both. The refinement lemma gives

$$U(\mathcal{P},f) - L(\mathcal{P},f) < \left(I + \tfrac{\varepsilon}{2}\right) - \left(I - \tfrac{\varepsilon}{2}\right) = \varepsilon.$$

($\Leftarrow$) Suppose that condition is met. For every $\varepsilon > 0$ there is $\mathcal{P}$ with $U(\mathcal{P},f) - L(\mathcal{P},f) < \varepsilon$, so that

$$0 \le \overline{I}(f) - \underline{I}(f) \le U(\mathcal{P},f) - L(\mathcal{P},f) < \varepsilon.$$

The number $\overline{I}(f) - \underline{I}(f)$ is nonnegative and smaller than every $\varepsilon > 0$, so that number is zero. The two integrals are equal, and f is integrable. □

The criterion turns a question about infinitely many partitions into a single inequality. A direct consequence: if the difference is small on one partition, that difference stays small on every refinement of it, since the refinement lemma can only narrow it.

Before looking for classes of functions that pass the criterion, we record the properties of the integral that will be used throughout. All of them are consequences of the definition of upper and lower sums, not new assumptions.

Proposition 13.4 (Basic properties of the integral)**.** *Let f and g be Riemann integrable on $[a,b]$. Then: (a) $f+g$ is integrable and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$; (b) if $f \le g$ on $[a,b]$ then $\int_a^b f \le \int_a^b g$; (c) if $a < c < b$ then f is integrable on $[a,c]$ and on $[c,b]$, and $\int_a^b f = \int_a^c f + \int_c^b f$; (d) $|f|$ is integrable and $\left|\int_a^b f\right| \le \int_a^b |f|$; (e) for every constant k, $\int_a^b k\,dx = k(b-a)$.*

Proof. (a) On every slice, $\sup(f+g) \le \sup f + \sup g$ and $\inf(f+g) \ge \inf f + \inf g$. Hence $U(\mathcal{P},f+g) \le U(\mathcal{P},f) + U(\mathcal{P},g)$ and $L(\mathcal{P},f+g) \ge L(\mathcal{P},f) + L(\mathcal{P},g)$. Take $\varepsilon > 0$ and choose $\mathcal{P}$ that makes $U - L$ less than $\varepsilon/2$ for f and g at once. Then

$$U(\mathcal{P},f+g) - L(\mathcal{P},f+g) \le \big[U(\mathcal{P},f) - L(\mathcal{P},f)\big] + \big[U(\mathcal{P},g) - L(\mathcal{P},g)\big] < \varepsilon,$$

so that $f+g$ is integrable. Its value is obtained by bracketing from both sides, since ε is arbitrary.

(b) If $f \le g$, then $M_i(f) \le M_i(g)$ on every slice, so that $U(\mathcal{P},f) \le U(\mathcal{P},g)$ for all $\mathcal{P}$. Take the infimum of both sides.

(c) If $\mathcal{P}$ contains c, the sum splits into the parts $[a, c]$ and $[c, b]$, with $L(\mathcal{P}, f) = L(\mathcal{P}_1, f) + L(\mathcal{P}_2, f)$ and the same for U. The supremum gives $\underline{I}(f) \le \underline{I}_{[a,c]}(f) + \underline{I}_{[c,b]}(f)$, the infimum gives the opposite direction. For a partition without c, add c first; the refinement lemma preserves the inequality.

(d) Since $-|f| \le f \le |f|$, parts (b) and (a) give $-\int |f| \le \int f \le \int |f|$. The integrability of $|f|$ follows from $\big||f(x)|-|f(y)|\big| \le |f(x)-f(y)|$, which gives $M_i(|f|)-m_i(|f|) \le M_i(f)-m_i(f)$.

(e) On every slice, $M_i = m_i = k$, so $U = L = k(b-a)$ for all partitions. □

Part (c) is the reason we may cut an interval at any point; without it the accumulation function could not be defined cleanly.

13.3 Classes of Integrable Functions

Level: intermediate

The criterion in the previous section is a test. Now we look for classes of functions that pass that test without having to find a special partition for each case. The two most useful classes are continuous functions and monotone functions.

Corollary 13.5 (Continuity guarantees integrability)**.** *If f is continuous on $[a, b]$, then f is Riemann integrable on $[a, b]$.*

Proof. Take $\varepsilon > 0$. Since $[a, b]$ is compact, f is uniformly continuous there (Chapter 11): there is $\delta > 0$ such that $|f(x) - f(t)| < \varepsilon/(b-a)$ when $|x - t| < \delta$. Choose a partition with mesh less than δ. On every slice, for all x, t in that slice we have $|f(x) - f(t)| < \varepsilon/(b-a)$, so that $M_i - m_i \le \varepsilon/(b-a)$. Then

$$U(\mathcal{P}, f) - L(\mathcal{P}, f) = \sum_{i=1}^{n} (M_i - m_i)\,\Delta x_i \le \frac{\varepsilon}{b-a} \sum_{i=1}^{n} \Delta x_i = \varepsilon.$$

The integrability criterion closes the proof. □

Note the role of uniform continuity: one δ must work for all slices at once, whereas ordinary continuity only gives a δ that depends on the point. This is where Chapter 11 is used, not as decoration.

The second class does not require continuity: a monotone function may jump, as long as its jumps are finite in number.

Theorem 13.6 (Monotonicity guarantees integrability)**.** *If f is monotone on $[a, b]$, then f is Riemann integrable on $[a, b]$.*

Proof. Suppose f is increasing. Divide $[a, b]$ into n slices of equal width, $\Delta x = (b-a)/n$. On the i-th slice, the smallest value is at the left endpoint and the largest value at the right endpoint, so that $M_i - m_i = f(x_i) - f(x_{i-1})$. Then

$$U(\mathcal{P}, f) - L(\mathcal{P}, f) = \Delta x \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big) = \frac{(b-a)\big(f(b) - f(a)\big)}{n},$$

because the sum in the middle telescopes. Choose $n > (b-a)(f(b)-f(a))/\varepsilon$ and the criterion is satisfied. If f is decreasing, apply this result to $-f$ and use linearity. □

That proof uses no δ: all it uses is that the sum of all the jumps does not exceed $f(b)-f(a)$, so a monotone function is easy to integrate even though its graph is full of corners.

Those two classes can be combined: a function that is continuous except at finitely many points is still integrable (Rudin, Theorem 6.10, not repeated here).

Now the other side: bounded functions that fail. The cleanest example is the Dirichlet function, equal to 1 at rational numbers and to 0 at irrational numbers.

Example 13.1 (The Dirichlet function is not integrable). Let $D(x) = 1$ if x is rational and $D(x) = 0$ if x is irrational, on $[0, 1]$. Every slice contains both (Chapter 3, density of $\mathbb{Q}$), so that $M_i = 1$ and $m_i = 0$ for every i, and

$$U(\mathcal{P}, D) = \sum_{i=1}^{n} 1 \cdot \Delta x_i = 1, \qquad L(\mathcal{P}, D) = 0$$

for *every* partition. The difference is always 1, never smaller than $\varepsilon = 1/2$, so by the criterion D is not integrable even though it is bounded.

That failure is not because D is broken at many points, but because its jumps are spread everywhere. The following function shows that the number of discontinuity points is not an obstacle.

Theorem 13.7 (Thomae's function)**.** *Define T on $[0, 1]$ by $T(x) = 1/q$ if $x = p/q$ in lowest terms with $q \geq 1$, and $T(x) = 0$ if x is irrational. Then T is integrable in the Riemann sense and $\int_0^1 T = 0$.*

Proof. Take $\varepsilon > 0$. The set $S = \{x : T(x) \geq \varepsilon\}$ consists of fractions with denominator $q \leq 1/\varepsilon$; for each q there are only $q + 1$ such fractions in $[0, 1]$, so that S is finite. Cover S with finitely many open intervals whose total length is less than ε.

Form a partition that contains all endpoints of those covering intervals. The slices that contain S have total length less than ε and there $M_i - m_i \leq 1$, so their contribution is less than ε. The other slices satisfy $M_i \leq \varepsilon$ and $m_i \geq 0$, so their contribution is at most ε. Hence $U(\mathcal{P}, T) - L(\mathcal{P}, T) < 2\varepsilon$, and the criterion closes the proof. The value is zero because every slice contains an irrational number, so that $L(\mathcal{P}, T) = 0$. □

So T is broken at every rational number, which is dense in $[0, 1]$, yet T is still integrable. What decides is not the number of discontinuity points but the amount of space they occupy: Lebesgue's criterion states that a bounded function is Riemann integrable if and only if its discontinuity points have measure zero (Rudin, Theorem 11.33(b); Abbott, Chapter 7). We quote it without proof because measure demands measure theory.

Challenge (problem solver)

Define h on $[0, 1]$ by $h(x) = 1$ if $x = p/q$ in lowest terms and the denominator q is odd, and $h(x) = 0$ for all other x, including every irrational number. Is h Riemann integrable on $[0, 1]$? If so, what is the value of the integral? If not, prove it using the integrability criterion, not by counting points.
Hint 1: do not count the number of points where h equals one. The number is infinite, and the criterion does not ask for that count.
Hint 2: take any slice $[x_{i-1}, x_i]$ with $x_{i-1} < x_i$. That slice contains fractions with odd denominator, so $M_i = 1$, and it contains irrational numbers, so $m_i = 0$. Thus $U - L = 1$ for every partition.
If you get stuck, go back to the section Classes of Integrable Functions, the Dirichlet example.

Riemann integrability (*keterintegralan*) — The property of a function whose lower and upper estimates can be brought together until their difference vanishes.

For younger readers (age 10 and up)

Have you ever looked at a car's speedometer and then asked how much distance has been covered? Distance is not displayed there, only instantaneous speed. But if we record the speed every second, we can add it up: one second at ten meters per second means 10 meters, the next second 12 meters, and so on. The sum of all those slices is the distance. The problem is that the speed changes in the middle of a second. A car can speed up and then slow down within the same second. If we use the speed at the start of the second, the distance we compute is too small. If we use the speed at the end of the second, the result is too large. The right value lies between them.

If we record ten times per second, the two estimates grow closer; a hundred times, closer still. Mathematicians ask further: if the recording can be as dense as we like, do the two estimates head toward one and the same number? The answer is not always yes, and in this chapter we shall learn when it is yes and why.

13.4 The Integral as an Accumulation Function

Level: intermediate

Up to here the integral is only a number attached to an interval. We can view it as a function by letting the upper endpoint move: the integral becomes a report on how much has accumulated up to that point.

Definition 13.3 (Accumulation function)**.** Let f be integrable on $[a, b]$. For $x \in [a, b]$ define

$$F(x) = \int_a^x f(t)\,dt.$$

For $x < a$ we use the convention $\int_a^x f = -\int_x^a f$, and for $x = a$ the value is zero. The function F is called the *accumulation function* of f on $[a, b]$.

With that convention the additive property holds for any ordering of points, so that

$$F(y) - F(x) = \int_x^y f(t)\,dt \tag{13.2}$$

for all $x, y \in [a, b]$. This is the form used in all the proofs that follow: the increase of savings between two times is the integral of the income rate in between.

The accumulation function is always better than the function it comes from: it is continuous even when f is not, and differentiable at every point where f is continuous.

Proposition 13.8 (Accumulation is always continuous)**.** *If f is bounded on $[a, b]$ with $|f| \le M$ and integrable, then the F that is defined above satisfies*

$$|F(y) - F(x)| \le M\,|y - x|$$

for all $x, y \in [a, b]$. In particular F is continuous on $[a, b]$.

Proof. For $x < y$, monotonicity and part (d) of the basic properties proposition give

$$|F(y) - F(x)| = \Big|\int_x^y f\Big| \le \int_x^y |f| \le M(y - x).$$

For $y < x$ the same result with the opposite sign. Hence F satisfies a Lipschitz condition with constant M, and every Lipschitz function is continuous. □

Now the first direction of the derivative-integral relationship. In everyday words: the rate of growth of savings at a given moment equals the income rate at that moment.

Theorem 13.9 (Fundamental theorem of calculus, first part)**.** *Let f be integrable on $[a,b]$ and continuous at the point $x_0 \in (a,b)$. Then the accumulation function F is differentiable at x_0 and*

$$F'(x_0) = f(x_0).$$

Proof. Take $h \neq 0$ with $x_0 + h \in [a,b]$. From the increment form,

$$\frac{F(x_0+h) - F(x_0)}{h} - f(x_0) = \frac{1}{h}\int_{x_0}^{x_0+h} (f(t) - f(x_0))\,dt,$$

because the integral of the constant function $f(x_0)$ over an interval of length $|h|$ equals $f(x_0)h$.

Take $\varepsilon > 0$. Since f is continuous at x_0, there is $\delta > 0$ such that $|t - x_0| < \delta$ implies $|f(t) - f(x_0)| < \varepsilon$. For $0 < |h| < \delta$ the whole interval of integration lies inside $(x_0 - \delta, x_0 + \delta)$, so that, by monotonicity and part (d) of the basic properties proposition,

$$\left|\frac{1}{h}\int_{x_0}^{x_0+h} (f(t) - f(x_0))dt\right| \leq \frac{1}{|h|}\left|\int_{x_0}^{x_0+h} |f(t) - f(x_0)|\,dt\right| \leq \frac{1}{|h|}\,\varepsilon\,|h| = \varepsilon.$$

Since ε is arbitrary, the difference quotient tends to $f(x_0)$. □

That theorem demands continuity at only one point. Where f jumps, the derivative of F may fail to exist, and that is not a weakness of the proof: for $f(t) = \operatorname{sgn} t$ on $[-1,1]$ we have $F(x) = \int_0^x \operatorname{sgn} t\,dt = |x|$, which is not differentiable at 0, exactly where the sign of f changes.

One important consequence: every continuous function has an antiderivative, and its antiderivative can be written as an integral. Chapter 12 derived the differentiation rules but did not answer whether every continuous function is the derivative of something. Now that question is answered, and the answer is constructive.

Corollary 13.10 (An antiderivative always exists)**.** *If f is continuous on $[a,b]$, then $F(x) = \int_a^x f$ is differentiable on $[a,b]$ with $F' = f$. Hence F is an antiderivative of f on $[a,b]$.*

Proof. The continuity of f gives integrability (the previous corollary) and continuity at every point, so that the first part of the theorem applies at every $x_0 \in (a,b)$. At the endpoints one-sided derivatives are used. □

The chain rule gives one further flexibility: if the upper limit depends on x,

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f\big(g(x)\big)\,g'(x),$$

provided g is differentiable. For example, $\frac{d}{dx}\int_0^{x^2} e^{-t^2}dt = 2x\,e^{-x^4}$.

The two directions are easy to confuse, so it is worth drawing them once. Figure 13.2 places both in one diagram: the upper arrow accumulates thin slices, the lower arrow differentiates. What the theorem guarantees is that differentiating the result of accumulating returns the original function, not a tidy algebraic inverse, since an antiderivative is fixed only up to a constant.

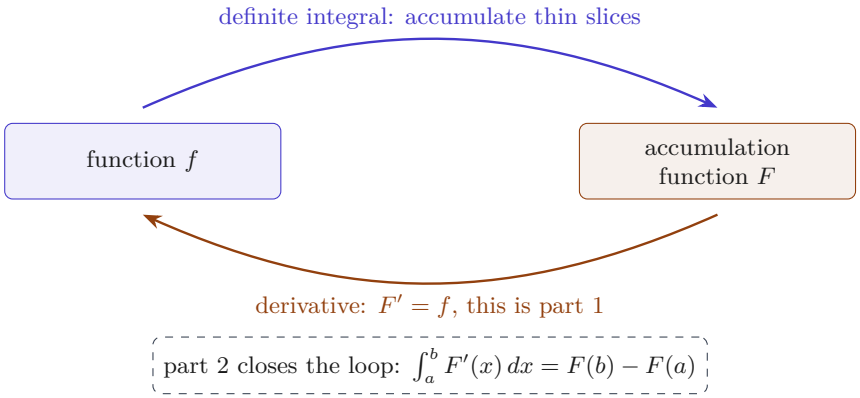


Figure 13.2: The two directions of the fundamental theorem of calculus as back-and-forth arrows. Upper arrow: accumulating thin slices produces $F(x) = \int_a^x f$. Lower arrow: differentiating F returns f, and that is the first part. The dashed box below is the second part, which uses an antiderivative to compute integrals. For $f(x) = x^2$ on $[0, 1]$, $F(x) = x^3/3$, and the second part gives $F(1) - F(0) = 1/3$, the same number as the bracketing in Figure 13.1.

13.5 The Fundamental Theorem of Calculus, Second Part

Level: advanced

The first part runs from the integral to the derivative. The second part runs the other way: if we already hold an antiderivative, the integral can be computed with one subtraction. In everyday words: total change equals the final value minus the initial value.

Theorem 13.11 (Fundamental theorem of calculus, second part)**.** *Let F be continuous on $[a, b]$, differentiable on (a, b), and let F' be Riemann integrable on $[a, b]$. Then*

$$\int_a^b F'(x)\,dx = F(b) - F(a).$$

Proof. Take a partition $\mathcal{P} = \{x_0, \ldots, x_n\}$ of $[a, b]$. On every subinterval, F is continuous on $[x_{i-1}, x_i]$ and differentiable inside it, so the mean value theorem (Chapter 12) gives $t_i \in (x_{i-1}, x_i)$ with

$$F(x_i) - F(x_{i-1}) = F'(t_i)\,\Delta x_i.$$

Since $m_i \le F'(t_i) \le M_i$, summing over i gives

$$L(\mathcal{P}, F') \le \sum_{i=1}^{n} (F(x_i) - F(x_{i-1})) \le U(\mathcal{P}, F').$$

The sum in the middle telescopes and equals $F(b) - F(a)$, so that number is trapped between the lower sum and the upper sum of every partition. Since F' is integrable, both brackets head toward the same value, namely $\int_a^b F'$. □

Step-by-step recipe (procedural learner)

1. Find F with $F' = f$, then check by differentiating F. If the derivative does not return to f, the candidate is wrong and the following steps are wasted.
2. Write F at the upper limit, write F at the lower limit, then subtract: $F(b) - F(a)$. Do not insert $+C$; the constant cancels.
3. Fix the sign if the interval is reversed: $\int_b^a f = -\int_a^b f$, and $\int_a^a f = 0$.
4. Test with one coarse slice, a trapezoid or `quad`. If the result is far off, the error is

in step 1 or 2, not in step 3.

Warning about order: checking the antiderivative must happen before subtracting. Two values of an arbitrary function still give a tidy and plausible difference, so that a mistake in step 1 is not visible from the shape of the answer.

Likewise the two parts do not replace each other: the first part differentiates the accumulation function without requiring an antiderivative, whereas the second part requires an antiderivative and integrability of its derivative.

The second part explains why the result is called net change, not area. For $F(x) = \frac{x^4}{4} - \frac{x^2}{2}$ on $[-1, 1]$, $F'(x) = x^3 - x$, so that

$$\int_{-1}^{1} (x^3 - x)\,dx = F(1) - F(-1) = \Big(\frac{1}{4} - \frac{1}{2}\Big) - \Big(\frac{1}{4} - \frac{1}{2}\Big) = 0.$$

The region above the axis has area $1/4$, the region below the axis also $1/4$, so the total area is $1/2$ whereas the integral is zero because the signs are opposite. Figure 13.3 shows both.

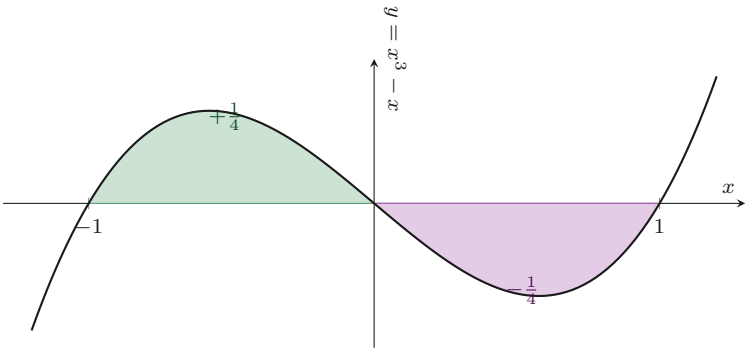


Figure 13.3: Signed area for $F'(x) = x^3 - x$ on $[-1, 1]$. The region above the x-axis contributes $+1/4$, the region below the axis contributes $-1/4$, so that $\int_{-1}^{1}(x^3 - x)\,dx = F(1) - F(-1) = 0$. The geometric area is still $1/2$; what is zero is the signed sum, and that is what the second part of the theorem computes.

The integrability hypothesis in the second part is not excessive. There are differentiable functions whose derivative is bounded but not Riemann integrable; the classical example was found by Volterra in the nineteenth century, and we do not prove it here. The second part of the theorem is not an unconditional statement.

13.6 Substitution and Integration by Parts

Level: advanced

The two most frequently used rules for computing integrals are not new axioms: both are differentiation rules read backwards, and their proofs use only the fundamental theorem of calculus. The first one reverses the chain rule.

Proposition 13.12 (Substitution)**.** *Let φ be differentiable with φ' continuous on $[a, b]$, and let f be continuous on an interval that contains $\varphi([a, b])$. Then*

$$\int_a^b f(\varphi(t))\,\varphi'(t)\,dt = \int_{\varphi(a)}^{\varphi(b)} f(u)\,du.$$

Proof. Since f is continuous, it has an antiderivative $G(u) = \int_c^u f$, with c inside the range of φ; by the first part of the theorem, $G' = f$. The chain rule (Chapter 12) gives

$$\frac{d}{dt} G(\varphi(t)) = G'(\varphi(t))\varphi'(t) = f(\varphi(t))\varphi'(t).$$

The right-hand side is continuous, hence integrable. The second part of the theorem applied to the function $G \circ \varphi$ gives

$$\int_a^b f(\varphi(t))\varphi'(t)\,dt = G(\varphi(b)) - G(\varphi(a)) = \int_{\varphi(a)}^{\varphi(b)} f(u)\,du.$$

□

The second rule reverses the product rule.

Proposition 13.13 (Integration by parts)**.** *Let F and G be differentiable on $[a,b]$ with F' and G' continuous. Then*

$$\int_a^b F(x)\,G'(x)\,dx = F(b)G(b) - F(a)G(a) - \int_a^b F'(x)\,G(x)\,dx.$$

Proof. Let $H = FG$. The product rule gives $H' = F'G + FG'$, and the right-hand side is continuous so that H' is integrable. The second part of the theorem applied to H gives

$$F(b)G(b) - F(a)G(a) = \int_a^b H' = \int_a^b F'G + \int_a^b FG'.$$

Move the first term to the left-hand side. □

antiderivative (*antiturunan*) — A function whose derivative is the function we are integrating, like the way home on a road: it is useful only when the way out is already known.

As a numerical check, take $\int_0^1 x\,e^{x^2}dx$. The substitution $u = x^2$, $du = 2x\,dx$ gives

$$\int_0^1 x\,e^{x^2}dx = \frac{1}{2}\int_0^1 e^u du = \frac{e-1}{2} = 0.8591409142295225.$$

`scipy.integrate.quad` gives 0.8591409142295227, a difference of 2×10^{-16} from machine rounding. For the second rule, $F(x) = x$ and $G'(x) = e^x$ give $\int_0^1 xe^x dx = e - (e-1) = 1$, and `quad` returns 1.0 with estimated error 1.1×10^{-14}.

13.7 The Mean Value Theorem for Integrals

Level: intermediate

An average is usually computed by adding and then dividing by the number of data. For a function that changes continuously, the sum is replaced by an integral and the divisor by the length of the interval: the average is the height of the rectangle whose area equals the area under the curve.

Definition 13.4 (Average value)**.** For f integrable on $[a,b]$, the *average value* of f on that interval is

$$\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Theorem 13.14 (Mean value for integrals)**.** *If f is continuous on $[a,b]$, then there is $c \in [a,b]$ with*

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Proof. Since f is continuous on a closed bounded interval, it attains its smallest value m and its largest value M there (Chapter 11). Monotonicity gives

$$m(b-a) \le \int_a^b f \le M(b-a),$$

so that $\bar{f} \in [m, M]$. By the intermediate value theorem (Chapter 11), the continuous function f takes every value between m and M, so there is $c \in [a, b]$ with $f(c) = \bar{f}$. □

As a number, for $f(x) = x^2$ on $[0, 1]$ the average is $1/3$, and the point that the theorem reaches is $c = 1/\sqrt{3} = 0.5773502691896258$, since $c^2 = 0.3333333333333334$. That point is not the midpoint of the interval: the theorem guarantees existence, not location.

13.8 Worked Examples

Level: intermediate

Now we work through three things: the upper and lower sums of $f(x) = x^2$ on $[0, 1]$, a comparison of five quadrature rules, and one integral without an elementary antiderivative. Fractions are computed with `fractions.Fraction`, without rounding.

Start from n slices of equal width with $x_i = i/n$ and $\Delta x = 1/n$, so that

$$U_n = \sum_{i=1}^{n} \Big(\frac{i}{n}\Big)^2 \frac{1}{n} = \frac{1}{n^3} \sum_{i=1}^{n} i^2 = \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2},$$

using the sum-of-squares formula from Chapter 1. In the same way $L_n = \frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2}$, so that

$$U_n - L_n = \frac{1}{n}. \tag{13.3}$$

That result is not an approximation. For every n the difference is exactly $1/n$; since $1/n$ can be made smaller than any ε by taking $n > 1/\varepsilon$, the integrability criterion is satisfied. Hence the following table is at the same time a proof that $\int_0^1 x^2 dx = 1/3$: the two sums bracket $1/3$, and their distance tends to zero.

n	U_n	L_n	$U_n - L_n$
1	1.0000000000	0.0000000000	1
4	0.4687500000	0.2187500000	1/4
16	0.3652343750	0.3027343750	1/16
64	0.3411865234	0.3255615234	1/64
256	0.3352890015	0.3313827515	1/256
1024	0.3338217735	0.3328452110	1/1024

At $n = 1024$ the difference is 9.77×10^{-4} and the distance from U_n to $1/3$ is half of that. Each time n is doubled the difference is exactly halved, and that is why upper and lower sums are not an efficient way to reach high accuracy.

Do it by hand (kinesthetic learner)

Materials: one sheet of graph paper, scissors, a ruler, tape, a pencil.
Steps. (1) On the graph paper draw a curve that rises faster and faster: one square from the left edge has height one square, two squares have height four squares, three squares have height nine squares. (2) Cut the region under the curve into vertical strips one square wide. (3) Lay those strips side by side on the table, from shortest to longest, with level bases. (4) Look at their upper edge: what appears is a jagged staircase, not a

smooth line. (5) Repeat with strips half a square wide, then a quarter square wide. What we should see: the serration on the upper edge grows finer as the strips grow thinner, whereas the total area does not change, since cutting and rearranging neither adds nor removes paper. Hence the area of that curved region equals the area of the jagged figure, and that jagged figure increasingly resembles one rectangle. The height of that rectangle is what we shall later call the average value of the function on that interval.

Now compare five quadrature rules on the same integral: left endpoint, right endpoint, midpoint $x_i = (i - \tfrac{1}{2})/n$, trapezoid, and Simpson with weights $1, 4, 2, 4, \dots, 4, 1$ divided by $3n$. The error is computed against the exact value $1/3$.

n	left	right	midpoint	trapezoid	Simpson
10	4.833×10^{-2}	5.167×10^{-2}	8.333×10^{-4}	1.667×10^{-3}	1.1×10^{-16}
100	4.983×10^{-3}	5.017×10^{-3}	8.333×10^{-6}	1.667×10^{-5}	0
1000	4.998×10^{-4}	5.002×10^{-4}	8.333×10^{-8}	1.667×10^{-7}	0

Three patterns stand out. The error of the left and right rules drops by a factor of ten each time n grows by a factor of ten: order $\mathcal{O}(1/n)$, about $1/(2n)$ because f is increasing. The error of the midpoint and trapezoid rules drops by a factor of a hundred, order $\mathcal{O}(1/n^2)$, with values $1/(12n^2)$ and $1/(6n^2)$; the two have opposite signs and the trapezoid is twice as large, so that $(2M + T)/3$ cancels the term. Simpson is exact to machine rounding, since the rule is exact for polynomials of degree at most three.

The third integral has no elementary antiderivative:

$$\int_0^1 e^{-x^2}\,dx.$$

That absence is a consequence of Liouville's theorem, which we quote without proof. The reference is the error function $\tfrac{\sqrt{\pi}}{2}\operatorname{erf}(1) = 0.746824132812427$; `quad` gives the same number with estimated error 8.3×10^{-15}. Simpson's rule gives an error of 3.563×10^{-4} at $n = 2$, 1.988×10^{-6} at $n = 8$, and 7.795×10^{-9} at $n = 32$.

For $f(x) = x^2$ on $[0, 1]$, derive the formula $U_n - L_n = 1/n$ yourself from the sum-of-squares formula, then compute U_{10} and L_{10} as fractions. Check that their average $(U_{10} + L_{10})/2$ equals $1/3 + 1/(6 \cdot 100)$, not $1/3$. Explain why that average is still not the best estimate, even though its error is far smaller than the error of U_{10} itself.

13.9 Improper Integrals and Their Relation to Series

Level: advanced

The interval of integration has so far always been bounded and the function bounded. Many real computations are not like that: we want the area of a region that extends without end, or the integral of a function that blows up at the endpoint of the interval. The method is not to widen the Riemann definition, but to write that integral as a limit of ordinary integrals.

Definition 13.5 (Improper integral)**.** If f is Riemann integrable on $[a, R]$ for every $R > a$, define

$$\int_a^{\infty} f = \lim_{R \to \infty} \int_a^R f,$$

provided that limit exists and is finite. If f is integrable on $[c, b]$ for every $c \in (a, b]$ but is unbounded near a, define

$$\int_a^b f = \lim_{c \to a^+} \int_c^b f.$$

In both cases, the integral is called *convergent* if the limit exists, and *divergent* if not.

The most important example is the family $f(x) = x^{-p}$ on $[1, \infty)$. For $p \neq 1$,

$$\int_1^R x^{-p} dx = \frac{R^{1-p} - 1}{1 - p},$$

so that the limit exists only when $p > 1$, with value $1/(p-1)$; for $p = 1$ the result is $\ln R$, which tends to infinity. Hence $\int_1^\infty x^{-2} dx = 1$, and the partial integrals head toward that value at rate $1/R$: 0.9, 0.99, 0.999 for $R = 10, 100, 1000$.

The link with the series of Chapter 5 is no coincidence: for f positive and decreasing, the areas of rectangles of height $f(n)$ bracket the area under the curve, so that

$$\sum_{n=N+1}^{\infty} f(n) \;\le\; \int_N^\infty f(x)\,dx \;\le\; \sum_{n=N}^{\infty} f(n).$$

Hence the integral test holds: $\sum f(n)$ converges if and only if $\int_1^\infty f$ converges. For $f(x) = 1/x^2$ both converge, with $\sum_{n=1}^\infty 1/n^2 = \pi^2/6 = 1.6449340668482264$. The left inequality also gives an error bound, and the larger N is the closer the error comes to that bound:

N	error of the partial sum	$1/N$
10	9.516634×10^{-2}	0.1
100	9.950167×10^{-3}	0.01
1000	9.995002×10^{-4}	0.001

It must be stressed: an improper integral is not a Riemann integral, but its limit. Hence substitution and integration by parts apply only when the limit exists, and for functions that change sign their convergence need not be absolute.

13.10 Computational Lab

Level: advanced

The following code is run with Python 3, `scipy` 1.18.0, and the module `fractions` for exact arithmetic. All partitions are of equal width on $[0, 1]$.

Computational lab

```
import math
from fractions import Fraction as F
from scipy.integrate import quad

def both_sums(n):
    U = sum(F(i * i, n ** 3) for i in range(1, n + 1))
    L = sum(F(i * i, n ** 3) for i in range(n))
    return U, L

print("n        U_n          L_n          U_n - L_n")
for n in (1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024):
    U, L = both_sums(n)
    print(f"{n:5d}  {float(U):.10f}  {float(L):.10f}  {float(U - L):.2e}")

def rules(f, n):
```

```
    h = 1 / n
    left = sum(f(i * h) for i in range(n)) * h
    right = sum(f(i * h) for i in range(1, n + 1)) * h
    mid = sum(f((i + 0.5) * h) for i in range(n)) * h
    trap = (f(0) / 2 + sum(f(i * h) for i in range(1, n)) + f(1) / 2) * h
    simp = (f(0) + f(1) + sum((4 if i % 2 else 2) * f(i * h)
                              for i in range(1, n))) * h / 3
    return left, right, mid, trap, simp

f = lambda x: x * x
print()
print("error |approximation - 1/3|, integral of x^2 on [0,1]")
print("   n      left     right       mid  trapezoid    Simpson")
for n in (10, 100, 1000):
    err = [f"{abs(v - 1 / 3):.3e}" for v in rules(f, n)]
    print(f"{n:4d}  " + "  ".join(err))

g = lambda x: math.exp(-x * x)
v, e = quad(g, 0, 1)
ref = math.sqrt(math.pi) / 2 * math.erf(1)
print()
print(f"quad exp(-x^2) = {v:.15f}   estimated error {e:.1e}")
print(f"erf reference  = {ref:.15f}   difference {v - ref:+.1e}")
print(f"quad x^2       = {quad(f, 0, 1)[0]:.17f}")
print()
print("error |Simpson - reference| for exp(-x^2)")
for n in (2, 4, 8, 16, 32):
    print(f"  n={n:3d}  {abs(rules(g, n)[4] - ref):.3e}")
print()
print("error |trapezoid - reference| for exp(-x^2)")
for n in (10, 100, 1000):
    print(f"  n={n:4d}  {abs(rules(g, n)[3] - ref):.3e}")
```

The output obtained, as it stands:

```
n        U_n          L_n          U_n - L_n
    1  1.0000000000  0.0000000000  1.00e+00
    2  0.6250000000  0.1250000000  5.00e-01
    4  0.4687500000  0.2187500000  2.50e-01
    8  0.3984375000  0.2734375000  1.25e-01
   16  0.3652343750  0.3027343750  6.25e-02
   32  0.3491210938  0.3178710938  3.12e-02
   64  0.3411865234  0.3255615234  1.56e-02
  128  0.3372497559  0.3294372559  7.81e-03
  256  0.3352890015  0.3313827515  3.91e-03
  512  0.3343105316  0.3323574066  1.95e-03
 1024  0.3338217735  0.3328452110  9.77e-04

error |approximation - 1/3|, integral of x^2 on [0,1]
   n      left     right       mid  trapezoid    Simpson
  10  4.833e-02  5.167e-02  8.333e-04  1.667e-03  1.110e-16
 100  4.983e-03  5.017e-03  8.333e-06  1.667e-05  0.000e+00
1000  4.998e-04  5.002e-04  8.333e-08  1.667e-07  0.000e+00

quad exp(-x^2) = 0.746824132812427   estimated error 8.3e-15
erf reference  = 0.746824132812427   difference +1.1e-16
quad x^2       = 0.33333333333333337

error |Simpson - reference| for exp(-x^2)
  n=  2  3.563e-04
  n=  4  3.125e-05
  n=  8  1.988e-06
```

```
  n= 16  1.246e-07
  n= 32  7.795e-09

error |trapezoid - reference| for exp(-x^2)
  n=  10  6.133e-04
  n= 100  6.131e-06
  n=1000  6.131e-08
```

Four things stand out. The difference halves each time n doubles, consistent with $1/1024$ in the last row. The left and right errors drop by a factor of ten, midpoint and trapezoid by a factor of a hundred. The Simpson column contains 1.110×10^{-16} and then zero, a residue of rounding, and Simpson's error for e^{-x^2} drops by about a factor of sixteen each time n is doubled.

That code does not prove a single theorem. It checks that the formulas of this chapter give the numbers stated. The proof of integrability of x^2 is in the computation $U_n - L_n = 1/n$ in the previous section: the output of a program is data, not an argument.

Note for instructors

The teaching order I suggest: start from the two different answers in the opening, then the definition of a partition and compute U, L for x^2 with four slices on the board. After that the refinement lemma, the criterion, and only then the classes of functions. Do not postpone the fundamental theorem of calculus: students need to see the hard work of the criterion bear fruit as a computational tool.

Two misconceptions that appear often. First, the integral is taken to be the same as the antiderivative, so that $F' = f$ is taken to hold by definition, whereas it is a theorem. Second, a function with many discontinuity points is taken to be certainly not integrable, because the Dirichlet example is too often used as the only example. Thomae's function removes that assumption in a single page.

Blackboard activity: ask students to guess $\int_0^1 x^2 dx$ only from U_n and L_n for $n = 1, 2, 4$, then to say how many n are needed for the error to be below 10^{-3} without computing the integral. The answer $n > 1000$ comes out of $1/n$, and that moment explains why an antiderivative is more valuable than quadrature. Sign of readiness to move on: students can explain the difference in direction of the two parts of the fundamental theorem of calculus, and can name one integrable function that is not continuous.

13.11 Summary

Level: basic

Summary.

- The Riemann integral is the only number trapped between all lower sums and all upper sums; when the two brackets meet, that value is called the integral.
- The integrability criterion replaces the examination of infinitely many partitions with one inequality $U(\mathcal{P}, f) - L(\mathcal{P}, f) < \varepsilon$.
- Continuous functions and monotone functions are integrable; the Dirichlet function is not, because every slice of it contains rational and irrational numbers, whereas Thomae's function is integrable even though it is broken at every rational number.
- The accumulation function $F(x) = \int_a^x f$ is always continuous and its derivative is f at every point where f is continuous, so every continuous function has an antiderivative.
- The second part states $\int_a^b F' = F(b) - F(a)$; its proof is the mean value theorem plus telescoping, with the condition that F' be integrable.

- Substitution and integration by parts are the chain rule and the product rule read backwards. An improper integral is a limit of Riemann integrals, and the integral test links it to the series of Chapter 5.

13.12 Exercises

Level: intermediate

Exercise 13.1.

1. Compute U_n and L_n for $f(x) = x$ on $[0, 1]$ with a uniform partition, express both as fractions in n, then determine $U_n - L_n$. Conclude that f is integrable and compute the integral.
2. Prove that f integrable on $[a, b]$ implies $|f|$ integrable and $\left|\int_a^b f\right| \le \int_a^b |f|$. Show that the converse is false with one example.
3. Assess this claim: "Every bounded function on $[a, b]$ is integrable, since its upper sum and lower sum are always finite." True, false, or not yet enough evidence? Name which part is supported by proof and which part is not.
4. Prove that an increasing function on $[a, b]$ is integrable by showing $U(\mathcal{P}, f) - L(\mathcal{P}, f) \le (b - a)(f(b) - f(a))/n$ for a uniform partition into n parts. Explain why that bound does not hold for a function that is not monotone.
5. Compute $\int_0^1 (3x^2 - 2x + 1)\,dx$ with the fundamental theorem of calculus, then check the result using Simpson's rule with $n = 100$ and the function `scipy.integrate.quad`. Report all three numbers and their differences.
6. Compute $\int_1^{10} x^{-2}dx$ and $\int_1^{\infty} x^{-2}dx$. Report the difference between them, then explain why that difference equals $1/10$ without recomputing.
7. Prove that if f is continuous, $f \ge 0$ on $[a, b]$, and $\int_a^b f = 0$, then $f(x) = 0$ for every $x \in [a, b]$. Also show that the statement fails if f is only integrable.
8. Compute $\int_0^{\pi/2} \sin x \cos x\,dx$ in two ways: the substitution $u = \sin x$, and the identity $\sin 2x = 2 \sin x \cos x$. Compare the results with the numerical value from `quad`.
9. Show that the Dirichlet function on $[0, 1]$ satisfies $U(\mathcal{P}, D) - L(\mathcal{P}, D) = 1$ for every partition, then explain why that fact is enough to conclude that it is not integrable.
10. Estimate $\sum_{n=101}^{\infty} 1/n^2$ with an integral bound and compare with the true value 9.950167×10^{-3}; what is the relative error?

Hints. Item 3: the claim contains two different statements, about the boundedness of the sums and about the equality of the two integrals. Item 7: use continuity at one point to make the integral small on a narrow interval around that point. Item 10: use $\sum_{n=N+1}^{\infty} 1/n^2 \le \int_N^{\infty} x^{-2}dx = 1/N$ with $N = 100$.

For researchers: limits and next steps

What is genuinely proved in this chapter: the integrability criterion with both of its directions, the integrability of continuous functions (using uniform continuity from Chapter 11), of monotone functions, of Thomae's function, both parts of the fundamental theorem of calculus, the mean value theorem for integrals, and the status of substitution and integration by parts as consequences, not axioms.

What is quoted without proof and should be called so in class: Lebesgue's criterion (Rudin, Theorem 11.33(b); Abbott, Chapter 7), Volterra's example of a bounded derivative that is not integrable, and the absence of an elementary antiderivative for e^{-x^2} (Liouville's theorem).

Primary sources: Rudin, *Principles of Mathematical Analysis*, Chapter 6, the section Definition and Existence of the Integral (Theorems 6.4, 6.6, 6.8, 6.9, 6.10) and the section

Integration and Differentiation (6.19–6.22). Rudin sets it up for the Riemann-Stieltjes integral; the Riemann integral is the case $\alpha(x) = x$. Abbott, *Understanding Analysis*, Chapter 7, is slower but more readable; Stewart only for the computational rules.
Open question: how sharp the quadrature error bounds are when only $|f''|$ is known, and how to estimate the error without any derivative at all; the standard answers are adaptive quadrature and the difference of two levels of refinement. Limits of validity: bounded functions on a closed bounded interval, improper integrals as limits, without measure theory, and vector-valued integrals are not touched.

Map to source texts: Chapter 13 follows Rudin, Chapter 6: the section Definition and Existence of the Integral (partitions, upper and lower sums, the refinement lemma as Theorem 6.4, the criterion as 6.6, continuous and monotone functions as 6.8 and 6.9) and the section Integration and Differentiation (6.19 change of variables, 6.21 the fundamental theorem of calculus, 6.22 integration by parts). In Abbott, Chapter 7, from The Definition of the Riemann Integral to Lebesgue's Criterion. In Stewart: The Substitution Rule and Average Value of a Function (Chapter 5), Integration by Parts and Improper Integrals (Chapter 7), and 11.3 The Integral Test, the bridge to Chapter 5 of this book.

Chapter 14

Series of Functions and the Exchange of Limits

A small numerical library computes $\ln(1+x)$ with the series $\sum_{k\ge 1}(-1)^{k+1}x^k/k$. Its author tested the function at $x=0.5$ with twelve terms and compared it against a standard library: the error was 6.4×10^{-6}. The test passed, the documentation states that the series is accurate to six digits, and the code went into production. The first user called it with $x=0.99$. Twelve terms give 0.6529204447, whereas $\ln 1.99 = 0.6881346387$. The error is 3.5×10^{-2}, about five thousand times larger, for the same function with the same number of terms.

That damage is not a typo. For every x in $(-1,1)$ the series converges to $\ln(1+x)$; what changes is only how many terms are needed: 16 terms at $x=0.5$, 89 at $x=0.9$, 721 at $x=0.99$, 5248 at $x=0.999$, all of them for an error below 10^{-6}. No single number of terms suffices for all x, because x may lie as close as we please to 1.

Compare with a closed subinterval $[-r,r]$, $0<r<1$. There one estimate holds simultaneously for all x: the remainder of an alternating series is bounded by $r^{N+1}/(N+1)$, and that bound does not mention x. For $r=0.5$ and $N=12$ the bound is 9.4×10^{-6}, the actual error 6.4×10^{-6}. This is the difference that this entire chapter is about.

Chapter 5 dealt with series of numbers, and each x gives a different series of numbers. Chapters 11 and 13 define continuity and the integral point by point, so neither of them has a way to move a limit past an integral sign without extra conditions. The tool still missing is a distance between two functions; once that distance exists, the whole family of series becomes a single sequence in a metric space.

For younger readers (age 10 and up)

Imagine a class with thirty students, and the teacher writes a single number on the board: one hundred. Each student's task is to guess that number, and after every guess the teacher announces whether the guess is getting closer. The first student is very fast: within five guesses he is off by less than one. The second student is slow: he needs twenty guesses. The third is very slow: he needs a hundred. But all three, given enough time, end up off by less than one.

Now the teacher changes the question. Not "does every student eventually get close", but "is there a moment when all students are already close at the same time". For that class the answer is no: the slowest student is always lagging, and one can always add a student who is slower still.

The same question arises for the formulas a computer uses. A formula may be correct for every value put into it, yet the number of terms needed can blow up without bound as the value approaches the edge. If a program takes "correct for every value" to mean

"one formula suffices for all values", that program will be far off at the most troublesome values, and the error will not show up in the test its author chose.
In this chapter we shall see why those two questions are not a matter of words, but a matter of whether two operations may be exchanged in order: limit with integral, limit with derivative, and summation with both.

14.1 Pointwise and uniform convergence

Level: basic

Take a family of functions $f_1, f_2, \dots$ all defined on a single set $E \subseteq \mathbb{R}$. For one point x that we hold fixed, the values $f_1(x), f_2(x), \dots$ are nothing but a sequence of numbers, and Chapter 4 already said what it means for that sequence to converge. Hence the first definition almost writes itself.

Definition 14.1. Suppose f_n and f are defined on $E \subseteq \mathbb{R}$.

1. We say $f_n \to f$ *pointwise* on E if for every $x \in E$ and every $\varepsilon > 0$ there is N such that $n \geq N$ implies $|f_n(x) - f(x)| < \varepsilon$. Here N may depend on x and on ε.
2. We say $f_n \to f$ *uniformly* on E if for every $\varepsilon > 0$ there is N such that $n \geq N$ implies $|f_n(x) - f(x)| < \varepsilon$ for *all* $x \in E$ at once. Here N depends only on ε.
3. If h is bounded on E, the supremum norm of h is $\|h\|_\infty = \sup_{x \in E} |h(x)|$.

uniform convergence (*konvergensi seragam*) — all points must already be close at the same time, not one by one on their own schedules.

One quantifier changes place, and the whole content of this chapter hangs on that move. In the first definition the order is "for every point there is a threshold". In the second the order is "there is one threshold for all points". The first sentence is true for the series in the chapter opening, the second is false, and the difference between them is the difference between 16 terms and 5248 terms.

Read it aloud (auditory learner)

Two voices take turns.
First voice: "Does this series get there?" Second voice: "It gets there. Slowly, but it gets there." First voice: "Does it get there for this point?" Second voice: "For this point. For the point next door, another time, at its own pace."
That is the first promise. Every point may be slow, as long as it has one tempo of its own. A point far from the edge arrives within a dozen steps; a point pressed against the edge needs thousands.
Now the first voice asks in other words: "Is there one tempo for all points at once?"
The second voice stays silent longer. "On a closed interval that stops before the edge, yes. On the whole open interval, no. And what is missing there is not a failure of counting; it merely demands a tempo that moves together with the point."
Repeat this line until it sits on the tongue: all points may be slow, as long as one tempo. If the tempo must differ for each point, its name is not uniform.

Example 14.1. Suppose $f_n(x) = x^n$ on $E = [0, 1]$. For $0 \leq x < 1$ we have $x^n \to 0$, whereas $x^n = 1$ for all n if $x = 1$. So f_n converges pointwise to

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

That convergence is not uniform, since $\sup_{x\in[0,1]} |x^n - f(x)| = 1$ for every n: at $x = 1$ the difference is exactly 1, and for x close enough to 1 from the left x^n approaches 1 while $f(x) = 0$.

That example shows two things. First, the pointwise limit of continuous functions need not be continuous: every x^n is continuous, its limit jumps at $x = 1$. Second, uniformity depends on the set, not only on the formula of the function.

That dependence can be measured. On $[0, r]$ with $r < 1$ the convergence is uniform, since $\sup_{[0,r]} |x^n| = r^n \to 0$. What is interesting is when the right endpoint also moves with n. On $[0, 1 - 1/n]$ the supremum is $(1 - 1/n)^n \to 1/e = 0.3678794412$; so shrinking by $1/n$ does not yet rescue uniformity. On $[0, 1 - 1/\sqrt{n}]$ the supremum is $(1 - 1/\sqrt{n})^n$, which collapses like $e^{-\sqrt{n}}$: 2.2×10^{-2} at $n = 10$, 2.7×10^{-5} at $n = 100$, 2.2×10^{-44} at $n = 10^4$. The rate of shrinking, not the formula of the function, is what decides.

The diagonal test yields sharper additional information. Take the points $x_n = 1 - 1/n$ chasing the break point. Then $f_n(x_n) = (1 - 1/n)^n \to 1/e$, while $f(x_n) = 0$ for all n. So not only does the supremum fail to shrink: the values at well-chosen points also refuse to approach the pointwise limit. This pattern is used repeatedly in this chapter. To refute uniformity, find one sequence of points that keeps the difference large.

Two simple families separate the two symptoms. In the first, g_n is a triangle of height 1 with base $1/n$ and apex at $1 - 1/n$: the apex moves right, the height never drops, and the area is $1/(2n)$. A point $x < 1$ is touched by the apex only while $1/(2(1 - x)) < n < 3/(2(1 - x))$, so at $x = 0.9$ only nine values of n give a positive value, at $x = 0.99$ ninety-nine, and once the last apex has passed the value is zero forever. The limit is zero at every point, whereas $\|g_n\|_\infty = 1$. In the second, h_n has height $1/n$ with base 1 in the same place: the apex shrinks instead of moving, and $\|h_n\|_\infty = 1/n \to 0$ (Figure 14.1).

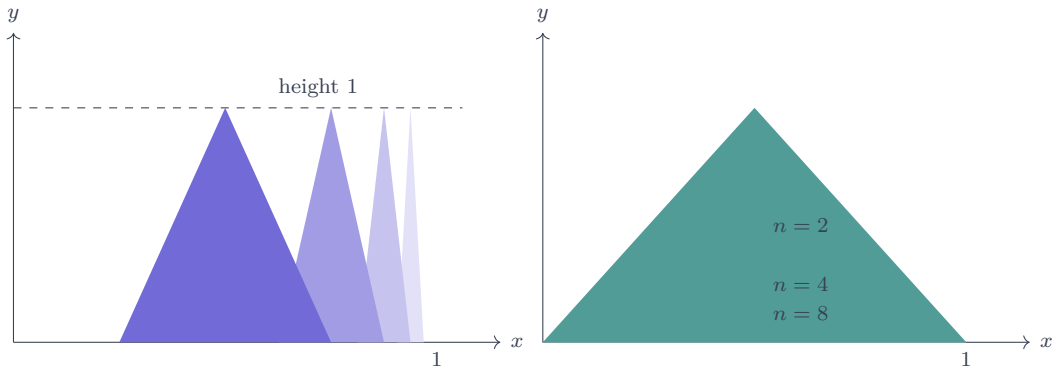


Figure 14.1: Apexes that move and apexes that shrink. Left: g_n has constant height 1, base $1/n$, and apex at $1 - 1/n$; the height does not drop, so $\|g_n\|_\infty = 1$ even though $g_n(x) \to 0$ at every point. Right: h_n has height $1/n$ with base 1 in the same place; its whole graph fits inside the band of height $1/n$, so it converges uniformly.

Lemma 14.1. *$f_n \to f$ uniformly on E if and only if $\|f_n - f\|_\infty \to 0$.*

Proof. Uniform convergence means: for every $\varepsilon > 0$ there is N such that for $n \geq N$ we have $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in E$, that is, $\sup_x |f_n(x) - f(x)| \leq \varepsilon$. That is precisely the statement that the sequence of real numbers $\|f_n - f\|_\infty$ tends to zero. The only technical difference is that the definition uses $< \varepsilon$ whereas the norm uses $\leq \varepsilon$; the two are equivalent because ε is arbitrary. □

That lemma turns a question about functions into a question about a single sequence of numbers: to check uniformity one supremum suffices, to reject it it suffices to show that this supremum does not tend to zero, as with x^n.

Uniformity has an exact picture, namely a band. Take $\varepsilon = 0.1$. If the whole graph of f_n for $n \geq N$ lies between $f - \varepsilon$ and $f + \varepsilon$, then that single N works for all points. What is checked is not one point, but whether any part of the graph escapes the band of width 0.2. For g_n that band is always pierced by the apex; the hoped-for situation is visible in Figure 14.2.

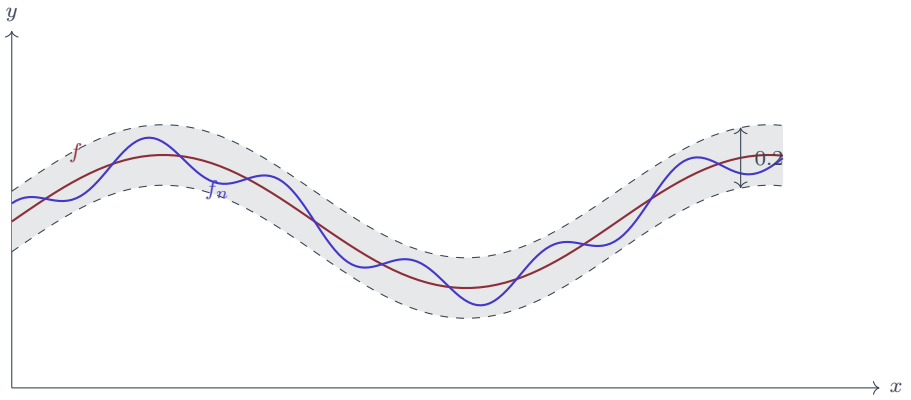


Figure 14.2: The uniform band of width 0.2 around f, that is between $f - 0.1$ and $f + 0.1$. The wavy graph of f_n still lies entirely inside the band, so $\|f_n - f\|_\infty \leq 0.1$ and one and the same N works for all points at once.

Do it by hand (kinesthetic learner)

Take a two-meter rope, tape, and two people. Stretch the rope straight on the floor, then tape five marks at equal spacing. The first person holds the left end and lifts it by just 5 cm. Watch: only that end rises, the other four marks still touch the floor. That is pointwise convergence.

Now both people lift both ends together on the same count: 5 cm, then 3 cm, then 1 cm, then half a centimeter. Watch the whole length of the rope. No point is left on the floor, and the greatest distance from rope to floor is the same at all points, because both ends rise together. That is uniform convergence.

Finally, the first person lifts his end while walking toward the middle, while the second lowers hers. The rope stays tilted, and there is always some part 5 cm above the floor: every point eventually comes down, but the greatest distance never shrinks. That is g_n in Figure 14.1.

Lemma 14.2 (Uniform Cauchy criterion)**.** *The sequence f_n converges uniformly on E if and only if for every $\varepsilon > 0$ there is N such that $n, m \geq N$ implies $|f_n(x) - f_m(x)| < \varepsilon$ for all $x \in E$. This last condition is called the uniform Cauchy criterion.*

Proof. ($\Rightarrow$) If $f_n \to f$ uniformly, take N from the definition of convergence for $\varepsilon/2$. For $n, m \geq N$ and every x,

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)| < \varepsilon.$$

($\Leftarrow$) For every fixed x, the sequence $f_n(x)$ is a Cauchy sequence in $\mathbb{R}$, so it converges to a number that we call $f(x)$. This defines a function f point by point. To obtain uniformity, fix $n \geq N$ in the Cauchy condition and let $m \to \infty$. Since $f_m(x) \to f(x)$, we get $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in E$ at once, and the N used does not depend on x. □

The proof of the second direction uses only the completeness of $\mathbb{R}$: the space of functions is complete with respect to the supremum norm, and that completeness is what lets a series of functions be summed without fear that the limit leaves the class of functions we use.

Note 14.1. Uniform convergence is a property of a pair: a sequence and a set. The sentence "x^n converges uniformly" is incomplete without naming the set. On $[0,1]$ the answer is no, on $[0,0.99]$ yes: the supremum norm of the error is 0.99^n, namely 0.37 at $n=100$ and 4.3×10^{-5} at $n=1000$. Therefore every claim of uniformity in this chapter names its set.

14.2 The Weierstrass test: uniformity from a single series of numbers

Level: intermediate

A series of functions is a series whose terms are functions. Its partial sums $S_N=\sum_{n=1}^{N} f_n$ are functions, so a series of functions is nothing but a particular sequence of functions, and all the definitions of the previous section apply as soon as we replace f_n by S_N.

Definition 14.2. The series $\sum_{n\ge1} f_n$ *converges pointwise* on E if for every $x\in E$ the series of numbers $\sum_n f_n(x)$ converges. That series *converges uniformly* on E if its sequence of partial sums converges uniformly on E.

The following test is the cheapest way to obtain uniformity, and almost every example in this book uses it. The idea: if each term is bounded by a number whose series converges, then the tail of the series of functions is bounded by the tail of that series of numbers, and the numerical tail knows nothing about x.

Theorem 14.3 (Weierstrass M-test)**.** *Suppose f_n is defined on E and there are numbers $M_n\ge0$ with $|f_n(x)|\le M_n$ for all $x\in E$ and all n. If $\sum M_n$ converges, then $\sum f_n$ converges uniformly on E. If E is a closed bounded interval and every f_n is continuous, then the sum is continuous on E.*

Proof. Write $S_N=\sum_{i=1}^{N} f_i$. For $n>m$ and every $x\in E$,

$$|S_n(x)-S_m(x)|\le\sum_{i=m+1}^{n}|f_i(x)|\le\sum_{i=m+1}^{n}M_i.$$

Since $\sum M_n$ converges, the tail $\sum_{i=m+1}^{n}M_i$ can be made smaller than ε provided m,n are large enough, and that choice does not mention x. The uniform Cauchy criterion closes the proof of uniformity. For continuity, each S_N is continuous as a finite sum of continuous functions, and the uniform limit of continuous functions is continuous; that last statement is proved in the next section. $\square$

Example 14.2. On $E=[-1,1]$ the series $\sum_{n\ge1}x^n/n^2$ converges uniformly, since $|x^n/n^2|\le 1/n^2$ and $\sum 1/n^2=\pi^2/6$ converges. Its sum is continuous on $[-1,1]$, and its value at $x=1$ is 1.6449340668. The sum of two million terms gives 1.6449335668; the difference is 5.0×10^{-7}, proportional to the tail bound $1/N=5\times10^{-7}$. At $x=-1$ the series converges to $-\pi^2/6$, since the terms alternate in sign with decreasing absolute value tending to zero.

Example 14.3. On $E=\mathbb{R}$ the series $\sum_{n\ge0}2^{-n}\sin(2^n x)$ converges uniformly, since $|2^{-n}\sin(2^n x)|\le 2^{-n}$ and $\sum 2^{-n}=1$. Every term is continuous, so the sum is continuous at every point. What is surprising is that the sum is not differentiable at any point: the classical result of Weierstrass (1872), extended by Hardy up to ratio 1, which we quote without proof. Uniformity guarantees continuity, but it guarantees nothing about derivatives.

Note 14.2. The M-test gives a sufficient condition, not a necessary one, and its failure is not a proof of anything. The series $\sum_{n\ge1}(-1)^n(1-x)x^n$ on $[0,1]$ converges uniformly: the remainder after the N-th term is bounded by $(1-x)x^{N+1}$, whose maximum on $[0,1]$ is

$\frac{1}{N+2}\left(\frac{N+1}{N+2}\right)^{N+1} \to 0$, namely 3.20×10^{-2} at $N = 10$ and 3.67×10^{-4} at $N = 1000$. But $\sup_{[0,1]} |f_n| = n^n/(n+1)^{n+1}$, proportional to $1/(en)$, so that $\sum_n \sup |f_n|$ diverges: 0.88 up to $N = 10$ and 2.53 up to $N = 1000$. There is no convergent comparison series, so the M-test does not apply even though the uniformity is genuine.

For younger readers (age 10 and up)

There are two ways to work through a list of chores. The first: finish all the chores one by one, then add up the total. The second: keep a running total while working, then finish the rest. For ordinary addition both ways give the same result, and we are used to swapping them without thinking.

But some tasks do not like being swapped. Imagine you measure the length of a pole's shadow every hour for a day, compute the average, then look for the hour when that average is longest. Reverse the order: first find the longest hour on each day, then average that hour over the month. The two results can differ widely, not because the arithmetic is wrong, but because "longest" and "average" genuinely do not commute.

The same thing happens with the formulas used to add up infinitely many terms, and then integrate or differentiate them. Sometimes the order may be swapped and the result is exactly the same; sometimes not, and the difference is large. In this chapter we shall know the warning signs before the result goes wrong.

14.3 When limits may be exchanged

Level: intermediate

The three operations we habitually use on functions are taking a limit at a point, integrating, and differentiating. Each of the three is itself a limit: continuity at x_0 means $\lim_{x\to x_0} f(x) = f(x_0)$, the integral is the limit of Riemann sums, the derivative is the limit of a difference quotient. So moving the limit of a sequence past one of those operations means exchanging two limits, and that exchange is not free: its price is one hypothesis, namely uniformity.

Theorem 14.4. *If f_n is continuous on E and $f_n \to f$ uniformly on E, then f is continuous on E.*

Proof. Take $x_0 \in E$ and $\varepsilon > 0$. Since the convergence is uniform, there is n with $\|f_n - f\|_\infty < \varepsilon/3$. Since f_n is continuous at x_0, there is $\delta > 0$ such that $|x - x_0| < \delta$ implies $|f_n(x) - f_n(x_0)| < \varepsilon/3$. Then for $x \in E$ with $|x - x_0| < \delta$,

$$|f(x) - f(x_0)| \le |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)| < \varepsilon.$$

□

Note the logical order: n is chosen first, then δ, and δ may depend on n. What is not allowed is choosing a δ that must work for all n at once.

Challenge (problem solver)

The series $\sum_{n\ge 0}(x^{2n} - x^{2n+1})$ has continuous terms on $[0,1]$ and converges at every one of its points. Determine its sum, then show that this sum is not continuous at $x = 1$. Why does this result not contradict the theorem above? A correct answer must name one number that may depend on ε but must not depend on x.

Hint 1: the partial sums have a closed form, and that form contains x^{2N+2}.

Hint 2: for $x < 1$ the sum is $1/(1+x)$, whereas at $x = 1$ every term equals zero; so the value jumps from $1/2$ to 0.
If you get stuck, go back to the section on the diagonal test and ask what happens at $x_N = 1 - 1/N$.

Theorem 14.5 (Term-by-term integration)**.** *If f_n is Riemann integrable on $[a,b]$ and $f_n \to f$ uniformly on $[a,b]$, then f is integrable on $[a,b]$ and*

$$\int_a^b f = \lim_{n\to\infty} \int_a^b f_n.$$

Proof. For integrability, take $\varepsilon > 0$ and choose n with $\|f_n - f\|_\infty < \varepsilon$. Since f_n is integrable, there is a partition $\mathcal{P}$ with $U(\mathcal{P}, f_n) - L(\mathcal{P}, f_n) < \varepsilon(b-a)$. On each subinterval, $M_i(f) \le M_i(f_n) + \varepsilon$ and $m_i(f) \ge m_i(f_n) - \varepsilon$, so that $U(\mathcal{P}, f) \le U(\mathcal{P}, f_n) + \varepsilon(b-a)$ and $L(\mathcal{P}, f) \ge L(\mathcal{P}, f_n) - \varepsilon(b-a)$. Those three inequalities give $U(\mathcal{P}, f) - L(\mathcal{P}, f) < 3\varepsilon(b-a)$, and the integrability criterion of Chapter 13 closes the argument. For the value of the integral, it suffices to note that

$$\left| \int_a^b f_n - \int_a^b f \right| = \left| \int_a^b (f_n - f) \right| \le \int_a^b |f_n - f| \le (b-a)\, \|f_n - f\|_\infty,$$

and the right-hand side tends to zero. □

Corollary 14.6 (Integration of series)**.** *If $\sum_{n\ge 1} f_n$ converges uniformly on $[a,b]$ and every f_n is integrable there, then*

$$\int_a^b \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} \int_a^b f_n.$$

This corollary is applied to the partial sums, with uniformity usually checked through the M-test. For example, the series $\sum_{n\ge 1} x^n/n^2$ converges uniformly on $[0,1]$, so that

$$\int_0^1 \sum_{n=1}^{\infty} \frac{x^n}{n^2}\, dx = \sum_{n=1}^{\infty} \frac{1}{n^2(n+1)} = \frac{\pi^2}{6} - 1 = 0.6449340668.$$

That last number is also obtained by summing the first two million terms of the series on the right-hand side, which gives 0.6449340668 at machine precision.

Example 14.4 (Failure without uniformity). Suppose $f_n(x) = nx(1-x^2)^n$ on $[0,1]$. For every $x \in (0,1]$ the factor $(1-x^2)^n$ decreases geometrically, so $n(1-x^2)^n \to 0$ and $f_n(x) \to 0$; at $x = 0$ every f_n equals 0. So the pointwise limit is the zero function. Its integral can be computed exactly by the substitution $u = 1 - x^2$:

$$\int_0^1 nx(1-x^2)^n\, dx = \frac{n}{2} \int_0^1 u^n\, du = \frac{n}{2(n+1)}.$$

That sequence tends to $1/2$, not to 0. Numerical quadrature gives 0.4545454545 for $n = 10$ and 0.4950495050 for $n = 100$, matching $n/(2(n+1))$ to the last digit. So $\lim_n \int_0^1 f_n = 1/2$ whereas $\int_0^1 \lim_n f_n = 0$.

That failure has a geometric shape. The function f_n is zero at both ends and peaks at $x = 1/\sqrt{2n+1}$, with height 1.3396685 at $n = 10$ and 4.2834707 at $n = 100$: the peak grows taller and narrower, so that its area stays near $1/2$ while its height at every point tends to zero.

Theorem 14.7 (Term-by-term differentiation)**.** *Suppose f_n is differentiable on $[a, b]$, the sequence f_n' converges uniformly on $[a, b]$, and there is one point $x_0 \in [a, b]$ where $f_n(x_0)$ converges. Then f_n converges uniformly to some f that is differentiable on $[a, b]$, and*

$$f'(x) = \lim_{n\to\infty} f_n'(x)$$

for every $x \in [a, b]$.

Sketch of proof. The heart of it is the mean value theorem applied to the difference of two terms: for m, n and every $x, y \in [a, b]$,

$$|(f_n - f_m)(x) - (f_n - f_m)(y)| \le \|f_n' - f_m'\|_\infty \, |x - y|.$$

Since f_n' is uniformly Cauchy, the right-hand side can be made small simultaneously for all pairs x, y once m, n are large. Together with convergence at one point x_0, that inequality makes f_n uniformly Cauchy, so $f_n \to f$ uniformly. For the derivative, divide by $|x - y|$, let $y \to x$, then exchange the two limits that appear; the same inequality guarantees that the exchange is legitimate. The technical details are in Rudin, Theorem 7.17. □

The condition "there is one point where $f_n(x_0)$ converges" cannot be dropped: f_n and $f_n + 1$ have the same sequence of derivatives, but only one of them converges at x_0. Conversely, uniformity of f_n alone is also not enough to conclude anything about the derivative.

Example 14.5 (Derivatives that do not converge). Suppose $f_n(x) = \sin(nx)/n$ on $[0, \pi]$. Since $\|f_n\|_\infty = 1/n$, the sequence converges uniformly to the zero function; its value is 0.1 at $n = 10$ and 0.01 at $n = 100$. But $f_n'(x) = \cos(nx)$: at $x = 0$ that sequence equals 1 for all n, whereas at $x = \pi/2$ its values are $0, -1, 0, 1, 0, -1, \ldots$ according to the parity of n, which does not converge. So f_n' fails to converge pointwise anywhere except at special points, let alone uniformly. Differentiation does not forgive small high-frequency oscillations: f_n shrinks to zero, but its slope does not.

Figure 14.3 shows three of those derivative curves on $[0, \pi]$. At $x = \pi/2$ they take the values 1, 0, and -1: three different values of one sequence at one point, and that is already enough to say the limit does not exist.

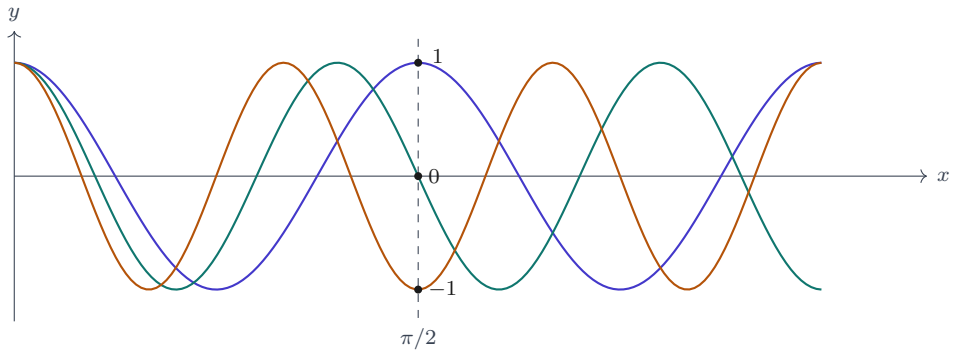


Figure 14.3: The derivatives $f_n' = \cos(nx)$ on $[0, \pi]$ for $n = 4$ (indigo), $n = 5$ (teal), and $n = 6$ (orange). At $x = \pi/2$ they take the values 1, 0, and -1: three different values of one sequence at one point, so the limit of the derivatives does not exist, even though $f_n = \sin(nx)/n$ converges uniformly to zero with amplitude 1/4, 1/5, and 1/6.

Note 14.3. In machine learning, the risk function is often written as an expectation over the data distribution, $\mathbb{E}_{(x,y)\sim D}[\ell(f(x), y)]$, and then differentiated with respect to the model parameters. If the distribution is finite discrete, as on the training set, that expectation

is a finite sum and exchanging the derivative with the sum is always legitimate, with no conditions. If it is continuous, the expectation is an integral, and moving the derivative inside it demands a domination condition more delicate than uniformity; the tool is the dominated convergence theorem of measure theory, which is not discussed here. “Differentiate inside the integral” is an operation with a license, not an operation that is always correct.

14.4 Power series and the radius of convergence

Level: advanced

A power series is a series of functions of the most rigid shape: the n-th term is a fixed coefficient times x^n. That rigidity bears fruit. The set on which the series converges is always an interval symmetric about its center, so that all the information about where the series is useful is contained in one number.

Definition 14.3. A series of the form $\sum_{n\ge 0} a_n x^n$ with $a_n \in \mathbb{R}$ is called a power series centered at 0. The number

$$R = \sup\Big\{ r \ge 0 : \sum_{n\ge 0} a_n r^n \text{ converges} \Big\}$$

is called the radius of convergence of that series.

power series (*deret pangkat*) — a series whose terms are fixed coefficients times rising powers of one variable, like the same recipe for every value.

radius of convergence (*jari-jari konvergensi*) — the furthest distance from the center at which that recipe still produces a finite number.

Theorem 14.8. *Suppose $\sum a_n x^n$ is a power series with radius R. Then the series converges absolutely for every $|x| < R$ and diverges for every $|x| > R$. Moreover*

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n},$$

with the conventions $1/0 = \infty$ and $1/\infty = 0$.

Proof. Fix $x \neq 0$ and write $|a_n x^n|^{1/n} = |x| \cdot |a_n|^{1/n}$. By the root test of Chapter 5, the series $\sum a_n x^n$ converges absolutely if $\limsup_n |a_n x^n|^{1/n} < 1$, that is if $|x| < 1/\limsup_n |a_n|^{1/n}$, and diverges if that limit superior is greater than 1, since its terms do not tend to zero. Since x is arbitrary, the boundary between the two regimes is exactly $R = 1/\limsup_n |a_n|^{1/n}$, and the set of convergence lies inside $[-R, R]$. □

At $|x| = R$ the root test gives no verdict, and indeed there is no general verdict: $\sum x^n/n^2$ converges at both ends, $\sum x^n/n$ only at $x = -1$, and $\sum x^n$ diverges at both.

Corollary 14.9. *If $a_n \neq 0$ for n large enough and $|a_{n+1}/a_n| \to L$, then $R = 1/L$ with the same conventions as above.*

The reason is the chain of inequalities

$$\liminf |a_{n+1}/a_n| \le \liminf |a_n|^{1/n} \le \limsup |a_n|^{1/n} \le \limsup |a_{n+1}/a_n|,$$

which is proved as an exercise in Chapter 5. If the ratio has a limit, all four of those numbers are equal, so the ratio formula and the root formula give the same R. In hand computation, the ratio formula is almost always faster.

Now the part that makes power series useful: they may be differentiated and integrated term by term, and the result holds on the whole interval of convergence, not only on a small piece of it.

Theorem 14.10 (Term-by-term differentiation and integration)**.** *Suppose $\sum a_n x^n$ has radius $R > 0$. Then for every x with $|x| < R$,*

$$\frac{d}{dx}\sum_{n\geq 0} a_n x^n = \sum_{n\geq 1} n\, a_n x^{n-1}, \qquad \int_0^x \sum_{n\geq 0} a_n t^n\, dt = \sum_{n\geq 0} \frac{a_n}{n+1} x^{n+1},$$

and both new series also have radius R.

Proof. Take r with $0 < r < R$ and choose ρ with $r < \rho < R$. The series $\sum |a_n|\rho^n$ converges, so there is C with $|a_n|\rho^n \leq C$ for all n. For $|x| \leq r$,

$$|n\, a_n x^{n-1}| \leq n\, \frac{C}{\rho}\left(\frac{r}{\rho}\right)^{n-1},$$

and the series $\sum_n n(r/\rho)^{n-1}$ converges since $r/\rho < 1$. The M-test therefore gives uniformity of the differentiated series on $[-r, r]$, and the term-by-term differentiation theorem gives equality there. Every point $|x| < R$ lies in some $[-r, r]$ with $r < R$, so the equality holds on all of $(-R, R)$. For the integral, the same route is used on $\sum |a_n| r^{n-1}$, which also converges. The radius does not change because the coefficients of the differentiated series are na_n, and

$$\limsup_n (n|a_n|)^{1/n} = \limsup_n |a_n|^{1/n} \cdot \lim_n n^{1/n} = \limsup_n |a_n|^{1/n},$$

since $n^{1/n} \to 1$. The same argument applies to the coefficients $a_n/(n+1)$, because $(n+1)^{-1/n} \to 1$. □

That proof uses the whole content of the chapter: the M-test for uniformity, the differentiation theorem to move the limit, and the radius formula to make sure the new series does not shrink. Uniformity is claimed only on $[-r, r]$, $r < R$; in the opening example, $\ln(1+x)$ is uniform on every $[-r, r]$ with $r < 1$ but not on $(-1, 1)$.

Example 14.6. For $|x| < 1$, the geometric series gives $\sum_{n\geq 0} x^n = 1/(1-x)$. Differentiating term by term,

$$\sum_{n\geq 1} n\, x^{n-1} = \frac{1}{(1-x)^2},$$

and at $x = 1/2$ the right-hand side is 4. The sum of the first two hundred terms of the series on the left gives 4.000000 at machine precision. Integrating term by term from 0 to x gives

$$\sum_{n\geq 1} \frac{(-1)^{n+1} x^n}{n} = \ln(1+x), \qquad |x| < 1,$$

which is the series from the chapter opening. Both results are not numerical coincidences: both are consequences of the theorem above, because $1/2$ and the x used lie inside the radius.

With that theorem, the opening question of the chapter is answered. The series for $\ln(1+x)$ converges uniformly on every $[-r, r]$ with $r < 1$, since $|(-1)^{n+1}x^n/n| \leq r^n/n$ and $\sum r^n/n$ converges. On $(-1, 1)$ uniformity fails, and the price shows in the number of terms: 16 at $x = 0.5$, 5248 at $x = 0.999$. At the endpoint $x = 1$ the series still converges, since it becomes an alternating harmonic series. That is not a consequence of the radius but of Abel's theorem: if $\sum a_n R^n$ converges, then $\sum a_n x^n$ converges uniformly on $[0, R]$. We quote it; the proof is in Rudin, chapter "Some Special Functions", Theorem 8.2.

One practical warning: uniform bounds are designed for the worst case. For the series $\ln(1+x)$ with $N = 12$ and $r = 0.5$, the bound $r^{13}/13 = 9.4 \times 10^{-6}$ is only three times the actual error at $x = 0.5$, but at $x = 0.1$ it is loose by about ten thousand times.

Step-by-step recipe (procedural learner)

1. Write down the difference, $M_N = \sup_{x \in E} |f_N(x) - f(x)|$, or estimate it with the value at the worst point.
2. Find the point x_N that maximizes that difference; usually a peak, or a point chasing the edge of the interval.
3. Compute M_N as an expression containing only N, then check $M_N \to 0$.

A warning about order: find x_N first for each N, then compute its supremum. Reversing that order, that is choosing N first and then looking for one x that represents all points, yields a wrong verdict of uniformity for g_n in Figure 14.1, whose apex actually shifts together with N.

14.5 Worked numerical examples

LEVEL: INTERMEDIATE

The three groups of numbers below are recomputed in the computational lab. The first measures the failure of uniformity of x^n on $[0, 1]$.

n	$\sup_{[0,1]} x^n$	$(1 - \frac{1}{n})^n$	$(1 - \frac{1}{\sqrt{n}})^n$	0.9^n
10	1	0.3486784401	2.234×10^{-2}	0.3486784401
100	1	0.3660323413	2.656×10^{-5}	2.656×10^{-5}
1000	1	0.3676954248	1.108×10^{-14}	1.748×10^{-46}
10000	1	0.3678610464	2.249×10^{-44}	$< 10^{-450}$

The second and fifth columns are two different rates. At the fixed point $x = 0.9$ the error is 0.9^n, falling by a factor of 0.9 per term: 132 terms are needed to reach 10^{-6}, whereas at $x = 0.5$ 20 terms suffice and at $x = 0.99$ it takes 1375. Every point has its own exponential rate, with base x. The second column does not move: the uniform error on $[0, 1]$ stays 1 for all n, so its rate of decrease is zero. What is being compared is therefore not two numbers, but something that tends to zero and something that does not.

The second group of numbers is the error of approximating e^x by a Taylor series at $x = 1$. The remainder of the series is bounded by $\frac{e}{(N+1)!}$ for $x \in [0, 1]$, since the $(N+1)$-st derivative of e^x is $e^x \le e$.

N	$S_N(1) = \sum_{k=0}^{N} \frac{1}{k!}$	$e - S_N(1)$	bound $\frac{e}{(N+1)!}$
5	2.716666666667	1.615×10^{-3}	3.775×10^{-3}
10	2.718281801146	2.731×10^{-8}	6.810×10^{-8}
15	2.718281828459	5.063×10^{-14}	1.299×10^{-13}
20	2.718281828459	0 (below precision)	5.320×10^{-20}

The bound is always correct and larger than the actual error, by a factor of about 2.3 to 2.6: the remainder is roughly $\frac{1}{(N+1)!}\left(1 + \frac{1}{N+2} + \cdots\right)$, and e appears because its terms are summed at $x = 1$. At $N = 20$ the actual error is no longer measurable, since 5.3×10^{-20} is far below one unit in the last place of a quantity of size 2.718, namely about 4.4×10^{-16}.

The third group of numbers is three radii of convergence, computed in two different ways at term 1000.

series	R from the formula	$\lvert a_{1000}/a_{1001} \rvert$	$\lvert a_{1000} \rvert^{1/1000}$
$\sum x^n/n^2$	1	1.0020010000	0.9862794856
$\sum \frac{n!}{n^n} x^n$	$e = 2.7182818285$	2.7169239322	0.3694916635
$\sum \frac{2^n}{n} x^n$	0.5	0.5005000000	1.9862320968

The third column estimates R from the ratio of two consecutive terms, the fourth from the reciprocal of the root estimate. For the second series, $|a_n/a_{n+1}| = (1+1/n)^n$ rises to e; at $n = 1000$ its value is 2.7169239, off by 0.0014. For the first series, $(1+1/n)^2$ decreases to 1 from above, so the estimate is 1.0020, and that error 0.0020 is a sign that we are only at term 1000. The fourth column approaches its limit more slowly, since $|a_n|^{1/n}$ approaches 1 at rate $|\ln a_n|/n$; at $n = 1000$ the difference is 0.0016, proportional to $\ln(2\pi n)/(2n)$ times $1/e$.

Summary of numbers. The uniform error of x^n on $[0,1]$ is 1 for every n, while the error at a fixed point x is x^n: 132 terms for 10^{-6} at $x = 0.9$, 1375 terms at $x = 0.99$. The series for e^x at $x = 1$ reaches an error of 2.7×10^{-8} with 10 terms, below the bound 6.8×10^{-8}. The radii of the three series are 1, e, and 0.5; the integral $\int_0^1 nx(1-x^2)^n dx = n/(2(n+1))$ equals 0.4545 at $n = 10$, while the limit of the functions is zero.

14.6 Computational lab

Level: advanced

The following code runs as it stands on Python 3.14.4 with NumPy 2.5.2 and SciPy 1.18.0. It recomputes the three groups of numbers above, plus quadrature for the counterexample on integration.

Computational lab

```
import math
import numpy as np
from scipy.integrate import quad

print("n      sup[0,1]   (1-1/n)^n   (1-1/sqrt n)^n     0.9^n")
for n in (10, 100, 1000, 10000):
    a = (1 - 1/n)**n
    b = (1 - 1/math.sqrt(n))**n
    print(f"{n:6d} {1.0:9.6f} {a:11.6f} {b:17.3e} {0.9**n:10.3e}")

print()
print("terms of ln(1+x) for error < 1e-6")
for x in (0.5, 0.9, 0.99, 0.999):
    N = 1
    while x**N / N > 1e-6:
        N += 1
    s = sum(((-1)**(k+1)) * x**k / k for k in range(1, N + 1))
    print(f"x={x:6.3f}  N={N:5d}  err={abs(math.log(1+x) - s):.3e}")

print()
print("N      S_N              e - S_N        e/(N+1)!")
for N in (5, 10, 15, 20):
    s = math.fsum(1/math.factorial(k) for k in range(N + 1))
    b = math.e / math.factorial(N + 1)
    print(f"{N:3d} {s:.12f} {math.e - s:14.3e} {b:13.3e}")

print()
print("radius from the ratio of terms 1000 and 1001 (log-space)")
la = lambda n: -2 * math.log(n)
lb = lambda n: math.lgamma(n + 1) - n * math.log(n)
lc = lambda n: n * math.log(2) - math.log(n)
for name, g in (("a", la), ("b", lb), ("c", lc)):
    R = math.exp(g(1000) - g(1001))
    H = math.exp(g(1000) / 1000)
    print(f"series {name}: R = {R!r}   |a_n|^(1/n) = {H!r}")
print("e =", repr(math.e))
```

```
print()
print("f_n(x) = n x (1-x^2)^n on [0,1]")
for n in (10, 100):
    val, _ = quad(lambda x, n=n: n * x * (1 - x*x)**n, 0, 1)
    print(f"n={n:4d}  integral={val!r}  n/(2(n+1))={n/(2*(n+1))!r}")
```

The output obtained:

```
n      sup[0,1]   (1-1/n)^n   (1-1/sqrt n)^n     0.9^n
    10  1.000000    0.348678         2.234e-02  3.487e-01
   100  1.000000    0.366032         2.656e-05  2.656e-05
  1000  1.000000    0.367695         1.108e-14  1.748e-46
 10000  1.000000    0.367861         2.249e-44  0.000e+00

terms of ln(1+x) for error < 1e-6
x= 0.500  N=   16  err=3.049e-07
x= 0.900  N=   89  err=4.478e-07
x= 0.990  N=  721  err=4.915e-07
x= 0.999  N= 5248  err=4.993e-07

N      S_N              e - S_N        e/(N+1)!
  5 2.716666666667      1.615e-03     3.775e-03
 10 2.718281801146      2.731e-08     6.810e-08
 15 2.718281828459      5.063e-14     1.299e-13
 20 2.718281828459      0.000e+00     5.320e-20

radius from the ratio of terms 1000 and 1001 (log-space)
series a: R = 1.0020010000000001   |a_n|^(1/n) = 0.9862794856312105
series b: R = 2.7169239322350776   |a_n|^(1/n) = 0.36949166347196
series c: R = 0.5004999999999973   |a_n|^(1/n) = 1.9862320968418674
e = 2.718281828459045

f_n(x) = n x (1-x^2)^n on [0,1]
n=  10  integral=0.4545454545454544  n/(2(n+1))=0.45454545454545453
n= 100  integral=0.4950495049504951  n/(2(n+1))=0.49504950495049505
```

Four things must be read from this output. First, the second column stays at 1.000000 for all four values of n, while the fifth column is already 3.5×10^{-1} at $n = 10$: the failure of uniformity shows up as a number. Second, the last entry of the fifth column prints $0.000e+00$ not because the error is zero, but because $0.9^{10000} = 10^{-457.6}$ lies below the smallest normal `float64` bound: underflow.

Third, the fourth column gives numerical evidence for two limits already proved: $(1 - 1/n)^n \to 1/e = 0.3678794$ and $(1 - 1/\sqrt{n})^n \to 0$, which collapses far faster.

Fourth, SciPy's quadrature gives 0.4545454545454544 for $n = 10$, whereas the formula $n/(2(n+1))$ gives 0.45454545454545453; a difference of one unit in the last place of rounding. Without a closed form, we would have no way to make sure the quadrature is not measuring something else.

One thing that does not show in the output: every bound in this chapter is an upper bound. The row N that the code reports is chosen by checking x^N/N, so that N is guaranteed to suffice, not to be minimal; the actual errors are 3.0×10^{-7} to 5.0×10^{-7}, below 10^{-6}.

Note for instructors

The teaching order I suggest: start from the failure of the $\ln(1+x)$ library in the opening, then the pointwise and uniform definitions with x^n as the single example. Do not rush to the M-test. Spend one full session making sure the students can say precisely why x^n fails to be uniform and why $(1 - 1/\sqrt{n})^n$ succeeds. After that the M-test, then the three exchange theorems, each with its own counterexample.

Two misconceptions appear most often. First, students think "every f_n continuous and $f_n \to f$" already suffices to conclude that f is continuous; they forget that what is needed is uniformity, and x^n is the cure. Second, students think uniformity is inherited downward: if $f_n \to f$ uniformly then $f_n' \to f'$. The example $\sin(nx)/n$ breaks that in one line; it is best written on the board before the differentiation theorem appears.
A board activity that works: write four unnamed sequences of functions, namely x^n, x^n/n, $nx(1-x^2)^n$, and $\sin(nx)/n$, then ask the class to decide for each whether it converges pointwise, whether uniformly, and whether its integral and derivative may be exchanged. The answers form a 4×4 table that can be filled in twenty minutes. Students are ready to go on when they can explain why a power series is uniform only on $[-r, r]$.

14.7 Summary

Level: basic

Summary.

- Pointwise convergence gives every point its own threshold; uniform convergence demands one threshold for all points at once, and that difference is equivalent to $\|f_n - f\|_\infty \to 0$.
- The Weierstrass M-test concludes uniformity from a single comparison series of numbers. It is a sufficient condition, not a necessary one; its failure proves nothing.
- The uniform limit of continuous functions is continuous, and its integral may be exchanged with the limit. Differentiation demands uniformity of the sequence of derivatives plus convergence at one point.
- Those three hypotheses cannot be removed: x^n, $nx(1-x^2)^n$, and $\sin(nx)/n$ each break one of them.
- A power series converges absolutely inside its radius and diverges outside; inside the radius it may be differentiated and integrated term by term without changing its radius.
- Uniformity is always local: $\ln(1+x)$ is uniform on every $[-r, r]$, $r < 1$, but not on $(-1, 1)$, and the price shows in the number of terms, from 16 at $x = 0.5$ to 5248 at $x = 0.999$.

14.8 Exercises

Level: intermediate

Exercise 14.1.

1. Compute in Python, for $n = 10, 100, 1000$, the value $\sup_{x \in [0,1]} |x^n|$ and the value of x^n at $x = 0.9$. Explain why the first column is flat and the second is not.
2. Prove: if $f_n \to f$ uniformly on E and every f_n is bounded on E, then f is bounded on E and the sequence $\|f_n\|_\infty$ is bounded. Show by example that pointwise convergence is not enough.
3. Suppose $f_n(x) = x/(1+nx^2)$ on $\mathbb{R}$. Show that $f_n \to 0$ pointwise, determine $\|f_n\|_\infty$, and prove that the convergence is uniform. What is the smallest n such that $\|f_n\|_\infty < 0.01$?
4. Judge this claim: true, false, or not enough evidence. "Since $x^n \to 0$ for every $x \in [0, 1)$, the sequence $f_n(x) = x^n$ converges uniformly on $[0, 1)$." State exactly where that reasoning breaks, and give one sequence of points that proves it.
5. Compute the error $|e - S_N(1)|$ for $N = 5, 10, 15, 20$ and compare it with the bound $e/(N+1)!$. What is the smallest N for which that bound is below 10^{-16}? Can the actual error still be measured at that N?

6. Prove that the series $\sum_{n\geq 0} x^n(1-x)$ converges pointwise on $[0,1]$ but not uniformly, by computing its sum first and then checking its continuity.
7. Compute the radii of convergence of $\sum x^n/n^2$, $\sum (n!/n^n)x^n$, and $\sum 2^n x^n/n$ from the ratio of terms 1000 and 1001, then compare with the exact values. Which series has the slowest estimate, and why?
8. Prove that if $f_n \to f$ uniformly on E and g is bounded on E, then $f_n g \to fg$ uniformly on E. Show by example that the boundedness of g cannot be removed.
9. For $f_n(x) = nx(1-x^2)^n$ on $[0,1]$, compute the peak point, the peak height, and $\int_0^1 f_n$ for $n = 10, 100$. Compare the peak height with $\sqrt{n/(2e)}$ and explain why the area does not tend to zero.
10. Differentiate $\sum_{n\geq 0} x^n = 1/(1-x)$ term by term and use the result to compute $\sum_{n\geq 1} n/2^n$. How many terms are needed for the partial sum to lie within 10^{-10} of the exact value?

Hints for the three hardest items. Item 4: the uniform property demands one N for all x, so it suffices to show that for every n there is an x with $x^n \geq \frac{1}{2}$; take $x = 2^{-1/n}$. Item 6: sum the geometric series first; the result is a function that differs from its limit at one point, and the uniform limit of continuous functions must be continuous. Item 8: write $|f_n g - fg| = |f_n - f|\,|g|$ and use $\|g\|_\infty$; for the counterexample take $g(x) = 1/x$ on $(0,1)$ and an f_n that decreases uniformly but not fast enough.

For researchers: limits and next steps

What is proved with complete proof in this chapter: the equivalence of uniformity with the supremum norm, the uniform Cauchy criterion, the Weierstrass M-test, the three exchange theorems together with their counterexamples, the radius formula through the limit superior, and the invariance of the radius under term-by-term differentiation and integration. Two results are quoted without proof: the non-differentiability of the Weierstrass function (1872, extended by Hardy 1916) and Abel's theorem. The primary sources are Rudin, *Principles of Mathematical Analysis*, chapters "Sequences and Series of Functions" (7.1–7.17) and "Some Special Functions" (8.1–8.2); Abbott, *Understanding Analysis*, chapter "Sequences and Series of Functions"; Stewart, *Calculus*, the section "Power Series".

What is often popularized beyond its due is the idea that uniformity removes all problems. It removes three particular problems, on the interval where it holds, and no more. For Fourier series uniformity is in fact rare, and the exchange of integral remains legitimate for other reasons. In neural-network approximation, what is used is convergence on a finite set.

Open question: if f_n is known only through its values on a finite grid, how large is the error of estimating $\|f_n\|_\infty$ by the maximum on the grid? Without regularity assumptions there is no general answer, and for narrow peaks a grid that looks dense can miss them entirely. Limits of validity: all functions here are real-valued on subsets of $\mathbb{R}$, with 64-bit arithmetic in Python 3.14.4, NumPy 2.5.2, and SciPy 1.18.0.

Map to source texts: Chapter 14 in Rudin, Principles of Mathematical Analysis*, chapters "Sequences and Series of Functions" (7.1–7.17) and "Some Special Functions" (8.1–8.2); in Abbott,* Understanding Analysis*, chapter "Sequences and Series of Functions"; in Stewart,* Calculus*, the sections "Power Series" and "Taylor and Maclaurin Series". Its basis is the second half of the chapter* Integral, Series of Functions, and Exchange of Limits *of the first edition of this book. This chapter uses the definitions of limit and the Cauchy criterion from Chapter 4, the root test of Chapter 5, uniform continuity of Chapter 11, and the integrability criterion of Chapter 13, all of them referred to, not repeated.*

Chapter 15

Multiple Integrals, Change of Variables, and Vector Field Theorems

A cooling plate is a disk of radius one meter. Its thickness is uniform, but its density falls off from the center: $e^{-(x^2+y^2)}$ kilograms per square meter. Its mass is the sum of the density over the whole disk: a numerical library reports 1.9858653038 kilograms after 11907 function evaluations.

A colleague writes a different version of the same computation. He moves to polar coordinates and writes

$$\int_0^{2\pi} \int_0^{1} e^{-r^2} \, dr \, d\theta,$$

and gets 4.6924344183 kilograms. His program runs without a warning. That number is 2.36 times larger than the first, and not one of the programs complains. The mistake is one letter: the factor r is missing, so the area of every small cell is counted too large.

Damage of that size also shows up on the other side of this chapter. Compute the work of $F(x, y) = (-y, x)/(x^2 + y^2)$ around the unit circle: the result is 6.2831853072, even though the curl of that field is zero at every point where it is defined. If zero curl meant zero closed work, one of those two computations would lie. What lies is not the computation but the hypothesis that was left unchecked: the field is not defined at the origin, and the origin lies inside the circle.

Chapter 13 built the integral on an interval, and Chapter 12 gave the Jacobian as the matrix of derivatives. Neither is enough yet: there is still no way to add up over a region of dimension greater than one, no rule for what happens to the area of a cell when the coordinates are changed, and no meaning for the sign on a boundary. This chapter fills in all three, then shows that Green, Gauss, and Stokes are in fact one statement.

For younger readers (age 10 and up)

Picture a tailor with a shirt pattern on squared paper. The pattern is square and regular; the cloth to be cut curves to follow the body. If that square pattern is pasted on as it is, the shirt is tight on one side and loose on the other, even though the number of cells stays the same.

An experienced tailor does something else: every small cell will stretch or shrink by a certain ratio as it is carried onto the cloth. Cells near the waist stretch a little; cells near the shoulder shrink. He records that ratio, multiplies the area of the cell by it, and only then adds up.

The same work appears with curved shapes: many objects, areas, or weights. If the stretch ratio is forgotten, the total is still a tidy number, and precisely because it is tidy it is hard to distrust. In this chapter we shall learn why that ratio has to be there, and how to compute it for round shapes.

15.1 From rectangles to general regions

Level: basic

We begin with the simplest object: a rectangle, and a number we want to add up over its whole surface. The method is the same as in Chapter 13, only the pieces are now cells rather than segments.

Definition 15.1 (The double integral)**.** Let $R = [a,b] \times [c,d]$ be a closed rectangle. A *partition* $\mathcal{P}$ of R is a division of $[a,b]$ into intervals and of $[c,d]$ into intervals; the result is mn small rectangles R_{ij}. The *mesh* of the partition is the longest diagonal among those cells, and is written $\|\mathcal{P}\|$.

For f bounded on R, write $M_{ij} = \sup_{R_{ij}} f$ and $m_{ij} = \inf_{R_{ij}} f$, then

$$U(\mathcal{P}, f) = \sum_{i,j} M_{ij} \operatorname{luas}(R_{ij}), \qquad L(\mathcal{P}, f) = \sum_{i,j} m_{ij} \operatorname{luas}(R_{ij}).$$

The function f is called *integrable* on R when the supremum of all L equals the infimum of all U; that common value is $\iint_R f \, dA$.

This definition says nothing about the order of integration. If f is continuous, the upper and lower sums meet because the largest gap of f on a single cell shrinks with the mesh: on a compact set, a continuous function is uniformly continuous (Chapter 11).

The next question is practical. To add up over cells means computing something two-dimensional, whereas our hands can only compute one-variable integrals. The bridge: slice along one direction first, then add up the results of those slices.

Theorem 15.1 (Fubini)**.** *If f is continuous on $R = [a,b] \times [c,d]$, then*

$$\iint_R f \, dA = \int_a^b \int_c^d f(x,y) \, dy \, dx = \int_c^d \int_a^b f(x,y) \, dx \, dy.$$

Proof. A full proof demands matching the two-dimensional upper and lower sums with the one-dimensional integral uniformly, and it is in Spivak, the chapter *Integration*, Theorem 3-10. The idea runs like this.

For $f \geq 0$, the double integral is the volume of the solid under the graph of f. Slice that solid with a plane of constant x. The slice is a region under the curve $y \mapsto f(x,y)$, so its area is

$$A(x) = \int_c^d f(x,y) \, dy.$$

The volume of the solid is the integral of the area of the slice:

$$\iint_R f \, dA = \int_a^b A(x) \, dx = \int_a^b \int_c^d f(x,y) \, dy \, dx.$$

Because f is continuous on a compact set, A is continuous. For f of any sign, split $f = f^+ - f^-$; the other direction comes from slices of constant y. □

What matters in this theorem is not the formula but the permission it grants: the order may be swapped. That permission lapses for functions that are not integrable, and that is discussed later.

The solid sliced in that proof appears in Figure 15.1: each column contributes $f(x,y)\,\Delta A$.

Read it aloud (auditory learner)

— How much sand is in this pile? — Add up its thin columns. Stand on the north side, slice the pile into strips, measure each strip, then add. — What do you get? — Eighteen buckets. — Now stand on the east side, slice in the other direction, measure, add. — What do you get? — Eighteen buckets as well. — The north-south strips and the east-west strips are not the same at all. Why is the answer the same? — Because the pile is one. We are not adding two piles; we are slicing the same object with two knives in different directions. — So the direction of the slice may be swapped? — As long as the object is measurable. If the pile has a hole in the middle, the two directions still agree; what fails later is the hole, not the order.
Say it once more, in an ordinary voice: the direction of the slice changes, the volume stays.

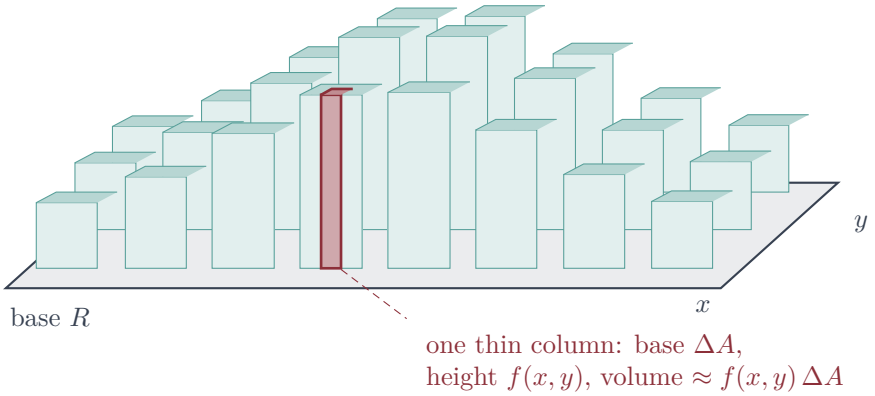


Figure 15.1: The solid under the graph of f over a rectangle R, cut into thin columns. Each column is almost a box with base ΔA and height $f(x,y)$, so its volume is $f(x,y)\,\Delta A$; adding up all the columns gives $\iint_R f\,dA$. The column marked in red is one term in that sum. Slicing along the x direction and along the y direction gives different columns, but the total volume is the same.

The regions that appear in problems are rarely rectangles. We handle them with one extra step: extend the function by zero outside the region.

Lemma 15.2 (A region between two graphs)**.** *Let $g_1, g_2 \colon [a,b] \to \mathbb{R}$ be continuous with $g_1 \le g_2$, and*

$$D = \{(x,y) : a \le x \le b,\ g_1(x) \le y \le g_2(x)\}.$$

If f is continuous on D, then

$$\iint_D f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx.$$

Proof. Take a rectangle R that contains D, and extend f by the value 0 on $R \setminus D$. The boundary of D can be covered by finitely many rectangles of total area less than ε: on the graph, uniform continuity on a compact interval gives a partition whose oscillation of g is small on each piece. A set that can be covered that way does not change the value of the integral, so

$$\iint_D f\,dA = \iint_R f \cdot \mathbf{1}_D\,dA.$$

By Fubini, the integral on the right is computed by integrating x first; for fixed x, the integrand $\mathbf{1}_D f$ equals f only on $g_1(x) \le y \le g_2(x)$. □

A type I region is enough for almost every problem in this chapter; a boundary that crosses a vertical line more than twice is split and then added up. The most important case is the disk: in Cartesian coordinates it demands two pieces, in polar only one.

For younger readers (age 10 and up)

Now take a closed room with several stoves inside it. Each stove gives off heat. The question: how much heat leaves that room every second, added up over all the walls, the floor, and the roof?

The first way to compute it is hard: walk around every wall, measure the heat flow at each point, then add up. The second way is much easier: count how much heat all the stoves inside produce. If no heat escapes through other gaps, the two numbers must agree: the amount inside equals the amount through the skin.

That sentence sounds too good to be true. The agreement holds only if the skin of the room is really closed and the direction "out" is fixed unambiguously. If one small hole is not counted, or if we mislabel which side is outside and which is inside, the numbers miss even though all our additions are correct.

In this chapter we shall learn why the amount inside can be read off from the skin, for any curved shape, and how one small hole can ruin the whole conclusion.

15.2 Change of variables and the Jacobian determinant

Level: intermediate

Changing coordinates means changing cells: a square cell in the (u, v) plane becomes a curved cell in the (x, y) plane, and its area changes. We need one number at each point.

Definition 15.2 (Change of variables)**.** Let D^* be an open region in $\mathbb{R}^2$ and $T\colon D^* \to \mathbb{R}^2$, $T(u, v) = (x(u, v), y(u, v))$, a function with continuous partial derivatives. T is called a *valid change of variables* if T is one-to-one on the interior of D^* and its Jacobian determinant

$$\det J_T(u, v) = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

is nonzero there. The matrix J_T itself we already met in Chapter 12.

In everyday words: it is the number that tells how many times a small cell grows when it is carried by T. A value of 3 means the cell becomes three times larger in area, $1/2$ means it shrinks by half, and zero means the cell is flattened into a line and its information is lost.

Figure 15.2 shows one cell and its image: area 0.04 becomes 0.064, and the number 1.6 is $|\det J|$ at the center of the cell.

Do it by hand (kinesthetic learner)

Take a square sheet of paper 12 cm by 12 cm, a ruler, a pencil, scissors, and tape. Draw 1 cm cells over the whole sheet; there are 144 cells. Cut the paper into twelve vertical strips, each 1 cm by 12 cm. On every strip, mark the point 3 cm to the right on the top edge, draw a line from the bottom left corner to that point, then cut along the line. Each strip is now two slanted pieces. Arrange all the pieces into one tight parallelogram, then tape them together.

Watch what happens: the outer frame is 15 cm by 12 cm, wider by 3 cm than the original sheet, but the parallelogram inside still has base 12 cm and height 12 cm. Measure both with the ruler, perpendicular, not along the slanted side: the result is 144 square centimeters, the same as the area of the sheet before it was cut. If your number is

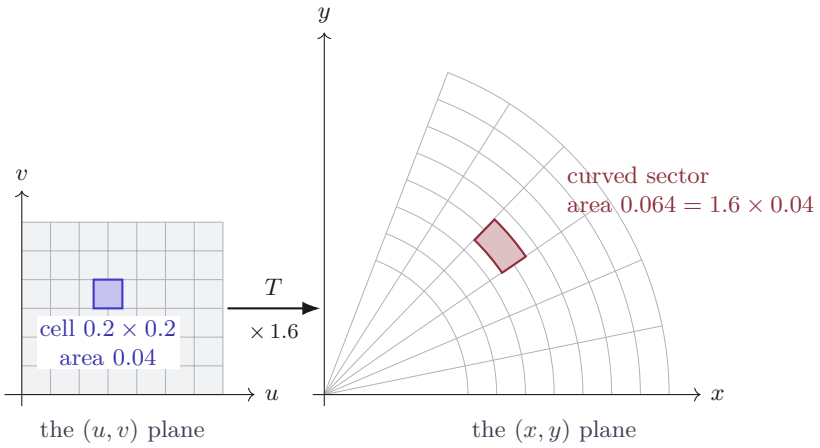


Figure 15.2: One small cell in the (u, v) plane and its image in the (x, y) plane under $T(u, v) = (u\cos v,\ u\sin v)$. The cell 0.2×0.2 of area 0.04 becomes a sector of area 0.064, that is 1.6 times as large, exactly the value $|\det J| = u$ at the center of the cell ($u = 1.6$). Note that its sides curve as well: what changes is not only the size but also the shape, and one number at each point is enough to record the change.

different, the height is almost certainly what went wrong.

Theorem 15.3 (Change of variables in a double integral)**.** *Let T be a valid change of variables from D^* to $D = T(D^*)$, and f continuous on D. Then*

$$\iint_D f(x,y)\,dx\,dy = \iint_{D^*} f\big(T(u,v)\big)\,\big|\det J_T(u,v)\big|\,du\,dv.$$

Proof. We prove the idea; the full proof is in Spivak, the chapter *Integration*, the section on change of variables, and in Rudin, the chapter *Integration of Differential Forms*.

Partition D^* into small cells. On a cell around (u_0, v_0), the map T is almost linear: $T(u_0 + h, v_0 + k) \approx T(u_0, v_0) + J_T(u_0, v_0)(h, k)$. That cell is mapped onto a parallelogram with sides $h\,\partial T/\partial u$ and $k\,\partial T/\partial v$, whose area is computed with a determinant (Chapter 7):

$$\text{luas} = \Big|h\,k\,\det\Big(\frac{\partial T}{\partial u}, \frac{\partial T}{\partial v}\Big)\Big| = |h\,k|\,|\det J_T(u_0, v_0)|.$$

Each cell contributes $f(T(u_0, v_0))$ times its original area times $|\det J_T|$, and the sum is the Riemann sum for the integral on the right-hand side; because the derivative of T is continuous on a compact set, the error of the linear approximation shrinks uniformly with the mesh. □

The absolute value on the determinant is not decoration: a determinant can be negative, an area cannot. The negative sign stores an orientation; in the double integral that orientation is discarded, but in the line and surface integrals later that sign decides the answer.

Now the determinants for the two coordinate systems used most often, beginning with polar.

Proposition 15.4 (Polar coordinates)**.** *With $x = r\cos\theta$, $y = r\sin\theta$, $r > 0$, we have*

$$\det J = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r,$$

so that $\iint_D f\,dx\,dy = \iint_{D^*} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$.

That factor r is the one missing from the program in the opening of the chapter. At $r = 1$ it is 1, but at $r = 0.1$ it is 0.1: a cell near the center contributes ten times less than a guess that ignores it, and the total error can be more than a factor of two.

Proposition 15.5 (Cylindrical and spherical coordinates)**.** *For cylindrical coordinates $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, the Jacobian determinant is r. For spherical coordinates*

$$x = \rho\sin\varphi\cos\theta, \quad y = \rho\sin\varphi\sin\theta, \quad z = \rho\cos\varphi,$$

with $\rho \geq 0$, $0 \leq \varphi \leq \pi$, $0 \leq \theta < 2\pi$, the Jacobian determinant is $\rho^2\sin\varphi$.

Proof. For the cylinder, the third column $(0, 0, 1)$ leaves the determinant as the 2×2 polar block determinant, namely r.

For the sphere, compute the 3×3 determinant by cofactor expansion. With $s_\varphi = \sin\varphi$, $c_\varphi = \cos\varphi$, $s_\theta = \sin\theta$, $c_\theta = \cos\theta$, the matrix is

$$J = \begin{pmatrix} s_\varphi c_\theta & \rho c_\varphi c_\theta & -\rho s_\varphi s_\theta \\ s_\varphi s_\theta & \rho c_\varphi s_\theta & \rho s_\varphi c_\theta \\ c_\varphi & -\rho s_\varphi & 0 \end{pmatrix}.$$

The entry in the third row and third column is zero, so expansion along the third column leaves two terms:

$$\det J = \rho^2 s_\varphi s_\theta^2 + \rho^2 s_\varphi c_\theta^2 = \rho^2 s_\varphi (s_\theta^2 + c_\theta^2) = \rho^2 s_\varphi.$$

□

The factor $\rho^2\sin\varphi$ is zero at the poles $\varphi = 0$ and $\varphi = \pi$, because there the whole sphere of radius ρ collapses to a single point and the map stops being one-to-one; that is why the poles are excluded from D^*. It is largest at the equator, where a small change in φ carries a point farthest.

Step-by-step recipe (procedural learner)

1. Write T explicitly: $x = x(u, v)$, $y = y(u, v)$, and check that T is one-to-one on the interior of D^*.
2. Compute the four partial derivatives, assemble the 2×2 matrix, then compute $\det J(u, v)$. Keep its sign; the absolute value is put in later.
3. Translate the boundary: draw D in the (x, y) plane, ask which values of (u, v) land there, then write D^*.
4. Replace every x, y in f by $x(u, v), y(u, v)$, multiply by $|\det J(u, v)|$, then close with $du\,dv$ in the order of the limits.

A warning about order: steps 2 and 3 may not be swapped. The determinant must already be in terms of (u, v) before the limits are put in. If the limits are written first, that factor is left in x and y and the integral cannot be computed.

Jacobian determinant (*determinan Jacobian*) — The number that says how many times a small cell stretches when the coordinates are changed.

double integral (*integral lipat dua*) — Adding up values on small cells of a surface, then narrowing the cells.

orientation (*orientasi*) — An agreement about which direction of rotation counts as positive.

15.3 Line integrals and conservative fields

Level: intermediate

Now we stop adding up over a surface and start adding up along a path. Picture a small object dragged along a curved path by a force whose direction and size change at every point. The work of that force is the sum of the component of the force along the path times the piece of length of the path.

Definition 15.3 (Line integral)**.** Let $\gamma\colon [a,b] \to \mathbb{R}^2$ be a curve with piecewise continuous derivative, and $F = (P, Q)$ a continuous vector field on its trace. The *line integral* of F along γ is

$$\int_\gamma F \cdot d\mathbf{r} = \int_a^b \Big(P(\gamma(t))\,\gamma_1'(t) + Q(\gamma(t))\,\gamma_2'(t) \Big)\, dt.$$

If γ is closed, meaning $\gamma(a) = \gamma(b)$, that integral is called the *circulation* of F around γ.

That formula does not choose a starting point, only a direction. Changing the parameter to a different speed does not change the value, because substitution (Chapter 13) absorbs the speed factor; reversing the direction changes the sign. So a line integral is oriented, and that is its first difference from the double integral.

Figure 15.3 shows such a rotating field, with one closed curve for measuring how much rotation is caught.

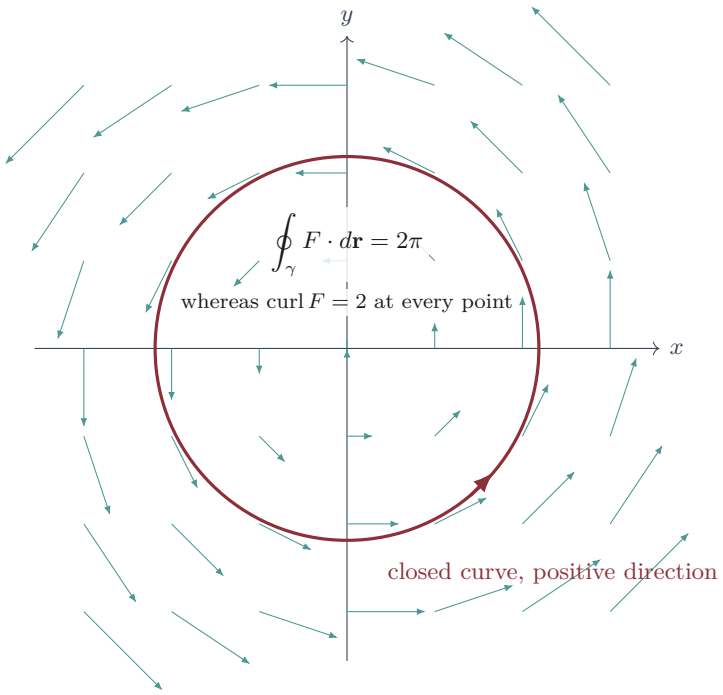


Figure 15.3: The field $F(x, y) = (-y, x)$ with an arrow at every grid point. The arrows turn around the origin, and their length grows with the distance of the point from the center. The circulation around a circle of radius 1 is 2π, equal to the area π times curl $F = 2$: this is Green for this field. The direction of the arrows on the curve is the positive direction, counterclockwise.

If a force field comes from a height function, the work depends only on the starting point and the ending point. This is the curve version of the fundamental theorem of calculus, and its name follows from that.

Theorem 15.6 (The fundamental theorem for line integrals)**.** *Let $f\colon U \to \mathbb{R}$ have continuous partial derivatives on an open set $U \subseteq \mathbb{R}^2$, and let γ be a curve from A to B inside U. Then*

$$\int_{\gamma} \nabla f \cdot d\mathbf{r} = f(B) - f(A).$$

Proof. Write $g(t) = f(\gamma(t))$. By the multivariable chain rule (Chapter 12),

$$g'(t) = \nabla f(\gamma(t)) \cdot \gamma'(t),$$

provided γ is differentiable at t. So the integrand in the line integral is the derivative of g, and the fundamental theorem of calculus (Chapter 13) gives $\int_a^b g'(t)\,dt = g(b) - g(a) = f(B) - f(A)$. For a curve that is only piecewise differentiable, add up over each piece; the terms at the joining points cancel each other. □

A field of the form ∇f is called *conservative*, and f is its potential. A direct consequence of the theorem: for a conservative field, the circulation around a closed path is always zero.

The converse is also true.

Theorem 15.7 (Converse)**.** *Let F be continuous on a connected open set $U \subseteq \mathbb{R}^2$, and let the line integral of F be independent of the path inside U. Then $F = \nabla f$ for some f, namely*

$$f(\mathbf{x}) = \int_{\mathbf{x}_0}^{\mathbf{x}} F \cdot d\mathbf{r},$$

with $\mathbf{x}_0 \in U$ fixed.

Proof. Take $\mathbf{x} \in U$ and h small enough that the segment from $\mathbf{x}$ to $\mathbf{x} + h\mathbf{e}_1$ stays inside U. Because the path does not matter, we may travel from $\mathbf{x}_0$ to $\mathbf{x}$ and then add that short segment, so that

$$f(\mathbf{x} + h\mathbf{e}_1) - f(\mathbf{x}) = \int_0^1 F_1(\mathbf{x} + th\mathbf{e}_1)\, h\, dt.$$

Divide by h and use continuity: the average value of the integrand tends to $F_1(\mathbf{x})$ as $h \to 0$, so $\partial f/\partial x_1 = F_1$, and the same argument gives $\partial f/\partial x_2 = F_2$. Both partial derivatives are continuous, hence $F = \nabla f$. □

Given F, how do we know it is conservative without looking for a potential? In the plane there is a quick test: $F = (P, Q)$ has a potential only if $\partial P/\partial y = \partial Q/\partial x$, because both sides are second derivatives of f and Clairaut (Chapter 12) makes them equal. The other direction demands a domain without holes and is proved through Green later. The field from the opening of the chapter is the counterexample: its curl is zero, but it is not the gradient of anything.

Example 15.1 (Two ways to compute work). Take $F(x, y) = (2xy,\ x^2 + 3y^2)$ and the path $\gamma(t) = (t, 2t^2)$ for $0 \le t \le 1$, from $(0, 0)$ to $(1, 2)$.

First way, directly. Here $dx = dt$ and $dy = 4t\,dt$, so

$$\int_{\gamma} F \cdot d\mathbf{r} = \int_0^1 \Big(2t \cdot 2t^2 + (t^2 + 12t^4) \cdot 4t\Big) dt = \int_0^1 (8t^3 + 48t^5) dt = 2 + 8 = 10.$$

Second way, through the potential. First check that $\partial P/\partial y = 2x = \partial Q/\partial x$, then guess $f(x, y) = x^2 y + y^3$. Its derivatives are indeed $\partial f/\partial x = 2xy$ and $\partial f/\partial y = x^2 + 3y^2$. Therefore

$$\int_{\gamma} F \cdot d\mathbf{r} = f(1, 2) - f(0, 0) = (2 + 8) - 0 = 10.$$

Both ways give 10; adaptive quadrature differs by 1.78×10^{-15}.

15.4 Green's theorem

Level: intermediate

The idea in the box about the room and the stoves now becomes a theorem: what happens inside a region can be read off from its edge.

Before that, one agreement on direction.

Definition 15.4 (Positive orientation)**.** A simple closed curve γ is oriented *positively* if the region it bounds always lies on the left as we traverse it. For a circle, the positive direction is counterclockwise.

Theorem 15.8 (Green)**.** *Let D be a closed bounded region in $\mathbb{R}^2$ whose boundary ∂D consists of finitely many simple closed curves that are piecewise smooth and positively oriented. Let P, Q have continuous partial derivatives on an open set containing D. Then*

$$\oint_{\partial D} P\,dx + Q\,dy = \iint_D \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)\,dA.$$

Proof. It is enough to prove it for regions that are both type I and type II, because a general region is split into finitely many pieces of that kind and the integrals are added up; the adjacent sides of the pieces are traversed in opposite directions and therefore cancel each other.

Suppose D is type I with g_1, g_2 smooth. On the vertical sides $dx = 0$, so their contribution is zero and

$$\oint_{\partial D} P\,dx = \int_a^b P(x, g_1(x))\,dx + \int_b^a P(x, g_2(x))\,dx = \int_a^b \big[P(x, g_1) - P(x, g_2)\big]\,dx.$$

By the fundamental theorem of calculus, $P(x, g_1) - P(x, g_2) = -\int_{g_1(x)}^{g_2(x)} \partial P/\partial y\,dy$, so that

$$\oint_{\partial D} P\,dx = -\int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y}\,dy\,dx = -\iint_D \frac{\partial P}{\partial y}\,dA.$$

In the same way, on a D of type II, where $dy = 0$ on the flat sides, $\oint_{\partial D} Q\,dy = \iint_D \partial Q/\partial x\,dA$. □

Example 15.2 (Green on the unit square). Take $P = x^2 - y$, $Q = x + 2y$, and $D = [0,1] \times [0,1]$. Here $\partial Q/\partial x - \partial P/\partial y = 1 - (-1) = 2$, so the right-hand side of Green is 2 times the area of the square, namely 2.

We compute the left-hand side on the four sides, counterclockwise:

$$\begin{aligned}
\text{bottom } (y = 0,\ dy = 0): &\quad \int_0^1 x^2\,dx = \tfrac{1}{3},\\
\text{right } (x = 1,\ dx = 0): &\quad \int_0^1 (1 + 2y)\,dy = 2,\\
\text{top } (y = 1,\ dy = 0): &\quad \int_1^0 (x^2 - 1)\,dx = \tfrac{2}{3},\\
\text{left } (x = 0,\ dx = 0): &\quad \int_1^0 2y\,dy = -1.
\end{aligned}$$

The sum is $\frac{1}{3} + 2 + \frac{2}{3} - 1 = 2$, the same as the right-hand side. Adaptive quadrature for the four sides gives 2.000000000000000, and the double integral also gives 2.000000000000000.

Now the sign error we asked for. Repeat the same computation clockwise. Every side is traversed backwards, so every term changes sign and the sum becomes -2; the right-hand side of Green does not change because the region is the same. The answer that should be 2 becomes -2: not slightly off, but wrong in sign entirely, with a difference of 4 that cannot be blamed on rounding.

So orientation is not a polite convention: Green's formula is correct only for one direction of rotation, and that direction must be checked before computing.

With Green in hand, the converse direction of the curl test from the previous section can be proved.

Corollary 15.9 (The curl test on a domain without holes)**.** *Let $U \subseteq \mathbb{R}^2$ be open and* simply connected*, meaning that every closed curve in U can be contracted to a single point inside U. Let P, Q have continuous partial derivatives on U and $\partial P/\partial y = \partial Q/\partial x$ throughout U. Then $F = (P, Q)$ is conservative on U.*

Proof. Take a simple closed curve γ in U and the region D that it bounds. Because U is simply connected, D stays inside U, so Green applies and

$$\oint_\gamma P\,dx + Q\,dy = \iint_D \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big) dA = 0.$$

A closed curve that crosses itself is split into simple arcs, and each piece gives zero. All circulations are zero, so the line integral is independent of path, and the converse theorem above supplies a potential. □

That hypothesis cannot be dropped. On a region with a hole, Green demands that the inner boundary also be counted; that is where the field from the opening of the chapter escapes the curl test.

15.5 Gauss's and Stokes' theorems

Level: advanced

Green talks about a region in the plane and its edge. In three-dimensional space there are two versions, and both use the divergence and the curl from Chapter 12.

Definition 15.5 (Divergence and curl)**.** For a field $F = (F_1, F_2, F_3)$ with continuous partial derivatives,

$$\operatorname{div} F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}, \qquad \operatorname{curl} F = \Big(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z},\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x},\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\Big).$$

The divergence measures how much field is born at a point: positive means a source, negative a sink. The curl measures the rotation of the field around that point.

Theorem 15.10 (Gauss, or the divergence theorem)**.** *Let $E \subseteq \mathbb{R}^3$ be a closed bounded region whose boundary ∂E is a piecewise smooth surface, and let $\mathbf{n}$ be the unit normal vector pointing outward. If F has continuous partial derivatives on an open set containing E, then*

$$\iiint_E \operatorname{div} F\,dV = \iint_{\partial E} F \cdot \mathbf{n}\,dA.$$

Proof. We prove it for a box first.

Take $E = [a,b]\times[c,d]\times[e,f]$. For the third term,

$$\iiint_E \frac{\partial F_3}{\partial z}\,dV = \iint_{[a,b]\times[c,d]} \big[F_3(x,y,f) - F_3(x,y,e)\big]\,dx\,dy,$$

by Fubini and the fundamental theorem of calculus in the z direction. That right-hand side is exactly the flux through the top face (normal $+\mathbf{k}$) and the bottom face (normal $-\mathbf{k}$), because there $\mathbf{n}\,dA = \pm\mathbf{k}\,dx\,dy$. The other terms give the flux through the two other pairs of faces.

For a general region, cut it into small boxes. An interior face appears twice with opposite normals and cancels, so all that remains are the faces that touch the outer boundary; the limit as the mesh goes to zero gives the statement for a curved boundary. The full proof is in Spivak, Theorem 5-8. □

Example 15.3 (Gauss on the unit ball). Take $F(x,y,z) = (x^3, y^3, z^3)$ and E the unit ball. Here $\operatorname{div} F = 3(x^2+y^2+z^2) = 3\rho^2$, so with the Jacobian determinant of the sphere

$$\iiint_E 3\rho^2\,dV = \int_0^{2\pi}\!\!\int_0^{\pi}\!\!\int_0^{1} 3\rho^4 \sin\varphi\,d\rho\,d\varphi\,d\theta = 3\cdot 2\pi\cdot 2\cdot\frac{1}{5} = \frac{12\pi}{5}.$$

The surface side: on the unit ball $\mathbf{n} = (x,y,z)$, so $F\cdot\mathbf{n} = x^4+y^4+z^4$ and

$$\iint_{\partial E}(x^4+y^4+z^4)\,dA = 3\iint_{\partial E} z^4\,dA = 3\int_0^{2\pi}\!\!\int_0^{\pi} \cos^4\varphi\sin\varphi\,d\varphi\,d\theta = 3\cdot 2\pi\cdot\frac{2}{5} = \frac{12\pi}{5},$$

with the factor 3 from symmetry. The value is 7.5398223686155, and quadrature on the surface gives the same number to the last digit.

The second theorem talks about an open surface whose boundary is a curve. Its formula requires the right-hand rule.

Theorem 15.11 (Stokes)**.** *Let S be an oriented piecewise smooth surface with normal $\mathbf{n}$, and let ∂S be its boundary curve with the orientation given by the right-hand rule. If F has continuous partial derivatives on an open set containing S, then*

$$\iint_S (\operatorname{curl} F)\cdot\mathbf{n}\,dA = \oint_{\partial S} F\cdot d\mathbf{r}.$$

Proof. Parametrize S by $\mathbf{r}(u,v)$ over a region D in the (u,v) plane, with the matching orientation. The chain rule turns the integrand into derivatives of the components of F with respect to u and v, and after rearrangement the surface integral becomes

$$\iint_D \Big(\frac{\partial \tilde{Q}}{\partial u} - \frac{\partial \tilde{P}}{\partial v}\Big)\,du\,dv,$$

for two functions $\tilde{P}, \tilde{Q}$ built from F and $\mathbf{r}$. That is exactly the right-hand side of Green, so Green gives the line integral on ∂D, which is nothing other than the line integral of F on ∂S. The full algebra is in Spivak, Theorem 5-9. □

If S is flat in the xy plane, Stokes becomes Green; if S is only an interval, it becomes the fundamental theorem of calculus.

Example 15.4 (Stokes on a hemisphere). Take $F = (z, x, y)$, whose curl is $(1,1,1)$, and S the upper half of the unit ball with outward normal, so that its boundary is the unit circle in the xy plane counterclockwise.

The surface side:

$$\iint_S (1,1,1)\cdot\mathbf{n}\,dA = \iint_S (x+y+z)\,dA = \iint_S z\,dA = \int_0^{2\pi}\!\!\int_0^{\pi/2} \cos\varphi\sin\varphi\,d\varphi\,d\theta = 2\pi\cdot\frac{1}{2} = \pi,$$

because the x and y terms cancel each other by the symmetry of the hemisphere. The boundary side, with $\mathbf{r}(t) = (\cos t, \sin t, 0)$:

$$\oint_{\partial S} F \cdot d\mathbf{r} = \int_0^{2\pi} \cos^2 t \, dt = \pi.$$

Both are 3.1415926535898; quadrature gives the same number on the two sides.

Now the sign error in Stokes. Turn the normal of the surface inward: the surface side becomes $-\pi$, the boundary side does not change as long as the direction of rotation stays the same, and the equality fails with a difference of 2π. The right-hand rule is what binds the two sides; reversing both together restores the equality.

15.6 Differential forms: one sentence for all of them

Level: advanced

We have now seen three theorems with three faces. There is a language that writes them as one statement, and that language is not mere tidiness: it explains why the sign of orientation appears everywhere.

Definition 15.6 (Differential forms)**.** A *k-form* on an open set $U \subseteq \mathbb{R}^n$ is a sum of terms of the form $a(x)\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$, with a a smooth function. The wedge product $\wedge$ is antisymmetric:

$$dx \wedge dy = -dy \wedge dx, \qquad dx \wedge dx = 0.$$

The exterior derivative d acts on the coefficients and adds one wedge:

$$d(a\, dx_I) = da \wedge dx_I, \qquad da = \sum_j \frac{\partial a}{\partial x_j} dx_j.$$

In everyday words: a differential form is an object that is ready to be integrated over a surface of a given dimension, and d raises the dimension by one level. Antisymmetry makes reversing the orientation reverse the sign automatically; the sign difficulties from before are built into that one rule.

Theorem 15.12 (General Stokes)**.** *Let M be an oriented manifold of dimension k with boundary, and let ω be a $(k-1)$-form with smooth coefficients. Then*

$$\int_{\partial M} \omega = \int_M d\omega,$$

with the orientation of ∂M induced by the orientation of M.

The complete proof is skipped: Spivak, *Integration on Chains*, Theorem 4-13 for chains, and Theorem 5-5 in *Integration of Manifolds* for manifolds; also Rudin, the chapter *Integration of Differential Forms*. There Green, Gauss, and classical Stokes are derived as special cases.

Its translation in three-dimensional space:

Language of forms	Vector field	Theorem
0-form f	scalar function	df gives the gradient
1-form $P\,dx + Q\,dy + R\,dz$	$F = (P, Q, R)$	$d\omega$ gives $\operatorname{curl} F$
2-form $P\,dy \wedge dz + \cdots$	$F = (P, Q, R)$	$d\eta$ gives $\operatorname{div} F$
$\int_{\partial M} \omega = \int_M d\omega$		Green, Gauss, Stokes, FTC

One small consequence: because $d(d\omega) = 0$, the exterior derivative taken twice is always zero. Translated, $\operatorname{curl} \nabla f = 0$ and $\operatorname{div} \operatorname{curl} F = 0$. A symbolic check for a general function and a general field gives zero in all six components, so those two identities that look like coincidences are one statement.

15.7 When these theorems fail

Level: advanced

All the theorems in this chapter are conditional equalities. This section collects four ways those equalities break, with numbers.

The first failure is a singularity. Take $F(x,y) = (-y,x)/(x^2+y^2)$ on the plane without the origin. Compute its curl:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{(x^2+y^2)-2x^2}{(x^2+y^2)^2} + \frac{(x^2+y^2)-2y^2}{(x^2+y^2)^2} = 0.$$

A finite difference with $h = 10^{-5}$ at $(1,0)$ gives -2.01×10^{-10}, zero up to the truncation error. But the circulation around the unit circle is $2\pi = 6.2831853072$, because there $F \cdot \gamma' = 1$ at every point. So zero curl does not mean zero circulation.

What is violated is the hypothesis of Green: P and Q must have continuous partial derivatives on an open set containing D, whereas both are not defined at the origin inside D, and D is not simply connected. On the annulus $1 \le r \le 2$ Green still applies, with two boundaries of opposite direction, and the integral is $2\pi - 2\pi = 0$; that zero is correct because there the field has the potential θ, the polar angle.

The second failure is swapping the order without integrability. Take

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}$$

on the square $(0,1)\times(0,1)$. For fixed $y > 0$,

$$\int_0^1 f(x,y)\,dx = \Big[-\frac{x}{x^2+y^2}\Big]_{x=0}^{x=1} = -\frac{1}{1+y^2},$$

so the outer integral gives $-\pi/4 = -0.7853981634$; in the reverse order the result is $+\pi/4$. Both iterated integrals exist but they differ, so by Fubini the double integral does not exist. In polar coordinates $f = \cos 2\theta / r^2$, and on the unit quarter disk

$$\iint |f|\,dA \ge \int_0^{\pi/2} |\cos 2\theta|\,d\theta \int_0^1 \frac{dr}{r} = \infty,$$

because $\int_0^1 dr/r$ diverges at $r = 0$. Fubini is not wrong: its hypothesis is not satisfied.

The third failure is an unmanageable boundary. Gauss demands that E be closed and bounded with a piecewise smooth boundary. On an unbounded set, the boundary can run off to infinity and the surface integral stops meaning anything; Spivak places this as an exercise after Theorem 5-5, which fails without compactness. Finitely many corners are still safe, because the normal jumps only at finitely many points; a boundary with endlessly jagged teeth, a coastline for instance, is not.

The fourth failure is a surface that cannot be oriented. On a Möbius strip, walking around the strip brings the normal back to the opposite side, so $\mathbf{n}$ in Stokes' formula is not defined. The theorem is not wrong; its statement is simply incomplete for objects of that kind.

Challenge (problem solver)

Take $T(u,v) = (u^2, v)$ on $D^* = (-2,2)\times(0,1)$. This map seems to widen things: the point $u = 2$ lands at $x = 4$, and its derivative $2u$ grows without bound as u moves away from zero. Compute the area of $T(D^*)$ from the picture, then compute it with the change-of-variables formula. The two numbers differ by exactly a factor 2. Explain why,

and name the line on which the Jacobian determinant is zero.
Hint 1: check whether T is one-to-one. There are two values of u that land at $x = 1$.
Hint 2: the line $u = 0$ is folded onto a single point, so every image point is counted twice by the integral; $|\det J| = |2u|$ does not know that the fold is only one sheet.
If you are stuck, go back to the definition of a valid change of variables and check the one-to-one condition.

One closing warning, which is not in the textbooks. Numerical quadrature never checks the hypotheses. It gives 6.2831853072 with seven correct digits for a field that violates the hypothesis of Green, and no error message. A tidy number is not proof that the hypothesis holds; it is only proof that the arithmetic ran.

15.8 Worked examples

Level: intermediate

Now three computations that form the backbone of this chapter. All three can be checked with Python, and all three have gone wrong in the author's own hands.

One integral, two coordinate systems.

Compute

$$I = \iint_D e^{-(x^2+y^2)}\,dA, \qquad D = \{(x,y) : x^2 + y^2 \le 1\}.$$

In polar coordinates, D becomes $0 \le r \le 1$, $0 \le \theta \le 2\pi$, and $dA = r\,dr\,d\theta$. The integral separates into two factors:

$$I = \int_0^{2\pi}\int_0^1 e^{-r^2} r\,dr\,d\theta = 2\pi \cdot \Big[-\frac{1}{2}e^{-r^2}\Big]_0^1 = 2\pi \cdot \frac{1 - e^{-1}}{2} = \pi(1 - e^{-1}).$$

Its value is 1.9858653038. The antiderivative with respect to r exists precisely because of the factor r: without that factor, the antiderivative involves the error function.

In Cartesian coordinates, the region demands two pieces:

$$I = \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} e^{-(x^2+y^2)}\,dy\,dx.$$

The inner integral has no elementary antiderivative, so we have to use quadrature. The results of the two forms, at the same tolerance:

$$\begin{aligned} \text{cartesian} &: 1.9858653037988705 \quad (11907 \text{ evaluations}),\\ \text{polar} &: 1.9858653037988712 \quad (441). \end{aligned}$$

The errors are 8.9×10^{-16} and 2.2×10^{-16}; the difference in accuracy does not matter, but the difference in cost is a factor of 27.

The midpoint rule on the same uniform Cartesian grid lowers the error more slowly: 1.141×10^{-3} on 50 cells per side, and 4.667×10^{-5} on 400. Each time the number of cells is doubled, the error drops by about a factor of 2.9, not 4, because the height of the columns near the edge behaves like $\sqrt{1 - x^2}$, so the error is of order $h^{3/2}$.

Volume of a ball with the Jacobian determinant.

The volume of a ball of radius R is

$$V = \int_0^{2\pi}\int_0^{\pi}\int_0^{R} \rho^2 \sin\varphi\,d\rho\,d\varphi\,d\theta = 2\pi \cdot 2 \cdot \frac{R^3}{3} = \frac{4\pi R^3}{3}.$$

For $R = 1$ its value is 4.1887902048. Three-dimensional quadrature on the ball gives

$$4.188790204786397 \quad (\text{difference } 6.2 \times 10^{-15}),$$

and a Monte Carlo simulation with 10^6 points in the cube $[-1,1]^3$ gives 4.192824 ± 0.004.

Compare that with dropping the Jacobian determinant. Without the factor $\rho^2 \sin\varphi$,

$$\int_0^{2\pi}\int_0^{\pi}\int_0^{1} 1 \, d\rho \, d\varphi \, d\theta = 2\pi^2 = 19.7392088022,$$

which is $3\pi/2$ times too large, and the result is still a number that looks plausible for a volume.

The following table collects every number in this chapter.

Quantity	Exact	Numerical	Error
$\iint_D e^{-(x^2+y^2)}dA$, polar	1.985865	1.985865	2.2×10^{-16}
$\iint_D e^{-(x^2+y^2)}dA$, Cartesian	1.985865	1.985865	8.9×10^{-16}
$\int_\gamma F \cdot d\mathbf{r}$	10	10.000000	1.8×10^{-15}
$\oint_{\partial D}$ Green, square	2	2.000000	$< 10^{-15}$
$\iint_S (\operatorname{curl} F)\cdot \mathbf{n}\, dA$	3.141593	3.141593	$< 10^{-13}$

Test yourself in three steps. (1) Recompute I by hand using the antiderivative $-\frac{1}{2}e^{-r^2}$, then compare with 1.9858653038; a difference in the last digit is a rounding error, not a formula error. (2) Run `scipy.integrate.dblquad` on both forms and count how many times the function is called; the numbers 11907 and 441 will reappear, provided the tolerance is the same for both. (3) Change D to the disk of radius 2 and check that the exact answer becomes $\pi(1 - e^{-4})$, namely 3.0841. If the third step gives a different number, what is wrong is almost certainly the limit on r, not the formula.

15.9 Computational lab

Level: advanced

The following code checks five things at once: the cost of the two coordinate systems, Green, Gauss, Stokes, and a field with a singularity. The numbers in the previous table come from this program.

Computational lab

```
import numpy as np
from scipy import integrate

# 1. One double integral, two coordinate systems
def fc(y, x):
    return np.exp(-(x*x + y*y))

cnt = [0]
def fc_count(y, x):
    cnt[0] += 1
    return fc(y, x)

cart = integrate.dblquad(fc_count, -1, 1,
                         lambda x: -np.sqrt(1-x*x),
                         lambda x: np.sqrt(1-x*x),
                         epsabs=1e-12, epsrel=1e-12)
ncart = cnt[0]
cnt[0] = 0
```

```
def fp(r, th):
    cnt[0] += 1
    return np.exp(-r*r)*r

pol = integrate.dblquad(fp, 0, 2*np.pi, 0, 1,
                        epsabs=1e-12, epsrel=1e-12)
value = np.pi*(1 - np.exp(-1))
print("cartesian %.16f error %.2e eval %d"
      % (cart[0], abs(cart[0]-value), ncart))
print("polar     %.16f error %.2e eval %d"
      % (pol[0], abs(pol[0]-value), cnt[0]))
print("exact     %.16f" % value)

# 2. Green: (x^2-y)dx + (x+2y)dy on the unit square
P = lambda x, y: x*x - y
Q = lambda x, y: x + 2*y
edges = [integrate.quad(lambda t: P(t, 0.0), 0, 1)[0],
         integrate.quad(lambda t: Q(1.0, t), 0, 1)[0],
         integrate.quad(lambda t: P(t, 1.0), 1, 0)[0],
         integrate.quad(lambda t: Q(0.0, t), 1, 0)[0]]
dbl = integrate.dblquad(lambda y, x: 2.0, 0, 1, 0, 1)[0]
print("Green edges %.15f  double %.15f" % (sum(edges), dbl))

# 3. Gauss: F=(x^3,y^3,z^3) on the unit ball
def flux(phi, th):
    sp, cp = np.sin(phi), np.cos(phi)
    x, y, z = sp*np.cos(th), sp*np.sin(th), cp
    return (x**4 + y**4 + z**4)*sp

F = integrate.dblquad(flux, 0, 2*np.pi, 0, np.pi,
                      epsabs=1e-11, epsrel=1e-11)[0]
print("Gauss flux %.13f  12pi/5 %.13f" % (F, 12*np.pi/5))

# 4. Stokes: F=(z,x,y) on the upper hemisphere of the unit ball
def curl_n(phi, th):
    sp, cp = np.sin(phi), np.cos(phi)
    x, y, z = sp*np.cos(th), sp*np.sin(th), cp
    return (x + y + z)*sp

S = integrate.dblquad(curl_n, 0, 2*np.pi, 0, np.pi/2)[0]
L = integrate.quad(lambda t: np.cos(t)**2, 0, 2*np.pi)[0]
print("Stokes surface %.13f  boundary %.13f" % (S, L))

# 5. A field with a singularity: F=(-y,x)/(x^2+y^2)
def circle(t):
    x, y = np.cos(t), np.sin(t)
    return (-y*(-np.sin(t)) + x*np.cos(t))/(x*x + y*y)

def Pg(x, y): return -y/(x*x + y*y)
def Qg(x, y): return x/(x*x + y*y)

h = 1e-5
curl_01 = ((Qg(1+h, 0) - Qg(1-h, 0))/(2*h)
           - (Pg(1, h) - Pg(1, -h))/(2*h))
C = integrate.quad(circle, 0, 2*np.pi)[0]
print("circle %.13f  2pi %.13f" % (C, 2*np.pi))
print("curl(1,0) %.2e" % curl_01)
```

The actual output, as it came out:

```
cartesian 1.9858653037988705 error 8.88e-16 eval 11907
polar     1.9858653037988712 error 2.22e-16 eval 441
```

```
exact     1.9858653037988714
Green edges 2.000000000000000  double 2.000000000000000
Gauss flux 7.5398223686155  12pi/5 7.5398223686155
Stokes surface 3.1415926535898  boundary 3.1415926535898
circle 6.2831853071796  2pi 6.2831853071796
curl(1,0) -2.01e-10
```

Four things deserve a reading. First, the two opening lines agree to the fifteenth digit at a cost that differs by a factor of 27: the right coordinates do not change the answer, only its price. Second, Green matches to the last digit, but that is only proof that the implementation is consistent with its formula. Third, Gauss and Stokes are computed from different sides and still meet; that is where verification has teeth. Fourth, quadrature gives 6.2831853071796 for a field that violates the hypothesis of Green, without a single error message.

Note for instructors

The teaching order I use: the density of the plate and the program that forgets the factor r, then Fubini, then the Jacobian determinant as the area of a stretching cell, then line integrals and their fundamental theorem, and only then Green. Green is best proved in full for type I regions at the board, because that proof uses only two tools the students already have. Gauss and Stokes are enough to prove for a box and for a parametrized surface.

Two misconceptions come up most often. First, students think $\partial P/\partial y = \partial Q/\partial x$ is enough for conservative, because they have only tested fields on a domain without holes. The field $(-y, x)/(x^2 + y^2)$ has to appear at the board, not on the reading list. Second, they treat the Jacobian determinant as a notational formality and write r or $\rho^2 \sin\varphi$ as something to memorize without checking the limits. Compare $4\pi/3$ with $2\pi^2$; that ratio of 4.712 times is more impressive than a hundred warnings.

One classroom activity: four groups, each computing one side of Green, Gauss, and Stokes for the same field, then matching numbers. The group with the sign error is found out within five minutes, and that is better than being told. The sign of readiness for what comes next: students can explain why orientation changes the sign without looking at the formula.

15.10 Summary

Summary.

- The double integral is defined through upper and lower sums over cells; Fubini allows the order of integration to be swapped when the function is integrable.
- Changing coordinates multiplies the area of a cell by $|\det J|$; polar gives r, spherical gives $\rho^2 \sin\varphi$. Without that factor the result is tidy but wrong.
- For a conservative field, the line integral depends only on the endpoints.
- Green, Gauss, and Stokes state one thing: the integral of a derivative over a region equals the integral of the field over its boundary, and orientation decides the sign.
- All those theorems are conditional: singularities, unmanageable boundaries, surfaces that cannot be oriented, and non-integrable functions make them fail without a warning.

15.11 Exercises

Exercise 15.1.

1. Compute $\iint_D (x^2 + y^2)\, dA$ on the unit disk in Cartesian and polar coordinates, check it with `dblquad`, and state the number of evaluations for each method.
2. Prove that if $T(u, v) = A(u, v)$ is linear and $\det A \neq 0$, the change-of-variables formula becomes $|\det A| \iint_D f \circ T\, dA$.

3. Repeat the Green example on the square in the clockwise direction: show that the result is -2 and state which side contributes which sign.
4. Compute the volume of the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 \le 1$ with the change of variables $x = au$, $y = bv$, $z = cw$, then check the formula on $a = b = c = 1$.
5. Assess this claim: true, false, or not enough evidence. "Because $\operatorname{curl} F = 0$ at every point where F is defined, the field $F = (-y, x)/(x^2+y^2)$ is conservative on $\mathbb{R}^2 \setminus \{(0,0)\}$." Include a reason and one supporting number.
6. Prove that the line integral is independent of path if and only if the circulation of every closed curve is zero, by using two paths from A to B that together form one closed curve.
7. Compute both sides of Stokes for $F = (-y, x, z)$ on the paraboloid $z = 1 - x^2 - y^2$, $z \ge 0$, with the normal pointing upward, then check with quadrature.
8. Compute both sides of Gauss for $F = (x, y, z)/(x^2+y^2+z^2)^{3/2}$ on the unit ball. Explain why the result does not contradict the theorem, and which hypothesis is violated. (Hint: compute $\operatorname{div} F$ first.)
9. Show that the Jacobian determinant of spherical coordinates is zero at $\varphi = 0$ and $\varphi = \pi$, then explain why the map may still be used on $0 < \varphi < \pi$. (Hint: check where the map stops being one-to-one.)

For researchers: limits and next steps

What is really proved in this chapter, with a written proof: the lemma on a region between two graphs, the fundamental theorem for line integrals, its converse through the integral potential, Green for type I and type II regions, the consequence for the curl test on a simply connected domain, and the sketch of the proof of Gauss through boxes and of Stokes through a parametrization. What I quote without proof: Fubini (Spivak, the chapter *Integration*, Theorem 3-10), the full change of variables (Spivak's same chapter; Rudin, the chapter *Integration of Differential Forms*), Gauss (Spivak, Theorem 5-8), and Stokes (Spivak, Theorem 5-9); general Stokes is Theorem 4-13 and 5-5.

What is overpopularized is the phrase "zero curl means conservative". What is correct: zero curl means closed form, and a closed form is exact only when the first cohomology group of the domain is zero. For $\mathbb{R}^3$ that holds, for $\mathbb{R}^2 \setminus \{0\}$ it does not, and the difference is measured by the integral on the circle, namely 2π here.

The open question that is genuinely open for me: can quadrature give a certificate that the hypotheses of a theorem are satisfied, and not merely an estimate of the value? The output in the lab shows that a numerical answer correct to fifteen digits can come from a field that violates the hypotheses, with no warning sign. For a field that is known only through samples on a grid, there is no general way to be sure that there is no singularity between the sample points.

Limits of validity: all functions here are real-valued and piecewise smooth; all numbers come from Python 3.14.4, NumPy 2.5.2, SciPy 1.18.0 with a tolerance of 10^{-12} unless stated otherwise, and spherical surfaces are parametrized with $\mathbf{n}$ pointing outward.

Map to source texts: Chapter 15 follows Stewart, Calculus: Early Transcendentals*, the chapter "Multiple Integrals" (15.1–15.3 and 15.6–15.9) and the chapter "Vector Calculus" (16.2 line integrals, 16.3 the fundamental theorem, 16.4 Green, 16.5 curl and divergence, 16.8 Stokes, 16.9 Gauss); on Spivak,* Calculus on Manifolds*, the chapter "Integration" (Fubini, Theorem 3-10, change of variables), the chapter "Integration on Chains" and the chapter "Integration of Manifolds" (Theorems 5-5, 5-8, 5-9); on Rudin,* Principles of Mathematical Analysis*, the chapters "Functions of Several Variables" and "Integration of Differential Forms"; on Active Calculus, chapters 11 and 12 for a more gradual*

order. The basis is the chapter Integral, Series of Functions, and Exchange of Limits *(the integral part) and the chapter* Continuity and Derivatives *(partial derivatives, Jacobian) of the first edition of this book. This chapter uses the Riemann integral and the fundamental theorem of calculus from Chapter 13, uniform continuity and compact sets from Chapter 11, determinants from Chapter 7, and gradients and Jacobian matrices from Chapter 12, all of them referred to and not repeated.*

Chapter 16

Differential Equations and Dynamical Systems

The pendulum clock in the living room swings with an amplitude of sixty degrees, and the length of its arm is one meter. The textbook gives the period

$$T = 2\pi\sqrt{L/g} = 2.006409 \text{ seconds},$$

and that formula is indeed correct for a very small swing. Measured at an amplitude of sixty degrees, the same pendulum needs 2.153242 seconds for one full swing: 7.32 percent longer. A clock set with the small-angle formula falls behind by 5892 seconds, about ninety-eight minutes, every day.

The formula is not a miscalculation. It is the exact answer for a different equation. The actual swing satisfies

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta,$$

whereas the small-period formula comes from replacing $\sin\theta$ with θ. At $\theta = 60^\circ$ that replacement turns 0.866025 into 1.047198, a difference of seventeen percent at the farthest point, and the period error left behind is 7.32 percent.

Chapter 12 taught us to differentiate, Chapter 14 series of functions, Chapter 15 vector fields. Not one of them answers the question here: if the rate of change is known, how do we find the function, is it unique, and what happens in the long run. This chapter fills those four gaps: how to solve, conditions for existence and uniqueness, the phase plane, and the stability test.

For younger readers (age 10 and up)

Imagine the water tank in a bathroom. The tap above runs at a steady rate, the drain below runs out, and the higher the water, the faster it leaves. If we open the tap and then walk away, tomorrow's water level cannot be computed from the size of the tap alone: what determines it is the water level now. High water pushes out faster, the rise slows down, and finally it stops when the water coming in equals the water going out.

Other things work the same way. Hot coffee cools faster while it is still very hot. Germs multiply faster when there are many of them, then slow down when the food runs out. What we know is the rule of change, not the final result.

Now the question that makes this work hard. Two tanks the same, taps the same, drains the same, but one starts empty and one starts half full: do they end at the same level? Sometimes yes, sometimes no. And there is a wicked drain, partly clogged and then suddenly open: two different stories can come out of the same initial state.

In this chapter we shall learn why the rule of change sometimes suffices to determine the whole future, and sometimes does not.

16.1 A rate that depends on itself

Level: basic

A differential equation sets the rate of change of a function equal to something that contains the function itself. To solve it means to find a function that satisfies that sentence on an entire interval. The derivatives of Chapter 12 were computed at a single point; the solution here is a function on an interval, and that difference is the one most often missed.

Definition 16.1 (Ordinary differential equation)**.** The equation

$$F(t, y, y', y'', \dots, y^{(n)}) = 0$$

is called an *ordinary differential equation* (ODE) of order n for a function y of the variable t if F contains the n-th derivative and contains no higher derivative. The ODE is called *linear* if F has the form

$$a_n(t)\, y^{(n)} + \cdots + a_1(t)\, y' + a_0(t)\, y = b(t),$$

and is called *autonomous* if F does not contain t explicitly.

All the examples in this chapter are of first or second order, because those two already contain every important idea.

Definition 16.2 (Solution and initial value problem)**.** A function y is called a *solution* of the ODE on an interval I if y has derivatives up to order n on I and satisfies the equation at every point of I. If the condition $y(t_0) = y_0$ is added, the combination of the two is called an *initial value problem*, and y_0 is called the initial value.

One example to make sure the definition works. Take the population model

$$y' = y(1 - y), \qquad y(0) = \tfrac{1}{2},$$

with y denoting the fraction of the population relative to the carrying capacity. The candidate solution is

$$y(t) = \frac{1}{1 + e^{-t}}.$$

Its derivative is $y'(t) = e^{-t}/(1 + e^{-t})^2$, and the right-hand side $y(1 - y) = e^{-t}/(1 + e^{-t})^2$ is exactly the same. The initial value $y(0) = 1/2$ matches as well. So the function is a solution, and that check uses only the quotient rule of Chapter 12.

What deserves attention: $y(t) = 1$ and $y(t) = 0$ are also solutions, both of them constant, and both appear when we ask when the right-hand side is zero. So one equation can have many solutions; the initial value selects one of them. The collection of all solutions is called the *general solution*, and it contains one constant for each order.

The interval on which a solution lives is part of the answer. Take

$$y' = y^2, \qquad y(0) = 1.$$

Separation of variables gives $y(t) = 1/(1 - t)$, and that formula makes sense only on $t < 1$. At $t = 0.9$ its value is 10; at $t = 0.99$ its value is 100; at $t = 0.999$ its value is 1000. There is no finite number at $t = 1$, and no solution continues that function past it. The solution

blows up in finite time, and a program that reports a value at $t = 2$ does not find a solution; it finds rounding.

differential equation (*persamaan diferensial*) — A sentence that ties the rate of change of a quantity to the size of that quantity itself.

general solution (*solusi umum*) — The family of all functions satisfying the equation, usually containing one free constant per order.

Read it aloud (auditory learner)

— If I let this ball go on the slope, it will move downhill. How fast? — It depends on where the ball is now. Near the top, slowly. In the middle of the slope, quickly. — So its speed is not a single fixed number? — No. It is a rule: at each point, the speed is determined by that point. — If the rule already knows everything, is the path already determined? — Not yet. The rule gives a direction at every point; the path is born from the starting point you choose. — Two balls, two starting points, one rule? — One rule, two paths. — If two paths meet at the same point, how do we know which one came from where? — We cannot. That is why this chapter demands one starting point for one path.
Repeat it once, in a normal voice: same rule, different starting point, different path.

16.2 Two families that can be solved by hand

Level: basic

Not every differential equation has a solution formula, but two large families do, and both appear in almost every first-level application. The first family separates variables; the second is the first-order linear equation.

16.2.1 Separation of variables

If the equation can be written $y' = f(t)g(y)$, we move all y to one side and all t to the other side:

$$\frac{dy}{g(y)} = f(t)\,dt, \qquad \int \frac{dy}{g(y)} = \int f(t)\,dt.$$

That step is not a notational trick but the chain rule run backwards: if H is an antiderivative of $1/g$, then $\frac{d}{dt}H(y(t)) = H'(y)y' = y'/g(y)$, so the original equation becomes $\frac{d}{dt}H(y(t)) = f(t)$ and Chapter 13 gives $H(y) = \int f\,dt + C$. The result is usually implicit, and that is a legitimate answer.

The logistic model above is used throughout the chapter. With $g(y) = y(1-y)$ and partial fractions,

$$\int \frac{dy}{y(1-y)} = \int \Big(\frac{1}{y} + \frac{1}{1-y}\Big)dy = \ln\Big|\frac{y}{1-y}\Big|,$$

so $\ln|y/(1-y)| = t + C$ and

$$y(t) = \frac{1}{1 + Ae^{-t}}, \qquad A = \frac{1-y_0}{y_0}.$$

For $y_0 = 1/2$ we obtain $A = 1$, which is the formula we checked a moment ago. Note that A is determined by the initial value, and that $A = 0$ gives the constant solution $y = 1$ which the division above does not cover: when separating variables we divide by $g(y_0)$, and that division is valid only if $g(y_0) \neq 0$.

16.2.2 Linear equations and the integrating factor

The standard form of a first-order linear equation is

$$y' + p(t)\,y = q(t).$$

The left-hand side is almost the derivative of a product, only one factor short. The missing factor is exactly what the following theorem looks for.

Theorem 16.1 (Integrating factor)**.** *Suppose p and q are continuous on an open interval I containing t_0, and suppose*

$$\mu(t) = \exp\Big(\int_{t_0}^{t} p(s)\,ds\Big).$$

Then the initial value problem $y' + p(t)y = q(t)$, $y(t_0) = y_0$ has exactly one solution on I, namely

$$y(t) = \frac{1}{\mu(t)}\Big(y_0 + \int_{t_0}^{t} \mu(s)\,q(s)\,ds\Big).$$

Proof. Since p is continuous, μ is defined, positive, and differentiable on I, with $\mu' = p\mu$ by the fundamental theorem of calculus (Chapter 13). Multiply the equation by μ:

$$\mu y' + \mu p\,y = \mu q \quad\Longleftrightarrow\quad (\mu y)' = \mu q,$$

because the product rule gives $(\mu y)' = \mu' y + \mu y' = \mu(py + y')$. Integrate both sides from t_0 to t and use the fundamental theorem of calculus:

$$\mu(t)y(t) - \mu(t_0)y(t_0) = \int_{t_0}^{t} \mu(s)q(s)\,ds.$$

Since $\mu(t_0) = 1$, division by $\mu(t) > 0$ gives the formula requested. Uniqueness: if y_1, y_2 are two solutions, their difference $w = y_1 - y_2$ satisfies $w' + pw = 0$ with $w(t_0) = 0$; multiplying by μ gives $(\mu w)' = 0$, so μw is constant, and since its value is zero at t_0 it is zero everywhere. □

That proof guesses nothing: μ is found by demanding $\mu' = p\mu$, an equation easier than the original. This pattern repeats: one problem is solved by turning it into a simpler problem.

Example 16.1 (One linear equation, two checks). Solve $y' + 2y = 4t$ with $y(0) = 1$. Here $p = 2$ and $q = 4t$, so $\mu = e^{2t}$ and

$$(e^{2t}y)' = 4t\,e^{2t}.$$

Integration by parts gives

$$\int 4t\,e^{2t}\,dt = \Big(2t - 1\Big)e^{2t} + C,$$

because the derivative of $(2t-1)e^{2t}$ is $2e^{2t} + 2(2t-1)e^{2t} = (4t)e^{2t}$. Thus

$$y(t) = 2t - 1 + Ce^{-2t},$$

and the condition $y(0) = 1$ gives $C = 2$, that is $y(t) = 2t - 1 + 2e^{-2t}$.

Check: substitution gives $y' + 2y = 4t$, and $y(1) = 1 + 2e^{-2} = 1.2706705665$; an adaptive integrator with tolerance 10^{-12} gives 1.270670566473.

16.2.3 The RC circuit: the simplest real example

A voltage source V_s is connected in series with a resistor R and a capacitor C. Kirchhoff's law of voltages gives $V_s = Ri + V_C$, and the law of the capacitor gives $i = C\,dV_C/dt$. The two together give

$$RC\,\frac{dV_C}{dt} + V_C = V_s, \qquad \text{or} \qquad \frac{dV_C}{dt} + \frac{1}{RC}V_C = \frac{V_s}{RC}.$$

This is a linear equation with $p = 1/(RC)$ constant. By the theorem above, with $V_C(0) = 0$,

$$V_C(t) = V_s(1 - e^{-t/\tau}), \qquad \tau = RC.$$

The number τ is called the time constant. For $R = 1\,\text{k}\Omega$ and $C = 1\,\mu\text{F}$, $\tau = 10^{-3}$ seconds. With $V_s = 5$ volts, the voltage at $t = \tau$ is 3.160603 volts and at $t = 5\tau$ it is 4.966310 volts, that is 99.33 percent of 5 volts. To reach 90 percent requires $t = \tau \ln 10 = 2.302585$ milliseconds.

Step-by-step recipe (procedural learner)

1. Write the equation in standard form $y' + p(t)y = q(t)$: divide by the coefficient of y' so that the coefficient is exactly one.
2. Compute $\mu(t) = \exp\big(\int p(t)\,dt\big)$, with no constant of integration.
3. Multiply the whole equation by μ, then write the left-hand side as $(\mu y)'$. If the left-hand side refuses to become a single derivative, step 1 is wrong.
4. Integrate both sides with respect to t, then divide by μ.
5. Use the initial value to determine the constant, and write down the interval of validity.

Warning: step 1 must not be skipped, and step 3 must not be done before step 2. Computing μ from coefficients that have not yet been normalized gives a wrong integrating factor, and the error only becomes visible when the solution is tested.

integrating factor (*faktor integrasi*) — A multiplier that turns the left-hand side of a linear equation into a single derivative of a product.

16.3 Existence, uniqueness, and Picard iteration

Level: intermediate

The example $y' = y^2$ shows that a solution can stop existing. Given f and one initial point, does the solution exist and is it only one? The answer depends on one property of f with respect to y.

Definition 16.3 (Lipschitz condition)**.** A function $f(t, y)$ satisfies the Lipschitz condition with respect to y on a region $D \subseteq \mathbb{R}^2$ if there is a constant L such that

$$|f(t, y_1) - f(t, y_2)| \le L\,|y_1 - y_2|$$

for every $(t, y_1), (t, y_2) \in D$. The smallest possible constant is called the Lipschitz constant.

Its everyday sentence: f must not change too abruptly when y shifts a little. If f has a partial derivative with respect to y that is continuous and bounded by L on D, that condition holds automatically, because the mean value theorem (Chapter 12) gives $f(t, y_1) - f(t, y_2) = \partial_y f(t, \xi)(y_1 - y_2)$.

Theorem 16.2 (Picard–Lindelöf)**.** *Suppose f is continuous on the closed rectangle $D = [t_0 - a, t_0 + a] \times [y_0 - b, y_0 + b]$ and satisfies the Lipschitz condition with respect to y on D with constant L. Suppose $M = \max_D |f|$ and $h = \min(a, b/M)$. Then the initial value problem $y' = f(t, y)$, $y(t_0) = y_0$ has exactly one solution on the interval $|t - t_0| \le h$.*

The full proof uses the Banach fixed point theorem on the space of continuous functions with the maximum norm, which is complete because uniform continuity is discussed in Chapter 11 and Chapter 14. I give the sketch, because the sketch contains the whole idea and can be turned into an algorithm.

Write the equation as an integral equation, which is equivalent because both sides are continuous:

$$y(t) = y_0 + \int_{t_0}^{t} f\big(s, y(s)\big)\, ds.$$

Define the map T by that formula, starting from the constant $y_0(t) = y_0$, then iterate:

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f\big(s, y_n(s)\big)\, ds.$$

Every y_n lies inside D as long as $|t - t_0| \le h$, because $|y_{n+1} - y_0| \le M|t - t_0| \le Mh \le b$. The difference of two successive iterations satisfies

$$|y_{n+1}(t) - y_n(t)| \le L\Big|\int_{t_0}^{t} |y_n(s) - y_{n-1}(s)|\, ds\Big| \le \frac{(L|t - t_0|)^n}{n!}\, \|y_1 - y_0\|,$$

and induction on n gives that bound. Since $\sum_n (Lh)^n/n!$ converges (ratio test, Chapter 5), the series $y_0 + \sum (y_{n+1} - y_n)$ converges uniformly, and its limit is the solution. Uniqueness comes from the same inequality: two different solutions would satisfy $|y_1 - y_2| \le L|t - t_0|\, \|y_1 - y_2\|$, which for $Lh < 1$ is possible only if the difference is zero; the case $Lh \ge 1$ is handled by shrinking h, and that is why the theorem promises only a short interval.

Run that iteration on $y' = y$, $y(0) = 1$. Starting from $y_0 = 1$:

$$y_1 = 1 + t, \quad y_2 = 1 + t + \frac{t^2}{2}, \quad y_3 = 1 + t + \frac{t^2}{2} + \frac{t^3}{6},$$

so y_n is the partial sum of the exponential series from Chapter 5. At $t = 1$, the eighth iteration gives 2.718278769841, whereas $e = 2.718281828459$; the error is 3.06×10^{-6}.

The Lipschitz condition must not be thrown away. Take

$$y' = \sqrt{|y|}, \qquad y(0) = 0.$$

The right-hand side is continuous, so a solution exists; but it does not satisfy the Lipschitz condition near $y = 0$, because its slope is unbounded there. And indeed there are two solutions: the zero function, $y(t) \equiv 0$, and

$$y(t) = \frac{t^2}{4} \quad (t \ge 0).$$

The check is one line: $y' = t/2$ and $\sqrt{|y|} = \sqrt{t^2/4} = t/2$ for $t \ge 0$, whereas $y(0) = 0$. Two different histories from one and the same initial state, exactly like the drain that opens suddenly in the opening box of the chapter.

Lipschitz is a *sufficient* condition, not a necessary one. Existence without uniqueness is guaranteed by Peano's theorem, which uses compactness (Chapter 11) and Arzelà–Ascoli; see Coddington and Levinson, chapters 1–2.

uniqueness (*ketunggalan*) — The guarantee that one initial state gives birth to only one history, with no branch.

16.4 Systems, the phase plane, and oscillators

Level: intermediate

One second-order equation is not a new animal; it is two first-order equations running together. By naming $v = \theta'$, the pendulum equation becomes

$$\theta' = v, \qquad v' = -\frac{g}{L}\sin\theta.$$

This pair is autonomous: its right-hand side does not contain t, so the initial state determines the whole history (provided the Lipschitz condition holds) and we may forget time for a moment in order to ask what happens on the plane (θ, v).

Definition 16.4 (Autonomous systems and the phase plane)**.** The system

$$\mathbf{x}' = \mathbf{f}(\mathbf{x}), \qquad \mathbf{x} \in \mathbb{R}^n,$$

is called *autonomous*. For $n = 2$, the plane on which the point $\mathbf{x}$ lives is called the *phase plane*; the curve $\{\mathbf{x}(t)\}$ is called a *trajectory*, and the picture of all the arrows $\mathbf{f}(\mathbf{x})$ is called the *direction field*. A point $\mathbf{x}^*$ with $\mathbf{f}(\mathbf{x}^*) = 0$ is called an *equilibrium point* or a fixed point.

The horizontal axis of the phase plane represents position, the vertical axis represents velocity, and one point means "here, with this speed". Time is not drawn; it is only the parameter that walks along the curve. That is its advantage: the whole history, for all initial points at once, appears as a single picture.

The equilibrium points of the pendulum are easy to find. $\theta' = 0$ demands $v = 0$, and $v' = 0$ demands $\sin\theta = 0$, so $\theta = k\pi$. At $\theta = 0$ the pendulum hangs straight down; at $\theta = \pi$ it stands straight up. The first is stable in the everyday sense, the second is not: the slightest touch brings it down.

For younger readers (age 10 and up)

Go to the park, find a swing, and sit down. Ask a friend to push you gently, only a little. Count ten swings by counting inside your head: one, two, three, up to ten. Now ask for a push that is much higher, until the swing rises almost as high as the upper bar, then count ten swings again.

What you will see: the high swing takes longer for one full round, even though you feel as though the high swing is faster. That feeling cheats because the distance traveled is much longer, while the time for one round is almost the same.

Now look at the top of the swing. When you are at the highest point, there is a short moment when you almost stop and hang. At the lowest point, you move fastest. A swing never stops in the middle, and is never fastest at the ends. Those two things always change places.

In this chapter we shall learn why a high swing is slower, and why a pendulum standing straight up is almost impossible to keep there.

Now the exact period formula. Multiply $v' = -(g/L)\sin\theta$ by $v = \theta'$ and integrate:

$$\frac{d}{dt}\Big(\frac{v^2}{2} + \frac{g}{L}(1 - \cos\theta)\Big) = 0.$$

That bracket is the energy per unit mass, and it is constant along the trajectory. With E denoting its value,

$$\frac{v^2}{2} + \frac{g}{L}(1 - \cos\theta) = E,$$

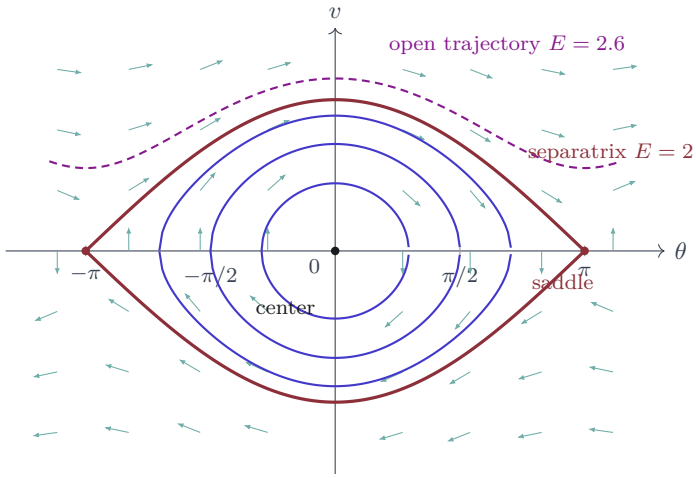


Figure 16.1: The phase plane of the pendulum in units $g/L = 1$, with the horizontal axis the angle θ and the vertical axis the velocity v. The small arrows show the direction field $(\theta', v') = (v, -\sin\theta)$. The dark blue curves are three closed trajectories with energy $E = 0.4$, 1.0, and 1.6; the larger E, the wider the turn. The red curve is the separatrix $E = 2$, the trajectory that reaches exactly $\theta = \pm\pi$ with zero velocity; it separates closed swings from open rotations. The dashed curve is the open trajectory $E = 2.6$, where the pendulum rotates fully without ever stopping. The point at $\theta = 0$ is a center, the points at $\theta = \pm\pi$ are saddles.

so $v = \pm\sqrt{2E - 2(g/L)(1 - \cos\theta)}$. The trajectories in the phase plane are those energy curves; in Figure 16.1 they are drawn for the unit $g/L = 1$.

That figure contains the entire qualitative behavior of the pendulum without a single equation being solved; all that is needed is the sign of $\sin\theta$ and the symmetry of the energy.

For small amplitude, $\sin\theta \approx \theta$ and the system becomes the harmonic oscillator

$$\theta' = v, \qquad v' = -\omega_0^2\theta, \qquad \omega_0 = \sqrt{g/L},$$

with solution $\theta(t) = \theta_0 \cos(\omega_0 t)$ and period $2\pi/\omega_0 = 2.006409$ seconds for $L = 1$ meter. Its trajectories are the ellipses $v^2 + \omega_0^2\theta^2 = \text{constant}$, and those are the three small rings near the center in the figure above.

The exact period comes from energy conservation, not from linearization. With $E = (g/L)(1 - \cos\theta_0)$ for a pendulum released from rest at θ_0, the time to fall from θ_0 to zero is

$$\frac{T}{4} = \sqrt{\frac{L}{2g}} \int_0^{\theta_0} \frac{d\theta}{\sqrt{\cos\theta - \cos\theta_0}} = \sqrt{\frac{L}{g}}\, K(k), \qquad k = \sin\frac{\theta_0}{2},$$

with K the complete elliptic integral of the first kind. For $\theta_0 = 60^\circ$ we obtain $k = 0.5$ and K at the parameter $m = k^2 = 0.25$ has the value 1.6857503548, so $T = 2.153242$ seconds. Quadrature on that integral gives the same number to within 9×10^{-13}; two different roads, one result. For $\theta_0 = 5^\circ$ the ratio is 1.00047617, that is an error of less than one part in two thousand: that is where the small formula deserves to be used, and where its limit of validity lies.

Do it by hand (kinesthetic learner)

Take a string one meter long, a metal nut as the weight, and a stopwatch on the phone. Tie the nut to the end of the string, stick the other end to the door frame with tape, and measure the length of the string from the suspension point to the center of the nut: try to make it exactly one meter. Pull the nut aside only about five centimeters, release it,

and record the time for ten full swings. Divide by ten; the result is about two seconds. Now pull much higher, roughly half a meter above the lowest point, and repeat. Compare the two numbers. What you will see: the second number is larger, but not twice as large. The difference is about seven percent, and that agrees with the computation above if the angle is about sixty degrees.
Repeat with a string a quarter of a meter long: the period becomes half, because the period depends on the square root of the length.

16.5 Stability: eigenvalues, linearization, and Lyapunov

Level: advanced

The question most often asked about a system is not "what is its value at $t = 7$" but "does it come back after being disturbed". To answer it we do not need the solution; the derivatives around the equilibrium point are enough.

Definition 16.5 (Stability)**.** Let $\mathbf{x}^*$ be an equilibrium point of $\mathbf{x}' = \mathbf{f}(\mathbf{x})$. That point is called *stable* if for every $\varepsilon > 0$ there is $\delta > 0$ such that $\|\mathbf{x}(0) - \mathbf{x}^*\| < \delta$ guarantees $\|\mathbf{x}(t) - \mathbf{x}^*\| < \varepsilon$ for every $t \geq 0$. The point is called *asymptotically stable* if it is stable and there is $\delta > 0$ such that $\mathbf{x}(t) \to \mathbf{x}^*$ as $t \to \infty$. If it is not stable, it is called *unstable*.

Theorem 16.3 (Linearization)**.** *Suppose* $\mathbf{f}$ *has continuous partial derivatives near the equilibrium point* $\mathbf{x}^*$ *and suppose* $A = J_{\mathbf{f}}(\mathbf{x}^*)$ *is the Jacobian matrix from Chapter 12. If no eigenvalue of* A *has real part zero (a* hyperbolic *equilibrium point), then the qualitative behavior of the nonlinear system near* $\mathbf{x}^*$ *is the same as that of the linear system* $\mathbf{u}' = A\mathbf{u}$*: the type of stability is the same, and nearby trajectories can be mapped one-to-one and continuously onto linear trajectories.*

This is the Hartman–Grobman theorem; its proof is long and uses the fixed point theorem on a space of functions, so I cite it: Hartman, chapter 9, Perko, section 2.8. The practical consequence: local stability is read off from the eigenvalues.

For a two-dimensional system, every possibility is read from the trace $\tau = \operatorname{tr} A$ and the determinant $\Delta = \det A$; the eigenvalues are the roots of $\lambda^2 - \tau\lambda + \Delta = 0$. If $\Delta < 0$, the roots are real and of opposite sign: a *saddle*. If $\Delta > 0$ and $\tau^2 \geq 4\Delta$, both are real and of the same sign: a *node*, stable if $\tau < 0$. If $\tau^2 < 4\Delta$, the roots are conjugates with real part $\tau/2$: a *spiral*, stable if $\tau < 0$, and a center if $\tau = 0$.

Example: the matrix $A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$ has $\tau = -3$ and $\Delta = 2$, so its roots are $\lambda^2 + 3\lambda + 2 = 0$, namely -1 and -2: a stable node. The matrix $B = \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix}$ has $\tau = 0$ and $\Delta = 2 > 0$, its roots are $\pm i\sqrt{2}$: a center.

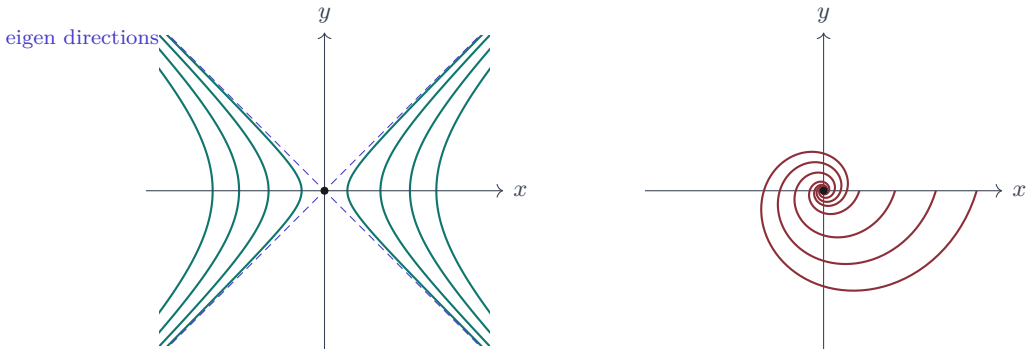
Figure 16.2 shows the two portraits that differ most in character: the saddle, where almost every initial point moves away, and the stable spiral, where everything spirals inward.

One warning that is expensive to learn. If $\tau = 0$ and $\Delta > 0$, linearization gives a center, but the nonlinear system at that point is not necessarily a center. Take

$$x' = -y + x(x^2 + y^2), \qquad y' = x + y(x^2 + y^2).$$

Its Jacobian at the origin is a rotation matrix, its eigenvalues are $\pm i$: linearization says center. But in polar coordinates $r^2 = x^2 + y^2$, the cross multiplication gives

$$\frac{1}{2}\frac{d}{dt}r^2 = xx' + yy' = r^2(x^2 + y^2) = r^4,$$



(a) saddle: $\lambda = 1, -1$ (b) stable spiral: $\lambda = -0.3 \pm i$

Figure 16.2: Two linear phase portraits of different types. (a) The system $x' = y$, $y' = x$: trajectories satisfy $x^2 - y^2 =$ constant, hyperbolas with asymptotes $y = \pm x$, and the dashed lines are the eigen directions, one in and one out, so the origin is a saddle and is unstable. (b) The system $x' = -0.3x + y$, $y' = -x - 0.3y$: the eigenvalues are $-0.3 \pm i$, so trajectories spiral while shrinking by the factor $e^{-0.3t}$ and the origin is a stable spiral. The four trajectories in (b) differ only in their initial radius, because the system is linear.

so $r' = r^3 > 0$ for $r > 0$: trajectories spiral outward, and the origin is not stable. Linearization is not wrong; it is only insufficient when there is an eigenvalue on the imaginary axis. That is where the theorem above demands a hyperbolic equilibrium point.

For the non-hyperbolic case, a second tool is used: a decreasing energy function, which we compose ourselves.

Definition 16.6 (Lyapunov function)**.** A function V that is continuous and has continuous derivatives on a neighborhood U of $\mathbf{x}^*$ is called a *Lyapunov function* if $V(\mathbf{x}^*) = 0$ and $V(\mathbf{x}) > 0$ for $\mathbf{x} \in U \setminus \{\mathbf{x}^*\}$. Its derivative along a trajectory is written

$$\dot{V}(\mathbf{x}) = \nabla V(\mathbf{x}) \cdot \mathbf{f}(\mathbf{x}).$$

That artificial hill is V, and $\dot{V}$ is the steepness of its descent.

Theorem 16.4 (Lyapunov test)**.** *Suppose V is a Lyapunov function near the equilibrium point $\mathbf{x}^*$.*

1. *If $\dot{V} \le 0$ on U, then $\mathbf{x}^*$ is stable.*
2. *If $\dot{V} < 0$ on $U \setminus \{\mathbf{x}^*\}$, then $\mathbf{x}^*$ is asymptotically stable.*

Proof. For the first statement, take $\varepsilon > 0$ small enough that the ball $\|\mathbf{x} - \mathbf{x}^*\| \le \varepsilon$ stays inside U. On that ball V is positive and the ball is compact, so $m = \min_{\|\mathbf{x}-\mathbf{x}^*\|=\varepsilon} V > 0$. Choose $\delta > 0$ such that $V(\mathbf{x}) < m$ for $\|\mathbf{x} - \mathbf{x}^*\| < \delta$; this is possible because V is continuous and $V(\mathbf{x}^*) = 0$. A trajectory starting in the δ ball satisfies $V(\mathbf{x}(t)) \le V(\mathbf{x}(0)) < m$, since $\dot{V} \le 0$, so it cannot cross the ε ball; hence it is stable.

For the second statement, $V(\mathbf{x}(t))$ decreases and is bounded below by zero, so it has a limit $L \ge 0$. Suppose $L > 0$; the trajectory then stays in the compact set $\{V \ge L\}$, and there $-\dot{V}$ has a positive minimum, so V would have to fall through L in finite time. Contradiction. Hence $L = 0$, and because V vanishes only at $\mathbf{x}^*$, the trajectory heads there. □

An example is the damped pendulum, $\theta' = v$, $v' = -(g/L)\sin\theta - bv$ with $b > 0$, and $V = v^2/2 + (g/L)(1 - \cos\theta)$, which is the mechanical energy, positive except at the equilibrium point. Its derivative is

$$\dot{V} = v\Big(-\frac{g}{L}\sin\theta - bv\Big) + \frac{g}{L}\sin\theta\, v = -bv^2 \le 0.$$

So the lower equilibrium point is stable and the energy never increases. This test does not give asymptotic stability, because $\dot V$ is zero along the whole line $v = 0$; the LaSalle version adds that the only invariant set inside $\{\dot V = 0\}$ is the equilibrium point, so all trajectories head there.

Linearization gives the same answer. At $\theta = 0$ the Jacobian is $\begin{pmatrix} 0 & 1 \\ -g/L & -b \end{pmatrix}$, with $\tau = -b$ and $\Delta = g/L > 0$; for $b = 0.5$ and $g/L = 9.80665$ the roots are $-0.25 \pm 3.121562\,i$, that is a stable spiral. At $\theta = \pi$ a $+g/L$ appears in the lower right corner, so $\Delta = -g/L < 0$: a saddle with eigenvalues ± 3.131557. A pendulum standing straight up is indeed a saddle, and not as a figure of speech: there is exactly one direction that keeps it up there, and every disturbance carries it away.

asymptotically stable (*stabil asimtotik*) — Stays close after a disturbance, then actually arrives back at the equilibrium point.

Lyapunov function (*fungsi Lyapunov*) — An artificial hill used as a ruler: as long as the system descends it, the system cannot run far away.

16.6 Euler, Runge–Kutta, and stiffness

Level: advanced

Almost no equation can be solved by a formula, so numerical methods are not a supplement but the main road. The idea goes back to Chapter 12: if we know the value and the derivative at one point, we can estimate the value a little to its right.

Euler's method does the simplest thing. From $y' = f(t, y)$ and the initial value,

$$y_{n+1} = y_n + h\, f(t_n, y_n), \qquad t_{n+1} = t_n + h.$$

It is exactly the first two terms of the Taylor series around t_n, so its one-step error is the term $h^2 y''(\xi)/2$, of order h^2. That local error accumulates into a global error of order h: doubling the accuracy demands twice as many steps, not four times.

Numbers for $y' = y$, $y(0) = 1$, at $t = 1$ with $h = 0.1$: Euler gives $(1.1)^{10} = 2.5937424601$, whereas $e = 2.7182818285$. The error is 0.1245, that is 4.58 percent, and the error of the first single step is 5.171×10^{-3}.

The fourth-order Runge–Kutta method repairs this by taking four samples of the slope at every step:

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4),$$

with $k_1 = f(t_n, y_n)$, $k_2 = f(t_n + h/2,\, y_n + hk_1/2)$, $k_3 = f(t_n + h/2,\, y_n + hk_2/2)$, and $k_4 = f(t_n + h,\, y_n + hk_3)$. Its local error is of order h^5 and its global error of order h^4: half the step means the error falls by about sixteen times.

The following table comes from the program in the lab section, on the logistic model with $y(0) = 1/2$ and $t = 2$; the exact value is 0.8807970780.

h	Euler error	RK4 error	Euler ratio
0.400	1.958×10^{-2}	1.245×10^{-5}	—
0.200	9.423×10^{-3}	7.237×10^{-7}	2.08
0.100	4.630×10^{-3}	4.359×10^{-8}	2.04
0.050	2.296×10^{-3}	2.674×10^{-9}	2.02
0.025	1.143×10^{-3}	1.656×10^{-10}	2.01

The last column is the proof of first order: every time h is halved, the Euler error falls by almost exactly two; for RK4 the ratio is about 16, in keeping with fourth order. Figure 16.3 displays the same numbers on a logarithmic scale on both axes.

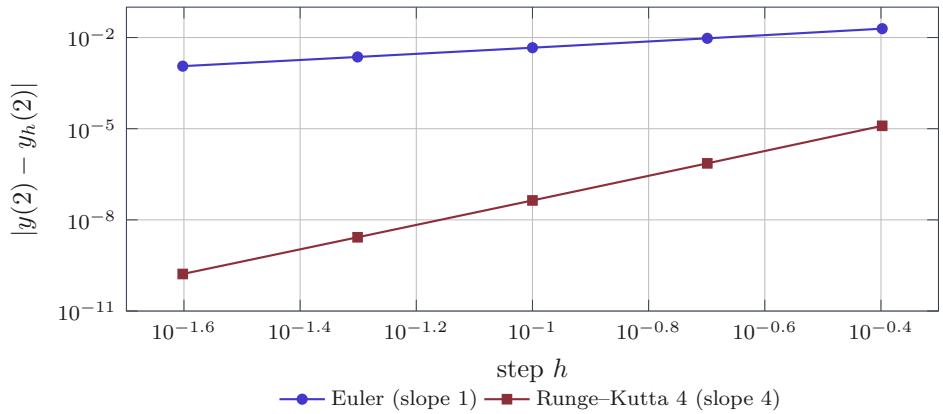


Figure 16.3: The error at $t = 2$ for the logistic model as a function of the step h, on a logarithmic scale on both axes. The Euler line falls with slope one: the error is proportional to h. The Runge–Kutta line falls with slope four: the error is proportional to h^4. At $h = 0.4$ the two methods are about three orders apart, and that distance widens when h is made smaller. The numbers in the figure are the same numbers as in the table in the text.

Now the property that deceives many people: stiffness. A system is called *stiff* if inside it there is a process that moves far faster than the process we are interested in. Its practical signature: explicit methods demand a very small step in order to remain *stable*, even when the accuracy we ask for demands only a large step.

The example is small and honest: $y' = -50(y - \cos t) - \sin t$, $y(0) = 1$, whose exact solution is $y = \cos t$. Here y follows $\cos t$ slowly, but every deviation is dragged back at the rate 50 per second. Euler is stable only if $|1 - 50h| < 1$, that is $h < 0.04$.

Results at $t = 3$, with exact value $\cos 3 = -0.989992$: Euler with $h = 0.02$ gives an error of 1.98×10^{-4}, with $h = 0.039$ it gives 7.97×10^{-4}, and with $h = 0.05$ it gives -1.839×10^{7}. One step up from 0.039 to 0.05 changes the answer from almost right to meaningless. RK4 holds out longer, because its stability region is wider: at $h = 0.05$ its error is 1.11×10^{-3}, but at $h = 0.1$ it gives -1.32×10^{31}; its practical limit is $2.785/50 = 0.0557$.

The lesson to carry away: error and stability are two different things. The local error on the first step is small and warns us of nothing; what explodes is stability, and that is a matter of eigenvalues.

The antidote is implicit methods, the simplest being backward Euler:

$$y_{n+1} = y_n + h\, f(t_{n+1}, y_{n+1}),$$

which demands solving an equation at every step, but is stable for large h on stiff equations; for the example above, backward Euler with $h = 0.5$ gives an answer close to $\cos t$. Reference: Hairer and Wanner, chapter IV.

Challenge (problem solver)

Linearization gives a center, but a nonlinear system need not be a center. Construct a two-dimensional system whose origin is *not* a center even though its linearization is a center, determine the character of that point (stable, asymptotically stable, or unstable), and include an explicit formula, the eigenvalues of the Jacobian at the origin, and a brief proof of your claim.

Hint 1: linearization sees only terms of degree one. Add a degree-three term that is symmetric under rotation, then write the system in polar coordinates.

Hint 2: in polar coordinates, the rotation matrix becomes $\theta' = 1$, and the radial disturbance becomes $r' = cr^3$ for some constant c. The sign of c decides everything, and

$V = r^2$ is a Lyapunov function that works for one of the two signs.
If you get stuck, look again at the example $r' = r^3$ in the Stability section.

16.7 Worked examples

Level: intermediate

The four computations below are the backbone of this chapter; all of them can be checked again with a calculator or with the program in the lab section, and all of them have gone wrong in my own hands.

Logistic model: one formula, two methods. Take $y' = y(1 - y)$ with $y(0) = 1/2$ and ask for the value at $t = 2$. The formula from the separation of variables section gives $y(2) = 1/(1 + e^{-2}) = 0.8807970780$. With $h = 0.1$ there are twenty steps: Euler gives 0.8854273272, an error of 4.630×10^{-3} or 0.53 percent, whereas Runge–Kutta with the same step gives 0.8807970344, an error of 4.359×10^{-8}. The cost is four times as much, the accuracy about one hundred thousand times.

Note also the direction of the error. Euler is too high here, because $y(1 - y)$ is concave down on that interval, so a straight line always lies above the curve. Checking the sign of the error is a cheap way to find out whether our program works as it should.

Variable coefficient: $y' + y/t = t^2$, $y(1) = 0$. Here $p(t) = 1/t$, so

$$\mu(t) = \exp\Big(\int \frac{dt}{t}\Big) = t$$

on $t > 0$. Multiplying the equation by t gives

$$t\,y' + y = t^3 \quad \Longleftrightarrow \quad (t\,y)' = t^3,$$

so $t\,y = t^4/4 + C$ and

$$y(t) = \frac{t^3}{4} + \frac{C}{t}.$$

The condition $y(1) = 0$ gives $C = -1/4$, so

$$y(t) = \frac{t^3}{4} - \frac{1}{4t} = \frac{t^4 - 1}{4t}, \qquad y(2) = \frac{15}{8} = 1.875.$$

Check: $y' = 3t^2/4 + 1/(4t^2)$ and $y/t = t^2/4 - 1/(4t^2)$, their sum is t^2, exactly the right-hand side. An adaptive integrator gives 1.875000000000.

Without $\mu = t$, the left-hand side $y' + y/t$ does not look like the derivative of anything; after multiplication by t, it becomes the derivative of ty.

How large a swing can be before the small formula misses by one percent. The ratio of the exact period to the small period is

$$\frac{T(\theta_0)}{T_0} = \frac{2}{\pi} K\Big(\sin^2 \frac{\theta_0}{2}\Big),$$

with K the complete elliptic integral. Some of its values:

θ_0	T/T_0	error	remark
5°	1.000476	0.048 percent	pendulum clock
10°	1.001907	0.191 percent	still safe
20°	1.007669	0.767 percent	nearing the limit
22.81°	1.010000	1.000 percent	the one-percent limit
30°	1.017409	1.741 percent	correction needed
60°	1.073182	7.318 percent	clock off by an hour

So the small formula is fit to be used up to about twenty degrees, and 22.81° is the one-percent error threshold; the error grows like the square of the amplitude, not like the amplitude. Doubling the amplitude from 10° to 20° raises the error from 0.19 to 0.77 percent, that is four times, in keeping with $1 + (\theta_0^2/16)$ as the first two terms of that estimate.

Test yourself in three steps. (1) Work the logistic example again with $h = 0.2$; you should get an Euler error of 9.423×10^{-3} and an RK4 error of 7.237×10^{-7}. (2) Derive the elliptic period formula yourself from energy conservation, then check that K at the parameter 0.25 has the value 1.6857503548. (3) Find the threshold h for Euler on $y' = -100y$; the answer is $h < 0.02$, and at $h = 0.02$ exactly, the method oscillates between $+1$ and -1 without either decaying or exploding.

16.8 Computational lab

Level: advanced

The following code computes all the numbers above: the Euler and RK4 errors on the logistic model, the pendulum period by two different roads, and the stiffness threshold. It uses only NumPy and SciPy.

Computational lab

```
import numpy as np
from scipy import integrate, special

def euler(f, y0, T, h):
    y, t, n = y0, 0.0, int(round(T / h))
    for _ in range(n):
        y += h * f(t, y); t += h
    return y

def rk4(f, y0, T, h):
    y, t, n = y0, 0.0, int(round(T / h))
    for _ in range(n):
        k1 = f(t, y); k2 = f(t + h/2, y + h*k1/2)
        k3 = f(t + h/2, y + h*k2/2); k4 = f(t + h, y + h*k3)
        y += h * (k1 + 2*k2 + 2*k3 + k4) / 6; t += h
    return y

logi = lambda t, y: y * (1 - y)
ex = lambda t: 1 / (1 + np.exp(-t))
print("logistic y(2) exact %.10f" % ex(2.0))
for h in (0.4, 0.2, 0.1, 0.05, 0.025):
    ee = abs(euler(logi, 0.5, 2, h) - ex(2))
    er = abs(rk4(logi, 0.5, 2, h) - ex(2))
    print("h=%.3f euler %.3e rk4 %.3e" % (h, ee, er))

g, L = 9.80665, 1.0
T0 = 2 * np.pi * np.sqrt(L / g)
th0 = np.radians(60.0)
k = np.sin(th0 / 2)
Tq = 4 * np.sqrt(L / (2*g)) * integrate.quad(
        lambda th: 1/np.sqrt(np.cos(th) - np.cos(th0)), 0, th0, limit=200)[0]
Te = 4 * np.sqrt(L / g) * special.ellipk(k**2)
print("T0 %.6f T60quad %.6f T60elliptic %.6f" % (T0, Tq, Te))
print("ratio %.6f percent %.4f" % (Te / T0, 100 * (Te / T0 - 1)))

stiff = lambda t, y: -50 * (y - np.cos(t)) - np.sin(t)
for h in (0.02, 0.039, 0.05):
    print("stiff euler h=%.3f y(3)=%.3e" % (h, euler(stiff, 1.0, 3, h)))
for h in (0.05, 0.1):
```

```
    print("stiff rk4 h=%.3f y(3)=%.3e" % (h, rk4(stiff, 1.0, 3, h)))
print("exact cos(3) %.10f" % np.cos(3))
```

The actual output, verbatim:

```
logistic y(2) exact 0.8807970780
h=0.400 euler 1.958e-02 rk4 1.245e-05
h=0.200 euler 9.423e-03 rk4 7.237e-07
h=0.100 euler 4.630e-03 rk4 4.359e-08
h=0.050 euler 2.296e-03 rk4 2.674e-09
h=0.025 euler 1.143e-03 rk4 1.656e-10
T0 2.006409 T60quad 2.153242 T60elliptic 2.153242
ratio 1.073182 percent 7.3182
stiff euler h=0.020 y(3)=-9.902e-01
stiff euler h=0.039 y(3)=-9.908e-01
stiff euler h=0.050 y(3)=-1.839e+07
stiff rk4 h=0.050 y(3)=-9.883e-01
stiff rk4 h=0.100 y(3)=-1.317e+31
exact cos(3) -0.9899924966
```

Three things deserve reading. First, the Euler error ratios 2.08, 2.04, 2.02, 2.01 and RK4 heading to $16 = 2^4$: first and fourth order are visible as numbers. Second, the two roads to the pendulum period meet to six digits, even though quadrature and the elliptic integral do not use the same formula. Third, the stiff lines: $h = 0.039$ is still reasonable, $h = 0.05$ is meaningless, and not one of those lines warns us.

The program does not check the hypotheses of the theorems: whether f satisfies the Lipschitz condition, whether the equilibrium point is hyperbolic. That remains our job.

Note for instructors

The teaching order I use: start from the pendulum clock and the 7.32 percent difference, then the definition of a solution along with the blow-up example $1/(1-t)$, then separation of variables and the integrating factor. The integrating factor is best proved in full at the board, because its proof uses only the product rule and the fundamental theorem of calculus, two tools they have had since Chapter 13. Picard–Lindelof is enough as a sketch with one iteration carried out; what must remain is the meaning of the Lipschitz condition, not the fixed point technique. Phase plane before eigenvalues: draw first, compute later.

Two misconceptions almost always appear. First, students take "solution" to mean a closed formula, then conclude that an equation without a formula has no solution; the example $y' = \sqrt{|y|}$ helps, because there two solutions exist and no single formula does. Second, they treat a small error as proof of correctness, and are surprised when Euler with $h = 0.05$ gives -1.8×10^7. Show them that output line; do not merely warn them about it.

One blackboard activity: draw the pendulum phase plane empty, then ask the students to mark where the velocity is zero, where the acceleration is zero, and where the energy is highest. The class will argue about the region between the two saddles, and that argument is exactly the content of the theorem. A sign of readiness to go further: students can guess the type of an equilibrium point from the trace and the determinant without computing the roots.

16.9 Summary

Summary.

- A differential equation sets the rate of change equal to the state; its solution is a function on an interval, and that interval is part of the answer.

- Two families are solvable by hand: separation of variables and the first-order linear equation with integrating factor $\mu = \exp(\int p\,dt)$.
- Picard–Lindelöf guarantees a unique solution on a short interval if f is continuous and satisfies the Lipschitz condition with respect to y; without that condition, one initial state can give birth to two histories.
- The phase plane replaces time with a picture: equilibrium points, closed trajectories, separatrices, and open trajectories are read without any formula for the solution.
- Local stability is read from the eigenvalues of the Jacobian, except when an eigenvalue lies on the imaginary axis; there linearization does not decide, and a Lyapunov function is used.
- Euler is first order globally and RK4 is fourth order; both are explicit and therefore fail on stiff systems, where stability, not accuracy, sets the size of the step.

16.10 Exercises

Exercise 16.1.

1. Solve $y' = 2ty$, $y(0) = 1$, then compute $y(1)$ and compare it with e. Also compute Euler with $h = 0.1$ and state its error. (My result: 2.33463336, error 3.84×10^{-1}.)
2. Prove that $y' + p(t)y = q(t)$ with p, q continuous has at most one solution, without an explicit formula: the difference of two solutions satisfies $w' = -p(t)w$ and $w(t_0) = 0$.
3. True, false, or not enough information: "because linearization at the origin gives eigenvalues $\pm i$, the origin is a stable center". Give one supporting example or one counterexample.
4. Prove that $E = v^2/2 + \omega_0^2\theta^2/2$ is constant for the harmonic oscillator, then explain why its trajectories are ellipses and its period is $2\pi/\omega_0$.
5. An RC circuit with $R = 2.2\,\mathrm{k\Omega}$, $C = 10\,\mu\mathrm{F}$, $V_s = 12$ volts starts from zero. Compute τ, the voltage at $t = \tau$, and the time to reach 6 volts. (22 milliseconds; 7.585447 volts; 15.249 milliseconds.)
6. Show that $v = \pm 2\cos(\theta/2)$ satisfies $E = 2$, then explain why that curve joins $(-\pi, 0)$ to $(\pi, 0)$ and is called a separatrix.
7. A pendulum with $L = 0.25$ meters is released from $30°$. Compute the small period and the exact period, then state the percentage difference. (1.003205 seconds; 1.020669 seconds; 1.74 percent.)
8. For $y' = -100y$ with $y(0) = 1$, determine the threshold h that keeps explicit Euler stable, then explain what happens exactly at that threshold. Check it with the program.
9. Prove that the scalar autonomous equation $y' = f(y)$ with f continuous and satisfying the Lipschitz condition cannot have a nonconstant periodic solution.

Hints for the three hardest items. Item 3: linearization sees only terms of degree one; add a symmetric cubic term as in the example $r' = r^3$ and compare it with $r' = -r^3$. Item 6: use $\cos\theta = 2\cos^2(\theta/2) - 1$ and check the value of v at $\theta = \pm\pi$. Item 9: suppose y is periodic and nonconstant; at a maximum $y' = 0$, so $f(y_{\max}) = 0$, and uniqueness forces that trajectory to be constant.

For researchers: limits and next steps

What is genuinely proved in this chapter, with written proofs: the integrating factor formula together with its uniqueness, the complete Picard–Lindelof sketch with the error bounds of the iteration, uniqueness of solutions of the linear equation through $w' = -pw$, the Lyapunov test for both of its statements, and all the energy identities for the

oscillator and the pendulum. What I cite without proof: the Hartman–Grobman theorem (Hartman, *Ordinary Differential Equations*, chapter 9; Perko, *Differential Equations and Dynamical Systems*, section 2.8), Peano's theorem and Arzelà–Ascoli, and the LaSalle invariance principle.

What is overpopularized is the phrase "the eigenvalues decide everything". What is true: eigenvalues determine local behavior only when none of them has real part zero; on the imaginary axis, degree-three terms can turn a center into a spiral, as in the example $r' = r^3$ here. For high-dimensional systems, local "asymptotic stability" does not even mean safe from large disturbances, because the basin of attraction can be very small.

An open question: how far can qualitative properties be certified numerically? A small error on a single trajectory says nothing about stability; Euler with $h = 0.039$ is right to four digits, whereas $h = 0.05$ gives 10^7. For a system known only through samples of its field, there is no general way to make sure that no stiff region hides between two sample points.

Limits of validity: all functions here are real and smooth enough; all numbers are from Python 3.14.4, NumPy 2.5.2, and SciPy 1.18.0, with quadrature at tolerance 10^{-13} unless stated otherwise; $g = 9.80665\ \mathrm{m/s^2}$ and friction is neglected in the period computations.

Map to source texts: Chapter 16 in Boyce and DiPrima, Elementary Differential Equations and Boundary Value Problems*, chapters 1 (introduction), 2 (first-order equations), 3 (second-order linear equations), 4 (first-order systems and the phase plane), and 9 (nonlinear equations and stability); in Strogatz,* Nonlinear Dynamics and Chaos*, chapters 2 (flows on the line), 5 (two-dimensional systems), 6 (phase planes), 7 (oscillators), and 8 (bifurcations); in Arnold,* Ordinary Differential Equations*, chapters 1–4 for existence, uniqueness, and the classification of equilibrium points; in Coddington and Levinson, chapters 1–2 for Picard–Lindelof and Peano; in Hartman, chapter 9, and Perko, section 2.8, for Hartman–Grobman; in Hairer and Wanner,* Solving Ordinary Differential Equations II*, chapter IV, for stiffness and implicit methods. The basis for the computations is the chapter* Derivatives: One and Several Variables *(partial derivatives and the Jacobian), the chapter* Series of Functions and Exchange of Limits *(uniform limits, which make Picard iteration converge), and the chapter* Multiple Integrals *(vector fields) in the first edition of this book; the eigenvalues of Chapter 9 are used directly without being repeated, and the floating-point error analysis of Chapter 22 is the continuation of the numerical part of this chapter.*

Part IV

Probability, Statistics, and Computer Arithmetic

This part prepares two things that look unrelated: how to measure uncertainty and how a computer stores numbers. The prerequisites are Chapters 2, 5, and 10. After six chapters you can compute probabilities, estimate parameters, test claims, fit a regression, and explain why $0.1 + 0.2$ does not equal 0.3.

Chapter 17

Probability and Conditional Probability

A clinic advertises a screening test for a rare disease: only one person in a thousand in that region has it. The test is good. Out of every hundred sick people, ninety-nine come back positive. Out of every hundred healthy people, only one is wrongly accused of being positive. The advertisement writes both numbers as "99 percent accuracy".

Now the positive result is in our hands. What is the probability that we are really sick? In a population of 100,000 people, about 100 have the disease and 99 of them are positive. The rest, 99,900 healthy people, contribute 999 false positives. So there are 1.098 positive results and only 99 that are right: $99/1.098 = 11/122$, that is 0.0902. About one in eleven, not ninety-nine in a hundred.

Both numbers are right and both answer different questions. The advertisement gives the probability of a positive result if the person is sick. The patient wants to know the probability that the person is sick if the result is positive. Swapping those two directions is not an arithmetic mistake; it is a mistake in naming the quantity, and the consequence is often a factor of ten.

Chapter 2 already gave us a language for events: union, intersection, complement. Chapter 5 gave us a way to sum infinitely many terms. Neither is enough, because a set carries no weight: $A \cup B$ is the same set whether A is almost certain or almost certainly not. What we need is three things: a weight function with rules that must not be broken, a way to count weights, and a rule for updating weights when new information arrives. This chapter builds the three, then checks them with a seeded simulation.

For younger readers (age 10 and up)

Imagine the guard at a school gate checking bags with a device that beeps. The device is good: out of a hundred bags that really carry a forbidden object, ninety-nine make it beep. But the device also beeps for one out of a hundred bags whose contents are perfectly ordinary. Now suppose that at that school only one bag in a thousand really carries a forbidden object.

The question is simple and surprising. If the device beeps, is that bag certainly dangerous? It turns out not. Dangerous bags are indeed rare, while ordinary bags are very many, so the false beeps from ordinary bags far outnumber the true ones. A guard who looks only at the red light will hold back many innocent children.

The question that makes us curious: what fraction of all those beeps are right? The answer can be computed without guessing, as long as we know how many bags there are of each kind. In this chapter we shall learn why a device that is "ninety-nine percent

accurate” can turn into a false accusation in nine cases out of ten, and why the direction of the question determines the answer.

17.1 Probability axioms and their consequences

Level: basic

Before a single number is computed, we have to agree on what object is being weighted. In everyday language: probability is the weight we place on each possibility, under the rule that the total weight of all possibilities is one and no weight is ever negative. That rule does not explain the philosophical meaning of probability; it fixes the arithmetic that every computation has to obey.

probability (*peluang*) — The weight placed on each possibility, with the total weight of all possibilities equal to one

Definition 17.1 (Probability space)**.** Let Ω be the set of outcomes, $\mathcal{F}$ a collection of events, that is, the subsets of Ω that contain Ω and are closed under complement and countable union, and $P\colon \mathcal{F} \to \mathbb{R}$. The function P is called a *probability* if

1. $P(A) \geq 0$ for every $A \in \mathcal{F}$;
2. $P(\Omega) = 1$;
3. if $A_1, A_2, \ldots$ are pairwise disjoint, then $P\big(\bigcup_{i \geq 1} A_i\big) = \sum_{i \geq 1} P(A_i)$.

The third axiom, countable additivity, separates modern probability theory from the gambling computations of the seventeenth century. Without it, a probability series such as $\sum_{n \geq 1} 2^{-n} = 1$ has no foundation, even though that series is exactly what appears when we ask on which toss the first success occurs. This axiom is also what allows probability to be used in limits, the subject of Chapter 4 and Chapter 5.

Three immediate consequences are used over and over, so they are worth proving once.

Proposition 17.1 (Consequences of the axioms)**.** *For events A and B in a probability space we have $P(A^c) = 1 - P(A)$, $P(\varnothing) = 0$, and if $A \subseteq B$ then $P(A) \leq P(B)$. Moreover $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.*

Proof. Since A and A^c are disjoint and their union is Ω, the second and third axioms give $1 = P(\Omega) = P(A) + P(A^c)$. For the second claim, take $A = \varnothing$ in the first result. For the third, write B as the disjoint union of A and $B \setminus A$; then $P(B) = P(A) + P(B \setminus A) \geq P(A)$ because probability is not negative. For the last, split $A \cup B$ into three disjoint parts $A \setminus B$, $A \cap B$, and $B \setminus A$. In the same way A and B each split into two parts, so that

$$P(A \cup B) = P(A \setminus B) + P(A \cap B) + P(B \setminus A) = P(A) + P(B) - P(A \cap B). \qquad \square$$

That last property, inclusion–exclusion, is the reason “or” may not simply be added. For example, in a class of 30 students, 18 like coffee and 12 like tea, and 8 like both. Then the probability that a randomly chosen student likes coffee or tea is

$$\frac{18 + 12 - 8}{30} = \frac{22}{30} = 0.7333,$$

not $30/30 = 1$ as the naive sum $18 + 12 = 30$ would give. Eight people are counted twice, and inclusion–exclusion subtracts them once.

For three events the formula extends with alternating signs:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

The proof applies the two-event version twice, and that is the first exercise of this chapter. The pattern of signs is not decoration: whatever was subtracted too often has to be added back.

One warning for Chapter 18. In a finite space where every outcome has positive probability, $P(A) = 0$ means that A is empty; in a continuous space it does not, since a single point can have probability zero without being impossible. This chapter works in finite spaces.

Example 17.1. Two dice are thrown. What is the probability that the sum of the faces is 8 or that both faces are equal? With 36 equally likely outcomes, the first event contains 5 outcomes, the second contains 6, and their intersection is one outcome, namely $(4, 4)$. So the probability is $(5 + 6 - 1)/36 = 10/36 = 0.2778$. Without subtracting the intersection the answer is $11/36 = 0.3056$, too large by $1/36$.

17.2 Counting and equally likely outcomes

Level: intermediate

If all outcomes are equally likely, the probability of an event is just the ratio of the number of favorable outcomes to the number of all outcomes. In everyday language: divide the number of ways we want by the number of ways there are. That definition is not a law of nature; it is an assumption about symmetry, and the assumption has to be checked, not presupposed.

combination (*kombinasi*) — A choice of several objects without regard to their order, like taking five cards from one deck

Proposition 17.2 (Product rule and counting)**.** *If $|A| = m$ and $|B| = n$, then $|A \times B| = mn$. The number of sequences of length k from n distinct objects without repetition is $n!/(n-k)!$, and the number of subsets of size k is*

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$

In particular $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

Proof. For the product, every pair (a, b) is determined by one choice from m and one choice from n. For sequences there are n choices for the first place, $n - 1$ for the second, and so on up to $n - k + 1$. For subsets, every set of size k can be ordered in $k!$ ways, so the number of sequences divided by $k!$ gives the number of sets. The last identity counts all subsets in two ways: summing by size, or choosing in or out for every object. □

Neat counting often shortens a long computation. A full house in poker, three cards of one rank plus a pair of another rank, is a good example: choose the rank of the triple (13 ways), choose three of its four suits ($\binom{4}{3} = 4$), choose the rank of the pair (12 ways), choose two of its four suits ($\binom{4}{2} = 6$). The total is $13 \cdot 4 \cdot 12 \cdot 6 = 3.744$ out of $\binom{52}{5} = 2{,}598{,}960$ hands, so the probability is 0.001441, that is one hand in 694.2. In 694 hands the expected number of full houses is 0.9998, so we wait about seven hundred hands to see the first one.

Counting also explains events that feel miraculous. Take the birthday problem: k people in one room, every birthday equally likely among 365 days, and birthdays independent between people. The probability that no two people share a birthday is

$$P(\text{no pair}) = \frac{365 \cdot 364 \cdots (365 - k + 1)}{365^k},$$

so the probability that there is a pair is its complement.

k	P(no pair)	P(a pair)
10	0.8831	0.1169
23	0.4927	0.5073
41	0.0968	0.9032
50	0.0296	0.9704
57	0.0099	0.9901

At $k = 23$ the probability passes one half, and at $k = 50$ it is already 0.9704. The source of the confusion is the wrong unit: what collides is not people but pairs of people, and 23 people form $\binom{23}{2} = 253$ pairs. Every pair matches with probability $1/365$, so the expected number of pairs is $253/365 = 0.69315$. If collisions are treated as rare and independent, the probability of no collision approaches $e^{-0.69315} = 0.5000$, differing by 0.0073 from the exact value 0.4927. That approximation is not a proof; it explains why the threshold falls near twenty-three.

The counting above assumes every birthday equally likely. In real populations that inequality actually raises the probability of a collision; the symmetry assumption here is an approximation that has to be checked.

Counting cannot answer the clinical question from before. The outcomes "sick" and "healthy" are not equally likely: their ratio is one to nine hundred ninety-nine. What is needed is not a way to count outcomes but a way to use the information that someone has already been declared positive. That is conditional probability.

17.3 Conditional probability and the product rule

Level: intermediate

New information narrows the world. If we know that B occurred, the relevant world is no longer Ω but B, and all weights have to be read again inside that narrower world. In everyday language: conditional probability is the probability recomputed after part of the possibilities has been crossed out.

conditional probability (*peluang bersyarat*) — The probability recomputed after we know that part of the possibilities no longer applies

Definition 17.2 (Conditional probability)**.** If A and B are events with $P(B) > 0$, the probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Dividing by $P(B)$ is the choice that makes conditional probabilities sum to one over B. If we also demand $P(B \mid B) = 1$ and that ratios between events inside B do not change, that division is the only possible choice; hence this definition is not an arbitrary convention.

The narrowing can be seen directly in a cross table. Figure 17.1 arranges the clinical population into four cells. Before any result, the denominator is all 100,000 people and $P(S) = 100/100{,}000 = 0.001$. After a positive result is known, only the + column, which holds 1.098 people, is still counted, and $P(S \mid +) = 99/1.098 = 0.0902$. The person does not change; only the set that may still be counted changes.

Proposition 17.3 (Conditional probability is a probability)**.** *For fixed B with $P(B) > 0$, the function $A \mapsto P(A \mid B)$ satisfies the three axioms of probability on $\mathcal{F}$.*

Proof. Nonnegativity comes from $P(A \cap B) \geq 0$. Normality comes from $P(\Omega \mid B) = 1$, because $P(B)/P(B) = 1$. For additivity, if $A_1, A_2, \ldots$ are pairwise disjoint then $A_i \cap B$ are also pairwise disjoint, so that

$$P\Big(\bigcup_i A_i \mid B\Big) = \frac{P\big(\bigcup_i (A_i \cap B)\big)}{P(B)} = \frac{\sum_i P(A_i \cap B)}{P(B)} = \sum_i P(A_i \mid B). \qquad \square$$

test result	+	−	total
sick S	99	1	100
healthy S^c	999	98,901	99,900
total	1.098	98,902	100,000

once + is known, only this column is counted:
99/1.098 = 11/122 = 0.0902

Figure 17.1: Cross table of a screening test on 100,000 people. Rows are the true state, columns are the test result. Once a positive result is known, the sample space narrows from the whole table to the + column, which holds 1.098 people, and 99 of them are really sick.

Hence all consequences of the axioms hold for conditional probability: complement, monotonicity, and inclusion–exclusion remain valid when every P is replaced by $P(\cdot \mid B)$. What does not hold is swapping the direction of the condition.

From the definition the product rule is born at once

$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A),$$

which extends to $P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$. This rule reads a staged experiment as a product of probabilities at each stage.

A single deck of 52 cards gives an example of a direction that is easy to swap. Let A be the event that the first card is a heart and B the event that the second card is red. Then

$$P(A \cap B) = \frac{13 \cdot 25}{52 \cdot 51} = \frac{25}{204},$$

so that

$$P(A \mid B) = \frac{25/204}{1/2} = \frac{25}{102} = 0.2451, \qquad P(B \mid A) = \frac{25/204}{1/4} = \frac{25}{51} = 0.4902.$$

Two numbers that differ by a factor of two, from the same data. The mistake is not in the arithmetic but in the placement of the events to the left and right of the vertical bar.

Read it aloud (auditory learner)

Two voices. The first does not know yet, the second already knows. Read them in turn.
First voice: “My world is as wide as a playing field. Every door is possible. I guess with my eyes closed.”
Second voice: “My world is one room. One door is already open, and I count only what is inside it.”
First voice: “If I know something, does my probability change?”
Second voice: “It is not your probability that changes. It is the world you count that changes. The world shrinks after we learn something: weight that was spread out now gathers in the places that are still possible. The denominator narrows, and the number that comes out sounds new even though the arithmetic has not changed since the beginning.”
First voice: “So it is the denominator that moves?”
Second voice: “The denominator moves, and what is left to be recounted is the remainder that may still take part.”

Repeat together, three times, more and more slowly: "The world shrinks, the probability does not betray us. Only what may still be counted changes."

The second tool splits the population into several groups that do not overlap. If $B_1, \dots, B_k$ are pairwise disjoint and their union is Ω, then

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)\, P(B_i).$$

Proposition 17.4 (Law of total probability)**.** *If $B_1, \dots, B_k$ are pairwise disjoint, their union is Ω, and every $P(B_i) > 0$, then $P(A) = \sum_i P(A \mid B_i) P(B_i)$ for every event A.*

Proof. The events $A \cap B_1, \dots, A \cap B_k$ are pairwise disjoint and their union is A. So $P(A) = \sum_i P(A \cap B_i)$, and every term is rewritten with the product rule as $P(A \mid B_i)P(B_i)$. □

This law looks like a formality until it is used on the clinical example: it is what turns two numbers from an advertisement into one useful number, and that is the content of the next section.

Before that, one example showing that ordinary sentences and mathematical events do not always correspond. A family has two children, each child a boy or a girl with equal probability, and the sexes of the two children are independent. If we know that *at least one* child is a boy, the probability that both are boys is $1/3 = 0.3333$: of the four equally likely arrangements three remain, and only one fits. If what we know is that *the older child* is a boy, the probability is $1/2 = 0.5$. Two similar sentences give different answers, because the first excludes one arrangement and the second two.

17.4 Bayes' rule and the interpretation of test results

Level: intermediate

Bayes' rule reverses the direction of the condition: from the probability of a result given sickness it computes the probability of sickness given the result. In everyday language: of all the ways a result can appear, what fraction comes through the path we are asking about.

Theorem 17.5 (Bayes' rule)**.** *If $B_1, \dots, B_k$ are pairwise disjoint with union Ω and $P(B_i) > 0$, then for every event A with $P(A) > 0$,*

$$P(B_j \mid A) = \frac{P(A \mid B_j)\, P(B_j)}{\sum_{i=1}^{k} P(A \mid B_i)\, P(B_i)}.$$

Proof. By definition, $P(B_j \mid A) = P(A \cap B_j)/P(A)$. The numerator is $P(A \mid B_j)P(B_j)$ by the product rule, and the denominator is $P(A)$ by the law of total probability. So the formula contains no new assumption; it is the definition of conditional probability written twice, in two different directions. □

Hence there is no "Bayesian method" standing on top of the axioms; what there is, is bookkeeping. The numerator is the path we are asking about, the denominator is the sum of all paths that produce the same result.

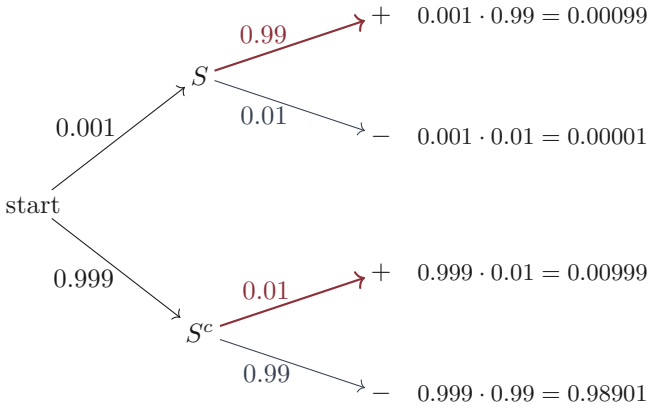
Step-by-step recipe (procedural learner)

1. First name two things: the hypothesis H we are asking about and the evidence E that has already been seen. Write the question as $P(H \mid E)$, not $P(E \mid H)$.

2. Write the numerator as a single path: $P(E \mid H)\,P(H)$. That is the weight of the world in which H is true.
3. Write the denominator as the sum of all paths that produce E, including the path through H^c: $P(E \mid H)P(H) + P(E \mid H^c)P(H^c)$.
4. Divide, then check for sense: the result lies between 0 and 1, and is smaller than $P(E \mid H)$ when H is rare.

A warning about order: numerator and denominator both use $P(E \mid \cdot)$. Writing $P(\cdot \mid E)$ in the denominator is the mistake that happens most often, and the size of the number does not tell us that the direction has been swapped.

The tree in Figure 17.2 writes out every road to a positive result. Each leaf is the product of the probabilities on its branch, and the two leaves marked + sum to $P(+) = 0.01098$. The Bayes numerator is just the first leaf, while the denominator is the sum of those two leaves.



$$P(+) = 0.00099 + 0.00999 = 0.01098,$$
$$P(S \mid +) = 0.00099/0.01098 = 0.0902$$

Figure 17.2: Two-stage tree of a screening test. The first branch separates sick from healthy, the second separates positive results from negative ones; the product at each leaf is the probability of that path, and the two positive leaves sum to $P(+) = 0.01098$.

Back to the clinic. Write S for the event of being sick and + for a positive result. From the advertisement and the prevalence of the disease we have $P(S) = 0.001$, $P(+ \mid S) = 0.99$, and $P(+ \mid S^c) = 0.01$. The law of total probability gives

$$P(+) = 0.99 \cdot 0.001 + 0.01 \cdot 0.999 = 0.00099 + 0.00999 = 0.01098,$$

so that

$$P(S \mid +) = \frac{0.00099}{0.01098} = \frac{11}{122} = 0.0902.$$

The false positive term, 0.00999, is ten times larger than the true detection term, 0.00099; their ratio is $999/99 = 10.09$. Not because the test is bad: healthy people are 999 times more numerous, 99,900 against 100.

A change of prevalence changes the answer far more strongly than a change in the quality of the test. With sensitivity and specificity fixed at 0.99, the following table shows $P(S \mid +)$ for several prevalences.

$P(S)$	$P(S \mid +)$	$1/P(S \mid +)$	reading
0.0001	0.0098	102.0	one in a hundred and two
0.001	0.0902	11.09	one in eleven
0.010	0.5000	2.00	one in two
0.100	0.9167	1.09	eleven in twelve

At a prevalence of 1 percent the same test gives $P(S \mid +) = 1/2$ exactly, not 0.99. At a prevalence of 10 percent, $P(S \mid +) = 11/12 = 0.9167$. These numbers explain a health policy that is often criticized: mass screening for a rare disease produces many false positives, and for that reason a positive result is usually followed by a second test rather than by treatment.

The second test uses the same rule, with the prior already updated. If the second test is independent of the first and of the same quality, then

$$P(S \mid +,+) = \frac{0.99 \cdot (11/122)}{0.99 \cdot (11/122) + 0.01 \cdot (111/122)} = \frac{10.89}{10.89 + 1.11} = \frac{363}{400} = 0.9075.$$

Note that 0.9075 is neither 0.99 nor 0.9999. Two positive tests raise the probability from 0.0902 to 0.9075, almost ten times, but they do not yet give certainty. What changes is only the prevalence inside the group that is already positive; the quality of the test is unchanged. With this quality of 0.99, the posterior exceeds 0.99 only when the initial prevalence is already past 1/2.

One technical point: $P(S \mid +) = 0.0902$ applies to a population with prevalence 1/1000, not to a patient who arrives with symptoms. Symptoms change the prevalence first, and that is why physicians ask about history before interpreting a laboratory result.

The last example of this section releases Bayes from the clinical test. Three bags hold ten marbles each: bag A one red, B five red, C nine red. We choose a bag at random, draw one marble, and the one that comes out is red. Before the draw the three bags each have probability 1/3; afterwards they have 0.0667, 0.3333, and 0.6, as in Figure 17.3. No weight is lost; weight merely moves to the bag that more easily produces that evidence.

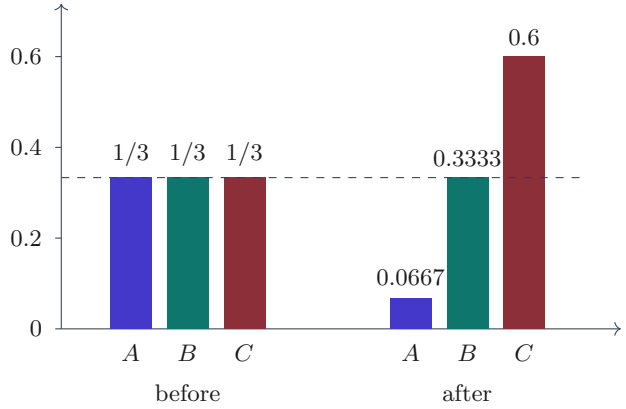


Figure 17.3: Bayes update on three bags. Before any evidence the three bags share the weight 1/3; after one red marble is seen, the weights move to 0.0667, 0.3333, and 0.6. The bag that most easily produces that evidence rises the highest.

17.5 Independence, and why disjoint is not independent

Level: advanced

Two events are called disjoint if they cannot occur together. Two events are called independent if knowing one does not change the probability of the other. The two sentences sound similar, and swapping them is the most expensive mistake in this chapter. In everyday language: disjoint means never together; independent means no news from one to the other.

independent (*independen*) — Events that send each other no news: knowing one does not change the probability of the other

For younger readers (age 10 and up)

There are two things that are often swapped, and the difference can be seen without numbers. At one school, two students are never in class at the same time: one studies in the morning, one studies in the afternoon. If we know that Andi has come in, we know that Budi has not. They are never together, and precisely for that reason news about one gives news about the other.

Now two other students. Both attend the same class, but they sit far apart and work on different exams. Their grades send each other no news: knowing Sari's grade does not help us guess Tono's.

The first group is called never together. The second group is called no news to each other. Both sound like "no relation", even though the first is in fact the strongest relation: news about one determines the other. In this chapter we shall learn why two things that are never together cannot possibly send each other news, and why swapping the two makes the computation miss by more than a little.

Definition 17.3 (Independence of two events)**.** Events A and B are *independent* if

$$P(A \cap B) = P(A)\,P(B).$$

This definition is symmetric, and when $P(B) > 0$ it is equivalent to $P(A \mid B) = P(A)$. Note that the definition does not mention cause and effect: two events can be independent without one causing the other, and can be dependent without one causing the other.

Proposition 17.6 (Disjoint is not independent)**.** *If A and B are disjoint and $P(A) > 0$ and $P(B) > 0$, then A and B are not independent. More precisely $P(A \cap B) = 0 < P(A)P(B)$, and $P(A \mid B) = 0 < P(A)$.*

Proof. Disjoint means $A \cap B = \varnothing$, so $P(A \cap B) = P(\varnothing) = 0$. Since $P(A)P(B) > 0$, the equality in the definition of independence fails. For the conditional statement, $P(A \mid B) = 0/P(B) = 0$, which is smaller than $P(A) > 0$. □

The mistake moves in two directions, and both directions can be computed.

Take one die and two pairs of events with the same marginals but a different relation. First pair: $A = \{1, 2\}$ and $B = \{3, 4\}$, disjoint, with $P(A) = P(B) = 1/3$. Second pair: $C = \{1, 2, 3\}$ and $D = \{1, 4\}$, with $P(C) = 1/2$ and $P(D) = 1/3$. For the first pair,

$$P(A)P(B) = \frac{1}{9} = 0.1111, \qquad P(A \cap B) = 0, \qquad P(A \mid B) = 0,$$

so the gap in the definition of independence is 0.1111 and the gap in conditional probability is 0.3333, because $P(A) = 1/3$. For the second pair,

$$P(C)P(D) = \frac{1}{6} = 0.1667 = P(C \cap D), \qquad P(C \mid D) = \frac{1}{2} = P(C),$$

so both gaps are zero.

What matters is not the number $1/6$ but the fact that two pairs can have the same marginals with a completely opposite relation. The pair $E = \{1, 2, 3\}$ and $F = \{4, 6\}$ also has marginals $1/2$ and $1/3$ but is disjoint. Marginals do not determine the relation; only the intersection determines it, so every claim of independence is an additional claim about the world.

For three events, pairwise independence is not enough. The events A, B, C are called mutually independent if $P(A \cap B) = P(A)P(B)$, $P(A \cap C) = P(A)P(C)$, $P(B \cap C) = P(B)P(C)$, and $P(A \cap B \cap C) = P(A)P(B)P(C)$.

Example 17.2 (Pairwise but not mutually independent). Two coins are tossed. Let A be heads on the first coin, B heads on the second coin, and C both coins the same; each has probability $1/2$. Every pair intersects in one outcome, so $P(A \cap B) = P(A \cap C) = P(B \cap C) = 1/4$, which equals $1/2 \cdot 1/2$. But $A \cap B \cap C$ occurs only when both are heads, so its probability is $1/4$, whereas $P(A)P(B)P(C) = 1/8$. So the three are pairwise independent and not mutually independent; the gap is 0.125.

A final warning: independence does not survive conditioning. If A and B are independent with probability $1/2$, then given $A \cup B$, $P(A \mid A \cup B) = 2/3$ while $P(A \cap B \mid A \cup B) = 1/3 \neq 4/9$. Combining two independent events and then conditioning on their union actually creates dependence, so the sentence "A and B are independent" is always relative to the probability space in use.

Do it by hand (kinesthetic learner)

Materials: twelve cards, six pairs of equal rank (ace through six), each pair one red and one black.
Step one, make the pairs. Lay the twelve cards face up in two rows; put each pair of equal rank side by side, red on the left, black on the right. Check with your eyes: rank sends no news about color, since every rank has a red and a black. Those two properties send each other no news.
Step two, filter. Shuffle, cover them all, then turn over one card: say the four of hearts. Give one hint, "this card's mate is black", then set aside every red card that is still face down. Of the eleven face-down cards, only six are black.
What you must see: the denominator changes from eleven to six, while the cards do not change at all. The probability of guessing its mate rises from one in eleven to one in six. Two properties that sent each other no news now send each other news, and that is what one hint does.

17.6 Two paradoxes that demand a computation

Level: advanced

Paradoxes in probability are almost always a collision between intuition and definition, and the collision is settled only by computing. The two examples below show different things: the answer depends on the rules of the game, and on the unit that is counted.

Monty Hall. Three closed doors; one holds a car, two hold goats. We choose one door. The host, who knows what is behind all the doors, opens one other door with a goat, then offers the chance to switch to the door that is still closed. Does switching raise the probability?

Compute with the law of total probability over the three possible locations of the car, each with probability $1/3$, and suppose that we first choose door 1. If the car is behind door 1, switching loses. If the car is behind door 2, the host must open door 3, so switching wins. If the car is behind door 3, he opens door 2, and switching wins. So switching wins in two of the three cases:

$$P(\text{win by switching}) = \frac{2}{3}, \qquad P(\text{win by staying}) = \frac{1}{3}.$$

The gap is $1/3$, not zero.

The argument "two doors left, so $1/2$" is wrong because it ignores that the host does not choose a door freely. He is forbidden to open the door with the car and forbidden to open the door we chose, so his action carries information about the door he does not open.

That information disappears when the rule changes. Suppose the host opens one of the other two doors at random, without looking at what is behind it, and the game is cancelled

if the car is revealed. Given that the game runs as far as the offer,

$$P(\text{car behind door 1} \mid \text{goat revealed}) = \frac{1/3}{1/3 + 1/6 + 1/6} = \frac{1}{2},$$

because if the car is behind door 2 or 3, he has only probability $1/2$ of showing a goat. Under that rule switching does not help: "switching has probability $2/3$" is not a property of the doors but a property of the protocol.

Birthdays. The previous section already computed 0.5073 for 23 people and 0.9704 for 50 people. The paradox is not in the numbers but in the unit: we compare it with the probability that one particular person matches us, whereas what is computed is 253 pairs at once. The same mistake appears in hash tables and in DNA match tests.

The prosecutor's fallacy is the same directional mistake in a criminal case. Suppose the probability that a feature appears by coincidence is one in a million, and the feature is found in a suspect taken from a database of a million people with no other evidence. Then $P(\text{match} \mid \text{innocent}) = 10^{-6}$, but

$$P(\text{guilty} \mid \text{match}) = \frac{10^{-6} \cdot 1}{10^{-6} \cdot 1 + (1 - 10^{-6}) \cdot 10^{-6}} = 0.5000003,$$

that is practically one half, not near certainty. The two numbers 10^{-6} are different quantities: the first is the probability of the feature by coincidence, the second is the prevalence of the suspect inside the database. Swapping them is another face of the same mistake, this time in a courtroom.

Challenge (problem solver)

Three cards on a table. The first card is red on both sides, the second is black on both sides, the third is red on one side and black on the other. We shuffle the three face down, take one, and lay it on the table. The visible side is red. What is the probability that the side underneath is also red?
Almost everyone answers one half: they say only two cards have a red side, so there are two possibilities. The correct answer is not one half.
Hint 1: do not count cards, count sides. How many red sides are there in total, and how many of them sit next to a red side?
Hint 2: the visible red side can come from three different red sides, and two of them are on the red-red card.
If you are stuck, go back to the section on conditional probability and rewrite the sample space as six sides, not three cards.

17.7 Worked computations

Level: intermediate

The three computations below are carried out with exact fractions, so that no rounding hides a step. All of them have been checked with Python, and the numbers that appear are the same numbers used in the previous sections.

First, two-step Bayes. With $P(S) = 1/1000$, $P(+ \mid S) = 99/100$, and $P(+ \mid S^c) = 1/100$, the law of total probability gives

$$P(+) = \frac{99}{100} \cdot \frac{1}{1000} + \frac{1}{100} \cdot \frac{999}{1000} = \frac{99 + 999}{100{,}000} = \frac{1098}{100{,}000}.$$

The numerator is two separate groups: 99 sick people who are detected and 999 healthy people who are wrongly accused. Hence

$$P(S \mid +) = \frac{99/100{,}000}{1098/100{,}000} = \frac{99}{1098} = \frac{11}{122} = 0.090164.$$

A second test of the same quality uses 11/122 as the new prevalence and gives 0.9075, just as in the computation of the previous section. The denominator is 1200, not a number in the millions: the factor 1/122 appears in both terms and cancels, a sign that what changed is only the prevalence, not the quality of the test.

Second, disjoint versus independent, with matched marginals.

pair	P(first)	P(second)	P(intersection)	P(first | second)
$C = \{1,2,3\}$, $D = \{1,4\}$	0.5	0.3333	0.1667	0.5
$E = \{1,2,3\}$, $F = \{4,6\}$	0.5	0.3333	0	0

In the first row $P(C)P(D) = 1/6 = 0.1667$ equals the intersection, so the gap is 0. In the second row $P(E)P(F) = 1/6$ while the intersection is 0, so the gap is 0.1667; the conditional probability is 0 while $P(E) = 0.5$. Anyone who takes "cannot happen together" to mean "does not influence each other" will report 0.1667 where 0 belongs.

Third, Monty Hall with the protocol stated. If the host always opens a goat door, the probability of winning by switching is $2/3 = 0.6667$ and by staying $1/3 = 0.3333$. If he opens at random and the game is cancelled when the car is revealed, the game runs with probability 2/3, and given that, both strategies are worth $1/2 = 0.5$. Two protocols, one set of doors, two answers.

For a test with prevalence 1/2000, sensitivity 0.95, and specificity 0.90, compute $P(S \mid +)$ with the 100,000 table as above, then compute it again with specificity 0.99. Write both answers as fractions and as decimals with four digits, and explain which part changes the most: the numerator or the denominator. After that compute $P(S \mid +, +)$ for a second test of the same quality, and check whether the result is still below 0.95.

17.8 Computational lab: Monte Carlo as a verification tool

Level: advanced

All the numbers in this chapter are derived, not measured. Simulation does not replace a derivation, but it catches the mistakes that happen most often: a swapped direction of the condition, a wrong denominator, and an independence assumption used without grounds. Generate the same experiment many times, compute the ratio of successes, then compare it with the exact value. By the law of large numbers (Chapter 19), that ratio tends to the true probability; what remains is only how fast, measured by the standard error $\sqrt{p(1-p)/m}$ with m the number of trials that produce the event.

A fixed seed is used so that the output can be checked again word for word. A fixed seed does not make the result more correct; it only makes it repeatable.

Computational lab

```
import numpy as np

rng = np.random.default_rng(20261024)

print("== 1. diagnostic test: prevalence 0.1%, sens 99%, spec 99% ==")
n = 1_000_000
sick = rng.random(n) < 0.001
positive = np.where(sick, rng.random(n) < 0.99, rng.random(n) < 0.01)
```

```
m = int(positive.sum())
ppv = sick[positive].mean()
err = np.sqrt(ppv * (1 - ppv) / m)
exact = 0.99 * 0.001 / (0.99 * 0.001 + 0.01 * 0.999)
print("sick=%d  positive=%d  true_positive=%d"
      % (sick.sum(), m, int((sick & positive).sum())))
print("PPV empirical=%.6f  exact=%.6f  std error=%.6f  z=%.3f"
      % (ppv, exact, err, (ppv - exact) / err))

print("== 2. Monty Hall, 10^5 games ==")
N = 100_000
car = rng.integers(0, 3, N)
pick = rng.integers(0, 3, N)
opened = np.where(pick == car,
                  (pick + 1 + rng.integers(0, 2, N)) % 3,
                  3 - pick - car)
switch = 3 - pick - opened
win_stay = float((pick == car).mean())
win_switch = float((switch == car).mean())
se_switch = np.sqrt(win_switch * (1 - win_switch) / N)
print("win without switch=%.4f  win switch=%.4f  std error=%.5f"
      % (win_stay, win_switch, se_switch))
print("exact 1/3=%.6f  2/3=%.6f" % (1/3, 2/3))

print("== 3. host opens a door at random ==")
open_random = (pick + 1 + rng.integers(0, 2, N)) % 3
valid = open_random != car
switch_random = 3 - pick - open_random
print("valid games=%d of %d (%.4f)" % (valid.sum(), N, valid.mean()))
print("win switch (valid)=%.4f  without switch (valid)=%.4f"
      % (float((switch_random[valid] == car[valid]).mean()),
         float((pick[valid] == car[valid]).mean())))

print("== 4. birthday probability, 20,000 rooms ==")
for k in (23, 50):
    days = rng.integers(0, 365, size=(20000, k))
    days.sort(axis=1)
    dup = (np.diff(days, axis=1) == 0).any(axis=1)
    q = 1.0
    for i in range(k):
        q *= (365 - i) / 365
    f = float(dup.mean())
    se = np.sqrt(f * (1 - f) / 20000)
    print("k=%2d  empirical=%.4f  exact=%.4f  std error=%.5f  z=%+.2f"
          % (k, f, 1 - q, se, (f - (1 - q)) / se))

print("== 5. disjoint versus independent, 10^6 die throws ==")
N = 1_000_000
die = rng.integers(1, 7, N)
A = np.isin(die, [1, 2])
B = np.isin(die, [3, 4])
C = np.isin(die, [1, 2, 3])
D = np.isin(die, [1, 4])
print("disjoint:  P(A)=%.4f P(B)=%.4f P(AnB)=%.4f P(A|B)=%.4f gap=%.4f"
      % (A.mean(), B.mean(), (A & B).mean(), (A & B).sum() / B.sum(),
         A.mean() * B.mean() - (A & B).sum() / B.sum()))
print("independent:  P(C)=%.4f P(D)=%.4f P(CnD)=%.4f P(C|D)=%.4f gap=%.4f"
      % (C.mean(), D.mean(), (C & D).mean(), (C & D).sum() / D.sum(),
         abs(C.mean() - (C & D).sum() / D.sum())))
print("exact disjoint: 1/3, 1/3, 0, 0, 1/9=%.6f" % (1 / 9))
print("exact independent: 1/2, 1/3, 1/6, 1/2, 0")
```

The actual output, exactly as it came out:

```
== 1. diagnostic test: prevalence 0.1%, sens 99%, spec 99% ==
sick=963  positive=11094  true_positive=952
PPV empirical=0.085812  exact=0.090164  std error=0.002659  z=-1.637
== 2. Monty Hall, 10^5 games ==
win without switch=0.3325  win switch=0.6675  std error=0.00149
exact 1/3=0.333333  2/3=0.666667
== 3. host opens a door at random ==
valid games=66472 of 100000 (0.6647)
win switch (valid)=0.4998  without switch (valid)=0.5002
== 4. birthday probability, 20,000 rooms ==
k=23  empirical=0.5019  exact=0.5073  std error=0.00354  z=-1.54
k=50  empirical=0.9701  exact=0.9704  std error=0.00120  z=-0.19
== 5. disjoint versus independent, 10^6 die throws ==
disjoint:  P(A)=0.3333 P(B)=0.3335 P(AnB)=0.0000 P(A|B)=0.0000 gap=0.1112
independent:  P(C)=0.5001 P(D)=0.3334 P(CnD)=0.1667 P(C|D)=0.5000 gap=0.0001
exact disjoint: 1/3, 1/3, 0, 0, 1/9=0.111111
exact independent: 1/2, 1/3, 1/6, 1/2, 0
```

Four things deserve reading from that output, and two of them actually show the limits of simulation.

First, the first line of part 1 reports 963 sick people out of a million, whereas the expectation is 1,000 with standard deviation $\sqrt{10^6(0.001)(0.999)} = 31.6$. So 963 lies 1.17 standard deviations below the expectation: a normal swing, not a programming oversight. The expected number of positive results becomes $963 \cdot 0.99 + 999{,}037 \cdot 0.01 = 10{,}943.7$, and the observed 11,094 is 1.4 standard deviations away. The empirical conditional probability 0.0858 differs from the exact 0.0902 by 0.0044, that is 1.64 standard errors. That gap is far smaller than the distance from 0.09 to 1, so the interpretation does not change.

Second, Monty Hall gives 0.6675 and 0.3325 with standard error 0.0015, while the expected gap between the strategies is 1/3. Agreement to three digits is not luck; the most useful form of verification is indeed not confirming the third digit but confirming that the gap we claim exists and has the right sign.

Third, parts 3 and 5 show that the protocol and the relations between events enter the numbers. A host who opens doors at random lets only 66.472 of 100,000 games run as far as the offer, and among those switching wins 0.4998 while staying wins 0.5002; that gap of 0.0004 is a quarter of a standard error. The disjoint pair gives $P(A)P(B) - P(A \cap B) = 0.1112$, matching 1/9 to three digits, while the independent pair gives 0.0001, that is zero within random swing. The number 0.1112 is the size of the mistake made by anyone who thinks two disjoint events must be independent.

What the program does not prove: not one of the five parts proves a formula. The program only checks formulas that have already been proved, with a sample far larger than the reach of the hand; Chapter 19 explains why that is legitimate.

Note for instructors

The teaching order I suggest: start from the 100,000 table, not from the axioms. Have the students fill in four cells: sick and positive, sick and negative, healthy and positive, healthy and negative. The numbers 99, 1, 999, and 98,901 appear on their own, and from there 99/1.098 becomes the conclusion of the class, not the result of a lecture. After that the axioms, their consequences, counting, conditional probability, Bayes, then independence. Monty Hall is best placed last, after the students can write the host's protocol as a conditional event.

Two misconceptions appear most often. First, "disjoint means independent". The symptom shows up in the arithmetic: students add the probabilities of two events that

should be multiplied, or the other way round, and then cannot explain why the result exceeds one. Second, "sensitivity equals the probability of being sick given a positive result". The symptom is the answer 0.99 for the clinical problem, without using the prevalence already given in the problem.
A blackboard activity that works: two students play the host with different written protocols, one always opening a goat door and one opening at random; the class records twenty games and then compares them with 2/3 and 1/2. Students are ready to go further when they can give one example of two independent events and one example of disjoint events with the same marginals, and compute a conditional probability from a two-way table without reversing the direction.

17.9 Summary

Summary.

- Probability is a function on events that is nonnegative, takes the value one on the whole space, and is countably additive on disjoint events; complement, monotonicity, and inclusion–exclusion follow from that.
- When outcomes are equally likely, probability is a ratio of counts, with the product rule and $\binom{n}{k}$; the assumption of equal likelihood has to be checked.
- Conditional probability $P(A \mid B) = P(A \cap B)/P(B)$ recomputes inside B and satisfies the three axioms; the product rule and the law of total probability are its consequences.
- Bayes' rule is not a new assumption but a definition read in two directions. Prevalence determines the answer more strongly than the quality of the test: with sensitivity and specificity 0.99, a prevalence of 0.1 percent gives $P(S \mid +) = 0.0902$.
- Disjoint and independent are different; when both probabilities are positive, disjoint events cannot be independent, and only the intersection determines the relation.
- The answer to a paradox depends on the protocol and the unit of counting: Monty Hall 2/3 under a fixed protocol and 1/2 under a random one; birthdays 0.5073 for 23 people because what is counted is 253 pairs. A seeded simulation checks, it does not prove.

17.10 Exercises

Exercise 17.1.

1. Prove the inclusion–exclusion formula for three events by applying the two-event version to $A \cup B$ and C. Show at which step the term $P(A \cap B \cap C)$ appears, and why its sign is positive.
2. Assess this claim: "Because $P(A \mid B) = 0$, A and B are disjoint." True, false, or not enough evidence? Give an example with $P(A \mid B) = 0$ but $A \cap B \neq \varnothing$, or explain why that is impossible.
3. A test has prevalence 2 percent, sensitivity 0.95, specificity 0.90. Compute $P(S \mid +)$ from a table of 10.000 people, then repeat with specificity 0.99; state both as fractions.
4. Prove that if A and B are independent, then A and B^c are also independent. Use $P(A) = P(A \cap B) + P(A \cap B^c)$ and explain why the same result holds for A^c and B^c.
5. Compute the probability that a pair with the same birthday exists for 30 people, compare it with $1 - e^{-\binom{30}{2}/365}$, then explain the source of the gap.
6. Prove the law of total probability for $k = 2$ directly from the axioms, without using the law for general k: write $A = (A \cap B) \cup (A \cap B^c)$ and check that the two parts are disjoint.
7. Four doors, one car, three goats. A host who knows the contents opens two goat doors among the three other doors and offers a switch to the one door that remains. Compute

the probability of winning by switching, then check it with a simulation of 10^4 games.

8. Assess this claim: "Two independent tests with sensitivity and specificity 0.99 give certainty 0.9999 when both are positive, as long as the prevalence is 0.1 percent." Compute the correct number and state the assumptions.
9. Is pairwise independence for three events enough to conclude $P(A \cap B \cap C) = P(A)P(B)P(C)$? Refute it with an example from two coin tosses, and compute both values.

Hints for the three hardest items. Item 2: in a finite space where every outcome has positive probability, $P(A \mid B) = 0$ forces $A \cap B = \varnothing$, so for a die throw the claim is true; a counterexample has to come from a continuous space, for instance $A = \{1/2\}$ and $B = (0, 1)$ with uniform probability, where $P(A) = 0$ even though A contains a point. Item 7: at first the car is behind the chosen door with probability 1/4; if not, the only door the host does not open must hold the car, so the probability of switching is 3/4. Item 9: take A heads on the first coin, B heads on the second coin, and C both coins the same; all three pairs give 1/4 while the intersection is 1/4, not 1/8.

For researchers: limits and next steps

What is genuinely proved in this chapter: the three axioms together with their consequences; Bayes' rule as a theorem derived from the definition of conditional probability; the impossibility of two disjoint events with positive probability also being independent; and the dependence of the Monty Hall answer on the protocol, an analysis stated sharply by Morgan, Chaganty, Dahiya, and Doviak, *The American Statistician* 1991, in the article "Let's Make a Deal: The Player's Dilemma". What is popular but not proved here is the claim that the natural frequency format improves prevalence reasoning in humans; that is an empirical claim about readers, not a theorem, and its evidence lies in the psychology of decision making literature.

Primary sources: Blitzstein and Hwang, *Introduction to Probability*, Chapter 1 (Why probability, Naive definition of probability, How to count, Story proofs, Non-naive definition of probability) and Chapter 2 (The importance of thinking conditionally, Definition and intuition, Bayes' rule and the law of total probability, Conditional probabilities are probabilities, Independence of events, Coherency of Bayes' rule); as well as Wasserman, *All of Statistics*, Chapter 1 in the section Probability for the formulation of the axioms and Chapter 2 in the section Independent Events.

Limits of validity: the whole chapter works in finite spaces where every outcome has positive probability, where $P(A) = 0$ is equivalent to $A = \varnothing$. In uncountable spaces, conditioning on an event of probability zero is not defined by the ratio formula and requires a regular conditional distribution; the Borel paradox shows that the answer can depend on how the limit is taken. Independence is not stable either: it can disappear or appear under conditioning, and in graphical models it appears as selection bias.

An open question: how far must the prevalence be known before Bayes' rule is used honestly, and what can be said when the prevalence is known only from a small sample. The formal answer is Bayesian inference with a prior, which brings in the problem of choosing the prior; the frequentist answer is the hypothesis test of Chapter 20, which does not use a prior but answers a different question. This chapter does not choose either.

Map to source texts: Chapter 17 in Blitzstein and Hwang, Introduction to Probability*: Chapter 1 (Probability and counting, including How to count and the birthday problem example) for the axioms, counting, and combinatorial probability; Chapter 2 (Conditional probability, Bayes' rule and the law of total probability, Independence of events) for conditional probability, Bayes, and independence. In Wasserman,* All of Statistics*: Chapter 1 in the section Probability and Chapter 2 in the section*

Independent Events. Sets and complements are used from Chapter 2 of this book; the series underlying countable additivity from Chapter 5; the law of large numbers that supports the simulation is proved in Chapter 18. Random variables in Chapter 17 move the weight from events to numbers, while the hypothesis tests in Chapter 19 use Bayes' rule and the law of total probability for a different question.

Chapter 18

Random Variables and Distributions

A messaging service receives an average of three requests per second at peak hours. That number is printed on the dashboard, and it does not answer the pressing question: how often does a single second hold eight requests or more, so that the queue overflows? Two systems can have exactly the same average and behave very differently: one almost never overflows, the other several times a day. The average alone is not enough, and not because the counting is insufficiently careful.

Chapter 17 gave us a language for events: the probability of one outcome, the probability of a union, conditional probability. That language stops working as soon as what we measure is a number. "The number of requests in one second" is not one event but a family of infinitely many events, one for each possible value. For counts, the probability of each value can still be added up one by one. For a quantity that is continuous, that way collapses: a waiting time can be 0.317 seconds or 0.3170001 seconds, and there is no way to state "the probability of exactly 0.317 seconds" without assigning weight zero to every value.

So two things are needed: a way to move probability onto the number line, and a way to replace summation by integration when the values are continuous. Once both exist, the average, the spread, and the shape of the tail can be computed.

For younger readers (age 10 and up)

Imagine a bakery. Every hour, the seller writes down how many loaves were sold: sometimes 3, sometimes 11, sometimes none at all. Those records are whole numbers, and cannot take the value one half. If we want to know the probability of selling exactly 7 loaves, we can compute it: count how many hours sold 7 loaves, divide by the number of hours recorded. All the probabilities can be added one by one, because the options are few and can be listed.

Now weigh one loaf on a digital scale. The result is not a whole number: 250.3 grams, or 249.8 grams. However many decimal places we take, there is always another possibility closer still. Hence the question "what is the probability that the weight is exactly 250.0000 grams" is useless, whereas "what is the probability that the weight lies between 249 and 251 grams" is very useful. Probability here belongs not to a single point but to an interval.

In this chapter we will learn why those two ways differ, and why both can still give a sensible average.

18.1 From Outcomes to Numbers

Level: basic

A measuring instrument does not record events; it records numbers. The bridge is a function that takes each outcome of the experiment and assigns one real number. The name "random variable" is misleading: it is not a variable, and it is not random. The randomness comes from the probability on the outcome space, whereas the function is ordinary.

Definition 18.1 (Random variable)**.** Let Ω be the outcome space and $\mathcal{F}$ a collection of events. The mapping $X : \Omega \to \mathbb{R}$ is called a *random variable* if for every $x \in \mathbb{R}$ the set

$$\{X \le x\} = \{\omega \in \Omega : X(\omega) \le x\}$$

is an event, that is, a member of $\mathcal{F}$.

That last condition guarantees that $P(X \le x)$ has a meaning. For almost every function that appears in practice it holds automatically, and it fails only for set indicators that are not measurable.

As an example, toss three fair coins and let X be the number of heads that appear. Its values are $0, 1, 2, 3$ with probabilities $1/8, 3/8, 3/8, 1/8$. Note that X maps eight outcomes onto four numbers, so several outcomes share the same value. That is why the probability of a value is the probability of a union of events.

We need one tool that answers "how far have the values moved already" for all thresholds at once, because the threshold is chosen by the reader, not by the author of the book.

Definition 18.2 (Cumulative distribution function)**.** The cumulative distribution function of a random variable X is $F_X(x) = P(X \le x)$ for $x \in \mathbb{R}$.

Proposition 18.1 (Properties of the distribution function)**.** *For every random variable X the following hold:*

1. *F_X is nondecreasing: if $a \le b$ then $F_X(a) \le F_X(b)$;*
2. *$\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to +\infty} F_X(x) = 1$;*
3. *F_X is right-continuous, that is, $F_X(x) = F_X(x^+)$ for every x;*
4. *$P(a < X \le b) = F_X(b) - F_X(a)$ for every $a < b$.*

Proof. For (1), the event $\{X \le a\}$ is contained in $\{X \le b\}$ when $a \le b$, and probability is monotone with respect to inclusion.

For (2), let $A_n = \{X \le -n\}$. These events are decreasing, and $\bigcap_n A_n = \varnothing$ because every outcome yields a finite value. Continuity of probability from above, which follows from countable additivity, gives $F_X(-n) \to 0$. The opposite direction uses $B_n = \{X \le n\}$, which increase to Ω, so that $F_X(n) \to 1$.

For (3), take a decreasing sequence $y_n \downarrow x$ and $C_n = \{X \le y_n\}$. Then $\bigcap_n C_n = \{X \le x\}$, hence $F_X(y_n) \to F_X(x)$ by continuity from above. Since the sequence is arbitrary, the right-hand limit equals the value at x.

For (4), the event $\{X \le b\}$ splits into $\{X \le a\}$ and $\{a < X \le b\}$, so that $F_X(b) = F_X(a) + P(a < X \le b)$. □

For two tosses of a fair coin with X the number of heads,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1/4, & 0 \le x < 1, \\ 3/4, & 1 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

This function is defined on the whole number line, even though X takes only the values $0, 1, 2$. That is why $F_X(1.4) = 3/4$: the question is not "is X equal to 1.4", but "is X at most 1.4".

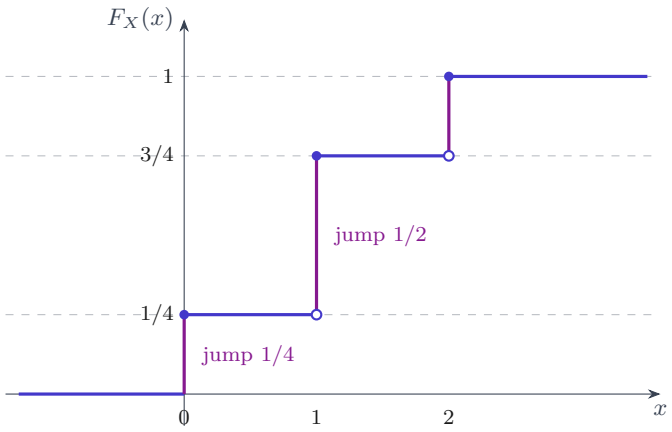


Figure 18.1: The cumulative distribution function of the number of heads in two tosses of a coin. The graph is a flat staircase between the possible values, then jumps exactly at those values. The jumps of $1/4$ at $x = 0$, $1/2$ at $x = 1$, and $1/4$ at $x = 2$ are the probability masses $p_X(0)$, $p_X(1)$, and $p_X(2)$. The filled dot at the left end of every step marks that F_X is right-continuous, so that $F_X(1) = 3/4$ and not $1/4$.

Every jump of the graph is the probability mass at a single value. Figure 18.1 draws that staircase: three horizontal steps, three vertical jumps, and the height of each jump equals the probability mass at the point of the jump.

random variable (*variabel acak*) — a function that attaches one number to each outcome of the experiment, like a clerk who records one number for every event

cumulative distribution function (*fungsi distribusi kumulatif*) — a rising staircase: its height at the point x states how much probability has already accumulated up to that threshold

18.2 Mass Functions and Density Functions

Level: basic

There are two ways of weighting the values of a random variable. For a variable whose values can be listed, the weight attaches to points; for one whose values fill an interval, the weight attaches to length. The most common mistake in this topic is to use the first way to read the outcome of the second.

Definition 18.3 (Discrete variables and probability mass function)**.** X is called *discrete* if its set of values is countable, say $\{x_1, x_2, \dots\}$. The *probability mass function* of X is $p_X(x) = P(X = x)$. This function satisfies $p_X(x) \geq 0$, $p_X(x) \leq 1$, and $\sum_i p_X(x_i) = 1$, as well as $F_X(x) = \sum_{x_i \leq x} p_X(x_i)$.

For X binomial with $n = 2$ and $p = 1/2$, the probability mass is $1/4, 1/2, 1/4$ at $x = 0, 1, 2$ and zero elsewhere. Probability mass is never greater than one, because it is a probability after all.

Definition 18.4 (Continuous variables and probability density function)**.** X is called *continuous* if there exists a function $f_X \geq 0$ such that

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

for every $a \leq b$, with $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$. The function f_X is called the *probability density function*.

Density is not probability, and the difference is not a matter of terminology. Take X with density $f_X(x) = 5$ on $[0, 1/5]$ and zero outside it. Then $f_X(0.1) = 5$, a number that is impossible for a probability, whereas $P(X \le 0.1) = \int_0^{0.1} 5\,dx = 0.5$. Density may also be unbounded: $f_X(x) = \frac{2}{3}x^{-1/3}$ on $(0, 1)$ integrates to one even though it blows up near zero. The units differ: probability per unit of length, just as population density is people per square kilometer. Figure 18.2 puts the two ways side by side: the bars on the left are added by height, the triangle on the right by area, and both total one.

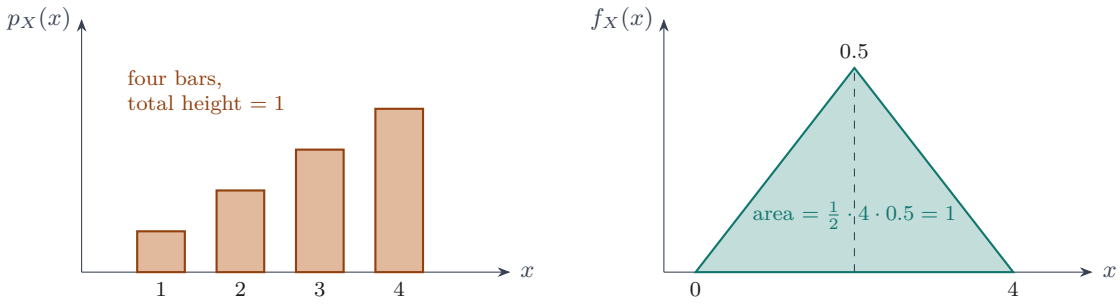


Figure 18.2: Two ways of weighting that both total one. Left: four bars of height 0.1, 0.2, 0.3, 0.4 at the values $1, 2, 3, 4$, added by height. Right: a triangle-shaped density with peak 0.5 on $[0, 4]$, added by area, and $\frac{1}{2} \cdot 4 \cdot 0.5 = 1$. The height 0.5 in the right-hand panel is not a probability of anything; what has probability is the area of the slice beneath it.

Read it aloud (auditory learner)

Read aloud in pairs, one person taking the Height voice and one the Weight voice. Start slowly, then speed up.
Height: I am the height of the curve. I tower at one point, and everyone points at me.
Weight: I am the weight under the curve. I pile up along the interval, and everyone weighs me. Height: My number may be five. May be ten. May be unbounded. Weight: My number never exceeds one, because I am the whole of probability. Height: Without me, you do not know where the most frequent value lies. Weight: Without me, you do not know whether that peak contains anything. Together: Height points to the place. Weight weighs the content.
Repeat together three times, more slowly each time: probability is weight, not height.

Theorem 18.2. *If X is continuous with density f_X, then $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$, and $f_X(x) = F_X'(x)$ at every point x where F_X is differentiable.*

Proof. The first equation is the definition of density with $a = -\infty$ and $b = x$. The second equation is the first part of the fundamental theorem of calculus: the derivative of the accumulation function equals f_X at every point where f_X is continuous. □

The question that most often causes confusion: if X is continuous then $P(X = x) = 0$ for every x, because $\int_x^x f_X = 0$. Does that mean that the event $X = x$ is impossible? No. The set $\{\omega : X(\omega) = x\}$ still exists and can still contain outcomes; what is zero is its probability, not its existence.

The difficulty appears when we try to add up all the points. The interval $[0, 1]$ is an uncountable union of points of probability zero, yet $P(0 \le X \le 1) = 1$. There is no contradiction: the third axiom of Chapter 17 guarantees additivity only for *countable* unions, and the integral measures length, it does not count points.

There is a practical layer. NumPy's built-in generator forms values as multiples of 2^{-53}, so there are only 2^{53} possible outputs on $[0, 1)$, and the probability that one particular value

appears in 10^7 draws is about $10^7/2^{53} = 1.1 \times 10^{-9}$. On a computer "probability zero" is really $2^{-53} \approx 1.1 \times 10^{-16}$; the zero we write is the idealization.

Example 18.1. Let X be uniform on $(0, 1)$, with $f_X(x) = 1$ for $0 < x < 1$. Then

$$P(0.49 < X < 0.51) = \int_{0.49}^{0.51} 1 \, dx = 0.02, \qquad P(X = 0.5) = 0.$$

The density around 0.5 is as large as the density around 0.01; what distinguishes them is only the length of the interval. That is why the right question for a continuous variable is always an interval.

Do it by hand (kinesthetic learner)

Materials: a tape measure, scissors, card stock, and ten blank cards.
Measure the heights of ten friends to the nearest centimeter, then write one height on each card. Make five boxes on the floor from strips of card stock five centimeters wide: 150–154, 155–159, up to 170–174. Put each card in the box that matches, stacking them upward. What we must see: those stacks have a shape, and the number of cards in all the boxes is still ten.
Cut the card stock again into boxes ten centimeters wide, arrange the cards again, and see what changes. The stacks become almost half as high, but not a single card is lost. The width of the box changes the height of the stack; the number of cards does not change. That is the difference between height and weight, and that is why a density must be multiplied by the width of the interval before it may be called a probability.

For younger readers (age 10 and up)

Imagine a football field full of people. We want to know where the people are most numerous. We cannot answer that by asking "how many people are exactly at this point". The answer is always zero, because a point has no area. What can be answered is "how many people are there per square meter". That number is called density, and it differs from place to place: dense at the entrance, sparse in the corner of the field.
High density does not mean that there are many people at one point there. High density means that around that place, within a small region, people are packed together. If we multiply the density by the area of the region, only then does the number of people come out. The same holds for probability: density times the length of the interval gives probability, and density itself is not probability.
This is why two curves of very different shape can give the same total, namely one. What matters is not the height of the curve at one point, but the area beneath it. In this chapter we will learn why a question about points must be replaced by a question about intervals.

18.3 Expectation and Variance

Level: intermediate

If the whole distribution has to be compressed into a single number, the number chosen is its balance point: the location of the center of mass when probability is treated as weight.

Definition 18.5 (Expectation)**.** The expectation of a random variable X is

$$\mathbb{E}[X] = \sum_i x_i \, p_X(x_i) \qquad \text{or} \qquad \mathbb{E}[X] = \int_{-\infty}^{\infty} x \, f_X(x) \, dx,$$

depending on whether X is discrete or continuous. The expectation exists only if $\mathbb{E}[|X|] < \infty$; otherwise the expectation of X is said not to exist.

That absolute-value condition is not excessive caution. For the Cauchy distribution with density $f(x) = 1/(\pi(1+x^2))$, $\int |x| f(x) dx$ diverges, so that $\mathbb{E}[X]$ does not exist even though the distribution is symmetric about zero; the sample mean does not concentrate anywhere, however much data we add. Chapter 19 uses this example as a boundary of the law of large numbers.

As short computations, for X Bernoulli with $P(X=1) = p$, $\mathbb{E}[X] = 0 \cdot (1-p) + 1 \cdot p = p$. For X uniform on $(0,1)$, $\mathbb{E}[X] = \int_0^1 x\,dx = 1/2$.

What is often needed is not the expectation of X but the expectation of a function of X, for example $\mathbb{E}[X^2]$. The direct route is to find the density of X^2 first, then integrate. There is a much shorter route.

Theorem 18.3 (Expectation of a function of a random variable)**.** *For a function r and a random variable X,*

$$\mathbb{E}[r(X)] = \sum_i r(x_i)\, p_X(x_i) \qquad \text{or} \qquad \mathbb{E}[r(X)] = \int_{-\infty}^{\infty} r(x)\, f_X(x)\, dx.$$

Proof. For the discrete case, group the values according to $y = r(x)$:

$$\sum_i r(x_i) p_X(x_i) = \sum_y y \sum_{i:\, r(x_i)=y} p_X(x_i) = \sum_y y\, P(r(X) = y),$$

and the last expression is the definition of $\mathbb{E}[r(X)]$ because $r(X)$ is discrete with probability mass $P(r(X)=y)$. For the continuous case, the proof is the change-of-variables formula for integrals in Chapter 15; the same grouping applies with integrals replacing sums. □

Proposition 18.4 (Linearity of expectation)**.** *For random variables X, Y and constants a, b, $\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$.*

Proof. For the discrete case with joint mass $p(x,y)$,

$$\mathbb{E}[aX+bY] = \sum_{x,y} (ax+by) p(x,y) = a \sum_x x \sum_y p(x,y) + b \sum_y y \sum_x p(x,y) = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

The continuous case is identical with integrals replacing sums. □

What is not used here is independence: linearity holds for any variables, including strongly dependent ones. Hence it is safe for sums of indicators, but it may not be used for products: $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ requires independence.

Expectation gives the center, but it does not give the width; the most widely used measure of width is the average squared distance to the center.

Definition 18.6 (Variance)**.** The variance of X with $\mu = \mathbb{E}[X]$ is $\operatorname{Var}(X) = \mathbb{E}[(X-\mu)^2]$, and its standard deviation is $\sigma_X = \sqrt{\operatorname{Var}(X)}$.

Proposition 18.5. *If the variance exists, then $\operatorname{Var}(X) = \mathbb{E}[X^2] - \mu^2$ and $\operatorname{Var}(aX+b) = a^2 \operatorname{Var}(X)$.*

Proof. Expand the square and use linearity:

$$\mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - 2\mu\,\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2.$$

For the second property, $\mathbb{E}[aX+b] = a\mu + b$, so that $\operatorname{Var}(aX+b) = \mathbb{E}[(aX+b-a\mu-b)^2] = a^2 \mathbb{E}[(X-\mu)^2]$. □

For X Bernoulli, $\mathbb{E}[X^2] = p$ because $X^2 = X$ for the values 0 and 1, so that $\mathrm{Var}(X) = p - p^2 = p(1-p)$. For X uniform on $(0,1)$, $\mathbb{E}[X^2] = 1/3$, so that $\mathrm{Var}(X) = 1/3 - 1/4 = 1/12 \approx 0.083333$. Variance has squared units, whereas the standard deviation has the same units as X. One costly warning: $\mathbb{E}[X^2] \neq \mathbb{E}[X]^2$, and the difference is precisely the variance; for the uniform variable above, 0.25 against 0.333333.

Step-by-step recipe (procedural learner)

Materials: a table of values x_i and probabilities $p_X(x_i)$.

1. Add up all the $p_X(x_i)$. If the result is not one, the table is not yet valid.
2. Multiply each x_i by $p_X(x_i)$, then add: that is $\mathbb{E}[X]$.
3. Square each x_i first, multiply by $p_X(x_i)$, then add: that is $\mathbb{E}[X^2]$.
4. Subtract the square of the result of step 2 from the result of step 3, namely $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$; its square root is the standard deviation.

Example: for $x = 0, 1, 2, 3$ with $p = 0.1$; 0.2; 0.3; 0.4, step 2 gives 2, step 3 gives 5, and step 4 gives 1.
Warning about the order: square x_i before multiplying by the probability, and subtract μ^2 after $\mathbb{E}[X^2]$ has been fully added up. Swapping that order produces a number that measures no width at all.

expectation (*harapan*) — the balance point of a loaded ruler: one number that marks where the values pile up

variance (*variansi*) — a measure of how far typical values stray from the balance point, computed from squared distances so that plus and minus signs do not cancel each other

18.4 Four Discrete Distributions

Level: intermediate

Four families of discrete distributions suffice for most practical work: one binary trial, the number of binary trials, the waiting time until the first success, and rare events in a time interval.

Bernoulli. X Bernoulli with parameter $p \in [0,1]$ takes the value 1 with probability p and 0 with probability $1-p$, so that $p_X(x) = p^x(1-p)^{1-x}$ for $x \in \{0,1\}$. From those two points, $\mathbb{E}[X] = p$ and $\mathrm{Var}(X) = p(1-p)$; the variance is largest at $p = 1/2$, namely $1/4$, exactly when the two possibilities are balanced.

Binomial. If n independent Bernoulli trials with success probability p are run, the number of successes X is binomial with

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Its mean and variance are derived most easily by splitting X into indicators, not from that formula.

Theorem 18.6. *If X is binomial with parameters n and p, then $\mathbb{E}[X] = np$ and $\mathrm{Var}(X) = np(1-p)$.*

Proof. Write $X = \sum_{i=1}^{n} X_i$ with $X_i = 1$ if trial i succeeds and 0 if it fails. Each X_i is Bernoulli with parameter p, hence $\mathbb{E}[X_i] = p$ and $\mathrm{Var}(X_i) = p(1-p)$. Linearity gives $\mathbb{E}[X] = \sum_i \mathbb{E}[X_i] = np$ without using independence at all.

For the variance, independence is needed for the first time. Because the trials are independent, the pairs X_i, X_j with $i \neq j$ are independent too, so that $\mathrm{Cov}(X_i, X_j) = p^2 - p^2 = 0$. Hence

$$\mathrm{Var}(X) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j) = np(1-p).$$

□

The variance formula is more fragile than the mean formula: the former requires independence, the latter does not. For $n = 20$ and $p = 0.35$, $\mathbb{E}[X] = 7$ and $\mathrm{Var}(X) = 4.55 = 91/20$, with standard deviation 2.133073; the mass around the center is $p_X(7) = 0.184401$, $p_X(10) = 0.068614$, $p_X(12) = 0.013564$. The tail is small but not zero: $P(X \geq 12) = 0.019579$ and $P(X \leq 4) = 0.118197$, so the rule "reject a box if it contains twelve defects or more" rejects about two percent of boxes.

Geometric. If independent Bernoulli trials are repeated until the first success, the number of trials needed is X geometric with $p_X(k) = p(1-p)^{k-1}$ for $k = 1, 2, \ldots$ Its total mass is one because of the geometric series: $\sum_{k\geq 1} p(1-p)^{k-1} = p/(1-(1-p)) = 1$. With $q = 1-p$,

$$\sum_{k\geq 1} k q^{k-1} = \frac{d}{dq}\frac{1}{1-q} = \frac{1}{(1-q)^2} = \frac{1}{p^2},$$

so that $\mathbb{E}[X] = p \cdot p^{-2} = 1/p$ and, in a similar way, $\mathrm{Var}(X) = (1-p)/p^2$ from $\mathbb{E}[X^2] = (2-p)/p^2$. For $p = 0.2$: $\mathbb{E}[X] = 5$, $\mathrm{Var}(X) = 20$, $P(X = 5) = 0.081920$ and $P(X > 10) = 0.10737418$; a simulation of 10^5 draws gives mean 4.99176 with standard error 0.01414.

Poisson. For X Poisson with parameter $\lambda > 0$,

$$p_X(k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

Its total mass is one because of the exponential series: $e^{-\lambda}\sum_k \lambda^k/k! = 1$. Its mean and variance are equal, and both come out of a single trick. Because $k/k! = 1/(k-1)!$,

$$\mathbb{E}[X] = \sum_{k\geq 1} k\, e^{-\lambda}\frac{\lambda^k}{k!} = \lambda e^{-\lambda}\sum_{k\geq 1}\frac{\lambda^{k-1}}{(k-1)!} = \lambda.$$

Writing $k(k-1)/k! = 1/(k-2)!$ gives $\mathbb{E}[X(X-1)] = \lambda^2$, hence $\mathrm{Var}(X) = \lambda$. One parameter governs the center and the width at once, and that restricts the Poisson: data whose variance far exceeds their mean (called *overdispersion*) do not fit the Poisson model.

For $\lambda = 3$: $p_X(0) = 0.049787$, $p_X(3) = 0.224042$, $P(X \geq 8) = 0.011905$. That last number answers the queueing question from the opening of the chapter: with three requests per second, about 1.2 percent of seconds contain eight requests or more.

The Poisson is the limit of the binomial when n is large, p is small, and $np \to \lambda$. For $n = 1000$ and $p = 0.003$, the binomial gives $P(0) = 0.049563083$ whereas the Poisson gives $e^{-3} = 0.049787068$, a relative difference of -4.5×10^{-3}. The error is small but not zero and its sign changes with k; the Poisson is used as an approximation, not as an identity.

18.5 Three Continuous Distributions

Level: intermediate

Uniform. X is uniform on (a, b) if its density is $1/(b-a)$ inside that interval and zero outside. Its distribution function rises linearly from 0 to 1, and

$$\mathbb{E}[X] = \frac{a+b}{2}, \qquad \operatorname{Var}(X) = \frac{(b-a)^2}{12}.$$

The factor 12 comes from $\int_a^b (x - \frac{a+b}{2})^2 \frac{dx}{b-a} = (b-a)^2/12$; for $(0,1)$ the result is $1/12 \approx 0.083333$, the same as the computation through $\mathbb{E}[X^2]$ in the previous section.

Exponential. X exponential with parameter $\beta > 0$ has density $f_X(x) = \beta^{-1} e^{-x/\beta}$ for $x > 0$, with $F_X(x) = 1 - e^{-x/\beta}$, $\mathbb{E}[X] = \beta$, and $\operatorname{Var}(X) = \beta^2$. The parameter β is the mean waiting time. The property that distinguishes it from every other continuous distribution is the absence of memory.

Proposition 18.7 (Memoryless)**.** *For $s, t \geq 0$, $P(X > s + t \mid X > s) = P(X > t)$.*

Proof. $P(X > u) = e^{-u/\beta}$, so that

$$P(X > s+t \mid X > s) = \frac{P(X > s+t)}{P(X > s)} = \frac{e^{-(s+t)/\beta}}{e^{-s/\beta}} = e^{-t/\beta} = P(X > t).$$

□

A component that has already lasted five years is no more likely to last one more year than a new component: sensible for a process without aging, absurd for a lamp that wears out. For $\beta = 2$, $P(X > 3) = 0.22313016$ and a simulation of 10^5 draws gives 0.22237; conditional on $X > 2$, the empirical probability of $X > 5$ is 0.22028 against the theoretical 0.22313. The exponential is the continuous twin of the geometric: if arrivals follow a Poisson process with rate λ, the gaps between arrivals are exponential with $\beta = 1/\lambda$.

Normal. X normal with parameters μ and $\sigma^2 > 0$ has density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} \right), \qquad x \in \mathbb{R}.$$

The constant in front is not a choice: it is the only value that makes the area under the curve equal to one. That the area is one is proved with polar coordinates in Chapter 15: write I^2 as the double integral of $e^{-(x^2+y^2)/2}$, change to polar, and obtain $I^2 = 2\pi$.

Proposition 18.8. *If $X \sim N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma \sim N(0, 1)$.*

Proof. $F_Z(z) = P(X \leq \mu + \sigma z) = F_X(\mu + \sigma z)$. Differentiate with respect to z by the chain rule: $f_Z(z) = \sigma f_X(\mu + \sigma z) = \varphi(z)$, the standard normal density. □

The function $\Phi(z) = P(Z \leq z)$ has no elementary form; it is computed numerically or read from a table. Three numbers worth memorizing: $P(|Z| \leq 1) = 0.682689492137$, $P(|Z| \leq 2) = 0.954499736104$, $P(|Z| \leq 3) = 0.997300203937$. For a probability that is not symmetric, subtract two values of Φ: $P(1 < Z < 2) = 0.135905122$. Its quantile, $\Phi^{-1}(0.975) = 1.959964$, appears in every 95 percent confidence interval in Chapter 20.

The normal is not a more correct distribution than the others. It matters for two different reasons: many natural quantities approach it, and the sum of many independent variables tends to approach it whatever their original shape. The second reason is the central limit theorem, proved in Chapter 19.

normal distribution (*distribusi normal*) — the symmetric bell curve: almost all values pile up near the center, and the tails thin out very quickly

exponential distribution (*distribusi eksponensial*) — waiting time in a process without aging: one that has waited a long time is no closer to the next event

18.6 Joint Distributions, Covariance, and Correlation

Level: advanced

Two separate distributions do not store information about whether the two quantities tend to rise together. The tool for that is the joint distribution.

Example 18.2. Let $X, Y \in \{0, 1\}$ with joint mass $P(0,0) = 0.1$, $P(0,1) = 0.2$, $P(1,0) = 0.3$, $P(1,1) = 0.4$. Adding up rows and columns gives the marginals $P(X = 1) = 0.7$ and $P(Y = 1) = 0.6$. If X and Y were independent, the mass at $(1,1)$ would have to be 0.42, whereas the actual value is 0.4; so the two are not independent, even though the difference is small.

In the continuous case, the joint distribution is described by the joint density $f_{X,Y} \geq 0$ with $\iint f_{X,Y} = 1$, and the marginals are obtained by integrating one variable out: $f_X(x) = \int f_{X,Y}(x, y)\, dy$. Independence means $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ for all x, y.

Definition 18.7 (Covariance and correlation)**.** With $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$,

$$\operatorname{Cov}(X, Y) = \mathbb{E}\big[(X - \mu_X)(Y - \mu_Y)\big], \qquad \rho(X, Y) = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Positive covariance means the two variables tend to deviate in the same direction; negative means in opposite directions. Correlation is the unitless version of covariance.

Proposition 18.9. $\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ *and* $-1 \leq \rho(X, Y) \leq 1$.

Proof. Expand the product and use linearity. For the bound on correlation, consider the inner product $\langle U, V \rangle = \mathbb{E}[UV]$ on centered variables; the Cauchy-Schwarz inequality of Chapter 10 gives $|\langle U, V \rangle|^2 \leq \langle U, U \rangle \langle V, V \rangle$, namely $\operatorname{Cov}(X, Y)^2 \leq \sigma_X^2 \sigma_Y^2$. □

Theorem 18.10. $\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y)$.

Proof. Write $X + Y - \mu_X - \mu_Y = (X - \mu_X) + (Y - \mu_Y)$, square it, and then use linearity on the three terms that appear; the cross term is exactly twice the covariance. □

Back to the table above: $\mathbb{E}[X] = 7/10$, $\mathbb{E}[Y] = 3/5$, $\mathbb{E}[XY] = 2/5$, so that $\operatorname{Cov}(X, Y) = -1/50 = -0.02$. With $\operatorname{Var}(X) = 21/100$ and $\operatorname{Var}(Y) = 6/25$, the correlation is -0.089087 and $\operatorname{Var}(X + Y) = 0.41$.

Zero covariance is often mistaken for independence, whereas the implication runs only one way: independence implies zero covariance, not the reverse. Take X taking the values $-1, 0, 1$ with equal probability and $Y = X^2$; then $\mathbb{E}[X] = 0$ and $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0$, so that the correlation is zero, even though Y is determined completely by X. Covariance captures only dependence in the form of a straight line.

covariance (*kovarians*) — a measure of whether two quantities tend to rise and fall together, computed from the product of their deviations

18.7 Transformations of Random Variables

Level: advanced

This work comes up again and again: prices converted to logarithms, lengths to areas, waiting times to rates. The question is always the same, namely how the distribution changes when $Y = r(X)$.

In the discrete case the answer is simple: mass is moved from the old values to the new values. Let $P(X = -1) = 1/4$, $P(X = 0) = 1/2$, $P(X = 1) = 1/4$, and $Y = X^2$. Then $P(Y = 0) = 1/2$ and $P(Y = 1) = P(X = -1) + P(X = 1) = 1/2$; Y has only two values, fewer than X, because $r(x) = x^2$ is not one-to-one.

In the continuous case the mass is zero at every point, so what is moved is length, and length changes scale. Three standard steps answer everything: determine $A_y = \{x : r(x) \le y\}$, compute $F_Y(y) = \int_{A_y} f_X(x)\,dx$, then differentiate $f_Y = F_Y'$.

Theorem 18.11 (Monotone transformation)**.** *If r is strictly monotone and differentiable, with inverse $s = r^{-1}$, then*

$$f_Y(y) = f_X(s(y)) \left| \frac{ds(y)}{dy} \right|.$$

Proof. Suppose r is strictly increasing. Then $r(x) \le y$ is equivalent to $x \le s(y)$, so that $F_Y(y) = F_X(s(y))$, and the chain rule gives $f_Y(y) = f_X(s(y))s'(y)$ with $s'(y) > 0$. If r is strictly decreasing, the inequality reverses: $F_Y(y) = 1 - F_X(s(y))$, so that $f_Y(y) = -f_X(s(y))s'(y) = f_X(s(y))|s'(y)|$. □

The factor $|ds/dy|$ is local stretching: if r stretches the scale, the density decreases, because the same probability is spread over a longer interval. In more than one dimension that factor becomes the Jacobian determinant, whose general form is in Chapter 15. Figure 18.3 shows that stretching for $Y = X^2$: four intervals of equal length land on intervals that grow longer as y approaches one, while the density of Y rises near zero, where the intervals shorten.

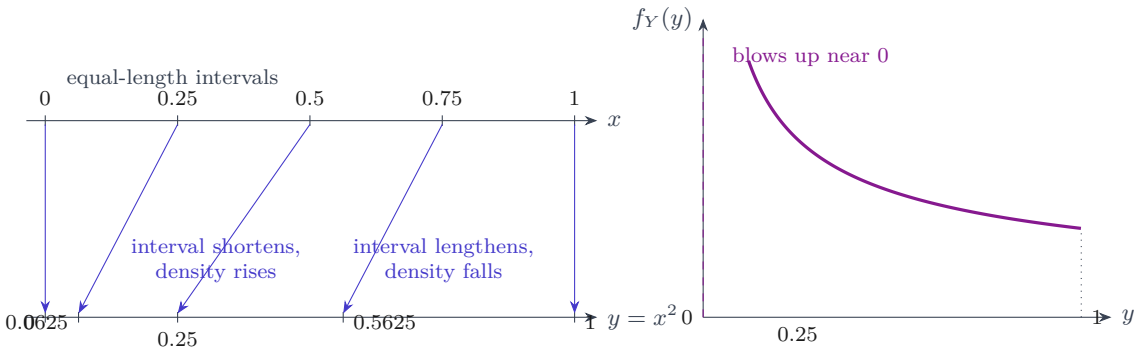


Figure 18.3: The transformation $Y = X^2$ for X uniform on $(0, 1)$, drawn on two axes. The point $x = 0.25$ lands at $y = 0.0625$, $x = 0.5$ at $y = 0.25$, and $x = 0.75$ at $y = 0.5625$; an interval of length 0.25 near zero shortens to 0.0625, whereas the same interval near one lengthens to 0.4375. The right-hand panel is the density $f_Y(y) = 1/(2\sqrt{y})$: its value is 2 at $y = 0.0625$, 1 at $y = 0.25$, and 0.5 at $y = 1$. The density rises exactly where the interval shortens, so that the total area stays one.

If r is not monotone, the theorem above does not apply directly, but the three standard steps still work with A_y a union of intervals.

Example 18.3. Let X be uniform on $(-1, 3)$, so that $f_X(x) = 1/4$ inside it, and $Y = X^2$ taking values in $(0, 9)$. For $0 < y < 1$ the set $A_y = [-\sqrt{y}, \sqrt{y}]$ lies entirely inside $(-1, 3)$, so $F_Y(y) = \frac{1}{4} \cdot 2\sqrt{y} = \sqrt{y}/2$; for $1 \le y < 9$ that set becomes $[-1, \sqrt{y}]$, so $F_Y(y) = (\sqrt{y} + 1)/4$.

Its derivative is

$$f_Y(y) = \begin{cases} \dfrac{1}{4\sqrt{y}}, & 0 < y < 1, \\ \dfrac{1}{8\sqrt{y}}, & 1 < y < 9, \\ 0, & \text{elsewhere.} \end{cases}$$

The two branches add up to one, $\frac{1}{2}$ and $\frac{1}{2}$. The density of Y blows up near zero: x^2 compresses the intervals there.

If X is uniform on $(0,1)$ and $Y = X^2$, the result is tidier: $F_Y(y) = \sqrt{y}$ and $f_Y(y) = 1/(2\sqrt{y})$, which is checked numerically in the Lab. If $Y = -\ln X$, then $F_Y(y) = 1 - e^{-y}$, exponential with $\beta = 1$: taking the logarithm of a uniform number produces an exponential waiting time, and the converse is what computers use. If U is uniform and F is the distribution function we want, then $F^{-1}(U)$ has distribution F.

Challenge (problem solver)

A set of lecture notes proposes the waiting-time density

$$f(x) = \frac{1}{2\sqrt{x}}, \qquad 0 < x < 4,$$

on the grounds that it is "positive, decreasing, and similar in shape to the example above". From that density the author computes $P(X \le 4) = 1$ and then concludes that the distribution is already valid.
Examine that claim. Determine the total area under f, repair the density in two different valid ways, and explain why the repair is not unique.
Hint 1: what decides the matter is not the value of f at a single point, but the whole region under its curve.
Hint 2: compute $\int_0^4 \frac{1}{2}x^{-1/2}\,dx$; the result is 2, not 1. For the total to be one, the constant in front must be halved to $1/4$, or the interval must be narrowed to $(0,1)$ with the constant unchanged.
If you get stuck, go back to the section "Mass Functions and Density Functions".

For younger readers (age 10 and up)

Imagine a lamp on the floor and a ball between the lamp and the wall. The shadow of the ball on the wall is much larger than the ball, and the farther the ball is from the lamp, the larger the shadow. Nothing increases in amount: the paint on the ball stays the same, it is merely spread over a larger area.
Now move the ball a little when it is already far from the lamp. Its shadow moves a long way. When the ball is still close to the lamp, the same movement shifts the shadow only a little. That is, one centimeter near the lamp becomes many centimeters on the wall, whereas one centimeter far from the lamp becomes very many centimeters.
Changing a random variable works like that. Some regions are compressed and some are stretched, and the resulting density has to adjust so that the total stays one. If it does not adjust, probability will appear out of thin air or vanish somewhere.

18.8 A Worked Example

Level: intermediate

A production line fills boxes containing 20 fuses. Each fuse is defective with probability 0.35, independently of the others, and a box is rejected if it contains twelve defects or more.

With $n = 20$ and $p = 0.35$, the rejection rule gives $P(X \geq 12) = 0.019579$ from the binomial tail, not from a normal approximation. Even the most likely value, $X = 7$, has probability only 18.4 percent, so that one box does not tell us much.

A simulation of 10^5 boxes with `default_rng(20261024)` gives mean 7.005260 and variance 4.552918. The standard error of the mean is 0.006745, so the deviation is $+0.78$ standard errors; the standard error of the variance is about 0.020348, so the deviation is $+0.14$. Both are close, and that is to be expected: a simulation does not prove a formula, it only fails to refute it.

The second check concerns the total probability mass of the Poisson distribution. The sum $e^{-\lambda}\lambda^k/k!$ up to a sufficiently large K must equal one. It is computed in log space so that nothing overflows:

λ	K	sum	$1 - \text{sum}$
0.5	68	1.00000000000000000	0
1	73	1.00000000000000000	0
3	83	1.00000000000000022	-2.2×10^{-16}
10	107	1.00000000000000311	-3.1×10^{-15}
50	194	0.99999999999999778	$+2.2 \times 10^{-15}$
100	280	1.00000000000003886	-3.9×10^{-14}

The largest deviation appears at $\lambda = 100$, namely 3.9×10^{-14}, about 175 times the machine epsilon 2.2×10^{-16}. That is not a sign that the formula is wrong, but a sign that adding 281 terms of very different sizes loses a few of the last digits; the first fourteen digits are still correct.

The third check concerns the total of the normal density, because the constant $1/\sqrt{2\pi}$ is easily mistyped. Adaptive quadrature on $[-8, 8]$ gives 0.99999999999999878, so the shortfall is 1.22×10^{-15}. Theory predicts that shortfall as exactly the tail outside the interval, namely $2\Phi(-8) = 1.24 \times 10^{-15}$. The two numbers agree to two significant figures: what is missing is indeed the tail, not an error in the constant. On the whole number line the result is 0.99999999999999978 with a printed error estimate of 1.0×10^{-8}, a conservative estimate.

The fourth check, the transformation: the density $1/(2\sqrt{y})$ has integral 0.9999999999999992, with a remainder of 8×10^{-16} from the end at 0 where the density blows up. The simulation gives $F_Y(0.09) = 0.29944$, $F_Y(0.25) = 0.50003$, $F_Y(0.49) = 0.69944$, $F_Y(0.81) = 0.90057$ against the theoretical 0.3, 0.5, 0.7, 0.9. The standard error of each number is about 0.00145 at $p = 0.3$ and 0.00095 at $p = 0.9$; all four deviations are below one standard error.

Check by hand. Compute $\binom{20}{7}0.35^7 0.65^{13}$ and compare it with 0.184401; explain why $\sum_{k=0}^{20} p_X(k) = 1$ follows from the binomial theorem, not from measurement. Finally: the sample mean 7.005260 differs from 7 by 0.00526. Is that a bug in the program? Answer with the standard error 0.006745.

18.9 Computational Lab

Level: advanced

The following code is run with Python 3, NumPy 2.5.2, and SciPy 1.18.0. All the numbers in the table and in the preceding paragraphs come from this run; the seed is fixed at 20261024, so that the results can be repeated.

Computational lab

```
import numpy as np
from math import exp, lgamma, log, sqrt, pi
```

```
from scipy import stats
from scipy.integrate import quad

rng = np.random.default_rng(20261024)
R = 100_000

# 1. binomial: exact formula vs Monte Carlo
n, p = 20, 0.35
mu, var = n * p, n * p * (1 - p)
sd = sqrt(var)
x = rng.binomial(n, p, size=R)
se = sd / sqrt(R)
print("Binomial n=20 p=0.35")
print(f"  theory E={mu:.6f} Var={var:.6f} sd={sd:.6f}")
print(f"  MC     mean={x.mean():.6f} Var={x.var(ddof=1):.6f}")
print(f"  SE mean={se:.6f} z={(x.mean()-mu)/se:+.4f}")
print(f"  SE Var  ={var*sqrt(2/(R-1)):.6f}")
print(f"  P(X>=12)={stats.binom.sf(11, n, p):.6f}")

# 2. Poisson pmf sum, computed in log space
print("\nPoisson pmf sum, k=0..K")
for lam in (0.5, 1.0, 3.0, 10.0, 50.0, 100.0):
    K = int(lam + 12 * sqrt(lam) + 60)
    s = sum(exp(-lam + k*log(lam) - lgamma(k+1)) for k in range(K+1))
    print(f"  lam={lam:6.1f} K={K:4d} sum={s:.17f}"
          f" 1-sum={1-s:+.2e}")

# 3. total normal density must be 1
phi = lambda t: exp(-t*t/2) / sqrt(2*pi)
print("\nStandard normal density")
print(f"  phi(0)={phi(0):.15f} 1/sqrt(2pi)={1/sqrt(2*pi):.15f}")
print(f"  quad [-8,8]  ={quad(phi, -8, 8)[0]:.17f}")
print(f"  remainder={1-quad(phi, -8, 8)[0]:.2e}"
      f"  2*Phi(-8)={2*stats.norm.cdf(-8):.2e}")
v, e = quad(phi, -np.inf, np.inf)
print(f"  quad (-inf,inf)={v:.17f} reported error={e:.1e}")

# 4. transformation X~Unif(0,1)
u = rng.uniform(0, 1, R)
y = u * u
print("\nY=X^2, X~Unif(0,1)")
print(f"  integral of f_Y={quad(lambda t: 1/(2*sqrt(t)), 0, 1)[0]:.16f}")
for yy in (0.09, 0.25, 0.49, 0.81):
    print(f"  F_Y({yy:.2f}) MC={np.mean(y<=yy):.5f} theory={sqrt(yy):.5f}")
z = -np.log(u)
print("Y=-ln(X), should be Exp(1)")
print(f"  mean={z.mean():.5f} SE={1/sqrt(R):.5f}"
      f" Var={z.var(ddof=1):.5f}")
print(f"  P(Y>2) MC={np.mean(z>2):.5f} theory={exp(-2):.5f}")

# 5. geometric and exponential
g = rng.geometric(0.2, R)
print("\nGeometric p=0.2: E=5 Var=20")
print(f"  mean={g.mean():.5f} SE={sqrt(20/R):.5f}"
      f" Var={g.var(ddof=1):.5f}")
w = rng.exponential(2.0, R)
print("Exponential beta=2: E=2 Var=4")
print(f"  mean={w.mean():.5f} SE={2/sqrt(R):.5f}"
      f" Var={w.var(ddof=1):.5f}")
print(f"  P(X>3) MC={np.mean(w>3):.5f} theory={exp(-1.5):.5f}")
print(f"  P(X>5|X>2) MC={np.mean(w[w>2]>5):.5f} theory={exp(-1.5):.5f}")
```

The output obtained, as it stands:

```
Binomial n=20 p=0.35
  theory E=7.000000 Var=4.550000 sd=2.133073
  MC     mean=7.005260 Var=4.552918
  SE mean=0.006745 z=+0.7798
  SE Var  =0.020348
  P(X>=12)=0.019579

Poisson pmf sum, k=0..K
  lam=   0.5 K=  68 sum=1.00000000000000000 1-sum=+0.00e+00
  lam=   1.0 K=  73 sum=1.00000000000000000 1-sum=+0.00e+00
  lam=   3.0 K=  83 sum=1.00000000000000022 1-sum=-2.22e-16
  lam=  10.0 K= 107 sum=1.00000000000000311 1-sum=-3.11e-15
  lam=  50.0 K= 194 sum=0.99999999999999778 1-sum=+2.22e-15
  lam= 100.0 K= 280 sum=1.00000000000003886 1-sum=-3.89e-14

Standard normal density
  phi(0)=0.398942280401433 1/sqrt(2pi)=0.398942280401433
  quad [-8,8]  =0.99999999999999878
  remainder=1.22e-15  2*Phi(-8)=1.24e-15
  quad (-inf,inf)=0.99999999999999978 reported error=1.0e-08

Y=X^2, X~Unif(0,1)
  integral of f_Y=0.9999999999999992
  F_Y(0.09) MC=0.29944 theory=0.30000
  F_Y(0.25) MC=0.50003 theory=0.50000
  F_Y(0.49) MC=0.69944 theory=0.70000
  F_Y(0.81) MC=0.90057 theory=0.90000
Y=-ln(X), should be Exp(1)
  mean=0.99811 SE=0.00316 Var=0.99623
  P(Y>2) MC=0.13478 theory=0.13534

Geometric p=0.2: E=5 Var=20
  mean=4.99176 SE=0.01414 Var=19.92571
Exponential beta=2: E=2 Var=4
  mean=1.99895 SE=0.00632 Var=3.98400
  P(X>3) MC=0.22237 theory=0.22313
  P(X>5|X>2) MC=0.22028 theory=0.22313
```

Four things can be read from that output. First, every sample mean deviates by less than one standard error: $z = +0.78$ for the binomial, whereas 0.99811, 4.99176, and 1.99895 against 1, 5, and 2 for the other three cases. A deviation of that size has to appear; if all of them were zero, something would be wrong.

Second, the sample variances are slightly below the theoretical value in the long-tailed case, 19.92571 against 20, because a single very large draw dominates the sum of squares. Third, the error of the Poisson summation grows with λ, from zero to 10^{-14} as the number of terms increases from 74 to 281. Fourth, the resulting density integrates to one to 16 digits, and F_Y matches $\sqrt{y}$ at four test points; the second is more important, because an integral of one only checks the constant, whereas agreement at four points checks the shape of the function.

That code does not prove a single theorem in this chapter; it checks that the formulas written down give the numbers stated. The proof of $\mathbb{E}[X] = np$ is in the indicator argument, the proof of $f_Y(y) = 1/(2\sqrt{y})$ in the computation of the distribution function; the output of a program is data, not an argument.

Note for instructors

The order I suggest: two systems with the same mean but different tails, then the definition of a random variable, then straight to the cumulative distribution function. Do one small table on the board, draw the staircase, ask for $F(1.4)$, and only after that the density. Resist the temptation to mention the integral before students see why summation is no longer enough.

The two misconceptions that appear most often. First, $f_X(x)$ is read as a probability, so that a density of value 5 feels impossible and $P(X = x)$ is taken to equal $f_X(x)$. Second, $P(X = x) = 0$ is concluded to mean impossible, so that $P(0 \le X \le 1) = 1$ becomes a paradox. The answer is countable additivity: an uncountable union is not added up.

Sign of readiness to move on: students can explain the difference between mass and density without reading their notes, and can compute $\mathbb{E}[X^2]$ without finding the density of X^2.

18.10 Summary

Level: basic

Summary.

- A random variable is a function from the outcome space to the real numbers, not a random number. The distribution function $F_X(x) = P(X \le x)$ carries the whole information of the distribution: nondecreasing, right-continuous, with limits 0 and 1.
- A mass function weights points and its values are at most one; a density function weights length, may exceed one, and $P(X = x) = 0$ at every point. Probability zero is not impossibility, because only countable unions are added up.
- Expectation is the balance point and is linear without any condition of independence; $\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$ is the mean squared deviation.
- Binomial $\mathbb{E} = np$, $\operatorname{Var} = np(1-p)$, derived through indicators; geometric $\mathbb{E} = 1/p$, $\operatorname{Var} = (1-p)/p^2$; Poisson $\mathbb{E} = \operatorname{Var} = \lambda$, the binomial limit for rare events.
- The normal is standardized to $N(0,1)$ and read through Φ; the exponential is memoryless, the continuous twin of the geometric.
- Covariance measures only linear dependence: zero does not mean independent, and a monotone transformation changes the density by the factor $|ds/dy|$.

18.11 Exercises

Level: intermediate

Exercise 18.1.

1. Let $p_X(-2) = 0.1$, $p_X(0) = 0.4$, $p_X(3) = c$. Determine c, then compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\operatorname{Var}(X)$.
2. Prove $P(a < X \le b) = F_X(b) - F_X(a)$, then explain why $P(a \le X \le b)$ can differ from $P(a < X \le b)$ for a discrete variable and why the two are equal for a continuous variable.
3. Assess this claim: “because $P(X = x) = 0$ for every x, the event $X = x$ is impossible.” Name the part that is correct, the part that is wrong, and the role of countable additivity.
4. Derive $\mathbb{E}[X^2] = (2-p)/p^2$ for the geometric by differentiating $\sum k^2 q^{k-1}$ twice, then conclude $\operatorname{Var}(X)$ and check for $p = 0.2$.
5. Let $X \sim N(3, 4)$. Compute $P(X > 1)$, $P(1 < X < 5)$, and the 0.2 quantile of X; compare with `scipy.stats.norm`.

6. Compute $\sum_{k=0}^{40} e^{-2.5}2.5^k/k!$ and $P(X \geq 8)$ for the Poisson with $\lambda = 2.5$, then compare with the binomial $n = 1000$, $p = 0.0025$; explain the direction of the difference.
7. Prove that the exponential is memoryless, then explain why the geometric is also memoryless and why the uniform on the integers is not.
8. For X uniform on $(-1, 3)$ and $Y = X^2$, derive f_Y, check $\int f_Y = 1$, then compute $\mathbb{E}[Y]$ through f_Y and through $\mathbb{E}[X^2]$.
9. Construct two variables whose covariance is zero but which are not independent. Compute Cov and ρ exactly, and exhibit a joint probability that proves the dependence.
10. How many repetitions R are needed for the standard error of the mean of a binomial with $n = 20$, $p = 0.35$ to be at most 0.005? And for the standard error of the variance estimate to be at most 0.02?

Hints. Item 3: separate the statement about probability from the statement about the existence of a set. Item 4: differentiate $\sum_{k\geq 0} q^k = 1/(1-q)$ twice with respect to q, and remember that $\sum k^2 q^{k-1}$ contains $\sum k(k-1)q^{k-1}$. Item 8: for $0 < y < 1$ the set A_y has two branches, for $y \geq 1$ only one.

For researchers: limits and next steps

What is genuinely proved in this chapter: the properties of the distribution function together with their proofs, the relation of the density to the distribution function through the fundamental theorem of calculus, the formula for the expectation of a function of a random variable in the discrete case, linearity of expectation, the variance formula, the mean and variance of the binomial through indicators, the geometric and Poisson means through series, memorylessness of the exponential, and the monotone transformation formula.

What is quoted, not proved: that a function satisfying the three properties of a distribution function always comes from some probability space (Wasserman, Theorem 2.8, whose direction uses the Caratheodory theorem), that the distribution function determines every probability (Wasserman, Theorem 2.7), the normalization of the normal through polar coordinates (Chapter 15), and the correlation bound through Cauchy-Schwarz (Chapter 10).

Primary sources: Wasserman, *All of Statistics*, Sections 2.1–2.4 (including Lemma 2.15 and the warning that a density may exceed one) and 2.11 (Example 2.47 for $Y = X^2$ on $U(-1, 3)$), as well as 3.1–3.4. Blitzstein and Hwang, *Introduction to Probability*, Chapters 3–5 (LOTUS in 4.5, Poisson in 4.7, exponential in 5.5) and 7.3 for covariance.

Limits of validity: the density is not unique, so every statement about f_X is a statement about the same class of functions almost everywhere; the Jacobian formula requires monotonicity and differentiability; the Cauchy expectation does not exist, so every "mean" of very heavy-tailed data must be suspected first. Open question: how well a density can be estimated from data without assuming a shape, and whether the tails of real distributions are thin enough to justify the use of variance.

Map to source texts: Chapter 18 follows Wasserman, All of Statistics*, Chapter 2 (2.1–2.4, Lemma 2.15, and 2.11 for transformation) and Chapter 3 (3.1–3.4). In Blitzstein and Hwang,* Introduction to Probability*, Chapter 3 (3.6–3.7), Chapter 4 (4.2, 4.5, 4.7), Chapter 5 (5.1–5.5), and 7.3. The integrals used are the Riemann integrals of Chapter 13; change of variables in more than one dimension is in Chapter 15.*

Chapter 19

The Law of Large Numbers and the Central Limit Theorem

A short program computes the average of $\exp(U)$ over ten thousand random numbers U uniform on $(0, 1)$. The result it prints is 1.722206. The true value is $e - 1 = 1.7182818285$, so the error of that run is $+0.003924$. The program is run again with four times as many numbers: 1.717502, error -0.000780. With ten times as many: 1.717767, error -0.000515. The pattern is not tidy. Four times the data cuts the error by more than a factor of five, whereas ten times the data cuts it by only 7.6 times. The naive guess "ten times the data, one tenth of the error" does not hold for a single run. One run of a program carries no rate at all; what has a rate is the distribution over many runs.

That distribution can be measured. With $n = 10^4$ the standard deviation of this estimator is 0.00492; with $n = 4 \cdot 10^4$ it becomes 0.00246; with $n = 10^5$ it becomes 0.00156. Every time n grows fourfold, that deviation is exactly halved. Here we have two different statements that are often confused. First, the average really does approach the true value: that is a tendency, and its name is the law of large numbers. Second, the remaining error has a definite size and shape: that is fluctuation, and its name is the central limit theorem. This chapter supplies names, proofs, and limits of validity for both.

Chapter 18 has already prepared the ingredients: expectation, variance, and the distribution of a sum through convolution. Convolution is useful only when n is small and the original shape is simple; for most distributions no closed form exists. What we need is not the exact distribution at one n, but a statement that holds for every large n. The practical questions are two: how close may the average be trusted, and how large an error is reasonable. The answers come from examining how the sequence of distributions behaves, not from computing one distribution.

For younger readers (age 10 and up)

Imagine a class is asked to guess the number of marbles in a large jar. The children's guesses miss badly in both directions. One child guesses three hundred, another guesses two thousand, when the jar holds eight hundred. The teacher then writes all the guesses on the board and computes their average. The number turns out to be very close to eight hundred.

Now the teacher adds students. With thirty guessers the average still wobbles. With three hundred guessers the average is calmer. With three thousand guessers the average hardly shifts any more. There are two strange things here. First, not one of the guessers has become better; only the number of guessers changed. Second, adding guessers from thirty to three hundred helps far more than adding from three hundred to three thousand.

The help keeps growing, but more and more slowly.
The question that makes us curious: if each guess can miss by a great deal, why can the average be trusted? And why does the help slow down in a pattern we can predict, rather than at random? In this chapter we will find out why, and at the same time find out when the habit of trusting the average deceives us.

19.1 Three Meanings of Approaching for a Random Variable

Level: basic

In everyday language: a sequence of random variables can "approach" in three different senses, and the loosest sense says nothing at all about the values coming closer.

A sequence of real numbers has only one sense of approaching, namely that the distance $|x_n - L|$ becomes small. For random variables that distance is itself a random variable, so we have to choose which quantity is made small. The first choice is the probability. This is the choice that is weakest intuitively but most useful practically.

convergence in probability (*konvergensi dalam peluang*) — a promise that the probability of missing by more than a tolerance shrinks to zero, not a promise that the value comes closer on a single run

Definition 19.1 (Convergence in probability)**.** A sequence of random variables $X_1, X_2, \ldots$ *converges in probability* to a random variable X if for every $\varepsilon > 0$

$$P(|X_n - X| > \varepsilon) \to 0 \qquad (n \to \infty).$$

Note where the quantifier sits. The value of ε is chosen first, and the probability may shrink only after that. This definition does not say that $|X_n - X|$ shrinks for one particular outcome of the experiment; it says that the event "missing by more than ε" becomes rarer and rarer. On the same experiment, X_n may still miss by a lot, provided that event is rare.

convergence in distribution (*konvergensi dalam distribusi*) — what resembles is the shape of the distribution, not the value itself; two variables can share a shape without ever coming close

Definition 19.2 (Convergence in distribution)**.** Let F_n be the distribution function of X_n and F the distribution function of X. We write $X_n \rightsquigarrow X$ when $F_n(t) \to F(t)$ at every point t at which F is continuous. That limit is called the *limit in distribution*.

The condition "at continuity points" is not a technical whim. Take X_n equal to $1/n$ with probability one, and X equal to 0 with probability one. Then $F_n(0) = 0$ for every n, whereas $F(0) = 1$, so the sequence does not converge at $t = 0$. At every $t \neq 0$ the sequence does converge, and $t = 0$ is the only point of discontinuity of F. Without that condition, a correct conclusion would be rejected because of a single point.

The third sense is stronger and more literal: almost every outcome of the experiment, as one point ω of the sample space, produces a convergent sequence of real numbers.

Definition 19.3 (Almost sure convergence)**.** X_n *converges almost surely* to X when

$$P(\{\omega : X_n(\omega) \to X(\omega)\}) = 1.$$

Here the sequence of real numbers $X_1(\omega), X_2(\omega), \ldots$ really does converge in the sense of Chapter 4, and the set of outcomes that fail to converge may only have probability zero. Therefore almost sure convergence speaks about one long run, whereas convergence in probability speaks about one n across many runs at once.

The three senses are not equivalent, but there is a fixed direction of implication.

Theorem 19.1 (Relations among the three modes)**.** *For random variables X_n and X: (a) $X_n \to X$ almost surely implies $X_n \to X$ in probability; (b) $X_n \to X$ in probability implies $X_n \rightsquigarrow X$; (c) if $X_n \rightsquigarrow c$ and c is a constant, then $X_n \to c$ in probability.*

Proof. For (a), fix $\varepsilon > 0$ and let $A_n = \{\sup_{k \geq n} |X_k - X| > \varepsilon\}$. The sets A_n decrease, and every ω in $\bigcap_n A_n$ satisfies $|X_k(\omega) - X(\omega)| > \varepsilon$ for infinitely many k, so the sequence does not converge to $X(\omega)$. Hence $\bigcap_n A_n$ is contained in a set of probability zero. Since probability is continuous from above on sets of finite probability, $P(A_n) \to P(\bigcap_n A_n) = 0$. The event $\{|X_n - X| > \varepsilon\}$ is contained in A_n, so its probability also tends to zero.

For (b), take t a continuity point of F and $\varepsilon > 0$. From $\{X_n \leq t\} \subseteq \{X \leq t + \varepsilon\} \cup \{|X_n - X| > \varepsilon\}$ we obtain

$$F_n(t) \leq F(t + \varepsilon) + P(|X_n - X| > \varepsilon).$$

In the same way, from $\{X \leq t - \varepsilon\} \subseteq \{X_n \leq t\} \cup \{|X_n - X| > \varepsilon\}$ we obtain $F(t - \varepsilon) \leq F_n(t) + P(|X_n - X| > \varepsilon)$. The two inequalities give

$$F(t - \varepsilon) - \delta_n \leq F_n(t) \leq F(t + \varepsilon) + \delta_n, \qquad \delta_n = P(|X_n - X| > \varepsilon).$$

Since $\delta_n \to 0$, the limit inferior and the limit superior of $F_n(t)$ are trapped between $F(t - \varepsilon)$ and $F(t + \varepsilon)$. Let $\varepsilon \downarrow 0$ and use the continuity of F at t; both tend to $F(t)$.

For (c), take $\varepsilon > 0$. Since F is the point-mass distribution at c, the points $c - \varepsilon$ and $c + \varepsilon$ are both continuity points of F, so

$$P(|X_n - c| > \varepsilon) \leq P(X_n < c - \varepsilon) + P(X_n > c + \varepsilon) \leq F_n(c - \varepsilon) + 1 - F_n(c + \varepsilon) \to 0 + 1 - 1 = 0.$$

□

The three reverse implications fail, and the counterexamples are worth remembering because each one shows what is actually lost.

Example 19.1 (In probability but not almost surely). Split $(0, 1)$ into k pieces of equal length, for $k = 1, 2, 3, \ldots$, then arrange all those pieces into one list. Define $X_n(\omega) = 1$ if ω lies in the n-th piece and 0 otherwise. The n-th piece comes from the split into k_n pieces, so $P(X_n = 1) = 1/k_n \to 0$ and $X_n \to 0$ in probability. But for every $\omega \in (0, 1)$, within every block of splits there is exactly one n with $X_n(\omega) = 1$. That value one appears infinitely often, so $X_n(\omega)$ does not converge, for every ω. Hence there is no almost sure convergence. This sequence even converges in L^1, because $\mathbb{E}|X_n| = 1/k_n \to 0$, so L^1 strength does not close that gap either.

Example 19.2 (In distribution but not in probability). Take $X \sim N(0, 1)$ and set $X_n = -X$ for every n. The distribution of X_n is exactly the distribution of X, so $F_n(t) = F(t)$ and $X_n \rightsquigarrow X$ vacuously. But $P(|X_n - X| > \varepsilon) = P(2|X| > \varepsilon) > 0$ for every ε, and that probability does not depend on n. Two variables of the same shape need not be close: X and $-X$ can be far apart on the same run.

The lesson of those two examples: convergence in distribution touches the distributions one at a time, whereas the other two modes touch the relation between variables on the same sample space. That is why part (c) of the theorem matters practically: in asymptotic statistics the limit is almost always a constant, and there the three modes meet.

One further warning has to be recorded from the start, because the mistake is expensive.

Note 19.1. Convergence in probability does not carry convergence of expectations with it. Suppose $P(X_n = n^2) = 1/n$ and $P(X_n = 0) = 1 - 1/n$. For every $\varepsilon > 0$, $P(|X_n| > \varepsilon) = 1/n \to 0$, so $X_n \to 0$ in probability. But $\mathbb{E}[X_n] = n$, so the expectation tends to infinity. An average that concentrates is no guarantee that its expectation concentrates.

19.2 Chebyshev's Inequality and the Weak Law of Large Numbers

Level: intermediate

In everyday language: the probability of straying far from the mean is bounded by the variance divided by the square of that stray distance.

To prove the law of large numbers we need only one inequality, and that inequality needs only one simple observation about expectation.

Lemma 19.2 (Markov)**.** *If $Y \geq 0$ and $a > 0$, then $P(Y \geq a) \leq \mathbb{E}[Y]/a$.*

Proof. Write $Y = Y\mathbf{1}_{\{Y<a\}} + Y\mathbf{1}_{\{Y\geq a\}}$. The first term is nonnegative. On the event $Y \geq a$ we have $Y \geq a$, so the expectation of the second term is at least $a\,P(Y \geq a)$. Hence $\mathbb{E}[Y] \geq 0 + a\,P(Y \geq a)$. □

Corollary 19.3 (Chebyshev's inequality)**.** *If $\mathbb{E}[X] = \mu$ and $\operatorname{Var}(X) = \sigma^2 < \infty$, then for every $a > 0$*

$$P\big(|X - \mu| \geq a\big) \leq \frac{\sigma^2}{a^2}.$$

Proof. Apply Markov's lemma to $Y = (X - \mu)^2$ with threshold a^2. The event $\{Y \geq a^2\}$ is the same as $\{|X - \mu| \geq a\}$, and $\mathbb{E}[Y] = \sigma^2$. □

This inequality is weak, and its weakness is not a hidden defect but its strength. It knows nothing about the shape of any distribution, only two moments. Therefore it holds for every distribution with finite variance, and precisely for that reason it is enough to prove a theorem that holds in general.

law of large numbers (*hukum bilangan besar*) — a promise that the average of many observations sticks to the true mean, with a probability of missing that shrinks

Theorem 19.4 (Weak law of large numbers)**.** *Suppose $X_1, X_2, \ldots$ are independent and identically distributed with $\mathbb{E}[X_i] = \mu$ and $\operatorname{Var}(X_i) = \sigma^2 < \infty$. Then for every $\varepsilon > 0$*

$$P\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \to 0, \qquad \bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Proof. Since expectation is linear, $\mathbb{E}[\bar{X}_n] = \mu$. Since the variables are independent, the variance of a sum is the sum of the variances, so $\operatorname{Var}(\bar{X}_n) = \sigma^2/n$. Chebyshev's inequality with $a = \varepsilon$ gives

$$P\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

The right-hand side tends to zero for each fixed $\varepsilon > 0$. □

Note what has just been proved and what has not. What is proved: for a tolerance ε that we choose ourselves, the probability that the average misses by more than ε tends to zero. What is not proved: how fast, and what the shape of the remaining error is. Those are two different questions.

Figure 19.1 draws the content of that statement for dice. Only two things change from $n = 4$ to $n = 64$: the width shrinks and the peak rises, while the center stays at 3.5. The law of large numbers is not a promise that one run is more accurate, but that the distribution of possibilities contracts to a single point.

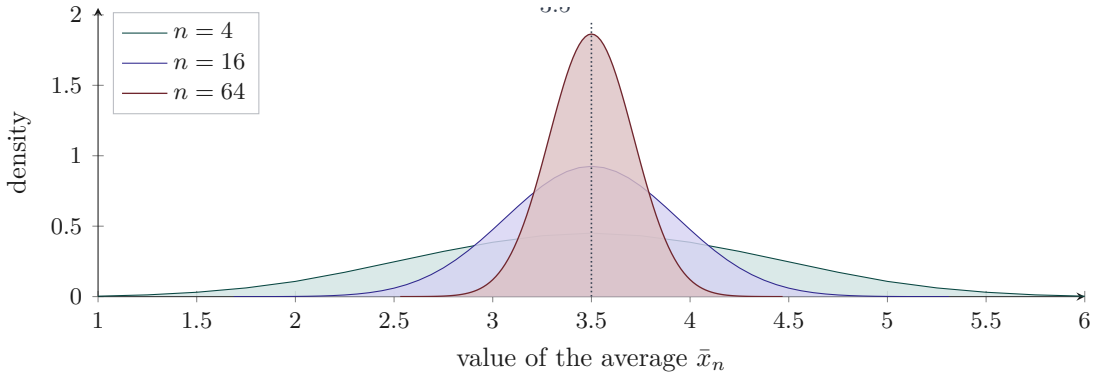


Figure 19.1: The three distributions of the average of n die rolls, drawn on the same scale: $n = 4$, 16, and 64. Their widths are 0.853913, 0.426956, and 0.213478, so each time n is multiplied by four the width is halved. The peak rises from 0.4506 to 0.9251 and 1.8641, while the center stays at 3.5.

The numbers make that difference tangible. For a coin toss with $p = 1/2$ and tolerance $\varepsilon = 0.1$, Chebyshev gives $\sigma^2/(n\varepsilon^2) = 0.25/(0.01n) = 25/n$. For the probability of missing to be at most 0.3, it suffices that $n \geq 25/0.3 = 83.33$, that is $n = 84$. At $n = 84$ that bound equals 0.297619, so Chebyshev guarantees that the probability of landing between 0.4 and 0.6 is at least 0.702381. The true probability, computed from the binomial distribution, is 0.937028, and the normal approximation gives 0.933197. The guarantee is correct but too low by about 23 percentage points.

The same weakness becomes severe when σ is large. For a fair die, $\mathrm{Var}(X) = 35/12 = 2.916667$ and $\sigma = 1.707825$. At $n = 100$ with $\varepsilon = 0.1$, Chebyshev gives the bound 2.916667, a number larger than one and therefore meaningless. To obtain an error of 0.01 with 95 percent confidence, the normal formula demands at least 112043 observations, whereas Chebyshev demands far more still.

This is where the problem that sets up the next section lies. Chebyshev answers "may the average be trusted" correctly but bluntly. It cannot answer "how close, with what probability" in a usable way. A sharp answer demands knowledge of the shape of the error distribution, not only of its size.

Read it aloud (auditory learner)

Two voices, taking turns. The first names the number of tosses; the second names the distance off that is still reasonable.

— Ten tosses. The miss is still wide. — A hundred tosses. Already narrower. — A thousand tosses. Narrower still.

Now quadruple, do not add ten: ten becomes forty, a hundred becomes four hundred, a thousand becomes four thousand. The miss is half. Not a quarter. Half. Every time fourfold, exactly half.

Then reverse the direction, and read it more slowly. What is the cost of one more digit after the decimal point? A hundred times the tosses. A hundred times the work, one digit. The average is impatient: it demands four times, not two, and it offers no discount. The line repeated three times: four times the data, half the error.

For younger readers (age 10 and up)

Imagine a machine that fills bags of rice. Each bag misses the requested weight by a little, sometimes over and sometimes under, and the size of the miss differs from bag to

bag; weighing one bag tells us nothing about the direction of its miss.
Now the machine fills a hundred bags, and we weigh all of them at once. The misses upward and downward cancel each other, so the total weight is much closer to a hundred times the requested weight. If there are ten thousand bags, the total is almost exact.
The surprising part is that we do not need to know the shape of the distribution of the miss. Whether the miss has a bell shape, takes only two values, or has any odd shape at all, the distribution of the total still resembles the same bell, provided there are enough bags and the miss does not turn into a giant too often. This is what lets gamblers, bridge builders, and weather researchers use the same formula.
The question that makes us curious: why does the original shape of the miss no longer matter once there are many bags? And why can one bag with a giant miss ruin the whole conclusion? In this chapter we will find out why.

19.3 The Strong Law: What It Actually Adds

Level: advanced

In everyday language: with confidence one, the average along one long run eventually settles, rather than merely coming close often.

The weak law speaks about each n separately: at a sufficiently large n, the probability of missing is small. It does not forbid the average from missing badly at $n = 10^6$, then calming down, and then missing again at $n = 10^9$, provided such events are rare at each n. The strong law closes that gap.

Theorem 19.5 (Strong law of large numbers)**.** *Suppose $X_1, X_2, \dots$ are independent and identically distributed with $\mathbb{E}|X_1| < \infty$ and $\mathbb{E}[X_1] = \mu$. Then $\bar{X}_n \to \mu$ almost surely.*

The difference is subtle but sharp. The sequence X_n of the previous section converges in probability to zero, yet at each ω its value is one infinitely often. The weak law cannot reject it; the strong law does, because it demands convergence at every ω outside a set of probability zero.

Note the hypothesis as well. The strong law demands only $\mathbb{E}|X_1| < \infty$, with no variance. The weak law we proved above demands $\sigma^2 < \infty$ only because its proof uses Chebyshev; that hypothesis is not the natural hypothesis of the theorem. For the weak law, a necessary and sufficient condition for an iid sequence is $n\,P(|X_1| > n) \to 0$, which is weaker than $\mathbb{E}|X_1| < \infty$. We leave those details aside, and the reader can find them in Feller, *An Introduction to Probability Theory and Its Applications*, volume 2, in the chapter on the law of large numbers.

We do not carry out the proof of the strong law here. Its classical form truncates the variables at the level n, applies Borel–Cantelli to the tails of the truncation, and then uses the Cauchy criterion for almost sure convergence; Etemadi gives a proof that avoids the variance assumption and needs only pairwise independence. Where the proofs live: Blitzstein and Hwang, *Introduction to Probability*, chapter 10, section 10.2; Wasserman, *All of Statistics*, appendix to chapter 5, section 5.7.1. What must be remembered from that statement: the strong law gives no rate. It says that the error is eventually small and stays small, not how fast.

Do it by hand (kinesthetic learner)

Take one coin and a sheet of lined paper. Heads counts one, tails counts zero.
Toss forty times, write 1 or 0 in a row. On the second line, write the running average under each toss: the sum so far divided by the number of tosses so far.
What we should see: the second line swings most wildly over the first five tosses, then

grows calmer. At the fortieth toss the number need not be 0.5; 0.475 or 0.525 is still normal.
Now repeat that experiment four times and draw the four paths on one graph with different pencils. All four paths narrow toward 0.5, and the width of their spread is halved each time the number of tosses is multiplied by four.
Tape the sheet to the wall. If someone says the results must balance, point to the fortieth toss, not the tenth.

19.4 The Central Limit Theorem

Level: advanced

In everyday language: the error of the average, once magnified by a factor $\sqrt{n}$, always has the same bell shape, whatever the original shape of the data.

The law of large numbers tells us that $\bar{X}_n$ sticks to μ. It does not tell us how wide that sticking is. The central limit theorem answers that, and the answer is surprising because it does not depend on the original distribution.

central limit theorem (*teorema limit pusat*) — the bell shape that appears in the error of the average, not in the data, and only after that error is magnified by the square root of the number of data points

Theorem 19.6 (Central limit, Lindeberg–Levy form)**.** *Suppose* $X_1, X_2, \dots$ *are independent and identically distributed with* $\mathbb{E}[X_i] = \mu$ *and* $0 < \mathrm{Var}(X_i) = \sigma^2 < \infty$*. Then*

$$Z_n = \frac{\sqrt{n}\,(\bar{X}_n - \mu)}{\sigma} \;\rightsquigarrow\; Z \sim N(0,1),$$

meaning that $P(Z_n \le z) \to \Phi(z)$ *for every* $z \in \mathbb{R}$*.*

Note where $\sqrt{n}$ sits. The standard deviation of $\bar{X}_n$ is $\sigma/\sqrt{n}$, not σ/n. That is why cutting the error in half requires four times the data, and adding one digit of precision requires a hundred times the data. This is bad news that cannot be dodged: precision comes at a quadratic price.

Figure 19.2 shows where the bell shape comes from. The left panel is the distribution of a single die: flat, with no bell at all. The right panel is the distribution of the average of twelve dice after standardization. The raw material is not a bell; the result is a bell.

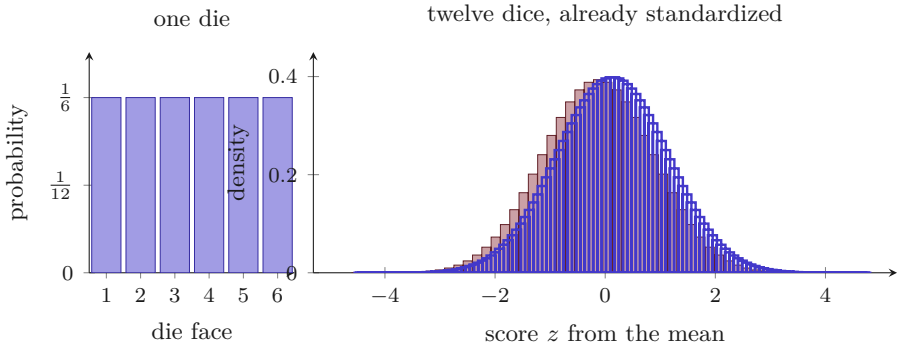


Figure 19.2: From flat to bell. Left: the distribution of one die, flat over six values. Right: the distribution of the average of twelve dice after standardization, stacked with the standard normal curve. The supremum distance between the two is 0.03384, less than half the distance at $n = 2$, which is 0.08333. The raw material is not a bell, the result is a bell.

The following proof uses moment generating functions. It is complete for the case in which the third moment is finite, and we state that limitation honestly at the end.

Lemma 19.7 (Convergence through moment generating functions)**.** *Suppose* $\psi_n(t) = \mathbb{E}[e^{tZ_n}]$ *and* $\psi(t) = \mathbb{E}[e^{tZ}]$ *are finite on some open interval around* $t = 0$*. If* $\psi_n(t) \to \psi(t)$ *for every* t *in that interval, then* $Z_n \rightsquigarrow Z$*.*

That lemma is the continuity theorem for moment generating functions. Its proof uses the fact that the distribution function is determined by its moment generating function on an interval around zero, and we do not repeat it here; see Wasserman, *All of Statistics*, chapter 5, section 5.7.2, Lemma 5.20.

Proof of the central limit theorem. Set $Y_i = (X_i - \mu)/\sigma$, so that $\mathbb{E}[Y_i] = 0$, $\operatorname{Var}(Y_i) = 1$, and $Z_n = n^{-1/2}\sum_{i=1}^{n} Y_i$. Let $\psi(t) = \mathbb{E}[e^{tY_1}]$ be the moment generating function of Y_1, which is finite on the interval $|t| \le \delta$. Since the variables are independent, the moment generating function of Z_n is

$$\xi_n(t) = \Big[\psi(t/\sqrt{n})\Big]^n.$$

Derivatives at zero give $\psi(0) = 1$, $\psi'(0) = \mathbb{E}[Y_1] = 0$, and $\psi''(0) = \mathbb{E}[Y_1^2] = 1$. The second-order Taylor expansion with third-order remainder, valid when $\mathbb{E}|Y_1|^3 < \infty$, reads

$$\psi(s) = 1 + \frac{s^2}{2} + R(s), \qquad |R(s)| \le C|s|^3 \quad (|s| \le \delta).$$

For n large enough we have $|t|/\sqrt{n} \le \delta$, so

$$\xi_n(t) = \Big(1 + \frac{t^2}{2n} + R\big(t/\sqrt{n}\big)\Big)^n = \Big(1 + \frac{t^2}{2n} + O\big(n^{-3/2}\big)\Big)^n.$$

Take logarithms. With $\log(1 + a) = a - a^2/2 + \dots$ and $a_n = t^2/(2n) + O(n^{-3/2})$,

$$\log \xi_n(t) = n\Big(a_n - \frac{a_n^2}{2} + \dots\Big) = \frac{t^2}{2} + O\big(n^{-1/2}\big) - \frac{t^4}{8n} + \cdots \longrightarrow \frac{t^2}{2}.$$

Hence $\xi_n(t) \to e^{t^2/2}$ for every fixed t. That last function is the moment generating function of $N(0,1)$, so the lemma above gives $Z_n \rightsquigarrow N(0,1)$. □

There is one further hypothesis that slips in: the third-order Taylor remainder demands $\mathbb{E}|Y_1|^3 < \infty$. For distributions with very heavy tails that hypothesis can fail even though the variance is finite, and the proof above has to be replaced. The usual route is to use the characteristic function $\varphi(t) = \mathbb{E}[e^{itX}]$, which always exists because $|e^{itX}| = 1$, and then to apply Levy's continuity theorem. The skeleton of the proof is the same: the logarithm of the characteristic function of Z_n tends to $-t^2/2$. The version without the third-moment assumption is in Wasserman, chapter 5, section 5.7.2, and in Blitzstein and Hwang, chapter 10, section 10.3.

This theorem does not say that the error is small. It says that the error has a bell shape once it is divided by $\sigma/\sqrt{n}$. Its size is still given by $\sigma/\sqrt{n}$, and σ is a quantity we usually do not know. In practice σ is replaced by the sample standard deviation s, and that replacement does not destroy the limit; the statement becomes $\sqrt{n}(\bar{X}_n - \mu)/s_n \rightsquigarrow N(0,1)$. The quantity $\sigma/\sqrt{n}$ is called the standard error (*galat baku*), and it is the basis of the confidence intervals in Chapter 20.

How fast that approximation improves can also be bounded.

Theorem 19.8 (Berry–Esseen inequality)**.** *If* $\rho = \mathbb{E}|X_1 - \mu|^3 < \infty$*, then*

$$\sup_z |P(Z_n \le z) - \Phi(z)| \le \frac{C\rho}{\sigma^3\sqrt{n}},$$

where C *is a universal constant; the value* $C = 0.4748$ *suffices for all distributions.*

The shape of this inequality matters: the distance shrinks like $1/\sqrt{n}$, with a constant that depends on the skewness of the original distribution through ρ/σ^3. A strongly skewed distribution will need a far larger n than a symmetric one. The numbers in the lab section show this quantitatively for the exponential distribution, whose third moment is $\rho = 2.414553$ and whose bound is therefore $1.1464/\sqrt{n}$.

Step-by-step recipe (procedural learner)

Three steps to standardize a sample average. The order must not be swapped.

1. **Subtract the center.** Compute $\bar{x} - \mu$. The bell shifts so that its peak falls at zero.
2. **Divide by the width.** Divide the result by the standard error $\sigma/\sqrt{n}$. The shape of the bell does not change; only its scale becomes one unit.
3. **Read the table.** The remaining number z is read off $\Phi(z)$; for the right tail use $1 - \Phi(z)$.

A warning about order: dividing before subtracting the center shifts the result by $\mu\sqrt{n}/\sigma$. For dice with $\mu = 3.5$, $\sigma = 1.707825$, and $n = 12$, that shift is 7.10; the number is already outside any normal table.

19.5 The Normal Approximation and the Continuity Correction

Level: intermediate

In everyday language: to compute the probability of a sum we may use the normal table, but when the sum can only take integer values, the threshold is shifted by one half.

The normal approximation for sums works most simply on discrete distributions, and there a small correction appears whose effect is large. It is called the continuity correction. Take $S \sim \text{Binomial}(100;\, 0.5)$, so that $\mu = 50$ and $\sigma = \sqrt{100 \cdot 0.5 \cdot 0.5} = 5$. The event $\{S \le 40\}$ is the same event as $\{S < 40.5\}$, because S only takes integer values. Hence the correct threshold to standardize is 40.5, not 40.

Event	Exact	No correction	With correction
$P(S \le 40)$	0.028444	0.022750	0.028717
$P(S \le 55)$	0.864373	0.841345	0.864334
$P(S = 60)$	0.010844	0.010798	0.010852

Read that table line by line. In the first row, the approximation without the correction misses by 0.005694, whereas with the correction it misses by 0.000273, almost twenty-one times closer. In the third row both are already close, because near the center of the distribution a width of one unit is already small compared with $\sigma = 5$. The general lesson: the continuity correction helps most in the tails and when σ is small. The size of the shift in z that it produces is $1/(2\sigma)$, namely 0.1 in this example.

Recompute the first row of this table by hand. $z = (40 - 50)/5 = -2$, $\Phi(-2) = 0.022750$; then $z = (40.5 - 50)/5 = -1.9$, $\Phi(-1.9) = 0.028717$. The exact value is $\sum_{k=0}^{40} \binom{100}{k} 2^{-100} = 0.028444$. Check also that the error without the correction 0.005694 is about 21 times the error with the correction 0.000273.

This correction is not a new formula, only a rereading of the event. For a discrete variable with spacing h between values, the threshold is shifted by $h/2$ inward. For a continuous variable nothing needs shifting, because there are no jumps.

One bad habit deserves mention here. Many books write that the normal approximation "may be used when $n \geq 30$". That is not a theorem but a habit. The numbers in the lab section show why: for the exponential average, the supremum distance to the normal at $n = 30$ is 0.02429, and the 95 percent quantile of Z_{30} is 1.7419 whereas the normal gives 1.6449. That difference of 0.097 means a confidence interval that is too narrow by about six percent. What decides is not n alone, but the skewness and kurtosis of the original distribution.

19.6 When the Central Limit Theorem Misleads: Heavy Tails

Level: advanced

In everyday language: when one observation can occasionally be far larger than all the others, the average no longer settles, and the bell does not appear.

heavy tail (*ekor berat*) — a probability of huge values that does not shrink fast enough, so that one observation can outweigh all the rest

All the theorems above use one hypothesis that is easy to pass over: finite variance. The Cauchy distribution, with density

$$f(x) = \frac{1}{\pi(1 + x^2)}, \qquad x \in \mathbb{R},$$

does not have it. Its tail shrinks like $1/x$, which is too slow. The probability of exceeding ten is $1 - (2/\pi)\arctan 10 = 0.063451$, and the probability of exceeding one hundred is still 0.006366. Compare the standard normal, whose probability of exceeding ten is practically zero. Even $\mathbb{E}|X|$ is not finite: the expectation truncated at level M equals $(2/\pi)\ln(1 + M^2)$, namely 2.9381 for $M = 10$, 8.7952 for $M = 10^3$, and 17.5905 for $M = 10^6$. That number grows like a logarithm, so there is no limit.

The consequence is blunt, and not merely "the CLT is less accurate".

Proposition 19.9. *If $X_1, \ldots, X_n$ are iid Cauchy, then $\bar{X}_n$ has a Cauchy distribution for every n.*

Proof. The characteristic function of the Cauchy distribution is $\varphi(t) = e^{-|t|}$. Since the variables are independent, the characteristic function of $\bar{X}_n$ is $\left[e^{-|t|/n}\right]^n = e^{-|t|}$, which is the Cauchy characteristic function again. Since the characteristic function determines the distribution, $\bar{X}_n$ has a Cauchy distribution. □

So the average of 10^5 Cauchy observations is no more informative than one observation. The law of large numbers fails, the central limit theorem fails, and "standard error" means nothing. The numbers in the lab section show all three symptoms at once. The median of $|\bar{X}_n|$ stays around one for all n (1.004, 1.008, 1.040, 0.973), whereas for a distribution with finite variance it must shrink like $1/\sqrt{n}$. The probability $|\bar{X}_n| > 10$ stays around 0.0635, its theoretical value, for all n. The empirical standard deviation jumps about: 33.04, 863.61, 23.87, 379.53 for $n = 10$ up to 10^4. Those jumps are not a bug in the program. The sample standard deviation of Cauchy data is itself a random variable with no typical value.

Figure 19.3 shows the difference on one run. The running average of uniform data sticks to 0.5 after a few dozen draws. The running average of Cauchy data jumps from -0.0866 to 4.4572 at the twentieth draw, because one value 90.79 happened to arrive, and then wanders again. One giant value is enough to cancel the entire history before it.

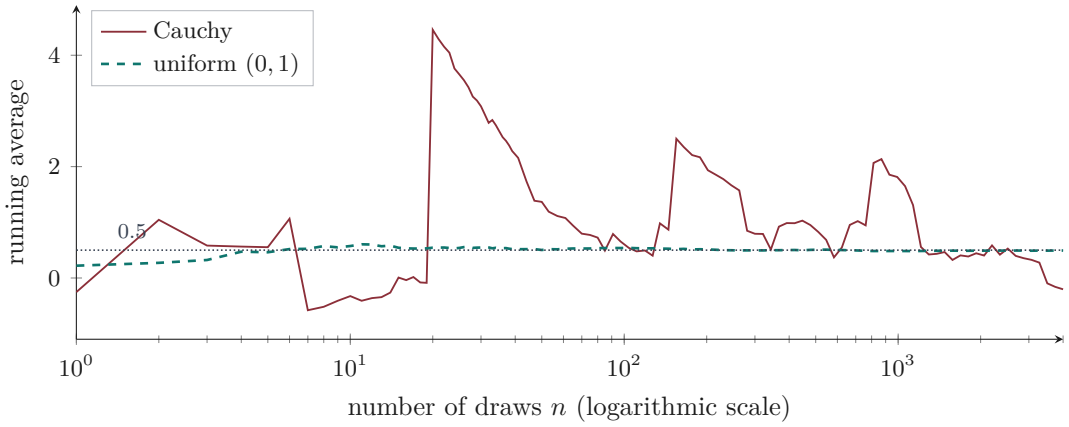


Figure 19.3: Running averages over 4000 draws. Dashed line: uniform on $(0, 1)$, settling near 0.5 and ending at 0.4930. Solid line: Cauchy, jumping from -0.0866 to 4.4572 at the twentieth draw when one value 90.79 arrived, then wandering again and ending at -0.203. The horizontal axis is logarithmic in width, so that the early jump is not hidden.

What replaces the central limit theorem is the family of stable distributions. For iid variables with tails that shrink like $x^{-\alpha}$ with $0 < \alpha < 2$, there are constants a_n such that

$$\frac{S_n - a_n}{n^{1/\alpha}} \rightsquigarrow Y_\alpha,$$

with Y_α the stable distribution of index α. For $\alpha = 1$ we get Cauchy and $n^{1/1} = n$, exactly as in the proposition above. For $\alpha \in (1, 2)$ the average still converges to μ, but the error shrinks like $n^{1-1/\alpha}$, slower than $1/\sqrt{n}$, and its shape is not a bell. This general statement is the result of Gnedenko and Kolmogorov; the details are in Feller volume 2, in the chapter on stable distributions.

If $\alpha > 2$ but the third moment does not exist, the central limit theorem still holds because the variance is finite. What is lost is the guarantee of a rate: the Berry–Esseen bound becomes infinite, so there is no longer any handle on the claim that the error shrinks like $1/\sqrt{n}$ with a controlled constant. In practice this is what bites most often: the distribution is not extreme like Cauchy, but it has enough tail to make confidence intervals too narrow at realistic sample sizes.

Challenge (problem solver)

A distribution with no mean. Take X with $P(X = 2^k) = P(X = -2^k) = 2^{-k-1}$ for $k = 1, 2, 3, \ldots$

Show two things. First, $\mathbb{E}|X| = \infty$, so there is no μ that may be approached. Second, the sample average does not converge in probability to any constant: the law of large numbers fails, it does not merely slow down.

Hint 1: what decides is not the average of all the terms, but the largest term. Ask how often one term exceeds the threshold n.

Hint 2: compute $n\,P(|X| > n)$ for n between two powers of two, then use the necessary condition $n\,P(|X_1| > n) \to 0$ stated in the section on the strong law.

If you get stuck, go back to the section on heavy tails and to the paragraph on the

necessary and sufficient condition in the section on the strong law.

For younger readers (age 10 and up)

In a village there are a hundred people with ordinary incomes, and one person who is extremely rich. If we compute the average income of that village, the number is decided almost entirely by that rich person. If the rich person happens to be away, the average drops drastically. If that person comes back, the average jumps up.

Now add inhabitants. Another hundred people, another thousand; as long as the newcomers are ordinary people, the average changes little. But if among the newcomers there is one more giant, the average jumps again. Adding many people does not help, because what decides is not the number of people but who is the largest.

This is what happens with data whose tails are heavy: one observation can outweigh all the others. The average becomes unstable, and the bell that is usually used to compute "how confident" does not appear at all.

The question that makes us curious: how can we know, before computing, whether our data are of the calm kind or of the kind ruled by giants? In this chapter we will find out why that question cannot be answered just by looking at the average.

19.7 Computational Lab: Five Checks

Level: advanced

The five claims of this chapter can be checked with one script: the $1/\sqrt{n}$ rate in the law of large numbers, the skewness and kurtosis of the exponential average, the supremum distance to the normal, the Monte Carlo standard error, and total failure for Cauchy. All the numbers below are real output from an execution with NumPy 2.5.2 and SciPy 1.18.0, generator `default_rng(20261024)`.

Computational lab

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(20261024)

print("== 1. law of large numbers: dice average ==")
for k in range(1, 6):
    n = 10**k
    m = rng.integers(1, 7, size=(2000, n)).mean(axis=1)
    print("n=%6d  mean=%.4f  rms=%.4f  sd/sqrt n=%.4f"
          % (n, m.mean(), np.sqrt(np.mean((m-3.5)**2)),
             np.sqrt(35/12/n)))

print("== 2. CLT: exponential average ==")
for n in (2, 5, 10, 30, 100, 500):
    z = (rng.exponential(size=(50000, n)).mean(axis=1)
         - 1)*np.sqrt(n)
    ks = stats.kstest(z, "norm")
    print("n=%4d skew=%+.3f (%+.3f) kurt=%+.3f (%+.3f)"
          % (n, stats.skew(z), 2/np.sqrt(n),
             stats.kurtosis(z), 6.0/n))
    print("      KS D=%.4f p=%.3f" % (ks.statistic, ks.pvalue))

print("== 3. sup distance to normal, exact via gamma ==")
rho = 2 + 2*(6/np.e - 2)          # E|X-1|^3 for X ~ Exp(1)
```

```
for n in (2, 5, 10, 30, 100, 500):
    zg = np.linspace(-4, 4, 40001)
    Fn = stats.gamma.cdf(n*(1+zg/np.sqrt(n)), a=n)
    D = np.max(np.abs(Fn - stats.norm.cdf(zg)))
    print("n=%4d sup|Fn-Phi|=%.5f  D*sqrt n=%.3f  BE=%.3f"
          % (n, D, D*np.sqrt(n), 0.4748*rho/np.sqrt(n)))

print("== 4. Monte Carlo standard error for E[exp(U)] ==")
for k in (2, 3, 4, 5):
    n = 10**k
    est = np.exp(rng.random(size=(2000, n))).mean(axis=1)
    print("n=%6d  sd=%.5f  sd*sqrt n=%.5f"
          % (n, est.std(), est.std()*np.sqrt(n)))

print("== 5. Cauchy: the average does not concentrate ==")
for k in (1, 2, 3, 4):
    n = 10**k
    m = rng.standard_cauchy(size=(4000, n)).mean(axis=1)
    print("n=%5d  median|m|=%.3f P(|m|>10)=%.4f sd=%.2f"
          % (n, np.median(np.abs(m)), np.mean(np.abs(m) > 10),
             m.std()))
print("theory P(|mean|>10) = %.4f"
      % (1 - 2/np.pi*np.arctan(10)))
```

Its output, verbatim:

```
== 1. law of large numbers: dice average ==
n=    10  mean=3.5014  rms=0.5440  sd/sqrt n=0.5401
n=   100  mean=3.4992  rms=0.1698  sd/sqrt n=0.1708
n=  1000  mean=3.4986  rms=0.0527  sd/sqrt n=0.0540
n= 10000  mean=3.5001  rms=0.0171  sd/sqrt n=0.0171
n=100000  mean=3.4997  rms=0.0053  sd/sqrt n=0.0054
== 2. CLT: exponential average ==
n=   2 skew=+1.440 (+1.414) kurt=+3.130 (+3.000)
      KS D=0.0929 p=0.000
n=   5 skew=+0.919 (+0.894) kurt=+1.393 (+1.200)
      KS D=0.0602 p=0.000
n=  10 skew=+0.631 (+0.632) kurt=+0.589 (+0.600)
      KS D=0.0480 p=0.000
n=  30 skew=+0.355 (+0.365) kurt=+0.181 (+0.200)
      KS D=0.0219 p=0.000
n= 100 skew=+0.196 (+0.200) kurt=+0.038 (+0.060)
      KS D=0.0167 p=0.000
n= 500 skew=+0.112 (+0.089) kurt=-0.003 (+0.012)
      KS D=0.0062 p=0.043
== 3. sup distance to normal, exact via gamma ==
n=   2 sup|Fn-Phi|=0.09446  D*sqrt n=0.134  BE=0.811
n=   5 sup|Fn-Phi|=0.05963  D*sqrt n=0.133  BE=0.513
n=  10 sup|Fn-Phi|=0.04211  D*sqrt n=0.133  BE=0.363
n=  30 sup|Fn-Phi|=0.02429  D*sqrt n=0.133  BE=0.209
n= 100 sup|Fn-Phi|=0.01330  D*sqrt n=0.133  BE=0.115
n= 500 sup|Fn-Phi|=0.00595  D*sqrt n=0.133  BE=0.051
== 4. Monte Carlo standard error for E[exp(U)] ==
n=   100  sd=0.04928  sd*sqrt n=0.49281
n=  1000  sd=0.01586  sd*sqrt n=0.50156
n= 10000  sd=0.00495  sd*sqrt n=0.49538
n=100000  sd=0.00155  sd*sqrt n=0.49056
== 5. Cauchy: the average does not concentrate ==
n=   10  median|m|=1.004 P(|m|>10)=0.0630 sd=33.04
n=  100  median|m|=1.008 P(|m|>10)=0.0680 sd=863.61
n= 1000  median|m|=1.040 P(|m|>10)=0.0653 sd=23.87
n=10000  median|m|=0.973 P(|m|>10)=0.0580 sd=379.53
```

```
theory P(|mean|>10) = 0.0635
```

Part 1 shows the law of large numbers at work, and shows its price. The rms column matches $\sigma/\sqrt{n}$ to two decimal places for $n \geq 100$: 0.1698 against 0.1708, 0.0171 against 0.0171, 0.0053 against 0.0054. But look at the magnitudes: at $n = 10^5$ the average still misses by about 0.005, which is 0.15 percent of 3.5. The average of 10^5 tosses is not "practically exact"; it is exact to two decimal places.

Parts 2 and 3 show two ways of measuring closeness to the normal. Skewness and kurtosis follow the theoretical prediction neatly from $n = 10$ on: 0.631 against $2/\sqrt{10} = 0.632$ and 0.589 against $6/10 = 0.600$. At $n = 2$ and $n = 5$ the kurtosis estimator is still noisy, because its deviation depends on the eighth moment of that distribution, not only on the number of repetitions. Part 3 computes the supremum distance exactly, without simulation, by using the Gamma distribution for the exponential average: 0.04211 at $n = 10$ and 0.00595 at $n = 500$. The column $D\sqrt{n}$ is nearly constant at 0.133, direct evidence that the rate is $1/\sqrt{n}$. The Berry–Esseen bound $1.1464/\sqrt{n}$ holds, with the numbers 0.209 against 0.02429 at $n = 30$; that bound is about 8.6 times looser than reality, which is normal for a bound that is uniform over all distributions.

The Kolmogorov–Smirnov test on fifty thousand draws rejects normality with $p = 0.000$ for all n up to 100, then gives $p = 0.043$ at $n = 500$. That is not a sign that the program is wrong: the normal approximation is never exact for finite n, and the distance merely shrinks until a test that large can no longer see it. The right question is not "is it normal" but "how large an error can we still accept", and part 3 answers it with numbers.

Part 4 separates two quantities that are often confused. The standard deviation of the estimator falls from 0.04928 to 0.00155 when n grows a thousandfold, whereas $\mathrm{sd} \cdot \sqrt{n}$ is nearly constant at 0.49. That constant value is the σ of $\exp(U)$, namely $\sqrt{(e^2-1)/2 - (e-1)^2} = 0.491971$. This is the $1/\sqrt{n}$ law in its barest form: for a precision of 0.001 at 95 percent confidence we need $(1.96 \cdot 0.491971/0.001)^2 = 929803$ draws.

Part 5 is the counterweight. The median of $|\bar{X}_n|$ does not shrink; the probability of exceeding ten does not shrink; and the empirical standard deviation jumps from 33.04 to 863.61 and then down to 23.87. Theory says 0.0635, and all five simulation numbers hover around it. There is no law of large numbers here, and no central limit theorem. The only way to rescue the analysis is to change the question: estimate the tail index, not the mean.

Note for instructors

The teaching order I suggest: start from the simulation table, not from the definitions. Show the class that the error of one run follows no pattern at all, whereas the distribution of the error does. After that, the three modes of convergence, with a counterexample for each direction that fails; do not postpone those examples, because that is where students understand that "approaching" needs to be made precise. Only then Markov, Chebyshev, the weak law, the strong law as a statement, and then the central limit theorem with the moment generating function proof.

Two misconceptions appear most often. First, students read "converges in probability" as "the value becomes fixed", and are then surprised when a simulation shows big jumps at large n; the split $(0, 1)$ example in this chapter is the antidote. Second, they read the central limit theorem as "the average becomes normal", whereas what becomes normal is the error divided by $\sigma/\sqrt{n}$; as a result they write $\bar{X}_n \approx N(\mu, \sigma^2)$ and forget $\sqrt{n}$.

A blackboard activity that works: ask two students to play the gambler and the bookmaker. The gambler names a tolerance, the bookmaker names the number of tosses that guarantees it, and the class computes with Chebyshev and then compares against the exact numbers from SciPy. The gap is always large, and that is the doorway into the

normal approximation. Sign that the class is ready to move on: they can explain why Chebyshev is still useful although it is loose, and they can name one distribution that makes the central limit theorem fail.

19.8 Summary

Summary.

- There are three different modes of convergence: almost sure, in probability, and in distribution. Almost sure implies in probability, in probability implies in distribution, and the reverse directions fail unless the limit is a constant.
- Markov's and Chebyshev's inequalities use only moments, which is why they hold for every distribution with finite variance, and why they are loose.
- Weak law of large numbers: $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \to 0$. The strong law adds that $\bar{X}_n \to \mu$ almost surely when $\mathbb{E}|X_1| < \infty$, without demanding a variance.
- Central limit theorem: $\sqrt{n}(\bar{X}_n - \mu)/\sigma \rightsquigarrow N(0,1)$ when the variance is finite. The rate is $1/\sqrt{n}$, hence four times the data for half the error, and a hundred times the data for one extra digit.
- The normal approximation for discrete sums requires a continuity correction of half a unit; in the binomial example of this chapter that correction cuts the error in the tail by about a factor of twenty.
- If the variance is infinite, the central limit theorem fails, it does not merely slow down. The average of n Cauchy observations is still Cauchy, and what replaces the bell is a stable distribution of index $\alpha < 2$.

19.9 Exercises

Exercise 19.1.

1. Prove: if $X_n \to X$ and $Y_n \to Y$ in probability, then $X_n + Y_n \to X + Y$ in probability. Write down the set inequality you use, then explain why $\varepsilon/2$ appears.
2. Prove that Chebyshev's inequality is sharp: for $0 < \sigma^2 \leq a^2$, take X with $P(X = a) = P(X = -a) = \sigma^2/(2a^2)$ and $P(X = 0) = 1 - \sigma^2/a^2$, then compute $\operatorname{Var}(X)$ and $P(|X| \geq a)$.
3. Evaluate this claim: "Since the central limit theorem holds for all distributions, the average of thirty observations may always be treated as normal." True, false, or not enough evidence? Name the hypothesis that is violated or that has not been checked, and use numbers from the lab section.
4. For a coin with $p = 1/2$ and tolerance $\varepsilon = 0.05$, find the smallest n that Chebyshev guarantees for a probability of missing of at most 0.05. Then find the smallest n that really satisfies that condition, by computing the binomial distribution in Python. Compare the two numbers.
5. Prove that convergence in probability does not guarantee convergence of expectations, using $P(X_n = n^2) = 1/n$ and $P(X_n = 0) = 1 - 1/n$. Compute $\mathbb{E}[X_n]$ and $P(|X_n| > \varepsilon)$ for fixed ε.
6. For $S \sim \text{Binomial}(200;\, 0.3)$, approximate $P(S \leq 50)$ with the normal without and with the continuity correction, then compare both with the exact value from SciPy. Report both errors.
7. Generate 4000 averages of n Cauchy numbers for $n = 10, 10^2, 10^3, 10^4$, then report the median of $|\bar{X}_n|$, the probability $|\bar{X}_n| > 10$, and the standard deviation. Compare with $1 - (2/\pi)\arctan 10$ and explain why the standard deviation column does not shrink.

8. Prove with characteristic functions that the average of n iid Cauchy variables has a Cauchy distribution. Then explain why a report of the form "average $\pm$ two standard errors" on such data cannot be interpreted.
9. How many Monte Carlo draws are needed for the estimator $\mathbb{E}[\exp(U)]$ to have a 95 percent interval half-width of 0.0005? Use $\sigma = 0.491971$ from the lab section, then compare with the n needed for a half-width of 0.001. State the ratio and explain where it comes from.
10. Prove that if $X_n \rightsquigarrow c$ with c a constant, then $X_n \to c$ in probability, without using part (c) of the theorem on the relations among the three modes at the start of the chapter, then show by example that the hypothesis "constant" cannot be dropped.

Hints for the three hardest items. Item 1: write $\{|(X_n + Y_n) - (X + Y)| > \varepsilon\}$ as contained in the union of two events, then use the union bound. Item 8: the characteristic function of a sum is the product of the characteristic functions, and the Cauchy characteristic function $e^{-|t|}$ is special because its form does not change when it is raised to a power. Item 10: write $P(X_n \le t) \le P(X_n \le c - \varepsilon) + P(|X_n - c| > \varepsilon)$ for t to the left of c, and remember that a point-mass distribution jumps exactly at c.

For researchers: limits and next steps

What is really proved in this chapter: the implications almost sure $\Rightarrow$ in probability $\Rightarrow$ in distribution, the weak law via Chebyshev, and the central limit theorem via moment generating functions under the assumption $\mathbb{E}|Y_1|^3 < \infty$. What is popular but not proved here: the strong law (the statement only, Etemadi's proof is skipped), the version of the central limit theorem without a third moment (it needs characteristic functions and Levy's continuity theorem), and the whole theory of stable distributions (the Gnedenko–Kolmogorov statement, without proof).

Primary sources I used: Wasserman, *All of Statistics*, chapter 5, sections 5.2 (types of convergence and the implication diagram), 5.3 (the weak law), 5.4 (the central limit theorem and Berry–Esseen), plus appendix 5.7.1 and 5.7.2 for the strong law and the proof of the central limit theorem. Blitzstein and Hwang, *Introduction to Probability*, chapter 10, sections 10.1 (inequalities), 10.2 (the law of large numbers), 10.3 (the central limit theorem). For the heavy-tail side: Feller volume 2.

Limits of validity that must be mentioned. First, all the theorems here assume independence and identical distribution; for dependent data, for instance time series or iterates of a Markov chain, what holds are martingale or mixing versions, and the rate can be far slower. Second, the optimal constant in Berry–Esseen is not known; 0.4748 suffices, and the best lower bound for that constant is still below 0.5. Third, the rate $1/\sqrt{n}$ is the worst rate that can be guaranteed uniformly; for particular distributions the true rate is better, and for heavy tails with $\alpha \in (1, 2)$ the true rate is worse. Open questions that are still alive: how sharp the Berry–Esseen bound is for classes of distributions with bounded moments, and how to estimate the tail index with valid confidence intervals at small n.

Map to source texts: Chapter 19 in Wasserman, All of Statistics*: chapter 5 (Convergence of Random Variables), sections 5.2 Types of Convergence, 5.3 The Law of Large Numbers, 5.4 The Central Limit Theorem, and appendix 5.7.1 Almost Sure and L1 Convergence as well as 5.7.2 Proof of the Central Limit Theorem. In Blitzstein and Hwang,* Introduction to Probability*: chapter 10 (Inequalities and limit theorems), sections 10.1 Inequalities, 10.2 Law of large numbers, 10.3 Central limit theorem. For stable distributions: Feller,* An Introduction to Probability Theory and Its Applications*, volume 2. The basic distributions used here are built in Chapter 17, and the confidence intervals that rest on this chapter are discussed in Chapter 19.*

Chapter 20

Estimation, Confidence Intervals, and Hypothesis Testing

A team tests an attack detector on 50 labeled examples, and the detector catches 33 of them. The report written that night reads: recall 66 percent. A second detector, tested on 100 examples, catches 53: recall 53 percent. The morning meeting decides to use the first, because 66 is larger than 53, and that decision is finished in one minute.

The arithmetic is correct: $33/50 = 0.66$. What is wrong is the assumption that this number points at one number. If those fifty examples are replaced by fifty other examples from the same collection, the result is no longer 33; it wobbles, and the wobble is not small. After two hours of writing the program, the team is in fact holding two sentences of very different strength. The first sentence, "on this data we see 66 percent", says almost nothing. The second, "the true catch rate lies somewhere inside this interval", says something usable, provided we know how wide that interval is and what that width means.

Chapter 19 already gave two tools: the law of large numbers, which says the sample mean concentrates at the true value, and the central limit theorem, which says the error spreads normally with standard deviation $\sigma/\sqrt{n}$. Both are statements about many trials at once, and neither answers the question of the morning meeting: from one trial already finished, what sentence may we utter about a parameter we have never seen, and how often is that sentence wrong if the procedure is repeated. This chapter builds the answer, from estimators to p-values.

For younger readers (age 10 and up)

A farmer wants to know how many weeds there are in his field. The whole field is too large to inspect, so he takes twenty small plots spread around, then counts the weeds in each plot. From those twenty numbers he guesses the state of the whole field. The trouble is that a guess is always a single number, whereas the state of a field is never that simple.

Now the part that makes one curious. In those twenty plots he finds no weeds at all. May he go home and say that his field is clean? Certainly not. It may be that there are really no weeds, and it may be that he merely happened to choose plots made of bare soil. All he can say is: if the weeds were plentiful, twenty empty plots would be almost impossible. From that he can compute an honest upper bound, for instance "at most three in a hundred plots".

The same holds when a teacher grades a few students' quizzes and when a computer program is tested on a few examples. In this chapter we shall learn why one number is never enough, why the honest answer is always an interval, and why the interval widens

when the evidence is thin.

20.1 Estimators, bias, and standard error

Level: basic

Every statistical conclusion begins with one small decision: which quantity we compute from the data. The mean, a proportion, the sample standard deviation, all are functions of the data and require no knowledge of the parameter being sought. In everyday language: an estimator is a surveyor who guesses the height of a tree from the length of its shadow, and we need to know whether that surveyor is systematically off or merely unsteady.

estimator (*penaksir*) — a surveyor who guesses the height of a tree from the length of its shadow, rather than from the tree itself

Definition 20.1 (Statistic and estimator)**.** Let $X_1, \dots, X_n$ be a random sample from a distribution that depends on a parameter θ. A *statistic* is a function of $X_1, \dots, X_n$ that does not contain θ. A statistic used to guess θ is called an *estimator* and written $\hat{\theta}$; the value obtained on one sample is called an *estimate*.

An estimator is a random variable, because it is a function of random variables. Hence it has an expectation and a variance, and those two quantities determine its quality. The expectation answers the question "does the guess keep missing in one direction", while the variance answers "how far does the guess wobble from sample to sample".

Definition 20.2 (Bias and unbiased estimator)**.** The bias of the estimator $\hat{\theta}$ for θ is $\operatorname{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$. An estimator is called *unbiased* if its bias is zero, namely $\mathbb{E}[\hat{\theta}] = \theta$ for every possible value of θ.

The sample mean is unbiased for μ, and that was already used in the proof of the law of large numbers in Chapter 19, so we do not repeat it here. More interesting is the estimator used for the variance. With divisor n, the estimator $S_n^2 = \frac{1}{n}\sum (X_i - \bar{X}_n)^2$ is biased downward; with divisor $n-1$, the estimator $S^2 = \frac{1}{n-1}\sum (X_i - \bar{X}_n)^2$ is unbiased. The reason is that $\sum (X_i - \bar{X}_n)^2$ loses one degree of freedom because $\bar{X}_n$ is itself computed from the same data, so its expectation is $(n-1)\sigma^2$ and not $n\sigma^2$. The divisor $n-1$ is a correction for that loss, not mere tidiness.

The quality of an estimator is usually summarized by one number, the mean squared error. In everyday language: how large an error we expect, computed in a way that punishes large errors more heavily.

Definition 20.3 (Mean squared error)**.** $\operatorname{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$.

Theorem 20.1 (MSE decomposition)**.** *For an estimator $\hat{\theta}$ with finite variance,*

$$\operatorname{MSE}(\hat{\theta}) = \operatorname{bias}(\hat{\theta})^2 + \operatorname{Var}(\hat{\theta}).$$

Proof. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}\hat{\theta}) + (\mathbb{E}\hat{\theta} - \theta)$. Square it and take the expectation:

$$\mathbb{E}[(\hat{\theta} - \theta)^2] = \mathbb{E}[(\hat{\theta} - \mathbb{E}\hat{\theta})^2] + 2(\mathbb{E}\hat{\theta} - \theta)\,\mathbb{E}[\hat{\theta} - \mathbb{E}\hat{\theta}] + (\mathbb{E}\hat{\theta} - \theta)^2.$$

The middle term vanishes because $\mathbb{E}[\hat{\theta} - \mathbb{E}\hat{\theta}] = 0$, while the first term is the variance and the third is the bias squared. $\square$

That identity explains why "unbiased" is not a championship title. An unbiased estimator may have a large variance, and a slightly biased estimator may have a far smaller variance so that its MSE is lower. To estimate μ from n observations, the statistic X_1 is unbiased with variance σ^2, whereas $\bar{X}_n$ is unbiased with variance σ^2/n; at $n = 20$ the difference is a factor of twenty.

The second example shows a bias that must be computed, because it is used every day for proportions. If $K \sim \mathrm{Bin}(n, p)$ and $\hat{p} = K/n$, then $\hat{p}(1-\hat{p})$ is the natural estimator for $p(1-p)$, that is the variance of one trial. Its expectation is

$$\mathbb{E}[\hat{p}(1-\hat{p})] = \mathbb{E}[\hat{p}] - \mathbb{E}[\hat{p}^2] = p - \Big(p^2 + \frac{p(1-p)}{n}\Big) = p(1-p)\Big(1 - \frac{1}{n}\Big),$$

so its bias is $-p(1-p)/n$. For $p = 0.3$ and $n = 20$ the value is $p(1-p) = 0.21$ and the bias is -0.0105, that is five percent downward. At $n = 200$ the bias is only -0.00105. A bias that shrinks like $1/n$ is typical: it disappears at infinity, but it is present in every finite sample.

The second property demanded of an estimator is that it actually heads for the target as the data grow. In everyday language: with enough evidence, a wrong guess becomes increasingly impossible, not merely increasingly rare.

Definition 20.4 (Consistent)**.** A sequence of estimators $\hat{\theta}_n$ is *consistent* for θ if for every $\varepsilon > 0$ we have $P(|\hat{\theta}_n - \theta| \geq \varepsilon) \to 0$ as $n \to \infty$.

Proposition 20.2 (MSE going to zero suffices for consistency)**.** *If* $\mathrm{MSE}(\hat{\theta}_n) \to 0$*, then* $\hat{\theta}_n$ *is consistent for* θ*.*

Proof. By Chebyshev's inequality, already proved in Chapter 19,

$$P(|\hat{\theta}_n - \theta| \geq \varepsilon) \leq \frac{\mathbb{E}[(\hat{\theta}_n - \theta)^2]}{\varepsilon^2} = \frac{\mathrm{MSE}(\hat{\theta}_n)}{\varepsilon^2}.$$

For fixed ε the right-hand side goes to zero, and the inequality closes the proof. □

Note that Chebyshev uses the MSE, not the variance. The bias term is squared, so a biased estimator is still consistent provided its bias goes to zero. Conversely, the estimator X_1 is unbiased for every n but never uses the rest of the data, so it is not consistent.

The next quantity is used continually until the end of the chapter. The variance of an estimator says how far the guess wobbles, and its square root has a name of its own.

standard error (*galat baku*) — the range of wobble of one guess, not the width of the whole data set

Definition 20.5 (Standard error)**.** The *standard error* of the estimator $\hat{\theta}$ is $\mathrm{se}(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$. If the formula contains an unknown parameter, the *estimated* standard error is obtained by replacing that parameter with its estimate.

For a proportion, $\mathrm{se}(\hat{p}) = \sqrt{p(1-p)/n}$, and the estimated version is $\sqrt{\hat{p}(1-\hat{p})/n}$. For the data 33 out of 50, $\hat{p} = 0.66$ and

$$\sqrt{\frac{0.66 \cdot 0.34}{50}} = \sqrt{0.004488} = 0.066993.$$

For twelve measurements of delay time (in milliseconds),

$$18.2,\ 19.1,\ 17.8,\ 20.3,\ 18.9,\ 19.7,$$
$$18.4,\ 21.0,\ 17.5,\ 19.9,\ 18.6,\ 20.1,$$

the mean is 19.125 and the sample standard deviation 1.082191. The standard error is $1.082191/\sqrt{12} = 0.312401$. That number does not say that the true mean is 0.312 away from 19.125; it says that guesses like this one, if repeated, wobble on a scale of about 0.31. The confidence interval is the way to turn the last sentence into a sentence we can use.

20.2 Confidence intervals for a mean

Level: intermediate

A single number without a measure of wobble is easily misused. What is needed is a pair of numbers that, with high probability, traps the parameter. In everyday language: we do not shoot at one point, we set a net and promise that the net catches the fish we are after on most throws.

confidence interval (*interval kepercayaan*) — a net set with a certain width, and a promise that the net catches the target on most throws

Definition 20.6 (Confidence interval and coverage)**.** Let $C(X_1, \dots, X_n)$ be an interval computed from the data. C is called a *confidence interval* of level $1-\alpha$ for θ if $P_\theta(\theta \in C) \geq 1-\alpha$ for every θ. That probability is called the *coverage* of the procedure.

Note where the probability sits: P_θ is computed before the data are taken, and what is random is C, not θ. This is where misinterpretation most often occurs, and we return there once the formula exists.

Theorem 20.3 (The z-interval)**.** *Let $X_1, \dots, X_n$ be independent and distributed $N(\mu, \sigma^2)$ with σ known. For $z = z_{1-\alpha/2}$, the interval*

$$C = \Big[\bar{X}_n - z\frac{\sigma}{\sqrt{n}},\ \bar{X}_n + z\frac{\sigma}{\sqrt{n}}\Big]$$

has coverage exactly $1-\alpha$. For other distributions with finite variance, its coverage tends to $1-\alpha$ as $n \to \infty$.

Proof. Let $Z = \sqrt{n}(\bar{X}_n - \mu)/\sigma$. For a normal sample, $Z \sim N(0,1)$ exactly; for other distributions, Z tends to $N(0,1)$ by the central limit theorem in Chapter 19. The inequality $\mu \in C$ is equivalent to $|Z| \leq z$, because

$$|\bar{X}_n - \mu| \leq z\frac{\sigma}{\sqrt{n}} \iff \frac{\sqrt{n}\,|\bar{X}_n - \mu|}{\sigma} \leq z.$$

So $P(\mu \in C) = P(|Z| \leq z) = 1-\alpha$ in the normal case, and tends to $1-\alpha$ in the general case. □

If σ is unknown, as in almost all real data, replacing σ by s adds uncertainty and thickens the tails. The result uses the t distribution with $n-1$ degrees of freedom, published by Gosset in 1908 under the pen name Student:

$$C = \Big[\bar{X}_n - t_{n-1,1-\alpha/2}\frac{s}{\sqrt{n}},\ \bar{X}_n + t_{n-1,1-\alpha/2}\frac{s}{\sqrt{n}}\Big],$$

with coverage exactly $1-\alpha$ if the sample is normal. The proof requires the joint distribution of $\bar{X}_n$ and s to be independent, and it is in Wasserman, *All of Statistics*, Chapter 6. For the $n = 12$ delay-time data above, $t_{11,0.975} = 2.200985$, so the half-width is $2.200985 \cdot 0.312401 = 0.687591$ and the interval is $(18.4374;\ 19.8126)$. With $z = 1.96$ the half-width is 0.612296: admitting that σ is unknown widens the interval by twelve percent at $n = 12$, and by one percent at $n = 100$.

Now the interpretation, and here we must go slowly. Once the data are taken, C is no longer a random variable: it is the fixed interval $(18.4374;\ 19.8126)$. The true mean μ is also a fixed number, only unknown. The sentence "the probability that μ lies inside $(18.4374;\ 19.8126)$ equals 0.95" is therefore meaningless in this framework: no probability is left, because nothing is random. What has probability 0.95 is *the procedure*, and that is a statement about a trial not yet run.

Figure 20.1 runs that trial twenty times, with $\mu = 19.125$ and $\sigma = 1$ known: nineteen intervals contain μ, one does not.

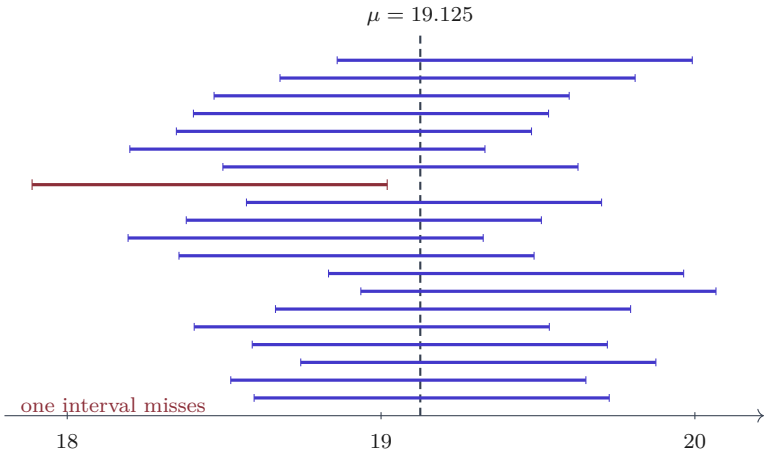


Figure 20.1: Twenty 95 percent confidence intervals for μ, each from one sample of size $n = 12$ with $\sigma = 1$ known, so the half-width is $1.96/\sqrt{12} = 0.5658$ and the width is 1.1316. The vertical dashed line is the true $\mu = 19.125$. Nineteen intervals contain it; the interval in row 13, namely (17.8880; 19.0196), does not. One failure in twenty is a reasonable number for a 95 percent promise, not a sign of a broken procedure.

Note 20.1. Three sentences that are true, and one that is false. True: "this procedure catches μ on 95 percent of repeated trials". True: "we are 95 percent confident that μ lies between 18.44 and 19.81", provided "confident" is read as a summary of repeated trials. True: "this procedure is wrong in about five cases out of a hundred". False: "the probability that μ is in this interval is 0.95". The last sentence pretends to know something about μ that the data do not provide.

The width of an interval shrinks like $1/\sqrt{n}$, not $1/n$: halving it demands four times the data, making it a tenth demands a hundred times. This rule can be computed before the experiment starts, since a half-width $z\sigma/\sqrt{n} \le d$ is equivalent to $n \ge (z\sigma/d)^2$; with $\sigma = 1$, $z = 1.96$, and $d = 0.1$, we need $n \ge 385$.

Do it by hand (kinesthetic learner)

Prepare twenty cards, write the numbers 1 to 20 on the cards, one number per card. Shuffle, draw five cards without looking, add them up, then divide by five. Write the result on one line of paper. Return the five cards, shuffle again, repeat. Do this thirty times, always five cards, and arrange those numbers as points on one number line.
What you should see: the points do not pile up at one number. The mean of all the cards is really 10.5, and your points scatter around it, some 8, some 13. Now repeat the whole job with ten cards per draw, thirty times more. The second spread is narrower, but not by half: it narrows by roughly the square root of two, because the number of cards doubled. Paste the two rows of points side by side, and see for yourself that doubling the work only trims the wobble by about thirty percent.

20.3 Proportions: Wald, Wilson, and the rule of three

Level: intermediate

For proportions, the most widely used formula is the simplest version of the previous theorem: replace $\bar{X}_n$ by $\hat{p}$ and $\sigma/\sqrt{n}$ by the standard error of the proportion. In everyday language: we set a symmetric interval around the observed proportion, twice its standard error wide.

Definition 20.7 (Wald interval)**.** The Wald interval of level $1-\alpha$ for a proportion is

$$\hat{p} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \qquad z = z_{1-\alpha/2}.$$

That formula is reasonable as long as $\hat{p}$ is far from zero and one. The difficulty appears exactly where the data are scarcest, and its shape is not smooth. If $\hat{p} = 0$, the formula gives the interval $[0; 0]$, of zero width: a claim of complete confidence that the event rate is zero, when all we have is the absence of events in n trials. If $\hat{p} = 1/20$, the lower limit is -0.0455, a number that is not a probability.

The cause is not rounding. Wald uses $\hat{p}$ for two jobs at once: as the center point and as the variance estimator. When $\hat{p}$ approaches zero, the estimated variance shrinks with it, although the real uncertainty is precisely large on that scale.

The way out is to invert the question: instead of assuming that $\hat{p}$ spreads normally around the true p, we ask which values of p are still plausible given the observed data. The criterion is that the normalized distance between $\hat{p}$ and p does not exceed z, with the variance in the denominator using p, not $\hat{p}$.

Wilson interval (*interval wilson*) — an interval found by asking which values are still plausible, rather than by setting a symmetric interval around the guess

Theorem 20.4 (Wilson interval)**.** *For $0 \le k \le n$, $\hat{p} = k/n$, and $z > 0$, the set of $p \in [0,1]$ satisfying*

$$n(\hat{p}-p)^2 \le z^2 p(1-p)$$

is a closed interval with endpoints

$$\frac{\hat{p} + \dfrac{z^2}{2n} \;\pm\; z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}.$$

Both endpoints always lie in $[0,1]$, and the interval is never empty.

Proof. Rearrange the inequality into $(n+z^2)p^2 - (2n\hat{p} + z^2)p + n\hat{p}^2 \le 0$. Its discriminant is

$$(2n\hat{p}+z^2)^2 - 4(n+z^2)n\hat{p}^2 = 4z^2\Big(n\hat{p}(1-\hat{p}) + \frac{z^2}{4}\Big) \ge 0,$$

so the roots are real and the quadratic formula gives the form in the theorem statement after numerator and denominator are divided by $2n$.

For the location, let $f(p) = n(\hat{p}-p)^2 - z^2p(1-p)$. The coefficient of p^2 is $n + z^2 > 0$, so $\{f \le 0\}$ is the interval $[r_-, r_+]$. Because $f(0) = n\hat{p}^2 \ge 0$, the point 0 is not in the interior of that interval, so $r_- \ge 0$; because $f(1) = n(1-\hat{p})^2 \ge 0$, neither is the point 1, so $r_+ \le 1$. Finally $f(\hat{p}) = -z^2\hat{p}(1-\hat{p}) \le 0$ places $\hat{p}$ inside the interval, so the interval is not empty. □

Now compare the two on the same data. The following table uses $z = 1.96$ and 95 percent confidence intervals; every number is computed with the formulas above and rechecked with Python.

data	$\hat{p}$	Wald 95 percent	Wilson 95 percent	Wilson width
33/50	0,660	(0.5287; 0.7913)	(0.5215; 0.7756)	0,254
53/100	0,530	(0.4322; 0.6278)	(0.4329; 0.6249)	0,192
13/20	0,650	(0.4410; 0.8590)	(0.4329; 0.8188)	0,386
1/20	0,050	(−0.0455; 0.1455)	(0.0089; 0.2361)	0,227
0/20	0,000	(0; 0)	(0; 0.1611)	0,161

The first two rows barely differ: when $\hat{p}$ is near one half, both are legitimate and agree to two decimal places. The fourth and fifth rows show the separation. At 1/20 the Wald interval is *narrower* than Wilson, 0.191 against 0.227, and that narrowness is the sign of an error: its lower limit falls outside $[0, 1]$. At 0/20 Wald gives zero width, whereas Wilson leaves room up to 0.1611. The conclusion is not that Wilson is always better, but that the width of an interval cannot be judged without its coverage.

For the zero-event case there is a special result older than Wilson and easier to remember. In everyday language: no events at all is not proof that the event rate is zero, and the upper limit is about three divided by the number of trials.

rule of three (*aturan tiga*) — if no event occurs in a given number of trials, the event rate may still be as high as three divided by the number of trials

Theorem 20.5 (Rule of three)**.** *Let $K \sim \mathrm{Bin}(n, p)$ and suppose $K = 0$ is observed. The one-sided upper limit at the 95 percent level is $p_{\mathrm{hi}} = 1 - 0.05^{1/n}$, and*

$$p_{\mathrm{hi}} \leq \frac{\ln 20}{n} < \frac{3}{n}.$$

Proof. The probability of observing zero events is $(1 - p)^n$. Setting that probability equal to 0.05 gives $(1 - p_{\mathrm{hi}})^n = 0.05$, that is $p_{\mathrm{hi}} = 1 - 0.05^{1/n} = 1 - e^{-c/n}$ with $c = \ln 20 = 2.9957\ldots$ Because $1 - e^{-x} \leq x$ for $x \geq 0$, we obtain $p_{\mathrm{hi}} \leq c/n < 3/n$. □

For $n = 100$, the rule of three gives $3/100 = 0.03$, the exact limit 0.0295130, and the two-sided Wilson upper limit at $k = 0$ is $z^2/(n + z^2) = 0.0369935$. All three numbers answer the same question with three different promises; the rule of three is the exact one-sided limit, Wilson a two-sided approximation. What is not allowed is to quote a number without naming the procedure.

20.4 Hypothesis tests and p-values

Level: intermediate

Up to here the question took the form “what is its value”. The next question takes the form “is its value equal to a particular number”. In everyday language, we file a charge that there is no effect, then check whether the data are surprising enough to overturn that charge.

Definition 20.8 (Null hypothesis, alternative, and level)**.** The *null hypothesis* H_0 is a statement about the parameter to be tested; the *alternative hypothesis* H_1 is the set of values held to be true if H_0 is rejected. The *level* α is the probability of rejecting H_0 when H_0 is in fact true. The rejection rule is determined by one test statistic T and one rejection region R: we reject H_0 if $T \in R$.

The level α is chosen before the data are seen, and therein lies the whole credibility of the procedure: we are willing to wrongly accuse in a small fraction of trials, and we state how large that fraction is. The p-value moves the same idea to a single data set.

Definition 20.9 (p-value)**.** For data x and test statistic T, the p-value is the probability of obtaining a result at least as extreme as the observed one if H_0 is true,

$$\text{p-value} p = \sup_{\theta \in H_0} P_\theta\big(T \text{ as far as } T(x)\big).$$

We reject H_0 at level α if the p-value $\leq \alpha$.

The phrase "at least as far" needs to be fixed for a two-sided test, and the usual convention is to add the two tails: for a symmetric z statistic, the p-value $= 2(1-\Phi(|z|))$. That definition makes the p-value a random variable, and its behavior under H_0 can be stated exactly.

Theorem 20.6 (The p-value is uniform under H_0)**.** *If H_0 is true and T has a continuous distribution with distribution function F_0, then $P_{H_0}(p\text{-value} \le \alpha) = \alpha$ for every $\alpha \in (0,1)$. Consequently $P_{H_0}(p\text{-value} \le 0.05) = 0.05$: a test with the rule "reject if the p-value $\le \alpha$" has level exactly α.*

Proof. For a right one-sided test, the p-value $= 1 - F_0(T)$, so

$$P_{H_0}(1 - F_0(T) \le \alpha) = P_{H_0}(F_0(T) \ge 1 - \alpha) = P_{H_0}(T \ge F_0^{-1}(1-\alpha)) = \alpha,$$

because $F_0(T)$ is uniformly distributed on $(0,1)$ when F_0 is continuous. For a two-sided test, the p-value depends on $|T|$ and the proof uses the same step on the distribution of $|T|$. □

That uniformity has a consequence often missed: under H_0 the p-value spreads evenly, so a p-value $= 0.04$ appears in four percent of trials whose null hypothesis is true. What distinguishes two trials is not the single p-value but the power of each.

Read it aloud (auditory learner)

Two voices, taking turns.
"The p-value is point zero four. So the probability that there is no effect is four percent."
"No. The p-value counts the data, not the charge. If there were no effect, and this trial were repeated a hundred times, about four times the data would look this far out."
"So it is the data that get counted."
"Yes, the data. The charge is only a place to stand."
"And if the p-value is large?"
"Then the data are not surprising. The silence of a scale is not proof that the goods are light."
Say it once more, slowly: the p-value is the probability of the data, not the probability of the charge. Repeat that line until it feels natural in the mouth: probability of the data, not probability of the charge.

Step-by-step recipe (procedural learner)

1. Write H_0 and H_1 as statements about one parameter, before computing anything.
2. Fix α and its rejection region, then pause. Do not look at the data yet.
3. Compute the test statistic with the standard error under H_0, then the p-value.
4. Compare the p-value with the α already fixed, and report the interval together with that decision.

A warning about order: step 2 must be finished before step 3. If α is chosen after the data are seen, α is no longer a promise but a record of the result, and step 4 loses its meaning. What may not be swapped: step 2 and step 3.

Now an example with numbers. For the data 33 out of 50 and $H_0: p = 0.5$, the z statistic uses the standard error under H_0, namely $\sqrt{0.25/50} = 0.070711$:

$$z = \frac{0.66 - 0.5}{0.070711} = 2.2627, \qquad \text{p-value}p = 2(1 - \Phi(2.2627)) = 0.0237.$$

The exact binomial test on the same rule, $P(X \ge 33) + P(X \le 17)$ for $X \sim \text{Bin}(50; 0.5)$, gives 0.0328. Both numbers reject H_0 at the five percent level, but the normal approximation

reports a p-value almost thirty percent smaller. At $n = 50$ a difference that large is enough to flip the decision when the level is 0.03; a good report states which test was used, not only its p-value.

Figure 20.2 places those two limits on the distribution of z under H_0: two tails of area 0.0250 each, and the observed statistic 2.2627 inside the right tail.

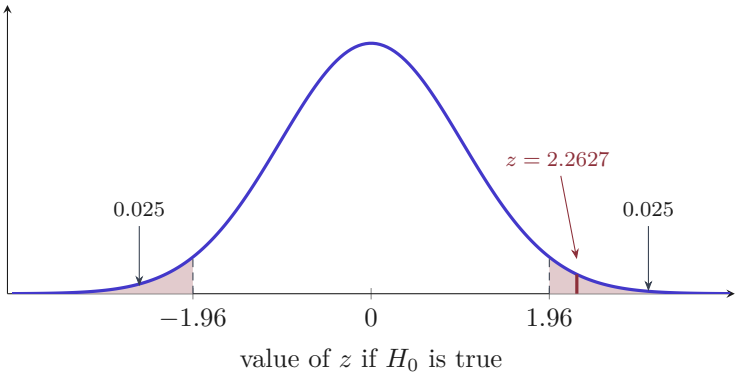


Figure 20.2: The distribution of the statistic $z = \sqrt{n}(\hat{p} - p_0)/\sqrt{p_0(1-p_0)}$ when H_0 is true, for the data 33 out of 50 with $H_0 : p = 0.5$. The two dashed lines at ± 1.96 cut off the two shaded tails, each of area $1 - \Phi(1.96) = 0.0250$, so the two-sided level is 0.05. The thick line at 2.2627 is the statistic observed on that data; the right tail from that point has area 0.0118, and twice that area is the p-value 0.0237 that the normal test reports.

A confidence interval and a hypothesis test are not two different procedures; they are two ways of reading the same set.

Proposition 20.7 (Equivalence of interval and test)**.** *For the Wald interval and the two-sided z-test that uses the standard error $\hat{p}$, we have that p_0 is contained in the Wald interval of level $1 - \alpha$ if and only if $H_0 : p = p_0$ is not rejected at level α. More generally, the Wilson interval is the set of p_0 not rejected by the score test at level α.*

Proof. For the Wald pair, both statements use the same inequality, namely $|\hat{p} - p_0| \le z\sqrt{\hat{p}(1-\hat{p})/n}$: the first reads it as "p_0 lies inside the interval", the second as "the test statistic does not exceed z". For Wilson, the set $\{p_0 : n(\hat{p} - p_0)^2 \le z^2 p_0(1-p_0)\}$ is the definition of the score test turned around, and the Wilson theorem has already computed its endpoints. □

20.5 Type I error, type II error, and power

Level: advanced

Every binary decision can be wrong in two ways, and the two are not symmetric. In everyday language: a fire alarm can sound when there is no fire, and can stay silent when there is one. The first mistake sends us out of the house for nothing; the second is far more costly.

Definition 20.10 (Type I error, type II error, and power)**.** A *type I error* is rejecting H_0 when H_0 is true; its probability is the level α. A *type II error* is not rejecting H_0 when H_1 is true; its probability is written β. The *power* of a test is $1 - \beta$, the probability of rejecting H_0 when the effect sought is really there.

Figure 20.3 arranges those four possibilities as a decision box, with probabilities from the exact computation at $n = 20$, $p_0 = 0.5$, and $p_1 = 0.75$ that appears in the computational lab: $\alpha = 0.0414$, power 0.6172, so $\beta = 0.3828$.

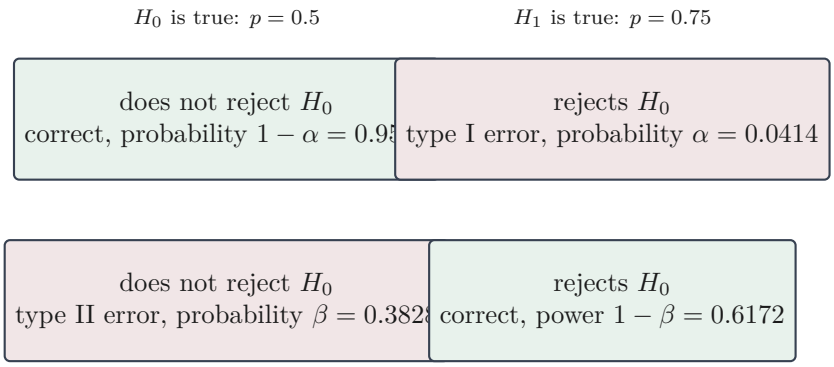


Figure 20.3: The two kinds of error as four decision boxes. Top row: the actual state H_0 is true; bottom row: H_1 is true with $p = 0.75$. Left column: the test does not reject; right column: the test rejects. The boxes in red are the two errors that cannot be suppressed together: shrinking $\alpha = 0.0414$ enlarges $\beta = 0.3828$, and conversely. The numbers are computed from the exact binomial distribution at $n = 20$ with the rejection rule $k \le 5$ or $k \ge 15$.

Power is not a property of the test alone, but of the test together with the sample size and the size of the effect. Hence it must be computed for one value p_1 deemed important, not in general. For a two-sided test with $H_0 \colon p = 0.5$, $n = 100$, and the rule to reject when $|z| > 1.96$, the rejection region is $k \le 40$ or $k \ge 60$. The exact binomial computation gives size 0.0569, not 0.05, and power 0.8750 at $p_1 = 0.65$, whereas the normal approximation gives 0.8508. The normal approximation, namely

$$1 - \beta \approx \Phi\Big(\frac{(p_1 - p_0)\sqrt{n}}{\sqrt{p_0(1 - p_0)}} - z\Big),$$

is close enough for planning, but the two-and-a-half-point difference is real and comes from the discreteness of the data.

First lesson: the nominal level is seldom attained exactly. At $n = 20$ the same rule rejects when $k \le 5$ or $k \ge 15$, with exact size 0.0414; at $n = 100$ the size is 0.0569. A discrete test cannot put a threshold between two whole numbers, so the actual level hangs on n and p_0. Second lesson: at $n = 20$ the power against $p_1 = 0.75$ is only 0.6172, so almost four in ten trials of size twenty fail to find it.

That last number opens the most costly misinterpretation: reading "p-value > 0.05" as "no effect". First, the p-value is $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$; both are conditional probabilities with the conditions swapped, and we already met that swap in the diagnostic tests of Chapter 17. Second, a large p-value is a statement about the data, not about the world: scanty data cannot surprise anyone. Third, failure to reject can occur precisely because the test is weak, as at $n = 20$ above.

A small example makes all three concrete. Ten successes out of twenty trials against $H_0 \colon p = 0.5$ give a two-sided p-value of exactly 1: that data set sits precisely in the middle. Is that proof that $p = 0.5$? No. The 95 percent Wilson interval for the same data is $(0.2993;\ 0.7007)$. So the true rate may be 0.30, may be 0.70, and this data set cannot tell it apart from 0.50. A p-value $= 1$ only means "no direction of deviation", and that is far less than what people usually read into it.

Challenge (problem solver)

Two versions of a detector are tested on 100 000 examples each. Version A catches 66 000, version B catches 65 000. A two-proportion test gives a p-value of about 0.0000026, and the team writes in the report: "A is better, $p < 0.001$". Does that sentence have the

support of the data? Answer with one interval, not with a p-value.
Hint 1: the p-value measures surprise, not the size of the effect. Compute the 95 percent interval for the difference of the two proportions, and mind the units.
Hint 2: the standard error of the difference is 0.002126, so the interval runs from 0.0058 to 0.0142, that is zero point six to one point four percentage points. Compare it with the practical threshold the team itself uses: three percentage points.
If you get stuck, go back to the section Type I error, type II error, and power.

For younger readers (age 10 and up)

A line judge is tasked with raising the flag when the ball goes out of the field. He raises the flag only when he sees the ball go out himself, clearly enough. If he does not raise the flag, does that mean the ball was always in? Certainly not. The ball may have gone out, but his view was blocked by another player, or the ball was very fast.
The way he works is similar to the way a hypothesis test works. He is given one rule that must be settled before the match: how sure he must be before raising the flag. If the rule is too loose, he often raises the flag when the ball is in, and the crowd is angry. If the rule is too strict, he almost never raises the flag, and big fouls slip by. The two mistakes are different, and he cannot suppress both at once by carefulness alone.
In this chapter we shall learn why "no evidence" is not the same as "no foul", why the number of observations matters as much as the rule chosen, and why a judge who never raises the flag is not a more careful judge.

20.6 Worked example: 33 out of 50 and 53 out of 100

Level: intermediate

We return to the two detectors in the chapter opening and compute both intervals all the way. Every number below uses $z = 1.96$, so $z^2 = 3.8416$.

For A, $\hat{p} = 0.66$ and $n = 50$. The first term inside the Wilson root is $0.66 \cdot 0.34/50 = 0.2244/50 = 0.004488$; the second term $z^2/(4n^2) = 3.8416/10000 = 0.00038416$; their sum is 0.00487216 and its root 0.069801. The center is

$$\frac{0.66 + 3.8416/100}{1 + 3.8416/50} = \frac{0.698416}{1.076832} = 0.648584,$$

and the half-width is $1.96 \cdot 0.069801/1.076832 = 0.127048$. So the Wilson interval for A is $(0.5215;\ 0.7756)$, of width 0.2541. The Wald interval for the same data uses standard error 0.066993 and half-width $1.96 \cdot 0.066993 = 0.131303$, hence $(0.5287;\ 0.7913)$ of width 0.2626.

For B, $\hat{p} = 0.53$ and $n = 100$. The first term $0.53 \cdot 0.47/100 = 0.002491$, the second term $3.8416/40000 = 0.00009604$, the sum 0.00258704 and its root 0.050863. The center $(0.53 + 0.019208)/1.038416 = 0.528890$, the half-width $1.96 \cdot 0.050863/1.038416 = 0.096003$, hence $(0.4329;\ 0.6249)$ of width 0.1920. Wald gives $(0.4322;\ 0.6278)$ of width 0.1956.

Now compare. The intervals A and B overlap on $(0.5215;\ 0.6249)$, an extent of 0.1034: about 41 percent of the length of A and 54 percent of the length of B. The normal-based two-proportion test gives $z = 1.5175$ and a two-sided p-value of 0.1291. Although the raw numbers differ by thirteen points, this data set is not enough to conclude that A is better; the interval for A is also wider, because it is built from fifty examples.

Recompute part A without looking at the text: $0.66 \cdot 0.34$, divide by 50, add $3.8416/10000$, take its root, then divide by 1.076832 after multiplying by 1.96. Check that the result is 0.127048. Then answer: how many examples are needed for the width of the interval for A to fall to 0.15, if $\hat{p}$ stays 0.66? Hint: the width is proportional to $1/\sqrt{n}$, and you already know the width at $n = 50$.

20.7 Computational lab: coverage, type I error, and power

LEVEL: ADVANCED

Every claim above can be checked with a computer, and there is one claim that can be checked only that way: the actual coverage of a procedure. The key is knowing the true p, something that never happens with real data. Hence we work with simulation and with exact binomial computations.

Computational lab

```
import numpy as np
from math import sqrt
from scipy.stats import norm, binom

z = norm.ppf(0.975)


def wald(k, n):
    p = k / n
    se = sqrt(p * (1 - p) / n)
    return p - z * se, p + z * se


def wilson(k, n):
    p = k / n
    z2 = z * z
    mid = (p + z2 / (2 * n)) / (1 + z2 / n)
    half = z * sqrt(p * (1 - p) / n + z2 / (4 * n * n)) / (1 + z2 / n)
    return mid - half, mid + half


# 1. width of the 95% interval for several proportions
print("== 1. width of the 95% interval ==")
print("   n    k   phat  Wald width    Wilson width")
for n in (20, 100):
    for p in (0.00, 0.05, 0.25, 0.50, 0.66):
        k = round(p * n)
        a = wald(k, n)
        b = wilson(k, n)
        print("%4d %4d  %5.2f   %9.4f    %9.4f"
              % (n, k, k / n, a[1] - a[0], b[1] - b[0]))

# 2. exact coverage at p = 0.05
print("== 2. exact coverage, p=0.05 ==")
for n in (20, 50, 100):
    for nama, f in (("Wald", wald), ("Wilson", wilson)):
        c = 0.0
        for k in range(n + 1):
            lo, hi = f(k, n)
            if lo <= 0.05 <= hi:
                c += binom.pmf(k, n, 0.05)
        print("n=%3d %-6s coverage=%.4f" % (n, nama, c))

# 3. simulated coverage with a fixed seed
rng = np.random.default_rng(20261024)
R = 20000
print("== 3. simulated coverage, R=%d ==" % R)
for n in (20, 100):
    X = rng.binomial(n, 0.05, R)
    for nama, f in (("Wald", wald), ("Wilson", wilson)):
        ok = 0
```

```
        w = 0.0
        for k in X:
            lo, hi = f(int(k), n)
            ok += (lo <= 0.05 <= hi)
            w += hi - lo
        print("n=%3d %-6s coverage=%.4f  width=%.4f" % (n, nama, ok / R, w / R))

# 4. type I error and power for H0: p=0.5
print("== 4. type I error and power, H0 p=0.5 ==")


def tolak(k, n, p0=0.5):
    return abs((k / n - p0) / sqrt(p0 * (1 - p0) / n)) > z


for n in (20, 100):
    rej = [k for k in range(n + 1) if tolak(k, n)]
    lo = max(k for k in rej if k < n / 2)
    hi = min(k for k in rej if k > n / 2)
    print("n=%3d reject if k<=%d or k>=%d" % (n, lo, hi))
    print("      exact size at p=0.5 : %.4f"
          % sum(binom.pmf(k, n, 0.5) for k in rej))
    for p1 in (0.65, 0.75):
        ex = sum(binom.pmf(k, n, p1) for k in rej)
        zc = (p1 - 0.5) / sqrt(0.25 / n)
        ap = norm.cdf(zc - z) + norm.cdf(-zc - z)
        print("      power at p=%.2f: exact %.4f  approx %.4f"
              % (p1, ex, ap))

# 4b. simulated type I error rate when H0 is true
print("== 4b. simulated type I error, R=%d ==" % R)
for n in (20, 100):
    X0 = rng.binomial(n, 0.5, R)
    tolak0 = np.abs((X0 / n - 0.5) / sqrt(0.25 / n)) > z
    print("n=%3d p=0.5: reject=%.4f" % (n, tolak0.mean()))

# 5. t-interval for 12 measurements
x = np.array([18.2, 19.1, 17.8, 20.3, 18.9, 19.7,
              18.4, 21.0, 17.5, 19.9, 18.6, 20.1])
n = len(x)
m = x.mean()
s = x.std(ddof=1)
from scipy.stats import t
tq = t.ppf(0.975, n - 1)
h = tq * s / sqrt(n)
print("== 5. t-interval, n=12 ==")
print("mean=%.6f  s=%.6f  se=%.6f" % (m, s, s / sqrt(n)))
print("t=%.6f  half-width=%.6f  CI=(%.6f, %.6f)" % (tq, h, m - h, m + h))
print("half-width with z=1.96: %.6f" % (z * s / sqrt(n)))
```

The actual output, verbatim:

```
== 1. width of the 95% interval ==
   n    k   phat  Wald width    Wilson width
  20    0   0.00      0.0000       0.1611
  20    1   0.05      0.1910       0.2272
  20    5   0.25      0.3795       0.3568
  20   10   0.50      0.4383       0.4014
  20   13   0.65      0.4181       0.3860
 100    0   0.00      0.0000       0.0370
 100    5   0.05      0.0854       0.0902
 100   25   0.25      0.1697       0.1676
```

```
 100   50   0.50      0.1960       0.1923
 100   66   0.66      0.1857       0.1826
== 2. exact coverage, p=0.05 ==
n= 20 Wald   coverage=0.6389
n= 20 Wilson coverage=0.9245
n= 50 Wald   coverage=0.9199
n= 50 Wilson coverage=0.9622
n=100 Wald   coverage=0.8775
n=100 Wilson coverage=0.9659
== 3. simulated coverage, R=20000 ==
n= 20 Wald   coverage=0.6430  width=0.1472
n= 20 Wilson coverage=0.9224  width=0.2193
n=100 Wald   coverage=0.8783  width=0.0829
n=100 Wilson coverage=0.9667  width=0.0884
== 4. type I error and power, H0 p=0.5 ==
n= 20 reject if k<=5 or k>=15
      exact size at p=0.5 : 0.0414
      power at p=0.65: exact 0.2457  approx 0.2687
      power at p=0.75: exact 0.6172  approx 0.6088
n=100 reject if k<=40 or k>=60
      exact size at p=0.5 : 0.0569
      power at p=0.65: exact 0.8750  approx 0.8508
      power at p=0.75: exact 0.9997  approx 0.9988
== 4b. simulated type I error, R=20000 ==
n= 20 p=0.5: reject=0.0430
n=100 p=0.5: reject=0.0554
== 5. t-interval, n=12 ==
mean=19.125000  s=1.082191  se=0.312401
t=2.200985  half-width=0.687591  CI=(18.437409, 19.812591)
half-width with z=1.96: 0.612296
```

Four things deserve reading from that output. First, parts 2 and 3 agree to almost three digits, and that agreement is itself a check, since a simulation with $R = 20000$ repetitions can approach the exact computation only to about $\sqrt{0.25/20000} = 0.0035$. At $p = 0.05$ and $n = 20$ the difference is 0.0041 for Wald and 0.0021 for Wilson. An honest simulation must report R.

Second, the coverage of Wald at $n = 20$ and $p = 0.05$ is 0.6389, not 0.95. The cause can be computed by hand: the probability of observing zero events is 0.3585, and at $k = 0$ the Wald interval is $[0; 0]$, which does not contain 0.05; that single point alone consumes almost thirty-six points of coverage. Wilson fails only when $k \geq 3$, whose probability is 0.0755, and its failure is not total because its interval stays wide. That is why it reaches 0.9245. Note also the width column: the mean Wald width 0.1472 is smaller than the Wilson 0.2193, and precisely therein lies the point. A narrower interval is not a better interval if it is narrow because it sits in the wrong place.

Third, part 4 separates the size of a test from the nominal level: 0.0414 at $n = 20$ and 0.0569 at $n = 100$, both for the rule $|z| > 1.96$; part 4b repeats it by simulation and lands close, 0.0430 and 0.0554. Power moves far more drastically: 0.2457 at $n = 20$ and 0.8750 at $n = 100$ for $p_1 = 0.65$. The normal approximation is not always on the safe side: at $n = 20$ it reports 0.2687 while the truth is 0.2457.

Fourth, part 5 merely repeats the t-interval already computed in the text, with a difference in half-width between t and z of 0.075295.

The program does not prove any one of all those things. It checks claims already proved, and in two places it shows its own limits: exact coverage uses the distribution we assume to be true, so if the model is wrong, the number 0.9245 means nothing; and simulation only measures how often the procedure succeeds under that model, not how right the model is.

Note for instructors

The teaching order I suggest: start from the morning meeting in the chapter opening, not from the definition of an estimator. Write 33/50 and 53/100 on the board, ask the class to choose, then ask what happens if those fifty examples are replaced. Out of that unease the definitions of estimator, bias, and standard error emerge as the answer. The MSE decomposition is proved at the board with one line of algebra. The interpretation of the confidence interval comes after that, as material for a lecture of its own, not a footnote.

Two misconceptions appear most often. First, students read "95 percent" as a conditional probability on data already in hand, then write "the probability that μ is in this interval is 0.95"; the symptom shows when they cannot explain what is random about an interval already computed. Second, they take a large p-value as proof that H_0 is true, and will compute the p-value for 10/20 and then conclude that $p = 0.5$ is established.

A blackboard activity that works: split the class into two groups, one computing the Wald interval and one the Wilson interval for 1/20, then have both defend their result in front of the class. The Wald group will find a negative lower limit and have to explain it. The sign that the class is ready to go on: they can name one example of a sample size that makes the p-value large although the effect is large, and can explain why that happens.

20.8 Summary

Level: basic

Summary.

- An estimator is a statistic that guesses a parameter, so it is a random variable with an expectation and a variance. Its quality is measured by MSE $=$ bias$^2 +$ Var; unbiasedness alone guarantees nothing, and MSE going to zero guarantees consistency through Chebyshev.
- The standard error is the square root of the variance of the estimator, not the width of the data. For a proportion its value is $\sqrt{\hat{p}(1-\hat{p})/n}$, shrinking like $1/\sqrt{n}$, so four times the data only cuts it in half.
- A 95 percent confidence interval is a statement about the procedure, not about the parameter: the procedure catches the true value on 95 percent of repeated trials. Once the data are taken, the interval is fixed and no longer carries a probability.
- The Wald interval collapses near 0 and 1: at 0/20 its width is zero, and at $p = 0.05$ with $n = 20$ its coverage is only 0.6389. The Wilson interval, the result of inverting the score test, reaches 0.9245 in the same setting.
- If no event occurs in n trials, the 95 percent upper limit is about $3/n$; for $n = 100$ the exact limit is 0.0295130 and the two-sided Wilson limit 0.0369935.
- The p-value is the probability of data at least that extreme if H_0 is true, and under H_0 it is uniformly distributed. A large p-value is not proof that H_0 is true: at $n = 20$ the power against $p = 0.75$ is only 0.6172.

20.9 Exercises

Level: intermediate

Exercise 20.1. 1. Assess this claim: "Because the p-value equals 0.42, the null hypothesis $p = 0.5$ is proved true." True, false, or not enough evidence? Name the part that is

right, the part that is wrong, and one additional computation that would strengthen your answer.

2. Prove that $\mathbb{E}[\hat{p}(1-\hat{p})] = p(1-p)(n-1)/n$ for $\hat{p} = K/n$ with $K \sim \text{Bin}(n,p)$, then determine the unbiased estimator of $p(1-p)$ obtained from it. Check the result at $p = 0.3$ and $n = 20$.
3. Prove that if $\text{bias}(\hat{\theta}_n) \to 0$ and $\text{Var}(\hat{\theta}_n) \to 0$, then $\hat{\theta}_n$ is consistent for θ. Explain why that condition is weaker than $\text{MSE} \to 0$ yet sufficient for the same conclusion.
4. Compute the 95 percent Wilson and Wald intervals for $7/40$, then compare their widths with $13/20$. Explain why the smaller width at $7/40$ is not a sign of higher precision, and check both intervals with Python.
5. For $n = 30$ and $H_0 \colon p = 0.5$ with a normal-based two-sided test, determine the rejection region, compute its exact size, then compute its power at $p_1 = 0.7$. Compare that power with $n = 100$ at the same p_1.
6. Prove that the Wald interval and the two-sided z-test using the standard error $\hat{p}$ really are equivalent, by writing the two inequalities side by side. Explain why the equivalence fails when the test uses the standard error under H_0.
7. A report writes: "there is no significant difference, $p = 0.31$, so the two methods are equally good". Rewrite that sentence honestly, naming the interval, the level, and the power. You may invent the numbers as long as they are consistent.
8. For $n = 100$ with not a single event, compute the rule of three, the exact one-sided limit, and the Wilson upper limit. Explain why the three numbers differ, then state which one may be used for the sentence "at most 5 percent".
9. Compute the 90 percent t-interval for the twelve delay-time data in this chapter, then compare it with the 95% interval. How much data is needed for the width of the 95% interval to fall by half, if s does not change?

Hints for the three hardest items. Item 3: write $\hat{\theta}_n - \theta = \text{bias}_n + (\hat{\theta}_n - \mathbb{E}\hat{\theta}_n)$, then use the triangle inequality and Chebyshev on the second term. Item 6: the first inequality is $|\hat{p} - p_0| \le z\sqrt{\hat{p}(1-\hat{p})/n}$, and a test that uses $\sqrt{p_0(1-p_0)/n}$ uses a different denominator, so the set of p_0 not rejected is no longer the Wald interval but a more complicated set. Item 5: the rejection region is $|k - 15| > 1.96\sqrt{30 \cdot 0.25}$, so $k \le 9$ or $k \ge 21$, and the power is computed by summing the binomial pmf at $p = 0.7$ over those two tails.

For researchers: limits and next steps

What is really proved in this chapter: the MSE decomposition; consistency from MSE going to zero; exact coverage of the z-interval under the normal model; the closed form of the Wilson interval and its location in $[0,1]$; the rule of three as an exact one-sided limit together with the inequality $p_{\text{hi}} < \ln 20/n$; the uniformity of the p-value under H_0 for a continuous statistic; and the equivalence of interval and test in the pairs mentioned. What is merely popular, not a theorem: that 95% means a conditional probability on data already observed; that a large p-value supports H_0; and that a smaller interval width is always better, when a comparison of widths without coverage means nothing. The subnominal coverage of Wilson at some (n,p) is also a computational fact, not a theorem.

Primary source: Wasserman, *All of Statistics*, Chapter 6 (Models, Statistical Inference, and Learning) for bias, MSE, and the interpretation of intervals; Chapter 9 (Parametric Inference) for intervals and the plug-in method; Chapter 10 (Hypothesis Testing and p-values) for p-values, type I and II errors, power, and the relation of tests to intervals. The Wilson interval comes from E. B. Wilson, "Probable Inference, the Law of Succession, and Statistical Inference" (1927); the rule of three from B. D. Jovanovic and P. S. Levy, "A

Look at the Rule of Three" (1997). Neither is discussed by Wasserman, so the references are cited directly, not through an intermediate textbook.
Limits of validity: the whole chapter works with independent and identically distributed samples, one parameter, and a known normal or binomial approximation. The z-interval is exact only under the normal model with σ known; beyond that it is an asymptotic statement, and at small n with heavy-tailed distributions its coverage can fall far below nominal. The Wilson interval is a score approximation, not an exact interval; exact intervals for proportions are computed from the binomial distribution and are wider. Every reported coverage is conditional on the binomial model being correct.
Open questions worth a reader's work: how well proportion intervals behave when n is not fixed but stops adaptively, as in sequential sampling; and how to build intervals for classification metrics computed from dependent data, for instance when one example appears in several cross-validation folds. Both are still research material.

Map to source texts: Chapter 20 in Wasserman, All of Statistics*: Chapter 6 (Models, Statistical Inference, and Learning: estimators, bias, MSE, standard error, interpretation of intervals), Chapter 9 (Parametric Inference: confidence intervals and the plug-in method), Chapter 10 (Hypothesis Testing and p-values: p-values, type I and II errors, power, equivalence of tests and intervals). The Wilson interval and the rule of three are taken from their original papers, Wilson (1927) and Jovanovic–Levy (1997), because Wasserman discusses neither. The law of large numbers, the central limit theorem, and Chebyshev's inequality used here are proved in Chapter 18; conditional probability and the error of swapping the conditions are discussed in Chapter 16. Chapter 25 continues this chapter for classification metrics, including the rule of three in the reporting of model evaluation.*

Chapter 21

Regression and Applied Statistics

A small café records nine trading days. Two numbers are written every evening: the number of online orders in tens, and the kitchen's overtime hours. The café owner needs one number for one question: if orders rise by ten, how many extra overtime hours must she prepare for tomorrow?

day	1	2	3	4	5	6	7	8	9
orders (tens)	1	2	3	4	5	6	7	8	9
overtime (hours)	10	12	13	14	16	18	19	20	22

A quick answer is available without any tool: from the first day to the last, orders rise by eight tens and overtime rises by twelve hours, so roughly 1.5 hours per ten orders. Least squares, which we will build in this chapter, gives 1.466667. The two numbers are close, and precisely because they are close we are tempted to treat the matter as settled. Three important questions are left unanswered by both. How far does that number move if those nine days are replaced by nine other days? If tomorrow's orders are eighty, how many hours must be prepared? And if one day turns out to be misrecorded, how much damage does that do?

Chapter 10 already gave the picture: least squares is the orthogonal projection of the observation vector onto the column space of the design matrix, and the normal equations say that the residual is perpendicular to that space. What is missing there is uncertainty. Chapter 20 already gave confidence intervals and hypothesis tests, but for a single parameter from observations that are independent and share one distribution. Here the parameter sought is a slope, a ratio of two sums that are both random, and the observations are no longer on equal footing: a point at the end of the range carries more influence than a point in the middle. This chapter joins the two and stops at an honest place: how far a conclusion may be carried.

For younger readers (age 10 and up)

A grandmother sells coconut ice in front of her house. She never records anything neatly, but her memory is good enough: on a hot day her stock sells out before the afternoon. On a cloudy day half the ice melts in the cooler. One day her grandchild arrives with an exercise book and records nine days in a row: roughly how hot it was, and how many cups were sold.

From those nine records the grandchild draws one straight line: the hotter it gets, the more is sold. The grandmother nods, but then asks, "if tomorrow is a little hotter, how many cups should I prepare?" The grandchild falls silent. The line gives the direction, not the number. Besides, on one day it rained hard although it was hot, and the sales were dreadful; that point sits far from the line.

In this chapter we will learn why the best-fitting line is still not a sure forecast, why one strange day can shift the line further than five ordinary days, and why anyone who sells numbers must state how wide their uncertainty is.

21.1 Nine days and one line

Level: basic

Nine pairs of numbers, two unknowns, nine equations. We met a system like this in Chapter 8: more equations than unknowns, so an exact solution exists only if all nine points happen to be collinear. No line passes exactly through every point above, and that is not a measurement failure but the usual state of affairs. All we can look for is the closest line, and "closest" has to be defined first.

In everyday language: we look for the one line whose total error is as small as possible, with the error counted as the perpendicular distance from the point to the line.

regression (*regresi*) — drawing the one line closest to all the points, rather than a line that passes through all of them

Definition 21.1 (Simple linear regression model)**.** For observations $i = 1, \dots, n$,

$$y_i = a + b\,x_i + \varepsilon_i,$$

with x_i the observed explanatory variable, y_i the explained variable, a the intercept, b the slope, and ε_i the unobserved error. The model is called *linear* because it is linear in the parameters a and b, not because the line is straight in x.

One small distinction governs the whole chapter: ε_i and the residual are not the same object. The error ε_i is the difference between the true value and the population line that we never see; the residual $e_i = y_i - \hat{y}_i$ is the difference between the observed value and the line we fitted from the data itself. A residual can be computed and drawn, an error can only be assumed. Whenever a book states an assumption about errors and then checks it through a picture of residuals, there is one jump that must not be forgotten: what is inspected is a stand-in, not the thing itself.

21.2 Least squares as a projection

Level: intermediate

Stack $y \in \mathbb{R}^n$ as the observation vector and let X be the $n \times 2$ matrix whose columns are one and x. Finding the best line means choosing $\hat{y}$ in the column space of X that is closest to y: the problem of Chapter 10, with the addition that we need the explicit form of the coefficients in order to compute their uncertainty.

Definition 21.2 (Least-squares estimator)**.** $\hat{\beta} = (\hat{a}, \hat{b})$ is the least-squares estimator if it minimizes

$$\mathrm{RSS}(\beta) = \|y - X\beta\|^2 = \sum_{i=1}^{n} (y_i - a - b x_i)^2.$$

least squares (*kuadrat terkecil*) — adding up all the errors after squaring them, then choosing the line whose total is smallest

Theorem 21.1 (The normal equations)**.** *$\hat\beta$ minimizes RSS if and only if*

$$X^\top (y - X\hat\beta) = 0, \qquad \text{that is} \qquad X^\top X\hat\beta = X^\top y.$$

If the columns of X are linearly independent, $\hat\beta = (X^\top X)^{-1}X^\top y$ is unique.

Proof. Write $f(\beta) = \|y - X\beta\|^2 = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta$. Its gradient is $-2X^\top y + 2X^\top X\beta$, so the stationary condition is exactly the normal equations. Since $X^\top X$ is positive definite when its columns are linearly independent, f is strictly convex and that stationary point is the global minimum.

The converse direction is more informative. Suppose $X^\top(y - X\hat\beta) = 0$. For any β,

$$\|y - X\beta\|^2 = \|y - X\hat\beta\|^2 + \|X(\hat\beta - \beta)\|^2 + 2(\hat\beta - \beta)^\top X^\top (y - X\hat\beta),$$

and the cross term vanishes. Hence $\mathrm{RSS}(\beta) \geq \mathrm{RSS}(\hat\beta)$ for every β, with equality only when $X\beta = X\hat\beta$. □

That proof contains the whole geometric content: the residual is perpendicular to every column of X. Since the first column is the vector of ones and the second is x, we get $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$, and those two identities are what support the decomposition of sums of squares in the next section.

Proposition 21.2 (Hat matrix)**.** *With $H = X(X^\top X)^{-1}X^\top$ we have $\hat y = Hy$, $H^\top = H$, $H^2 = H$, $\operatorname{tr} H = 2$, and $0 \leq h_{ii} \leq 1$ with $\sum_i h_{ii} = 2$. The value h_{ii} is called the* leverage *of point i.*

Proof. Symmetry and idempotence follow directly from the definition:

$$H^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = H.$$

Hence H is an orthogonal projection matrix, so $h_{ii} = e_i^\top H e_i = \|He_i\|^2 \geq 0$ and $h_{ii} \leq \|e_i\|^2 = 1$. Finally $\operatorname{tr} H = \operatorname{tr}\big((X^\top X)^{-1}X^\top X\big) = \operatorname{tr} I_2 = 2$. □

Now the numbers, because the whole chapter rests on one computation that can be rechecked. For the café data,

$$X^\top X = \begin{pmatrix} 9 & 45 \\ 45 & 285 \end{pmatrix}, \qquad X^\top y = \begin{pmatrix} 144 \\ 808 \end{pmatrix}, \qquad \det(X^\top X) = 9 \cdot 285 - 45^2 = 540.$$

Therefore

$$\hat\beta = \frac{1}{540}\begin{pmatrix} 285 & -45 \\ -45 & 9 \end{pmatrix}\begin{pmatrix} 144 \\ 808 \end{pmatrix} = \frac{1}{540}\begin{pmatrix} 4680 \\ 792 \end{pmatrix} = \begin{pmatrix} 8.666667 \\ 1.466667 \end{pmatrix}.$$

The number 808 is worth checking by hand: $\sum x_i y_i = 10{+}24{+}39{+}56{+}80{+}108{+}133{+}160{+}198$. So is 4680: $285 \cdot 144 - 45 \cdot 808 = 41040 - 36360$.

One more theorem is needed before we may speak about the quality of an estimator. That theorem explains why least squares is used in almost every field, and at the same time why it is not a sacred choice.

Theorem 21.3 (Gauss–Markov)**.** *Assume (A1) $y = X\beta + \varepsilon$; (A2) $\mathbb{E}[\varepsilon \mid X] = 0$; (A3) $\operatorname{Var}(\varepsilon \mid X) = \sigma^2 I$; (A4) the columns of X are linearly independent. Then $\hat\beta$ is unbiased with $\operatorname{Var}(\hat\beta) = \sigma^2(X^\top X)^{-1}$, and among all linear unbiased estimators, $\hat\beta$ has the smallest variance. Such an estimator is called* BLUE *(*best linear unbiased estimator*).*

Proof. Unbiasedness: $\mathbb{E}[\hat\beta \mid X] = (X^\top X)^{-1}X^\top X\beta = \beta$, using (A2). Variance: $\hat\beta - \beta = (X^\top X)^{-1}X^\top\varepsilon$, so $\mathrm{Var}(\hat\beta) = (X^\top X)^{-1}X^\top(\sigma^2 I)X(X^\top X)^{-1} = \sigma^2(X^\top X)^{-1}$.

The third part is not proved in full here. The sketch: every linear unbiased estimator has the form $\tilde\beta = Cy$ with $CX = I$, so $\tilde\beta - \beta = C\varepsilon$ and $\mathrm{Var}(\tilde\beta) = \sigma^2 CC^\top$. Writing $C = (X^\top X)^{-1}X^\top + D$ gives $DX = 0$ and

$$CC^\top - (X^\top X)^{-1} = DD^\top,$$

which is positive semi-definite, so every component of $\tilde\beta$ has variance no smaller than the matching component of $\hat\beta$. The full proof is in Seber and Lee, *Linear Regression Analysis*, Chapter 3, and in Greene, *Econometric Analysis*, Chapter 3. □

21.3 Coefficients and their units

Level: basic

The slope 1.466667 means nothing until its unit is named. The unit is hours per ten orders: every additional ten orders raises the forecast overtime by 1.466667 hours, that is 88 minutes, or 8.8 minutes per order. The café owner can use that sentence; the sentence "the coefficient is 1.47" she cannot.

The intercept 8.666667 hours is the forecast for $x = 0$, a day with no online orders at all. No such day occurs in the data, and outside the range 1 to 9 the line is not guaranteed to hold. Calling that number "fixed overtime cost" is extrapolation wrapped in a name: we are using the line outside the place where it was fitted. What is safe: the intercept is the number that makes the line pass through $(\bar x, \bar y)$, because $\hat a = \bar y - \hat b\bar x$; it is useful for computation, not for interpretation.

A change of units does not change the conclusion. If x is written in orders rather than tens, the slope is 0.146667 hours per order and its standard error is 0.004714, both divided by ten; t and R^2 do not move. Multiplying one variable by a constant c multiplies the coefficient and its standard error by c or $1/c$, while their ratio stays put, so a report that quotes t without units cannot be read.

Do it by hand (kinesthetic learner)

Prepare graph paper, nine coins, and a piece of string as long as the side of the paper. Place the coins on the grid according to the nine pairs of numbers at the start of the chapter: horizontally orders, vertically overtime, one square to one unit. Tie the ends of the string to the two edges of the paper so that the string can be slid as a straight line. Slide and rotate the string until the sum of the perpendicular distances from all the coins to the string feels smallest; do not measure, just compare by eye, then lock it with tape. What should be visible: two people will put the string in slightly different places, and the difference is largest when you shift the ends of the string. Now lift the leftmost coin by one square and notice how far the string has to turn. Put it back, then lift the middle coin by one square: almost nothing changes. That is leverage, and you have just seen it with your hands before meeting the formula.

21.4 R squared and its correlation

Level: intermediate

Once the line exists, people ask "how good is it". The most common answer is one number that compares the residual with the original spread of y. In everyday language: what share of the wobble in y the line manages to explain, and what share is left over.

Definition 21.3 (R^2)**.** With $\mathrm{SST} = \sum_i (y_i - \bar{y})^2$, $\mathrm{SSE} = \sum_i e_i^2$, and $\mathrm{SSR} = \sum_i (\hat{y}_i - \bar{y})^2$, the coefficient of determination is

$$R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}} = \frac{\mathrm{SSR}}{\mathrm{SST}}.$$

coefficient of determination (*koefisien determinasi*) — the share of the wobble that the line manages to explain, expressed as a fraction of the whole

Theorem 21.4 (Decomposition of sums of squares)**.** *If X contains a column of ones, then* $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ *and* $0 \le R^2 \le 1$.

Proof. Write $y - \bar{y}\mathbf{1} = (\hat{y} - \bar{y}\mathbf{1}) + e$. Since $\sum e_i = 0$, the vector e is perpendicular to $\bar{y}\mathbf{1}$, and by the normal equations e is also perpendicular to $\hat{y}$. Hence the two terms on the right are perpendicular, and Pythagoras gives $\|y - \bar{y}\mathbf{1}\|^2 = \|\hat{y} - \bar{y}\mathbf{1}\|^2 + \|e\|^2$. □

For the café data, $\mathrm{SST} = 130$, $\mathrm{SSE} = 0.933333$, so $R^2 = 1 - 0.933333/130 = 0.992821$. In a simple regression, R^2 equals the squared correlation: $r = 88/\sqrt{60 \cdot 130} = 0.996404$ and $r^2 = 0.992821$. That equality is special to one explanatory variable; as soon as there are two, R^2 is no longer the square of any correlation.

The number 0.992821 easily makes people stop thinking, so let us tame it. What is explained is 99.28 percent of the *wobble* in overtime, and that wobble is measured against its own mean. The residuals still have variance $s^2 = 0.933333/7 = 0.133333$, that is a standard deviation $s = 0.365148$ hours or 21.9 minutes. A café owner who prepares overtime on the basis of this line can still miss by more than twenty minutes. R^2 is relative; s is absolute, and what people feel is s.

For younger readers (age 10 and up)

Imagine two students. The first runs a hundred-meter race and finishes 0.2 seconds behind the winner. The second runs a marathon and finishes twenty minutes behind the winner. Both lose, but not in the same sense. Two tenths of a second is very close; twenty minutes is very far. If the teacher writes only “second place” for both, we lose the whole story.

Numbers of that kind occur in this chapter. There is a number that measures the *share* guessed correctly, and there is a number that measures the *size* of the error left over. The first can look almost perfect while the second stays large. Someone who reads only the first number will prepare too little, then wonder why the forecast misses.

Proposition 21.5 (Adding a variable never lowers R^2)**.** *If a column block X_2 is added to X_1 so that its column space contains the column space of X_1, then* SSE *does not increase and R^2 does not decrease.*

Proof. The set of vectors $X_1\beta_1$ is contained in the set $X_2\beta_2$, so the minimum distance to y taken over the larger set cannot be larger. Hence R^2 cannot fall, even when the added variable has nothing to do with y. □

The practical consequence matters: comparing models by raw R^2 always favors the fatter model. The usual correction is $R^2_{\mathrm{adj}} = 1 - (1 - R^2)(n - 1)/(n - p)$ with p the number of parameters. For the multiple model of the next section, $R^2 = 0.997951$ falls to 0.997496 when $p = 3$ and $n = 12$; at small n with many variables the correction can reverse the ranking.

21.5 Multiple regression and collinearity

Level: advanced

So far, one explanatory variable. In reality almost every problem demands more than one, and that is where its characteristic disease appears: two explanatory variables carrying almost the same information.

Twelve branches of a food stall are recorded: daily sales y in millions of rupiah, the number of employees x_1, and the seating area x_2 in square meters.

branch	1	2	3	4	5	6	7	8	9	10	11	12
x_1	6	7	7	8	8	9	9	10	10	11	12	12
x_2	18	22	20	26	24	28	26	32	30	34	36	38
y	11.2	12.4	12.0	14.1	13.6	15.2	14.8	17.0	16.4	18.1	19.4	19.9

Least squares gives

$$\hat{y} = 2.442991 + 0.686916\,x_1 + 0.239252\,x_2,$$

with standard errors 0.207583, 0.135366, and 0.042101; the t values are 11.7687, 5.0745, and 5.6828 on 9 degrees of freedom; $s = 0.143607$ and $R^2 = 0.997951$. All of it looks healthy. Then we fit one explanatory variable at a time: y on x_1 gives 1.446019, y on x_2 gives 0.450075, both nearly twice their multiple coefficients.

Not because of a miscalculation. Employees and floor area almost determine each other: their correlation is $r_{12} = 0.986801$. Once x_2 is in, x_1 carries almost no new information, so its coefficient shrinks to its additional effect only, namely 0.687 million per employee at fixed area. The single coefficient 1.446 mixes the employee effect with the area effect that usually accompanies it, and the gap between the two is a measure of how far the two variables overlap.

Definition 21.4 (Variance inflation factor)**.** For explanatory variable j, let R_j^2 be the coefficient of determination of the regression of x_j on all the other explanatory variables. Its variance inflation factor is

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}.$$

variance inflation factor (*varians inflasi*) — how much the standard error of a coefficient is blown up by other explanatory variables that carry similar information

Proposition 21.6 (VIF with two explanatory variables)**.** *If there are only two explanatory variables,* $\mathrm{Var}(\hat{b}_1) = \sigma^2/(S_{11}(1 - r_{12}^2))$*, with* $S_{11} = \sum_i (x_{i1} - \bar{x}_1)^2$*. Hence the standard error of* $\hat{b}_1$ *is inflated by exactly* $\sqrt{\mathrm{VIF}_1}$ *compared with the case of mutually orthogonal explanatory variables.*

Proof. For two explanatory variables, inverting the 2×2 matrix of the centered block $X^\top X$ gives first element $S_{22}/(S_{11}S_{22} - S_{12}^2)$, and $S_{12} = r_{12}\sqrt{S_{11}S_{22}}$. Dividing numerator and denominator by S_{22} yields $\sigma^2/(S_{11}(1 - r_{12}^2))$. □

The numbers can be checked: $S_{11} = 42.9167$, $r_{12}^2 = 0.973775$, so VIF $= 38.132232$ and $\sqrt{\mathrm{VIF}} = 6.1751$. The standard error of $\hat{b}_1$ is indeed 0.135366, while $\sigma/\sqrt{S_{11}}$ is only 0.021921; the ratio is 6.1751, matching the root of the VIF to four digits.

Collinearity does not damage the forecast; it damages the interpretation. The value $R^2 = 0.997951$ shows that the line is still close to the points, and a forecast at $x_1 = 9$, $x_2 = 28$ is still useful. What cannot be trusted is the sentence "one extra employee raises sales by 0.687 million": its 95 percent interval (0.380697; 0.993135) is wide, and if the data shift a little that coefficient can change a lot without the line changing much.

21.6 Intervals, the t test, and prediction intervals

Level: advanced

The next question is the same as in Chapter 20, only the object changes: not a mean, but a slope. We need one quantity whose distribution is known when the model is correct, then we invert it into an interval.

In everyday language: we measure how far the observed slope sits from zero, in units of the slope's own typical wobble.

prediction interval (*selang prediksi*) — an interval for one day that has not happened yet, always wider than the interval for the mean because it has to bear the surprise of that day as well

Theorem 21.7 (The t test and an interval for one coefficient)**.** *Assume (A1) to (A4) and normal errors. With p the number of parameters, $s^2 = \mathrm{SSE}/(n-p)$, and $\mathrm{se}(\hat b_j) = s\sqrt{[(X^\top X)^{-1}]_{jj}}$, we have*

$$\frac{\hat b_j - \beta_j}{\mathrm{se}(\hat b_j)} \sim t_{n-p},$$

so the $1-\alpha$ interval for β_j is $\hat b_j \pm t_{n-p,1-\alpha/2}\,\mathrm{se}(\hat b_j)$, and the test of $H_0\colon \beta_j = 0$ rejects at level α when $|t| > t_{n-p,1-\alpha/2}$.

Proof. $\hat b_j$ is a linear combination of y with fixed coefficients, so it is normal with expectation β_j and variance $\sigma^2[(X^\top X)^{-1}]_{jj}$ by Gauss–Markov. Moreover $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p}$ and is independent of $\hat\beta$, because the residual is perpendicular to the column space while $\hat\beta$ lies entirely inside it. The ratio of a standard normal variable to the root of a chi-square variable divided by its degrees of freedom follows a t distribution. The full proof is in Wasserman, *All of Statistics*, Chapter 13. □

For the café data, $\hat b = 1.466667$, $s = 0.365148$, $S_{xx} = 60$, so $\mathrm{se}(\hat b) = 0.047140$. With $t_{7,0.975} = 2.364624$,

$$\hat b \pm t\,\mathrm{se}(\hat b) = 1.466667 \pm 0.111470 = (1.355197;\ 1.578136),$$

and $t = 31.1127$ with a p-value below 10^{-6}. Rejecting $\beta = 0$ here is no achievement: with residuals as tight as 22 minutes and an x range this wide, almost any data would reject it. What deserves reporting is the interval, because it says that the extra overtime per ten orders lies between 1.36 and 1.58 hours.

Now distinguish two questions that are often confused. What is the average overtime on a day with eighty orders? And what is the overtime on one particular day with eighty orders? The first answer narrows like $1/\sqrt{n}$; the second never narrows below the standard deviation of the residuals, because a single day brings its own surprise.

Proposition 21.8 (Two kinds of forecast variance)**.** *For fixed x_0 and a new observation $y_0 = x_0^\top\beta + \varepsilon_0$ independent of the old data,*

$$\mathrm{Var}(\hat y_0) = \sigma^2 x_0^\top (X^\top X)^{-1} x_0, \qquad \mathrm{Var}(y_0 - \hat y_0) = \sigma^2\bigl(1 + x_0^\top (X^\top X)^{-1} x_0\bigr).$$

Proof. The first already appeared in Gauss–Markov. For the second, write $y_0 - \hat y_0 = \varepsilon_0 - x_0^\top(\hat\beta - \beta)$; the two terms are independent because ε_0 is independent of the old data, so the variance is the sum of the two variances. □

For $x_0 = 8$ in the café data, $x_0^\top (X^\top X)^{-1} x_0 = 1/9 + 9/60 = 0.261111$. The half-width of the mean interval is $2.364624\sqrt{0.133333 \cdot 0.261111} = 0.441209$, so $(19.9588;\ 20.8412)$ with width 0.882. The half-width of the prediction interval is $2.364624\sqrt{0.133333 \cdot 1.261111} = 0.969635$,

so (19.4304; 21.3696) with width 1.939. The ratio of widths is $\sqrt{1.261111/0.261111} = 2.198$: the prediction interval is more than twice as wide, and no amount of data can remove that excess. Figure 21.1 shows both bands over the whole range of x.

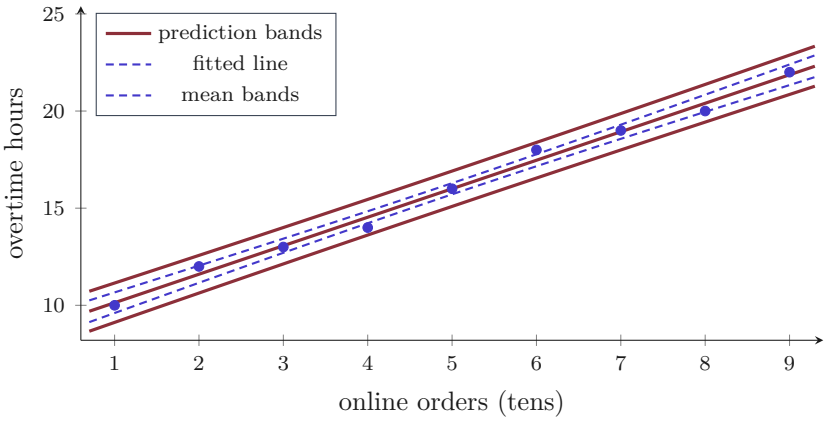


Figure 21.1: The 95 percent confidence interval for mean overtime (dashed line) and the 95 percent prediction interval for one new day (thick line), both around the fitted line $\hat{y} = 8.666667 + 1.466667x$. At $x = 8$ the mean interval (19.9588; 20.8412) has width 0.882, while the prediction interval (19.4304; 21.3696) has width 1.939. Both bands widen at the ends of the range, where the line is less supported by data; the ratio of their widths, 2.198, does not depend on n.

21.7 Residual diagnostics

Level: intermediate

Every interval above is conditional on the model being right. Hence the first step after fitting a line is not to compute p-values but to inspect the residuals. In everyday language: we look at whether what is left over from the line still has a pattern; a residual with a pattern means the line has the wrong shape, not that chance is at work.

Four patterns deserve recognition: a curve against x, sign of a non-linear relation; a funnel that widens with x, a sign that (A3) fails; one faraway point on its own, a suspect misrecording; and a pattern by time order, a sign that observations are not independent.

For the café data, the residuals in order are

$$-0.1333,\ 0.4,\ -0.0667,\ -0.5333,\ 0,\ 0.5333,\ 0.0667,\ -0.4,\ 0.1333.$$

There is no evident curvature, and the sum is zero as it should be. What is interesting is their size: the largest residual is 0.5333 hours, about 32 minutes, on the fourth and sixth days, both near the center of the data. Those points are in fact *not* very influential, because their leverage is only 0.1278.

The leverage of point i in a simple regression has a closed form

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}},$$

so on this data $h_{11} = h_{99} = 1/9 + 16/60 = 0.3778$, while $h_{55} = 0.1111$. An end point holds 3.4 times the weight of a middle point, and that is not a designer's choice but a consequence of geometry: a line rotated about the center of the data moves furthest at the ends. Actual influence combines the size of the residual with leverage, and the usual measure is Cook's distance

$$D_i = \frac{e_i^2}{p\, s^2} \cdot \frac{h_{ii}}{(1 - h_{ii})^2},$$

which here is largest, 0.2870, on the second and eighth days, both with residual 0.4 and leverage 0.2611. Values below one are generally considered not worrying, but that threshold is a convention, not a law.

One warning that is rarely written: with $n = 9$ diagnostics have almost no power. A pattern in nine points easily arises from chance, and a pattern that is really there is easily invisible. Diagnostics offer suspicion, not verdict.

Read it aloud (auditory learner)

Two voices.
"The R squared is zero point nine nine. So the model is almost perfect."
"Perfect for what? The residuals are still twenty-two minutes."
"But ninety-nine percent of the wobble is already explained."
"Explained by what? By orders. Not by a kitchen that suddenly slowed down."
"So a high R squared is not a guarantee."
"A high R squared only means the line is tight to the points that exist. It knows nothing about a day that has not come yet."
Say it once more, slowly: tight to the old points is not a promise about a new day. Repeat it until it feels natural in the mouth: tight to the old points, not a promise.

Step-by-step recipe (procedural learner)

1. Plot y against x before computing anything, and ask whether a straight line makes sense.
2. Fit the line by least squares, then write the equation complete with the unit of every coefficient.
3. Inspect the residuals against x and against time order; look for curvature, a funnel, and isolated points.
4. Report s together with R^2, then an interval for the coefficient that carries the main conclusion.
5. If the aim is to forecast one event, use a prediction interval; if the aim is a mean, use a mean interval.

A warning about order: step 1 must not be swapped with step 2. Plotting after fitting the line makes the eye look for justification, not for faults.

21.8 When the answer is yes or no

Level: advanced

So far y has been a number that can be measured. Many important questions are binary: a customer returns or not, a transaction is fraudulent or not, a patient recovers or not. It is still tempting to fit an ordinary straight line to 0 and 1, then read it as a probability. Two things go wrong at once, and both can be computed.

The following data: 26 customers, the distance of their home from the stall x in kilometers, and $y = 1$ if they returned within thirty days. Of the 26, 15 returned. A linear regression on the binary y, called the *linear probability model*, gives $\hat{y} = 1.346154 - 0.096154x$. At $x = 2$ the forecast is 1.153846: a probability greater than one, and that is not a rounding error but a defect of form. At $x = 20$, still outside the data range but only seven kilometers outside, the forecast is -0.576923.

The second failure is subtler. For binary y, the variance of the residual is really $p(1 - p)$ and therefore depends on x: at $x = 2$ its value is 0.007225, at $x = 14$ it is 0.028890. The

linear probability model assumes one variance for all points, namely $s^2 = 0.124199$ on this data, so its standard errors are wrong exactly where it matters most. Ordinary least squares, which works well when the variance is constant, here relies on an assumption that plainly does not hold.

Definition 21.5 (Logistic model)**.** For $p_i = P(y_i = 1 \mid x_i)$,

$$\log \frac{p_i}{1 - p_i} = a + b\, x_i, \qquad \text{that is} \qquad p_i = \frac{1}{1 + e^{-(a + b x_i)}}.$$

The parameters are estimated by maximum likelihood, not by least squares, so there is no closed form and the computation is iterative.

odds ratio (*rasio peluang*) — how many times the odds multiply for one additional unit, not by how many points the probability rises

For that data, maximum likelihood gives $\hat{a} = 6.315432$ and $\hat{b} = -0.699945$ with standard errors 2.383413 and 0.258594. The odds ratio is $e^{-0.699945} = 0.496613$: each extra kilometer cuts the odds of returning to about a half, in the sense of a ratio, not a difference in points. The estimated probabilities are 0.992722 at $x = 2$ and 0.029777 at $x = 14$, both legitimate as probabilities. The 95 percent interval for b is $(-1.233657;\ -0.166233)$; it is entirely negative, so the direction is clear even if the magnitude is not.

One costly misreading: a logistic coefficient is not the marginal effect on probability. The value $b = -0.699945$ acts on the log-odds scale, and its effect on probability is largest around $p = 0.5$ and smallest at the ends. A good report states probabilities at several meaningful values of x, not just the coefficient.

21.9 Correlation is not causation

Level: basic

Now the part most often violated in reports that look tidy. Two groups of employees attend training of different length, and the number of errors per hundred transactions is recorded.

morning shift	training hours	1	2	3	4	5
	errors	4.0	3.6	3.2	2.9	2.5
night shift	training hours	5	6	7	8	9
	errors	8.0	7.5	7.0	6.4	6.0

Within the morning shift, the slope is -0.36 errors per training hour. Within the night shift, -0.474286. Both are negative: longer training goes with fewer errors. But when the two groups are pooled, the slope is $+0.359036$ with $R^2 = 0.226121$. The sign flips, and not one of the numbers is wrong.

The cause is not magic but the addition of two sources:

$$b_{\text{pooled}} = \frac{\sum_g S_{xy,g} + \sum_g n_g(\bar{x}_g - \bar{x})(\bar{y}_g - \bar{y})}{\sum_g S_{xx,g} + \sum_g n_g(\bar{x}_g - \bar{x})^2}.$$

The morning group contributes $S_{xy} = -6.3$ and $S_{xx} = 17.5$; the night group $S_{xy} = -8.3$ and $S_{xx} = 17.5$; the between-group term is $6 \cdot (-2)(-1.85) + 6 \cdot (2)(1.85) = 44.4$; the denominator is 83. Hence $b_{\text{pooled}} = (-6.3 - 8.3 + 44.4)/83 = 29.8/83 = 0.359036$. The between-group term 44.4 defeats the sum of within-group terms -14.6, and that is the whole secret.

The interpretation: training hours are not assigned at random. Night-shift employees, who make more errors because of working hours and the kinds of transactions, also receive more training, so what is measured as the effect of training is mostly the difference between shifts. Figure 21.2 places everything on one plane.

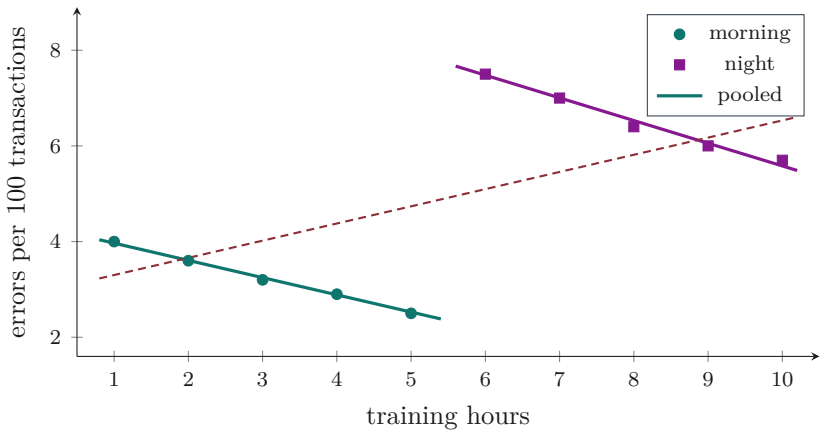


Figure 21.2: Simpson's paradox in the training data. Within each shift the line falls: -0.36 errors per hour for the morning and -0.4743 for the night. The pooled line, computed from all twelve points regardless of shift, instead rises with slope $+0.3590$. The change of sign comes from the difference between groups, not from one odd point; the between-group term 44.4 defeats the within-group term -14.6.

For younger readers (age 10 and up)

Two classes take part in a reading contest. Class A reads thin books, many pages, and gets good marks. Class B reads thick books, fewer pages, and gets poor marks. If all the pupils are pooled, it looks as if the fewer pages read, the higher the mark. Yet inside each class, the more pages, the higher the mark.

What is wrong is not the numbers but the pooling. Class B really was given thicker books, and class B really is weaker at reading. The two facts blend into one deceptive line.

In this chapter we will learn why pooling two groups can reverse the direction of a conclusion, why the question "does training reduce errors" cannot be answered just by plotting all the points, and why anyone changing policy must know which groups they are really comparing.

Challenge (problem solver)

Build a table with two groups, five points each, such that both within-group slopes are positive but the pooled slope is negative, and the difference between the group means of y is as small as still possible. How small is that difference, and why can it not be smaller?

Hint 1: the sign of the pooled slope is decided by the between-group term. Make the first group have small x and small y, the second large x and large y, then reverse the direction inside each group.

Hint 2: with five points and within-group slope $+1$ at $S_{xx} = 10$, each group contributes $S_{xy} = +10$ and the between-group term must be less than -20. Since $n_g(\bar{x}_g - \bar{x})(\bar{y}_g - \bar{y})$ is negative only when one of the mean differences has the opposite sign to the other, the mean difference must pass the point where the two terms cancel.

If you get stuck, go back to the section Correlation is not causation and recompute 29.8/83 step by step.

21.10 Honest reporting

Level: intermediate

A regression analysis can be done correctly and reported so as to mislead without a single number being falsified. Four habits close most of that gap.

First, state how many observations and how many parameters, because the reader needs both to compute the degrees of freedom and to judge whether s can be trusted. On the café data, $n = 9$ and $p = 2$, so s is built from seven degrees of freedom; hiding that number makes s look sturdier than it is.

Second, separate the data used to choose the model from the data used to judge it. Choosing the model form, choosing the variables, and computing the error on the same set always produces numbers that are too good. At $n = 9$ that separation is impossible, and precisely for that reason the conclusion must be called weak, not reinforced with $R^2 = 0.992821$.

Third, state how many models were tried. If ten possibilities are tested and the best is reported, the chance that at least one of them looks meaningful is $1 - 0.95^{10} = 0.401263$ even though none of the effects is real. That number is not the probability of being wrong, but the probability of picking something that is a coincidence.

Fourth, report magnitudes with units and intervals, not just stars. The sentence "$\hat{b} = 0.687$ (95 percent: 0.381 to 0.993 million per employee)" can be checked; the sentence "significantly affects" cannot. If the conclusion is about causation, say why: randomization, or adjustment for the named confounders.

21.11 A worked example from end to end

Level: intermediate

We repeat the whole chain on the café data, with the intermediate numbers written out, so that every step can be checked with a calculator.

The means are $\bar{x} = 5$ and $\bar{y} = 16$. The squared deviations are $S_{xx} = \sum (x_i - 5)^2 = 60$ and $S_{yy} = 130$; their cross product is $S_{xy} = \sum (x_i - 5)(y_i - 16) = 808 - 9 \cdot 5 \cdot 16 = 88$. Hence $\hat{b} = 88/60 = 1.466667$ and $\hat{a} = 16 - 1.466667 \cdot 5 = 8.666667$.

The forecasts at the nine points are 10.1333; 11.6; 13.0667; 14.5333; 16; 17.4667; 18.9333; 20.4; 21.8667. The residuals are -0.1333; 0.4; -0.0667; -0.5333; 0; 0.5333; 0.0667; -0.4; 0.1333, and their sum of squares is SSE = 0.933333. With $n - p = 7$, $s^2 = 0.133333$ and $s = 0.365148$ hours.

The standard error of the slope is $s/\sqrt{S_{xx}} = \sqrt{0.133333/60} = \sqrt{0.00222222} = 0.047140$. The statistic is $t = 1.466667/0.047140 = 31.1127$, and $t_{7,0.975} = 2.364624$, so the half-width of the interval is 0.111470 and the interval is (1.355197; 1.578136).

Repeat the same computation for one change of data. The ninth day is recorded with 22 hours of overtime; suppose the correct record is 26 hours. Recompute $\hat{b}$, s, se($\hat{b}$), and its 95 percent interval, then compare with the table above. The correct answers: $\hat{b} = 1.733333$, $s = 1.306880$, se($\hat{b}$) = 0.168718, interval (1.3344; 2.1323). The width is 0.798 against 0.223, that is 3.6 times. Explain why one point at the end can do that, when it is only one of nine.

21.12 Computational lab

Level: advanced

Every number in this chapter can be recomputed in one short file. The following program uses $X^\top X$ directly, so it also checks the algebra of the second section.

Computational lab

```
import numpy as np
```

```
from math import sqrt
from scipy import stats

x = np.arange(1, 10, dtype=float)
y = np.array([10, 12, 13, 14, 16, 18, 19, 20, 22], dtype=float)
n = len(y)
X = np.column_stack([np.ones(n), x])

beta = np.linalg.solve(X.T @ X, X.T @ y)
fit = X @ beta
e = y - fit
s2 = (e ** 2).sum() / (n - 2)
sst = ((y - y.mean()) ** 2).sum()

print("X'X =", (X.T @ X).tolist(), " X'y =", (X.T @ y).tolist())
print("beta = %.6f %.6f" % tuple(beta))
sse = (e ** 2).sum()
print("SSE = %.6f  R2 = %.6f  s = %.6f"
      % (sse, 1 - sse / sst, sqrt(s2)))
print("residuals =", np.round(e, 4).tolist())

cov = s2 * np.linalg.inv(X.T @ X)
se = np.sqrt(np.diag(cov))
tq = stats.t.ppf(0.975, n - 2)
print("se = %.6f %.6f  t = %.4f %.4f"
      % (se[0], se[1], beta[0] / se[0], beta[1] / se[1]))
lo, hi = beta[1] - tq * se[1], beta[1] + tq * se[1]
print("CI95 slope = (%.6f, %.6f)" % (lo, hi))

for x0 in (5.0, 8.0):
    c = np.array([1.0, x0])
    y0 = c @ beta
    v = s2 * (c @ np.linalg.inv(X.T @ X) @ c)
    print("x0=%.0f  yhat=%.4f  mean +/-%.4f  prediction +/-%.4f"
          % (x0, y0, tq * sqrt(v), tq * sqrt(v + s2)))

h = np.einsum("ij,jk,ik->i", X, np.linalg.inv(X.T @ X), X)
cook = e ** 2 / (2 * s2) * h / (1 - h) ** 2
print("h_ii =", np.round(h, 4).tolist())
print("D_Cook =", np.round(cook, 4).tolist())

x1 = np.array([6, 7, 7, 8, 8, 9, 9, 10, 10, 11, 12, 12], dtype=float)
x2 = np.array([18, 22, 20, 26, 24, 28, 26, 32, 30, 34, 36, 38], dtype=float)
yb = np.array([11.2, 12.4, 12.0, 14.1, 13.6, 15.2, 14.8, 17.0, 16.4, 18.1,
               19.4, 19.9])
Xb = np.column_stack([np.ones(12), x1, x2])
bb = np.linalg.solve(Xb.T @ Xb, Xb.T @ yb)
eb = yb - Xb @ bb
s2b = (eb ** 2).sum() / 9
seb = np.sqrt(np.diag(s2b * np.linalg.inv(Xb.T @ Xb)))
r12 = np.corrcoef(x1, x2)[0, 1]
print("multiple beta = %.4f %.4f %.4f  s = %.4f"
      % (bb[0], bb[1], bb[2], sqrt(s2b)))
print("multiple se   = %.4f %.4f %.4f" % tuple(seb))
print("r12 = %.4f  VIF = %.2f  sqrt VIF = %.2f"
      % (r12, 1 / (1 - r12 ** 2), 1 / sqrt(1 - r12 ** 2)))


def ols(xx, yy):
    A = np.column_stack([np.ones(len(xx)), xx])
    return np.linalg.solve(A.T @ A, A.T @ yy)[1]
```

```
xp = np.array([1, 2, 3, 4, 5, 6], dtype=float)
yp = np.array([4.0, 3.6, 3.2, 2.9, 2.5, 2.2])
xm = np.array([5, 6, 7, 8, 9, 10], dtype=float)
ym = np.array([8.0, 7.5, 7.0, 6.4, 6.0, 5.7])
print("Simpson: morning %.4f  night %.4f  pooled %.4f"
      % (ols(xp, yp), ols(xm, ym), ols(np.r_[xp, xm], np.r_[yp, ym])))

xd = np.array([2, 2, 3, 3, 4, 4, 5, 5, 6, 6, 7, 7, 8, 8, 9, 9, 10, 10,
               11, 11, 12, 12, 13, 13, 14, 14], dtype=float)
yd = np.array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0,
               1, 0, 0, 0, 0, 0, 0], dtype=float)
A = np.column_stack([np.ones(len(yd)), xd])
lpm = np.linalg.solve(A.T @ A, A.T @ yd)
print("LPM: %.4f %+.4f  prediction at x=2: %.4f  at x=20: %.4f"
      % (lpm[0], lpm[1], lpm @ [1, 2], lpm @ [1, 20]))


def nll(th):
    z = A @ th
    return float(np.sum(np.log1p(np.exp(-z)) + (1 - yd) * z))


g = np.array([0.0, 0.0])
for _ in range(40):
    p = 1 / (1 + np.exp(-(A @ g)))
    grad = A.T @ (p - yd)
    Hm = A.T @ (A * (p * (1 - p))[:, None])
    g = g - np.linalg.solve(Hm, grad)
se_g = np.sqrt(np.diag(np.linalg.inv(Hm)))
p = 1 / (1 + np.exp(-(A @ g)))
print("logit: %.4f %+.4f  se %.4f %.4f" % (g[0], g[1], se_g[0], se_g[1]))
print("deviance = %.4f  p(x=2) = %.4f  p(x=14) = %.4f"
      % (2 * nll(g), p[0], p[-1]))
print("odds ratio per km = %.4f" % np.exp(g[1]))
```

The output, exactly as it comes:

```
X'X = [[9.0, 45.0], [45.0, 285.0]]  X'y = [144.0, 808.0]
beta = 8.666667 1.466667
SSE = 0.933333  R2 = 0.992821  s = 0.365148
residuals = [-0.1333, 0.4, -0.0667, -0.5333, 0.0, 0.5333, 0.0667, -0.4, 0.1333]
se = 0.265274 0.047140  t = 32.6706 31.1127
CI95 slope = (1.355197, 1.578136)
x0=5  yhat=16.0000  mean +/-0.2878  prediction +/-0.9101
x0=8  yhat=20.4000  mean +/-0.4412  prediction +/-0.9696
h_ii = [0.3778, 0.2611, 0.1778, 0.1278, 0.1111, 0.1278, 0.1778, 0.2611, 0.3778]
D_Cook = [0.0651, 0.287, 0.0044, 0.1792, 0.0, 0.1792, 0.0044, 0.287, 0.0651]
multiple beta = 2.4430 0.6869 0.2393  s = 0.1436
multiple se   = 0.2076 0.1354 0.0421
r12 = 0.9868  VIF = 38.13  sqrt VIF = 6.18
Simpson: morning -0.3600  night -0.4743  pooled 0.3590
LPM: 1.3462 -0.0962  prediction at x=2: 1.1538  at x=20: -0.5769
logit: 6.3154 -0.6999  se 2.3834 0.2586
deviance = 17.8843  p(x=2) = 0.9927  p(x=14) = 0.0298
odds ratio per km = 0.4966
```

Two things must be read from that output. First, $\mathrm{X}^{\top}\mathrm{X}$ and $\mathrm{X}^{\top}y$ are printed exactly as computed by hand, and $\hat{\beta}$ comes out of solving the system; agreement of this kind is not proof that the theory is true, but it removes arithmetic slips, which are the most frequent cause. Second, the last three parts show why interpretation is more fragile than computation:

0.6869 against 1.4460 in the single regression, VIF 38.13, and a pooled slope whose sign reverses. Not one of the three is a programming error.

Note for instructors

The teaching order I suggest: start from the table of nine days, ask the class to guess the slope in two ways, by the difference of the ends and by "the closest line". The two answers differ only slightly, and that is exactly the entry point: the real question is not the number but its wobble. Only after the class is restless, move to the normal equations and show that $\sum x_i e_i = 0$ is a geometric statement, not an algebraic trick.

The two misconceptions that appear most often. First, students read R^2 as "the probability that the forecast is right", then are surprised when the prediction interval is two hours wide at $R^2 = 0.99$; the symptom shows when they cannot explain why s does not shrink along with R^2. Second, they interpret multiple coefficients like single ones, even though on the twelve-branch data the number moves from 1.446 to 0.687 only because another explanatory variable enters.

A blackboard activity that works: have two groups compute the slope on the training data, one group using all twelve points, one separating the shifts. Write both results side by side, then ask each group to defend theirs. The sign that the class is ready to move on: they can name one situation in which a multiple coefficient changes sign, and can explain why the forecast stays good.

21.13 Summary

Level: basic

Summary.

- Regression fits the line that minimizes the sum of squared residuals. By the normal equations the residual is perpendicular to every column of X; on the café data $\hat{b} = 88/60 = 1.466667$ and $\hat{a} = 8.666667$.
- A coefficient is meaningful only together with its unit: 1.466667 hours per ten orders equals 8.8 minutes per order. Changing the scale of x changes the coefficient and its standard error, not t or R^2.
- R^2 is relative, s is absolute. On the café data $R^2 = 0.992821$ but $s = 0.365148$ hours, that is 21.9 minutes.
- Under (A1)–(A4) Gauss–Markov guarantees that least squares is BLUE with $\mathrm{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$; that guarantee lapses when the error variance is not constant or the explanatory variables are collinear.
- Collinearity does not damage the forecast but it damages the interpretation: on the twelve branches, $r_{12} = 0.986801$ gives VIF $= 38.13$ and standard errors 6.18 times larger, while R^2 stays at 0.997951.
- A prediction interval is always wider than a mean interval; at $x_0 = 8$ the ratio is 2.198 and does not shrink with n. The sign of a slope can reverse when two groups are pooled: -0.36 and -0.474286 within groups, $+0.359036$ once pooled.

21.14 Exercises

Level: intermediate

Exercise 21.1. 1. Evaluate this claim: "Because $R^2 = 0.99$, this line can forecast a single day with an error of at most twenty minutes." True, false, or not enough evidence?

Name the part that is right, the part that is wrong, and the additional number that settles the assessment.

2. Prove that with a column of ones in X we have $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$, then explain why both fail when the line is forced through the origin.
3. Prove $\mathrm{Var}(\hat{b}) = \sigma^2 / S_{xx}$ from $\hat{b} = \sum_i (x_i - \bar{x}) y_i / S_{xx}$ without matrix formulas.
4. Compute $\hat{a}$, $\hat{b}$, s, $\mathrm{se}(\hat{b})$, and the 95 percent interval for $(1, 2)$, $(2, 2.5)$, $(3, 4)$, $(4, 3.5)$, $(5, 5)$, then check with Python. Why is the interval much wider than the interval for the café data?
5. Compute the 95% prediction interval for $x_0 = 2$ and $x_0 = 9$ on the café data. Why do the widths differ when n is the same, and why does the interval never shrink to zero when n is increased?
6. Show that adding explanatory variables cannot raise SSE, then explain why R^2_{adj} can fall. Compute R^2_{adj} for the café data if p rises from 2 to 3 at the same R^2.
7. For $S_{11} = 50$, $S_{22} = 200$, $r_{12} = 0.9$, and $\sigma^2 = 2$, compute the VIF and the standard error of $\hat{b}_1$ with and without x_2 in the model.
8. Build a table with two groups with within-group slope -0.2 and pooled slope $+0.4$, then check the result with the decomposition formula.
9. Rewrite this sentence so that it is honest: “the best model of the twelve we tried gives $p = 0.03$, so the effect is real”.

Hints for the three hardest items. Item 3: write $\hat{b} - \beta = \sum_i (x_i - \bar{x}) \varepsilon_i / S_{xx}$, take the variance, use independence and $\mathrm{Var}(\varepsilon_i) = \sigma^2$. Item 6: compare the column spaces of the two models, then remember the divisor $n - p$ that shrinks. Item 7: $\mathrm{VIF} = 1/(1 - 0.81) = 5.263158$, so the standard error is $\sqrt{5.263158}$ times larger.

For researchers: limits and next steps

What is genuinely proved here: the normal equations as a necessary and sufficient condition for the minimum, together with uniqueness when X has full column rank; the properties of the hat matrix and $\operatorname{tr} H = p$; unbiasedness and $\mathrm{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$; the Gauss–Markov inequality within the class of linear unbiased estimators; the decomposition $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ when a column of ones is present; the non-monotonicity of R^2 under added columns; the t distribution for the coefficient ratio under normal errors; and the variance of the forecast error.

What is merely popular: that R^2 measures the quality of a model, that multiple coefficients always represent effects, and that the thresholds $\mathrm{VIF} > 10$ or $D_i > 1$ are derived results. Primary sources: Seber and Lee, *Linear Regression Analysis*, Chapter 3; Greene, *Econometric Analysis*, Chapter 3; Wasserman, *All of Statistics*, Chapter 13; Simpson (1951) for the reversal of direction in contingency tables; Hosmer, Lemeshow, and Sturdivant, *Applied Logistic Regression*, Chapters 1–3.

Limits of validity: every interval is conditional on the model being right, the errors being independent, and the variance being constant. On grouped data, time series, or repeated observations, independence fails and the reported standard errors are too small. Open question: how far post-selection intervals depart from their nominal level, and how influence diagnostics are constructed when n is far smaller than p.

Map to source texts: Chapter 21 in Wasserman, All of Statistics*: Chapter 13 (Linear and Logistic Regression: least squares, the t distribution for coefficients, and logistic regression) and Chapter 6 (Models, Statistical Inference, and Learning: the interpretation of intervals and standard errors). The geometry of least squares as an orthogonal projection is set out in Chapter 10 of this book, where the normal equations first appear as a consequence of orthogonality, and is used again in Chapter 24 for*

regularization. Confidence intervals, p-values, and the interpretation of levels are taken from Chapter 20 and not repeated here. The Gauss–Markov theorem and its conditions are referred to Seber and Lee Chapter 3 and Greene Chapter 3, because Wasserman does not prove it. Simpson's paradox is referred to Simpson's (1951) paper and used again in Chapter 25 when discussing data separation and evaluation leakage, and in Chapter 26 when discussing causal claims that must be proved, not inferred from correlation.

Chapter 22

Floating-Point Arithmetic and Error Analysis

A program solves $x^2 + 10^8 x + 1 = 0$ with the quadratic formula we have used since school, $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$. For $a = 1$, $b = 10^8$, $c = 1$ it reports the roots -10^8 and $-7.450580596923828 \times 10^{-9}$. The first root is correct to the last digit. The second is wrong by a quarter: the true value is -10^{-8}, and its relative error is 0.255. The formula is not mistyped. Neither is the machine broken. The coefficients are exact, the discriminant is exact, and the subtraction is even exact according to the theorem we shall prove below.

What collapses is the algebraic form. The numbers $b = 10^8$ and $\sqrt{b^2 - 4ac}$ differ by only one ulp, that is one smallest step the machine can still tell apart at that scale: the difference is 1.49×10^{-8}, or 1.5×10^{-16} times the size of the numbers themselves. A difference that small means nothing as long as we regard numbers of size 10^8. As soon as the two are subtracted, the result has size 10^{-8} and that difference becomes the whole answer: what remains is error, not a number.

Chapters 4 and 5 build limits, continuity, and derivatives over $\mathbb{R}$ whose every point is exact. Chapter 20 uses the variance formula and the square root under the same assumption. A computer does not work in $\mathbb{R}$. It works in a finite set of numbers that has rules of its own: every operation is rounded, addition is not associative, and two algebraically equivalent routes can give answers far apart. This chapter builds a way of reading those rules, of measuring the damage before the program runs, and of choosing a form of the formula that does not collapse.

For younger readers (age 10 and up)

Imagine a digital kitchen scale whose display holds only five digits. We weigh a sack of rice together with its container, then weigh the empty container, then subtract the two numbers to get the weight of the rice. If container and rice each weigh about twenty kilograms, the display can only show numbers to the nearest tens of grams. The difference of two large numbers that are nearly equal is correct as arithmetic, but its last digits no longer mean anything. Rice that actually weighs eight hundred grams can show up as seven hundred grams, or even as an absurd number.

There is a second, subtler problem. The same scale cannot weigh a single grain of rice: the needle does not move at all. If we pour in a million grains one at a time, not one of them is recorded, even though a million grains have weight. A calculating machine works in exactly that way: some numbers are too small to change anything, and some subtractions throw away almost every digit.

The question: if the machine is like that, how can we know when an answer can still

be trusted? In this chapter we shall learn why that error is not a random event but something whose bound can be computed before a single line of the program runs.

22.1 Numbers without a representative

Level: basic

Between 0 and 1 there are infinitely many real numbers. Inside a computer, all numbers are stored in 32 or 64 bits, so only finitely many are available. The compromise adopted from the first machines down to the IEEE-754 standard is binary scientific notation with fixed field widths: one sign, one exponent, one mantissa.

floating-point system (*sistem floating-point*) — Binary scientific notation with a fixed number of digits, like writing 1.01×2^{7} in limited room.

Definition 22.1 (Floating-point number system)**.** Let $\beta = 2$ be the base, p the precision, and $e_{\min} \le e \le e_{\max}$ the exponent bounds. A normal number is

$$x = \pm\, (1.d_1 d_2 \ldots d_{p-1})_2 \times 2^{e},$$

with $d_i \in \{0, 1\}$. Subnormal numbers use $e = e_{\min}$ with leading digit 0. We call the set of all representable numbers, together with zero, the two infinities, and NaN, $\mathbb{F}$.

The leading digit is always 1 for a normal number, so that digit is not stored; that is why the mantissa field has $p - 1$ bits. One bit saved doubles the precision: 23 stored bits give 24 significant bits.

Theorem 22.1 (Number of machine numbers)**.** *For every exponent e, the number of normal numbers in the interval $[2^{e}, 2^{e+1})$ is 2^{p-1}. The set $\mathbb{F}$ is finite, hence countable.*

Proof. On that interval every number has the form $M \cdot 2^{e-p+1}$ with M an integer and $2^{p-1} \le M < 2^{p}$. The number of such M is exactly 2^{p}, minus 2^{p-1}, that is 2^{p-1}. There are finitely many allowed exponents, so $\mathbb{F}$ is a finite union of finite sets. $\square$

For binary64, $2^{p-1} = 2^{52} = 4\,503\,599\,627\,370\,496$. That is a lot, but Chapter 2 already showed that $\mathbb{R}$ is uncountable. So almost every real number has no representative: representation always means rounding, and rounding always means error. The question is not whether there is error, but how large it is and where it goes.

Read it aloud (auditory learner)

Two voices, read alternately with a friend. Tap the table every time the word “piece” is said.
“One divided by three, how much?” “Zero point three three three, on and on, without end.” “Does the machine write it out whole?” “No. It stops where there is room.” “So what is stored?” “A piece. Not the value, the piece.” “And if the pieces are added?” “What grows is the pieces, not the value.” “And if the pieces are subtracted?” “The leading digits are used up, dust is what remains.”
Repeat the three middle lines, faster and faster, with the same tapping: a piece is not the value, a piece is not the value, a piece is not the value.
The second voice closes quietly: the machine does not lie, it only cuts. We are the ones who go wrong if we read the piece as the value.
The line the reader must repeat, once, in a clear voice: a computer stores pieces, not values, so every computed result must be checked, not trusted.

The IEEE-754 standard fixes two widths that are used everywhere. This table contains all the numbers; nothing needs memorizing except p and the bias.

	binary32	binary64
sign / exponent / mantissa bits	1/8/23	1/11/52
precision p	24	53
exponent bias	127	1023
$e_{\min},\ e_{\max}$	$-126,\ 127$	$-1022,\ 1023$
$\text{eps} = 2^{-p+1}$	1.1920929×10^{-7}	$2.2204460 \times 10^{-16}$
$u = 2^{-p}$	5.9604645×10^{-8}	$1.1102230 \times 10^{-16}$
smallest normal	$1.1754944 \times 10^{-38}$	$2.2250739 \times 10^{-308}$
smallest subnormal	$1.4012985 \times 10^{-45}$	$4.9406565 \times 10^{-324}$
largest	3.4028235×10^{38}	$1.7976931 \times 10^{308}$

Exponents are stored with a bias so that no separate sign is needed: a stored value of 123 means an actual exponent of $123 - 127 = -4$. This trick makes the bit patterns from zero up to positive infinity increase monotonically, so that two positive numbers can be compared as integers.

Let us take apart three bit patterns that Python actually produces. In binary32, 0.1 prints as `0 01111011 10011001100110011001101` with $e = 123 - 127 = -4$; 6 becomes `0 10000001 10000000000000000000000`, that is 1.5×2^2; and -12.375 becomes `1 10000010 10001100000000000000000`, that is -1.546875×2^3. Two boundary cases complete the picture:

smallest subnormal	`0 00000000`	`00000000000000000000001`
largest	`0 11111110`	`11111111111111111111111`

The first has the value 2^{-149}: all its exponent bits are zero and only the last mantissa bit is on. The second has the value 3.4028235×10^{38}, with stored exponent 254 and a full mantissa.

A more readable form for binary64 is provided by `float.hex()`. Here is the result for several numbers:

0.1	`0x1.999999999999ap-4`
6.0	`0x1.8000000000000p+2`
10^{-8}	`0x1.5798ee2308c3ap-27`
5×10^{-324}	`0x0.0000000000001p-1022`

That form shows three things. The digits after the hexadecimal point are always thirteen, and $13 \times 4 = 52$, the width of the binary64 mantissa. The last digit of 0.1 is `a`, that is 1010, a rounding up from the repeating pattern 1001. The last row has a zero before the point: the mark of a subnormal number.

Special values complete this system: $\pm\infty$ for results beyond $e_{\max}$, signed zero, subnormal numbers below $2^{e_{\min}}$, and NaN (*not a number*) for indeterminate forms such as $0/0$. One property of NaN deserves remembering because it often surprises: it is not equal to itself, so that the comparison `x == x` is false precisely when x is NaN.

22.2 The only bridge: rounding

Level: basic

No operation other than rounding connects $\mathbb{R}$ with $\mathbb{F}$. That is why the whole of error analysis rests on one question: how far is the nearest machine number from the original number, and how does that distance scale.

IEEE-754 lays down the rule *round to nearest, ties to even*: choose the nearest machine number, and if two candidates are equally far, choose the one whose last mantissa digit is even. The second rule is no theoretical luxury. Without that rule, rounding at a tie point always leans one way, and the error of a million operations piles up in the same direction instead of canceling out.

Tie points really do occur, and we can create one deliberately. In binary64, the distance between 1 and the number after it is 2^{-52}. Hence $1 + 2^{-53}$ falls exactly halfway, and the even rule chooses 1 because its mantissa ends in zero. Python gives

`1 + 2.0**-53 = 1.0`, whereas `1 + 3*2.0**-53 = 1.0000000000000004`.

The first case rounds down, the second up, although both are exactly halfway. Only the parity of the last digit distinguishes them. This is why $1 + 10^{-16} = 1$ in binary64: not because 10^{-16} is ignored carelessly, but because there is no room to store it.

machine epsilon (*epsilon mesin*) — The distance between 1 and the next machine number: the size of one step at scale one.

Two numbers are often confused, and the confusion changes an error bound by a factor of two. The machine epsilon $\text{eps} = 2^{-p+1}$ is the distance between 1 and the number after it, namely 2^{-52} for binary64. The unit roundoff $u = 2^{-p} = 2^{-53}$ is half that distance, and only u appears in error bounds. Python calls the first `numpy.finfo(float).eps`; the second we must compute ourselves.

The distance between numbers is not uniform. Near 1 it is 2^{-52}, near 10^{10} it is 1.9×10^{-6}. What is uniform is that distance relative to the size of the number, and that is the content of the following lemma.

Lemma 22.2 (Ulp)**.** *If $2^e \le |x| < 2^{e+1}$, then the distance between x and the next machine number is* $\text{ulp}(x) = 2^{e-p+1}$.

Proof. On that interval all normal numbers have the form $M \cdot 2^{e-p+1}$ with M an integer and $2^{p-1} \le M < 2^p$; the number of such M is 2^{p-1} and the gap between two consecutive M is one unit 2^{e-p+1}. The next interval uses $e + 1$ so that the gap doubles, and that is the only point where the density changes. □

ulp (*unit in the last place*) — The smallest step the machine can tell apart at the scale of a number.

The lemma explains two things at once. $\text{ulp}(1) = 2^{-52} = 2.220446049250313 \times 10^{-16}$, whereas $\text{ulp}(10^{10}) = 2^{-52} \cdot 2^{33} \approx 1.9 \times 10^{-6}$. So adding 1 to 10^{16} changes nothing in binary64, because $\text{ulp}(10^{16}) = 2$: there really is no machine number between the two.

Figure 22.1 shows the consequence of that lemma over the first four octaves: each octave holds 2^{23} points, but its width doubles, so that the distance between points doubles as well.

There is one subtlety worth naming before we use the ulp as a unit of error. The largest rounding error is half an ulp, but expressed as a relative error that figure is not one number but an interval. For x exactly equal to a power of two, half an ulp is 2^{-53} relative; for x that nearly reaches the next power of two, the value approaches 2^{-54}. So the relative error of one rounding can swing by a factor of two. Goldberg named that swing *wobble*, and that is why the bound u is always used as a worst case, not as a typical value.

Theorem 22.3 (Relative rounding error)**.** *For every $x \in \mathbb{R}$ in the normal range, correct rounding satisfies $\text{fl}(x) = x(1 + \delta)$ with $|\delta| \le u = 2^{-p}$.*

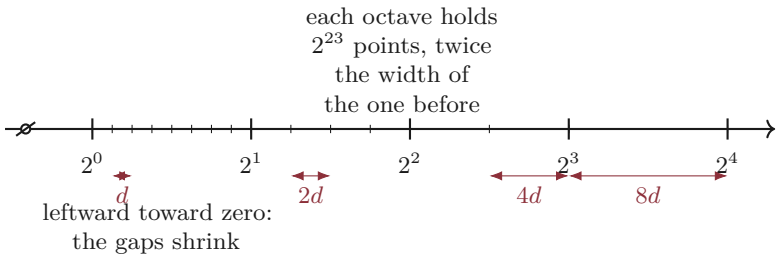


Figure 22.1: The first four octaves of binary32 machine numbers. Each octave holds 2^{23} points, but its width is twice that of the octave before, so the spacing between points doubles: $d = 2^{-23}$ on $[1, 2)$, $2d$ on $[2, 4)$, $4d$ on $[4, 8)$, and $8d$ on $[8, 16)$. The tail does not hold fewer points, only sparser ones; toward zero the spacing shrinks down to the subnormal limit.

Proof. Take $2^e \le |x| < 2^{e+1}$. The nearest machine number is at most half an ulp away, that is 2^{e-p}, so that $|\,\mathrm{fl}(x) - x| \le 2^{e-p} = u \cdot 2^e \le u|x|$. Dividing by $|x|$ gives $|\delta| \le u$, and the sign of δ follows the direction of rounding. □

The form $x(1+\delta)$ is the beginning of everything: it turns the question "how wrong" into "how many factors $(1+\delta)$ pile up". The operations $+, -, \times, \div, \sqrt{\ }$ in IEEE-754 are required to return the correctly rounded exact result, so every operation contributes one such factor and no more. This guarantee is what makes error analysis possible.

22.3 Error, condition number, and cancellation

Level: intermediate

Now we have a unit. What we do not yet have is a measure of how much the error of a computation exceeds the error of its input. Two problems can both receive input that is wrong by 10^{-16} and come out with an answer that is wrong by 10^{-16} or by 10^{-1}, depending on the problem itself.

condition number (*bilangan kondisi*) — The multiplier showing how many times a computation amplifies the error of its input.

Definition 22.2 (Absolute, relative, and backward error)**.** Let $\tilde{x}$ be an approximation to x. Its absolute error is $|\tilde{x} - x|$, its relative error is $|\tilde{x} - x|/|x|$ for $x \neq 0$. If $\tilde{x}$ is the exact answer to a problem with data $x + \Delta x$, then $\|\Delta x\|/\|x\|$ is called the backward error, and an algorithm that always gives a small backward error is called *backward stable*.

The last distinction is the most useful in practice. A backward stable algorithm may be badly wrong, provided its answer is right for a slightly different problem: the error belongs to the problem. A large backward error is the mistake of the person writing the code.

For a function of one variable, that amplification is computed directly with the derivative: the condition number is the slope of the function at that point, multiplied by the size of the point, divided by the size of its value.

Theorem 22.4 (Condition number)**.** *Let f be differentiable on an interval containing x and $f(x) \neq 0$. If $\tilde{x} = x(1+\delta)$ with $|\delta| \le u$, then*

$$\frac{|f(\tilde{x}) - f(x)|}{|f(x)|} \;\le\; \kappa_f(x)\, u + O(u^2), \qquad \kappa_f(x) = \frac{|x\, f'(x)|}{|f(x)|}.$$

Proof. By the mean value theorem there is ξ between x and $\tilde{x}$ with $f(\tilde{x}) - f(x) = f'(\xi)(\tilde{x} - x)$. Since $|\tilde{x} - x| \le u|x|$ and f' is continuous at x, we have $|f'(\xi)| = |f'(x)| + O(u)$. Hence $|f(\tilde{x}) - f(x)| \le u|x|\,|f'(x)| + O(u^2|x|)$. Dividing by $|f(x)|$ gives the claimed inequality. □

The number κ_f does not depend on the machine, only on the problem. For $f(x) = \sqrt{x}$ it is $1/2$: a square root never makes the relative error worse. For $f(x) = \log x$ it is $1/|\log x|$, which is large when x is near 1 because there $\log x$ is almost zero. For $f(x) = x - c$ with c a large constant, it is $|x|/|x - c|$, which grows without bound as x approaches c. That last pattern is what becomes the source of the whole trouble in this chapter.

Subtracting two numbers that are close together is the most dangerous operation in machine arithmetic, and the danger is often blamed on the wrong thing. A common misunderstanding runs: subtraction loses digits, so the operation of subtraction is itself inaccurate. The truth is subtler and more reassuring: when the two numbers are close together, the subtraction is in fact exact.

Theorem 22.5 (Sterbenz)**.** *If $a, b \in \mathbb{F}$ and $b/2 \le a \le 2b$ with $b > 0$, then $a - b$ is represented exactly in $\mathbb{F}$.*

Proof. Let $2^e \le b < 2^{e+1}$. Every machine number on that interval is an integer multiple of 2^{e-p+1}, so a is as well, and hence $a - b$ is a multiple of 2^{e-p+1}. From $b/2 \le a \le 2b$ we get $0 \le a - b \le b/2 < 2^e$. A multiple of 2^{e-p+1} that lies in $[0, 2^e)$ has the form $M \cdot 2^{e-p+1}$ with $0 \le M < 2^p$, and all of them are represented with p digits. Hence the result is exact. □

So when the result of a subtraction looks bad, what is bad is the input. The digits that "disappear" had already disappeared before the subtraction was carried out: a and b are each known only to relative error u, and an absolute error of size $u|a|$ does not shrink along with the result when the two numbers cancel each other. Only the size of the result shrinks, so that the relative error explodes.

Proposition 22.6 (Condition number of subtraction)**.** *Let $x = a - b \neq 0$ and let the machine work with $\tilde{a} = a(1 + \delta_a)$, $\tilde{b} = b(1 + \delta_b)$, $|\delta_a|, |\delta_b| \le u$. Then*

$$\frac{|\operatorname{fl}(\tilde{a} - \tilde{b}) - x|}{|x|} \;\le\; \kappa\, u + u, \qquad \kappa = \frac{|a| + |b|}{|a - b|}.$$

Proof. Since $\operatorname{fl}(\tilde{a} - \tilde{b})$ is the correctly rounded version of $\tilde{a} - \tilde{b}$, the additional error is at most $u|\tilde{a} - \tilde{b}|$. Furthermore $|\tilde{a} - \tilde{b} - (a - b)| \le |a||\delta_a| + |b||\delta_b| \le u(|a| + |b|)$, and dividing by $|x| = |a - b|$ gives $u\kappa$. Adding the two errors with the triangle inequality completes the proof. □

The number κ is never less than one and grows without bound as $|a - b|$ shrinks. For $a = 12345.678$ and $b = 12345.671$ we get $a - b = 0.007$ and $\kappa \approx 3.53 \times 10^6$, so that the relative error of the result can reach 3.9×10^{-10} even though each input is off by only 10^{-16}. Since every factor of ten in κ erases one significant digit, $\log_{10} \kappa \approx 6.5$ means six and a half of the sixteen machine digits are gone, and the result 0.007 leaves only about ten digits that can be trusted.

One note that saves us: Sterbenz's rule applies to differences, not to sums. For the sum $a + b$ the condition number is $(|a| + |b|)/|a + b|$, which approaches one when a and b have the same sign. Adding large numbers is not dangerous. Subtracting two large numbers that are nearly equal is dangerous, and that is the only habit that needs watching.

22.4 Catastrophic cancellation: two faces

Level: intermediate

We already know the mechanism. Now let us give it a name, because two similar kinds of cancellation are often taken for the same thing although their consequences differ widely.

For younger readers (age 10 and up)

If we want to know the thickness of a sheet of paper, an ordinary ruler will not help: one sheet is far thinner than a millimeter, so the needle seems not to move. The way is not to look for a finer ruler but to change what is measured: stack five hundred sheets, measure the thickness of the stack, then divide by five hundred. The result is far more trustworthy, even though the ruler is the same ruler.

It turns out that a formula can be treated in the same way. Some formulas work by subtracting two large numbers that are nearly equal, and that is the same as weighing a single sheet of paper with a ruler. Other formulas are mathematically exactly the same, but order their steps so that the small number is measured directly rather than recovered from a difference. The results can differ widely, even though the paper has not changed. The question: if two formulas that are equally correct give different answers, which one should we trust? In this chapter we shall learn why the wrong formula is not the one whose arithmetic is mistaken but the one that chooses the step that throws information away, and how to recognize it before the program runs.

catastrophic cancellation (*pembatalan katastrofik*) — The loss of significant digits when two nearly equal numbers are subtracted, so that only error is left.

Definition 22.3 (Catastrophic and benign cancellation)**.** Let x and y be machine numbers computed from earlier data. The subtraction $x - y$ is called *benign cancellation* if x and y are exact, so that the result is exact by Sterbenz's theorem. It is called *catastrophic cancellation* if x and y themselves already carry rounding error, so that the digits the two share disappear as well and what is left is dominated by that error.

A benign example: $1.0000000000000002 - 1$ in binary64. Both are machine numbers, the difference is 2^{-52} and is represented exactly. The result is small, but not one digit disappears: we really do know both inputs to the last bit.

The catastrophic example is the quadratic formula from the opening of the chapter. There b^2 and $4ac$ are each the result of a multiplication, so both are rounded. For $a = 1$, $b = 10^8$, $c = 1$ both happen to be exact, and it is precisely that specialness which makes this example worth examining down to the bit: the collapse does not come from the multiplications but from the rounding of the square root. Python computes

discriminant $b^2 - 4ac$	`9999999999999996.0`
$\sqrt{b^2 - 4ac}$	`99999999.99999999`
$b - \sqrt{b^2 - 4ac}$	`1.4901161193847656e-08`
$\mathrm{ulp}(b)$	`1.4901161193847656e-08`

The true value of $b - \sqrt{b^2 - 4ac}$ is 2×10^{-8}, that is 1.34 ulp. What is computed is exactly 1 ulp, and half an ulp of error in the square root is enough for that: the square root is correct to 0.46 ulp at the scale 10^8, but a difference that ought to be 1.34 ulp can only land on 1 or 2 ulp, so its information is down to one binary choice. The final result is -7.45×10^{-9}, that is -2^{-27}, whereas the true root is -10^{-8}. Relative error 0.255.

Note the division of labor. The square root works with relative error 5×10^{-17}, far below u. The problem itself is well conditioned: shifting b by one ulp shifts the small root by one ulp, so $\kappa \approx |b|/\sqrt{b^2 - 4ac} \approx 1$. What is bad is the *form* we have chosen: the condition number of the subtraction at that step is

$$\kappa = \frac{|b| + \sqrt{b^2 - 4ac}}{\left|\sqrt{b^2 - 4ac} - |b|\right|} = \frac{2 \times 10^8}{1.49 \times 10^{-8}} \approx 1.34 \times 10^{16},$$

so that $\kappa u \approx 1.49$: the relative error bound passes one, which means not a single digit is guaranteed.

Figure 22.2 separates two things that are often mixed up at that step: the size of the numbers, and how many digits still mean something after the subtraction.

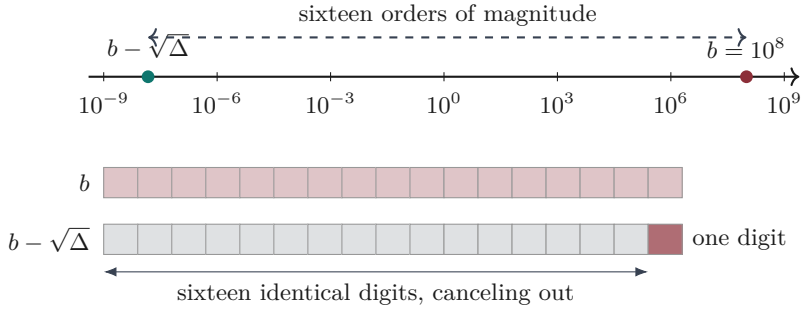


Figure 22.2: Top: b and $\sqrt{b^2 - 4ac}$ on a log scale, sixteen orders of magnitude away from their difference. Bottom: the boxes are significant digits, seventeen go in and sixteen of them are the same so that they cancel out. What remains is one digit, namely the difference 1.49×10^{-8} itself; since half an ulp of error in the square root is enough to shift the last digit, the small root comes out with a relative error of 0.255.

The fix is not more precision but choosing which root to compute first. The product of the two roots is c/a, so once the large root r_1 is known, the small root is obtained by a division, not by a subtraction:

$$r_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \qquad r_2 = \frac{c}{a\, r_1} = \frac{2c}{-b - \sqrt{b^2 - 4ac}}.$$

The second form never subtracts two numbers that are close together. For the example above it gives -10^{-8} exactly, that is the correctly rounded root. The same formula with the signs arranged according to the sign of b is used in almost every library that solves quadratic equations, and the reason is stability, not tidiness.

The second face of catastrophic cancellation appears in a place that looks harmless: the difference $1 - \cos x$ for small x. For small x,

$$\frac{1 - \cos x}{x^2} \longrightarrow \frac{1}{2},$$

because $\cos x = 1 - x^2/2 + x^4/24 - \cdots$. The formula is right, and it is also the most natural one to write. But $\cos x$ is stored with relative error u, and because $|\cos x| \le 1$, its absolute error is about u. After division by x^2, that error becomes a relative error of about $2u/x^2$. For $x = 10^{-8}$ that factor is 2.2: every digit is wiped out.

Here are the results of an actual run for ten values of x, compared with a reference computed at 50 digits and with the Taylor series $0.5 - x^2/24 + x^4/720$; for $x \le 10^{-3}$ the two comparators agree to the last digit. The "naive" column is $(1 - \cos x)/x^2$, the "identity" column is $\frac{1}{2}\,(\sin(x/2)/(x/2))^2$, which comes from $1 - \cos x = 2\sin^2(x/2)$.

x	naive	relative error	identity
10^{-1}	0.499583472197418	1.1×10^{-14}	0.499583472197423
10^{-2}	0.499995833347366	2.9×10^{-13}	0.499995833347222
10^{-3}	0.499999958325503	1.6×10^{-11}	0.499999958333335
10^{-4}	0.499999996961265	5.2×10^{-9}	0.499999999583333
10^{-5}	0.500000041370185	8.3×10^{-8}	0.499999999995833
10^{-6}	0.500044450291171	8.9×10^{-5}	0.499999999999958
10^{-7}	0.499600361081320	8.0×10^{-4}	0.500000000000000
10^{-8}	0.000000000000000	1.0	0.500000000000000
10^{-9}	0.000000000000000	1.0	0.500000000000000
10^{-10}	0.000000000000000	1.0	0.500000000000000

The pattern is typical and worth memorizing. As long as $x^2/2$ is far above u, the naive formula is nearly right. When $x^2/2$ shrinks to the scale of u, the significant digits run out and the error grows like $2u/x^2$: each tenfold decrease of x adds two digits of loss. At $x = 10^{-8}$, $1 - \cos x$ is smaller than half an ulp around 1, so that $\cos x$ is rounded exactly to 1 and the result is zero. That breakdown point is no mystery: $x^2/2 \approx u$, that is $x \approx \sqrt{2u} \approx 1.5 \times 10^{-8}$.

The last column shows that the problem is well conditioned. The double-angle identity gives an answer correct to the last digit for every x in the table, because it never subtracts two numbers that are close together. Error is not a property of the problem. It is a property of the formula chosen.

Step-by-step recipe (procedural learner)

Four steps for choosing a formula form that is safe from cancellation:

1. Mark every subtraction $a - b$ inside the formula, then compute $\kappa = (|a| + |b|)/|a - b|$ on the worst data that can reasonably arrive.
2. Translate κ into lost digits: $\kappa u \approx 10^{-d}$ means the last d digits cannot be trusted.
3. Look for an equivalent form that contains no such difference: multiply by the conjugate, use the double-angle identity, factor, or call `log1p`, `expm1`, and `hypot`.
4. Repeat the computation at higher precision. The difference between the two results should be comparable to κu; if it is far larger, the new form still hides a subtraction.

A warning about order: step 1 must not be swapped with step 3. Rewriting a formula before knowing which subtraction is dangerous only moves the problem, as in the one-pass form of the variance that trades one dangerous subtraction for another.

22.5 Summation: error that grows with n

Level: advanced

Summation is the operation most often taken for granted, and yet it is where error is easiest to compute. Before we come to the bound, here is a demonstration that dismantles the assumption that order does not matter:

```
(0.1 + 0.2) + 0.3 = 0.6000000000000001,   0.1 + (0.2 + 0.3) = 0.6.
```

The difference is 1.11×10^{-16}, exactly one ulp at the scale 0.6. No digit is written down wrongly: only three numbers, each with no finite binary representative, summed in a different order. In mathematics, addition is associative. Inside the machine, it is associative only when no differing rounding happens to occur.

Theorem 22.7 (Error of sequential summation)**.** *Let $s_1 = x_1$ and $s_k = \mathrm{fl}(s_{k-1} + x_k)$ for $k = 2, \dots, n$ with $x_i \in \mathbb{F}$. Then*

$$\left| s_n - \sum_{i=1}^{n} x_i \right| \;\le\; (n-1)\,u \sum_{i=1}^{n} |x_i| + O(u^2).$$

Proof. Every step uses one rounding, so $s_k = (s_{k-1} + x_k)(1 + \delta_k)$ with $|\delta_k| \le u$. Expanding the result recursively gives

$$s_n = \sum_{i=1}^{n} x_i \prod_{k=i}^{n} (1 + \delta_k),$$

where the first term carries no δ_1 because $s_1 = x_1$ has not been rounded. Since $\prod(1+\delta_k) = 1+\theta$ with $|\theta| \le (1+u)^{n-1} - 1 \le (n-1)u + O(u^2)$, the triangle inequality gives the bound claimed. $\square$

That bound is honest but loose: it assumes every rounding reaches its worst error and that all of them point the same way, which in practice never happens. The following table shows the gulf between bound and reality for the harmonic series $H_n = \sum_{i=1}^{n} 1/i$ in binary32, with the exact value computed by `math.fsum`, which sums exactly.

n	sequential	exact	bound $(n-1)u\sum\lvert x_i\rvert$
10^2	5.1873779297	5.1873775176	3.1×10^{-5}
10^3	7.4854784012	7.4854708606	4.5×10^{-4}
10^4	9.7876129150	9.7876060360	5.8×10^{-3}
10^5	12.0908508301	12.0901461299	7.2×10^{-2}
10^6	14.3573579788	14.3927267229	8.6×10^{-1}

The actual error at $n = 10^6$ is 3.54×10^{-2}, about twenty-five times smaller than the bound, but it has already erased two and a half significant digits. Nor does the error rise smoothly: it dips at $n = 10^4$ because roundings in the same and in opposite directions happened to cancel. What can be relied on is not the size of the error at one n, but its tendency to grow, at a worst-case rate proportional to n.

The cause is not large rounding but an accumulator that moves away from zero. Once s is around 14, the term $1/i$ for large i lies below half an ulp of s and every addition loses all its information. The extreme case is easy to construct: add 1 to a million terms of value 10^{-16}; exactly the result is 1.0000000001, whereas sequential summation stays at 1 for however many terms there are. What is lost is not one term but the whole contribution.

Figure 22.3 summarizes the same pattern for the five values of n in the table above, this time with Kahan summation as the comparison.

Do it by hand (kinesthetic learner)

Materials: eleven blank cards, a marker, and one rule of the game. Every time two cards are added, write the result on a new card with at most four significant figures, then throw the old cards away.

Write 1.000 on one card and 0.0001 on ten other cards. First route: put down the big card, then add the ten small cards one at a time. Each time $1.000 + 0.0001 = 1.0001$ is cut down to 1.000, so that the small card disappears without a trace and the pile stays 1.000.

Shuffle the cards again, second route: first add up the ten small cards into 0.0010, then add the big card. The final pile is 1.001.

What must be visible: the same eleven cards, the same cutting rule, two different results in the last digit. Only the order changed, and the machine repeats that order a million times.

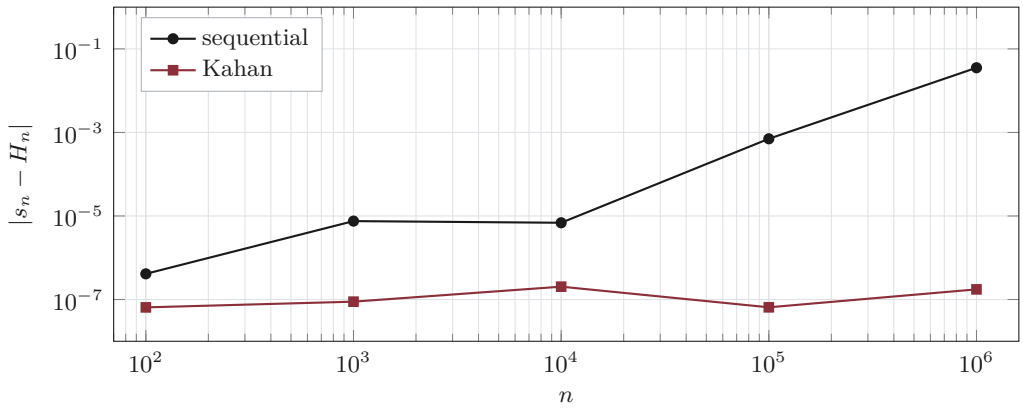


Figure 22.3: Absolute error of the harmonic sum in binary32 at five sizes n, log-log scale. Sequential summation rises from 4.1×10^{-7} at $n = 10^2$ to 3.54×10^{-2} at $n = 10^6$; Kahan holds at order 10^{-7} across the whole range, so that at $n = 10^6$ the two are 2×10^5 times apart. The data, the precision, and the number of operations are the same; only the order of summation differs.

Kahan summation attacks that problem directly. The idea is simple: every time a term is lost to rounding, keep the lost remainder in a compensation variable, then put it back in at the next step.

```
s = 0.0; c = 0.0
for x in data:
   y = x - c
   t = s + y
   c = (t - s) - y
   s = t
```

The line `c = (t - s) - y` is the heart of it, and it works precisely because of Sterbenz: $t - s$ is the difference of two numbers that are close together and hence exact, and $(t - s) - y$ is therefore exactly the error that was just discarded. The result no longer piles up error proportional to n.

Theorem 22.8 (Kahan summation bound)**.** *With Kahan's algorithm above, the computed sum satisfies the inequality*

$$s = \sum_{i=1}^{n} x_i(1 + \delta_i) + O(nu^2) \sum_{i=1}^{n} |x_i|, \qquad |\delta_i| \le 2u.$$

We do not repeat the proof of this theorem here. It is Theorem 8 in Goldberg's survey, and its proof is in the appendix of that work, taken from Knuth. What is worth reading from the statement: the relative error of each term is bounded by $2u$, twice the error of a single rounding, but the factor n is gone from the leading term. For small n this bound is in fact looser than the naive one, because $2u > u$. Its advantage appears only when n is large, and that is where its whole use lies.

The real numbers on the harmonic series with $n = 10^6$: sequential summation in binary32 gives an error of 3.54×10^{-2}, whereas Kahan gives 1.75×10^{-7}, that is 0.18 ulp from the correct answer. In binary64 the difference is just as sharp: sequential is off by 7.35×10^{-13}, or 414 ulp, whereas Kahan produces a figure identical to `math.fsum`.

Two other improvements deserve mention. Pairwise summation, which `numpy.sum` uses, splits the data recursively so that the error grows like $\log n$; it is nearly free and suits data in memory. `math.fsum` uses an exact accumulator and gives a result that is correctly rounded for the whole list, at a higher cost in time. Summing from the smallest term to the largest is the cheapest improvement, but it only reduces the error.

22.6 Practical guidance

Level: basic

Everything we have built can be compressed into a few habits that are cheap to keep and expensive to neglect.

First, choose precision from the condition number, not from habit. The relative error of the result is roughly κu. With binary64, a computation with condition 10^6 still leaves ten digits; with binary32, the same computation leaves only one digit. For 10^8 sequential operations the crude bound is nu, and in binary32 that already passes one, which means the answer is meaningless.

Second, do not test fractional numbers for equality with `==`. Two paths that are mathematically identical may produce different bits, like $0.1 + 0.2$ above. The right test uses a scaled tolerance, for instance $|x - y| \le 10^{-12} \max(1, |x|, |y|)$, and that tolerance must come from error analysis, not from experimenting until the test passes.

Third, distrust every formula that contains a difference of two nearly equal quantities. The standard form of the variance, $(1 - \cos x)/x^2$, and $\log(1 + x)$ for small x can all be written differently. The standard library already does so: `log1p`, `expm1`, and `hypot` exist precisely because their naive forms are not stable.

Fourth, for a long sum, or one with terms of very different sizes, use `math.fsum` or pairwise summation, and keep the accumulator in higher precision where available. Sort from the smallest term if there is no other choice.

Fifth, do not conclude truth from agreement between numbers. Two programs that agree to twelve digits are fairly convincing; ones that agree to three digits only tell us that both use the same formula. A better test is to repeat the computation at a different precision: if the result shifts far beyond the prediction κu, something is wrong with the formula, not with the machine.

Challenge (problem solver)

Compute $10^8\left(\sqrt{10^{16}+1}-10^8\right)$ in binary64. The true value is 0.5, but a program that writes the formula down as it stands reports 0.0.
Explain why the result is zero, then compute the form that gives 0.5 without raising the precision at all.
Hint 1: ask what $\operatorname{ulp}(10^{16})$ is, then whether $10^{16}+1$ still has room on the grid of machine numbers.
Hint 2: multiply numerator and denominator by the conjugate $\sqrt{10^{16}+1}+10^8$; the dangerous subtraction becomes a division.
If you are stuck, go back to the section Catastrophic cancellation: two faces.

22.7 Worked example

Level: intermediate

Every number in this section can be checked with a calculator, `struct`, and ordinary fractions. We begin by taking apart one bit pattern, namely 0.1 in binary32:

$$\underbrace{0}_{1}\ \underbrace{01111011}_{8}\ \underbrace{10011001100110011001101}_{23}.$$

The stored exponent is 123, so after subtracting the bias 127 we get $e=-4$. The 23-bit integer in the mantissa is 5033165, so that the significand is $1+5033165/2^{23}$. Hence

$$x=\left(1+\frac{5033165}{2^{23}}\right)2^{-4}=\frac{13421773}{134217728}.$$

The numerator comes from $2^{23}+5033165=13421773$. Meanwhile the denominator is 134217728, and a tenth of it is 13421772.8, so that the error is exactly $0.2/134217728$, that is 1.4901161×10^{-9}, and the relative error is 1.49×10^{-8}, that is $u_{32}/4$. That figure makes sense: the number $1/10$ equals $1.1001100110011\ldots_2\times2^{-4}$, which needs a periodic mantissa and does not terminate, so that correct rounding cuts it at the 24th bit. The pattern that is stored ends in `...1101`, not `...1100`, because the remainder behind it is greater than half an ulp and the rounding goes up.

Take apart the pattern `0 01111100 01000000000000000000000` by hand: determine the sign, the exponent after subtracting the bias, the value of the mantissa, then write the number as a fraction and as a decimal. Compute the ulp of that number, then decide the result of adding 2^{-27} and $3\cdot2^{-27}$ under the *tie to even* rule. Check your answer with `struct.unpack` and `numpy.float32`. This number was chosen so that its mantissa stops after two digits, so that all of its arithmetic can be done without a calculator.

Now the case $0.1+0.2$ in binary64, which can be computed entirely with fractions. All three numbers involved have power-of-two denominators:

$$\mathrm{fl}(0.1)=\frac{3602879701896397}{2^{55}},\qquad \mathrm{fl}(0.2)=\frac{3602879701896397}{2^{54}},$$

$$\mathrm{fl}(0.3)=\frac{5404319552844595}{2^{54}}.$$

The sum of the first two before rounding is $10808639105689191/2^{55}$, whereas $\mathrm{fl}(0.3)=10808639105689190/2^{55}$. So the exact sum lies exactly half an ulp above $\mathrm{fl}(0.3)$: a tie point, not something near one. The last digit of its mantissa is 1, that is odd, so that *tie to even* rounds up: $0.30000000000000004440892098500 6\ldots$

The difference from fl(0.3) is therefore exactly one ulp, $2^{-54} = 5.551115123125783 \times 10^{-17}$, not a random error. The subtraction that follows is exact by Sterbenz, so that the figure the program prints is that figure precisely.

The same numbers explain why the conclusion depends on the precision: in binary32, $0.1 + 0.2$ and 0.3 produce identical bit patterns because the two rounding errors happen to cancel each other. So `0.1+0.2 != 0.3` is not a law of nature but the consequence of a coincidence at one precision.

Now the square root from before, with ulp arithmetic. The discriminant $b^2 - 4ac$ is exactly $10^{16} - 4$, because 10^{16} is still below 2^{54} and that number is a multiple of 4. Its square root is correct to 0.46 ulp, so that it lands one ulp below b because $\mathrm{ulp}(b) = 2^{-26}$. The true difference is $2 \times 10^{-8} = 1.34$ ulp, the computed difference 1 ulp; that error of 0.34 ulp at the scale 10^8, divided by two, becomes the whole error of the small root:

$$r_2^{\text{naive}} = -\frac{2^{-26}}{2} = -2^{-27} = -7.450580596923828 \times 10^{-9}, \qquad r_2 = -10^{-8}.$$

The stable form $2c/(-b - \sqrt{b^2 - 4ac})$ suffers no dangerous subtraction, and its result is -10^{-8} exactly, the correctly rounded root.

22.8 Computational lab

Level: advanced

Computational lab

```
import math, struct
import numpy as np

def bit32(x):
    return ".join(f'{b:08b}' for b in struct.pack('>f', float(x)))

print("0.1     :", bit32(0.1))
print("6.0     :", bit32(6.0))
print("-12.375 :", bit32(-12.375))
print("hex     :", (0.1).hex(), (6.0).hex(), (1e-8).hex())
print("eps     :", np.finfo(np.float32).eps, np.finfo(np.float64).eps)
print("ulp     :", math.ulp(1.0), math.ulp(1e10))
print("tie     :", 1 + 2.0**-53, 1 + 3*2.0**-53)

s = (0.1 + 0.2) - 0.3
print("0.1+0.2-0.3 :", repr(s), "=", s / math.ulp(0.3), "ulp")
print("associative :", repr((0.1 + 0.2) + 0.3), repr(0.1 + (0.2 + 0.3)))

n = 10**6
exact = math.fsum(1.0 / i for i in range(1, n + 1))

def jumlah(prec, kahan):
    s, c = prec(0), prec(0)
    for i in range(1, n + 1):
        y = prec(1) / prec(i)
        if kahan:
            y = y - c
            t = s + y
            c = (t - s) - y
            s = t
        else:
            s = s + y
    return s
```

```
print("H_n exact   :", f"{exact:.12f}")
for prec, nama in ((np.float32, "float32"), (np.float64, "float64")):
    for kahan in (False, True):
        v = float(jumlah(prec, kahan))
        print(f"{nama} {'Kahan' if kahan else 'naive':7} {v:.12f}"
              f"  error {abs(v - exact):.2e}")

print(f"{'x':>8} {'naive':>19} {'error':>9} {'identity':>19} {'error':>9}")
for k in range(1, 11):
    x = 10.0**-k
    lugu = (1 - math.cos(x)) / x**2
    stabil = 2 * math.sin(x / 2)**2 / x**2
    acuan = 0.5 - x**2/24 + x**4/720 - x**6/40320
    print(f"{x:8.0e} {lugu:19.15f} {abs(lugu-acuan)/acuan:9.1e}"
          f" {stabil:19.15f} {abs(stabil-acuan)/acuan:9.1e}")

a, b, c = 1.0, 1e8, 1.0
d = b*b - 4*a*c
sq = math.sqrt(d)
print("discriminant :", d, " root:", sq)
print("b - root    :", b - sq, " ulp(b):", math.ulp(b))
print("naive root  :", (-b + sq) / (2*a))
print("stable root :", 2*c / (-b - sq))
```

The actual output:

```
0.1     : 0011110111001100110011001100110011001101
6.0     : 0100000011000000000000000000000000
-12.375 : 1100000101000110000000000000000000
hex     : 0x1.999999999999ap-4 0x1.8000000000000p+2 0x1.5798ee2308c3ap-27
eps     : 1.1920929e-07 2.220446049250313e-16
ulp     : 2.220446049250313e-16 1.9073486328125e-06
tie     : 1.0 1.0000000000000004
0.1+0.2-0.3 : 5.551115123125783e-17 = 1.0 ulp
associative : 0.6000000000000001 0.6
H_n exact   : 14.392726722866
float32 naive   14.357357978821  error 3.54e-02
float32 Kahan   14.392726898193  error 1.75e-07
float64 naive   14.392726722865  error 7.35e-13
float64 Kahan   14.392726722866  error 0.00e+00
       x               naive     error            identity     error
   1e-01   0.499583472197418   5.6e-15   0.499583472197423   5.4e-15
   1e-02   0.499995833347366   2.9e-13   0.499995833347222   1.1e-16
   1e-03   0.499999958325503   1.6e-11   0.499999958333335   0.0e+00
   1e-04   0.499999996961265   5.2e-09   0.499999999583333   1.1e-16
   1e-05   0.500000041370185   8.3e-08   0.499999999995833   1.1e-16
   1e-06   0.500044450291171   8.9e-05   0.499999999999958   0.0e+00
   1e-07   0.499600361081320   8.0e-04   0.500000000000000   1.1e-16
   1e-08   0.000000000000000   1.0e+00   0.500000000000000   0.0e+00
   1e-09   0.000000000000000   1.0e+00   0.500000000000000   0.0e+00
   1e-10   0.000000000000000   1.0e+00   0.500000000000000   0.0e+00
discriminant : 9999999999999996.0  root: 99999999.99999999
b - root    : 1.4901161193847656e-08  ulp(b): 1.4901161193847656e-08
naive root  : -7.450580596923828e-09
stable root : -1e-08
```

The first three lines confirm that the table at the beginning of the chapter is not invented: the bit patterns of 0.1, 6, and -12.375 come out exactly as in the hand decomposition, sign bit included. The `hex` line shows thirteen hexadecimal digits. The `tie` line is the plainest demonstration of the even rule: adding half an ulp to 1 changes nothing, adding one and a

half ulp gives $1+2^{-51}$. The `0.1+0.2-0.3` line prints $5.551115123125783\times10^{-17}$ and declares itself to be 1.0 ulp; the `associative` line gives two different results, differing by 1.11×10^{-16}.

The next four lines are the core of this chapter in its most compact form: at $n=10^6$, binary32 sequential loses 3.54×10^{-2} whereas Kahan loses only 1.75×10^{-7}; binary64 sequential 7.35×10^{-13} whereas Kahan zero, the same bits as `math.fsum`. That error of 7.35×10^{-13} is equivalent to 414 ulp, so double precision only moves the problem to a smaller scale.

The table for $(1-\cos x)/x^2$ shows the pattern already computed on paper. At $x=10^{-1}$ both methods differ from the reference by about 5×10^{-15} because the reference itself is truncated at the x^6 term; from $x=10^{-2}$ on that reference is far more accurate than the machine, and the error of the naive formula rises from 2.9×10^{-13} to 1.0 as x falls to 10^{-8}, whereas the identity column holds at 10^{-16}. The zero at $x=10^{-8}$ is not lazy rounding: $\cos(10^{-8})$ really equals 1.

The last block checks the square root: discriminant $10^{16}-4$, root 10^8-2^{-26}, a difference of one ulp of b, namely 2^{-26}, naive root -2^{-27}, and stable root 10^{-8}, which is correctly rounded. The last two lines are 25.5 percent apart, even though the formulas are algebraically equivalent and the data are exactly the same.

Note for instructors

A teaching order that has worked in practice: begin with one failure that can be recomputed, namely the quadratic formula at the opening of the chapter, and only then go into representation. If representation comes first, students memorize 1/8/23 and 1/11/52 without knowing what for. After that failure, the order that works is: the number of machine numbers (so that it is clear rounding cannot be avoided), the rounding rule, the ulp lemma, the $(1+\delta)$ bound, then the condition number. Summation is better placed after the condition number, because the bound $(n-1)u\sum|x_i|$ only reads as "error that grows" once κ is familiar.

Two typical misconceptions. First, students believe that rounding error is random and therefore cancels out in long sums. The harmonic table contradicts that sharply: the error moves in one direction, and at $n=10^6$ it reaches thirty-five thousand ulp. Second, they believe that subtraction always damages digits, and then avoid all subtraction. Sterbenz's theorem shows the opposite: the subtraction of two nearby machine numbers is in fact exact, and what does the damage is the error already present in the input. The rule that needs repeating: what is dangerous is not the minus sign, but a small difference of two numbers that have each already been rounded.

A cheap and effective whiteboard activity: ask the whole class to compute $1+2^{-53}$ and $1+3\cdot2^{-53}$ with the even rule, then compare with Python. After that, ask two volunteers to work out $(0.1+0.2)+0.3$ and $0.1+(0.2+0.3)$ by writing the 52-bit mantissa as a fraction. The hubbub usually starts when they discover that both are right.

A sign of readiness to move on: students can guess, without running a program, that $\mathrm{fl}(1+10^{-16})=1$, can explain why the form $2c/(-b-\sqrt{\Delta})$ is better than $(-b+\sqrt{\Delta})/(2a)$, and can estimate how many digits are lost from a given κ. If they say the words "backward error" without being prompted, they are ready for Chapter 25, where the gradient of a network is computed with a similar rule.

22.9 Summary

Level: basic

Summary.

- $\mathbb{F}$ is finite: 2^{p-1} normal numbers per power of two, so almost every real number has no representative and every act of storage means rounding.

- IEEE rounding is *round to nearest, ties to even*. Consequently $\mathrm{fl}(x) = x(1+\delta)$ with $|\delta| \le u = 2^{-p}$, that is 1.11×10^{-16} for binary64 and 5.96×10^{-8} for binary32.
- $\mathrm{eps} = 2^{-p+1}$ is the distance from 1 to the number after it; $\mathrm{ulp}(x) = 2^{e-p+1}$ for $2^e \le |x| < 2^{e+1}$. The error of one rounding is half an ulp, and expressed as a relative error it swings by a factor of two.
- The condition number $\kappa_f = |x f'(x)|/|f(x)|$ measures the amplification of input error. For subtraction, $\kappa = (|a| + |b|)/|a - b|$, which grows without bound as $|a - b|$ shrinks.
- Subtracting two nearby machine numbers is exact (Sterbenz). Catastrophic cancellation happens because the inputs already carry error, not because of the subtraction.
- The naive quadratic formula computes the small root through a subtraction of two numbers of size $|b|$; the form $2c/(-b - \sqrt{\Delta})$ avoids it and gives the correctly rounded root.
- Sequential summation has a worst-case error of $(n-1)u\sum |x_i|$ and is not associative. Kahan moves the error into a compensation variable so that the factor n disappears from the leading term.

22.10 Exercises

Level: intermediate

Exercise 22.1.

1. Determine the ulp of 1; 1024; 0.1; and 3×10^8 in binary64, then compare with $|x|\,u$. How many binary64 numbers lie in $[1, 2)$?
2. Show that $1 + 2^{-53} = 1$ and $1 + 2^{-52} > 1$ in binary64, then determine the smallest positive number x with $1 + x > 1$ in binary32. Express your answer as 2^{-k} and compare it with the table in the opening section.
3. Prove that for every normal x we have $|\,\mathrm{fl}(x) - x| \le \frac{1}{2}\,\mathrm{ulp}(x)$, then explain why that inequality cannot be replaced by equality for all x.
4. Assess this claim: *because every IEEE-754 operation is correctly rounded, the relative error of the final result of a program that uses only $+, -, \times, \div$ is always at most a few u.* Decide whether it is true, false, or not supported by evidence, and name one example from this chapter that supports your decision.
5. Compute by hand $(0.1 + 0.2) + 0.3$ and $0.1 + (0.2 + 0.3)$ as fractions with power-of-two denominators, then show that the difference is one ulp. Check with `math.fsum` and with `float.hex()`. (Hint: write each number as $m/2^{55}$ or $m/2^{54}$ and compute the numerators as integers.)
6. Repeat the computation of $(1 - \cos x)/x^2$ for $x = 10^{-7}$ with the double-angle identity, then compare with $0.5 - x^2/24 + x^4/720$. What relative error remains, and where does it come from?
7. For $a = 1$, $b = 10^4$, $c = 1$, compute both roots with the naive formula and with the stable form in binary64. Determine the condition number of the subtraction, then predict how many digits are lost before running the program. Check your prediction with Python. (Hint: $\sqrt{10^8 - 4} \approx 10^4 - 2 \times 10^{-4}$; compare with $\mathrm{ulp}(10^4) = 2^{-39}$.)
8. Prove that pairwise summation over $n = 2^k$ terms has a worst-case error proportional to $ku\sum |x_i|$, not $nu\sum |x_i|$, by expanding the summation tree and counting the depth of the tree.
9. Let $x_i = 10^{-16}$ for $i = 1, \dots, 10^6$ and $x_0 = 1$. Compute sequential summation and Kahan summation both by hand (a calculator may be used), then explain why the Kahan answer differs even though each of its steps is also rounded. (Hint: in sequential summation every term lies below half an ulp of the accumulator; in Kahan, the

compensation c keeps that remainder and adds it back at the next step.)

10. A program computes the variance with $\frac{1}{n}\sum x_i^2 - \left(\frac{1}{n}\sum x_i\right)^2$ on data that are all around 999. Determine the condition number of that form, predict how many digits are lost in binary32, and compare with the two-pass form $\frac{1}{n}\sum(x_i - \bar{x})^2$. Explain why the second form is better even though it requires two passes over the data.

For researchers: limits and next steps

What is really proved in this chapter are standard statements that do not depend on any particular machine: the bound $\mathrm{fl}(x) = x(1+\delta)$ with $|\delta| \le u$, the ulp lemma, Sterbenz's theorem, the error bound for sequential summation, and the condition-number inequality via the mean value theorem. The primary reference is Goldberg, *What Every Computer Scientist Should Know About Floating-Point Arithmetic* (ACM Computing Surveys 23(1), 1991), sections "Relative Error and Ulps", "Cancellation", "Exactly Rounded Operations", and "Errors in Summation". The Kahan bound is Theorem 8 of that work; its proof is in the appendix, taken from Knuth, and is not repeated here. For a systematic treatment of stability, the further reference is Higham, *Accuracy and Stability of Numerical Algorithms*, the chapters on error analysis and summation.

Three things are more often popularized than proved. First, the claim that "rounding error is random and cancels out". That is true only as a statistical model for uncorrelated rounding, and it fails precisely in the most important case: accumulation in long sums is systematic, as the harmonic table shows. Second, the claim that the error can always be reduced with higher precision. For catastrophic cancellation, raising the precision only moves the breakdown point, it does not remove it: in binary64 that point lies at $x \approx 1.5 \times 10^{-8}$, and in binary128 it lies around 10^{-17}. Third, the claim that the ulp is an unambiguous unit of error. Goldberg shows that the ulp may be defined as the distance between numbers or as half that distance, and the difference between the two definitions appears as a factor of two in every bound. The figure "0.46 ulp" in the discussion of the square root in this chapter uses the first definition.

The limits of validity of the results here must be stated. The whole analysis assumes a model of correct rounding and no *fused multiply-add* that changes the order of operations, no rearrangement by the compiler, and no rounding mode other than *to nearest*. On real hardware, vectorization and FMA instructions can change results bit for bit; that is not a violation of the standard, but it makes comparisons between machines non-deterministic. Error analysis also does not touch model error, data error, or truncation error: this chapter discusses only rounding error.

Honest open questions. First, classical error analysis gives worst-case bounds, whereas what practitioners need is a tight estimate for the typical case; the probabilistic treatment of rounding is still an actively researched tool and not a finished theory. Second, for iterative algorithms and the solution of linear systems, rounding error interacts with the condition number of the matrix in such a way that naive bounds are far too pessimistic; backward analysis, which is the central idea of Higham and Wilkinson, gives a sharper framework but is not discussed here. Third, practical decisions such as "is binary32 enough" depend on the distribution of the data, and no answer holds in general.

Map to source texts: This chapter follows Goldberg, What Every Computer Scientist Should Know About Floating-Point Arithmetic *(ACM Computing Surveys, 1991), sections "Introduction", "Relative Error and Ulps", "Cancellation", "Exactly Rounded Operations", "Formats and Operations", "Special Quantities", and "Errors in Summation", with Theorem 8 for the Kahan summation bound. For the real-analysis side of the limit ideas and the mean value theorem used in the proof of the condition number, see Abbott,* Understanding Analysis*, sections 2.2 (Limit of a Sequence), 5.3 (The Mean Value*

Theorem), and 6.6 (Taylor Series). The example $0.1 + 0.2$*, the decomposition of bit patterns, the analysis of* $(1 - \cos x)/x^2$*, and all the numbers in the tables and the lab are computations of our own run with Python 3.14 and NumPy 2.5 and re-verified with ordinary fractions and 50-digit arithmetic. For an advanced treatment of stability and backward error, the reference commonly used is Higham,* Accuracy and Stability of Numerical Algorithms*.*

Part V

Optimization

This part turns “find the best” into a problem with a stopping condition. The prerequisites are Chapters 6 and 12. After two chapters you can separate a local minimum from a global one, use duality to prove bounds, and recognise when a linear problem can be solved in polynomial time and when it cannot.

Chapter 23

Convex Optimization and Duality

A paint mixer works with two pigments. Each pigment has a proportion the machine prefers most: two parts for the first pigment, two parts for the second. But the recipe demands that the two amounts total exactly two parts, and the solubility of the first pigment does not allow more than half a part. We want the mixture as close as possible to the machine's preference, under two demands that cannot be violated. Without the demands the answer is two and two; with the sum demand alone, one and one. The solubility demand shifts the answer again, and it is no longer clear in which direction.

That shape of problem shows up where nothing smells of pigment. Two model weights have best values from an earlier fit, a calibration condition demands that their sum stay fixed, and one device limit caps one of them. What are the best weights that still satisfy both, and what is the price of each demand in units of the objective function? The second question is the one engineers use to decide which demand is worth relaxing.

Chapter 10 gave least squares, but only without constraints. Chapter 12 gave the gradient, but the gradient speaks only about neighbors: a point with zero gradient may be a minimum, a maximum, or a saddle. Chapter 9 gave eigenvalues, which are used here, but eigenvalues do not answer the most important question: if an algorithm stops at some point, is that the best answer over the whole feasible region? For a convex function, yes. For a general problem, no, and the fourth section shows how duality supplies a valid lower bound without examining the entire feasible region.

For younger readers (age 10 and up)

Have you ever put a marble inside a bowl? The marble always stops at the bottom, in the same place, wherever it is released, because the bottom curves upward like the letter U. Now imagine the same marble released on a rocky hillside. It stops in the first hollow that is deep enough, and that hollow is not necessarily the lowest one. From where it stops, you cannot tell whether another valley is deeper without walking far and looking. In this chapter we shall learn why a bowl is so friendly and a hillside so deceptive, and why we can tell the two apart before computing anything. There is a second, subtler idea: sometimes we do not need to know where the bottom of the valley lies, only the lowest bound that can possibly be reached. If that bound equals the place where we stand, we are sure that nothing is better, without examining a single point. Such bounds are sold by a price list for each demand, and that price list is what makes many optimization problems trustworthy without being solved to the end.

23.1 Convex sets and convex combinations

Level: basic

In everyday language the word “convex” means bulging outward, like the back of a turtle. In optimization it means almost the opposite: a set is called convex when it has no inward dent, so that the line segment between any two of its members stays inside. The bowl above is convex; the region between the two horns of a crescent moon is not.

There is one rule of the game: mix two points with nonnegative weights that add up to one, and the mixture must still be a member of the set. Weight one means choosing the first point, weight zero the second point, weight one half exactly halfway.

convex combination (*kombinasi konveks*) — A recipe for mixing several points with proportions that add up to exactly one, like mixing paint: the result always lies among the ingredients.

Definition 23.1 (Convex set and convex combination)**.** A set $C \subseteq \mathbb{R}^n$ is called *convex* when for every $\mathbf{x}, \mathbf{y} \in C$ and every $t \in [0,1]$ we have $t\mathbf{x} + (1-t)\mathbf{y} \in C$. That point is called a *convex combination* of $\mathbf{x}$ and $\mathbf{y}$. In general, $\sum_{i=1}^{k} \theta_i \mathbf{x}_i$ is called a convex combination when $\theta_i \geq 0$ and $\sum_i \theta_i = 1$. *The convex hull* $\operatorname{conv} S$ is the set of all finite convex combinations of points in S.

The definition with two points suffices for all points: if C contains every mixture of two members, it contains every finite mixture of its members, by induction on the number of points.

Four important examples. A *hyperplane* $\{\mathbf{x} : \mathbf{a}^\top \mathbf{x} = b\}$ is convex, because $\mathbf{a}^\top(t\mathbf{x} + (1-t)\mathbf{y}) = tb + (1-t)b = b$. A *half-space* $\{\mathbf{x} : \mathbf{a}^\top \mathbf{x} \leq b\}$ is convex for the same reason, using the inequality. A *ball* $\{\mathbf{x} : \|\mathbf{x} - \mathbf{c}\| \leq r\}$ is convex because the norm satisfies the triangle inequality (Chapter 10):

$$\|t\mathbf{x} + (1-t)\mathbf{y} - \mathbf{c}\| \leq t\|\mathbf{x} - \mathbf{c}\| + (1-t)\|\mathbf{y} - \mathbf{c}\| \leq tr + (1-t)r = r.$$

A *polyhedron*, that is a finite intersection of hyperplanes and half-spaces, is convex for the following reason.

Proposition 23.1 (Intersections preserve convexity)**.** *If $C_1, \dots, C_k$ are convex, then $\bigcap_{i=1}^{k} C_i$ is convex.*

Proof. Take $\mathbf{x}, \mathbf{y}$ in the intersection and $t \in [0,1]$. For every i, both points are members of C_i, so $t\mathbf{x} + (1-t)\mathbf{y} \in C_i$ because C_i is convex. Hence the mixture lies in every C_i, that is in the intersection. □

Intersection is generous; union is not. Take $C_1 = \{(x_1, 0) : x_1 \in [0,1]\}$ and $C_2 = \{(x_1, 1) : x_1 \in [0,1]\}$, both convex. The points $(0,0)$ and $(0,1)$ lie in the union, but their halfway mixture $(0, \tfrac{1}{2})$ does not. The same happens with $\{(x_1, x_2) : x_1 x_2 \geq 1\}$: the points $(-1,-1)$ and $(1,1)$ are both members, whereas the origin is not, because $0 < 1$. That set is two separate regions.

The convex hull is the cheapest patch for a hole of that kind: the smallest convex set that contains S. It always exists, because the intersection of all convex sets containing S is convex by the proposition above and nonempty, since $\mathbb{R}^n$ itself is convex.

23.2 Convex functions and their two tests

Level: intermediate

Convex sets speak about regions; convex functions speak about the shape of the surface above that region. What is demanded: the value at a mixture is no greater than the mixture

of the values. In the left panel of Figure 23.1 the chord from $(-1,1)$ to $(1,1)$ lies above the graph of $f(t)=t^2$; in the right panel the chord cuts through it.

convex function (*fungsi konveks*) — A surface that never bends upward like a roof; the chord between any two of its points always lies above the surface.

Definition 23.2 (Convex function)**.** Let $C \subseteq \mathbb{R}^n$ be convex. A function $f : C \to \mathbb{R}$ is called *convex* when

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \le t\, f(\mathbf{x}) + (1-t)\, f(\mathbf{y})$$

for all $\mathbf{x}, \mathbf{y} \in C$ and $t \in [0,1]$. It is *strictly convex* when that inequality is strict for $\mathbf{x} \neq \mathbf{y}$ and $t \in (0,1)$, and *concave* when $-f$ is convex.

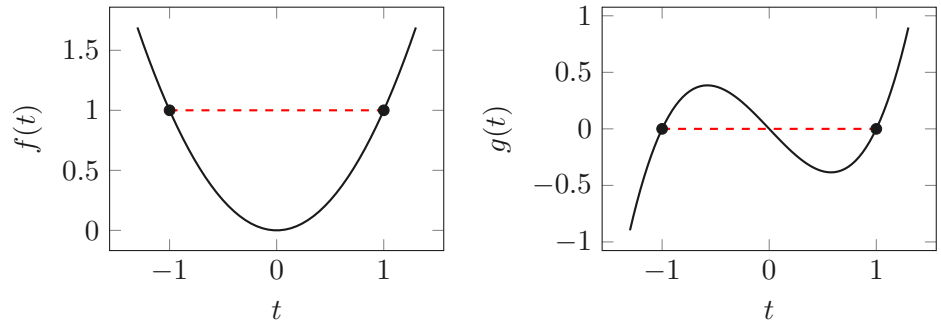


Figure 23.1: The chord between two points of the graph. Left: $f(t)=t^2$, the chord from $(-1,1)$ to $(1,1)$ has value 1 at $t=0$ whereas the graph has 0, so the chord lies above the graph. Right: $g(t)=t^3-t$, the chord from $(-1,0)$ to $(1,0)$ crosses the graph at $t=-1$, 0, 1; at $t=-0.5$ the graph has value 0.375 whereas the chord has 0.

That definition is equivalent to a geometric statement about the *epigraph*, that is the set of points above the graph: f is convex exactly when $\operatorname{epi} f = \{(\mathbf{x}, t) : f(\mathbf{x}) \le t\}$ is convex. The equivalence is purely algebraic, and it turns a question about functions into a question about sets.

For differentiable functions there are two far faster tests, and both are necessary-and-sufficient, not merely sufficient.

Theorem 23.2 (First-order test)**.** *Let f be differentiable on an open convex set C. The function f is convex if and only if*

$$f(\mathbf{y}) \ge f(\mathbf{x}) + \nabla f(\mathbf{x})^{\top}(\mathbf{y}-\mathbf{x})$$

for all $\mathbf{x}, \mathbf{y} \in C$.

Proof. Suppose f is convex, take $\mathbf{x} \neq \mathbf{y}$ and $t \in (0,1)$. From the definition,

$$\frac{f(\mathbf{x} + t(\mathbf{y}-\mathbf{x})) - f(\mathbf{x})}{t} \le f(\mathbf{y}) - f(\mathbf{x}).$$

As $t \downarrow 0$ the left side tends to the directional derivative $\nabla f(\mathbf{x})^{\top}(\mathbf{y}-\mathbf{x})$, so the required inequality holds in the limit. For the converse direction, suppose that inequality holds and take $\mathbf{z} = t\mathbf{x} + (1-t)\mathbf{y}$ with $t \in [0,1]$. Apply the inequality at $\mathbf{z}$ against $\mathbf{x}$ and against $\mathbf{y}$:

$$f(\mathbf{x}) \ge f(\mathbf{z}) + \nabla f(\mathbf{z})^{\top}(\mathbf{x}-\mathbf{z}), \qquad f(\mathbf{y}) \ge f(\mathbf{z}) + \nabla f(\mathbf{z})^{\top}(\mathbf{y}-\mathbf{z}).$$

Multiply the first by t, the second by $1-t$, then add. The gradient terms vanish because $t(\mathbf{x}-\mathbf{z}) + (1-t)(\mathbf{y}-\mathbf{z}) = \mathbf{0}$, and what remains is $t\, f(\mathbf{x}) + (1-t) f(\mathbf{y}) \ge f(\mathbf{z})$. $\square$

The practical reading: the graph lies above every tangent plane. A single tangent plane already gives a global lower bound, and that is why the gradient at any point becomes an honest tool rather than a local forecast.

Theorem 23.3 (Second-order test)**.** *Let f be twice differentiable on an open convex set C. The function f is convex if and only if the Hessian $\nabla^2 f(\mathbf{x})$ is positive semidefinite for every $\mathbf{x} \in C$, that is $\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} \geq 0$ for all $\mathbf{v}$.*

Proof. If f is convex, then for every $\mathbf{x}$ and $\mathbf{v}$ the one-variable function $g(s) = f(\mathbf{x} + s\mathbf{v})$ is convex on an interval around zero, because composition with an affine map preserves convexity. The first-order test on g shows that g' is increasing, so $g''(0) \geq 0$; and $g''(0) = \mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v}$.

Conversely, if the Hessian is positive semidefinite on all of C, Taylor's theorem with integral remainder (Chapter 12) gives

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \int_0^1 (1 - s)\, (\mathbf{y} - \mathbf{x})^\top \nabla^2 f(\mathbf{x} + s(\mathbf{y} - \mathbf{x}))\, (\mathbf{y} - \mathbf{x})\, ds.$$

The integrand is nonnegative because the Hessian is positive semidefinite and $(1 - s) \geq 0$, so the integral term is nonnegative. The first-order inequality holds, and f is convex. $\square$

Two warnings about the second-order test. First, a Hessian that is positive *definite* suffices for strict convexity, but is not a necessary condition: $f(x) = x^4$ is strictly convex although $f''(0) = 0$. Second, a positive semidefinite Hessian gives convexity only, not strictness; $f(x_1, x_2) = x_1^2$ has Hessian $\operatorname{diag}(2, 0)$ and its minimum is the entire line $x_1 = 0$.

Read it aloud (auditory learner)

Two voices, read alternately with a friend. The first voice asks, the second answers.
"I drop a marble into the bowl. Where does it go?" "To the bottom. Always to the same bottom." "And if I release it from the other side?" "To that bottom too." "So there is only one lowest place?" "Only one, and every descending path finds it."
"Now I release it on a stone slope. Where does it go?" "Into the first hollow that is deep enough." "Is that the deepest one?" "Not necessarily. That hollow holds it, it does not convince it."
Repeat alternately, faster and faster, with drumming on the table: a bowl has no traps, a hillside has traps.
The line the reader must repeat, once, in a clear voice: if every descending path ends at the same point, there are no traps there, and that is what convex means.

Several operations preserve convexity. The sum of two convex functions stays convex, because the inequalities are added; a positive multiple does too. The composition $f(\mathbf{A}\mathbf{x} + \mathbf{b})$ is convex when f is convex, because an affine map sends mixtures to mixtures. The pointwise maximum is convex as well: if $f = \max_i f_i$, the inequality for each f_i holds for the maximum. That explains why $\max\{|x_1|, |x_2|\}$ has a crease and is still convex.

Step-by-step recipe (procedural learner)

Four steps to check convexity, from the cheapest.

1. Check the region first: take two points in C and make sure the segment between them stays in C. If that fails, stop, because the next tests assume that C is convex.
2. Test through the definition with a general pair $\mathbf{x}, \mathbf{y}$: write $f(t\mathbf{x} + (1 - t)\mathbf{y})$ and compare it with $t\, f(\mathbf{x}) + (1 - t) f(\mathbf{y})$ for $t \in [0, 1]$. One pair that reverses the inequality is enough to reject.

3. Compute the Hessian $\nabla^2 f(\mathbf{x})$ as a function of $\mathbf{x}$, not as one matrix of numbers at one point.
4. Decide the sign: nonnegative eigenvalues, or all leading principal minors from 1×1 up to $n \times n$ nonnegative. One negative eigenvalue closes the case.

A warning about the order: step 4 must not be taken before step 1. A Hessian that is positive definite at one point speaks only about that point, whereas the definition speaks about all of C.

23.3 Standard form and why a local optimum becomes global

Level: intermediate

Up to here we have two separate ideas: a region with no holes and a surface that does not bend upward. Optimization unites the two through a single way of writing. Every optimization problem can be written in the same way: one function to be made small, a set of inequality conditions, and a set of equality conditions.

standard form (*bentuk baku*) — A way of writing every optimization problem with one pattern: one objective, conditions of the form "no more than zero", and conditions of the form "equals".

Definition 23.3 (Optimization problem in standard form)**.** An *optimization problem in standard form* is

$$\begin{aligned} \text{minimize} \quad & f_0(\mathbf{x}) \\ \text{subject to} \quad & f_i(\mathbf{x}) \le 0, \qquad i = 1, \dots, m, \\ & h_j(\mathbf{x}) = 0, \qquad j = 1, \dots, p, \end{aligned}$$

with variable $\mathbf{x} \in \mathbb{R}^n$, objective function f_0, inequality constraint functions f_i, and equality constraint functions h_j. Its domain is $D = \bigcap_i \operatorname{dom} f_i \cap \bigcap_j \operatorname{dom} h_j$, its feasible region is $X = \{\mathbf{x} \in D : f_i(\mathbf{x}) \le 0,\ h_j(\mathbf{x}) = 0\}$, and the *optimal value* is $p^* = \inf_{\mathbf{x} \in X} f_0(\mathbf{x})$, possibly $-\infty$ when the problem is unbounded below. The problem is called a *convex problem* when $f_0, \dots, f_m$ are convex and $h_1, \dots, h_p$ are affine.

That every constraint can be brought to this pattern is easy to check. The condition $f(\mathbf{x}) \ge b$ becomes $b - f(\mathbf{x}) \le 0$; a linear equality stays an equality; maximizing f is the same as minimizing $-f$. What is not free is convexity alone: flipping the sign of an inequality flips the convexity of the function, so the direction of an inequality must not be changed without checking again.

Corollary 23.4 (Convex feasible region)**.** *In a convex problem, the feasible region X is convex.*

Proof. The sublevel set $S_i = \{\mathbf{x} : f_i(\mathbf{x}) \le 0\}$ is convex for every i: if $f_i(\mathbf{x}) \le 0$ and $f_i(\mathbf{y}) \le 0$, convexity gives $f_i(t\mathbf{x} + (1-t)\mathbf{y}) \le t\, f_i(\mathbf{x}) + (1-t) f_i(\mathbf{y}) \le 0$. The set $\{\mathbf{x} : h_j(\mathbf{x}) = 0\}$ is a hyperplane because h_j is affine, hence convex. The feasible region is the intersection of all of them, and an intersection of convex sets is convex by the proposition in the first section. □

That corollary is what makes this whole chapter make sense: two feasible points can always be joined without leaving the feasible region. The simplest polyhedron is the feasible region of a two-variable linear program in Figure 23.2: a quadrilateral, and its linear objective is optimal at a vertex.

Now the statement promised by the chapter opening, and its proof is short precisely because the feasible region is convex.

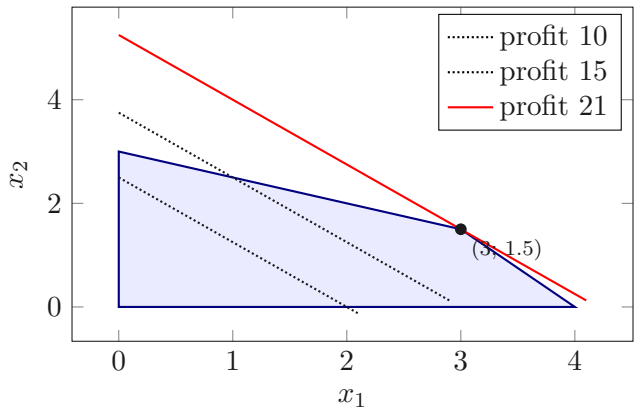


Figure 23.2: The feasible region of the workshop linear program of Chapter 24: $6x_1 + 4x_2 \le 24$ (plywood), $x_1 + 2x_2 \le 6$ (hours), $x_1, x_2 \ge 0$, objective $5x_1 + 4x_2$. The vertices $(0;0)$, $(4;0)$, $(3;1.5)$, $(0;3)$ have values 0, 20, 21, 12. The level line 21 touches only the vertex $(3;1.5)$; the line 20 still cuts the feasible region.

Theorem 23.5 (A local minimum is a global minimum)**.** *Let f_0 be convex on a convex set X, and let $\mathbf{x}^* \in X$ satisfy $f_0(\mathbf{x}^*) \le f_0(\mathbf{x})$ for every $\mathbf{x} \in X$ close enough to $\mathbf{x}^*$. Then $f_0(\mathbf{x}^*) \le f_0(\mathbf{x})$ for every $\mathbf{x} \in X$. The set of all minimum points is convex as well.*

Proof. Suppose there is $\mathbf{y} \in X$ with $f_0(\mathbf{y}) < f_0(\mathbf{x}^*)$. For $t \in (0,1]$ the point $\mathbf{z}_t = (1-t)\mathbf{x}^* + t\mathbf{y}$ lies in X and concentrates at $\mathbf{x}^*$ as $t \to 0$, so for t small enough it enters the neighborhood where $\mathbf{x}^*$ is a local minimum. But convexity gives

$$f_0(\mathbf{z}_t) \le (1-t) f_0(\mathbf{x}^*) + t\, f_0(\mathbf{y}) < f_0(\mathbf{x}^*),$$

contrary to local minimality. For the second part, if $\mathbf{x}^*, \mathbf{y}^*$ are both minimum with value p^*, then $f_0(\mathbf{z}_t) \le p^*$, whereas p^* is the smallest possible value, so $f_0(\mathbf{z}_t) = p^*$ and $\mathbf{z}_t$ is a minimum too. □

Corollary 23.6 (Stationary condition without constraints)**.** *If f is convex and differentiable on an open set, then $\nabla f(\mathbf{x}^*) = \mathbf{0}$ if and only if $\mathbf{x}^*$ is a global minimum.*

Proof. If $\nabla f(\mathbf{x}^*) = \mathbf{0}$, the first-order test gives $f(\mathbf{y}) \ge f(\mathbf{x}^*)$ for every $\mathbf{y}$. Conversely, if $\mathbf{x}^*$ is a minimum, then for every direction $\mathbf{v}$ and small $t > 0$ we have $f(\mathbf{x}^* + t\mathbf{v}) \ge f(\mathbf{x}^*)$, hence the directional derivative $\nabla f(\mathbf{x}^*)^{\top}\mathbf{v} \ge 0$. Since $\mathbf{v}$ may also be replaced by $-\mathbf{v}$, it must be that $\nabla f(\mathbf{x}^*) = \mathbf{0}$. □

Do it by hand (kinesthetic learner)

Materials: two meters of string, one small ring, a blindfold, and a friend.
Tie the string between two posts of equal height, then stretch it straight. Ask the friend to close their eyes, to hang the ring at the middle of the string, and to slide it slowly left and right with one finger. The ring slides to the lowest point, wherever it is released. Ask the friend to mark the stopping point without opening their eyes; repeat five times from different starting positions. All the marks must fall on one point.
Now tie a small knot in the string, a little to the left, so that the string forms two hollows. Repeat the experiment. The ring stops in the left hollow when released near there, and in the middle when released to the right of the knot: two stopping places, two valleys.
What must be visible: a string without a knot has only one lowest point, a string with a knot has two, and a finger that stops in the left hollow does not know what happens on the right, exactly like an algorithm that stops at the first stationary point.

Without convexity, that corollary is false. Take $f(x_1, x_2) = x_1 x_2$ on the square $[-1, 1]^2$. Its gradient (x_2, x_1) vanishes at the origin, whereas $f(\mathbf{0}) = 0$ while $f(1, -1) = -1$. Its Hessian is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ with eigenvalues 1 and -1, so the second-order test rejects convexity: the origin is a saddle, not a valley. Chapter 24 gives a more deceptive example: rounding a continuous answer gives a profit of 19, whereas the best integer answer gives 20.

23.4 The Lagrangian and lower bounds from the dual

Level: advanced

A constraint always has the form "something must not be positive", so we can put a price on violating it: every time $f_i(\mathbf{x})$ is positive we pay a fine λ_i per unit, and every deviation $h_j(\mathbf{x})$ from zero is paid or received at ν_j per unit. The resulting function is no larger than the original objective, provided the price is nonnegative. A bad price gives a loose bound; a good price gives an exact bound.

Definition 23.4 (Lagrangian and dual function)**.** For a problem in standard form, its *Lagrangian* is

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i f_i(\mathbf{x}) + \sum_{j=1}^{p} \nu_j h_j(\mathbf{x}),$$

with $\boldsymbol{\lambda} \in \mathbb{R}^m$ and $\boldsymbol{\nu} \in \mathbb{R}^p$ called *multipliers* or *dual variables*. The *dual function* is its infimum over the original variable,

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x} \in D} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}),$$

and the *dual problem* is to maximize g under the condition $\boldsymbol{\lambda} \geq \mathbf{0}$. A pair $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ is called *dual feasible* when $\boldsymbol{\lambda} \geq \mathbf{0}$ and $g(\boldsymbol{\lambda}, \boldsymbol{\nu}) > -\infty$. Its optimal value is written d^*.

Proposition 23.7 (The dual function is always concave)**.** *g is the infimum of affine functions in $(\boldsymbol{\lambda}, \boldsymbol{\nu})$, because for every fixed $\mathbf{x}$ the function $L(\mathbf{x}, \cdot, \cdot)$ is affine. Hence g is concave, and the dual problem is always a convex problem, whatever the shape of the original problem.*

Proof. Take two dual points $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ and $(\boldsymbol{\lambda}', \boldsymbol{\nu}')$ and $t \in [0, 1]$. For every $\mathbf{x}$, linearity gives

$$L(\mathbf{x},\, t\boldsymbol{\lambda} + (1-t)\boldsymbol{\lambda}',\, t\boldsymbol{\nu} + (1-t)\boldsymbol{\nu}') = t\, L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) + (1-t)\, L(\mathbf{x}, \boldsymbol{\lambda}', \boldsymbol{\nu}').$$

Take the infimum of both sides over $\mathbf{x}$; the infimum of a sum is no less than the sum of the infima, so $g \geq t\, g(\boldsymbol{\lambda}, \boldsymbol{\nu}) + (1-t) g(\boldsymbol{\lambda}', \boldsymbol{\nu}')$, that is g is concave. □

A pleasant corollary: maximizing a concave function under the constraint $\boldsymbol{\lambda} \geq \mathbf{0}$ is always a convex problem. Every problem, even one full of holes, has a convex twin; the twin does not always give the same answer, but it always gives a bound.

Theorem 23.8 (Weak duality)**.** *For every dual feasible $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ and every primal feasible $\mathbf{x}$ we have $L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f_0(\mathbf{x})$. Hence $g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq p^*$, that is $d^* \leq p^*$.*

Proof. Take $\mathbf{x} \in X$. Every $f_i(\mathbf{x}) \leq 0$ and $\lambda_i \geq 0$, so $\lambda_i f_i(\mathbf{x}) \leq 0$. Every $h_j(\mathbf{x}) = 0$, so $\nu_j h_j(\mathbf{x}) = 0$ whatever ν_j is. Therefore $L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f_0(\mathbf{x})$. That inequality holds for every feasible $\mathbf{x}$, hence also for the infimum over all feasible $\mathbf{x}$; the infimum over all of D is no larger, so $g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq p^*$. The last inequality is taken over all dual feasible pairs. □

That proof uses neither convexity nor differentiability, so weak duality holds for integer problems, combinatorial problems, and any problem at all in standard form. That is why dual lower bounds are used in branch and bound in Chapter 24: the bound is valid even though the relaxation is not the answer.

The difference $p^* - d^*$ is called the *duality gap*. Since $d^* \le p^*$, the gap is never negative. When the gap is zero, we say that *strong duality* holds; the next section discusses when that happens.

Before that, one corollary used to check computed results. If there is a feasible point $\mathbf{x}$ and a dual feasible pair $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ with $f_0(\mathbf{x}) = g(\boldsymbol{\lambda}, \boldsymbol{\nu})$, then $d^* \ge g = f_0(\mathbf{x}) \ge p^* \ge d^*$, so all the inequalities become equalities: the point is optimal, the pair is optimal, and the gap is zero. Equality of values is a certificate of optimality that can be checked without knowing the answer in advance.

For younger readers (age 10 and up)

Have you ever haggled in a market? You name one price, the seller names another, and if the bargaining stops halfway, neither of you knows the true price yet. But you already know one thing: the seller will not go below a certain number, and you will not go above a certain number. Those two numbers cage the true price, even though nobody has ever spoken it.

The idea of duality works exactly like that. Instead of looking for the answer, we attach a price to each demand, then ask: with this price list, what is the worst outcome still possible? Every sensible price list gives one number that the true answer cannot exceed. If some price list gives a number equal to the answer we found, the matter is settled.

In this chapter we shall learn why such a price list always exists, why it sometimes fails to close the gap, and why the gap that remains can be measured by a number.

23.5 Strong duality and Slater's condition

Level: advanced

Convexity turns a local minimum into a global one, but it does not yet guarantee a zero duality gap. What is missing is a condition about the *geometry of feasibility*: if the feasible region only touches the boundary of a constraint without entering it, there is no sufficiently clean separating plane. That condition is named after Slater.

Theorem 23.9 (Slater's condition)**.** *Let the problem be convex, $f_0, \dots, f_m$ differentiable, and $h_j(\mathbf{x}) = \mathbf{a}_j^{\top}\mathbf{x} - b_j$. If there is a feasible point $\mathbf{x} \in \operatorname{ri} D$ with $f_i(\mathbf{x}) < 0$ for every i and $h_j(\mathbf{x}) = 0$ for every j, then strong duality holds, $d^* = p^*$, and the optimal value of the dual is attained. If some of the inequality constraints turn out to be affine, only the constraints that are not affine need be satisfied strictly.*

A point with $f_i(\mathbf{x}) < 0$ is called *strictly feasible*. The existence of one such point is enough, and the condition demands nothing about the objective function.

The proof is not carried out here. The route is this: the set $\mathcal{A} = \{(u, t) : \exists \mathbf{x},\ f_i(\mathbf{x}) \le u_i,\ h_j(\mathbf{x}) = 0,\ f_0(\mathbf{x}) \le t\}$ is convex, and the point $(0, p^*)$ lies on its boundary; the hyperplane separation theorem (Chapter 10 uses a similar idea for projection) gives a supporting plane with coefficients $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ and value d^*. The full proof is in Boyd and Vandenberghe section 5.3.2, and that theorem is what justifies the step "solve KKT and then trust the result".

Slater is sufficient, not necessary, and the difference shows in the following small example. Take $\min x$ with constraint $x^2 \le 0$. Its feasible region is a single point, $x = 0$, so there is no strictly feasible point and Slater fails. Its optimal value is $p^* = 0$. The Lagrangian

$L = x + \lambda x^2$ gives $g(\lambda) = -1/(4\lambda)$ for $\lambda > 0$, because the minimum occurs at $x = -1/(2\lambda)$, so $\sup_{\lambda \geq 0} g(\lambda) = 0 = d^*$. Strong duality still holds, but that value is only approached, not attained: at every finite λ, $g(\lambda) < 0$. So what is lost when Slater fails is not always equality of values, but the guarantee that the dual optimum is attained.

Conversely, Slater guarantees something that strong duality alone does not give: optimal multipliers exist, so that the KKT equations have a solution to look for.

23.6 The KKT conditions: necessary, and sufficient when convex

Level: advanced

If the multipliers are already optimal, the optimal primal point must minimize the Lagrangian without constraints, and there the gradient vanishes. Expanding that condition component by component gives four groups of equations: the Karush-Kuhn-Tucker conditions.

Lagrange multiplier (*pengali Lagrange*) — The price per unit of violating a constraint; zero means that constraint is not pressing, positive means that it is pressing.

Definition 23.5 (KKT conditions)**.** A point $(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ satisfies the *KKT conditions* when

$$
\begin{aligned}
& f_i(\mathbf{x}) \leq 0, \quad h_j(\mathbf{x}) = 0, && \text{(primal feasibility)} \\
& \lambda_i \geq 0, && \text{(dual feasibility)} \\
& \lambda_i f_i(\mathbf{x}) = 0, \quad i = 1, \dots, m, && \text{(complementary slackness)} \\
& \nabla f_0(\mathbf{x}) + \textstyle\sum_i \lambda_i \nabla f_i(\mathbf{x}) + \sum_j \nu_j \nabla h_j(\mathbf{x}) = \mathbf{0}. && \text{(stationarity)}
\end{aligned}
$$

The third group causes the most misunderstanding, so let us translate it: a multiplier may be nonzero only for a constraint that is really pressing. A loose constraint is paid nothing, because loosening it changes nothing. A zero multiplier on a pressing constraint is also possible, and that is a sign of degeneracy.

The geometric side of the last group is visible in Figure 23.3.

Now the statement that gives this section its title, and which must be read carefully because its two directions have different conditions.

Theorem 23.10 (KKT is sufficient for optimality)**.** *Let the problem be convex, $f_0, \dots, f_m$ differentiable, and h_j affine. If $(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ satisfies the KKT conditions, then $\mathbf{x}$ is primal optimal, $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ is dual optimal, and the duality gap is zero.*

Proof. Because $\lambda_i \geq 0$ and f_i is convex, every term $\lambda_i f_i$ is convex; because h_j is affine, every term $\nu_j h_j$ is affine. So $L(\cdot, \boldsymbol{\lambda}, \boldsymbol{\nu})$ is convex in $\mathbf{x}$. The stationarity condition says that its gradient vanishes at $\mathbf{x}$, so by the corollary in the third section that point minimizes L globally: $g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$. Primal feasibility and complementary slackness remove all the constraint terms:

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_i \lambda_i f_i(\mathbf{x}) + \sum_j \nu_j h_j(\mathbf{x}) = f_0(\mathbf{x}).$$

Because $g \leq d^* \leq p^* \leq f_0(\mathbf{x})$ by weak duality, all those inequalities become equalities. Hence $\mathbf{x}$ is primal optimal, the pair is dual optimal, and $p^* - d^* = 0$. □

The opposite direction is different in nature, and the difference is not technical.

Theorem 23.11 (KKT is necessary under a qualification condition)**.** *Let the problem be convex and differentiable, satisfy Slater's condition, and let $\mathbf{x}^*$ be primal optimal with p^* finite. Then there exists $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ such that $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ satisfies the KKT conditions.*

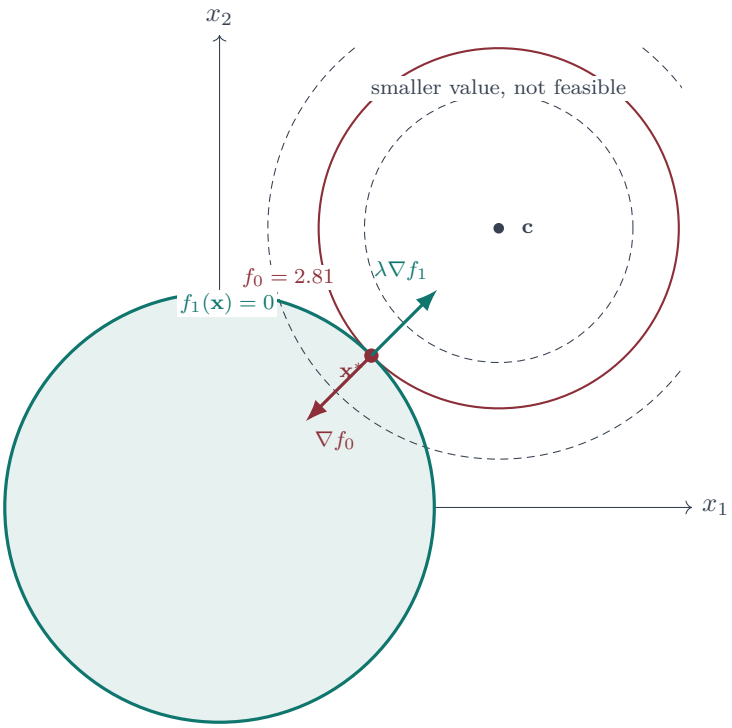


Figure 23.3: The contours of the objective $f_0(\mathbf{x}) = (x_1 - 2.6)^2 + (x_2 - 2.6)^2$, concentric around the point $\mathbf{c} = (2.6; 2.6)$, which is the unconstrained minimum and falls outside the feasible region $f_1(\mathbf{x}) = x_1^2 + x_2^2 - 4 \leq 0$. The contour $f_0 = 2.81$ only touches the constraint circle at $\mathbf{x}^* = (\sqrt{2}; \sqrt{2})$; the deeper dashed contours have smaller values but lie entirely outside the feasible region, whereas the outer ones still cut the feasible region with larger values. At that point of tangency ∇f_0 and ∇f_1 are parallel and oppositely directed, with a positive multiplier.

The proof uses the hyperplane separation theorem on the achievable set, and is not repeated here; see Boyd and Vandenberghe section 5.5.3. What must be remembered is the role of Slater in it: without a qualification condition, an optimal point may fail KKT altogether, so that "solve KKT" is not a legitimate route for finding the answer.

The smallest counterexample is $\min x$ with constraint $x^2 \leq 0$. Its optimal point is $x^* = 0$ and $\nabla f_0 = 1$, whereas $\nabla f_1 = 2x = 0$ at that point. Stationarity demands $1 + \lambda \cdot 0 = 0$, which is impossible. So that optimal point does not satisfy KKT, and what fails is only the constraint qualification: the feasible region is one point, there is no direction to move, there is no supporting plane.

The logical conclusion, and it is often quoted wrongly: in a convex problem, KKT is *sufficient* without extra conditions, but *necessary* only when a constraint qualification is present. The statement "$\mathbf{x}$ is optimal if and only if there are multipliers satisfying KKT" demands both directions, so Slater is part of it.

One interpretation for remembering the multipliers. Suppose constraint i is relaxed to $f_i(\mathbf{x}) \leq u_i$, and the optimal value becomes $p^*(u)$. If p^* is differentiable at $u = 0$, then

$$\lambda_i^* = -\frac{\partial p^*}{\partial u_i}(0).$$

The multiplier is a shadow price: how much the objective value falls when that constraint is relaxed by one unit. In the worked example later, relaxing the solubility limit from one half to 0.6 lowers the deviation from 2.5 to 2.32, whereas the shadow price predicts $2.5 - 2(0.1) = 2.30$. The difference 0.02 is a second-order term, and its direction is always the same: the linear forecast is a lower bound, so the improvement it promises is never exceeded.

The mechanical interpretation, already used by Lagrange himself: two blocks are connected by a spring and must not pass through each other or through the wall. Potential energy is the objective function; at equilibrium, stationarity is the balance of forces, and the multiplier is the contact force of the wall. A contact force only pushes, never pulls, that is $\lambda_i \geq 0$; it appears only when there is contact, that is complementary slackness (Boyd and Vandenberghe section 5.5.4).

Note 23.1 (Subgradients). If f is convex but not differentiable, the gradient is replaced by a *subgradient*: $\mathbf{s}$ is a subgradient of f at $\mathbf{x}$ when $f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{s}^\top(\mathbf{y} - \mathbf{x})$ for every $\mathbf{y}$; the set of all subgradients is written $\partial f(\mathbf{x})$. For $f(x) = |x|$ at $x = 0$ that set is the interval $[-1, 1]$, because every slope between -1 and 1 gives a line below the graph. The optimality condition without constraints becomes $\mathbf{0} \in \partial f(\mathbf{x}^*)$, and KKT still holds with ∇f_i replaced by a subgradient (Rockafellar section 23).

23.7 Worked examples

Level: intermediate

The following three computations are carried out to the last digit, with simple fractions so that they can be checked by hand.

First, a convexity test through eigenvalues. Take $f(x_1, x_2) = x_1^2 + 4x_1x_2 + 5x_2^2$. Its Hessian is constant,

$$\nabla^2 f = \begin{pmatrix} 2 & 4 \\ 4 & 10 \end{pmatrix}, \qquad \operatorname{tr} = 12, \quad \det = 20 - 16 = 4.$$

Its eigenvalues are $\lambda_{1,2} = 6 \pm \sqrt{36 - 4} = 6 \pm 4\sqrt{2}$, that is 0.343146 and 11.656854. Both are positive, so the Hessian is positive definite and f is strictly convex. For comparison, $g(x_1, x_2) = x_1^2 - x_2^2$ has Hessian $\operatorname{diag}(2, -2)$ with eigenvalues 2 and -2: not convex, and its critical point at the origin is a saddle. The function $h(x_1, x_2) = x_1^2$ has eigenvalues 0 and 2: convex, not strict, and its minimum is the entire line $x_1 = 0$, so the answer is not unique.

Challenge (problem solver)

Take $f(x_1, x_2) = x_1^2 x_2^2$. Hold $x_2 = c$ fixed: the graph is a parabola opening upward, so every horizontal slice looks convex, and the same holds for every vertical slice. Is f convex on $\mathbb{R}^2$? If not, exhibit two points that prove it, then explain why checking slice by slice is never enough.

Hint 1: do not move parallel to an axis. Move both coordinates at once, then check the midpoint.

Hint 2: compute $f(1, 0)$, $f(0, 1)$, and the value of f at the midpoint of that segment; or compute $\det \nabla^2 f$ at $(1, 1)$, which equals $-12x_1^2x_2^2$.

If you get stuck, go back to the section Convex functions and their two tests.

Second, the paint mixer problem through KKT. With x_1, x_2 the amounts of the two pigments, the problem is

$$\text{minimize } (x_1 - 2)^2 + (x_2 - 2)^2 \quad \text{subject to} \quad x_1 + x_2 = 2, \quad x_1 \leq \tfrac{1}{2}.$$

Write $g(\mathbf{x}) = x_1 - \frac{1}{2} \leq 0$ and $h(\mathbf{x}) = x_1 + x_2 - 2 = 0$. Stationarity gives

$$2(x_1 - 2) + \lambda + \nu = 0, \qquad 2(x_2 - 2) + \nu = 0,$$

and complementary slackness gives $\lambda(x_1 - \frac{1}{2}) = 0$. If $\lambda = 0$, the two equations give $x_1 = x_2$, and with $x_1 + x_2 = 2$ we obtain $x_1 = x_2 = 1$, which violates $x_1 \leq \frac{1}{2}$. So $\lambda > 0$: $x_1 = \frac{1}{2}$ and $x_2 = \frac{3}{2}$.

The second equation gives $\nu = 2(2 - \frac{3}{2}) = 1$, the first gives $\lambda = 4 - 2x_1 - \nu = 4 - 1 - 1 = 2$. Since $\lambda = 2 \geq 0$, that point is globally optimal by the sufficiency theorem, with

$$p^* = \left(\tfrac{1}{2} - 2\right)^2 + \left(\tfrac{3}{2} - 2\right)^2 = \tfrac{9}{4} + \tfrac{1}{4} = \tfrac{5}{2} = 2.5.$$

Checking through the dual, not through trust. The dual function is computed by completing the square in each variable:

$$\min_{x_1}\Big[(x_1 - 2)^2 + (\lambda + \nu)x_1\Big] = 4 - \frac{(\lambda + \nu - 4)^2}{4}, \qquad \min_{x_2}\Big[(x_2 - 2)^2 + \nu x_2\Big] = 4 - \frac{(\nu - 4)^2}{4},$$

so that

$$g(\lambda, \nu) = 8 - \frac{(\lambda + \nu - 4)^2}{4} - \frac{(\nu - 4)^2}{4} - \frac{\lambda}{2} - 2\nu.$$

At $(\lambda, \nu) = (2, 1)$ we get $g = 8 - \frac{1}{4} - \frac{9}{4} - 1 - 2 = \frac{5}{2} = p^*$: the gap is zero, as the strong duality guaranteed by Slater requires, because (0.4; 1.6) is strictly feasible. The partial derivatives give $-\frac{1}{2}(\lambda + \nu - 4) - \frac{1}{2}(\nu - 4) - 2 = 0$ and $-\frac{1}{2}(\lambda + \nu - 4) - \frac{1}{2} = 0$; from the second $\lambda + \nu = 3$, then $\nu = 1$ and $\lambda = 2$. That pair is the only maximizer.

The shadow prices can be checked with numbers. If the solubility limit is relaxed to 0.6, the answer is $x_1 = 0.6$, $x_2 = 1.4$, and $p^* = 1.96 + 0.36 = 2.32$. The linear forecast $p^* - \lambda \cdot 0.1 = 2.5 - 0.2 = 2.30$ is off by 0.02, and $0.02 = \frac{1}{2} \cdot 4 \cdot (0.1)^2$ exactly, because the second derivative of p^* with respect to that limit is 4. If the demanded sum is raised to 2.1 with the limit unchanged, the answer is $x_1 = \frac{1}{2}$, $x_2 = 1.6$, and $p^* = 2.25 + 0.16 = 2.41$; the forecast $2.5 - \nu \cdot 0.1 = 2.40$ is off by $0.01 = \frac{1}{2} \cdot 2 \cdot (0.1)^2$. Both prices are slopes that can be used to decide, with an error that can be computed.

Third, the duality gap on a nonconvex problem. Take one variable,

$$\text{minimize } (x - 2)^2 \quad \text{subject to} \quad x^3 - x \leq 0.$$

That constraint is equivalent to $x(x-1)(x+1) \leq 0$, so the feasible region is $(-\infty, -1] \cup [0, 1]$: two separate pieces, not convex. On $[0, 1]$ the objective decreases, so its smallest value is 1 at $x = 1$. On $(-\infty, -1]$ the objective increases to the left from $x = -1$, which has value 9. So $p^* = 1$, attained at $x = 1$.

The dual side is very different. The Lagrangian is $L = (x - 2)^2 + \lambda(x^3 - x)$. For $\lambda > 0$ the cubic term dominates as $x \to -\infty$ and carries a negative sign, so $L \to -\infty$ and $g(\lambda) = -\infty$: there is no lower bound to collect. Only $\lambda = 0$ leaves a finite value, and there $g(0) = \min_x (x - 2)^2 = 0$ at $x = 2$. Hence $d^* = 0$ and

$$p^* - d^* = 1 - 0 = 1,$$

a gap of 100% of the optimal value itself. The lower bound from the dual here is not a loose bound but a bound in the wrong place: $x = 2$ is the unconstrained answer, and the constraint is what determines the answer.

The same example shows why convexity cannot be removed from the sufficiency theorem. Check the point $x = -1$, which is feasible because $(-1)^3 - (-1) = 0$. There $\nabla f_0 = 2(x - 2) = -6$ and $\nabla g = 3x^2 - 1 = 2$, so stationarity demands $-6 + 2\lambda = 0$, that is $\lambda = 3$; complementary slackness holds because the constraint is active. So $(x, \lambda) = (-1, 3)$ satisfies all four groups of KKT conditions, even though $f_0(-1) = 9$ while $p^* = 1$.

Repeat the second computation with the limit $x_1 \leq 1$. Before computing, decide whether that constraint is still pressing, then check whether its multiplier is still 2. After that compute the optimal value and compare it with 2.5. As an extra exercise, find the value of the limit c at which the constraint stops pressing, and explain why the multiplier must be zero exactly at that c.

23.8 Computational lab

Level: advanced

The two blocks below were run with Python 3.14.4, numpy 2.5.2, and scipy 1.18.0. The output is copied verbatim.

Computational lab

```
import numpy as np
from scipy.optimize import minimize

np.set_printoptions(precision=8, suppress=True, linewidth=78)

# mixer problem: min (x1-2)^2+(x2-2)^2  s.t. x1+x2=2, x1<=0.5
f  = lambda x: (x[0]-2)**2 + (x[1]-2)**2
df = lambda x: np.array([2*(x[0]-2), 2*(x[1]-2)])
cons = [{"type": "eq",   "fun": lambda x: x[0]+x[1]-2,
         "jac": lambda x: np.array([1., 1.])},
        {"type": "ineq", "fun": lambda x: 0.5-x[0],
         "jac": lambda x: np.array([-1., 0.])}]
r = minimize(f, [0., 0.], jac=df, constraints=cons, method="SLSQP",
             options={"ftol": 1e-15, "maxiter": 500})
xs = r.x
print("x* =", xs, " f(x*) =", f(xs))
print("multipliers (scipy):", r.multipliers)

# multipliers from stationarity, using the active set {eq, ineq}
A = np.column_stack([[1., 0.], [1., 1.]])   # columns: grad g, grad h
lam, nu = np.linalg.solve(A, -df(xs))
print("lambda, nu =", lam, nu)

# KKT residuals
print("h(x*)          =", xs[0]+xs[1]-2)
print("g(x*)          =", 0.5-xs[0])
print("lambda*g(x*)   =", lam*(0.5-xs[0]))
print("||grad L||     =", np.linalg.norm(df(xs) + lam*np.array([1.,0.])
                                         + nu*np.array([1.,1.])))

# dual function and its maximization
g = lambda lam, nu: (8 - (lam+nu-4)**2/4 - (nu-4)**2/4
                     - 0.5*lam - 2*nu)
print("g(2,1)         =", g(2., 1.))
rd = minimize(lambda z: -g(z[0], z[1]), [0.5, 0.5], method="SLSQP",
              bounds=[(0, None), (None, None)], options={"ftol": 1e-15})
print("dual argmax    =", rd.x, " d* =", -rd.fun)
print("gap p*-d*      =", f(xs) + rd.fun)

# Hessian eigenvalue test
for name, H in [("x1^2+4x1x2+5x2^2", np.array([[2.,4.],[4.,10.]])),
                ("x1^2-x2^2",        np.array([[2.,0.],[0.,-2.]])),
                ("x1^2",             np.array([[2.,0.],[0.,0.]]))]:
    print(name, "eig =", np.linalg.eigvalsh(H))
```

Actual output:

```
x* = [0.5 1.5]  f(x*) = 2.500000000000001
multipliers (scipy): [-1.  2.]
lambda, nu = 2.0000000000000027 0.9999999999999987
h(x*)          = 0.0
g(x*)          = 6.106226635438361e-16
lambda*g(x*)   = 1.2212453270876738e-15
||grad L||     = 0.0
```

```
g(2,1)         = 2.5
dual argmax    = [1.99999997 0.99999999]  d* = 2.499999999999999
gap p*-d*      = 1.7763568394002505e-15
x1^2+4x1x2+5x2^2 eig = [ 0.34314575 11.65685425]
x1^2-x2^2 eig = [-2.  2.]
x1^2 eig = [0. 2.]
```

Computational lab

```
import numpy as np
from scipy.optimize import minimize

# nonconvex: min (x-2)^2  s.t. x^3 - x <= 0
f3 = lambda x: (x[0]-2)**2
r3 = minimize(f3, [0.5], constraints=[{"type": "ineq",
              "fun": lambda x: -(x[0]**3 - x[0])}],
              method="SLSQP", options={"ftol": 1e-14, "maxiter": 500})
print("x* =", r3.x, " p* =", f3(r3.x))

# Lagrangian for several lambda, evaluated far to the left
for lam in (0.0, 1e-3, 1e-2, 0.1):
    L = lambda x: (x[0]-2)**2 + lam*(x[0]**3 - x[0])
    print(f"lam={lam:<6} L(-1e4)={L(np.array([-1e4])):.3e}",
          f" L(-1e6)={L(np.array([-1e6])):.3e}")

# points that satisfy KKT but are not the answer
for xv in (-1.0, 1.0, 0.0):
    f0p = 2*(xv-2); gp = 3*xv*xv - 1
    lam = -f0p/gp if gp != 0 else float("nan")
    print(f"x={xv:>4}  f0'={f0p:>5}  g'={gp:>4}  lam={lam:>6}"
          f"  f0={ (xv-2)**2 }  feasible={xv**3-xv <= 0}")
```

Actual output:

```
x* = [1.]  p* = 0.999999999995369
lam=0.0    L(-1e4)=1.000e+08  L(-1e6)=1.000e+12
lam=0.001  L(-1e4)=-9.000e+08  L(-1e6)=-9.990e+14
lam=0.01   L(-1e4)=-9.900e+09  L(-1e6)=-9.999e+15
lam=0.1    L(-1e4)=-9.990e+10  L(-1e6)=-1.000e+17
x=-1.0  f0'= -6.0  g'= 2.0  lam=   3.0  f0=9.0  feasible=True
x= 1.0  f0'= -2.0  g'= 2.0  lam=   1.0  f0=1.0  feasible=True
x= 0.0  f0'= -4.0  g'=-1.0  lam=  -4.0  f0=4.0  feasible=True
```

Four things must be read. First, the reported x^* is (0.5; 1.5) with value 2.500000000000001; the difference of 10^{-15} from 2.5 is binary fraction rounding. Second, the KKT residuals computed separately are very small: the inequality constraint 6.1×10^{-16}, the product λg about 1.2×10^{-15}, and the norm of the Lagrangian gradient zero, so that point really satisfies the conditions, not almost satisfies them.

Third, the easiest to trip over, `scipy` reports the multipliers $(-1, 2)$, whereas stationarity gives $\lambda = 2.0000000000000027$ and $\nu = 0.9999999999999987$, that is $(\nu, \lambda) = (1, 2)$ to fourteen digits. The second entry matches; the first has the opposite sign. The sign convention for multipliers in SQP routines differs between libraries, so the sign must not be guessed: check again through the stationarity equations. The number 2 on the inequality constraint happens to match, and that masks the sign error.

Fourth, $d^* = 2.499999999999999$ and the gap $p^* - d^* = 1.78 \times 10^{-15}$: zero to fifteen digits. The dual maximizer returns $(\lambda, \nu) = (1.99999997,\ 0.99999999)$, that is $(2, 1)$ to eight digits, with a difference from the `ftol` tolerance. Agreement of three routes, KKT by hand, KKT numerically, and dual maximization, is what makes the number 2.5 worth trusting.

The second output points the other way. The primal gives $x^* = 1$ with $p^* \approx 1$, as in the hand computation. The Lagrangian with $\lambda = 0.001$ is already worth -9×10^8 at $x = -10^4$ and -9.99×10^{14} at $x = -10^6$; adding λ makes it worse in proportion to the cube, so no $\lambda > 0$ is dual feasible and $d^* = 0$ from $\lambda = 0$ is the only bound available. Its three critical points: $x = -1$ with $\lambda = 3$ satisfies KKT and has value 9; $x = 1$ is the true answer with value 1; $x = 0$ fails because $\lambda = -4$.

Note for instructors

The sequence that works: start from the bowl and the marble, then the definition of a convex set, then straight to the local minimum theorem. The first-order and second-order tests are better placed after the students have seen why convexity is used; otherwise both feel like two formulas to memorize.

Two typical misconceptions. First, "convex means bulging", so students reject the bowl as convex; what helps is the line segment between two points inside the bowl. Second, KKT is taken as necessary and sufficient without conditions, and then students are disappointed when the example $x^3 - x \leq 0$ gives a KKT point that is not the answer. That example is worth writing on the board early, so that sufficiency appears as a consequence of convexity.

Board activity: give one two-variable QP with one inequality constraint, ask the class to guess whether the constraint is pressing, then solve KKT by branching on $\lambda = 0$ and $\lambda > 0$, then compute its dual function and show that the values agree. Students who can explain why the $\lambda = 0$ branch fails here are ready for duality.

A sign of readiness for the advanced part: a student can name one convex and one nonconvex problem from their own reading, and state what changes in their optimality guarantees.

23.9 Summary

Summary.

- A convex set contains every segment between two of its members. Intersection preserves convexity, union does not. Balls, hyperplanes, half-spaces, and polyhedra are convex; the set $\{x_1 x_2 \geq 1\}$ is not.
- A convex function lies below its chords. First-order test: the graph lies above every tangent plane. Second-order test: the Hessian is positive semidefinite, necessary and sufficient. A positive definite Hessian is sufficient for strict convexity, but not necessary.
- In standard form with f_i convex and h_j affine, the feasible region is convex, so every local minimum is a global minimum. Without convexity, a stationary point may be a saddle, such as $x_1 x_2$ at the origin.
- The Lagrangian puts a price on the constraints, and the dual function is its infimum over the original variable. The dual function is always concave, so the dual problem is always convex, and weak duality $d^* \leq p^*$ holds without any condition at all.
- Strong duality $d^* = p^*$ needs Slater's condition, that is one strictly feasible point, which at the same time guarantees that optimal multipliers exist. Slater is sufficient, not necessary: for $\min x$ with $x^2 \leq 0$ strong duality holds without the dual optimum being attained.
- KKT is sufficient for optimality in a convex problem without extra conditions, but necessary only when there is a constraint qualification. The multipliers are shadow prices, and the duality gap can be positive, even infinite, in nonconvex problems: $\min (x-2)^2$ with $x^3 - x \leq 0$ gives $p^* = 1$ and $d^* = 0$.

23.10 Exercises

Exercise 23.1.

1. Decide whether $f = x_1^2 + x_1x_2 + x_2^2$, $g = x_1x_2$, and $h = 2x_1^2 - 4x_1x_2 + 3x_2^2$ are convex, concave, or neither. State the Hessian and its eigenvalues; for the one that is positive semidefinite but not definite, explain the consequence for the uniqueness of the minimum.
2. Prove that the intersection of two convex sets is convex, then give two convex sets whose union is not convex, with two concrete points and one mixture point that fails.
3. Prove that if f_1, f_2 are convex then $\max\{f_1, f_2\}$ is convex. Check $f(x_1, x_2) = \max\{x_1^2, x_2\}$ at $(0, 0)$ and $(1, 1)$: does the gradient exist, and does the subdifferential at the crease contain zero? (Hint: use the definition of a subgradient and the first-order test for nonsmooth functions.)
4. Solve by KKT: minimize $x_1^2 + x_2^2$ subject to $x_1 + 2x_2 = 4$ and $x_1 \geq 1$. Determine the optimal point, both multipliers, and the optimal value, then check against `scipy.optimize.minimize`. (Hint: solve first without $x_1 \geq 1$; if that answer violates it, the constraint is pressing and $x_1 = 1$.)
5. Compute the dual function of $\min x_1^2 + x_2^2$ subject to $x_1 + x_2 \geq 2$, determine its maximizer, and compare it with the primal optimal value 2. Why is the gap zero even though there is no equality constraint?
6. Assess this claim: *strong duality guarantees that the primal optimum is attained.* True, false, or not enough evidence? Include a counterexample if needed.
7. Assess this claim: *complementary slackness states that the multiplier of an active constraint is always positive.* True, false, or not enough evidence? Show what that condition actually states.
8. In the paint mixer problem, change the limit to $x_1 \leq 1$ and compute again. Determine the value of the limit c at which the constraint stops pressing, and show that its multiplier is zero at that c. (Hint: use the condition that the answer without the limit already satisfies $x_1 \leq c$.)
9. For the nonconvex problem $\min (x - 2)^2$ with $x^3 - x \leq 0$, prove that $g(\lambda) = -\infty$ for every $\lambda > 0$ by choosing a sequence $x_k \to -\infty$, then explain why $d^* = 0$. Compute p^* as well by examining both pieces of the feasible region.
10. Prove that if f is convex and differentiable and $\nabla f(\mathbf{x}^*) = \mathbf{0}$, then $\mathbf{x}^*$ is a global minimum, without using strict convexity. Why does that proof fail for $f(x) = x^3$, and at which point is its condition violated?

For researchers: limits and next steps

What is really proved in this chapter: the closure of convexity under intersection; the equivalence of convexity with both derivative tests, in both directions; the convexity of the feasible region; the theorem that a local minimum is a global one; the concavity of the dual function; weak duality without any condition at all; the sufficiency of KKT without a constraint qualification. Two things are borrowed: Slater's theorem and the necessity of KKT, both through the hyperplane separation theorem, with proofs in Boyd and Vandenberghe sections 5.3.2 and 5.5.3.

What is popularized beyond its proof. "Convex means easy": convexity guarantees that every KKT point is global, but it gives neither a rate nor a guarantee that an algorithm reaches it. "The multiplier is always a price": that interpretation is valid only when p^* is differentiable with respect to the right-hand side of the constraint. "The gap measures nonconvexity": the example $x^3 - x$ shows that the dual can be weaker than the minimum over the convex hull of the feasible region.

Primary source: Boyd and Vandenberghe chapters 2 to 5; Nesterov, *Lectures on Convex Optimization*, chapters 1 to 3; Rockafellar, *Convex Analysis*, sections 2 and 23.
Limits of validity: finite dimension, differentiable functions except in the note on subgradients, and explicitly stated constraints. For implicit constraints, for instance ones defined by a simulation, KKT cannot be written down and what remains are derivative-free methods. Open question: the rate of first-order methods on large convex problems is still under investigation.

Map to source texts: This chapter follows Boyd and Vandenberghe, Convex Optimization*: sections 2.1 to 2.3 (convex sets, important examples, operations that preserve them), 3.1 (basic properties of convex functions, including the first- and second-order tests), 4.1 and 4.2 (optimization problems and convex problems), 5.1 and 5.2 (the Lagrangian, weak and strong duality, Slater's condition), 5.5 (KKT and the mechanical interpretation). The viewpoint of the dual certificate is taken from Bertsimas and Tsitsiklis,* Introduction to Linear Optimization*, chapters 1 and 4; for subgradients, Rockafellar,* Convex Analysis*, sections 2 and 23.*

Chapter 24

Linear and Integer Programming

A woodworking shop receives 24 sheets of plywood and 6 hours of assembly-machine time for the day. One bookshelf uses 6 sheets of plywood and 1 hour of assembly. One bench uses 4 sheets and 2 hours. A shelf earns a profit of 5 units, a bench 4 units. How many shelves and how many benches should it make?

The first instinct picks shelves, because the profit per unit is higher. That instinct is wrong: one shelf consumes as much plywood as one and a half benches. The best answer of the continuous model is 3 shelves and 1.5 benches, with profit 21, and half a bench cannot be sold. Rounding down gives 19; 4 shelves without benches gives 20. So the best integer answer is not the rounding of the best fractional answer, and the difference is an error of reasoning, not an error of arithmetic.

The tools of the previous chapters are not enough. Chapter 8 taught us to solve $A\mathbf{x} = \mathbf{b}$, but here there are fewer equations than variables, so there are infinitely many answers and we must choose the best one. Chapter 23 gives the KKT conditions, but in this problem the gradient of the objective is constant and never vanishes: the answer clings to the boundary of the feasible region, not to a stationary point. The guarantee of Chapter 21 that a local optimum is also global does not show where the optimum is, and it disappears as soon as the variables are required to be integers.

For that reason this chapter takes three steps: writing the problem as a linear program and recognizing that its answer lies at a corner of the feasible region; putting a price on every constraint through duality, so that optimality can be checked without sweeping the whole region; then facing the requirement of integrality, when the continuous model gives a valid bound that is not necessarily attained.

For younger readers (age 10 and up)

Imagine you have 24 sheets of colored paper and 6 hours to make two kinds of classroom decoration. The first decoration uses 6 sheets and 1 hour, the second uses 4 sheets and 2 hours. The first decoration is in greater demand, so its price is higher. How many of each should you make?

If you try things out, there are many mixtures that fit. What is surprising is that the most profitable mixture turns out to use half a decoration. Half a decoration cannot be sold, and this is where common sense begins to deceive: if you round down, the result is not the best one. There is another mixture, entirely whole, that is in fact more profitable, even though it does not look like the rounding of anything.

The question that makes one curious is this. If there are thousands of possible mixtures, how do we know that one mixture is the best without trying them all? And if your friend offers one extra sheet of paper, what is a fair price for that sheet? In this chapter we

will learn why the best answer always lies at a corner of the possible region, why every material has a price that can be computed, and why a fractional answer must be rounded with care, not with instinct.

24.1 From a production plan to standard form

Level: basic

A production plan is a pair of numbers: how many shelves and how many benches. A plan is called feasible when its use of materials does not exceed the supply. For x_1 shelves and x_2 benches, the conditions are

$$6x_1 + 4x_2 \le 24, \qquad x_1 + 2x_2 \le 6, \qquad x_1 \ge 0, \quad x_2 \ge 0.$$

Each condition cuts the plane with one line and leaves one side; the intersection is a quadrilateral with four corners. The profit is $5x_1 + 4x_2$, and we want that number as large as possible.

The pattern of this kind of problem is called a linear program. The name is apt: everything involved is a straight line, with no curves and no powers.

linear program (*program linear*) — A problem of finding the best value of a quantity computed by a weighted sum, when all the rules take the shape of straight lines.

Definition 24.1 (Linear program)**.** A linear program is the problem of maximizing or minimizing a linear function $\mathbf{c}^{\top}\mathbf{x}$ over variables $\mathbf{x} \in \mathbb{R}^n$ subject to constraints in the form of linear inequalities or equalities. The set of points satisfying all the constraints is called the *feasible region*. If the feasible region is empty, the program is called *infeasible*; if the objective value can be made as large as we please, the program is called *unbounded*.

Software and textbooks use one uniform written pattern, and that is the pattern used throughout this chapter.

Definition 24.2 (Standard form)**.** A linear program in *standard form* is

$$\text{maximize } \mathbf{c}^{\top}\mathbf{x} \quad \text{with} \quad A\mathbf{x} = \mathbf{b}, \qquad \mathbf{x} \ge \mathbf{0},$$

where A is $m \times n$ of rank m, $\mathbf{b} \in \mathbb{R}^m$, and $\mathbf{c} \in \mathbb{R}^n$. Every linear program can be brought to this form without changing its optimal value.

Four translations do all that work. A constraint $\mathbf{a}^{\top}\mathbf{x} \le b$ becomes an equality by adding a variable that records the unused remainder; a constraint $\mathbf{a}^{\top}\mathbf{x} \ge b$ by subtracting one. A variable that may take negative values is replaced by the difference of two nonnegative variables, and maximizing $\mathbf{c}^{\top}\mathbf{x}$ is the same as minimizing $-\mathbf{c}^{\top}\mathbf{x}$.

slack variable (*variabel kendur*) — A number recorded as the unused remainder, so that “no more than” turns into “equal to”.

For the shop above, the standard form uses s_1 as the plywood remainder and s_2 as the remainder of assembly hours:

$$\begin{aligned} 6x_1 + 4x_2 + s_1 &= 24, \\ x_1 + 2x_2 + s_2 &= 6, \end{aligned} \qquad x_1, x_2, s_1, s_2 \ge 0.$$

Here $m = 2$ and $n = 4$, so A is 2×4. The two slack variables are not decoration: later both appear as columns in the simplex tableau and as a sign of which constraint is still loose.

Standard form also demands $\mathbf{b} \ge \mathbf{0}$, and that can always be arranged by multiplying one row by -1. What cannot be arranged is the rank: rows of A that are linearly dependent must be discarded first, since otherwise the basis matrix chosen may be singular.

24.2 Geometry: why the answer is at a corner

Level: intermediate

The four corners of that quadrilateral are $(0;0)$, $(4;0)$, $(3;1.5)$, and $(0;3)$, with profits 0, 20, 21, and 12. The best is at a corner, and that is no coincidence: in a linear program the best value can always be found at a corner, which is why the simplex method only travels from corner to corner.

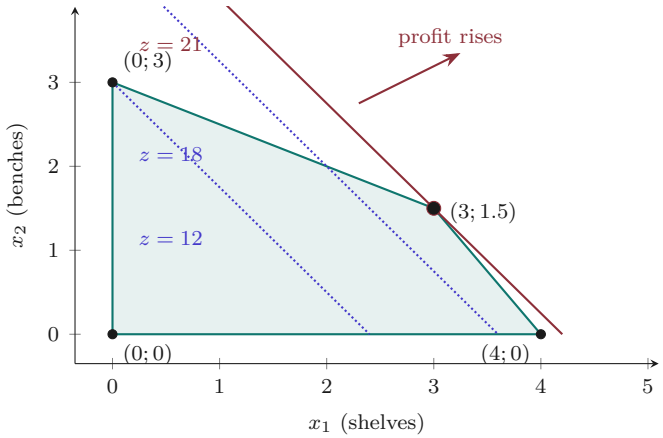


Figure 24.1: The feasible region of the shop and three lines of equal profit. The polygon has four vertices, $(0;0)$, $(4;0)$, $(3;1.5)$, $(0;3)$, with values 0, 20, 21, 12. The lines $z = 12$ and $z = 18$ still cut the feasible region; the line $z = 21$ only touches the vertex $(3;1.5)$, so no feasible point has a higher value.

Figure 24.1 sharpens that coincidence. The three lines there are lines of equal profit, $5x_1 + 4x_2 = 12$, 18, and 21. A higher profit line is farther from the origin, and shifting it in that direction moves it out of the feasible region; the line 21 only touches the vertex $(3;1.5)$.

A corner has a more precise name, and the name does not refer to a picture.

Definition 24.3 (Vertex)**.** Let $P \subseteq \mathbb{R}^n$ be a convex set. A point $\mathbf{x} \in P$ is called a *vertex* or an *extreme point* if there are no $\mathbf{y}, \mathbf{z} \in P$ with $\mathbf{y} \neq \mathbf{z}$ and $t \in (0,1)$ such that $\mathbf{x} = t\mathbf{y} + (1-t)\mathbf{z}$. In other words, $\mathbf{x}$ is not the midpoint of a segment between two other points of P.

In the quadrilateral above, the point $(3;1.5)$ is a vertex: every segment inside the quadrilateral that passes through it must end at that point itself. Conversely $(2;1)$ is not a vertex, since it is the midpoint between $(1;1)$ and $(3;1)$.

vertex (*verteks*) — A corner of the feasible region, namely a point that never becomes the midpoint between two other feasible points.

The first step of the proof turns the negation into a direction that can be used.

Lemma 24.1 (A point that is not a vertex gives a line)**.** *If $\mathbf{x} \in P$ is not a vertex, then there is $\mathbf{d} \neq \mathbf{0}$ such that $\mathbf{x} + \mathbf{d} \in P$ and $\mathbf{x} - \mathbf{d} \in P$.*

Proof. Since $\mathbf{x}$ is not a vertex, there are $\mathbf{y} \neq \mathbf{z} \in P$ and $t \in (0,1)$ with $\mathbf{x} = t\mathbf{y} + (1-t)\mathbf{z}$. Set $s = \frac{1}{2}\min\{t, 1-t\} > 0$ and $\mathbf{d} = s(\mathbf{y} - \mathbf{z})$. Then $\mathbf{x} + \mathbf{d} = (t+s)\mathbf{y} + (1-t-s)\mathbf{z}$ and $\mathbf{x} - \mathbf{d} = (t-s)\mathbf{y} + (1-t+s)\mathbf{z}$. Since $0 < t \pm s < 1$, both points are convex combinations of $\mathbf{y}$ and $\mathbf{z}$, hence members of P because P is convex. Also $\mathbf{d} \neq \mathbf{0}$, since $\mathbf{y} \neq \mathbf{z}$ and $s > 0$. □

Now the statement that makes the simplex method reasonable. In everyday language: if the feasible region is enclosed and the objective does not grow in any direction, then the highest summit is at one of the corners, not in the middle of the plane.

Theorem 24.2 (The optimum is attained at a vertex)**.** *Let $P = \{\mathbf{x} : A\mathbf{x} \le \mathbf{b}\}$ be nonempty and bounded, and suppose $\mathbf{c}^{\top}\mathbf{x}$ is bounded above on P. Then there is a vertex $\mathbf{x}^{*}$ of P with $\mathbf{c}^{\top}\mathbf{x}^{*} = \max_{\mathbf{x}\in P} \mathbf{c}^{\top}\mathbf{x}$.*

Proof. The optimal set $F = \{\mathbf{x} \in P : \mathbf{c}^{\top}\mathbf{x} = p^{*}\}$ is nonempty, closed, bounded, and convex, so F is also a polytope. Write the constraints of P as $\mathbf{a}_i^{\top}\mathbf{x} \le b_i$, $i = 1, \dots, k$. Choose $\mathbf{x}^{*} \in F$ that maximizes the number of active constraints, namely the constraints with $\mathbf{a}_i^{\top}\mathbf{x}^{*} = b_i$.

Suppose $\mathbf{x}^{*}$ is not a vertex. By the lemma there is $\mathbf{d} \neq \mathbf{0}$ with $\mathbf{x}^{*} \pm \mathbf{d} \in F$. Every constraint active at $\mathbf{x}^{*}$ is also active at both of those points, so $\mathbf{a}_i^{\top}\mathbf{d} = 0$ for all active i. Consider the line $\mathbf{x}^{*} + s\mathbf{d}$ for $s \in \mathbb{R}$. Its intersection with P is a closed interval $[s_{-}, s_{+}]$ that contains $[-1, 1]$, and because P is bounded both endpoints are finite. The endpoint s_{+} is determined by the constraint with $\mathbf{a}_i^{\top}\mathbf{d} > 0$ that is reached first; that constraint is active at s_{+} but not at $\mathbf{x}^{*}$, so the new point is feasible and active at more constraints. Its value stays p^{*} because $\mathbf{c}^{\top}(\mathbf{x}^{*} + s_{+}\mathbf{d}) = p^{*} + s_{+}\mathbf{c}^{\top}\mathbf{d} = p^{*}$, and $\mathbf{c}^{\top}\mathbf{d} = 0$ since both points $\mathbf{x}^{*} \pm \mathbf{d}$ have value p^{*}. That new point is a member of F with more active constraints, contradicting the choice of $\mathbf{x}^{*}$; hence $\mathbf{x}^{*}$ is a vertex. $\square$

Read it aloud (auditory learner)

Two voices, read alternately with a friend. Tap the table once each time the word “summit” is spoken.

“Where is the best answer?” “At the summit.” “Why not in the middle?” “Because in the middle there is a neighbor on two sides.” “If there is a neighbor, why is that wrong?” “Because there is still a direction to move.” “Move in which direction?” “In the direction that climbs.” “If there is no climbing direction?” “Then we are already standing at the summit.”

Repeat the three middle lines, slower and slower, with the same taps: in the middle there is a neighbor, in the middle one can still move, a summit has no climbing direction.

The line the reader must repeat, once, in a clear voice: the best answer is at the summit, not in the middle, and a summit is recognized by the absence of a climbing direction.

That proof uses the boundedness of P. If P is unbounded but the objective is bounded above, the conclusion is still true; the route goes through the decomposition of P into the convex hull of its vertices plus the cone of recession directions, and the complete proof is in `WSPrinted.pdf`, in the section on the structure of polyhedral sets.

The number of vertices is finite, but not always small. In standard form, a vertex corresponds to a choice of m linearly independent columns of A: the chosen columns determine $\mathbf{x}_B = B^{-1}\mathbf{b}$, the other columns are zero, and the point is feasible when $\mathbf{x}_B \ge \mathbf{0}$. There are $\binom{n}{m}$ candidates, which for $n = 100$ and $m = 50$ exceeds 10^{29}; checking them all is not a sensible plan. This is where the simplex method comes in: it walks from a vertex to a better neighboring vertex.

Do it by hand (kinesthetic learner)

Materials: a tiled floor, two lengths of string or paper tape, and a handful of buttons.

Stretch two straight strings as two boundaries; one for plywood, one for assembly hours. Stand on a corner tile and call that tile the origin: each tile to the right adds one shelf, each tile forward adds one bench.

Walk along the strings and stop only at tiles where two strings meet. Put a button there, and only there. What must be visible: all the buttons cling to the turns of the fence, not one of them stands in the middle of the yard. After that, count the profit at each button on your fingers, then move one button to the most profitable corner.

Shift one string one tile inward, then repeat: the buttons move, the winner is still the button at the corner.

24.3 The simplex method: one tableau, two pivots

Level: intermediate

The idea is simple. Stand at one vertex, check whether any edge direction raises the profit; if one does, walk to the vertex at its end. If none does, stop: no neighbor is better, which in a linear program means no other point is better.

That edge direction is determined by the *reduced cost*: how much the profit increases when a variable that is now zero is raised by one unit, at the optimal expense of the basis variables.

basic feasible solution (*solusi layak basis*) — A point obtained by setting some variables to zero and then solving for the rest, which happens to still satisfy all the bounds; such a point is always a vertex.

For the shop above, start from the vertex $(0;0)$, namely $s_1 = 24$ and $s_2 = 6$. Its first tableau contains the profit row z and one row for each constraint:

basis	x_1	x_2	s_1	s_2	right-hand side
z	-5	-4	0	0	0
s_1	6	4	1	0	24
s_2	1	2	0	1	6

The z row is written as $z - 5x_1 - 4x_2 = 0$; a negative coefficient means that raising that variable raises z, so the column with the most negative coefficient enters, here x_1 with -5.

The ratio test decides who leaves: divide the right-hand side by the positive entries in the entering column, then take the smallest ratio. $24/6 = 4$ and $6/1 = 6$, so s_1 leaves and the pivot is 6. Dividing the pivot row by 6 and then clearing that column from the other rows gives

$$\begin{aligned} x_1 + \tfrac{2}{3}x_2 + \tfrac{1}{6}s_1 &= 4, \\ \tfrac{4}{3}x_2 - \tfrac{1}{6}s_1 + s_2 &= 2, \\ z - \tfrac{2}{3}x_2 + \tfrac{5}{6}s_1 &= 20. \end{aligned}$$

The point is now $(4;0)$ with profit 20, and a negative coefficient remains: x_2 with $-2/3$. The ratio test gives $4/(2/3) = 6$ and $2/(4/3) = 1.5$, so s_2 leaves and the pivot is $4/3$. After one more round,

$$\begin{aligned} x_1 + \tfrac{1}{4}s_1 - \tfrac{1}{2}s_2 &= 3, \\ x_2 - \tfrac{1}{8}s_1 + \tfrac{3}{4}s_2 &= \tfrac{3}{2}, \\ z + \tfrac{3}{4}s_1 + \tfrac{1}{2}s_2 &= 21. \end{aligned}$$

The z row now has no negative coefficient, so no direction raises the profit: $(3;1.5)$ is optimal with profit 21. All the arithmetic above was rechecked with exact fractions, so that no decimal rounding slipped in.

That last row is more than a stopping sign. The coefficients of s_1 and s_2 there, $3/4$ and $1/2$, are the prices of the two materials. Those numbers come from the basis matrix, $\mathbf{y}^\top = \mathbf{c}_B^\top B^{-1}$, and $\mathbf{b}^\top \mathbf{y} = 24 \cdot \tfrac{3}{4} + 6 \cdot \tfrac{1}{2} = 21$, the same as the profit. That equality is duality, and the next part explains why it is itself already a proof of optimality.

Two remarks. The most-negative-coefficient rule does not guarantee termination, since there are small examples with an endless cycle; Bland's rule, which picks the smallest index, guarantees termination. Modern software uses the dual simplex or interior-point methods, but reports the same vertices and the same prices.

24.4 Duality: a price for every constraint

Level: advanced

Every scarce material has a hidden price. One extra sheet of plywood raises the profit, and the size of that rise is the price of the sheet to the shop. That price is not written in the original problem, but it can be computed, and duality is the way to compute it.

Multiply the plywood constraint by $y_1 \geq 0$ and the hours constraint by $y_2 \geq 0$, then add them. The result is

$$(6y_1 + y_2)x_1 + (4y_1 + 2y_2)x_2 \leq 24y_1 + 6y_2.$$

If $6y_1 + y_2 \geq 5$ and $4y_1 + 2y_2 \geq 4$, the left-hand side is at least $5x_1 + 4x_2$, which is the profit. So $24y_1 + 6y_2$ becomes an upper bound on the profit for every price satisfying those two conditions, and the best bound is made as small as possible.

Definition 24.4 (Primal and dual pair)**.** For the linear program

$$\text{(P)} \quad \text{maximize } \mathbf{c}^\top \mathbf{x} \ \text{ with } A\mathbf{x} \leq \mathbf{b},\ \mathbf{x} \geq \mathbf{0},$$

its dual program is

$$\text{(D)} \quad \text{minimize } \mathbf{b}^\top \mathbf{y} \ \text{ with } A^\top \mathbf{y} \geq \mathbf{c},\ \mathbf{y} \geq \mathbf{0}.$$

The variables $\mathbf{y}$ are called *dual variables* or *shadow prices*.

For the shop above, the dual minimizes $24y_1 + 6y_2$ subject to $6y_1 + y_2 \geq 5$, $4y_1 + 2y_2 \geq 4$, and $y_1, y_2 \geq 0$. Its solution is $y_1 = 3/4$, $y_2 = 1/2$, with value 21, the same as the best profit.

That the upper bound from the dual never falls below the primal value is the first thing to hold on to, and its proof is three lines.

Theorem 24.3 (Weak duality)**.** *For every feasible* $\mathbf{x}$ *of (P) and every feasible* $\mathbf{y}$ *of (D) we have* $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$.

Proof. Since $\mathbf{y} \geq \mathbf{0}$ and $\mathbf{b} - A\mathbf{x} \geq \mathbf{0}$, we have $\mathbf{b}^\top \mathbf{y} \geq \mathbf{y}^\top A\mathbf{x}$. Since $\mathbf{x} \geq \mathbf{0}$ and $A^\top \mathbf{y} - \mathbf{c} \geq \mathbf{0}$, we have $\mathbf{y}^\top A\mathbf{x} \geq \mathbf{c}^\top \mathbf{x}$. The two chains together give the required result. □

The practical consequence is large: a pair of feasible points, one primal and one dual, with the same value already proves both optimal without inspecting any other point. In the shop above, $(3; 1.5)$ and $(0.75;\ 0.5)$ give the value 21, so 21 can be neither improved nor lowered.

What weak duality does not give is a guarantee that the two values always meet. That guarantee is the content of the next theorem.

Theorem 24.4 (Strong duality)**.** *If (P) or (D) has a finite optimal value, then both have optimal values and those values are equal.*

The proof is not carried out here. There are two routes: through Farkas' lemma on systems of linear inequalities, or through the fact that the simplex stops at a tableau with no negative coefficient, and that tableau at the same time gives a feasible dual $\mathbf{y}$ of the same value. The complete proof is in `WSPrinted.pdf`, in the section on linear programming duality, and in Boyd and Vandenberghe, section 5.2.

From that equality comes the condition used to check a computation quickly. In everyday language: a constraint that is still loose does not need to be paid for, and a variable that is used must pay its price exactly.

Corollary 24.5 (Complementary slackness)**.** *If* $\mathbf{x}$ *and* $\mathbf{y}$ *are both optimal, then*

$$y_i(b_i - \mathbf{a}_i^\top \mathbf{x}) = 0 \ \text{ for every } i, \qquad x_j((A^\top \mathbf{y})_j - c_j) = 0 \ \text{ for every } j.$$

Conversely, if $\mathbf{x}$ *is feasible for (P),* $\mathbf{y}$ *is feasible for (D), and both groups of equalities hold, then both are optimal.*

Proof. Since both are optimal, $\mathbf{c}^\top \mathbf{x} = \mathbf{b}^\top \mathbf{y}$, and the chain in the proof of weak duality becomes a chain of equalities. Every term summed inside it is nonnegative, so every term is zero; those terms are exactly $y_i(b_i - \mathbf{a}_i^\top \mathbf{x})$ and $x_j((A^\top \mathbf{y})_j - c_j)$. The converse direction: if all terms are zero, the same chain gives equality of values, and by weak duality that value is optimal for both problems. □

Now the shadow prices themselves. If the right-hand side is perturbed, the optimal value changes, and the rate of change is the component of $\mathbf{y}$. Adding one sheet of plywood raises the profit from 21 to 21.75; adding one assembly hour raises it to 21.5. So a sheet of plywood is worth 0.75 units of profit, an assembly hour 0.5.

That price does not hold forever, and its range of validity can be computed. As long as the set of active constraints does not change, $\mathbf{x}_B = B^{-1}\mathbf{b}$ is linear in $\mathbf{b}$, so its value is linear too. For the shop above,

$$B^{-1} = \begin{pmatrix} \frac{1}{4} & -\frac{1}{2} \\ -\frac{1}{8} & \frac{3}{4} \end{pmatrix}, \qquad \mathbf{x}_B = \begin{pmatrix} \frac{1}{4}b_1 - \frac{1}{2}b_2 \\ -\frac{1}{8}b_1 + \frac{3}{4}b_2 \end{pmatrix}.$$

With $b_2 = 6$ fixed, the condition $\mathbf{x}_B \geq \mathbf{0}$ gives $b_1 \in [12, 36]$; outside that interval one basis variable becomes negative and the slope changes. With $b_1 = 24$ fixed, the interval is $b_2 \in [4, 12]$; a shadow price is therefore always reported together with its interval, not as a bare number.

One warning is useful from the start: shadow prices are an idea of the continuous model. In an integer model, one extra sheet of plywood may raise the profit by 1 or not at all, and there is no single price that summarizes both.

24.5 The LP relaxation: a valid bound, a deceptive rounding

Level: intermediate

If shelves and benches may be fractional, the problem above is finished: 3 shelves and 1.5 benches. In reality they may not, and the relation between the continuous problem and the integer problem must be stated with care.

Definition 24.5 (Integer program, relaxation, and integrality gap)**.** An *integer program* is a linear program in which some or all of the variables are required to be integer. If all the variables are required to be integer, it is called a pure integer program. Its *LP relaxation* is the same linear program without the integrality requirement. For a maximization problem with optimal values z^*_{LP} and z^*_{IP}, the *integrality gap* is $z^*_{\text{LP}} - z^*_{\text{IP}}$.

integrality gap (*celah integralitas*) — The distance between the bound that the continuous model promises and the actual integer answer; that distance measures how far the continuous model misleads.

The reason the relaxation is useful is one sentence: the set of feasible integer points is part of the set of feasible continuous points, so maximizing over the larger set cannot give a smaller value.

Proposition 24.6 (The relaxation gives a valid bound)**.** *For a maximization problem, $z^*_{\text{IP}} \leq z^*_{\text{LP}}$. For a minimization problem, the inequality reverses.*

Proof. Every integer point that is feasible for the integer program is also feasible for its relaxation, since the relaxation only removes conditions, it does not add any. So the feasible region of the integer program is contained in the feasible region of the relaxation. A maximum over a larger set is not smaller than the maximum over its subset. □

The bound is valid, but not always tight, and the difference can be very misleading when it is used for rounding. Rounding fails in two ways, and both appear in our small example.

The first way: the rounded result is infeasible. The point $(3; 1.5)$ rounded up becomes $(3; 2)$. Its profit is 23, higher than 21, but its plywood is $6 \cdot 3 + 4 \cdot 2 = 26$ sheets, exceeding 24: that is not a production plan, only a row of numbers.

The second way, subtler: the rounded result is feasible but not the best. Rounding down gives $(3; 1)$ with profit 19, although $(4; 0)$ has profit 20 and is entirely feasible. Rounding the coordinates one by one in the nearest direction therefore gives a worse answer than another integer point that is farther from $(3; 1.5)$.

Both failures are a consequence of the shape of the feasible region. The set of feasible integer points is not convex: $(3; 1)$ and $(4; 0)$ are members, but their midpoint $(3.5;\ 0.5)$ is not, because it is not integer. On a nonconvex set the vertex-optimum theorem does not apply: no vertex is guaranteed to contain the answer, and no neighbor can be relied upon.

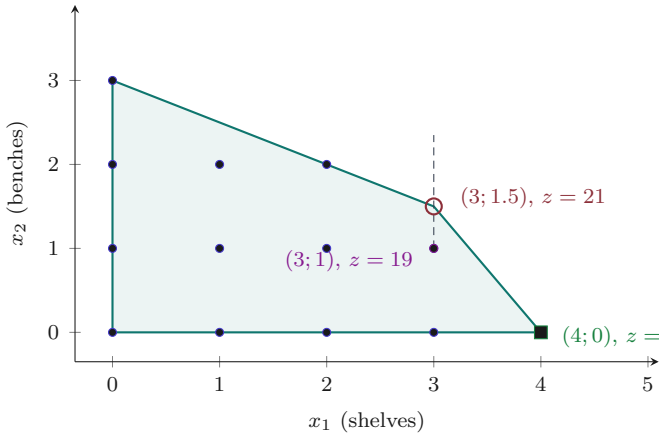


Figure 24.2: The integer lattice versus the relaxation point. There are 13 feasible integer points, marked with filled dots. The relaxation gives $(3; 1.5)$ with value 21, which stands midway between $(3; 1)$ with value 19 and $(3; 2)$, which is infeasible since $6 \cdot 3 + 4 \cdot 2 = 26 > 24$. The best integer answer $(4; 0)$ with value 20 is in fact farther from the relaxation point.

Figure 24.2 contains everything: thirteen feasible integer points, and one relaxation point standing between two points. The one in between is not an integer point, and the nearest is not necessarily the best.

Rounding also has no single meaning in many dimensions. With two variables, “nearest integer point” can still be pictured; with a hundred variables, the nearest is not necessarily feasible, and the feasible one is not necessarily the best. The practice is not to round, but to use the relaxation bound to prune the search.

Challenge (problem solver)

Maximize $6x_1 + 5x_2$ subject to $2x_1 + 3x_2 \le 12$, $4x_1 + x_2 \le 13$, and $x_1, x_2 \ge 0$ integer. The relaxation gives $(2.7;\ 2.2)$ with value 27.2. Rounding to the nearest integer, $(3; 2)$, gives value 28: higher than the value of any relaxation point that is still feasible. Show why 28 cannot be trusted, then determine the correct integer answer together with its value.

Hint 1: check feasibility, not only the value. A higher number may come from a point that is not a plan.

Hint 2: squeeze x_1. The largest integer that is still feasible is 3; for $x_1 = 3$, compute the bound on x_2 from both constraints and take the smaller one.

If you get stuck, go back to the section The LP relaxation: a valid bound, a deceptive

rounding.

24.6 Why integer programs are harder

Level: intermediate

Up to here the question is fair: if the relaxation only adds one requirement, why is an integer program not solved by solving its relaxation and then rounding? The short answer: the difficulty is not in finding an answer, but in making sure that the answer is the best one.

Three differences explain it.

First, there is no simple certificate of optimality. In a linear program, a pair $\mathbf{x}$ and $\mathbf{y}$ of equal value is a finished proof that can be checked in time proportional to the size of the data. In an integer program, the prices from the relaxation give only a bound, not an integer point that attains it: the dual of the relaxation gives a lower bound, not an answer.

Second, the difference between bound and answer can be large. In the vertex cover of the complete graph on n vertices, the relaxation allows $x_v = 1/2$ for all v with value $n/2$, whereas the integer answer is $n - 1$; the ratio $2(n-1)/n$ approaches 2. The complete computation is in the computational lab and in Exercise 10.

Third, the computational cost is of a different class, and the number of candidates grows exponentially. A linear program is solved in time polynomial in the input size; Khachiyan's result through the ellipsoid method and Karmarkar's through interior-point methods are the two landmarks. The decision version of the integer program, "is there a feasible integer point of value at least K", is NP-complete, and Karp's list of twenty-one problems contains knapsack, which by itself has 2^n subsets. No polynomial algorithm is known for integer programming, and if one existed, many other things would collapse with it.

For younger readers (age 10 and up)

Have you ever laid tiles on a floor? You measure the length of the room, then divide by the size of the tile. The result is rarely a whole number. If you round down, there is a gap at the end of the floor. If you round up, you cut a tile and throw away the rest. And sometimes the best answer is neither of the two, but a laying pattern that is altogether different: start from the middle, or use cut tiles on both sides.

Someone who looks only at the single number that comes out of the division will not find that pattern. He needs two bounds: at most how many tiles are used, and at least how many. With those two numbers he stops thinking about patterns that are plainly too wasteful, and concentrates on the ones that may still win.

In this chapter we will learn why the bound from the continuous model is always valid, why rounding can give an answer that is feasible but not the best, and how a computer uses that bound to prune millions of possibilities without inspecting them one by one.

24.7 Branch and bound

Level: advanced

If the relaxation answer is fractional, we do not round it, but split the problem into two and ask again with the relaxation on each piece. Every integer answer satisfies one of two mutually exclusive possibilities, $x_2 \le 1$ or $x_2 \ge 2$, so solving both is the same as solving the original problem.

The procedure has four steps. Solve the relaxation at one node. If the result is integer, record it as a candidate answer and do not continue with that node. If it is fractional, choose

one fractional variable, $x_j = v$, then create two children: $x_j \le \lfloor v \rfloor$ and $x_j \ge \lceil v \rceil$. Before inspecting a node, compare its relaxation bound with the best candidate; if it is not better, discard that node together with all its descendants. Stop when no node remains.

That discarding rule is valid, and the reason is one line.

Proposition 24.7 (Pruning discards no answer)**.** *Let $z_{\mathrm{LP}}(N)$ be the relaxation value at node N and z_{best} the best candidate already recorded. If $z_{\mathrm{LP}}(N) \le z_{\mathrm{best}}$, then no feasible integer point at N is better than z_{best}.*

Proof. Every integer point that is feasible at N is also feasible for the relaxation of N, so its value is at most $z_{\mathrm{LP}}(N)$ by the relaxation-bound proposition. If $z_{\mathrm{LP}}(N) \le z_{\mathrm{best}}$, all those points have value at most z_{best}, so there is nothing left to inspect. □

The tree for the shop above is small, and can be written entirely as a table.

node	added constraint	relaxation **x**	z_{LP}
root	–	(3; 1.5)	21
left	$x_2 \le 1$	(10/3; 1)	$20.\overline{6}$
right	$x_2 \ge 2$	(2; 2)	18
left-left	$x_2 \le 1,\ x_1 \le 3$	(3; 1)	19
left-right	$x_2 \le 1,\ x_1 \ge 4$	(4; 0)	20

The root gives 21 with x_2 fractional. Branching on x_2 gives two children: left $20.\overline{6}$ with $x_1 = 10/3$ still fractional, right 18 and already integer. The right node is recorded as a candidate. At the left node, branching on x_1 gives $(3; 1)$ with value 19 and $(4; 0)$ with value 20; the second updates the candidate. No node remains, and the answer is 20.

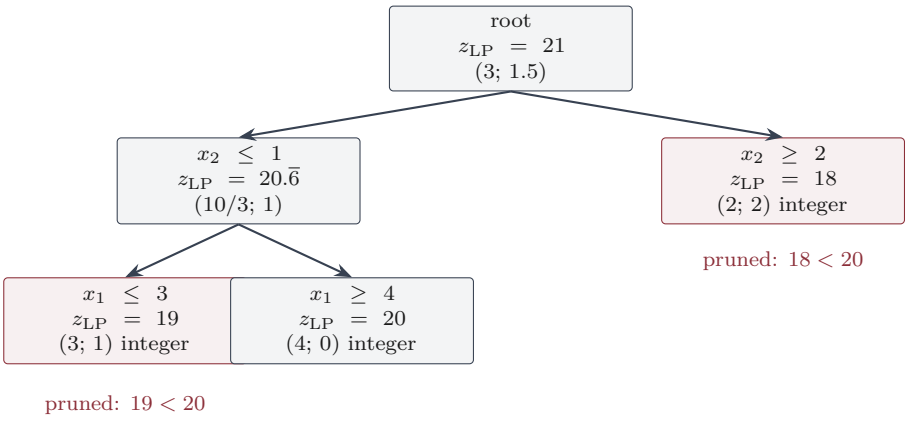


Figure 24.3: The branch-and-bound tree for the shop: five nodes. The root gives $z = 21$ with x_2 fractional. The branch $x_2 \ge 2$ ends at (2; 2) with value 18; the branch $x_2 \le 1$ branches again into (3; 1) with value 19 and (4; 0) with value 20. The two nodes labeled integer whose values are below 20 have a red frame: their bounds are no longer better than the best candidate, so both are pruned without further inspection.

Figure 24.3 draws the same tree. Two nodes there have a red frame, namely (2; 2) with value 18 and (3; 1) with value 19: both are integer, but their bounds are below the best candidate 20, so both are discarded.

The right node with value 18 would not need further inspection if it were fractional: the bound 18 is already below the candidate 20, so its whole branch can be discarded. On an example this small the saving is not noticeable; on a problem with hundreds of variables, that rule is what separates finishing in a minute from not finishing at all, since without good bounds branch and bound degenerates into 2^n nodes.

Two things must be said honestly. The worst-case time is still exponential, and there is no guarantee that the tree is small. The choice of branching variable and the order of the nodes also strongly affect the size of the tree; software uses heuristics, for example the variable closest to one half. Both explain why an integer program can be far slower than a linear program of the same size.

24.8 Worked example

Level: intermediate

Now all the numbers are carried to the end, so that every claim above can be rechecked by hand or with Python.

Step-by-step recipe (procedural learner)

Four steps solve a two-variable linear program graphically:

1. Turn each constraint into a line by replacing the inequality sign with an equals sign, then draw the line through two axis intercepts.
2. Determine the feasible side by testing one point, usually the origin; shade the wrong side so that the feasible region stays clean.
3. Intersect every pair of lines, discard the intersection points that do not satisfy all the constraints, and what remains are the vertices.
4. Compute the objective value at every vertex, then take the largest for maximization or the smallest for minimization.

A warning about order: step 3 must not be carried out before step 2. Intersection points computed earlier often lie outside the feasible region and their values are in fact higher than the true answer; a nice number that appears before feasibility is checked is hard to throw away.

First, the linear program of the shop. Its feasible region is a quadrilateral with four vertices, and all four are inspected:

vertex	active constraints	profit $5x_1 + 4x_2$
$(0;\,0)$	$x_1 = 0,\ x_2 = 0$	0
$(4;\,0)$	$6x_1 + 4x_2 = 24,\ x_2 = 0$	20
$(3;\,1.5)$	$6x_1 + 4x_2 = 24,\ x_1 + 2x_2 = 6$	21
$(0;\,3)$	$x_1 + 2x_2 = 6,\ x_1 = 0$	12

The point $(3; 1.5)$ is computed from the two equations: $x_1 = 6 - 2x_2$ substituted into $6x_1 + 4x_2 = 24$ gives $36 - 8x_2 = 24$, so $x_2 = 3/2$, $x_1 = 3$, and the profit is $15 + 6 = 21$.

Second, the dual and strong duality. The dual minimizes $24y_1 + 6y_2$ subject to $6y_1 + y_2 \geq 5$ and $4y_1 + 2y_2 \geq 4$. Since $x_1, x_2 > 0$, complementary slackness requires both dual constraints to be active; solving both as equalities gives $y_1 = 3/4$ and $y_2 = 1/2$. The value is $18 + 3 = 21$, the same as the profit. A cross-check: $6 \cdot \frac{3}{4} + \frac{1}{2} = 5$ and $4 \cdot \frac{3}{4} + 2 \cdot \frac{1}{2} = 4$, exactly the right-hand sides, so the gap is zero.

Third, shadow prices and their range of validity. Perturbing the right-hand side gives the following table, with z recomputed from scratch in every row.

perturbation	optimal $\mathbf{x}$	z	change
$b = (25;\,6)$	$(3.25;\,1.375)$	21.75	$+0.75$
$b = (24;\,7)$	$(2.5;\,2.25)$	21.50	$+0.50$
$b = (12;\,6)$	$(0;\,3)$	12.00	-9.00
$b = (24;\,4)$	$(4;\,0)$	20.00	-1.00

The third row reduces the plywood to 12, with change $-9 = 0.75 \cdot (-12)$; the fourth row reduces the hours by 2, with change $-1 = 0.5 \cdot (-2)$. Both are still inside the intervals $b_1 \in [12, 36]$ and $b_2 \in [4, 12]$, so the linearity holds all the way to the end. Outside them the slope changes: with $b_1 = 40$, the relaxation gives (0; 3) with value 12, not $21 + 0.75 \cdot 16 = 33$.

Fourth, the integer program of the shop. The enumeration of the two directions above gave the answer 20 at $(4; 0)$, whereas its relaxation is 21 at $(3; 1.5)$. The gap is 1, namely $1/21 = 4.76\%$ of the relaxation value. Rounding down gives 19, so it loses 1 unit of profit, or 5% of the true answer, and that happens on a problem with only two variables.

Fifth, a knapsack with four items. Four items with weights $5, 4, 3, 2$ and values $10, 7, 5, 3$ are loaded into a container of capacity 8. Their value densities are 2, 1.75, 1.667, 1.5, so the greedy rule takes the first item and then $\frac{3}{4}$ of the second: weight $5 + 3 = 8$, value 15.25. The best integer answer is the first and the third item, weight 8 exactly full, value 15. Its integrality gap is 0.25, only 1.64% of the bound. But rounding down throws the second item away entirely and leaves the value 10, namely 33.3% below the true answer: a small gap does not mean that rounding is safe, since the two measure different things.

For the shop above, suppose the assembly machine is available for only 5 hours, not 6. Recompute the vertices, determine the optimal continuous profit, then determine the optimal integer profit with a small enumeration. Before computing, write down your guess: does the integrality gap grow or shrink? After that, inspect the shadow price of assembly hours at $b_2 = 5$ and compare it with 0.5. As an extra, find the b_2 at which the best integer answer moves away from $(4; 0)$.

24.9 Computational lab

Level: advanced

The two blocks below are run with Python 3.14.4, numpy 2.5.2, and scipy 1.18.0. The first block solves the linear program, its dual, and the perturbations of the right-hand side. The second block solves the branch-and-bound tree, the knapsack, and the vertex cover of the complete graph on five vertices. The output is copied as it stands.

Computational lab

```
import numpy as np
from scipy.optimize import linprog

np.set_printoptions(precision=6, suppress=True, linewidth=78)

c = np.array([5., 4.])              # profit per shelf, per bench
A = np.array([[6., 4.], [1., 2.]])  # plywood, assembly hours
b = np.array([24., 6.])

r = linprog(-c, A_ub=A, b_ub=b, bounds=[(0, None)]*2, method="highs")
print("LP   x* =", r.x, "  z* =", -r.fun)
print("residual b - Ax =", b - A @ r.x)
print("marginals (scipy):", r.ineqlin.marginals)

rd = linprog(b, A_ub=-A.T, b_ub=-c, bounds=[(0, None)]*2, method="highs")
print("dual y* =", rd.x, "  dual value =", rd.fun)
print("duality gap =", rd.fun + r.fun)

print("B^-1 =", np.linalg.inv(A), "  x_B =", np.linalg.inv(A) @ b)

for nama, db in [("b=(25, 6)", (1., 0.)), ("b=(24, 7)", (0., 1.)),
                 ("b=(12, 6)", (-12., 0.)), ("b=(24, 4)", (0., -2.))]:
    rr = linprog(-c, A_ub=A, b_ub=b + np.array(db), bounds=[(0, None)]*2,
                 method="highs")
```

```
    print(f"{nama:<10} x = {np.round(rr.x, 6) + 0.0}  z = {-rr.fun:7.4f}"
          f"  delta z = {-rr.fun + r.fun:+.4f}")
```

The actual output:

```
LP   x* = [3.  1.5]   z* = 21.0
residual b - Ax = [0. 0.]
marginals (scipy): [-0.75 -0.5 ]
dual y* = [0.75 0.5 ]   dual value = 21.0
duality gap = 0.0
B^-1 = [[ 0.25  -0.5  ]
 [-0.125  0.75 ]]   x_B = [3.  1.5]
b=(25, 6)  x = [3.25  1.375]  z = 21.7500  delta z = +0.7500
b=(24, 7)  x = [2.5  2.25]  z = 21.5000  delta z = +0.5000
b=(12, 6)  x = [0. 3.]  z = 12.0000  delta z = -9.0000
b=(24, 4)  x = [4. 0.]  z = 20.0000  delta z = -1.0000
```

Computational lab

```
import itertools
import numpy as np
from scipy.optimize import linprog, milp, LinearConstraint, Bounds

c = np.array([5., 4.])
A = np.array([[6., 4.], [1., 2.]])
b = np.array([24., 6.])


def lp_bengkel(tambahan_A=None, tambahan_b=None):
    Aub = A if tambahan_A is None else np.vstack([A, tambahan_A])
    bub = b if tambahan_b is None else np.concatenate([b, tambahan_b])
    r = linprog(-c, A_ub=Aub, b_ub=bub, bounds=[(0, None)]*2, method="highs")
    if r.status != 0:
        return (None, "infeasible")
    return (-r.fun, np.round(r.x, 6) + 0.0)


print("- branch and bound tree -")
simpul = [("root",        None, None),
          ("x2<=1",       [[0., 1.]],  [1.]),
          ("x2>=2",       [[0., -1.]], [-2.]),
          ("x2<=1,x1<=3", [[0., 1.], [1., 0.]],  [1., 3.]),
          ("x2<=1,x1>=4", [[0., 1.], [-1., 0.]], [1., -4.])]
for nama, ta, tb in simpul:
    z, x = lp_bengkel(None if ta is None else np.array(ta),
                      None if tb is None else np.array(tb))
    print(f"{nama:<14} z_LP = {z:8.4f}  x = {x}")

rb = milp(-c, constraints=LinearConstraint(A, -np.inf, b),
          integrality=[1, 1], bounds=Bounds(0, np.inf))
print("IP optimum   z_IP =", -rb.fun, " x =", np.round(rb.x, 6) + 0.0,
      " gap =", 21.0 + rb.fun)

print("- knapsack, 4 items, capacity 8 -")
w = np.array([5., 4., 3., 2.])
v = np.array([10., 7., 5., 3.])
for mask in range(16):
    S = [i + 1 for i in range(4) if mask >> i & 1]
    bw, bv = w[[i - 1 for i in S]].sum(), v[[i - 1 for i in S]].sum()
    print(f"{str(S):<12} weight {bw:4.0f}  value {bv:4.0f}"
          f"  {'feasible' if bw <= 8 else 'infeasible'}")
rk = milp(-v, constraints=LinearConstraint(w, -np.inf, 8.),
```

```
          integrality=[1]*4, bounds=Bounds(0, 1))
print("enumeration/MILP z_IP =", -rk.fun, " x =", np.round(rk.x, 6) + 0.0)
rl = linprog(-v, A_ub=[w], b_ub=[8.], bounds=[(0, 1)]*4, method="highs")
print("relaxation      z_LP =", -rl.fun, " x =", np.round(rl.x, 6) + 0.0)
print("integrality gap =", -rl.fun + rk.fun)

print("- vertex cover of K5: LP vs IP -")
n = 5
E = list(itertools.combinations(range(n), 2))
M = np.zeros((len(E), n))
for k, (i, j) in enumerate(E):
    M[k, i] = M[k, j] = 1.
rv = linprog(np.ones(n), A_ub=-M, b_ub=-np.ones(len(E)),
             bounds=[(0, 1)]*n, method="highs")
rc = milp(np.ones(n), constraints=LinearConstraint(M, 1, np.inf),
          integrality=[1]*n, bounds=Bounds(0, 1))
print("LP =", rv.fun, " x =", np.round(rv.x, 6) + 0.0)
print("IP =", rc.fun, " x =", np.round(rc.x, 6) + 0.0)
```

The actual output:

```
- branch and bound tree -
root           z_LP =  21.0000  x = [3.  1.5]
x2<=1          z_LP =  20.6667  x = [3.333333 1.      ]
x2>=2          z_LP =  18.0000  x = [2. 2.]
x2<=1,x1<=3    z_LP =  19.0000  x = [3. 1.]
x2<=1,x1>=4    z_LP =  20.0000  x = [4. 0.]
IP optimum   z_IP = 20.0  x = [4. 0.]  gap = 1.0
- knapsack, 4 items, capacity 8 -
[]           weight    0  value    0  feasible
[1]          weight    5  value   10  feasible
[2]          weight    4  value    7  feasible
[1, 2]       weight    9  value   17  infeasible
[3]          weight    3  value    5  feasible
[1, 3]       weight    8  value   15  feasible
[2, 3]       weight    7  value   12  feasible
[1, 2, 3]    weight   12  value   22  infeasible
[4]          weight    2  value    3  feasible
[1, 4]       weight    7  value   13  feasible
[2, 4]       weight    6  value   10  feasible
[1, 2, 4]    weight   11  value   20  infeasible
[3, 4]       weight    5  value    8  feasible
[1, 3, 4]    weight   10  value   18  infeasible
[2, 3, 4]    weight    9  value   15  infeasible
[1, 2, 3, 4] weight   14  value   25  infeasible
enumeration/MILP z_IP = 15.0  x = [1. 0. 1. 0.]
relaxation      z_LP = 15.25  x = [1.   0.75 0.   0.  ]
integrality gap = 0.25
- vertex cover of K5: LP vs IP -
LP = 2.5  x = [0.5 0.5 0.5 0.5 0.5]
IP = 4.0  x = [1. 1. 1. 1. 0.]
```

Three things must be read from the first output. The reported $\mathbf{x}^*$, (3; 1.5), with z^* exactly 21 and no residual in either constraint, matches complementary slackness. `scipy` reports the multipliers $(-0.75, -0.5)$, the negative of the shadow prices, because the routine minimizes $-\mathbf{c}^\top\mathbf{x}$; the true numbers are 0.75 and 0.5, and the way to be sure is to recompute through B^{-1}. The duality gap comes out exactly 0, not 10^{-15}, because 21 is represented exactly in binary; on decimal data a residual of order 10^{-12} is normal.

The second output matches the hand computation: five nodes with the same values to four decimals, the integer answer 20 at (4; 0), the knapsack relaxation 15.25, and its integer

answer 15 with gap 0.25. The enumeration of all sixteen subsets confirms that 15 is the best: every subset of higher value exceeds the capacity.

The last part of the second output checks the factor-two claim above. For the complete graph on five vertices, the relaxation is 2.5 whereas the integer answer is 4; the ratio is 1.6, and for n vertices the ratio $2(n-1)/n$ approaches 2 from below. The relaxation bound here can be off by nearly twice.

Note for instructors

The order that works: start from the shop and the half bench, then standard form, then vertices, and only then the simplex. Do not start from the tableau. Students who see the tableau first memorize the pivot step without knowing why that step may stop; those who have already seen that the answer is at a corner will treat a pivot as a way of moving from corner to corner, and the optimality test as a way of knowing that no corner is better.

Two typical misconceptions. First, "the vertices are all the intersections of the constraint lines". Many intersections lie outside the feasible region, for example $(4; 3)$ in our example, which cuts both lines but violates both; checking the feasibility of every candidate must be done on the board. Second, "the integer answer is the rounding of the fractional answer". The shop gives 19 against 20, the knapsack gives 10 against 15; both are enough to break that guess.

Class activity: write the initial tableau on the board, ask the class to choose the entering column and compute the ratio test, then predict the profit before the pivot. After the final tableau, ask for the profit when the plywood becomes 25 sheets; 21.75 opens the way to the interval $[12, 36]$.

Sign of readiness to go on: students can explain why the simplex method does not need to inspect every vertex, and can name one reason why the relaxation must not be rounded.

24.10 Summary

Summary.

- A linear program demands an objective and constraints of straight-line shape. The standard form $\max \mathbf{c}^\top \mathbf{x}$ with $A\mathbf{x} = \mathbf{b}$ and $\mathbf{x} \geq \mathbf{0}$ covers every case, with slack variables to turn inequalities into equalities.
- If the feasible region is bounded and the objective is bounded above, the best value is attained at a vertex. In standard form, a vertex is the same as a basic feasible solution, and their number is $\binom{n}{m}$, so inspecting all of them is not a sensible plan.
- The simplex method moves from vertex to vertex through columns with negative reduced cost and a ratio test, and stops when there is no negative reduced cost; at that moment the z row already contains the dual prices.
- Weak duality holds without conditions; strong duality states that the two values are equal when both are finite; complementary slackness states that a loose constraint has price zero.
- The shadow price y_i is the rate of change of the optimal value with respect to the right-hand side, and it holds only as long as the set of active constraints does not change; in the shop 0.75 per sheet and 0.5 per hour, valid for $b_1 \in [12, 36]$ and $b_2 \in [4, 12]$.
- The LP relaxation gives a valid bound, not an answer. Rounding can give an infeasible point, such as $(3; 2)$, or a feasible point that is not the best, such as $(3; 1)$ with value 19 against 20.
- Branch and bound uses the relaxation bound to discard branches. The shop tree has five nodes, its integer answer is 20 with gap 1. The worst-case time is still exponential,

and the decision version is NP-complete.

24.11 Exercises

Exercise 24.1.

1. A bakery uses 10 kg of flour and 8 oven hours. Cake A uses 2 kg and 1 hour with profit 30; cake B uses 1 kg and 2 hours with profit 20. Write the linear program, draw its feasible region, and compute the profit at every vertex. Then determine the best integer answer by enumeration, and compare it with the result of rounding down the continuous answer.
2. Prove that the set $\{\mathbf{x} : A\mathbf{x} \le \mathbf{b}\}$ is convex, then explain why that proof does not apply to the set $\{\mathbf{x} \in \mathbb{Z}^n : A\mathbf{x} \le \mathbf{b}\}$.
3. Change the profit of the shop to $x_1 + 2x_2$. Compute its value at the four vertices, state which vertex is optimal, and show that every point of the segment between the two optimal vertices is also optimal. Explain why a point in the middle of that segment is not a vertex, then connect the result with the shadow price of the hours constraint.
4. Solve with the simplex tableau: maximize $3x_1 + 2x_2$ subject to $x_1 + x_2 \le 4$, $x_1 \le 3$, $x_2 \le 3$, $x \ge 0$. Write every intermediate tableau, name the entering column and the reason for it, then read the dual prices from the final tableau. (Hint: two pivots suffice; compare the final result with the vertex obtained from the intersection of $x_1 + x_2 = 4$ and $x_1 = 3$.)
5. Write the dual of the program in exercise 1, solve it, and check strong duality numerically. Explain the meaning of every shadow price in one sentence.
6. Assess this claim: *rounding down the answer of the relaxation always gives a feasible integer point.* Decide true, false, or not enough evidence, and exhibit one counterexample with numbers.
7. Assess this claim: *a positive shadow price means that the constraint must be binding.* Decide true, false, or not enough evidence, then explain the actual relation between shadow prices and complementary slackness. (Hint: inspect what happens to a constraint with $y_i > 0$ according to $y_i(b_i - \mathbf{a}_i^{\top}\mathbf{x}) = 0$.)
8. For the knapsack with weights $(5, 4, 3, 2)$, values $(10, 7, 5, 3)$, and capacity 8, compute its LP relaxation by hand using the density order, then compare it with the enumeration result. Compute the integrality gap and state it as a percentage of the bound. (Hint: sort by value per unit of weight, then fill the container to the top; the last item enters partly.)
9. Repeat exercise 8 with capacity 7. Determine the relaxation, the integer answer, and the gap, then explain why the gap can change although only one data value changed.
10. For the vertex cover of the complete graph on n vertices, prove that the relaxation has value $n/2$ and the integer answer $n - 1$, then compute the ratio of the two for $n = 4, 5, 10$. Explain why that result shows that rounding is not a general way to solve integer programs.

For researchers: limits and next steps

What is really proved in this chapter: weak duality for a primal-dual pair in standard form, without any condition; the converse direction of complementary slackness as an optimality test; optimality at a vertex for a bounded feasible region, with a proof through the active-constraint argument; and the validity of pruning in branch and bound. What is borrowed: strong duality, whose usual proof goes through Farkas' lemma or through the termination of the simplex; both are recorded in `WSPrinted.pdf`, in the section on

linear programming duality, and in Boyd and Vandenberghe, section 5.2.
What is popularized beyond its proof is quite a lot, and three things must be named. "The LP relaxation is nearly right": its worst-case approximation factor can grow without bound on certain classes of problems, and even for the vertex cover it reaches 2. "The shadow price is the derivative of the optimal value": that is true only when the optimal value is differentiable with respect to the right-hand side; at a degenerate point, where two optimal bases share the same vertex, the derivative does not exist and what remains is only a subgradient. "Branch and bound is always fast": its worst-case time is exponential, and slowness on a particular instance is not a sign of a bad implementation.
Open questions: how strong the Chvatal-Gomory cutting-plane hierarchy is for particular classes of polytopes, and whether a superpolynomial lower bound exists for the size of extended formulations of the traveling salesman polytope; the results of Fiorini and coauthors in 2012 show an exponential bound for that case, while the search for similar bounds for other problems continues. Limits of validity of this chapter: rational data, exact arithmetic for proofs, floating-point tolerance in practice; all statements about shadow prices hold for the continuous model, not for the integer model.

Map to source texts: This chapter follows Introduction to Linear Optimization *in the file* `WSPrinted.pdf`*: the sections on formulation, the geometry of polyhedral sets (vertices and extreme points), systems of linear inequalities and duality (complementary slackness, sensitivity analysis), and the simplex method (basic feasible solutions, reduced costs, the optimality test). The integer-programming side follows Wolsey,* Integer Programming*, chapter 1 (formulation) and chapter 2 (optimality, relaxation, bounds), which places the relaxation as the source of bounds and branching as the way to use them. The convex viewpoint in the second and fourth parts is Boyd and Vandenberghe,* Convex Optimization*, chapters 4 and 5. Relation to other chapters: standard form and the feasible region continue Chapter 21, solving linear systems uses Chapter 8, and the integrality gap is used again in Chapter 25.*

Part VI

Machine Learning, Security, and Privacy

This part closes the book with the question most often avoided: when is a machine that looks clever actually fragile, and how is its fragility proved with numbers. The prerequisites are Chapters 10, 12, 18, and 23. After eight chapters you can train and evaluate a model honestly, attack it, prove limits on attacks, and explain cryptography and differential privacy well enough to read the original papers.

Chapter 25

Neural Networks and Backpropagation

An intrusion prevention system observes four quantities on every connection: the number of requests per second, the connection duration, the number of rejected packets, and the average payload size. An analyst trying to separate malicious traffic from ordinary traffic finds a pattern that is awkward to read. Connections with many requests but no rejected packets are usually legitimate, because that is what fetching a large file looks like. Connections with many rejected packets but few requests are also usually legitimate, because that is what a failed scan looks like. The combination of the two is almost always an attack. Each feature by itself separates nothing, whereas their combination separates sharply.

A linear threshold cannot draw such a separator, and the first part of this chapter shows that the obstacle is not the amount of data. What must change is the shape of the function. One hidden layer is already enough for that pattern. Once the shape has changed, the next question appears, and it is more serious: how do we tune the thirteen parameters of that small network without deriving a new formula for every architecture?

The answer is the chain rule of Chapter 12, run backward on the computational graph. This chapter carries that rule down to the numbers: one forward step with given weights, all thirteen gradients computed by hand and then checked with NumPy, one gradient-descent step, and the gradient with respect to the input that later becomes a weapon in Chapter 29. What this chapter promises is not a clever network, but a differentiation engine that can be checked line by line.

For younger readers (age 10 and up)

Imagine an assembly line with several workers. The first worker works on the raw material, the second works on the output of the first, and so on until the last worker hands the finished item to an inspector. The inspector weighs the item, then says: too heavy, two hundred grams over.

Now the question: who has to change the way they work? The first worker? The last worker? If the foreman can change only one setting and then rerun the whole line, he needs two tries per setting. With thirteen settings, that is twenty-six runs of the line for a single fix.

It turns out there is a far cleverer way. After one run, we can walk back from the inspector to the last worker, then to the previous worker, all the way to the first. At each stop we compute one number: how much the final error changes if this person changes their output a little. With a single backward pass, every worker knows their own share, and nothing has to be run again.

In this chapter we will learn why information can flow backward even though the goods only flow forward, and why that backward pass gives an answer exactly as accurate as running the line twenty-six times.

25.1 Understanding bridge: from story to symbol

Level: basic

The formulas in this chapter are easier to read if the order is reversed: a story without symbols, then arithmetic on paper, then symbols as shorthand for steps we have already performed ourselves.

25.1.1 Plain analogy: how much, not who

The assembly-line analogy answered "who should change". It did not answer "how much". Picture three taps in a row above one bucket, with a target line on the wall. The third tap feels closest, but it can only pass on what the second tap sends; the first tap matters most, yet its effect depends on the two taps between it and the bucket. What we need is not turning every tap and re-measuring, but one number per tap: how many millimetres does the water rise if this tap moves a little? One pass, traced backwards from the bucket.

25.1.2 Compute by hand first

Example 25.1. One neuron, four numbers: weight $w = 0.5$, bias $b = 0.1$, input $x = 2.0$, correct answer $y = 1$, learning rate $\eta = 0.1$. Follow the steps in order; every number here can be checked on an ordinary calculator.

1. Raw score: $wx = 0.5 \times 2.0 = 1.0$, plus the bias gives $z = 1.1$.
2. Probability: $e^{-1.1} = 0.33287$, so $\sigma = 1/(1 + 0.33287) = 0.7502601$.
3. Loss: because $y = 1$ only one term survives, $L = -\ln 0.7502601 = 0.2873353$.
4. Gap between probability and truth: $\sigma - y = 0.7502601 - 1 = -0.2497399$.
5. Chain rule to the weight: multiply that gap by the input, $-0.2497399 \times 2.0 = -0.4994798$. The bias is not multiplied by anything, so it keeps -0.2497399.
6. Move: $w' = 0.5 - 0.1(-0.4994798) = 0.5499480$ and $b' = 0.1 - 0.1(-0.2497399) = 0.1249740$. The new score is $z' = 1.2248699$, the new probability 0.7729194, the new loss 0.2575805: down 0.0297549 from 0.2873353.

Mind the sign in step 5: the gradient is negative, and because we *subtract* it, the weight rises. The minus sign in the update rule is not decoration; it reverses the direction.

One check you must not skip: replace the analytic derivative by a finite difference, that is, evaluate the loss at $w + 0.000001$ and $w - 0.000001$ and divide the difference by 0.000002. The answer is -0.4994798, equal to seven digits. When the hand result and the numeric result disagree, the sign is almost always what is wrong.

25.1.3 Symbol dictionary

The symbols below appear in any book on neural networks. Nothing here needs memorising: reading each one aloud takes a single sentence, and that sentence is what we computed above.

$\partial L/\partial w$ read "del L del w": if weight w moves a little, how much does the loss L change? In our example the answer is -0.4994798 (step 5).

$z = Wx + b$ read "z equals W times x plus b": the raw score before it is squeezed into a probability. The one-number version of this formula is step 1.

$\sigma(z)$ read "sigma of z": the S-shaped function that keeps the score between zero and one. In step 2 it gives 0.7502601.

$\nabla_w L$ read "nabla w L": every derivative of the loss with respect to every weight at once; its arrow answers "which way does the loss fall fastest". Its entries are numbers like the one in step 5, only many.

25.1.4 Why does this matter for security?

The same derivative that repairs a model also breaks it. An attacker needs one number: if one input feature moves a little, how much does the loss rise? In our example x has derivative -0.1248699, so shrinking x raises the loss. Moving it by 1.9 sends x to 0.1, the probability drops to 0.5374298, and the loss rises from 0.2873353 to 0.6209570, a factor of 2.16. One feature, one direction, one step.

For the defender the same question runs backwards: which feature has the largest derivative? That feature is the cheapest to fake, so it most needs rounding, clipping, or sanity checks. Chapter 29 uses this recipe in full, with perturbation size ϵ and the formula $x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x L)$.

Note 25.1. A warning that is easily missed: on real systems an attacker rarely computes the gradient exactly; it is estimated by changing the input many times. That estimate is more expensive and noisier, but it still works. A closed model is therefore not a defence.

25.1.5 Hand computation drills

Work with a calculator, write on paper, then compare. Answers sit in parentheses.

Note 25.2.

1. Repeat the whole example with $b = 0.3$, everything else unchanged. Compute z, σ, L, $\partial L/\partial w$, $\partial L/\partial b$. ($z = 1.3$; $\sigma = 0.7858350$; $L = 0.2410085$; -0.4283300; -0.2141650.)
2. Repeat the original example with $\eta = 0.5$. What are w', b', and the new loss? ($w' = 0.7497399$; $b' = 0.2248699$; $L' = 0.1640633$, down 0.1232720.)
3. Repeat the original example with $y = 0$, everything else unchanged. What is $\partial L/\partial w$ now, and which way does the weight move? (0.7502601 and 1.5005202; positive, so the weight falls.)
4. In the original example, move x by $\epsilon = 0.5$ in the direction that raises the loss. What are the new x and the new loss? ($x = 1.5$; $L = 0.3558651$.)

25.1.6 From paper to code

The code below runs the same six steps, with variable names that point at their step. Nothing here is new: every line is one of the lines above.

Computational lab

```
import numpy as np

# One example, the same numbers as in "Compute by hand first".
x, y = 2.0, 1.0
w, b, eta = 0.5, 0.1, 0.1

# Step 1: z = w*x + b  (raw score)
z = w*x + b
```

```
print("Step 1  z        =", round(z, 7))

# Step 2: sigma(z) = 1/(1+e^-z)  (squeeze the score into 0..1)
s = 1/(1+np.exp(-z))
print("Step 2  sigma    =", round(s, 7))

# Step 3: L = -ln(s)  (loss; y=1 leaves a single term)
L = -np.log(s)
print("Step 3  L        =", round(L, 7))

# Step 4: dL/dz = sigma - y  (probability minus reality)
dLdz = s - y
print("Step 4  dL/dz    =", round(dLdz, 7))

# Step 5: chain rule down to weight and bias
dLdw, dLdb = dLdz*x, dLdz
print("Step 5  dL/dw    =", round(dLdw, 7), " dL/db =", round(dLdb, 7))

# Step 6: one gradient-descent step
w2, b2 = w - eta*dLdw, b - eta*dLdb
s2 = 1/(1+np.exp(-(w2*x + b2)))
print("Step 6  w', b'   =", round(w2, 7), round(b2, 7),
      " L' =", round(-np.log(s2), 7))

# Check: analytic gradient against a finite difference.
def loss(ww, bb):
    return -np.log(1/(1+np.exp(-(ww*x + bb))))
h = 1e-6
print("finite difference dL/dw =", round((loss(w+h, b)-loss(w-h, b))/(2*h), 7),
      " dL/db =", round((loss(w, b+h)-loss(w, b-h))/(2*h), 7))

# Gradient with respect to the INPUT: this is what attacks a model (Chapter 29).
dLdx = dLdz*w
xa = x + 1.9*np.sign(dLdx)          # the sign of the gradient sets the direction
sa = 1/(1+np.exp(-(w*xa + b)))
print("dL/dx =", round(dLdx, 7), " x_adv =", round(xa, 4),
      " sigma =", round(sa, 7), " L =", round(-np.log(sa), 7))
```

The output must match the paper: 1.1, 0.7502601, 0.2873353, -0.2497399, -0.4994798, then 0.2575805. If one number is off, do not patch the code: find the step that differs on paper.

25.2 Linear thresholds and the perceptron

Level: basic

The oldest form of classifier is the linear threshold. For an input $\mathbf{x} \in \mathbb{R}^d$, weights $\mathbf{w} \in \mathbb{R}^d$, and bias $b \in \mathbb{R}$, we compute a single score and decide the class from the sign of that score.

perceptron (*perceptron*) — A classifier that decides only from which side of a single separating line or plane a point falls.

Definition 25.1 (Linear threshold)**.** A function $f : \mathbb{R}^d \to \{-1, +1\}$ is called a linear threshold if there are $\mathbf{w}$ and b such that $f(\mathbf{x}) = \operatorname{sign}(\mathbf{w}^\top \mathbf{x} + b)$. The set $\{\mathbf{x} : \mathbf{w}^\top \mathbf{x} + b = 0\}$ is the separating hyperplane, and that hyperplane is the only form of separator this model can learn.

Perceptron training works by correcting the weights only when there is a mistake. For a label $y \in \{-1, +1\}$ and an example $\mathbf{x}$, if $y\,\mathbf{w}^\top \mathbf{x} \le 0$ then we update

$$\mathbf{w} \leftarrow \mathbf{w} + y\,\mathbf{x}.$$

Note that there is no learning rate here: the size of the step is determined by the size of the example, and training stops by itself as soon as all examples are correct. Does it always stop?

Theorem 25.1 (Perceptron convergence)**.** *Suppose every example satisfies* $\|\mathbf{x}_i\| \le 1$ *and there is a unit vector* $\mathbf{w}^*$ *and* $\gamma > 0$ *such that* $y_i\, \mathbf{w}^{*\top}\mathbf{x}_i \ge \gamma$ *for every* i*. Then the perceptron algorithm starting from* $\mathbf{w}_0 = \mathbf{0}$ *makes at most* $\lfloor 1/\gamma^2 \rfloor$ *updates.*

Proof. Suppose an update occurs k times, on the examples $\mathbf{x}_1, \dots, \mathbf{x}_k$. Every update occurs only when that example is misclassified, so $y_j\, \mathbf{w}_{j-1}^{\top}\mathbf{x}_j \le 0$. Hence

$$\|\mathbf{w}_j\|^2 = \|\mathbf{w}_{j-1}\|^2 + 2y_j\mathbf{w}_{j-1}^{\top}\mathbf{x}_j + \|\mathbf{x}_j\|^2 \le \|\mathbf{w}_{j-1}\|^2 + 1.$$

By induction, $\|\mathbf{w}_k\|^2 \le k$.

On the other side, the inner product with $\mathbf{w}^*$ grows by at least γ each time:

$$\mathbf{w}^{*\top}\mathbf{w}_j = \mathbf{w}^{*\top}\mathbf{w}_{j-1} + y_j\mathbf{w}^{*\top}\mathbf{x}_j \ge \mathbf{w}^{*\top}\mathbf{w}_{j-1} + \gamma,$$

so $\mathbf{w}^{*\top}\mathbf{w}_k \ge k\gamma$. Since $\mathbf{w}^*$ is a unit vector, the Cauchy-Schwarz inequality gives

$$k\gamma \le \mathbf{w}^{*\top}\mathbf{w}_k \le \|\mathbf{w}_k\| \le \sqrt{k}.$$

After dividing by $\sqrt{k}$ and squaring, $k\gamma^2 \le 1$, that is $k \le 1/\gamma^2$. □

That bound does not depend on the dimension or on the number of examples, and that is its appeal. But when γ does not exist, this theorem says nothing, and the perceptron really does oscillate forever; it is also not a statement about generalization.

Now a more fundamental bound. Four points in the plane are already enough to show that not every labeling can be separated by a single hyperplane. The points $(0,0)$ and $(1,1)$ have label 0, the points $(1,0)$ and $(0,1)$ have label 1. This pattern is usually called XOR.

Proposition 25.2. *The four XOR points are not linearly separable.*

Proof. Suppose there are w_1, w_2, b that separate them, with label 1 on the positive side. Label 0 at $(0,0)$ gives $b \le 0$. Label 1 at $(1,0)$ gives $w_1 + b > 0$, hence $w_1 > -b \ge 0$. By the same route, $w_2 > -b \ge 0$. Therefore $w_1 + w_2 > -2b$, and

$$w_1 + w_2 + b > -b \ge 0.$$

But label 0 at $(1,1)$ demands $w_1 + w_2 + b \le 0$. Contradiction. □

This contradiction holds for every choice of $\mathbf{w}$ and b, not only for some particular training outcome. Adding examples, changing the learning rate, or normalizing the features will not help. What must change is the shape of the function, and the smallest change available is to insert one layer in between.

hidden layer (*lapisan tersembunyi*) — A layer of units that the outside world does not see: its input comes from the previous layer, its output goes to the next layer.

A hidden layer works in two steps. First, compute $\mathbf{z}_1 = \mathbf{W}_1\mathbf{x} + \mathbf{b}_1$ with $\mathbf{W}_1 \in \mathbb{R}^{2\times 4}$, then apply $\mathrm{ReLU}(t) = \max\{t, 0\}$ to every component, and write the result $\mathbf{h}$. Second, compute a linear score from $\mathbf{h}$. For XOR, with

$$\mathbf{W}_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathbf{b}_1 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \qquad \mathbf{w}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \qquad b_2 = 0,$$

the four points give: $(0,0)$ produces $\mathbf{z}_1 = (0,-1)$ and $\mathbf{h} = (0,0)$ with score 0; the points $(1,0)$ and $(0,1)$ both produce $\mathbf{z}_1 = (1,0)$ and $\mathbf{h} = (1,0)$ with score 1; the point $(1,1)$ produces

$\mathbf{z}_1 = (2, 1)$ and $\mathbf{h} = (2, 1)$ with score $1(2) - 2(1) = 0$. With a threshold of 0.5 all four points are classified correctly.

What makes this work is not magic but simple arithmetic. The first unit computes $\max\{x_1 + x_2, 0\}$ and the second computes $\max\{x_1 + x_2 - 1, 0\}$. Both are linear, but their difference is a piecewise function that turns at two places. A linear combination of linear functions is still linear; a linear combination of piecewise functions is not.

Such a boundary is made of line segments. Each unit in the first layer contributes one straight line, and the second layer chooses which side is used, so the final boundary is a polygon; Figure 25.1 shows its shape.

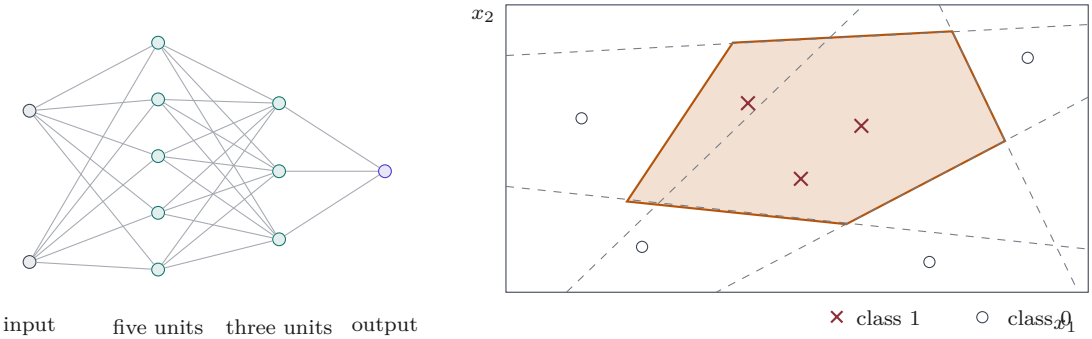


Figure 25.1: A sketch of how two hidden layers turn one line into a fold. Left: each unit in the first layer pulls one straight line across the input plane, and the second layer combines the sides of those lines. Right: the decision boundary is a five-sided polygon, drawn thick; the dashed lines are the extension of each side, namely the line belonging to one unit. Crosses are class 1 inside the polygon, circles class 0 outside. The figure is a sketch, not a training result: the number of sides follows the number of units in the first layer.

25.3 Activation functions and two diseases of the inner layers

Level: intermediate

Without a nonlinear function between the layers, the whole network collapses into a single matrix. If ReLU is replaced by the identity, the output is $\mathbf{W}_2\mathbf{W}_1\mathbf{x}$, that is, a single matrix product $\mathbf{W}\mathbf{x}$ with $\mathbf{W} = \mathbf{W}_2\mathbf{W}_1$. Every hidden layer therefore needs a nonlinear activation, and the choice of activation determines which disease will appear when the network is made deep.

activation function (*fungsi aktivasi*) — The gate that determines how much of a unit's signal is passed on, and at the same time the one that fixes the shape of its derivative.

Definition 25.2 (ReLU, sigmoid, and tanh)**.** For $t \in \mathbb{R}$,

$$\mathrm{ReLU}(t) = \max\{t, 0\}, \qquad \sigma(t) = \frac{1}{1 + e^{-t}}, \qquad \tanh(t) = \frac{e^{t} - e^{-t}}{e^{t} + e^{-t}}.$$

The derivative of ReLU is 1 for $t > 0$ and 0 for $t < 0$; at $t = 0$ the derivative does not exist, and this book uses the value 0 as a convention. The derivative of tanh satisfies $\tanh'(t) = 1 - \tanh^2(t)$.

The value 0 at the break point of ReLU is a convention: the set where the derivative fails is only $\{0\}$, of measure zero, so it does not affect the derivative almost everywhere. The practical consequence: a unit whose input is always negative receives zero gradient and will never change.

Sigmoid has a neater property, and that property is the source of the trouble.

Lemma 25.3 (Derivative of sigmoid)**.** *For every t we have $\sigma'(t) = \sigma(t)\big(1-\sigma(t)\big)$, hence $0 < \sigma'(t) \le \tfrac{1}{4}$.*

Proof. Write $\sigma(t) = (1+e^{-t})^{-1}$. Then

$$\sigma'(t) = -(1+e^{-t})^{-2}\cdot(-e^{-t}) = \frac{e^{-t}}{(1+e^{-t})^2} = \frac{1}{1+e^{-t}}\cdot\frac{e^{-t}}{1+e^{-t}} = \sigma(t)\big(1-\sigma(t)\big).$$

Since $0 < \sigma(t) < 1$, the product is positive. Writing $s = \sigma(t)$ gives $s(1-s) = \tfrac{1}{4} - (s-\tfrac{1}{2})^2 \le \tfrac{1}{4}$, and equality is attained when $s = \tfrac{1}{2}$. □

The bound $\tfrac{1}{4}$ is the one to hold on to. The following table computes σ, σ', and tanh at several values; the numbers were produced by NumPy and are written to six decimals.

t	$\sigma(t)$	$\sigma'(t)$	$\tanh(t)$	$1-\tanh^2(t)$
0	0.500000	0.250000	0.000000	1.000000
1	0.731059	0.196612	0.761594	0.419974
2	0.880797	0.104994	0.964028	0.070650
5	0.993307	0.006648	0.999909	0.000181
10	0.999955	0.000045	0.999999	0.000002

The fourth and fifth columns show what is called saturation: once t moves away from zero, the output barely changes with the input, and the derivative tends to zero. At $t = 10$ the derivative is only 4.54×10^{-5}, about one twenty-thousandth of its maximum.

The consequence can be computed. If the error propagates backward through L consecutive layers whose derivative is at most $\tfrac{1}{4}$, the remaining signal is at most 4^{-L}: for $L = 5$ about 9.77×10^{-4}, and for $L = 10$ about 9.54×10^{-7}. A layer far from the output receives a signal that is only one millionth, so it practically does not learn. This symptom is called the vanishing gradient. ReLU avoids it for a different reason: its derivative is 1 on the positive side, so there is no shrinkage, but its derivative is 0 on the negative side, so there is unit death. Both diseases are not implementation errors; both are consequences of the shape of the function.

One more statement is often quoted beyond its content: one hidden layer is enough for all continuous functions.

Theorem 25.4 (Universal approximation, statement)**.** *Suppose φ is bounded, monotonically increasing, and has different limits at $-\infty$ and $+\infty$. For every compact set $K \subset \mathbb{R}^d$, every continuous function $f : K \to \mathbb{R}$, and every $\varepsilon > 0$, there are N and parameters $\mathbf{w}_i, b_i, v_i$ such that*

$$\sup_{\mathbf{x}\in K}\Big| f(\mathbf{x}) - \sum_{i=1}^{N} v_i\,\varphi(\mathbf{w}_i^{\top}\mathbf{x} + b_i)\Big| < \varepsilon.$$

The proof of this statement is not included here; the original proof is in Cybenko (1989) and a more general version in Hornik (1991), and the extension to ReLU is in Leshno et al. (1993). What is important to read carefully is what the theorem does *not* say. It states that such a set of functions is dense, not that a particular network equals f. It does not say how large N must be for error ε in dimension d; in the worst case N grows exponentially with d (Goodfellow et al., section 6.4.1). It also does not say that those parameters can be found by gradient descent, and it says nothing about data not yet seen. The sentence “a network can approximate anything” is therefore true as a statement of density and misleading as a practical promise.

25.4 From score to loss

Level: intermediate

The network's output must be compared with the label through a single number that can be minimized, and the choice of that number determines how much signal reaches the lower layers. For binary classification with label $y \in \{0, 1\}$ and output $a = \sigma(z_2)$, the loss used is the binary cross-entropy

$$L(a, y) = -\big[y \ln a + (1 - y) \ln(1 - a)\big].$$

Cross-entropy is the negative logarithm of the confidence the model gives to the correct class, so a confident mistake is punished far more heavily than a hesitant one.

cross-entropy (*entropi silang*) — A measure of surprise: the more confident the model is in a wrong answer, the larger the penalty, and the penalty grows without bound.

Proposition 25.5 (Gradient of cross-entropy)**.** *For $a = \sigma(z)$ and L the binary cross-entropy we have $\dfrac{\partial L}{\partial z} = a - y$.*

Proof. Differentiate L with respect to a:

$$\frac{\partial L}{\partial a} = -\frac{y}{a} + \frac{1 - y}{1 - a} = \frac{a - y}{a(1 - a)}.$$

By the chain rule and the lemma on the derivative of the sigmoid, $\dfrac{\partial a}{\partial z} = a(1 - a)$, so

$$\frac{\partial L}{\partial z} = \frac{a - y}{a(1 - a)} \cdot a(1 - a) = a - y.$$

Numerator and denominator cancel, and that is the point. □

Compare with the squared loss $L = \frac{1}{2}(a - y)^2$, which gives $\partial L/\partial z = (a - y)\, a(1 - a)$. The factor $a(1 - a)$ does not cancel; it shrinks exactly when the model is most wrong with most confidence. For $z = 10$ and $y = 0$, $a(1 - a) = 4.5396 \times 10^{-5}$, so the signal to the weights is only 4.5394×10^{-5} even though the error is almost one. With cross-entropy the signal stays $a - y = 0.999955$. The ratio of the two numbers is about 2.2×10^4; this is why the sigmoid output is paired with cross-entropy, not with the square.

For K classes, the same pairing holds, and this time we prove it.

softmax (*softmax*) — Turns a list of scores into a division of probabilities that sums to one, by comparing every score with the sum of them all.

Proposition 25.6 (Softmax and cross-entropy)**.** *Suppose $\mathbf{p} = \mathrm{softmax}(\mathbf{z})$ with $p_k = e^{z_k} / \sum_j e^{z_j}$, and $L = -\sum_k y_k \ln p_k$ with $\sum_k y_k = 1$. Then $\dfrac{\partial L}{\partial z_i} = p_i - y_i$.*

Proof. The component derivative of softmax is $\dfrac{\partial p_k}{\partial z_i} = p_k(\delta_{ki} - p_i)$, with δ_{ki} the Kronecker delta. Hence

$$\frac{\partial L}{\partial z_i} = -\sum_k \frac{y_k}{p_k} \cdot p_k(\delta_{ki} - p_i) = -\sum_k y_k \delta_{ki} + p_i \sum_k y_k = -y_i + p_i,$$

because $\sum_k y_k = 1$. □

For two classes, $\mathrm{softmax}((t, 0))_1 = \sigma(t)$, so this formula returns to $a - y$ as before. Note also that adding a constant c to all z_k does not change $\mathbf{p}$, because the factor e^c cancels; only differences of scores are meaningful, which is also why adding a large constant to all logits damages floating-point accuracy (Chapter 22).

For younger readers (age 10 and up)

Have you ever played a number-guessing game where, every time you are wrong, you are only told "too big" or "too small", without being told by how much? If your guess is nine hundred and the answer is one hundred, the remark "too big" is exactly the same as if your guess were one hundred and one.

A neural network has the same problem, but more complicated. It has thirteen knobs, and every knob affects the answer through several intermediaries. If we only know that the answer is wrong, we do not know which knob to turn, in which direction, or by how much.

Fortunately, a wrong answer has a degree: how far off, and in which direction. With one measure called the loss, we can ask every knob: if you were turned a little, does the loss go up or down, and how fast? The answer is one number per knob.

In this chapter we will learn why a single loss measure is enough to give thirteen answers at once, and why those answers can be trusted.

25.5 Backpropagation

Level: advanced

A network is a composition of functions, so its derivative is computed with the chain rule of Chapter 12. What makes it cheap is the order of work: from the output backward to the input, not from the input forward to the output. For every intermediate quantity u in the computational graph, we assign one number stating how much the loss changes if u changes.

adjoint (*adjoint*) — The image of how much the final error is affected by one quantity in the middle of the network; one number per quantity.

Definition 25.3 (Adjoint)**.** Suppose C is the loss on one example and u one of the intermediate quantities. The adjoint of u is $\bar{u} = \partial C / \partial u$.

Proposition 25.7 (Adjoint recursion)**.** *If u affects C only through $v_1, \dots, v_m$, then* $\bar{u} = \sum_{i=1}^{m} \bar{v}_i \dfrac{\partial v_i}{\partial u}$.

Proof. This is the chain rule for functions of several variables (Chapter 12) applied to C as a function of $v_1, \dots, v_m$, with each v_i as a function of u. Since u does not affect C in any other way, all other terms in that chain rule are zero. □

That recursion turns the work into a single pass. Starting from $\bar{C} = 1$, we visit the intermediate quantities in the reverse of the forward order. Every quantity receives contributions from all the quantities that use it, so nothing is missed or counted twice. This is backpropagation, and it is not a new algorithm: it is only the chain rule run on the graph in the right order.

Figure 25.2 places both directions on one graph. Full arrows carry the forward values; dashed arrows carry the adjoints, and the numbers on those arrows are taken from the worked example in the next section.

Read it aloud (auditory learner)

Two voices. First voice: "The loss is too large." Second voice: "Who is at fault?" First voice: "Not one person. The fault is at the end, and the fault is divided backward, little by little."

Listen once more, slowly. The error is born at the end, in the last number. From there

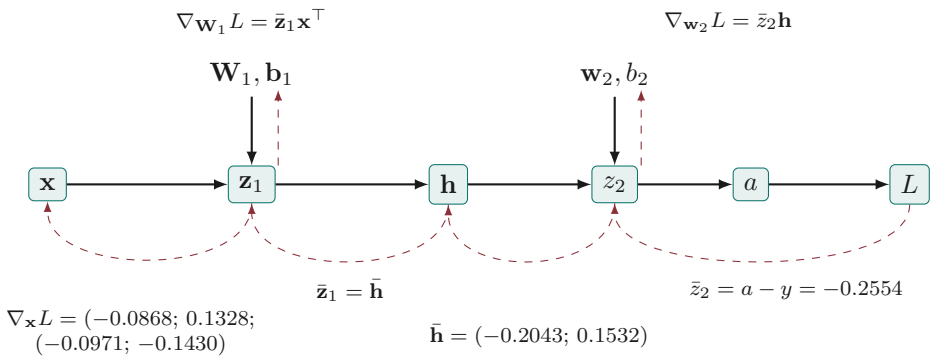


Figure 25.2: The computational graph of the 4–2–1 network on the worked example. Thick full arrows are the forward direction: input and weights become $\mathbf{z}_1$, the ReLU gate becomes $\mathbf{h}$, then the score z_2, the output a, and the loss L. Dashed arrows are the backward direction that carries the adjoints: $\bar{z}_2 = a - y = -0.2554$ enters the output weights, $\bar{\mathbf{h}} = (-0.2043;\ 0.1532)$ enters the gate, and $\nabla_{\mathbf{x}} L = (-0.0868;\ 0.1328;\ -0.0971;\ -0.1430)$ exits at the input. One backward pass produces all thirteen gradients at once.

it walks backward, stops at every junction, and asks: through here, how much is your share? The one closest to the end answers largest. The one far behind answers smallest, because the answer has already been filtered twice on the way home.
Second voice: "So the ones in front are not at fault?" First voice: "They are at fault to the extent of the share that reaches them. No more. That is why we do not accuse, we compute."
Repeat three times until it sounds right: fault at the end, divided backward, little by little, along the path.

Now derive it in full for the 4–2–1 network with ReLU in the hidden layer and sigmoid at the output. The forward computation is

$$\mathbf{z}_1 = \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1, \qquad \mathbf{h} = \mathrm{ReLU}(\mathbf{z}_1), \qquad z_2 = \mathbf{w}_2^\top \mathbf{h} + b_2, \qquad a = \sigma(z_2),$$

then L is the binary cross-entropy of a and y. Every link in the chain has its own local derivative, and we already have them all:

$$\frac{\partial L}{\partial z_2} = a - y \text{ (previous proposition)}, \qquad \frac{\partial z_2}{\partial h_i} = w_2[i],$$

$$\frac{\partial h_i}{\partial z_1[i]} = \mathbb{1}[z_1[i] > 0], \qquad \frac{\partial z_1[i]}{\partial W_1[i,j]} = x_j.$$

Multiplying backward along that chain gives, one by one,

$$\bar{z}_2 = a - y, \qquad \nabla_{\mathbf{w}_2} L = \bar{z}_2\, \mathbf{h}, \qquad \frac{\partial L}{\partial b_2} = \bar{z}_2,$$

then the signal that reaches the hidden layer

$$\bar{\mathbf{h}} = \bar{z}_2\, \mathbf{w}_2, \qquad \bar{\mathbf{z}}_1 = \bar{\mathbf{h}} \odot \mathbb{1}[\mathbf{z}_1 > 0],$$

with $\odot$ the componentwise product. Note that $\bar{\mathbf{z}}_1$ is $\bar{\mathbf{h}}$ shaved by the ReLU gate: components whose gate is closed become exactly zero, not small. After that

$$\nabla_{\mathbf{W}_1} L = \bar{\mathbf{z}}_1 \mathbf{x}^\top, \qquad \nabla_{\mathbf{b}_1} L = \bar{\mathbf{z}}_1, \qquad \nabla_{\mathbf{x}} L = \mathbf{W}_1^\top \bar{\mathbf{z}}_1.$$

The outer-product form of $\nabla_{\mathbf{W}_1} L$ is no accident: the derivative $\partial z_1[i]/\partial W_1[i,j] = x_j$ does not depend on i, so the whole i-th row is $\bar{z}_1[i]$ times the vector $\mathbf{x}$. Every weight receives a gradient whose size is proportional to the input it multiplies, and that is why large-scale features automatically receive large steps.

None of those formulas depends on $d = 4$ or on two hidden units; the pattern is the same for any layer.

The cost is comparable: one forward step needs one matrix-vector product per layer, one backward step one transposed product and one outer product per layer. A centered finite difference for P parameters needs $2P$ forward steps, whereas backpropagation needs about two. For $P = 13$ in this chapter, twenty-six against two; for a million parameters, two million against two, and that is the whole difference. Nielsen names this observation as the reason backpropagation is regarded as a great discovery, not merely an algebraic trick.

Do it by hand (kinesthetic learner)

Take three sheets of paper and a pencil, and label them: INPUT, WEIGHTS, OUTPUT. On the INPUT sheet write four numbers: 1; 0.5; −1; 2. On the WEIGHTS sheet write the two weights to the output: 0.8 and −0.6. On the OUTPUT sheet write 0.7446, the probability the network gives, and remember that the true label is 1.
Lay the three sheets in a row from left to right: INPUT, WEIGHTS, OUTPUT. Now work backward, always from right to left. On the OUTPUT sheet, subtract 0.7446 from 1 and write −0.2554. Move to the WEIGHTS sheet: multiply −0.2554 by 0.8, write −0.2043; multiply by −0.6, write 0.1532. Move to the INPUT sheet: multiply −0.2043 by each number on that sheet, then write the row: −0.2043; −0.1022; 0.2043; −0.4086. What you must see: every time a sheet moves one step to the left, the sign of the number can flip, and its size follows the contents of the sheet you are holding. If you reverse the direction, from INPUT to OUTPUT, the order of work is wrong and the result means nothing.

25.6 Initialization and gradient descent

Level: intermediate

Once the gradient exists, the next step is simple: move a little in the direction that lowers the loss. For all parameters $\boldsymbol{\theta}$,

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \, \nabla_{\boldsymbol{\theta}} L,$$

with η the learning rate. For B examples and the average loss $L = \frac{1}{B}\sum_i L_i$, the gradient is also the average of the per-example gradients; the backpropagation formula does not change, only how many backward passes. The conditions and bounds on η are in Chapter 23; because the network is nonconvex, the guarantees there are only a benchmark.

stochastic gradient descent (*penurunan gradien stokastik*) — Walking down a slope using a slope estimate from a small part of the data, not from all of the data.

Three variants deserve mention. SGD takes a small B and updates the parameters every time; the gradient is noisy, and the spread of that noise shrinks like $1/\sqrt{B}$ because the variance of the average of B independent variables is σ^2/B. Momentum keeps a velocity,

$$\mathbf{v} \leftarrow \beta \mathbf{v} + \mathbf{g}, \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \mathbf{v},$$

so that consistent directions accumulate and oscillating directions cancel. Adam normalizes every coordinate with an estimate of its second moment,

$$\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1-\beta_1)\mathbf{g}, \qquad \mathbf{v} \leftarrow \beta_2 \mathbf{v} + (1-\beta_2)\mathbf{g}^2,$$

$$\hat{\mathbf{m}} = \frac{\mathbf{m}}{1-\beta_1^t}, \qquad \hat{\mathbf{v}} = \frac{\mathbf{v}}{1-\beta_2^t}, \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\frac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}}}+\epsilon},$$

with t the step number. The factors $1-\beta_1^t$ and $1-\beta_2^t$ are called bias correction, and their purpose is to remove the tendency of the moments toward zero in the early steps. On the first step, $\hat{m}_1 = g$ and $\hat{v}_1 = g^2$, so the update is exactly $-\eta\, g/(|g|+\epsilon)$: its size is almost exactly η, however large the gradient. That property makes Adam scale-insensitive, and at the same time explains its weakness: because its step does not shrink when the gradient shrinks, Adam is not guaranteed to converge on all convex problems, and the counterexample can be read in Reddi, Kale, and Kumar (2018).

What remains, and what is most often underestimated, is the starting point.

Proposition 25.8 (Symmetry that never breaks)**.** *If two rows of $\mathbf{W}_1$ are identical and $b_1[1] = b_1[2]$ at the start of training, then both stay identical after every gradient update, for every data set and every learning rate.*

Proof. We prove it by induction on the update step. Suppose the first and second rows of $\mathbf{W}_1$ are the same and $b_1[1] = b_1[2]$. Then for every input $z_1[1] = z_1[2]$, so both ReLU gates make the same decision and $\bar{z}_1[1] = \bar{z}_1[2]$. Because $\nabla_{\mathbf{W}_1} L = \bar{\mathbf{z}}_1 \mathbf{x}^\top$, those two rows of the gradient are identical, and so are the two components of $\nabla_{\mathbf{b}_1} L$. An update with the same rate preserves the equality. For a set of examples, the total gradient is the sum of the per-example gradients, so the equality survives that too. □

The consequence is blunt: eight units initialized uniformly behave like one unit, however long it is trained.

Zero initialization is worse still. If all of $\mathbf{W}_1$ and $\mathbf{b}_1$ are zero, then $\mathbf{z}_1 = \mathbf{0}$, and because the gate is $\mathbb{1}[z > 0]$, all gates are closed: the gradient with respect to $\mathbf{W}_1$ is exactly zero, and the weights stay zero forever. The only thing that can still learn is b_2, and the result is a constant prediction equal to the class proportion. On the lab data, the final loss is 0.690923, exactly the label entropy. Eight units, zero benefit.

Challenge (problem solver)

Design a 2–2–1 network with ReLU in the hidden layer whose gradient is exactly zero on all six weights and both biases of the first layer from the first step, while $\mathbf{W}_1$ is not zero, and the network is still wrong on at least one point. Only b_2 may move. Show why your choice works.

Write your answer as a table: unit, $z_1[i]$ for each example, its gate state, and the gradient that comes in.

Hint 1: what freezes the weights is not the value of the weights, but the ReLU gate closing for all examples.

Hint 2: use data in the nonnegative quadrant, then take $\mathbf{W}_1 = -\mathbf{1}\mathbf{1}^\top$ and $\mathbf{b}_1 = \mathbf{0}$. Then $z_1[i] = -(x_1 + x_2) \le 0$, so $\mathbf{h} = \mathbf{0}$, $\bar{\mathbf{z}}_1 = \mathbf{0}$, and $\nabla_{\mathbf{W}_1} L = \mathbf{0}$ even though $\mathbf{W}_1 \neq \mathbf{0}$.

If you get stuck, go back to the section Initialization and gradient descent.

Random initialization is therefore not decoration, but a condition for those units to explore different functions. The next question: how large. The answer comes from computing variances. If the components of $\mathbf{x}$ are independent with mean zero and variance σ_x^2, and the weights are independent with variance σ_w^2, then

$$\operatorname{Var}(z_1[i]) = \sum_{j=1}^{n_{\text{in}}} \operatorname{Var}(W_1[i,j]x_j) = n_{\text{in}}\,\sigma_w^2\sigma_x^2.$$

In order for that variance not to depend on n_{in}, we need $\sigma_w^2 \propto 1/n_{\text{in}}$. The He initialization uses $\sigma_w^2 = 2/n_{\text{in}}$, and the number 2 is not decoration: for symmetric z, ReLU cuts away half

the distribution, so $\mathbb{E}[h^2] = \mathrm{Var}(z)/2 = \sigma_x^2$, and that variance survives layer after layer. The Xavier initialization uses $\sigma_w^2 = 2/(n_{\text{in}} + n_{\text{out}})$, a compromise between preserving the forward signal and the backward signal. That compromise is exact only when $n_{\text{in}} = n_{\text{out}}$.

For sigmoid layers, an initialization that is too large punishes immediately: on a two-layer network with standard deviation 1.0, the initial loss reaches 2.223899, whereas with standard deviation 0.5 it is only 1.220409. The reverse is more serious: with standard deviation 0.5 and eight layers, training stops at accuracy 0.5333. Initialization is not an administrative detail; it determines whether the signal is still there when it reaches the first layer.

25.7 Worked example

Level: intermediate

Step-by-step recipe (procedural learner)

1. Start from the end: $\bar{z}_2 = a - y$ for sigmoid with cross-entropy, or $\mathbf{p} - \mathbf{y}$ for softmax. Without this number there is nothing to divide.
2. Output weights: $\nabla_{\mathbf{w}_2} L = \bar{z}_2 \mathbf{h}$ and $\partial L / \partial b_2 = \bar{z}_2$.
3. Signal to the previous layer: $\bar{\mathbf{h}} = \bar{z}_2 \mathbf{w}_2$, then shave with the gate $\bar{\mathbf{z}}_1 = \bar{\mathbf{h}} \odot \mathbb{1}[\mathbf{z}_1 > 0]$.
4. First-layer weights: $\nabla_{\mathbf{W}_1} L = \bar{\mathbf{z}}_1 \mathbf{x}^\top$ and $\nabla_{\mathbf{b}_1} L = \bar{\mathbf{z}}_1$; if needed, the input gradient $\nabla_{\mathbf{x}} L = \mathbf{W}_1^\top \bar{\mathbf{z}}_1$.

A warning about order: step 3 must not come before step 1. $\bar{\mathbf{z}}_1$ is computed from $\bar{z}_2$, and the gate uses $\mathbf{z}_1$ from the forward computation that has already been stored. Swapping the two steps kills the entire gradient of the first-layer weights.

We use one example and compute everything with exact numbers. The input is $\mathbf{x} = (1.0;\ 0.5;\ -1.0;\ 2.0)$, the label is $y = 1$, and the initial parameters are

$$\mathbf{W}_1 = \begin{pmatrix} 0.5 & -0.5 & 0.25 & 1.0 \\ 0.1 & 0.2 & -0.3 & 0.4 \end{pmatrix}, \quad \mathbf{b}_1 = \begin{pmatrix} 0.1 \\ -0.2 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 0.8 \\ -0.6 \end{pmatrix}, \quad b_2 = 0.05.$$

That is all the parameters.

Forward step. The first unit receives $0.5(1) - 0.5(0.5) + 0.25(-1) + 1(2) + 0.1 = 0.5 - 0.25 - 0.25 + 2 + 0.1 = 2.1$. The second unit receives $0.1(1) + 0.2(0.5) - 0.3(-1) + 0.4(2) - 0.2 = 0.1 + 0.1 + 0.3 + 0.8 - 0.2 = 1.1$. Both are positive, so $\mathbf{h} = (2.1;\ 1.1)$ without clipping. The output score is $z_2 = 0.8(2.1) - 0.6(1.1) + 0.05 = 1.68 - 0.66 + 0.05 = 1.07$, so $a = \sigma(1.07) = 0.7445969158$ and the loss is $L = -\ln a = 0.2949122596$.

Backward step. The output adjoint is $\bar{z}_2 = a - y = -0.2554030842$. From there

$$\nabla_{\mathbf{w}_2} L = \bar{z}_2 \mathbf{h} = (-0.5363464768;\ -0.2809433926), \qquad \frac{\partial L}{\partial b_2} = -0.2554030842.$$

The signal to the hidden layer is $\bar{\mathbf{h}} = \bar{z}_2 \mathbf{w}_2 = (-0.2043224673;\ 0.1532418505)$; both gates are open, so $\bar{\mathbf{z}}_1 = \bar{\mathbf{h}}$. The outer product gives

$$\nabla_{\mathbf{W}_1} L = \begin{pmatrix} -0.2043224673 & -0.1021612337 & 0.2043224673 & -0.4086449347 \\ 0.1532418505 & 0.0766209253 & -0.1532418505 & 0.3064837010 \end{pmatrix},$$

and $\nabla_{\mathbf{b}_1} L = (-0.2043224673;\ 0.1532418505)$. Two quick checks: the second row is exactly -0.75 times the first, because $-0.6/0.8 = -0.75$; and every column is proportional to x_j, so the fourth column is twice the first, the second is half of it, and the third is its negative. The Euclidean norm of the whole gradient is 0.9796964023.

One gradient-descent step with $\eta = 0.5$ updates every parameter according to $\theta \leftarrow \theta - \eta\, \partial L/\partial\theta$, giving

$$\mathbf{W}_1' = \begin{pmatrix} 0.6021612337 & -0.4489193832 & 0.1478387663 & 1.2043224673 \\ 0.0233790747 & 0.1616895374 & -0.2233790747 & 0.2467581495 \end{pmatrix},$$

$$\mathbf{b}_1' = (0.2021612337;\, -0.2766209253), \qquad \mathbf{w}_2' = (1.0681732384;\, -0.4595283037),$$

$$b_2' = 0.1777015421.$$

After that update $\mathbf{z}_1' = (2.8406689;\, 0.5444983)$ and $a' = 0.9508189702$, so the loss falls to 0.0504315919; the difference is -0.2444806677, or about 83% of the initial loss.

The number 83% invites a misreading. What is measured is a single example, with no test data, on a landscape that is not convex. The valid conclusion is not "the rate 0.5 is good", but "this gradient direction lowers the loss on this example".

Check one row of the lab table by hand. Take $W_1[2,1]$ and compute $\bar{z}_1[2]\, x_1$ to get 0.1532418505. Then compute the centered finite difference for that parameter yourself: run one forward step with $W_1[2,1] + 10^{-6}$ and one forward step with $W_1[2,1] - 10^{-6}$, divide the difference of the losses by 2×10^{-6}, and compare with the analytic number above. After that, explain why the result is not exactly the same, and why the sign of the difference can change if h is made smaller, 10^{-12}.

25.8 Computational lab

Level: advanced

The three blocks are run with Python 3.14.4 and NumPy 2.5.2, without machine-learning libraries; the output is copied as it stands.

The first block checks the hand computation above, then compares every analytic gradient with a centered finite difference at $h = 10^{-6}$.

Computational lab

```
import numpy as np
np.set_printoptions(precision=10, suppress=True, linewidth=76)

# 4-2-1 network: ReLU hidden, sigmoid output
x  = np.array([1.0, 0.5, -1.0, 2.0])
y  = 1.0
W1 = np.array([[0.5, -0.5, 0.25, 1.0],
               [0.1,  0.2, -0.3,  0.4]])
b1 = np.array([0.1, -0.2])
w2 = np.array([0.8, -0.6])
b2 = 0.05

def forward(x, W1, b1, w2, b2):
    z1 = W1 @ x + b1
    h  = np.maximum(z1, 0.0)
    o  = w2 @ h + b2
    p  = 1.0/(1.0+np.exp(-o))
    return z1, h, o, p, -(y*np.log(p) + (1-y)*np.log(1-p))

def backward(x, W1, b1, w2, b2):
    z1, h, o, p, L = forward(x, W1, b1, w2, b2)
    d   = p - y                    # dL/dz2 for sigmoid + cross-entropy
    dz1 = d*w2*(z1 > 0)            # ReLU gate
    return dict(W1=np.outer(dz1, x), b1=dz1, w2=d*h, b2=d,
                x=W1.T @ dz1), L, p, z1, h, dz1
```

```
g, L, p, z1, h, dz1 = backward(x, W1, b1, w2, b2)
print("z1 =", z1, "  h =", h)
print("p = %.12f  L = %.12f" % (p, L))
print("dz1 =", dz1)
print("g_w2 =", g["w2"], " g_b2 = %.12f" % g["b2"])
for i in range(2):
    print("g_W1[%d] =" % i, np.array2string(g["W1"][i], precision=10,
                                            suppress_small=False))
print("g_x =", g["x"])

# centered finite-difference check, one parameter at a time
eps = 1e-6
def loss_at(nama, idx, delta):
    W, B, V, c = W1.copy(), b1.copy(), w2.copy(), b2
    if nama == "W1":  W[idx] += delta
    if nama == "b1":  B[idx] += delta
    if nama == "w2":  V[idx] += delta
    if nama == "b2":  c += delta
    return forward(x, W, B, V, c)[4]

daftar = [("W1", i, j) for i in range(2) for j in range(4)]
daftar += [("b1", i) for i in range(2)] + [("w2", i) for i in range(2)]
print("\nparameter     analytic        numeric     relative error")
for nama, *idx in daftar + [("b2",)]:
    idx = tuple(idx)
    ana = {"W1": g["W1"], "b1": g["b1"], "w2": g["w2"], "b2": np.array(g["b2"])}
    a = ana[nama][idx] if idx else ana[nama]
    n = (loss_at(nama, idx, eps) - loss_at(nama, idx, -eps))/(2*eps)
    rel = abs(a-n)/max(abs(a), abs(n), 1e-300)
    print("%-10s %15.10f %15.10f %12.2e" %
          (nama + "".join("[%d]" % i for i in idx), a, n, rel))

# one gradient-descent step with rate 0.5
eta = 0.5
W1n, b1n = W1 - eta*g["W1"], b1 - eta*g["b1"]
w2n, b2n = w2 - eta*g["w2"], b2 - eta*g["b2"]
z1b, hb, ob, pb, Lb = forward(x, W1n, b1n, w2n, b2n)
print("\none step: p = %.12f  L = %.12f  difference = %.12f"
      % (pb, Lb, L - Lb))
for i in range(2):
    print("W1'[%d] =" % i, np.array2string(W1n[i], precision=10))
print("w2' =", w2n, " b2' = %.12f" % b2n)

# gradient with respect to the input: analytic vs finite difference
E = np.eye(4)
gx = np.array([(forward(x + eps*E[j], W1, b1, w2, b2)[4]
              - forward(x - eps*E[j], W1, b1, w2, b2)[4])/(2*eps)
               for j in range(4)])
print("\ng_x analytic    =", g["x"])
print("g_x finite diff =", gx)
print("relative error   =", ["%.1e" % v for v in np.abs(gx-g["x"])/np.abs(g["x"])])
```

The actual output:

```
z1 = [2.1 1.1]   h = [2.1 1.1]
p = 0.744596915827  L = 0.294912259605
dz1 = [-0.2043224673  0.1532418505]
g_w2 = [-0.5363464768 -0.2809433926]  g_b2 = -0.255403084173
g_W1[0] = [-0.2043224673 -0.1021612337  0.2043224673 -0.4086449347]
g_W1[1] = [ 0.1532418505  0.0766209253 -0.1532418505  0.306483701 ]
g_x = [-0.0868370486  0.1328096038 -0.097053172  -0.1430257271]

parameter     analytic        numeric     relative error
```

```
W1[0][0]     -0.2043224673   -0.2043224674     1.17e-10
W1[0][1]     -0.1021612337   -0.1021612337     2.53e-10
W1[0][2]      0.2043224673    0.2043224674     1.17e-10
W1[0][3]     -0.4086449347   -0.4086449347     4.95e-11
W1[1][0]      0.1532418505    0.1532418505     7.22e-11
W1[1][1]      0.0766209253    0.0766209252     8.33e-10
W1[1][2]     -0.1532418505   -0.1532418505     7.22e-11
W1[1][3]      0.3064837010    0.3064837010     1.09e-10
b1[0]        -0.2043224673   -0.2043224674     1.17e-10
b1[1]         0.1532418505    0.1532418506     6.16e-10
w2[0]        -0.5363464768   -0.5363464768     9.80e-11
w2[1]        -0.2809433926   -0.2809433925     2.41e-10
b2           -0.2554030842   -0.2554030842     3.59e-11

one step: p = 0.950818970185  L = 0.050431591881  difference = 0.244480667724
W1'[0] = [ 0.6021612337 -0.4489193832  0.1478387663  1.2043224673]
W1'[1] = [ 0.0233790747  0.1616895374 -0.2233790747  0.2467581495]
w2' = [ 1.0681732384 -0.4595283037]  b2' = 0.177701542086

g_x analytic    = [-0.0868370486  0.1328096038 -0.097053172  -0.1430257271]
g_x finite diff = [-0.0868370486  0.1328096037 -0.097053172  -0.1430257272]
relative error   = ['7.7e-10', '5.0e-10', '1.4e-10', '9.8e-11']
```

That table is the proof that was asked for. All thirteen parameters agree to nine or ten digits, with the largest relative error 8.33×10^{-10} at $W_1[1,2]$. That number should not be zero: the centered finite difference carries a truncation error of order h^2 and a rounding error of order $\varepsilon_{\text{mach}}/h$ (Chapter 22), so there is a best h that cannot make the error zero. For $h = 10^{-6}$ an order of 10^{-10} is exactly what is expected.

The gradient with respect to the input agrees to 7.7×10^{-10} on the first component. That number means: raising the second feature slightly raises the loss fastest, whereas raising the fourth feature lowers it fastest. In Chapter 29 this direction is used in reverse, to raise the loss as fast as possible on inputs the attacker is allowed to change.

The second block trains a 2–8–1 network from scratch on two-dimensional data with an XOR pattern, then compares it with a linear baseline and with two defective initializations.

Computational lab

```
import numpy as np

rng = np.random.default_rng(7)
n = 240
X = rng.uniform(-1.0, 1.0, size=(n, 2))
y = (X[:, 0]*X[:, 1] > 0).astype(float)        # label 1 when the signs agree
print("examples =", n, " class 1 =", int(y.sum()))

def sig(o):
    return 1.0/(1.0+np.exp(-o))

def bce(p, t):
    p = np.clip(p, 1e-12, 1-1e-12)
    return float(-np.mean(t*np.log(p) + (1-t)*np.log(1-p)))

def init(mode, rs, nh=8):
    if mode == "nol":
        return np.zeros((nh, 2)), np.zeros(nh), np.zeros(nh), 0.0
    if mode == "seragam":
        return np.full((nh, 2), 0.5), np.zeros(nh), np.full(nh, 0.5), 0.0
    return (rs.normal(0, np.sqrt(2/2), (nh, 2)), np.zeros(nh),
            rs.normal(0, np.sqrt(2/nh), nh), 0.0)          # He
```

```
def fwd(X, W1, b1, w2, b2):
    Z1 = X @ W1.T + b1
    H  = np.maximum(Z1, 0.0)
    return Z1, H, sig(H @ w2 + b2)

def grads(X, t, W1, b1, w2, b2):
    Z1, H, P = fwd(X, W1, b1, w2, b2)
    d  = (P - t)/len(t)
    dZ = np.outer(d, w2)*(Z1 > 0)
    return dZ.T @ X, dZ.sum(0), H.T @ d, d.sum()

def latih(mode, langkah=3000, lr=0.5, seed=0):
    W1, b1, w2, b2 = init(mode, np.random.default_rng(seed))
    jejak = []
    for t in range(langkah + 1):
        gW1, gb1, gw2, gb2 = grads(X, y, W1, b1, w2, b2)
        W1, b1 = W1 - lr*gW1, b1 - lr*gb1
        w2, b2 = w2 - lr*gw2, b2 - lr*gb2
        if t % 500 == 0:
            jejak.append((t, bce(fwd(X, W1, b1, w2, b2)[2], y)))
    Z1, H, P = fwd(X, W1, b1, w2, b2)
    mati = int(np.sum(np.all(H <= 0, axis=0)))
    return jejak, float(np.mean((P > 0.5) == (y > 0.5))), bce(P, y), mati

jejak, akur, L, mati = latih("he")
for t, v in jejak:
    print("step %4d   L = %.6f" % (t, v))
print("He      : accuracy = %.4f  dead units = %d/8" % (akur, mati))
for mode in ("nol", "seragam"):
    j, a, l, m = latih(mode)
    print("%-8s: L final = %.6f  accuracy = %.4f  dead units = %d/8"
          % (mode, l, a, m))

w, b = np.zeros(2), 0.0             # garis dasar linear, tanpa lapisan
for t in range(4000):
    p = sig(X @ w + b)
    w -= 0.5*(X.T @ (p - y))/n
    b -= 0.5*np.mean(p - y)
pl = sig(X @ w + b)
print("linear  : L = %.6f  accuracy = %.4f"
      % (bce(pl, y), float(np.mean((pl > 0.5) == (y > 0.5)))))
```

The actual output:

```
examples = 240  class 1 = 112
step    0   L = 0.648815
step  500   L = 0.060245
step 1000   L = 0.036445
step 1500   L = 0.027164
step 2000   L = 0.022039
step 2500   L = 0.018717
step 3000   L = 0.016349
He      : accuracy = 1.0000  dead units = 0/8
nol     : L final = 0.690923  accuracy = 0.5333  dead units = 8/8
seragam : L final = 0.539475  accuracy = 0.7167  dead units = 0/8
linear  : L = 0.686130  accuracy = 0.7208
```

Its loss curve decreases at every sampling, from 0.648815 to 0.016349. Training accuracy reaches 1.0000 with eight units and zero dead units. The linear baseline, which uses the same optimization and the same data but no hidden layer, stops at 0.7208. That difference of 0.28 is the price of a single separating plane, and it agrees with the XOR proposition: this pattern is not linearly separable.

Figure 25.3 plots those seven points.

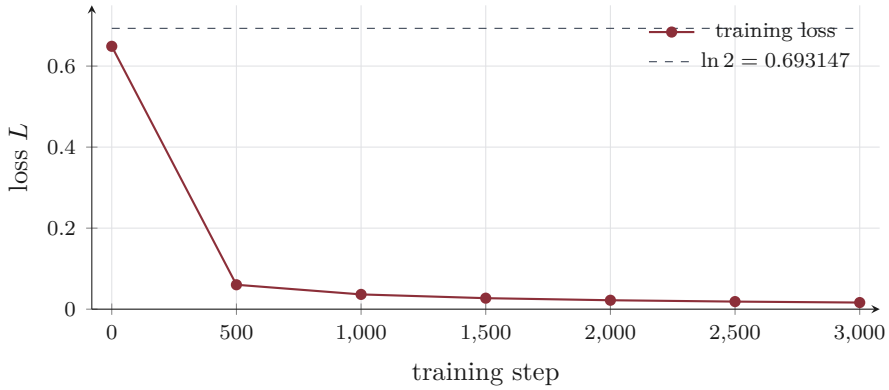


Figure 25.3: The training loss curve of the 2–8–1 network in the second block, from 0.648815 at step 0 to 0.016349 at step 3000. The dashed line is $\ln 2 = 0.693147$, the loss of a uniform guess; the first point is already below it because it is measured after the first update.

Zero initialization teaches a harsher lesson. Its final loss is 0.690923, equal to the label entropy to six decimals, and its accuracy is 0.5333, namely the proportion of the majority class; all eight units are dead. Uniform initialization gives 0.7167, practically the same as the linear baseline.

The third block measures two things: how many layers a sigmoid network can still learn through, and how fast the signal shrinks from the output to the first layer.

Computational lab

```
import numpy as np

rng = np.random.default_rng(7)
n = 240
X = rng.uniform(-1.0, 1.0, size=(n, 2))
y = (X[:, 0]*X[:, 1] > 0).astype(float)

def sig(o):
    return 1.0/(1.0+np.exp(-o))

def bce(p, t):
    p = np.clip(np.ravel(p), 1e-12, 1-1e-12)
    t = np.ravel(t)
    return float(-np.mean(t*np.log(p) + (1-t)*np.log(1-p)))

def bangun(dalam, std, seed=0):
    rs = np.random.default_rng(seed)
    ukuran = [2] + [16]*(dalam-1) + [1]
    W = [rs.normal(0, std, (ukuran[i+1], ukuran[i])) for i in range(dalam)]
    return W, [np.zeros(ukuran[i+1]) for i in range(dalam)]

def maju(W, b, X):
    A = [X]
    for Wl, bl in zip(W[:-1], b[:-1]):
        A.append(sig(A[-1] @ Wl.T + bl))
    return A, sig(A[-1] @ W[-1].T + b[-1])

def mundur(W, b, X, t):
    A, P = maju(W, b, X)
    d = (P - t[:, None])/len(t)
    gW, gb = [None]*len(W), [None]*len(b)
    gW[-1], gb[-1] = d.T @ A[-1], d.sum(0)
```

```
    delta = d @ W[-1]
    for l in range(len(W)-1, 0, -1):
        delta = delta*A[l]*(1 - A[l])
        gW[l-1], gb[l-1] = delta.T @ A[l-1], delta.sum(0)
        if l > 1:
            delta = delta @ W[l-1]
    return gW, gb, bce(P, t), P

print("depth  std   L(start)   L(final)  accuracy  ||gW1||/||gWlast||")
for dalam in (1, 2, 4, 8):
    for std in (0.5, 1.0):
        W, b = bangun(dalam, std)
        gW, gb, L0, P = mundur(W, b, X, y)
        rasio = np.linalg.norm(gW[0])/np.linalg.norm(gW[-1])
        for t in range(2000):
            gW, gb, L, P = mundur(W, b, X, y)
            for l in range(dalam):
                W[l] -= 0.5*gW[l]
                b[l] -= 0.5*gb[l]
        akur = float(np.mean((np.ravel(P) > 0.5) == (y > 0.5)))
        print("%5d %5.1f %9.6f %9.6f %8.4f %14.2e"
              % (dalam, std, L0, bce(P, y), akur, rasio))
```

The actual output:

```
depth  std   L(start)   L(final)  accuracy  ||gW1||/||gWlast||
    1   0.5  0.693190  0.686130   0.7208       1.00e+00
    1   1.0  0.693951  0.686130   0.7208       1.00e+00
    2   0.5  1.220409  0.332913   0.8417       1.66e-02
    2   1.0  2.223899  0.299519   0.8542       2.16e-02
    4   0.5  0.691597  0.147887   0.9458       5.77e-02
    4   1.0  0.701977  0.023491   0.9958       2.05e-01
    8   0.5  0.739920  0.690867   0.5333       1.26e-04
    8   1.0  0.875314  0.002383   1.0000       3.76e-03
```

The first three rows of that table are a promise and a warning at once. Depth one gives 0.7208, the same as the linear baseline in the previous block; depth two gives 0.8417 and depth four 0.9458. But the last row with standard deviation 0.5 gives 0.5333 with loss 0.690867: eight sigmoid layers with a small initialization practically do not learn. The gradient-norm ratio in the last column explains why: 1.26×10^{-4}, four orders of magnitude smaller than at depth four. The same row with standard deviation 1.0 succeeds perfectly. So the failure is not about depth alone, but about the initial scale.

Two limits of these lab results must be stated. The data is only 240 self-made two-dimensional examples, with no test set, so all of it is training accuracy; generalization is the business of Chapter 26 and Chapter 27. All experiments use a single seed, so variation between seeds is invisible. What may be concluded is only what was checked directly: backpropagation agrees with finite differences, the loss decreases, zero and uniform initialization fail in ways that can be explained, and a deep network with a small initialization loses its signal.

Note for instructors

The order that has worked: start from one separating line and four XOR points, then show the 2–2–1 construction on the board, and only after that the general definition. The proof that XOR is impossible is usually accepted easily; what stops students is the backward step, so set aside the most time for backpropagation, not for activation functions.

Two typical misconceptions. First, “backpropagation is another algorithm, not the chain rule”, so that students memorize formulas without being able to derive them for another

architecture; the cure is to ask them to re-derive the 4–2–1 network with three hidden units on the board. Second, "a relative error of 10^{-10} means the implementation is proved correct"; a small error only checks agreement at one point, for one set of parameters, and does not check the sign, the scaling, or the case of a closed gate.
The most helpful board activity: give a network with one hidden unit whose $z_1 < 0$, ask the class to compute the gradient of the whole row $\mathbf{W}_1$, then ask what happens to that unit after a hundred steps. The answer "nothing happens, ever" usually opens the discussion about unit death and about why a large learning rate is dangerous.
Sign of readiness to go on: students can write the backpropagation for a network with two hidden layers without looking at their notes, and can explain why zero initialization fails not because of "too little data" but because of symmetry.

25.9 Summary

Summary.

- A linear threshold can only separate sets that are linearly separable, and the four XOR points are not one of them. The proof does not depend on the algorithm, so adding data does not help. One hidden layer with ReLU is already enough for that pattern.
- Without a nonlinear activation, a layered network collapses into a single matrix. Sigmoid and tanh shrink the signal because their derivatives are at most $\frac{1}{4}$ and 1; through L sigmoid layers the remaining signal is at most 4^{-L}, namely 9.54×10^{-7} for $L = 10$. ReLU does not shrink, but it kills units whose input is always negative.
- Cross-entropy is paired with sigmoid or softmax so that the output adjoint is exactly $a - y$ or $\mathbf{p} - \mathbf{y}$. With the squared loss, the factor $a(1 - a)$ is left behind and the signal shrinks exactly when the model is most confidently wrong; at $z = 10$ the ratio is about 2.2×10^4.
- Backpropagation is the chain rule of Chapter 12 run on the computational graph in reverse order. For 4–2–1, all the gradients are: $\bar{z}_2 = a - y$, $\nabla_{\mathbf{w}_2} L = \bar{z}_2 \mathbf{h}$, $\bar{\mathbf{z}}_1 = \bar{z}_2 \mathbf{w}_2 \odot \mathbb{1}[\mathbf{z}_1 > 0]$, $\nabla_{\mathbf{W}_1} L = \bar{\mathbf{z}}_1 \mathbf{x}^\top$, $\nabla_{\mathbf{x}} L = \mathbf{W}_1^\top \bar{\mathbf{z}}_1$. Its cost is about two forward steps, not $2P$. On the 4–2–1 example, those thirteen gradients agree with centered finite differences to a relative error of 8.33×10^{-10}.
- Initialization determines whether the network can learn. Zero closes all gates and leaves a prediction equal to the class proportion; uniform preserves the symmetry so that the effective capacity is one unit; too small shrinks the signal before the first layer; too large saturates the sigmoid. He uses $\sigma_w^2 = 2/n_{\text{in}}$, Xavier $2/(n_{\text{in}} + n_{\text{out}})$.
- Gradient descent, SGD, momentum, and Adam differ only in how they use the same gradient. Adam's first step has a size of almost exactly η however large the gradient, and that property explains both its robustness and its failure on some problems.

25.10 Exercises

Exercise 25.1.

1. For data with margin $\gamma = 0.2$ and $\|\mathbf{x}_i\| \le 1$, what is the upper bound on the number of perceptron updates according to the convergence theorem, and what is it for $\gamma = 0.05$? Explain why that bound does not depend on the number of examples and why it is useless when γ is small.
2. Prove that a two-layer network with identity activation in the hidden layer is no stronger than a linear threshold, that is, that $\mathbf{W}_2 \,\mathrm{id}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$ can be written as a single linear threshold. Explain the consequence for further identity layers.

3. Compute $\nabla_{\mathbf{W}_1} L$ for the network in the worked example if $\mathbf{x} = (0.5; -1.0; 0.25; 2.0)$ and $y = 0$, with the same parameters. Determine which unit's gate is closed, and why that row has zero gradient without meaning that the loss is minimal.
4. Assess this claim: *because the largest relative error is only* 8.33×10^{-10}*, the backpropagation implementation in this chapter is proved correct.* Decide true, false, or not enough evidence, and name the additional checks required.
5. Assess this claim: *the universal approximation theorem guarantees that a network with one hidden layer can reach an arbitrarily small error on test data.* Decide true, false, or not enough evidence, and name two things the theorem does not state.
6. With $\bar{z}_2 = -0.2554030842$, compute one Adam step for $b_2 = 0.05$ with $\eta = 0.01$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, at $t = 1$. Show that the step size is practically η, then compute the same step if the gradient is 100 times larger.
7. Show that for two ReLU units in $\mathbb{R}^2$ the output is a piecewise linear function, and that the input plane is divided into at most four regions with a linear formula each. How many regions for k ReLU units in $\mathbb{R}^2$ in general position? (Hint: each unit gives one line $z_1[i] = 0$, and inside each cell all gates stay fixed.)
8. Compute the gradient with respect to the input for the worked example, then determine the perturbation $\boldsymbol{\delta}$ with $\|\boldsymbol{\delta}\|_\infty \leq 0.05$ that raises the loss most according to the linearization, namely $\boldsymbol{\delta} = 0.05\,\mathrm{sign}(\nabla_{\mathbf{x}} L)$. Compute the loss at $\mathbf{x} + \boldsymbol{\delta}$ and compare it with the linear prediction $L + \nabla_{\mathbf{x}} L^\top \boldsymbol{\delta}$.
9. Without running a program, predict what happens if all of $\mathbf{b}_1$ is initialized to $+10$ while $\mathbf{W}_1$ stays small and random, using the saturation table; then if $\mathbf{b}_1 = -10$. Check your prediction by running the second lab block after changing one line.

Hints for the three hardest items. Item 2: write $\mathbf{W} = \mathbf{W}_2\mathbf{W}_1$ and $\mathbf{b} = \mathbf{W}_2\mathbf{b}_1 + \mathbf{b}_2$, then compare with the definition of a linear threshold. Item 7: compute the number of regions for $k = 1, 2, 3$ (they are $2, 4, 7$), then look for the pattern $1 + k + k(k-1)/2$. Item 9: the second lab block uses ReLU, so $\mathbf{b}_1 = +10$ saturates nothing; what saturates is $\mathbf{b}_1 = -10$, when every unit is dead and its derivative zero. In the third block, which uses the sigmoid, $z \approx 10$ makes the derivative of the hidden layer about 4.5×10^{-5}, so the gradient to $\mathbf{W}_1$ shrinks proportionally, while $\mathbf{b}_1$ still receives the full gradient.

For researchers: limits and next steps

What is really proved in this chapter: the impossibility of linear separation for XOR; the bound $\lfloor 1/\gamma^2 \rfloor$ on the perceptron, with a complete proof; the derivative of the sigmoid and the bound $\frac{1}{4}$; the equality $\partial L/\partial z = a - y$ for sigmoid with cross-entropy, and $\mathbf{p} - \mathbf{y}$ for softmax, both with proofs; the adjoint recursion as the chain rule; and the proposition on initialization symmetry. What is borrowed without proof: the universal approximation theorem, with the original proofs in Cybenko (1989) and Hornik (1991), both using the representation theorem for continuous linear functionals. What is popularized beyond its proof: "a network can approximate anything" as a practical promise, when it is a statement about the density of functions, not about the reachability of parameters by optimization; and "Adam is always better", when the counterexample is in Reddi, Kale, and Kumar (2018).

The primary readings that shaped this chapter: Nielsen sections 2.4 and 2.8 (the four backpropagation equations and why it is cheap) and section 3.1 (cross-entropy, softmax); Goodfellow et al. section 6.4.1 (universal approximation) and 6.5 (backpropagation). For initialization, Glorot and Bengio (2010) and He et al. (2015); for Adam, Kingma and Ba (2015).

Limits of validity: all computations in this chapter are for one example on a network

with one hidden layer, with the convention that the ReLU derivative is zero at the break point. There is no statement about generalization here; that is the content of Chapter 26. Open questions: why wide networks are easier to train than narrow ones at the same error, why a network with far more parameters than examples still generalizes, and whether there is an initialization that is optimal for any depth.

Map to source texts: This chapter follows Nielsen, Neural Networks and Deep Learning*: sections 1.1 (perceptron), 1.2 (sigmoid neuron), 1.5 (gradient descent), 2.1 to 2.5 (matrix notation, the two assumptions about the loss function, the Hadamard product, the four backpropagation equations with their proofs), 2.8 (why backpropagation is fast), 3.1 (cross-entropy, softmax), and 3.3 (weight initialization). For the general formulation and for the side Nielsen does not prove, Goodfellow, Bengio, and Courville,* Deep Learning *is used: section 6.2 (gradient-based learning, loss functions, and output units), 6.3 (hidden units and the choice of activation), 6.4.1 (universal approximation and depth), 6.5 (backpropagation and computational graphs), 8.3 (basic algorithms: SGD, momentum), 8.4 (parameter initialization strategies), and 8.5 (algorithms with adaptive learning rates, including Adam). For the gradient with respect to the input as a basis for attacks, see the discussion of FGSM in Chapter 27.*

Chapter 26

Generalization, Regularization, and Training

A security team builds an intrusion detector from two raw quantities: how many requests per second and the average payload size. From those two quantities they derive twenty-one features by multiplying and raising both to the fifth power, then train a logistic classifier on sixty labeled connections. On the training data, the detector is wrong on zero of the sixty connections. On twenty thousand new connections it has never seen, it is wrong on thirty-two percent. The same model, the same weights, two numbers far apart.

That zero is neither a coincidence nor an achievement. With twenty-one features and sixty examples, a separating plane can always be found that cuts every training example correctly, including the example whose label is wrong. What the model learns is not the shape of the correct separator but the position of those sixty points. As soon as the points change, the plane is in the wrong place.

Chapter 25 taught us to compute gradients and follow them down until the training error is small; Chapter 20 taught us to read a single number together with its uncertainty. Neither answers the question that decides matters here: which quantity must be made small so that the number in the field shrinks as well, and how to choose training settings without deceiving ourselves.

For younger readers (age 10 and up)

A student is given sixty practice problems before an exam. He studies by memorizing: every problem together with its answer, down to the letter. On the practice problems he is always right, a hundred out of a hundred, and he goes home proud. On the exam the problems are different: different numbers, different sentences, different order. His grade collapses, and the teacher is puzzled, because this student looked like the most diligent one.

A second student learns more slowly. He does not memorize answers; he looks for the pattern: when to divide, when to multiply, when an answer makes sense. On the practice problems he gets several wrong, and he is not proud. On the exam he does better.

Now the part that makes one curious. Suppose the first student were given two thousand practice problems instead of sixty. Is memorizing still bad? It turns out that it is not always: if there are enough problems and they vary enough, memorizing becomes an expensive way of learning that works. So the problem is not memorizing but memorizing from few examples. In this chapter we will learn why the number of examples and the complexity of the learning method must be balanced, why adding complexity actually makes the exam grade worse, and how to measure that balance without waiting for the

exam.

26.1 Two errors and one gap

Level: basic

Every trained model has two numbers, and both are computed in exactly the same way: the proportion of mistakes on a set of examples. The only difference is which set is used. In everyday language: the grade on a test at home differs from the grade on a test at school, even though the problems are of the same kind.

generalization (*generalisasi*) — the ability to work on new problems, not on problems that have already been memorized

Definition 26.1 (Training error, population error, and the gap)**.** Suppose the examples (x_i, y_i) are drawn from an unknown distribution $\mathcal{D}$. For a hypothesis h and 0-1 error, the *population error* is $R(h) = \mathbb{P}_{(x,y)\sim\mathcal{D}}\big(h(x) \neq y\big)$, whereas the *training error* on n examples is $\hat{R}_n(h) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{h(x_i) \neq y_i\}$. The difference $R(h) - \hat{R}_n(h)$ is called the *generalization gap* for that h.

Training error can be computed; population error cannot, because the distribution is unknown. All we can do is estimate it on another set that is not used for training, and that estimate itself wobbles like every proportion in Chapter 20. So every "validation error" number in this chapter is an estimate, not the truth.

Now two opposite diseases. In everyday language: a model that is too simple cannot capture the pattern, whereas a model that is too complex memorizes coincidence.

Definition 26.2 (Underfitting and overfitting)**.** A model *underfits* when its training error and its population error are both high: the model is too rigid for the actual shape. A model *overfits* when its training error is low but its generalization gap is large: the model adapts itself to the coincidence in the training data, not to the actual shape.

The two diseases are not symmetric in practice. Underfitting is visible: the training error is high, and adding features helps. Overfitting deceives, because the only number that is easily available, namely the training error, is precisely the one that looks perfect. The model in the chapter opening is the purest example: its largest weight is 406.204, a sign that its score depends on small differences between points, not on a plausible scale.

What governs both is *capacity*: how rich the set of functions is that the model may choose from. For a linear classifier over d features, capacity rises with d; for a neural network, with the number of units, the depth, and the number of training steps. Capacity is not merely a parameter count: a large model with strong regularization can behave like a small model.

Read it aloud (auditory learner)

Two voices, alternating.
“A hundred out of a hundred. That means he understands, doesn’t it?”
“Not necessarily. A hundred out of a hundred only means there was no new problem.”
“So memorizing is wrong?”
“Memorizing is not wrong. Memorizing from few examples is.”
“Where is the difference?”
“In the problems that are not in the notes. Memorization answers what has already been seen. Understanding answers what has not been seen.”
“And if there are two thousand problems instead of sixty?”

"Then memorization starts to help. Not because memorization turns into understanding, but because the stored coincidences are used again more often."
Say it once more, slowly: memorization answers what is already there, understanding answers what is not yet there. Repeat that line until it feels natural in the mouth: already there, not yet there.

The generalization gap shrinks as examples are added. In the capacity experiment with a fixed step budget, the gap is 0.1338 at $n = 60$, 0.0274 at $n = 150$, 0.0193 at $n = 600$, and 0.0097 at $n = 2400$, whereas the training error barely moves, staying in the range 0.17 to 0.21.

26.2 The bias-variance decomposition

Level: intermediate

A sharper question: where the gap comes from. In everyday language: a shooter can miss because the sights are shifted to one side, or because his hands shake, and the two causes call for different remedies.

bias-variance (*bias-variansi*) — a systematic miss versus shaking; both contribute error, and pressing one down sometimes raises the other

Theorem 26.1 (Bias-variance decomposition)**.** *Suppose $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$, and suppose $\hat{f}$ is trained on a random training set D that is independent of ε. For every fixed x,*

$$\mathbb{E}_{D,\varepsilon}\big[(y - \hat{f}(x))^2\big] = \sigma^2 + \big(\mathbb{E}_D \hat{f}(x) - f(x)\big)^2 + \mathrm{Var}_D\big(\hat{f}(x)\big).$$

Proof. Write $y - \hat{f}(x) = \varepsilon + \big(f(x) - \mathbb{E}_D\hat{f}(x)\big) + \big(\mathbb{E}_D\hat{f}(x) - \hat{f}(x)\big)$, namely three terms: noise that cannot be reduced, bias, and the deviation from the mean. Square it and take the expectation. The cross terms containing ε vanish because $\mathbb{E}[\varepsilon] = 0$ and ε is independent of D; the cross term between bias and deviation vanishes because $\mathbb{E}_D[\hat{f}(x) - \mathbb{E}_D\hat{f}(x)] = 0$. What remains is exactly the three terms of the theorem's statement. □

The first term cannot be reduced by any model; it belongs to the data, is called the irreducible error, and on the data of the chapter opening its value is about 0.21 because the labels are random. The second and third terms can be traded: adding capacity lowers bias and raises variance, reducing capacity does the opposite. Regularization is the name for that trade.

The cheapest example that contains both terms at once is the shrinkage estimator. To measure μ from n noisy observations with variance σ^2, people usually use $\bar{X}_n$, which is unbiased. We may use $c\bar{X}_n$ for $c \in (0, 1)$, and that estimator is deliberately biased. Its mean squared error

$$\mathrm{MSE}(c) = (c - 1)^2\mu^2 + c^2\frac{\sigma^2}{n},$$

is minimized at $c^* = \mu^2/(\mu^2 + \sigma^2/n)$. For $\mu = 2$, $\sigma = 1$, and $n = 25$ we obtain $c^* = 0.990099$: the best estimator is in fact slightly shrunk.

c	bias^2	variance	MSE
0.70	0.360000	0.019600	0.379600
0.90	0.040000	0.032400	0.072400
0.99	0.000400	0.039204	0.039604
1.00	0.000000	0.040000	0.040000
1.10	0.040000	0.048400	0.088400
1.30	0.360000	0.067600	0.427600

Three things must be read from that table. The gain from shrinking is small: MSE falls from 0.040000 to 0.039604, about one percent. The loss from deviating in the wrong direction is far larger: $c = 1.3$ gives 0.4276, whereas $c = 0.7$ gives only 0.3796. And the minimum is flat, so choosing a slightly wrong c is not a disaster.

The theorem does not give a formula for an $\hat{f}$ as complicated as a neural network. Its decomposition holds formally, but neither term can be computed from data alone; what is observed is only their sum, namely the validation error. For that reason the rest of this chapter works with observed numbers, not with the true bias and variance.

26.3 L2 and L1 regularization

Level: intermediate

The most direct way to press down variance is to forbid the weights from growing large. In everyday language: we add a fine to the objective, so that the model has to pay for every large weight, and will pay only if the gain on the training data is worth it.

regularization (*regularisasi*) — a fine that the model pays for large weights, so that it does not memorize the training data

Definition 26.3 (The regularized objective)**.** For an original objective $J(\mathbf{w})$ and a penalty Ω, the objective being minimized is $\tilde{J}(\mathbf{w}) = J(\mathbf{w}) + \lambda\Omega(\mathbf{w})$ with $\lambda \geq 0$ as a hyperparameter. For the L2 penalty, $\Omega = \frac{1}{2}\|\mathbf{w}\|_2^2$; for the L1 penalty, $\Omega = \|\mathbf{w}\|_1 = \sum_j |w_j|$.

For L2, the gradient of the objective is $\nabla J + \lambda\mathbf{w}$, so one gradient-descent step with rate η has the form

$$\mathbf{w} \leftarrow \mathbf{w} - \eta(\nabla J + \lambda\mathbf{w}) = (1 - \eta\lambda)\mathbf{w} - \eta\nabla J.$$

The factor $(1 - \eta\lambda)$ is the origin of the name *weight decay*: at every step, before the gradient is used, the weights are multiplied by a number slightly less than one. If the gradient is zero, the weights decay toward zero geometrically with half-life $\ln 2/(\eta\lambda)$ steps.

Corollary 26.2 (Weight decay is the same as an L2 penalty)**.** *For a rate η and an L2 penalty with coefficient λ, gradient descent on the regularized objective is identical to gradient descent on the original objective followed by multiplying the weights by $1 - \eta\lambda$. Thus λ and η appear only as the product $\eta\lambda$ in the decay step, so doubling the rate and halving λ give the same decay trajectory.*

The L2 penalty also has a Bayesian reading: it is equivalent to a Gaussian prior on the weights, so the solution is a MAP estimate, not a maximum-likelihood one. That explains why L2 believes small weights are more likely, and why it never forces a single weight to zero.

L1 behaves differently, and the whole difference lies in the shape of its penalty near zero. In everyday language: a quadratic penalty pulls ever more weakly once the weight is small, whereas an absolute-value penalty pulls with constant strength until the weight is exactly zero.

Lemma 26.3 (Soft thresholding)**.** *For $a \in \mathbb{R}$ and $t > 0$, the minimizer of $u \mapsto \frac{1}{2}(u-a)^2 + t|u|$ is*

$$u^* = \operatorname{sign}(a)\max\{|a| - t, 0\},$$

so that $u^* = 0$ exactly when $|a| \leq t$.

For $u > 0$ the function is $\frac{1}{2}(u - a)^2 + tu$ with derivative $u - a + t$, so its critical point is $u = a - t$, valid only when $a > t$. For $u < 0$ the derivative is $u - a - t$ with critical point $u = a + t$, valid only when $a < -t$. At $u = 0$ the subgradient is the interval $[-a - t, -a + t]$,

which contains zero exactly when $|a| \le t$; in that case zero is the minimizer because the function is convex. The three cases join into the formula above.

That lemma explains the whole behavior of L1. With a quadratic approximation around the unpenalized solution w^* and curvature H, the objective is approximately $\frac{H}{2}(w-w^*)^2+\lambda|w|$, so that $w = \operatorname{sign}(w^*)\max\{|w^*| - \lambda/H, 0\}$: every weight below λ/H becomes exactly zero, not merely small. For L2 the solution is $w = w^*/(1 + \lambda/H)$, which is never zero unless w^* itself is zero.

Proposition 26.4 (Why L1 produces true zeros)**.** *Suppose J is differentiable and convex around zero. The point $\mathbf{w} = \mathbf{0}$ is a minimizer of $J(\mathbf{w}) + \lambda\|\mathbf{w}\|_1$ if and only if $|\partial J/\partial w_j| \le \lambda$ for every j. For the L2 penalty, the same condition is $\partial J/\partial w_j = 0$ for every j.*

Proof. The subgradient of $\|\cdot\|_1$ at zero is the box $[-1, 1]^d$, so the condition $0 \in \nabla J(\mathbf{0}) + \lambda[-1, 1]^d$ is equivalent to $|(\nabla J)_j| \le \lambda$ for all j; for a convex function, that condition is necessary and sufficient. For the L2 penalty, the gradient is $\nabla J + \lambda\mathbf{w}$, which at $\mathbf{0}$ equals ∇J, so the condition is $\nabla J(\mathbf{0}) = \mathbf{0}$. □

The practical consequence matters for reading results. L1 gives a flat region of width 2λ around zero, so that weights that are "almost useless" are switched off entirely; L2 only shrinks them. For that reason L1 is used as a feature selector, and for the same reason the number of zero weights under L1 is not a delicate measure of importance: it jumps, it does not shift slowly.

Now the numbers. The same logistic classifier, the same data, twenty-one features on sixty training examples, everything computed with the script in the lab section.

λ	$\max_j \lvert w_j \rvert$	training error	validation error
0	406.204	0.0000	0.3262
0.001	3.056	0.1000	0.2927
0.003	1.814	0.1000	0.2688
0.010	1.018	0.1167	0.2475
0.030	0.631	0.1167	0.2433
0.100	0.399	0.1500	0.2406
0.300	0.268	0.2333	0.2452
1.000	0.142	0.2500	0.2695
3.000	0.062	0.3333	0.3205
10.000	0.022	0.3333	0.3395
100.000	0.003	0.3333	0.3395

Figure 26.1 plots the third and fourth columns of that table as a single valley. Read that table as three regions, not eleven rows. Without regularization the training error is zero and the validation error is 0.3262, with weights up to the hundreds because nothing holds them back. In the region of λ between 0.001 and 0.1, the training error rises to 0.1500 while the validation error falls to 0.2406. In the region $\lambda \ge 3$ both are bad: 0.3333 and 0.3395.

The last row deserves emphasis. With $\lambda = 10$ the validation error is 0.3395, *worse* than the 0.3262 of the model without regularization. So regularization is not a medicine that always helps; it helps at a certain dose and does damage beyond it. At $\lambda = 10$ the model practically always guesses the majority class, whose error is 0.3419: its weights are so small that the score is almost entirely determined by the bias.

The L1 penalty gives a similar picture with a different shape. At $\lambda = 0.05$ its validation error is 0.2334 with eleven of the twenty-one weights alive; at $\lambda = 0.2$ only one weight remains, with error 0.3395. The count of live weights does not fall monotonically: $\lambda = 0.005$ leaves eleven, $\lambda = 0.01$ leaves fourteen. That number is stable against a fourfold increase in iterations, so it is not an optimizer failure; the L1 trajectory for the logistic loss really does

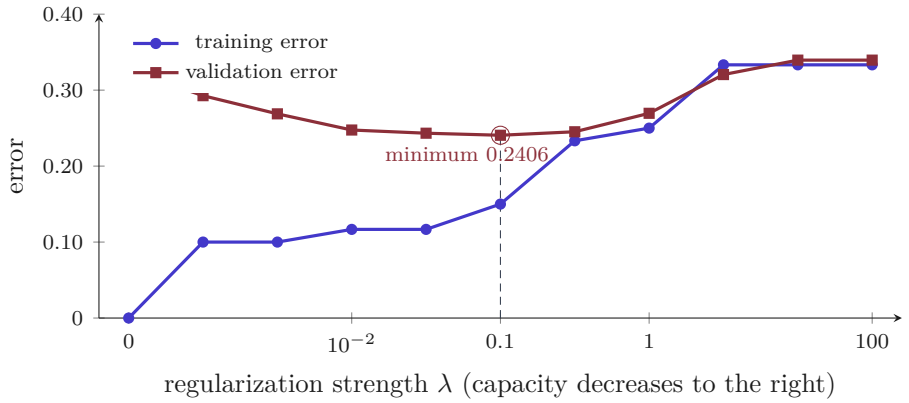


Figure 26.1: Training error and validation error against the regularization strength λ; the direction to the right means capacity decreases, because the weights are held back ever more tightly. The eleven points are the rows of the previous table. The training error rises monotonically from 0.0000 to 0.3333, whereas the validation error forms a valley: 0.3262 at $\lambda = 0$, a minimum of 0.2406 at $\lambda = 0.1$, then 0.3395 at $\lambda = 10$ and above. The bottom of the valley is wide, from 0.2433 at $\lambda = 0.03$ to 0.2452 at $\lambda = 0.3$.

not guarantee a monotone count of zeros, and that is one more reason not to read it as a measure of importance.

The final comparison: L1 reaches 0.2334 and L2 0.2406. The difference 0.0072 is smaller than the wobble between partitions, about 0.02, so this data is not enough to conclude that L1 is better; all that can be concluded is that both improve on the model without regularization.

For younger readers (age 10 and up)

Imagine a class that has to guess a friend's body weight. The teacher gives two methods. Method one: every guess that is too far from the answer is fined an amount that grows the farther away it is. Method two: every guess that is not exactly the answer is fined a fixed amount, no matter whether the difference is small or large.

Method one makes all guesses shift slightly toward the middle, but none of them becomes zero. Method two rounds guesses that are already close straight down to zero, because the fine is fixed and no longer worth the improvement that might be gained. That is the difference between two fines that look alike: one shrinks, the other switches off.

The question: if we have twenty-one features and only sixty examples, what kind of fine makes the model stop memorizing? The answer is not "a large fine", because a fine that is too large switches everything off, including what is correct. In this chapter we will learn why there is a dose, why that dose must be searched for, and why a dose that is searched for on the wrong data will deceive us neatly.

26.4 Early stopping, dropout, and augmentation

Level: intermediate

Three other ways of pressing down variance without touching the formula of the objective. In everyday language: we can stop learning before coincidence begins to be memorized, we can switch off part of the memory at random, and we can multiply examples by changing the examples we already have.

Early stopping is the cheapest form of regularization, because it adds not a single line to the objective: it only stops the trajectory. During training, the training error almost always

falls, but the validation error falls and then rises. The lowest point of the validation error is the one taken, and the number of steps becomes a hyperparameter to be searched for.

epoch	training error	validation error	$\max_j \lvert w_j \rvert$
1	0.1333	0.2481	0.15
5	0.1333	0.2429	0.38
10	0.1333	0.2432	0.48
50	0.1167	0.2454	0.77
200	0.1167	0.2541	1.38
800	0.1000	0.2879	2.85
2400	0.0833	0.3109	4.81

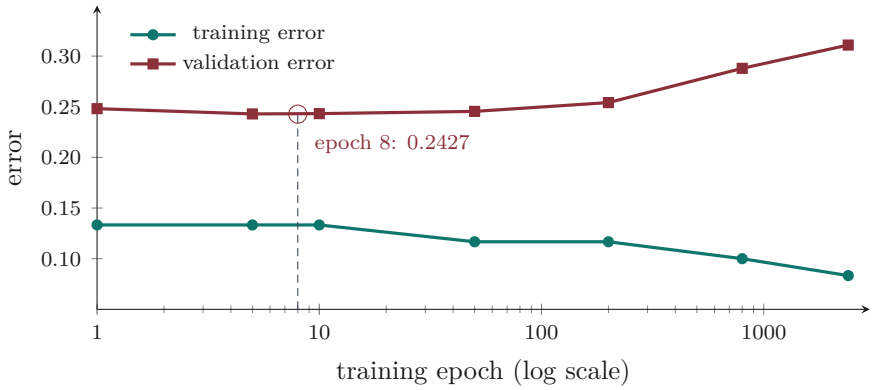


Figure 26.2: Both errors along one training trajectory, horizontal axis on a log scale. The training error falls without ever rising, from 0.1333 at epoch 1 to 0.0833 at epoch 2400. The validation error falls to 0.2427 at epoch 8 and then rises to 0.3109. The dashed line marks the best stopping point: stopping there gives an error 0.0682 lower than training all the way to epoch 2400, and the curves begin to separate just after that point.

Figure 26.2 draws those two columns. The training error falls along the trajectory, from 0.1333 to 0.0833, without ever rising, whereas the validation error falls until epoch 8 with a best value of 0.2427 and then rises to 0.3109 at epoch 2400. The last column explains why: what grows throughout training is the size of the weights, and that is the variance we are paying for.

Early stopping is not merely a tactic. For a quadratic objective and a small rate, stopping at step τ is nearly equivalent to an L2 penalty with coefficient about $1/(\tau\eta)$; that is an approximation, not an exact identity. For that reason the number of steps is a capacity hyperparameter, not a matter of "training until finished".

Step-by-step recipe (procedural learner)

1. Split three sets before a single training step is run: train, validation, test. The test set is sealed.
2. Train on the training set only; every few epochs compute the error on the validation set, and record the three values (epoch, validation error, training error).
3. Take the epoch with the smallest validation error, then return the weights to that point; do not use the weights of the last epoch.
4. Only now open the test set, compute its error once, and report it together with the chosen epoch and the width of the validation curve.

A warning about order: step 1 must be finished before step 2, and step 4 must not be used to shift the choice made in step 3. If the test set is looked at while the epoch is

being chosen, it is no longer a test set but a second validation set, and its numbers stop being an honest estimate.

Dropout takes another route: during training every unit is switched off at random with probability $1-p$, and the ones that survive are divided by p so that the expectation does not shift. In everyday language: at every step some members of the team do not show up, so that no member can be relied on alone.

Lemma 26.5 (Expectation and variance of the dropout mask)**.** *Suppose $m = B/p$ with $B \sim \text{Bernoulli}(p)$ and $0 < p \le 1$. Then $\mathbb{E}[m] = 1$ and $\text{Var}(m) = (1-p)/p$. For a unit with inputs $z_1, \dots, z_k$ and independent masks $m_1, \dots, m_k$, its output $s = \sum_j m_j w_j z_j$ satisfies $\mathbb{E}[s] = \sum_j w_j z_j$ and $\text{Var}(s) = \frac{1-p}{p} \sum_j (w_j z_j)^2$.*

Proof. $\mathbb{E}[m] = p/p = 1$ and $\text{Var}(m) = \mathbb{E}[m^2] - 1 = p/p^2 - 1 = (1-p)/p$. For s, linearity of expectation gives the first term, and independence of the masks gives the variance of the sum, namely $\sum_j (w_j z_j)^2 \,\text{Var}(m_j)$. □

That variance formula explains two things that are often forgotten. Dropout noise is proportional to the sum of squared contributions, so units with large weights receive relatively more noise, and dropout points in the same direction as an L2 penalty. And that noise disappears entirely at test time, so that the model that is used differs from the model that was trained; on very small networks that price can exceed the benefit.

Data augmentation multiplies examples by changing the examples that already exist, for instance shifting them or adding small noise, provided that the label does not change. For a linear model, its relation to L2 can be computed exactly.

Proposition 26.6 (Gaussian noise on the input is the same as an L2 penalty)**.** *Suppose $\hat{y} = \mathbf{w}^\top \mathbf{x}$ and $\boldsymbol{\delta} \sim N(\mathbf{0}, \sigma^2 I)$ is added to the input. Then*

$$\mathbb{E}_{\boldsymbol{\delta}}\big[(\mathbf{w}^\top(\mathbf{x}+\boldsymbol{\delta}) - y)^2\big] = (\mathbf{w}^\top \mathbf{x} - y)^2 + \sigma^2 \|\mathbf{w}\|_2^2.$$

Proof. Expand: the linear term is $2(\mathbf{w}^\top \mathbf{x} - y)\,\mathbb{E}[\mathbf{w}^\top \boldsymbol{\delta}] = 0$, and the quadratic term is $\mathbb{E}[(\mathbf{w}^\top \boldsymbol{\delta})^2] = \mathbf{w}^\top \mathbb{E}[\boldsymbol{\delta}\boldsymbol{\delta}^\top]\mathbf{w} = \sigma^2 \|\mathbf{w}\|_2^2$. □

So adding small Gaussian noise to the input is the same as adding an L2 penalty with $\lambda = \sigma^2$: the two are one idea written twice. The numerical check, for $\mathbf{w} = (1, 2)$, $\mathbf{x} = (1, 0)$, $y = 3$, and $\sigma = 0.1$: the original objective is 4, the formula predicts $4 + 0.01 \cdot 5 = 4.05$, and a simulation of one million noises gives 4.049107. That difference of 0.0009 is comparable to the random error of the simulation, not to a mistake in the formula.

A warning for the security field: augmentation is valid only when the label is genuinely invariant to the change that is made. Adding small noise to traffic features is allowed, but reversing the order of packets or shifting arrival times can turn an attack into not-an-attack. Augmentation with a wrong label is not regularization but a lie that will keep looking good as long as the validation data is changed in the same way.

26.5 SGD, momentum, and Adam

Level: advanced

Up to here we have treated "training" as a single box. The contents of that box determine how fast the error falls and, on a limited step budget, how low an error is reached. In everyday language: the right direction is not enough; what decides is how the step is taken and how large it is.

Stochastic gradient descent replaces the full gradient by the gradient over one minibatch of size b. That estimate is unbiased, and its variance shrinks proportionally to $1/b$: $\mathrm{Var}(\hat{\mathbf{g}}_b) = \sigma_g^2/b$. So a large minibatch reduces the wobble, but does not change the mean direction. What changes the direction is momentum and Adam.

Definition 26.4 (Momentum and Adam)**.** Momentum keeps a running average of the gradient: $\mathbf{v}_t = \beta \mathbf{v}_{t-1} + \mathbf{g}_t$ and $\mathbf{w}_t = \mathbf{w}_{t-1} - \eta \mathbf{v}_t$, with $0 \le \beta < 1$. Adam keeps two moments, $\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\mathbf{g}_t$ and $\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)\mathbf{g}_t^2$ (squared per coordinate), then uses bias correction and the update $\mathbf{w}_t = \mathbf{w}_{t-1} - \eta\, \hat{\mathbf{m}}_t / (\sqrt{\hat{\mathbf{v}}_t} + \epsilon)$ with $\hat{\mathbf{m}}_t = \mathbf{m}_t/(1-\beta_1^t)$ and $\hat{\mathbf{v}}_t = \mathbf{v}_t/(1-\beta_2^t)$.

That bias correction is not decoration: without it $\mathbf{m}_1 = (1-\beta_1)\mathbf{g}_1$ is very small, so the first step is almost zero. Division by $\sqrt{\hat{\mathbf{v}}}$ matters even more for badly scaled problems: it makes the step size in each coordinate roughly η, no matter whether the gradient is large or small. Momentum only smooths the direction over time.

The numbers on the following quadratic show the difference. Take $f(\mathbf{w}) = \frac{1}{2}\mathbf{w}^\top H \mathbf{w}$ with $H = \mathrm{diag}(1, 10^4)$ and $\mathbf{w}_0 = (1, 0)$; the condition number of H is 10^4, which means the curve is narrow in one direction and very flat in the other. We count steps until $f \le 10^{-8} f(\mathbf{w}_0)$.

method	rate	steps to 10^{-8}
gradient descent	10^{-4}	92099
momentum ($\beta = 0.9$)	10^{-4}	9132
Adam	0.1	82

Gradient descent needs 92099 steps because its largest rate is limited by the largest curvature, while the flat direction moves only at that rate. Momentum cuts one digit, in line with the estimate that acceleration changes the dependence on the condition number κ into about $\sqrt{\kappa}$. Adam cuts three digits, because per-coordinate division means the flat direction is no longer punished by the steep one. That is not a general promise: on well-scaled problems the advantage of Adam shrinks.

On the logistic problem of this chapter, with minibatch eight and a target training error of 0.40, the median of five random seeds gives SGD with rate 0.05 a hundred steps, whereas Adam with rate 0.1 needs only ten. The conclusion must be careful: the advantage is about one digit and partly comes from the larger rate, not from a better direction.

The learning rate is usually not constant. Two common schedules are step decay, namely the rate divided by ten at certain epochs, and cosine decay $\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min})(1 + \cos(\pi t/T))$: large steps at the start so as to approach quickly, small steps at the end so as not to jump around in the valley. With 3000 minibatch steps and $\lambda = 0.1$, a constant rate of 0.2 gives a median training error of 0.3866, whereas a cosine from 0.5 to 0.01 gives 0.3549.

26.6 Cross validation, hyperparameter selection, and leakage

Level: advanced

All the validation numbers in this chapter come from a set that is not used for training. That is already correct, but it is not enough: as soon as we choose λ with an eye that sees the validation error, that validation error stops being an unbiased estimate of the population error. In everyday language: if we choose the exam questions by the grade we get, that grade no longer measures our ability.

cross validation (*validasi silang*) — rotating the roles of the data in turn, so that every example has once been an exam for a model that did not see it

Definition 26.5 (k-fold cross validation)**.** The training data is split into k folds that are disjoint and of similar size. For each j, the model is trained on $k-1$ folds and tested on fold j; the cross-validation error is the average error over those k folds, and the hyperparameter that is chosen is the one that minimizes it. For $k = n$ this procedure is called *leave-one-out*.

Cross validation answers one weakness of a single split: at $n = 60$ the validation set contains twelve examples, and the error on twelve examples wobbles with a standard deviation of about 0.12. With five folds every example is tested once, but the wobble does not disappear.

λ	CV error (average of 5 partitions)	spread between partitions
0	0.3233	0.1245
0.003	0.2400	0.0253
0.030	0.2467	0.0217
0.100	0.2467	0.0274
1	0.2967	0.0298

The first row shows two things at once: without regularization the CV error is both the worst and the least stable, with a spread of 0.1245, four times that of the other rows. The second to fourth rows are almost indistinguishable: 0.2400, 0.2467, and 0.2467, with a spread of about 0.025, and across five different partitions the winner keeps changing. This CV curve is flat around its minimum, so that choosing its exact value is an illusion of precision.

The consequence is real: CV selects $\lambda = 0.003$ with validation error 0.2688, whereas the unselected $\lambda = 0.1$ gives 0.2406; that difference of 0.0282 is not far from one standard deviation of the CV curve. The lesson is not that cross validation is broken, but that with sixty examples, hyperparameter selection carries an uncertainty as large as the effect being measured. Reporting "the best λ is 0.003" without mentioning the width of the curve is the subtlest lie in this chapter.

Do it by hand (kinesthetic learner)

Materials: ten cards, a pencil, paper. Write one two-digit number on each card, shuffle, then lay out seven cards face up. Build a "rule" in the form of a list: position one is worth this much, position two is worth that much, through position seven.
Cover one card with your palm and guess. Because your list contains that card, your guess is exact. Repeat for all seven positions: seven out of seven correct, and that is your training score.
Now gather the ten cards, shuffle, lay out seven new cards, and use the same list. Cover one card, guess, uncover. What must be visible: your list does not contain the new cards, so your guess is pure luck. Repeat ten times with new layouts and record how often you are exact; that is your exam score. Finally write a second rule that uses only one property, for example "the card on the left is not larger than the card on its right", then compare the two scores.

There is a source of error larger than that wobble and invisible in any number: leakage, namely information from the test data entering the training process. Its most common form is standardizing features using the mean and standard deviation of all the data before the split is made.

The numbers, with an artificial covariate shift: if validation is shifted by 0.5, standardization with training statistics only gives 0.3765, whereas pooled statistics give 0.3361; at a shift of 1.0 the numbers are 0.5387 against 0.3819. Pooled statistics always look better, and that is precisely the problem: they use information about where the test data lies that is not available in the field. Without a shift the two are equal, 0.2452 against 0.2463, a sign that the earlier difference came from leakage.

The honest protocol is layered: separate train, validation, and test from the start; compute every quantity, including means, standard deviations, decision thresholds, and feature selection, inside the training folds only; choose hyperparameters on the validation set or through cross validation inside the training data; touch the test set once, then report it as it is.

The relation between capacity and the amount of data can be made quantitative. For a finite hypothesis class, the generalization gap is controlled by the square root of the logarithm of the number of hypotheses divided by the number of examples.

Theorem 26.7 (A uniform bound for finite classes)**.** *Suppose $\mathcal{H}$ is finite with $|\mathcal{H}| = m$, its error is 0-1, and n examples are drawn independently from $\mathcal{D}$. With probability at least $1 - \delta$ over the drawing of the examples, it holds simultaneously for all $h \in \mathcal{H}$*

$$|R(h) - \hat{R}_n(h)| \le \sqrt{\frac{\ln m + \ln(2/\delta)}{2n}}.$$

Proof. For a fixed h, $\hat{R}_n(h)$ is the average of n Bernoulli variables scaled to $[0, 1]$, so Hoeffding's inequality gives $\mathbb{P}(|R(h) - \hat{R}_n(h)| > t) \le 2e^{-2nt^2}$. Combine over all h with the union bound: the probability that some h exceeds t is at most $2m\, e^{-2nt^2}$. Setting that expression equal to δ gives $t = \sqrt{(\ln m + \ln(2/\delta))/(2n)}$, and its complement is the statement of the theorem. □

For the class that "memorizes", namely a model that gives one label to every binary pattern over d features, $m = 2^d$. With $d = 21$ and $\delta = 0.05$ the bound is $\sqrt{(21 \ln 2 + \ln 40)/(2n)}$, namely 0.3899 at $n = 60$ and 0.0617 at $n = 2400$. The measured gap is 0.1338 and 0.0097 at the same n, so the bound holds but is loose; that looseness is not a defect, because the bound must hold for every distribution and every algorithm.

What matters is not the numbers but their shape. The gap shrinks proportionally to $1/\sqrt{n}$, whereas the number of hypotheses enters through a logarithm. Quadrupling the data only halves the gap, and increasing capacity tenfold only raises the bound by the square root of the logarithmic increment. This is the honest reason why large models are still useful: as long as their gap is controlled by regularization, rising capacity does not automatically mean a rising gap.

26.7 Worked example: closed-form ridge on four points

Level: intermediate

All the formulas above can be checked by hand on a small example: four points with one feature, $\mathbf{x} = (1, 2, 3, 4)$ and $\mathbf{y} = (2, 4, 6, 9)$, a model without an intercept $\hat{y} = wx$, and the objective $\frac{1}{2}\sum_i (y_i - wx_i)^2 + \frac{\lambda}{2} w^2$. Setting its derivative to zero gives

$$w(\lambda) = \frac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda} = \frac{64}{30 + \lambda},$$

since $\sum x_i^2 = 30$ and $\sum x_i y_i = 2 + 8 + 18 + 36 = 64$.

λ	w	training SSE	$\lvert w \cdot 5 - 10 \rvert$
0	2.133333	0.466667	0.666667
1	2.064516	0.608741	0.322581
2	2.000000	1.000000	0.000000
5	1.828571	3.253061	0.857143
10	1.600000	9.000000	2.000000
30	1.066667	34.600000	4.666667
100	0.492308	81.255621	7.538462

Figure 26.3 plots both weight trajectories against λ on this data: L2 shrinks without end, L1 stops exactly at zero. The last column uses one test point at $x = 5$ with true value 10, which does not take part in determining w. That is where the curve takes a U shape and its minimum falls exactly at $\lambda = 2$: there $w = 2$, so that the prediction at $x = 5$ is 10 and the error is zero. Without regularization, $w = 64/30 = 2.1333$ is more attractive on the training data (SSE 0.466667 against 1.000000) but predicts 10.667, off by 0.667. This is the bias-variance trade in its barest form: the training SSE doubles, the test error becomes zero.

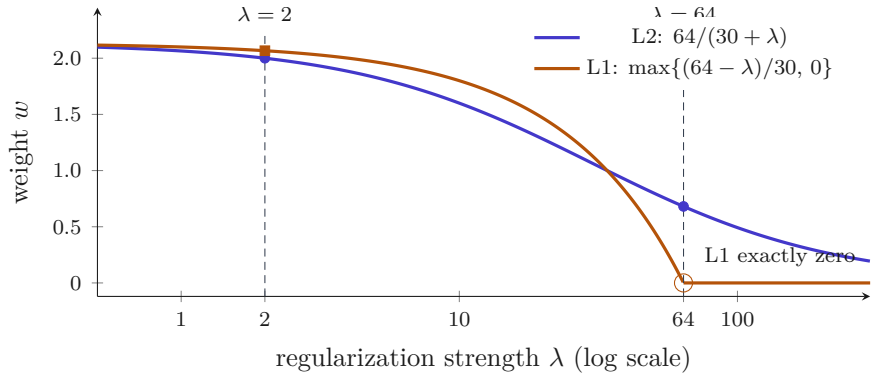


Figure 26.3: Weight trajectories against regularization strength on the worked example with four points, with $\sum x_i^2 = 30$ and $\sum x_i y_i = 64$. The L2 penalty gives $w = 64/(30 + \lambda)$, which shrinks but never touches zero. The L1 penalty gives $w = \max\{(64 - \lambda)/30, 0\}$, which reaches zero exactly at $\lambda = 64$. At $\lambda = 2$ the two are nearly the same, 2.0000 against 2.0667; at $\lambda = 64$ L2 is still 0.6809 whereas L1 is already 0.0000. The dashed lines mark the two values of λ that are compared.

Note the right-hand wall. At $\lambda = 100$, w is only 0.4923, the training SSE explodes to 81.255621, and the test error is 7.538462, worse than a model that learns nothing at all. This example also shows why searching for λ on the training data is misleading: the training SSE curve rises monotonically, so that it always chooses $\lambda = 0$.

Challenge (problem solver)

A model reports a training error of 0.0000 on 60 examples and a test error of 0.4100. A colleague proposes three changes: (i) train longer, (ii) add features up to degree seven, (iii) raise λ from 0 to 0.1. For each proposal, predict the sign of the change in training error and in test error, then state which one cannot possibly help, with the reason.
Hint 1: first ask in which direction capacity moves under each proposal.
Hint 2: two of the three proposals enlarge capacity, and this model is already at zero training error, so that enlarging capacity only widens the gap. What helps is what narrows it, and that must raise the training error.
If you get stuck, go back to the Definition of underfitting and overfitting in the first section.

Check the second column without a calculator: $64/32 = 2$ exactly, $64/31 = 2.064516$, and $64/40 = 1.6$. Then compute the SSE at $\lambda = 2$ yourself: the residuals are $(0, 0, 0, 1)$, so SSE $= 1$. Finally answer: which λ makes $w = 1.28$, and what is the prediction at $x = 5$ then? Hint: solve $64/(30 + \lambda) = 1.28$.

26.8 Computational lab

Level: advanced

Every number in this chapter comes from a single numpy script with no machine-learning library. The data is deliberately small so that it can be rechecked within minutes: two raw

features, twenty-one polynomial features up to degree five, sixty training examples, and twenty thousand validation examples from the same distribution. The labels are not deterministic, so that there is an irreducible error of about 0.21 which no model can go below.

Computational lab

```
import numpy as np

def data(n, rng):
    x = rng.uniform(-1, 1, size=(n, 2))
    p = 1/(1 + np.exp(-4.0*(0.9 - x[:,0]**2 - x[:,1]**2)))
    return x, (rng.random(n) < p).astype(float)

def poly(x, deg=5):
    return np.column_stack([x[:,0]**i * x[:,1]**j
                            for i in range(deg+1) for j in range(deg+1-i)])

def sig(z):
    return 1/(1 + np.exp(-np.clip(z, -400, 400)))

def soft(t, l):
    return np.sign(t)*np.maximum(np.abs(t) - l, 0.0)

def er(Z, y, w, b):
    return float(np.mean((sig(Z@w + b) > 0.5) != y))

def fit(Z, y, lam, iters=30000, lr=0.05):
    n, d = Z.shape
    w = np.zeros(d); b = 0.0
    mw = np.zeros(d); vw = np.zeros(d); mb = 0.0; vb = 0.0
    for t in range(1, iters+1):
        p = sig(Z@w + b); g = (p - y)/n
        gw = Z.T@g + lam*w; gb = g.sum()
        mw = 0.9*mw + 0.1*gw; vw = 0.999*vw + 0.001*gw*gw
        mb = 0.9*mb + 0.1*gb; vb = 0.999*vb + 0.001*gb*gb
        w -= lr*(mw/(1-0.9**t))/(np.sqrt(vw/(1-0.999**t)) + 1e-8)
        b -= lr*(mb/(1-0.9**t))/(np.sqrt(vb/(1-0.999**t)) + 1e-8)
    return w, b

def fit_l1(Z, y, lam, iters=150000):
    n, d = Z.shape
    w = np.zeros(d); b = 0.0
    st = 0.5/(0.25*np.linalg.eigvalsh(Z.T@Z/n).max())
    for _ in range(iters):
        p = sig(Z@w + b); g = (p - y)/n
        w = soft(w - st*(Z.T@g + lam*w), st*lam); b -= st*g.sum()
    return w, b
```

The first part computes the error for eleven values of λ; the second part repeats that with the L1 penalty through proximal gradient descent, namely a gradient step followed by the soft threshold from the lemma above; the third part stops early; the fourth and fifth parts compare the optimizers; the sixth to eighth parts examine cross validation, leakage, and capacity. The actual output, cut only where it repeats:

```
d=21  n_train=60  n_validation=20000
==1 L2: lam  |w|max  train  validation
  0.0000 406.204  0.0000  0.3262
  0.0010   3.056  0.1000  0.2927
  0.0030   1.814  0.1000  0.2688
  0.0100   1.018  0.1167  0.2475
  0.0300   0.631  0.1167  0.2433
```

```
  0.1000   0.399  0.1500  0.2406
  0.3000   0.268  0.2333  0.2452
  1.0000   0.142  0.2500  0.2695
  3.0000   0.062  0.3333  0.3205
 10.0000   0.022  0.3333  0.3395
100.0000   0.003  0.3333  0.3395
==2 L1: lam  zeros/total  train  validation
  0.0020 16/21  0.1167  0.2630
  0.0050 11/21  0.1167  0.2455
  0.0100 14/21  0.1167  0.2412
  0.0200 14/21  0.1500  0.2369
  0.0500 11/21  0.1833  0.2334
  0.1000  7/21  0.2667  0.2492
  0.2000  1/21  0.3333  0.3395
==3 early stopping: epoch  train  validation  |w|max
ep=   1  0.1333  0.2481  0.15
ep=   5  0.1333  0.2429  0.38
ep=  10  0.1333  0.2432  0.48
ep=  50  0.1167  0.2454  0.77
ep= 200  0.1167  0.2541  1.38
ep= 800  0.1000  0.2879  2.85
ep=2400  0.0833  0.3109  4.81
best 0.2427 at epoch 8
==4 quadratic diag(1,1e4): steps to error 1e-8
GD=1e-4  92099 | momentum  9132 | Adam=0.1  82
==5 logistic minibatch: steps to training error 0.40 (median of 5 seeds)
sgd  lr=0.05 [100, 100, 90, 90, 100] median=100
sgd  lr=0.2  [30, 50, 40, 40, 20] median=40
adam lr=0.03 [30, 20, 20, 20, 30] median=20
adam lr=0.1  [20, 10, 20, 10, 10] median=10
==6 CV 5 partitions x 5 folds: lam  CV error (mean +- sd between partitions)
 0.000  0.3233 +- 0.1245
 0.003  0.2400 +- 0.0253
 0.030  0.2467 +- 0.0217
 0.100  0.2467 +- 0.0274
 1.000  0.2967 +- 0.0298
==7 standardization: shift  training stats  pooled stats
0.0  0.2452  0.2463
0.5  0.3765  0.3361
1.0  0.5387  0.3819
==8 capacity vs data (d=21, lam=0, nested sets)
n=   60 train=0.1667 validation=0.3005 diff=+0.1338 bound=0.3899
n=  150 train=0.2133 validation=0.2407 diff=+0.0274 bound=0.2466
n=  600 train=0.2067 validation=0.2260 diff=+0.0193 bound=0.1233
n= 2400 train=0.2058 validation=0.2155 diff=+0.0097 bound=0.0617
```

Three things need to be read from that output, and all three disappoint ordinary expectations.

First, the training error is nearly useless as a guide: it moves in the opposite direction to the validation error. The validation curve is also not symmetric. From $\lambda = 0$ to 0.1 its error falls by 0.086; from 0.1 to 10 it rises by 0.099 until it passes its starting point. When in doubt between two values, the smaller one is safer.

Second, selection through cross validation is not automatically correct. CV chooses $\lambda = 0.003$, whereas validation chooses 0.1; the difference between their CV errors is 0.0006, far below the spread between partitions of 0.025, even though on the validation data the two differ by 0.0282.

Third, the theoretical bound is loose but its direction is right. The capacity experiment here trains the model with a budget of sixty thousand steps on nested sets, so that its training error is not zero. At $n = 60$ the bound is 0.3899 whereas the gap is 0.1338; at $n = 2400$ the

bound is 0.0617 whereas the gap is 0.0097. The ratio is about three, so that the bound is useful for comparing settings, not for predicting numbers.

Note for instructors

The teaching order I suggest: start from the opening numbers of the chapter, sixty examples with training error zero and validation error 0.3262, then ask whether that model "learned". The answer of the class usually splits in two, and that split is the best lecture material there is. After that the formal definitions, and let the bias-variance decomposition be derived on the board in five lines. L1 regularization is better introduced through the soft-threshold lemma than through a picture of the L1 ball, because students can check that lemma themselves.

The two misconceptions that appear most often. Students treat a low training error as proof of a good model, then are surprised when its validation error is high; another symptom is that they suggest "training longer" as the cure, when in the early-stopping table that is precisely what does the damage. And they treat regularization as always helpful, then install a large λ with the conviction that it is "safer". The row $\lambda = 10$ with error 0.3395 is the cure for both.

A blackboard activity that works: ask the class to guess, before it is computed, where the minimum of the validation curve lies, then compute just three rows together; the guess usually misses toward a λ that is too small. The sign that the class is ready to go on: they can explain why choosing λ with the validation data makes that number optimistic, and they can name one leakage that is invisible in the numbers.

26.9 Summary

Level: basic

Summary.

- Training error and population error are computed in the same way on different sets; a training error of 0.0000 with a validation error of 0.3262 signals memorization.
- The population error decomposes into irreducible noise, squared bias, and variance; regularization trades bias for variance, and that trade pays only as long as the fall in variance is larger than the rise in bias.
- The L2 penalty adds $\lambda \mathbf{w}$ to the gradient, equivalent to multiplying the weights by $1 - \eta\lambda$ at every step; L1 uses soft thresholding so that weights below λ / H become exactly zero. Both have a dose: $\lambda = 0.1$ gives 0.2406, whereas $\lambda = 10$ gives 0.3395, worse than without regularization.
- Early stopping is regularization through the number of steps: 0.2427 at epoch 8 against 0.3109 at epoch 2400. Dropout adds noise with variance $(1 - p)/p$ per unit, and Gaussian-noise augmentation with variance σ^2 is equivalent to an L2 penalty of scale σ^2.
- Optimizers differ on badly scaled problems: on a quadratic with condition number 10^4, gradient descent needs 92099 steps, momentum 9132, Adam 82; a cosine schedule lowers the training error from 0.3866 to 0.3549.
- Cross validation reduces the wobble without removing it: the CV curve is flat with a spread of 0.025, and its choice misses by 0.0282. Standardization leakage makes the numbers look better (0.3361 against 0.3765) without being really better, and the generalization gap shrinks proportionally to $1/\sqrt{n}$: from 0.1338 at $n = 60$ to 0.0097 at $n = 2400$.

26.10 Exercises

Level: intermediate

Exercise 26.1.

1. Assess this claim: "Because the training error of our model is zero, the model is already good enough." True, false, or not enough evidence? Name the part that is mistaken and one additional number that you need.
2. Prove the bias-variance decomposition for $y = f(x) + \varepsilon$ with $\mathrm{Var}(\varepsilon) = \sigma^2$: write the three terms, show that both cross terms vanish, and state where the independence of D from ε is used.
3. Derive $w(\lambda) = 64/(30 + \lambda)$ for the data in the worked example, then show that w decreases monotonically in λ but is never zero. Compare it with the L1 penalty on the same one-dimensional problem, and determine the λ that makes $w = 0$ exactly.
4. Compute the validation error for $c = 0.95$ and $c = 1.05$ in the shrinkage example with $\mu = 2$, $\sigma = 1$, $n = 25$, then compare them with $c = 1$. Explain why two deviations of equal size do not give losses of equal size.
5. A report writes: "We chose the best λ through cross validation, so that cross-validation number is an estimate of the error in the field." Fix that sentence, and name two additional procedures that are needed.
6. Prove the soft-threshold lemma for $a < 0$ in full, then explain why the result is symmetric in the sign of a. Hint: treat the cases $u < 0$ and $u = 0$ separately.
7. Compute the number of weight-decay steps needed for a weight to become a quarter of its initial value if $\eta = 0.1$ and $\lambda = 0.02$, using the half-life $\ln 2/(\eta\lambda)$. Check with a numpy simulation for a zero gradient.
8. On the data of this chapter, $\lambda = 0.1$ gives a training error of 0.1500 and a validation error of 0.2406. Compute the gap, compare it with the gap at $\lambda = 0$ and $\lambda = 10$, then explain why a small gap is not enough to conclude that a model is better.
9. Derive the uniform bound $\sqrt{(\ln m + \ln(2/\delta))/(2n)}$ from Hoeffding's inequality and the union bound, then compute it for $d = 10$, $n = 200$, and $\delta = 0.01$. How many times must the data be doubled for the bound to halve?

Hints for the three hardest items. Item 3: its one-dimensional objective is $\frac{1}{2}\sum_i (y_i - wx_i)^2 + \lambda|w|$; at $w = 0$ its subgradient contains zero when $\lambda \geq |\sum_i x_i y_i| = 64$, so that $w = 0$ exactly at $\lambda = 64$. Item 6: at $u = 0$ the subgradient is the interval $[-a - t, -a + t]$, which contains zero when $|a| \leq t$. Item 9: with $m = 2^{10}$ and $\delta = 0.01$ the bound is $\sqrt{(10 \ln 2 + \ln 200)/400} = 0.1749$; halving it demands four times the data.

For researchers: limits and next steps

What is proved in this chapter: the bias-variance decomposition together with its conditions; the equivalence of the weight-decay step with an L2 penalty; the closed form of the minimizer of $\frac{H}{2}(w - w^*)^2 + \lambda|w|$ and the subgradient condition that makes L1 produce true zeros; the equivalence of Gaussian-noise augmentation with an L2 penalty for linear models; the formulas for the expectation and variance of the dropout mask; and the uniform bound for finite hypothesis classes. What is only popular: that regularization always helps; that Adam is always faster; that early stopping "is the same as" L2 (it is an approximation for a quadratic objective with a small rate); that the count of zeros under L1 measures feature importance.

Primary sources: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 5 (capacity, bias-variance), Chapter 7 (L1 and L2 penalties, augmentation, noise, early stopping, dropout), Chapter 8 (SGD, momentum, Adam, learning-rate schedules, hyperparameter selection). Wasserman, *All of Statistics*, Chapter 13 and Chapter 20. The uniform bound

is a direct consequence of Hoeffding's inequality and the union bound.
Limits of validity: every number comes from one synthetic set with a single 60/20000 split, so that each validation error carries an uncertainty of order 0.01 to 0.02. For that reason the comparison of L1 and L2 here is not a recommendation. The uniform bound is only for finite classes and 0-1 error.
Open questions: how large the leakage of batch normalization is, since it computes statistics from the minibatch and so mixes information between examples; how to choose λ when the cost of mistakes is asymmetric; and whether a flat CV curve can be turned into a valid interval, not merely a spread between partitions.

Map to source texts: Chapter 26 in Goodfellow, Bengio, and Courville, Deep Learning*: Chapter 5 (capacity, overfitting and underfitting, the bias-variance decomposition), Chapter 7 (L1 and L2 penalties, weight decay, data augmentation, noise as regularization, early stopping, dropout), Chapter 8 (SGD, momentum, Adam, learning-rate schedules, hyperparameter selection). In Wasserman,* All of Statistics*: Chapter 13 (model selection and cross validation), Chapter 20 (bias-variance in smoothing). Backpropagation and network training are covered in Chapter 23; honest reading of evaluation metrics, including leakage and data splitting, continues in Chapter 25; the convex optimization underlying penalties and subgradients is in Chapter 21.*

Chapter 27

Honest Model Evaluation

A short report lands on a security analyst's desk. It holds one table: a new intrusion detector, tested on network traffic captured from a campus network over a month, reaches an accuracy of 99 percent. That number is framed in bold type and closed by a single sentence: "ready to deploy".

The data contains one attack for every thousand connections. On the same collection, a three-line program that always answers "clean" obtains an accuracy of 99.9 percent: it reads no feature at all and never raises a false accusation. The detector trained for hours loses, and its report contains not one number that leaks that defeat.

What is wrong is the summary: a single number stands in for four numbers with different roles, on data whose composition is never stated. Chapter 26 explained why test error differs from training error; this chapter closes three gaps: which metrics are honest about what they ignore, how much those numbers wobble, and whether the test data is truly unseen by the procedure that chose the model.

For younger readers (age 10 and up)

Imagine a metal detector at a school gate. Every morning it beeps. Yesterday it beeped a thousand times, and the guard searched a thousand bags. Only sixty of those bags really carried a forbidden object. The other nine hundred and forty students had to open their bags for no reason.

Now an inspector from the district office arrives and asks: what percentage of the device's answers were right? The guard answers with pride: ninety-nine percent. He is not lying; out of a hundred thousand bags, the device was wrong on only a thousand. But that answer does not answer the question that upsets the students: they are upset not because the device is often wrong, but because almost every time it beeps, it is wrong.

The two questions sound alike and their answers are far apart: one asks about all the bags, the other about the bags that were beeped. In this chapter we will learn why one device can excel at the first question and collapse on the second, why a number of a hundred percent is sometimes not good news, and why an honest report always carries more than one number.

27.1 Four numbers that are not symmetric

Level: basic

A binary classifier, however sophisticated its interior, leaves the same trace: four numbers. What separates one evaluation report from another is whether all four are reported. In everyday language: every decision has two ways to be right and two ways to be wrong, and

closing off one of them is the same as hiding.

confusion matrix (*matriks konfusi*) — a four-column ledger that records every decision, right or wrong, together with the direction of the error

Definition 27.1 (Confusion matrix)**.** Suppose every example has a true label $y \in \{0, 1\}$ and a prediction $\hat{y} \in \{0, 1\}$, with 1 meaning positive (attack, disease, harmful message). The four numbers

$$\mathrm{TP} = \#\{y = 1,\ \hat{y} = 1\}, \qquad \mathrm{FP} = \#\{y = 0,\ \hat{y} = 1\},$$
$$\mathrm{FN} = \#\{y = 1,\ \hat{y} = 0\}, \qquad \mathrm{TN} = \#\{y = 0,\ \hat{y} = 0\}$$

together with their arrangement are called the *confusion matrix*. The total number of examples is $n = \mathrm{TP} + \mathrm{FP} + \mathrm{FN} + \mathrm{TN}$.

From those four numbers we derive metrics that answer different questions. Precision asks: when the model says positive, how likely is it to be right. Recall asks: of all the positive cases, what fraction was caught.

Definition 27.2 (Precision, recall, specificity)**.** With $\mathrm{TP} + \mathrm{FP} > 0$ and $\mathrm{TP} + \mathrm{FN} > 0$,

$$\mathrm{prec} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad \mathrm{rec} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad \mathrm{spec} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}.$$

Accuracy is $(\mathrm{TP} + \mathrm{TN})/n$, and F_1 is $2\,\mathrm{TP}/(2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$.

Let us use one worked example: a detector is tested on 1000 labeled connections, 100 of them attacks. It flags 62 attacks correctly, misses 38, and accuses 17 clean connections of being attacks.

	predicted positive	predicted negative	total
actually positive	TP = 62	FN = 38	100
actually negative	FP = 17	TN = 883	900
total	79	921	1000

Its precision is $62/79 = 0.784810$: about four out of five alarms point to an attack. Its recall is $62/100 = 0.620000$: almost four out of ten attacks go unseen. Its specificity is $883/900 = 0.981111$. Its accuracy is $945/1000 = 0.945000$, and $F_1 = 124/179 = 0.692737$. All five numbers are true about the same model, and not one of them may be used on its own. A check with `sklearn.metrics` gives the same five numbers to six decimal places.

Figure 27.1 puts the four numbers in their places, together with the row and column totals; every number in the rest of the chapter must be traceable to one of the cells of that figure.

The roles of the four numbers are not symmetric. Precision answers the operator's question: when the alarm sounds, how likely is a real attack. Recall answers the policy designer's question: of all attacks, what fraction is caught. Both depend on the threshold, so a metric without a threshold is not a complete statement: the same model gives precision 0.25 with recall 0.45, or precision 0.013 with recall 0.86.

F_1 tries to compress precision and recall into a single number, and that is exactly where it is easiest to misuse.

Proposition 27.1. *F_1 is the harmonic mean of precision P and recall R, that is $F_1 = 2PR/(P + R)$, so that $F_1 \le (P + R)/2$ with equality holding exactly when $P = R$.*

	predicted positive	predicted negative	total
actually positive (100)	TP = 62	FN = 38	100
actually negative (900)	FP = 17	TN = 883	900
total	79	921	1000

accuracy = (62 + 883)/1000 = 0.945; precision = 62/79; recall = 62/100

Figure 27.1: The confusion matrix of the worked example on 1000 connections: 62 attacks caught, 38 missed, 17 false alarms, and 883 clean connections correctly rejected. The rows sum to 100 positive and 900 negative, the columns to 79 predicted positive and 921 predicted negative, and everything to 1000. The two green cells are the two ways to be right, the two red cells the two ways to be wrong. Accuracy 0.945 adds the two green cells and therefore cannot distinguish the 62 that were caught from the 38 that were missed.

Proof. Write $a = \mathrm{TP}$, $b = \mathrm{FP}$, $c = \mathrm{FN}$. Then $P = a/(a+b)$, $R = a/(a+c)$, and

$$\frac{2PR}{P+R} = \frac{2a^2/\big((a+b)(a+c)\big)}{a(2a+b+c)/\big((a+b)(a+c)\big)} = \frac{2a}{2a+b+c} = F_1.$$

The harmonic–arithmetic inequality $4PR \le (P+R)^2$ is equivalent to $(P-R)^2 \ge 0$, and equality occurs only when $P = R$. □

The harmonic property pulls the value down toward the smaller component: precision 0.9 with recall 0.2 gives $F_1 \approx 0.327$, not 0.55; in our example $F_1 = 0.6927$, slightly below the midpoint 0.7024.

Three defects of the summary must be named one by one, because the remedies differ. Precision is blind to attacks that slip through: a model that flags every connection obtains precision 0.1, namely its own prevalence, but recall 1. F_1 ignores TN entirely: two models with the same precision and recall but ten times as many false alarms report an identical F_1, even though their operating costs are far apart. F_1 also weighs both components as equal, even though one missed attack is usually far more expensive than one false alarm.

Challenge (problem solver)

A report carries only three numbers: “accuracy 0.9720 on 5000 test connections; precision 0.000; model ready to deploy”. Determine the whole confusion matrix that is consistent with that report, explain why its recall is zero and not some other number, then find the largest prevalence that is still possible.

Hint 1: accuracy only adds the two diagonal cells, so $5000(1 - 0.9720) = 140$ errors are free to be split between the two remaining cells.

Hint 2: precision 0.000 means $\mathrm{TP} = 0$, so FN equals the number of positive examples and FP is the rest. But precision with $\mathrm{TP} = \mathrm{FP} = 0$ is $0/0$, not 0; the figure 0.000 in that report therefore demands at least one false alarm.

If you get stuck, go back to the section Deceptive accuracy on unbalanced data in this chapter.

27.2 Deceptive accuracy on unbalanced data

Level: basic

Accuracy is the most often reported and the easiest to deceive: it counts all errors at the same price on a collection whose composition is never stated. In everyday language: accuracy is a report card computed as if the two groups were balanced.

Definition 27.3 (Balanced accuracy)**.** Let π be the prevalence of the positive class. *Balanced accuracy* is $(\mathrm{rec} + \mathrm{spec})/2$, that is the average of the correct rates on the two classes. This quantity does not change when the numbers of positive and negative examples are changed without changing those two rates.

Ordinary accuracy has no such property, and the consequence can be written exactly.

Proposition 27.2. *A classifier that always answers negative obtains accuracy* $1 - \pi$*. Hence a model whose accuracy is below* $1 - \pi$ *loses to a baseline that learns nothing, and a report that states an accuracy without stating* π *cannot be judged.*

Proof. Of n examples, πn are labeled positive and all of them are wrong, while $(1 - \pi)n$ are labeled negative and all of them are right. The accuracy is $(1 - \pi)n/n = 1 - \pi$. Any other model with smaller accuracy therefore makes more errors than that baseline. □

Read it aloud (auditory learner)

Two voices, read alternately with a friend. Tap the table every time the word “zero” is spoken.
“A hundred cards. How many red?” “Ten.” “How many white?” “Ninety.” “What does the machine answer?” “White. All white.” “How many does it get right?” “Ninety.” “So its accuracy?” “Ninety percent.” “And the red ones?” “Zero. Not a single one.” “Was that number lying?” “No. It answered only one question.” “Which question?” “How many of everything there is did I guess right.” “Which question went unanswered?” “Of the truly dangerous ones, how many did I find.”
Repeat the three middle lines, faster and faster, with the same taps: what is counted is everything, what is lost goes unmentioned, what is lost goes unmentioned.
The second voice closes softly: the small class drowns easily inside the average of the large class, and that average never names who drowned.
The line the reader must repeat, once, in a clear voice: accuracy adds up what is right, not what matters, so on an unbalanced class ninety percent can mean zero.

In our worked example $\pi = 0.1$, so the baseline is 0.9 and the model with 0.945 really is better. Now change only the prevalence, without changing the quality of the model, and watch what happens to the same number.

Example 27.1 (One attack for every thousand connections)*.* Take 100 000 connections with 100 attacks. Scores on clean connections are normally distributed with mean 0 and standard deviation 1; scores on attacks with mean 2.6 and standard deviation 1. The threshold is set at 2.3263, the 0.99 quantile of the negative score distribution, so by design about one percent of clean connections will be flagged. One run with a fixed seed gives:

$$\mathrm{TP} = 62, \quad \mathrm{FP} = 1013, \quad \mathrm{FN} = 38, \quad \mathrm{TN} = 98\,887.$$

Its accuracy is $(62 + 98\,887)/100\,000 = 0.989490$. The baseline is $99\,900/100\,000 = 0.999000$. Its precision is $62/1075 = 0.057674$, its recall $62/100 = 0.620000$, and its balanced accuracy 0.804930.

That number 0.9895, rounded, becomes “99 percent”. It is not merely uninformative: it is lower than the baseline by 0.009510, nearly a full percentage point. The model makes ten

times as many errors as the program that always answers “clean”, 1051 against 100, and all of them are false alarms, 1013 of them, or about 16.3 per attack caught.

Here rounding adds damage: 0.9895 and 0.9990 both become “99 percent”, even though they are separated by the whole difference that matters. A good report therefore writes three or four decimal places, states n, and states the prevalence; without all three, the reader does not know whether 99 percent is an achievement or a failure.

Should accuracy then not be used at all? It is legitimate when the two errors cost the same, when the prevalence is near one half, and when the threshold is fixed. Outside that it must be accompanied by precision and recall, or replaced by balanced accuracy: 0.8049 is more honest than 0.9895, because the minority class is caught only 62 percent of the time.

Do it by hand (kinesthetic learner)

Materials: 100 cards, 90 white-faced and 10 red-faced, a marker, and one sheet of tally paper with two columns.
Write “clean” on the ninety white cards and “attack” on the ten red ones. Shuffle, then deal them into ten piles. First rule: every pile is answered “clean” without turning over a single card. Record the correct answers, then turn all the cards over: ninety right, ten red ones through. Write on the board: accuracy ninety percent, recall zero percent.
Now the second rule. Put a dot in the corner of six red cards, and without telling the player, put the same dot in the corner of four white cards. Answer “attack” only when that dot is visible. Count again: six red caught, four red through, four white accused, eighty-six white right.
What must become visible: accuracy moves only from ninety to ninety-two percent, while recall jumps from zero to sixty percent. The number that changes most is precisely the number missing from the first report.

27.3 The ROC curve, AUC, and prevalence

Level: intermediate

Every threshold gives one point on two different rates; choosing a threshold without looking at all the possibilities is guessing. In everyday language: we sweep the threshold from loose to strict, then draw the trail the model leaves behind.

receiver operating characteristic (*kurva roc*) — the trail a model leaves when its threshold is swept from loose to strict, drawn on two rates that do not depend on prevalence

Definition 27.4 (ROC curve and AUC)**.** For a threshold t, the true positive rate and the false positive rate are

$$\mathrm{TPR}(t) = P(S > t \mid y = 1), \qquad \mathrm{FPR}(t) = P(S > t \mid y = 0),$$

with S the model score. The ROC curve is the set of points $(\mathrm{FPR}(t), \mathrm{TPR}(t))$ for all t. The area under that curve is called the AUC, and for finite data it is computed as the trapezoid area between the points actually observed.

The AUC has a far simpler meaning than its name, and that meaning is what made it popular.

Proposition 27.3. *For data without tied scores, the AUC equals the probability that the score of one randomly drawn positive example exceeds the score of one randomly drawn negative example. When there are tied scores, each tied pair counts as one half.*

Proof. Sort all scores from large to small and sweep the threshold downward. Each time the threshold passes one positive example, TPR rises by $1/n_+$ at the height of FPR at that moment, namely the fraction of negative examples already passed. Each time the threshold passes a negative example, FPR rises at constant TPR and adds no area. The total area is therefore

$$\mathrm{AUC} = \frac{1}{n_+} \sum_{i:\ y_i=1} \mathrm{FPR}_i = \frac{1}{n_+ n_-} \#\{(i,j) :\ y_i = 1,\ y_j = 0,\ S_i > S_j\},$$

which is the probability claimed, and that last term is what the Mann–Whitney statistic computes. Tied scores add no area, so each such pair contributes one half. □

The AUC does not see prevalence: if every negative example is duplicated ten times, the ordering of scores does not change and the AUC stays. How to compute it on paper is shown in the Worked examples section.

Now the part that makes the ROC curve dangerous: sweep the threshold in a one-in-a-thousand simulation.

threshold	recall	FPR	precision	accuracy
3.0000	0.450	0.00132	0.254	0.99813
2.8000	0.510	0.00263	0.162	0.99688
2.6000	0.550	0.00467	0.105	0.99488
2.3263	0.620	0.01014	0.058	0.98949
2.0000	0.740	0.02269	0.032	0.97707
1.5000	0.860	0.06729	0.013	0.93264

On the ROC curve those rows move from $(0.001; 0.45)$ to $(0.067; 0.86)$: a trail that rises steeply near the vertical axis, exactly what every practitioner wants to see. Its AUC is 0.966448. But the precision column says something else. Even at the strictest threshold, where only 132 clean connections are flagged, the precision is 0.254: three out of four alarms are still wrong. When recall is raised to 0.86, the precision is 0.013.

Figure 27.2 draws the six rows of that table on one threshold axis: recall climbs while precision falls, and no point gives both at once.

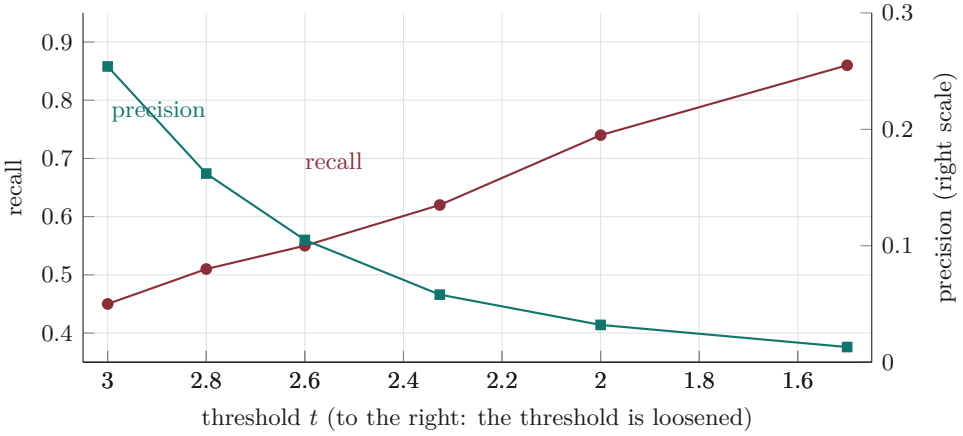


Figure 27.2: Precision and recall at the six thresholds of the previous table, drawn on one threshold axis that is turned so that to the right the threshold is loosened. Recall climbs from 0.450 at $t = 3.0$ to 0.860 at $t = 1.5$, while precision falls from 0.254 to 0.013: the two curves move in opposite directions and there is not one point that gives both at once. The precision scale is on the right, 0 to 0.30; at $t = 2.3263$, the threshold used in the simulation, precision 0.058 with recall 0.620.

The two readings do not contradict each other: they measure different things.

Proposition 27.4. *The ROC curve and the AUC depend only on the conditional score distributions $S \mid y = 1$ and $S \mid y = 0$, so they do not change when the prevalence π is changed. Precision at a fixed threshold does change, and follows*

$$\operatorname{prec}(t) = \frac{\pi \operatorname{TPR}(t)}{\pi \operatorname{TPR}(t) + (1 - \pi) \operatorname{FPR}(t)}.$$

Proof. By definition, TPR and FPR are probabilities conditional on y, so neither contains π; a curve assembled from the two contains no π either. For precision, the law of total probability gives $P(\hat{y} = 1) = \pi \operatorname{TPR} + (1 - \pi)\operatorname{FPR}$, while $P(y = 1, \hat{y} = 1) = \pi \operatorname{TPR}$. Bayes' rule from Chapter 17 gives the form above. □

The consequence can be measured. If prevalence is changed to one half without changing the conditional score distributions, the AUC stays at 0.967061 (the difference 0.0006 is only Monte Carlo error), while *average precision* jumps from 0.287636 to 0.967544. At a prevalence of one in a thousand the baseline is $\pi = 0.001$, so 0.288 is still far better than guessing, but far closer to operational reality than 0.966.

The rule of thumb is simple: when the positive class is rare, report a precision–recall curve, not only the ROC curve, and state the prevalence. The ROC curve remains useful for comparing two models on the same population because it is not polluted by composition; it misleads as soon as it is used to conclude usefulness on another population or at a particular threshold. The AUC is also blind to calibration: a model whose scores are twice too large has exactly the same AUC, because the ordering does not change. Those two defects carry us to the next section.

27.4 Calibration: does the number mean a probability

Level: intermediate

An operator who receives an alarm needs to know one thing that neither precision nor the AUC answers: when the model writes 0.8, are eight of ten such cases really positive. In everyday language: calibration checks whether the weather forecast that says "seventy percent rain" is really right on seven of ten such days.

calibration (*kalibrasi*) — the agreement between the number a model promises and what actually happens in the group sharing that same promise

Definition 27.5 (Perfect calibration)**.** A score $S \in [0, 1]$ is called perfectly calibrated when $P(y = 1 \mid S = s) = s$ for every possible s. On finite data, calibration is checked by grouping scores into intervals, then comparing the mean score in that interval with the observed positive rate. The absolute difference, weighted by the size of the interval, is called the *expected calibration error*.

A worked example with 1000 predictions shows the shape of the check.

score interval	count	positive	mean score	positive rate
$[0.5; 0.6)$	40	21	0.55	0.525
$[0.6; 0.7)$	60	36	0.65	0.600
$[0.7; 0.8)$	120	90	0.75	0.750
$[0.8; 0.9)$	200	176	0.85	0.880
$[0.9; 1.0]$	580	551	0.95	0.950

The largest difference is in the second interval, 0.05, and the overall calibration error is

$$\tfrac{40}{1000}(0.025) + \tfrac{60}{1000}(0.050) + \tfrac{120}{1000}(0) + \tfrac{200}{1000}(0.030) + \tfrac{580}{1000}(0) = 0.010.$$

The direction of the error differs between intervals, but its size is small: one percentage point. The cross-check is free: the counts in the five intervals add to 1000, and the positives to 874.

Calibration is measured by squared error, and there is a neat identity there.

Proposition 27.5 (Brier for calibrated scores)**.** *Let S be perfectly calibrated and $B = \mathbb{E}[(S-y)^2]$ be the Brier score. Then*

$$B = \mathbb{E}\big[S(1-S)\big].$$

In particular the constant score $S = \pi$ is perfectly calibrated with $B = \pi(1-\pi)$, and that constant score is the least sharp calibrated forecaster.

Proof. Conditional on $S = s$, the variable y takes the value 1 with probability s. Then

$$\mathbb{E}\big[(s-y)^2 \mid S = s\big] = s(s-1)^2 + (1-s)s^2 = s(1-s)\big[(1-s)+s\big] = s(1-s).$$

Taking the expectation over S gives the first form. For constant $S = \pi$, the calibration condition demands $P(y=1) = \pi$, which is indeed true, and the formula gives $\pi(1-\pi)$. □

That identity explains two things: calibration alone is not enough, since the constant score π is perfectly calibrated but distinguishes no examples; and for rare events the Brier baseline is very low, so a model that looks good on accuracy can lose badly on Brier. In our simulation $\pi(1-\pi) = 0.000999$, while the hard labels of the model give a Brier equal to its error rate, $1051/100\,000 = 0.010510$. The Brier skill score, $1 - B/B_{\text{garis dasar}}$, is -9.52: the model is ten times worse than forecasting the constant prevalence.

That negative number is no oddity: hard labels throw away the information about uncertainty, and for rare events that information is the most valuable of all. A model that outputs 0.06 is saying something true, and a report that carries only accuracy throws that sentence away.

27.5 Uncertainty of metrics

Level: advanced

All the numbers above are computed from a limited number of examples, so all of them wobble when the data is replaced. In everyday language: a metric without an interval is a single shot with no indication of how wide the target is.

Chapter 20 already derived the Wilson interval and proved its properties, so we do not repeat it here. What must be remembered is its form for a proportion k/n:

$$\frac{\hat p + \dfrac{z^2}{2n} \pm z\sqrt{\dfrac{\hat p(1-\hat p)}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}, \qquad z = 1.96.$$

For younger readers (age 10 and up)

Two goalkeepers practice stopping penalties. The first stops nine out of ten shots. The second stops nine hundred out of a thousand shots. The newspaper prints the same headline for both: ninety percent.

But we feel that the two sentences are not equally strong, and that feeling is right. The first keeper has faced only ten shots. Perhaps he really is excellent, perhaps he was merely lucky ten times in a row. The second keeper has faced a thousand shots, and luck

over that long a run is almost impossible. The same number, a different strength. The question that makes one curious: if we may write only one sentence about the first keeper, may "ninety percent" still be written? Yes, provided the sentence is completed. In this chapter we will learn why the same number means "between fifty-nine and ninety-eight percent" on ten shots, but only "between eighty-eight and ninety-two percent" on a thousand shots, and why multiplying the shots a hundredfold narrows the answer only to a tenth.

The following table applies that formula to accuracy, with $z = 1.96$.

data	$\hat{p}$	Wilson interval	width
9/10	0.900	(0.5959; 0.9821)	0.3863
900/1000	0.900	(0.8798; 0.9171)	0.0372
10/10	1.000	(0.7225; 1.0000)	0.2775
0/1000	0.000	(0.0000; 0.0038)	0.0038

The first two rows answer the two goalkeepers. The point estimate is the same, 0.9, but the first interval is 0.3863 wide and the second 0.0372: ten times narrower, not a hundred times.

The third row is the case that appears most often: not a single error in 10 trials, and then "accuracy 100 percent" is written. The honest Wilson interval, $(0.7225; 1)$, is 0.2775 wide: the error rate may still be 27.75 percent. At the other end, zero events in 1000 trials gives an upper bound of 0.003827, while the rule of three from Chapter 20 gives 0.003; either is acceptable, as long as the procedure is named.

Comparing two models by checking whether their intervals overlap is a wrong habit: both are tested on the same data, and that dependence sharpens the comparison rather than blurring it.

Proposition 27.6 (The difference of two paired models)**.** *Let b be the number of examples that model A gets right and model B gets wrong, and c the reverse, on the same n examples. Their accuracy difference is $d = (b - c)/n$ with variance*

$$\operatorname{Var}(d) = \frac{1}{n}\Big(p_b + p_c - (p_b - p_c)^2\Big),$$

so that its estimated standard error is $\sqrt{b + c - (b - c)^2/n}\,/\,n$.

Proof. Every example falls in one of four cells: right for both, wrong for both, right only for A, right only for B. Since the examples are independent, (b, c) is multinomial with probabilities p_b, p_c. Then $\operatorname{Var}(b - c) = n(p_b + p_c - (p_b - p_c)^2)$, and dividing by n^2 gives the formula above. □

At $b = 37$, $c = 15$, and $n = 1000$, the difference is 0.022 with standard error 0.007177, so the 95 percent interval is $(0.007932; 0.036068)$, entirely above zero, and the exact McNemar test gives a p-value of 0.003185. The naive treatment, which assumes the two models independent, gives the interval $(-0.004296; 0.048296)$, which contains zero. The two models can therefore be declared different, provided the comparison is paired and the test uses the difference, not two separate intervals.

Finally, the size of the test determines how small a difference can be seen. To detect a difference δ at level α and power $1 - \beta$,

$$n \approx \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 \,(p_1(1 - p_1) + p_0(1 - p_0))}{\delta^2}.$$

At $p_0 = p_1 = 0.9$, level 0.05, and power 0.8, a difference of one point demands $n \approx 14\,128$ examples per model, a difference of two points demands 3532. A test set of a thousand examples cannot separate two models one point apart.

27.6 Data leakage and nested cross-validation

Level: advanced

All the computations above assume one thing that must not be violated: the test number is measured on data the model has never seen, and never seen either by the procedure that chose the model. In everyday language: an exam score means something only if the questions were never leaked, including by the teacher who wrote them.

data leakage (*kebocoran data*) — exam answers that slip accidentally into the study material, so that the score goes up without any skill

Definition 27.6 (Data leakage)**.** Leakage occurs when information that would not be available at the moment the model is used enters training, model selection, or reporting, so that the evaluation number measures the ability to memorize, not the ability to generalize.

Its forms are various, and the most frequent ones are invisible. Computing a preprocessing mean or median from all the data, test data included, is leakage: that statistic should come from training data only. Selecting the 50 best features by correlation with the label before separating training from test is also leakage, since the feature filter has already seen the test labels. Tuning hyperparameters on the test set and reporting the best number is the leakage most often committed knowingly. On grouped data, a random split places the same group on both sides; on time series, it leaks the future into the past. The hardest to detect: a feature that is a consequence of the label, for example a column "action taken" filled in after the incident is known.

Feature-selection leakage can be measured, and its size is startling. Take 200 randomly labeled examples and 2000 features of normal noise, with no signal at all. Select the 20 features with the largest absolute correlation with the label, then cross-validate with five folds using a nearest centroid classifier. If the selection is done once on all the data before the folds are formed, the accuracy is 0.7800; if the selection is repeated inside each fold using only its training rows, 0.4600. On three other seeds the numbers are 0.775 and 0.415, 0.820 and 0.545: the leaky version is always high, the honest one always near 0.5. That number 0.78 is an illusion from beginning to end, and not one line of its code is mistyped.

The protocol is one sentence: separate first, then compute anything. Split the data into training, validation, and test; compute preprocessing statistics from training only; tune hyperparameters on validation; touch test once, at the end, for one number. When cross-validation is used, the preprocessing lives inside the fold. For grouped data, the folds follow the groups; for time data, they follow the time order.

Figure 27.3 draws that order as three boxes and one forbidden arrow.

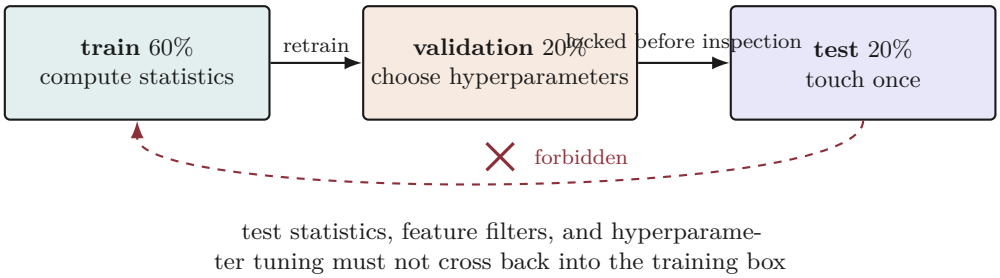


Figure 27.3: Three separate data sets, with proportions 60/20/20; on 1000 connections that means 600, 200, and 200. Preprocessing statistics are computed from the training box, hyperparameters are chosen in the validation box, and the test box is touched once at the end. The dashed arrow from test back to training is leakage: that single arrow is enough to raise accuracy from 0.4600 to 0.7800 on data that is pure noise.

What remains is one subtle problem. When cross-validation is used both to choose hyperparameters and to report quality, the reported number is the best of many trials on the same data, so it is too optimistic. The way out is two layers of folds.

nested cross-validation (*validasi silang bersarang*) — two layers of folds, the outer layer to judge and the inner layer to choose, like two separate examination rooms

Definition 27.7 (Nested cross-validation)**.** Let Λ be a set of hyperparameters. The data is split into K outer folds. For each outer fold k: train on the remaining data with every $\lambda \in \Lambda$ using J inner folds, choose the best λ_k, retrain on the whole training part of fold k with λ_k, then measure on fold k. The reported number is the average of those K numbers. Judging and selection therefore use different data.

Its cost is K times higher, and that is why this protocol is often skipped; without the outer layer, not one number in the report can be read as an estimate of quality on new data.

27.7 Why one test is not enough

Level: intermediate

The sentence "we tested the model on 1000 examples and obtained an accuracy of 0.900" sounds conclusive, yet it carries three uncertainties. In everyday language: one weighing does not tell you whether the scale wobbles, whether the goods changed, or whether we picked the most pleasant result.

The first is statistical and already computed: at $n = 1000$ the 95 percent interval is 0.037 wide, and separating two models one point apart demands 14 128 examples. The conclusion "A is better" from a difference that small is a conclusion about a coin toss.

The second is selective, and its name is *winner's curse*. Training 20 variants of identical quality and then reporting the best means reporting the maximum of 20 wobbles: in a simulation with true accuracy 0.90 and $n = 1000$, the mean of each variant is 0.899987 while the mean of their maximum is 0.917337, so 1.7 points of that number are luck. When the variants are dependent, the wobble is smaller, but its direction is always upward.

The third is procedural: a test set used repeatedly stops being a test set. If each comparison has a 5 percent chance of deceiving, the chance that at least one of 20 comparisons deceives is $1 - 0.95^{20} = 0.6415$, larger than the chance of being right.

The remedies match: report intervals, separate the data used for selection from the data used for reporting, and lock the test set before the experiment begins. If the test has already been used repeatedly, the next number is called an exploratory number.

27.8 A reporting checklist

Level: basic

These seven lines separate a report that can be checked from one that merely convinces. In everyday language: a receipt does not make the goods arrive, but without a receipt nothing can be traced.

1. The size of each class and its prevalence, not only the total n.
2. The four numbers of the confusion matrix, then precision and recall at the threshold actually used, together with the value of that threshold.
3. A confidence interval for every main metric, with the method named; for small n use Wilson, not Wald.
4. A precision–recall curve when the positive class is rare, together with average precision and its baseline π.

5. Calibration: the score interval bounds, the number of examples per interval, and the calibration error.
6. The separation protocol: what was computed from which data, how many times the test data was touched, and how many variants were tried; the best number without that denominator is a selected number, not a quality number.
7. The spread of results across seeds or across folds, together with the code and the random seed.

What must not be there: accuracy without prevalence, AUC without a curve, and the phrase “ready to deploy”.

Step-by-step recipe (procedural learner)

Four steps for reporting evaluation results that do not mislead:

1. Separate the data first: training, validation, test, with their proportions and the splitting key, written down before a single computation is run.
2. Compute preprocessing and select features from training only, tune hyperparameters on validation only, then save test for a single touch.
3. Report all four cells of the confusion matrix together with n, π, and the threshold used, then precision, recall, and a confidence interval for every metric.
4. State how many variants were tried and the spread of results across seeds or across folds, so that the reader knows how many degrees of freedom have already been spent.

A warning about order: step 1 must not come after step 2 or step 3. Statistics computed before the data is separated have already moved test information into training, and after that no number can repair it; the whole report then measures memory.

27.9 Worked examples

Level: intermediate

The three computations below can be repeated on paper.

First, the confusion matrix from the author’s own working notes, on 35 069 test connections: TP $= 23\,279$, FP $= 997$, FN $= 590$, TN $= 10\,203$.

$$\text{prec} = \frac{23\,279}{24\,276} = 0.958931, \qquad \text{rec} = \frac{23\,279}{23\,869} = 0.975282,$$
$$F_1 = \frac{46\,558}{48\,145} = 0.967037, \qquad \text{acc} = \frac{33\,482}{35\,069} = 0.954746,$$

and its balanced accuracy is 0.943132. One thing in those notes is inconsistent: one sentence states 11 200 attacks and 23 869 normal, while the matrix places 23 869 in the positive row. The arithmetic is right, the class labels are swapped, and the prevalence must be read from the matrix, not from the sentence.

Second, the AUC of a small data set. Positives scoring 0.92, 0.81, 0.55 against negatives scoring 0.70, 0.41, 0.35, 0.22 give 12 pairs; 0.92 and 0.81 beat all four, 0.55 beats three, so $11/12 = 0.916667$. If one positive score equals one negative score, that pair counts as one half and the result is 0.875; `sklearn` gives the same number when y and the scores are fed in as vectors.

Third, the effect of prevalence on precision when sensitivity 0.90 and specificity 0.95 are held fixed. With $\text{prec} = \pi(0.9)/(\pi(0.9) + (1-\pi)(0.05))$:

π	prec	true alarms	false alarms
0.001	0.017699	1	56
0.010	0.153846	1	5.5
0.100	0.666667	2	1
0.500	0.947368	18	1

The last column is the reciprocal of precision minus one, namely the number of false alarms for every true alarm. At a prevalence of one in a thousand, one true alarm arrives with 55.5 false alarms; at a prevalence of one half, with 0.056. The same model, the same quality, and a completely different experience for the operator.

Take the matrix of 1000 predictions from earlier and recompute its five metrics without looking at the text. Then answer: how many FP must be removed, without changing TP and FN, for precision to become 0.900? Hint: precision $62/(62+x) = 0.9$ gives $x = 6.89$, so FP must fall from 17 to at most 6. Check that the accuracy after that change is 0.956.

27.10 Computational lab: one number, four ways to read it

Level: advanced

Every number in this chapter is computed by one short program, which also shows two things a formula cannot show: how far leakage raises a number, and how far the winner among many variants is flattered. The code is run as it stands with `numpy` and `scipy`.

Computational lab

```
import numpy as np
from math import sqrt
from scipy.stats import norm

z = norm.ppf(0.975)


def wilson(k, n):
    p, z2 = k / n, z * z
    mid = (p + z2 / (2 * n)) / (1 + z2 / n)
    half = z * sqrt(p * (1 - p) / n + z2 / (4 * n * n)) / (1 + z2 / n)
    return mid - half, mid + half


def roc_pr(spos, sneg):
    s = np.concatenate([spos, sneg])
    y = np.concatenate([np.ones(len(spos)), np.zeros(len(sneg))])
    o = np.argsort(-s)
    s, y = s[o], y[o]
    tp, fp = np.cumsum(y), np.cumsum(1 - y)
    tpr, fpr = tp / len(spos), fp / len(sneg)
    prec = tp / (tp + fp)
    auc = np.sum(np.diff(np.concatenate([[0], fpr])) *
                 (tpr + np.concatenate([[0], tpr[:-1]])) / 2)
    ap = np.sum(np.diff(np.concatenate([[0], tpr])) * prec)
    return auc, ap


rng = np.random.default_rng(20261025)
Nneg, Npos = 99900, 100
sneg = rng.normal(0.0, 1.0, Nneg)
spos = rng.normal(2.6, 1.0, Npos)
t = 2.3263
tp = int((spos > t).sum()); fp = int((sneg > t).sum())
```

```
fn, tn = Npos - tp, Nneg - fp
n = Nneg + Npos
print("TP=%d FP=%d FN=%d TN=%d" % (tp, fp, fn, tn))
print("accuracy %.6f  baseline %.6f" % ((tp + tn) / n, Nneg / n))
print("precision %.6f  recall %.6f" % (tp / (tp + fp), tp / Npos))
print("balanced accuracy %.6f" % ((tp / Npos + tn / Nneg) / 2))
print("brier hard labels %.6f  brier constant %.6f"
      % ((fp + fn) / n, 0.001 * 0.999))
a1, p1 = roc_pr(spos, sneg)
print("pi=0.001: AUC=%.6f  AP=%.6f" % (a1, p1))
a2, p2 = roc_pr(rng.normal(2.6, 1.0, 50000), rng.normal(0.0, 1.0, 50000))
print("pi=0.500: AUC=%.6f  AP=%.6f" % (a2, p2))
for k, m in ((9, 10), (900, 1000), (10, 10), (0, 1000)):
    lo, hi = wilson(k, m)
    print("%4d/%-4d (%.6f, %.6f) width %.6f" % (k, m, lo, hi, hi - lo))
b, c = 37, 15
d = (b - c) / 1000
se = sqrt(b + c - (b - c) ** 2 / 1000) / 1000
print("difference %.6f  se %.6f  CI (%.6f, %.6f)"
      % (d, se, d - z * se, d + z * se))
rt = np.random.default_rng(11)
a = rt.binomial(1000, 0.90, (20000, 20)) / 1000
print("20 variants: mean %.6f  mean max %.6f"
      % (a.mean(), a.max(axis=1).mean()))

rl = np.random.default_rng(4)
nn, dd, kk = 200, 2000, 20
X = rl.normal(size=(nn, dd))
y = rl.permutation(np.repeat([0, 1], nn // 2))
corr = np.array([abs(np.corrcoef(X[:, j], y)[0, 1]) for j in range(dd)])
sel_all = np.argsort(-corr)[:kk]


def cv(pick):
    idx = rl.permutation(nn)
    ok = 0
    for f in np.array_split(idx, 5):
        tr = np.setdiff1d(idx, f)
        cols = pick(tr)
        m1 = X[np.ix_(tr[y[tr] == 1], cols)].mean(0)
        m0 = X[np.ix_(tr[y[tr] == 0], cols)].mean(0)
        d1 = ((X[np.ix_(f, cols)] - m1) ** 2).sum(1)
        d0 = ((X[np.ix_(f, cols)] - m0) ** 2).sum(1)
        ok += (np.where(d1 < d0, 1, 0) == y[f]).sum()
    return ok / nn


def top_in(tr):
    c = np.array([abs(np.corrcoef(X[tr, j], y[tr])[0, 1]) for j in range(dd)])
    return np.argsort(-c)[:kk]


print("feature selection: leaky %.4f  honest %.4f"
      % (cv(lambda tr: sel_all), cv(top_in)))
```

The actual output, as it stands:

```
TP=62 FP=1013 FN=38 TN=98887
accuracy 0.989490  baseline 0.999000
precision 0.057674  recall 0.620000
balanced accuracy 0.804930
brier hard labels 0.010510  brier constant 0.000999
```

```
pi=0.001: AUC=0.966448  AP=0.287636
pi=0.500: AUC=0.967061  AP=0.967544
   9/10   (0.595850, 0.982124) width 0.386274
 900/1000 (0.879848, 0.917091) width 0.037243
  10/10   (0.722467, 1.000000) width 0.277533
   0/1000 (0.000000, 0.003827) width 0.003827
difference 0.022000  se 0.007177  CI (0.007932, 0.036068)
20 variants: mean 0.899987  mean max 0.917330
feature selection: leaky 0.7800  honest 0.4600
```

Four things must be read from that output. First, the two numbers on the second line must appear side by side: 0.989490 and 0.999000. The direction of the difference decides the whole interpretation: the model under test is below the baseline.

Second, the AUC barely moves, 0.966448 against 0.967061, while AP moves from 0.287636 to 0.967544, a factor of 3.4: numerical evidence for the previous proposition.

Third, the width of the interval shrinks proportionally to $1/\sqrt{n}$, from 0.386274 at $n = 10$ to 0.037243 at $n = 1000$, and 0/1000 gives an upper bound of 0.003827.

Fourth, the last two lines carry two diseases that are invisible in a single number: the paired difference 0.022 with interval $(0.007932; 0.036068)$ does not contain zero, yet $(0.8798; 0.9171)$ and $(0.8563; 0.8969)$ overlap widely; and the winner among 20 identical variants is reported as 0.917337 although the true quality is 0.899987.

The program proves not one proposition in this chapter; it checks claims that have already been proved. The simulation uses distributions we assume, and honest cross-validation on noise data gives 0.4600, not 0.5000: even a number without signal wobbles.

Note for instructors

The teaching order I suggest: start from the vendor report in the chapter opening, write 99 percent and 99.9 percent on the board, then ask the class to decide which is better. Almost the whole class picks the detector, and out of that unease the confusion matrix appears as an answer, not as a definition to be memorized. Once precision and recall exist, then F_1 and balanced accuracy can enter. The ROC curve is best drawn by hand for the six points of the threshold table, because there the students see for themselves that a point which looks good on the ROC curve can have precision 0.013.

Two typical misconceptions. First, the AUC is taken to be a context-free measure of quality, so the threshold is chosen at random; the symptom shows when students cannot explain the precision at the threshold in use. Second, cross-validation is taken to be always honest, even though feature selection or standardization outside the fold is enough to raise the number by 30 points on data that is pure noise.

Board activity: split the class into two groups, one computing precision for $\pi = 0.001$ and one for $\pi = 0.5$ with the same sensitivity and specificity, then ask both to defend their results. The first group finds 0.018 and must explain why the same model feels different. Sign that the class is ready to go on: they can name one situation in which accuracy 0.99 is worse than not using a model at all.

27.11 Summary

LEVEL: BASIC

Summary.

- The four numbers of the confusion matrix play unsymmetric roles: precision answers the operator's question, recall the policy designer's question, and both depend on the threshold; F_1 is their harmonic mean and ignores TN.

- The baseline accuracy is $1 - \pi$: at $\pi = 0.001$ a model with 0.989490 loses to a program that always answers negative (0.999000), while balanced accuracy 0.804930 is more honest because it does not depend on composition.
- The AUC equals the probability that a random positive score exceeds a random negative score, so it depends only on the conditional score distributions; it does not change when prevalence changes, whereas precision and average precision do, from 0.288 to 0.968.
- Calibration checks whether a score of 0.8 means eight out of ten. For calibrated scores, Brier equals $\mathbb{E}[S(1 - S)]$; the constant score π is perfectly calibrated but not sharp. Hard labels here give 0.010510 against 0.000999.
- The width of the interval shrinks proportionally to $1/\sqrt{n}$: 0.3863 at $n = 10$ and 0.0372 at $n = 1000$. To compare two models, use the paired difference: $b = 37$, $c = 15$ gives $(0.007932; 0.036068)$, whereas the unpaired treatment gives an interval that contains zero.
- One test is not enough, and leakage raises numbers falsely: 0.7800 when features are selected outside the fold, 0.4600 when inside; the winner among 20 identical variants is flattered by 1.7 points; and 20 comparisons at the 5 percent level give a chance of 0.6415 that one of them deceives.

27.12 Exercises

Level: intermediate

Exercise 27.1.

1. Assess this claim: "Model A reaches an accuracy of 99.1 percent and model B 98.7 percent on the same test set, so A is better." True, false, or not enough evidence? Name the part that is wrong and two additional numbers that would strengthen it.
2. Prove that the Brier score for hard 0/1 labels equals the error rate, $(\mathrm{FP} + \mathrm{FN})/n$. Why does that equality fail for scores that are probabilities?
3. Prove that a constant score $S = s$ minimizes the Brier score exactly when $s = \pi$, with minimum value $\pi(1 - \pi)$.
4. Given TP = 180, FP = 45, FN = 20, TN = 1755. Compute precision, recall, F_1, accuracy, specificity, and balanced accuracy to six decimal places, then check with `sklearn.metrics`.
5. Compute the 95 percent Wilson interval for 95/100 and 950/1000. Compare the two widths, then explain why that comparison must not be used to conclude which model is better.
6. Compute the AUC from five positives scoring 0.95, 0.88, 0.70, 0.60, 0.40 and five negatives scoring 0.75, 0.65, 0.50, 0.30, 0.10. Write down the number of correct pairs, then check with `roc_auc_score`.
7. A model has sensitivity 0.80 and specificity 0.99. Compute its precision at $\pi = 0.002$ and $\pi = 0.2$, then determine the minimum prevalence for precision to be at least 0.5.
8. You have 6000 records from 300 hosts, about 20 records each, and 40 hyperparameter settings. Design an evaluation protocol without leakage: state the data split, where preprocessing is computed, where hyperparameters are chosen, and how many times test is touched.
9. Assess this claim: "Because the AUC of this model is 0.95, the model is useful at any threshold we choose." True, false, or not enough evidence? Include one supporting computation.

Hints for the three hardest items. Item 2: write $(s - y)^2$ for $s \in \{0, 1\}$; its value is 1 exactly on an error. Item 3: differentiate $\mathbb{E}[(s - y)^2] = s^2 - 2s\pi + \pi$ with respect to s,

find its minimum, then use the Brier proposition in this chapter. Item 7: its precision is $0.8\pi/(0.8\pi + 0.01(1-\pi))$; solve $0.8\pi \geq 0.5(0.8\pi + 0.01(1-\pi))$ for π.

For researchers: limits and next steps

What is proved in this chapter: the accuracy of a constant classifier is $1-\pi$; F_1 is the harmonic mean of precision and recall with $F_1 \leq (P+R)/2$; the AUC equals the probability of a correct ranking together with its Mann–Whitney form; ROC and AUC are invariant to prevalence while precision is not; for calibrated scores Brier equals $\mathbb{E}[S(1-S)]$; and the variance formula for the difference of two paired proportions.

What is popular but not a theorem: that "AUC 0.9 means a good model"; that the threshold 0.5 is neutral; that cross-validation is always honest; and that overlapping intervals mean there is no difference. The last three are mathematically wrong, not merely incomplete. *Average precision* is also not an unbiased estimate of the area under the population precision–recall curve: it is a summary of the observed curve, and its value depends on the sample prevalence.

Primary sources: Wasserman, *All of Statistics*, Chapter 6, Chapters 9–10, Chapter 13 in the Model Selection part, and Chapter 22 in the part "Assessing Error Rates and Choosing a Good Classifier". Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 11 in the part "Performance Metrics", give the one-in-a-million disease example and the notion of *coverage*. Athalye, Carlini, and Wagner (2018) show that weak evaluation produces false security claims: seven of nine defenses without a certificate at ICLR 2018 relied on obfuscated gradients, and six fell once their evaluation was fixed.

Limits of validity: all simulations use assumed normal distributions, so they measure the behavior of the procedure, not the truth of the model. The Wilson interval is a score approximation, and McNemar assumes the discordant pairs are symmetric under the null hypothesis. The leakage result depends on n, d, and the number of features selected: 0.78 is not a constant of nature but one point on a curve that rises with d/n.

Open questions: honest intervals for metrics from mutually dependent folds; calibration tests without the assumption of uniform score intervals; and reporting uncertainty when the test set is chosen adaptively.

Map to source texts: Chapter 27 of Wasserman, All of Statistics*: Chapter 6 (estimators, bias, standard error), Chapters 9–10 (intervals and tests), Chapter 13 in the Model Selection part, and Chapter 22 in the part "Assessing Error Rates and Choosing a Good Classifier". The Wilson interval and the rule of three are referred from Chapter 20 of this book and from Wilson (1927) and Jovanovic–Levy (1997). The example of deceptive accuracy follows Goodfellow, Bengio, and Courville,* Deep Learning*, Chapter 11; the limits of adversarial-defense evaluation follow Athalye, Carlini, and Wagner (2018), discussed in Chapters 27 and 28. Bayes' rule for precision is proved in Chapter 16; data splitting in Chapter 24.*

Chapter 28

Formal Methods with Lean 4

A file named `Sertifikat.lean` arrived together with a single sentence of introduction: “everything has already been proved by machine”. It contained forty one-line theorems, almost all of them ending with `by native_decide`, and at the bottom there was one summary constant that mentioned the threshold 10^{-3}. Not one of its lines can we inspect with our eyes in a reasonable amount of time: what is being compared are sixteen-digit integers that come from turning binary fractions into exact form. That introductory sentence is true in a narrow sense and misleading in an important sense. The machine does check every line, but it does not check whether what is being checked is what was meant.

Testing and proving promise different things, and the difference is not a question of degree of confidence. Testing takes a number of inputs, runs the program, and then reports what happened on those inputs. The conclusion always has the form “for 10^4 test samples, nothing failed”. A formal proof promises more: one statement quantified over all inputs, together with an object that a third party can check without trusting its author. That promise is kept only when three conditions hold at once, namely that the statement is written correctly, that the object being checked really is a proof of that statement, and that the foundation of the checking has not been misused. This chapter takes those three conditions apart on one concrete system, Lean 4.

The tools of the previous chapter are not enough for this work. Chapter 27 taught us how to report what is actually measured: keep the training data apart from the test data, state the distribution, report the variance across seeds. That is a discipline about *empirical claims*. Here we face a claim of a different shape: not an average over samples, but a statement quantified as $\forall x,\ P(x)$. For a claim of that kind no confidence interval helps, because a single counterexample is enough to bring it down, and the absence of a counterexample is not a proof. What is needed is an object that can be rechecked by someone who does not believe us.

For younger readers (age 10 and up)

Imagine a very meticulous chess referee. He never asks whether the game is interesting, whether both players play well, or whether the rules of chess are in fact the rules that ought to be used. His task is only one: every time a move appears, he checks whether that move is legal according to the rule book. If all the moves are legal, he signs the board and says “this game is valid”. He does not care who wins.

Now imagine someone hands over a board full of legal moves, then says “this proves that I can beat anyone”. The referee will answer honestly: what I checked is only that the moves are legal, not that your sentence is true. If the sentence is wrong, a legal board does not help at all, because what is wrong is not the moves but the question.

Have you ever given a correct answer to the wrong problem? Or done your homework neatly when what was asked for was a different page? In this chapter we will find out why a computer can check a proof down to the smallest step, why that checking does not in the least guarantee that the question is right, and how people write the right question in a language that a machine can check.

28.1 Informal proof and formal proof

Level: intermediate

A proof in a mathematics book is a text that convinces a trained reader. It contains jumps that are agreed on in common: “clearly”, “without loss of generality”, “analogous for the other cases”. Those jumps are not cheating; they are a compression accepted by the community. The trouble is that such compression cannot be checked by a machine, and cannot be separated from the knowledge of its reader.

A formal proof is another version of the same object. Its statement is written in a language with a strict grammar and strict types, and its proof is a *term*, namely an expression whose type is that statement. The checking is carried out by the kernel of the system, a small piece of code that knows only a few rules and knows nothing at all about mathematics. Because the kernel is small, its checking can be trusted far more strongly than any large program around the system.

formal verification (*verifikasi formal*) — Step-by-step checking by a clerk who is never bored and never trusts the author.

Definition 28.1 (Formal proof)**.** Let P be a statement written in the formal language of the system. A *formal proof* of P is a term p such that $p : P$ can be derived by the kernel of the system from definitions, theorems that have already been checked, and axioms declared open. The system accepts P if and only if a term of that kind is found and passes the type check. indexformal verification

That definition contains three words that are easy to pass over. “Can be derived” means that there is a mechanical procedure, not that there is a person who agrees. “Checked” means that the outcome is binary: accepted or rejected, with no degrees of confidence. “Declared open” means that the axiom is recorded, not hidden; we will use that recording later as our main instrument.

The difference that is most often misunderstood is the difference between *truth* and *provability*. Lean does not know whether a statement is true in the real world. It knows only whether there is a term whose type is that statement. If the statement is false, that term will never be found, but the absence of a term is not a proof that the statement is false; it may simply be that we have not found it yet.

Read it aloud (auditory learner)

First voice: “I have checked it, everything is correct.” Second voice: “Show me how.” First voice: “I read it again three times.” Second voice: “That is a habit, not a method.” First voice: “I am sure.” Second voice: “Certainty is not proof. Give me the steps, one by one.”
That is what a conversation with a machine is like. It does not ask whether we are good people, whether we have had breakfast, or whether we graduated with high marks. It asks one thing, patiently, every time, with no angry tone: show me the steps. If the steps are there, the door opens. If the steps are not there, the door stays shut, as tightly as before, even if we repeat the request a hundred times. The machine does not judge our

intentions; it only checks the form.
Say it together, slowly, then repeat it once more: the machine does not believe, it checks.

28.2 Propositions as types, proofs as terms

Level: intermediate

In Lean there is no gulf between a statement and its proof. Both are terms, checked by the same rules. Every sentence that can take the value true or false is given the type `Prop`; a proof of that sentence is a term whose type is that sentence. The statement "P is proved" therefore does not mean that P is true in some model, but that there is a term of type P that passes the check.

proposition (*proposisi*) — A sentence that can be judged true or false, like an exam question whose answer must be proved, not merely chosen.

Definition 28.2 (Inhabitant of a type)**.** A type T is called *inhabited* when there is a term t with $t : T$; the notation $t : T$ is read "t has type T". If $P : \texttt{Prop}$, then P is called a *proposition* and $t : P$ is called a *proof term* for P.
indexproof term Two terms are called *definitionally equal* when both reduce to an identical form by symbolic computation.
indexproposition-type

This dictionary, namely the correspondence between logic and types, is known as Curry–Howard.
indexCurry-Howard Implication becomes a function type, because a proof of $P \Rightarrow Q$ must turn an arbitrary proof of P into a proof of Q, and that is the work of a function. The universal quantifier becomes a dependent function, because a proof of $\forall x,\ P(x)$ must produce a proof of $P(a)$ for every a that is given. Conjunction becomes a pair, because a proof of $P \wedge Q$ contains both of its proofs at once. The last row of the table below is the most important: $\bot$ has no inhabitant.

Logic	Type in Lean	How to build the term
$P \Rightarrow Q$	$P \to Q$	`fun h => ...`
$P \wedge Q$	`And P Q`	`And.intro`
$P \vee Q$	`Or P Q`	`Or.inl`, `Or.inr`
$\top$	`True`	`True.intro`
$\bot$	`False`	no inhabitant
$\forall x,\ P(x)$	$\forall x,\ P\,x$	`fun x => ...`
$\exists x,\ P(x)$	`Exists`	`Exists.intro`

proof term (*term bukti*) — A building frame whose shape follows the blueprint of its statement exactly; if the shape does not fit, the building is rejected.

The following terms are the smallest useful proofs: the identity function as a proof of the identity implication, and the composition of two functions as a proof of the transitivity of implication. Both are checked without using any knowledge at all about the contents of P, Q, and R.

```
- proof of the identity implication, term style
def bukti_id {P : Prop} : P -> P := fun h => h

- composition of two implications
def bukti_komposisi {P Q R : Prop} (h1 : P -> Q) (h2 : Q -> R) : P -> R :=
```

```
  fun hp => h2 (h1 hp)

#check bukti_id
#check bukti_komposisi
#check Prop
```

The output of the last command, as it stands, is `Prop : Type`: the set of propositions is itself a type. The two lines before it print the types of the two definitions. Note that `#check` does not prove anything; it only asks for the type, and that is also what the kernel of the system does when it checks.

Proposition 28.1. *For every* P : `Prop` *we have* $P \to P$. *For* P, Q, R : `Prop` *we have* $(P \to Q) \to (Q \to R) \to P \to R$.

Proof. The term $\lambda h.\, h$ has type $P \to P$: in the context $h : P$, the variable h itself is a proof of P, so abstraction over h produces a function from P to P. For the second statement, let $h_1 : P \to Q$, $h_2 : Q \to R$, and $h_p : P$. The application $h_1\, h_p$ has type Q, then the application $h_2\, (h_1\, h_p)$ has type R. Three abstractions in succession over h_p, h_2, and h_1 give a term of type $(P \to Q) \to (Q \to R) \to P \to R$. □

The tree in Figure 28.1 writes that second proof as a derivation tree: three leaves, two application stems, one root. The rightmost leaf is a hypothesis that is given, not a result of derivation.

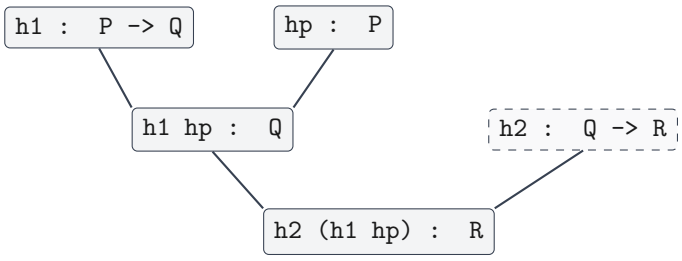


Figure 28.1: The type tree for the term `h2 (h1 hp)`: three leaves and two application stems, with a root of type R. The dashed boxes are hypotheses that are given, not results of derivation.

There is one property that must be mentioned because it explains why proofs are used as certificates rather than as data. The Lean kernel treats any two proofs of the same proposition as definitionally equal. This property is called *proof irrelevance*, and its consequence is that a proof carries no information other than "this statement is true". For that reason a proof can be thrown away once it has been checked, and what remains is only the status of the statement.

One warning is installed from the start. The Curry–Howard correspondence holds in full for constructive logic. As soon as we use the law of excluded middle, `Classical.em` : $P \vee \neg P$, we add an axiom, and a proof that uses it stops being computational: it remains valid, but it is no longer read as a program that produces a witness. Many theorems of the large libraries carry the axiom of choice in their list of axioms, and the last part of this chapter shows how to ask Lean for that list directly.

28.3 Quantifiers: introduction and elimination rules

Level: intermediate

An interesting statement is almost always quantified.
indexquantifier “For every natural number” and “there is a natural number” are not decorations of language; both change the kind of object that must be supplied. In the type view, that difference becomes the difference between a function and a pair.

quantifier (*kuantor*) — The words “every” and “there is”; the first demands a recipe for all cases, the second demands one example together with its proof.

The rule for the universal quantifier is entirely the same as the rule for a function. To prove $\forall x,\ P(x)$, we take an arbitrary x and prove $P(x)$ for that x; that is abstraction. To use the hypothesis $\forall x,\ P(x)$, we apply that hypothesis to an object; that is application. The rule for the existential quantifier is the rule for a pair: to prove $\exists x,\ P(x)$ we supply one witness a together with a proof of $P(a)$; to use an existential hypothesis we unpack that pair into the witness and its proof.

Form of the statement	How to prove it	How to use it
$\forall x,\ P(x)$	`intro x`	apply to an object
$\exists x,\ P(x)$	`exact Exists.intro a h`	`cases` on the hypothesis
$P \wedge Q$	`constructor`	`h.1`, `h.2`
$P \vee Q$	`left`, `right`	`cases` on the hypothesis
$\neg P$	`intro h` then reach $\bot$	apply to a proof of P

Do it by hand (kinesthetic learner)

Prepare ten blank cards and one marker. On five cards write introduction rules: “take an arbitrary”, “supply a witness”, “pair two proofs”, “choose left”, “choose right”. On the other five cards write use rules: “apply to an object”, “unpack the pair”, “take the first component”, “split into cases”, “use it on the opponent’s proof”.
Lay one goal card at the end of the table, for example “every natural number is zero or a successor”. Build a chain toward the goal card: the first card must be an introduction rule whose shape fits the goal, not a use rule. Every following card may only work on the part that is still open. A card that has no remainder to work on is pushed to the edge of the table; in Lean that card will never be used.
What you must see: the chain gets shorter toward the goal, and there are never two cards working on the same remainder. If that happens, your chain branches, and the machine demands two proofs, not one.

The following example shows both at work on the natural numbers. The first statement is a tautology that requires no reasoning at all; the second statement says that every natural number is zero or the successor of something, and its proof requires case splitting on the shape of the number.

```
- universal quantifier: take an arbitrary n, then compute
theorem nol_kanan : forall n : Nat, n + 0 = n := by
  intro n
  rfl

- existential quantifier: supply a witness, then prove its property
theorem ada_tiga : exists n : Nat, n = 3 := by
  exact Exists.intro 3 rfl

- every natural number is zero or a successor
theorem nol_atau_suksesor (n : Nat) : n = 0 \/ exists m : Nat, n = m + 1 := by
  cases n with
  | zero => left; rfl
  | succ k => right; exact Exists.intro k rfl
```

The three theorems are checked without any additional axiom, and that can be confirmed with the command `#print axioms`, which we will use many times. Note also that `Exists.intro k rfl` uses `rfl` as a proof of equality: in the case `succ k`, the goal is $k + 1 = k + 1$, which is true by definition.

Note 28.1. The reverse direction does not hold automatically. From $\exists x,\ P(x)$ we may not conclude $\forall x,\ P(x)$, and Lean will reject every attempt in that direction. The mistake of swapping the order of quantifiers is the most frequent specification error of all, and it slips through testing without complaint because all test samples satisfy it. The part on limits at the end of this chapter discusses its consequences.

28.4 Tactics: turning the goal into a term

Level: basic

Writing proof terms directly quickly becomes unreadable, because a deep term has to be read from the inside out. That is why Lean provides a tactic mode.
indextactic After the word `by`, we do not write a term, but a sequence of commands that change the *goal state*. The goal state is printed as a list of hypotheses, then a separator line, then the target.

tactic (*taktik*) — A step-by-step cooking instruction, not a complete recipe that has to be memorized all at once.

```
- the goal state is printed after every tactic; the line "|-" reads "therefore"
theorem imp_trans {P Q R : Prop} (h1 : P -> Q) (h2 : Q -> R) : P -> R := by
  intro hp
  trace_state          - P Q R : Prop, h1 : P -> Q, h2 : Q -> R, hp : P, |- R
  apply h2
  trace_state          - the target becomes |- Q
  exact h1 hp
```

The output of `trace_state` on this test machine is the following two blocks. Lean prints the arrow as a Unicode character; here everything is rewritten in ASCII so as to be consistent with the whole chapter.

```
P Q R : Prop
h1 : P -> Q
h2 : Q -> R
hp : P
|- R

P Q R : Prop
h1 : P -> Q
h2 : Q -> R
hp : P
|- Q
```

The tactic `intro` takes a target of implication shape and moves its antecedent into the hypotheses; that is exactly function abstraction. The tactic `exact` closes the target with a term whose type fits exactly. The tactic `apply` matches the conclusion of a lemma against the target, then leaves the remaining arguments as new goals. These three tactics are already enough for all the pure-logic proofs in this chapter, and all three are surface forms of the same rule.

Figure 28.2 compresses those two states into two boxes. What changes is only the line below the bar: the target shrinks from R to Q. The list of hypotheses does not grow by a single line, because `apply` does not create a new proof; it only lowers the demand.

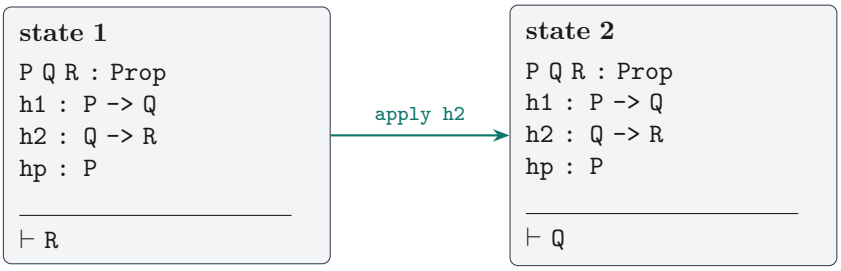


Figure 28.2: The two goal states in the proof of `imp_trans`: the hypotheses stay at four lines, the target shrinks from R to Q after one tactic. The second box is still open; it is closed by `exact h1 hp`.

Tactic	What it does	When to use it
`intro`	antecedent becomes a hypothesis	target of shape $P \to Q$
`exact`	closes the target with a term	the term is already available
`apply`	lemma conclusion matches target	the lemma ends the same way
`rw`	rewrites with an equation	there is a hypothesis $a = b$
`simp`	simplifies repeatedly	many trivial steps
`cases`	splits the target by constructor	case analysis is needed
`induction`	base case and inductive step	recursive target over `Nat`

The tactic `rw` rewrites the target using an equation. It works in the direction that is given, and that direction matters: rewriting in the wrong direction will not find the pattern that is being looked for. Note the two attempts below, one of which succeeds and one of which is rejected.

```
theorem geser (a b : Nat) (h : a = b) : a + 1 = b + 1 := by
  rw [h]        - the target becomes b + 1 = b + 1, then it is closed by definition
  rfl

- commutativity of addition does NOT hold by definition, so rfl fails
theorem tukar_salah (n : Nat) : n + 1 = 1 + n := by
  rfl           - error: The rfl tactic failed

- the correct version: rewrite with the commutativity lemma
theorem tukar_benar (n : Nat) : n + 1 = 1 + n := by
  rw [Nat.add_comm]
```

The failure in the middle is a lesson, not an accident. Addition on the natural numbers is defined by recursion on the second argument, so $n + 1$ reduces to $\mathtt{succ}\, n$ whereas $1 + n$ reduces to $\mathtt{succ}\, (0 + n)$, and the two stop at different forms. What we call the "commutative property" at school is a theorem, not a definition, and Lean refuses to treat it as if it were a definition. This is the most noticeable difference between reading and writing formal proofs: every thing that we usually skip over has to be named.

The tactic `cases` splits the target according to the constructors of the type. On the natural numbers it gives two cases, zero and successor. The tactic `simp` closes a goal by rewriting with a collection of lemmas that are marked; it is useful but also dangerous, because its output depends on the collection of lemmas that is currently active. A proof that begins with `simp` without arguments can stop working when the library changes, and that is why in this chapter `simp` is always mentioned together with the list of lemmas it uses.

One warning about tactics that are not here. Names such as `linarith`, `ring`, `norm_num`, and `induction'` come from the Mathlib library, not from the Lean kernel. All the code fragments in this chapter were run with Lean 4.10.0 without Mathlib, so those tactics are not available and are not used; in their place we use `omega` (which is in the kernel) and `simp` with

an explicit list of lemmas. This difference is not a matter of style: the library adds thousands of lemmas and several tactics, and every addition also determines what can be closed in one line.

For younger readers (age 10 and up)

Take a pile of small stones. You arrange one stone into the smallest square. Then you add three stones around it, and the result is a larger square. Add five more stones, and the result is a larger square still. Add seven, nine, eleven, and so on, always increasing by two, and every time the result is still a tidy square.
Now the question: will this go on being so if we add up to a hundred times, up to a million times? You cannot try all of them. Nor do you need to. If you can show that the pattern of adding stones always closes one corner and one side of the square neatly, then one small explanation holds for every size, including sizes that you will never be able to arrange by hand.
The lazy computer does not want to try a million cases, and the overconfident computer does not want to check anything. In this chapter we will find out why there is a third way: explain the pattern once, then ask the machine to check that explanation down to its last letter.

28.5 A complete inductive proof: the sum of odd numbers

Level: advanced

We are going to prove one arithmetic statement that can be checked entirely by machine, namely that the sum of the first n odd numbers equals n^2. This statement was chosen for three reasons: the formula can be computed with Python without Lean, the proof requires induction, and the algebraic step is small enough that every tactic can be accounted for.

induction (*induksi*) — Explaining a pattern with only two things: the smallest case, and how to climb one step from any case.

Before writing code, we state the statement and its proof in ordinary language. The function is defined recursively: the sum of the first zero odd numbers is zero, and the sum of the first $n+1$ odd numbers is the sum of the first n odd numbers plus the odd number number $n+1$, namely $2n+1$.

Definition 28.3 (Sum of odd numbers)**.** The function $S : \mathbb{N} \to \mathbb{N}$ is defined by $S(0) = 0$ and $S(n+1) = S(n) + (2n+1)$ for every $n \in \mathbb{N}$.
indexinduction

The number $2n+1$ is not an arbitrary choice: the nth odd number really does have that form, and $2n+1$ is also the difference between $(n+1)^2$ and n^2. That last equality is the heart of the proof.

Lemma 28.2. *For every $k \in \mathbb{N}$ we have $(k+1)^2 = k^2 + 2k + 1$.*

Proof. $(k+1)^2 = (k+1)(k+1) = k(k+1) + (k+1) = k^2 + k + k + 1 = k^2 + 2k + 1$. Each step uses distributivity and associativity of addition on the natural numbers. □

Theorem 28.3. *For every $n \in \mathbb{N}$ we have $S(n) = n^2$.*

Proof. By induction on n. For $n = 0$, both sides have the value 0. Suppose the statement is true for k, namely $S(k) = k^2$. Then $S(k+1) = S(k) + (2k+1) = k^2 + 2k + 1 = (k+1)^2$, with the lemma above at the last step. So the statement is true for $k+1$. By the principle of induction, the statement is true for every $n \in \mathbb{N}$. □

The proof above is a complete informal proof, in the sense that no step is skipped. Now we write it for the machine. Note that the recursive definition and the theorem are almost word for word what we have just written, whereas the proof turns into a sequence of tactics.

```
- recursive definition: two equations, without ":=" on the first line
def oddSum : Nat -> Nat
  | 0     => 0
  | n + 1 => oddSum n + (2 * n + 1)

theorem oddSum_sq (n : Nat) : oddSum n = n * n := by
  induction n with
  | zero => rfl
  | succ k ih =>
    rw [oddSum]        - unfold the definition: oddSum (k+1) = oddSum k + (2*k+1)
    rw [ih]            - use the induction hypothesis: oddSum k = k*k
    simp [Nat.succ_eq_add_one, Nat.mul_succ, Nat.add_mul, Nat.one_mul]
    omega              - close the linear remainder
```

The line `induction n with` generates two goals. The first goal is labelled `zero` and reads $S(0) = 0 \cdot 0$, and it is closed by `rfl` because both sides reduce to zero. The second goal is labelled `succ k` and carries one extra hypothesis named `ih`, namely $S(k) = k \cdot k$. The name `ih` is chosen by us; its role is what matters.

The line `rw [oddSum]` unfolds the definition on the left-hand side. After that line the goal reads $S(k) + (2k+1) = (k+1)(k+1)$. The line `rw [ih]` replaces $S(k)$ with $k \cdot k$, so the goal becomes $k \cdot k + (2k+1) = (k+1)(k+1)$.

At this point we stand exactly where the lemma above does its work, but the machine does not yet know that $(k+1)(k+1)$ can be written as $k \cdot k + 2k + 1$. `simp` with four explicit lemmas changes the shape of the right-hand side into $k \cdot k + k + (k+1)$, which is a shape that is entirely linear in k. The last line, `omega`, closes a linear goal of that kind.

It must be stressed that `omega` is not magic and is not a hidden axiom either. It is a decision procedure for linear arithmetic over the integers, and it accepts only goals all of whose operations are linear. If we call it before `simp`, the goal still contains both $k \cdot k$ and $(k+1)(k+1)$. On this test machine, the temporary experiment file named `odd.lean` produces a rejection:

```
odd.lean:11:4: error: omega could not prove the goal:
a possible counterexample may satisfy the constraints
  a - b + 2*c <= -2
  ...
```

That message is honest and useful: it names a possible counterexample, and precisely because of that we know that what fails is the shape of the goal, not the truth of the statement. The order of tactics is therefore not a matter of taste; it changes the kind of goal that the decision procedure faces.

Note 28.2. If `simp` above is run without `omega`, the remaining goal has the shape $k \cdot k + (1 + 2k) = k + (k + (k \cdot k + 1))$. Mathematically that is true, but it is no longer linear, and the simplifier does not close it. On this test machine that experiment ends with `unsolved goals`. This is the most common shape of a failed formal proof: not that the statement is false, but that the shape of the goal does not fit the tool in use.

Corollary 28.4. *For every $n \in \mathbb{N}$, the sum of the first n odd numbers is a perfect square.*

That corollary is only a repetition of the theorem in other words, and that is precisely its use: it restates the result in the language used in this part, so that a reader who meets the formula elsewhere can recognize it.

28.6 Structures, typeclasses, and what is really guaranteed

Level: advanced

A certificate is rarely a single loose theorem. Usually it is a collection of data together with claims about that data: thresholds, tables, file names, and several statements. Lean provides two tools for grouping things of this kind, namely *structure* and *typeclass*.

typeclass (*type class*) — A list of capabilities that a type must have in order to be usable in a certain place, like the licensing requirement for driving a particular kind of vehicle.

Definition 28.4 (Structure and typeclass)**.** A *structure* is a type whose fields are named, so that a value of it is built by naming the fields. A *typeclass* is a structure that is used to look for an *instance* automatically: when a definition requires a certain capability from a type, Lean searches for a matching instance in the environment that is currently active.
indextypeclass

The practical difference is simple. A structure is used when the data itself is what matters, for example a certificate with a claim field and a threshold field. A typeclass is used when what matters is the capability: something that can be used as a proof, something whose norm can be computed, something that can be compared. In both, the checking remains equally strict: every field must be filled, and every instance must pass the type check.

```
- structure: data with named fields
structure Sertifikat where
  klaim  : String
  ambang : Nat

- typeclass: a capability that is looked up automatically
class Terperiksa (a : Type) where
  jumlah_teorema : a -> Nat

instance : Terperiksa Sertifikat where
  jumlah_teorema _ := 1

def contoh : Sertifikat := { klaim := "oddSum", ambang := 100 }

#eval (Terperiksa.jumlah_teorema contoh)   - prints 1
```

Now the part that is most often skipped in popular talk about Lean, namely the question of what the acceptance of a theorem actually guarantees. Four things are guaranteed, and four other things are not.

Figure 28.3 puts those three stages on one line: the statement is written, the term is submitted, the kernel decides. There are only two outputs; there is no third box that says “perhaps”.

What is guaranteed: (i) there is a term whose type is exactly the statement that was written; (ii) that term is checked by the kernel, not by the tactic that produced it; (iii) the checking uses only definitions, other theorems that have already been checked, and recorded axioms; (iv) the list of axioms that are used can be asked for at any time with `#print axioms`. Point (iv) is the most useful instrument in this whole chapter, because it turns “just believe me” into “check it yourself”.

What is not guaranteed: (i) that the statement says what we mean; (ii) that the definitions model objects in the real world; (iii) that the kernel, the parser, and the hardware are free of errors; (iv) that the tactics in use do not add hidden assumptions. Point (iv) is not a theoretical flaw: it can be checked, and the result is often surprising.

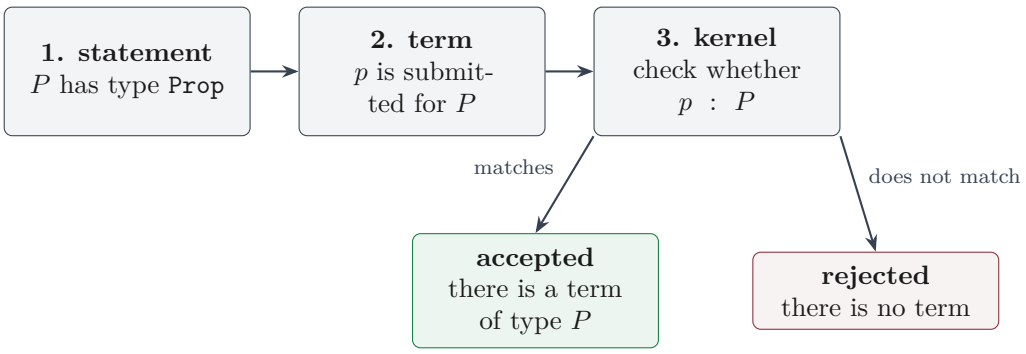


Figure 28.3: The checking flow: three stages (statement, term, kernel) and two outputs, accepted or rejected. There is no third output; a file that stops in the middle is reported as a goal that is not yet closed, not as "almost correct".

Example 28.1 (Two ways to prove the same thing, two lists of axioms). Consider two proofs of the statement $S(10) = 100$, both of them correct. The first uses `decide`, which runs the computation inside the kernel. The second uses `native_decide`
indexnative-decide, which compiles the computation into machine code and runs it. The two statements are the same, but their lists of axioms differ, and that difference is visible from the command `#print axioms`.

```
def oddSum : Nat -> Nat
  | 0     => 0
  | n + 1 => oddSum n + (2 * n + 1)

theorem cek_native : oddSum 10 = 100 := by native_decide
theorem cek_kernel : oddSum 10 = 100 := by decide

#print axioms cek_native
#print axioms cek_kernel
```

The output on this test machine, as it stands:

```
'cek_native' depends on axioms: [Lean.ofReduceBool]
'cek_kernel' does not depend on any axioms
```

The first line is the heart of the whole discussion about `native_decide`. The computation is not carried out by the kernel, but by compiled code, and Lean records that trust as an axiom named `Lean.ofReduceBool`. In practice this is not a disaster: the Lean compiler is used to compile Lean itself every day. In principle it is a large difference, because the boundary of trust moves from the small kernel to a long chain of compiler, linker, and processor. A certificate built on top of `native_decide` is therefore weaker than a certificate built on top of `decide`, even though both print "accepted" without complaint.

In the induction theorem above, the list of axioms is not empty either. The command `#print axioms` for `oddSum_sq` prints `[propext, Quot.sound]`. Both are standard axioms used for classical reasoning and for quotients, and both appear because `omega` proves the goal by contradiction inside the integers. This is not an error; it is a fact that ought to be mentioned when a certificate is used as the basis of a claim.

28.7 Limits: rejecting wrong proofs, `sorry`, and axioms

Level: advanced

The ability to reject is as important as the ability to accept. The following file deliberately contains errors of different kinds, and the output is Lean's answer to each of them.

```
- File: Salah.lean
theorem salah_arit : 2 + 2 = 5 := by decide

theorem salah_rfl (n : Nat) : n + 1 = 1 + n := by rfl

axiom magic : 1 = 2
theorem from_magic : 1 = 2 := magic
#print axioms from_magic

theorem belum_selesai (n : Nat) : n + 0 = n := by sorry
```

The actual output, cut at the long part:

```
Salah.lean:2:37: error: tactic 'decide' proved that the proposition
  2 + 2 = 5
is false
Salah.lean:4:50: error: The rfl tactic failed. Possible reasons:
- The goal is not a reflexive relation ...
- The arguments of the relation are not equal.
...
n : Nat
|- n + 1 = 1 + n
'from_magic' depends on axioms: [magic]
Salah.lean:10:8: warning: declaration uses 'sorry'
```

Three kinds of rejection appear here, and all three differ in nature. The first is the strongest rejection: `decide` does not merely fail to close the goal, it proves that the proposition is false. For a finite statement, failure and refutation are the same thing. The second is a subtler rejection: `rfl` refuses because the two sides are not equal by definition, even though the statement is true. Here Lean does not blame the statement, it blames the tool in use; the message even suggests an explicit reflexivity lemma. The third is not a rejection at all.

The line `axiom magic : 1 = 2` is accepted without complaint. Lean does not check whether the content of an axiom is reasonable; it only records it.
indexaxiom After that, `from_magic` is accepted as well, and the whole of arithmetic can be taken apart with `False.elim` from there. The only danger sign is the `#print axioms` line, which mentions `[magic]` calmly, without a warning and without the color red. This is the practical reason why inspecting the list of axioms is a compulsory habit: the system will not reproach us.

axiom (*aksioma*) — A rule that we ask to be accepted without proof, like an opening assumption in a debate; legitimate, but it has to be named.

The last line of that file uses `sorry`, and its output is only the warning `declaration uses 'sorry'`.
indexsorry The file is still processed, the theorem still "exists", and anyone who reads the output carefully will see the word `warning`, not `error`. In a large library, one `sorry` left behind among thousands of theorems is almost invisible. Lean provides the answer, and the answer is the same as the answer for axioms: `#print axioms` prints `[sorryAx]`. Lean's mathematics library, namely Mathlib, uses a stricter policy: a file that contains `sorry` is rejected by the automatic check before it is accepted. That policy deserves to be copied by anyone who publishes certificates.

sorry (*sorry*) — A note saying "I will do it later" that makes the work look finished; the machine accepts it, but records it as a debt.

Now the most important limit, and the one least often mentioned. Formal verification guarantees that the term fits the statement. It does not guarantee that the statement fits the intention. If the specification is too weak, the proof can be perfect and useless.
indexspecification The following example shows this in its most naked form: the first statement

says “there is a small perturbation that does not change the output”, the second statement says “every small perturbation does not change the output”. The first can be proved for *all* functions, including the worst function imaginable, because the witness $\delta = 0$ is always available.

```
def identitas (n : Nat) : Nat := n

- weak specification: THERE IS a delta that does not change the output (for all f)
theorem ada_delta (f : Nat -> Nat) (x : Nat) :
    exists delta : Nat, f (x + delta) = f x := by
  exact Exists.intro 0 rfl

- the strong version is false for f = identity, so it cannot be proved
theorem ada_penyangkal :
    exists x : Nat, exists delta : Nat,
      Not (identitas (x + delta) = identitas x) := by
  exact Exists.intro 0 (Exists.intro 1 (by decide))
```

Both theorems pass without additional axioms, and that is exactly the problem. The first theorem says nothing about f; it only says that $\delta = 0$ is a perturbation that changes nothing, and that holds for every function, including a classifier that is wrong one hundred percent of the time. A certificate of this kind passes every check and protects nothing. The second theorem shows the other side: the statement we actually want, namely the universally quantified version, is false for the identity function, so no proof can possibly be found.

Challenge (problem solver)

A certificate contains one line that looks impossible to be wrong: for every natural number a and b we have $(a - b) + b = a$. On paper the reason is simple: subtract, add again, and you are back where you started. Write that line in Lean and close it with `omega`; the checker rejects it, and the rejection is not because the tactic is not clever enough.

Two questions. Why is that statement not true for all a and b? And what additional condition makes it true while keeping it checkable by machine?

Hint 1: do not trust the parentheses; ask what happens when b is larger than a. Hint 2: subtraction on the natural numbers stops at zero, so $0 - 1 = 0$, hence the left-hand side becomes $0 - 1 + 1 = 1$ whereas the right-hand side becomes 0.

If you get stuck, go back to the section “Tactics: turning the goal into a term”, where $n + 1$ and $1 + n$ stop at different forms.

Before reading further, try to answer on your own. The output of this test machine, as it stands, is the comparison for your answer: the second file contains the same statement with the hypothesis $b \leq a$.

```
$ lean t.lean
t.lean:1:49: error: omega could not prove the goal:
a possible counterexample may satisfy the constraints
  d >= 0
  c >= 0
  c - d >= 1
where
 c := b
 d := a
$ lean t2.lean
'potong_benar' depends on axioms: [propext, Quot.sound]
```

The first output mentions the constraint $b - a \geq 1$, that is a counterexample with $b > a$; at $a = 0$ and $b = 1$ the left-hand side has the value 1, the right-hand side 0. The hypothesis

$b \le a$ makes the same line pass, with the two standard axioms that we already know from `omega`. The shape of the repair is not an additional tactic, but a condition.

> The claim that may be made from this chapter: the Lean code fragments that are shown have been run with Lean 4.10.0 without Mathlib on the machine of writing, and their output is copied as it stands; Unicode output is reprinted in ASCII and cuts are marked with three dots. What was not run: the Mathlib tactics (`linarith`, `ring`, `norm_num`, `induction'`) and the Mathlib library itself, because it is not installed. What is not proved here and is not claimed: any certificate about neural networks, language models, or robustness to attack. Chapter 28 does not contain a single proof of that kind.

These limits have a direct connection with Chapter 27 and Chapter 29. An honest evaluation of a model demands that we state the distribution in which the numbers are measured; formal verification demands that we state the proposition that is proved. Both are the same disease in a different form: a claim that does not state its conditions. The certificate "this model is proved safe" without a formal statement behind it is as useless as "99 percent accuracy" without a note on the test data. In Chapter 29 and Chapter 30, when we discuss attacks and defenses, the first question that must be put to every robustness certificate is: how many quantifiers does it have, and in what space do those quantifiers run?

28.8 Worked example

Level: intermediate

The numbers in this part are computed, not estimated. The first column is n, the second column is the nth odd number that is added at that step, the third column is $S(n)$, and the fourth column is n^2.

n	term n	$S(n)$	n^2
0	–	0	0
1	1	1	1
2	3	4	4
5	9	25	25
10	19	100	100
20	39	400	400
100	199	10000	10000

The value $S(100) = 10000$ is worth checking by hand once, because it shows that the formula is not a coincidence for small numbers: the sum of a hundred odd numbers, from 1 to 199, equals $100 \cdot 100$. A check with Python over all values $n < 5000$ gives `True`, and a check of the algebraic identity $(k+1)^2 = k^2 + 2k + 1$ for all $k < 100000$ also gives `True`. Neither check is a proof, but both rule out the possibility that the formula being proved has a typo.

There is one more number that is more informative than the table above, namely the size of the proof after elaboration. The tactic script for `oddSum_sq` is eight lines long. When Lean prints the term that results from elaborating it with `#print`, the result is 10,783 characters in 149 lines. That means one line of tactics expands on average into about nineteen lines of term. No human reads that term; what reads it is the kernel, and that is the division of labor that is the heart of the whole method.

The last number concerns the limit of direct checking. In the default setting, the recursion-depth limit has the value 1000. With that limit, the tactic `decide` succeeds in proving $S(n) = n^2$ only for $n \le 122$. The first experiment that fails is $n = 123$, and the message that appears mentions that the maximum recursion depth has been reached. If the limit is raised to 100000, the value $n = 2000$ is proved successfully in 0.48 seconds. At $n = 20000$ the check fails because the recursion is too deep for the stack space. So direct checking is not an alternative that scales, whereas an inductive proof does not care about the size of n.

28.9 Computational lab

Level: advanced

This lab runs two things: the Lean file that is the same as the one discussed above, and a cross-check with Python. The Lean file is written without `import`, so it uses only the kernel.

Computational lab

```
- BilanganGanjil.lean : Lean 4.10.0, without Mathlib
def oddSum : Nat -> Nat
  | 0     => 0
  | n + 1 => oddSum n + (2 * n + 1)

theorem oddSum_sq (n : Nat) : oddSum n = n * n := by
  induction n with
  | zero => rfl
  | succ k ih =>
    rw [oddSum]
    rw [ih]
    simp [Nat.succ_eq_add_one, Nat.mul_succ, Nat.add_mul, Nat.one_mul]
    omega

#eval oddSum 0
#eval oddSum 5
#eval oddSum 10
#eval oddSum 100
#print axioms oddSum_sq
```

```
def odd_sum(n):
    return sum(2 * i + 1 for i in range(n))

print([odd_sum(n) for n in range(11)])
print(odd_sum(20), odd_sum(100))
print(all(odd_sum(n) == n * n for n in range(5000)))
print(all((k + 1) * (k + 1) == k * k + 2 * k + 1 for k in range(100000)))
```

The commands and the output, as they stand:

```
$ lean -version
Lean (version 4.10.0, x86_64-unknown-linux-gnu, commit c375e19f6b65, Release)
$ time lean BilanganGanjil.lean
0
25
100
10000
'oddSum_sq' depends on axioms: [propext, Quot.sound]
time 0.14 s
$ python3 BilanganGanjil.py
[0, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
400 10000
True
True
```

The first four lines of the Lean output are the results of `#eval`, and all four match the table in the previous part. The fifth line is the list of axioms, and its contents must be read carefully: `propext` and `Quot.sound` are not an error, but a consequence of the way `omega` works. The Python output confirms the same thing from the other direction: the list of the first eleven values is a list of perfect squares, $S(100) = 10000$, and both exhaustive checks print `True`. The time 0.14 seconds is the time of this test machine, and a number of that

kind depends on the hardware; what is meaningful is not the number, but the fact that the check finishes before we have time to read the file.

A reader who wants to repeat this lab needs two things. First, Lean 4, which can be installed through `elan`, the official version manager of the Lean project; after installation, the command `lean -version` must print a version. Second, the file above is saved as `BilanganGanjil.lean` and run with `lean BilanganGanjil.lean` from the same directory; no `lake` project is needed because there is no import. To use `linarith` or `ring`, the reader needs a project with Mathlib, and from that moment `lean` alone is not enough; the command becomes `lake env lean`. This difference is often the first source of confusion for a beginner, and mentioning it is better than letting it be discovered through an `unknown tactic` message.

Step-by-step recipe (procedural learner)

1. Create a plain text file named `Uji.lean`; that extension is what makes the command `lean` recognize it as a source file.
2. Write all `import`s on the topmost lines, one per line, before any declaration.
3. Write the statement first, complete with the type of each variable, then the proof after `:= by`.
4. Run `lean Uji.lean` from the directory of that file and read the output from the first line; the first error is usually the real one.
5. Ask for `#print axioms name_of_theorem` before calling the result proved.

A warning about order: `import` must not be preceded by a declaration. One `def` slipped above it makes Lean reject the whole file with the message `invalid 'import' command`, and the file is not checked at all. The correct order is not the author's taste; it is a condition for the file to be checkable.

Note for instructors

The teaching order that works in class: start from the failure of `rfl` on $n+1 = 1+n$, not from the definition. Students who see that something "obvious" is rejected will ask, and that question is the doorway into Curry–Howard. Only after that come the definition of a proposition as a type, then the first three tactics, then induction. Leave one full session for `#print axioms`; that is where students change from users into checkers.

Two misconceptions appear most often. First, "Lean proves that my program is correct". What is true: Lean checks that a term has a certain type of statement. If the statement does not mention the important thing, for example a quantifier over perturbations, the certificate is empty. Second, "`simp` is a proof". `simp` is a procedure whose output depends on the active collection of lemmas, so a proof that depends on it can stop working when the library changes; that is why we train students to write the list of lemmas explicitly.

A blackboard activity that works: ask two groups to write a specification for "a classifier that is robust to perturbation". One group will almost always write an existential quantifier. Put both formulations on the board, then prove the weak one for a constant function. The class will see for itself that the sentence that is proved and the sentence that is meant are two different sentences.

A sign of readiness to go on: students can explain why $n + 1 = 1 + n$ requires a lemma, can write an inductive proof without copying, and can name one axiom that their own theorem uses.

28.10 Summary

Summary.

- An informal proof convinces a trained reader; a formal proof is a term whose type is its statement, checked by a small kernel that knows no mathematics.
- According to Curry–Howard, implication is a function type, conjunction is a pair, the universal quantifier is a dependent function, and $\bot$ is uninhabited.
- Tactics change the goal state; `intro`, `exact`, `apply`, `rw`, `cases`, and `induction` are enough for basic arithmetic.
- Induction proves a statement for all n with two goals, not by checking all n; `decide` by contrast stops scaling very quickly.
- `#print axioms` is the main instrument of inspection. The tactic `native_decide` leaves behind the axiom `Lean.ofReduceBool`. The tactic `sorry` leaves behind `sorryAx`. A homemade axiom leaves behind its own name.
- Verification guarantees that the term fits the statement, not that the statement fits the intention; a weak specification produces an empty certificate.

28.11 Exercises

Exercise 28.1.

1. Assess this claim: *if a Lean file is accepted without an error message, then all the statements inside it are true.* Decide true, false, or not enough evidence, and name one command that can be used to check that claim for a particular file.
2. Prove in Lean, without Mathlib, that for every n we have $2 \cdot S(n) = n \cdot (2n)$ with S as in this chapter. State first the auxiliary lemma that you need.
3. Prove that the sum of the first n even numbers, namely $0 + 2 + \cdots + (2n - 2)$, equals $n(n - 1)$. Write the informal proof, then write the Lean version and name the tactic that is used at every step.
4. Compute with Python: the sum of the first 37 odd numbers, the sum of the first 250 odd numbers, and check that both are perfect squares. Include the code and its output.
5. Compute with Python how many cases must be checked if one wants to prove $S(n) = n^2$ by direct checking for all $n \leq 1000$, then compare it with the number of goals that an inductive proof generates.
6. Explain why `rfl` closes the goal $n + 0 = n$ but does not close $0 + n = n$. The hint lies in the way addition on the natural numbers is defined recursively; name which argument is reduced first.
7. Write a weak specification and a strong specification for the claim "this classifier does not change under a perturbation of size at most one". Prove the weak version in Lean for an arbitrary function, then explain why the strong version cannot be proved for the identity function.
8. Inspect with `#print axioms` three theorems that you wrote yourself: one using `decide`, one using `omega`, one using `sorry`. Report the three lists of axioms and explain the origin of every name.
9. A certificate contains `theorem aman : exists delta : Nat, f (x + delta) = f x` with `delta` as a free variable in the context. Explain two errors that may be present in that statement, and write the version that you consider correct.
10. Prove in Lean that $\neg(P \wedge \neg P)$ for every proposition P. Hint: start with `intro h`, then take the components with `h.1` and `h.2`, and use the second component on the first one.

Hints for the three hardest items. Item 2: the auxiliary lemma that is needed has the shape $2\,(k \cdot k + (2k + 1)) = (k + 1) \cdot (2(k + 1))$, and after `rw` on the definition and on the induction hypothesis, the remaining goal is linear so that `omega` is enough. Item 7: in the weak

version, the witness $\delta = 0$ makes both sides equal by definition, so `rfl` closes it immediately; in the strong version, the identity function gives a counterexample with $\delta = 1$. Item 9: note that the free `delta` can be chosen by the reader of the statement, not by the attacker, so that statement asserts no robustness at all; the second error is the absence of a quantifier over x.

For researchers: limits and next steps

What is really proved in this chapter is only small arithmetic statements: $S(n) = n^2$ for all natural numbers, a few propositional tautologies, and two statements about specifications that are too weak. All of them are checked by the Lean 4.10.0 kernel without Mathlib, and their lists of axioms are reported. What is popularized far beyond that: the phrase "formally verified" is often used for "checked by an automatic tool" or even for "tested exhaustively". Only the first produces an object that a third party can recheck.

Primary sources. Avigad, de Moura, Kong, and Ullrich, *Theorem Proving in Lean 4*, the sections "Propositions and Proofs" (propositions as types, proof irrelevance, the axiom warning), "Tactics" (the goal state), and "Induction and Recursion" (`induction`, `sorryAx`). Avigad and Massot, *Mathematics in Lean*, Chapters 2, 3.1, 5.2, and 8; the Indonesian edition uses the same section numbering.

Open questions. This chapter does not contain a formal proof about neural networks. An interesting robustness specification has the shape $\forall x\, \forall \delta,\ \|\delta\| \le \varepsilon \to f(x+\delta) = f(x)$, that is a statement over a continuous space with a function with hundreds of thousands of parameters. Formalizing it demands real arithmetic that is decidable at that scale, and the Lean libraries today do not provide it for trained models. The claim "we verified the robustness of the model" in a paper usually refers to verification with interval abstraction or with SMT, not to a Lean proof; both are legitimate, but their guarantees differ and must not be called by the same word.

Limits of validity. All the claims in this chapter hold for Lean 4.10.0 with its default kernel, and the details of the output can change between versions. What does not change is the structure of the argument: statement, term, kernel, list of axioms.

Map to source texts: This chapter follows Avigad, de Moura, Kong, and Ullrich, Theorem Proving in Lean 4*, the sections "Propositions and Proofs" (propositions as types, Curry–Howard, proof irrelevance, the axiom warning), "Quantifiers and Equality", "Tactics" (the goal state, `intro`, `apply`, `exact`), and "Induction and Recursion" (`induction`, `sorry` and `sorryAx`). The counterpart in Avigad and Massot,* Mathematics in Lean*, is in Chapter 2 (Basics, in particular 2.3 Using Theorems and Lemmas and 2.4 More Examples Using apply and rw), Chapter 3.1 (Implications and Universal Quantifiers), Chapter 5.2 (Induction and Recursion), and Chapter 8 (Hierarchies) for typeclasses; the Indonesian edition keeps the same section numbering. For the connection with evaluation practice, see Chapter 27 of this book; for attacks that make robustness specifications nontrivial, see Chapter 27 and Chapter 28.*

Chapter 29

Adversarial Machine Learning: Attacks and Defenses

In 2014 Goodfellow, Shlens, and Szegedy showed one picture of a panda. The GoogLeNet network called it a panda with confidence 57.7 percent. They added a perturbation that the eye cannot see, of size 0.007 on the scale $[0, 1]$, which is about two units out of 255 gray levels. The same network then called that picture a gibbon with confidence 99.3 percent. Nothing changed for a human. What changed was only the direction of the perturbation: every pixel was shifted a little, and the direction of that shift was chosen from the sign of the derivative of the error with respect to the pixel.

Numbers like that are not an oddity of one model. In this chapter we compute for ourselves a small case that can be rechecked. A two-layer network with 64 inputs reaches accuracy 97.99 percent on 597 test images of digits. With a perturbation budget 0.15 per pixel, an iterative attack of 40 steps lowers that accuracy to 10.72 percent; at the same budget, a random perturbation without a direction only lowers it to 95.48 percent. So the direction that the gradient chooses is not decoration: it buys the attacker about eighty points of accuracy.

Chapter 25 taught us to compute the derivative of the error with respect to the weights, and Chapter 27 taught us to measure accuracy honestly. Neither is enough. Accuracy is an average over one data distribution; security is a question about the worst point inside one allowed set. A high average and a bad worst point can live together, and that is what makes an ordinary evaluation report misleading. This chapter builds the vocabulary for that worst-case question, computes its first attack, then separates the defenses that raise the standard from those that only hide the gradient.

For younger readers (age 10 and up)

Imagine a building guard who recognizes people by small habits: how many buttons of the shirt are fastened, the bag in the left hand or the right, and hat or no hat. So far that method has worked. Everyone entitled to enter really does differ in those three things. Then someone comes who knows exactly how the guard sees. He undoes one button, moves the bag to the other hand, and puts on a hat. His face is the same, his name is the same, he is the same person. The guard still lets him in, because according to his notes a person with those three marks is the cleaning staff.

The question that makes this story important is not “how do we fool the guard”, but “how do we know that the guard can be fooled”. If we test only with random people in random clothes, the guard will pass almost always. The error appears only when someone deliberately chooses those three marks. Testing with chance is not the same as

testing with malice.
A computer that recognizes images works like that guard, only with thousands of small features at once. In this chapter we will find out why a feature that looks trivial can be chosen deliberately, why a change that the eye cannot see can flip the answer, and how to tell a defense that is truly strong from a defense that only leaves the attacker confused.

29.1 Threat model: what the attacker may do

Level: basic

The sentence "the attacker changes the input a little" cannot be tested before we agree on the meaning of "a little". Every robustness claim is always relative to that choice, so that choice is part of the claim, not a footnote.

threat model (*model ancaman*) — The complete list of what the attacker may do: how large the perturbation is, what it knows, and what its goal is.

Definition 29.1 (Adversarial example and threat model)**.** Let $f : \mathbb{R}^d \to \{1, \dots, K\}$ be a classifier and $(\mathbf{x}, y)$ a correctly classified example. An *adversarial example* is $\mathbf{x}' = \mathbf{x} + \boldsymbol{\delta}$ with $f(\mathbf{x}') \neq y$ but $\boldsymbol{\delta}$ small according to an agreed measure. The threat model is determined by four things:

1. the set of allowed perturbations $S(\mathbf{x}) = \{\mathbf{x} + \boldsymbol{\delta} : \|\boldsymbol{\delta}\| \le \varepsilon\}$, where $\|\cdot\|$ is usually the ℓ_∞ norm and $\varepsilon > 0$ is the *budget*;
2. the knowledge of the attacker: full access to f and its gradient (*white box*) or only to the output of f (*black box*);
3. the goal: any misclassification (*untargeted*) or a particular class (*targeted*);
4. other constraints, for example the perturbation must still be a valid object.

The attack succeeds when there is a $\boldsymbol{\delta} \in S(\mathbf{x})$ that changes the label.

The ℓ_∞ norm is chosen because it bounds every coordinate separately: $\|\boldsymbol{\delta}\|_\infty \le \varepsilon$ means every pixel shifts by at most ε. For an 8-bit image with values 0 to 255, the budget $\varepsilon = 1/255$ means every pixel may change by at most one level. The ℓ_2 norm bounds the total length of the vector and allows one pixel to change a lot as long as the others barely change. The two answer different questions, and a robustness claim that does not name its norm cannot be compared with another claim.

The most honest measure of robustness is the distance to the nearest misclassification point,

$$r(\mathbf{x}) = \inf\{\|\boldsymbol{\delta}\| : f(\mathbf{x} + \boldsymbol{\delta}) \neq y\}.$$

This quantity does not depend on the algorithm of any attacker. Unfortunately r is hard to compute: finding it means maximizing a nonconvex function on a large set. For a network with ReLU units, r can be computed exactly by turning the problem into a mixed-integer linear program, and its cost grows quickly with the number of units. That is why almost every report in this field names what a particular attacker managed to achieve, not the value of r. The gap between the two is the source of most conclusions that are too optimistic.

One fact deserves to be recorded from the start. On tabular data, the ℓ_∞ ball contains impossible points: negative ages, non-integer packet counts, combinations of TCP flags that never appear in a real network. Examples like that fool the model but cannot be realized, so their success tells us nothing about the real risk. A budget that is too loose produces attacks that are spectacular and meaningless.

29.2 Why the sign of the gradient is enough

Level: intermediate

Now we attack. Let the error of the model be expressed by a function $L(\mathbf{x}, y)$, for example cross-entropy. We want to raise that error as much as possible while keeping every coordinate inside the box. For a linear function, the answer can be written in closed form.

fast gradient sign method (*metode tanda gradien*) — A one-step attack that shifts every coordinate by the budget, with the direction following the sign of the derivative of the error.

Lemma 29.1 (Maximum of a linear form on the ℓ_∞ ball)**.** *For $\mathbf{w} \in \mathbb{R}^d$ and $\varepsilon > 0$,*

$$\max_{\|\boldsymbol{\delta}\|_\infty \le \varepsilon} \mathbf{w}^\top \boldsymbol{\delta} = \varepsilon \|\mathbf{w}\|_1,$$

and the maximum is attained at $\delta_j = \varepsilon \operatorname{sign}(w_j)$ for every coordinate with $w_j \neq 0$.

Proof. Write $\mathbf{w}^\top \boldsymbol{\delta} = \sum_j w_j \delta_j$. Because $|\delta_j| \le \varepsilon$,

$$\sum_j w_j \delta_j \le \sum_j |w_j|\,|\delta_j| \le \varepsilon \sum_j |w_j|.$$

If $\delta_j = \varepsilon \operatorname{sign}(w_j)$, then $w_j \delta_j = \varepsilon |w_j|$ for every j, so that bound is attained. Coordinates with $w_j = 0$ do not affect the value, so their choice is free. □

For a smooth function, we use the first-order approximation of L around $\mathbf{x}$ together with the lemma above. The result is the recipe that Goodfellow and colleagues call the *fast gradient sign method*:

$$\mathbf{x}^{\mathrm{adv}} = \mathbf{x} + \varepsilon \operatorname{sign}\big(\nabla_{\mathbf{x}} L(\mathbf{x}, y)\big).$$

In everyday language it goes like this. The gradient tells us in which direction the error rises fastest, but we may only walk a distance ε; only the sign of each component is used, while the size is fixed by the budget. That is why this method is called the sign attack. For a linear function, that step is exactly optimal; Figure 29.1 shows the result on a model with two features.

Theorem 29.2 (Linearization and the remainder)**.** *Let $L : \mathbb{R}^d \to \mathbb{R}$ be differentiable and let its gradient be M-Lipschitz on the segment between $\mathbf{x}$ and $\mathbf{x} + \boldsymbol{\delta}$. For every $\|\boldsymbol{\delta}\|_\infty \le \varepsilon$,*

$$\Big| L(\mathbf{x} + \boldsymbol{\delta}) - L(\mathbf{x}) - \nabla L(\mathbf{x})^\top \boldsymbol{\delta} \Big| \le \frac{M}{2} \|\boldsymbol{\delta}\|_2^2 \le \frac{M d}{2} \varepsilon^2.$$

Proof. With the integral form of the mean value theorem,

$$L(\mathbf{x} + \boldsymbol{\delta}) - L(\mathbf{x}) = \int_0^1 \nabla L(\mathbf{x} + t\boldsymbol{\delta})^\top \boldsymbol{\delta}\, dt.$$

Subtract $\nabla L(\mathbf{x})^\top \boldsymbol{\delta} = \int_0^1 \nabla L(\mathbf{x})^\top \boldsymbol{\delta}\, dt$, then use the Cauchy–Schwarz inequality and the Lipschitz property:

$$\left| \int_0^1 \big(\nabla L(\mathbf{x} + t\boldsymbol{\delta}) - \nabla L(\mathbf{x}) \big)^\top \boldsymbol{\delta}\, dt \right| \le \int_0^1 M t \|\boldsymbol{\delta}\|_2^2\, dt = \frac{M}{2} \|\boldsymbol{\delta}\|_2^2.$$

Because $\|\boldsymbol{\delta}\|_2^2 \le d\,\varepsilon^2$, the second inequality follows. □

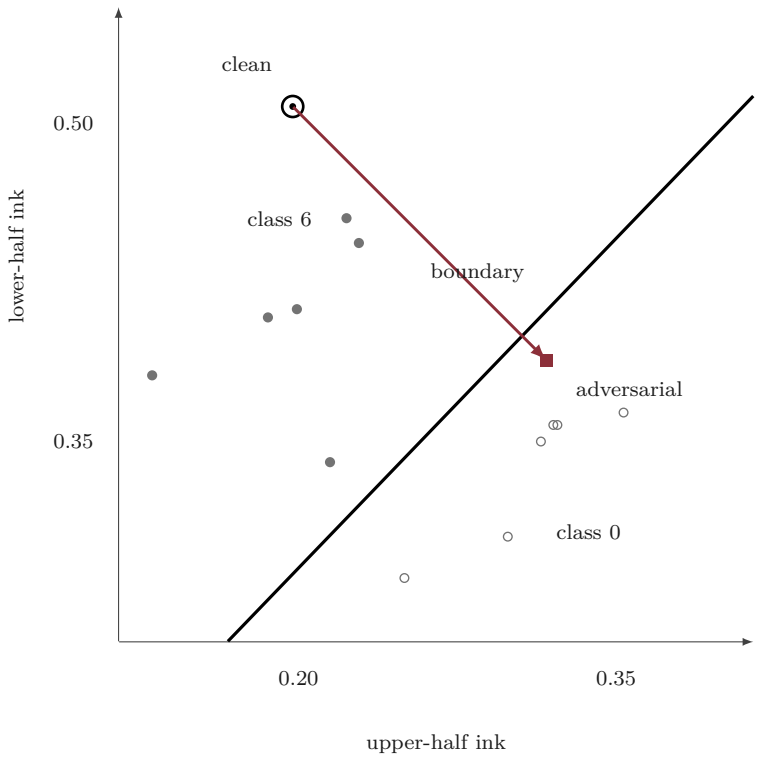


Figure 29.1: Decision boundary of a model with two features on the digits 0 against 6 (236 training data, 123 test, accuracy 98.37 percent). Thick circle: a clean example $(0.197; 0.508)$ with score $+1.936$. Red square: the result of FGSM with $\varepsilon = 0.12$ at $(0.317; 0.388)$ with score -0.208. Its arrow is the perturbation $(0.12; -0.12)$, $\|\boldsymbol{\delta}\|_\infty = 0.12$ and $\|\boldsymbol{\delta}\|_2 = 0.1697$.

For a linear model, that remainder is zero and FGSM is not an approximation but the exact answer. Goodfellow and colleagues use the case of logistic regression to show this. There the sign of the gradient of the error with respect to the input is $-\operatorname{sign}(\mathbf{w})$ times the label, so that $\mathbf{w}^\top \operatorname{sign}(\mathbf{w}) = \|\mathbf{w}\|_1$ and the margin of the model shrinks by exactly $\varepsilon\|\mathbf{w}\|_1$. That is why adversarial training on logistic regression can be written as a penalty $\zeta\big(y(\varepsilon\|\mathbf{w}\|_1 - \mathbf{w}^\top\mathbf{x} - b)\big)$, that is a penalty resembling an ℓ_1 regularizer that enters the margin, not the cost; this exact form is used in the worked example later.

What needs to be stated honestly: the theorem above justifies the *direction* of the step, not the *distance* of the step. The remainder $M d\varepsilon^2/2$ depends on the Lipschitz constant of the gradient, and for a ReLU network that constant can be very large. In our experiment, the predicted rise in the error from the linear term at $\varepsilon = 0.15$ is 0.0008 nats, whereas the rise that actually occurs is 4.862 nats. So linearization does not explain the size of the damage; it only explains why the chosen direction makes sense.

The other side of the story is stronger instead. A perturbation of the same size but with random signs only lowers accuracy from 97.99 to 95.48 percent, whereas FGSM lowers it to 14.91 percent. So what is valuable is the pattern of signs, even though the error at the starting point is almost flat, so that the length of the gradient and its linear forecast are small.

So the sign of the gradient is enough not because its linear term is large, but because the sign is more resistant to saturation of the error. The number 0.0008 against 4.862 is a warning that linearity holds only as an argument about direction.

Read it aloud (auditory learner)

Two voices, read alternately with a friend. Tap the table every time the word “a little” is said.
“Shift it a little.” “How little?” “A fingertip. No more.” “Does the answer change?” “It changes completely. Not doubtful, not off target: changed.” “What do our eyes see?” “Nothing. Exactly the same picture.” “And if it is shifted that far in another direction?” “The answer becomes correct again.” “So what is doing the work?” “The direction. Not the size.” “Where does it learn the direction?” “It reads the slope, asks every pixel: up or down? Then all the answers are used at once.” “And if it misreads the slope?” “Nothing happens. The picture stays, the answer stays.”
Repeat three times, faster and faster: what works is the direction, not the size.

29.3 Iterative attacks and the min-max game

Level: intermediate

One step is a compromise. If the gradient may be computed many times, we can climb further: a small step in the direction of the sign of the gradient, then pull back into the box.

Definition 29.2 (Iterative attack)**.** Let Π be the projection onto $S(\mathbf{x})$. The *projected gradient descent* attack (PGD) computes

$$\mathbf{x}_{t+1} = \Pi\Big(\mathbf{x}_t + \alpha\ \operatorname{sign}\big(\nabla_{\mathbf{x}} L(\mathbf{x}_t, y)\big)\Big), \qquad \mathbf{x}_0 = \mathbf{x},$$

with $\alpha > 0$ the step size. If $\mathbf{x}_0$ is taken at random inside $S(\mathbf{x})$, the attack is called PGD with a *random start*.

Step-by-step recipe (procedural learner)

Four steps to run a bounded iterative attack:

1. Fix the norm, ε, α, and the number of steps; record the clean accuracy before a single perturbation is made.
2. Start from $\mathbf{x}$ or from a random point inside the ball, compute the gradient of the error with respect to the input, then take only its sign.
3. Step: $\mathbf{x}_{t+1} = \mathbf{x}_t + \alpha\ \operatorname{sign}(\mathbf{g})$, project onto the ε ball, then onto $[0,1]$.
4. Repeat, then report the accuracy at several numbers of steps and several ε; compare with FGSM, which uses $\alpha = \varepsilon$.

Warning about order: the projection is always with respect to the original $\mathbf{x}$, not to the previous step point, and always after the gradient step. If the reference is swapped, the ball moves along and the perturbation exceeds the reported budget.

The projection onto the ℓ_∞ box cut to $[0,1]^d$ has a very simple form: clip every coordinate, one by one.

Proposition 29.3 (Projection onto the box)**.** *For* $S = \{\mathbf{z} : \|\mathbf{z} - \mathbf{x}\|_\infty \le \varepsilon\} \cap [0,1]^d$,

$$\Pi(\mathbf{z})_j = \min\{\max\{z_j,\ x_j - \varepsilon,\ 0\},\ x_j + \varepsilon,\ 1\}$$

for every j.

Proof. The objective function $\|\mathbf{y} - \mathbf{z}\|_2^2$ separates per coordinate, and the allowed set also separates: every y_j must lie in the interval $[\max\{0, x_j - \varepsilon\}, \min\{1, x_j + \varepsilon\}]$. For one coordinate,

minimizing $(y_j - z_j)^2$ on a closed interval means taking the nearest point, that is z_j if it is inside the interval, and the nearest end of the interval if it is outside. That is exactly the clipping formula above. □

With a projection of this kind, PGD adds only two things compared with FGSM: repeated small steps and the pull back into the box. The increase in cost is proportional to the number of steps, because every step requires one backward pass. Figure 29.2 shows three such steps on the same model.

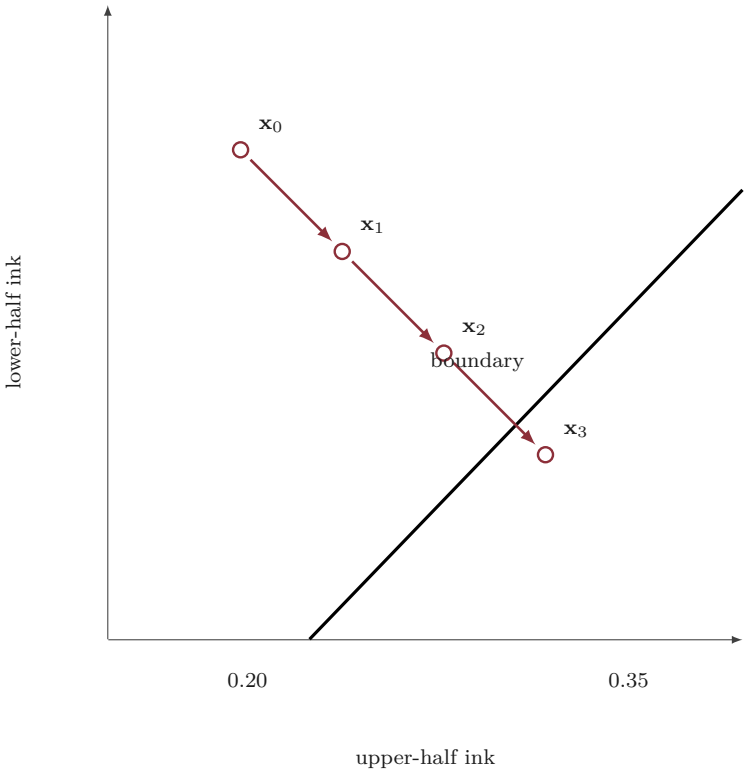


Figure 29.2: Three sign-of-gradient steps from $\mathbf{x}_0$, each step $\alpha = 0.04$. The $\|\boldsymbol{\delta}\|_\infty$ norm: 0.04, 0.08, 0.12; the $\|\boldsymbol{\delta}\|_2$ norm: 0.0566, 0.1131, 0.1697. The score of class 6: +1.936, +1.221, +0.506, −0.208; the boundary is crossed at the third step.

There is one warning that is often missed. FGSM is PGD with one step and $\alpha = \varepsilon$, not PGD with one step and $\alpha = \varepsilon/10$. In our experiment, one step with $\alpha = \varepsilon/10$ only lowers accuracy to 94.64 percent, whereas FGSM lowers it to 14.91 percent. So “one step” is not the name of one algorithm; it only means something once its step size is named. A fair comparison uses the same step size; the following table is computed at $\varepsilon = 0.15$ with a random start.

steps	accuracy (%)
1	94.64
2	88.78
5	66.67
10	24.79
20	11.22
40	10.72

That curve falls sharply up to ten steps, then is almost flat; adding steps after that point rarely changes the conclusion, and that is why PGD is used as the default attacker. Madry

and colleagues read it as a sign that the inner problem is easy enough to solve with first-order methods, even though in general it is nonconcave.

Do it by hand (kinesthetic learner)

Materials: ten cards, a marker, and a ruler. On each card write one number between zero and one, two digits after the decimal point. Make two cards whose numbers differ by only 0.01, for example 0.62 and 0.63.
Extra rule: when two numbers differ by less than 0.02, the higher card in the pile wins. Place the two nearly tied cards in the top two positions, then call the topmost card the answer of the model.
Now shift the card 0.62 up by one millimeter, until it passes the card 0.63. No number changes, no card is swapped. Look at the topmost card: the answer changes, because the tie-breaking rule now chooses the other card.
What must be visible: the answer changes without a single number changing, because only the difference decides, and a thin difference can be flipped by a shift that the eye cannot see.

From the point of view of training, attack and defense are two sides of one problem. If the attacker maximizes the error inside the box, then the careful trainer minimizes that worst error:

$$\min_{\boldsymbol{\theta}} \ \mathbb{E}_{(\mathbf{x},y)\sim\mathcal{D}}\Big[\max_{\boldsymbol{\delta}\in S(\mathbf{x})} L(\boldsymbol{\theta}, \mathbf{x}+\boldsymbol{\delta}, y)\Big].$$

This formulation is a saddle problem: a maximum inside, a minimum outside. The inner term is the problem of the attacker, the outer term is the problem of ordinary training. This is *adversarial training*. If the inner maximum is approximated with FGSM, we get cheap training; if it is approximated with PGD of several steps, we get training that is more expensive and usually more robust.

What that formulation can and cannot promise must be separated. If the saddle value is really attained and its value is zero, no perturbation inside the box can fool the model, so we have a guarantee. But that saddle problem is not concave inside and not convex outside, so any algorithm only gives an approximation: a small saddle value is not necessarily attained, and the value that is obtained is not necessarily the saddle value. That is why adversarial training produces a model that is *more resistant to the attacker used during training*, not a model that is proved resistant to all attackers.

Our numbers show both sides. Model D, trained with FGSM at $\varepsilon = 0.15$, loses 1.51 points of clean accuracy: from 97.99 to 96.48 percent. In exchange, at $\varepsilon = 0.15$ its accuracy under FGSM is 66.67 percent and under PGD of 40 steps 62.65 percent, whereas the ordinary model is left with 14.91 and 10.72 percent. The trade is real, but 62.65 percent is also not security; it only means the attacker must work harder.

29.4 Transfer, black box, and targeted attacks

Level: advanced

Up to here our attacker knows everything. In the real world it can often only send an input and look at the answer. What saves us when there is no gradient is not the secrecy of the model, but the fact that the direction of attack tends to carry over between models.

transferability (*transferabilitas*) — The property of an adversarial example that is built for one model but also fools another model that was trained separately.

This phenomenon was reported by Szegedy and colleagues in 2014: an example that fools one network often fools another network with a different architecture and different training

data. Goodfellow and colleagues record its quantitative version on MNIST: an example built from an old model gives an error of 19.6 percent on a model trained adversarially, whereas an example from the new model gives an error of 40.9 percent on the old model. The same direction, a different strength.

In our experiment, model A (64 hidden units) and model B (128 units) are trained on the same training data but with different initialization. At $\varepsilon = 0.15$, PGD of 40 steps built on A lowers the accuracy of A itself to 10.72 percent, and lowers the accuracy of B to 23.79 percent. FGSM built on A gives 14.91 on A and 25.96 on B. So the attack transfers, but it weakens: about half of its strength is lost when it crosses models.

The most economical explanation is still the one of this chapter: at a small scale the two models learn functions that are almost linear, and a direction that raises the error on one linear function also raises it on another linear function whose normal is close by. That is why transfer works best for one-step attacks. Our numbers are consistent: the gap between white box and transfer is smaller for FGSM (11 points) than for PGD of 40 steps (13 points).

For a black-box attacker, transfer offers a cheap path: train a substitute model, attack that model, then send the result to the victim model. Another route is to estimate the gradient from differences of output over many queries, which is accurate but expensive and easy to detect. The black box is weaker because its information is less, not because its attacker is less clever.

The goal of the attacker also changes the numbers drastically. An untargeted attack only has to make the label change; a targeted attack has to make the label become a particular class. At $\varepsilon = 0.15$, our PGD of 40 steps reaches the target on 33.00 percent of the examples, whereas the untargeted attack succeeds on 89.06 percent of the examples (accuracy is left at 10.72 percent). With $\varepsilon = 0.30$, the target is reached on 99.50 percent of the examples. That is why the number "the attack succeeds 90 percent" without a note on whether it is targeted cannot be compared with another number. For a defense evaluator, a targeted test is harder for the attacker, whereas an untargeted test is harder for the defender.

29.5 Defenses: what works and what only appears to work

Level: advanced

The defender has three families of answers. The first changes the data: add adversarial examples to the training. The second changes the model: make its surface smoother, for example with quantized inputs or smoothed outputs. The third adds tools outside the model: detect strange inputs, or refuse answers that are not confident. Only the first family has a clear basis, namely the saddle formulation above; the second and third families often give good numbers for the wrong reason.

The most frequently repeated mistake has a name.

Definition 29.3 (Gradient obfuscation)**.** A defense performs *gradient obfuscation* when it makes the gradient that the attacker uses useless: missing, pointing the wrong way, or random. Such a defense lowers the success of gradient-based attacks without raising the distance to the decision boundary.

In everyday language: that guard does not become more observant, he starts using a form that the attacker cannot read. Athalye, Carlini, and Wagner distinguish three ways obfuscation happens. First, a *shattered* gradient: an operation that is nondifferentiable or numerically unstable makes the gradient absent or wrong. Second, a *random* gradient: a defense that adds randomness makes one sample of the gradient not the true gradient. Third, a gradient that *explodes or vanishes*: a defense that evaluates the network many

times in sequence, as in repeated quantization, behaves like a very deep network when it is differentiated.

Why is obfuscation not a defense? Because it changes nothing that a persistent attacker sees. If the gradient is useless, the attacker uses a substitute gradient from a smoother model; if randomness is the obstacle, he averages the gradient over many samples; if the gradient explodes, he uses a finite-difference approximation.

We build the small example ourselves. Take model A and put input quantization in front of it: every pixel is rounded to a multiple of $1/8$. The derivative of rounding is zero almost everywhere, so an attacker who differentiates through that path gets a zero vector and does not move. The result looks convincing: clean accuracy 97.82 percent, and after "PGD of 40 steps" its accuracy is 96.98 percent. Almost no damage. But if the attacker uses the gradient of the model without quantization at the point that has already been quantized, its accuracy falls to 13.23 percent, only a little higher than 10.72 percent on the model without quantization. That defense does not close the road; it only closes the most naive road. The shape of the picture is in Figure 29.3.

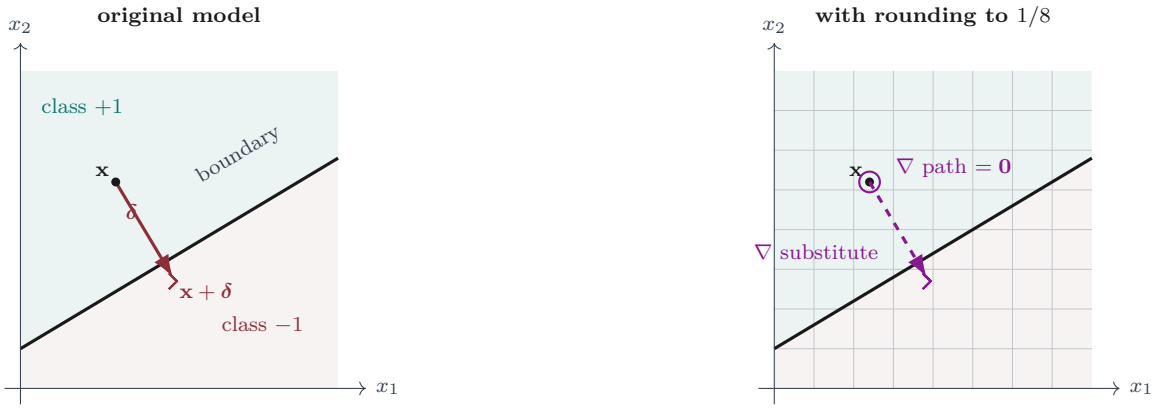


Figure 29.3: Input quantization on the same model. Left: the true gradient gives a direction $\boldsymbol{\delta}$ that takes $\mathbf{x}$ across to the other side of the boundary, so that its label changes. Right: rounding every feature to a multiple of $1/8$ makes the derivative of the path zero at almost all points, so the reported gradient arrow stops; the decision boundary itself does not shift, and the gradient of the model without rounding still takes it across. The square marks the point after the perturbation.

Athalye and colleagues examined nine white-box defenses at ICLR 2018. Seven of them depend on gradient obfuscation, and adapted attacks succeeded in breaking six completely and one partially, under the threat model each of them claimed. This is not a small error in one paper; it is a recurring pattern. That is why there is a list of symptoms that can be used before trusting a robustness number. If a one-step attack is stronger than an iterative attack, something is wrong: an iterative attack should always be stronger in the white box. If a black-box attack is stronger than a white-box one, something is wrong: the black box is a smaller set of information. If an attack without a bound on the perturbation does not reach one hundred percent success, something is wrong. If a random search finds an adversarial example while the gradient attack does not, something is wrong. If enlarging the budget does not raise success, something is wrong.

Challenge (problem solver)

A service refuses to answer when the absolute value of its score margin is below 0.50, and answers only when it is confident enough. In the linear example of the previous part, the clean margin is 1.500 and $\|\mathbf{d}\|_1 = 6$. An attack at $\varepsilon = 0.25$ pushes the margin to zero, exactly at the threshold of reversal, so it is refused, and the service looks safe. Determine

the smallest budget that keeps the attacker served, that is the one that switches the refusal off on the wrong side. Is that rule a defense or obfuscation?
Hint 1: the refusal looks only at the size of the margin, not at its sign.
Hint 2: solve $1.500 - 6\varepsilon = -0.500$.
If you get stuck, go back to the section Worked example: a linear model you can compute by hand.

What does a correct evaluation look like? It is not one number, but one protocol. Fix the threat model first: norm, ε, white box or black box, targeted or not. Use the strongest attacker you can afford, and adapt it to the defense: if the defense quantizes, attack through a substitute gradient; if it adds randomness, average over several samples; if it filters the input, attack that filter as well. Report the accuracy at several ε, not at one point, and check that the curve falls monotonically. Report more than one random seed and include transfer attacks from another model. Report the cost: the number of backward passes per example, the time, and the hardware. If you want to talk about guarantees, use certification, not testing; that is another path that is not discussed here.

For younger readers (age 10 and up)

Think about a multiple-choice exam that is graded by machine. The machine reads the answer from the position of the bubble that is filled in. If we test the machine only with answers that are filled in neatly, the machine will always be right. Then comes a student who knows how the machine reads: he fills one other bubble a little more thickly, and the machine reads it as a different answer. The paper is still the same paper, the writing is still the same writing.
Now the interesting part. One school tries to fix that machine by scanning the paper twice and taking the answer that appears most often. The same student just fills two bubbles, and the machine can still be fooled. Another school adds a ruler so that the position of the bubbles shifts a little; the patient student just tries a few positions. All those additions make the cheating more troublesome, but they do not make the machine able to tell an honest answer.
In this chapter we will find out why making things harder for the attacker is not the same as closing the hole, and how to test a fix so that we are not fooled by a fix that only appears to work.

29.6 Worked example: a linear model you can compute by hand

Level: intermediate

Now we leave the networks and return to a model that can be computed without a computer: a linear classifier with two classes and four features. Every number in this part can be checked with a calculator.

Let class A use the weights $\mathbf{w}_A = (1; -2; 0.5; 1.5)$ and class B use $\mathbf{w}_B = (0; 1; 0.5; -0.5)$, without a bias. The example that is examined is $\mathbf{x} = (1; 0.5; 1; 1)$. The scores of the two classes are

$$s_A = \mathbf{w}_A^\top \mathbf{x} = 1 - 1 + 0.5 + 1.5 = 2.0, \qquad s_B = \mathbf{w}_B^\top \mathbf{x} = 0 + 0.5 + 0.5 - 0.5 = 0.5.$$

Its margin is $m = s_A - s_B = 1.5$, so the model is confident but not excessively so. The difference of the weights is $\mathbf{d} = \mathbf{w}_A - \mathbf{w}_B = (1; -3; 0; 2)$, with $\|\mathbf{d}\|_1 = 1 + 3 + 0 + 2 = 6$ and $\|\mathbf{d}\|_2 = \sqrt{1 + 9 + 0 + 4} = \sqrt{14} \approx 3.7417$.

An untargeted attack has to lower the margin. By the lemma, the largest decrease is attained by $\boldsymbol{\delta} = -\varepsilon \operatorname{sign}(\mathbf{d}) = \varepsilon(-1; 1; 0; -1)$, and the size of the decrease is $\varepsilon\|\mathbf{d}\|_1 = 6\varepsilon$. So the margin after the attack is $1.5 - 6\varepsilon$, and the label changes when

$$\varepsilon > \varepsilon^* = \frac{1.5}{6} = 0.25.$$

At $\varepsilon = 0.25$ the margin is exactly zero, so the two classes tie and arg max chooses the first class; a clear reversal happens only a little above 0.25. The following table contains the whole course of it, and every row can be checked by multiplying $\mathbf{d}$ by $\mathbf{x} + \boldsymbol{\delta}$.

ε	s_A	s_B	margin
0	2.000	0.500	+1.500
0.05	1.775	0.575	+1.200
0.10	1.550	0.650	+0.900
0.15	1.325	0.725	+0.600
0.20	1.100	0.800	+0.300
0.25	0.875	0.875	0.000
0.30	0.650	0.950	−0.300

If the attacker is bounded by the ℓ_2 norm instead of ℓ_∞, the answer changes. The maximum of $\mathbf{d}^\top \boldsymbol{\delta}$ on the ℓ_2 ball of radius ε is $\varepsilon\|\mathbf{d}\|_2$, attained at $\boldsymbol{\delta} = \varepsilon\mathbf{d}/\|\mathbf{d}\|_2$. So $\varepsilon^* = 1.5/\sqrt{14} \approx 0.4009$. The ℓ_2 budget that gives damage of the same size as $\varepsilon = 0.25$ on ℓ_∞ is 0.4009; both move the example to the same point, namely the point of intersection $\mathbf{x} - \varepsilon \operatorname{sign}(\mathbf{d})$. This is a small example of a general rule: the number ε cannot be compared across norms.

Now the part that is often asked: is PGD stronger on this model? Every PGD step adds a decrease of the margin of $\alpha\|\mathbf{d}\|_1 = 6\alpha$, but the whole trajectory stays inside the ball, so the total decrease never exceeds 6ε. For a linear model, the worst point inside the ball really is the point that FGSM chooses, so iteration cannot beat it. Hand computation shows it: with $\varepsilon = 0.10$ and 40 steps of $\alpha = 0.01$, the final margin 0.900 is exactly the same as the result of FGSM at $\varepsilon = 0.10$.

Repeat the example above for $\mathbf{w}_A = (2; 1; -1; 0.5)$, $\mathbf{w}_B = (0; 1; 1; -0.5)$, and $\mathbf{x} = (1; 1; 0.5; 1)$. Compute s_A, s_B, $\mathbf{d}$, $\|\mathbf{d}\|_1$, and ε^* for the ℓ_∞ norm. Then compute ε^* for the ℓ_2 norm and explain why the two numbers differ. Check with a three-line Python snippet. After that decide, without running a program, whether PGD of 20 steps with $\alpha = \varepsilon/10$ can change the label when ε is only half of ε^*.

29.7 Computational lab

Level: advanced

This lab trains three small networks on handwritten digit data provided by the `scikit-learn` library: 1797 examples, 64 features, values 0 to 16 which we divide by 16 so that they fall in $[0, 1]$. Twelve hundred examples are used for training and 597 for test. Model A has one hidden layer of 64 units, model B 128 units, and model D is the same as A but trained with FGSM at $\varepsilon = 0.15$ (the reason for using D rather than A is that the comparison of clean and robust accuracy should come from models of the same size).

All the gradients are computed by hand with NumPy, without an automatic differentiation library, so that every step can be checked. The function `grad_x` is the backward pass of Chapter 25 that stops at the input, not at the weights.

Computational lab

```
import numpy as np, time
from sklearn.datasets import load_digits

X, y = load_digits(return_X_y=True)
X = X / 16.0                        # 0..16 -> 0..1
perm = np.random.default_rng(0).permutation(len(X))
tr, te = perm[:1200], perm[1200:]
Xtr, ytr, Xte, yte = X[tr], y[tr], X[te], y[te]
K = 10
Ytr, Yte = np.eye(K)[ytr], np.eye(K)[yte]

def init(dims, seed):
    r = np.random.default_rng(seed); W, b = [], []
    for i in range(len(dims) - 1):
        W.append(r.normal(0, np.sqrt(2.0 / dims[i]),
                          (dims[i], dims[i + 1])))
        b.append(np.zeros(dims[i + 1]))
    return W, b

def fwd(W, b, X):
    acts, pre, h = [X], [], X
    for i in range(len(W) - 1):
        z = h @ W[i] + b[i]; pre.append(z)
        h = np.maximum(z, 0); acts.append(h)
    z = h @ W[-1] + b[-1]; pre.append(z)
    z = z - z.max(1, keepdims=True); p = np.exp(z)
    return p / p.sum(1, keepdims=True), acts, pre

def loss_grad(W, b, X, Y):
    p, acts, pre = fwd(W, b, X); n = len(X)
    loss = -np.mean(np.sum(Y * np.log(p + 1e-12), 1))
    d = p - Y; gW = [None] * len(W); gb = [None] * len(W)
    for i in range(len(W) - 1, -1, -1):
        gW[i] = acts[i].T @ d / n + 1e-4 * W[i]; gb[i] = d.mean(0)
        if i > 0:
            d = (d @ W[i].T) * (pre[i - 1] > 0)
    return loss, gW, gb

def grad_x(W, b, X, Y):             # gradient of the error wrt the input
    p, acts, pre = fwd(W, b, X)
    d = (p - Y) / len(X)
    for i in range(len(W) - 1, 0, -1):
        d = (d @ W[i].T) * (pre[i - 1] > 0)
    return d @ W[0].T

def train(dims, seed, Xa, Ya, epochs=350, lr=0.01, adv_eps=None):
    W, b = init(dims, seed)
    mW = [np.zeros_like(w) for w in W]
    vW = [np.zeros_like(w) for w in W]
    mb = [np.zeros_like(v) for v in b]
    vb = [np.zeros_like(v) for v in b]
    for t in range(1, epochs + 1):
        if adv_eps is None:
            Xb = Xa
        else:
            g = grad_x(W, b, Xa, Ya)
            Xb = np.clip(Xa + adv_eps * np.sign(g), 0, 1)
        _, gW, gb = loss_grad(W, b, Xb, Ya)
        for i in range(len(W)):
            mW[i] = 0.9 * mW[i] + 0.1 * gW[i]
            vW[i] = 0.999 * vW[i] + 0.001 * gW[i] ** 2
            mb[i] = 0.9 * mb[i] + 0.1 * gb[i]
```

```
            vb[i] = 0.999 * vb[i] + 0.001 * gb[i] ** 2
            mh = mW[i] / (1 - 0.9 ** t)
            vh = vW[i] / (1 - 0.999 ** t)
            W[i] -= lr * mh / (np.sqrt(vh) + 1e-8)
            mh = mb[i] / (1 - 0.9 ** t)
            vh = vb[i] / (1 - 0.999 ** t)
            b[i] -= lr * mh / (np.sqrt(vh) + 1e-8)
    return W, b

def acc(W, b, X, y):
    return float(np.mean(fwd(W, b, X)[0].argmax(1) == y))

def fgsm(W, b, X, y, eps):
    g = grad_x(W, b, X, np.eye(K)[y])
    return np.clip(X + eps * np.sign(g), 0, 1)

def pgd(W, b, X, y, eps, steps, Ylab=None, arah=1):
    alpha = eps / 10
    r = np.random.default_rng(1)
    xa = np.clip(X + r.uniform(-eps, eps, X.shape), 0, 1)
    Y = np.eye(K)[y] if Ylab is None else Ylab
    for _ in range(steps):
        g = arah * np.sign(grad_x(W, b, xa, Y))
        xa = np.clip(np.clip(xa + alpha * g, X - eps, X + eps), 0, 1)
    return xa
```

The rest is calls and printing. We train A, B, and D, then sweep ε from 0.02 to 0.30.

```
WA, bA = train([64, 64, 10], 1, Xtr, Ytr)
WB, bB = train([64, 128, 10], 2, Xtr, Ytr)
WD, bD = train([64, 64, 10], 3, Xtr, Ytr, epochs=120, adv_eps=0.15)

print(f"clean test accuracy: A {acc(WA,bA,Xte,yte):.4f}"
      f"  B {acc(WB,bB,Xte,yte):.4f}  D {acc(WD,bD,Xte,yte):.4f}")
print(f"{'eps':>5} {'A clean':>9} {'A FGSM':>7} {'A PGD40':>8}"
      f" {'D FGSM':>7} {'D PGD40':>8}")
for eps in [0.02, 0.05, 0.10, 0.15, 0.20, 0.30]:
    xf = fgsm(WA, bA, Xte, yte, eps)
    xp = pgd(WA, bA, Xte, yte, eps, 40)
    xd = fgsm(WD, bD, Xte, yte, eps)
    xq = pgd(WD, bD, Xte, yte, eps, 40)
    print(f"{eps:5.2f} {acc(WA,bA,Xte,yte):9.4f} {acc(WA,bA,xf,yte):7.4f}"
          f" {acc(WA,bA,xp,yte):8.4f} {acc(WD,bD,xd,yte):7.4f}"
          f" {acc(WD,bD,xq,yte):8.4f}")

r = np.random.default_rng(7)
rs = np.sign(r.normal(size=Xte.shape))          # control: random signs
print("random signs eps=0.15:",
      f"{acc(WA,bA,np.clip(Xte+0.15*rs,0,1),yte):.4f}")
print("PGD steps, eps=0.15, alpha=eps/10, random start")
for k in [1, 2, 5, 10, 20, 40]:
    print(f"  {k:2d} steps:",
          f"{acc(WA,bA,pgd(WA,bA,Xte,yte,0.15,k),yte):.4f}")
print("  FGSM (1 step, alpha=eps):",
      f"{acc(WA,bA,fgsm(WA,bA,Xte,yte,0.15),yte):.4f}")
```

The actual output, as it stands, for the block above:

```
clean test accuracy: A 0.9799  B 0.9849  D 0.9648
  eps  A clean  A FGSM  A PGD40  D FGSM  D PGD40
 0.02    0.9799  0.9548   0.9548  0.9497   0.9497
 0.05    0.9799  0.8476   0.8425  0.9162   0.9129
```

```
 0.10    0.9799  0.4992   0.4590  0.8241   0.8174
 0.15    0.9799  0.1491   0.1072  0.6667   0.6265
 0.20    0.9799  0.0184   0.0050  0.4858   0.3719
 0.30    0.9799  0.0000   0.0000  0.0687   0.0201
random signs eps=0.15: 0.9548
PGD steps, eps=0.15, alpha=eps/10, random start
   1 steps: 0.9464
   2 steps: 0.8878
   5 steps: 0.6667
  10 steps: 0.2479
  20 steps: 0.1122
  40 steps: 0.1072
  FGSM (1 step, alpha=eps): 0.1491
```

The first reading of that table: clean accuracy 97.99 percent, and at $\varepsilon = 0.15$ only 14.91 percent is left.

The second reading: the gap between FGSM and PGD widens in the middle, from zero at 0.02 to 4.19 points at 0.15. At a small budget one step already touches the wall of the box; at a large budget several steps are needed to find a worse direction.

The third reading, and the most important one for this chapter: the row of random signs. If a perturbation of size $\varepsilon = 0.15$ is given with random signs on every pixel, accuracy only falls to 95.48 percent. FGSM with the same size lowers it to 14.91 percent. So the choice of signs by the gradient is worth about eighty points of accuracy: it is the direction that works, not the size.

The fourth reading: at $\varepsilon = 0.30$, A is already at zero percent while D is still at 6.87 percent under FGSM and 2.01 percent under PGD. Adversarial training does not make the model immune; it shifts the curve to the right.

The next block measures two things that are not visible in the table: whether the gradient we compute really is the gradient, and how much it costs.

```
i0 = int(np.argmax(np.abs(grad_x(WA, bA, Xte, Yte)).sum(1)))
x0, y0 = Xte[i0:i0 + 1], Yte[i0:i0 + 1]
g0 = grad_x(WA, bA, x0, y0)
L1 = lambda z: loss_grad(WA, bA, z, y0)[0]
print(f"finite difference at sample i={i0}, h=1e-6")
for j in np.argsort(-np.abs(g0[0]))[:3]:
    e = np.zeros_like(x0); e[0, j] = 1e-6
    fd = (L1(x0 + e) - L1(x0 - e)) / 2e-6
    print(f"  j={j:2d} analytic {g0[0,j]:+.6e} finite-diff {fd:+.6e}"
          f" error {abs(fd-g0[0,j])/abs(g0[0,j]):.0e}")
g_all = grad_x(WA, bA, Xte, Yte)
lin = 0.15 * np.abs(g_all).sum(1).mean()
lc = loss_grad(WA, bA, Xte, Yte)[0]
la = loss_grad(WA, bA, fgsm(WA, bA, Xte, yte, 0.15), Yte)[0]
print(f"mean |g|_1 {np.abs(g_all).sum(1).mean():.4f};"
      f" linear prediction {lin:.4f} nats; actual {la-lc:.3f} nats")

t0 = time.perf_counter(); fgsm(WA, bA, Xte, yte, 0.15)
t1 = time.perf_counter(); pgd(WA, bA, Xte, yte, 0.15, 40)
t2 = time.perf_counter()
print(f"time for 597 examples: FGSM {t1-t0:.4f} s  PGD40 {t2-t1:.4f} s")
```

Its output:

```
finite difference at sample i=145, h=1e-6
  j=53 analytic -6.391833e+00 finite-diff -6.391833e+00 error 2e-11
  j=38 analytic +6.211518e+00 finite-diff +6.211518e+00 error 1e-10
  j=13 analytic -6.175321e+00 finite-diff -6.175321e+00 error 8e-11
mean |g|_1 0.0056; linear prediction 0.0008 nats; actual 4.862 nats
time for 597 examples: FGSM 0.0006 s  PGD40 0.0302 s
```

The first three lines confirm that our backward pass is correct: the analytic gradient and the finite difference with $h = 10^{-6}$ agree to about ten digits, with relative errors 10^{-10} and 10^{-11}. For a network that uses ReLU, agreement that smooth is reasonable: at the scale 10^{-6} no unit changes sign, so the function really is linear around that point. If these numbers differed a lot, what would be wrong is the backward-pass code, not the attack.

The fourth line is the surprising part. The average $\|\nabla L\|_1$ is only 0.0056, so the linear prediction for $\varepsilon = 0.15$ is 0.0008 nats. The actual rise in the error is 4.862 nats. So the linear term underestimates the damage by about four thousand times. This is not a sign that the gradient is wrong (the three lines above prove the opposite), but a sign that L is strongly curved at the distance 0.15: ReLU units change sign, and the first-order approximation loses its meaning. What remains useful is the sign, as the row of random signs showed.

The last line is about time: for 597 examples, FGSM needs 0.6 milliseconds and PGD of 40 steps 30 milliseconds, about fifty times more expensive. This number depends on the machine and on vectorization, so what is worth remembering is its order of magnitude, not the exact number.

Finally, a demonstration of gradient obfuscation. We add input quantization to multiples of 1/8 in front of model A, then attack in two ways.

```
def q(z): return np.round(z * 8) / 8
accq = lambda W, b, Z, yy: float(np.mean(fwd(W, b, q(Z))[0]
                                         .argmax(1) == yy))
def pgd_naif(W, b, Z, yy, eps, steps):
    r = np.random.default_rng(1)
    z = np.clip(Z + r.uniform(-eps, eps, Z.shape), 0, 1)
    for _ in range(steps):          # gradient = zero, so it stands still
        z = np.clip(np.clip(z, Z - eps, Z + eps), 0, 1)
    return z
def pgd_bpda(W, b, Z, yy, eps, steps):
    r = np.random.default_rng(1)
    z = np.clip(Z + r.uniform(-eps, eps, Z.shape), 0, 1)
    for _ in range(steps):
        g = np.sign(grad_x(W, b, q(z), np.eye(K)[yy]))
        z = np.clip(np.clip(z + eps / 10 * g, Z - eps, Z + eps), 0, 1)
    return z
print("inputs quantized to multiples of 1/8")
print(f"  clean {accq(WA,bA,Xte,yte):.4f}"
      f"  PGD40 naive {accq(WA,bA,pgd_naif(WA,bA,Xte,yte,0.15,40),yte):.4f}"
      f"  PGD40 BPDA"
      f" {accq(WA,bA,pgd_bpda(WA,bA,Xte,yte,0.15,40),yte):.4f}")
```

Its output:

```
inputs quantized to multiples of 1/8
  clean 0.9782  PGD40 naive 0.9698  PGD40 BPDA 0.1323
```

This is the most concise form of a false defense. Its clean number is only 0.17 points below the original model, and the naive attack does not move at all: if that were all that was reported, this defense would look almost perfect. The substitute gradient brings it down to 13.23 percent, almost the same as 10.72 percent on the model without quantization. What changes above all is the readability of its gradient.

29.8 Cost, limitations, and how to report

Level: basic

The computational budget determines which attack is honestly used. FGSM requires one backward pass per example. PGD of k steps requires k passes. In our experiment, the cost of

PGD of 40 steps is about fifty times that of FGSM, for a result that, at small ε, is almost the same. For the assessment of a defense, that cost is reasonable: a robustness number measured with a weak attack is meaningless. For training, that cost multiplies again: every epoch demands a new attack.

Adversarial training also has an accuracy price. Model D loses 1.51 points of clean accuracy in exchange for a rise of about fifty points at $\varepsilon = 0.15$. Madry and colleagues report that larger models give better robustness, which shows that robustness is not a free property of the architecture, but capacity that has to be paid for.

There are three limitations before the numbers in this chapter are used.

First, scale. Our data has only 64 features and 10 classes, with 597 test examples. On large images, a meaningful budget is much smaller than 0.15; on ImageNet, Goodfellow and colleagues used 0.007 and already flipped the classification. The number ε cannot be moved between datasets, because its meaning depends on the scale of the features. On our data, the pixels are already quantized to multiples of $1/16$, so that $\varepsilon = 0.05$ means less than one quantization step per pixel.

Second, the coverage of attacks. What we run is only two families: FGSM and PGD with ℓ_∞. A model that resists both is not necessarily resistant to an ℓ_2 attack, to an attack that changes a single pixel, or to an attack that sticks an object into the real world. Every robustness claim holds only for the family of attacks that was tested, and that is why the evaluation protocol in the previous part demands several kinds of attack at once.

Third, there is no guarantee. All the numbers here are results of testing, not a proved lower bound. Model D, which reaches 62.65 percent under PGD of 40 steps, may still be broken by a better attack; we do not know, because what was not found is not proof that it does not exist. For a stronger statement, certification is needed: for example with randomized smoothing or with formal verification on a small network. Both give a guarantee on a certain radius, at the price of far higher accuracy and computation cost, and both are outside the scope of this chapter.

An honest way of reporting contains five things: the complete threat model (norm, ε, the knowledge of the attacker, targeted or not), the attacker used together with the number and size of its steps, the clean accuracy and the accuracy at several ε, the number of seeds and the spread of the results, and the computational cost. Without that, the number “90 percent robust” cannot be checked and cannot be compared. Chapter 27 has already taught us that metrics can deceive; here the metric deceives in a subtler way, because its denominator is right and only its numerator is chosen.

Note for instructors

The teaching order that has worked: start from the table of accuracy against ε, not from the definition. Students see 97.99 and 14.91 on the same model, then ask why. Only after that come the definition of the threat model, the lemma on the maximum of a linear form, and FGSM. Only after FGSM runs do we enter PGD and the saddle formulation. The reverse order, that is definitions first and then attacks, makes students memorize notation without ever seeing the damage.

Two typical misconceptions. First, students think an adversarial attack is the same as adding noise, so they conclude that a model resistant to noise will be resistant to attacks. The row of random signs in the lab refutes that with one number: noise of size $\varepsilon = 0.15$ only lowers accuracy to 95.48 percent, whereas FGSM with the same size lowers it to 14.91 percent. Second, they think one accuracy number under an attack is already evidence of robustness. The quantization demonstration in the lab shows the opposite: 96.98 percent looks safe until someone uses a substitute gradient and gets 13.23 percent.

A cheap blackboard activity: write the four-feature linear example from the worked

example part, then ask the class to compute ε^* for ℓ_∞ and ℓ_2. After that ask them to predict, without computing, whether adding PGD steps changes the result on that linear model. The correct answer, together with its reason, forces them to understand that iteration is useful only because f is not linear.
A sign of readiness to go on: students can explain why only the sign of the gradient is used, can compute ε^* for a linear model, can name one symptom of gradient obfuscation, and can design one attack adapted to a defense that quantizes the input. If they demand numbers at several ε before believing, they are ready to enter Chapter 30.

29.9 Summary

Summary.

- A robustness claim is always relative to the threat model: the norm, the budget ε, the knowledge of the attacker, and whether the goal is targeted or not. Without that, two numbers cannot be compared.
- FGSM shifts every coordinate by ε following the sign of the gradient of the error. On a linear model it is exact and optimal, because $\max_{\|\boldsymbol{\delta}\|_\infty \le \varepsilon} \mathbf{w}^\top \boldsymbol{\delta} = \varepsilon \|\mathbf{w}\|_1$.
- Linearization justifies the direction, not the distance. In our experiment the linear prediction is 0.0008 nats, whereas the actual rise is 4.862 nats; what remains useful is the pattern of signs.
- PGD adds iteration and projection onto the box. At $\varepsilon = 0.15$ accuracy falls from 94.64 percent (1 small step) to 10.72 percent (40 steps), and is flat after about twenty steps.
- Adversarial training is an approximation of the saddle problem $\min_\theta \mathbb{E} \max_\delta L$. It raises robustness at the price of clean accuracy: our model loses 1.51 points to rise from 10.72 to 62.65 percent.
- Gradient obfuscation lowers the success of attacks without raising the distance to the decision boundary. Our example: 96.98 percent under the naive attack, 13.23 percent under the attack with a substitute gradient.
- An honest evaluation uses the strongest attack that can be afforded, adapted to the defense, at several ε and several seeds, with the cost reported.

29.10 Exercises

Exercise 29.1.

1. Compute ε^* for the linear model with $\mathbf{w}_A = (1; -1; 2)$, $\mathbf{w}_B = (0; 1; 1)$ and $\mathbf{x} = (1; 2; 1)$ under the ℓ_∞ norm, then under the ℓ_2 norm. Check both answers with Python. (Hint: compute $\mathbf{d} = \mathbf{w}_A - \mathbf{w}_B$ first, then compare $\|\mathbf{d}\|_1$ with $\|\mathbf{d}\|_2$.)
2. Prove that $\max_{\|\boldsymbol{\delta}\|_2 \le \varepsilon} \mathbf{w}^\top \boldsymbol{\delta} = \varepsilon \|\mathbf{w}\|_2$, and determine all $\boldsymbol{\delta}$ that attain the maximum. Explain why the answer has a different form from the ℓ_∞ case.
3. Prove that on a linear classifier, PGD with any number of steps never produces an error larger than FGSM with the same ε. (Hint: the error is a linear function of $\boldsymbol{\delta}$ on a convex set; the maximum of a linear function on a compact convex set is attained at an extreme point. Show that an extreme point of the ℓ_∞ ball has the form $\pm\varepsilon$ in every coordinate, then compare with the FGSM point.)
4. Assess this claim: *because a model reaches accuracy* 98 *percent on test data and stays at* 95 *percent under random perturbation of size ε, that model is resistant to adversarial attacks.* Decide true, false, or not enough evidence, and name one number from this chapter that supports your decision.
5. From the lab table, compute the success rate of the untargeted attack for model A at $\varepsilon = 0.05$, 0.10, and 0.15, then for model D at the same budgets. State the results in

percent and explain why the gap between A and D narrows at large ε.

6. Show that the projection onto $\{\mathbf{y} : \|\mathbf{y} - \mathbf{x}\|_\infty \le \varepsilon\} \cap [0,1]^d$ may be done by clipping first to the ε box and then to $[0,1]$, or the other way round, and the result is the same. (Hint: both sets are products of intervals, so it is enough to check one coordinate.)
7. On model D, the clean accuracy is 96.48 percent and the accuracy under PGD of 40 steps at $\varepsilon = 0.15$ is 62.65 percent. On model A, the numbers are 97.99 and 10.72. Compute the price of robustness in points of clean accuracy per point of robustness gained, then explain why that ratio is not a complete measure.
8. A defense rounds the probability output to 0 or 1 before the decision is taken, so that the gradient with respect to the input becomes zero. Design one evaluation protocol to test its robustness claim, and name which symptom from the list in this chapter is most easily seen if that defense only obfuscates the gradient.
9. Given two models with clean accuracy 97.99 and 98.49 percent. At $\varepsilon = 0.15$, PGD of 40 steps built on the first model lowers the accuracy of the first model to 10.72 percent and of the second model to 23.79 percent. Compute the gap between the success of the white-box attack and of the transfer attack, then explain why that gap is not evidence that the second model is safer. (Hint: an attack built specifically for the second model is also available; see row B in the lab output.)
10. Explain why a targeted attack is harder than an untargeted attack, then use the numbers 33.00 percent at $\varepsilon = 0.15$ and 99.50 percent at $\varepsilon = 0.30$ to show that the difficulty depends on the budget, not only on the kind of attack.

For researchers: limits and next steps

What is proved in this chapter is little. The lemma on the maximum of a linear form is a one-line proof. The linearization inequality with remainder $M\|\boldsymbol{\delta}\|_2^2/2$ is the mean value theorem in integral form, valid for every differentiable function with a Lipschitz gradient. The exactness of FGSM on logistic regression is a result of Goodfellow, Shlens, and Szegedy (ICLR 2015), the section "Adversarial Training of Linear Models versus Weight Decay". The saddle formulation and the claim that PGD can be relied on come from Madry, Makelov, Schmidt, Tsipras, and Vladu (ICLR 2018), sections 2 and 3; the latter is an empirical claim, not a theorem, and its own authors call the inner problem nonconcave.

What is more often popularized than proved is the claim that linearity explains adversarial examples. As an argument about direction it is useful; as an explanation of the size of the damage it fails in our experiment: a prediction of 0.0008 nats against an actual 4.862 nats, with a gradient that has been verified to 10^{-10}. What works is the pattern of signs, and that is why FGSM beats random signs (14.91 against 95.48 percent). A second claim that is exaggerated: adversarial training gives robustness to the attacker and the budget used during training, at the price of clean accuracy, without a guarantee for other attacks.

Limits of validity: 64 features, 597 test examples, one hidden layer, one seed, two families of ℓ_∞ attacks, without certification. The numbers here demonstrate a mechanism, not a ranking of robustness. Open questions: how large a radius can be certified without erasing the benefit in accuracy; why transfer works that well; and what threat model makes sense for real systems such as intrusion detectors, which is the subject of Chapter 30.

Map to source texts: This chapter follows Goodfellow, Shlens, and Szegedy, Explaining and Harnessing Adversarial Examples *(ICLR 2015), sections 3 ("The Linear Explanation of Adversarial Examples"), 4 ("The Fast Gradient Sign Method"), and 5 ("Adversarial Training of Linear Models versus Weight*

Decay") for the derivation of FGSM, its exactness on logistic regression, and the MNIST and CIFAR-10 numbers. The saddle formulation, the PGD algorithm, and the MNIST and CIFAR-10 robustness numbers come from Madry, Makelov, Schmidt, Tsipras, and Vladu, Towards Deep Learning Models Resistant to Adversarial Attacks *(ICLR 2018), sections 2, 3, and 4, including Algorithm 1. The discussion of gradient obfuscation follows Athalye, Carlini, and Wagner,* Obfuscated Gradients Give a False Sense of Security *(ICML 2018), sections 2 to 4, including the three kinds of obfuscation and the five symptoms used here. The phenomenon of transferability was first reported by Szegedy and colleagues,* Intriguing Properties of Neural Networks *(ICLR 2014). All the numbers in the tables, the lab, and the worked example are our own computation with Python 3.14, NumPy 2.5, and the* `load_digits` *dataset from scikit-learn 1.9: Adam training for 350 epochs, a train-test split of 1200/597 with seed 0, and input gradients verified with finite differences.*

Chapter 30

The Limits of Attack: Obfuscated Gradients and NIDS Evasion

A group of students added one line of rounding to the model input, then reported that their own attack failed completely: the success rate fell from seventeen percent to zero point zero two percent. That number went into the executive summary, then into the slides, then into the abstract. Two weeks later an examiner asked for one thing: prove that the gradient the attacker used still points in the right direction. It did not. Rounding is flat almost everywhere, so its derivative is zero, so the attacker takes a zero step, so the attack that failed is really an attack that never happened. When the examiner replaced the derivative with a reasonable approximation, the success rate returned to seventeen percent within minutes.

This chapter is about that mistake. The first mistake is simple: rounding erases the gradient, and erasing the gradient means blinding the attacker, not making the model strong. The second mistake is more interesting: there are times when obfuscating the gradient really does raise the cost of attack, so part of that robustness is real. Telling the two apart demands experiment, not rhetoric.

The second half of the chapter carries the same problem to a much noisier place: network intrusion detection systems, or NIDS. There packets must still arrive, protocols must remain valid, decisions are made in milliseconds, and the model is retrained every day. All of that makes an evasion attack that is beautiful on paper impractical in the field. But “impractical” is not “impossible”.

30.1 Gradients that vanish, and the sense of safety born from them

Level: basic

Chapter 29 taught two attacks that rest on one object: the gradient of the loss with respect to the input. The one-step attack takes the direction of the sign of the gradient and steps by ε,

$$x' = x + \varepsilon \, \operatorname{sign}\big(\nabla_x L(x, y)\big), \tag{30.1}$$

whereas the many-step attack repeats a small step and projects the result back onto the ℓ_∞ ball of radius ε,

$$x_{t+1} = \Pi_{B(x,\varepsilon)}\Big(x_t - \alpha \, \operatorname{sign}\big(\nabla_x L(x_t, y)\big)\Big). \tag{30.2}$$

Both work for one reason that already appeared in the chapter on linearization: for a small perturbation the loss can be approximated by a linear function, so the direction that

raises the loss fastest is the direction of the gradient. If that gradient is correct, the attack works at the cost of one backward pass. If that gradient is wrong, the attacker goes blind. Those two sentences are not equivalent to "if the gradient is wrong, the model is safe". A blind attacker can still guess, and a guess without a gradient, for example random noise on packets, can still succeed in some cases. What is lost is only the cheapest road, not every road.

The term *obfuscated gradient* names the situation in which the gradient an attacker sees does not give enough information about the direction that actually raises the loss, not because the model is strong, but because of the way the model is wrapped. Athalye, Carlini, Wagner and colleagues gave this symptom a sharper name: a false sense of security. In their review of nine defenses from a machine learning conference, seven depended on an obfuscated gradient; six were broken completely and one partially, at an attack cost roughly the same as an ordinary attack.

For younger readers (age 10 and up)

Imagine a teacher who does not want her students to cheat, so she seals the notebook shut with duct tape. The honest student does have trouble. But the creative student does not stop: he notices that the answer to question one is always larger than the answer to question two, then guesses the whole sheet from that pattern. The tape does not erase the answers; it only blocks one road to them.

A model that rounds its input works like that tape. The numbers that enter the model are cut to a fixed multiple, for instance two decimal places, so that small changes are invisible altogether. A program that tries to find the "raise it a little at a time" direction will find that nothing changes, then give up and report that the model is immune. That report is wrong, and not because the program is stupid. The program simply asked in a way that does not fit.

Now the part that matters most to remember. The same model can still be wrong if the numbers are changed enough to cross a cut boundary. So the right question is not "does a small attack succeed", but "how large a change is needed". This chapter teaches how to ask that question, and how to recognize a report that never asks it.

Why does rounding erase the gradient? Because rounding is a step function. Take $q > 0$ and define $Q_q(t) = q\,\mathrm{round}(t/q)$, the nearest multiple of q. On the interval $kq - q/2 \le t < kq + q/2$ this function is constant at kq, so its derivative is zero inside the interval and undefined at the boundary points. The set of boundary points has measure zero, so for any smooth function f, the composition $f \circ Q_q$ has zero gradient almost everywhere.

Proposition 30.1. *Let $Q_q \colon \mathbb{R}^d \to \mathbb{R}^d$ act coordinate by coordinate as above, and let $f \colon \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable. Then $\nabla(f \circ Q_q)(x) = 0$ for almost every $x \in \mathbb{R}^d$.*

Proof. Partition $\mathbb{R}^d$ into half-open boxes of side q, bounded by the hyperplanes $x_j = k_j q \pm q/2$. In the interior of every box, Q_q is constant, so $f \circ Q_q$ is also constant and its gradient is zero. The complement of the union of all those interiors is the union of the boundary hyperplanes, which has measure zero. So the gradient is zero almost everywhere. □

Corollary 30.2. *A one-step attack that uses the exact gradient of $f \circ Q_q$ produces $x' = x$ for every $\varepsilon > 0$. The measurable success rate is therefore exactly zero, and the reported "accuracy under attack" is precisely the clean accuracy.*

This corollary is the heart of the whole chapter. The statement is not about the strength of the model; it is about the arithmetic of the attacker, and the zero in the results table does not measure robustness, but the absence of a step. If the "accuracy under attack" column is

identical to the clean accuracy down to two decimal places, you are not looking at a defense that worked, but at an attacker that stopped.

Read it aloud (auditory learner)

Two voices. The first carries a map, the second carries a flashlight.
— Look. I have made the map I drew blurry. Every uphill road I erased, there are no arrows, no numbers. — Then the mountain is gone. — No. The mountain is still there. — So what is gone is only the picture? — What is gone is only the picture. — Someone who can smell the soil can still climb. — Still can. — Someone who throws a pebble and listens to the echo can also still climb. — Also still can. — So your map is not a wall. — Your map is not a wall. Your map is only a map that refuses to answer.
Repeat the last line three times, slower and slower: a map that refuses to answer is not a wall. Not a wall. Not a wall.

Athalye and colleagues divided the mechanism into three families. The first family, shattered gradients, appears when part of the model is non-differentiable or differentiable but useless, such as rounding, median filtering, and compression. The second family, stochastic gradients, appears when the model randomizes something on every call, so that the gradient of one call does not hold for the next. The third family, vanishing or exploding gradients, appears when the defense suppresses the gradient during training, so that the gradient becomes too small or too large to use for a step.

All three share one feature: they make the measurement look better without making the model better, and because the measurement is usually the only evidence that is reported, this error spreads fast. This chapter does not say that all such defenses are useless; some raise the cost of attack, and that cost is real. What is wrong is concluding robustness from a broken measurement.

30.2 Seven ways to obfuscate the gradient

Level: intermediate

The table below summarizes the seven techniques that appear most often, their mechanism family, and the symptom that ought to make you suspicious. The numbers in the last column are not a diagnosis, but a trigger for a question.

No	Technique	Family	Visible symptom
1	Input or output quantization	shattered	zero gradient; attack accuracy equals clean accuracy
2	Non-differentiable preprocessing: median, blur, compression	shattered	zero gradient, or spikes with no pattern
3	Sample filtering and rejection	shattered	discrete decisions; no gradient
4	Random transforms: resize, pad, shift	stochastic	a different gradient on every call
5	Random classifiers and random ensembles	stochastic	one sample points the wrong way
6	Gradient suppressed during training	vanishing	small gradient despite a large loss
7	Multi-step optimization defenses	vanishing	exploding gradient; attack does not converge

The first technique, quantization, has already been proved above, and its effect is exact: not approximately zero, but zero. The second technique works the same way but more dirtily. Median filtering replaces every value with the median of its neighborhood, so the function

stays flat in most places, but the boundaries between its pieces are no longer evenly spaced. Compression such as JPEG discards high-frequency components; the derivative exists but is unstable, and a finite difference gradient can change sign when the step is shifted. The symptom is characteristic: the gradient is not zero, but large-scale garbage.

The third technique is the crudest form of obfuscation. The model refuses to answer for inputs it considers strange, for example those whose distance to the data center is too large, then returns the label "unknown". For the attacker this is not a bad gradient but no gradient at all: the function is an indicator function, whose derivative is zero everywhere except on the decision surface. A defense of this kind also rejects legitimate new data, and legitimate data is the reason the system was installed.

The fourth and fifth techniques are the stochastic family. In the fourth, the input is randomly transformed before it enters the model: the image is enlarged, shifted by one pixel, or some of its features are dropped. In the fifth, what is randomized is the model: one ensemble is chosen at random on every call, or the prediction is taken from the average of many models. In both, the gradient of one call is the gradient of one realization, whereas what the attacker faces is the distribution of realizations. A bad gradient sample is not due to bias but to variance: the sign of a direction that keeps changing is not a good direction.

The sixth technique attacks from the inside. Training deliberately makes the gradient small around the data points, for example by penalizing the gradient norm or by using saturating activations, so that the loss surface is very flat around the examples and steep elsewhere. An attacker that steps by ε in a flat region goes nowhere, and the gradient really is small, but not because the model is right. The seventh technique is its relative: the input is first cleaned by a multi-step optimization procedure before being classified, and the gradient of that pipeline easily explodes.

Figure 30.1 shows the shape of the surface that quantization produces on two coordinates. Every flat plateau there is one step. Inside one plateau, the loss does not change at all, so an attacker that takes a small step stops where it is — the three short arrows in that figure. An attacker that is allowed to step far enough can cross to the next plateau, and the dashed arrow shows that crossing.

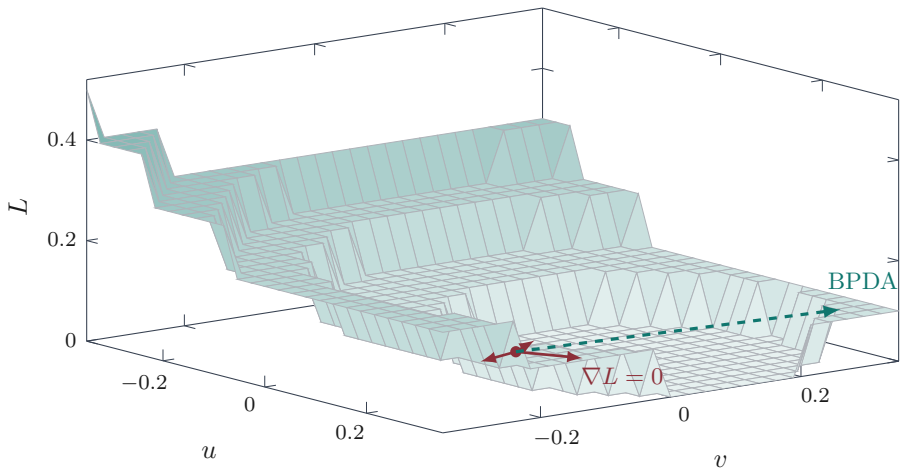


Figure 30.1: A loss surface made flat by quantization: $L(u, v) = \frac{1}{10}\,\mathrm{round}(10\,s)$ with $s = (u - 0.15)^2 + (v - 0.10)^2$. Every plateau of height 0.1 is one step: inside it $\nabla L = 0$, so the three short arrows from the red point do not change the height at all. The dashed arrow crosses to the next step, and only an attacker willing to step as far as the distance between boundaries reaches it.

Do it by hand (kinesthetic learner)

Materials: one long ruler, one coin, one friend, one table.
Prop up one end of the ruler with a book so that its surface is tilted, then place the coin in the middle. Close your eyes and hold the coin with one finger. Your friend taps the table slowly, irregularly, and your task is to guess in which direction the ruler rises — only from the vibration that reaches your finger.
Repeat ten times and record right or wrong. Then ask your friend to tap harder, and repeat ten times. Then ask him to stop tapping altogether, and repeat again.
What you should see: harder tapping does not make your guesses right. Tapping that stops makes your guesses pure guesses, about half right, no better than tossing a coin. The vibration tells you that something is moving; it does not tell you the direction. That is the difference between a bad gradient and a gradient that does not exist.

Those seven techniques produce seven symptoms. The first five are the list of Athalye and colleagues; the last two we add from the same typology.

1. **The attack is unbounded.** The report allows a perturbation of any size, then concludes robustness from an attack that ignores the bound.
2. **The attack uses a single random sample.** Against a defense that randomizes, the attacker draws one realization then reports its failure. What is measured is one throw of the die, not the probability of winning.
3. **The attack is only one step.** A one-step attack is easily stopped by an uneven loss surface, and the robustness it implies disappears as soon as the step is repeated.
4. **Increasing the distortion does not increase success.** If the budget is enlarged but the success does not move, what is broken is usually the gradient.
5. **The gradient is reported as zero or NaN.** This is the most naked symptom: a gradient that is entirely zero on a non-constant model signals a non-differentiable piece, not a flat surface.
6. **Accuracy under attack equals clean accuracy.** Our addition. Equality down to several decimal places at several different ε is the mark of an attack that does not move.
7. **There is no adaptive attack.** Our addition. The defense is tested only with attacks designed before that defense existed, so it has not been tested by an attacker who knows that he is being tested.

obfuscated gradient (*gradien dikaburkan*) — the gradient given to the attacker does not point in the right direction, usually because there is a non-differentiable or random piece inside the pipeline, not because the model is strong

quantization (*kuantisasi*) — replacing every value with the nearest multiple of a fixed distance; the result is a step function whose derivative is zero almost everywhere

adaptive attack (*serangan adaptif*) — an attack designed with knowledge of the defense being tested, including knowing that its gradient may be broken

That list of symptoms is useful for rejecting weak claims but not for accepting strong ones: a defense can pass all seven and still be fragile, and conversely it can trigger one symptom and still raise the cost of attack.

30.3 Breaking through: BPDA, EOT, and reparameterization

Level: advanced

If the gradient that is available is wrong, the attacker has three options. Replace the wrong gradient with the gradient of a function that approximates it; average the gradient over the randomness that makes it change; or move the work to another space, where the defense obfuscates nothing. All three have names, and all three work for different reasons.

The first option is called *backward pass differentiable approximation*, abbreviated BPDA. The idea is as simple as it sounds: on the forward pass use the actual function, including the rounding or the filtering; on the backward pass skip that piece as if it were the identity. If the wrapped model is $h(x) = f\big(g(x)\big)$, the attacker uses

$$\widehat{\nabla}_x h(x) \;=\; \nabla f\big(g(x)\big), \tag{30.3}$$

that is, the gradient of f evaluated at the point after wrapping, not the gradient of the composition. For rounding, this means the attacker stops asking "what is the derivative of rounding" and starts asking "if the numbers are already rounded, in which direction does f rise fastest".

Why is this approximation reasonable? Because the error is bounded by how far g shifts the point, not by how strange the shape of g is.

Proposition 30.3. *Let f be continuously differentiable with a Lipschitz gradient, that is, $\|\nabla f(u) - \nabla f(v)\| \le L\|u - v\|$ for all u, v, and let g satisfy $\|g(x) - x\| \le \delta$. Then*

$$\big\|\nabla f\big(g(x)\big) - \nabla f(x)\big\| \;\le\; L\,\delta. \tag{30.4}$$

Proof. Write $v = g(x) - x$ and use the mean value theorem in integral form,

$$\nabla f(x + v) - \nabla f(x) = \int_0^1 H_f(x + tv)\, v \, dt, \tag{30.5}$$

with H_f the Hessian matrix. Because the spectral norm of H_f is bounded by L,

$$\|\nabla f(x + v) - \nabla f(x)\| \le \int_0^1 \|H_f(x + tv)\|\, \|v\|\, dt \le L\|v\| \le L\delta. \tag{30.6}$$

□

This proposition explains when BPDA can be trusted. If rounding only shifts the input by $q/2$ and f is smooth enough at that scale, the gradient error is small and the attack step still moves in the right direction. If q is large so that the shift is comparable to the scale of the curvature of f, the bound $L\delta$ becomes large and nothing is guaranteed. This proposition does not say that $\widehat{\nabla}$ is the gradient of h; that gradient does not exist. It only says that the direction used is close to the correct direction for f at the point that has already been wrapped.

The second option is *expectation over transformation*, abbreviated EOT. Against a stochastic defense, the attacker does not face one function but a distribution of functions. What makes sense to maximize is not the loss on one realization, but the expected loss over realizations,

$$\overline{L}(x) \;=\; \mathbb{E}_{T \sim \mathcal{T}}\, L\big(f_T(x), y\big), \tag{30.7}$$

with T denoting the randomness the defense uses, for example a feature mask or a transformation parameter.

Lemma 30.4. *If $\mathcal{T}$ is finite and $\nabla_x L\big(f_T(x), y\big)$ is uniformly bounded in T, then*

$$\nabla_x \overline{L}(x) \;=\; \mathbb{E}_T\, \nabla_x L\big(f_T(x), y\big). \tag{30.8}$$

Moreover, the average of m independent samples, $\frac{1}{m}\sum_{i=1}^{m} \nabla_x L\big(f_{T_i}(x), y\big)$, is an unbiased estimator whose variance shrinks proportionally to $1/m$.

Proof. Because $\mathcal{T}$ is finite, the expectation is a finite sum, and the derivative may enter the sum because the derivative is a linear operator. The sample average has the same expectation as each of its terms, so it is unbiased, and its variance is the variance of one term divided by m because the samples are independent. □

This lemma answers an objection that comes up often: if the defense is random, is not any attack just luck? No, as long as the attacker may call the model repeatedly. Randomness raises the cost, because the attacker needs m calls per step, but it does not change the direction being pursued; what changes is only the precision of the estimate. This is the first example of a pattern that recurs throughout the chapter: a defense that raises the cost is real, whereas one that only makes measurement harder is illusory.

The third option is reparameterization. Instead of attacking x, the attacker attacks the variable that produces x through a transformation that is not obfuscated. If the defense wraps the model with a function g that has an inverse, the attacker works in the space z with $x = g^{-1}(z)$ and optimizes z. This is useful when g is a rotation or another linear transformation that does not destroy information, and it is not useful when g destroys information: no z can recover numbers that have already been thrown away.

In practice, the three options are used together, plus two habits. First: start from several random initial points inside the ε ball, because a loss surface that is not convex makes a single initial point easy to trap. Second: do not stop at one value of ε. Robustness measured at a single budget is robustness that is not measured.

Step-by-step recipe (procedural learner)

Four steps for testing a defense claim, in this order.

1. **Write the budget first.** Fix ε or the byte limit, the number of queries, and the time. These numbers must not change after the first result comes out.
2. **Attack by transfer, without queries.** Train a substitute model on the data you have, build adversarial examples there, then send them to the victim model. This step measures whether the defense merely blocks one model.
3. **Spend the remaining queries adaptively.** Only now use BPDA or EOT on the victim, and record the queries per step, not just the final result.
4. **Report both.** Transfer success, adaptive success, and the cost of each, always together with clean accuracy.

A warning about order: step 2 must be finished before step 3. If you measure transfer after seeing the adaptive results, you already know which examples succeed, and your transfer numbers leak from the future.

BPDA (*backward pass differentiable approximation*) — a technique that replaces the gradient of a non-differentiable piece with the gradient of the function at the point after that piece, as if the piece were the identity

EOT (*expectation over transformation*) — an attack that maximizes the expected loss over the randomness of the defense, usually by averaging gradients over several realizations

Note for instructors

This part is usually taught in the wrong order, namely the definition of BPDA first. A better order is the reverse: show a model whose accuracy does not move at all under attack, ask the students to guess why, then show that its gradient is zero. Almost all of them will guess “the model is strong”, and the silence that follows is a teaching moment in itself.

Two mistakes need to be anticipated. First, concluding that BPDA "removes" the defense; what is true is that BPDA restores access to an approximately correct direction, and its usefulness depends on how large δ is relative to the scale of the model curvature. Second, concluding that stochastic defenses are useless because EOT exists; what is true is that a stochastic defense raises the per-step cost linearly in m, so that the defense changes the attacker's budget, not the direction of the attack. The exercises at the end of the chapter force students to compute that cost, not merely name it.

30.4 Worked example: one point on a toy model

Level: intermediate

For younger readers (age 10 and up)

A shopkeeper has an odd habit: every price is rounded to the nearest rupiah before it is recorded. If a customer quietly shifts a price by one cent, the shopkeeper's record does not change at all. That customer can conclude that there is no point in shifting prices, then stop trying. The conclusion is wrong, but the reasoning is sensible: what he measured really did not move.

Now imagine another customer. She does not ask whether the record changes because of a one-cent shift; she asks in which direction the record rises fastest, then shifts the price far enough that the rounding changes. The shopkeeper's record changes, and she succeeds without breaking any rule. What separates the two is not cleverness, but the question that was asked.

A model that uses rounding works like that shopkeeper. A program that only tries small shifts will stop and report that the model is safe; a program that understands rounding will step further. So when you read a report that a model cannot be defeated, ask first: did the attacker stop because it lost, or because it only tried one way?

Before trusting the table, let us compute one point by hand. The model is made as simple as possible: eight features with all weights equal to one, so the score is the sum of the eight features. The data come from two normal distributions centered at -0.5 and $+0.5$, with clean accuracy around 92 percent. The defense is rounding to multiples of $q = 0.05$ on every feature, plus random filtering with drop probability r that multiplies the rest by $1/(1-r)$. The attacker budget is $\varepsilon = 0.25$ in the ℓ_∞ norm.

Take one example that is correctly labeled $+1$,

$$x_0 = (0.30;\ 0.20;\ 0.15;\ 0.10;\ 0.05;\ 0.00;\ -0.10;\ -0.08). \tag{30.9}$$

The sum is 0.62, so the model answers $+1$ with a thin margin. After rounding, the vector that enters the classifier is $(0.30; 0.20; 0.15; 0.10; 0.05; 0.00; -0.10; -0.10)$ and the score is 0.60. Note that one coordinate shifts from -0.08 to -0.10: that is the work of rounding.

Now compare two attackers. The naive attacker computes the derivative of the actual function, that is, the derivative of the composition that contains the rounding. Because rounding is flat around x_0, the derivative is zero in all eight coordinates, so its step is $x_0 + \varepsilon \cdot 0 = x_0$. The score stays 0.60 and the label stays $+1$. That attacker's report will read: attack failed, robust accuracy one hundred percent of clean accuracy.

The second attacker uses BPDA. It does not ask about the derivative of rounding; it evaluates the gradient of the classifier at the point that has already been rounded, that is, the vector $(1, 1, \ldots, 1)$, then steps against the label direction by ε in each coordinate,

$$x_1 = x_0 - 0.25 \cdot (1, 1, \ldots, 1) = (0.05;\ -0.05;\ -0.10;\ -0.15;\ -0.20;\ -0.25;\ -0.35;\ -0.33). \tag{30.10}$$

After rounding, that vector becomes

$$(0.05;\ -0.05;\ -0.10;\ -0.15;\ -0.20;\ -0.25;\ -0.35;\ -0.35)$$

with score -1.40. The label flips to -1. One step, the cost of one gradient computation, and the model that was just reported immune turns out to be wrong with a margin that is not thin.

This example shows something sharper than "BPDA works". The second attacker does not use information that the first does not have: both know the architecture, the weights, and q. What differs is one decision, namely whether to respect the non-differentiable piece or skip over it. So the "robustness" reported by the first attacker is not a property of the model, but a property of that attacker's notation.

Example 30.1. On the same model, with $q = 0.05$, $r = 0.5$, and a fixed seed, a finite difference gradient with step $h = 10^{-5}$ is zero in 15 998 of 16 000 entries; the two remaining entries are exactly $q/h = 5\,000$, because their points happen to sit at a step boundary. The median finite difference gradient norm is zero, whereas the norm of the BPDA gradient for one mask that keeps four features is 4. But the direction of a single mask is still far from the direction of the weights, with a cosine between 0.35 and 0.94 over 200 masks. The average gradient over 200 masks gives a vector of norm 2.738 and cosine 0.9982 with respect to the weight vector.

Those numbers illustrate three things. First, a shattered gradient is not merely small; it can also be large and wrong, and two entries that spike will dominate every step so that the naive attack fails in a way that looks random. Second, BPDA restores the direction almost perfectly: cosine 0.9982 means an angle of about three degrees. Third, the norm of the BPDA gradient for a single mask, namely 4, differs from its expected norm, 2.738; that difference in scale does not hurt an attack that uses the sign, but it does hurt one that uses the magnitude of the gradient, and that is why EOT is still needed.

Recompute the example above without looking at the text. Sum x_0, round to multiples of 0.05, then sum again; check that the result is 0.60. Subtract 0.25 from each coordinate, round again, sum, and check that the result is -1.40. After that, answer: what is the smallest value of q that makes this one-step BPDA attack fail to flip the label, and why does that value depend on the initial margin 0.60 and not on ε alone?

30.5 Computational lab: accuracy at every level of obfuscation

Level: advanced

Now we repeat that experiment on two thousand examples, five seeds, and several levels of obfuscation, with `numpy` alone. What matters is not the code itself, but one decision inside it: the function `fgsm` computes the gradient by finite differences on the function that has already been wrapped, whereas `pgd` uses the gradient of the classifier at the point after wrapping. The results, swept over seven values of ε, are in Figure 30.2.

Computational lab

```
import numpy as np

D, N, EPS = 8, 2000, 0.25
W = np.ones(D)
```

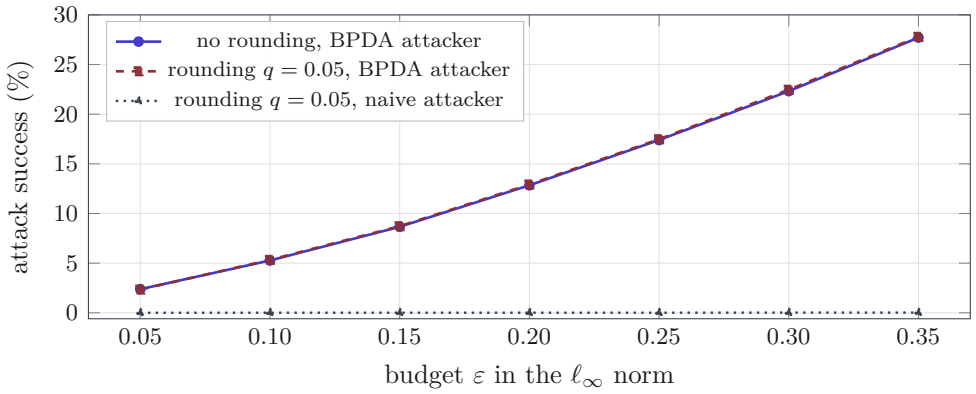


Figure 30.2: Attack success against ε on the eight-feature toy model, 2 000 examples, average of five seeds. At $\varepsilon = 0.25$ the three curves give 17.40, 17.48, and 0.01 percent; at $\varepsilon = 0.35$ they give 27.71, 27.79, and 0.02 percent. The first two curves coincide within the seed-to-seed deviation, which is 0.35 to 1.08 percentage points.

```
def data(rng, n, mu=0.5):
    y = 2 * rng.integers(0, 2, n) - 1
    return rng.normal(mu * y[:, None], 1.0, (n, D)), y

def quant(x, q):
    return x if q == 0 else np.round(x / q) * q

def mask(shape, r, rng):
    if r == 0:
        return np.ones(shape)
    return (rng.random(shape) >= r) / (1 - r)

def logit(x, q, m):
    return (quant(x, q) * m) @ W

def fgsm(X, y, q, r, rng):
    m, h = mask(X.shape, r, rng), 1e-5
    G = np.zeros_like(X)
    for j in range(D):
        e = np.zeros(D)
        e[j] = h
        G[:, j] = (logit(X + e, q, m) - logit(X - e, q, m)) / (2 * h)
    return X - EPS * y[:, None] * np.sign(G)

def pgd(X, y, q, r, rng, m_eot=1, steps=40):
    Xa, a = X.copy(), EPS / 8
    for _ in range(steps):
        G = sum(W * mask(X.shape, r, rng) for _ in range(m_eot))
        Xa = np.clip(Xa - a * y[:, None] * np.sign(G / m_eot),
                     X - EPS, X + EPS)
    return Xa

def run(seed, q, r):
    rng = np.random.default_rng(seed)
    X, y = data(rng, N)
    def pred(Z):
        return np.sign(logit(Z, q, mask(Z.shape, r, rng)))
    ok = pred(X) == y
    def succ(Z):
        return 100 * np.mean(pred(Z)[ok] != y[ok])
    return (100 * ok.mean(), succ(fgsm(X, y, q, r, rng)),
            succ(pgd(X, y, q, r, rng)), succ(pgd(X, y, q, r, rng, 20)))
```

```
print("  q     r    clean  naive   bpda1   eot")
for q, r in ((0.00, 0.0), (0.01, 0.0), (0.05, 0.0),
             (0.05, 0.5), (0.20, 0.0)):
    M = np.array([run(s, q, r) for s in (1, 2, 3, 4, 5)]).mean(0)
    print("%5.2f %5.2f  %6.2f  %5.2f  %5.2f  %5.2f"
          % (q, r, M[0], M[1], M[2], M[3]))
```

The output of that program, as it stands:

```
  q     r    clean  naive   bpda1   eot
 0.00  0.00   92.02  17.40  17.40  17.40
 0.01  0.00   92.00   0.02  17.40  17.40
 0.05  0.00   91.97   0.01  17.48  17.48
 0.05  0.50   82.87  12.16  25.48  25.67
 0.20  0.00   91.77   0.00  17.65  17.65
```

Read that table line by line. The first line is the baseline without obfuscation: clean accuracy 92.02 percent, one-step attack success 17.40 percent. That number is small because $\varepsilon = 0.25$ really is small compared with the spread of the data, not because the attack is weak.

The second line is the heart of this chapter. With $q = 0.01$, clean accuracy barely changes, 92.00 percent, but the naive success falls from 17.40 to 0.02 percent, about 870 times smaller. The attacker that uses BPDA gets 17.40 percent, exactly the baseline. That defense changes nothing that an attacker who computes correctly can observe; it only changes what an attacker who computes incorrectly sees.

The third line repeats the same thing at $q = 0.05$: naive 0.01 percent, BPDA 17.48 percent, clean accuracy 91.97 percent. Raising q fivefold does not change the BPDA success, but it makes the naive report look better. That is the fourth symptom from the list in section two, in numerical form.

The fourth line adds random filtering with $r = 0.5$, and two things happen at once. Clean accuracy falls from 91.97 to 82.87 percent, because discarding half the features and multiplying the rest by two raises the variance of the score. BPDA success rises to 25.48 percent, above the baseline, not because the defense is worse but because the examples that are still correct after obfuscation have a smaller margin.

The fifth line shows that coarse quantization does not save the model. With $q = 0.20$, comparable to ε, BPDA success is 17.65 percent, slightly above the baseline: a point that sits near a step boundary needs only a small push to cross.

The variance across seeds is small but not zero: at $q = 0.05$ with $r = 0$ the standard deviation is about 0.6 percentage points, and on the line with random filtering about 1.4 points. The standard error of a single proportion at 2 000 examples and a rate of 0.17 is about 0.8 points, so both numbers are consistent with ordinary sampling.

There are four things this experiment does not measure. First, the model is linear, so the BPDA gradient is almost perfect and the cosine 0.9982 is too good for a real model. Second, the attacker knows q and r. Third, only one value of ε was tested. Fourth, there is no search over attack hyperparameters. All of that makes the numbers above a lower bound on attack success, not an upper bound.

The third shortcoming can be closed, and the result is in Figure 30.2. The same program is run at seven values of ε from 0.05 to 0.35, five seeds, without random filtering. The curve of the model without rounding and the curve of the model with rounding $q = 0.05$ almost coincide: at $\varepsilon = 0.25$ the two give 17.40 and 17.48 percent, and at $\varepsilon = 0.35$ the two give 27.71 and 27.79 percent. The curve of the naive attacker stays below 0.02 percent across that whole range. This defense therefore does not shift the curve and does not break it; it merely adds a third curve that is flat.

30.6 Case study: NIDS, and why evasion remains impractical

Level: advanced

Up to here we have been dealing with feature vectors that may be changed at will. A network intrusion detection system, or NIDS, is a different world. A NIDS reads a packet stream, extracts features such as flow duration, packet count, byte count, TCP flags, and inter-arrival times, then decides whether that flow is an attack. The decision must be made fast, often in milliseconds, for thousands of flows per second.

Figure 30.3 puts that chain on one line. What matters is not the boxes, but the red mark between flow reconstruction and feature extraction. That is where the attacker works: he does not touch the model, he touches the packets, and the packets are what determine the feature values that enter the model. Everything to the right of that mark sees only the consequence; everything to the left is a cause that he can control, within the protocol limits discussed below.

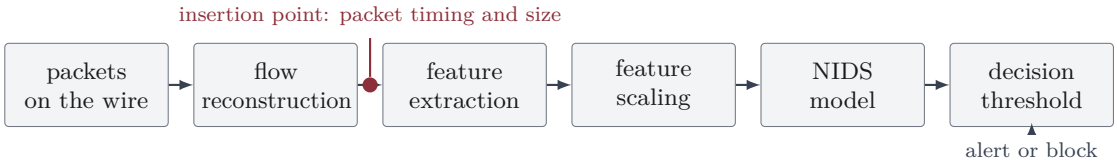


Figure 30.3: The NIDS processing chain, from the wire to the decision. The red mark is one insertion point that an attacker can use: features such as packet count, byte count, and inter-arrival times are all determined to the left of that mark, whereas the model on the right only reads the consequence. A defense that adds obfuscation to the right of the mark does not touch the cause, and that is the limit discussed in this section.

NIDS (*network intrusion detection system*) — a system that monitors network traffic and decides whether a packet flow is an attack

problem space (*ruang masalah*) — the space of real objects that an attacker may change, for example packets and timing, as distinct from the feature space in which the model works

The first difficulty is the distance between the two spaces. An attack is usually formulated in feature space, whereas the attacker sends packets: every feature change must be translated into a packet change that is still valid under the protocol and still carries out its malicious function, and that translation does not always exist. Some features cannot even be changed by the sender, because the packet and byte counts in the reverse direction are determined by the receiving system.

The second difficulty is the correlation between features. Duration is tied to packet count and inter-arrival times, and TCP flags are tied to the transport protocol, so an attacker cannot raise the duration while lowering both inter-arrival times; discrete features such as protocol number and packet length add further constraints.

The third difficulty is time. An attack that requires ten thousand queries means real traffic that passes and is logged, and a trace is the first thing an analyst looks at. A real cost is a real defense; the absence of a gradient is not.

Numbers from the literature make that distance concrete. On the CSE-CIC-IDS2018 data with fifty filtered features, three models were trained and tested on twenty percent of the data, then attacked with FGSM, PGD, and BIM. Before the attack all three were almost perfect: accuracy 0.997 for ANN, 0.998 for SVM, and 0.997 for CNN. On the day of the attack, with full knowledge, FGSM dropped ANN to 0.756 with F1 0.677, CNN to 0.842 with F1 0.863, and SVM to 0.150 with F1 0.000. PGD was slightly stronger on CNN, 0.584 with F1 0.689, and BIM gave 0.688 with F1 0.751.

Those numbers look like a victory for the attacker, and on a static model they are; but the same experiment continues with a model retrained every day. With knowledge of a model that is one day old, SVM returns to 0.998, ANN to 0.938, and CNN to 0.957 with F1 0.959, and on the second day the numbers are between 0.966 and 0.998. That retraining does not use adversarial examples, so the recovery is not the result of an adversarial defense but a consequence of the fact that the gradient computed on yesterday's model no longer points in the right direction.

This is the NIDS version of the whole chapter: deliberate gradient obfuscation is fragile, because an attacker can break it with BPDA, whereas obfuscation that comes from model change lasts longer, since the attacker must update his knowledge, and updating knowledge means gaining access, and access leaves a trace. The difference is not in the technique, but in who pays the cost.

The authors of that review built an attack tree and a taxonomy of practicality problems, then concluded that only three paths are viable. The first path is blind perturbation: adding random bytes to UDP packets, or to TCP packets with the PSH flag, while recomputing the checksum so that the protocol stays valid. This path needs no knowledge of the model and no queries, so it is immune to retraining, but it has no direction. The second path is querying through a side channel, which demands continuous network access and depends on indicators that change with network load and hardware. The third path is a substitute model trained on data captured from one's own botnet, which works through transferability but demands training data that is still fresh.

All three paths are real, and the authors stress that the distance between research and practice is wide and that the threat model assumptions in many papers are too generous to the attacker.

The authors' own work, which we will call a working note and not a published result, moves on the other side of the same problem. On one network flow dataset, a model reaches F1 0.967037 from 46 558 out of 48 145 test examples. The question is not "does the attack succeed", but "how many features must be changed for an example to cross the decision boundary while remaining valid under the protocol". With a mixed-integer linear program, the answer depends sharply on the number of features that may be changed: at seven features, not one of fifty examples could be moved across; at twenty features, thirty-three of fifty could. The ratio of distance to the boundary between successful and control examples is about 3.28 for the linear model and 2.13 for the neural network.

Two more numbers show the caution involved. Of twenty thousand candidates tested for protocol consistency, none passed; with the rule of three, the upper bound at 95 percent for that rate is about 0.019 percent. On the model with an enlarged safety margin, the condition takes the form $\varepsilon + 40\delta$ with $\delta = 1.245 \times 10^{-7}$, so that the effective budget is 0.150004981 and not 0.15.

The limits of that working note must be stated together with its results. It uses one dataset, fifty test examples, and two model families. It verifies the absence of an attack that meets the conditions, not the discovery of an attack. The number 0 out of 20 000 is a statement about the candidate generator that was used, not about all possible packets, and the result says nothing about dynamic NIDSs.

Now the conclusion that must be stated clearly. The toy model in section five does not prove anything about real NIDSs; it shows that rounding that erases the gradient produces a misleading report, and that the mechanism can be broken at a reasonable cost. It does not measure the cost of an attack in packet space. What can be carried away from this chapter is a habit: every time a defense reports zero success, ask about its gradient, then ask who pays the cost.

Challenge (problem solver)

A paper reports a defense that lowers attack success from 96 percent to 3 percent, with clean accuracy that does not change at all. Their attacker uses one-step FGSM, $\varepsilon = 8/255$, the exact gradient, one initial point, and one seed. The question is: is that robustness real, and what is the cheapest experiment that most quickly shows that the 3 percent figure is an artifact?
First hint: look at what is not being paid for. A defense that genuinely raises the cost almost always pays something, either accuracy, or latency, or a number of parameters.
Second hint: if the gradient is computed through a non-differentiable piece, the 3 percent figure is not attack success, but the residue of examples that happen to lie near the decision boundary. Repeat with forty-step PGD, BPDA, and three initial points; then sweep ε and report the curve.
If you get stuck, go back to section three.

For researchers: limits and next steps

Three things in this section are often misquoted. First, the categories shattered, stochastic, and vanishing are the taxonomy of the obfuscated gradients paper; the list of seven techniques in section two is our pedagogical decomposition. In that paper there are three categories, nine defenses tested, seven of which depend on an obfuscated gradient, and six broken completely plus one broken partially.
Second, BPDA is not a general solution. The error bound $L\delta$ holds only when f is smooth and δ is small relative to the scale of the curvature; for rounding with q comparable to ε that guarantee is empty. Conversely, EOT is not a cure for an attack that points in the wrong direction: averaging wrong directions still produces a wrong direction.
Third, the NIDS results we quote come from feature-space experiments with full knowledge and daily retraining, not from packet-space attacks that were actually run. They support a narrow claim: retraining restores accuracy against attacks that use the gradient of an old model. They do not support the claim that NIDSs are immune to an attacker who may change packets.

30.7 A checklist before writing that an attack succeeded

Level: basic

This section is a summary that can be used directly. Every item is a question that must be answerable before a claim about an attack or a defense enters the abstract.

1. **The threat model is written in full.** What does the attacker know: architecture, weights, training data, probabilities, or only labels? How many queries may he make, and for how long?
2. **The attack is bounded by a reasonable budget.** State ε and its norm, or the byte and packet limits if the work is in problem space. An attack without a bound measures something that nobody faces.
3. **The attack is adaptive.** The attacker must know the defense being tested, including the possibility that its gradient is broken. A defense tested only by attacks from an earlier paper has not been tested.
4. **The gradient is inspected before it is trusted.** Report the gradient norm, the fraction of entries that are zero, and their correlation with the correct direction. A zero gradient on a non-constant model is a finding, not a result.

5. **Success is reported together with clean accuracy.** A defense that lowers clean accuracy from 92 to 60 percent must not be compared directly with one that keeps 92 percent.
6. **Several seeds, with mean and deviation.** One seed is one throw of the die; a difference between two lines that is smaller than their deviation is not a difference.
7. **A curve, not a single point.** Sweep the value of ε, the number of steps, the number of EOT samples, and the number of queries. A good defense shifts the curve; an illusory one shifts a single point.
8. **The attacker's compute budget is stated:** how many times the model is called, how many hours, how many GPUs? An attack that succeeds after a million queries is not the same attack as one that succeeds after a hundred.
9. **Code and weights are available.** A claim that cannot be repeated cannot be checked. If release is impossible, the procedure and hyperparameters must be detailed enough to reconstruct.
10. **Negative results are reported as negative results.** A defense that fails is information; a report that succeeds because its attack was stopped before time is an expensive mistake.

rule of three (*aturan tiga*) — if no event occurs in n independent trials, the upper bound at 95 percent for the probability of that event is about $3/n$

Summary.

- An obfuscated gradient blinds the attacker, it does not make the model strong. The seven techniques produce the same symptom: the measurement looks better without anything getting better.
- For a step function such as rounding, the gradient of the composition is zero at almost every point, so a one-step attack produces $x' = x$ and "accuracy under attack" is precisely the clean accuracy.
- BPDA replaces the gradient of a non-differentiable piece with the gradient of the classifier at the point after the piece. Its error is bounded by $L\delta$: it makes sense when the shift is small relative to the scale of the curvature, and it gives no guarantee when it is not.
- EOT maximizes the expected loss over the randomness of the defense; the average of m samples is unbiased with variance proportional to $1/m$. Randomness raises the per-step cost, it does not change the direction.
- On the eight-feature toy model with $\varepsilon = 0.25$, rounding $q = 0.01$ drops naive success from 17.40 to 0.02 percent, whereas BPDA stays at 17.40 percent; $q = 0.20$ does not hold BPDA back either.
- On NIDS, FGSM drops SVM accuracy from 0.998 to 0.150, but daily retraining restores it to 0.998. The durable defenses there come from the cost of access and protocol constraints, not from gradient obfuscation.

30.8 Exercises

Level: intermediate

Exercise 30.1.

1. A paper writes: "Our defense lowers attack success from 98 to 0 percent at $\varepsilon = 8/255$." State three checks before accepting that sentence, and what to expect from each check if the defense only obfuscates the gradient.
2. Derive the BPDA error bound for $f(x) = w \cdot x + b$, show that the error is zero, then explain why that result does not carry over to a two-layer model with ReLU activations.

3. On the toy model of section four, determine the smallest value of q that makes one BPDA step fail to flip the label of x_0, express it as a function of the initial margin and ε, then check it with the program.
4. You have 1 000 queries. The defense randomizes a feature mask with $r = 0.5$ and refuses to answer if more than half the features are dropped. Compute how many PGD steps you can run with $m = 20$ and with $m = 1$, then explain what you gain and what you pay.
5. Repeat the experiment of section five with a two-layer model with ReLU activations. Report clean accuracy, naive success, and BPDA success for three values of q, then explain why the cosine is no longer close to one.
6. A NIDS reads fifty features, seven of which are binary flags and three protocol numbers. Compute how many features an attacker can actually change, state your assumptions, and compare with a 32×32 image that has 3 072 values.
7. Explain why daily retraining restores accuracy against attacks that use gradients but not against blind perturbation. Connect this with the EOT lemma and the definition of an adaptive attack.
8. Design an experiment that separates two hypotheses: (a) the defense raises the cost of attack, (b) the defense only corrupts the measurement. State the metrics, the number of seeds, the shape of the curve, and what results would make you accept (a).

Map to source texts: Chapter 30 of Athalye et al., Obfuscated Gradients Give a False Sense of Security *(2018): section 3 for the three families of obfuscated gradients, section 4 for BPDA and adaptive attacks, section 6 for the checklist. In Madry et al.,* Towards Deep Learning Models Resistant to Adversarial Attacks *(2018): section 3 for PGD and random initial points. In elShehaby and Matrawy,* Evasion Adversarial Attacks Remain Impractical Against ML-based Network Intrusion Detection Systems*, for the attack tree, the practicality taxonomy, and the retraining experiment. In Chapter 20 of this book for the rule of three, and in Chapter 27 for FGSM and PGD.*

Chapter 31

Cryptography: Modular Integers to the Quantum Threat

A student submits an assignment containing a cipher of his own making. Each letter is replaced by its number, that number is multiplied by three, and the remainder after division by twenty-six is used as the new letter. To open the message, he multiplies again by nine. When asked where the number nine came from, his answer is nearly right: "because there is no division here, I multiply by the inverse." The examiner then replaces the multiplier three with the multiplier four. The assignment is never finished. With the multiplier four, the first letter and the fourteenth letter produce the same symbol, so two different messages produce an identical cipher.

The question that arises from that failure is the question of this whole chapter: when may a number be divided inside remainder arithmetic? The answer is short: exactly when that number shares no factor with its modulus. That short answer, pressed hard, yields the three things every browser uses: key exchange, public keys, and digital signatures.

The previous three chapters are used explicitly. Chapter 2 gives the language for bijections and partitions; here a bijection proves that multiplication by a unit only shuffles the order, and partitions count the order. Chapter 3 gives the completeness property of the real numbers, and that property does not cross over, because $\mathbb{Z}_n$ has no order compatible with multiplication; what replaces it is finiteness, since on a finite set an injective map is automatically surjective. Chapter 28 explains why "secure" cannot be a theorem without conditions.

One warning applies to every page here. All the moduli are small, chosen so that they can be checked by hand and by Python in a matter of seconds. The largest modulus, $1\,050\,589$, is factored by a six-line script. Not one cipher here is safe to use.

For younger readers (age 10 and up)

Imagine the wall clock in a classroom. Its hand moves from the number one to twelve, then back to one. That clock never shows the number thirteen or twenty. If we turn the hand five steps, then five steps again, then five steps again, we do not need to know how much five plus five plus five is; we only look at where the hand stops.

Now imagine a combination lock with thirteen positions. We may turn it as far as we like, and only the final position is recorded. If we turn three steps, how many times must we turn three steps again to get back to where we started? The answer is nine, and that is not a coincidence, but because three and thirteen share no factor. Try the same thing with four steps on a twelve-position lock: however many times we turn, we will never land on all positions, because four and twelve share a factor.

In this chapter we will understand why that small difference decides whether a message

can be opened again or is lost forever, and why two people who have never met can agree on one secret number in front of a third person who hears everything.

31.1 Residue classes: arithmetic that goes around

Level: basic

We begin by fixing the notion of “remainder” already used in the first paragraph. Two integers are called congruent modulo n if their difference is divisible by n.

Definition 31.1. Let $n \geq 1$ be an integer. For $a, b \in \mathbb{Z}$ we write $a \equiv b \pmod{n}$ if n divides $a - b$. The set of equivalence classes under this relation is called $\mathbb{Z}_n$, and the class containing a is written $[a]_n$.

We owe two things: that this really is an equivalence relation, and that the remainder is unique.

Lemma 31.1. *Congruence modulo n is an equivalence relation on $\mathbb{Z}$, and every class $[a]_n$ contains exactly one representative in $\{0, 1, \dots, n-1\}$, so that $|\mathbb{Z}_n| = n$.*

Proof. Reflexive: n divides 0. Symmetric: if $n \mid (a-b)$ then $n \mid (b-a)$. Transitive: if $n \mid (a-b)$ and $n \mid (b-c)$ then $n \mid (a-c)$ because $(a-c) = (a-b) + (b-c)$. For uniqueness, take a and consider $S = \{a - qn \geq 0 : q \in \mathbb{Z}\}$. That set is nonempty by the Archimedean property (Chapter 3), so it has a least element r. If $r \geq n$ then $r - n \in S$ is smaller than r, a contradiction; so $0 \leq r < n$. If r' also satisfies $0 \leq r' < n$, then $r - r'$ is a multiple of n whose value lies between $-(n-1)$ and $n-1$, hence must be zero. $\square$

Operations on $\mathbb{Z}_n$ are defined through their representatives, and a definition of this kind is dangerous: if the result depends on the representative, the operation is not a function. The next lemma closes that hole with one line of algebra.

Lemma 31.2. *If $a \equiv a'$ and $b \equiv b'$ modulo n, then $a + b \equiv a' + b'$ and $ab \equiv a'b'$ modulo n.*

Proof. Write $a' = a + kn$ and $b' = b + ln$. Then $a' + b' = (a + b) + (k + l)n$ and $a'b' = ab + (al + bk + kln)n$. $\square$

Addition and multiplication are therefore well defined, and both inherit associativity and distributivity from $\mathbb{Z}$. What is lost is order: there is no way to put a sign “$<$” on $\mathbb{Z}_n$ such that $a < b$ implies $a + c < b + c$ and $ac < bc$ for positive c. The supremum tool of Chapter 3 stops at the entrance, and what works inside is counting.

Division need not exist. The class $[3]_{26}$ has an inverse, namely $[9]_{26}$, since $3 \cdot 9 = 27 \equiv 1$. The class $[2]_{26}$ has none: every multiple of two is even, whereas 27 is odd, so $2b \equiv 1$ has no solution. Consequently multiplication by two modulo twenty-six is not a bijection, because $[0]$ and $[13]$ both go to $[0]$.

residue class (*kelas sisa*) — all the numbers that stop at the same clock position, like three in the morning and three in the afternoon

Definition 31.2. A class $[a]_n$ is called a unit if there is $[b]_n$ with $[a]_n[b]_n = [1]_n$. The set of all units modulo n is written $U(n)$, and its number of members is written $\varphi(n)$.

For $n = 26$ there are twelve units, namely the classes of $1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25$. The program in the lab counts them again; that number reappears in part three under a different name.

31.2 Common factors, Euclid, and when division exists

Level: basic

The question "when does $[a]_n$ have an inverse" turns out to be a question about the greatest common factor, which has an algorithm more than two thousand years old.

Definition 31.3. For $a, b \in \mathbb{Z}$ not both zero, $\gcd(a, b)$ is the largest natural number that divides a and b.

The key to Euclid's algorithm is one equality of sets: the common divisors of a and b are exactly the common divisors of b and $a - qb$.

Lemma 31.3. *For every $q \in \mathbb{Z}$, $\gcd(a, b) = \gcd(b, a - qb)$.*

Proof. If d divides a and b, then d divides $a - qb$. Conversely, if d divides b and $a - qb$, then d divides $(a - qb) + qb = a$. So the two pairs have the same set of common divisors, hence the same greatest one. □

Replacing (a, b) by $(b, a \bmod b)$ therefore does not change the answer but shrinks b, and the process stops because natural numbers do not shrink without bound.

Theorem 31.4 (Bezout)**.** *For a, b not both zero, there are $x, y \in \mathbb{Z}$ with $ax + by = \gcd(a, b)$.*

Proof. Consider $S = \{ax + by > 0 : x, y \in \mathbb{Z}\}$. This set is nonempty because it contains $|a|$ or $|b|$, and every nonempty subset of the natural numbers has a least element; this is the discrete form of the infimum property in Chapter 3. Let $d = ax_0 + by_0$ be that least element. Divide a by d: $a = qd + r$ with $0 \le r < d$. Then $r = a(1 - qx_0) + b(-qy_0)$, so $r \in S$ if $r > 0$. Since d is least, $r > 0$ is impossible; so $r = 0$ and $d \mid a$, and in the same way $d \mid b$. Hence $d \le \gcd(a, b)$. Conversely $\gcd(a, b)$ divides every expression $ax + by$, in particular d, so that $\gcd(a, b) \le d$. □

Theorem 31.5. *The class $[a]_n$ is a unit if and only if $\gcd(a, n) = 1$.*

Proof. If $[a]_n$ is a unit, there is b with $ab \equiv 1$, that is $ab - 1 = kn$. Every common divisor of a and n divides ab and kn, hence divides 1; so $\gcd(a, n) = 1$. Conversely, if $\gcd(a, n) = 1$, Bezout's theorem gives $ax + ny = 1$. Read modulo n, that equality says $ax \equiv 1$, so $[x]_n$ is the inverse of $[a]_n$. □

That proof also gives a recipe: run Euclid, then build the combination backward. For $a = 1071$ and $b = 462$ the steps are three divisions,

$$1071 = 2 \cdot 462 + 147, \qquad 462 = 3 \cdot 147 + 21, \qquad 147 = 7 \cdot 21 + 0,$$

so that $\gcd = 21$. Building backward gives $21 = 1071 \cdot (-3) + 462 \cdot 7$, and since the common factor is 21 rather than 1, the class $[462]_{1071}$ is not a unit.

How long does this algorithm run? Every two steps the remainder at least halves, so the number of divisions is proportional to the number of digits. The sharp bound was given by Lamé in 1844: no more than $\log_\phi(b) + 1$ divisions with $\phi = (1 + \sqrt{5})/2$, and the pairs that attain it are consecutive Fibonacci pairs. The pair $(F_{47}, F_{46}) = (2\,971\,215\,073,\ 1\,836\,311\,903)$ needs 45 divisions, whereas $\log_\phi(F_{46}) = 44.33$. For two thousand random thirty-two-bit pairs, the average is 18.75 with maximum 32.

modular inverse (*invers modular*) — the number that, multiplied by the original number, gives remainder one, like the "undo" button on a cash register

31.3 Fermat's little theorem and Euler

Level: intermediate

This chapter needs one fact about powers, and that fact is born from counting: multiply every unit by one fixed unit; the result is the same list in a different order.

Lemma 31.6. *If* $\gcd(a, n) = 1$*, the map* $[x]_n \mapsto [ax]_n$ *is a bijection from* $U(n)$ *to* $U(n)$*.*

Proof. The result lies in $U(n)$ because the product of two units is a unit: if $xx' \equiv 1$ and $yy' \equiv 1$ then $(xy)(x'y') \equiv 1$. The map is injective: if $ax \equiv ay$, multiply both sides by the inverse of a, which exists by the unit theorem in part two. The set is finite, and on a finite set an injective map from the set to itself is automatically surjective (Chapter 2). So it is a bijection. □

Theorem 31.7 (Euler)**.** *If* $\gcd(a, n) = 1$*, then* $a^{\varphi(n)} \equiv 1 \pmod{n}$*.*

Proof. Let $u_1, \dots, u_k$ be all the units modulo n, with $k = \varphi(n)$, and $P = u_1 \cdots u_k$. By the lemma above, the list $au_1, \dots, au_k$ contains every unit exactly once, so

$$a^k P \equiv (au_1)(au_2)\cdots(au_k) \equiv u_1 u_2 \cdots u_k = P \pmod{n}.$$

Since P is itself a unit, it has an inverse, and both sides may be multiplied by it. The result is $a^k \equiv 1$. □

Corollary 31.8 (Fermat's little)**.** *If* p *is prime and* $p \nmid a$*, then* $a^{p-1} \equiv 1 \pmod{p}$*. For all* a *we have* $a^p \equiv a \pmod{p}$*.*

The second form follows from the first: if $p \mid a$ both sides are zero, if not, multiply $a^{p-1} \equiv 1$ by a. The name "little" distinguishes it from Fermat's last theorem, which is far harder.

The number twelve from part one now has two names: $\varphi(26) = 12$, and $3^{12} \equiv 1 \pmod{26}$; the program checks that 5^{12} and 7^{12} are also $\equiv 1$. Large powers can therefore be trimmed: for $7^{222} \bmod 13$ write $222 = 12 \cdot 18 + 6$, so that $7^{222} \equiv 7^6 \equiv 9 \cdot 10 \equiv 12$ because $7^2 \equiv 10$ and $7^4 \equiv 9$.

The converse does not hold, and that matters for the honesty of this chapter. The number $561 = 3 \cdot 11 \cdot 17$ is not prime, yet $a^{560} \equiv 1 \pmod{561}$ for every a that shares no factor with it; the program checks all a without finding a counterexample. Numbers of this kind are called Carmichael, and for that reason practical primality tests use Miller–Rabin.

Euler totient function (*fungsi phi*) — the number of positions on a combination lock that can reach every other position, that is the positions that share no factor with the number of positions

31.4 Cyclic groups, order, and primitive roots

Level: intermediate

If we multiply one unit repeatedly, the result goes around and finally returns to one. The useful question is how long that circuit is.

Definition 31.4. For $\gcd(a, n) = 1$, the order of a modulo n is the least natural number k with $a^k \equiv 1 \pmod{n}$, written $\operatorname{ord}_n(a)$.

An order always exists: the sequence $a^0, a^1, \dots$ has only $\varphi(n)$ distinct values, so two of them are equal and $a^{i-j} \equiv 1$ for some $i > j$. That bound can be sharpened: the order divides $\varphi(n)$.

Lemma 31.9. *If* $\gcd(a, n) = 1$*, then* $\mathrm{ord}_n(a)$ *divides* $\varphi(n)$*.*

Proof. Let $k = \mathrm{ord}_n(a)$ and $H = \{a^0, \dots, a^{k-1}\}$, whose members are all distinct, so $|H| = k$. For every unit x, the set $xH = \{xa^0, \dots, xa^{k-1}\}$ also has k members, because multiplication by a unit is a bijection. The sets xH form a partition of $U(n)$: two of them are equal or disjoint, since xH and yH intersect if and only if $y^{-1}x \in H$ (Chapter 2). So $\varphi(n)$ is k times the number of parts. $\square$

Definition 31.5. If $\mathrm{ord}_n(g) = \varphi(n)$, then g is called a primitive root modulo n. For a unit h, the number k with $g^k \equiv h$ is called the discrete logarithm of h to the base g.

Modulo 13, $2, 6, 7, 11$ have full order 12 and are therefore primitive roots, whereas 3 and 9 have order 3, 5 and 8 have order 4, 4 and 10 have order 6, and 12 has order 2. All those numbers divide 12.

For 2, the circuit visits all twelve nonzero classes before returning: $2, 4, 8, 3, 6, 12$, then $11, 9, 5, 10, 7, 1$. Figure 31.1 shows it as a clock with thirteen positions. For 3, the circuit is only $3, 9, 1$ and then repeats.

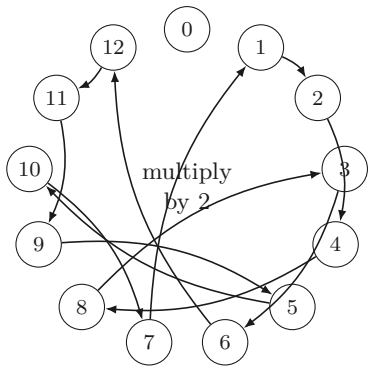


Figure 31.1: Multiplication by 2 modulo 13 goes around through all twelve nonzero classes before returning to 1. The length of that circuit is the order, and full order means 2 is a primitive root modulo 13.

The existence of a primitive root for every prime p is Gauss's theorem, and we skip its proof. What must be carried away: for a prime p, the units are a single circuit of $p - 1$ positions, and every unit is a power of one primitive root. If $h = g^k$, knowing k means knowing h, but not conversely; that asymmetry is the raw material of public-key cryptography.

primitive root (*akar primitif*) — one starting position that, if turned continuously, passes through every other position before returning

31.5 Diffie-Hellman key exchange

Level: intermediate

Two people want to agree on one secret number, and the only channel they have is overheard by a third party. In 1976 Diffie and Hellman showed that this is possible, and the mechanism uses only two formulas.

They first agree on a prime p and a primitive root g modulo p; both may be announced. Alice chooses a secretly and sends $A = g^a \bmod p$. Bob chooses b secretly and sends $B = g^b \bmod p$. Alice computes $K = B^a \bmod p$, Bob computes $K = A^b \bmod p$. Both obtain the same number, because

$$B^a \equiv (g^b)^a \equiv g^{ab} \equiv (g^a)^b \equiv A^b \pmod{p}.$$

What the eavesdropper sees is only p, g, A, B. To obtain K he needs g^{ab}, and the easiest known way is to find a or b, that is to solve the discrete logarithm.

Read it aloud (auditory learner)

Read it as a dialogue for two voices, alternating, one line per breath.
First voice: "Agree first, announce two." Second voice: "Keep your own, never say it." First voice: "Raise, take the remainder, send the result." Second voice: "Receive the delivery, raise it again." First voice: "The last remainder, that is the key." Together: "Two directions, one number, never crossed."
Repeat three times, faster and faster. The line that makes the tongue stumble is the step not yet understood, not the line that needs to be memorized harder. Once it flows, close the book and name four things in order: what is announced, what is kept, what is sent, what is computed. Naming the secret number first cancels the whole game.
The line repeated three times aloud: "what is sent is the power, what is kept is the exponent."

The concrete numbers, all checked by the program in the lab part: $p = 10\,007$, $g = 5$ has order $10\,006$ and is therefore a primitive root, $a = 1234$, $b = 5678$. Then $A = 5^{1234} \bmod p = 4873$ and $B = 5^{5678} \bmod p = 3189$. Alice computes $3189^{1234} \bmod p = 5674$; Bob computes $4873^{5678} \bmod p = 5674$. Their shared key is 5674, and that number never crosses the channel. Figure 31.2 shows the flow together with what the eavesdropper hears.

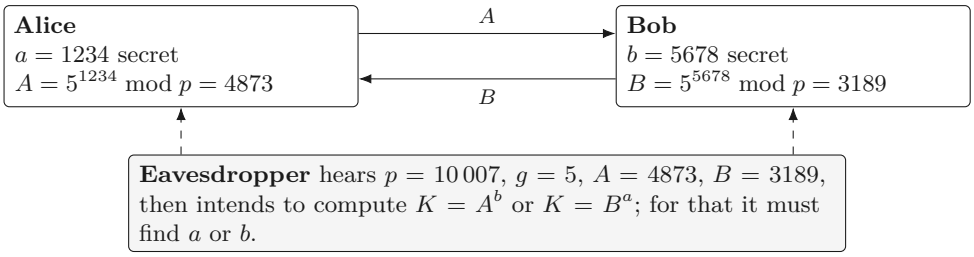


Figure 31.2: The flow of Diffie-Hellman key exchange on the numerical example in the text. The values A and B cross the open channel; what does not cross are the exponents a and b, and it is exactly those two that the eavesdropper needs.

That equality only states that the protocol is consistent, not that it is secure. All that can be proved is a conditional statement: if the discrete logarithm modulo p is hard, then computing g^{ab} from A and B is also hard. That condition is not a theorem; there are groups in which the discrete logarithm is easy, so security is a property of particular groups.

Do it by hand (kinesthetic learner)

Materials: yellow watercolor, one secret color per person (red and blue), four plastic cups, one spoon, two sheets of white paper.
First step, in front of the table: pour one spoonful of yellow paint into each cup. The yellow plays the role of the prime and the base, and both are announced. Second step, behind your back: each person adds two spoonfuls of their secret color, stirs, then hands over the mixed cup. Third step: each adds two spoonfuls of their own secret color to the cup just received, stirs, then puts one drop on the paper.
What must be visible: the two drops are the same color, even though neither of the two people ever handed over their secret color. The first mixed cup held by the third person cannot be separated again; he sees only two different dull colors.

Two parameter traps deserve mention. First, g must generate a large subgroup of prime order: the modulus 10 007 is a safe prime with $p-1 = 2 \cdot 5003$, and the program records that $5^{5003} \bmod p = 10\,006$ has order only 2. If Alice uses that base, their shared key is only 1 or 10 006, and the eavesdropper need not compute any logarithm at all. Second, this protocol names no one's identity, so an eavesdropper can stand in the middle and read everything. Figure 31.2 depicts only a passive eavesdropper.

About size: the modulus 10 007 can be broken in less than one second by the index calculus algorithm, and we use it because the powers fit on one page. Real practice uses 2048 bits or more, and even that is a time limit, not a guarantee.

For younger readers (age 10 and up)

Imagine two people who want to agree on one secret color, even though all their conversation is overheard. They first agree on a base color, say yellow. Alice chooses her secret color, mixes it with yellow, then sends the mixture. Bob does the same with his own secret color. Now each mixes the color received with their own secret color. Both get the same color, and the eavesdropper sees only two mixtures that cannot be separated again. That separation is what is considered hard: mixing colors is easy, separating a mixture without one of its ingredients is considered not. Cryptography replaces colors with numbers and "considered not easy" with a mathematical sentence.

In this chapter we will find out how numbers replace colors, and why the security depends on one "considered" sentence that nobody has ever proved.

31.6 RSA: from key to message and back

Level: intermediate

Diffie-Hellman agrees on a key, but does not send a message. A year later Rivest, Shamir, and Adleman showed that the same property can be used for two different keys: one to lock, one to open.

Choose two large primes p and q, compute $n = pq$ and $\varphi(n) = (p-1)(q-1)$. Choose e with $\gcd(e, \varphi(n)) = 1$; the unit theorem gives its inverse d modulo $\varphi(n)$. The public key is (n, e), the secret key d. For a message $0 \le m < n$, the cipher is $c = m^e \bmod n$ and its opening is $m = c^d \bmod n$.

Why does opening return m? Because $ed = 1 + k\varphi(n)$, so that $c^d = m^{ed} = m\,(m^{\varphi(n)})^k$. If $\gcd(m, n) = 1$, Euler's theorem makes that bracket one. The remaining case explains why n must be the product of two distinct primes.

Theorem 31.10. *Let $n = pq$ with p, q distinct primes, $ed \equiv 1 \pmod{(p-1)(q-1)}$, and $0 \le m < n$. Then $m^{ed} \equiv m \pmod n$.*

Proof. If $p \nmid m$, Fermat's little theorem gives $m^{p-1} \equiv 1 \pmod p$, so that $m^{ed} = m\,(m^{p-1})^{k(q-1)} \equiv m \pmod p$. If $p \mid m$, both sides are congruent to zero modulo p. So the congruence holds modulo p for every m, and for the same reason modulo q. Since p and q both divide $m^{ed} - m$ and $\gcd(p, q) = 1$, their product also divides that number: if $m^{ed} - m = pt$ and $q \mid pt$ with q prime and $q \nmid p$, then $q \mid t$, so that $pq \mid m^{ed} - m$. □

The numerical example is small and entirely checked by the program. Take $p = 1019$, $q = 1031$, then $n = 1\,050\,589$ and $\varphi(n) = 1018 \cdot 1030 = 1\,048\,540$. Choose $e = 65\,537$; Euclid gives $d = 60\,493$, and indeed $65\,537 \cdot 60\,493 = 3\,964\,529\,741 = 1 + 3781 \cdot 1\,048\,540$. The message $m = 424\,242$ produces the cipher $c = 552\,241$, and $c^d \bmod n$ returns 424 242. That modulus is only twenty-one bits and is factored by a six-line script.

Plain RSA, the formula above with no addition of any kind, is not secure; its three reasons can be shown with the same modulus. First, it is deterministic: the same message always produces the same cipher, so that repeated messages can be recognized. Second, it is multiplicative: $E(m_1)E(m_2) \equiv E(m_1 m_2)$, and the program shows $E(1234) \cdot E(567) \bmod n = 551\,173$ opened to 699 678, exactly $1234 \cdot 567$. Third, small messages are not protected: with $e = 3$ and $m = 42$, the cipher is 74 088, coincidentally equal to 42^3 because the result has not yet passed n, so that an ordinary cube root recovers the message.

All three are closed by padding: the message receives a random addition before exponentiation, so that the result is full length, always different, and no longer multiplicative. RSA-OAEP, the standard used today, has a security proof against chosen-ciphertext attacks in the random oracle model under the assumption that RSA holds. In practice RSA wraps a 256-bit symmetric key and then the data is encrypted with AES; this pattern is called hybrid encryption, because RSA is slow: its time is proportional to the cube of the modulus length.

padding (*bantalan*) — extra contents put into a package before shipping, like bubble wrap that makes every package a different shape and size

31.7 Hash functions, MACs, and signatures

Level: intermediate

Encryption hides the contents. Sometimes what is needed is not hiding but summarizing: one short fingerprint that changes when the contents change.

Definition 31.6. A cryptographic hash function maps a message of any length to an output of fixed size. It must be preimage resistant (from h, it is hard to find m with $H(m) = h$), second-preimage resistant (from m, it is hard to find $m' \neq m$ with $H(m') = H(m)$), and collision resistant (it is hard to find a distinct pair m, m' with the same hash).

Those three properties are not equivalent; the weakest is the one most often misunderstood. Collision resistance is limited by birthdays: if the output has N values, a collision appears after about $1.177\sqrt{N}$ trials, not after N; for 128 bits that number is $2.17 \cdot 10^{19}$ and for 64 bits $5.06 \cdot 10^{9}$. The program in the lab tests that prediction on SHA-256 truncated to 16 bits: an average of 362.7 samples against a prediction of 320.8, with a spread from 100 to 743. So a hash with an n-bit output gives only $n/2$ bits of collision resistance.

A MAC adds a key to the hash: two parties who share a key can check that a message came from the other party and did not change, because without the key there is no way to compute the correct tag. The limit: a MAC is useful only between two parties who both hold the key, so that a third party cannot be a witness.

A signature uses a key pair in the opposite direction: the signer uses the secret key, the verifier uses the public key. In RSA, the signature on m is $s = H(m)^d \bmod n$, and verification compares $s^e \bmod n$ with $H(m)$. The need for H is not a matter of convenience: without a hash, RSA signatures are multiplicative like its encryption, so that from the signatures of two documents someone can assemble a valid signature for their product without holding the secret key. The hash breaks that property, because $H(m_1 m_2)$ is unrelated to $H(m_1)H(m_2)$. The security requirement is resistance to existential forgery under chosen-message attack, and once again that is an assumption.

digital signature (*tanda tangan digital*) — a wax seal on a document that anyone can check but only the holder of the stamp can make

31.8 What the security model actually assumes

Level: advanced

Every sentence with the word “hard” in this chapter is a promise about what the known algorithms cannot do, not a statement about what is impossible.

Definition 31.7. A function f is called one-way if there is a polynomial algorithm to compute $f(x)$ from x, but for random y in the range, every polynomial algorithm fails to find x with $f(x) = y$ except with negligible probability.

The direction is depicted in Figure 31.3.

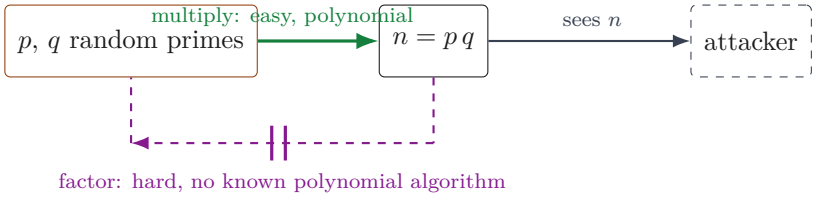


Figure 31.3: Two directions that are not equally expensive. From p and q to n one multiplication suffices, whereas from n back to p and q is a problem that to this day has no known polynomial algorithm; that gap is what RSA and Diffie-Hellman use. The bar on the return arrow marks a direction assumed closed, not a direction already proved impossible.

That definition does not name a single function, and that is not an oversight. To this day there is no proof that one-way functions exist; their existence would imply $P \neq NP$, which is also still open. Public-key cryptography therefore stands on a conjecture, and the chain ends in an assumption that does not turn into a theorem by being used for a long time.

The factoring assumption says: for n the product of two random primes of size k bits, there is no polynomial algorithm in k that factors it. The best known algorithm, the number field sieve, runs in time $L_n[1/3, 1.923]$, which grows subexponentially. The number RSA-250, 250 digits long or about 829 bits, was factored in 2020 at a cost of about 2 700 core-years.

The strength equivalences recorded by NIST in SP 800-57 are useful as a rough map.

Strength (bits)	RSA or DH modulus (bits)	Symmetric cipher
112	2048	3DES three keys
128	3072	AES-128
192	7680	AES-192
256	15360	AES-256

The numbers in the middle column are cost estimates for the best algorithm at the time the table was compiled, so a table of this kind has an expiry date.

Formal security models close the gap in the rhetoric with a game: the challenger generates keys, answers encryption queries, then chooses two messages and sends the encryption of one of them; the scheme is secure if the attacker cannot guess which one far above the chance of one half. Security is proved as a reduction: an attacker who wins is turned into an algorithm that factors n, so that all the doubt moves to one place that is narrow but does not disappear.

31.9 Shor, and what collapses with it

Level: advanced

The whole building above depends on one difficulty, and in 1994 Peter Shor showed that a quantum computer solves it in polynomial time.

The idea splits in two, and the first part is entirely classical. To factor n, choose a with $\gcd(a, n) = 1$ and compute $r = \operatorname{ord}_n(a)$. If r is odd, repeat with another a. If r is even and $a^{r/2} \not\equiv -1 \pmod{n}$, then $\gcd(a^{r/2} - 1, n)$ is a proper factor of n: because $a^r \equiv 1$, the number

$a^{r/2}$ is a square root of one that is not ± 1, and a root of that kind appears only when the modulus splits. The rest of the work is finding the order, and this is the quantum part.

The classical part alone is not enough. For $n = 1019 \cdot 1031$ we tried 2 996 random units and only 319, about 10.6 percent, satisfied the condition and so gave a factor; success therefore comes from repeating with different a. What is missing without a quantum computer is a fast way to find r: no known classical algorithm runs in polynomial time in $\log n$.

A quantum computer finds r with the quantum Fourier transform, and that is where the speedup comes from. The consequence is not only that RSA collapses: Shor's algorithm also solves the discrete logarithm, both over $\mathbb{Z}_p$ and over elliptic curves, so that Diffie-Hellman, DSA, ECDH, and ECDSA collapse too, including the elliptic curves with the shortest moduli because they are immune to index calculus.

What does not collapse is symmetric ciphers and hash functions. Grover gives a quadratic speedup in key search, so that AES-128 drops from 2^{128} to about 2^{64} quantum operations, whereas AES-256 stays at about 2^{128}; the collision resistance of hashes is practically unchanged. The answer for symmetric data is therefore simple: use keys twice as long.

Two things that are often mixed together in the news must be distinguished. Shor proves that the factorization can be computed in polynomial time on a quantum computer without error; building the machine is another technical matter. The estimate of Gidney and Ekera (2021) mentions about twenty million physical qubits and eight hours for a 2048-bit modulus, almost all of it for error correction. That number has dropped by several orders of magnitude in one decade, but no machine has come close. What applies now is the pattern "harvest now, open later": today's messages can be stored and opened when the machine exists.

31.10 Post-quantum cryptography: lattices and Kyber

Level: advanced

If factorization falls, what is needed is another problem that does not fall with it. The strongest candidates come from lattices, that is sets of points generated by integer combinations of a few vectors.

Learning with errors sets up the problem like this. Given a random $m \times n$ matrix A over $\mathbb{Z}_q$ and a vector $b = As + e \bmod q$, with s secret and e a small error, find s. Without e this is only a system of linear equations solved by Gaussian elimination (Chapter 8); with e small but not zero, the system is almost consistent and finding s is considered hard. Regev's reduction (2005) adds to its appeal: attacking the random scheme is as hard as solving the worst case of the lattice problem, not an average case chosen by the attacker.

Kyber, which NIST standardized in August 2024 as ML-KEM in FIPS 203, is a module variant of that idea with $n = 256$, $k = 3$, and $q = 3329$. Its public key is 1 184 bytes and its ciphertext 1 088 bytes, about 37 and 34 times the 32-byte symmetric key it wraps, and that space cost is real. Post-quantum signatures have a similar story: ML-DSA (Dilithium) is also lattice-based, whereas SLH-DSA (SPHINCS+) takes the conservative hash-based road.

An honesty that is obligatory here: the lattice assumptions are much younger than the factoring assumption. Factorization has been attacked since Gauss; LWE is only twenty years old, and the history of cryptography is full of schemes that looked strong and then broke after ten years of analysis. The recommended transition is therefore not replacement but hybrid: the session key is formed from Kyber and from Diffie-Hellman exchange at once.

31.11 When output becomes an oracle: padding and models

Level: advanced

There is one family of attacks that does not attack the mathematics, but the way the program answers. Bleichenbacher in 1998 showed that RSA with PKCS#1 v1.5 padding can be opened completely if the server answers differently for “bad padding” and “good padding but wrong contents”. Every answer leaks one bit about the message, and with a few thousand queries the message is read. The mathematics of RSA is not violated; what is violated is the assumption that the attacker only sees the ciphertext.

The threat pattern is the same as the one discussed in Chapter 29 and Chapter 30. A machine learning model that returns probabilities, or error messages that differ, is an oracle; an attacker who may ask many times uses those small differences to infer what he should not know, including whether an example is in the training data. Chapter 30 adds a warning that applies in both worlds: a measurement that seems to improve is not necessarily improving because the defense works.

The defensive lesson is therefore the same. Output must be made uniform: one error message for all failures, response times that do not depend on secrets, refusals that do not distinguish the cause. For models, this means limiting the number of queries and disguising scores. All of that reduces leakage; it does not remove it.

Challenge (problem solver)

A library uses AES-GCM with the same key K for two messages, but its nonce counter is reset so that the same 96-bit nonce is used twice. The attacker holds both ciphers, both tags, and the contents of the first message, namely a login page that is fixed. Show how he obtains the authentication key H, then why H is enough to forge a tag for a message of his own making without knowing K.
Hint 1: the same nonce gives the same stream key, so that the difference of the two ciphers removes that key.
Hint 2: the tag is a polynomial in H with known coefficients; two tags with the same nonce give two equations for one H, and their difference becomes a polynomial whose roots can be computed.
If you get stuck, go back to the section “Hash functions, MACs, and signatures” and reread the paragraph about MACs.

31.12 Worked example

Level: intermediate

We compute one inverse by hand, because the backward steps are where mistakes happen most often. Take $\varphi(n) = 1\,048\,540$ from the toy modulus in part six and $e = 65\,537$. Forward Euclid gives

$$1\,048\,540 = 15 \cdot 65\,537 + 65\,485, \qquad 65\,537 = 1 \cdot 65\,485 + 52,$$

$$65\,485 = 1259 \cdot 52 + 17, \qquad 52 = 3 \cdot 17 + 1, \qquad 17 = 17 \cdot 1 + 0.$$

The last nonzero remainder is 1, so that gcd $= 1$ and the inverse exists. Build backward starting from the remainder one:

$$1 = 52 - 3 \cdot 17 = 52 - 3(65\,485 - 1259 \cdot 52) = 3778 \cdot 52 - 3 \cdot 65\,485.$$

Continue with $52 = 65\,537 - 65\,485$ and $65\,485 = 1\,048\,540 - 15 \cdot 65\,537$:

$$1 = 3778 \cdot 65\,537 - 3781 \cdot 65\,485 = 3778 \cdot 65\,537 - 3781(1\,048\,540 - 15 \cdot 65\,537).$$

Group: $1 = (3778 + 15 \cdot 3781) \cdot 65\,537 - 3781 \cdot 1\,048\,540 = 60\,493 \cdot 65\,537 - 3781 \cdot 1\,048\,540$. So $d = 60\,493$, and the negative sign disappears because it is read modulo $\varphi(n)$. The check is one line: $65\,537 \cdot 60\,493 = 3\,964\,529\,741$, whereas $3781 \cdot 1\,048\,540 = 3\,964\,529\,740$; the difference is one.

Two other computations in this chapter can be checked in the same way: the trimming of 7^{222} mod 13 in part three, which uses $222 = 12 \cdot 18 + 6$ and gives 12, and the small key exchange $p = 23$, $g = 5$, $a = 4$, $b = 3$ which gives $A = 4$, $B = 10$, and key 18 from both directions.

Step-by-step recipe (procedural learner)

Four steps to find the inverse of a modulo n.

1. Write (a, n). Run Euclid: replace (a, n) by $(n, a \bmod n)$ until the remainder is zero; record each line $n_0 = q\,n_1 + r$.
2. The last nonzero remainder is gcd. If it is not 1, stop; the inverse does not exist.
3. From the last line with remainder 1, write $1 = r - q\,s$, then replace s and r by their forms from the line above, one line up for each step.
4. Collect the coefficient of a; that is the inverse. If it is negative, add n until it lies in $0, \ldots, n-1$, then check $a \cdot x \bmod n = 1$.

Warning about order: build backward from the smallest remainder upward, not from the first line. Swapping the roles of a and n also changes the answer: the inverse of a modulo n is not the inverse of n modulo a.

Repeat the first computation without looking at the text. Write five lines of forward Euclid for 65 537 and 1 048 540, then build backward until you obtain d. Check by multiplying e and d and dividing by $\varphi(n)$: the remainder must be one. After that, answer: why can the same steps not be used to open RSA without the secret key, even though this whole computation needs only five divisions?

31.13 Computational lab

Level: advanced

The following program uses only the standard library and runs in under one second. It checks four claims in the body of the chapter: the units modulo 26, the statistics of Euclid's steps, the equality of the Diffie-Hellman keys, and the looseness of plain RSA. The last part tests the birthday prediction on SHA-256 truncated to two bytes.

Computational lab

```
import hashlib, math, random

def egcd(a, b):
    if b == 0:
        return a, 1, 0
    g, x, y = egcd(b, a % b)
    return g, y, x - (a // b) * y

def invers(a, n):
    g, x, _ = egcd(a, n)
    return None if g != 1 else x % n

def langkah(a, b):
    k = 0
    while b:
```

```
        a, b = b, a % b
        k += 1
    return k

def orde(a, n):
    k, y = 1, a % n
    while y != 1:
        y = y * a % n
        k += 1
    return k

unit = [a for a in range(26) if invers(a, 26) is not None]
print("units mod 26:", len(unit), unit)
print("inverse 3 mod 26 =", invers(3, 26), " inverse 2 mod 26 =", invers(2, 26))

g0, x0, y0 = egcd(1071, 462)
print("gcd(1071,462) =", g0, " x,y =", x0, y0, " steps =", langkah(1071, 462))

p, g, a, b = 10007, 5, 1234, 5678
A, B = pow(g, a, p), pow(g, b, p)
print("DH: A =", A, " B =", B, " K =", pow(B, a, p), pow(A, b, p))
print("order(5) =", orde(5, p), " order(5^5003) =", orde(pow(5, 5003, p), p))

random.seed(29)
L = [langkah(random.getrandbits(32), random.getrandbits(32))
     for _ in range(20000)]
print("euclid 32-bit: mean %.3f, max %d, min %d" % (sum(L) / len(L), max(L), min(L)))

p, q, e = 1019, 1031, 3
n, phi = p * q, (p - 1) * (q - 1)
d = invers(e, phi)
print("RSA: n =", n, " bits =", n.bit_length(), " phi =", phi, " d =", d)
for m in (424242, 42):
    c = pow(m, e, n)
    print("  m =", m, " c =", c, " D(c) =", pow(c, d, n), " m^3 =", m ** 3)
kali = pow(1234, e, n) * pow(567, e, n) % n
print("E(1234)E(567) mod n =", kali, " D(kali) =", pow(kali, d, n),
      " 1234*567 mod n =", 1234 * 567 % n)

def tabrakan():
    terlihat, k = {}, 0
    while True:
        m = random.getrandbits(32)
        h = hashlib.sha256(m.to_bytes(4, "big")).digest()[:2]
        k += 1
        if h in terlihat:
            return k
        terlihat[h] = m

T = [tabrakan() for _ in range(20)]
print("16-bit collisions: mean %.1f, min %d, max %d" % (sum(T) / 20, min(T), max(T)))
print("prediction sqrt(pi/2 * 2^16) =", round(math.sqrt(math.pi / 2 * 2**16), 1))
```

The output, as it stands:

```
units mod 26: 12 [1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25]
inverse 3 mod 26 = 9  inverse 2 mod 26 = None
gcd(1071,462) = 21  x,y = -3 7  steps = 3
DH: A = 4873  B = 3189  K = 5674 5674
order(5) = 10006  order(5^5003) = 2
euclid 32-bit: mean 18.750, max 32, min 6
RSA: n = 1050589  bits = 21  phi = 1048540  d = 699027
```

```
  m = 424242  c = 618000  D(c) = 424242  m^3 = 76355615883580488
  m = 42  c = 74088  D(c) = 42  m^3 = 74088
E(1234)E(567) mod n = 551173  D(kali) = 699678  1234*567 mod n = 699678
16-bit collisions: mean 362.7, min 100, max 743
prediction sqrt(pi/2 * 2^16) = 320.8
```

The first line confirms part one: twelve units modulo 26, with 1 and 25 their own inverses and the remaining ten paired; that number 12 is the same as $\varphi(26)$.

The second and third lines show two faces of Euclid: three divisions for 1071 and 462 with Bezout coefficients $(-3, 7)$, and an average of 18.75 divisions with maximum 32 for two thousand random 32-bit pairs. The fourth and fifth lines are the promised key exchange: 4873 and 3189 on the channel, 5674 at both ends, whereas 5^{5003} has order 2 and is therefore the trap of part five.

The three RSA lines show two opposite things. For $m = 424\,242$ everything works: the cipher 618 000 returns to 424 242. For $m = 42$ with $e = 3$, the cipher is 74 088, exactly 42^3, so that no key needs to be found, only a cube root; the next line shows the multiplicative property: the product of two ciphers opens the product message.

The last four lines test the birthday prediction: an average of 362.7 samples against a prediction of 320.8, with the wide spread expected from an exponential distribution.

Note for instructors

The teaching order I suggest: start from the student's cipher in the opening, then the well-definedness lemma and the unit theorem in the first meeting; leave Diffie-Hellman and RSA for the second meeting, and put Shor and post-quantum as reading for the third. Do not teach Euclid before asking "when does division exist".

Two misconceptions almost always appear. First, students think $\varphi(n) = n - 1$ for all n, then are startled when $\varphi(26) = 12$; cure it by asking them to list the units modulo 26 on the board before any formula is mentioned. Second, they think RSA is secure because n is hard to factor, without mentioning the size of n and without mentioning that this difficulty is an assumption; ask them to write the security claim in the form "if ... then ...".

A blackboard activity: two groups compute 3^{12} mod 26, one by repeated multiplication and one by Fermat's little theorem, then compare time and results. A sign of readiness: they can explain without notes why $[2]_{26}$ has no inverse.

31.14 Summary

Summary.

- Arithmetic modulo n works on residue classes, and division exists only for classes that share no factor with n; Euclid gives the inverse in time proportional to the number of digits.
- The shuffling lemma, namely that multiplication by a unit is only a shuffling of the order, gives Euler's theorem in five lines and Fermat's little theorem as a corollary. The order divides $\varphi(n)$, and a primitive root is an element of full order.
- Diffie-Hellman agrees on a key through the equality g^{ab} from two directions, with security conditional on the difficulty of the discrete logarithm; without authentication it falls to the man-in-the-middle attack.
- RSA reverses the direction: the public key locks, the secret key opens, and the proof uses Fermat's little theorem on both prime factors. Plain RSA is deterministic, multiplicative, and does not protect small messages; padding closes all three.

- A hash function gives a fingerprint, a MAC gives authenticity between two parties who share a key, a signature gives evidence that anyone can check; collision resistance is only half the output length, and the numbers 362.7 against 320.8 in the lab show that the prediction works.
- Every guarantee in this chapter is conditional. Shor brings down RSA and the whole of discrete logarithms if a large-scale quantum computer exists; symmetric ciphers and hashes only need to double the key length, and lattice schemes such as ML-KEM are the possible replacements.

31.15 Exercises

Exercise 31.1.

1. Assess this claim: "Because $2^{100} \equiv 1 \pmod{101}$, the number 101 is proved prime." Decide true, false, or not enough evidence, and name what that congruence does prove.
2. Prove that the product of two units modulo n is also a unit, then conclude that $U(n)$ is closed under multiplication; show also that if a is not a unit, there is $b \neq 0$ with $ab \equiv 0 \pmod{n}$.
3. Prove that for prime p, $a^2 \equiv 1 \pmod{p}$ is satisfied only by $a \equiv 1$ or $a \equiv -1$; explain why the same fails for modulus 8, and name all square roots of one modulo 8.
4. Prove $\varphi(pq) = \varphi(p)\varphi(q)$ for distinct primes p, q by counting the classes $0 \leq a < pq$ that share a factor with pq, then check it on $p = 1019$ and $q = 1031$.
5. Compute $17^{-1} \bmod 3120$ with backward Euclid, then check it with Python. Repeat for $65\,537^{-1} \bmod 1\,048\,540$ and compare the number of divisions of the two.
6. Compute $5^{100} \bmod 101$, $3^{200} \bmod 101$, and $2^{1000} \bmod 101$ using Fermat's little theorem, without computing the full power. Check all three with `pow(a,k,m)`.
7. On modulus 23 with base 5, Alice uses $a = 7$ and Bob $b = 15$. Compute A, B, and the shared key, then compute the key obtained by an active eavesdropper who replaces A with 11 in front of Bob, and explain why two different keys appear.
8. Explain why RSA with $e = 3$ and a message $m < n^{1/3}$ can be opened without the secret key, how long the longest message affected on the modulus $1\,050\,589$ is, and what padding does to prevent it.
9. Assess this claim: "ML-KEM is already proved secure, so we can stop using RSA now." Name two things that must be true for the first sentence to make sense, and one reason why the second sentence remains wrong even if the first is true.
10. A system uses a hash with a 48-bit output. How many samples are needed for a collision probability of one in two, and how many if the output is truncated to 32 bits?

Hints for the three hardest items. Item 3: write $a^2 - 1 = (a-1)(a+1)$ and use Euclid's lemma; for modulus 8, find the roots by trying $a = 0, \ldots, 7$. Item 7: compute $11^7 \bmod 23$ as the key that Bob sees, then compare it with $11^{15} \bmod 23$ computed by Alice. Item 9: distinguish "proved secure" from "secure against known attacks".

For researchers: limits and next steps

What is really proved in this chapter is its mathematical part: Bezout's theorem, the unit theorem, Euler's theorem, the order lemma, and the RSA theorem all have complete proofs and do not depend on any conjecture. What is only popularized is the word "secure". No theorem states that RSA-2048 is hard; what exists are cost estimates for the best known algorithms, and those estimates change whenever cryptanalysis makes progress.

The primary sources: Diffie and Hellman (1976), sections 3–4; Rivest, Shamir, and Adleman (1978), sections 2–4; Shor (1997), section 5 for the reduction of factorization to order finding; Regev (2005), section 3 for the LWE reduction; Bleichenbacher (1998); Gidney and Ekera (2021); FIPS 203 and 204.
Limits of validity that must be mentioned: security reductions usually produce loose bounds, the attack models do not contain side channels, and padding attacks show that the difference of one error message is enough to bring down a scheme that is proved secure in the model. Open questions: how tight the reduction for ML-KEM is, whether there is a faster quantum algorithm for lattice problems, and how to prove the security of schemes used together with secret data without leaking access patterns.

Map to source texts: Chapter 31 follows Diffie and Hellman (1976), section 3 for key exchange; Rivest, Shamir, and Adleman (1978), sections 2–4 for RSA and signatures; Shor (1997), sections 5–6 for the reduction of factorization to order finding; Regev (2005), section 3 for the LWE reduction; Bleichenbacher (1998); Gidney and Ekera (2021); FIPS 203 and 204. On Chapter 2 for bijections and partitions, Chapter 3 for the Archimedean property and the discrete form of the infimum property, Chapter 8 for Gaussian elimination, Chapter 26 for the reason a security claim cannot be a theorem without conditions, and Chapters 29–30 for the oracle threat pattern.

Chapter 32

Differential Privacy and the Mathematical Foundations of QML

A dental clinic collects the records of ten thousand patients, trains a model to predict who is at high risk, and then publishes that model as a file of weights. The following month a student takes one name from the patient list that leaked through another channel, feeds that person's medical record into the model, and reads the output score. The score is far higher than the scores of other people with similar medical records who are not patients of that clinic. The student did not break into the database and did not use anyone's password. He only asked the model, again and again, and the model answered too confidently for someone it had seen during training.

A parallel to that incident sits at the other end of this book. A paper reports "quantum advantage" because its circuit prepares a state with 2^{50} amplitudes, a number that no computer can store. The number is correct. What the paper does not say: the amplitudes cannot be read, and the measurement result is only a sample.

The two stories have a defect of the same shape: the claim runs faster than the mathematics that supports it. The first half of the chapter builds the definition that makes the sentence "this model is private" testable, namely *differential privacy*, and then shows why that definition takes the form of a comparison of two worlds rather than a prohibition on remembering. The second half builds the foundation for assessing the sentence "quantum models are superior": complex amplitudes, unitary gates, the Born rule, and the cost of simulation.

differential privacy (*privasi diferensial*) — a promise that the result of a procedure barely changes, in the sense of probability, when one person enters or leaves the training data

For younger readers (age 10 and up)

Imagine an ice cream shop that records the average amount spent by all customers that day, and then sticks that number on the door. If Sari is the only customer who bought a large birthday cake, the average that is posted goes up too, and a neighbor who knows that it is Sari's birthday can guess that Sari is on that day's list of customers. The number is not a name, but the number still tells a story about one person.

Now imagine a more careful shop. Before posting the number, it adds a small random jitter, say thirty cents up or down, with equal probability. Whether Sari is included or not, the number that is posted is nearly the same, because the jitter is much larger than the influence of one person. Other customers can still learn that day's average spending accurately enough, but there is no longer any way to determine whether Sari is on the list.

The question that makes one curious: if the jitter is too small, the secret leaks; if it is

too large, the number is useless. Where is the honest boundary between them, and how can we prove that the boundary is right? In this chapter we will learn why a privacy promise has to take the form of a comparison of two nearly identical worlds, and not a promise that the computer will forget.

32.1 What leaks from a model trained on personal data

Level: basic

A model does not store a copy of the training data. It stores parameters, and those parameters are chosen so as to fit the data. Precisely because of that choice the data leave a trace, and the trace can be read in three ways of different difficulty.

The first way, *membership inference*. The attacker has one medical record and wants to know whether that record is in the training data. He does not need to open the model; it is enough to send the record and read the output. A model trained on that record usually answers with higher confidence, because the parameters have already been adjusted to include it. Shokri, Stronati, Song, and Shmatikov showed in 2017 that this difference in confidence is enough to be separated by a second classifier trained on "shadow models", namely copies of the model trained on data with and without a particular example. The attack works on a model that can only be queried, with no weights and no gradients.

The second way, *attribute inference*. The attacker does not ask about membership, but about a property that is not in the data: whether this person smokes, whether she is pregnant. A model trained on a particular population can reveal correlations that should not be used to decide anything about one person.

The third way, *reconstruction*: the attacker recovers part of the contents of the data. Dinur and Nissim showed in 2003 that a sequence of aggregate queries with small noise still leaks most of the contents of a database when the queries are numerous and interdependent. Zhu, Liu, and Han showed in 2019 that gradients in distributed training can be used to recover training images almost exactly.

Those three channels explain why the sentence "we publish only parameters, not data" is not enough. What has to be promised is not "nothing was copied", but "the output of this procedure barely depends on one person". To state that mathematically we need two definitions of distance: the distance between two datasets that differ in one person, and the distance between two output distributions.

Definition 32.1 (Neighboring datasets)**.** Two datasets D and D' over the same domain are called *neighboring* when the two are identical except in one record: D' is obtained from D by adding or removing one record, or by replacing one record with another. The set of all such pairs is written $\mathcal{N}$.

The choice "add or remove" is called unbounded adjacency, "replace" is called bounded adjacency, and the two give different sensitivity numbers. A privacy claim that does not state the choice is incomplete.

32.2 The definition of differential privacy, and why not an average

Level: intermediate

"Nearly the same" is measured with a ratio of probabilities, not with a difference, because the ratio does not change when the probabilities are very small, and it is precisely very small probabilities that attackers most often use to infer something.

Definition 32.2 ((ε, δ) differential privacy)**.** Let $\varepsilon \geq 0$ and $\delta \in [0, 1]$. A randomized procedure M that maps datasets to an output space $\mathcal{Y}$ satisfies (ε, δ) differential privacy when for every neighboring pair $(D, D') \in \mathcal{N}$ and every measurable set $S \subseteq \mathcal{Y}$ we have

$$P[M(D) \in S] \;\leq\; e^{\varepsilon}\, P[M(D') \in S] + \delta. \tag{32.1}$$

If $\delta = 0$, the procedure satisfies pure ε differential privacy.



Figure 32.1: The output distribution of the Laplace mechanism at sensitivity 1 for two neighboring databases that differ in one record, with a different vertical scale in the two panels. At $\varepsilon = 8$ the two curves are far apart, so one output is already enough to guess which data were used. At $\varepsilon = 0.25$ the two almost coincide: the ratio of their ordinates at every point is at most $e^{0.25} \approx 1.284$, so one output hardly favors either dataset.

A new reader usually asks two things: why a comparison of probabilities rather than an average, and why there is a δ at all if ε already measures the leakage.

The answer to the first question is the core of the first half of the chapter. Note the order of the quantifiers: *for every* S, *for every* neighboring pair. That promise has to hold for any test S, including a test that the attacker picks after seeing the mechanism. A definition in the form of an average puts the quantifier inside: "there is a small expectation". The two forms are not equivalent; Figure 32.1 shows the difference at two values of ε.

Proposition 32.1. *Let M satisfy $(\varepsilon, 0)$ differential privacy. Then for every attacker algorithm A whose output is only 0 or 1, and every neighboring pair (D, D'),*

$$P[A(M(D)) = 1] \;\leq\; e^{\varepsilon}\, P[A(M(D')) = 1]. \tag{32.2}$$

Proof. The set $\{y : A(y) = 1\}$ is measurable, so it may be used as S in the definition. Direct substitution gives the result. □

That proposition looks trivial, and its use is precisely to close the door that is often used to defend a weak mechanism. If the promise holds only for the *average* output, an attacker can average the results of many repetitions so that the random components cancel one another and what remains is a signal about one person. With a per-event promise, every repetition pays its own privacy budget.

Read it aloud (auditory learner)

Read alternately, two voices. The first voice asks, the second voice answers.
"If one person leaves the list, does your answer change?"
"Slightly."
"How slightly?"
"Slightly enough that nobody can be certain."
Repeat that conversation three times, faster and faster, until it feels like a recitation.

Then the first voice asks again, more slowly:
"If I repeat that question a hundred times, is a hundred times slight still slight?"
Stay silent for one breath. Answer together: something slight repeated a hundred times is no longer slight. Then say together, once: one person must not be able to change the conclusion. Repeat that line once more, more slowly, and then once again while pointing at one empty chair in the room.

Example 32.1 (Randomized response). A respondent is asked whether he has ever used an illegal drug. The procedure: toss a coin; if side A comes up, answer honestly; if side B comes up, toss again and answer "yes" if heads, "no" if tails. The probability of answering "yes" if he really uses the drug is $1/2 + 1/2 \cdot 1/2 = 3/4$, and if he does not use it the probability is $1/2 \cdot 1/2 = 1/4$. The ratio of the two is 3, so $\varepsilon = \ln 3 = 1.0986$. No answer proves anything about one person, and that is what the definition promises.

Do it by hand (kinesthetic learner)

Materials: one coin per person, paper, pencil. Ten friends each write one yes/no secret on a slip of paper and fold it.
The rule is one. Toss the coin: heads, answer honestly; tails, toss again, heads means "yes" and tails means "no". Each person says only one word, and the recorder writes only two groups of tally marks.
Now ask one person to change his secret, then repeat the whole round. Look at the tally marks: they shift by at most one, and nobody can point to anyone. Conclusions about the group survive. A conclusion about one person never appears.

Now δ, namely the failure probability: with probability at most δ the ratio promise may be violated entirely. Hence δ is not a tolerance that can be chosen at will. If δ is comparable to $1/n$ for n records, an attacker can target one person and let the rest serve as camouflage; that is why δ must be much smaller than $1/n$. The number ε is not a probability either: $\varepsilon = 1$ means a probability ratio of at most $e \approx 2.718$, whereas $\varepsilon = 8$ means a ratio up to 2 981.

32.3 Global sensitivity and the Laplace mechanism

Level: intermediate

The only thing that stops us from adding noise arbitrarily is the question: how far can one record shift the answer?

Definition 32.3 (Global sensitivity)**.** For f that maps datasets to $\mathbb{R}^d$, its global sensitivity is

$$\Delta f \;=\; \sup_{D \sim D'} \bigl\| f(D) - f(D') \bigr\|_1, \tag{32.3}$$

with the supremum taken over all neighboring pairs.

A count query, $f(D) = |D|$, has $\Delta f = 1$. A sum query over values in $[0, R]$ has $\Delta f = R$. An average query over n records has $\Delta f = R/n$, because one person shifts the average by only R/n; here it becomes visible that an average is safer than a sum, but only as long as n is known and fixed. Sensitivity also depends on the choice of adjacency: if D' may replace, the count query has $\Delta f = 2$.

Theorem 32.2 (Laplace mechanism)**.** *Let f have global sensitivity Δf, and let Y be distributed according to a Laplace distribution with scale $b = \Delta f / \varepsilon$, that is with density $p_b(y) = \frac{1}{2b} e^{-|y|/b}$. Then $M(D) = f(D) + Y$ satisfies $(\varepsilon, 0)$ differential privacy.*

Proof. Take $S \subseteq \mathbb{R}$ measurable and $D \sim D'$. It is enough to show that for every y,

$$\frac{p_b(y - f(D))}{p_b(y - f(D'))} \;\le\; e^{\varepsilon}. \tag{32.4}$$

With $a = f(D)$ and $a' = f(D')$, the left-hand side is $\exp((|y - a'| - |y - a|)/b)$. The triangle inequality gives $|y - a'| - |y - a| \le |a - a'|$, so the left-hand side is at most $\exp(|a - a'|/b) \le \exp(\Delta f / b) = e^{\varepsilon}$, because Δf is the supremum. Hence $p_b(y - a) \le e^{\varepsilon} p_b(y - a')$ for every y; integrating both sides over S gives $P[M(D) \in S] \le e^{\varepsilon} P[M(D') \in S]$, and because there is no remainder term, $\delta = 0$. □

A numerical example goes like this. For a count query with $\varepsilon = 0.5$, the scale is $b = 1/0.5 = 2$; the probability that the noise exceeds 5 in absolute value is $e^{-5/2} = 0.0821$, so about one answer in twelve misses by more than five people. Conversely, with probability 0.95 the error does not exceed $b \ln 20 = 2 \times 2.9957 = 5.991$. Here the trade-off that recurs throughout the chapter becomes visible: shrinking ε enlarges b, and enlarging b worsens precision linearly.

Step-by-step recipe (procedural learner)

1. Fix f and clip every value to the range $[0, R]$ before anything is computed.
2. Compute Δf from the worst case of one record on the clipped data.
3. Choose $\varepsilon > 0$, compute $b = \Delta f / \varepsilon$, add Laplace noise of scale b to the answer.
4. Record ε in the budget book; the remaining budget is what has not been spent.

A warning about order: clipping after the noise has been added changes the output distribution and cancels the guarantee, because Δf was computed from the data before clipping.

global sensitivity (*sensitivitas global*) — how far one person can shift the answer of a query in the worst case; this is what determines the amount of noise that is needed

Sometimes Laplace noise does not fit, for instance because the output is high-dimensional. There Gaussian noise is used: for f with ℓ_2 sensitivity $\Delta_2 f$, the mechanism $M(D) = f(D) + N(0, \sigma^2 \Delta_2 f^2 I)$ satisfies (ε, δ) when $\sigma \ge \sqrt{2 \ln(1.25/\delta)}/\varepsilon$. The proof is skipped here; it compares two shifted normals with a concentration inequality, and the full version is in Dwork and Roth (2014), appendix A. That condition is sufficient, not necessary: for small σ it is far looser than the true bound, and the tight bound is given by the analytic Gaussian mechanism of Balle and Wang (2018).

32.4 Composition: the privacy budget and why it runs out

Level: advanced

A single mechanism is rarely used once. A model is trained with hundreds of steps, each leaking a little, and what is needed is the total.

Theorem 32.3 (Basic composition)**.** *Let $M_1, \dots, M_T$ be chosen adaptively, that is M_t may depend on the outputs $M_1, \dots, M_{t-1}$, and let every M_t satisfy $(\varepsilon_t, \delta_t)$ for all possible inputs. Then the joint output $(M_1, \dots, M_T)$ satisfies $(\sum_t \varepsilon_t, \sum_t \delta_t)$.*

Proof. It is enough to take $T = 2$; the rest is induction. By the law of total probability,

$$\begin{aligned} P[(M_1, M_2) \in S_1 \times S_2] &= \int_{S_1} P[M_1(D) \in dy_1]\, P[M_2(D, y_1) \in S_2 \mid M_1(D) = y_1] \\ &\le \int_{S_1} e^{\varepsilon_1} P[M_1(D') \in dy_1]\, \left(e^{\varepsilon_2} P[M_2(D', y_1) \in S_2] + \delta_2\right) + \delta_1 \\ &\le e^{\varepsilon_1 + \varepsilon_2} P[(M_1, M_2)(D') \in S_1 \times S_2] + \delta_1 + \delta_2. \end{aligned} \tag{32.5}$$

The first step uses the guarantee for M_2 at fixed y_1, the second step uses the guarantee for M_1 and $P[M_1(D') \in S_1] \le 1$. □

Basic composition adds up, and that addition is what makes the budget run out quickly: a hundred steps with $\varepsilon = 0.1$ per step already spend $\varepsilon = 10$. There is a sharper bound when the leakage per step is small.

Challenge (problem solver)

An office publishes the average salary of its team with Laplace noise of scale $b = \Delta f / \varepsilon$, with $\varepsilon = 1$ and $\Delta f = 1$. It says, “one answer already satisfies $\varepsilon = 1$, so one person is safe.” How many times may that answer be published before the average of those answers is sharp enough to guess the salary of one person, and why?
Hint 1: random noise has mean zero; watch what happens to the average of the noise when the answer is repeated.
Hint 2: the standard error of the average of k answers is $b\sqrt{2}/\sqrt{k}$. With $b = 1$ and $k = 100$ its value is 0.1414, whereas one person can shift the answer by 1.
If you get stuck, go back to the section on Composition.

Theorem 32.4 (Advanced composition)**.** *For every $\varepsilon' > 0$, the composition of T procedures that each satisfy (ε, δ) satisfies $(\varepsilon_{\mathrm{tot}}, \delta_{\mathrm{tot}})$ with*

$$\varepsilon_{\mathrm{tot}} = \sqrt{2T \ln(1/\delta')}\, \varepsilon + T\, \varepsilon (e^{\varepsilon} - 1), \qquad \delta_{\mathrm{tot}} = T\delta + \delta'. \tag{32.6}$$

The proof is skipped: it is obtained by bounding the moments of the probability ratio, or with a moments *accountant*. The references are Dwork and Roth section 3.5, and Kairouz, Oh, and Viswanath (2015) for optimal composition. Its first term grows like $\sqrt{T}$, not like T.

However, advanced composition is not always better, and this is often misquoted. The second term, $T\varepsilon(e^{\varepsilon} - 1)$, blows up when ε is not small. In the experiment in section eleven, with $T = 200$ and $\delta = 10^{-5}$, basic composition gives $\varepsilon = 36.5$ at $\sigma = 32$ whereas advanced composition gives 20.6; at $\sigma = 2$ the two give 583.7 and 11 302.8. So advanced composition wins when ε per step is small and loses when it is large.

One more result makes the budget go further: when several procedures work on disjoint parts of the data, their leakage is not added but the largest is taken, because one record is in only one part. Splitting the data among several users, each with its own budget, is far more economical than one user who runs all the queries.

32.5 DP-SGD: clip, add noise, and the price that is paid

Level: advanced

Ordinary training satisfies no definition at all, because one example can shift the gradient by any amount: there is no upper bound on Δf, so the noise that is needed is infinite. The repair consists of two steps that support each other.

The first step, clipping: every per-example gradient is clipped so that its norm does not exceed C,

$$\operatorname{clip}_C(g) \;=\; g \cdot \min\Bigl(1,\; \frac{C}{\|g\|_2}\Bigr). \tag{32.7}$$

After clipping, one example contributes at most C to the sum, so the sensitivity of the average over a batch of size B is C/B. The second step, Gaussian noise with standard deviation $\sigma C/B$ is added to that average,

$$w_{t+1} = w_t - \eta\Bigl(\frac{1}{B}\sum_{i \in \mathcal{B}_t} \operatorname{clip}_C(\nabla \ell_i(w_t)) + \frac{\sigma C}{B}\,\xi_t\Bigr), \qquad \xi_t \sim N(0, I). \tag{32.8}$$

This algorithm is what is called DP-SGD, introduced by Abadi and colleagues in 2016 together with the moments *accountant* that makes its budget computable for hundreds of thousands of steps.

Theorem 32.5. *Every DP-SGD step satisfies (ε_0, δ) with*

$$\varepsilon_0 = \sqrt{2\ln(1.25/\delta)}\,/\,\sigma, \tag{32.9}$$

and after T steps the whole procedure satisfies $(T\varepsilon_0, T\delta)$.

Proof. The ℓ_2 sensitivity of the average of clipped gradients is C/B, because replacing one example shifts the average by at most C/B. Apply the Gaussian mechanism from section three, then basic composition from section four. □

The numbers from our experiment explain why the noise does not immediately erase the learning. With $n = 200$, $C = 1$, and $\sigma = 32$, the standard deviation of the noise per coordinate is $32/200 = 0.16$, whereas the norm of the average gradient at the starting point is 0.83: the noise is about one fifth of the signal. But the noise changes sign at random, so its contribution to the final position grows like $\sqrt{T}$, whereas the signal grows like T. After two hundred steps the signal wins, and the model still reaches 93.71 percent.

What has to be paid is also visible. At $\sigma = 128$ the accuracy falls to 85.51 percent with a spread of 6.16 across seeds; at $\sigma = 512$ the accuracy is 70.39 percent with a spread of 17.26. That growing spread is not an oddity: at large noise the result of training depends on the noise draws that happen to occur, so a single model is no longer a sample that represents the procedure. Reporting one seed at large σ would be misleading.

Four limits have to be mentioned together with that result. First, clipping introduces bias: examples with large gradients, usually the hard or rare ones, are clipped harder, so the model pays them less attention. Second, DP-SGD demands more steps, and that computational cost is real. Third, the privacy guarantee is uniform but its utility cost is not: Bagdasaryan, Poursaeed, and Shmatikov showed in 2019 that minority groups can suffer a much larger drop in accuracy at the same budget. Fourth, and most important: differential privacy guarantees something about *individuals*, not about the *truth of the model*. A model with $\varepsilon = 0.1$ can still be completely wrong, can still be discriminatory, and can still be used for unfair decisions. Privacy is a promise about how much one person can be learned from the output; it is not in the least a promise that the output is any good.

privacy budget (*anggaran privasi*) — the total amount of ε that a system may spend; every query or training step spends some of it, and once it is exhausted there is no guarantee left

32.6 Why amplitudes use complex numbers

Level: basic

Now the second half, starting from a question that is often taken as technical although its answer is conceptual. Why is a quantum state represented by complex numbers, and not by real numbers?

The short answer: because two paths to the same result have to be able to cancel each other, and cancellation of that kind demands a sign that can rotate, not merely positive and negative. With real numbers an amplitude has only two signs, and interference can only add or subtract. With complex numbers an amplitude has a phase that rotates continuously, so two paths can meet at any angle and the result varies smoothly.

The complex number $z = a + bi$ with $i^2 = -1$ has modulus $|z| = \sqrt{a^2 + b^2}$, conjugate $\bar{z} = a - bi$, and phase θ in the notation $z = |z|e^{i\theta} = |z|(\cos\theta + i\sin\theta)$. Modulus and phase play different roles and cannot be exchanged: the modulus determines the size of the probability, the phase determines the sign of the interference. Multiplying two complex numbers multiplies the moduli and adds the phases, and that property makes the phase useful as a coordinate of the state.

probability amplitude (*amplitudo*) — a complex number whose modulus squared is the probability; different from a probability because amplitudes can cancel one another

The state space of one qubit is $\mathbb{C}^2$, and the state space of n qubits is $\mathbb{C}^{2^n}$ with the inner product $\langle \phi | \psi \rangle = \sum_k \overline{\phi_k} \psi_k$. In bra-ket notation, $|\psi\rangle$ is a column vector and $\langle \psi|$ is its conjugate transpose. A state is always normalized, $\langle \psi | \psi \rangle = 1$, because the sum of all probabilities has to be one.

Definition 32.4 (Finite-dimensional Hilbert space)**.** A finite-dimensional Hilbert space is a finite-dimensional complex vector space equipped with a complete inner product. In this book the only example that is used is $\mathbb{C}^d$ with the standard inner product, usually $d = 2^n$.

Two properties of this notation have to be held from the start. First, the global phase has no physical meaning: $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ give the same probability for every measurement outcome, because $|e^{i\theta} a_k|^2 = |a_k|^2$. Second, the *relative* phase has full meaning: changing the phase of one component alone changes the result of interference, and that is what quantum algorithms use. The difference between the invisible global phase and the visible relative phase becomes clear in the worked example in section nine.

Proposition 32.6. *If $|\psi\rangle$ and $|\phi\rangle$ are both of norm one, then $|\langle \phi | \psi \rangle| \leq 1$.*

Proof. The Cauchy-Schwarz inequality gives $|\langle \phi | \psi \rangle|^2 \leq \langle \phi | \phi \rangle \langle \psi | \psi \rangle = 1$, then take the square root. □

32.7 Gates as unitary matrices, and why they must be unitary

Level: intermediate

A change of state due to a gate is a matrix multiplication, $|\psi'\rangle = U|\psi\rangle$. Not every matrix may be used. The condition is one: the total probability has to stay one for every initial state, so $\langle U\psi | U\psi \rangle = \langle \psi | \psi \rangle$ for all ψ.

Definition 32.5 (Unitary matrix)**.** A square matrix U is called unitary when $U^\dagger U = U U^\dagger = I$, with $U^\dagger$ the conjugate transpose of U.

Theorem 32.7. *For a square matrix U over $\mathbb{C}^d$, the following statements are equivalent: (i) U is unitary; (ii) $\langle U\phi | U\psi \rangle = \langle \phi | \psi \rangle$ for all ϕ, ψ; (iii) $\|U\psi\| = \|\psi\|$ for all ψ.*

Proof. (i) ⇒ (ii): $\langle U\phi|U\psi\rangle = \phi^\dagger U^\dagger U\psi = \phi^\dagger\psi$. (ii) ⇒ (iii): take $\phi = \psi$. (iii) ⇒ (i): from $\langle U\psi|U\psi\rangle = \langle\psi|\psi\rangle$ for all ψ, use the polarization identity to conclude $\langle\phi|U^\dagger U\psi\rangle = \langle\phi|\psi\rangle$ for all ϕ, ψ; therefore $U^\dagger U = I$, and for a square matrix in finite dimension a left inverse means a right inverse. □

That theorem explains three things. First, a gate must be unitary because probability must be conserved; no other reason is needed. Second, every gate has an inverse $U^\dagger$ that is also unitary, so a circuit can be reversed and no information is lost while the gates act. Third, the only non-unitary operation is measurement, and that is where information is lost. That is why quantum computation is not computation that erases information, but computation that postpones erasure until the end.

Proposition 32.8 (No cloning)**.** *There is no unitary U on two qubits such that $U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$ for all $|\psi\rangle$.*

Proof. Suppose there is one, and take $|\psi\rangle, |\phi\rangle$ of norm one. Because U is unitary, the inner products of the two sides are equal: $\langle\phi|\psi\rangle = \langle\phi|\psi\rangle^2$. The equation $x = x^2$ has only the solutions 0 and 1, so every pair of states has to be orthogonal or identical; that is impossible for more than two states. □

unitary (*uniter*) — a property of a matrix that preserves the length of a vector; in everyday language, a gate that neither adds nor removes total probability

32.8 Fourier as a change of basis, and its connection to convolution

Level: advanced

The Fourier transform is a change of basis, and that is the most useful way to see it here. In a space of dimension N, the standard basis is replaced by the basis $|f_k\rangle$ whose components are $(|f_k\rangle)_j = \omega^{jk}/\sqrt{N}$ with $\omega = e^{2\pi i/N}$. The matrix is $F_{jk} = \omega^{jk}/\sqrt{N}$.

Proposition 32.9. *The Fourier matrix F is unitary.*

Proof. Compute $(F^\dagger F)_{jk} = \frac{1}{N}\sum_{l=0}^{N-1} \omega^{-lj}\omega^{lk} = \frac{1}{N}\sum_{l=0}^{N-1} \omega^{l(k-j)}$. If $k = j$ every term has the value one, so the result is 1. If $k \neq j$, this is a geometric series with ratio $\omega^{k-j} \neq 1$ and its sum is $\frac{1-\omega^{N(k-j)}}{1-\omega^{k-j}} = 0$ because $\omega^N = 1$. Hence $F^\dagger F = I$. □

Because it is unitary, F may legitimately be used as a gate, and its n-qubit version is called the quantum Fourier transform. What makes it interesting is not only that it is a legitimate gate, but that it turns convolution into pointwise multiplication.

Theorem 32.10 (Convolution becomes multiplication)**.** *For $x, y \in \mathbb{C}^N$, define the cyclic convolution $(x * y)_k = \sum_{j=0}^{N-1} x_j\, y_{(k-j) \bmod N}$. Then*

$$F(x * y) \;=\; \sqrt{N}\,(Fx) \odot (Fy), \tag{32.10}$$

with $\odot$ the componentwise product.

Proof. The k-th component of the left-hand side is $\frac{1}{\sqrt{N}}\sum_m \omega^{km}\sum_j x_j y_{m-j}$. Exchange the order of summation and write $r = m - j$:

$$\frac{1}{\sqrt{N}}\sum_j x_j\omega^{kj}\sum_r \omega^{kr}y_r = \sqrt{N}\,(Fx)_k\,(Fy)_k.$$

The exchange of the order of summation is legitimate because both sums are finite. □

That exchange of two finite sums looks trivial, but the infinite version of the same step is the content of Chapter 14: a Fourier series may be differentiated term by term only when its convergence is uniform enough. The connection with Chapter 15 is more conceptual: a change of basis is a change of variable, and a unitary matrix is a change of variable whose Jacobian determinant has modulus one, so volumes and probabilities do not change.

The connection with quantum computation lies in its cost. A discrete Fourier transform over $N = 2^n$ points is computed classically in $\mathcal{O}(N \log N)$ operations with the FFT. Its quantum version is built from $\mathcal{O}(n^2)$ gates. The difference is real and is the basis of Shor's algorithm. But there is a condition that is often left out: once the transform is finished, we cannot read the N amplitudes, and all we can do is take samples. Speed in preparing a state does not automatically become speed on the task one wants to finish; it has to be paired with a procedure that extracts the answer from only a few samples. Many claims of quantum advantage in machine learning have no procedure of that kind.

32.9 The Born rule, and why the output is always probabilistic

Level: intermediate

Up to here everything is deterministic: given a state and a gate, the next state is unique. Probability enters at one step that cannot be avoided, namely measurement.

Definition 32.6 (Born rule)**.** If the state $|\psi\rangle = \sum_k a_k |k\rangle$ is measured in the orthonormal basis $\{|k\rangle\}$, the probability of obtaining outcome k is

$$p(k) \;=\; \big|\langle k|\psi\rangle\big|^2 \;=\; |a_k|^2, \tag{32.11}$$

and after the outcome k is obtained, the state becomes $|k\rangle$.

Why must it be so, and not merely our ignorance of the state that is "really" there? The strongest answer is not philosophical but experimental: Bell's inequality, tested by Aspect, Grangier, and Roger in 1982, shows that the correlations between two separated measurements violate a bound that every theory storing hidden definite values has to obey. So probabilism here is not a way of saying "we do not know yet"; it is the structure of the theory. The practical consequence: one measurement gives one outcome, not the whole state.

Born rule (*aturan Born*) — the rule that turns an amplitude into a probability by squaring its modulus; the only place where probability enters the theory

The consequence for machine learning is often underestimated. To estimate $p(k)$ with standard error γ one needs about $p(1-p)/\gamma^2$ shots. In our experiment, the state $R_y(60^\circ)|0\rangle$ has true probabilities 0.75 and 0.25; with a thousand shots the recorded counts were 727 and 273, the estimate 0.727, differing by 0.023 from the true value. Its theoretical standard error is $\sqrt{0.75 \cdot 0.25/1000} = 0.0137$, so that difference is about 1.7 standard errors, an ordinary event. Three digits after the decimal point need about a million shots, and every shot means one state preparation and one measurement. This is where the factor $\sqrt{N}$ appears, and this is where many "quadratic speeds" turn into ordinary speeds once the cost of readout is counted.

32.10 Worked example: one and two qubits by hand

Level: intermediate

For younger readers (age 10 and up)

Two children toss coins. The first child tosses one coin, then closes his hand and asks: heads or tails? The result is one, and nobody can know anything before that coin is opened. The second child prepares two coins in a strange way: he tells the coins to "consult" first, so that the two always agree, without ever agreeing in advance whether they will both be heads or both be tails. When the first coin is opened and shows heads, the second coin is certainly heads. When the first coin shows tails, the second coin is certainly tails.

What makes it strange: there is no way to make two ordinary coins behave like that. If both were fixed from the start, whoever made the coins would know both. In coins that consult there is no record of that kind, and real experiments show that no hidden record explains the outcome.

That last example is what is called a Bell state, and this chapter uses it to show why "zero" and "very small" are two different things.

All the numbers in this section are rechecked with NumPy in section eleven, and the reader can check them with an ordinary calculator.

The Hadamard gate on one qubit is $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. On $|0\rangle = (1, 0)$ it gives

$$H|0\rangle = \frac{1}{\sqrt{2}}(1, 1) = (0.7071;\ 0.7071),$$

so the probabilities of the outcomes 0 and 1 are 0.5 each.

Now the circuit H, then $Z = \mathrm{diag}(1, -1)$, then H again. The first step gives $(0.7071; 0.7071)$; the second step flips the sign of the second component to $(0.7071; -0.7071)$; the third step multiplies by H:

$$H\begin{pmatrix} 0.7071 \\ -0.7071 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 0.7071 - 0.7071 \\ 0.7071 + 0.7071 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The result is $|1\rangle$ with probability one. The amplitude of the $|0\rangle$ component becomes zero not because something was thrown away, but because two paths meet with opposite signs. This is interference, and this is why the relative phase matters: if Z is replaced by the identity, the result is $|0\rangle$ again.

The third example uses a rotation. The gate $R_y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$ rotates the state in the real plane. With $\theta = 60^\circ$ and initial state $|0\rangle$,

$$R_y(60^\circ)|0\rangle = (\cos 30^\circ;\ \sin 30^\circ) = (0.8660;\ 0.5),$$

so the probabilities are $\cos^2 30^\circ = 0.75$ and $\sin^2 30^\circ = 0.25$. Note that $|a_0|^2 + |a_1|^2 = 0.75 + 0.25 = 1$.

The fourth example is two qubits. Start from $|00\rangle$, apply H to the first qubit, then CNOT with the first qubit as control and the second as target. After H the state is $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$. CNOT flips the second qubit only when the first qubit is 1, so

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (0.7071;\ 0;\ 0;\ 0.7071).$$

The probabilities of 00 and 11 are 0.5 each; the probabilities of 01 and 10 are zero. The two amplitudes in the middle are zero because the two paths that lead there cancel one another,

not because both are impossible. Consequently, if both qubits are measured, the outcomes always agree: 00 or 11, never a mixture.

Check the examples above without looking at the text. (a) Compute $HZH|0\rangle$ by writing out every matrix multiplication, then explain why the result is $|1\rangle$ and not $|0\rangle$. (b) Replace $R_y(60°)$ with $R_y(90°)$, compute both probabilities, then explain why two different states can give the same probabilities. (c) For the Bell state above, compute the probability of obtaining 01 and explain why that number is zero, and not "very small".

32.11 Why classical simulation explodes, and what that means for claims of advantage

Level: advanced

The state of n qubits is a vector in $\mathbb{C}^{2^n}$, so it has 2^n complex amplitudes. For $n = 10$ that is 1 024 numbers, no larger than a small image. For $n = 50$ it is $2^{50} = 1.1259 \times 10^{15}$ numbers; with 16 bytes per amplitude, what is needed is 2^{54} bytes, that is exactly 16 pebibytes. It is a limit that doubles every time one qubit is added.

The experiment in section eleven measures the consequence. Preparing a state with one Hadamard gate on every qubit takes 0.2 milliseconds for $n = 10$, 2.3 for $n = 14$, 81.4 for $n = 18$, and 1 357 milliseconds for $n = 22$. That time column is the noisiest in the whole chapter: two runs on the same machine can differ by almost a factor of two. What does not change is the pattern, namely a straight line on a logarithmic scale against 2^n.

What may and may not be concluded? What may: a general quantum state cannot be stored classically for large n, so an algorithm that genuinely uses all 2^n amplitudes cannot be imitated directly. What may not: that every quantum circuit is faster. There are three reasons.

The first reason, the state cannot be read. One measurement gives one outcome, whereas a classical simulation that stores the whole vector can compute the probability of every outcome. So more information is available in the simulation than in the hardware, and the inability to copy a state is not evidence of advantage.

The second reason, the circuits that can be simulated include many circuits that are used in practice: small depth, limited entanglement, product states, and Clifford gates have efficient classical imitations. Goguen, Clarke, Martens, and Selman showed that family of circuits can be computed classically in polynomial time, and that family is wide enough to include many variational ansatzes.

The third reason, advantage can be removed by data structure. Ewin Tang showed in 2019 that a quantum recommendation algorithm that was thought to give an exponential speedup has a classical counterpart with a polynomial speedup, because the structure of its data matrix can be exploited without quantum mechanics. Huang, Kueng, and Preskill (2021) give a strict separation, but only for a particular family of data distributions.

barren plateau (*plateau tandus*) — a region of the loss landscape of a variational circuit where the gradients shrink exponentially in the number of qubits, so that training does not move

An honest statement about QML in 2026 therefore takes this form. There is evidence of separation for particular tasks in oracle models, and there are theoretical claims of speedup for learning tasks under particular distributional assumptions. There is no established evidence of practical advantage for machine learning tasks on classical data, in the sense of an experiment repeated on real hardware that beats the best classical method at a meaningful data size. Cerezo and colleagues (2021, 2022) call the *barren plateau*, namely gradients that vanish exponentially in deep circuits, an unresolved obstacle rather than a technical detail.

32.12 Computational lab: privacy budget, accuracy, and the size of a state

Level: advanced

The following program runs the two experiments above with no library other than NumPy: DP-SGD on two-class logistic regression with five seeds per noise level, then several quantum states and the cost of simulating them. All the numbers in this chapter come from the output below, except those marked as hand computations.

Computational lab

```
import numpy as np
import time

D, N, T, C, DELTA, LR = 8, 200, 200, 1.0, 1e-5, 0.5


def data(rng, n, mu=0.6):
    y = 2 * rng.integers(0, 2, n) - 1
    return rng.normal(mu * y[:, None], 1.0, (n, D)), y


def sig(t):
    return 1.0 / (1.0 + np.exp(-t))


def clip(G):
    nrm = np.linalg.norm(G, axis=1, keepdims=True)
    return G * np.minimum(1.0, C / np.maximum(nrm, 1e-9))


def dp_sgd(X, y, sigma, seed):
    rng = np.random.default_rng(seed)
    w = np.zeros(D)
    for _ in range(T):
        G = -(1.0 - sig(y * (X @ w)))[:, None] * (y[:, None] * X)
        w -= LR * (clip(G).mean(0) + rng.normal(0.0, sigma * C / len(X), D))
    return w


def eps_basic(s):
    return T * np.sqrt(2 * np.log(1.25 * T / DELTA)) / s


def eps_adv(s):
    d = DELTA / 2
    e = np.sqrt(2 * np.log(1.25 * T / d)) / s
    return np.sqrt(2 * T * np.log(1 / d)) * e + T * e * (np.exp(e) - 1)


rng = np.random.default_rng(11)
Xn, yn = data(rng, N)
Xe, ye = data(rng, 3000)
print("sigma  accuracy     eps_basic  eps_adv")
A0 = [100 * np.mean(np.sign(Xe @ dp_sgd(Xn, yn, 1e-9, k)) == ye) for k in range(5)]
print("%5s  %5.2f+-%4.2f  %8s  %9s" % ("0", np.mean(A0), np.std(A0), "-", "-"))
for s in (2.0, 8.0, 32.0, 128.0, 512.0):
    A = [100 * np.mean(np.sign(Xe @ dp_sgd(Xn, yn, s, k)) == ye) for k in range(5)]
    print("%5.0f  %5.2f+-%4.2f  %8.1f  %9.1f"
          % (s, np.mean(A), np.std(A), eps_basic(s), eps_adv(s)))
```

```
I2 = np.eye(2)
Z = np.diag([1.0, -1.0])
S = np.diag([1.0, 1.0j])
H = np.array([[1.0, 1.0], [1.0, -1.0]]) / np.sqrt(2)
CNOT = np.array([[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 0, 1], [0, 0, 1, 0]], float)


def ry(t):
    c, s = np.cos(t / 2), np.sin(t / 2)
    return np.array([[c, -s], [s, c]])


def kron1(U, n, q):
    out = np.array([[1.0]])
    for j in range(n):
        out = np.kron(out, U if j == q else I2)
    return out


s0 = np.array([1.0, 0.0])
print()
print("H|0>        ", np.round(H @ s0, 4))
print("HZH|0>      ", np.round(H @ Z @ (H @ s0), 4))
print("HSH|0>      ", np.round(H @ S @ (H @ s0), 4))
print("Ry(60)|0>   ", np.round(ry(np.pi / 3) @ s0, 4),
      " probability", np.round((ry(np.pi / 3) @ s0) ** 2, 4))
psi = CNOT @ (kron1(H, 2, 0) @ np.kron(s0, s0))
print("Bell        ", np.round(psi, 4), " probability", np.round(psi ** 2, 4))
p1 = (ry(np.pi / 3) @ s0) ** 2
sh = np.bincount(np.random.default_rng(4).choice(2, 1000, p=p1), minlength=2)
print("1000 shots", sh, " estimate", np.round(sh / 1000, 3))
print()
print("  n   amplitudes    memory (GB)   time (ms)")
for n in (10, 14, 18, 22):
    best = 1e9
    for _ in range(3):
        v = np.zeros(2 ** n, dtype=complex)
        v[0] = 1.0
        t0 = time.perf_counter()
        for q in range(n):
            v = v.reshape([2] * n)
            v = np.moveaxis(np.tensordot(H, v, axes=([1], [q])), 0, q).reshape(-1)
        best = min(best, 1000 * (time.perf_counter() - t0))
    print("%3d  %11.3e  %12.3e  %10.1f" % (n, 2.0 ** n, 2.0 ** n * 16 / 2 ** 30, best))
```

The output of that program, as it stands:

```
sigma  accuracy     eps_basic  eps_adv
    0  95.20+-0.00         -          -
    2  95.13+-0.05     583.7    11302.8
    8  95.03+-0.13     145.9      216.5
   32  93.71+-0.81      36.5       20.6
  128  85.51+-6.16       9.1        3.7
  512  70.39+-17.26      2.3        0.8

H|0>         [0.7071 0.7071]
HZH|0>       [0. 1.]
HSH|0>       [0.5+0.5j 0.5-0.5j]
Ry(60)|0>    [0.866 0.5  ]  probability [0.75 0.25]
Bell         [0.7071 0.     0.     0.7071]  probability [0.5 0.  0.  0.5]
1000 shots [727 273]  estimate [0.727 0.273]
```

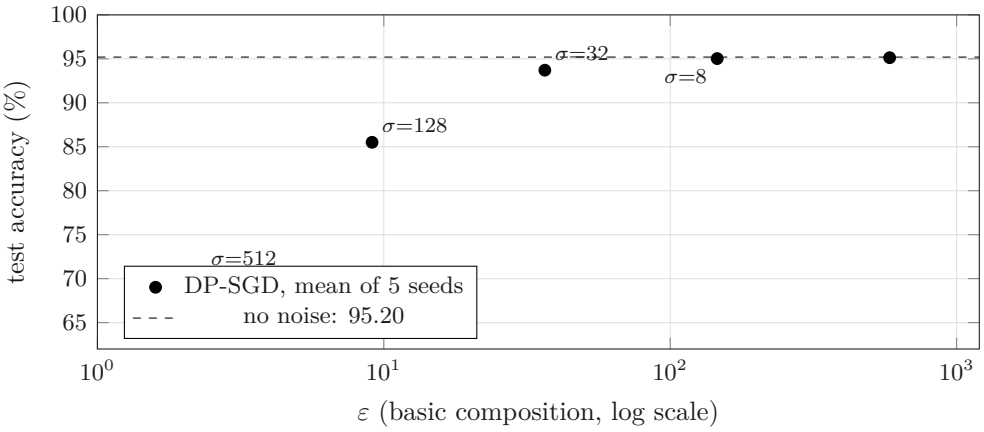


Figure 32.2: Test accuracy against the privacy budget in the Computational lab experiment ($n = 200$, $d = 8$, $T = 200$, $\delta = 10^{-5}$, five seeds). Every point is the result of a real execution, paired with the `eps_basic` column in the output. The dashed line is the accuracy without noise. The horizontal axis is logarithmic: lowering ε from 36.5 to 2.3 costs 23 points of accuracy, whereas going from 583.7 to 145.9 is almost free.

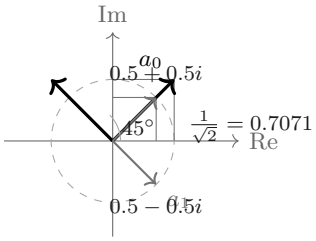


Figure 32.3: Amplitudes as arrows in the complex plane. The thick arrows are the state $H|0\rangle$ with $a_0 = a_1 = 1/\sqrt{2}$; the thin arrows are $HSH|0\rangle$ with $S = \mathrm{diag}(1, i)$, namely $0.5 + 0.5i$ and $0.5 - 0.5i$. The modulus stays 0.7071, so the probability of both outcomes stays 1/2, but the phase shifts by 45° in opposite directions. A measurement does not distinguish the two states; the next gate can. The arrow coordinates are taken from the output of the Computational lab, not drawn as an illustration.

```
 n   amplitudes    memory (GB)   time (ms)
10    1.024e+03     1.526e-05        0.2
14    1.638e+04     2.441e-04        2.3
18    2.621e+05     3.906e-03       81.4
22    4.194e+06     6.250e-02     1356.6
```

The zero row is the baseline: 95.20 percent with zero spread. The row $\sigma = 2$ gives $\varepsilon = 583.7$, a guarantee that is practically empty because the ratio e^{583} constrains nothing; the row $\sigma = 8$ lowers ε to 145.9 with no measurable loss of accuracy, so a little noise is almost free. The price only begins to be felt at $\sigma = 32$: ε drops to 36.5 or 20.6 at a cost of 1.49 points, then 85.51 percent at $\sigma = 128$, and 70.39 percent at $\sigma = 512$ with a spread across seeds of 17.26 points, so that a single model no longer represents the procedure.

The second block verifies the hand computations of section nine. $HZH|0\rangle$ gives $(0; 1)$, $R_y(60°)$ gives $(0.8660; 0.5)$, and the Bell state appears as $(0.7071; 0; 0; 0.7071)$ with two exact zeros. The row $HSH|0\rangle$ gives $(0.5 + 0.5i;\ 0.5 - 0.5i)$, and that pair is used in Figure 32.3.

The third block measures the cost of simulation. Memory grows by exactly a factor of two per qubit, from 1.5×10^{-5} GB at $n = 10$ to 6.25×10^{-2} GB at $n = 22$. Time grows faster than memory over this range for architectural reasons, not mathematical ones; on a machine with a larger cache the jump shifts to a higher n, but the slope stays.

Figure 32.2 is the quantitative shape of the privacy-utility trade-off: the curve is not

linear, and the most expensive part is at the left end, in the region that is most often praised in abstracts. Figure 32.3 is the geometric shape of the sentence “the modulus determines the probability, the phase determines the interference”.

Note for instructors

The teaching order I recommend: start from Figure 32.2, not from the definition. Show that the curve has an expensive left end, then ask why. Students usually guess “because the noise is large”, which is true but not yet enough; what has to come out is that the privacy budget is summed along the steps, so shrinking ε per step is paid twice, in noise and in the number of steps.
Two misconceptions almost always appear. First, students read “(ε, δ)-DP” as “this model is safe”. What is true is that it is a promise about the influence of one record on the output distribution, and the model can still be completely wrong. Second, students think the quantum phase is a decoration of notation because “probability depends only on the modulus”. The example $HZH|0\rangle$ breaks that in three lines.
A blackboard activity that worked in our class: ask two students to compute $HZH|0\rangle$ and $HHH|0\rangle$ separately, then compare. Both use three gates, they differ in only one matrix in the middle, and the results are $|1\rangle$ against $|0\rangle$. The class is ready to go on when a student can explain why $HSH|0\rangle$ gives the same probabilities as $H|0\rangle$ but a different state.

32.13 Summary

Summary.

- A model trained on personal data leaks information through three channels: membership, attributes, and reconstruction, and the third works from gradients.
- (ε, δ) differential privacy compares two output distributions for data that differ in one person, with the quantifier “for every output set”. That quantifier is what makes the promise hold against an attacker, not merely against an average.
- The Laplace mechanism with scale $b = \Delta f/\varepsilon$ satisfies $(\varepsilon, 0)$; its proof only compares the ratio of two densities. The Gaussian mechanism uses $\sigma \geq \sqrt{2\ln(1.25/\delta)}/\varepsilon$ and gives $\delta > 0$.
- Basic composition adds up ε; advanced composition gives the better $\sqrt{T}$ term only when ε per step is small: at $\sigma = 32$ the results are 36.5 versus 20.6, whereas at $\sigma = 2$ they are 583.7 versus 11 302.8.
- DP-SGD clips per-example gradients at norm C and then adds noise $\sigma C/B$. Accuracy falls from 95.20 percent without noise to 93.71 percent at $\varepsilon = 36.5$ and 70.39 percent at $\varepsilon = 2.3$.
- Differential privacy guarantees something about individuals, not the truth of the model; it does not remove bias, and its utility cost can be uneven across groups.
- A quantum state lives in $\mathbb{C}^{2^n}$; a gate must be unitary because probability must be conserved, and measurement is the only non-unitary step. The output is therefore always a sample.
- Classical simulation needs 2^n amplitudes: $n = 50$ means 16 pebibytes. That fact limits imitation, but it does not prove advantage, and so far there is no established evidence of practical advantage for QML.

32.14 Exercises

Level: intermediate

Exercise 32.1.

1. Assess this claim: “Our model satisfies differential privacy with $\varepsilon = 0.5$, so no personal information can leak from it.” State true, false, or not enough evidence, and name two things that must hold for the claim to apply.
2. Prove that the Laplace mechanism with scale $b = \Delta f/\varepsilon$ satisfies $(\varepsilon, 0)$ when sensitivity is measured with the ℓ_1 norm on an output of dimension $d > 1$. Show where the one-dimensional argument has to be replaced.
3. Prove parallel composition: if M_1 works on D_1 and M_2 on D_2 with $D_1 \cap D_2 = \varnothing$, and each is $(\varepsilon_i, \delta_i)$, then their output satisfies $(\max_i \varepsilon_i, \max_i \delta_i)$.
4. Compute ε for randomized response when the probability of answering honestly is p and the rest is answered at random with a fair coin. State ε as a function of p, then determine the p that gives $\varepsilon = 2$.
5. For a count query with $\varepsilon = 0.5$ on $n = 10\,000$, compute the probability that its absolute error exceeds 10 people, then compare it with the same probability when the query is an average with $R = 1$. Check both numbers by simulation.
6. Compute $H\,S\,H|0\rangle$ with $S = \mathrm{diag}(1, i)$ step by step, then compare its probabilities with those of $H|0\rangle$. Explain why the measurement results are the same although the states differ, and which gate distinguishes them.
7. For the Bell state $|\Phi^+\rangle$, compute the probability of the outcome 01 when the second qubit is measured first and the result is 0; repeat for the result 1. Explain why both answers show perfect correlation.
8. Show that $\begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix}$ is unitary for every φ, then explain why that gate does not change probabilities in the computational basis but can change the result of the next gate.
9. Compute the number of amplitudes and the memory requirement for $n = 40$ and $n = 60$ qubits with 16 bytes per amplitude. State them in sensible units, then compare them with the storage capacity that you have.
10. A paper writes: “Our circuit prepares a state with 2^{60} amplitudes, so it beats classical simulation for machine learning.” Name three questions that must be answered before that sentence is accepted, and for each question state what kind of result would make you accept it.

Hints for the three hardest items. Item 2: write the ratio of densities as a product of per-coordinate ratios, then use $\sum_j |a_j - a'_j| \le \Delta f$. Item 3: at every step only one of the two mechanisms can change. Item 10: whether all the amplitudes are used, whether the output can be extracted with a number of measurements that is not exponential, and whether there is a classical comparison on the same task and data.

For researchers: limits and next steps

Three things in the first half are often quoted without their conditions. First, the (ε, δ) definition here is the definition of Dwork, McSherry, Nissim, and Smith (2006) in the form of the book by Dwork and Roth; the condition $\delta \ll 1/n$ comes from the camouflage argument for one individual, not from analytical convenience. Second, the Gaussian bound $\sigma \ge \sqrt{2\ln(1.25/\delta)}/\varepsilon$ is loose for small σ; the tight bound is given by the analytic Gaussian mechanism of Balle and Wang (2018), and the difference can reach a factor of two at $\varepsilon < 1$. Third, the moments *accountant* of Abadi and colleagues (2016) uses subsampling, so an ε that is reported without the subsampling rate and the number of steps cannot be compared with other numbers.

In the second half, what is really proved is a conditional statement: complexity separations for particular families of tasks, and the impossibility of classical simulation for general

states. What is popular but unproved is the claim that variational circuits give an advantage on classical data. Tang (2019) and Huang, Kueng, Preskill (2021) limit that claim from two directions: data structure can remove the speedup, and the separations that exist depend on distributional assumptions. McClean and colleagues (2018) show gradients that vanish exponentially in deep circuits, so variational training has an unresolved obstacle.

The most important question is not "is there a speedup", but "what kind of speedup survives once the cost of readout, state preparation, and error correction is counted". Until there is an answer measured on hardware with real data, every claim of practical advantage for QML should be read as a hypothesis.

Map to source texts: The privacy half follows Dwork and Roth, The Algorithmic Foundations of Differential Privacy *(chapters 2–3 and appendix A), Abadi et al.,* Deep Learning with Differential Privacy *(2016) for DP-SGD, Shokri et al. (2017) for membership inference, Dinur and Nissim (2003) for reconstruction from aggregate queries, and Zhu, Liu, Han (2019) for reconstruction from gradients. The quantum half follows Nielsen and Chuang,* Quantum Computation and Quantum Information *(chapters 2 and 4), Cerezo et al. (2021, 2022) for the limits of variational algorithms, Tang (2019) for dequantization, and Huang, Kueng, Preskill (2021) for data-based separation. The prerequisites are in Chapter 10 (SVD and orthogonality), Chapter 16 (conditional probability), and Chapter 17 (random variables), whereas the connection to Fourier and convolution is in Chapter 14 and Chapter 15.*

Appendix A

Table of Notation

The following notation is used throughout the book. When a chapter uses a symbol that is not listed here, that symbol is introduced where it is used. The number at the end of each entry refers to the chapter that uses it.

Sets, logic, and numbers (Chapters 1–5, 22)

$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$
: the natural, integer, rational, real, and complex numbers.

$\forall$, $\exists$, $\neg$
: for all, there exists, and not.

$\Rightarrow$, $\Leftrightarrow$
: implies, and if and only if.

$\in$, $\notin$, $\subseteq$, $\cup$, $\cap$, $\setminus$
: membership, not an element of, subset, union, intersection, difference (Chapter 2).

$\mathcal{P}(S)$, $|S|$, $S \times T$
: power set, cardinality, Cartesian product (Chapter 2).

$\sup S$, $\inf S$
: least upper bound and greatest lower bound (Chapter 3).

$a_n \to a$
: the sequence (a_n) converges to a (Chapter 4).

$\sum_{n=1}^{\infty} a_n$
: series and its sum (Chapter 5).

$\mathrm{fl}(x)$, ulp
: the rounded value of a real number and the smallest unit that can be represented (Chapter 22).

Linear algebra (Chapters 6–10)

$\mathbf{x}$, $\mathbf{x}^\top$, $\|\mathbf{x}\|$
: column vector, transpose, and Euclidean norm.

$\operatorname{span} S$, $\dim V$
: span of a set of vectors and dimension of a vector space.

$\mathcal{L}(V, W)$
: the space of all linear transformations from V to W (Chapter 7).

$\ker T$, $\operatorname{im} T$
: kernel and image of a transformation (Chapter 7).

$\det A$, $\operatorname{tr} A$, A^{-1}
: determinant, trace, and inverse of a matrix (Chapter 8).

λ, $\mathbf{v}$ eigenvalue and eigenvector (Chapter 9).

σ_i, $A = U\Sigma V^{\top}$
: singular values and the singular value decomposition (Chapter 10).

A^{+}, $\kappa(A)$
: pseudoinverse and condition number (Chapters 10, 20).

Analysis and calculus (Chapters 11–16)

ε, δ small positive numbers in the definition of limit, continuity, and uniform continuity.

$f'(x)$, ∇f
: ordinary derivative and gradient.

J_f, H_f
: Jacobian and Hessian matrices (Chapter 12).

$\int_a^b f$ Riemann integral on $[a, b]$ (Chapter 13).

$\sum_{\mathcal{P}} M_i \Delta x_i$
: upper sum for the partition $\mathcal{P}$ (Chapter 13).

$f_n \rightrightarrows f$
: sequence of functions converging uniformly (Chapter 14).

$\dfrac{\partial(x, y)}{\partial(u, v)}$
: Jacobian determinant of a change of variables (Chapter 15).

$x' = f(t, x)$, $\dot{x} = F(x)$
: ordinary differential equation and autonomous system, where $'$ and $\dot{}$ both denote the derivative with respect to time (Chapter 16).

$\varphi_t(x_0)$
: flow: the state at time t when starting from x_0 (Chapter 16).

x^*, $J_F(x^*)$
: equilibrium point and the Jacobian matrix at that point; the sign of the real part of its eigenvalues determines stability (Chapter 16).

Probability and statistics (Chapters 17–21, 26)

$P(A)$, $P(A \mid B)$
: probability of the event A and conditional probability.

$\mathbb{E}[X]$, $\operatorname{Var} X$
: expectation and variance.

$\mathrm{Cov}(X, Y)$
: covariance of two random variables (Chapter 18).

$\bar{X}_n$, se
: sample mean and its standard error.

$\mathrm{CI}_{1-\alpha}$
: confidence interval with level $1 - \alpha$ (Chapter 20).

TP, FP, FN, TN
: the four cells of a confusion matrix (Chapter 27).

$\hat{\beta}$, $\hat{y} = X\hat{\beta}$
: least squares regression coefficients and predicted values (Chapter 21).

$e_i = y_i - \hat{y}_i$, R^2
: residual and the proportion of variance explained by the model (Chapter 21).

$\mathrm{se}(\hat{\beta}_j)$, VIF_j
: standard error of a coefficient and variance inflation factor (Chapter 21).

Optimization, machine learning, and verification (Chapters 23–30)

$\min f_0(x)$ **with** $f_i(x) \le 0$
: standard form of an optimization problem (Chapter 23).

$L(x, \lambda, \nu)$
: Lagrangian; λ, ν multipliers (Chapter 23).

$c^\top x$, $Ax \le b$, $x \in \mathbb{Z}^n$
: LP form and integer program (Chapter 24).

$\sigma(\cdot)$, softmax
: activation function and exponential normalization (Chapter 25).

$\nabla_{\mathbf{x}} \ell$
: gradient of the loss function with respect to the input; the fuel for attacks (Chapters 25, 29).

L_2, L_1, $\lambda_{\mathbf{reg}}$
: kinds of regularization and their strength (Chapter 26).

$\|\delta\|_\infty \le \varepsilon$
: perturbation bound in adversarial attacks (Chapters 29, 30).

$\vdash$, `by`, `sorry`
: provability, the tactic block opener, and deferred proof in Lean 4 (Chapter 28).

Appendix B

Glossary

The terms below are used with fixed technical meanings throughout the book. Indonesian equivalents are given in parentheses, because the Indonesian edition of this book uses those terms and much of the local literature follows it.

Absolute convergence
a series that still converges when all its terms are forced positive (*konvergensi mutlak*).

Absolute value
the distance of a number from zero on the number line, regardless of its sign (*nilai mutlak*).

Accumulation point
a point every neighborhood of which contains a member of the set other than itself (*titik akumulasi*).

Activation function
a nonlinear mapping on the output of a unit; it decides how much signal is passed on and the shape of its derivative (*fungsi aktivasi*).

Adaptive attack
an attack designed with knowledge of the defense being tested, including knowledge that its gradient is manipulated (*serangan adaptif*).

Adjoint
in the backward pass, the number that says how strongly the final loss is affected by one quantity in the middle of the network; one adjoint per quantity (*adjoint*).

Adjugate
the cofactor matrix arranged in transpose; used to write the inverse as a single formula, $A^{-1} = \operatorname{adj}(A)/\det A$ (*adjoin*).

Adversarial example
a small disturbance to the input, deliberately made to change the output of a model (*serangan adversarial*).

Adversarial learning
the study of examples deliberately designed to fool a model, and of how to defend against them (*pembelajaran adversarial*).

Antiderivative
a function whose derivative is the function being integrated; the way home from a derivative (*antiturunan*).

Asymptotically stable
stays close after being disturbed, and then actually returns to the equilibrium point (*stabil asimtotik*).

Axiom
a statement accepted without proof as a starting point; the whole content of a chapter rests on it (*aksioma*).

Backpropagation
an algorithm that computes the gradient of every parameter by applying the chain rule from the output back to the input (*turunan balik*).

Backward gradient
see backpropagation (*gradien terbalik*).

Backward pass differentiable approximation
a stand-in for the gradient on a piece that is not differentiable, using the gradient of the function at the point just after it, so that an attack can still be computed (*BPDA*).

Barren plateau
a region of the loss landscape of a variational circuit where the gradient shrinks exponentially with the number of qubits (*plateau tandus*).

Basic feasible solution
a point obtained by setting some variables to zero and then solving for the rest, and still satisfying every constraint (*solusi layak basis*).

Basis
a set of vectors that is linearly independent and spans the whole space (*basis*).

Bias-variance tradeoff
the split of model error into error from being too simple and error from being too sensitive to the data (*bias-variansi*).

Bijection
a perfect pairing of two sets: injective and surjective at once, with nothing left over on either side (*bijeksi*).

Bootstrap
estimating the distribution of a statistic by resampling from the data itself, many times, with replacement (*bootstrap*).

Born rule
the rule that turns an amplitude into a probability by squaring its modulus (*aturan Born*).

Calibration
the agreement between the numbers a model promises and the actual frequency in the group promised the same (*kalibrasi*).

Catastrophic cancellation
the loss of significant digits when two nearly equal numbers are subtracted, so that only error is left (*pembatalan katastrofik*).

Cauchy sequence
a sequence whose terms crowd closer and closer together, without needing to name the value they approach (*barisan Cauchy*).

Central limit theorem
the average of many independent random variables approaches the normal distribution once it is standardized (*Teorema Limit Pusat*).

Chain rule
the rate of change of a composed function equals the product of the rates at each link in the chain (*aturan rantai*).

Characteristic polynomial
$\det(A - \lambda I)$; its roots are the eigenvalues of A (*polinomial karakteristik*).

Coefficient of determination
the share of variation explained by the model, stated as a fraction of the whole; written R^2 (*koefisien determinasi*).

Combination
a choice of several members without regard to their order; there are $\binom{n}{k}$ of them (*kombinasi*).

Compact
a closed and bounded set; every sequence in it has a subsequence that converges inside that set (*kompak*).

Completeness axiom
the promise that every nonempty set bounded above in $\mathbb{R}$ has a supremum; this is what leaves the number line with no holes (*aksioma kelengkapan*).

Condition number
a measure of how much the relative error of the output can grow compared with the relative error of the input (*bilangan kondisi*).

Conditional probability
the probability of an event given that another event is known to have occurred; written $P(A \mid B)$ (*peluang bersyarat*).

Confidence interval
a range of values that, if the sampling procedure were repeated, contains the true parameter with a given probability (*interval kepercayaan*).

Confusion matrix
a four-cell table that compares predictions with reality (*matriks konfusi*).

Continuous
a function whose graph can be drawn without lifting the pen at that point, that is, $\lim_{x \to c} f(x) = f(c)$ (*kontinu*).

Convergence in distribution
its distribution function tends to the limit distribution function at every point of continuity; what comes to resemble is the shape of the distribution, not its value (*konvergensi dalam distribusi*).

Convergence in probability
$P(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$ (*konvergensi dalam peluang*).

Convex combination
a weighted mixture of vectors with nonnegative weights that sum to one (*kombinasi konveks*).

Convex function
a function whose graph always lies below the chord joining two of its points; every local minimum of it is a global minimum (*fungsi konveks*).

Coordinate vector
the list of coefficients that expresses a vector in a basis (*koordinat*).

Correlation
a measure of the strength of the linear relation between two variables; it does not state cause and effect (*korelasi*).

Countable
a set whose members can be numbered with nothing skipped; $\mathbb{Q}$ is countable, $\mathbb{R}$ is not (*terhitung*).

Countable set
a set whose members can be listed as first, second, third, and so on with nothing left behind (*himpunan terhitung*).

Counterexample
one example that breaks a statement of the form $\ddot{\text{f}}$or all$\ddot{(}$ (*lawan contoh*).

Covariance
a measure of two quantities moving together, computed from the product of their deviations (*kovarians*).

Covariance
a measure of how far two variables move together; its unit is the product of their units (*kovariansi*).

Cross-entropy
a measure of surprise $-\sum_i y_i \log \hat{p}_i$; the more confident the model is on a wrong answer, the larger the penalty (*entropi silang*).

Cross-validation
assessing a model by dividing the data in turn into training and test parts (*validasi silang*).

Cumulative distribution function
the function $F(x) = P(X \le x)$; a rising staircase whose height is the probability already accumulated up to x (*fungsi distribusi kumulatif*).

Data leakage
information from the test data that leaks into the training process without anyone intending it, so that test results look better than reality (*kebocoran data*).

Data split
the division of data into training, validation, and test sets that do not leak into one another (*pemisahan data*).

Defective
a matrix whose eigenvalues repeat but whose eigenspace is too narrow to be a basis (*defective*).

Dense set
a set every point of which can be approached by members of that set other than the point itself (*himpunan padat*).

Derivative
the limit of the difference quotient; the slope of the tangent line and at the same time the instantaneous rate of change (*turunan*).

Determinant
one number that measures the factor by which volume changes and at the same time tests invertibility (*determinan*).

Diagonalization
a change of basis until the matrix becomes diagonal, so that the work is reduced to multiplying numbers (*diagonalisasi*).

Differential equation
a sentence that ties the rate of change of a quantity to the size of that quantity itself (*persamaan diferensial*).

Differential privacy
the promise that one person's participation in the data changes the output distribution only slightly, measured by ε (*privasi diferensial*).

Digital signature
a value that only the holder of the private key can create but anyone can check with the public key (*tanda tangan digital*).

Dimension
the number of vectors in a basis; the number belongs to the space, not to one basis that happens to be chosen (*dimensi*).

Direct sum
the sum of two subspaces that meet only at the zero vector, so that every vector has one way of being written (*jumlah langsung*).

Double integral
the sum of values over small cells of a region, then its limit as the cells shrink (*integral lipat dua*).

Duality
formulating an optimization problem from the other side, whose optimal value gives a lower bound for the original problem (*dualitas*).

Eigenspace
all the eigenvectors for one eigenvalue together with the zero vector; it is a subspace (*ruang eigen*).

Eigenvalue
a scalar λ such that $A\mathbf{v} = \lambda\mathbf{v}$ for some nonzero vector (*nilai eigen*).

Elementary matrix
an identity matrix that has had one row operation applied to it, so that it becomes a record of that operation (*matriks elementer*).

Elimination
replacing a system by a simpler system without changing its solution set (*eliminasi*).

Epsilon-delta definition
the style of defining a limit by naming the output tolerance first, then the input bound that guarantees it (*epsilon-delta*).

Equilibrium point
a state of a dynamical system that does not change when it is not disturbed (*titik setimbang*).

Estimator
a rule that turns a sample into a guess at a parameter; the guess it produces is called the estimate (*penaksir*).

Euler method
a numerical solution of a differential equation that steps along the tangent line; its error is first order in the step (*metode Euler*).

Euler totient function
the function $\varphi(n)$: the count of numbers between 1 and n that are coprime with n (*fungsi phi*).

Expectation
the expected value $\mathbb{E}[X]$: the probability-weighted average, the balance point of a distribution (*harapan*).

Expectation over transformation
an attack that maximizes the expected loss over the randomness of the defense, usually by averaging gradients over several random copies (EOT).

Exponential distribution
the distribution of waiting time in a process without aging, with density $\lambda e^{-\lambda t}$ (*distribusi eksponensial*).

Fast gradient sign method
a one-step attack (FGSM) that shifts every coordinate by the budget, in the direction of the sign of the loss derivative (*metode tanda gradien*).

Field
a set with two operations that is closed, has identities and inverses, and satisfies the distributive law (*lapangan*).

Floating-point system
binary scientific notation with a fixed number of digits; a number is stored as sign, exponent, and mantissa, so that the spacing between representable numbers is not uniform: dense near zero, sparse at the top end, and two different numbers can fall into the same place (*sistem floating-point*).

Formal verification
a proof that a program or formula meets its specification, checked by a machine (*verifikasi formal*).

Functional limit
$f(x)$ can be forced as close as we like to L provided x is close enough to c but not equal to c (*limit fungsi*).

General solution
the family of all solutions of a differential equation, depending on arbitrary constants (*solusi umum*).

Generalization
the ability to do problems never seen before, not problems already memorized (*generalisasi*).

Global sensitivity
the largest shift in the answer to a query that one person can cause in the worst case (*global sensitivity*).

Gradient
the vector of partial derivatives; the direction of steepest ascent of a function (*gradien*).

Gradient descent
the parameter update rule that moves against the direction of the gradient (*PGC, penurunan gradien*).

Harmonic series
the series $1 + 1/2 + 1/3 + \cdots$; its terms go to zero but its sum diverges (*deret harmonik*).

Heavy tail
a distribution whose probability of enormous values shrinks too slowly, so that one observation can outweigh all the rest (*ekor berat*).

Hessian
the matrix of second partial derivatives; the concavity of a function is decided by its sign (*Hessian*).

Hidden layer
a layer of units that touches neither the input nor the output of the network directly (*lapisan tersembunyi*).

Hypothesis test
a procedure for deciding, with a stated risk of error, whether the data are enough to reject an initial claim (*uji hipotesis*).

Implication
a statement of the form if P then Q; it is false only when P is true and Q is false (*implikasi*).

Independent
two events that give each other no news, $P(A \cap B) = P(A)P(B)$; knowing one does not change the probability of the other (*independen*).

Independent and identically distributed
samples that are mutually independent and share one distribution; the most often used assumption, and the most often violated (*i.i.d.*).

Induction
the domino pattern of proof: the base case holds, then each case knocks down the next (*induksi*).

Inner product
a mapping of two vectors to one scalar that measures length and angle; in $\mathbb{R}^n$ its value is $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$ (*hasilkali dalam*).

Integrality gap
the distance between the best value of the LP relaxation and the best value of the integer answer; it measures how far a continuous model misleads (*celah integralitas*).

Integrating factor
a multiplier that turns the left side of a linear differential equation into a single derivative of a product (*faktor integrasi*).

Interpolation
finding a function that passes through every data point; different from regression, which does not demand that (*interpolasi*).

Isomorphism
a bijective mapping that preserves structure; two isomorphic spaces cannot be told apart algebraically (*isomorfisme*).

Jacobian
the matrix of first partial derivatives of a vector mapping; its determinant measures the local volume scale factor (*Jacobian*).

Jacobian determinant
the local volume scale factor when coordinates are changed; its absolute value multiplies area or volume in a multiple integral (*determinan Jacobian*).

Kamus (glossary box)
a small box inside a chapter that explains one new term together with its analogy (*kamus*).

Karush-Kuhn-Tucker conditions
necessary conditions for optimality in a problem with inequality constraints (*kondisi KKT*).

Kernel, null space
the set of all vectors mapped to zero by a linear transformation (*inti*).

Kernel, null space
the set of all inputs mapped to zero by a linear transformation; the same as the null space (*kernel*).

Lagrange multiplier
an auxiliary variable that puts a price on a constraint; its value shows how much the objective could improve if the constraint were loosened (*pengali Lagrange*).

Law of large numbers
the promise that the average of many observations sticks to the true mean, with the chance of missing it shrinking (*hukum bilangan besar*).

Least squares
the choice of parameters that minimizes the sum of squared residuals when there is no exact solution (*kuadrat terkecil*).

Linear map
a mapping that preserves addition and scalar multiplication (*transformasi linear*).

Linear program
the problem of maximizing or minimizing a linear function with linear constraints (*program linear*).

Lipschitz condition
the condition that the rate of change of a function is bounded by one constant; this is what guarantees a unique solution of a differential equation (*Lipschitz*).

Logical connective
an operation that combines statements: and, or, not, if-then, if-and-only-if (*penghubung logika*).

Logistic regression
a model for binary output that maps a linear combination to a probability through the logistic function (*regresi logistik*).

Loss function
a number we want to make small while the model is trained; it measures how wrong a prediction is, not how right (*fungsi kerugian*).

LP relaxation
replacing the integer condition by a real condition so that an integer problem can be solved as an LP (*relaksasi LP*).

Lyapunov function
a positive-valued function that decreases along the trajectories of a system; its existence proves stability without solving the equations (*fungsi Lyapunov*).

Machine epsilon
the distance between 1 and the next machine number; the size of one step at scale one (*epsilon mesin*).

Modular inverse
a number b such that $ab \equiv 1 \pmod{n}$; the undo button of clock arithmetic (*invers modular*).

Monotone sequence
a sequence that only rises or only falls, never turning back (*barisan monoton*).

Multicollinearity
the situation in which explanatory variables are nearly linearly dependent on one another, so that regression coefficients become unstable (*multikolinearitas*).

Multiplicity
the number of times an eigenvalue occurs as a root of the characteristic polynomial, which is not always equal to the dimension of its eigenspace (*multiplisitas*).

Necessary condition
a condition that must hold for a statement to be true; without it the statement is impossible, but meeting it is not yet enough (*syarat perlu*).

Nested cross-validation
two layers of folds: the outer layer assesses, the inner layer selects, so that model selection does not leak test data (*validasi silang bersarang*).

Network intrusion detection system
a system that watches network traffic and decides whether a flow of packets is an attack (*NIDS*).

Norm
a measure of the length of a vector; the L_∞ norm takes the largest component (*norma*).

Normal distribution
the symmetric bell distribution determined entirely by its mean and its variance (*distribusi normal*).

Null hypothesis
the initial claim under test, usually "there is no effect"; it is rejected only when the data fit it very badly (*hipotesis nol*).

Obfuscated gradient
a gradient handed to the attacker that does not point in the right direction, usually because a piece is not differentiable or is random (*gradien dikaburkan*).

Odds ratio
the ratio of the probability of success to the probability of failure; the exponential of a logistic regression coefficient states how many times that ratio multiplies per unit (*rasio peluang*).

Ordered field
a field whose members can be ordered and whose arithmetic rules respect that order (*lapangan terurut*).

Ordinary differential equation
an equation that relates a function to its derivative with respect to one independent variable (*persamaan diferensial biasa*).

Orientation
an agreement about which direction of rotation counts as positive; reversing it reverses the sign of an integral (*orientasi*).

Orthogonal
mutually perpendicular in the sense that their inner product is zero (*ortogonal*).

Orthogonal matrix
a square matrix whose columns are orthonormal; it preserves length and angle (*matriks ortogonal*).

Orthonormal
an orthogonal set each of whose members has length one (*ortonormal*).

Out-of-sample
data not used when the model was trained, so that they are honest for judging it (*luar sampel*).

Overfitting
memorizing the training data including its noise, so that performance on new data falls (*overfitting*).

Padding
extra content added to a packet before it is sent so that every packet has a fixed size, so that packet size no longer leaks its content (*bantalan*).

Partition
a division of a set into pieces that do not overlap and that cover the whole set (*partisi*).

Perceptron
the simplest linear classifier: the sign of the inner product of the weights and the input (*perceptron*).

Poles and zeros
values of a complex variable at which a function blows up or vanishes; used in the analysis of dynamical systems and control (*kutub dan nol*).

Polynomial time
the class of algorithms whose number of steps grows no faster than a polynomial in the size of the input (*waktu polinomial*).

Power series
a series of the form $\sum a_n(x-c)^n$; it holds only inside its radius of convergence (*deret pangkat*).

Power set
the set of all subsets, including the empty set and the set itself; its cardinality is 2^n (*himpunan kuasa*).

Prediction interval
an interval for one observation that has not yet occurred; always wider than the interval for a mean (*selang prediksi*).

Primitive root
a member that, raised repeatedly to powers, passes through every other member of the cyclic group before returning to one; in cryptography, a starting position that reaches every position of the key (*akar primitif*).

Privacy budget
the amount ε a system may spend; every query or training step draws down what is left (*anggaran privasi*).

Probability
a weight on every possibility, nonnegative and summing to one (*peluang*).

Probability amplitude
a complex number whose squared modulus gives a probability; different from a probability because two amplitudes can cancel each other (*amplitudo*).

Problem space
the space of real objects an attacker may change, for example packets and timing; different from the feature space in which the model works (*ruang masalah*).

Projection
the point in a subspace closest to a vector, that is, its perpendicular image (*proyeksi*).

Proof term
an expression that is at once a proof and a value; checking a proof is the same as checking a type (*term bukti*).

Proposition
a sentence that is either true or false, not a question or a command; in this book also the name of a theorem environment (*proposisi*).

Pseudoinverse
a generalized inverse A^{+} that still works when some directions have been flattened: what can be inverted is inverted, what is lost is left alone (*pseudoinvers*).

P-value
the probability of obtaining a result at least as extreme as the one observed if the null hypothesis is true; not the probability that the null hypothesis is true (*nilai-p*).

Quantifier
a marker for all ($\forall$) or there exists ($\exists$); the order in which they are used changes the meaning of a statement (*kuantor*).

Quantization
replacing every value by the nearest multiple of a fixed step; the result is a staircase function whose derivative is zero almost everywhere (*kuantisasi*).

Quantum computing era
the era when quantum computers are large enough to run Shor's algorithm on the keys used today (*zaman kuantum*).

Radius of convergence
the greatest distance from the center at which a power series still converges (*jari-jari konvergensi*).

Random variable
a function that maps the outcome of an experiment to a number (*variabel acak*).

Rank
the number of independent directions that survive after mapping, that is, $\dim \operatorname{im} T$ (*rank*).

Rank-nullity theorem
$\dim \ker T + \dim \operatorname{im} T = \dim V$ for a linear transformation on a finite-dimensional space (*rank-nullity*).

Ratio test
a series converges absolutely if $\lim |a_{n+1}/a_n| < 1$, and diverges if that limit is greater than one (*uji rasio*).

Rearrangement
swapping the order of the terms of a series; in a series that converges absolutely the sum does not change, in one that does not, the sum can change (*susunan ulang*).

Regression
drawing the function closest to all the points, not the function that passes through all of them (*regresi*).

Regression line
the line (or plane) that minimizes the sum of squared residuals; equivalent to the orthogonal projection onto the column space (*garis regresi*).

Regularization
a penalty on the size of the parameters so that the model is not too sensitive to the training data (*regularisasi*).

Residual
the difference between an observed value and the value predicted by the model (*residu*).

Residue class
the set of numbers that leave the same remainder when divided by n, like three in the morning and three in the afternoon (*kelas sisa*).

Riemann integrability
the property of a function whose upper and lower sums can be squeezed until their difference vanishes (*keterintegralan*).

Riemann integral
the limit of the sum of rectangle areas as the width of the partition shrinks (*integral Riemann*).

ROC curve
a plot of the true rate against the false positive rate over all thresholds; the area under it is called AUROC (*kurva ROC*).

Rounding
replacing a real number by the nearest machine number (*pembulatan*).

Rounding error
the difference between a real number and its machine representative, which piles up in long sums (*kesalahan pembulatan*).

Rule of three
if no event occurs in n trials, the event rate may still be as high as about $3/n$ at 95 percent confidence (*aturan tiga*).

Runge-Kutta
a family of numerical methods that evaluate the derivative several times per step for a smaller error (*Runge-Kutta*).

Sample complexity
the number of examples needed for a model to generalize with a given confidence (*kompleksitas sampel*).

Sensitivity
in differential privacy, the largest change in output that one person can cause (*sensitivitas*).

Sequence
a row of numbers given a running order $a_1, a_2, \ldots$; the order is part of the definition, not something added later (*barisan*).

Series
the sum of the terms of a sequence without stopping, whose value is read as the limit of the partial sums (*deret*).

Similar
two matrices that represent the same linear mapping in two different bases, $B = P^{-1}AP$ (*serupa*).

Singular value
the square root of an eigenvalue of $A^{\top}A$; it measures the length of a principal axis of the mapping by the matrix A (*nilai singular*).

Singular value decomposition
writing a matrix as $U\Sigma V^{\top}$ with U, V orthogonal (*dekomposisi nilai singular, SVD*).

Slack variable
an extra variable that records the unused remainder, so that a no more than constraint becomes an equation (*variabel kendur*).

Softmax
a mapping of a row of scores into probabilities that sum to one, $e^{z_i} / \sum_j e^{z_j}$ (*softmax*).

Sorry
a Lean 4 marker that a proof has been postponed; the machine accepts it but records it as an axiom (*sorry*).

Stability
: the property of an equilibrium point or dynamical system that stays close to its state after a small disturbance (*kestabilan*).

Standard error
: the standard deviation of an estimator, not the standard deviation of the data; it measures the wobble of one guess (*galat baku*).

Standard form
: a uniform way of writing an optimization problem: one objective, constraints in the form no more than zero, and nonnegative variables (*bentuk baku*).

Stochastic gradient descent
: gradient descent that uses a small fraction of the data at every step; cheaper and noisier (*penurunan gradien stokastik*).

Strong induction
: induction that may use every earlier case, not only the nearest one (*induksi kuat*).

Subsequence
: a sequence obtained by discarding some terms without changing the order of the remaining terms (*subbarisan*).

Subspace
: a subset of a vector space that is closed under addition and scalar multiplication (*subruang*).

Summary box
: a box of bullet points at the end of a chapter that can be read on its own without the body of the chapter (*ringkasan*).

Supremum, least upper bound
: the least upper bound; the smallest number that is not smaller than every member of the set (*supremum*).

Tactic
: a step-by-step command in Lean 4 that changes the goal of a proof (*taktik*).

Taylor series
: writing a function as a sum of derivative terms at one point, together with a remainder that measures the truncation error (*deret Taylor*).

Threat model
: the complete list of an attacker's abilities: the size of the disturbance, what the attacker knows, and what the attacker wants (*model ancaman*).

Transferability
: the property of an adversarial example made for one model that also fools another model trained separately (*transferabilitas*).

Truncation error
: the difference that appears because an infinite process is replaced by a finite piece (*galat pemotongan*).

Type class
: a list of abilities a type must have in order to be usable in a given place (*typeclass*).

Uniform continuity
: continuity with one δ that works for every point of the set, not one δ per point (*kekontinuan seragam*).

Uniform convergence
: a sequence of functions that approaches its limit function at the same rate over the whole domain (*kekonvergenan seragam*).

Uniform convergence
: one rate for the whole domain, not one schedule per point (*konvergensi seragam*).

Uniqueness
: the guarantee that one initial state gives rise to only one solution trajectory (*ketunggalan*).

Unit in the last place
: the smallest step a machine can distinguish at the scale of one number (*ULP*).

Unitary
: a matrix that preserves the inner product, $U^*U = I$; the complex version of an orthogonal matrix (*uniter*).

Upper limit
: the largest value that a sequence still visits arbitrarily often near its end, written $\limsup_{n\to\infty} a_n$ (*limit superior*).

Upper sum
: a deliberately enlarged estimate of an integral, using the largest value on each piece (*jumlah atas*).

Variance
: a measure of spread around the mean; its square root is the standard deviation (*variansi*).

Variance inflation factor
: how much the variance of a coefficient is inflated by other explanatory variables carrying similar information; abbreviated VIF (*varians inflasi*).

Vector space
: a set with addition and scalar multiplication that is closed and satisfies eight axioms (*ruang vektor*).

Vertex
: a corner of the feasible region, that is, a point that is never the midpoint between two other feasible points (*verteks*).

Wilson interval
: a confidence interval for a proportion found by inverting a test, so that it stays sensible when the proportion approaches zero or one (*interval Wilson*).

Wilson interval
: a confidence interval for a proportion that stays sensible when the proportion approaches zero or one (*Wilson*).

Appendix C

Guidelines and Grading Criteria for Exercises

The answer key is in Appendix D, but its form is deliberately incomplete: final numbers and sketches of the steps, not full proofs. The reason is consistency. A final number without a trace of checking will teach the reader the very habit this book wants to remove, and a proof served up whole removes the only useful exercise, namely finding for yourself where an argument does not yet close. So use this key as a last comparison, not as a place to start. What is provided here is a way to check your own work before you open Appendix D.

Three kinds of exercises and how to check them

Type	What is assessed	How to check
Proof	every step must start from a definition or a result already proved; no leaps are allowed	swap drafts with a classmate; ask her to point to the sentence she cannot derive herself
Computation	the final number and the intermediate steps	redo it with Python; compute the relative error against the exact result; check whether the answer makes sense in order of magnitude
Evaluating a claim	the quality of the argument, not a binary true-false	first write down what must be true for the claim to hold; look for one example that breaks it; only then decide “true”, “false”, or “not enough evidence”

Verification tools per part

Chapter	Tool	Typical check
1–5	`python3`	table of values, factorization, partial sums; compare with an analytic estimate (e.g. $H_n \approx \ln n + \gamma$)
6–10	`numpy`, `scipy.linalg`	rank, null space (`null_space`), eigenvalues, SVD, least squares (`lstsq`); check $\|A\mathbf{v} - \lambda\mathbf{v}\|$
11–16	`scipy.integrate`, finite differences	quadrature vs antiderivative; rate of error at several h or several n
17–21	`numpy.random`, `scipy.stats`	Monte Carlo simulation with a fixed *seed*; compare the empirical probability with the exact probability and report its standard error
22	`struct`, `float.hex`	bitfields, ULP, machine epsilon; compare the naive algorithm with Kahan
23–24	`scipy.optimize`	`linprog`, `minimize`, `milp`; check the KKT conditions and strong duality numerically
25–26	`numpy`	gradient check by finite differences; training vs validation error curves at several regularization settings
27–30	`numpy`, `scikit-learn` when available	confusion matrix from stored predictions; Wilson interval for proportions; accuracy under attack for several ε and several *seed*

How to grade your own work

1. First write down what is given and what is to be proved. If neither is clear, the difficulty is not in the mathematics.
2. For computation exercises, guess the answer before computing. A wrong guess is still useful: it shows where intuition misses.
3. Compare the result by two different methods. One method can be right by accident; two methods that agree are rarely an accident.
4. If the results differ greatly, look for the mistake in a simple case first (one variable, one step, $n = 1$).
5. Write down the limits of validity of the answer. An answer that is correct for a special case but presented as a general answer is the mistake that most often slips through.

Appendix D

Answer Keys

Every number in this appendix was recomputed with Python before it was written down, and the checking script ships with the manuscript sources, so a reader can run it independently. For proof exercises only a sketch of two to five lines is given; that is enough to tell whether the direction of your work is right, but not enough to be copied as a proof. Items whose statement is itself flawed are marked plainly as ambiguous instead of being forced to have an answer.

Chapter 1: The Language of Mathematics and the Art of Proof

1.1 $\exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N :\ |a_n - L| \geq \varepsilon$. In words: there is one tolerance that is never attained, because however large N is chosen, there is always a later term that deviates by at least that ε.

1.2 The negation of "every prime > 2 is odd" is "some prime > 2 is even". That negation is *false*, so the original statement is true and there is no counterexample. For the second claim, the negation is "for every n, $n^2 + n + 41$ is prime"; the counterexample is $n = 40$ with $1681 = 41^2$.

1.3 $n^2 + n = n(n+1)$. Cases: $n = 2k$ gives $2k(2k+1)$, $n = 2k+1$ gives $(2k+1)(2k+2) = 2(2k+1)(k+1)$. Contraposition: if $n(n+1)$ is odd then n and $n+1$ are both odd, yet two consecutive numbers cannot both be odd.

1.4 Check: $n = 1$ gives 1, $n = 2$ gives 5, $n = 3$ gives 14, matching $n(n+1)(2n+1)/6$. Induction step: add $(k+1)^2$ to both sides, then $\frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$.

1.5 Strong induction on n. For $n = 1$: $k = 0$, $m = 1$. For even n write $n = 2j$ with $j < n$; the induction hypothesis gives $j = 2^k m$, so that $n = 2^{k+1} m$ with m still odd. The strong hypothesis is used exactly here: $j < n$, not $j = n - 1$.

1.6 (a) True, the ordinary induction principle. (b) False: take $P(n)$ to mean $n \leq 10^6$; the counterexample is $n = 10^6 + 1$. (c) False: $n = 40$ gives $1681 = 41^2$. (d) False: $n = 3$ is divisible by 3 but not by 6.

1.7 The script finds 87 primes out of 101 values for $n = 0$ through 100. It fails at 40, 41, 44, 49, 56, 65, 76, 81, 82, 84, 87, 89, 91, 96. Only 5 of the 14 have the form $p - 1$ or p: $40, 41$ via $p = 41$, 82 via 83, 89, 96 via 97.

1.8 $2^1 = 2 > 1$; $2^2 = 4 = 4$; $2^3 = 8 < 9$; $2^4 = 16 = 16$; $2^5 = 32 > 25$. Hence $n_0 = 5$. Induction: if $2^k > k^2$ and $k \geq 5$, then $2^{k+1} > 2k^2 \geq (k+1)^2$ because $k^2 - 2k - 1 > 0$.

1.9 The claim is false, because $n = 40$ gives $40^2 + 40 + 41 = 1\,681 = 41^2$, which is not prime. Forty data points only support it, they do not prove it: one counterexample is enough to refute "for every $n \geq 0$".

1.10 False: $n = 2$ gives $4 \mid 4$ but $4 \nmid 2$. The correct claim is: $4 \mid n^2$ implies $2 \mid n$. Proof by

contraposition: if $n = 2k + 1$ then $n^2 = 4k^2 + 4k + 1 \equiv 1 \pmod 4$.

Chapter 2: Sets, Relations, Functions, and Cardinality

2.1 $x \in A \cap (B \cup C) \iff x \in A \wedge (x \in B \vee x \in C) \iff (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C)$, using distributivity of $\wedge$ over $\vee$. Second: $x \in A \setminus (B \setminus C) \iff x \in A \wedge (\neg x \in B \vee x \in C) \iff x \in (A \setminus B) \cup (A \cap C)$.

2.2 False. The smallest counterexample: $A = \{1\}$ with $R = \varnothing$; the empty relation is vacuously symmetric and transitive, but $(1, 1) \notin R$. What is missing is reflexivity for elements that occur in no pair at all.

2.3 Take $S_1, S_2 \subseteq A$ with $f(S_1) = f(S_2)$. For $s \in S_1$ there is $t \in S_2$ with $f(s) = f(t)$, and injectivity gives $s = t \in S_2$. Without injectivity: f constant on $\{1, 2\}$ makes $f(\{1\}) = f(\{2\})$.

2.4 The direction ($\Leftarrow$) is trivial. For ($\Rightarrow$): if $a = b$, the equality $\{\{a\}\} = \{\{c\}, \{c, d\}\}$ forces $\{c\} = \{c, d\}$, so $c = d = a$. If $a \neq b$, then $\{a\}$ must equal $\{c\}$, since $\{c, d\}$ has size two; so $a = c$ and then $b = d$.

2.5 Define $g : f(A) \to A$ by choosing one preimage for each $y \in f(A)$; g is injective, so $|f(A)| \leq |A|$ by the previous lemma. If f is not injective, two elements of A give one value, so $|f(A)| < |A|$.

2.6 $\binom{5}{2} = 10$ partitions of type $2 + 3$ and 5 of type $1 + 4$, altogether $15 = 2^4 - 1 = S(5, 2)$. The script counts partitions directly and also gives 15; the total number of equivalence relations on 5 elements is 52.

2.7 $\pi(7, 5) = T_{12} + 5 = 78 + 5 = 83$. For 1000: $T_{44} = 990 \leq 1000 < T_{45} = 1035$, so $s = 44$, $n = 10$, $m = 34$; the check $\pi(34, 10) = 990 + 10 = 1000$.

2.8 Cumulative through $k = 9$ is 27; level $k = 10$ contains 1/9, 3/7, 7/3, 9/1 at positions 28–31, so 7/3 sits at position 30. Because $\varphi(11) = 10$, position 41 is the last term of level 11, namely 10/1.

2.9 Arrange $A_n = \{a_{n1}, a_{n2}, \dots\}$ and then run the Cantor diagonal over the pairs (n, m): every member of $\bigcup A_n$ appears at least once, and discarding the repeated appearances does not increase the count.

2.10 $g(x) = \tan\bigl(\pi(x - \tfrac{1}{2})\bigr)$ is a bijection from $(0, 1)$ onto $\mathbb{R}$. Its domain must be restricted because tan is infinite at $\pm\tfrac{1}{2}\pi$; on $[0, 1]$ that formula is undefined at both ends.

Chapter 3: Axioms of the Real Numbers and Supremum

3.1 $a < b$ means $b - a > 0$. Add $-a - b$ to both sides (closure under addition): $(-a) - (-b) = b - a > 0$, that is $-b < -a$. The reverse direction is obtained by adding $a + b$. Hence multiplying by -1 reverses the direction.

3.2 Since $a < b$, we have $2a < a + b$ and $a + b < 2b$; multiply by $\tfrac{1}{2} > 0$ and we get $a < \tfrac{a+b}{2} < b$. Only the order axioms are used. Conversely $\sup\{q : q^2 < 2\}$ does not exist in $\mathbb{Q}$, so completeness is required.

3.3 $\sup S = 2$: 2 is an upper bound, and $2 - \varepsilon$ is not an upper bound because $2 - 1/n > 2 - \varepsilon$ for $n > 1/\varepsilon$. $\inf S = 1$ is attained at $n = 1$, so a minimum exists; no maximum exists because $2 \notin S$.

3.4 True. The premise is the Archimedean property: for every $x > 0$ there is n with $1/n < x$. Consequently, if $m > 0$ were the least, then $m/2 > 0$ is smaller than m, a contradiction.

3.5 $q = 1$: $h = 1/4$, so $q + h = 5/4$ with $(5/4)^2 = 1.5625 < 2$. $q = 7/5$: $h = 1/120$, so

$q+h = 169/120 \approx 1.4083$ with square $1.9834 < 2$. The second increment is 30 times smaller: the closer q is to $\sqrt{2}$, the shorter the correction.

3.6 $|x^3 - y^3| = |x-y|\,|x^2+xy+y^2| \le |x-y|(|x|^2+|x||y|+|y|^2) \le 12\,|x-y|$, because each term is at most 4. The triangle inequality is used in the second step to split the absolute value of a sum.

3.7 If S is open and x is a limit point of S^c with $x \in S$, then there is $B(x,r) \subseteq S$, so that $B(x,r) \cap S^c = \varnothing$: a contradiction. Conversely, if S is closed and $x \in S^c$, the absence of a ball inside S^c makes x a limit point of $S \subseteq S$, a contradiction.

3.8 $x_1 = 7/4$; $x_2 = 97/56 \approx 1.732143$; $x_3 = 18817/10864 \approx 1.732050810$; x_4 and x_5 agree to 16 digits, 1.732050807568877. Relative error after five steps: 1.0363×10^{-2}, 5.3145×10^{-5}, 1.4121×10^{-9}, 9.97×10^{-19}, 4.97×10^{-37}. Ratios e_{k+1}/e_k^2: 0.4330, 0.4949, 0.5000, tending to $1/2$.

3.9 0 lies in every $(-1/n, 1/n)$. For $x \neq 0$ choose $n > 1/|x|$, then $x \notin (-1/n, 1/n)$. A finite intersection of open sets stays open, whereas this infinite intersection equals $\{0\}$, which is not open.

3.10 By density of $\mathbb{Q}$, in the interval $(a/\sqrt{2},\, b/\sqrt{2})$ there is $q \in \mathbb{Q}$ with $q \neq 0$. Then $a < q\sqrt{2} < b$, and $q\sqrt{2}$ is irrational because if it were rational then $\sqrt{2}$ would be rational.

Chapter 4: Sequences and Limits

4.1 $|x_n - \tfrac{2}{3}| = \left|\tfrac{3(2n-1)-2(3n+4)}{3(3n+4)}\right| = \tfrac{11}{9n+12}$. The condition is $n > \tfrac{11/\varepsilon - 12}{9}$, so $N(\varepsilon) = \lfloor (11/\varepsilon - 12)/9 \rfloor$. For $\varepsilon = 10^{-3}$: $N = 1220$, while the smallest term that qualifies is $n = 1221$ with error 9.9991×10^{-4}.

4.2 $\big||x_n| - |L|\big| \le |x_n - L|$ (the reverse triangle inequality), so the same ε bound applies. The converse is false: $x_n = (-1)^n$ gives $|x_n| = 1$ convergent, but x_n diverges.

4.3 For x_n: even terms $2 + 1/n \to 2$, odd terms $1/n \to 0$; hence $\limsup = 2$, $\liminf = 0$ (the maximum 2.5 at $n = 2$ is not the limsup). For y_n: even $1 + 1/n \to 1$, odd $-(1 + 1/n) \to -1$; hence $\limsup = 1$, $\liminf = -1$. Numerically at $n = 1000$: 1,001 and $-1{,}001$.

4.4 Partly right, wrong conclusion. It is true that $(1 + 1/n)^n$ increases and is bounded by 3, so its limit exists and is at most 3. But an upper bound is not the value of the limit: the limit is $e \approx 2.7183$.

4.5 $3 = (3^n)^{1/n} \le (2^n + 3^n)^{1/n} \le 3 \cdot 2^{1/n}$, and $2^{1/n} \to 1$ because $(1+\delta)^n \ge 1 + n\delta$ (Bernoulli) gives $\delta \le 2^{1/n} - 1 \le 1/n$. Both fences go to 3.

4.6 The error is $< 10^{-6}$ first at $n = 12$ (2.6146×10^{-7}) and $< 10^{-12}$ at $n = 21$. The ratio of successive errors settles at 0.250000, consistent with $|x_{n+1} - 2| = |x_n - 2|/(\sqrt{2 + x_n} + 2) \to |x_n - 2|/4$.

4.7 A Cauchy sequence is bounded: take $\varepsilon = 1$ and get N; all terms after N lie in $(x_N - 1, x_N + 1)$, and the rest are finitely many. A bounded monotone sequence converges by the monotone convergence theorem, and a convergent sequence is Cauchy.

4.8 $\varepsilon = 10^{-3}$ gives the threshold $5/\varepsilon - 2 = 4998$, so $N = 4998$ is valid. Errors: $n = 4998$ gives exactly 10^{-3} (not yet qualifying), $n = 4999$ gives 9.9980×10^{-4}, $n = 5000$ gives 9.9960×10^{-4}. $N - 1 = 4997$ is not valid because $n = 4998 > 4997$ still has error exactly ε, not less.

4.9 Suppose $x_n \not\to L$; then there is $\varepsilon > 0$ and a subsequence (x_{n_k}) with $|x_{n_k} - L| \ge \varepsilon$. By Bolzano–Weierstrass that subsequence has a subsequence converging to L' with $|L' - L| \ge \varepsilon$, contradicting the hypothesis.

Chapter 5: Infinite Series

5.1 $s_{20} = 2.4999451576$, whereas the sum is $5/2 = 2.5$; the error is 5.4842×10^{-5}. The bound $\frac{r^{21}}{1-r}$ gives exactly the same figure, 5.4842×10^{-5}, because the tail is geometric with positive terms.

5.2 $\frac{1}{n(n+2)} = \frac{1}{2}(\frac{1}{n} - \frac{1}{n+2})$, so that $s_N = \frac{1}{2}(1 + \frac{1}{2} - \frac{1}{N+1} - \frac{1}{N+2})$; its limit is $\frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$.

5.3 Because $a_n \to 0$, for $n \geq N$ we have $0 \leq a_n \leq 1$, so $a_n^2 \leq a_n$; the comparison test closes it. The converse is false: $a_n = 1/n$ gives $\sum a_n^2$ convergent but $\sum a_n$ divergent.

5.4 False. The argument breaks at "smaller than a geometric term of ratio $\frac{1}{2}$ for n sufficiently large": $1/n < (1/2)^n$ in fact fails, because $(1/2)^n$ shrinks far faster. The comparison test needs an inequality on the whole tail. The harmonic series diverges.

5.5 (a) Convergent, ratio test: the ratio $\to \frac{1}{2} < 1$. (b) Divergent, ratio test: the ratio $2\big(\frac{n}{n+1}\big)^2 \to 2 > 1$. (c) Convergent, ratio test: $\big(\frac{n}{n+1}\big)^n \to \frac{1}{e} \approx 0.3679 < 1$. (d) Divergent, p-series with $p = \frac{1}{3} \leq 1$.

5.6 $s_{10^6} = 1.6449330668$ and $\pi^2/6 = 1.6449340668$; the difference is 9.9999950×10^{-7}, close to $\frac{1}{N}$. After the correction $\frac{1}{N} - \frac{1}{2N^2}$ the result is 1.6449340668 with error 2.2×10^{-16}. Correct decimal digits: 5 before, 15 after.

5.7 We need $\frac{1}{\sqrt{N+1}} < 10^{-3}$, that is $N + 1 > 10^6$, so $N = 10^6$. At that N the sum is 0.6043986435 while the limit is 0.6048986434; the actual error is 5.0000×10^{-4}, exactly half the bound. The bound is loose by about a factor of two.

5.8 $|b_n| \leq M$ gives $|a_n b_n| \leq M|a_n|$. Because $\sum M|a_n|$ converges, the comparison test concludes that $\sum |a_n b_n|$ converges.

5.9 After K blocks, the positive terms used run up to denominator $6K - 1$ and the negative terms up to $\frac{1}{2K}$, so $S_K = H_{6K} - \frac{1}{2}H_{3K} - \frac{1}{2}H_K \to \ln 6 - \frac{1}{2}\ln 3 = \ln(2\sqrt{3}) \approx 1.2425$. Numerically $K = 10^6$ gives 1.2424531. Its ratio to $\ln 2$ is 1.7925, so it is not a rational multiple of $\ln 2$.

5.10 Write $p_n = \max(a_n, 0)$ and $q_n = \max(-a_n, 0)$; then $a_n = p_n - q_n$ and $|a_n| = p_n + q_n$. If both of those series converged, $\sum |a_n|$ would converge, contradicting conditional convergence. Hence both diverge, and those two unbounded piles are the fuel of the Riemann rearrangement theorem.

Chapter 6: Vector Spaces and Subspaces

6.1 W is the solution set of the homogeneous system $x - 2y + z = 0$, $x + y = 0$, so it is closed under addition and scalar multiplication. From $x = -y$ and $z = 3y$ we get $W = \operatorname{span}\{(-1, 1, 3)\}$: a one-vector basis, $\dim W = 1$.

6.2 The set $D = \{A : a_{11} = a_{22} = 0\}$ is closed under addition and scalars, because both diagonal entries stay zero. Every one of its members is $\begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix} = bE_{12} + cE_{21}$, so $\{E_{12}, E_{21}\}$ is a basis and $\dim D = 2$.

6.3 $(1 + x) + (1 + x^2) = 2 + x + x^2$: the set is linearly dependent, hence $\dim \operatorname{span} S = 2$. The vector x is not in that span, because $a(1 + x) + b(1 + x^2)$ demands $a = 1$ and $a + b = 0$ at the same time. So $\{1 + x,\ 1 + x^2,\ x\}$ is a basis of P_2.

6.4 Collect the coefficients: $(c_1 + c_3)\mathbf{v}_1 + (c_1 + c_2)\mathbf{v}_2 + (c_2 + c_3)\mathbf{v}_3 = \mathbf{0}$. Independence of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ gives three homogeneous equations with determinant $2 \neq 0$, so $c_1 = c_2 = c_3 = 0$.

6.5 U is the xy-plane, W is the plane $x = y$. Their intersection is $U \cap W = \operatorname{span}\{(1, 1, 0)\}$, so $\dim(U \cap W) = 1$ and $\dim(U + W) = 2 + 2 - 1 = 3$. The formula fits: $U + W = \mathbb{R}^3$,

and numpy gives the rank of the four spanning vectors together as 3.

6.6 Elimination puts pivots in columns $1, 2, 4$, so the rank is 3; a basis of the column space is $(1, 2, 0)$, $(2, 4, 1)$, and $(4, 9, 1)$. Nullity $4 - 3 = 1$ with basis $\{(-1, -1, 1, 0)\}$; `matrix_rank` and `null_space` confirm both numbers.

6.7 False. In $\mathbb{R}^2$ take $U_1 = \operatorname{span}\{(1, 0)\}$, $U_2 = \operatorname{span}\{(0, 1)\}$, $U_3 = \operatorname{span}\{(1, 1)\}$. All pairwise intersections are $\{\mathbf{0}\}$, yet $(1, 1) = (1, 0) + (0, 1)$: the representation is not unique, and $\dim U_1 + \dim U_2 + \dim U_3 = 3 > 2 = \dim \mathbb{R}^2$.

6.8 Take a basis $\mathbf{u}_1, \ldots, \mathbf{u}_{n-1}$ of U. That set is linearly independent and has fewer than n elements, so it does not yet span V; there is some $\mathbf{v} \in V$ outside its span. The extension theorem makes $\mathbf{u}_1, \ldots, \mathbf{u}_{n-1}, \mathbf{v}$ a basis of V, so that $V = U \oplus \operatorname{span}\{\mathbf{v}\}$.

6.9 A polynomial of degree at most 4 that vanishes at 0 and 1 is divisible by $x(x - 1)$, so $p = x(x - 1)(a + bx + cx^2)$: three free parameters, dimension 3. Basis $\{x(x - 1),\ x^2(x - 1),\ x^3(x - 1)\}$. The evaluation matrix 2×5 has rank 2, so nullity $5 - 2 = 3$.

6.10 The coordinates satisfy $c_1 + c_2 + c_3 + c_4 = 4$, $c_2 + c_3 + c_4 = 2$, $c_3 + c_4 = -5$, $c_4 = 1$. From the bottom: $c_4 = 1$, $c_3 = -6$, $c_2 = 7$, $c_1 = 2$. So $[q]_B = (2, 7, -6, 1)$, the same as the result of `numpy.linalg.solve`.

Chapter 7: Basis, Dimension, and Linear Transformations

7.1 Solve $c_1(1, 1, 0) + c_2(1, 0, 1) + c_3(0, 1, 1) = (2, -1, 3)$, that is $c_1 + c_2 = 2$, $c_1 + c_3 = -1$, $c_2 + c_3 = 3$. The result is $c = (-1, 3, 0)$, so $[\mathbf{w}]_{B'} = (-1, 3, 0)$. Back: $-1(1, 1, 0) + 3(1, 0, 1) + 0 = (2, -1, 3)$.

7.2 Let φ be an isomorphism. Because φ is bijective, $\mathbf{b}_i = \varphi^{-1}(\mathbf{e}_i)$ is defined; because φ^{-1} is linear and bijective, $\{\mathbf{b}_i\}$ is a basis and $\varphi = \varphi_B$. It is not unique because each basis B gives a different isomorphism.

7.3 A has rank 2 (pivots in columns $2, 3$), nullity 1. Kernel: $x_2 = 0$, $2x_3 = 0$, so $\ker A = \operatorname{span}\{(1, 0, 0)\}$. Range $\operatorname{span}\{(1, 0, 0), (0, 2, 0)\}$. This is the derivative matrix on P_2 in the basis $(1, x, x^2)$: $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$.

7.4 $\operatorname{im}(S \circ T) \subseteq \operatorname{im} S$ gives $\operatorname{rank}(S \circ T) \leq \operatorname{rank} S$. Besides that, $\ker T \subseteq \ker(S \circ T)$, so $\operatorname{nullity}(S \circ T) \geq \operatorname{nullity} T$; rank-nullity applied to both maps gives $\operatorname{rank}(S \circ T) \leq \operatorname{rank} T$.

7.5 False. $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ both have rank 1, but the second is nilpotent (trace 0, square zero) while the first has trace 1. Rank measures only the dimension of the range.

7.6 T sends $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ to $(1, 0)$, $(3, 1)$, $(3, 2)$. Coordinates relative to C are $(x, y) \mapsto (y, x - y)$, so the columns are $(0, 1), (1, 2), (2, 1)$ and $[T]_{C \leftarrow B} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$. For $\mathbf{v} = (1, 2, 3)$: $[\mathbf{v}]_B = (-1, -1, 3)$ gives $(5, 0)$, the same as $[T(\mathbf{v})]_C$.

7.7 Take a basis $\mathbf{u}_1, \ldots, \mathbf{u}_m$ of U with $m = \dim V$. It is linearly independent and has $\dim V$ elements, so it spans V; hence every $\mathbf{v} \in V$ lies in U, that is $U = V$.

7.8 $B = P^{-1}AP$, so $B\mathbf{x} = \mathbf{0} \iff AP\mathbf{x} = \mathbf{0} \iff P\mathbf{x} \in \ker A$. The mapping $\mathbf{x} \mapsto P\mathbf{x}$ is an isomorphism from $\ker B$ onto $\ker A$, so the nullities agree. That equality is exactly the statement that the dimension of the kernel does not depend on the basis.

7.9 The third row is the sum of the first two, so rank 2 and nullity $4 - 2 = 2$. A nonzero kernel vector $(-2, -1, 1, 0)$: $F(-2, -1, 1, 0) = \mathbf{0}$. That direction shifts no feature at all, because $F(\mathbf{x} + t\mathbf{k}) = F(\mathbf{x})$ for all t.

Chapter 8: Linear Systems, Determinants, and Inverses

8.1 The pivots that appear are 1, 1, -1; no row needs swapping, so $\det A = 1 \cdot 1 \cdot (-1) = -1$. Backward: $z = 3$, $y = -1$, $x = 2$. Cramer gives $\det A_x = -2$, $\det A_y = 1$, $\det A_z = -3$; divided by -1 the result is $(2, -1, 3)$.

8.2 Cofactor expansion gives $\det M = 0$. The refuting vector is $\mathbf{v} = (1, -1, -1, 1)$: every row gives 0, for instance $0 - 1 - 2 + 3 = 0$. So $\det M = 0$ without any expansion. numpy gives 5.3×10^{-15}, not zero, because elimination subtracts nearly equal numbers and the rounding error is counted too.

8.3 Leibniz formula: $\det A = \sum_\sigma \operatorname{sgn}(\sigma) \prod_i a_{i\sigma(i)}$. If columns j and k agree, pair every σ with $\sigma \circ (j\ k)$; the products of the terms are the same but the signs are opposite, so the sum of all terms is zero.

8.4 Multiply $A \operatorname{adj} A = (\det A) I$ by the determinant: $\det A \cdot \det(\operatorname{adj} A) = (\det A)^n$, because $\det(cI) = c^n$. If $\det A \neq 0$, divide by $\det A$. If $\det A = 0$, both sides vanish for $n \geq 2$, so the equality still holds.

8.5 False as a general statement. Sensitivity is measured by $\kappa(A) = \|A\| \, \|A^{-1}\|$, not by $\det A$: $A = \varepsilon I$ has $\det A = \varepsilon^n$ very small yet $\kappa = 1$. A small determinant is dangerous only when it comes with a large $\|A^{-1}\|$.

8.6 $\|A\|_1 = 2 + \varepsilon$ and $A^{-1} = \frac{1}{\varepsilon} \begin{pmatrix} 1+\varepsilon & -1 \\ -1 & 1 \end{pmatrix}$ give $\|A^{-1}\|_1 = (2+\varepsilon)/\varepsilon$. So $\kappa_1(A) = (2+\varepsilon)^2/\varepsilon$. The condition $\kappa > 10^6$ is $\varepsilon < 4.000016 \times 10^{-6}$; at $\varepsilon = 4 \times 10^{-6}$ numpy gives 1.000004×10^6.

8.7 Adding a multiple of a row means multiplying on the left by $E = I + cE_{ij}$, and $\det E = 1$, so $\det B = \det A$. Because $B = EA$, we have $B^{-1} = A^{-1}E^{-1}$ with $E^{-1} = I - cE_{ij}$.

8.8 $A + AD = A(I + D)$, and $I + D$ is diagonal, so $\det(A + AD) = \det A \prod_i (1 + d_i)$. It is invertible exactly when $\det A \neq 0$ and no $d_i = -1$. Example $\det A = 5$ with $D = \operatorname{diag}(2, -\frac{1}{2}, 3)$: the left side is 30, the right side $5 \cdot 3 \cdot \frac{1}{2} \cdot 4 = 30$.

8.9 With $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$, $\det A = 1$ and

$$\operatorname{adj} A = A^{-1} = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix},$$

the same as `numpy.linalg.inv` up to machine error. For $\det A = 7$: $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 7 \end{pmatrix}^{-1}$ contains $1/7$ in the third entry, because division by 7 cannot be avoided.

Chapter 9: Eigenvalues and Diagonalization

9.1 $\operatorname{tr} A_1 = 5$ and $\det A_1 = 4$, so $\lambda = 4$ and $\lambda = 1$. Eigenvectors: $(1, 1)$ for $\lambda = 4$, $(1, -2)$ for $\lambda = 1$. Then $P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$ and $D = \operatorname{diag}(4, 1)$; $P^{-1} A_1 P = D$ is verified.

9.2 $\lambda = 5, 2$ with eigenvectors $(2, 1)$ and $(1, -1)$. For $k \geq 1$,

$$G^k = \frac{1}{3} \begin{pmatrix} 2 \cdot 5^k + 2^k & 2 \cdot 5^k - 2 \cdot 2^k \\ 5^k - 2^k & 5^k + 2 \cdot 2^k \end{pmatrix},$$

so that $G^{10} = \begin{pmatrix} 6510758 & 6509734 \\ 3254867 & 3255891 \end{pmatrix}$, the same as `matrix_power`.

9.3 $A\mathbf{v} = \lambda\mathbf{v}$ gives $A^2\mathbf{v} = \lambda^2\mathbf{v}$; induction gives $A^k\mathbf{v} = \lambda^k\mathbf{v}$. The converse fails: $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has $A^2 = -I$ with eigenvalue -1, whereas A itself has no real eigenvalue.

9.4 0 is an eigenvalue $\iff$ there is $\mathbf{v} \neq \mathbf{0}$ with $A\mathbf{v} = \mathbf{0}$ $\iff$ $\ker A \neq \{\mathbf{0}\}$ $\iff$ A is not invertible. If A is invertible and $A\mathbf{v} = \lambda\mathbf{v}$, then $\lambda \neq 0$ and $A^{-1}\mathbf{v} = \lambda^{-1}\mathbf{v}$.

9.5 False. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have characteristic polynomial $(\lambda-1)^2$, but the second is defective: $\operatorname{rank}(J - I) = 1$, so nullity $1 < 2$.

9.6 $\lambda I - A$ and its transpose have the same determinant, because $\det X = \det X^\top$; hence A and $A^\top$ have the same characteristic polynomial and the same eigenvalues. Their eigenvectors do not: A and $A^\top$ have different eigenspaces as soon as A is not symmetric.

9.7 $K - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ has rank 2, so nullity = 1. The algebraic multiplicity of $\lambda = 2$ is 2, the geometric one is 1; because $1 < 2$, K cannot be diagonalized.

9.8 $\lambda = 1$ and 0.5. The fixed point of $x = 4y$ with $x + y = 1$ is $(0.8; 0.2)$. The error shrinks like 0.5^k; because $0.5^{19} = 1.91 \times 10^{-6} > 10^{-6}$ and $0.5^{20} = 9.54 \times 10^{-7}$, it takes 20 steps.

9.9 The printed matrix gives $\operatorname{tr}(B^{10}) = (-19)+21+1024 = 1026$, and $1^{10}+1^{10}+2^{10} = 1026$: they agree. For a diagonalizable A, $A^k = PD^kP^{-1}$ yields $\operatorname{tr}(A^k) = \operatorname{tr}(D^k) = \sum_i \lambda_i^k$. For defective B that step is unavailable, because B has no P and D. Instead use the Schur triangular form $B = QTQ^{-1}$ with T upper triangular and diagonal $\lambda_i = 1, 1, 2$; then $B^k = QT^kQ^{-1}$, and T^k is still upper triangular with diagonal λ_i^k, so $\operatorname{tr}(B^k) = \operatorname{tr}(T^k) = \sum_i \lambda_i^k = 1026$. The trace is invariant under change of basis ($\operatorname{tr} XY = \operatorname{tr} YX$), which is why the result does not depend on Q. In short: diagonalization carries the proof only in the diagonalizable case; in the defective case the proof switches to the triangular form, the formula stays the same, and the factor k that appears in the entries of B^k never touches the diagonal.

Chapter 10: Orthogonality, SVD, and Least Squares

10.1 $(1, 0, 0)$ and $(0, 1, 1)$ are already orthogonal, so $\operatorname{proj}_W \mathbf{b} = 1 \cdot (1, 0, 0) + \frac{4}{2}(0, 1, 1) = (1, 2, 2) = \mathbf{b}$. Because $\mathbf{b} \in W$, the distance is 0; `qr` gives $QQ^\top\mathbf{b} = (1, 2, 2)$ as well.

10.2 $\mathbf{b} = \mathbf{p} + \mathbf{r}$ with $\mathbf{r} \perp W$, so $\mathbf{r} \perp \mathbf{p}$ and Pythagoras gives $\|\mathbf{b}\|^2 = \|\mathbf{p}\|^2 + \|\mathbf{r}\|^2$. If $\mathbf{p}$ is not an orthogonal projection, $\mathbf{r}$ is not perpendicular to $\mathbf{p}$; on an oblique basis in $\mathbb{R}^2$ a cross term $2\langle\mathbf{p}, \mathbf{r}\rangle \neq 0$ appears.

10.3 Orthonormal columns: $(A^\top A)_{ij} = \mathbf{a}_i^\top\mathbf{a}_j = \delta_{ij}$, so $A^\top A = I$. The normal equations $A^\top A\hat{\mathbf{x}} = A^\top\mathbf{b}$ immediately give $\hat{\mathbf{x}} = A^\top\mathbf{b}$. All singular values of A are 1, so $\kappa(A) = 1$.

10.4 False. $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ gives $A^\top A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, which is invertible ($\det = 3$), yet its rows are dependent: the third row is the sum of the first two. What invertibility of $A^\top A$ guarantees is only independence of the columns.

10.5 Singular values 2 and 0, so rank 1; $U = V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (up to sign). $A^+ = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, so $A^+\mathbf{b} = (1, 1)$ and the residual is $(-1, 1)$ with norm $\sqrt{2} = 1.4142$; `pinv` agrees.

10.6 Write $\mathbf{b} = A\hat{\mathbf{x}} + \mathbf{r}$ with $A\hat{\mathbf{x}} \in \operatorname{col} A$. For $\mathbf{h} \in \operatorname{col} A$ the function $t \mapsto \|\mathbf{b} - A(\hat{\mathbf{x}} + t\mathbf{h})\|^2$ is minimal at $t = 0$, hence $\langle\mathbf{r}, \mathbf{h}\rangle = 0$; this is the "necessary" direction. Conversely, if $\langle\mathbf{r}, \mathbf{h}\rangle = 0$ for every $\mathbf{h} \in \operatorname{col} A$, then $\|\mathbf{b} - A\mathbf{x}\|^2 = \|\mathbf{r}\|^2 + \|A(\hat{\mathbf{x}} - \mathbf{x})\|^2 \geq \|\mathbf{r}\|^2$, so $\hat{\mathbf{x}}$

minimises. Column independence is what makes $\hat{\mathbf{x}}$ unique.

10.7 $\mathbf{u}_1 = (1,1,1)/\sqrt{3}$; the remainder $(1,1,0) - \frac{2}{3}(1,1,1) = \frac{1}{3}(1,1,-2)$, so $\mathbf{u}_2 = (1,1,-2)/\sqrt{6}$; then $\mathbf{u}_3 = (1,-1,0)/\sqrt{2}$. $\|Q^\top Q - I\|$: 6.7×10^{-16} (by hand), 3.1×10^{-16} (`qr`). The signs may differ because each $\mathbf{u}_i$ is determined only up to $\pm$.

10.8 $AA^\top \mathbf{x} = \mathbf{0}$ gives $\|A^\top \mathbf{x}\|^2 = \mathbf{x}^\top AA^\top \mathbf{x} = 0$, so $A^\top \mathbf{x} = \mathbf{0}$; independence of the rows makes $\mathbf{x} = \mathbf{0}$. Hence $AA^\top$ is square and invertible. For $\mathbf{x} = A^\top (AA^\top)^{-1}\mathbf{b}$ we have $A\mathbf{x} = \mathbf{b}$ and $\mathbf{x} \in \operatorname{col} A^\top = (\ker A)^\perp$.

10.9 With the last reading 314.3: $\sum y = 1540.5$, $\sum Ty = 157161.7$, determinant 50. Cramer: $a = (52030 \cdot 1540.5 - 510 \cdot 157161.7)/50 = -5.04$ and $b = (5 \cdot 157161.7 - 510 \cdot 1540.5)/50 = 3.07$; `lstsq` agrees. The intercept shifts by -10.10, the slope by $+0.10$; the endpoint has leverage 0.6, the largest of the five.

10.10 B has rank 1, so B^+ gives $\mathbf{x}^+ = \frac{1}{70}(1,2) \approx (0.01429;\, 0.02857)$ and $\mathbf{k} = (-2,1)$. All solutions are $\mathbf{x}^+ + t\mathbf{k}$; because $\mathbf{x}^+ \perp \mathbf{k}$, $\|\mathbf{x}^+ + t\mathbf{k}\|^2 = \|\mathbf{x}^+\|^2 + t^2\|\mathbf{k}\|^2$ is minimal at $t = 0$.

Chapter 11: Continuity and Uniform Continuity

11.1 $|(2x-1)-5| = 2|x-3|$, so $\delta = \varepsilon/2$ is valid and largest: at $x = 3+\delta$ the value is exactly ε. The choice $\delta = \varepsilon/3$ is also valid, but it uses only a third of the room available.

11.2 $|x^2 - c^2| = |x-c|\,|x+c|$; with $|x-c| < 1$ we get $|x+c| \le 2|c|+1$, so $\delta = \min(1, \varepsilon/(2|c|+1))$. At $c = 2$: $\delta = 0.002$; at $c = 100$: $\delta = 4.975 \times 10^{-5}$.

11.3 Partly true. $1/x$ is indeed continuous at every point of $(0,1]$. But the uniform conclusion is wrong: $(0,1]$ is not compact and $1/x$ is unbounded near 0. The refuting pair $x_n = 1/n$, $y_n = 1/(n+1)$: $|x_n - y_n| \to 0$ but the difference of the values is 1.

11.4 Given ε, take δ_f and δ_g; with $\delta = \min(\delta_f, \delta_g)$, $|(f+g)(x) - (f+g)(y)| \le \varepsilon/2 + \varepsilon/2$. For fg: $f = g = x$ on $\mathbb{R}$ with $x_n = n$, $y_n = n + 1/n$ gives difference $2 + 1/n^2 \ge 2$.

11.5 $|\sqrt{x} - \sqrt{y}|^2 \le |x-y|$ (rationalize the denominator), so $\delta = \varepsilon^2$ works for all $x, y \ge 0$. Only $x, y \ge 0$ is used, so that the roots are defined; closedness is not needed because the proof never touches a limit point outside the domain.

11.6 $(1+\delta)^3 - 1 = \varepsilon$ gives $\delta = (1+\varepsilon)^{1/3} - 1 = 0{,}0033223$. The choice $\delta = \varepsilon/3 = 0{,}0033333$ is actually too large: $(1+\varepsilon/3)^3 - 1 - \varepsilon = \varepsilon^2/3 + \varepsilon^3/27 > 0$ for every $\varepsilon > 0$, so it violates the bound and is not valid.

11.7 With $\delta = \varepsilon/2$: the largest difference $2\delta - \delta^2 = 0.009975 < \varepsilon$, so it is safe but not optimal. With $\delta = 1 - \sqrt{1-\varepsilon} = 0.0050126$: the largest difference $1 - (1-\delta)^2 = \varepsilon$ exactly, attained at $x = 1 - \delta$.

11.8 $|f|$ is continuous on the compact $[a,b]$, so it attains a minimum m; because $f \neq 0$ everywhere, $m > 0$ and $|f| \ge m$. For $f(x) = x$ on $(0,1)$ the conclusion fails: the infimum is 0 and it is not attained.

11.9 At $c \neq 0$ take $q_n \to c$ rational and $r_n \to c$ irrational; the function values tend to c and $-c$, which differ, so f is not continuous. At 0: $|f(x)| = |x| \to 0 = f(0)$. The function $|f|$ is $x \mapsto |x|$, continuous everywhere; there is no contradiction because $|f|$ throws away the sign information.

Chapter 12: Derivatives: One and Several Variables

12.1 $((2+h)^3 - 8)/h = 12 + 6h + h^2 \to 12$. For $\sqrt{x}$: $(\sqrt{2+h} - \sqrt{2})/h = 1/(\sqrt{2+h} + \sqrt{2}) \to 1/(2\sqrt{2}) = 0.3535534$. In both cases h drops out of the denominator.

12.2 $f'(c) > 0$ makes $(f(c+h) - f(c))/h > f'(c)/2 > 0$ for h small enough, so $f(c+h) > f(c)$.

That is only a one-sided statement. The example $f(x) = x + 2x^2 \sin(1/x^2)$ has $f'(0) = 1 > 0$ but is not increasing on any neighbourhood of 0: its derivative changes sign arbitrarily close to 0.

12.3 Differentiate $1/g$ first: $(1/g)' = -g'/g^2$; here the continuity of g is used, so that $g(x+h) \to g(x) \neq 0$. Then $(f/g)' = f' \cdot (1/g) + f \cdot (1/g)'$, which simplifies to $(f'g - fg')/g^2$.

12.4 False. $f(x,y) = xy/(x^2+y^2)$ for $(x,y) \neq (0,0)$ and $f(0,0) = 0$: both partial derivatives at the origin exist and equal 0, but along $y = x$ the value is $\frac{1}{2}$, so f is not continuous there.

12.5 $\nabla f = (3x^2y - y^3,\ x^3 - 3xy^2)$, at $(2,1)$ equal to $(11,2)$. The steepest ascent direction is $(11,2)/\sqrt{125}$, the maximum rate $\sqrt{125} = 11.1803$. The central difference with $h = 10^{-4}$ gives relative error 9.1×10^{-10} and 1.0×10^{-8}, consistent with order h^2 in the second component.

12.6 By the mean value theorem, $f(b) - f(a) = f'(\xi)(b-a) = 0$ for every pair $a < b$ in the interval, so f is constant. On a disconnected set that step fails because ξ need not lie in the set; for example $f = 0$ on $(0,1)$ and $f = 1$ on $(2,3)$.

12.7 At $(0,1)$: $J_H = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$; at $H(0,1) = (1,1)$: $J_G = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$. The matrix product is $\begin{pmatrix} 3 & 4 \\ 0 & -1 \end{pmatrix}$, and the direct derivative gives the same matrix.

12.8 $f_x = 3x^2 - 3y$, $f_y = 3y^2 - 3x$; both zero gives $y = x^2$ and $x = y^2$, so $x(x^3 - 1) = 0$: the points $(0,0)$ and $(1,1)$. The Hessian $\begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}$: at $(0,0)$ its determinant is $-9 < 0$ (saddle), at $(1,1)$ its determinant is $27 > 0$ with $f_{xx} = 6 > 0$ (local minimum, $f = -1$). If the determinant is zero, that test says nothing.

12.9 $p_2(x) = 1 + \frac{1}{2}h - \frac{1}{8}h^2$ at $h = 0.04$ gives 1.0198, whereas $\sqrt{1.04} = 1.0198039$; the error is 3.9×10^{-6}, the Lagrange remainder $\frac{3}{8}\xi^{-5/2}h^3/6 \leq 4.0 \times 10^{-6}$. The bound is tight, not loose, because ξ is close to 1: the Lagrange remainder really does nearly equal the actual error.

Chapter 13: Riemann Integrals and the Fundamental Theorem of Calculus

13.1 $U_n = \frac{1}{n}\sum_{k=1}^{n} \frac{k}{n} = \frac{n+1}{2n}$ and $L_n = \frac{n-1}{2n}$, so $U_n - L_n = 1/n \to 0$: f is integrable and $\int_0^1 x\,dx = \frac{1}{2}$.

13.2 $\big||f(x)| - |f(y)|\big| \leq |f(x) - f(y)|$ (reverse triangle inequality), so integrability carries over. The bound comes from $-|f| \leq f \leq |f|$ and monotonicity of the integral. The converse is false: $f = 1$ on the rationals and -1 on the irrationals has $|f| = 1$, which is integrable.

13.3 False. The correct part: the upper and lower sums of a bounded function are always finite. What the proof does not support is the equality $\sup_{\mathcal{P}} L = \inf_{\mathcal{P}} U$. The Dirichlet function is bounded with $U - L = 1$ for every partition.

13.4 On a uniform partition each subinterval contributes oscillation $f(x_k) - f(x_{k-1})$, so that $U - L \leq \frac{b-a}{n}\sum_k \big(f(x_k) - f(x_{k-1})\big) = \frac{(b-a)(f(b)-f(a))}{n}$. For a non-monotone function that telescoping fails because the signs of the oscillations are not uniform.

13.5 $\int_0^1 (3x^2 - 2x + 1)\,dx = [x^3 - x^2 + x]_0^1 = 1$. Simpson with $n = 100$ gives 1.0000000000 and `quad` 1.0000000000 (error 1.1×10^{-14}); the three numbers differ by zero to machine precision.

13.6 $\int_1^{10} x^{-2}dx = 1 - \frac{1}{10} = 0.9$ and $\int_1^{\infty} x^{-2}dx = 1$. Their difference is $0.1 = \frac{1}{10}$ exactly, namely the tail $\int_{10}^{\infty} x^{-2}dx = 1/10$; there is no need to recompute it, because the missing part is exactly that tail.

13.7 If $f(x_0) > 0$, continuity gives $f > \frac{1}{2}f(x_0)$ on an interval of width 2δ around x_0 (of width δ when x_0 is an endpoint), so the integral is at least $\frac{1}{2}\delta f(x_0) > 0$: a contradiction. For f that is merely integrable the statement fails: $f = 1$ at a single point and 0 elsewhere has integral 0.

13.8 $u = \sin x$: $\int_0^1 u\,du = \frac{1}{2}$. Through the identity: $\frac{1}{2}\int_0^{\pi/2} \sin 2x\,dx = \frac{1}{2}\left[-\frac{1}{2}\cos 2x\right]_0^{\pi/2} = \frac{1}{2}$. `quad` gives 0.5 as well.

13.9 On every subinterval there is a rational (value 1) and an irrational (value 0), so $U(\mathcal{P}, D) = 1$ and $L(\mathcal{P}, D) = 0$; the difference is 1 for all $\mathcal{P}$. The Riemann criterion demands that this difference can be made as small as we please, so D is not integrable.

13.10 The integral bounds give $\frac{1}{101} < \sum_{n=101}^{\infty} n^{-2} < \frac{1}{100}$, that is 9.9010×10^{-3} up to 1.0000×10^{-2}. Their midpoint is 9.9505×10^{-3}, whereas the true value is 9.950167×10^{-3}; the relative error is 3.3×10^{-5}.

Chapter 14: Series of Functions and the Exchange of Limits

14.1 $\sup_{[0,1]}|x^n| = 1$ for all n, attained at $x = 1$. At $x = 0.9$: 0.3487 for $n = 10$, 2.656×10^{-5} for $n = 100$, 1.748×10^{-46} for $n = 1000$. The first column is flat because the peak is always at the endpoint; the second column shrinks because $0.9 < 1$.

14.2 For $n \geq N$, $\|f_n - f\|_\infty < \varepsilon$, so $\|f\|_\infty \leq \|f_N\|_\infty + 1$ and likewise for $\|f_n\|_\infty$. Pointwise convergence is not enough: $f_n = n\chi_{(0,1/n)}$ converges to 0 at every point but $\|f_n\|_\infty = n$.

14.3 The peak is at $x = 1/\sqrt{n}$ with $\|f_n\|_\infty = 1/(2\sqrt{n}) \to 0$, and that bound holds for all x at once, so the convergence is uniform. The condition $1/(2\sqrt{n}) < 0.01$ requires $\sqrt{n} > 50$, so $n = 2501$.

14.4 False. Pointwise reasoning gives one N for each x, not one N for all x. The counterexample is $x_n = 2^{-1/n}$: $x_n^n = \frac{1}{2}$ for every n, so $\sup_{[0,1)}|x^n| \geq \frac{1}{2}$ and it does not tend to 0.

14.5 The error $|e - S_N(1)|$ for $N = 5, 10, 15$: 1.615×10^{-3}, 2.731×10^{-8}, 5.077×10^{-14}; the bound $e/(N+1)!$ gives 3.775×10^{-3}, 6.810×10^{-8}, 1.299×10^{-13}, always larger. At $N = 20$ the bound is already 5.32×10^{-20}, far below machine precision, so the measured error stops at 4.44×10^{-16}. The bound first drops below 10^{-16} at $N = 18$, and there the true error is unmeasurable as well.

14.6 The sum is $(1 - x)/(1 - x) = 1$ for $x \in [0, 1)$, whereas at $x = 1$ the sum is 0. The limit is not continuous even though every partial sum is continuous; the uniform limit of continuous functions must be continuous, so the convergence is not uniform.

14.7 Radii: $R = 1$ for $\sum x^n/n^2$, $R = e$ for $\sum (n!/n^n)x^n$, $R = \frac{1}{2}$ for $\sum 2^n x^n/n$. The ratio of the 1001st and 1000th terms: 0.998003; 0.368063 (so $R \approx 2.7169$); 1.998002. The slowest is $\sum x^n/n^2$, because its ratio tends to 1 like $(1 - 1/n)^2$.

14.8 $|f_n g - fg| = |f_n - f|\,|g| \leq \|f_n - f\|_\infty \|g\|_\infty \to 0$. Without boundedness of g: $f_n = x + 1/n \to x$ uniformly on $(0, 1)$, but $(x + 1/n)/x = 1 + 1/(nx)$ is unbounded, so $f_n g$ does not converge uniformly.

14.9 The peak is at $x = 1/\sqrt{2n+1}$. For $n = 10$: height 1.3397 at $x = 0.2182$, $\int_0^1 f_{10} = \frac{5}{11} = 0.4545$. For $n = 100$: height 4.2835 at $x = 0.07053$, the integral $\frac{50}{101} = 0.4950$. The height follows $\sqrt{n/(2e)}$; the area tends to $\frac{1}{2}$ because the peak gets narrower.

14.10 Differentiating $\sum_{n\geq 0} x^n = 1/(1 - x)$ gives $\sum_{n\geq 1} nx^{n-1} = 1/(1 - x)^2$; at $x = \frac{1}{2}$ we get $\sum_{n\geq 1} n/2^n = 2$. The tail remainder is $(N+2)/2^N$, so $N = 39$ gives $7.46 \times 10^{-11} < 10^{-10}$.

Chapter 15: Multiple Integrals, Change of Variables, and Vector Field Theorems

15.1 Cartesian: $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}(x^2+y^2)\,dy\,dx = \pi/2$. Polar: $\int_0^{2\pi}\int_0^1 r^3\,dr\,d\theta = 2\pi\cdot\frac{1}{4} = \pi/2 = 1.5708$. `dblquad` gives 1.5707963268; in this test environment the Cartesian route requires 8379 evaluations, the polar route 441.

15.2 For $T(\mathbf{u}) = A\mathbf{u}$ linear, $DT = A$ is constant, so its Jacobian determinant is $\det A$ and $|\det A|$ is the area scale factor. If $\det A \neq 0$ the map is one-to-one on D, so $\iint_{T(D)} f = |\det A| \iint_D f \circ T$.

15.3 Clockwise, each side is traversed in reverse, so the four terms of the example change sign: $-\frac{1}{3} - 2 - \frac{2}{3} + 1 = -2$, whereas the right-hand side stays 2. The bottom and right sides that used to contribute $+\frac{1}{3}$ and $+2$ become negative; the left side that used to be -1 becomes $+1$.

15.4 $x = au$, $y = bv$, $z = cw$ gives Jacobian abc and the unit ball: $V = abc\cdot\frac{4\pi}{3} = \frac{4\pi abc}{3}$. At $a = b = c = 1$ the result is 4.1888, the familiar formula for the volume of a ball.

15.5 False. $\operatorname{curl} F = 0$ does hold at every point where F is defined, but the domain has a hole so the curl test does not apply. The supporting number: $\oint_{|\mathbf{r}|=1} F\cdot d\mathbf{r} = 2\pi \neq 0$, so F is not conservative.

15.6 If the integral does not depend on the path, two paths C_1, C_2 from A to B give the same integral, so the circulation around $C_1 - C_2$ is zero. Conversely, if every closed curve has zero circulation, $C_1 - C_2$ is closed and hence the two integrals agree.

15.7 $\operatorname{curl} F = (0, 0, 2)$; the flux through $z = 1 - x^2 - y^2$ with upward normal $= 2\cdot\pi = 2\pi$. The line integral along the boundary $x^2 + y^2 = 1$, $z = 0$: $\int_0^{2\pi}(-\sin t, \cos t, 0)\cdot(-\sin t, \cos t, 0)\,dt = 2\pi$. Both are 6.2832.

15.8 $\operatorname{div} F = 0$ away from the origin, so the multiple integral is 0, whereas the flux through the unit sphere is $4\pi = 12.5664$. Gauss's theorem requires a C^1 field on a region that contains the whole interior of the surface; that condition is violated at the enclosed origin.

15.9 $\det J = r^2\sin\varphi$, zero at $\varphi = 0$ and $\varphi = \pi$. At both poles the map stops being one-to-one: all θ give the same point. On $0 < \varphi < \pi$ the map is one-to-one and $\det J > 0$, so the change of variables formula remains valid even though the determinant is zero on the boundary.

Chapter 16: Differential Equations and Dynamical Systems

16.1 $y = Ce^{t^2}$ with $y(0) = 1$ gives $y = e^{t^2}$, so $y(1) = e = 2.718282$. Euler with $h = 0.1$ uses $y_{n+1} = (1 + 0.02n)y_n$; the result is 2.334633, error 3.8365×10^{-1}. That method lags behind because it uses the slope at the left end of each step.

16.2 The difference $w = y_1 - y_2$ satisfies $w' = -p(t)w$ and $w(t_0) = 0$. Multiply by $e^{\int p}$: $(we^{\int p})' = 0$, so $we^{\int p}$ is constant, and its value is zero at t_0. Hence $w \equiv 0$. No solution formula is used, only the product rule and uniqueness of initial values.

16.3 Not enough evidence. Linearization sees only the degree-one terms; the eigenvalues $\pm i$ say only that the linear part is a rotation. With $r' = r^3$ the origin is a center for the linearization but an outward spiral, whereas $r' = -r^3$ gives it an inward spiral. So $\pm i$ does not determine stability.

16.4 $E' = vv' + \omega_0^2\theta\theta' = v(-\omega_0^2\theta) + \omega_0^2\theta v = 0$, so E is constant. The equation $v^2/(2E) + \theta^2/(2E/\omega_0^2) = 1$ is an ellipse, and $\theta = A\cos(\omega_0 t + \varphi)$ closes one turn in $2\pi/\omega_0$.

16.5 $\tau = RC = 2.2 \times 10^3 \cdot 10^{-5} = 0.022$ seconds. At $t = \tau$: $V = 12(1 - e^{-1}) = 7.585447$

volts. For 6 volts, $t = -\tau \ln(1 - \frac{6}{12}) = 0.022 \cdot 0.693147 = 15.249$ milliseconds.

16.6 With $v = 2\cos(\theta/2)$: $v^2/2 = 2\cos^2(\theta/2)$ and $1 - \cos\theta = 2\sin^2(\theta/2)$, and their sum is $E = 2$. At $\theta = \pm\pi$ the value is $v = 0$, so the curve joins those two points; it separates oscillation from full rotation, and the time needed to travel it is infinite.

16.7 $T_0 = 2\pi\sqrt{0.25/9.80665} = 1.003205$ seconds. $T = T_0 \cdot \frac{2}{\pi}K(\sin 15°) = 1.020669$ seconds; the difference is 1.74 percent. An amplitude of 30° lengthens the period, it does not shorten it.

16.8 The factor per step is $1 + h\lambda = 1 - 100h$; the condition $|1 - 100h| < 1$ gives $0 < h < 0.02$. Exactly at $h = 0.02$ the factor is -1: the value changes sign with fixed magnitude 1, neither decaying nor blowing up. That is the marginal stability boundary, not stability.

16.9 Suppose y is periodic and not constant. At its maximum point $y' = 0$, so $f(y_{\max}) = 0$. The constant solution $y \equiv y_{\max}$ passes through that point, and uniqueness (Lipschitz) forces the solutions to be the same, contradicting non-constancy. So there is no non-constant periodic orbit.

Chapter 17: Probability and Conditional Probability

17.1 $P(A \cup B \cup C) = P(A \cup B) + P(C) - P\big((A \cup B) \cap C\big)$, then $P\big((A \cup B) \cap C\big) = P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$. The triple-overlap term is subtracted three times, once for each pair, and then added back once with a + sign.

17.2 False on general spaces. $P(A \mid B) = 0$ only means $P(A \cap B) = 0$. On a finite space with positive probabilities, that forces $A \cap B = \varnothing$. On a continuous space it does not: $A = \{1/2\}$ and $B = (0, 1)$ uniform give $P(A \cap B) = 0$ even though $1/2 \in A \cap B$.

17.3 From 10 000 people: 200 are sick, 190 correctly positive; with specificity 0.90 there are 980 false positives, so $190/1170 = 19/117 \approx 0.1624$. With 0.99: only 98 false positives remain, so $190/288 = 95/144 \approx 0.6597$: the precision rises fourfold.

17.4 $P(A) = P(A \cap B) + P(A \cap B^c)$, so $P(A \cap B^c) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B^c)$. For A^c, B^c use $P(A^c \cap B^c) = 1 - P(A \cup B)$ and inclusion–exclusion, or apply the result above twice.

17.5 $1 - \prod_{k=0}^{29}(1 - k/365) = 0.706316$, whereas $1 - e^{-435/365} = 0.696320$; the difference is 0.009996. The approximation uses $1 - x \approx e^{-x}$ for x up to $29/365$, and treats pair events as mutually independent when they are not.

17.6 $A = (A \cap B) \cup (A \cap B^c)$ and the two pieces are disjoint, because $B \cap B^c = \varnothing$. Additivity gives $P(A) = P(A \cap B) + P(A \cap B^c) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c)$, using the definition of conditional probability twice.

17.7 The probability of winning by switching is $3/4 = 0.75$: the initial choice wins only when the car really is there $(1/4)$. A simulation of 10^4 games gives about 0.75 with standard deviation $\sqrt{0.75 \cdot 0.25/10^4} \approx 0.0043$.

17.8 The correct figure is 0,9075, not 0,9999: with 0,1 percent prevalence only 1 person in 1000 is ill, while the false positives among two positive tests are $999 \cdot 0{,}01^2 \approx 0{,}1$. Hence $0{,}99^2 \cdot 0{,}001 / (0{,}99^2 \cdot 0{,}001 + 0{,}01^2 \cdot 0{,}999) = 0{,}9075$. The assumptions: the two tests are conditionally independent and both sensitivity and specificity are exactly 0,99.

17.9 Not enough. Take A the first coin shows heads, B the second coin shows heads, C both coins match. Every pair gives $1/4 = P \cdot P$, but $P(A \cap B \cap C) = 1/4$ whereas $P(A)P(B)P(C) = 1/8$; the difference is $1/8$.

Chapter 18: Random Variables and Distributions

18.1 $c = 1 - (0.1 + 0.4) = 0.5$; $\mathbb{E}[X] = -0.2 + 0 + 1.5 = 1.3$; $\mathbb{E}[X^2] = 0.4 + 0 + 4.5 = 4.9$; $\mathrm{Var}(X) = 4.9 - 1.69 = 3.21$.

18.2 $(-\infty, b] = (-\infty, a] \cup (a, b]$ and the two pieces are disjoint, so $F(b) = F(a) + P(a < X \le b)$. For discrete variables $P(X = a) > 0$ makes $P(a \le X \le b)$ larger; for continuous ones $P(X = a) = 0$ so the two agree.

18.3 It is true that $P(X = x) = 0$ for every x in a continuous distribution. The mistake is concluding "impossible": probability zero does not mean the empty set. Countable additivity gives probability zero to a countable union of points, but an interval contains uncountably many points and has probability one.

18.4 Differentiate $\sum q^k = 1/(1-q)$ twice: $\sum k(k-1)q^{k-2} = 2/(1-q)^3$. Since $\sum k^2 q^{k-1} = (1+q)/(1-q)^3$, with $q = 1 - p$ we get $\mathbb{E}[X^2] = p(1+q)/p^3 = (1+q)/p^2 = (2-p)/p^2$. Hence $\mathrm{Var} = (2-p)/p^2 - 1/p^2 = (1-p)/p^2$; for $p = 0.2$: $\mathbb{E}[X^2] = 45$ and $\mathrm{Var} = 20$.

18.5 $\sigma = 2$: $P(X > 1) = 1 - \Phi(-1) = 0.841345$; $P(1 < X < 5) = 2\Phi(1) - 1 = 0.682689$; the 0.2 quantile is $3 + 2(-0.841621) = 1.316758$. All three agree with `scipy.stats.norm` to six digits.

18.6 $\sum_{k=0}^{40} e^{-2.5} 2.5^k / k! = 1 - 5.39 \times 10^{-35}$, that is 1 to every printed digit. $P(X \ge 8) = 0.004247$, whereas the binomial with $n = 1000$, $p = 0.0025$ gives 0.004191; the binomial is smaller because its variance is $2.4938 < 2.5$ and it is bounded above by n.

18.7 $P(X > s + t \mid X > s) = e^{-\lambda(s+t)} / e^{-\lambda s} = e^{-\lambda t}$. Geometric: $P(X > k) = q^k$, so the ratio is $q^{s+t}/q^s = q^t$. Uniform on the integers fails because the tail probability runs out, and the ratio depends on s.

18.8 $f_Y(y) = \frac{1}{4\sqrt{y}}$ for $0 < y < 1$ (two preimages) and $\frac{1}{8\sqrt{y}}$ for $1 \le y < 9$ (one preimage). The integral is $\frac{1}{2} + \frac{1}{2} = 1$. $\mathbb{E}[Y] = \frac{1}{4}\int_{-1}^{3} x^2\,dx = 7/3 = 2.3333$, the same as $\mathbb{E}[X^2]$, and via f_Y: $\int_0^1 \frac{y}{4\sqrt{y}}\,dy + \int_1^9 \frac{y}{8\sqrt{y}}\,dy = \frac{1}{6} + \frac{13}{6} = \frac{7}{3}$.

18.9 Take X uniform on $\{-1, 0, 1\}$ and $Y = |X|$. Then $\mathbb{E}[XY] = \mathbb{E}[X] = 0$, so $\mathrm{Cov} = 0$ and $\rho = 0$. But $P(X = 0, Y = 0) = 1/3$ whereas $P(X = 0)P(Y = 0) = 1/9$: they are not independent.

18.10 $\mathrm{se}(\bar{X}) = \sqrt{p(1-p)/(nR)} \le 0{,}005$ gives $R \ge 0{,}2275/(20 \cdot 0{,}000025) = 455$. For the variance, $\mathrm{se}(\bar{s}^2) = \sigma^2\sqrt{2/((n-1)R)}$ (normal approximation) with $\sigma^2 = 4{,}55$ and $n = 20$, so $R \ge 2 \cdot 4{,}55^2/(19 \cdot 0{,}02^2) = 5\,448{,}03$, that is $R = 5\,449$. Using $2 \cdot 4{,}55^2/0{,}02^2$ alone drops the factor $n - 1$ and gives a figure nineteen times too large.

Chapter 19: The Law of Large Numbers and the Central Limit Theorem

19.1 $\{|(X_n + Y_n) - (X + Y)| > \varepsilon\}$ is contained in $\{|X_n - X| > \varepsilon/2\} \cup \{|Y_n - Y| > \varepsilon/2\}$. The union bound gives a probability of at most the sum of two terms, each going to 0. The $\varepsilon/2$ is needed so that two errors both below it add up to less than ε.

19.2 $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] = 2a^2 \cdot \sigma^2/(2a^2) = \sigma^2$, so $\mathrm{Var}(X) = \sigma^2$. Then $P(|X| \ge a) = \sigma^2/a^2$, exactly equal to the Chebyshev bound: the bound is attained, so it is sharp.

19.3 False. The central limit theorem demands finite variance and iid observations; it also speaks only about the limit, not about $n = 30$. In the lab section, the exponential with $n = 30$ still gives $D = 0.0219$ with $p = 0.000$, and Cauchy gives a median $|\bar{X}_n|$ that settles near 1.

19.4 Chebyshev: $n \ge \sigma^2/(\varepsilon^2\delta) = 0.25/(0.0025 \cdot 0.05) = 2000$. The exact binomial computation gives $n = 371$ (tail 0.0484). The Chebyshev bound is five times looser.

19.5 $\mathbb{E}[X_n] = n^2 \cdot (1/n) = n \to \infty$, whereas for fixed ε we have $P(|X_n| > \varepsilon) = 1/n \to 0$. So $X_n \to 0$ in probability but its expectation explodes: no domination is available.

19.6 Exact: 0.069545. Without the correction, $z = (50-60)/\sqrt{42} = -1.5430$ gives 0.061411, error 0.008134. With the correction at 50.5: 0.071340, error 0.001795. The continuity correction improves it fourfold.

19.7 Median $|\bar{X}_n|$: 1.004; 1.008; 1.040; 0.973 for $n = 10$ up to 10^4. $P(|\bar{X}_n| > 10)$: 0.0630; 0.0680; 0.0653; 0.0580, theory 0.0635. The standard deviation 33.0; 863.6; 23.9; 379.5 does not shrink because the Cauchy variance is infinite.

19.8 $\varphi_{\bar{X}_n}(t) = \big[\varphi(t/n)\big]^n = \big[e^{-|t|/n}\big]^n = e^{-|t|}$, so the average is again Cauchy for every n. Hence the se stays as large as the scale of the data itself, and "mean ± two se" does not narrow even as n grows.

19.9 $n = (1.96 \cdot 0.491971/0.0005)^2 = 3\,719\,214$; for half-width 0.001 it is enough to take 929 803. The ratio is 4, because $n \propto h^{-2}$: halve the width and the sample size rises fourfold.

19.10 For $t < c$ and $\varepsilon = c - t$: $P(X_n \le t) \le P(X_n \le c-\varepsilon) + P(|X_n - c| > \varepsilon) \to 0$; for $t > c$ use the complement. The constant condition cannot be dropped: take X Rademacher and $X_n = (-1)^n X$. Every X_n is still Rademacher (because X is symmetric), so $X_n \rightsquigarrow X$; but for odd n we have $|X_n - X| = 2$ with probability 1, so X_n does not converge to X in probability.

Chapter 20: Estimation, Confidence Intervals, and Hypothesis Testing

20.1 False. A p-value of 0.42 means the data are not enough to contradict $p = 0.5$; it does not prove $p = 0.5$. The supplementary computations: a 95 percent interval, for instance (0.28; 0.57), and the power of the test at the difference considered to matter.

20.2 $\mathbb{E}[\hat{p}(1-\hat{p})] = p - \big(p^2 + p(1-p)/n\big) = p(1-p)(n-1)/n$. The unbiased estimator is $\frac{n}{n-1}\hat{p}(1-\hat{p})$. For $p = 0.3$, $n = 20$: the expectation is $0.21 \cdot 19/20 = 0.1995$, and that estimator brings it back to 0.21.

20.3 Write $\hat{\theta}_n - \theta = \text{bias}_n + (\hat{\theta}_n - \mathbb{E}\hat{\theta}_n)$. The first term goes to 0; the second goes to 0 in probability by Chebyshev, because its variance goes to 0. The MSE = bias^2 + Var going to 0 is stronger, but only the two conditions above are used.

20.4 7/40: Wilson (0,0875; 0,3195), width 0,2320; Wald (0,0572; 0,2928), width 0,2355. 13/20: Wilson width 0,3860, Wald 0.4181. The 7/40 interval is narrower, but that is not a sign of higher precision. Partly it is $n = 40$; partly it is $\hat{p} = 0.175$ being far from 0.5, which shrinks $\hat{p}(1-\hat{p})$: at $n = 40$ with $\hat{p} = 0.5$ the Wilson width is 0.2960, still wider than 0.2320. (At equal n, 7/20 and 13/20 both give 0.3860, since $\hat{p}(1-\hat{p})$ is symmetric about 1/2.) What settles precision is not width but coverage: the Wald coverage at $p = 0.175$, $n = 40$ is only 0.921, whereas Wilson gives 0.943. A narrow interval near the edge is paid for with coverage below 95 percent.

20.5 $|k - 15| > 1.96\sqrt{7.5} = 5.3676$, so reject when $k \le 9$ or $k \ge 21$. Its exact size is 0.042774; its power at $p = 0.7$ is only 0.588816. With $n = 100$ (reject when $k \le 40$ or $k \ge 60$) the power rises to 0.987502.

20.6 Wald: $|\hat{p} - p_0| \le z\sqrt{\hat{p}(1-\hat{p})/n}$, which is equivalent to p_0 lying in $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$. A test that uses $\sqrt{p_0(1-p_0)/n}$ uses a different denominator, so the set of p_0 that are not rejected is not that Wald interval.

20.7 Suppose the observed difference is 0.87 hours with standard error 0.86 hours. "At the 5 percent level the data do not reject equality of the two methods ($p = 0.31$). The 95 percent interval for the difference is (−0.82; 2.56) hours, so a 2-hour difference is still a

plausible value and even a 0.5-hour difference cannot be ruled out. The power of this test against a difference of 0.5 hours is only 0.09: absence of evidence is not evidence of absence, and the honest conclusion is that these data do not decide the question."

20.8 Rule of three: $3/100 = 0.03$. The exact one-sided bound is $1 - 0.05^{1/100} = 0.029513$; the Wilson upper bound is 0.036993. The three differ because the rule of three is only an approximation and Wilson is two-sided. For the statement "at most 5 percent" to be valid at the 95 percent level: the exact bound is 0.0295.

20.9 $\bar{x} = 19.125$, $s = 1.082191$, se $= 0.312401$; $t_{11;0.95} = 1.795885$ gives (18.5640; 19.6860) with width 1.1221, whereas 95 percent gives width 1.3752. The width scales as $1/\sqrt{n}$, so halving it needs $n = 48$.

Chapter 21: Regression and Applied Statistics

21.1 Not enough evidence. $R^2 = 0.99$ says only that the variance is explained; s determines the error. On the cafe data $R^2 = 0.992821$ but $s = 0.365148$ hours $= 21.9$ minutes, and the half-width of the 95 percent prediction interval at $x_0 = 5$ is 0.9101 hours, that is 55 minutes. What should be quoted: s and a prediction interval.

21.2 The normal equations are $X^\top e = 0$. The first column of X holds ones, so $\sum e_i = 0$; the second column holds x_i, so $\sum x_i e_i = 0$. Without the ones column (a line through the origin), $\sum e_i = 0$ is no longer imposed and generally fails.

21.3 $\hat{b} - \beta = \sum_i (x_i - \bar{x})\varepsilon_i / S_{xx}$. The terms are independent and $\mathrm{Var}(\varepsilon_i) = \sigma^2$, so $\mathrm{Var}(\hat{b}) = \sigma^2 \sum (x_i - \bar{x})^2 / S_{xx}^2 = \sigma^2 / S_{xx}$. No matrix formula is used.

21.4 $\hat{b} = 0.7$, $\hat{a} = 1.3$, $S_{xx} = 10$, $s = 0.516398$, $\mathrm{se}(\hat{b}) = 0.163299$; with $t_3 = 3.182446$ the interval is (0.1803; 1.2197). It is wider than for the cafe data because $n = 5$, S_{xx} is small, and s is large.

21.5 In the cafe data $\bar{x} = 5$, $n = 9$, $S_{xx} = 60$, so the leverage is $c(x_0) = 1/9 + (x_0 - 5)^2/60$: $c(2) = 0.261111$ and $c(9) = 0.377778$. With $2.364624\sqrt{0.133333\,(1 + c)}$ the half-widths are 0.9696 for $x_0 = 2$ and 1.0135 for $x_0 = 9$; they differ because $(x_0 - \bar{x})^2 / S_{xx}$ is larger farther from the centre. That term shrinks like $1/n$, so the difference between the two intervals does vanish for large n: both tend to $1.96\sigma = 0.716$ half-width. What never vanishes is the width itself, because the $+1$ term (the error of one new observation) remains: a prediction interval does not shrink to zero, it shrinks to the spread of the error.

21.6 The column space of the big model contains the column space of the small model, so its projection is closest: the SSE cannot rise. R^2 may stay the same, but the divisor $n - p$ shrinks, so R^2_{adj} falls: from 0.991795 ($p = 2$) to 0.990428 ($p = 3$).

21.7 VIF $= 1/(1 - 0.81) = 5.263158$, so $\mathrm{se}(\hat{b}_1)$ grows by a factor $\sqrt{5.263158}$. With x_2: $\sqrt{2/(50 \cdot 0.19)} = 0.458831$. Without x_2: $\sqrt{2/50} = 0.2$. The ratio is 2.294157.

21.8 Group A: (1; 2), (2; 1,8), (3; 1,6); group B: (5; 4,0), (6; 3,8), (7; 3,6). Each group has slope $-0{,}2$, the pooled slope is $+0{,}4$. The valid decomposition adds contributions, not slopes: $b_{\mathrm{gab}} = (S_{xy}^{\mathrm{dalam}} + S_{xy}^{\mathrm{antar}})/S_{xx} = -0{,}8/28 + 12/28 = -0{,}028571 + 0{,}428571 = 0{,}4$, whereas the between-group slope on its own is $12/24 = 0{,}5$.

21.9 "Of the twelve models tried, the best gives $p = 0.03$. After correcting for twelve comparisons (Bonferroni: 0.36) the effect is no longer significant; therefore these data have not decided."

Chapter 22: Floating-Point Arithmetic and Error Analysis

22.1 $\mathrm{ulp}(1) = 2^{-52} = 2.220 \times 10^{-16}$; $\mathrm{ulp}(1024) = 2^{-42} = 2.274 \times 10^{-13}$; $\mathrm{ulp}(0.1) = 2^{-56} = 1.388 \times 10^{-17}$; $\mathrm{ulp}(3 \times 10^8) = 2^{-24} = 5.960 \times 10^{-8}$. $|x|u$ differs from the ulp by at most a factor of two, because the ulp is constant inside one binade. In $[1, 2)$ there are 2^{52} numbers.

22.2 2^{-53} is exactly half an ulp of 1, and rounding to even sends it back to 1: $1 + 2^{-53} = 1$. Since 2^{-52} is one full ulp, $1 + 2^{-52} > 1$. In binary32, $u = 2^{-24}$ and $\mathrm{ulp}(1) = 2^{-23}$, so the smallest positive number is $2^{-23} = 1.192 \times 10^{-7}$.

22.3 $\mathrm{fl}(x)$ is the nearest number, so $|\,\mathrm{fl}(x) - x| \le \frac{1}{2}\,\mathrm{ulp}(x)$ by the definition of rounding. Equality cannot hold for all x: when x is itself representable the error is 0, and $\frac{1}{2}$ ulp is attained only halfway between two numbers.

22.4 False. Every operation is rounded correctly, but the errors accumulate along the algorithm. Example from this chapter: the sequential sum $1 + 10^6 \cdot 10^{-16}$ stays 1, whereas Kahan gives 1.0000000001; the difference 10^{-10} is far above u.

22.5 With denominator 2^{54}: $(0.1+0.2)+0.3 = 10808639105689192/2^{54}$ and $0.1+(0.2+0.3) = 10808639105689190/2^{54}$. The difference is $2/2^{54} = 2^{-53}$, exactly one ulp. `fsum` gives 0.6; `float.hex` differs in the last digit.

22.6 The naive formula gives 0.4996003611; the double-angle identity gives 0.4999999999999996 and $0.5 - x^2/24 + x^4/720$ agrees. Its relative error is -7.99×10^{-4} and comes from cancellation in $1 - \cos x$: $\cos x$ discards the digits the difference needs.

22.7 Naive: the small root is $-1.0000000111 \times 10^{-4}$. The stable form $-2c/(b + \sqrt{b^2 - 4ac})$ gives $-1.0000000100 \times 10^{-4}$, relative error 1.12×10^{-9}. The condition number of the subtraction is $(2 \cdot 10^4)/(2 \cdot 10^{-4}) = 10^8$, so about eight digits are lost.

22.8 At level j there are 2^{k-j} additions, each with error at most u times the sum of the absolute values of its inputs; summed over the whole tree, the error is $\le ku \sum |x_i|$ with $k = \log_2 n$. That tree depth is what replaces n in the bound for the sequential sum.

22.9 Sequential: every 10^{-16} is below $\frac{1}{2}\,\mathrm{ulp}(1) = 1.11 \times 10^{-16}$, so the result stays 1. Kahan: 1.0000000001, the same as the exact value. Each Kahan step is also rounded, but the lost remainder is kept in c and returned on the next step.

22.10 $\kappa \approx 1 + \bar{x}^2/s^2$. For 1000 values $999 + ((k \bmod 7) - 3)/4$ with $k = 1, \dots, 1000$: $\bar{x} = 999.00075$, $s = 0.4997$ (population standard deviation, matching the $1/n$ divisor in the problem), $\kappa \approx 4.0 \times 10^6$. With $u_{32} = 5.96 \times 10^{-8}$, $\kappa u \approx 0.238$: almost all digits are lost. Measured: one pass errors by 25 percent, two passes by 4.4×10^{-8}.

Chapter 23: Convex Optimization and Duality

23.1 f: Hessian $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, eigenvalues 1 and 3, positive definite, strictly convex. g: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, eigenvalues ± 1, indefinite, neither of the two. h: $\begin{pmatrix} 4 & -4 \\ -4 & 6 \end{pmatrix}$, eigenvalues 0.8769 and 9.1231, positive definite. None is singular semidefinite; for $x_1^2 + 2x_1x_2 + x_2^2$ (eigenvalues 0 and 4) the minimum is not unique.

23.2 For $x, y \in C_1 \cap C_2$ and $t \in [0, 1]$, the mixed point lies in C_1 and in C_2, hence in their intersection. Union: $[0, 1] \cup [2, 3]$ contains 1 and 2 but not $(1 + 2)/2 = 1.5$.

23.3 $f_i(tx + (1-t)y) \le tf_i(x) + (1-t)f_i(y) \le tM(x) + (1-t)M(y)$ for each i, then take the maximum. For $\max\{x_1^2, x_2\}$ at $(0, 0)$ the gradient does not exist (a crease along $x_2 = x_1^2$), but its subdifferential $\mathrm{conv}\{(0, 0), (0, 1)\}$ contains $\mathbf{0}$: a minimum. At $(1, 1)$ the subdifferential is $\mathrm{conv}\{(2, 0), (0, 1)\}$ with no zero, not a minimum.

23.4 Without $x_1 \geq 1$: $x_1 = 0.8$, $x_2 = 1.6$, which violates it. So $x_1 = 1$, $x_2 = 1.5$, $p^* = 3.25$. The multipliers are $\nu = -1.5$ for the equality and $\lambda = 0.5 \geq 0$ for the inequality. `scipy.optimize.minimize` gives the same value.

23.5 $g(\lambda) = \min_x \left[x_1^2 + x_2^2 + \lambda(2 - x_1 - x_2)\right] = 2\lambda - \lambda^2/2$. Its derivative is $2 - \lambda$, so the maximum is at $\lambda = 2$ with $d^* = 2 = p^*$. The gap is zero because the constraint is affine and $x = (1.5; 1.5)$ is strictly feasible: Slater holds.

23.6 False. Strong duality only equates the values, it does not guarantee attainment. Example: minimize e^x over $\mathbb{R}$. $p^* = 0$ but it is not attained; the dual side is also 0, so the gap is zero while the primal optimum does not exist.

23.7 False. Complementary slackness states $\lambda_i g_i(\mathbf{x}^*) = 0$: an active constraint may have a zero multiplier, and an inactive constraint must have a zero multiplier. The implication runs one way, so "active ⇒ positive" is not valid.

23.8 Without the bound, $(x_1 - 2)^2 + (x_2 - 2)^2$ on $x_1 + x_2 = 2$ is minimized at $x_1 = x_2 = 1$. That satisfies $x_1 \leq c$ if and only if $c \geq 1$, so at $c = 1$ the multiplier is $\lambda = 0$ and $p^* = 2$; the constraint starts to bind only for $c < 1$ (at $c = 0.5$: $p^* = 2.5$, $\lambda = 2$).

23.9 For $\lambda > 0$ and $x_k \to -\infty$, the term λx_k^3 dominates and is negative, so $g(\lambda) = -\infty$. Only $\lambda = 0$ is finite: $g(0) = \min(x - 2)^2 = 0$, hence $d^* = 0$. Meanwhile $p^* = 1$ at $x = 1$ (the piece $[0, 1]$), because at $x = -1$ the value is 9. The gap is 1, one hundred percent.

23.10 The first-order inequality for a convex function gives $f(y) \geq f(x^*) + \nabla f(x^*)^\top (y - x^*) = f(x^*)$ for every y, without using strict convexity. That proof fails for $f(x) = x^3$ because f is not convex; the condition violated is precisely convexity of f, at $x = 0$ where $\nabla f = 0$ but it is not a minimum.

Chapter 24: Linear and Integer Programming

24.1 Maximize $30x_1 + 20x_2$ subject to $2x_1 + x_2 \leq 10$, $x_1 + 2x_2 \leq 8$, $x \geq 0$. The vertices $(0; 0)$, $(5; 0)$, $(0; 4)$, $(4; 2)$ give profit 0; 150; 80; 160. The continuous optimum is $(4; 2)$ with profit 160; integer enumeration also gives 160, so here rounding down happens to be exact and the gap is 0.

24.2 For $x, y \in P$ and $\lambda \in [0, 1]$: $A(\lambda x + (1 - \lambda)y) = \lambda Ax + (1 - \lambda)Ay \leq \lambda b + (1 - \lambda)b = b$, using A linear and $\lambda \geq 0$. On integer sets that argument collapses: a convex combination of two integer points need not be integer, for example 0 and 1 are feasible while their midpoint is not a member of $\mathbb{Z}^n$.

24.3 Profit $x_1 + 2x_2$ at the vertices: 0; 4; 6; 6. The optima $(0; 3)$ and $(3; 1{,}5)$ both equal 6, and the whole segment $x_1 + 2x_2 = 6$ is optimal because the objective is constant there; the midpoint (1,5; 2,25) is not a vertex. The dual is $y = (0; 1)$: the price of an hour is 1, of plywood 0.

24.4 Initial table with basis $s_1, s_2, s_3 = (4; 3; 3)$. First pivot: x_1 enters (reduced cost 3), ratios 4 and 3, so s_2 leaves; value 9. Second pivot: x_2 enters, s_1 leaves (ratio 1), optimum $(3; 1)$ with $z = 11$. The dual prices from the final table: $(2; 1; 0)$, and $4(2) + 3(1) + 3(0) = 11$.

24.5 The dual minimizes $10y_1 + 8y_2$ subject to $2y_1 + y_2 \geq 30$, $y_1 + 2y_2 \geq 20$, $y \geq 0$; the solution $y = (40/3;\ 10/3)$ has value 160, the same as the primal. Meaning: one extra kg of flour is worth 13,333 and one extra oven hour 3,333.

24.6 False. Rounding down is not always feasible once there are equality constraints or constraints with sign $\geq$. Example: maximize $x_1 + x_2$ with $2x_1 + 2x_2 = 5$; the continuous answer (2,5; 0) rounds to $(2; 0)$, which gives $4 \neq 5$ and is therefore infeasible.

24.7 True for the forward direction, and only for optimal pairs. Complementary slackness $y_i(b_i - \mathbf{a}_i^\top \mathbf{x}) = 0$ gives: $y_i > 0$ forces that constraint to be active. The converse does not hold; at a degenerate point there is an active constraint with $y_i = 0$, so a zero price

does not mean the constraint is slack.

24.8 Densities 2,0; 1,75; 1,667; 1,5. The first item fills up (5 kg, 10), then 3/4 of the second, total weight 8: the LP bound is 15,25. Enumeration gives 15 (items 1 and 3). Gap 0,25, that is 1,639 percent of the bound.

24.9 Capacity 7: the first item fills the sack, then 1/2 of the second, so the LP bound is 13,5; the best integer answer is 13 (items 1 and 4, weight 7). The gap is 0,5, i.e. 3,704 percent. The gap changes because capacity 7 rules out the pair $(1, 3)$, whose weight $5 + 3 = 8$ used to fit, so the best integer answer moves to items 1 and 4; the fractional and integer optima do not move in step.

24.10 $x_v = 1/2$ for every v is feasible ($x_u + x_v = 1$) and gives $n/2$; every cover must contain $n - 1$ vertices, since leaving out two vertices means the edge between them is uncovered. Ratio of LP to integer: 0,6667; 0,625; 0,5556 for $n = 4, 5, 10$. The ratio tends to 1/2, so the relaxation bound can be off by a factor of two and rounding is not a general method.

Chapter 25: Neural Networks and Backpropagation

25.1 The bound is $\lfloor 1/\gamma^2 \rfloor$: for $\gamma = 0{,}2$ we get 25, for $\gamma = 0{,}05$ we get 400. That bound only counts mistakes, and each mistake consumes margin, which is why n does not appear. For small γ the bound blows up like $1/\gamma^2$ and is therefore useless.

25.2 Write $\mathbf{W} = \mathbf{W}_2\mathbf{W}_1$ and $\mathbf{b} = \mathbf{W}_2\mathbf{b}_1 + \mathbf{b}_2$; then the output is $\mathbf{W}\mathbf{x} + \mathbf{b}$, that is, one linear threshold. Adding the next identity layer only multiplies matrices, so the capacity does not increase at all.

25.3 With $\mathbf{x} = (0{,}5; -1; 0{,}25; 2)$ and $y = 0$: $\mathbf{z}_1 = (2{,}9125; 0{,}375)$, $a = 0{,}8961350845$, $\bar{z}_2 = 0{,}8961350845$, hence $\bar{\mathbf{z}}_1 = (0{,}7169081; -0{,}5376810)$ and

$$\nabla_{\mathbf{W}_1} L = \begin{pmatrix} 0{,}3584540 & -0{,}7169081 & 0{,}1792270 & 1{,}4338161 \\ -0{,}2688405 & 0{,}5376810 & -0{,}1344203 & -1{,}0753621 \end{pmatrix},$$

that is, the second row equals $-0{,}75$ times the first row. This item is ambiguous: both gates are open, so there is no row with zero gradient.

25.4 Not enough evidence. Finite differences only check thirteen scalars at one point, and at $h = 10^{-6}$ an error of 10^{-10} is exactly what is expected; a shared bug in both computations would not show up. What is needed: several points, both gate states, the cases $y = 0$ and $y = 1$, gradients with respect to the input, and an automatic differentiation comparison.

25.5 False. The theorem is about density of continuous functions on a compact domain: there exist parameters that approximate, not that those parameters are reachable by optimization. Two things are not stated: how many units are needed, and the error on test data (generalization), not only on the target function.

25.6 At $t = 1$, $\hat{m} = g$ and $\hat{v} = g^2$, so the step is $\eta g/(|g| + \epsilon) = -0{,}01$ and b_2 becomes 0,06. With a gradient 100 times larger the step stays $-0{,}01$, because Adam only uses the ratio to the gradient scale; b_2 stays 0,06.

25.7 The output $w_2 \max(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1, 0)$ is a sum of two linear functions, each active on one half-plane, hence piecewise linear; two lines $z_1[i] = 0$ cut the plane into at most four cells, each with its own linear formula. For k units in general position, k lines cut the plane into $1 + k + k(k-1)/2$ regions, that is $2, 4, 7$ for $k = 1, 2, 3$ (not 8: the third line adds only three regions, not four).

25.8 $\nabla_{\mathbf{x}} L = (-0{,}0868370; 0{,}1328096; -0{,}0970532; -0{,}1430257)$, so $\boldsymbol{\delta} = 0{,}05\,\mathrm{sign}(\nabla) = (-0{,}05; 0{,}05; -0{,}05; -0{,}05)$. The loss at $\mathbf{x} + \boldsymbol{\delta}$ is 0,3186800, whereas the linear prediction

$0{,}2949123 + 0{,}0229863 = 0{,}3178985$; a difference of $7{,}81 \times 10^{-4}$ from curvature.

25.9 Prediction: the second lab block uses ReLU, and ReLU saturates only on the negative side. Saturation table (8 units, random He $\mathbf{W}_1$, $\mathbf{b}_2 = 0$):

	range of Z_1 (initial)	dead units	final L	accuracy
$\mathbf{b}_1 = 0$	$[-2.43; 2.40]$	0/8	0.0163	1.0000
$\mathbf{b}_1 = +10$	$[7.57; 12.41]$	0/8	0.3225	0.8375
$\mathbf{b}_1 = -10$	$[-12.43; -7.60]$	8/8	0.6909	0.5333

So $\mathbf{b}_1 = +10$ indeed does not kill the hidden units (Z_1 far above zero passes through unchanged), but it is not free: $H \cdot \mathbf{w}_2$ becomes very large, the output sigmoid sticks at 0 or 1, and L starts at 4.04; after 3000 steps it stops at $L = 0.3225$ with accuracy 0.8375, worse than $\mathbf{b}_1 = 0$. For $\mathbf{b}_1 = -10$ every unit is dead from the start (Z_1 below zero, the derivative exactly 0, not merely small), $\partial L/\partial b_1 = \mathbf{0}$ exactly, and training stops at $L = 0.6909$ – the label entropy – with majority-class accuracy 0.5333.

Chapter 26: Generalization, Regularization, and Training

26.1 Not enough evidence, and what is wrong is equating training error with generalization error. Zero training error only means the training data has been memorized. The additional numbers needed: validation (or test) error on data not used for training, together with its interval width.

26.2 With $\hat{f} = \hat{f}_D$ and $\bar{f} = \mathbb{E}_D \hat{f}$: $\mathbb{E}(y - \hat{f})^2 = \sigma^2 + (\bar{f} - f)^2 + \mathrm{Var}(\hat{f})$. The cross term vanishes because $\mathbb{E}[\varepsilon] = 0$ and ε is independent of D, so $\mathbb{E}[\varepsilon(\hat{f} - f)] = 0$; independence is used precisely at this step, on ε, not on $\hat{f}$. The second cross term vanishes for a different reason: $\mathbb{E}[\hat{f} - \bar{f}] = 0$ by the definition of $\bar{f}$, so $2(f - \bar{f})\,\mathbb{E}[\hat{f} - \bar{f}] = 0$ without needing any independence.

26.3 From $\frac{d}{dw}\left[\frac{1}{2}\sum(y_i - wx_i)^2\right] = 0$ we get $w(\lambda) = \sum x_i y_i / (\sum x_i^2 + \lambda) = 64/(30 + \lambda)$. Its derivative $-64/(30 + \lambda)^2 < 0$ but never zero, so w shrinks without touching zero. In L1, the subgradient at $w = 0$ contains zero when $\lambda \geq |\sum x_i y_i| = 64$; so $w = 0$ exactly at $\lambda = 64$.

26.4 $\mathrm{MSE}(c) = (c-1)^2\mu^2 + c^2\sigma^2/n$: for $c = 0{,}95$ we get $0{,}01 + 0{,}0361 = 0{,}0461$; for $c = 1{,}05$ we get $0{,}01 + 0{,}0441 = 0{,}0541$; for $c = 1$ we get 0,04. An equally large deviation is not an equally large loss, because the variance term grows like c^2, so exceeding the budget costs more than shrinking.

26.5 Fix: that cross-validation number is an estimate of the error of the selection procedure, not of the selected model's error in the field, and it is optimistically biased because λ was chosen on the same folds. Two things are needed: nested cross-validation (or a test set never seen during selection), and repetition with several partitions along with intervals.

26.6 For $a < 0$: with $u < 0$ the derivative is $u - a - t$, zero at $u = a + t$, which is valid only when $a < -t$. For $u = 0$ the subgradient is $[-a - t, -a + t]$, which contains zero when $|a| \leq t$, so $u = 0$ is optimal. Symmetric because the objective $\frac{1}{2}(u - a)^2 + t|u|$ is unchanged when u and a are both sign-flipped.

26.7 The half-life is $\ln 2/(\eta\lambda) = 0{,}693147/0{,}002 = 346{,}57$ steps; a quarter of the initial value means two half-lives, 693,15 steps, and $(1 - \eta\lambda)^{693} = 0{,}24973$. A numpy simulation with zero gradient gives the same numbers, because $w_k = w_0(1 - \eta\lambda)^k$.

26.8 The gap is $0{,}2406 - 0{,}1500 = 0{,}0906$; at $\lambda = 0$ the gap is $0{,}3262 - 0 = 0{,}3262$, at $\lambda = 10$ only $0{,}3395 - 0{,}3333 = 0{,}0062$. A small gap is not enough: at $\lambda = 10$ the gap is smallest precisely because both are equally bad, close to the majority-class error 0,3419.

26.9 With m hypotheses, $P(\exists i : |\hat{R}_i - R_i| > \epsilon) \le 2me^{-2n\epsilon^2}$, so $\epsilon = \sqrt{(\ln m + \ln(2/\delta))/(2n)}$. For $m = 2^{10}$, $n = 200$, $\delta = 0{,}01$: $\epsilon = 0{,}1749$. Halving it demands four times the n, namely 800.

Chapter 27: Honest Model Evaluation

27.1 Not enough evidence. A difference of 0,4 points on a single test set is not necessarily beyond noise; what is wrong is treating two paired proportions as independent. Two additional numbers: the size of the test set, and the count of paired disagreements (for McNemar's test) or the interval width.

27.2 For hard labels $s, y \in \{0, 1\}$ we have $(s - y)^2 = 1$ exactly at mistakes and 0 at correct predictions, so the average is $(\mathrm{FP} + \mathrm{FN})/n$. For scores that are probabilities, $(s - y)^2$ takes intermediate values, so the average is no longer a count of mistakes but a calibration penalty.

27.3 $\mathbb{E}[(s - y)^2] = s^2 - 2s\pi + \pi$ with $\pi = \mathbb{E}[y]$. The derivative is $2s - 2\pi$, zero at $s = \pi$; the minimum value is $\pi^2 - 2\pi^2 + \pi = \pi(1 - \pi)$, and the second derivative $2 > 0$ confirms that it is a minimum.

27.4 Precision $180/225 = 0{,}8$; recall $180/200 = 0{,}9$; $F_1 = 0{,}847059$; accuracy $1935/2000 = 0{,}9675$; specificity $1755/1800 = 0{,}975$; balanced accuracy $(0{,}9 + 0{,}975)/2 = 0{,}9375$. `sklearn.metrics` gives the same numbers to six decimals.

27.5 Wilson 95/100: [0,8882; 0,9785], width 0,0902; for 950/1000: [0,9347; 0,9619], width 0,0272, that is 3,318 times narrower even though n is ten times larger. Width is not a measure of model quality: it depends on n and on the data, and the two experiments are not paired.

27.6 Out of 25 pairs, 19 are correct: the positive 0,95 beats five negatives, 0,88 also five (it sits above 0,75), 0,70 four, 0,60 three, 0,40 two. Hence AUC $= 19/25 = 0{,}76$; `roc_auc_score` reports 0,76.

27.7 Precision $= 0{,}8\pi/(0{,}8\pi + 0{,}01(1 - \pi))$: at $\pi = 0{,}002$ we get 0,138169, at $\pi = 0{,}2$ we get 0,952381. The requirement precision $\ge 0{,}5$ gives $0{,}4\pi \ge 0{,}005 - 0{,}005\pi$, that is $\pi \ge 0{,}005/0{,}405 = 0{,}012346$.

27.8 Split by host, not by record: 60/20/20 becomes 180 training hosts, 60 validation, 60 test, each about 20 records. Preprocessing is computed inside the training fold only. The choice among 40 settings goes through nested cross-validation in the training part. The test set is touched once, at the end.

27.9 False. AUC measures ranking, not usefulness at a particular threshold. With sensitivity 0,8 and specificity 0,99, at prevalence 0,002 the precision is only 0,138169: of ten alarms, 8,62 are wrong, so about nine. The chosen threshold still has to be checked for precision and recall.

Chapter 28: Formal Methods with Lean 4

28.1 False. A file that is accepted means every theorem has a proof that is checked, but `sorry` passes with a warning, and `axiom` as well as `native_decide` extend the trusted base. The checking command: `#print axioms` on every theorem that is claimed.

28.2 Helper lemma: for every k, $k(2k) + 2(2k + 1) = (k + 1) \cdot 2(k + 1)$. That goal contains $k \cdot k$, so it is not a linear goal: `omega` will not close it on its own and a rewrite is needed first. Then `induction n with`: base case `rfl`; the `succ k ih` step uses `rw [oddSum, Nat.mul_add, ih]`, which leaves exactly that helper lemma, then `exact bantu k`. The helper lemma itself is closed by `simp only [Nat.mul_add,`

`Nat.add_mul, Nat.mul_one, Nat.one_mul]` followed by `omega`: `omega` cannot close $k \cdot (2k) + 2(2k+1) = (k+1) \cdot 2(k+1)$ on its own, because that goal contains $k \cdot k$ and is not linear (tried: `omega could not prove the goal`).

28.3 Informal: for $n = 0$ both sides are 0; if $\sum_{i<n} 2i = n(n-1)$, then adding $2n$ gives $n(n-1)+2n = (n+1)n$. The Lean version defines `evenSum : Nat -> Nat` recursively (`evenSum 0 = 0`, `evenSum (n+1) = evenSum n + 2*n`) and states the form without truncated subtraction, $E(n) + n = n \cdot n$. The steps: `induction`, base `rfl`, then `rw [evenSum]`. Note that `rw [evenSum, ih]` fails outright (`did not find instance of the pattern evenSum k + k`), because the goal afterwards reads $E(k)+2k+(k+1) = (k+1)(k+1)$. Insert the regrouping `have h : evenSum k + 2 * k + (k + 1) = (evenSum k + k) + (2 * k + 1) := by omega` first, continue with `rw [h, ih]`, and close with `simp only [Nat.mul_add, Nat.add_mul, Nat.mul_one, Nat.one_mul]` then `omega`.

28.4 The sum of the first 37 odd numbers $= 37^2 = 1369$; the sum of the first $250 = 250^2 = 62500$. Script: `sum(2*i+1 for i in range(n))` gives both numbers, and both are perfect squares.

28.5 A direct check for all $n \le 1000$ demands 1001 cases, namely $n = 0$ through $n = 1000$; the bound $n \ge 1$ would give 1000, but the exercise does not say that, and the `zero` base case above really does check $n = 0$. An induction proof generates only two goals, `zero` and `succ`, then concludes for all n; so 1001 against 2.

28.6 `Nat.add` is defined by recursion on the second argument, so $n + 0$ reduces straight to n and `rfl` closes it. On $0 + n$ the recursion gets stuck on the unknown n, so there is nothing to reduce and `induction n` is needed.

28.7 Weak: for fixed x, $\exists \delta,\ |\delta| \le 1 \wedge f(x+\delta) = f(x)$; the witness $\delta = 0$ makes both sides equal and `rfl` closes it. Strong: $\forall x\, \forall \delta,\ |\delta| \le 1 \to f(x+\delta) = f(x)$, and that is false for the identity f at $\delta = 1$.

28.8 `decide`: no axioms, pure kernel computation. `omega`: `propext` and `Quot.sound`, because it proves goals by contradiction inside the integers; `Classical.choice` does not appear, and this chapter prints the same list. `sorry`: `sorryAx`, the axiom added specifically to close a skipped goal.

28.9 First mistake: `delta` is already free in the context and then bound again by `exists`, so the reader of the statement determines its value, not the attacker; robustness is not implied. Second: there is no quantifier over x, so x is free too. Correct version: $\forall x,\ \exists \delta,\ \|\delta\| \le 1 \wedge f(x+\delta) = f(x)$.

28.10 `intro h` gives $h : P \wedge \neg P$. Its components `h.1` have type P and `h.2` have type $\neg P$; then `exact h.2 h.1` closes the goal, because $\neg P$ is $P \to$ false.

Chapter 29: Adversarial Machine Learning: Attacks and Defenses

29.1 $\mathbf{s}_A = 1$, $\mathbf{s}_B = 3$, so the margin is already -2 (the example leans toward B); $\mathbf{d} = (1; -2; 1)$ with $\|\mathbf{d}\|_1 = 4$ and $\|\mathbf{d}\|_2 = \sqrt{6} \approx 2{,}449$. Hence $\varepsilon^*_\infty = 2/4 = 0{,}5$ and $\varepsilon^*_2 = 2/2{,}449 = 0{,}8165$; the ratio is $\sqrt{8/3} = 1{,}633$.

29.2 The derivative of the objective with respect to $\boldsymbol{\delta}$ is $\mathbf{w}$, so the maximum is $\varepsilon\|\mathbf{w}\|_2$, attained uniquely at $\boldsymbol{\delta} = \varepsilon\mathbf{w}/\|\mathbf{w}\|_2$ (Cauchy–Schwarz, with equality when parallel). In ℓ_∞ the maximum is $\varepsilon\|\mathbf{w}\|_1$ at $\varepsilon\,\mathrm{sign}(\mathbf{w})$, a vertex; the form differs because the dual norm differs.

29.3 The error $\mathbf{d}^\top\boldsymbol{\delta}$ is linear on a compact convex set, so the maximum is at an extreme point. The extreme points of the ℓ_∞ ball are $\pm\varepsilon$ in each coordinate, and FGSM picks

exactly that point; hence the PGD path, which always stays inside the ball, cannot exceed it.

29.4 False. In this chapter a random signed perturbation at $\varepsilon = 0{,}15$ only drops accuracy to 95,48 percent, whereas PGD 40 steps brings it down to 10,72 percent. Robustness to random noise says nothing about an attack that chooses the direction.

29.5 Untargeted success (PGD with 40 steps, clean accuracy as the reference) for A: 14,02, 53,16, 89,06 percent at $\varepsilon = 0{,}05; 0{,}10; 0{,}15$. For D: 5,38, 15,28, 35,06 percent. Over that range the gap actually widens (8,64, 37,88, 54,00 percentage points), because A is already close to the 100 percent ceiling while D is still far; the narrowing only happens outside that range, at $\varepsilon \geq 0{,}20$, once both models run out of margin.

29.6 One coordinate suffices: projection onto an interval is clipping, and $[x_j - \varepsilon, x_j + \varepsilon] \cap [0, 1]$ equals $[\max(0, x_j - \varepsilon),\ \min(1, x_j + \varepsilon)]$. Clipping to the ε box and then to $[0, 1]$, or the other way round, yields the same interval because interval intersection is commutative; the Cartesian product then gives the same point.

29.7 D loses $97{,}99 - 96{,}48 = 1{,}51$ points of clean accuracy to gain $62{,}65 - 10{,}72 = 51{,}93$ points of robustness, so the price is $1{,}51/51{,}93 = 0{,}0291$ clean points per point of robustness. That ratio is not complete: it depends on one ε, on one attacker that is not an upper bound, and it ignores the clean accuracy of the comparison model.

29.8 Protocol: attack with PGD with more steps and several random starts, with EOT over the defense's randomness, then repeat with attacks without gradients (imitation/transfer, finite-difference estimation). Report the curve of accuracy against ε, not a single point. Easiest symptom: success jumps as soon as the gradient is replaced by BPDA or EOT, while naive attacks report zero.

29.9 White box 89,06 percent, transfer $1 - 0{,}2379/0{,}9849 = 75{,}85$ percent, a difference of 13,21 points. That is not evidence that B is safer: an attack built specifically for B will be stronger, and in this chapter the white-box versus transfer difference does shrink when the attacker is adapted. This item is ambiguous: the lab output does not contain a PGD row for model B.

29.10 Untargeted only needs a label change; targeted must reach a particular class, so the set of targets that must be met is far smaller. At $\varepsilon = 0{,}15$ the target is reached 33,00 percent of the time, whereas untargeted is 89,06 percent; with $\varepsilon = 0{,}30$ the target is reached 99,50 percent. So the difficulty is a matter of budget, not of attack type.

Chapter 30: The Limits of Attack: Obfuscated Gradients and NIDS Evasion

30.1 Three checks: (a) repeat with adaptive attacks (BPDA, EOT), (b) report the curve of accuracy against ε, not a single point, (c) check gradient quality (cosine against finite differences or against the weight direction). If what is happening is gradient masking: success collapses only for attacks based on gradients, while random search and transfer still succeed, and the curve is flat.

30.2 For $f(x) = w \cdot x + b$ the gradient is constant and the Hessian zero, so the Lipschitz constant is $L = 0$ and the bound $L\delta$ gives zero error: BPDA is exact. With two ReLU layers, f is piecewise linear so the gradient jumps at the kinks and the Lipschitz-gradient hypothesis fails; the bound is then meaningless.

30.3 As the program shows, the label flips up to $q = 0{,}6$ (score $-1{,}2$) and fails at $q \geq 0{,}70$, when every coordinate rounds to zero so the score is 0, a tie rather than a flip. Hence $q^* = 2 \max_j |x_j - \varepsilon| = 2(\varepsilon + |\min_j x_j|) = 0{,}70$. Analytic safety bound: the rounding error is at most $4q$, so the attack is only guaranteed to succeed when $q < (8\varepsilon - m)/4 = 0{,}345$;

at $q = 0{,}345$ the label still flips, so the direction of the inequality matters.

30.4 Probability of rejection = P(lebih dari 10 dari 20 hilang) = 0,411901, so what is answered is 588 of 1000 queries: $588/20 = 29$ PGD steps with $m = 20$. With $m = 1$ the rejection probability is 0,5, so 500 steps. What is paid: the gradient becomes noisier and its direction further from the true one.

30.5 What can be predicted without running: clean accuracy is about the same, naive success stays near zero, and BPDA success is near the baseline for $q = 0$ and $q = 0{,}05$ but falls for large q. The cosine no longer approaches one because the ReLU gate adds kinks on top of rounding, so the BPDA approximation only approaches the direction. The numbers for the third value of q must be taken from execution, not invented.

30.6 Binary flags and protocol numbers are assumed not to be changeable by the attacker: $7 + 3 = 10$ locked features, the remaining 40 features can really be shifted (out of 50). A 32×32 image with three channels has 3072 values, so $3072/40 = 76{,}8$ times more.

30.7 Daily retraining changes the parameters that a gradient-based attack uses, so old directions go stale quickly; a blind attack does not use those parameters, so it keeps working. According to the EOT lemma what must be checked is the expectation over transformations, and an adaptive attack must be defined against that defense itself, including its retraining schedule.

30.8 Metric: success of the adaptive attack against ε, with at least 5 seeds and an interval. The curve shape that accepts (a): the curve is shifted to the right or downward systematically at all ε, not only at one point. If after BPDA and EOT the success returns to the baseline, then only the measurement was broken, so (b).

Chapter 31: Cryptography: Modular Integers to the Quantum Threat

31.1 False. $2^{100} \equiv 1 \pmod{101}$ only says the order of 2 divides 100, that is, 101 passes the Fermat test to base 2; pseudoprimes (and Carmichael numbers) also pass. What that congruence proves is only $101 \nmid 2$ and $2^{100} \equiv 1$, not the absence of divisors.

31.2 If $\gcd(a, n) = \gcd(b, n) = 1$, Bezout's theorem gives $ax + ny = 1$ and $bz + nw = 1$; the product of the two gives $(ab)(xz) + n(\cdot) = 1$, so ab is a unit and $U(n)$ is closed. If a is not a unit, $d = \gcd(a, n) > 1$ and $b = n/d \neq 0$ satisfies $ab = n(a/d) \equiv 0$.

31.3 $a^2 - 1 = (a - 1)(a + 1) \equiv 0 \pmod{p}$; since p is prime, Euclid's lemma gives $p \mid a - 1$ or $p \mid a + 1$, so $a \equiv \pm 1$. For modulus 8 that lemma fails because 8 is not prime: the square roots of one are $1, 3, 5, 7$, since $3^2 = 5^2 = 7^2 = 49 \equiv 1$.

31.4 The classes $0 \leq a < pq$ that share a factor with pq are q multiples of p plus p multiples of q, and 0 is counted twice; so $\varphi(pq) = pq - p - q + 1 = (p - 1)(q - 1)$. For $p = 1019$, $q = 1031$: $\varphi = 1018 \cdot 1030 = 1\,048\,540$.

31.5 Extending Euclid backwards gives $17^{-1} \equiv 2753 \pmod{3120}$ (since $17 \cdot 2753 = 46\,801 = 15 \cdot 3120 + 1$), and $65\,537^{-1} \equiv 60\,493 \pmod{1\,048\,540}$. The division takes 4 steps against 5, so adding one digit to the modulus adds only a few steps.

31.6 Fermat's little theorem: $a^{100} \equiv 1 \pmod{101}$ for $101 \nmid a$. Hence $5^{100} \equiv 1$, $3^{200} = (3^{100})^2 \equiv 1$, and $2^{1000} = (2^{100})^{10} \equiv 1$. All three are checked by `pow(a,k,101)` and match.

31.7 $A = 5^7 \equiv 17$, $B = 5^{15} \equiv 19$, shared key $19^7 \equiv 15$ (the same as 17^{15}). If an eavesdropper replaces A with 11, Bob computes $11^{15} \equiv 10$ while Alice still holds 15; two different keys appear because the paired exponents are no longer a legitimate Diffie–Hellman pair.

31.8 With $e = 3$ and $m < n^{1/3}$ we have $m^3 < n$, so $c = m^3$ with no modular reduction and the integer cube root returns m. For $n = 1\,050\,589$ the bound is $n^{1/3} \approx 101{,}66$, so the longest message is $m = 101$. Random padding makes $m^3 > n$ so that integer root does not exist.

31.9 Two things must be true: (i) the parameters and implementation really meet the claimed security guarantee, that is, the reduction to a lattice problem that is still hard; (ii) the hardness assumption underneath still holds and the proof model covers the attacks that are actually used. Widespread adoption is not a condition for the claim to be true: a scheme used everywhere is still not proven secure if its assumption fails. The second sentence remains wrong: RSA is still embedded in legacy systems, and migration is done hybrid, so it takes years. (Post-quantum signatures already exist: ML-DSA, FIPS 204.)

31.10 About $1{,}1774\sqrt{2^{48}} = 1{,}1774 \cdot 2^{24}$, that is 19 753 662 samples for a one-half collision probability; for 32 bits $1{,}1774 \cdot 2^{16} = 77\,163$. The numbers 2^{24} and 2^{16} are rough square-root guesses, not the exact values.

Chapter 32: Differential Privacy and the Mathematical Foundations of QML

32.1 False as a sentence. $\varepsilon = 0{,}5$ bounds the probability ratio per record, it does not remove leakage; it is also a worst-case guarantee. Two things that must be true: the mechanism really satisfies (ε, δ) for the adjacency relation and sensitivity used, and all outputs (queries, budget remainder, post-processing) are counted within that budget.

32.2 The d-dimensional Laplace density is a product, so the ratio is $\prod_j \exp((|y_j - a'_j| - |y_j - a_j|)/b)$, and by the triangle inequality each factor is at most $\exp(|a_j - a'_j|/b)$. The product is $\exp(\sum_j |a_j - a'_j|/b) \le \exp(\Delta f/b) = e^{\varepsilon}$. The one-dimensional argument is replaced by the product of per-coordinate ratios, and ℓ_1 takes the place of the single absolute value.

32.3 Take $D \sim D'$ differing in one record. That record exists in only one of D_1, D_2, so only one mechanism has its input changed; its ratio is bounded by e^{ε_i}, while the others have ratio 1. Hence $\varepsilon = \max_i \varepsilon_i$ and $\delta = \max_i \delta_i$, not the sum.

32.4 The probability of answering "yes" is $p + (1-p)/2$ if the truth is "yes" and $(1-p)/2$ if the truth is "no"; the ratio is $(1+p)/(1-p)$. Hence $p = (e^{\varepsilon} - 1)/(e^{\varepsilon} + 1)$, and for $\varepsilon = 2$ we get $p = 0{,}761594$, that is $\tanh 1$.

32.5 Count query: $b = 2$, so $P(|Y| > 10) = e^{-5} = 0{,}006738$. Mean query with $R = 1$: $b = 10^{-4}/0{,}5 = 2 \times 10^{-4}$; an error of 10 people means 10^{-3} on the mean scale, which again gives e^{-5}, whereas an absolute error of 10 on that scale gives $e^{-50000} \approx 0$. This item is ambiguous about the units of the error.

32.6 $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$; S gives $(|0\rangle + i|1\rangle)/\sqrt{2}$; then H gives $\frac{1+i}{2}|0\rangle + \frac{1-i}{2}|1\rangle$. The probabilities are $1/2$ and $1/2$, the same as $H|0\rangle$, because measurement in the computational basis only sees the modulus of the amplitude. What distinguishes them is the relative phase, and the gate is $S^{\dagger}$ (it multiplies $|1\rangle$ by $-i$). Z after H is no substitute: $HZH = X$, which swaps $|0\rangle$ with $|1\rangle$ instead of rotating the phase.

32.7 $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. If the second qubit is measured first and the result is 0, the first qubit collapses to $|0\rangle$, so the probability of the outcome 01 is 0; if the result is 1, the first qubit collapses to $|1\rangle$, so the outcome 11 is certain and the probability of 01 is also 0. Both show perfect correlation: the outcomes of the two qubits are always the same.

32.8 $U^{\dagger}U = \operatorname{diag}(1, e^{-i\varphi}) \operatorname{diag}(1, e^{i\varphi}) = I$, so it is unitary for every φ. In the computational

basis $|1\rangle$ only picks up a phase, its amplitude modulus stays the same, so probabilities do not change; but that relative phase changes the interference at the next H gate, so the result can differ.

32.9 $n = 40$: $2^{40} = 1{,}0995 \times 10^{12}$ amplitudes, times 16 bytes, gives 16 TiB (not GiB: 2^{40} bytes is already one TiB). $n = 60$: $2^{60} = 1{,}1529 \times 10^{18}$ amplitudes, that is 16 EiB or $1{,}8447 \times 10^{19}$ bytes, far beyond any storage that exists.

32.10 Three questions: whether all the amplitudes are really used, whether the output can be extracted with a number of measurements that is not exponential, and whether there is a classical comparison on the same task and data. What makes me accept: a proven separation or measurable scaling on that task, a polynomial number of measurements, and an advantage over a seriously tuned classical baseline.

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Colophon

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This book is an original piece of writing: its sentences, its order of presentation, its worked examples, and its exercises were composed by the author. Its mathematical ideas come from the books in the References, and the reader is urged to read them. No part of the manuscript is copied verbatim from a source book; where a definition or a theorem follows a source closely, that source is named in the chapter map.

Every number inside a computing Lab comes from a real execution on the author's machine (Python 3, NumPy, SciPy) and can be repeated by the reader. The Lean 4 code fragments in Chapter 28 were run with Lean 4.10.0 without Mathlib and their output is quoted as it stood; library tactics such as `linarith` and `ring` were not run because Mathlib is not installed, and that chapter states the boundary.

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